

# Restriction, Keakeya, and Decoupling

A Graduate Pathway from Second-Year Analysis

Volumes 1–14

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# Preface to the Combined Edition

## What this is

This volume collects the first five parts of *Restriction, Keakeya, and Decoupling: A Graduate Pathway from Second-Year Analysis*. The series builds, from a second-year undergraduate background, towards the Fourier restriction conjecture, the Keakeya conjecture, and the decoupling theorems of Bourgain and Demeter. Parts 1–5 supply the analysis: measure and integration; the Fourier transform and interpolation; singular integrals; function spaces; and oscillatory integrals.

## How to read the numbering

Each Part is one Volume, and chapters restart at 1 in each. Every numbered item therefore carries its volume in its number:

$$\text{Theorem } \underbrace{3}_{\text{Volume}} \cdot \underbrace{2}_{\text{Chapter}} \cdot \underbrace{4}_{\text{item}}$$

so Theorem 3.2.3 is Volume 3, Chapter 2. Worked examples and drills are numbered in their own sequences within each chapter, so Worked Example 3.2.1 and Theorem 3.2.1 may both exist and are different objects.

## A note on cross-references

In the separate volumes, a reference to an earlier volume was a piece of text — “Volume 1, Theorem 7.3” — carrying a number written by hand. Producing this combined edition made every such reference a genuine cross-reference, resolved by the typesetter.

This was not merely cosmetic. Checking the hand-written numbers against the labels they were meant to name turned up a substantial number of errors,

in two waves: first a systematic failure caused by the source volumes being renumbered after the citations were drafted, and then a residue of individually wrong numbers that a check for *existence* could not detect, because the wrong number happened to name some other result. In the present edition no cross-reference carries a hand-written number, so that entire class of error is gone by construction. The episode is recorded in the project's feedback log as items F5 and F7.

## Conventions

The typographic system is uniform across the five volumes and is described in the Preface to Volume 1. In brief: worked examples come in three tiers, distinguished by rule weight and by a printed label; commentary — *Idea of the proof*, *Scratch work*, *Where we are*, *Looking ahead*, *Pitfall* — is set off by a rule in the margin and may be skipped without loss of continuity; and proofs longer than half a page are broken into numbered steps with stated goals.

Each volume opens with its own preface, stating what it assumes and what it will prove, and closes with two appendices: one listing every result quoted from earlier volumes together with the place it is used, and one giving notation and an index of results. Volumes 4 and 5 also record, in their second appendix, the results removed from the original plan, with reasons.

## What is not proved

The series holds itself to proving every result it states, with two classes of exception, both listed explicitly where they occur.

A *deferral* names the later volume that supplies the argument. There is exactly one outstanding in Parts 1–5: the existence of a Besicovitch configuration with pairwise disjoint translates, used in Volume 3 to prove Fefferman's ball multiplier theorem and proved in Volume 9.

A *plan revision* removes a result from the series altogether, and is announced at the point where it would have appeared. There are four, in Volumes 4 and 5; each removed statement is true, is attributed to its source, and is used nowhere in this series.

# Part 1

## Volume 1

Measure, Integration, and  $L^p$  Spaces





# Preface

## What this series is for

There is a cluster of problems in modern analysis — the Fourier restriction conjecture, the Kakeya conjecture, and the decoupling theorems of Bourgain and Demeter — which are among the most active and most beautiful in contemporary mathematics. They are also notoriously hard to approach. The research papers assume a great deal; the textbooks that cover them, Demeter’s *Fourier Restriction, Decoupling, and Applications* foremost among them, begin roughly where a strong first-year graduate student ends.

This series begins where a second-year undergraduate ends. Concretely, we assume the analysis of a Warwick second year: sequences and series, continuity and differentiation, the Riemann integral, uniform convergence, metric spaces, and multivariable calculus. We assume linear algebra. We do *not* assume Lebesgue measure, functional analysis, complex analysis, Fourier analysis, or algebraic geometry. Everything needed is built here, in order, with complete proofs.

The destination is the frontier: Guth’s polynomial partitioning method, the Bourgain–Demeter decoupling theorem, and the 2025 resolution of the three-dimensional Kakeya conjecture by Wang and Zahl. Fourteen volumes stand between here and there. This is the first.

## The pedagogical contract

Three commitments shape every page of this series.

**Complete proofs, with no gestures.** Every theorem we state, we prove. Where a proof is long, it is long; we do not write “it can be shown that” and move on. The one exception is clearly marked: in the final volume, results at the very edge of current research are given with their key lemmas proved and their global architecture explained, but not every technical estimate reproduced.

**Explanation before formalism.** A proof that is merely correct teaches very little. Before the significant arguments you will find an *Idea of the proof* box explaining what the argument is really doing, what the obstruction was, and why the natural first attempt fails. Failed approaches are discussed on purpose. In this subject, the reason a technique exists is almost always that something simpler broke, and a reader who has not seen the break will not understand the repair.

**Worked examples in place of exercise lists.** Rather than a long list of unsolved exercises, this series provides worked examples at three tiers. Each is set off from the text by rules, and the tier is named in the heading; the weight of the rules — thin, medium, doubled — repeats the same information for a reader scanning a page.

- **Tier A (calibration).** Short computations that build fluency. Roughly 30% of the examples.
- **Tier B (technique).** A method the reader must own, worked in full and often worked twice from different angles. Roughly 45%.
- **Tier C (research taste).** A genuine piece of the literature, self-contained, reachable from what has been proved so far. These are what make the series worth reading rather than one of the many good measure theory books already available. Roughly 25%.

Four kinds of commentary are set off by a single rule in the margin. *Idea of the proof* states what an argument is really doing before it is done. *Scratch work* shows how it was found. *Where we are* interrupts a long development to say what has been used and what remains. *Pitfall*, marked by a broken rule, names an error that readers actually make. None of these boxes contains anything the proofs depend on, so a reader in a hurry may skip them all; a reader trying to learn the subject should read them first.

A small number of **Drills** remain unsolved. These are deliberately mandatory. They are almost all computational, they concentrate in Volumes 1–6, and they exist because there are things that cannot be learned by reading someone else’s solution. Do them.

## How to read Volume 1

This volume covers measure theory. Many readers will be tempted to skip it, or to skim it having seen some measure theory elsewhere. Two chapters make that a mistake.

Chapter 1.7 develops covering lemmas and the Hardy–Littlewood maximal function. This is not a service chapter. The Hardy–Littlewood maximal function is the first object in the series with genuine *tube geometry* hiding inside it, and the Vitali and Besicovitch covering lemmas are the direct ancestors of the covering arguments that run through the Kakeya problem all the way to Wang–Zahl. A reader who understands why the Besicovitch covering lemma fails without a dimensional constant is already thinking the right way about Kakeya.

Chapter 1.8 develops  $L^p$  spaces and, crucially, weak  $L^p$ . Almost every theorem in this subject is an inequality between  $L^p$  norms, and the difference between strong and weak bounds is where a great deal of the real content lives.

Sections marked with a dagger (†) may be omitted on first reading without loss of continuity; they are cross-referenced from later volumes when needed.

## Notation

We write  $A \lesssim B$  to mean  $A \leq CB$  for some constant  $C$  independent of the parameters in play, and  $A \approx B$  when  $A \lesssim B \lesssim A$ . Subscripts indicate dependence:  $A \lesssim_\varepsilon B$  means the constant may depend on  $\varepsilon$ . This notation is used relentlessly in the research literature of this subject, and we adopt it from the very beginning so that it becomes invisible long before it becomes essential. A full notation index is Appendix B.

*A note on the constants.* Beginners often find the suppression of constants unsettling. It is worth saying plainly why analysts do it: in this subject the constants are almost never the point, and carrying them makes the structure of an argument invisible. When a constant *does* matter — and in the induction-on-scales arguments of Volume 13 it matters enormously, because a constant that degrades at each of  $\log(1/\delta)$  scales is fatal — we track it explicitly and say so.



## Chapter 1.1

# The Failure of Riemann Integration

### 1.1.1 What you already have, and what it cannot do

You arrive at this book with the Riemann integral. It is a fine object. It integrates every continuous function on a closed bounded interval, it satisfies the fundamental theorem of calculus, and it is defined by a construction — chop the domain into small intervals, sample the function, sum, refine — that is transparent enough to draw on a napkin.

It is also, for our purposes, unusable. Not merely inconvenient: unusable. The central objects of this series cannot be defined with it, let alone manipulated. This chapter is about why, and it is worth taking seriously, because the shape of the failure determines the shape of the repair. Lebesgue's theory is not an arbitrary generalisation that happens to be more powerful. It is the specific fix for a specific structural defect, and if you see the defect clearly then every definition in Chapters 1.2 and 1.3 will feel forced rather than invented.

We identify three failures. The first is the one every textbook mentions. The second is the one that actually matters. The third is the one that matters for *us*.

### 1.1.2 Failure I: limits do not pass through the integral

Suppose  $f_n \rightarrow f$  pointwise on  $[0, 1]$ , with every  $f_n$  Riemann integrable and every  $|f_n| \leq 1$ . Is  $f$  Riemann integrable? And if so, does  $\int f_n \rightarrow \int f$ ?

The answer to the second question is subtle and the answer to the first is simply *no*.

**WORKED EXAMPLE 1.1.1— TIER A A pointwise limit of Riemann integrable functions that is not Riemann integrable**

Enumerate the rationals in  $[0, 1]$  as  $q_1, q_2, q_3, \dots$  and set

$$f_n(x) = \mathbf{1}_{\{q_1, \dots, q_n\}}(x) = \begin{cases} 1 & x \in \{q_1, \dots, q_n\}, \\ 0 & \text{otherwise.} \end{cases}$$

Each  $f_n$  is Riemann integrable: it differs from the zero function at finitely many points, so for any partition into intervals of total length  $< \varepsilon/n$  containing those points, the upper sum is at most  $\varepsilon$  and the lower sum is 0. Hence  $\int_0^1 f_n = 0$  for every  $n$ .

The sequence increases pointwise to

$$f = \mathbf{1}_{\mathbb{Q} \cap [0, 1]},$$

the Dirichlet function. This is not Riemann integrable. For *any* partition  $P$  of  $[0, 1]$ , every subinterval of positive length contains both a rational and an irrational, so the upper sum  $U(f, P) = 1$  and the lower sum  $L(f, P) = 0$ . The upper and lower integrals are 1 and 0 and never meet.

So: a uniformly bounded, monotone increasing sequence of Riemann integrable functions, converging pointwise everywhere, with a limit that the theory cannot integrate at all. Note how mild the hypotheses were. Monotone *and* bounded *and* pointwise convergent everywhere is about as much as one could ask for, and it is not enough.

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The Lebesgue theory will integrate  $\mathbf{1}_{\mathbb{Q} \cap [0, 1]}$  without effort — the answer is 0, because  $\mathbb{Q}$  is countable and so has measure zero — and the Monotone Convergence Theorem (Chapter 1.5) will make the interchange  $\lim \int = \int \lim$  automatic under exactly the hypotheses above.

*Idea of the proof.* Look again at why the Dirichlet function defeats the Riemann integral. The obstruction is that we insisted on partitioning the *domain* into intervals. On every interval, however short,  $f$  takes both values 0 and 1; the function oscillates at every scale, so no refinement of an interval partition ever improves the estimate. But the function only takes two values. If we were allowed to group together *all* the points where  $f = 1$ , regardless of whether they form an interval, the integral would be obvious. Lebesgue's move is exactly this: partition the *range* instead of the domain, and then be forced to make sense of the sizes of the resulting, possibly hideous, level sets. That forced side-problem is measure theory.

### 1.1.3 Failure II: the space is not complete

The first failure is the famous one. This one is more serious, and it is the reason measure theory is not optional in analysis.

Equip the space  $C[0, 1]$  of continuous functions with the norm

$$\|f\|_{L^1} = \int_0^1 |f(x)| \, dx,$$

the integral being Riemann's. This is a perfectly good normed vector space. It is not complete, and the failure is not exotic.

**WORKED EXAMPLE 1.1.2— TIER B**  $C[0, 1]$  **is not complete in the  $L^1$  norm**

Define, for  $n \geq 2$ , the piecewise linear function

$$f_n(x) = \begin{cases} 0, & 0 \leq x \leq \frac{1}{2}, \\ n(x - \frac{1}{2}), & \frac{1}{2} < x < \frac{1}{2} + \frac{1}{n}, \\ 1, & \frac{1}{2} + \frac{1}{n} \leq x \leq 1. \end{cases}$$

Each  $f_n$  is continuous: a ramp that climbs from 0 to 1 over a window of width  $1/n$  just to the right of  $\frac{1}{2}$ .

*Step 1: the sequence is Cauchy.* For  $m > n$ , the functions  $f_n$  and  $f_m$  agree outside the interval  $(\frac{1}{2}, \frac{1}{2} + \frac{1}{n})$ , and on that interval both take values in  $[0, 1]$ , so  $|f_n - f_m| \leq 1$  there. Hence

$$\|f_n - f_m\|_{L^1} = \int_{1/2}^{1/2+1/n} |f_n - f_m| \leq \frac{1}{n} \xrightarrow{n \rightarrow \infty} 0.$$

So  $(f_n)$  is Cauchy in  $\|\cdot\|_{L^1}$ .

*Step 2: the limit is not continuous.* Pointwise,  $f_n(x) \rightarrow f(x)$  where  $f = \mathbf{1}_{(1/2, 1]}$ , a step function. And indeed  $\|f_n - f\|_{L^1} \leq 1/n \rightarrow 0$ . So if the sequence converges to anything in  $L^1$ , it converges to  $f$ .

*Step 3: no continuous function works.* Suppose  $g \in C[0, 1]$  satisfied  $\|f_n - g\|_{L^1} \rightarrow 0$ . Then  $\|f - g\|_{L^1} \leq \|f - f_n\|_{L^1} + \|f_n - g\|_{L^1} \rightarrow 0$ , so  $\int_0^1 |f - g| = 0$ . Now  $|f - g|$  is continuous on  $[0, \frac{1}{2})$  and on  $(\frac{1}{2}, 1]$  separately, and a nonnegative continuous function with zero integral on an interval vanishes identically there (if  $|f - g|(x_0) = c > 0$  at some point of one of these open intervals, then by continuity  $|f - g| > c/2$  on a subinterval of some length  $\ell > 0$ , forcing the integral to be at least  $c\ell/2 > 0$ ). Therefore  $g = 0$  on  $[0, \frac{1}{2})$  and  $g = 1$  on  $(\frac{1}{2}, 1]$ . Letting  $x \uparrow \frac{1}{2}$  gives  $g(\frac{1}{2}) = 0$ ; letting  $x \downarrow \frac{1}{2}$  gives  $g(\frac{1}{2}) = 1$ . Contradiction.

**WORKED EXAMPLE 1.1.3— TIER A The same phenomenon, seen as an infinite series**

Completeness failures are often easier to spot in series form. Consider  $\sum_k g_k$  where  $g_k = f_{k+1} - f_k$  with  $f_n$  as above. Then  $\sum_k \|g_k\|_{L^1} < \infty$  — the series is absolutely convergent in norm — yet the partial sums have no continuous limit. In a complete normed space, absolute convergence implies convergence; here it does not. We will prove the equivalence of these two statements as Chapter 1.8 develops, and it is worth noticing now that the ability to sum an absolutely convergent series of functions and land somewhere is the single most-used consequence of completeness in all of harmonic analysis.

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Why does this matter so much? Because completeness is what lets you *build* objects rather than merely inspect them. Almost every construction in later volumes — the Fourier transform on  $L^2$  (Volume 2), the wave packet decomposition of a function (Volume 12), the limiting arguments in induction-on-scales (Volume 13) — proceeds by producing a Cauchy sequence of approximations and taking a limit. Without completeness there is no limit to take.

**Looking ahead: Plancherel needs this**

The Fourier transform is defined by an integral formula  $\widehat{f}(\xi) = \int f(x)e^{-2\pi i x \cdot \xi} dx$  which requires  $f \in L^1$ , but its most important property — Plancherel’s theorem,  $\|\widehat{f}\|_{L^2} = \|f\|_{L^2}$  — lives in  $L^2$ , and  $L^2 \not\subset L^1$  on  $\mathbb{R}^n$ . The resolution is to define the transform on the dense subspace  $L^1 \cap L^2$  and extend by continuity, and *extension by continuity requires the target space to be complete*. The Riemann theory cannot even state Plancherel’s theorem. This is not an exaggeration for effect; it is why measure theory comes first.

**1.1.4 Failure III: there is no theory of size for the sets we care about**

The two failures above are standard. The third is specific to this series, and it is the one to keep in mind.

The problems this series is aimed at are, at bottom, questions about the *size of unions of thin tubes*. Here is the archetype, in a form you can already understand completely.

**Definition 1.1.1** (Besicovitch set). A *Besicovitch set* (or *Kakeya set*) in  $\mathbb{R}^n$  is a compact set containing a unit line segment in every direction: for every  $e \in \mathbb{S}^{n-1}$  there is  $x \in \mathbb{R}^n$  with  $\{x + te : t \in [0, 1]\} \subseteq K$ .



The obvious example in the plane is a disc of diameter 1, which has area  $\pi/4$ . One can do better with a Reuleaux triangle, better still with a three-cusped hypocycloid (the deltoid, area  $\pi/8$ ). The question Kakeya asked in 1917 was: how small can such a set be?

The answer, found by Besicovitch, is that it can have area zero. And note what kind of statement that is. It is a statement about *Lebesgue measure*. It cannot be phrased in terms of the Riemann integral at all, because the sets involved are not Jordan measurable — they are precisely the wild, everywhere-oscillating sets that defeated us in §1.2. The entire subject of this series begins with a theorem that the Riemann theory cannot express.

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**WORKED EXAMPLE 1.1.4— TIER C: RESEARCH TASTE   How a Besicovitch set gets small: the bisection trick**

We will not construct a measure-zero Besicovitch set here — that is Chapter 1.3, once we have the tools — but the mechanism is elementary and worth seeing now, because it is the first appearance of a phenomenon (*overlapping tubes at many scales*) that will occupy us for thirteen more volumes.

*Setup.* Let  $T$  be a triangle with base  $[0, 1]$  on the  $x$ -axis and apex at height  $h$ . Slide a needle around inside  $T$  and you can realise every direction in a certain angular range. Now cut  $T$  vertically through the apex into two triangles  $T_1$  (base  $[0, \frac{1}{2}]$ ) and  $T_2$  (base  $[\frac{1}{2}, 1]$ ), each of area  $|T|/2$ , so no saving yet.

*The trick.* Translate  $T_2$  horizontally to the left by an amount  $t \in [0, \frac{1}{2}]$ , obtaining  $T_2 - (t, 0)$ . Crucially, *translation does not change the set of directions of segments inside a triangle*. So the union  $T_1 \cup (T_2 - (t, 0))$  still realises every direction that  $T$  did. But its area has dropped, because the two triangles now overlap.

*The computation.* Take  $t = \frac{1}{2} - \frac{\alpha}{2}$  for a parameter  $\alpha \in (0, 1)$ , so that the two triangles are pushed together until their bases overlap in a segment of length  $\frac{1-\alpha}{2}$ . A direct similar- triangles calculation — the two triangles have parallel corresponding sides, and their intersection is a triangle similar to  $T$  — gives

$$|T_1 \cap (T_2 - (t, 0))| = (1 - \alpha)^2 \frac{|T|}{4},$$

so that

$$|T_1 \cup (T_2 - (t, 0))| = 2 \cdot \frac{|T|}{2} - (1 - \alpha)^2 \frac{|T|}{4} = |T| \left( 1 - \frac{(1 - \alpha)^2}{4} \right).$$

Taking  $\alpha \rightarrow 0$  (pushing the triangles fully together) saves a factor of  $3/4$ . One bisection, one quarter of the area gone, no directions lost.

*The iteration and where it goes.* Now bisect again: each of the two triangles splits into two, and we slide each pair together. Iterating  $k$  times produces  $2^k$  thin triangles, arranged in a spiky figure called a *Perron tree*, whose total area tends to 0 as  $k \rightarrow \infty$  while the set of directions is preserved throughout. Making this rigorous requires care — the naive product  $\prod_k (1 - \frac{1}{4})$  tends to 0 too fast to be honest, and the true argument must control the overlaps at *all* scales simultaneously — and that is precisely the point.

*Why this is the whole subject in miniature.* Notice what made the saving possible: many long thin sets, pointing in many different directions, can be translated so that they overlap far more than a random arrangement would suggest. The *Keakeya conjecture* (Volumes 9, 10, 14) asserts that this overlapping has a hard limit — that although the *measure* can be zero, the *dimension* cannot drop below  $n$ . The *restriction conjecture* (Volumes 6, 12) turns out to be a statement of the same type in disguise, about how much wave packets travelling in different directions can conspire to be large in the same place. Every technique in this series — covering lemmas, maximal functions, induction on scales, polynomial partitioning — is ultimately a device for controlling the overlap of tubes. You have just seen the first one.

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### Looking ahead: the same picture, thirteen volumes later

In Chapter 1.7 of this volume you will meet the Vitali covering lemma, which controls overlaps of *balls*. Balls are easy: two balls of comparable size either intersect a lot or not at all. Tubes are hard: two tubes of length 1 and width  $\delta$  crossing at angle  $\theta$  intersect in a set of measure  $\approx \delta^2/\theta$ , so the overlap depends on the *angle*, and an arrangement of tubes can be tuned to make many angles small at once. The gap between these two sentences is, more or less, the difference between a first course in measure theory and the Wang–Zahl theorem.

## 1.1.5 The plan of repair

Lebesgue’s programme, which occupies the rest of this volume, is:

1. Assign a size  $m(E) \in [0, \infty]$  to as many subsets  $E \subseteq \mathbb{R}^n$  as possible, agreeing with length/area/volume on boxes, and behaving well under *countable* unions rather than merely finite ones (Chapters 1.2 and 1.3).
2. Discover, unavoidably, that “as many as possible” is not “all” — non-measurable sets exist — and isolate the good class, the measurable sets (Chapter 1.3).

3. Define the integral by partitioning the range, using step 1 to measure the level sets (Chapter 1.4).
4. Harvest the convergence theorems that the Riemann theory lacked (Chapter 1.5), and the Fubini–Tonelli theorem that lets us integrate in  $\mathbb{R}^n$  one variable at a time (Chapter 1.6).
5. Develop the differentiation theory — covering lemmas and the maximal function — which is where measure theory stops being infrastructure and starts being harmonic analysis (Chapter 1.7).
6. Build the  $L^p$  spaces, prove they are complete, and meet weak  $L^p$  (Chapter 1.8).

## Drills

**Drill 1.1.1.** Show that  $f : [0, 1] \rightarrow \mathbb{R}$  defined by  $f(x) = 1/q$  if  $x = p/q$  in lowest terms and  $f(x) = 0$  if  $x$  is irrational is Riemann integrable with integral 0, despite being discontinuous at every rational. Contrast with the Dirichlet function and explain, in one sentence, what distinguishes the two.

**Drill 1.1.2.** Carry out the similar-triangles computation asserted in Worked Example 1.4: verify that the intersection of the two overlapping half- triangles is similar to  $T$  with ratio  $(1 - \alpha)/2$ , and hence has area  $(1 - \alpha)^2|T|/4$ .

**Drill 1.1.3.** Exhibit a sequence  $f_n \in C[0, 1]$  with  $\|f_n\|_{L^1} \rightarrow 0$  but  $f_n(x) \not\rightarrow 0$  for any  $x \in [0, 1]$ . (Such a sequence rules out the naive hope that  $L^1$  convergence implies pointwise convergence; we will see in Chapter 1.5 what survives.)

## Notes

The Dirichlet function dates to 1829. Besicovitch’s construction appeared in 1919 in a paper on Riemann integration — he was answering a question about whether a Riemann-integrable function of two variables always admits a coordinate system in which the iterated integrals behave, and the measure-zero Kakeya set was a by-product. The connection to Fourier analysis was not visible until Fefferman’s 1971 disproof of the ball multiplier conjecture, which we prove in Volume 5. Readers who want to see the destination early may look at Tao’s expository article on the Kakeya problem, though nothing there is needed here.



## Chapter 1.2

# Outer Measures and the Carathéodory Construction

### 1.2.1 The problem, stated honestly

We want a function  $m$  assigning to each subset  $E \subseteq \mathbb{R}^n$  a number  $m(E) \in [0, \infty]$ , such that:

- (M1)  $m(\text{box}) = \text{its elementary volume}$ ;
- (M2)  $m$  is invariant under translations;
- (M3)  $m(\bigcup_{k=1}^{\infty} E_k) = \sum_{k=1}^{\infty} m(E_k)$  whenever the  $E_k$  are pairwise disjoint (*countable additivity*).

It is a theorem, not an accident, that no such  $m$  exists on *all* subsets of  $\mathbb{R}^n$ . We prove this in Chapter 1.3 via Vitali's construction. So something must be given up, and the whole design of measure theory is a decision about *what* to give up.

The options are: weaken (M3) to finite additivity — this is what Banach measures do, and it destroys the convergence theorems, which were the entire point — or restrict the domain of  $m$  to a well-chosen family of sets. Lebesgue and Carathéodory take the second road, and the elegance of Carathéodory's formulation is that the good family is not chosen by hand: it *emerges* from a single condition.

*Idea of the proof.* The strategy is two-step and it is worth naming before we begin. First, define a crude size  $\mu^*$  on *all* sets by covering them efficiently with boxes. This  $\mu^*$  will be countably *subadditive* but not additive — covering is wasteful, and the waste is real. Second, identify the sets on which  $\mu^*$  stops being wasteful. Carathéodory's insight is that the right test

is not about the set itself but about how the set *cuts* other sets:  $E$  is good precisely when it splits every test set  $A$  cleanly, with no loss.

## 1.2.2 The language of $\sigma$ -algebras

Before constructing anything we fix vocabulary. A reader who has met measure theory before may skim this section; a reader who has not should read it carefully, because every term defined here is used without further comment for the remaining thirteen volumes.

**Definition 1.2.1** (Algebra and  $\sigma$ -algebra). Let  $X$  be a set. A family  $\mathcal{A} \subseteq \mathcal{P}(X)$  is an *algebra* if

$$(A1) \quad \emptyset \in \mathcal{A};$$

$$(A2) \quad E \in \mathcal{A} \implies E^c \in \mathcal{A};$$

$$(A3) \quad E, F \in \mathcal{A} \implies E \cup F \in \mathcal{A}.$$

It is a  $\sigma$ -*algebra* if (A3) is strengthened to

$$(A3') \quad E_1, E_2, \dots \in \mathcal{A} \implies \bigcup_{k=1}^{\infty} E_k \in \mathcal{A}.$$

By De Morgan's laws, an algebra is automatically closed under finite intersections and set differences, and a  $\sigma$ -algebra under countable intersections as well:  $\bigcap_k E_k = (\bigcup_k E_k^c)^c$ . Also  $X = \emptyset^c \in \mathcal{A}$  always.

*Scratch work.* The single letter of difference between (A3) and (A3') is the whole subject. Finite additivity is easy to arrange and useless; the convergence theorems of Chapter 1.5, which are the entire reason for building this machinery, are statements about countable operations and fail without (A3'). Whenever a proof in this volume looks as though it is working unreasonably hard, it is almost always working to upgrade a finite statement to a countable one — see Steps 4 and 5 of Theorem 1.2.11, where exactly this happens.

**Proposition 1.2.2** (Generated  $\sigma$ -algebra). *For any  $\mathcal{E} \subseteq \mathcal{P}(X)$  there is a smallest  $\sigma$ -algebra containing  $\mathcal{E}$ , denoted  $\sigma(\mathcal{E})$ .*

*Proof.* The family of  $\sigma$ -algebras containing  $\mathcal{E}$  is nonempty, since  $\mathcal{P}(X)$  is one. An arbitrary intersection of  $\sigma$ -algebras is a  $\sigma$ -algebra — each defining condition is preserved under intersection — so  $\sigma(\mathcal{E}) := \bigcap \{\mathcal{A} : \mathcal{A} \text{ a } \sigma\text{-algebra, } \mathcal{E} \subseteq \mathcal{A}\}$  works.  $\square$

**Pitfall:  $\sigma(\mathcal{E})$  has no constructive description**

It is tempting to imagine  $\sigma(\mathcal{E})$  as “everything obtainable from  $\mathcal{E}$  by countably many unions and complements”. That family is in general *not* a  $\sigma$ -algebra: one may take countable unions of sets each of which required countably many operations, and iterate, and the process does not terminate after  $\omega$  steps — it must be run through all countable ordinals. This is why Proposition 1.2.2 is proved from above, by intersecting, rather than from below by constructing. It is also why arguments about  $\sigma(\mathcal{E})$  almost always proceed by the “good sets” principle: show that the family of  $E$  with the desired property is a  $\sigma$ -algebra containing  $\mathcal{E}$ , and conclude it contains  $\sigma(\mathcal{E})$ . That move is used in Proposition 1.4.3, in Lemma 1.6.3, and a dozen times besides.

**Definition 1.2.3** (Borel sets;  $G_\delta$  and  $F_\sigma$ ). The *Borel  $\sigma$ -algebra*  $\mathcal{B}(X)$  of a metric space  $X$  is  $\sigma(\{\text{open sets}\})$ . Its members are *Borel sets*.

A  $G_\delta$  set is a countable intersection of open sets; an  $F_\sigma$  set is a countable union of closed sets. (The notation is German:  $G$  for *Gebiet*, region, with  $\delta$  for *Durchschnitt*, intersection;  $F$  for *fermé*, closed, with  $\sigma$  for *somme*, sum.) Both are Borel, and the two classes are exchanged by complementation.

*Remark 1.2.4* (The Borel hierarchy does not stop). Every open set is  $G_\delta$  (trivially) and also  $F_\sigma$ : in a metric space,  $U = \bigcup_k \{x : \text{dist}(x, U^c) \geq 1/k\}$ . But the classes do not exhaust  $\mathcal{B}$ . One may form countable unions of  $G_\delta$  sets ( $G_{\delta\sigma}$ ), then countable intersections of those, and so on, and the hierarchy is strictly increasing through all countable ordinals. So “Borel” is genuinely more generous than “ $G_\delta$  or  $F_\sigma$ ”.

This matters in Chapter 1.3. Theorem 1.3.6 will say that a Lebesgue measurable set is a  $G_\delta$  *minus a null set*, not that it is Borel outright — and the null set is not decoration. Worked Example 1.3.2(c) shows there are Lebesgue measurable sets that are not Borel at all, and indeed vastly more of the former than the latter.

**Definition 1.2.5** (Measure). Let  $\mathcal{A}$  be a  $\sigma$ -algebra on  $X$ . A *measure* is a map  $\mu : \mathcal{A} \rightarrow [0, \infty]$  with  $\mu(\emptyset) = 0$  which is *countably additive*: for pairwise disjoint  $E_1, E_2, \dots \in \mathcal{A}$ ,

$$\mu\left(\bigcup_{k=1}^{\infty} E_k\right) = \sum_{k=1}^{\infty} \mu(E_k).$$

The triple  $(X, \mathcal{A}, \mu)$  is a *measure space*. The measure is *finite* if  $\mu(X) < \infty$ ;  *$\sigma$ -finite* if  $X = \bigcup_k X_k$  with  $\mu(X_k) < \infty$  for each  $k$ ; and *complete* if every subset of every  $\mu$ -null set lies in  $\mathcal{A}$ .

**Proposition 1.2.6** (Elementary properties). *Let  $\mu$  be a measure on  $\mathcal{A}$  and  $E, F, E_k \in \mathcal{A}$ .*

- (i) (*monotonicity*)  $E \subseteq F \implies \mu(E) \leq \mu(F)$ , with  $\mu(F \setminus E) = \mu(F) - \mu(E)$  when  $\mu(E) < \infty$ ;
- (ii) (*countable subadditivity*)  $\mu(\bigcup_k E_k) \leq \sum_k \mu(E_k)$ ;
- (iii) (*continuity from below*)  $E_1 \subseteq E_2 \subseteq \dots$  implies  $\mu(\bigcup_k E_k) = \lim_k \mu(E_k)$ ;
- (iv) (*continuity from above*)  $E_1 \supseteq E_2 \supseteq \dots$  with  $\mu(E_1) < \infty$  implies  $\mu(\bigcap_k E_k) = \lim_k \mu(E_k)$ .

*Proof.* (i) Write  $F = E \sqcup (F \setminus E)$  and use additivity (padding with  $\emptyset$  to make the union countable).

(ii) Disjointify:  $F_k = E_k \setminus \bigcup_{j < k} E_j \in \mathcal{A}$  are disjoint with the same union, and  $F_k \subseteq E_k$ , so  $\mu(\bigcup_k E_k) = \sum_k \mu(F_k) \leq \sum_k \mu(E_k)$  by (i).

(iii) With  $F_k$  as above,  $\bigcup_{j \leq k} E_j = \bigsqcup_{j \leq k} F_j$ , so  $\mu(E_k) = \sum_{j \leq k} \mu(F_j)$ , whose limit is  $\sum_j \mu(F_j) = \mu(\bigcup_j E_j)$ .

(iv) Apply (iii) to  $E_1 \setminus E_k \uparrow E_1 \setminus \bigcap_k E_k$  and subtract, which is legitimate as  $\mu(E_1) < \infty$ .  $\square$

### **Pitfall: continuity from above needs finiteness**

On  $\mathbb{R}$  with Lebesgue measure,  $E_k = [k, \infty)$  decreases with  $\bigcap_k E_k = \emptyset$ , yet  $m(E_k) = \infty$  for every  $k$ . So (iv) fails without the hypothesis  $\mu(E_1) < \infty$ . This is not a technicality: it is exactly the hypothesis that fails in Theorem 1.5.9, and the reason Egorov's theorem is false on  $\mathbb{R}$ .

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#### WORKED EXAMPLE 1.2.1— TIER A **Computing a generated $\sigma$ -algebra**

(a) *A finite example.* Let  $X = \{1, 2, 3, 4\}$  and  $\mathcal{E} = \{\{1, 2\}, \{2, 3\}\}$ . Closing under complements and unions gives the *atoms*  $\{1\}, \{2\}, \{3\}, \{4\}$  — for instance  $\{2\} = \{1, 2\} \cap \{2, 3\}$  and  $\{1\} = \{1, 2\} \setminus \{2\}$  — so  $\sigma(\mathcal{E}) = \mathcal{P}(X)$ , all 16 subsets. In a finite setting  $\sigma(\mathcal{E})$  is always the family of unions of atoms, and this is the one case where a constructive description exists.

(b) *Generating the Borel sets of  $\mathbb{R}$ .* We claim  $\mathcal{B}(\mathbb{R}) = \sigma(\{(a, \infty) : a \in \mathbb{R}\})$ . One inclusion is clear since each  $(a, \infty)$  is open. Conversely, the right-hand family contains  $(-\infty, b] = (b, \infty)^c$ , hence  $(a, b] = (a, \infty) \cap (-\infty, b]$ , hence  $(a, b) = \bigcup_k (a, b - \frac{1}{k}]$ , and every open subset of  $\mathbb{R}$  is a countable union of open intervals with rational endpoints. So it contains all open sets, and therefore  $\mathcal{B}(\mathbb{R})$ .

*Why this is worth doing once.* The same argument, with the same three lines, shows that measurability of a function may be tested on the single family  $\{(a, \infty)\}$  rather than on all Borel sets — which is Lemma 1.4.2 and Proposition 1.4.3 in Chapter 1.4, and which is what makes measurability checkable in practice.

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### 1.2.3 Outer measures

**Definition 1.2.7** (Outer measure). An *outer measure* on a set  $X$  is a function  $\mu^* : \mathcal{P}(X) \rightarrow [0, \infty]$  such that

- (i)  $\mu^*(\emptyset) = 0$ ;
- (ii) (monotonicity)  $A \subseteq B \implies \mu^*(A) \leq \mu^*(B)$ ;
- (iii) (countable subadditivity)  $\mu^*(\bigcup_{k=1}^{\infty} A_k) \leq \sum_{k=1}^{\infty} \mu^*(A_k)$  for all  $A_k \subseteq X$ .

Two features of this definition deserve comment before we do anything with it.

First, an outer measure is defined on *every* subset of  $X$ . There is no restriction, no measurability hypothesis, no fine print. That is what will make outer measures easy to construct — Proposition 1.2.8 below builds one out of essentially nothing — and it is bought at the price of (iii) being an inequality rather than an equality.

Second, notice that (iii) is stated for arbitrary  $A_k$ , not merely disjoint ones. For subadditivity this costs nothing, since the disjoint case would imply the general one anyway (replace  $A_k$  by  $A_k \setminus \bigcup_{j < k} A_j$  and use monotonicity). The asymmetry with (M3) above, which was stated only for disjoint sets, is therefore not a weakening in disguise; it is exactly the gap between  $\leq$  and  $=$  that the rest of the chapter is about.

**Proposition 1.2.8** (Outer measure from a covering family). *Let  $X$  be a set and let  $\mathcal{E} \subseteq \mathcal{P}(X)$  be a family of subsets with the following two properties:  $\emptyset \in \mathcal{E}$ , and  $\mathcal{E}$  covers  $X$ , meaning that  $X = \bigcup_{j=1}^{\infty} E_j$  for some countable family  $E_j \in \mathcal{E}$ . Let  $\rho : \mathcal{E} \rightarrow [0, \infty]$  be any function with  $\rho(\emptyset) = 0$ . Define, for  $A \subseteq X$ ,*

$$\mu^*(A) = \inf \left\{ \sum_{j=1}^{\infty} \rho(E_j) : E_j \in \mathcal{E} \text{ for all } j, \quad A \subseteq \bigcup_{j=1}^{\infty} E_j \right\}. \quad (1.2.1)$$

*Then  $\mu^*$  is an outer measure on  $X$ .*

Before the proof, three remarks on the statement, since it is more slippery than it looks.

The hypothesis that  $\mathcal{E}$  covers  $X$  guarantees that the infimum in (1.2.1) is taken over a nonempty collection — every  $A$  has at least one admissible cover, namely the one covering all of  $X$ . Without this we would be taking  $\inf \emptyset = +\infty$ , which is harmless but makes the statement of (i) false unless  $\emptyset$  is separately handled.

The covers are indexed by  $\mathbb{N}$ , that is, they are *countable*. Finite covers are included: pad with copies of  $\emptyset$ , which contribute  $\rho(\emptyset) = 0$  to the sum. This

padding convention is why we required  $\emptyset \in \mathcal{E}$ , and we use it without comment from now on.

Finally,  $\rho$  is required to satisfy nothing except  $\rho(\emptyset) = 0$ . It need not be monotone, additive, or even finite. All the structure in (1.2.1) comes from the infimum-over-covers, not from  $\rho$ .

*Proof.* We verify the three properties of Definition 1.2.7 in turn.

**Step 1.**  $\mu^*(\emptyset) = 0$ .

Cover  $\emptyset$  by the countable family  $E_j = \emptyset$  for every  $j$ . This is admissible, since  $\emptyset \subseteq \bigcup_j \emptyset$ . Its cost is  $\sum_{j=1}^{\infty} \rho(\emptyset) = \sum_{j=1}^{\infty} 0 = 0$ . Since  $\mu^*$  is an infimum of nonnegative quantities and we have exhibited a cover of cost 0, we get  $0 \leq \mu^*(\emptyset) \leq 0$ .

**Step 2.** *Monotonicity.*

Suppose  $A \subseteq B$ . The point is a comparison of the two collections over which the infima are taken. If  $\{E_j\}$  is any admissible cover of  $B$  — that is,  $E_j \in \mathcal{E}$  and  $B \subseteq \bigcup_j E_j$  — then since  $A \subseteq B \subseteq \bigcup_j E_j$ , the same family is an admissible cover of  $A$ . So every competitor in the infimum defining  $\mu^*(B)$  is also a competitor in the infimum defining  $\mu^*(A)$ , with the same cost. The infimum over a larger collection is no larger, hence  $\mu^*(A) \leq \mu^*(B)$ .

▷ *Goal: countable subadditivity, which is the only substantial part.*

**Step 3.** *Setting up.*

Let  $A_1, A_2, \dots \subseteq X$  be arbitrary and put  $A = \bigcup_k A_k$ . We must show  $\mu^*(A) \leq \sum_k \mu^*(A_k)$ .

Dispose of a trivial case first: if  $\mu^*(A_k) = \infty$  for some  $k$ , the right-hand side is  $\infty$  and there is nothing to prove. So assume  $\mu^*(A_k) < \infty$  for every  $k$ . (We do *not* assume the sum is finite; if it diverges the inequality is again trivial, but the argument below does not need to distinguish the cases.)

Fix  $\varepsilon > 0$ .

**Step 4.** *Choosing good covers, one for each  $k$ .*

Since  $\mu^*(A_k)$  is an infimum and  $\mu^*(A_k) < \infty$ , the number  $\mu^*(A_k) + \varepsilon 2^{-k}$  is strictly larger than that infimum, so it is not a lower bound for the competing costs. Hence there exists an admissible cover  $\{E_{k,j}\}_{j \geq 1} \subseteq \mathcal{E}$  of  $A_k$  with

$$\sum_{j=1}^{\infty} \rho(E_{k,j}) \leq \mu^*(A_k) + \varepsilon 2^{-k}. \quad (1.2.2)$$

This uses the axiom of countable choice, once for each  $k$ ; we will not remark on such uses again.

**Step 5.** *Assembling a single cover of  $A$ .*

Consider the doubly-indexed family  $\{E_{k,j}\}_{k \geq 1, j \geq 1}$ . It is a countable family, being a countable union of countable families, so after fixing any bijection  $\mathbb{N} \rightarrow \mathbb{N} \times \mathbb{N}$  it is an admissible competitor in (1.2.1). And it covers  $A$ : given  $x \in A$ , we have  $x \in A_k$  for some  $k$ , and  $\{E_{k,j}\}_j$  covers  $A_k$ , so  $x \in E_{k,j}$  for some  $j$ .

Therefore, by the definition of  $\mu^*(A)$  as an infimum,

$$\mu^*(A) \leq \sum_{k=1}^{\infty} \sum_{j=1}^{\infty} \rho(E_{k,j}).$$

The rearrangement of the doubly-indexed sum into an iterated sum is legitimate because all terms are nonnegative, so the sum is independent of the order of summation (Tonelli for series, or elementary rearrangement of nonnegative series).

**Step 6.** *Cashing in the budget.*

Apply (1.2.2) to each inner sum:

$$\mu^*(A) \leq \sum_{k=1}^{\infty} \left( \mu^*(A_k) + \varepsilon 2^{-k} \right) = \sum_{k=1}^{\infty} \mu^*(A_k) + \varepsilon \sum_{k=1}^{\infty} 2^{-k} = \sum_{k=1}^{\infty} \mu^*(A_k) + \varepsilon.$$

This holds for every  $\varepsilon > 0$ . Since the left-hand side does not depend on  $\varepsilon$ , letting  $\varepsilon \downarrow 0$  gives  $\mu^*(A) \leq \sum_k \mu^*(A_k)$ .  $\square$

*Scratch work.* Step 4 contains the single most-used device in measure theory, and it is worth isolating. We had countably many approximations to make, each with its own error, and we needed the *total* error to be small. Distributing a fixed budget  $\varepsilon$  across the tasks as  $\varepsilon 2^{-1}, \varepsilon 2^{-2}, \varepsilon 2^{-3}, \dots$  does this, because the weights are summable. Any summable sequence of positive weights would serve;  $2^{-k}$  is merely the most convenient.

You will see this again in the proof of Theorem 1.3.6 (approximating a measurable set by an open one), in the construction of the Cantor function, in the proof of Egorov's theorem, and — in a form that looks quite different but is the same idea — in the  $\varepsilon$ -removal arguments of Volume 12, where the tasks being budgeted are dyadic spatial scales rather than sets. When a later proof opens with “fix  $\varepsilon > 0$  and choose, for each  $k$ ”, this is what is happening.

**WORKED EXAMPLE 1.2.2— TIER A Lebesgue outer measure on  $\mathbb{R}^n$** 

Take  $X = \mathbb{R}^n$ , let  $\mathcal{E}$  be the collection of open boxes  $Q = \prod_{i=1}^n (a_i, b_i)$  together

with  $\emptyset$ , and let  $\rho(Q) = \prod_{i=1}^n (b_i - a_i)$  be the elementary volume, written  $|Q|$ . The family  $\mathcal{E}$  covers  $\mathbb{R}^n$  (use the boxes  $(-k, k)^n$ ,  $k \in \mathbb{N}$ ), so Proposition 1.2.8 applies and

$$m^*(A) = \inf \left\{ \sum_j |Q_j| : A \subseteq \bigcup_j Q_j, Q_j \text{ open boxes} \right\}$$

is an outer measure, called *Lebesgue outer measure*.

*Singletons are null.* Fix  $x \in \mathbb{R}^n$  and  $\varepsilon > 0$ . The open box of side  $\varepsilon$  centred at  $x$  covers  $\{x\}$  and has volume  $\varepsilon^n$ , so  $m^*(\{x\}) \leq \varepsilon^n$ . As  $\varepsilon > 0$  was arbitrary and  $\varepsilon^n \rightarrow 0$ , we get  $m^*(\{x\}) = 0$ .

*Countable sets are null.* If  $A = \{x_1, x_2, \dots\}$  is countable then countable subadditivity gives  $m^*(A) \leq \sum_k m^*(\{x_k\}) = \sum_k 0 = 0$ . In particular  $m^*(\mathbb{Q}^n) = 0$ .

*What we have just done.* In Chapter 1.1 the Riemann integral was defeated by  $\mathbf{1}_{\mathbb{Q} \cap [0,1]}$ , and the reason was that  $\mathbb{Q}$  is dense, so no interval partition can separate it from its complement. We have now assigned  $\mathbb{Q}$  the size 0, using two lines and no cleverness. Once the integral is constructed in Chapter 1.4 it will follow immediately that  $\int \mathbf{1}_{\mathbb{Q} \cap [0,1]} = 0$ . The Riemann theory could not compute this integral because it could not express the question; the measure-theoretic answer is trivial because the right question was asked.

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**Pitfall:**  $m^*(Q) = |Q|$  is not obvious

It is tempting to regard the identity  $m^*(Q) = |Q|$  for a box  $Q$  as part of the definition. It is not. The inequality  $m^*(Q) \leq |Q|$  is trivial (cover  $Q$  by a slightly larger box). The reverse inequality says that *no* countable family of boxes can cover  $Q$  at total cost less than  $|Q|$ , and this genuinely requires an argument — specifically, a compactness argument, since a countable cover must first be reduced to a finite one before elementary volume comparisons apply. We prove it as Theorem 3.4. Until then, do not use it, and be alert to the fact that several plausible-looking arguments in this chapter would become circular if you did.

## 1.2.4 The Carathéodory criterion

### 1.2.4.1 Three attempts at a definition

We now need to say which sets are “good”. Since the whole difficulty is that  $\mu^*$  is subadditive but not additive, the good sets should be those which do not

create waste when we cut along them. Turning that sentence into a usable definition takes three attempts, and seeing the first two fail is the fastest way to understand why the third looks the way it does.

**Attempt 1:  $E$  is good if  $\mu^*(E) + \mu^*(E^c) = \mu^*(X)$ .** This is the most direct transcription of “no waste”. It fails immediately on  $X = \mathbb{R}^n$ , because  $\mu^*(\mathbb{R}^n) = \infty$ , and so the condition reads  $\infty = \infty$  and is satisfied by *every* set. Useless.

**Attempt 2:  $E$  is good if  $\mu^*(Q \cap E) + \mu^*(Q \cap E^c) = \mu^*(Q)$  for every box  $Q$ .** Localising repairs the previous failure — the condition is now non-vacuous — and this definition does in fact single out the right sets in  $\mathbb{R}^n$ . But it has a serious structural defect: try to prove from it that the good sets are closed under intersection. Given  $E$  and  $F$  good, one wants to test  $E \cap F$  against a box  $Q$ ; the natural move is to first cut  $Q$  by  $E$  and then cut the two pieces by  $F$ . But the pieces  $Q \cap E$  and  $Q \cap E^c$  are not boxes, so the hypothesis on  $F$  does not apply to them. The argument stalls at the first step.

**Attempt 3 (Carathéodory):  $E$  is good if it cuts *every* set cleanly.** The defect in Attempt 2 was that the family of admissible test sets was too small to be stable under the cutting operation we needed to iterate. The fix is to enlarge it as far as it will go.

**Definition 1.2.9** (Carathéodory measurability). Let  $\mu^*$  be an outer measure on  $X$ . A set  $E \subseteq X$  is  $\mu^*$ -measurable if for *every*  $A \subseteq X$ ,

$$\mu^*(A) = \mu^*(A \cap E) + \mu^*(A \cap E^c). \quad (1.2.3)$$

We write  $\mathcal{M}_{\mu^*}$  for the collection of  $\mu^*$ -measurable sets, and call  $A$  in (1.2.3) the *test set*.

*Idea of the proof.* The definition looks unmotivated on first meeting, and students routinely find it the least intuitive definition in the subject. The honest account is that it is *reverse-engineered from the proof it has to support*. The theorem we want is that  $\mathcal{M}_{\mu^*}$  is a  $\sigma$ -algebra; the proof of that theorem proceeds by cutting a test set repeatedly, and at each cut the test set changes into something we have no control over. Quantifying over all  $A$  is precisely what makes the induction go through. Attempt 2 fails because its induction breaks at step two.

There is a second thing to notice, which makes the definition much less alarming in practice: although (1.2.3) demands something of every  $A$ , it is usually verified by a single argument uniform in  $A$ , because the only tool available is an efficient cover of  $A$ , and efficient covers behave the same way

regardless of what  $A$  is. Worked Example 2.7 below is the model case.

### 1.2.4.2 Two standing reductions

Before proving anything, we record two simplifications that halve the work in every verification of (1.2.3).

**Lemma 1.2.10** (Reductions). *Let  $\mu^*$  be an outer measure and  $E \subseteq X$ . Then  $E \in \mathcal{M}_{\mu^*}$  if and only if*

$$\mu^*(A) \geq \mu^*(A \cap E) + \mu^*(A \cap E^c) \quad \text{for every } A \subseteq X \text{ with } \mu^*(A) < \infty. \quad (1.2.4)$$

*Proof.* Suppose (1.2.4) holds; we must recover (1.2.3) for all  $A$ .

First, the reverse inequality is automatic and costs nothing: for any  $A$  whatsoever,  $A = (A \cap E) \cup (A \cap E^c)$ , so finite subadditivity — which is contained in Definition 1.2.7(iii), padding with  $\emptyset$  — gives

$$\mu^*(A) \leq \mu^*(A \cap E) + \mu^*(A \cap E^c).$$

So (1.2.3) is equivalent to the single inequality  $\geq$ .

Second, the restriction to  $\mu^*(A) < \infty$  loses nothing: if  $\mu^*(A) = \infty$  then the inequality  $\mu^*(A) \geq \mu^*(A \cap E) + \mu^*(A \cap E^c)$  holds trivially, both sides being  $\infty$  (the left is  $\infty$  by assumption).  $\square$

#### Pitfall: which direction is the content

Beginners frequently produce “proofs” of measurability that establish  $\mu^*(A) \leq \mu^*(A \cap E) + \mu^*(A \cap E^c)$  and stop. That inequality is true for every set  $E$ , measurable or not; it is subadditivity. If your argument did not use any property of  $E$ , you have proved nothing.

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#### WORKED EXAMPLE 1.2.3— TIER B Half-spaces are Lebesgue measurable

Let  $m^*$  be Lebesgue outer measure on  $\mathbb{R}^n$  and fix  $a \in \mathbb{R}$ . We show that the half-space

$$H = \{x \in \mathbb{R}^n : x_1 > a\}$$

is  $m^*$ -measurable. This is the first non-trivial verification of (1.2.3) in the book, and the argument is the template for all later ones.

*What is available.* By the Pitfall box above we may not use  $m^*(Q) = |Q|$ . What we *may* use is the following purely elementary fact about elementary volume, proved by inspection of the defining product: if  $Q$  is an open box and we cut it by the hyperplane  $\{x_1 = a\}$  into  $Q' = Q \cap \{x_1 > a\}$  and

$Q'' = Q \cap \{x_1 < a\}$ , then  $Q'$  and  $Q''$  are again open boxes (possibly empty) and

$$|Q'| + |Q''| = |Q|. \quad (1.2.5)$$

Indeed, if  $Q = (b_1, c_1) \times R$  with  $R$  a box in  $\mathbb{R}^{n-1}$ , then for  $b_1 \leq a \leq c_1$  we have  $Q' = (a, c_1) \times R$  and  $Q'' = (b_1, a) \times R$ , so  $|Q'| + |Q''| = ((c_1 - a) + (a - b_1))|R| = |Q|$ ; and if  $a$  lies outside  $[b_1, c_1]$  then one of the two is empty and the identity is trivial. Note that the sliver  $Q \cap \{x_1 = a\}$  is discarded, and this is harmless precisely because we are working with *open* boxes.

*The verification.* By Lemma 1.2.10 it suffices to prove

$$m^*(A) \geq m^*(A \cap H) + m^*(A \cap H^c) \quad (1.2.6)$$

for every  $A \subseteq \mathbb{R}^n$  with  $m^*(A) < \infty$ .

Fix such an  $A$  and fix  $\varepsilon > 0$ . By the definition of  $m^*$  as an infimum, choose open boxes  $\{Q_j\}_{j \geq 1}$  with

$$A \subseteq \bigcup_j Q_j \quad \text{and} \quad \sum_j |Q_j| \leq m^*(A) + \varepsilon.$$

Cut each box: let  $Q'_j = Q_j \cap \{x_1 > a\}$  and  $Q''_j = Q_j \cap \{x_1 < a\}$ , both open boxes. Now observe two coverings:

$$A \cap H \subseteq \left( \bigcup_j Q_j \right) \cap H = \bigcup_j (Q_j \cap H) = \bigcup_j Q'_j,$$

and, writing  $H^c = \{x_1 \leq a\}$ ,

$$A \cap H^c \subseteq \bigcup_j (Q_j \cap H^c) \subseteq \bigcup_j (Q''_j \cup \{x_1 = a\}) \subseteq \left( \bigcup_j Q''_j \right) \cup P, \quad P := \{x_1 = a\}.$$

The hyperplane  $P$  is null. To see this, fix  $\eta > 0$  and, for each  $k \in \mathbb{N}$ , cover the bounded piece  $P \cap ((-k, k)^n)$  by the single open box

$$R_k = \left( a - \frac{\eta_k}{2}, a + \frac{\eta_k}{2} \right) \times (-k, k)^{n-1}, \quad \eta_k = \frac{\eta 2^{-k}}{(2k)^{n-1}},$$

whose elementary volume is  $|R_k| = \eta_k (2k)^{n-1} = \eta 2^{-k}$ . The family  $\{R_k\}_{k \geq 1}$  covers  $P$ , since every point of  $P$  lies in some  $(-k, k)^n$ , and its total cost is  $\sum_k \eta 2^{-k} = \eta$ . Hence  $m^*(P) \leq \eta$  for every  $\eta > 0$ , so  $m^*(P) = 0$ . (This is the  $\varepsilon 2^{-k}$  device of the Scratch box on page 19 again, now used to cover an unbounded set.)

Therefore, using monotonicity, then subadditivity, then  $m^*(P) = 0$ , then the definition of  $m^*$  applied to the two covers:

$$\begin{aligned}
 m^*(A \cap H) + m^*(A \cap H^c) &\leq m^*\left(\bigcup_j Q'_j\right) + m^*\left(\bigcup_j Q''_j\right) + m^*(P) \\
 &\leq \sum_j |Q'_j| + \sum_j |Q''_j| \\
 &= \sum_j (|Q'_j| + |Q''_j|) = \sum_j |Q_j| \leq m^*(A) + \varepsilon,
 \end{aligned}$$

where the second-to-last equality is (1.2.5). Letting  $\varepsilon \downarrow 0$  gives (1.2.6).

*What made it work.* The argument never inspected  $A$ . It took an arbitrary efficient cover of  $A$  and cut the *cover*, not the set, exploiting the fact that the cutting operation is exactly additive on the covering family. This is the general shape of every measurability proof: one shows that  $E$  interacts well with the covering family  $\mathcal{E}$  used to build  $\mu^*$ , and the “for all  $A$ ” quantifier then comes for free.

*Consequence.* Once Theorem 1.2.11 tells us  $\mathcal{M}_{m^*}$  is a  $\sigma$ -algebra, the measurability of all half-spaces gives us, by finite intersections, all open boxes; by countable unions, all open sets (every open subset of  $\mathbb{R}^n$  is a countable union of boxes); and hence all Borel sets. So this one worked example, plus the theorem below, delivers the entire Borel  $\sigma$ -algebra.

### 1.2.4.3 The main theorem

**Theorem 1.2.11** (Carathéodory). *Let  $\mu^*$  be an outer measure on a set  $X$ . Then:*

- (a)  $\mathcal{M}_{\mu^*}$  is a  $\sigma$ -algebra of subsets of  $X$ ;
- (b) the restriction  $\mu := \mu^*|_{\mathcal{M}_{\mu^*}}$  is a countably additive measure;
- (c)  $\mu$  is complete: if  $E \in \mathcal{M}_{\mu^*}$  with  $\mu(E) = 0$  and  $F \subseteq E$ , then  $F \in \mathcal{M}_{\mu^*}$  and  $\mu(F) = 0$ .

*Scratch work.* It is worth previewing the shape of the proof, because it is long and the reader should know where it is going.

To prove (a) we must show  $\mathcal{M}_{\mu^*}$  contains  $\emptyset$ , is closed under complements, and is closed under countable unions. The first two are one-line consequences of the symmetry of (1.2.3) in  $E$  and  $E^c$ . Countable unions is the whole content, and it is reached in three moves: first *finite* unions (Step



2), which makes  $\mathcal{M}_{\mu^*}$  an algebra; then a form of additivity that holds relative to an arbitrary test set (Step 3), which is the technical engine; then countable *disjoint* unions (Step 5), from which the general case follows by a standard disjointification (Step 4).

Part (b) then falls out of Step 5 by specialising the test set, and part (c) is independent and easy.

The one step to watch is Step 5. Everything before it is finitary. Step 5 is where a limit  $N \rightarrow \infty$  is taken, and where the “for every test set  $A$ ” quantifier in Definition 1.2.9 is finally spent: we apply measurability of the partial union  $F_N$  to the *same*  $A$  for every  $N$ , obtaining a sequence of inequalities that can be passed to the limit. Under Attempt 2 above,  $A$  would have had to be a box while  $F_N$  was not, and there would be no such sequence.

*Proof.* Throughout,  $A$  denotes an arbitrary test set. By Lemma 1.2.10 we only ever need to prove inequalities of the form  $\mu^*(A) \geq \mu^*(A \cap \cdot) + \mu^*(A \cap \cdot^c)$ .

**Step 1.**  $\emptyset \in \mathcal{M}_{\mu^*}$ , and  $\mathcal{M}_{\mu^*}$  is closed under complementation.

For  $E = \emptyset$  and any  $A$  we have  $A \cap \emptyset = \emptyset$  and  $A \cap \emptyset^c = A$ , so (1.2.3) reads  $\mu^*(A) = 0 + \mu^*(A)$ , which is true. For complementation, note that condition (1.2.3) is literally unchanged when  $E$  is replaced by  $E^c$ , since  $(E^c)^c = E$  and addition is commutative. Hence  $E \in \mathcal{M}_{\mu^*} \iff E^c \in \mathcal{M}_{\mu^*}$ .

**Step 2.**  $\mathcal{M}_{\mu^*}$  is closed under finite unions, hence is an algebra.

Let  $E, F \in \mathcal{M}_{\mu^*}$  and let  $A$  be a test set.

Apply measurability of  $E$  with test set  $A$ :

$$\mu^*(A) = \mu^*(A \cap E) + \mu^*(A \cap E^c). \quad (1.2.7)$$

Now apply measurability of  $F$  *twice*, once with test set  $A \cap E$  and once with test set  $A \cap E^c$  — this is legitimate precisely because (1.2.3) was demanded for all test sets:

$$\begin{aligned} \mu^*(A \cap E) &= \mu^*(A \cap E \cap F) + \mu^*(A \cap E \cap F^c), \\ \mu^*(A \cap E^c) &= \mu^*(A \cap E^c \cap F) + \mu^*(A \cap E^c \cap F^c). \end{aligned}$$

Substituting both into (1.2.7) decomposes  $\mu^*(A)$  into four pieces:

$$\mu^*(A) = \underbrace{\mu^*(A \cap E \cap F)}_{\text{(I)}} + \underbrace{\mu^*(A \cap E \cap F^c)}_{\text{(II)}} + \underbrace{\mu^*(A \cap E^c \cap F)}_{\text{(III)}} + \underbrace{\mu^*(A \cap E^c \cap F^c)}_{\text{(IV)}}. \quad (1.2.8)$$

Observe now that the set-theoretic identity

$$E \cup F = (E \cap F) \cup (E \cap F^c) \cup (E^c \cap F)$$

holds — the three pieces are the three ways a point can lie in  $E \cup F$  — and intersecting with  $A$  gives

$$A \cap (E \cup F) = (A \cap E \cap F) \cup (A \cap E \cap F^c) \cup (A \cap E^c \cap F).$$

Subadditivity of  $\mu^*$  therefore yields

$$\mu^*(A \cap (E \cup F)) \leq \text{(I)} + \text{(II)} + \text{(III)}. \quad (1.2.9)$$

Meanwhile, by De Morgan,  $A \cap (E \cup F)^c = A \cap E^c \cap F^c$ , so the fourth piece is exactly

$$\text{(IV)} = \mu^*(A \cap (E \cup F)^c). \quad (1.2.10)$$

Combining (1.2.8), (1.2.9) and (1.2.10),

$$\mu^*(A) = \text{(I)} + \text{(II)} + \text{(III)} + \text{(IV)} \geq \mu^*(A \cap (E \cup F)) + \mu^*(A \cap (E \cup F)^c),$$

which by Lemma 1.2.10 is what we needed. Hence  $E \cup F \in \mathcal{M}_{\mu^*}$ .

Together with Step 1,  $\mathcal{M}_{\mu^*}$  is an algebra: it is closed under complements and finite unions, hence (De Morgan again) under finite intersections and set differences.

**Where we are.** We have an algebra, and we have not yet used any countability. Everything so far would work verbatim for a merely finitely subadditive  $\mu^*$ . The next two steps are where countable subadditivity earns its place.

### Step 3. Additivity relative to a test set.

This is the technical engine of the proof. We claim: if  $E_1, \dots, E_N \in \mathcal{M}_{\mu^*}$  are *pairwise disjoint*, then for every test set  $A$ ,

$$\mu^*\left(A \cap \bigcup_{k=1}^N E_k\right) = \sum_{k=1}^N \mu^*(A \cap E_k). \quad (1.2.11)$$

Note the presence of  $A$ : this is stronger than additivity of  $\mu^*$  on  $\mathcal{M}_{\mu^*}$ , and the extra strength is exactly what Step 5 will need.

We argue by induction on  $N$ . The case  $N = 1$  is trivial. Assume (1.2.11) for  $N - 1$  and set  $F_N = \bigcup_{k \leq N} E_k$ , which lies in  $\mathcal{M}_{\mu^*}$  by Step 2. Apply measurability of  $E_N$  with test set  $A \cap F_N$ :

$$\mu^*(A \cap F_N) = \mu^*(A \cap F_N \cap E_N) + \mu^*(A \cap F_N \cap E_N^c).$$

Now use disjointness to simplify the two pieces. Since  $E_N \subseteq F_N$  we have  $F_N \cap E_N = E_N$ ; and since the  $E_k$  are pairwise disjoint,  $F_N \cap E_N^c = \bigcup_{k \leq N-1} E_k =:$

$F_{N-1}$ . Hence

$$\mu^*(A \cap F_N) = \mu^*(A \cap E_N) + \mu^*(A \cap F_{N-1}) = \mu^*(A \cap E_N) + \sum_{k=1}^{N-1} \mu^*(A \cap E_k)$$

by the inductive hypothesis, which is (1.2.11).

**Step 4.** *Disjointification.*

We record the standard device that reduces arbitrary countable unions to disjoint ones. Given any  $G_1, G_2, \dots \in \mathcal{M}_{\mu^*}$ , set

$$E_1 = G_1, \quad E_k = G_k \setminus \bigcup_{j=1}^{k-1} G_j \quad (k \geq 2).$$

Then: each  $E_k \in \mathcal{M}_{\mu^*}$ , because  $\mathcal{M}_{\mu^*}$  is an algebra (Step 2) and  $E_k$  is built from finitely many of the  $G_j$  by unions, complements and intersections; the  $E_k$  are pairwise disjoint, since for  $k < l$  we have  $E_l \cap G_k = \emptyset$  while  $E_k \subseteq G_k$ ; and  $\bigcup_k E_k = \bigcup_k G_k$ , since  $E_k \subseteq G_k$  gives  $\subseteq$ , while if  $x \in \bigcup_k G_k$  then  $x \in E_{k_0}$  for  $k_0$  the *least* index with  $x \in G_{k_0}$ .

**Step 5.**  $\mathcal{M}_{\mu^*}$  is closed under countable unions.

By Step 4 it suffices to treat a pairwise disjoint sequence  $(E_k)_{k \geq 1} \subseteq \mathcal{M}_{\mu^*}$ . Write

$$F_N = \bigcup_{k=1}^N E_k \in \mathcal{M}_{\mu^*}, \quad F = \bigcup_{k=1}^{\infty} E_k,$$

and let  $A$  be a test set. Since  $F_N \in \mathcal{M}_{\mu^*}$  by Step 2,

$$\mu^*(A) = \mu^*(A \cap F_N) + \mu^*(A \cap F_N^c).$$

We now estimate the two terms separately.

For the first, apply (1.2.11) from Step 3:  $\mu^*(A \cap F_N) = \sum_{k=1}^N \mu^*(A \cap E_k)$ .

For the second, note  $F_N \subseteq F$ , hence  $F^c \subseteq F_N^c$ , hence  $A \cap F^c \subseteq A \cap F_N^c$ , and so monotonicity gives  $\mu^*(A \cap F_N^c) \geq \mu^*(A \cap F^c)$ . This is the step that discards information, and it is the only inequality in the argument.

Combining,

$$\mu^*(A) \geq \sum_{k=1}^N \mu^*(A \cap E_k) + \mu^*(A \cap F^c) \quad \text{for every } N \in \mathbb{N}. \quad (1.2.12)$$

The left-hand side does not depend on  $N$ , and the right-hand side is nondecreasing in  $N$ , so we may let  $N \rightarrow \infty$ :

$$\mu^*(A) \geq \sum_{k=1}^{\infty} \mu^*(A \cap E_k) + \mu^*(A \cap F^c). \quad (1.2.13)$$

Finally, apply countable subadditivity to the decomposition  $A \cap F = \bigcup_k (A \cap E_k)$ , giving  $\sum_k \mu^*(A \cap E_k) \geq \mu^*(A \cap F)$ . Substituting into (1.2.13),

$$\mu^*(A) \geq \mu^*(A \cap F) + \mu^*(A \cap F^c),$$

so  $F \in \mathcal{M}_{\mu^*}$  by Lemma 1.2.10. With Step 1, this proves (a).

**Step 6.** *Countable additivity.*

Retain the notation of Step 5 and specialise the test set to  $A = F$ . Then  $F \cap E_k = E_k$  and  $F \cap F^c = \emptyset$ , so (1.2.13) becomes

$$\mu^*(F) \geq \sum_{k=1}^{\infty} \mu^*(E_k).$$

The reverse inequality is countable subadditivity. Hence  $\mu(F) = \sum_k \mu(E_k)$ , which is (b). Note that  $\mu(\emptyset) = \mu^*(\emptyset) = 0$ , so  $\mu$  is a bona fide measure.

**Step 7.** *Completeness.*

Suppose  $\mu^*(E) = 0$  and  $F \subseteq E$ . For any test set  $A$ , monotonicity gives

$$\mu^*(A \cap F) \leq \mu^*(F) \leq \mu^*(E) = 0, \quad \mu^*(A \cap F^c) \leq \mu^*(A),$$

and adding these,  $\mu^*(A \cap F) + \mu^*(A \cap F^c) \leq 0 + \mu^*(A) = \mu^*(A)$ . By Lemma 1.2.10,  $F \in \mathcal{M}_{\mu^*}$ , and  $\mu(F) = \mu^*(F) = 0$ . This is (c), and it did not use measurability of  $E$  at all — only  $\mu^*(E) = 0$ .  $\square$

**Where we are.** Three things are worth extracting from that proof.

*Where the quantifier was spent.* Step 2 applied the measurability of  $F$  to the test sets  $A \cap E$  and  $A \cap E^c$ , which are not of any prescribed form; Step 5 applied the measurability of  $F_N$  to a single fixed  $A$  for infinitely many  $N$ . Neither move is available under Attempt 2 of §2.4.1. This is the precise sense in which the strange-looking definition is the right one.

*Where countability was spent.* Steps 1–4 are finitary. Countable subadditivity of  $\mu^*$  entered only twice: at the very end of Step 5, and implicitly in Step 6.

*What we did not get.* Theorem 1.2.11 says nothing about whether  $\mathcal{M}_{\mu^*}$  is large. For all it tells us,  $\mathcal{M}_{\mu^*}$  could be  $\{\emptyset, X\}$ , and Drill 2.2 exhibits an outer measure for which it very nearly is. Establishing that the measurable sets include everything we care about is a separate task, carried out for  $m^*$  in Worked Example 2.7 above and, in the general metric setting, in §2.5 next.

## 1.2.5 Metric outer measures

Carathéodory's theorem produces a  $\sigma$ -algebra but, as just noted, gives no guarantee that it contains anything interesting. On a metric space there is a clean criterion ensuring that all Borel sets are measurable, and it will matter enormously in Volume 7 when we construct Hausdorff measure — where, unlike the Lebesgue case, there is no convenient covering family to run the argument of Worked Example 2.7.

**Definition 1.2.12.** An outer measure  $\mu^*$  on a metric space  $(X, d)$  is a *metric outer measure* if

$$\mu^*(A \cup B) = \mu^*(A) + \mu^*(B) \quad \text{whenever } \text{dist}(A, B) > 0,$$

where  $\text{dist}(A, B) = \inf\{d(x, y) : x \in A, y \in B\}$ .

The hypothesis asks for additivity only on pairs of sets that are *strictly separated* — not merely disjoint. That is a weak requirement, and it is typically easy to verify, because a covering set of small enough diameter simply cannot meet both.

**Theorem 1.2.13** (Carathéodory's criterion). *If  $\mu^*$  is a metric outer measure on  $(X, d)$ , then every Borel subset of  $X$  is  $\mu^*$ -measurable.*

*Idea of the proof.* Since the Borel  $\sigma$ -algebra is generated by the closed sets, and  $\mathcal{M}_{\mu^*}$  is a  $\sigma$ -algebra by Theorem 1.2.11, it is enough to show every closed set  $C$  is measurable.

The obstruction is immediate. Given a test set  $A$ , we would like to apply the metric hypothesis to the pair  $A \cap C$  and  $A \cap C^c$ , but these two sets can touch — a point of  $A \setminus C$  may lie arbitrarily close to  $C$  — so their distance is 0 and the hypothesis says nothing.

The fix is to retreat. Replace  $A \cap C^c$  by the smaller set  $B_j$  of points at distance at least  $1/j$  from  $C$ , which *is* strictly separated from  $A \cap C$ . Now the hypothesis applies and gives a clean inequality for each  $j$ . It remains to show that nothing is lost as  $j \rightarrow \infty$ , that is, that  $\mu^*(B_j) \rightarrow \mu^*(A \cap C^c)$ . This is the only real work, and it is done by decomposing the region between  $B_j$  and  $A \cap C^c$  into the dyadic-type annular shells  $D_j = B_{j+1} \setminus B_j$  and showing  $\sum_j \mu^*(D_j) < \infty$ , so that the tails vanish. The trick for summing them is that non-adjacent shells are strictly separated, so the metric hypothesis applies to the even-indexed shells and the odd-indexed shells separately.

This “peel the bad set into annuli around it” manoeuvre recurs constantly. You will see it again in Chapter 1.7 in the proof of the Hardy–Littlewood maximal inequality, where the annuli are dyadic in radius.

*Proof.* By the remarks above it suffices to fix a closed set  $C \subseteq X$  and a test set  $A$  with  $\mu^*(A) < \infty$ , and to prove

$$\mu^*(A) \geq \mu^*(A \cap C) + \mu^*(A \cap C^c). \quad (1.2.14)$$

**Step 1.** *The approximating sets.*

For  $j \geq 1$  define

$$B_j = \left\{ x \in A \setminus C : \text{dist}(x, C) \geq \frac{1}{j} \right\}.$$

Then  $B_1 \subseteq B_2 \subseteq \cdots$ , and

$$\bigcup_{j \geq 1} B_j = A \setminus C. \quad (1.2.15)$$

For (1.2.15): the inclusion  $\subseteq$  is clear. Conversely, if  $x \in A \setminus C$  then, because  $C$  is closed and  $x \notin C$ , we have  $\text{dist}(x, C) > 0$  — this is exactly where closedness is used — so  $\text{dist}(x, C) \geq 1/j$  for all large  $j$ .

**Step 2.** *The metric hypothesis applies to  $A \cap C$  and  $B_j$ .*

Every point of  $B_j$  is at distance  $\geq 1/j$  from every point of  $C$ , and  $A \cap C \subseteq C$ , so  $\text{dist}(A \cap C, B_j) \geq 1/j > 0$ . Hence

$$\mu^*(A \cap C) + \mu^*(B_j) = \mu^*((A \cap C) \cup B_j) \leq \mu^*(A),$$

the last step by monotonicity, since  $(A \cap C) \cup B_j \subseteq A$ . Thus

$$\mu^*(A \cap C) + \mu^*(B_j) \leq \mu^*(A) \quad \text{for every } j. \quad (1.2.16)$$

In view of (1.2.14) and (1.2.15), the proof will be complete once we show

$$\lim_{j \rightarrow \infty} \mu^*(B_j) = \mu^*(A \setminus C). \quad (1.2.17)$$

**Step 3.** *The shells and their summability.*

Set  $D_j = B_{j+1} \setminus B_j$  for  $j \geq 1$ , the set of points of  $A \setminus C$  whose distance to  $C$  lies in  $[\frac{1}{j+1}, \frac{1}{j})$ .

We claim that if  $|j - k| \geq 2$  then  $\text{dist}(D_j, D_k) > 0$ . Indeed, suppose  $k \geq j + 2$ , let  $x \in D_j$  and  $y \in D_k$ . Then  $\text{dist}(x, C) \geq \frac{1}{j+1}$  while  $\text{dist}(y, C) < \frac{1}{k} \leq \frac{1}{j+2}$ , so by the triangle inequality

$$d(x, y) \geq \text{dist}(x, C) - \text{dist}(y, C) > \frac{1}{j+1} - \frac{1}{j+2} = \frac{1}{(j+1)(j+2)} > 0,$$

a bound independent of  $x$  and  $y$ . (For the first inequality: for any  $c \in C$ ,  $d(x, c) \leq d(x, y) + d(y, c)$ , so  $\text{dist}(x, C) \leq d(x, y) + d(y, c)$  for all  $c$ , and taking the infimum over  $c$  gives  $\text{dist}(x, C) \leq d(x, y) + \text{dist}(y, C)$ .)

Consequently the sets  $D_2, D_4, D_6, \dots$  are pairwise strictly separated, so iterating the metric hypothesis and then using monotonicity gives, for each  $N$ ,

$$\sum_{j=1}^N \mu^*(D_{2j}) = \mu^*\left(\bigcup_{j=1}^N D_{2j}\right) \leq \mu^*(A),$$

since all the  $D_j$  lie in  $A$ . The same argument applies to the odd-indexed shells. Letting  $N \rightarrow \infty$  and adding,

$$\sum_{j=1}^{\infty} \mu^*(D_j) \leq 2\mu^*(A) < \infty. \quad (1.2.18)$$

This is the one place the hypothesis  $\mu^*(A) < \infty$  is used, and it is essential.

**Step 4.** *Passing to the limit.*

Since  $A \setminus C = B_j \cup \bigcup_{k \geq j} D_k$  — every point of  $A \setminus C$  not already in  $B_j$  lies in exactly one later shell — countable subadditivity gives, for every  $j$ ,

$$\mu^*(A \setminus C) \leq \mu^*(B_j) + \sum_{k \geq j} \mu^*(D_k).$$

By (1.2.18) the tail  $\sum_{k \geq j} \mu^*(D_k)$  tends to 0 as  $j \rightarrow \infty$ . Hence

$$\mu^*(A \setminus C) \leq \liminf_{j \rightarrow \infty} \mu^*(B_j).$$

The reverse inequality holds for every  $j$  by monotonicity, since  $B_j \subseteq A \setminus C$ . This proves (1.2.17).

**Step 5.** *Conclusion.*

Let  $j \rightarrow \infty$  in (1.2.16) and use (1.2.17):

$$\mu^*(A \cap C) + \mu^*(A \cap C^c) = \mu^*(A \cap C) + \mu^*(A \setminus C) \leq \mu^*(A),$$

which is (1.2.14). □

**WORKED EXAMPLE 1.2.4— TIER C: RESEARCH TASTE** **Hausdorff measure: the construction we will need in Volume 7**

The Kakeya conjecture asserts that a Besicovitch set in  $\mathbb{R}^n$  has *Hausdorff dimension*  $n$ , even though Chapter 1.1 promised us its Lebesgue measure can be 0. To make that statement one needs a family of measures finer than Lebesgue's, and Proposition 1.2.8 together with Theorem 1.2.13 delivers them immediately. We record the construction now, while the machinery is in front of us; the theory proper is Volume 7.

*The definition.* Fix  $s \geq 0$  and  $\delta > 0$ . For  $A \subseteq \mathbb{R}^n$  put

$$\mathcal{H}_\delta^s(A) = \inf \left\{ \sum_j (\text{diam } U_j)^s : A \subseteq \bigcup_j U_j, \text{diam } U_j \leq \delta \right\}.$$

By Proposition 1.2.8 — with  $\mathcal{E}$  the family of sets of diameter at most  $\delta$ , and  $\rho(U) = (\text{diam } U)^s$  — each  $\mathcal{H}_\delta^s$  is an outer measure. The hypotheses of the proposition hold:  $\emptyset$  has diameter 0, and  $\mathbb{R}^n$  is covered by countably many sets of diameter  $\leq \delta$ .

*Monotonicity in  $\delta$ , and the limit.* If  $\delta' < \delta$  then every  $\delta'$ -admissible cover is  $\delta$ -admissible, so the infimum defining  $\mathcal{H}_{\delta'}^s$  is over a *smaller* collection and therefore  $\mathcal{H}_{\delta'}^s(A) \geq \mathcal{H}_\delta^s(A)$ . So  $\mathcal{H}_\delta^s(A)$  increases as  $\delta$  decreases, and the limit

$$\mathcal{H}^s(A) = \lim_{\delta \rightarrow 0^+} \mathcal{H}_\delta^s(A) = \sup_{\delta > 0} \mathcal{H}_\delta^s(A)$$

exists in  $[0, \infty]$ . It is again an outer measure: properties (i) and (ii) pass to the limit trivially, and for (iii) one applies subadditivity of  $\mathcal{H}_\delta^s$  and then takes suprema, using that a supremum of sums of nondecreasing quantities is the sum of the suprema.

*Why the  $\delta \rightarrow 0$  limit is essential.* Without it the construction is worthless. If arbitrarily large covering sets were allowed, any bounded  $A$  could be covered by a single set of diameter  $\text{diam } A$ , giving  $\mathcal{H}_\infty^s(A) \leq (\text{diam } A)^s$  — a quantity that sees only the overall extent of  $A$  and nothing of its internal structure. Forcing the covering sets to be small is exactly what makes the quantity detect fine geometry. This is why the *Keakeya* problem is a problem about small scales: all the content is in the  $\delta \rightarrow 0$  asymptotics, and in Volume 9 the parameter  $\delta$  will be reincarnated as the width of a tube.

*$\mathcal{H}^s$  is a metric outer measure.* Suppose  $\text{dist}(A, B) = d > 0$  and take any  $\delta < d/2$ . Let  $\{U_j\}$  be a  $\delta$ -admissible cover of  $A \cup B$ . No single  $U_j$  can meet both  $A$  and  $B$ : if  $x \in U_j \cap A$  and  $y \in U_j \cap B$  then  $d \leq d(x, y) \leq \text{diam } U_j \leq \delta < d/2$ , absurd. So the cover splits into a subfamily covering  $A$  and a subfamily covering  $B$  (discarding any  $U_j$  meeting neither), whence  $\mathcal{H}_\delta^s(A) + \mathcal{H}_\delta^s(B) \leq \sum_j (\text{diam } U_j)^s$ . Taking the infimum over covers gives  $\mathcal{H}_\delta^s(A) + \mathcal{H}_\delta^s(B) \leq \mathcal{H}_\delta^s(A \cup B)$ , and the reverse is subadditivity. Letting  $\delta \rightarrow 0$  preserves the equality.

By Theorem 1.2.13, therefore, *every Borel set is  $\mathcal{H}^s$ -measurable* — which is precisely the licence we need in order to speak of the Hausdorff measure of a Besicovitch set, an object with no reason whatsoever to be nice.



*The dimension jump.* Suppose  $\mathcal{H}^s(A) < \infty$  and let  $t > s$ . Given a  $\delta$ -admissible cover  $\{U_j\}$  with  $\sum_j (\text{diam } U_j)^s \leq \mathcal{H}^s(A) + 1$ , we may estimate

$$\sum_j (\text{diam } U_j)^t = \sum_j (\text{diam } U_j)^{t-s} (\text{diam } U_j)^s \leq \delta^{t-s} \sum_j (\text{diam } U_j)^s \leq \delta^{t-s} (\mathcal{H}^s(A) + 1),$$

using  $\text{diam } U_j \leq \delta$  and  $t - s > 0$ . Hence  $\mathcal{H}_\delta^t(A) \leq \delta^{t-s} (\mathcal{H}^s(A) + 1) \rightarrow 0$  as  $\delta \rightarrow 0$ , so  $\mathcal{H}^t(A) = 0$ .

So the function  $s \mapsto \mathcal{H}^s(A)$  drops from  $\infty$  to 0 at a single critical exponent, and we may define

$$\dim_{\mathcal{H}} A = \inf\{s \geq 0 : \mathcal{H}^s(A) = 0\},$$

the *Hausdorff dimension* of  $A$ .

*The statement of the conjecture.* The Kakeya conjecture, proved in  $\mathbb{R}^3$  by Wang and Zahl in 2025 and open for  $n \geq 4$ , asserts:

every Besicovitch set  $K \subseteq \mathbb{R}^n$  satisfies  $\dim_{\mathcal{H}} K = n$ .

You now understand this statement completely: Definition 1.1.1 defined the sets, and the two pages above defined the dimension. Note how sharp the tension is with Chapter 1.1: such a  $K$  can have  $m(K) = 0$ , so  $\mathcal{H}^n(K) = 0$  as well, and the conjecture nonetheless asserts that  $\mathcal{H}^s(K) = \infty$  for every  $s < n$ . Proving it will take us twelve more volumes.

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## 1.2.6 Premeasures and the extension theorem<sup>†</sup>

This section makes the construction of Proposition 1.2.8 canonical: if one starts from a set function already additive on an algebra, the outer measure it generates extends it, and the extension is unique under a  $\sigma$ -finiteness hypothesis. The material is standard and is used in Volume 6 (construction of surface measure on a hypersurface) but nowhere before then, so it may be skipped on first reading.

**Definition 1.2.14.** Let  $\mathcal{A}$  be an algebra of subsets of  $X$ . A *premeasure* is a map  $\mu_0 : \mathcal{A} \rightarrow [0, \infty]$  with  $\mu_0(\emptyset) = 0$  which is countably additive on  $\mathcal{A}$ : whenever  $(E_k) \subseteq \mathcal{A}$  are pairwise disjoint and  $\bigcup_k E_k$  happens to lie in  $\mathcal{A}$ , then  $\mu_0(\bigcup_k E_k) = \sum_k \mu_0(E_k)$ .

**Theorem 1.2.15** (Carathéodory extension theorem). *Let  $\mu_0$  be a premeasure on an algebra  $\mathcal{A} \subseteq \mathcal{P}(X)$ , and let  $\mu^*$  be the outer measure it generates via Proposition 1.2.8 with  $\mathcal{E} = \mathcal{A}$ ,  $\rho = \mu_0$ . Then:*

- (i)  $\mu^*|_{\mathcal{A}} = \mu_0$ ;
- (ii) every  $E \in \mathcal{A}$  is  $\mu^*$ -measurable, so  $\mu = \mu^*|_{\mathcal{M}_{\mu^*}}$  is a measure extending  $\mu_0$  to a  $\sigma$ -algebra containing  $\sigma(\mathcal{A})$ ;
- (iii) if  $\mu_0$  is  $\sigma$ -finite, the extension to  $\sigma(\mathcal{A})$  is unique.

*Idea of the proof.* Part (i) says the extension really is an extension: the covering procedure does not lose information on sets where  $\mu_0$  was already defined. The inequality  $\leq$  is free, since  $E$  covers itself. For  $\geq$  one must show no cover of  $E$  by  $\mathcal{A}$ -sets is cheaper than  $\mu_0(E)$ , and this is exactly where countable additivity of the premeasure — as opposed to mere finite additivity — is spent.

Part (ii) is a verification of (1.2.3), and it follows the template of Worked Example 1.2.3 precisely: take an efficient cover of the test set, cut the *cover* rather than the set, and use that cutting is exactly additive on the covering family. There, that was the box-splitting identity (1.2.5); here it is finite additivity of  $\mu_0$ .

Part (iii) is the only place  $\sigma$ -finiteness is needed, and the mechanism is worth naming in advance: any competing extension  $\nu$  is automatically *dominated* by  $\mu$ , because  $\nu$  must respect covers. Upgrading domination to equality requires subtracting, and subtraction requires finiteness.

*Proof. Step 1. Monotonicity and subadditivity of  $\mu_0$  on  $\mathcal{A}$ .*

We record two consequences of the definition. If  $E \subseteq F$  with  $E, F \in \mathcal{A}$  then  $F = E \sqcup (F \setminus E)$  with both pieces in  $\mathcal{A}$ , so  $\mu_0(F) = \mu_0(E) + \mu_0(F \setminus E) \geq \mu_0(E)$ . And if  $E \subseteq \bigcup_k E_k$  with all sets in  $\mathcal{A}$  and  $E \in \mathcal{A}$ , then setting  $F_k = E \cap E_k \setminus \bigcup_{j < k} E_j \in \mathcal{A}$  gives pairwise disjoint sets with  $\bigcup_k F_k = E \in \mathcal{A}$  and  $F_k \subseteq E_k$ , whence by countable additivity and monotonicity

$$\mu_0(E) = \sum_k \mu_0(F_k) \leq \sum_k \mu_0(E_k). \quad (1.2.19)$$

**Step 2.** *Proof of (i):  $\mu^*|_{\mathcal{A}} = \mu_0$ .*

Let  $E \in \mathcal{A}$ . The family  $\{E, \emptyset, \emptyset, \dots\}$  is an admissible cover, so  $\mu^*(E) \leq \mu_0(E)$ . Conversely, any admissible cover  $\{E_k\} \subseteq \mathcal{A}$  of  $E$  satisfies  $\sum_k \mu_0(E_k) \geq \mu_0(E)$  by (1.2.19); taking the infimum,  $\mu^*(E) \geq \mu_0(E)$ .

**Step 3.** *Proof of (ii):  $\mathcal{A} \subseteq \mathcal{M}_{\mu^*}$ .*

Fix  $E \in \mathcal{A}$  and a test set  $A$  with  $\mu^*(A) < \infty$ ; by Lemma 1.2.10 we must show  $\mu^*(A \cap E) + \mu^*(A \cap E^c) \leq \mu^*(A)$ .

Let  $\varepsilon > 0$  and choose  $\{E_k\} \subseteq \mathcal{A}$  covering  $A$  with  $\sum_k \mu_0(E_k) \leq \mu^*(A) + \varepsilon$ . Since  $\mathcal{A}$  is an algebra, each  $E_k \cap E$  and  $E_k \cap E^c$  lies in  $\mathcal{A}$ , and they are disjoint with union  $E_k$ , so finite additivity gives

$$\mu_0(E_k) = \mu_0(E_k \cap E) + \mu_0(E_k \cap E^c).$$

The families  $\{E_k \cap E\}$  and  $\{E_k \cap E^c\}$  cover  $A \cap E$  and  $A \cap E^c$  respectively, so by the definition of  $\mu^*$  as an infimum,

$$\mu^*(A \cap E) + \mu^*(A \cap E^c) \leq \sum_k \mu_0(E_k \cap E) + \sum_k \mu_0(E_k \cap E^c) = \sum_k \mu_0(E_k) \leq \mu^*(A) + \varepsilon.$$

Let  $\varepsilon \downarrow 0$ . With Theorem 1.2.11,  $\mu = \mu^*|_{\mathcal{M}_{\mu^*}}$  is a complete measure, and since  $\mathcal{M}_{\mu^*}$  is a  $\sigma$ -algebra containing  $\mathcal{A}$  it contains  $\sigma(\mathcal{A})$  by Proposition 1.2.2. This proves (ii).

▷ *Goal: (iii), uniqueness on  $\sigma(\mathcal{A})$  under  $\sigma$ -finiteness.*

**Step 4.** *Any extension is dominated.*

Let  $\nu$  be a measure on  $\sigma(\mathcal{A})$  with  $\nu|_{\mathcal{A}} = \mu_0$ , and let  $E \in \sigma(\mathcal{A})$ . For any admissible cover  $\{E_k\} \subseteq \mathcal{A}$  of  $E$ , countable subadditivity of  $\nu$  (Proposition 1.2.6(ii)) gives

$$\nu(E) \leq \sum_k \nu(E_k) = \sum_k \mu_0(E_k).$$

Taking the infimum over covers,  $\nu(E) \leq \mu^*(E) = \mu(E)$ .

**Step 5.** *Agreement on sets of finite measure.*

Suppose  $\mu(E) < \infty$ . Fix  $\varepsilon > 0$  and choose a cover  $\{E_k\} \subseteq \mathcal{A}$  of  $E$  with  $\sum_k \mu_0(E_k) \leq \mu(E) + \varepsilon$ . Put  $F = \bigcup_k E_k \in \sigma(\mathcal{A})$ , so  $E \subseteq F$  and, by subadditivity,  $\mu(F) \leq \mu(E) + \varepsilon$ .

We claim  $\nu(F) = \mu(F)$ . Indeed the partial unions  $F_N = \bigcup_{k \leq N} E_k$  lie in  $\mathcal{A}$ , since  $\mathcal{A}$  is an algebra, so  $\nu(F_N) = \mu_0(F_N) = \mu(F_N)$ ; and  $F_N \uparrow F$ , so continuity from below (Proposition 1.2.6(iii)), applied to  $\nu$  and to  $\mu$  separately, gives  $\nu(F) = \lim_N \nu(F_N) = \lim_N \mu(F_N) = \mu(F)$ .

Now split  $F = E \sqcup (F \setminus E)$  for both measures and use Step 4 twice:

$$\mu(E) \leq \mu(F) = \nu(F) = \nu(E) + \nu(F \setminus E) \leq \nu(E) + \mu(F \setminus E) \leq \nu(E) + \varepsilon,$$

the last step because  $\mu(F \setminus E) = \mu(F) - \mu(E) \leq \varepsilon$ , a subtraction licensed by  $\mu(E) < \infty$ . Letting  $\varepsilon \downarrow 0$  gives  $\mu(E) \leq \nu(E)$ , and with Step 4, equality.

**Step 6.** *The  $\sigma$ -finite case.*

Write  $X = \bigcup_k X_k$  with  $X_k \in \mathcal{A}$  and  $\mu_0(X_k) < \infty$ ; replacing  $X_k$  by  $X_k \setminus \bigcup_{j < k} X_j$ , which stays in the algebra, we may take the  $X_k$  pairwise disjoint. For  $E \in \sigma(\mathcal{A})$  each  $E \cap X_k$  has  $\mu(E \cap X_k) \leq \mu_0(X_k) < \infty$ , so Step 5 applies and  $\mu(E \cap X_k) = \nu(E \cap X_k)$ . Summing over  $k$  by countable additivity gives  $\mu(E) = \nu(E)$ .  $\square$

**Where we are.** Notice that uniqueness did *not* require a monotone class or  $\pi$ - $\lambda$  argument: Steps 4 and 5 got there directly, because the outer measure is by construction the infimum over covers and any competing extension must respect covers. The monotone class lemma is genuinely needed elsewhere — it appears in Chapter 1.6 to build the product measure, where no covering description is available — and is proved there.

## Drills

**Drill 1.2.1.** Show that if  $\mu^*$  is an outer measure and  $\mu^*(E) = 0$ , then for every  $A \subseteq X$  we have  $\mu^*(A \cup E) = \mu^*(A) = \mu^*(A \setminus E)$ . Deduce Step 7 of Theorem 1.2.11 a second way.

**Drill 1.2.2.** Let  $X = \{1, 2, 3\}$  and define  $\mu^*(\emptyset) = 0$ ,  $\mu^*(X) = 2$ , and  $\mu^*(A) = 1$  for every other  $A \subseteq X$ . Verify that  $\mu^*$  is an outer measure, and compute  $\mathcal{M}_{\mu^*}$  explicitly. (You should find  $\mathcal{M}_{\mu^*} = \{\emptyset, X\}$ . This is the promised warning that Theorem 1.2.11 alone guarantees nothing whatsoever about the richness of the measurable family.)

**Drill 1.2.3.** In the notation of Step 3 of Theorem 1.2.13, show that the conclusion  $\text{dist}(D_j, D_k) > 0$  genuinely fails for adjacent shells  $k = j + 1$ , by exhibiting a closed set  $C \subseteq \mathbb{R}$  and points of  $D_j, D_{j+1}$  arbitrarily close together. This is why the even and odd shells had to be treated separately.

**Drill 1.2.4.** Show that  $\mathcal{H}^0$  is counting measure. Then show that on  $\mathbb{R}$ ,  $\mathcal{H}^1 = m^*$ . (The second direction requires the box lemma deferred to Chapter 1.3; do the first now and return to the second later.)

## Notes

Carathéodory's criterion dates to his 1914 paper on the theory of outer measure, and Theorem 1.2.13 to the same circle of ideas. The presentation here follows Folland in outline, with two deliberate departures. First, the metric outer measure section has been promoted from an aside to a main result, because of its role in Volume 7: for Hausdorff measure there is no analogue of the argument of Worked Example 2.7, and Theorem 1.2.13 is the only route to measurability of Borel sets. Second, we have given the failed Attempts 1 and 2 of §2.4.1 explicitly, which most treatments omit.

Readers who have seen measure theory built via the “open set approximation” definition of measurability, as in Stein–Shakarchi, should note that the two approaches agree on  $\mathbb{R}^n$ ; we prove this as Theorem 1.3.6. The Carathéodory

route is preferred here purely because it transfers to Hausdorff measure unchanged, whereas the open-set definition does not.



## Chapter 1.3

# Lebesgue Measure on $\mathbb{R}^n$

Chapter 1.2 was abstract. We built outer measures from arbitrary covering families, and we proved that the Carathéodory criterion always extracts a  $\sigma$ -algebra from them. But we proved almost nothing about the specific outer measure we care about. In particular we still do not know that  $m^*(Q) = |Q|$  for a box  $Q$  — the Pitfall box in §2.3 warned that this is a theorem, not a definition — and without it the entire construction could, for all we have shown, be identically zero.

This chapter repairs that, and then goes considerably further. By the end we will have: the box lemma; a complete structural description of the measurable sets; the behaviour of  $m$  under translations and linear maps; a proof that non-measurable sets exist, so that the restriction to  $\mathcal{M}_{m^*}$  was genuinely forced; and, in §3.6, the Perron tree — a construction which shows that a plane set containing a unit segment in every direction can have area less than any prescribed  $\varepsilon$ . That last result is the reason this series exists.

### 1.3.1 The box lemma

#### 1.3.1.1 Elementary volume on grids

Everything rests on a completely elementary fact about products, which we isolate because it is used three times and is invisible if left implicit.

**Lemma 1.3.1** (Grid additivity). *Let  $Q = \prod_{i=1}^n (a_i, b_i)$  be an open box. For each coordinate  $i$  choose a finite partition*

$$a_i = t_0^i < t_1^i < \cdots < t_{m_i}^i = b_i,$$

*and let  $\mathcal{R}$  be the resulting family of  $m_1 m_2 \cdots m_n$  open sub-boxes  $R_{k_1 \dots k_n} = \prod_{i=1}^n (t_{k_i-1}^i, t_{k_i}^i)$ . Then*

$$|Q| = \sum_{R \in \mathcal{R}} |R|.$$

*Proof.* For each  $i$ , the partition gives  $b_i - a_i = \sum_{k_i=1}^{m_i} (t_{k_i}^i - t_{k_i-1}^i)$ , since the sum telescopes. Multiply these  $n$  identities together and expand the product of sums by the distributive law:

$$|Q| = \prod_{i=1}^n (b_i - a_i) = \prod_{i=1}^n \left( \sum_{k_i=1}^{m_i} (t_{k_i}^i - t_{k_i-1}^i) \right) = \sum_{k_1=1}^{m_1} \cdots \sum_{k_n=1}^{m_n} \prod_{i=1}^n (t_{k_i}^i - t_{k_i-1}^i),$$

and the summand is exactly  $|R_{k_1 \dots k_n}|$ .  $\square$

*Remark 1.3.2.* The sub-boxes in Lemma 1.3.1 are open, so they do not cover  $Q$ : the grid hyperplanes are omitted. This is harmless here because we are computing elementary volumes, not measures, and the identity above is an identity between products of real numbers. It will become genuinely convenient in Lemma 1.3.3, where we want to talk about “the cells contained in a box” without worrying about shared faces.

### 1.3.1.2 The finite covering lemma

**Lemma 1.3.3** (Finite covers of a box). *Let  $K$  be a closed box and let  $Q_1, \dots, Q_N$  be open boxes with  $K \subseteq \bigcup_{j=1}^N Q_j$ . Then*

$$|K| \leq \sum_{j=1}^N |Q_j|.$$

*Scratch work.* The obstruction is that the  $Q_j$  overlap arbitrarily, so we cannot simply add their volumes and compare. The standard device is to *impose a common grid*: extend every face of every box (and of  $K$ ) to a full hyperplane. These finitely many hyperplanes chop  $\mathbb{R}^n$  into finitely many cells, and each of  $K, Q_1, \dots, Q_N$  becomes a union of cells. Volume is additive over cells by Lemma 1.3.1, so the comparison becomes a purely combinatorial statement about which cells lie in which boxes — and there the covering hypothesis does the work directly.

*Proof.* Write  $K = \prod_i [c_i, d_i]$  and  $Q_j = \prod_i (a_i^j, b_i^j)$ .

**Step 1.** *Build the grid.*

For each coordinate  $i \in \{1, \dots, n\}$ , let  $T_i \subseteq \mathbb{R}$  be the finite set consisting of  $c_i, d_i$ , and all the endpoints  $a_i^j, b_i^j$  for  $j = 1, \dots, N$ . List  $T_i$  in increasing order. The points of  $T_i$  partition the smallest interval containing them into finitely many open intervals, and taking products over  $i$  produces a finite family  $\mathcal{R}$  of open cells.



By construction, each of  $K$  and each  $Q_j$  has all of its faces on grid hyperplanes. Consequently, by Lemma 1.3.1 applied to each of them separately,

$$|K| = \sum_{R \in \mathcal{R}, R \subseteq K} |R|, \quad |Q_j| = \sum_{R \in \mathcal{R}, R \subseteq Q_j} |R|. \quad (1.3.1)$$

(For  $K$ , note that the interior of  $K$  is the union of the cells contained in  $K$  together with parts of grid hyperplanes, which contribute nothing to the elementary-volume identity of Lemma 1.3.1.)

**Step 2.** *Every cell inside  $K$  is inside some  $Q_j$ .*

Let  $R \in \mathcal{R}$  with  $R \subseteq K$ , and pick any point  $x \in R$ . Since  $R \subseteq K \subseteq \bigcup_j Q_j$ , there is some  $j$  with  $x \in Q_j$ . We claim  $R \subseteq Q_j$ .

Indeed, fix a coordinate  $i$  and write  $R$ 's  $i$ -th edge as  $(t_{k-1}, t_k)$  with  $t_{k-1}, t_k \in T_i$  consecutive. We know  $a_i^j < x_i < b_i^j$ , and  $a_i^j, b_i^j \in T_i$  while  $x_i \in (t_{k-1}, t_k)$ . Since  $t_{k-1}$  and  $t_k$  are *consecutive* in  $T_i$ , no element of  $T_i$  lies strictly between them; as  $a_i^j < x_i$  and  $a_i^j \in T_i$  we must have  $a_i^j \leq t_{k-1}$ , and similarly  $b_i^j \geq t_k$ . Hence  $(t_{k-1}, t_k) \subseteq (a_i^j, b_i^j)$ . As  $i$  was arbitrary,  $R \subseteq Q_j$ .

**Step 3.** *Count.*

By Step 2, each cell  $R \subseteq K$  appears in at least one of the sums  $\sum_{R \subseteq Q_j} |R|$ . Since all terms are nonnegative, allowing a cell to be counted several times only increases the total, so

$$|K| = \sum_{R \subseteq K} |R| \leq \sum_{j=1}^N \sum_{R \subseteq Q_j} |R| = \sum_{j=1}^N |Q_j|,$$

using (1.3.1) at both ends. □

### 1.3.1.3 The box lemma itself

**Theorem 1.3.4** (Box lemma). *For every box  $Q \subseteq \mathbb{R}^n$  — open, closed, or half-open —*

$$m^*(Q) = |Q|.$$

*Idea of the proof.* The inequality  $m^*(Q) \leq |Q|$  is a one-line consequence of the definition:  $Q$  is (nearly) a cover of itself.

For the reverse we must rule out the possibility that some ingenious countable family of boxes covers  $Q$  more cheaply than  $|Q|$ . The only handle we have on a *countable* family is Lemma 1.3.3, which handles finite ones, so the entire content of the proof is the reduction from countable to finite. That reduction is compactness — Heine–Borel — and it costs two  $\varepsilon$ 's: one to shrink  $Q$  to a compact box strictly inside it, and one to fatten each

covering box slightly so that a cover by arbitrary boxes becomes a cover by open sets. Neither  $\varepsilon$  can be avoided, and it is worth noticing that this is the first and only place in the construction of Lebesgue measure where a topological property of  $\mathbb{R}^n$  is used.

*Proof. Step 1. The upper bound.*

Suppose first  $Q$  is open. Then  $\{Q, \emptyset, \emptyset, \dots\}$  is an admissible cover of  $Q$  in the sense of (1.2.1), so  $m^*(Q) \leq |Q|$ . For general  $Q$  with edge lengths  $\ell_i = b_i - a_i$  and any  $\eta > 0$ , the open box  $Q^\eta = \prod_i (a_i - \eta, b_i + \eta)$  contains  $Q$ , so by monotonicity

$$m^*(Q) \leq m^*(Q^\eta) \leq |Q^\eta| = \prod_i (\ell_i + 2\eta) \xrightarrow{\eta \rightarrow 0} \prod_i \ell_i = |Q|.$$

▷ *Goal:*  $m^*(Q) \geq |Q|$ .

**Step 2.** *Reduce to a compact box.*

Fix  $\varepsilon > 0$ . Choose a closed box  $K \subseteq Q$  with

$$|K| \geq |Q| - \varepsilon; \tag{1.3.2}$$

concretely, if  $Q$  has edges  $(a_i, b_i)$  take  $K = \prod_i [a_i + \eta, b_i - \eta]$  for  $\eta$  small, which is legitimate provided  $|Q| > 0$  (if  $|Q| = 0$  there is nothing to prove). By monotonicity it suffices to bound  $m^*(K)$  from below, since  $m^*(Q) \geq m^*(K)$ .

**Step 3.** *Fatten an arbitrary cover.*

Let  $\{Q_j\}_{j \geq 1}$  be any countable family of open boxes with  $K \subseteq \bigcup_j Q_j$ . (The definition of  $m^*$  uses open boxes, so no fattening is strictly needed here; we record the step because the same proof is run later for covers by arbitrary boxes, where one replaces  $Q_j$  by an open box  $Q_j^{\eta_j} \supseteq Q_j$  with  $|Q_j^{\eta_j}| \leq |Q_j| + \varepsilon 2^{-j}$ , at a total cost of  $\varepsilon$ .)

**Step 4.** *Extract a finite subcover.*

$K$  is closed and bounded, hence compact by Heine–Borel, and  $\{Q_j\}$  is an open cover of  $K$ . So there is  $N$  with  $K \subseteq \bigcup_{j=1}^N Q_j$ .

**Step 5.** *Apply the finite lemma and unwind.*

By Lemma 1.3.3 and then (1.3.2),

$$\sum_{j=1}^{\infty} |Q_j| \geq \sum_{j=1}^N |Q_j| \geq |K| \geq |Q| - \varepsilon.$$

The left-hand side is the cost of an arbitrary admissible cover of  $K$ , so taking the infimum over all such covers gives  $m^*(K) \geq |Q| - \varepsilon$ , whence  $m^*(Q) \geq |Q| - \varepsilon$ . Let  $\varepsilon \downarrow 0$ .  $\square$

**Where we are.** We now know the construction is not degenerate:  $m^*$  assigns to boxes exactly their elementary volume. Combining with Worked Example 2.7 — half-spaces are measurable, hence so are all boxes, open sets and Borel sets — and with Theorem 1.2.11, we have a complete, countably additive, translation-invariant measure defined on a  $\sigma$ -algebra containing the Borel sets and assigning boxes their volume. Requirements (M1) and (M3) of §2.1 are met. Requirement (M2) is Proposition 1.3.8 below, and the price we paid — that  $\mathcal{M}_{m^*}$  is not all of  $\mathcal{P}(\mathbb{R}^n)$  — is shown to be unavoidable in §3.4.

**Notation 1.3.5.** From now on we write  $m(E)$  or  $|E|$  for  $m^*(E)$  when  $E \in \mathcal{M}_{m^*}$ , and call such  $E$  *measurable* without qualification. We write  $\mathcal{L}$  for  $\mathcal{M}_{m^*}$  and  $\mathcal{B}$  for the Borel  $\sigma$ -algebra, so  $\mathcal{B} \subseteq \mathcal{L}$ . Sets with  $m(E) = 0$  are called *null*, and “almost everywhere” (a.e.) means “off a null set”.

## 1.3.2 Regularity and the structure of measurable sets

Carathéodory’s definition is excellent for proofs but useless for intuition: it tells you nothing about what a measurable set *looks like*. The following theorem does, and it is the result one actually uses.

**Theorem 1.3.6** (Outer regularity and the structure theorem). *For  $E \subseteq \mathbb{R}^n$  the following are equivalent.*

- (i)  $E$  is measurable.
- (ii) For every  $\varepsilon > 0$  there is an open set  $U \supseteq E$  with  $m^*(U \setminus E) < \varepsilon$ .
- (iii) There is a  $G_\delta$  set  $G \supseteq E$  with  $m^*(G \setminus E) = 0$ .
- (iv) For every  $\varepsilon > 0$  there is a closed set  $C \subseteq E$  with  $m^*(E \setminus C) < \varepsilon$ .
- (v) There is an  $F_\sigma$  set  $F \subseteq E$  with  $m^*(E \setminus F) = 0$ .

*In words:* a measurable set is a Borel set modified by a null set.

**Pitfall:**  $m^*(U \setminus E)$ , **not**  $m^*(U) - m^*(E)$

Condition (ii) must be stated with the outer measure of the difference. The subtraction  $m^*(U) - m^*(E)$  is meaningless before we know  $E$  is measurable — outer measure is not additive — and it is also useless when  $m^*(E) = \infty$ . Getting this wrong makes (ii) trivially true for every set of infinite outer measure, which would prove that every such set is measurable, a statement refuted in §3.4.

*Proof.* We prove (ii) $\Rightarrow$ (iii) $\Rightarrow$ (i) $\Rightarrow$ (ii), and then obtain (iv), (v) by complementation.

**Step 1.** (ii)  $\Rightarrow$  (iii).

For each  $k \in \mathbb{N}$  apply (ii) with  $\varepsilon = 1/k$  to get an open  $U_k \supseteq E$  with  $m^*(U_k \setminus E) < 1/k$ . Put  $G = \bigcap_{k \geq 1} U_k$ , a  $G_\delta$  set containing  $E$ . For every  $k$  we have  $G \setminus E \subseteq U_k \setminus E$ , so monotonicity gives  $m^*(G \setminus E) < 1/k$ ; letting  $k \rightarrow \infty$ ,  $m^*(G \setminus E) = 0$ .

**Step 2.** (iii)  $\Rightarrow$  (i).

Write  $N = G \setminus E$ , so  $m^*(N) = 0$  and  $E = G \setminus N$ .

Now  $G$  is a  $G_\delta$ , hence Borel, hence measurable (Worked Example 2.7 plus Theorem 1.2.11). And  $N$  is measurable because  $m$  is a *complete* measure: Theorem 1.2.11(c) says every set of outer measure zero is measurable, with no further hypothesis. Since  $\mathcal{L}$  is a  $\sigma$ -algebra it is closed under set differences, so  $E = G \setminus N \in \mathcal{L}$ .

*Remark 1.3.7.* This step is where completeness of the Carathéodory construction earns its keep. Had we built Lebesgue measure only on the Borel  $\sigma$ -algebra, the set  $E$  here need not be Borel — there are subsets of the Cantor set (Worked Example 3.4) that are Lebesgue measurable but not Borel — and the theorem would be false as stated.

**Step 3.** (i)  $\Rightarrow$  (ii), *finite measure case*.

Assume  $E$  measurable with  $m(E) < \infty$ , and fix  $\varepsilon > 0$ . By the definition of  $m^*$  as an infimum, there are open boxes  $\{Q_j\}$  with

$$E \subseteq \bigcup_j Q_j =: U \quad \text{and} \quad \sum_j |Q_j| \leq m(E) + \varepsilon.$$

Then  $U$  is open,  $U \supseteq E$ , and by countable subadditivity together with the box lemma,  $m(U) \leq \sum_j |Q_j| \leq m(E) + \varepsilon$ .

Both  $U$  and  $E$  are measurable and  $E \subseteq U$ , so additivity applies to the disjoint decomposition  $U = E \cup (U \setminus E)$ :

$$m(U) = m(E) + m(U \setminus E).$$

Because  $m(E) < \infty$  we may subtract it, obtaining  $m(U \setminus E) = m(U) - m(E) \leq \varepsilon$ .

**Step 4.** (i)  $\Rightarrow$  (ii), *general case*.

Now let  $E$  be measurable of possibly infinite measure. Decompose  $\mathbb{R}^n$  into the disjoint annuli  $A_k = \{x : k-1 \leq |x| < k\}$ ,  $k \geq 1$ , and set  $E_k = E \cap A_k$ . Each  $E_k$  is measurable and bounded, hence of finite measure. By Step 3

choose open  $U_k \supseteq E_k$  with  $m(U_k \setminus E_k) < \varepsilon 2^{-k}$ , and put  $U = \bigcup_k U_k$ , an open set containing  $E$ .

The point is the inclusion

$$U \setminus E \subseteq \bigcup_{k \geq 1} (U_k \setminus E_k),$$

valid because if  $x \in U \setminus E$  then  $x \in U_k$  for some  $k$ , and  $x \notin E \supseteq E_k$ . Countable subadditivity now gives  $m(U \setminus E) \leq \sum_k \varepsilon 2^{-k} = \varepsilon$ .

**Step 5.** (iv) and (v).

Apply the equivalence already proved to  $E^c$ , which is measurable iff  $E$  is. Given  $\varepsilon > 0$  choose open  $U \supseteq E^c$  with  $m^*(U \setminus E^c) < \varepsilon$  and set  $C = U^c$ , a closed subset of  $E$ . Since  $E \setminus C = E \cap U = U \setminus E^c$ , we get  $m^*(E \setminus C) < \varepsilon$ , which is (iv). Taking a countable union of such  $C$  gives (v).  $\square$

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**WORKED EXAMPLE 1.3.1— TIER A   A dense null set, and an open dense set of small measure**

These two examples should be held in mind permanently: measure and topology measure completely different things, and neither controls the other.

(a) *A dense set of measure zero.*  $\mathbb{Q}^n$  is dense in  $\mathbb{R}^n$  and  $m(\mathbb{Q}^n) = 0$  (Worked Example 2.5). So a set can be everywhere and still be negligible.

(b) *An open dense set of measure  $< \varepsilon$ .* Enumerate  $\mathbb{Q} \cap [0, 1]$  as  $q_1, q_2, \dots$  and put

$$U = \bigcup_{k=1}^{\infty} (q_k - \varepsilon 2^{-k-1}, q_k + \varepsilon 2^{-k-1}).$$

Then  $U$  is open (a union of intervals) and dense in  $[0, 1]$  (it contains every rational). By countable subadditivity and the box lemma,

$$m(U) \leq \sum_{k=1}^{\infty} \varepsilon 2^{-k} = \varepsilon.$$

So  $[0, 1] \setminus U$  is a closed set with empty interior — it contains no interval, since  $U$  is dense — yet its measure is at least  $1 - \varepsilon$ . A set can be topologically tiny (nowhere dense) and measure-theoretically enormous.

*Why this matters later.* Every intuition of the form “a set of small measure must be small” is wrong, and the Kakeya problem is the definitive illustration: §3.6 produces a set of measure  $< \varepsilon$  that contains a unit segment pointing in *every* direction. If you find that impossible to picture, example (b) is the honest place to recalibrate — it already shows that a set of measure  $1 - \varepsilon$  can fail to contain a single interval.

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### 1.3.3 Translations, dilations, and linear maps

**Proposition 1.3.8** (Translation invariance). *For every  $E \subseteq \mathbb{R}^n$  and  $x \in \mathbb{R}^n$ ,  $m^*(E + x) = m^*(E)$ . Moreover  $E \in \mathcal{L}$  if and only if  $E + x \in \mathcal{L}$ , and then  $m(E + x) = m(E)$ .*

*Proof.* The map  $Q \mapsto Q + x$  is a bijection of the family of open boxes onto itself preserving elementary volume, since translating an interval  $(a_i, b_i)$  to  $(a_i + x_i, b_i + x_i)$  does not change its length. Hence  $\{Q_j\}$  is an admissible cover of  $E$  of cost  $c$  if and only if  $\{Q_j + x\}$  is an admissible cover of  $E + x$  of the same cost  $c$ . The two infima in (1.2.1) are therefore over identical sets of numbers.

For measurability, note that (1.2.3) for  $E + x$  with test set  $A$  is exactly (1.2.3) for  $E$  with test set  $A - x$ , and  $A$  ranges over all subsets iff  $A - x$  does.  $\square$

**Theorem 1.3.9** (Uniqueness of translation-invariant measure). *Let  $\nu$  be a measure on  $\mathcal{B}(\mathbb{R}^n)$  which is translation invariant and satisfies  $c := \nu([0, 1]^n) < \infty$ . Then  $\nu = cm$  on  $\mathcal{B}(\mathbb{R}^n)$ .*

*Proof. Step 1. Dyadic cubes.*

For  $k \geq 0$  let  $\mathcal{D}_k$  be the family of half-open dyadic cubes  $2^{-k}(m + [0, 1]^n)$ ,  $m \in \mathbb{Z}^n$ . The unit cube  $[0, 1]^n$  is the disjoint union of exactly  $2^{kn}$  cubes of  $\mathcal{D}_k$ , and all of these are translates of one another. By translation invariance they have equal  $\nu$ -measure, and by finite additivity that common value is  $c2^{-kn}$ . Since  $m(R) = 2^{-kn}$  for  $R \in \mathcal{D}_k$  by the box lemma, we conclude

$$\nu(R) = cm(R) \quad \text{for every dyadic cube } R. \quad (1.3.3)$$

**Step 2. Open sets.**

Every open  $U \subseteq \mathbb{R}^n$  is a countable disjoint union of dyadic cubes: run through  $k = 0, 1, 2, \dots$  and select each  $R \in \mathcal{D}_k$  with  $\overline{R} \subseteq U$  that is not already contained in a previously selected cube. Any two dyadic cubes are either nested or disjoint, so the selected family is disjoint; and it covers  $U$ , since each  $x \in U$  has a ball  $B(x, r) \subseteq U$  and the dyadic cube of generation  $k$  containing  $x$  has diameter  $\sqrt{n}2^{-k} < r$  for large  $k$ . Countable additivity of both  $\nu$  and  $m$ , together with (1.3.3), gives  $\nu(U) = cm(U)$ .

**Step 3. Borel sets.**

Both  $\nu$  and  $cm$  are Borel measures agreeing on open sets and finite on bounded sets. By outer regularity (Theorem 1.3.6(ii), which applies to  $cm$ , and the analogous statement for  $\nu$  obtained from Step 2 by the same argument) they agree on all Borel sets. Explicitly: for  $E$  Borel and bounded,  $\nu(E) = \inf\{\nu(U) : U \supseteq E \text{ open}\} = c \inf\{m(U)\} = cm(E)$ ; the unbounded case follows by exhausting with annuli as in Step 4 of Theorem 1.3.6.  $\square$

**Theorem 1.3.10** (Linear images). *Let  $T : \mathbb{R}^n \rightarrow \mathbb{R}^n$  be linear. Then  $T$  maps measurable sets to measurable sets, and for every  $E \in \mathcal{L}$ ,*

$$m(T(E)) = |\det T| m(E).$$

*Idea of the proof.* There are three moves. First, if  $T$  is singular then  $T(\mathbb{R}^n)$  lies in a hyperplane, which is null, and both sides vanish; so assume  $T$  invertible. Second,  $E \mapsto m(T(E))$  is a translation-invariant Borel measure, finite on compacta, so Theorem 1.3.9 says it equals  $c(T)m$  for some constant  $c(T) \geq 0$  depending only on  $T$ . The whole problem is thereby reduced to computing one number. Third,  $c$  is multiplicative —  $c(ST) = c(S)c(T)$ , immediately from the definition — so it suffices to compute  $c$  on a set of generators of  $GL_n(\mathbb{R})$ , and every invertible matrix is a product of elementary matrices of three types.

The only type requiring thought is the shear, and the classical trick there is to cut the sheared cube into two pieces and translate one of them back, which reassembles the original cube. That argument uses nothing but translation invariance and additivity, which is why we can run it here rather than waiting for the change-of-variables theorem in Chapter 1.6.

*Proof. Step 1. Singular  $T$ .*

If  $\det T = 0$  then  $T(\mathbb{R}^n) \subseteq H$  for some hyperplane  $H$ , and  $H$  is null (the argument in Worked Example 2.7 covered a hyperplane by boxes of total volume  $\eta$  for arbitrary  $\eta$ ). So  $T(E)$  is a subset of a null set, hence measurable with measure 0 by completeness, and both sides are 0.

**Step 2.** *Invertible  $T$  maps measurable sets to measurable sets.*

$T$  is a homeomorphism, so it maps open sets to open sets and hence Borel sets to Borel sets. It is also Lipschitz, with  $|Tx - Ty| \leq \|T\| |x - y|$ , which implies it maps null sets to null sets: if  $m^*(N) = 0$ , cover  $N$  by boxes of total volume  $< \eta$ ; each box  $Q$  has  $T(Q)$  contained in a box of volume at most  $(\sqrt{n}\|T\|)^n |Q|$ , so  $m^*(T(N)) \leq (\sqrt{n}\|T\|)^n \eta$ . Now use the structure theorem: a measurable  $E$  is  $F \cup N$  with  $F$  an  $F_\sigma$  and  $N$  null (Theorem 1.3.6(v)), so  $T(E) = T(F) \cup T(N)$  is Borel union null, hence measurable.

**Step 3.** *Reduction to a constant.*

Define  $\nu(E) = m(T(E))$  for  $E$  Borel. This is a measure:  $T$  is a bijection, so it preserves disjointness and countable unions. It is translation invariant, because  $T(E+x) = T(E)+Tx$  and  $m$  is translation invariant. And  $\nu([0, 1]^n) = m(T([0, 1]^n)) < \infty$ , since  $T([0, 1]^n)$  is bounded. By Theorem 1.3.9,  $\nu = c(T)m$  with  $c(T) = m(T([0, 1]^n))$ .

**Step 4. *Multiplicativity.***

For linear  $S, T$  we have  $c(ST)m(E) = m(S(T(E))) = c(S)m(T(E)) = c(S)c(T)m(E)$ ; taking  $E = [0, 1]^n$  gives  $c(ST) = c(S)c(T)$ . Since  $|\det|$  is also multiplicative, it suffices to verify  $c(T) = |\det T|$  for  $T$  in a generating set of  $GL_n(\mathbb{R})$ , and every invertible matrix is a finite product of elementary matrices of the following three types.

**Step 5. *Type 1: diagonal scaling.***

$T(x_1, \dots, x_n) = (\lambda_1 x_1, \dots, \lambda_n x_n)$  with all  $\lambda_i \neq 0$ . Then  $T([0, 1]^n)$  is a box with edge lengths  $|\lambda_i|$ , so by the box lemma  $c(T) = \prod_i |\lambda_i| = |\det T|$ .

**Step 6. *Type 2: coordinate permutation.***

$T$  permutes the coordinates. Then  $T([0, 1]^n) = [0, 1]^n$ , so  $c(T) = 1$ , while  $\det T = \pm 1$ . Agreement.

**Step 7. *Type 3: shear.***

Let  $T(x) = x + x_j e_i$  for fixed  $i \neq j$ , so  $\det T = 1$ ; we must show  $c(T) = 1$ , i.e. that the sheared unit cube  $P = T([0, 1]^n)$  has measure 1.

Since only the  $i$ -th and  $j$ -th coordinates are involved, and the others are carried along unchanged, it suffices to treat  $n = 2$  with  $i = 1, j = 2$ : the map  $(x_1, x_2) \mapsto (x_1 + x_2, x_2)$ . Then

$$P = \{(y_1, y_2) : 0 \leq y_2 < 1, y_2 \leq y_1 < 1 + y_2\},$$

a parallelogram. Split it along the line  $y_1 = 1$ :

$$P_{\text{left}} = P \cap \{y_1 < 1\}, \quad P_{\text{right}} = P \cap \{y_1 \geq 1\},$$

which are disjoint measurable sets with  $m(P) = m(P_{\text{left}}) + m(P_{\text{right}})$ .

Now translate the right piece by  $-e_1$ . We claim

$$P_{\text{left}} \cup (P_{\text{right}} - e_1) = [0, 1]^2,$$

and that the union is disjoint. Indeed a point  $(y_1, y_2)$  with  $0 \leq y_2 < 1$  lies in  $P_{\text{left}}$  iff  $y_2 \leq y_1 < 1$ , and lies in  $P_{\text{right}} - e_1$  iff  $(y_1 + 1, y_2) \in P$  with  $y_1 + 1 \geq 1$ , i.e. iff  $0 \leq y_1 < y_2$ . These two conditions are complementary within  $[0, 1)$  for  $y_1$ , which proves both claims.

By translation invariance and additivity,

$$1 = m([0, 1]^2) = m(P_{\text{left}}) + m(P_{\text{right}} - e_1) = m(P_{\text{left}}) + m(P_{\text{right}}) = m(P).$$

So  $c(T) = 1 = |\det T|$ , completing the proof.  $\square$



### 1.3.4 Non-measurable sets

We now show that the restriction to  $\mathcal{L}$  was not a failure of imagination.

**Theorem 1.3.11** (Vitali). *There exists a set  $V \subseteq [0, 1)$  which is not Lebesgue measurable. Consequently there is no translation-invariant, countably additive measure defined on all subsets of  $\mathbb{R}$  assigning to each interval its length.*

*Idea of the proof.* The proof manufactures a set that is simultaneously “one  $\mathbb{Q}$ -translate’s worth” of  $[0, 1)$  and is reproduced by countably many disjoint translates that tile  $[0, 1)$ . If such a set had a measure  $c$ , countable additivity would force  $1 = \sum_{k=1}^{\infty} c$ , and no value of  $c$  satisfies this:  $c = 0$  gives 0, and  $c > 0$  gives  $\infty$ . The whole construction exists to produce this impossible arithmetic, and the arithmetic is impossible precisely because countable additivity has no room for an intermediate answer.

*Proof. Step 1. The equivalence relation.*

Define  $x \sim y$  on  $[0, 1)$  by  $x - y \in \mathbb{Q}$ . This is an equivalence relation: reflexive since  $0 \in \mathbb{Q}$ , symmetric since  $\mathbb{Q}$  is closed under negation, and transitive since  $\mathbb{Q}$  is closed under addition.

**Step 2. Choice.**

By the axiom of choice, let  $V \subseteq [0, 1)$  contain exactly one element from each equivalence class. This is the only non-constructive step, and it is unavoidable: it is a theorem of Solovay that there are models of  $\mathbf{ZF} +$  dependent choice in which every subset of  $\mathbb{R}$  is Lebesgue measurable.

**Step 3. The translates.**

Enumerate  $\mathbb{Q} \cap [0, 1)$  as  $q_1, q_2, \dots$  and define the “translation modulo 1”

$$V_k = ((V \cap [0, 1 - q_k)) + q_k) \cup ((V \cap [1 - q_k, 1)) + q_k - 1) \subseteq [0, 1).$$

Each  $V_k$  is the union of two ordinary translates of disjoint pieces of  $V$ .

**Step 4. The  $V_k$  are pairwise disjoint.**

Suppose  $x \in V_j \cap V_k$ . Then  $x = v + q_j - m$  and  $x = w + q_k - m'$  for some  $v, w \in V$  and  $m, m' \in \{0, 1\}$ . Subtracting,  $v - w = (q_k - q_j) + (m - m') \in \mathbb{Q}$ , so  $v \sim w$ . Since  $V$  contains exactly one representative per class,  $v = w$ , whence  $q_j - q_k \in \mathbb{Z}$ ; as both lie in  $[0, 1)$  this forces  $q_j = q_k$ , i.e.  $j = k$ .

**Step 5. The  $V_k$  cover  $[0, 1)$ .**

Let  $x \in [0, 1)$  and let  $v \in V$  be the representative of its class, so  $x - v \in \mathbb{Q}$ . Since  $x, v \in [0, 1)$  we have  $x - v \in (-1, 1)$ , so exactly one of  $x - v$  and  $x - v + 1$  lies in  $[0, 1) \cap \mathbb{Q}$ ; call it  $q_k$ . Unwinding the two cases shows  $x \in V_k$ .

**Step 6.** *The contradiction.*

Suppose  $V$  were measurable. By Proposition 1.3.8 each of the two pieces in the definition of  $V_k$  is measurable, and by additivity

$$m(V_k) = m(V \cap [0, 1 - q_k]) + m(V \cap [1 - q_k, 1]) = m(V) =: c,$$

the same value for every  $k$ . By Steps 4 and 5 and countable additivity,

$$1 = m([0, 1]) = \sum_{k=1}^{\infty} m(V_k) = \sum_{k=1}^{\infty} c.$$

If  $c = 0$  the sum is  $0 \neq 1$ ; if  $c > 0$  the sum is  $+\infty \neq 1$ . Either way a contradiction, so  $V \notin \mathcal{L}$ .

For the final sentence of the theorem, observe that the argument used only translation invariance, countable additivity, and  $m([0, 1]) = 1$ .  $\square$

**WORKED EXAMPLE 1.3.2— TIER B Cantor sets: null but uncountable, and nowhere dense but fat**

The Cantor set is the standard source of counterexamples in this subject, and two variants are needed later.

(a) *The middle-thirds set.* Let  $C_0 = [0, 1]$  and obtain  $C_k$  from  $C_{k-1}$  by deleting the open middle third of each of its constituent intervals. Then  $C_k$  is a union of  $2^k$  closed intervals each of length  $3^{-k}$ , so by the box lemma and finite additivity

$$m(C_k) = 2^k 3^{-k} = (2/3)^k.$$

The sets  $C_k$  decrease, so  $C = \bigcap_k C_k$  is compact with  $m(C) \leq (2/3)^k$  for every  $k$ , hence  $m(C) = 0$ .

Yet  $C$  is uncountable: the map sending a sequence  $(a_k) \in \{0, 2\}^{\mathbb{N}}$  to  $\sum_k a_k 3^{-k}$  is an injection of  $\{0, 2\}^{\mathbb{N}}$  into  $C$  (in fact a bijection), and  $\{0, 2\}^{\mathbb{N}}$  has the cardinality of the continuum. So “measure zero” is much weaker than “countable”.

(b) *A fat Cantor set.* Fix  $\alpha \in (0, 1)$ . Repeat the construction, but at stage  $k$  delete from each of the  $2^{k-1}$  intervals an open middle interval of length  $\alpha 2^{-(2k-1)}$ . The total length deleted is

$$\sum_{k=1}^{\infty} 2^{k-1} \cdot \alpha 2^{-(2k-1)} = \alpha \sum_{k=1}^{\infty} 2^{k-1-2k+1} = \alpha \sum_{k=1}^{\infty} 2^{-k} = \alpha,$$

so the resulting compact set  $C_\alpha$  has  $m(C_\alpha) = 1 - \alpha > 0$ . It contains no interval, because at stage  $k$  it is covered by  $2^k$  intervals of total length  $1 - \alpha \sum_{j \leq k} 2^{-j} \leq 1$ , each therefore of length at most  $2^{-k}$ ; so  $C_\alpha$  is closed with empty interior, i.e. nowhere dense.

(c) *A measurable set that is not Borel.*  $C$  has measure zero, so by completeness *every* one of its  $2^{\mathfrak{c}}$  subsets is Lebesgue measurable. But there are only  $\mathfrak{c}$  Borel sets. Hence  $\mathcal{B} \subsetneq \mathcal{L}$ , strictly, and by a wide margin. This is the promised justification of the Remark in Step 2 of Theorem 1.3.6.

### 1.3.5 The Perron tree

We come to the theorem this volume has been pointing at since Chapter 1.1.

**Theorem 1.3.12** (Besicovitch–Perron). *Let  $\Delta \subseteq \mathbb{R}^2$  be a triangle with one side on the  $x$ -axis. For every  $\varepsilon > 0$  there exist  $k \in \mathbb{N}$  and horizontal translations  $\tau_1, \dots, \tau_{2^k}$  such that the  $2^k$  elementary triangles  $\Delta_1^{(k)}, \dots, \Delta_{2^k}^{(k)}$  obtained from  $\Delta$  by  $k$  successive bisections satisfy*

$$\left| \bigcup_{i=1}^{2^k} \tau_i(\Delta_i^{(k)}) \right| < \varepsilon.$$

*Consequently, for every  $\varepsilon > 0$  there is a compact set in  $\mathbb{R}^2$  of measure less than  $\varepsilon$  containing a unit segment in every direction.*

Here *bisection* means: cut a triangle by the segment joining its apex (the vertex off the axis) to the midpoint of its base, producing two triangles of equal area, each again with a side on the axis. After  $k$  bisections we have  $2^k$  triangles, in left-to-right order along the axis, each of area  $2^{-k}|\Delta|$ .

*Scratch work.* Two things must be arranged simultaneously, and they pull against each other.

We must not lose directions. A triangle with apex  $P$  and base  $[u, v]$  contains, for each direction  $\theta$  in the angular range subtended, a segment pointing in direction  $\theta$  — namely part of the segment from  $P$  to the appropriate base point. The elementary triangles of the  $k$ -th bisection subtend angular ranges that together cover the range of  $\Delta$ . Crucially, *translation does not change the set of directions of segments contained in a set*. So however we slide the elementary triangles, the union still meets every direction of  $\Delta$ . This is free, and it is the whole reason the construction is possible.

We must lose area. Sliding two adjacent triangles toward each other makes them overlap, and overlap is exactly area not paid for twice. The question is how much can be saved before the figures stop resembling triangles.

Lemma 1.3.13 below does the local computation for a single adjacent pair. The output is not simply “a smaller triangle”: it is a smaller triangle (the *trunk*) plus two leftover pieces sticking out of the top (the *sprouts*). If we

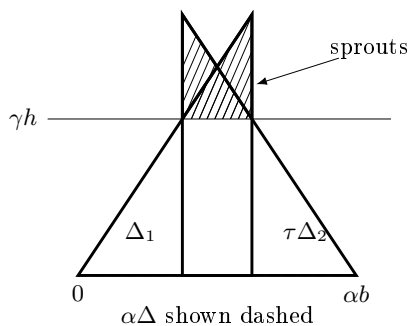
recurse on the trunks, the sprouts accumulate, and the entire difficulty of the proof is showing that the accumulated sprout area is summable. It is, because the trunks shrink geometrically.

**Lemma 1.3.13** (Two-triangle lemma). *Let  $\Delta$  be a triangle with base  $[0, b] \times \{0\}$  and apex  $P = (p, h)$ ,  $h > 0$ . Let  $\Delta_1, \Delta_2$  be its bisection, with bases  $[0, b/2]$  and  $[b/2, b]$ . Fix  $\alpha \in (1/2, 1)$  and let  $\tau$  be translation by  $-(1-\alpha)b$  in the  $x$ -direction. Then*

$$\Delta_1 \cup \tau(\Delta_2) \subseteq \alpha\Delta \cup S_1 \cup S_2,$$

where  $\alpha\Delta$  is the image of  $\Delta$  under the homothety of ratio  $\alpha$  centred at the origin, and  $S_1, S_2$  are triangles with

$$|S_1| + |S_2| = 4(1-\alpha)^2|\Delta|.$$



**Figure 1.3.1:** The two-triangle lemma (Lemma 1.3.13), drawn with  $\alpha = \frac{4}{5}$ . Bisecting  $\Delta$  and sliding the right half left by  $(1-\alpha)b$  makes the halves overlap. Below the height  $\gamma h$ ,  $\gamma = 2\alpha - 1$ , the union lies inside the shrunken triangle  $\alpha\Delta$ ; the left edge of  $\Delta_1$  and the right edge of  $\tau\Delta_2$  are *exactly* the edges of  $\alpha\Delta$ , which is what Step 1 of the proof verifies. What protrudes above  $\gamma h$  is the two hatched sprouts, of total area  $4(1-\alpha)^2|\Delta|$ .

*Proof.* Write  $\gamma = 2\alpha - 1 \in (0, 1)$  and split at the height  $\gamma h$ ; Figure 1.3.1 shows the configuration.

**Step 1.** *The right edge of  $\tau\Delta_2$  is the right edge of  $\alpha\Delta$ .*

$\alpha\Delta$  has vertices  $(0, 0)$ ,  $(\alpha b, 0)$ ,  $(\alpha p, \alpha h)$ . The triangle  $\tau\Delta_2$  has vertices  $((\alpha - \frac{1}{2})b, 0)$ ,  $(\alpha b, 0)$ ,  $(p - (1-\alpha)b, h)$ . Its right edge runs from  $(\alpha b, 0)$  to  $(p - (1-\alpha)b, h)$ ; at height  $y$  this edge is at

$$x = \alpha b + \frac{y}{h}(p - (1-\alpha)b - \alpha b) = \alpha b + \frac{y}{h}(p - b).$$

The right edge of  $\alpha\Delta$  runs from  $(\alpha b, 0)$  to  $(\alpha p, \alpha h)$  and is at  $x = \alpha b + \frac{y}{\alpha h}(\alpha p - \alpha b) = \alpha b + \frac{y}{h}(p - b)$ . The two are the same line.

Similarly the left edge of  $\Delta_1$  (from  $(0,0)$  to  $(p,h)$ ), at  $x = \frac{y}{h}p$  is the left edge of  $\alpha\Delta$  (from  $(0,0)$  to  $(\alpha p, \alpha h)$ ), at  $x = \frac{y}{\alpha h}\alpha p = \frac{y}{h}p$ .

**Step 2.** *Below height  $\gamma h$ , the union lies in  $\alpha\Delta$ .*

Consider a point of  $\Delta_1$  at height  $y$ . It lies to the right of  $\Delta_1$ 's left edge, which by Step 1 is  $\alpha\Delta$ 's left edge, so that side is fine. It lies to the left of  $\Delta_1$ 's right edge, which at height  $y$  is at  $x = \frac{b}{2} + \frac{y}{h}(p - \frac{b}{2})$ . We need this to be at most  $\alpha\Delta$ 's right edge, at  $x = \alpha b + \frac{y}{h}(p - b)$ :

$$\frac{b}{2} + \frac{y}{h}(p - \frac{b}{2}) \leq \alpha b + \frac{y}{h}(p - b) \iff \frac{b}{2}\left(1 - \frac{y}{h}\right) \leq b\left(\alpha - \frac{y}{h}\right) \iff \frac{y}{2h} \leq \alpha - \frac{1}{2} \iff y \leq \gamma h.$$

Symmetrically, a point of  $\tau\Delta_2$  at height  $y$  lies to the left of  $\alpha\Delta$ 's right edge by Step 1, and to the right of  $\tau\Delta_2$ 's left edge, which at height  $y$  is at  $x = (\alpha - \frac{1}{2})b + \frac{y}{h}(p - \frac{b}{2})$ ; requiring this to be at least  $\alpha\Delta$ 's left edge  $\frac{y}{h}p$  gives  $(\alpha - \frac{1}{2})b \geq \frac{y}{h} \cdot \frac{b}{2}$ , i.e. again  $y \leq \gamma h$ .

Hence  $(\Delta_1 \cup \tau\Delta_2) \cap \{y \leq \gamma h\} \subseteq \alpha\Delta$ .

**Step 3.** *The sprouts.*

Let  $S_1 = \Delta_1 \cap \{y > \gamma h\}$  and  $S_2 = \tau\Delta_2 \cap \{y > \gamma h\}$ . Since  $\Delta_1$  is a triangle with apex at height  $h$ , its portion above height  $\gamma h$  is a triangle similar to  $\Delta_1$  with ratio  $1 - \gamma = 2(1 - \alpha)$ , so

$$|S_1| = (2(1 - \alpha))^2 |\Delta_1| = 4(1 - \alpha)^2 \cdot \frac{1}{2} |\Delta| = 2(1 - \alpha)^2 |\Delta|,$$

and identically for  $S_2$ . Adding gives the stated total, and Steps 2–3 give the stated inclusion.  $\square$

*Proof of Theorem 1.3.12.* Normalise  $|\Delta| = 1$ , and fix  $\alpha \in (1/2, 1)$  to be chosen at the end.

**Step 1.** *The recursive scheme.*

Start with the  $2^k$  elementary triangles of the  $k$ -th bisection. Group them into  $2^{k-1}$  adjacent pairs; the two members of a pair are precisely the bisection of a single elementary triangle of the  $(k - 1)$ -st stage. Apply Lemma 1.3.13 to each pair. This replaces the pair by  $\alpha$ -scaled copy of its parent, plus two sprouts.

The resulting  $2^{k-1}$  trunks are the  $\alpha$ -scaled copies of the level- $(k - 1)$  triangles. They are therefore, up to translation, exactly the level- $(k - 1)$  elementary triangles of the scaled triangle  $\alpha\Delta$  — so the scheme can be applied to them again. Repeat  $k$  times in all. After stage  $j$  we hold  $2^{k-j}$  trunks which are the level- $(k - j)$  elementary triangles of  $\alpha^j\Delta$ , together with all sprouts produced so far.

Since every operation is a horizontal translation and the sprouts are retained,

$$\bigcup_{i=1}^{2^k} \tau_i(\Delta_i^{(k)}) \subseteq (\text{a translate of } \alpha^k \Delta) \cup \bigcup_{\text{all sprouts}} S. \quad (1.3.4)$$

**Step 2.** *Total sprout area.*

At stage  $j$  (for  $j = 1, \dots, k$ ) we apply Lemma 1.3.13 to each of the  $2^{k-j}$  pairs among the current trunks. The parents at that stage are the level- $(k-j)$  elementary triangles of  $\alpha^{j-1} \Delta$ , whose areas sum to

$$|\alpha^{j-1} \Delta| = \alpha^{2(j-1)}.$$

The lemma charges  $4(1-\alpha)^2$  times the parent's area to each pair, so the sprout area created at stage  $j$  is exactly  $4(1-\alpha)^2 \alpha^{2(j-1)}$ . Summing the geometric series over  $j = 1, \dots, k$  and extending to  $j = \infty$ ,

$$\sum_{\text{all sprouts}} |S| \leq 4(1-\alpha)^2 \sum_{j=0}^{\infty} \alpha^{2j} = \frac{4(1-\alpha)^2}{1-\alpha^2} = \frac{4(1-\alpha)}{1+\alpha} \leq 4(1-\alpha). \quad (1.3.5)$$

**Step 3.** *Conclusion.*

By (1.3.4), subadditivity, (1.3.5) and  $|\alpha^k \Delta| = \alpha^{2k}$ ,

$$\left| \bigcup_i \tau_i(\Delta_i^{(k)}) \right| \leq \alpha^{2k} + 4(1-\alpha). \quad (1.3.6)$$

Given  $\varepsilon > 0$ , first choose  $\alpha \in (1/2, 1)$  with  $4(1-\alpha) < \varepsilon/2$ ; then,  $\alpha$  being fixed and less than 1, choose  $k$  with  $\alpha^{2k} < \varepsilon/2$ . This proves the first assertion.

**Step 4.** *From the tree to a set of segments in all directions.*

The triangle  $\Delta$  contains, for each direction  $\theta$  in the angular range subtended by  $\Delta$  at its apex, a segment in direction  $\theta$ . Each elementary triangle  $\Delta_i^{(k)}$  subtends a sub-range, and these sub-ranges cover the whole range. Since translation preserves directions, the set  $P = \bigcup_i \tau_i(\Delta_i^{(k)})$  contains a segment in every direction of the original range.

Taking  $\Delta$  to be a triangle subtending an angular range of, say,  $\pi/3$ , and forming three rotated copies of the construction — rotations preserve Lebesgue measure by Theorem 1.3.10, since a rotation has determinant 1 — we obtain a set of measure  $< \varepsilon$  containing a segment in every direction of  $[0, \pi)$ . Rescaling so that all these segments have length at least 1, and taking closures (which adds nothing to the measure, each set being a finite union of triangles), gives the compact set claimed.  $\square$

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WORKED EXAMPLE 1.3.3— TIER C: RESEARCH TASTE **Reading the estimate** (1.3.6)

The bound  $\alpha^{2k} + 4(1 - \alpha)$  deserves to be stared at, because its shape recurs throughout this series.

*The two terms fight.* The first term,  $\alpha^{2k}$ , is the trunk; it wants  $\alpha$  *small*, so that each bisection compresses aggressively. The second,  $4(1 - \alpha)$ , is the accumulated sprout debt; it wants  $\alpha$  *close to 1*, so that each compression is gentle and leaks little. Neither term alone is the obstruction; the theorem works only because we may choose  $\alpha$  first, paying the sprout debt once, and then take  $k \rightarrow \infty$  to kill the trunk. Fix the order of quantifiers wrongly — choose  $k$  first, then optimise  $\alpha$  — and you get nothing.

*One bisection is not enough.* If we simply optimise  $\alpha^2 + 4(1 - \alpha)^2$ , the single-stage bound from Lemma 1.3.13, we find the minimum at  $\alpha = 4/5$ , with value  $4/5$ . A single bisection saves 20% and no more. All of the strength comes from iterating, and from the fact that the trunk shrinks geometrically while the debt is summable. This is an early instance of the phenomenon that dominates Volume 13: *a bound that is useless at one scale becomes decisive when applied at  $\log(1/\delta)$  scales, provided the loss per scale is summable rather than multiplicative*. In decoupling this discipline is called *induction on scales*, and (1.3.5) is its ancestor.

*What is not proved here.* Theorem 1.3.12 gives measure  $< \varepsilon$  for each  $\varepsilon$ , not a single set of measure zero. Passing to the limit requires a compactness argument in the space of compact subsets of a fixed disc — the Blaschke selection theorem for the Hausdorff metric — together with the observation that a limit of sets each containing a unit segment in direction  $\theta$  again contains one. We defer this to Volume 7, where the Hausdorff metric is developed properly; nothing in Volumes 1–6 requires the measure-zero refinement, and stating it here without the machinery would be a gesture rather than a proof.

*Why the conjecture is not thereby settled.* It is tempting to conclude that Besicovitch sets are “as small as possible” and that there is nothing further to say. The opposite is true. Lebesgue measure is simply the wrong instrument: it returns 0 and stops. The Hausdorff measures  $\mathcal{H}^s$  of Worked Example 2.12 form a continuum of finer instruments, and the Kakeya conjecture asserts that every one of them with  $s < n$  returns  $+\infty$ . So the Perron tree exhausts what can be squeezed out of *measure*, and the entire subject of this series begins on the other side of that exhaustion. Note also that Lemma 1.3.13 is irredeemably two-dimensional — it manipulates triangles with a base on a line — and that no analogue producing dimension loss is available in  $\mathbb{R}^3$ . That is not a defect of the exposition; it is the mathematical situation, and it is why the three-dimensional problem stayed open from 1917 until Wang and Zahl in 2025.

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**Looking ahead: tubes, and the quantity that replaces area**

In Volume 9 the elementary triangles of Theorem 1.3.12 are replaced by  $\delta$ -tubes: rectangles of dimensions  $1 \times \delta$ . The Perron construction shows that  $\delta^{-1}$  such tubes, one in each  $\delta$ -separated direction, can be arranged so that  $|\bigcup T| \rightarrow 0$  as  $\delta \rightarrow 0$  — but only *slowly*, like  $1/\log(1/\delta)$ . Making the rate of decay the object of study, rather than the mere fact of decay, is what converts Besicovitch’s curiosity into the Kakeya maximal function conjecture. You will meet that operator in Chapter 1.7, three chapters from now, and Worked Example 7.6 will show that everything proved in this volume is insufficient to control it.

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**Figure 1.3.1 (to be drawn).** *The Cantor set and a fat Cantor set*

**Content required:** Four generations of the middle-thirds construction, and beside it the same construction removing middle intervals of length  $4^{-k}$ , whose limit has positive measure. The two look alike and differ in measure; that is the point of Theorem 1.3.11 and the discussion following it.

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## Drills

**Drill 1.3.1.** Prove that  $m^*$  is not finitely additive on all subsets of  $\mathbb{R}$ , directly from Theorem 1.3.11, without reference to measurability: exhibit disjoint sets  $A, B$  with  $m^*(A \cup B) < m^*(A) + m^*(B)$ .

**Drill 1.3.2.** Let  $E \subseteq \mathbb{R}$  be measurable with  $m(E) > 0$ . Show that for every  $\lambda \in (0, 1)$  there is an interval  $I$  with  $m(E \cap I) > \lambda m(I)$ . (Use Theorem 1.3.6(ii) and decompose the approximating open set into its component intervals. This “density” statement is the seed of the Lebesgue differentiation theorem in Chapter 1.7.)

**Drill 1.3.3.** Verify the claim in Step 7 of Theorem 1.3.10 that it suffices to treat  $n = 2$ : write out the general shear  $T(x) = x + x_j e_i$  and check that the cut-and-translate decomposition goes through verbatim in the two relevant coordinates.

**Drill 1.3.4.** Compute, for  $\alpha = 4/5$ , the bound (1.3.6) for  $k = 1, 5, 20, 100$ . Then find, for each  $\varepsilon \in \{10^{-1}, 10^{-2}, 10^{-3}\}$ , the smallest  $k$  for which some admissible  $\alpha$  achieves  $\alpha^{2k} + 4(1 - \alpha) < \varepsilon$ . Observe how badly  $k$  grows, and reconcile this with the assertion in the Looking Ahead box that the decay is only logarithmic.



## Notes

The box lemma is due to Lebesgue; the grid argument in Lemma 1.3.3 is the standard one and appears in essentially every text. Theorem 1.3.11 is Vitali’s, from 1905. Solovay’s theorem, referred to in Step 2 of its proof, appeared in 1970 and shows that the appeal to the axiom of choice cannot be removed.

Besicovitch’s construction dates to 1919, and Perron’s simplification — the tree of §3.6 — to 1928. Our bookkeeping in (1.3.6) follows Perron’s scheme, with the recursion organised so that the sprout debt appears explicitly as a geometric series; this is the form in which the argument generalises. Schoenberg later gave a different construction via “sprouting” figures, and Kahane a third via products of Cantor sets, which has the advantage of producing the measure-zero set directly. Readers curious about the modern state of the Kakeya problem may consult Tao’s survey and Guth’s expository notes on the Wang–Zahl theorem, though neither is needed before Volume 9.



## Chapter 1.4

# Measurable Functions and the Lebesgue Integral

We have a measure. We now build the integral it was designed to support.

Recall the diagnosis of Chapter 1.1: the Riemann integral partitions the *domain*, and fails on functions that oscillate at every scale because no refinement of an interval partition ever separates the level sets. Lebesgue's remedy is to partition the *range* instead — to group together all the points where  $f$  takes a given value, whatever shape that set has — and the price is that one must then know the size of the resulting level sets. That price was paid in Chapters 1.2 and 1.3. This chapter collects the goods.

### 1.4.1 Measurable functions

Throughout,  $(X, \mathcal{M}, \mu)$  is a measure space. The reader may take  $X = \mathbb{R}^n$ ,  $\mathcal{M} = \mathcal{L}$ ,  $\mu = m$  on first reading; nothing in §§4.1–4.5 uses anything specific to  $\mathbb{R}^n$ .

**Definition 1.4.1.** A function  $f : X \rightarrow [-\infty, \infty]$  is  $\mathcal{M}$ -*measurable* (or just *measurable*) if

$$f^{-1}((a, \infty]) = \{x \in X : f(x) > a\} \in \mathcal{M} \quad \text{for every } a \in \mathbb{R}.$$

A complex-valued  $f$  is measurable if its real and imaginary parts are.

**Lemma 1.4.2** (Equivalent formulations). *For  $f : X \rightarrow [-\infty, \infty]$  the following are equivalent:  $\{f > a\} \in \mathcal{M}$  for all  $a$ ;  $\{f \geq a\} \in \mathcal{M}$  for all  $a$ ;  $\{f < a\} \in \mathcal{M}$  for all  $a$ ;  $\{f \leq a\} \in \mathcal{M}$  for all  $a$ .*

*Proof.*  $\{f \geq a\} = \bigcap_{k \geq 1} \{f > a - \frac{1}{k}\}$  and  $\{f > a\} = \bigcup_{k \geq 1} \{f \geq a + \frac{1}{k}\}$  express each of the first two families in terms of countable operations on the other, and  $\mathcal{M}$  is a  $\sigma$ -algebra. The remaining two are the complements of these.  $\square$

*Scratch work.* Notice how little was needed: only that  $\mathcal{M}$  is closed under countable unions and intersections. That is the whole reason Definition 1.4.1 tests against the single family  $\{(a, \infty]\}$  rather than against all Borel sets. The family  $\{(a, \infty] : a \in \mathbb{R}\}$  generates  $\mathcal{B}(\mathbb{R})$ , and preimage commutes with all set operations, so testing on a generating family is enough. Spelling this out gives the following, which is the form to remember.

**Proposition 1.4.3.**  *$f : X \rightarrow \mathbb{R}$  is measurable if and only if  $f^{-1}(B) \in \mathcal{M}$  for every Borel  $B \subseteq \mathbb{R}$ .*

*Proof.*  $(\Leftarrow)$  is immediate, since  $(a, \infty)$  is Borel. For  $(\Rightarrow)$ , let  $\mathcal{S} = \{B \subseteq \mathbb{R} : f^{-1}(B) \in \mathcal{M}\}$ . Because preimage commutes with complements and countable unions —  $f^{-1}(B^c) = f^{-1}(B)^c$  and  $f^{-1}(\bigcup_k B_k) = \bigcup_k f^{-1}(B_k)$  — and because  $\mathcal{M}$  is a  $\sigma$ -algebra,  $\mathcal{S}$  is a  $\sigma$ -algebra. By hypothesis it contains every  $(a, \infty)$ , and those generate  $\mathcal{B}(\mathbb{R})$ . Hence  $\mathcal{B}(\mathbb{R}) \subseteq \mathcal{S}$ .  $\square$

**Pitfall: measurable is not “preimages of measurable sets are measurable”**

It is tempting to define measurability of  $f : \mathbb{R} \rightarrow \mathbb{R}$  by requiring  $f^{-1}(E) \in \mathcal{L}$  for every  $E \in \mathcal{L}$ . This is a *strictly stronger* and wrong condition: the identity map fails nothing, but there exist continuous increasing  $f$  and  $E \in \mathcal{L}$  with  $f^{-1}(E) \notin \mathcal{L}$ , built from the Cantor function. The asymmetry — Borel sets on the target, measurable sets on the source — is deliberate and is what makes Proposition 1.4.4 below true.

**Proposition 1.4.4** (Stability). *Let  $f, g, f_1, f_2, \dots : X \rightarrow [-\infty, \infty]$  be measurable.*

- (i) *If  $\varphi : \mathbb{R} \rightarrow \mathbb{R}$  is Borel measurable (in particular, if  $\varphi$  is continuous) then  $\varphi \circ f$  is measurable.*
- (ii)  *$f + g, fg, |f|, \max(f, g), \min(f, g)$  are measurable (sums where defined).*
- (iii)  *$\sup_k f_k, \inf_k f_k, \limsup_k f_k, \liminf_k f_k$  are measurable.*
- (iv) *If  $f_k \rightarrow f$  pointwise then  $f$  is measurable.*

*Proof.* (i)  $(\varphi \circ f)^{-1}(B) = f^{-1}(\varphi^{-1}(B))$ , and  $\varphi^{-1}(B)$  is Borel, so Proposition 1.4.3 applies.

(iii) is the heart, and is a one-line computation:

$$\left\{ \sup_k f_k > a \right\} = \bigcup_{k \geq 1} \{f_k > a\}, \quad \left\{ \inf_k f_k < a \right\} = \bigcup_{k \geq 1} \{f_k < a\},$$

both countable unions of measurable sets. The first identity holds because a supremum exceeds  $a$  exactly when some term does. Then  $\limsup_k f_k = \inf_m \sup_{k \geq m} f_k$  and  $\liminf_k f_k = \sup_m \inf_{k \geq m} f_k$  are obtained by two further applications, and (iv) follows since a pointwise limit equals the  $\limsup$ .

(ii) For the sum, observe

$$\{f + g > a\} = \bigcup_{q \in \mathbb{Q}} \left( \{f > q\} \cap \{g > a - q\} \right),$$

a *countable* union: if  $f(x) + g(x) > a$  then  $f(x) > a - g(x)$ , and density of  $\mathbb{Q}$  provides a rational  $q$  with  $f(x) > q > a - g(x)$ . For the product use  $fg = \frac{1}{4}((f + g)^2 - (f - g)^2)$  together with (i) applied to  $\varphi(t) = t^2$ . Finally  $\max(f, g) = \sup$  of the two-term sequence and  $|f| = \max(f, -f)$ .  $\square$

**Where we are.** Compare this list with Chapter 1.1. The class of Riemann integrable functions is *not* closed under pointwise limits — that was Worked Example 1.1, the Dirichlet function. The class of measurable functions is, by (iv), and the proof took one line. We have not yet integrated anything; but the class of objects we intend to integrate is already stable under exactly the operation that broke the old theory.

## 1.4.2 Simple functions and approximation

**Definition 1.4.5.** A *simple function* is a measurable  $\varphi : X \rightarrow \mathbb{R}$  taking finitely many values. Writing its distinct nonzero values as  $c_1, \dots, c_N$  and  $E_j = \varphi^{-1}(\{c_j\}) \in \mathcal{M}$ , we have the *standard representation*

$$\varphi = \sum_{j=1}^N c_j \mathbf{1}_{E_j},$$

with the  $E_j$  pairwise disjoint.

**Theorem 1.4.6** (Simple approximation). *Let  $f : X \rightarrow [0, \infty]$  be measurable. Then there exist simple functions*

$$0 \leq \varphi_1 \leq \varphi_2 \leq \dots \leq f \quad \text{with} \quad \varphi_k \rightarrow f \text{ pointwise on } X,$$

*and the convergence is uniform on any set where  $f$  is bounded.*

*Idea of the proof.* Partition the range, exactly as Chapter 1.1 promised. At stage  $k$ , chop  $[0, k)$  into  $k2^k$  intervals of length  $2^{-k}$  and round  $f$  down to the nearest gridpoint, truncating at height  $k$ . Two features must be arranged: the grid at stage  $k + 1$  must *refine* that at stage  $k$ , which is why the mesh is

dyadic rather than, say,  $1/k$ ; and the truncation height must grow, which is what allows  $\varphi_k \rightarrow f$  even where  $f = \infty$ .

*Proof.* For  $k \geq 1$  define

$$\varphi_k = \sum_{j=0}^{k2^k-1} j2^{-k} \mathbf{1}_{E_{k,j}} + k \mathbf{1}_{F_k}, \quad E_{k,j} = f^{-1}\left(\left[\frac{j}{2^k}, \frac{j+1}{2^k}\right)\right), \quad F_k = f^{-1}([k, \infty)).$$

Each  $E_{k,j}$  and  $F_k$  is measurable by Proposition 1.4.3, and there are finitely many of them, so  $\varphi_k$  is simple. By construction  $0 \leq \varphi_k \leq f$ : on  $E_{k,j}$  we have  $\varphi_k = j2^{-k} \leq f$ , and on  $F_k$  we have  $\varphi_k = k \leq f$ .

*Monotonicity.* Fix  $x$  and suppose  $f(x) < k$ , so  $x \in E_{k,j}$  with  $j2^{-k} \leq f(x) < (j+1)2^{-k}$ . At stage  $k+1$  the interval  $[j2^{-k}, (j+1)2^{-k})$  is split into its two dyadic halves, and  $f(x)$  lies in one of them; in either case the corresponding gridpoint is  $2j \cdot 2^{-(k+1)} = j2^{-k}$  or  $(2j+1)2^{-(k+1)} > j2^{-k}$ . So  $\varphi_{k+1}(x) \geq \varphi_k(x)$ . If instead  $f(x) \geq k$  then  $\varphi_k(x) = k$ , while  $\varphi_{k+1}(x) \geq k$  since the stage- $(k+1)$  grid reaches height  $k+1$ .

*Convergence.* If  $f(x) < \infty$ , then for all  $k > f(x)$  we have  $0 \leq f(x) - \varphi_k(x) < 2^{-k}$ , so  $\varphi_k(x) \rightarrow f(x)$ ; and this estimate is uniform over any set where  $f \leq M$ , once  $k > M$ . If  $f(x) = \infty$  then  $\varphi_k(x) = k \rightarrow \infty = f(x)$ .  $\square$

### 1.4.3 The integral of a nonnegative function

**Definition 1.4.7.** For a simple  $\varphi = \sum_{j=1}^N c_j \mathbf{1}_{E_j} \geq 0$  in standard representation, define

$$\int_X \varphi \, d\mu = \sum_{j=1}^N c_j \mu(E_j) \in [0, \infty],$$

with the convention  $0 \cdot \infty = 0$ . For measurable  $f : X \rightarrow [0, \infty]$  define

$$\int_X f \, d\mu = \sup \left\{ \int_X \varphi \, d\mu : \varphi \text{ simple, } 0 \leq \varphi \leq f \right\}. \quad (1.4.1)$$

For  $E \in \mathcal{M}$  write  $\int_E f \, d\mu = \int_X f \mathbf{1}_E \, d\mu$ .

**Pitfall: the convention  $0 \cdot \infty = 0$**

This is not a dodge. It encodes the statement that a function which vanishes on a set of infinite measure contributes nothing there, which is exactly what we want:  $\int_{\mathbb{R}} \mathbf{1}_{\{0\}} \, dm = 0$  even though  $\mathbf{1}_{\{0\}}$  is supported in a set of measure zero and  $\int_{\mathbb{R}} 0 \, dm = 0$  even though  $m(\mathbb{R}) = \infty$ . The convention is used only inside Definition 1.4.7; it is never applied to limits, where  $0 \cdot \infty$  genuinely

is indeterminate — see the failure modes of Fatou's lemma in Chapter 1.5.

**Lemma 1.4.8** (Consistency and basic properties on simple functions). *Let  $\varphi, \psi \geq 0$  be simple and  $c \geq 0$ . Then*

- (i)  $\int \varphi d\mu$  is unambiguous: any representation  $\varphi = \sum_i a_i \mathbf{1}_{A_i}$  with the  $A_i$  disjoint and measurable gives the same value  $\sum_i a_i \mu(A_i)$ ;
- (ii)  $\int(\varphi + \psi) = \int \varphi + \int \psi$  and  $\int c\varphi = c \int \varphi$ ;
- (iii)  $\varphi \leq \psi \implies \int \varphi \leq \int \psi$ ;
- (iv)  $E \mapsto \int_E \varphi d\mu$  is a measure on  $\mathcal{M}$ .

*Proof.* (i) Refine: given two disjoint representations, the common refinement by intersections  $A_i \cap E_j$  is disjoint and measurable, and finite additivity of  $\mu$  shows both sums equal the sum over the refinement.

(ii) Pass to the common refinement  $\{A_i \cap E_j\}$ , on which  $\varphi + \psi$  is constant with value  $a_i + c_j$ ; then use finite additivity of  $\mu$  and (i). Homogeneity is immediate.

(iii) Write  $\psi = \varphi + (\psi - \varphi)$ , note  $\psi - \varphi \geq 0$  is simple, and apply (ii) together with nonnegativity.

(iv) Countable additivity of  $E \mapsto \int_E \varphi = \sum_j c_j \mu(E \cap E_j)$  follows from countable additivity of  $\mu$  applied to each of the finitely many terms.  $\square$

**Proposition 1.4.9** (Monotonicity of the integral). *If  $0 \leq f \leq g$  are measurable then  $\int f d\mu \leq \int g d\mu$ . If  $E \subseteq F$  are measurable and  $f \geq 0$  then  $\int_E f \leq \int_F f$ .*

*Proof.* Every simple  $\varphi$  with  $0 \leq \varphi \leq f$  also satisfies  $0 \leq \varphi \leq g$ , so the supremum in (1.4.1) for  $g$  is over a superset. The second claim is the first applied to  $f\mathbf{1}_E \leq f\mathbf{1}_F$ .  $\square$

**Where we are.** Monotonicity was free. *Additivity* — the statement  $\int(f + g) = \int f + \int g$  for general nonnegative measurable  $f, g$  — is not, and cannot be proved directly from (1.4.1): given simple  $\varphi \leq f + g$  there is no way to split  $\varphi$  into pieces below  $f$  and below  $g$  respectively. The tool that resolves this is the Monotone Convergence Theorem, which lets us approximate  $f$  and  $g$  separately by Theorem 1.4.6 and add in the limit. So additivity of the integral is deferred to Chapter 1.5, one chapter away, and everything below is stated so as not to use it prematurely.

### 1.4.4 Integrable functions and $L^1$

**Definition 1.4.10.** For measurable  $f : X \rightarrow [-\infty, \infty]$  write  $f^+ = \max(f, 0)$  and  $f^- = \max(-f, 0)$ , so that  $f = f^+ - f^-$  and  $|f| = f^+ + f^-$ , both nonnegative and measurable by Proposition 1.4.4. Say  $f$  is *integrable* if  $\int |f| d\mu < \infty$ , and then define

$$\int_X f d\mu = \int_X f^+ d\mu - \int_X f^- d\mu,$$

a difference of two finite numbers. For complex  $f = u + iv$ , integrability means  $\int |f| < \infty$  and  $\int f = \int u + i \int v$ . The set of integrable functions is denoted  $L^1(\mu)$ .

**Proposition 1.4.11** (Triangle inequality). *If  $f \in L^1(\mu)$  then  $|\int f d\mu| \leq \int |f| d\mu$ .*

*Proof.* For real  $f$ ,  $|\int f| = |\int f^+ - \int f^-| \leq \int f^+ + \int f^- = \int |f|$ , using additivity for the last step — which for the specific decomposition  $|f| = f^+ + f^-$  into functions with *disjoint supports* follows from Lemma 1.4.8(iv) and does not require the general additivity deferred above.

For complex  $f$ , choose  $\theta \in \mathbb{R}$  with  $e^{i\theta} \int f = |\int f|$ . Then

$$\left| \int f \right| = \int \operatorname{Re}(e^{i\theta} f) d\mu \leq \int |e^{i\theta} f| d\mu = \int |f| d\mu,$$

where the middle step uses  $\operatorname{Re}(z) \leq |z|$  and monotonicity. □

### 1.4.5 Almost everywhere

**Definition 1.4.12.** A property holds *almost everywhere* (a.e.) if the set where it fails is contained in a set of measure zero.

**Proposition 1.4.13.** *Let  $\mu$  be complete.*

- (i) *If  $f = g$  a.e. and  $f$  is measurable, so is  $g$ , and  $\int f = \int g$  whenever either side is defined.*
- (ii) *If  $f \geq 0$  is measurable and  $\int f d\mu = 0$ , then  $f = 0$  a.e.*
- (iii) *If  $f_k \rightarrow f$  a.e. and each  $f_k$  is measurable, then  $f$  is measurable.*

*Proof.* (i) Let  $N$  be null with  $f = g$  off  $N$ . Then  $\{g > a\} = (\{f > a\} \setminus N) \cup (\{g > a\} \cap N)$ ; the first set is measurable and the second is a subset of a null set, hence measurable by completeness. Equality of integrals holds because every simple  $\varphi \leq f$  may be modified on  $N$  without changing  $\int \varphi$ , by Lemma 1.4.8(iv).



(ii) Let  $E_k = \{f > 1/k\}$ . The simple function  $\frac{1}{k}\mathbf{1}_{E_k}$  satisfies  $0 \leq \frac{1}{k}\mathbf{1}_{E_k} \leq f$ , so  $\frac{1}{k}\mu(E_k) \leq \int f = 0$ , giving  $\mu(E_k) = 0$ . Then  $\{f > 0\} = \bigcup_k E_k$  is null by countable subadditivity.

(iii) is Proposition 1.4.4(iv) combined with (i).  $\square$

### Looking ahead: why completeness keeps mattering

Part (i) is false without completeness: on the Borel  $\sigma$ -algebra, a function equal a.e. to a Borel function need not be Borel, since Worked Example 3.14(c) produced Lebesgue measurable non-Borel subsets of the Cantor set. Since virtually every argument in later volumes modifies functions on null sets without comment — redefining a Fourier transform on a null set, adjusting a wave packet decomposition, discarding a bad set produced by a maximal inequality — we work throughout with  $\mathcal{L}$  rather than  $\mathcal{B}$ .

## 1.4.6 Comparison with the Riemann integral

We can now settle exactly which functions the old theory handled.

**Theorem 1.4.14** (Lebesgue's criterion). *Let  $f : [a, b] \rightarrow \mathbb{R}$  be bounded. Then  $f$  is Riemann integrable if and only if its set of discontinuities is Lebesgue null. In that case  $f$  is Lebesgue measurable and the two integrals agree.*

*Idea of the proof.* Attach to each point the *oscillation*  $\text{osc}(f, x) = \lim_{r \rightarrow 0} (\sup_{B(x, r)} f - \inf_{B(x, r)} f)$ , which is 0 exactly at points of continuity. Riemann integrability says upper and lower sums can be brought together, which says that the total length of subintervals carrying large oscillation can be made small — a statement about the measure of  $\{\text{osc}(f, \cdot) \geq \varepsilon\}$ . Running this in both directions is the whole proof.

*Proof.* Let  $D_\varepsilon = \{x : \text{osc}(f, x) \geq \varepsilon\}$  and  $D = \bigcup_k D_{1/k}$  the set of discontinuities. Each  $D_\varepsilon$  is closed: if  $\text{osc}(f, x) < \varepsilon$  then the same holds on a neighbourhood of  $x$ .

**Step 1.** *Riemann integrable  $\Rightarrow m(D) = 0$ .*

Fix  $\varepsilon, \eta > 0$  and choose a partition  $P$  with  $U(f, P) - L(f, P) < \varepsilon\eta$ . Let  $\mathcal{J}$  be the set of subintervals of  $P$  whose interior meets  $D_\varepsilon$ . On each such interval the oscillation of  $f$  is at least  $\varepsilon$ , so

$$\varepsilon \sum_{J \in \mathcal{J}} |J| \leq U(f, P) - L(f, P) < \varepsilon\eta,$$

giving  $\sum_{J \in \mathcal{J}} |J| < \eta$ . Since  $D_\varepsilon$  is covered by these intervals together with the finitely many partition points (a null set),  $m(D_\varepsilon) \leq \eta$  for every  $\eta$ , so  $m(D_\varepsilon) = 0$  and hence  $m(D) = 0$  by countable subadditivity.

**Step 2.**  $m(D) = 0 \Rightarrow$  *Riemann integrable*.

Fix  $\varepsilon > 0$ . Then  $D_\varepsilon$  is closed, bounded and null, hence compact and coverable by finitely many open intervals of total length  $< \varepsilon$ . Their complement in  $[a, b]$  is a finite union of closed intervals on each of which  $\text{osc}(f, \cdot) < \varepsilon$  pointwise; by compactness each may be partitioned so finely that  $f$  oscillates by at most  $2\varepsilon$  on every piece. Assembling, we get a partition  $P$  with

$$U(f, P) - L(f, P) \leq 2\varepsilon(b - a) + 2\|f\|_\infty\varepsilon,$$

the two terms coming from the good part and from the covering intervals. As  $\varepsilon$  was arbitrary,  $f$  is Riemann integrable.

**Step 3.** *Agreement*.

Take partitions  $P_k$  with  $\text{mesh} \rightarrow 0$  and let  $\underline{\varphi}_k, \overline{\varphi}_k$  be the associated lower and upper step functions. These are simple,  $\underline{\varphi}_k \leq f \leq \overline{\varphi}_k$ , and their integrals are the lower and upper Riemann sums. Both converge a.e. to  $f$  (namely at every point of continuity, which is a.e. by Step 1), so  $f$  is measurable by Proposition 1.4.13, and monotonicity of the Lebesgue integral traps  $\int f \, dm$  between the lower and upper sums, whose common limit is the Riemann integral.  $\square$

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WORKED EXAMPLE 1.4.1— TIER A **The two functions from Chapter 1, resolved**

(a) *Dirichlet*.  $f = \mathbf{1}_{\mathbb{Q} \cap [0, 1]}$  is discontinuous at every point of  $[0, 1]$ , so  $D = [0, 1]$  has measure  $1 \neq 0$ : not Riemann integrable, confirming Worked Example 1.1. But  $\mathbb{Q} \cap [0, 1]$  is measurable with measure 0, so  $f$  is simple and  $\int_0^1 f \, dm = 1 \cdot m(\mathbb{Q} \cap [0, 1]) = 0$ .

(b) *Thomae*.  $g(p/q) = 1/q$  in lowest terms,  $g = 0$  on irrationals. Here  $g$  is continuous at every irrational and discontinuous at every rational, so  $D = \mathbb{Q}$  is null and  $g$  is Riemann integrable, with integral 0. This is Drill 1.1, now with the general principle behind it: what matters is not how many discontinuities there are, but how much they measure. Both functions are discontinuous on a dense set; only one of them is discontinuous on a set of positive measure.

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WORKED EXAMPLE 1.4.2— TIER B **A measurable function that is not Borel, and why the distinction is harmless**

Let  $C$  be the Cantor set and  $A \subseteq C$  a subset that is not Borel; one exists by the cardinality argument of Worked Example 3.14(c). Then  $\mathbf{1}_A$  is Lebesgue measurable, since  $m^*(A) \leq m(C) = 0$  and null sets are measurable by completeness, but it is not Borel measurable, since  $\mathbf{1}_A^{-1}((\frac{1}{2}, \infty)) = A$ . Nevertheless  $\mathbf{1}_A = 0$  a.e., and  $\int \mathbf{1}_A \, dm = 0$ . This is the general situation: by Theorem 1.3.6(v) every Lebesgue measurable function agrees a.e. with a Borel function, and by Proposition 1.4.13(i) the integral cannot tell them apart. So

the enlargement from  $\mathcal{B}$  to  $\mathcal{L}$  costs nothing in the theory of integration while buying the convenience recorded in the Looking Ahead box above.

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**WORKED EXAMPLE 1.4.3— TIER C: RESEARCH TASTE   A first restriction-flavoured computation**

This example uses nothing beyond Definition 1.4.7 and the box lemma, yet it produces — in embryo — the single most important example in the theory of Fourier restriction. Read it now for the shape of the computation; the meaning will arrive in Volume 6.

*The set-up.* Let  $\delta \in (0, 1)$  and consider the “cap”

$$\kappa_\delta = \{(\xi_1, \xi_2) \in \mathbb{R}^2 : |\xi_1| \leq \delta, \xi_2 = \xi_1^2\},$$

a piece of the parabola of horizontal extent  $2\delta$ . Since  $\xi_1^2 \leq \delta^2$  on this range,  $\kappa_\delta$  is contained in the axis-parallel rectangle  $R_\delta$  of dimensions  $2\delta \times \delta^2$ . Note the shape: the cap is  $\delta$  long and  $\delta^2$  thick. *Curvature converts a length scale into a thickness scale by squaring*, and every phenomenon in this subject descends from that sentence.

*The computation.* Consider the function

$$F(x) = \int_{-\delta}^{\delta} e^{2\pi i(x_1 t + x_2 t^2)} dt,$$

which is the object Volume 6 will call the extension of the constant function from  $\kappa_\delta$ . We do not yet have the tools to evaluate  $F$ ; but we can bound the region where it is large, using only the triangle inequality (Proposition 1.4.11) and the box lemma.

If  $|x_1| \leq \frac{1}{100\delta}$  and  $|x_2| \leq \frac{1}{100\delta^2}$ , then for  $|t| \leq \delta$  the phase satisfies

$$|x_1 t + x_2 t^2| \leq \frac{1}{100\delta} \cdot \delta + \frac{1}{100\delta^2} \cdot \delta^2 = \frac{1}{50},$$

so  $e^{2\pi i(x_1 t + x_2 t^2)}$  stays within an arc of angle  $2\pi/50$  of 1; in particular its real part exceeds  $\frac{1}{2}$ . Hence on the rectangle

$$T_\delta = \{|x_1| \leq \frac{1}{100\delta}\} \times \{|x_2| \leq \frac{1}{100\delta^2}\}, \quad m(T_\delta) = \frac{4}{100^2} \cdot \frac{1}{\delta^3},$$

we have  $|F(x)| \geq \operatorname{Re} F(x) \geq \frac{1}{2} \cdot 2\delta = \delta$ .

*Reading the answer.*  $F$  is of size  $\approx \delta$  on a rectangle of dimensions  $\delta^{-1} \times \delta^{-2}$  — a long thin *tube*, dual to the cap in the precise sense that its side lengths

are the reciprocals of the cap's. So the total mass is

$$\int_{T_\delta} |F|^p dm \gtrsim \delta^p \cdot \delta^{-3}.$$

Comparing this with  $\|1\|_{L^p(\kappa_\delta)}^p \approx \delta$  is exactly how one shows that no inequality  $\|F\|_{L^p(\mathbb{R}^2)} \lesssim \|f\|_{L^\infty(\kappa)}$  can hold for  $p$  too small. This is the *Knapp example*, and the exponent it produces is conjectured to be sharp.

*What to take away now.* Three things, all of which recur permanently. First, curvature makes a  $\delta$ -cap sit inside a  $\delta \times \delta^2$  box. Second, the Fourier transform converts that box into the dual tube  $\delta^{-1} \times \delta^{-2}$ . Third, the entire argument was a lower bound obtained by finding a region where the phase does not oscillate — “stationary phase for beginners”. Volume 5 develops the systematic version. Note also the kinship with Chapter 1.3: there, tubes overlapped and we measured the union; here, a single tube appears from a curvature computation. Volume 6 is where the two stories meet.

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## Drills

**Drill 1.4.1.** Show that  $f : X \rightarrow \mathbb{R}$  is measurable if and only if  $f^{-1}(U) \in \mathcal{M}$  for every *open*  $U \subseteq \mathbb{R}$ . Deduce that every continuous  $f : \mathbb{R}^n \rightarrow \mathbb{R}$  is Borel measurable.

**Drill 1.4.2.** Let  $f \geq 0$  be measurable with  $\int_E f d\mu = 0$  for every  $E$  in some family generating  $\mathcal{M}$ . Must  $f = 0$  a.e.? Prove it or find a counterexample.

**Drill 1.4.3.** Carry out the estimate of Worked Example 4.14 in  $\mathbb{R}^n$ : show that the cap  $\{|\xi'| \leq \delta, \xi_n = |\xi'|^2\}$  on the paraboloid sits inside a box of dimensions  $\delta^{n-1} \times \delta^2$ , and identify the dual tube.

## Notes

Definition 1.4.1 and the range-partition construction of Theorem 1.4.6 are Lebesgue's, from his 1902 thesis. Theorem 1.4.14 is also his, and it is the cleanest possible epitaph for the Riemann theory: the old integral works exactly when the discontinuities are negligible in the sense of the new one.

The deferral of additivity to Chapter 1.5 is a deliberate exposition choice. Some texts define  $\int f$  via Theorem 1.4.6 and prove additivity at once; we prefer (1.4.1) because it makes monotonicity self-evident and because it makes visible *which* theorem is doing the work when additivity finally arrives.

## Chapter 1.5

# The Convergence Theorems

This chapter delivers what Chapter 1.1 promised. The Riemann integral failed because limits could not be passed through it; the Lebesgue integral admits three theorems — monotone convergence, Fatou, dominated convergence — which together make the interchange  $\lim \int = \int \lim$  available under hypotheses so mild that in practice one stops noticing them.

There is a second, less advertised payoff. Additivity of the integral was left unproved in Chapter 1.4, because (1.4.1) gives no way to split a simple minorant of  $f + g$  into minorants of  $f$  and of  $g$ . The Monotone Convergence Theorem removes the obstruction immediately, and we harvest it in §5.1 before doing anything else.

### 1.5.1 Monotone convergence

**Theorem 1.5.1** (Monotone Convergence Theorem). *Let  $f_k : X \rightarrow [0, \infty]$  be measurable with  $f_1 \leq f_2 \leq \dots$  and  $f_k \rightarrow f$  pointwise. Then  $f$  is measurable and*

$$\lim_{k \rightarrow \infty} \int_X f_k \, d\mu = \int_X f \, d\mu.$$

*Idea of the proof.* One inequality is free:  $f_k \leq f$  gives  $\int f_k \leq \int f$ , and the left side is nondecreasing, so  $\lim_k \int f_k \leq \int f$ .

For the reverse we must show that the limit of the  $\int f_k$  dominates  $\int \varphi$  for every simple  $\varphi \leq f$ . The natural attempt — “ $f_k$  eventually exceeds  $\varphi$ ” — is false: convergence is pointwise, not uniform, so no single  $k$  works everywhere. The repair is to weaken  $\varphi$  by a factor  $c < 1$  and to track the sets  $E_k = \{f_k \geq c\varphi\}$ , which increase to all of  $X$ . Continuity of the measure  $E \mapsto \int_E \varphi$  from below then does the rest, and  $c \uparrow 1$  at the end. The two-parameter structure — one parameter for the approximation, one

to be relaxed afterwards — is the same manoeuvre as the  $\alpha$  and  $k$  of the Perron tree in §3.6.

*Proof.* Measurability of  $f$  is Proposition 1.4.4(iv).

**Step 1.** *The easy inequality.*

Since  $f_k \leq f_{k+1} \leq f$ , Proposition 1.4.9 gives that  $(\int f_k)_k$  is nondecreasing and bounded above by  $\int f$ . So the limit  $L := \lim_k \int f_k$  exists in  $[0, \infty]$  and  $L \leq \int f$ .

▷ *Goal:*  $L \geq \int f$ , equivalently  $L \geq \int \varphi$  for every simple  $0 \leq \varphi \leq f$ .

**Step 2.** *The truncated sets.*

Fix a simple  $\varphi$  with  $0 \leq \varphi \leq f$  and fix  $c \in (0, 1)$ . Put

$$E_k = \{x \in X : f_k(x) \geq c\varphi(x)\}.$$

Each  $E_k$  is measurable, since  $f_k - c\varphi$  is measurable. The sets increase, because  $f_k \leq f_{k+1}$ . And they exhaust  $X$ : given  $x$ , either  $\varphi(x) = 0$ , in which case  $x \in E_1$  as  $f_1 \geq 0$ ; or  $\varphi(x) > 0$ , in which case  $f(x) \geq \varphi(x) > c\varphi(x)$ , and since  $f_k(x) \uparrow f(x)$  we get  $f_k(x) \geq c\varphi(x)$  for  $k$  large. Hence  $\bigcup_k E_k = X$ .

**Step 3.** *Estimate and pass to the limit.*

By monotonicity of the integral and the definition of  $E_k$ ,

$$\int_X f_k \, d\mu \geq \int_{E_k} f_k \, d\mu \geq c \int_{E_k} \varphi \, d\mu.$$

By Lemma 1.4.8(iv) the set function  $\nu(E) = \int_E \varphi \, d\mu$  is a measure, and measures are continuous from below along increasing sequences — apply countable additivity to the disjointification  $E_k \setminus E_{k-1}$ . Since  $E_k \uparrow X$ ,

$$\int_{E_k} \varphi \, d\mu \longrightarrow \int_X \varphi \, d\mu.$$

Letting  $k \rightarrow \infty$  in the display above gives  $L \geq c \int_X \varphi \, d\mu$ .

**Step 4.** *Release  $c$ .*

This holds for every  $c \in (0, 1)$ , so  $L \geq \int \varphi$ . Taking the supremum over simple  $\varphi \leq f$  and using (1.4.1) gives  $L \geq \int f$ .  $\square$

### 1.5.1.1 Immediate harvest

**Corollary 1.5.2** (Additivity). *For measurable  $f, g : X \rightarrow [0, \infty]$  and  $a, b \geq 0$ ,*

$$\int_X (af + bg) \, d\mu = a \int_X f \, d\mu + b \int_X g \, d\mu.$$

*The same holds for  $f, g \in L^1(\mu)$  and  $a, b \in \mathbb{C}$ , and  $L^1(\mu)$  is a vector space.*

*Proof.* Take simple  $\varphi_k \uparrow f$  and  $\psi_k \uparrow g$  by Theorem 1.4.6. Then  $a\varphi_k + b\psi_k$  is simple and increases to  $af + bg$ , so three applications of Theorem 1.5.1 together with Lemma 1.4.8(ii) give

$$\int (af + bg) = \lim_k \int (a\varphi_k + b\psi_k) = \lim_k \left( a \int \varphi_k + b \int \psi_k \right) = a \int f + b \int g.$$

For the signed and complex cases, decompose into positive and negative (real and imaginary) parts, noting  $|af + bg| \leq |a||f| + |b||g|$  so that integrability is preserved; the bookkeeping is routine once additivity is known for nonnegative functions.  $\square$

**Corollary 1.5.3** (Term-by-term integration). *If  $f_k : X \rightarrow [0, \infty]$  are measurable then  $\int \sum_{k \geq 1} f_k d\mu = \sum_{k \geq 1} \int f_k d\mu$ .*

*Proof.* The partial sums increase to the total sum; apply Corollary 1.5.2 to each partial sum and Theorem 1.5.1 to the sequence of partial sums.  $\square$

**Corollary 1.5.4.** *For measurable  $f \geq 0$ , the set function  $\nu(E) = \int_E f d\mu$  is a measure on  $\mathcal{M}$ .*

*Proof.*  $\nu(\emptyset) = 0$ , and for disjoint  $E_k$  with  $E = \bigcup_k E_k$  we have  $f\mathbf{1}_E = \sum_k f\mathbf{1}_{E_k}$  pointwise, so Corollary 1.5.3 applies.  $\square$

#### WORKED EXAMPLE 1.5.1— TIER B The layer-cake formula

For measurable  $f : X \rightarrow [0, \infty]$ ,

$$\int_X f d\mu = \int_0^\infty \mu(\{x : f(x) > t\}) dt, \quad (1.5.1)$$

and more generally, for  $0 < p < \infty$ ,

$$\int_X |f|^p d\mu = p \int_0^\infty t^{p-1} \mu(\{|f| > t\}) dt. \quad (1.5.2)$$

The function  $d_f(t) = \mu(\{|f| > t\})$  is called the *distribution function* of  $f$ ; it is nonincreasing, hence Borel measurable, so the right-hand sides make sense as Lebesgue integrals on  $(0, \infty)$ .

*Proof of (1.5.1) for simple  $f$ .* Let  $f = \sum_{j=1}^N c_j \mathbf{1}_{E_j}$  in standard representation with  $c_1 > c_2 > \cdots > c_N > 0$  and the  $E_j$  disjoint. For  $t \in [c_{j+1}, c_j]$  the superlevel set is  $\{f > t\} = E_1 \cup \cdots \cup E_j$ , of measure  $\sum_{i \leq j} \mu(E_i)$ , and for  $t \geq c_1$  it is empty. Hence, with  $c_{N+1} := 0$ ,

$$\int_0^\infty d_f(t) dt = \sum_{j=1}^N (c_j - c_{j+1}) \sum_{i=1}^j \mu(E_i) = \sum_{i=1}^N \mu(E_i) \sum_{j \geq i} (c_j - c_{j+1}) = \sum_{i=1}^N c_i \mu(E_i),$$

where the middle step exchanges the order of a finite double sum and the last telescopes. The right-hand side is  $\int f \, d\mu$ .

*Proof for general  $f \geq 0$ .* Take simple  $\varphi_k \uparrow f$  (Theorem 1.4.6). Then  $\{\varphi_k > t\} \uparrow \{f > t\}$  for each fixed  $t$  — the inclusion  $\subseteq$  is monotonicity, and if  $f(x) > t$  then  $\varphi_k(x) > t$  eventually — so continuity of  $\mu$  from below gives  $d_{\varphi_k}(t) \uparrow d_f(t)$  pointwise in  $t$ . Apply Theorem 1.5.1 *twice*: on  $X$  to get  $\int \varphi_k \rightarrow \int f$ , and on  $(0, \infty)$  to get  $\int_0^\infty d_{\varphi_k} \rightarrow \int_0^\infty d_f$ . The simple case equates the two sequences, so the limits agree.

*Proof of (1.5.2).* Apply (1.5.1) to  $|f|^p$  and substitute  $t = s^p$ , noting  $\{|f|^p > s^p\} = \{|f| > s\}$  and  $dt = ps^{p-1} ds$ .

*Why this matters.* Formula (1.5.2) converts every  $L^p$  norm into a statement about the *sizes of superlevel sets*, and superlevel sets are geometric objects. That conversion is why the subject of this series is geometric at all. In Volume 9 the superlevel sets of the Kakeya maximal function will be unions of tubes, and (1.5.2) is the bridge that turns the analytic question “how large is  $\|\mathcal{K}_\delta f\|_p$ ?” into the geometric question “how much can  $\delta$ -tubes overlap?”. It is also the definition behind weak  $L^p$  in Chapter 1.8, and the engine of the Marcinkiewicz interpolation theorem in Volume 2.

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## 1.5.2 Fatou’s lemma

**Theorem 1.5.5** (Fatou). *For measurable  $f_k : X \rightarrow [0, \infty]$ ,*

$$\int_X \liminf_{k \rightarrow \infty} f_k \, d\mu \leq \liminf_{k \rightarrow \infty} \int_X f_k \, d\mu.$$

*Proof.* Put  $g_m = \inf_{k \geq m} f_k$ , measurable by Proposition 1.4.4(iii). The sequence  $(g_m)$  is nondecreasing with  $g_m \uparrow \liminf_k f_k$ , by definition of  $\liminf$ . Also  $g_m \leq f_k$  for every  $k \geq m$ , so by monotonicity  $\int g_m \leq \inf_{k \geq m} \int f_k$ . Now apply Theorem 1.5.1 to  $(g_m)$ :

$$\int \liminf_k f_k = \lim_m \int g_m \leq \lim_m \inf_{k \geq m} \int f_k = \liminf_k \int f_k. \quad \square$$

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### WORKED EXAMPLE 1.5.2— TIER A The three ways mass escapes

Fatou’s inequality is typically strict, and there are exactly three mechanisms. Recognising which one is operating is a genuine skill — in Chapter 1.7 and again in Volume 13 the question “where did the mass go?” is the first thing to ask when an estimate fails — so each is worth a concrete example on  $\mathbb{R}$  with Lebesgue measure. In all three,  $\int f_k = 1$  for every  $k$  while  $f_k \rightarrow 0$  pointwise, so the left side of Fatou is 0 and the right side is 1.



(a) *Escape to horizontal infinity.*  $f_k = \mathbf{1}_{[k,k+1]}$ . The bump keeps its shape and marches off to the right. Nothing changes about the function except its location.

(b) *Escape to vertical infinity.*  $f_k = k \mathbf{1}_{[0,1/k]}$ . The bump grows taller and narrower, concentrating at a point. This is the mode that the convention  $0 \cdot \infty = 0$  of Chapter 1.4 does *not* rescue: the limit function is 0 a.e., but the mass is not lost gradually, it is lost in the limit.

(c) *Escape to width.*  $f_k = \frac{1}{k} \mathbf{1}_{[0,k]}$ . The bump flattens and spreads. Note that here  $f_k \rightarrow 0$  *uniformly*, which shows that even uniform convergence does not justify passing a limit through an integral over a set of infinite measure.

*The moral.* Modes (a) and (c) are impossible on a finite measure space with uniformly bounded functions; mode (b) is impossible when there is an integrable dominating function. That observation is precisely the Dominated Convergence Theorem, which we now prove.

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### 1.5.3 Dominated convergence

**Theorem 1.5.6** (Dominated Convergence Theorem). *Let  $f_k : X \rightarrow \mathbb{C}$  be measurable with  $f_k \rightarrow f$  a.e., and suppose there is  $g \in L^1(\mu)$  with  $|f_k| \leq g$  a.e. for every  $k$ . Then  $f \in L^1(\mu)$  and*

$$\lim_{k \rightarrow \infty} \int_X |f_k - f| d\mu = 0; \quad \text{in particular} \quad \lim_{k \rightarrow \infty} \int_X f_k d\mu = \int_X f d\mu.$$

*Idea of the proof.* Fatou's lemma applies to nonnegative functions, and  $|f_k - f|$  is nonnegative but converges to 0, so applying Fatou directly gives only  $0 \leq \liminf$ , which is vacuous. The trick is to apply Fatou to the nonnegative sequence  $2g - |f_k - f|$ , whose  $\liminf$  is  $2g$ . The dominating function is therefore not merely a technical hypothesis: it is the object that converts a useless application of Fatou into a decisive one, by supplying a fixed ceiling from which the deficit can be measured.

*Proof.* Discarding a null set, assume the hypotheses hold everywhere. Since  $|f| = \lim_k |f_k| \leq g$ , we get  $f \in L^1$  and also  $|f_k - f| \leq 2g$ , so  $2g - |f_k - f| \geq 0$ .

Apply Theorem 1.5.5 to this sequence. Since  $f_k \rightarrow f$  pointwise,  $\liminf_k (2g - |f_k - f|) = 2g$ , so

$$\int 2g \leq \liminf_k \int (2g - |f_k - f|) = \int 2g + \liminf_k \left( - \int |f_k - f| \right),$$

using Corollary 1.5.2. As  $\int 2g < \infty$  we may cancel it, obtaining

$$0 \leq \liminf_k \left( - \int |f_k - f| \right) = - \limsup_k \int |f_k - f|,$$

that is,  $\limsup_k \int |f_k - f| \leq 0$ . Since the integrals are nonnegative the limit is 0. The final claim follows from  $|\int f_k - \int f| \leq \int |f_k - f|$  (Proposition 1.4.11 and Corollary 1.5.2).  $\square$

**WORKED EXAMPLE 1.5.3— TIER B Differentiating under the integral sign**

Let  $I \subseteq \mathbb{R}$  be an open interval and  $f : X \times I \rightarrow \mathbb{C}$  satisfy:  $f(\cdot, s) \in L^1(\mu)$  for each  $s$ ;  $s \mapsto f(x, s)$  is differentiable for each  $x$ ; and there is  $g \in L^1(\mu)$  with  $|\partial_s f(x, s)| \leq g(x)$  for all  $x, s$ . Then  $F(s) = \int_X f(x, s) d\mu(x)$  is differentiable with

$$F'(s) = \int_X \partial_s f(x, s) d\mu(x).$$

*Proof.* Fix  $s$  and let  $h_k \rightarrow 0$  be any nonzero sequence. Put

$$q_k(x) = \frac{f(x, s + h_k) - f(x, s)}{h_k},$$

so  $q_k(x) \rightarrow \partial_s f(x, s)$  for every  $x$ . By the mean value theorem applied to  $t \mapsto f(x, t)$  on the interval between  $s$  and  $s + h_k$ , there is a point  $\xi$  there with  $q_k(x) = \partial_s f(x, \xi)$ , whence  $|q_k| \leq g$ . Theorem 1.5.6 now gives  $\int q_k \rightarrow \int \partial_s f(\cdot, s)$ , and the left side is  $(F(s + h_k) - F(s))/h_k$ . Since the sequence  $(h_k)$  was arbitrary,  $F'(s)$  exists and equals the stated integral.  $\square$

*The hypothesis that does the work* is the *uniform* domination of the  $s$ -derivative. Readers should note how the mean value theorem is what converts a bound on the derivative into a bound on the difference quotients — this is the step that fails for complex-valued  $f$  if applied naively, which is why one applies the argument to real and imaginary parts separately, or uses the integral form of the remainder.

*Why this is here.* Almost every smoothness statement about a Fourier transform is an application of this lemma. In Worked Example 1.5.4 below we use it to show  $\hat{f}$  is continuous; in Volume 2 the same argument shows  $\hat{f}$  is  $C^\infty$  when  $f$  decays rapidly, which is the starting point of the theory of the Schwartz space.

**WORKED EXAMPLE 1.5.4— TIER B Continuity of the Fourier transform, and Riemann–Lebesgue**

For  $f \in L^1(\mathbb{R}^n)$  define

$$\widehat{f}(\xi) = \int_{\mathbb{R}^n} f(x) e^{-2\pi i x \cdot \xi} dx.$$

The integral converges absolutely, since  $|f(x)e^{-2\pi i x \cdot \xi}| = |f(x)|$ , and  $\|\widehat{f}\|_\infty \leq \|f\|_1$  by Proposition 1.4.11. This is the first theorem of Fourier analysis in this series, and both halves are corollaries of Theorem 1.5.6.

(a)  *$\widehat{f}$  is continuous.* Let  $\xi_k \rightarrow \xi$ . The integrands  $f(x)e^{-2\pi i x \cdot \xi_k}$  converge pointwise to  $f(x)e^{-2\pi i x \cdot \xi}$  and are dominated by  $|f| \in L^1$ . Apply Theorem 1.5.6.

(b)  *$\widehat{f}(\xi) \rightarrow 0$  as  $|\xi| \rightarrow \infty$  (Riemann–Lebesgue).* The identity  $e^{-\pi i} = -1$  gives, after the substitution  $x \mapsto x + \frac{\xi}{2|\xi|^2}$ ,

$$\widehat{f}(\xi) = - \int_{\mathbb{R}^n} f\left(x + \frac{\xi}{2|\xi|^2}\right) e^{-2\pi i x \cdot \xi} dx,$$

so averaging this with the original formula,

$$|\widehat{f}(\xi)| \leq \frac{1}{2} \int_{\mathbb{R}^n} \left| f(x) - f\left(x + \frac{\xi}{2|\xi|^2}\right) \right| dx = \frac{1}{2} \|f - \tau_h f\|_1, \quad h = \frac{\xi}{2|\xi|^2}.$$

As  $|\xi| \rightarrow \infty$  we have  $h \rightarrow 0$ , so it suffices to know that translation is continuous in  $L^1$ :  $\|f - \tau_h f\|_1 \rightarrow 0$  as  $h \rightarrow 0$ . This holds for  $f$  continuous with compact support by uniform continuity and Theorem 1.5.6, and extends to all of  $L^1$  by density (Chapter 1.8) together with the estimate  $\|\tau_h\|_{L^1 \rightarrow L^1} = 1$ .

*Reading (b).* The translation trick is the whole content: shifting by half a period turns the integrand into its own negative, so the transform measures the failure of  $f$  to be invariant under a small translation — and integrable functions are, in the limit, invariant. Every decay estimate for oscillatory integrals in Volume 5 is a quantitative refinement of this one idea, with stationary phase supplying the rate.

## 1.5.4 Modes of convergence

We now compare the several senses in which a sequence of functions can converge.

**Definition 1.5.7.** Let  $f_k, f$  be measurable and finite a.e. We say  $f_k \rightarrow f$

- *almost everywhere* if  $\mu(\{x : f_k(x) \not\rightarrow f(x)\}) = 0$ ;
- *in  $L^1$*  if  $\int |f_k - f| d\mu \rightarrow 0$ ;

- *in measure* if  $\mu(\{|f_k - f| > \varepsilon\}) \rightarrow 0$  for every  $\varepsilon > 0$ ;
- *almost uniformly* if for every  $\eta > 0$  there is  $E$  with  $\mu(E) < \eta$  and  $f_k \rightarrow f$  uniformly on  $X \setminus E$ .

**Proposition 1.5.8** (The implications that hold).  *$L^1$  convergence implies convergence in measure. Almost uniform convergence implies both a.e. convergence and convergence in measure. Convergence in measure implies that some subsequence converges a.e.*

*Proof.*  $L^1 \Rightarrow$  *measure* is Chebyshev's inequality: the simple function  $\varepsilon \mathbf{1}_{\{|f_k - f| > \varepsilon\}}$  lies below  $|f_k - f|$ , so

$$\mu(\{|f_k - f| > \varepsilon\}) \leq \frac{1}{\varepsilon} \int |f_k - f| d\mu. \quad (1.5.3)$$

*Almost uniform  $\Rightarrow$  a.e.* For each  $m$  pick  $E_m$  with  $\mu(E_m) < 1/m$  and uniform convergence off  $E_m$ ; then  $f_k \rightarrow f$  everywhere off  $\bigcap_m E_m$ , which is null.

*Measure  $\Rightarrow$  a.e. along a subsequence.* Choose  $k_j$  with  $\mu(\{|f_{k_j} - f| > 2^{-j}\}) < 2^{-j}$  and let  $A_j = \{|f_{k_j} - f| > 2^{-j}\}$ . Since  $\sum_j \mu(A_j) < \infty$ , the set  $A = \bigcap_m \bigcup_{j \geq m} A_j$  of points lying in infinitely many  $A_j$  satisfies  $\mu(A) \leq \sum_{j \geq m} \mu(A_j) \rightarrow 0$ , so  $\mu(A) = 0$ . Off  $A$ , every point lies in only finitely many  $A_j$ , so  $|f_{k_j} - f| \leq 2^{-j}$  eventually.  $\square$

**WORKED EXAMPLE 1.5.5— TIER B The typewriter sequence: no arrow can be reversed**

Define on  $[0, 1]$ , for  $k = 2^m + j$  with  $0 \leq j < 2^m$ ,

$$f_k = \mathbf{1}_{[j2^{-m}, (j+1)2^{-m}]}$$

As  $k$  increases, the indicator sweeps across  $[0, 1]$ , then sweeps again at half the width, and so on — like the carriage of a typewriter.

*It converges in  $L^1$  and in measure.*  $\int f_k = 2^{-m} \rightarrow 0$ , so  $f_k \rightarrow 0$  in  $L^1$ , hence in measure by (1.5.3).

*It converges nowhere.* Fix  $x \in [0, 1]$ . For each  $m$  there is exactly one  $j$  with  $x$  in the corresponding interval, so  $f_k(x) = 1$  for at least one  $k$  in each dyadic block, while  $f_k(x) = 0$  for the others. Hence  $\limsup_k f_k(x) = 1$  and  $\liminf_k f_k(x) = 0$  for every  $x$ .

*What this rules out.*  $L^1$  convergence does not imply a.e. convergence, and neither does convergence in measure; only a subsequence survives, as Proposition 1.5.8 says — here  $f_{2^m} \rightarrow 0$  a.e. Combining with Worked Example 5.9: (a) shows a.e. convergence does not imply  $L^1$ ; (b) shows a.e. and even

convergence in measure do not imply  $L^1$ ; and this example shows the reverse implication fails too. The four notions are genuinely four.

**Theorem 1.5.9** (Egorov). *Let  $\mu(X) < \infty$  and let  $f_k \rightarrow f$  a.e., all measurable and finite a.e. Then  $f_k \rightarrow f$  almost uniformly.*

*Proof.* Discard a null set so that  $f_k \rightarrow f$  everywhere. For  $m, N \geq 1$  set

$$E_{m,N} = \bigcup_{k \geq N} \{x : |f_k(x) - f(x)| > \frac{1}{m}\}.$$

For fixed  $m$  these decrease in  $N$ , and  $\bigcap_N E_{m,N} = \emptyset$  by pointwise convergence. Because  $\mu(X) < \infty$ , continuity of  $\mu$  from above applies and  $\mu(E_{m,N}) \rightarrow 0$  as  $N \rightarrow \infty$ . Given  $\eta > 0$ , choose  $N_m$  with  $\mu(E_{m,N_m}) < \eta 2^{-m}$  and set  $E = \bigcup_m E_{m,N_m}$ , so  $\mu(E) < \eta$ . Off  $E$  we have  $|f_k - f| \leq 1/m$  for all  $k \geq N_m$ , which is uniform convergence.  $\square$

#### Pitfall: Egorov needs finite measure

On  $\mathbb{R}$  take  $f_k = \mathbf{1}_{[k,k+1]}$ . Then  $f_k \rightarrow 0$  everywhere, but for any set  $E$  of finite measure the convergence off  $E$  is not uniform, since  $\sup_{\mathbb{R} \setminus E} f_k = 1$  for infinitely many  $k$ . This is escape mode (a) of Worked Example 5.9 again. Continuity from above — the step that used  $\mu(X) < \infty$  — is exactly what fails.

**Theorem 1.5.10** (Lusin). *Let  $f : \mathbb{R}^n \rightarrow \mathbb{C}$  be measurable and finite a.e., supported in a set of finite measure. For every  $\eta > 0$  there is  $g \in C_c(\mathbb{R}^n)$  with  $m(\{f \neq g\}) < \eta$ .*

*Idea of the proof.* Three tools combine. Simple functions approximate  $f$  pointwise; each level set of a simple function is, by regularity, sandwiched between a compact set and an open set of almost the same measure, and Urysohn's lemma turns that sandwich into a *continuous* function agreeing with the indicator off a small set. That gives continuous  $g_k$  agreeing with  $\varphi_k$  off a small set. Egorov then upgrades pointwise convergence  $g_k \rightarrow f$  to uniform convergence off a further small set — and a uniform limit of continuous functions is continuous. Finally Tietze extends the resulting continuous function from a compact set to all of  $\mathbb{R}^n$ .

*Proof.* Let  $S$  be a set of finite measure outside which  $f$  vanishes, and fix  $\eta > 0$ .

**Step 1.** *Reduce to a bounded set.*

Since  $m(S) < \infty$  we may choose  $R$  with  $m(S \setminus B(0, R)) < \eta/5$ , and replace

$S$  by  $S \cap B(0, R)$ ; the discarded part is absorbed into the final exceptional set. So assume  $S$  is bounded.

**Step 2.** *Continuous approximants to a simple function.*

Let  $\varphi = \sum_{j=1}^N c_j \mathbf{1}_{E_j}$  be simple with all  $E_j \subseteq S$  measurable and disjoint. Fix  $\rho > 0$ . By Theorem 1.3.6(ii) and (iv) choose, for each  $j$ , a compact  $K_j \subseteq E_j$  and an open  $U_j \supseteq E_j$  with  $m(U_j \setminus K_j) < \rho/N$ . Urysohn's lemma (Appendix A) supplies  $h_j \in C_c(\mathbb{R}^n)$  with  $\mathbf{1}_{K_j} \leq h_j \leq \mathbf{1}_{U_j}$ . Then  $h_j = \mathbf{1}_{E_j}$  off  $U_j \setminus K_j$ , so

$$g := \sum_{j=1}^N c_j h_j \in C_c(\mathbb{R}^n) \quad \text{satisfies} \quad m(\{g \neq \varphi\}) \leq \sum_{j=1}^N m(U_j \setminus K_j) < \rho.$$

**Step 3.** *Apply this to an approximating sequence.*

By Theorem 1.4.6 (applied to positive and negative parts) choose simple  $\varphi_k \rightarrow f$  pointwise on  $S$ , each supported in  $S$ . By Step 2 choose  $g_k \in C_c(\mathbb{R}^n)$  with  $m(\{g_k \neq \varphi_k\}) < \eta 2^{-k}/5$ . Put  $A = \bigcup_k \{g_k \neq \varphi_k\}$ , so  $m(A) < \eta/5$ , and off  $A$  we have  $g_k = \varphi_k \rightarrow f$  pointwise.

**Step 4.** *Egorov.*

The set  $S$  has finite measure, so Theorem 1.5.9 applied to  $(g_k)$  on  $S \setminus A$  yields  $B$  with  $m(B) < \eta/5$  such that  $g_k \rightarrow f$  *uniformly* on  $S \setminus (A \cup B)$ .

**Step 5.** *Pass to a compact set.*

By inner regularity (Theorem 1.3.6(iv)) choose a compact  $K \subseteq S \setminus (A \cup B)$  with  $m((S \setminus (A \cup B)) \setminus K) < \eta/5$ . On  $K$  the functions  $g_k$  are continuous and converge uniformly to  $f$ , so  $f|_K$  is continuous.

**Step 6.** *Extend.*

By Tietze's extension theorem (Appendix A) there is  $g \in C_c(\mathbb{R}^n)$  with  $g = f$  on  $K$ . Collecting the five discarded pieces,

$$m(\{f \neq g\}) \leq m(S \setminus B(0, R)) + m(A) + m(B) + m((S \setminus (A \cup B)) \setminus K) < \eta,$$

with room to spare for the set where both vanish. □

**WORKED EXAMPLE 1.5.6— TIER C: RESEARCH TASTE Pigeonholing over scales**

The following exercise looks artificial and is not: it is the bookkeeping discipline of *induction on scales*, isolated in a setting where nothing else is difficult. Volume 13 consists largely of arguments of this shape, and readers who meet the pattern for the first time there usually find it opaque. There is no reason to wait.

*The set-up.* Let  $\delta \in (0, \frac{1}{2})$  and let  $F \subseteq [0, 1]$  be measurable with  $m(F) = \sigma > 0$ . Write  $M = \lceil \log_2(1/\delta) \rceil$  and decompose  $[0, 1]$  into the  $M$  dyadic scales

$$\mathcal{D}_m = \{\text{dyadic intervals of length } 2^{-m}\}, \quad m = 1, \dots, M.$$

Suppose we are given, for each  $m$ , a quantity  $A_m \geq 0$  with  $\sum_{m=1}^M A_m \leq K$ .

*The pigeonhole step.* There exists  $m^* \in \{1, \dots, M\}$  with

$$A_{m^*} \leq \frac{K}{M} \approx \frac{K}{\log(1/\delta)}.$$

This is trivial — the minimum is at most the average — and it is used constantly. The content is in what one does with it: a bound that is lossy by a factor  $\log(1/\delta)$  at a single scale becomes *lossless up to constants* once one is permitted to choose the scale.

*The reverse discipline, which is the real point.* Now suppose instead that at each scale we lose a factor:  $A_{m+1} \leq C A_m$ . Iterating  $M$  times gives  $A_M \leq C^M A_1 = \delta^{-\log_2 C} A_1$ , which is a *power* of  $1/\delta$  and therefore catastrophic. But if the loss is additive,  $A_{m+1} \leq A_m + c_m$  with  $\sum_m c_m < \infty$ , then  $A_M \leq A_1 + \sum c_m$  and nothing is lost at all.

*Where you have already seen this.* Precisely this dichotomy decided the Perron tree in §3.6. There the “loss per scale” was the sprout debt  $4(1 - \alpha)^2 \alpha^{2(j-1)}$ , and (1.3.5) showed it was summable; had it instead been multiplicative, the construction would have produced nothing. The estimate (1.3.6) is exactly an instance of the second display above.

*Where you will see it next.* In Chapter 1.7 the Hardy–Littlewood maximal function is bounded by summing over dyadic radii; in Volume 12 the  $\varepsilon$ -removal argument of Tao converts a local restriction estimate into a global one by paying  $\delta^{-\varepsilon}$  rather than  $\log(1/\delta)$  per scale; and in Volume 13 the entire Bourgain–Demeter theorem is the statement that a certain decoupling constant satisfies a recursion whose loss per scale is subpolynomial, hence harmless over  $\log(1/\delta)$  scales. The theorem is hard; the bookkeeping principle is the one on this page.

#### WORKED EXAMPLE 1.5.7— TIER B **Which convergence theorem to reach for**

Three theorems license passing a limit through an integral, and choosing among them is mechanical once the differences are clear.

(a) *Monotone convergence.*  $0 \leq f_k \uparrow f$ . No domination needed, and the conclusion allows  $\int f = \infty$ . Use it when the sequence is built by adding

non-negative pieces — as every construction of a measure from a sum does.

(b) *Dominated convergence.*  $f_k \rightarrow f$  pointwise with  $|f_k| \leq g \in L^1$ . The workhorse. The hypothesis to check is always the domination, and it is where the errors are.

(c) *Fatou.*  $f_k \geq 0$ , no other hypothesis, and the conclusion is only an inequality  $\int \liminf \leq \liminf \int$ . Use it when domination is unavailable and an inequality suffices — which is often, since most estimates are inequalities.

*The decision procedure.* Is the sequence monotone? Use (a). Is there a dominating function? Use (b). Neither, but an inequality in the right direction would do? Use (c). If none applies, the limit may genuinely fail to pass, and the standard counterexample  $f_k = k\mathbf{1}_{[0,1/k]}$  has  $\int f_k = 1$  for every  $k$  while  $f_k \rightarrow 0$  pointwise.

*Where each is used later.* Fatou appears in Volume 4's atomic argument, where domination is exactly what is missing; monotone convergence in every construction of an outer measure; dominated convergence throughout.

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## Drills

**Drill 1.5.1.** Show that the hypothesis  $f_k \geq 0$  cannot be dropped from Fatou's lemma, by finding  $f_k : \mathbb{R} \rightarrow \mathbb{R}$  measurable with  $\int \liminf f_k > \liminf \int f_k$ .

**Drill 1.5.2.** Prove the “generalised” dominated convergence theorem: if  $f_k \rightarrow f$  a.e.,  $|f_k| \leq g_k$  with  $g_k \rightarrow g$  a.e.,  $g_k, g \in L^1$  and  $\int g_k \rightarrow \int g$ , then  $\int |f_k - f| \rightarrow 0$ . (Follow the proof of Theorem 1.5.6, applying Fatou to  $g_k + g - |f_k - f|$ .)

**Drill 1.5.3.** Let  $f \in L^1(\mathbb{R})$ . Show  $d_f(t) = o(1/t)$  as  $t \rightarrow \infty$ , and exhibit  $f \in L^1$  for which  $d_f(t)$  decays no faster than  $1/(t \log^2 t)$ . (Use (1.5.1).)

**Drill 1.5.4.** In Worked Example 5.16, suppose  $A_{m+1} \leq (1 + \frac{1}{M})A_m$  for  $m = 1, \dots, M-1$ . Show  $A_M \leq e A_1$ , and explain why a loss of  $1 + \frac{1}{\log(1/\delta)}$  per scale is therefore harmless while a loss of  $1 + \frac{1}{100}$  per scale is not.

## Notes

The three convergence theorems are due to Beppo Levi (1906), Fatou (1906) and Lebesgue (1908) respectively, though all three are commonly attributed to Lebesgue. Egorov's theorem is from 1911 and Lusin's from 1912.



The proof of Theorem 1.5.6 given here — applying Fatou to  $2g - |f_k - f|$  — is the standard one and repays study, because the manoeuvre of subtracting from a fixed ceiling to convert a  $\liminf$  statement into a  $\limsup$  statement is used repeatedly in later volumes, notably in the concentration-compactness arguments that appear in the study of extremisers for Strichartz estimates.

Worked Example 5.16 has no counterpart in standard measure theory texts, for the good reason that it belongs to no standard course. We include it because the single most common obstacle for readers approaching the decoupling literature is not any individual theorem but the unfamiliar rhythm of multi-scale bookkeeping, and that rhythm can be learned early at almost no cost.



## Chapter 1.6

# Product Measures and Fubini–Tonelli

Integration in  $\mathbb{R}^n$  would be impractical if every integral had to be computed from (1.4.1). What makes it practical is the ability to integrate one variable at a time. This chapter establishes that, and then spends the resulting technology on two computations: the volume of a ball, and — in §6.5 — the measure of the intersection of two thin tubes, from which we deduce the two-dimensional Kakeya conjecture.

### 1.6.1 The product measure

Let  $(X, \mathcal{M}, \mu)$  and  $(Y, \mathcal{N}, \nu)$  be measure spaces, assumed  $\sigma$ -finite throughout:  $X = \bigcup_k X_k$  with  $\mu(X_k) < \infty$ , and similarly for  $Y$ . Lebesgue measure on  $\mathbb{R}^n$  is  $\sigma$ -finite, by exhaustion with balls.

**Definition 1.6.1.** A *measurable rectangle* is a set  $A \times B$  with  $A \in \mathcal{M}$ ,  $B \in \mathcal{N}$ . The *product  $\sigma$ -algebra*  $\mathcal{M} \otimes \mathcal{N}$  is the  $\sigma$ -algebra generated by the measurable rectangles.

**Definition 1.6.2.** For  $E \subseteq X \times Y$  and  $x \in X$ , the *slice* is  $E_x = \{y \in Y : (x, y) \in E\}$ . For  $f$  on  $X \times Y$ ,  $f_x(y) = f(x, y)$ .

**Lemma 1.6.3** (Slices are measurable). *If  $E \in \mathcal{M} \otimes \mathcal{N}$  then  $E_x \in \mathcal{N}$  for every  $x \in X$ . If  $f$  is  $\mathcal{M} \otimes \mathcal{N}$ -measurable then  $f_x$  is  $\mathcal{N}$ -measurable for every  $x$ .*

*Proof.* Let  $\mathcal{S} = \{E \subseteq X \times Y : E_x \in \mathcal{N} \ \forall x\}$ . Slicing commutes with all set operations —  $(E^c)_x = (E_x)^c$  and  $(\bigcup_k E_k)_x = \bigcup_k (E_k)_x$  — so  $\mathcal{S}$  is a  $\sigma$ -algebra. It contains every measurable rectangle, since  $(A \times B)_x$  is  $B$  if  $x \in A$  and  $\emptyset$  otherwise. Hence  $\mathcal{M} \otimes \mathcal{N} \subseteq \mathcal{S}$ . The statement for functions follows by applying this to  $\{f > a\}$ , whose slice is  $\{f_x > a\}$ .  $\square$

**Pitfall: slices of *Lebesgue* measurable sets**

Lemma 1.6.3 concerns  $\mathcal{M} \otimes \mathcal{N}$ . It is false for  $\mathcal{L}(\mathbb{R}^2)$ : if  $A \subseteq \mathbb{R}$  is non-measurable then  $\{0\} \times A$  is  $\mathcal{L}(\mathbb{R}^2)$ -null, hence measurable, but its slice at  $x = 0$  is  $A$ . The resolution is that  $\mathcal{L}(\mathbb{R}) \otimes \mathcal{L}(\mathbb{R}) \subsetneq \mathcal{L}(\mathbb{R}^2)$ , the latter being the completion of the former, and that for a set in the completion *almost every* slice is measurable. We state Fubini below for the product  $\sigma$ -algebra and note the a.e. caveat where it matters.

**Theorem 1.6.4** (Existence and uniqueness of the product measure). *There is a unique measure  $\mu \times \nu$  on  $\mathcal{M} \otimes \mathcal{N}$  with  $(\mu \times \nu)(A \times B) = \mu(A)\nu(B)$  for all measurable rectangles. Moreover, for every  $E \in \mathcal{M} \otimes \mathcal{N}$  the function  $x \mapsto \nu(E_x)$  is  $\mathcal{M}$ -measurable and*

$$(\mu \times \nu)(E) = \int_X \nu(E_x) d\mu(x). \quad (1.6.1)$$

*Idea of the proof.* Formula (1.6.1) is both the construction and the theorem: define the product measure by the right-hand side and check it works. The only issue is measurability of  $x \mapsto \nu(E_x)$ , which is obvious for rectangles and must be propagated to the generated  $\sigma$ -algebra. One cannot propagate along complements directly — the class of  $E$  for which  $\nu(E_x)$  is measurable is not obviously closed under complementation when  $\nu$  is infinite — so one works first with sets of finite measure and uses a monotone class argument. This is the sole reason  $\sigma$ -finiteness is assumed.

*Proof. Step 1. The monotone class lemma.*

We first prove the tool. A *monotone class* is a family  $\mathcal{C} \subseteq \mathcal{P}(X)$  closed under increasing unions and decreasing intersections.

**Lemma 1.6.5** (Monotone class lemma). *If  $\mathcal{A}$  is an algebra and  $\mathcal{C}$  is a monotone class with  $\mathcal{A} \subseteq \mathcal{C}$ , then  $\sigma(\mathcal{A}) \subseteq \mathcal{C}$ .*

*Proof.* An arbitrary intersection of monotone classes is a monotone class, so there is a smallest monotone class  $\mathcal{M}$  containing  $\mathcal{A}$ , and  $\mathcal{M} \subseteq \mathcal{C}$ . It suffices to show  $\mathcal{M}$  is a  $\sigma$ -algebra, for then  $\sigma(\mathcal{A}) \subseteq \mathcal{M}$ .

For  $E \in \mathcal{M}$  put

$$\mathcal{M}(E) = \{F \in \mathcal{M} : E \setminus F, F \setminus E, E \cap F \in \mathcal{M}\}.$$

Each  $\mathcal{M}(E)$  is a monotone class: if  $F_k \uparrow F$  with  $F_k \in \mathcal{M}(E)$ , then  $E \cap F_k \uparrow E \cap F$  and  $F_k \setminus E \uparrow F \setminus E$  and  $E \setminus F_k \downarrow E \setminus F$ , and  $\mathcal{M}$  is closed under both kinds of limit; decreasing sequences are symmetric.

Observe the symmetry  $F \in \mathcal{M}(E) \iff E \in \mathcal{M}(F)$ , immediate from the definition.

Now let  $E \in \mathcal{A}$ . For  $F \in \mathcal{A}$  all three sets  $E \setminus F, F \setminus E, E \cap F$  lie in  $\mathcal{A} \subseteq \mathcal{M}$ , so  $\mathcal{A} \subseteq \mathcal{M}(E)$ ; as  $\mathcal{M}(E)$  is a monotone class and  $\mathcal{M}$  is the smallest one containing  $\mathcal{A}$ , we get  $\mathcal{M}(E) = \mathcal{M}$ . By the symmetry, this says: for every  $F \in \mathcal{M}$  and every  $E \in \mathcal{A}$ ,  $E \in \mathcal{M}(F)$ . Hence  $\mathcal{A} \subseteq \mathcal{M}(F)$  for every  $F \in \mathcal{M}$ , and the same minimality argument gives  $\mathcal{M}(F) = \mathcal{M}$  for all  $F \in \mathcal{M}$ .

So  $\mathcal{M}$  is closed under differences and finite intersections, hence under finite unions, and contains  $X$ : it is an algebra. An algebra closed under increasing unions is a  $\sigma$ -algebra, since  $\bigcup_k E_k = \lim_N \uparrow \bigcup_{k \leq N} E_k$ .  $\square$

**Step 2.** *Finite measure case.*

Assume  $\mu(X), \nu(Y) < \infty$ . Let  $\mathcal{C}$  be the family of  $E \in \mathcal{M} \otimes \mathcal{N}$  for which  $x \mapsto \nu(E_x)$  is measurable. Rectangles lie in  $\mathcal{C}$ , and so does the algebra  $\mathcal{A}$  of finite disjoint unions of rectangles, since slices and  $\nu$  are finitely additive. If  $E_k \uparrow E$  with  $E_k \in \mathcal{C}$  then  $\nu((E_k)_x) \uparrow \nu(E_x)$  by continuity from below, so  $E \in \mathcal{C}$  as a pointwise limit of measurable functions; if  $E_k \downarrow E$  the same works using continuity from above, legitimate because  $\nu(Y) < \infty$ . So  $\mathcal{C}$  is a monotone class containing  $\mathcal{A}$ , hence, by Lemma 1.6.5, contains  $\sigma(\mathcal{A}) = \mathcal{M} \otimes \mathcal{N}$ .

**Step 3.** *Definition and countable additivity.*

Define  $(\mu \times \nu)(E)$  by (1.6.1). If  $E = \bigsqcup_k E_k$  then  $E_x = \bigsqcup_k (E_k)_x$ , so  $\nu(E_x) = \sum_k \nu((E_k)_x)$  by countable additivity of  $\nu$ , and Corollary 1.5.3 allows term-by-term integration. Hence  $\mu \times \nu$  is a measure, and on a rectangle (1.6.1) gives  $\int_X \nu(B) \mathbf{1}_A d\mu = \mu(A) \nu(B)$ .

**Step 4.**  *$\sigma$ -finite case and uniqueness.*

Write  $X \times Y = \bigcup_k X_k \times Y_k$  with each factor of finite measure and the union increasing; apply Steps 2–3 on each and pass to the limit. Uniqueness follows because two measures agreeing on the algebra  $\mathcal{A}$  and  $\sigma$ -finite along it agree on  $\sigma(\mathcal{A})$ , by the same monotone class argument.  $\square$

## 1.6.2 Tonelli and Fubini

**Theorem 1.6.6** (Tonelli). *Let  $f : X \times Y \rightarrow [0, \infty]$  be  $\mathcal{M} \otimes \mathcal{N}$ -measurable. Then  $x \mapsto \int_Y f_x d\nu$  is  $\mathcal{M}$ -measurable,  $y \mapsto \int_X f^y d\mu$  is  $\mathcal{N}$ -measurable, and*

$$\int_{X \times Y} f d(\mu \times \nu) = \int_X \left( \int_Y f(x, y) d\nu(y) \right) d\mu(x) = \int_Y \left( \int_X f(x, y) d\mu(x) \right) d\nu(y).$$

*Proof.* For  $f = \mathbf{1}_E$  this is exactly (1.6.1) and its mirror image. By linearity (Corollary 1.5.2) it holds for nonnegative simple functions. For general  $f \geq 0$

take simple  $\varphi_k \uparrow f$  (Theorem 1.4.6); then  $(\varphi_k)_x \uparrow f_x$  for each  $x$ , so by Theorem 1.5.1 applied on  $Y$ ,  $\int_Y (\varphi_k)_x d\nu \uparrow \int_Y f_x d\nu$ , which is therefore measurable in  $x$  as a pointwise limit. Apply Theorem 1.5.1 again on  $X$ , and once more on  $X \times Y$ , and equate.  $\square$

**Theorem 1.6.7** (Fubini). *Let  $f \in L^1(\mu \times \nu)$ . Then  $f_x \in L^1(\nu)$  for  $\mu$ -a.e.  $x$ , the a.e.-defined function  $x \mapsto \int_Y f_x d\nu$  is in  $L^1(\mu)$ , and the two iterated integrals both equal  $\int f d(\mu \times \nu)$ .*

*Proof.* Apply Theorem 1.6.6 to  $|f|$ : the function  $x \mapsto \int_Y |f_x| d\nu$  has finite integral, hence is finite  $\mu$ -a.e., which is the first claim. Split  $f = f^+ - f^-$  (or into four parts if complex), apply Tonelli to each, and subtract; the subtraction is legitimate because all quantities involved are finite a.e.  $\square$

**Where we are.** The practical protocol, used without further comment from here to the end of the series: *run Tonelli on  $|f|$  to license Fubini on  $f$* . Tonelli needs no integrability hypothesis but requires  $f \geq 0$ ; Fubini allows signs but requires integrability. Together they say that for nonnegative integrands one may always interchange, and for general integrands one may interchange as soon as one interchange of the absolute value gives a finite answer.

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**WORKED EXAMPLE 1.6.1— TIER A Two counterexamples**

(a) *Integrability is needed.* On  $\mathbb{N} \times \mathbb{N}$  with counting measure let  $f(m, n) = 1$  if  $m = n$ ,  $-1$  if  $m = n + 1$ , and  $0$  otherwise. Summing in  $n$  first gives  $0$  for every  $m$  except a boundary term; carefully,  $\sum_m \sum_n f = 1$  while  $\sum_n \sum_m f = 0$ . Here  $\sum |f| = \infty$ , so Fubini does not apply and Tonelli says nothing since  $f \not\geq 0$ .

(b)  *$\sigma$ -finiteness is needed.* Let  $X = Y = [0, 1]$ ,  $\mu$  Lebesgue and  $\nu$  counting measure, and  $f = \mathbf{1}_\Delta$  with  $\Delta$  the diagonal. Then  $\int_X \int_Y f d\nu d\mu = \int_X 1 d\mu = 1$ , while  $\int_Y \int_X f d\mu d\nu = \int_Y 0 d\nu = 0$ . Counting measure on  $[0, 1]$  is not  $\sigma$ -finite.

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## 1.6.3 Convolution and approximate identities

**Definition 1.6.8.** For measurable  $f, g$  on  $\mathbb{R}^n$ ,  $(f * g)(x) = \int_{\mathbb{R}^n} f(x - y)g(y) dy$  whenever the integral converges absolutely.

**Proposition 1.6.9** (Young’s inequality, the case  $q = 1$ ). *If  $f \in L^p(\mathbb{R}^n)$ ,  $1 \leq p \leq \infty$ , and  $g \in L^1(\mathbb{R}^n)$ , then  $(f * g)(x)$  converges absolutely for a.e.  $x$  and  $\|f * g\|_p \leq \|f\|_p \|g\|_1$ .*

*Proof.* For  $p = 1$ : by Tonelli and translation invariance,

$$\iint |f(x-y)| |g(y)| \, dy \, dx = \int |g(y)| \left( \int |f(x-y)| \, dx \right) \, dy = \|f\|_1 \|g\|_1 < \infty,$$

so the inner integral is finite a.e. and Fubini gives the bound. For  $p = \infty$  the estimate is immediate. The intermediate cases follow by Minkowski's integral inequality, proved as Theorem 1.8.4 (proved in Chapter 1.8, and independent of this section) in Chapter 1.8; the general Young inequality with  $1 + \frac{1}{r} = \frac{1}{p} + \frac{1}{q}$  requires interpolation and is deferred to Volume 2.  $\square$

**Proposition 1.6.10** (Approximate identity). *Let  $\phi \in L^1(\mathbb{R}^n)$  with  $\int \phi = 1$ , and set  $\phi_t(x) = t^{-n} \phi(x/t)$ . Then for  $f \in L^p(\mathbb{R}^n)$ ,  $1 \leq p < \infty$ ,*

$$\|f * \phi_t - f\|_p \longrightarrow 0 \quad \text{as } t \rightarrow 0^+.$$

*Proof.* By Theorem 1.3.10,  $\int \phi_t = 1$  for every  $t$ , so

$$(f * \phi_t)(x) - f(x) = \int (f(x-y) - f(x)) \phi_t(y) \, dy = \int (f(x-tz) - f(x)) \phi(z) \, dz,$$

substituting  $y = tz$ . Minkowski's integral inequality (Theorem 1.8.4) gives  $\|f * \phi_t - f\|_p \leq \int \|\tau_{tz} f - f\|_p |\phi(z)| \, dz$ . The integrand tends to 0 pointwise in  $z$  by continuity of translation in  $L^p$  (Worked Example 5.14 for  $p = 1$ ; the same density argument works for all  $p < \infty$ ), and is dominated by  $2\|f\|_p |\phi| \in L^1$ . Apply Theorem 1.5.6.  $\square$

## 1.6.4 Polar coordinates and surface measure

**Theorem 1.6.11** (Polar coordinates). *There is a unique Borel measure  $\sigma$  on  $\mathbb{S}^{n-1}$  such that for every measurable  $f \geq 0$  on  $\mathbb{R}^n$ ,*

$$\int_{\mathbb{R}^n} f(x) \, dx = \int_0^\infty \int_{\mathbb{S}^{n-1}} f(r\omega) r^{n-1} \, d\sigma(\omega) \, dr. \quad (1.6.2)$$

*Proof.* Define  $\sigma$  on Borel  $A \subseteq \mathbb{S}^{n-1}$  by

$$\sigma(A) = n m(\{r\omega : 0 < r < 1, \omega \in A\}),$$

the normalised measure of the cone over  $A$ . Countable additivity of  $\sigma$  follows from that of  $m$ , since disjoint  $A$  give disjoint cones. For  $f = \mathbf{1}_E$  with  $E = \{r\omega : a < r < b, \omega \in A\}$  the identity (1.6.2) reduces, using the scaling  $m(tE) = t^n m(E)$  from Theorem 1.3.10, to  $\frac{1}{n}(b^n - a^n)\sigma(A) = \int_a^b r^{n-1} \, dr \sigma(A)$ , which is calculus. Such sets generate the Borel  $\sigma$ -algebra of  $\mathbb{R}^n \setminus \{0\}$  and form a  $\pi$ -system, so the two Borel measures appearing on the sides of (1.6.2) agree; extend to simple  $f$  by linearity and to general  $f \geq 0$  by Theorem 1.5.1.  $\square$

---

**WORKED EXAMPLE 1.6.2— TIER B The volume of the unit ball, and what happens in high dimensions**

Let  $\omega_n = m(B(0, 1))$  in  $\mathbb{R}^n$ . Applying (1.6.2) to  $f(x) = e^{-\pi|x|^2}$  and using  $\sigma(\mathbb{S}^{n-1}) = n\omega_n$ ,

$$\int_{\mathbb{R}^n} e^{-\pi|x|^2} dx = n\omega_n \int_0^\infty e^{-\pi r^2} r^{n-1} dr.$$

The left side factors by Tonelli into  $\left(\int_{\mathbb{R}} e^{-\pi t^2} dt\right)^n = 1$ , using the classical Gaussian identity (Drill 6.2). Substituting  $s = \pi r^2$  on the right,

$$\int_0^\infty e^{-\pi r^2} r^{n-1} dr = \frac{1}{2} \pi^{-n/2} \Gamma(n/2),$$

so that

$$\omega_n = \frac{\pi^{n/2}}{\Gamma(\frac{n}{2} + 1)}.$$

*Reading the formula.* By Stirling,  $\omega_n \approx \left(\frac{2\pi e}{n}\right)^{n/2} (\pi n)^{-1/2}$ , which tends to 0 *superexponentially*. The unit ball in  $\mathbb{R}^{100}$  has volume about  $10^{-40}$ . Equivalently, almost all of the volume of the cube  $[-1, 1]^n$  lies outside the inscribed ball, and indeed within distance  $O(1/\sqrt{n})$  of the boundary sphere of a ball of the same volume.

*Why this is not a curiosity.* Constants of the form  $C^n$  or  $n^{n/2}$  appear in every covering argument in high dimensions, and they are the reason the Besicovitch covering lemma of Chapter 1.7 carries a dimension-dependent constant that cannot be removed. When Volume 12 proves multilinear restriction estimates in  $\mathbb{R}^n$  with constants depending on  $n$ , the dependence is of this kind and it is genuinely there.

---

## 1.6.5 The tube intersection estimate, and Kakeya in the plane

We now use Fubini for the purpose this series cares about. Everything in this section is a complete proof, and its conclusion — Theorem 1.6.14 — is the two-dimensional case of the Kakeya conjecture.

**Definition 1.6.12.** For  $\delta \in (0, 1)$ , a  $\delta$ -tube in  $\mathbb{R}^n$  is a set of the form

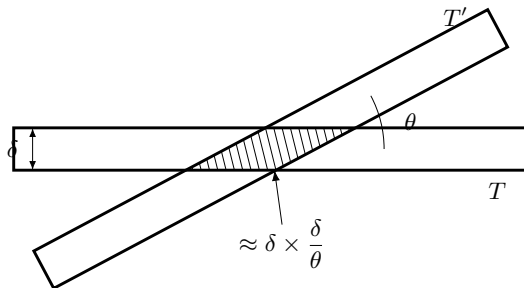
$$T = T_\delta^e(a) = \left\{x \in \mathbb{R}^n : |(x - a) \cdot e| \leq \frac{1}{2}, \quad |(x - a) - ((x - a) \cdot e)e| \leq \frac{\delta}{2}\right\},$$

the closed cylinder of length 1 and radius  $\delta/2$  with axis in direction  $e \in \mathbb{S}^{n-1}$  through the point  $a$ . Thus  $m(T) = c_n \delta^{n-1}$ .



**Lemma 1.6.13** (Tube intersection estimate). *Let  $T, T'$  be  $\delta$ -tubes in  $\mathbb{R}^n$  with directions  $e, e'$ , and let  $\theta = \angle(e, e') \in [0, \pi/2]$  be the angle between the axes. Then*

$$m(T \cap T') \lesssim_n \frac{\delta^n}{\theta + \delta}.$$



**Figure 1.6.1:** The tube intersection estimate (Lemma 1.6.13). Two  $\delta$ -tubes meeting at angle  $\theta$  cross in a parallelogram of width  $\delta$  and length  $\approx \delta/\theta$ : the smaller the angle, the longer they stay together. In  $\mathbb{R}^n$  the remaining  $n - 2$  transverse directions each contribute a further factor  $\delta$ , giving  $m(T \cap T') \lesssim \delta^n/(\theta + \delta)$ . The term  $+\delta$  records that once  $\theta \lesssim \delta$  the tubes are parallel to within their own width and the overlap can be the whole tube.

*Idea of the proof.* Two long thin cylinders crossing at a large angle meet in a small “parallelepiped” whose dimensions are  $\delta$  in each of the  $n - 1$  transverse directions and  $\delta/\theta$  along the direction where they graze past one another: the more nearly parallel they are, the longer the overlap. When  $\theta \lesssim \delta$  the tubes are parallel to within their own width and the overlap can be the whole tube, of measure  $\delta^{n-1}$ , which is what the  $+\delta$  in the denominator records. The proof is Fubini: slice by hyperplanes perpendicular to the plane spanned by  $e$  and  $e'$ , and estimate each slice.

*Proof.* By rotating (which preserves  $m$ , by Theorem 1.3.10) and translating, we may assume  $e = e_1$  and that  $e'$  lies in the  $(x_1, x_2)$ -plane, so  $e' = (\cos \theta, \sin \theta, 0, \dots, 0)$ .

Write  $x = (x_1, x_2, x'')$  with  $x'' \in \mathbb{R}^{n-2}$ . Both tubes constrain  $x''$ : a point of  $T$  has  $|x'' - a''| \leq \delta/2$ , and similarly for  $T'$ . So the set of admissible  $x''$  has measure  $\lesssim_n \delta^{n-2}$ , and by Tonelli

$$m(T \cap T') \lesssim_n \delta^{n-2} \cdot \sup_{x''} m_2((T \cap T')_{x''}), \quad (1.6.3)$$

where  $m_2$  is planar measure and the slice is taken in the  $(x_1, x_2)$ -variables.

Each slice is contained in the intersection of two planar strips:  $S$  of width  $\delta$  with direction  $e_1$ , and  $S'$  of width  $\delta$  with direction  $e'$ . It therefore suffices to bound  $m_2(S \cap S')$  where

$$S = \{|x_2 - c| \leq \tfrac{\delta}{2}\}, \quad S' = \{|-x_1 \sin \theta + x_2 \cos \theta - c'| \leq \tfrac{\delta}{2}\},$$

intersected with the bounded region  $|x_1| \leq 1$  coming from the tube length.

If  $\theta \geq \delta$ : for each fixed  $x_2$  in the  $\delta$ -range allowed by  $S$ , the constraint from  $S'$  reads  $|x_1 - \gamma(x_2)| \leq \delta/(2 \sin \theta)$  for some  $\gamma$ , an interval of length  $\delta/\sin \theta \lesssim \delta/\theta$  since  $\theta \leq \pi/2$ . By Tonelli in the plane,  $m_2(S \cap S') \lesssim \delta \cdot \frac{\delta}{\theta} = \delta^2/\theta$ .

If  $\theta < \delta$ : simply use  $m_2(S \cap S') \leq m_2(S \cap \{|x_1| \leq 1\}) \lesssim \delta$ , and note  $\delta \approx \delta^2/\delta$ .

In both cases  $m_2(S \cap S') \lesssim \delta^2/(\theta + \delta)$ ; substituting into (1.6.3) gives the lemma.  $\square$

**WORKED EXAMPLE 1.6.3— TIER C: RESEARCH TASTE** **Córdoba’s argument: the Kakeya conjecture in the plane**

We now prove a sharp lower bound on the measure of a union of tubes pointing in separated directions. It is due to Córdoba, it uses nothing beyond Lemma 1.6.13 and the Cauchy–Schwarz inequality, and it settles the planar case of a conjecture that took until 2025 to settle in  $\mathbb{R}^3$ .

**Theorem 1.6.14.** *Let  $\delta \in (0, \frac{1}{2})$  and let  $\mathbb{T}$  be a collection of  $\delta$ -tubes in  $\mathbb{R}^2$  whose directions are  $\delta$ -separated, with  $\#\mathbb{T} \approx \delta^{-1}$ . Then*

$$\left| \bigcup_{T \in \mathbb{T}} T \right| \gtrsim \frac{1}{\log(1/\delta)}.$$

*Step 1: the  $L^2$  computation.* Set  $F = \sum_{T \in \mathbb{T}} \mathbf{1}_T$ , the multiplicity function. Expanding the square and using Tonelli,

$$\|F\|_2^2 = \int_{\mathbb{R}^2} \left( \sum_T \mathbf{1}_T \right)^2 = \sum_{T, T' \in \mathbb{T}} |T \cap T'|.$$

Fix  $T$  with direction  $e$ . Because the directions are  $\delta$ -separated, for each dyadic value  $2^{-m}$  with  $\delta \leq 2^{-m} \leq 1$  the number of  $T'$  whose direction makes an angle  $\approx 2^{-m}$  with  $e$  is at most  $\lesssim 2^{-m}/\delta$ . Applying Lemma 1.6.13 with  $n = 2$ ,

$$\sum_{T' \in \mathbb{T}} |T \cap T'| \lesssim \sum_{\substack{m \geq 0 \\ 2^{-m} \geq \delta}} \frac{2^{-m}}{\delta} \cdot \frac{\delta^2}{2^{-m}} = \sum_{\substack{m \geq 0 \\ 2^{-m} \geq \delta}} \delta \lesssim \delta \log(1/\delta),$$

because there are  $\approx \log(1/\delta)$  admissible values of  $m$ . The contribution from  $T'$  nearly parallel to  $T$  (angle  $\lesssim \delta$ ) is  $\lesssim \delta$  as well, since there are  $O(1)$  such tubes each contributing  $\lesssim \delta$ . Summing over the  $\approx \delta^{-1}$  choices of  $T$ ,

$$\|F\|_2^2 \lesssim \delta^{-1} \cdot \delta \log(1/\delta) = \log(1/\delta). \quad (1.6.4)$$

*Step 2: Cauchy-Schwarz.* Let  $U = \bigcup_T T$ , so  $F$  is supported in  $U$ . Then

$$\sum_T |T| = \int_U F \leq |U|^{1/2} \|F\|_2,$$

and since each  $|T| \approx \delta$  and there are  $\approx \delta^{-1}$  tubes, the left side is  $\approx 1$ . Squaring and inserting (1.6.4),

$$1 \lesssim |U| \cdot \log(1/\delta),$$

which is the theorem. □

*Why the logarithm is not an artefact.* Theorem 1.3.12 produced a configuration of tubes in  $\delta$ -separated directions whose union has measure tending to 0. Theorem 1.6.14 says it cannot go to zero faster than  $1/\log(1/\delta)$ , and a refinement of the Perron construction shows this rate is attained. So the two theorems of this volume meet exactly: the upper and lower bounds match, and the planar problem is closed.

*The shape of the argument.* Every subsequent advance on Kakeya, up to and including Wang-Zahl, has the same two-step skeleton: (i) bound the  $L^2$  or  $L^p$  norm of the multiplicity function by summing a pairwise intersection estimate over the configuration; (ii) convert that into a lower bound for  $|U|$  by an inequality of Cauchy-Schwarz type. What changes is that in  $\mathbb{R}^n$ ,  $n \geq 3$ , the pairwise estimate of Lemma 1.6.13 is *insufficient*: a configuration can have every pair of tubes nearly transverse while the whole family conspires to concentrate near a low-dimensional surface. Wolff's hairbrush argument replaces pairwise information with information about triples through a common tube; the polynomial method replaces it with algebraic information about the whole family at once. Lemma 1.6.13 is where all of them start.

*What this does and does not prove.* Theorem 1.6.14 bounds the measure of a union of finitely many tubes. To deduce that a planar Besicovitch set has Hausdorff dimension 2 one applies the theorem at every scale  $\delta = 2^{-k}$  to the  $\delta$ -neighbourhood of the set and converts the resulting bound into a dimension statement — a routine argument, but one that needs the Hausdorff machinery of Volume 7. The Kakeya *maximal* function conjecture in the plane, which is the sharpest form, follows from the same  $L^2$  computation by duality and is proved in Chapter 1.7, Worked Example 7.7.

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*Remark 1.6.15* (A forward reference, and why it is not circular). Two proofs in this chapter invoke Minkowski’s integral inequality, Theorem 1.8.4, which is stated and proved in Chapter 1.8. This is the only place in Volume 1 where a chapter uses a result proved later, and it is worth saying explicitly why no circularity arises.

$L^p$  spaces cannot be introduced before Chapter 1.8, since their completeness rests on the convergence theorems of Chapter 1.5; so the inequality cannot be stated here. But its proof — by duality and Hölder — uses only Chapter 1.4 and Chapter 1.5, and in particular does not use Fubini or anything else from the present chapter. The reader may therefore turn to Theorem 1.8.4 at once without a gap.

The alternative — proving the special case needed here by hand — would duplicate the argument twice, and the duplication seemed the worse defect.

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**WORKED EXAMPLE 1.6.4— TIER B Fubini and Tonelli: which to cite**

The two theorems are constantly conflated and the difference matters.

*Tonelli.*  $f \geq 0$  measurable. The iterated integrals and the double integral all agree, possibly  $= +\infty$ . *No integrability hypothesis.*

*Fubini.*  $f \in L^1$  of the product. Same conclusion, and additionally the slices are integrable almost everywhere.

*The usual pattern.* Apply Tonelli to  $|f|$  to prove  $f \in L^1$ , then apply Fubini to  $f$  to exchange the order. Citing Fubini alone leaves the integrability unestablished, and that is the commonest gap.

*Where it bites in this series.* Volume 12’s Proposition 12.8.2 exchanges an integral over the sphere with a double integral against  $\mu \times \mu$ ; the justification is that the integrand is bounded and the measures finite — a Tonelli argument, stated there explicitly for that reason.

*The counterexample to keep.* On  $[0, 1]^2$  with  $f(x, y) = \frac{x^2 - y^2}{(x^2 + y^2)^2}$ , the two iterated integrals are  $\frac{\pi}{4}$  and  $-\frac{\pi}{4}$ . Tonelli does not apply ( $f$  changes sign) and Fubini does not ( $f \notin L^1$ ), and the conclusion genuinely fails.

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## Drills

**Drill 1.6.1.** Prove that  $\mathcal{L}(\mathbb{R}) \otimes \mathcal{L}(\mathbb{R}) \subsetneq \mathcal{L}(\mathbb{R}^2)$ , using the set  $\{0\} \times A$  of the Pitfall box.

**Drill 1.6.2.** Prove  $\int_{\mathbb{R}} e^{-\pi t^2} dt = 1$  by writing the square of the integral as a double integral over  $\mathbb{R}^2$  (Tonelli) and passing to polar coordinates (Theorem 1.6.11).

**Drill 1.6.3.** Verify the claim in Worked Example 1.6.3 that at most  $\lesssim 2^{-m}/\delta$  tubes of a  $\delta$ -separated family make angle  $\approx 2^{-m}$  with a fixed direction. Then redo the computation of (1.6.4) in  $\mathbb{R}^n$  and show it yields  $\|F\|_2^2 \lesssim \delta^{-(n-2)} \log(1/\delta)$  for  $\#\mathbb{T} \approx \delta^{-(n-1)}$ , hence  $|U| \gtrsim 1/\log(1/\delta)$  only when  $n = 2$ . Identify precisely which step degrades.

## Notes

Fubini's theorem is from 1907 and Tonelli's from 1909. The monotone class route taken in Theorem 1.6.4 is standard.

Lemma 1.6.13 is folklore and appears in essentially every paper on the Kakeya problem; the reader will find it quoted, usually without proof, as “the standard tube intersection estimate”. Theorem 1.6.14 is Córdoba's, from 1977. Drill 6.3 is worth doing carefully: the failure it exposes in dimensions  $n \geq 3$  is not a defect of the argument but the entire difficulty of the subject, and readers who have located it themselves will find Volumes 9 and 10 much easier going.



## Chapter 1.7

# Differentiation: Covering Lemmas and the Maximal Function

### 1.7.1 Why this is not a service chapter

Most measure theory courses treat differentiation theory as the last technical duty before the interesting material begins. For this series it is the opposite: this chapter is where measure theory stops being infrastructure and becomes harmonic analysis, and three things appear here for the first time that will never leave.

The first is an *operator*. Until now we have integrated functions; the Hardy–Littlewood maximal operator  $\mathcal{M}$  takes a function and returns another function, and the entire subject from here on consists of proving inequalities about such operators.

The second is a *weak-type inequality*. Theorem 1.7.3 does not bound  $\|\mathcal{M}f\|_1$  — no such bound is true — but bounds the measure of the set where  $\mathcal{M}f$  is large. Covering arguments produce weak bounds and only weak bounds, which is why weak  $L^p$  is a permanent inhabitant of the subject rather than a technicality.

The third is a *covering argument*, and this is the one that matters most. The Vitali lemma controls how much a family of balls can overlap. Balls are easy. Chapter 1.6 already showed that tubes are not, and Worked Example 1.7.2 below will show that the entire machinery of this chapter fails — provably, not merely as a gap in our ingenuity — when balls are replaced by tubes. That failure is the Kakeya problem.

## 1.7.2 The Vitali covering lemma

**Lemma 1.7.1** (Vitali, finite version). *Let  $B_1, \dots, B_M$  be a finite collection of open balls in  $\mathbb{R}^n$ . There is a pairwise disjoint subcollection  $B_{i_1}, \dots, B_{i_k}$  such that*

$$\left| \bigcup_{j=1}^M B_j \right| \leq 3^n \sum_{l=1}^k |B_{i_l}|.$$

*Idea of the proof.* Select greedily, largest first, always discarding every ball that meets something already chosen. A discarded ball  $B$  meets a selected ball  $B'$  that is at least as large, and elementary geometry then puts  $B$  inside the concentric triple  $3B'$ . So the selected balls, tripled, cover everything, and tripling costs a factor  $3^n$  in measure by Theorem 1.3.10.

*Proof.* Order the balls by decreasing radius,  $r_1 \geq r_2 \geq \dots \geq r_M$ . Let  $i_1 = 1$ . Having chosen  $i_1 < \dots < i_l$ , let  $i_{l+1}$  be the least index greater than  $i_l$  whose ball is disjoint from  $B_{i_1}, \dots, B_{i_l}$ ; stop when no such index exists. The selected balls are pairwise disjoint by construction.

Let  $B_j$  be any ball of the family. If  $j$  was not selected, then  $B_j$  meets some selected  $B_{i_l}$  with  $i_l < j$ , hence with  $r_{i_l} \geq r_j$ . Pick  $z \in B_j \cap B_{i_l}$  and let  $x \in B_j$ . Writing  $c_j, c_{i_l}$  for the centres,

$$|x - c_{i_l}| \leq |x - c_j| + |c_j - z| + |z - c_{i_l}| < r_j + r_j + r_{i_l} \leq 3r_{i_l},$$

so  $B_j \subseteq 3B_{i_l}$ , the concentric ball of triple the radius. Hence  $\bigcup_j B_j \subseteq \bigcup_l 3B_{i_l}$ , and by subadditivity together with  $|3B| = 3^n |B|$ ,

$$\left| \bigcup_j B_j \right| \leq \sum_l |3B_{i_l}| = 3^n \sum_l |B_{i_l}|. \quad \square$$

**Pitfall: the constant  $3^n$  grows with dimension, and must**

It is natural to hope for a dimension-free constant. There is none from this argument, and the growth is not an artefact of the tripling: as Worked Example 1.6.2 showed, volumes of balls collapse superexponentially in  $n$ , and covering arguments in high dimensions genuinely cost exponentially. This is worth registering now, because the same phenomenon — estimates whose constants degrade with dimension — recurs in the multilinear Kakeya and restriction theorems of Volume 12, where it is a substantive obstacle rather than a nuisance.



### 1.7.3 The Hardy–Littlewood maximal function

**Definition 1.7.2.** For  $f \in L^1_{\text{loc}}(\mathbb{R}^n)$  define

$$\mathcal{M}f(x) = \sup_{r>0} \frac{1}{|B(x, r)|} \int_{B(x, r)} |f(y)| \, dy.$$

$\mathcal{M}f$  is measurable: for each  $r$  the average is a continuous function of  $x$  by Theorem 1.5.6, so  $\{\mathcal{M}f > \lambda\}$  is a union over rational  $r$  of open sets, hence open.

**Theorem 1.7.3** (Hardy–Littlewood maximal inequality). *For every  $f \in L^1(\mathbb{R}^n)$  and every  $\lambda > 0$ ,*

$$m(\{x : \mathcal{M}f(x) > \lambda\}) \leq \frac{3^n}{\lambda} \|f\|_1. \quad (1.7.1)$$

Moreover  $\|\mathcal{M}f\|_\infty \leq \|f\|_\infty$ , and for  $1 < p \leq \infty$

$$\|\mathcal{M}f\|_p \leq C_{n,p} \|f\|_p. \quad (1.7.2)$$

*Proof. Step 1. The weak bound.*

Fix  $\lambda > 0$  and let  $E_\lambda = \{\mathcal{M}f > \lambda\}$ , an open set. Let  $K \subseteq E_\lambda$  be compact; by inner regularity (Theorem 1.3.6(iv)) it suffices to bound  $m(K)$ .

For each  $x \in K$  there is, by definition of the supremum, a radius  $r_x > 0$  with

$$\int_{B(x, r_x)} |f| > \lambda |B(x, r_x)|. \quad (1.7.3)$$

These balls cover  $K$ , so finitely many  $B_1, \dots, B_M$  do. Apply Lemma 1.7.1 to extract a disjoint subfamily  $B_{i_1}, \dots, B_{i_k}$  with  $m(\bigcup_j B_j) \leq 3^n \sum_l |B_{i_l}|$ . Then, using disjointness for the last step,

$$m(K) \leq \left| \bigcup_j B_j \right| \leq 3^n \sum_l |B_{i_l}| \stackrel{(1.7.3)}{<} \frac{3^n}{\lambda} \sum_l \int_{B_{i_l}} |f| \leq \frac{3^n}{\lambda} \int_{\mathbb{R}^n} |f|.$$

**Step 2. The  $L^\infty$  bound.**

Every average of  $|f|$  is at most  $\|f\|_\infty$ , so  $\mathcal{M}f \leq \|f\|_\infty$  pointwise.

**Step 3. The  $L^p$  bound by splitting at height  $\lambda$ .**

Let  $1 < p < \infty$  and  $f \in L^p$ . Fix  $\lambda > 0$  and decompose

$$f = f \mathbf{1}_{\{|f| > \lambda/2\}} + f \mathbf{1}_{\{|f| \leq \lambda/2\}} =: g_\lambda + b_\lambda.$$

Then  $\|b_\lambda\|_\infty \leq \lambda/2$ , so  $\mathcal{M}b_\lambda \leq \lambda/2$  by Step 2, and since  $\mathcal{M}$  is sublinear,  $\mathcal{M}f \leq \mathcal{M}g_\lambda + \lambda/2$ . Consequently

$$\{\mathcal{M}f > \lambda\} \subseteq \{\mathcal{M}g_\lambda > \lambda/2\},$$

and Step 1 applied to  $g_\lambda \in L^1$  (it is integrable because  $|g_\lambda| \leq |f|(2|f|/\lambda)^{p-1}$ ) gives

$$\lambda_{\mathcal{M}f}(\lambda) \leq \frac{2 \cdot 3^n}{\lambda} \int_{\{|f| > \lambda/2\}} |f| \, dm. \quad (1.7.4)$$

Now integrate using the layer-cake formula (1.5.2) and Tonelli:

$$\begin{aligned} \|\mathcal{M}f\|_p^p &= p \int_0^\infty \lambda^{p-1} \lambda_{\mathcal{M}f}(\lambda) \, d\lambda \leq 2 \cdot 3^n p \int_0^\infty \lambda^{p-2} \int_{\{|f| > \lambda/2\}} |f(x)| \, dx \, d\lambda \\ &= 2 \cdot 3^n p \int_{\mathbb{R}^n} |f(x)| \left( \int_0^{2|f(x)|} \lambda^{p-2} \, d\lambda \right) dx = \frac{2^p 3^n p}{p-1} \int_{\mathbb{R}^n} |f(x)|^p \, dx, \end{aligned}$$

the inner integral converging precisely because  $p > 1$ . □

**Where we are.** Note what Step 3 is: a hand-rolled proof of the Marcinkiewicz interpolation theorem in the only case we currently need. The general theorem — weak  $(1,1)$  plus  $(\infty, \infty)$  implies strong  $(p,p)$  for  $1 < p < \infty$  — is proved in Volume 2, and the proof is the argument above with  $\mathcal{M}$  replaced by an arbitrary sublinear operator. The constant blows up as  $p \rightarrow 1$ , and it must:  $\mathcal{M}f$  is never integrable unless  $f = 0$ , by Worked Example 7.5.

---

**WORKED EXAMPLE 1.7.1— TIER A  $\mathcal{M}f$  is never integrable**

Let  $f \in L^1(\mathbb{R}^n)$  be not identically 0, so there is a ball  $B(0, R)$  with  $\int_{B(0,R)} |f| = c > 0$ . For  $|x| > R$ , taking  $r = 2|x|$  gives  $B(0, R) \subseteq B(x, r)$ , hence

$$\mathcal{M}f(x) \geq \frac{1}{|B(x, 2|x|)|} \int_{B(0,R)} |f| = \frac{c}{\omega_n 2^n |x|^n}.$$

Since  $|x|^{-n}$  is not integrable at infinity — by polar coordinates (1.6.2),  $\int_{|x| > R} |x|^{-n} \, dx = \sigma(\mathbb{S}^{n-1}) \int_R^\infty r^{-1} \, dr = \infty$  — we get  $\mathcal{M}f \notin L^1$ .

So the weak bound (1.7.1) is the strongest statement available at  $p = 1$ , not a weaker substitute for something better. This is the standard situation throughout the subject.

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## 1.7.4 The Lebesgue differentiation theorem

**Theorem 1.7.4.** *Let  $f \in L^1_{\text{loc}}(\mathbb{R}^n)$ . Then for a.e.  $x$ ,*

$$\lim_{r \rightarrow 0^+} \frac{1}{|B(x, r)|} \int_{B(x, r)} f(y) \, dy = f(x),$$

and indeed  $\lim_{r \rightarrow 0^+} \frac{1}{|B(x,r)|} \int_{B(x,r)} |f(y) - f(x)| \, dy = 0$  for a.e.  $x$  (such  $x$  are called Lebesgue points).

*Idea of the proof.* The statement is obvious for continuous  $f$  and is a density statement in general. The architecture is worth naming, because it recurs so often:

$$(\text{density of a good class}) + (\text{a weak-type maximal inequality}) \\ \implies (\text{a.e. convergence}).$$

One proves the result on a dense subclass where it is easy, controls the error by a maximal operator, and uses the weak bound to show the error set is null. Nothing about balls or about differentiation is essential; the same template proves Carleson's theorem on a.e. convergence of Fourier series, and in Volume 13 it proves the a.e. convergence of solutions to the Schrödinger equation to their initial data — the Carleson problem — where the relevant maximal operator is so difficult that the sharp result was not obtained until 2017.

*Proof.* It suffices to treat  $f \in L^1(\mathbb{R}^n)$  supported in a large ball, since the statement is local. Define the *oscillation operator*

$$\Omega f(x) = \limsup_{r \rightarrow 0^+} \frac{1}{|B(x,r)|} \int_{B(x,r)} |f(y) - f(x)| \, dy.$$

We must show  $\Omega f = 0$  a.e.

**Step 1.** *The good class.*

If  $g$  is continuous then  $\Omega g = 0$  everywhere: given  $\varepsilon > 0$ , continuity at  $x$  gives  $|g(y) - g(x)| < \varepsilon$  for  $|y - x| < r$  small, so every average is at most  $\varepsilon$ .

**Step 2.** *Sublinearity and the maximal bound.*

For any  $f, g$ , the triangle inequality gives

$$\Omega f \leq \Omega(f - g) + \Omega g = \Omega(f - g),$$

when  $g$  is continuous, and

$$\Omega(f - g)(x) \leq \mathcal{M}(f - g)(x) + |(f - g)(x)|.$$

**Step 3.** *The error set is null.*

Fix  $\lambda > 0$ . By density of  $C_c(\mathbb{R}^n)$  in  $L^1$  (Chapter 1.8), choose continuous  $g$  with  $\|f - g\|_1 < \eta$ . Then

$$\{\Omega f > \lambda\} \subseteq \{\mathcal{M}(f - g) > \tfrac{\lambda}{2}\} \cup \{|f - g| > \tfrac{\lambda}{2}\},$$

and by (1.7.1) and Chebyshev (1.5.3),

$$m(\{\Omega f > \lambda\}) \leq \frac{2 \cdot 3^n}{\lambda} \eta + \frac{2}{\lambda} \eta.$$

As  $\eta > 0$  was arbitrary,  $m(\{\Omega f > \lambda\}) = 0$ ; taking  $\lambda = 1/k$  and a countable union,  $\Omega f = 0$  a.e.  $\square$

**Corollary 1.7.5** (Lebesgue density). *If  $E$  is measurable then for a.e.  $x \in E$ ,  $\lim_{r \rightarrow 0} \frac{|E \cap B(x, r)|}{|B(x, r)|} = 1$ , and for a.e.  $x \notin E$  the limit is 0.*

*Proof.* Apply Theorem 1.7.4 to  $f = \mathbf{1}_E$ .  $\square$

## 1.7.5 The Vitali covering theorem

Lemma 1.7.1 selects finitely many disjoint balls that capture a definite fraction of a union. For the fundamental theorem of calculus we need more: a disjoint family that captures a set *up to measure zero*. This is available provided the given balls come in arbitrarily small sizes at every point.

**Definition 1.7.6.** A family  $\mathcal{V}$  of closed balls is a *fine cover* of  $E \subseteq \mathbb{R}^n$  if for every  $x \in E$  and every  $\varepsilon > 0$  there is  $B \in \mathcal{V}$  with  $x \in B$  and  $0 < \text{diam } B < \varepsilon$ .

**Theorem 1.7.7** (Vitali covering theorem). *Let  $E \subseteq \mathbb{R}^n$  with  $m^*(E) < \infty$  and let  $\mathcal{V}$  be a fine cover of  $E$  by closed balls. Then there is a countable pairwise disjoint subfamily  $\{B_k\} \subseteq \mathcal{V}$  with*

$$m^*\left(E \setminus \bigcup_k B_k\right) = 0.$$

*Idea of the proof.* Select greedily, at each stage taking a ball disjoint from those already chosen and of radius more than half the largest available. Two facts then do everything. The chosen balls are disjoint inside a set of finite measure, so their radii tend to 0; and any point left uncovered after  $N$  stages is caught by a ball which must meet some *later* chosen ball of comparable size, hence lies in its five-fold dilate. The uncovered set is therefore inside  $\bigcup_{k > N} 5B_k$ , whose measure is a tail of a convergent series.

*Proof. Step 1. Localise.*

Choose an open  $U \supseteq E$  with  $m(U) < \infty$ ; this is possible because  $m^*(E) < \infty$ , by the definition of outer measure as an infimum over covers by open boxes. Discard from  $\mathcal{V}$  every ball not contained in  $U$ . The remaining family is still a fine cover of  $E$ : for  $x \in E \subseteq U$  with  $U$  open, all sufficiently small balls containing  $x$  lie in  $U$ .

**Step 2. Greedy selection.**

Suppose disjoint  $B_1, \dots, B_k$  have been chosen, and let

$$d_k = \sup \{ r(B) : B \in \mathcal{V}, B \cap B_i = \emptyset \text{ for all } i \leq k \}.$$

If no such  $B$  exists, then  $E \subseteq \bigcup_{i \leq k} B_i$  and we are done: a point  $x \in E$  outside the closed set  $\bigcup_{i \leq k} B_i$  would admit, by fineness, a small ball disjoint from it. Otherwise  $0 < d_k < \infty$  (finite because all balls lie in the bounded-measure set  $U$ ), and we choose  $B_{k+1} \in \mathcal{V}$  disjoint from  $B_1, \dots, B_k$  with

$$r(B_{k+1}) > \frac{1}{2}d_k. \quad (1.7.5)$$

**Step 3. Radii tend to zero.**

The  $B_k$  are pairwise disjoint and contained in  $U$ , so  $\sum_k m(B_k) \leq m(U) < \infty$ . Hence  $m(B_k) \rightarrow 0$  and so  $r_k \rightarrow 0$ , and by (1.7.5) also  $d_k \rightarrow 0$ .

**Step 4. The key inclusion.**

We claim that for every  $N$ ,

$$E \setminus \bigcup_{k \leq N} B_k \subseteq \bigcup_{k > N} 5B_k, \quad (1.7.6)$$

where  $5B$  is the concentric ball of five times the radius.

Let  $x \in E$  with  $x \notin \bigcup_{k \leq N} B_k$ . That union is closed, so by fineness there is  $B \in \mathcal{V}$  with  $x \in B$  and  $B$  disjoint from  $B_1, \dots, B_N$ . Now  $B$  must meet some chosen ball: if  $B$  were disjoint from every  $B_k$ , then  $r(B) \leq d_k$  for all  $k$ , forcing  $r(B) = 0$  by Step 3, which is excluded. Let  $k$  be the least index with  $B \cap B_k \neq \emptyset$ ; then  $k > N$ , and  $B$  is disjoint from  $B_1, \dots, B_{k-1}$ , so  $r(B) \leq d_{k-1} < 2r_k$  by (1.7.5).

Pick  $y \in B \cap B_k$  and let  $c_k$  be the centre of  $B_k$ . Then

$$|x - c_k| \leq |x - y| + |y - c_k| \leq 2r(B) + r_k < 4r_k + r_k = 5r_k,$$

so  $x \in 5B_k$ , proving (1.7.6).

**Step 5. Conclude.**

By (1.7.6), subadditivity and  $m(5B) = 5^n m(B)$ ,

$$m^*\left(E \setminus \bigcup_k B_k\right) \leq m^*\left(E \setminus \bigcup_{k \leq N} B_k\right) \leq 5^n \sum_{k > N} m(B_k),$$

a tail of the convergent series of Step 3, hence  $\rightarrow 0$  as  $N \rightarrow \infty$ .  $\square$

### Looking ahead: covering with bounded overlap, and why tubes admit none

There is a third covering theorem, Besicovitch's, which for balls centred at the points of a set produces subfamilies covering it with *overlap bounded by a dimensional constant*  $C_n$  — stronger than Vitali's dilation-by-3 and essential for differentiating a general Radon measure  $\mu$ , where  $\mu(3B)$  need not be comparable to  $\mu(B)$ . It is not needed anywhere in this volume, and is proved in Volume 7 where Radon measures are introduced.

What is worth recording here is that no analogue can exist for tubes, and the reason is Theorem 1.3.12. In the Perron configuration,  $\approx \delta^{-1}$   $\delta$ -tubes with  $\delta$ -separated directions are packed into a set of measure  $o(1)$ ; since the tubes have total measure  $\approx \delta^{-1} \cdot \delta = 1$ , the average multiplicity on the union is  $\approx 1/o(1) \rightarrow \infty$ . So no bound  $C$  independent of  $\delta$  can hold, for any covering scheme whatsoever.

Every theorem in this chapter rests on a covering lemma for balls. That is why none of it transfers to the Kakeya problem, and Worked Example 1.7.2 states the same obstruction analytically.

## 1.7.6 Differentiation of measures

**Theorem 1.7.8** (Lebesgue decomposition and Radon–Nikodym). *Let  $\mu, \nu$  be  $\sigma$ -finite measures on  $(X, \mathcal{M})$ . Then  $\nu$  decomposes uniquely as  $\nu = \nu_{ac} + \nu_s$  with  $\nu_{ac} \ll \mu$  and  $\nu_s \perp \mu$ , and there is a  $\mu$ -a.e. unique measurable  $h \geq 0$  with  $\nu_{ac}(E) = \int_E h \, d\mu$  for all  $E \in \mathcal{M}$ .*

*Proof for finite  $\mu, \nu$ ; the  $\sigma$ -finite case follows by exhausting  $X$  with sets of finite measure and* Let

$$\mathcal{F} = \left\{ h : X \rightarrow [0, \infty] \text{ measurable, } \int_E h \, d\mu \leq \nu(E) \, \forall E \in \mathcal{M} \right\},$$

which contains 0, and let  $s = \sup_{h \in \mathcal{F}} \int_X h \, d\mu \leq \nu(X) < \infty$ . If  $h_1, h_2 \in \mathcal{F}$  then  $\max(h_1, h_2) \in \mathcal{F}$ : split  $E$  into the part where  $h_1 > h_2$  and its complement, and apply the defining inequality on each. Take  $h_k \in \mathcal{F}$  with  $\int h_k \rightarrow s$  and replace  $h_k$  by  $\max(h_1, \dots, h_k)$ , so that  $h_k$  increases; let  $h = \lim_k h_k$ . By Theorem 1.5.1,  $h \in \mathcal{F}$  and  $\int h \, d\mu = s$ .

Define  $\nu_{ac}(E) = \int_E h \, d\mu$ , a measure by Corollary 1.5.4, and  $\nu_s = \nu - \nu_{ac} \geq 0$ . It remains to show  $\nu_s \perp \mu$ . If not, then for some  $\varepsilon > 0$  the signed measure  $\nu_s - \varepsilon\mu$  has a positive set  $P$  with  $\mu(P) > 0$  (Hahn decomposition, Appendix A). But then  $h + \varepsilon\mathbf{1}_P \in \mathcal{F}$ , since for any  $E$ ,

$$\int_E (h + \varepsilon\mathbf{1}_P) \, d\mu = \nu_{ac}(E) + \varepsilon\mu(E \cap P) \leq \nu_{ac}(E) + \nu_s(E \cap P) \leq \nu(E),$$

and its integral exceeds  $s$ , contradicting maximality. Uniqueness: if  $\int_E h = \int_E h'$  for all  $E$ , take  $E = \{h > h'\}$  to get  $h \leq h'$  a.e., and symmetrically.  $\square$

## 1.7.7 The fundamental theorem of calculus

**Definition 1.7.9.**  $F : [a, b] \rightarrow \mathbb{R}$  is *absolutely continuous* (AC) if for every  $\varepsilon > 0$  there is  $\delta > 0$  such that for every finite collection of pairwise disjoint subintervals  $(a_i, b_i)$  with  $\sum_i (b_i - a_i) < \delta$  one has  $\sum_i |F(b_i) - F(a_i)| < \varepsilon$ .

**Proposition 1.7.10** (Indefinite integrals are AC). *If  $f \in L^1([a, b])$  then  $F(x) = \int_a^x f$  is absolutely continuous, and  $F' = f$  a.e.*

*Proof.* Absolute continuity is the statement that  $\int_A |f| < \varepsilon$  whenever  $m(A) < \delta$ , which follows from Theorem 1.5.6 applied to  $|f|\mathbf{1}_{\{|f|>k\}}$ : choose  $k$  with  $\int |f|\mathbf{1}_{\{|f|>k\}} < \varepsilon/2$  and then  $\delta = \varepsilon/(2k)$ . For  $F' = f$  a.e., note that  $(F(x+r) - F(x))/r$  is the average of  $f$  over  $[x, x+r]$ , and one-sided averages are dominated by the two-sided averages of Theorem 1.7.4; at a Lebesgue point of  $f$  both tend to  $f(x)$ .  $\square$

The converse is the substantial half, and the Vitali covering theorem is what makes it available.

**Lemma 1.7.11** (Monotone functions are differentiable a.e.). *An increasing  $G : [a, b] \rightarrow \mathbb{R}$  is differentiable at a.e. point.*

*Proof.* For  $x \in (a, b)$  let  $\overline{DG}(x) = \limsup_{h \rightarrow 0} \frac{G(x+h) - G(x)}{h}$  and  $\underline{DG}(x)$  the corresponding lim inf, both in  $[0, \infty]$ . It suffices to show that  $\overline{DG} = \underline{DG} < \infty$  a.e.

Fix rationals  $0 \leq t < s$  and put  $E = \{x \in (a, b) : \underline{DG}(x) < t < s < \overline{DG}(x)\}$ , with  $\alpha = m^*(E)$ . Fix  $\varepsilon > 0$  and choose an open  $U \supseteq E$  with  $m(U) < \alpha + \varepsilon$ .

Every  $x \in E$  has arbitrarily short closed intervals  $[x, x+h] \subseteq U$  (or  $[x-h, x]$ ) with  $G$ -increment less than  $th$ ; these form a fine cover of  $E$ , so Theorem 1.7.7 gives disjoint intervals  $I_1, I_2, \dots \subseteq U$  with  $m^*(E \setminus \bigcup_k I_k) = 0$ . Choose  $N$  with  $m^*(E \cap \bigcup_{k \leq N} I_k) > \alpha - \varepsilon$ , and note

$$\sum_{k \leq N} (G(\text{right end of } I_k) - G(\text{left end})) < t \sum_{k \leq N} m(I_k) \leq t m(U) < t(\alpha + \varepsilon). \quad (1.7.7)$$

Now work inside  $V = \bigcup_{k \leq N} \text{int } I_k$ . Every  $x \in E \cap V$  has arbitrarily short intervals  $J \subseteq V$  containing it with  $G$ -increment greater than  $sm(J)$ ; another application of Theorem 1.7.7 yields disjoint  $J_1, \dots, J_M \subseteq V$  with

$\sum_{m \leq M} m(J_m) > \alpha - 2\varepsilon$  and

$$\sum_{m \leq M} (G(\text{right end of } J_m) - G(\text{left end})) > s(\alpha - 2\varepsilon). \quad (1.7.8)$$

Since the  $J_m$  are disjoint and each lies inside some  $I_k$ , and  $G$  is increasing, the left side of (1.7.8) is at most the left side of (1.7.7). Comparing,  $s(\alpha - 2\varepsilon) < t(\alpha + \varepsilon)$ ; letting  $\varepsilon \downarrow 0$  gives  $s\alpha \leq t\alpha$ , and since  $s > t$  we conclude  $\alpha = 0$ .

Taking the union over the countably many rational pairs  $t < s$  shows  $\overline{DG} = \underline{DG}$  a.e. Finiteness a.e. follows from the same computation with  $s$  replaced by an arbitrarily large constant: if  $\overline{DG} = \infty$  on a set of measure  $\alpha > 0$ , then (1.7.8) with  $s$  arbitrary forces  $s\alpha \leq G(b) - G(a)$  for all  $s$ , so  $\alpha = 0$ .  $\square$

**Theorem 1.7.12** (Fundamental theorem of calculus).  *$F : [a, b] \rightarrow \mathbb{R}$  is absolutely continuous if and only if  $F'$  exists a.e., lies in  $L^1([a, b])$ , and*

$$F(x) = F(a) + \int_a^x F'(t) dt \quad \text{for all } x \in [a, b].$$

*Proof.* ( $\Leftarrow$ ) is Proposition 1.7.10. For ( $\Rightarrow$ ), let  $F$  be AC.

**Step 1.**  *$F$  has bounded variation.*

Take  $\delta$  corresponding to  $\varepsilon = 1$  in the definition, and partition  $[a, b]$  into  $\lceil (b-a)/\delta \rceil$  intervals of length  $< \delta$ . On each, every finite sum  $\sum |F(b_i) - F(a_i)|$  over disjoint subintervals is at most 1, so the variation of  $F$  on each is at most 1, and the total variation is finite.

**Step 2.**  *$F'$  exists a.e. and is integrable.*

By Jordan's decomposition, a function of bounded variation is a difference of two increasing functions,  $F = G_1 - G_2$ ; by Lemma 1.7.11 each is differentiable a.e., hence so is  $F$ . For integrability, apply Fatou (Theorem 1.5.5) to the nonnegative difference quotients of  $G_i$ :

$$\int_a^b G'_i \leq \liminf_{h \rightarrow 0} \int_a^b \frac{G_i(x+h) - G_i(x)}{h} dx \leq G_i(b) - G_i(a) < \infty,$$

the last inequality because the integral of the difference quotient telescopes to an average of  $G_i$  near the endpoints. Hence  $F' \in L^1$ .

**Step 3.** *An AC function with vanishing derivative is constant.*

Suppose  $H$  is AC on  $[a, b]$  with  $H' = 0$  a.e., and let  $\varepsilon > 0$  with  $\delta$  as in the definition of absolute continuity. Let  $E = \{x : H'(x) = 0\}$ , so  $m(E) = b - a$ . For  $x \in E$  there are arbitrarily short intervals  $[x, x+h]$  with  $|H(x+h) - H(x)| \leq \varepsilon h$ ; these form a fine cover, so Theorem 1.7.7 produces disjoint



$I_1, I_2, \dots$  with  $m([a, b] \setminus \bigcup_k I_k) = 0$ . Choose  $N$  with  $\sum_{k>N} m(I_k) < \delta$ . The complement of  $\bigcup_{k \leq N} I_k$  in  $[a, b]$  is a finite union of intervals of total length  $< \delta$ , so absolute continuity bounds the sum of  $|H|$ -increments over them by  $\varepsilon$ ; and the increments over  $I_1, \dots, I_N$  total at most  $\varepsilon(b-a)$ . Hence  $|H(b) - H(a)| \leq \varepsilon + \varepsilon(b-a)$ , and letting  $\varepsilon \downarrow 0$  gives  $H(b) = H(a)$ . The same applies on every subinterval, so  $H$  is constant.

**Step 4. Conclude.**

Put  $H(x) = F(x) - \int_a^x F'$ . By Step 2 and Proposition 1.7.10 the integral term is AC with derivative  $F'$  a.e., so  $H$  is AC with  $H' = 0$  a.e.; by Step 3,  $H$  is constant, equal to  $H(a) = F(a)$ .  $\square$

## 1.7.8 The Kakeya maximal function

We close the volume's main line with the operator that the whole series is about.

**Definition 1.7.13** (Kakeya maximal operator). For  $f \in L^1_{\text{loc}}(\mathbb{R}^n)$ ,  $\delta \in (0, 1)$  and  $e \in \mathbb{S}^{n-1}$ ,

$$\mathcal{K}_\delta f(e) = \sup_{a \in \mathbb{R}^n} \frac{1}{m(T_\delta^e(a))} \int_{T_\delta^e(a)} |f| \, dm,$$

the supremum of averages over all  $\delta$ -tubes pointing in direction  $e$ . Thus  $\mathcal{K}_\delta f$  is a function on  $\mathbb{S}^{n-1}$ .

WORKED EXAMPLE 1.7.2— TIER C: RESEARCH TASTE **The Kakeya maximal conjecture, and why this chapter cannot touch it**

(a) *The trivial bounds.* Since every average of  $|f|$  is at most  $\|f\|_\infty$ ,

$$\|\mathcal{K}_\delta f\|_{L^\infty(\mathbb{S}^{n-1})} \leq \|f\|_\infty.$$

At the other end,  $m(T_\delta^e(a)) = c_n \delta^{n-1}$ , so crudely

$$\mathcal{K}_\delta f(e) \leq \frac{\|f\|_1}{c_n \delta^{n-1}} \quad \text{for every } e.$$

Both are immediate and both are sharp for the obvious examples: the first for  $f$  constant, the second for  $f$  concentrated in a single  $\delta$ -tube.

(b) *No  $\delta$ -uniform bound of any kind is possible.* Fix  $p < \infty$  and suppose there were a constant  $C$ , independent of  $\delta$ , with

$$\|\mathcal{K}_\delta f\|_{L^p(\mathbb{S}^{n-1})} \leq C \|f\|_{L^p(\mathbb{R}^n)}. \quad (1.7.9)$$

Let  $\varepsilon > 0$  and let  $K \subseteq \mathbb{R}^2$  be the compact set given by Theorem 1.3.12:  $m(K) < \varepsilon$  and  $K$  contains a unit segment in every direction. Let  $K_\delta$  denote its  $\delta$ -neighbourhood and take  $f = \mathbf{1}_{K_\delta}$ .

For every  $e \in \mathbb{S}^1$  there is a unit segment in direction  $e$  inside  $K$ , whose  $\delta/2$ -neighbourhood is a  $\delta$ -tube  $T$  contained in  $K_\delta$ . Hence  $\mathcal{K}_\delta f(e) = 1$  for every  $e$ , so the left side of (1.7.9) is  $\approx 1$ . But the right side is  $m(K_\delta)^{1/p}$ , and since  $K_\delta \downarrow K$  as  $\delta \rightarrow 0$  with  $m(K_1) < \infty$ , continuity from above gives  $m(K_\delta) \rightarrow m(K) < \varepsilon$ . Choosing  $\varepsilon$  small and then  $\delta$  small contradicts (1.7.9).

(c) *What survives, and what is conjectured.* A  $\delta$ -dependent loss is therefore unavoidable, and the conjecture is that it is as small as possible:

**Conjecture 1.7.14** (Kakeya maximal function conjecture). *For every  $\varepsilon > 0$ ,*

$$\|\mathcal{K}_\delta f\|_{L^n(\mathbb{S}^{n-1})} \lesssim_{n,\varepsilon} \delta^{-\varepsilon} \|f\|_{L^n(\mathbb{R}^n)}.$$

The exponent  $n$  is forced: the two trivial bounds in (a), interpolated, give  $L^p$  bounds with a  $\delta$ -loss that vanishes to first order exactly at  $p = n$ , and the Knapp-type example of a single tube shows no smaller  $p$  can work. This conjecture implies the Kakeya set conjecture — that every Besicovitch set has Hausdorff dimension  $n$  — and is strictly stronger.

(d) *Why the machinery of this chapter is powerless.* Every theorem in §§7.2–7.6 was proved by a covering argument for balls, and the Pitfall after the Looking Ahead box of §7.5 showed that covering lemmas simply fail for tubes. Part (b) above is the same fact stated analytically. So the reader should finish this volume holding a precisely formulated open problem, together with a proof that nothing they have been taught so far can address it. That is the honest position, and it is the correct starting point for Volume 2.

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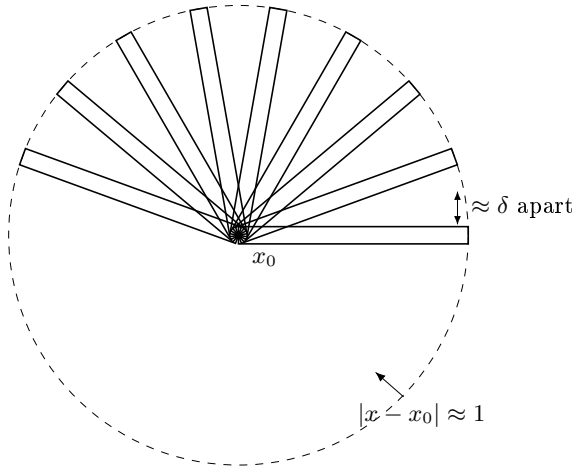
WORKED EXAMPLE 1.7.3— TIER C: RESEARCH TASTE **The bush argument**

Not quite nothing can be said. Here is a complete proof of a nontrivial lower bound, using only Lemma 1.6.13's underlying geometry and a counting argument. It was the state of the art for many years.

**Theorem 1.7.15** (Bush bound). *Let  $\mathbb{T}$  be a family of  $N \approx \delta^{-(n-1)}$   $\delta$ -tubes in  $\mathbb{R}^n$  with  $\delta$ -separated directions, and let  $U = \bigcup_{T \in \mathbb{T}} T$ . Then*

$$|U| \gtrsim_n \delta^{(n-1)/2}.$$

*Proof.* Let  $F = \sum_{T \in \mathbb{T}} \mathbf{1}_T$  be the multiplicity function and let  $\mu = \|F\|_{L^\infty}$  be the maximal multiplicity, attained (up to a constant) at some point  $x_0$  lying in  $\approx \mu$  tubes.



**Figure 1.7.1:** The bush (Theorem 1.7.15). Tubes through a common point  $x_0$ , with  $\delta$ -separated directions. Near  $x_0$  they all overlap; but at distance  $\approx 1$  from  $x_0$  two tubes whose directions differ by  $\theta \geq \delta$  are separated by  $\approx \theta \geq \delta$ , which is their own width, so they are essentially disjoint there. Hence a bush of  $\mu$  tubes occupies measure  $\gtrsim \mu \delta^{n-1}$  — the estimate that drives the whole argument.

*Step 1: a bush is disjoint away from its centre.* Consider the  $\approx \mu$  tubes through  $x_0$ , and work in the annulus

$$A = \{x : \tfrac{1}{2} \leq |x - x_0| \leq 1\}.$$

Suppose  $x \in A$  lies in two of them, with directions  $e \neq e'$ . Writing  $x - x_0 = ru$  with  $\frac{1}{2} \leq r \leq 1$  and  $|u| = 1$ , membership of the tube about  $x_0 + \mathbb{R}e$  forces  $\text{dist}(x, x_0 + \mathbb{R}e) \leq \delta$ , that is  $\angle(u, e) \lesssim \delta/r \leq 2\delta$ ; and likewise  $\angle(u, e') \leq 2\delta$ . By the triangle inequality for angles,

$$\angle(e, e') \leq \angle(e, u) + \angle(u, e') \lesssim 4\delta.$$

So if the directions are  $5\delta$ -separated the tubes are *pairwise disjoint on  $A$* . A  $\delta$ -separated family splits into  $O_n(1)$  subfamilies that are  $5\delta$ -separated, so after passing to one of them — losing only a dimensional constant — we may assume this.

Each tube through  $x_0$  meets  $A$  in a set of measure  $\gtrsim_n \delta^{n-1}$ , being a piece of the tube of length  $\frac{1}{2}$ . Summing over the  $\approx \mu$  pairwise disjoint pieces,

$$|U| \geq |U \cap A| \gtrsim_n \mu \delta^{n-1}. \quad (1.7.10)$$

Geometrically: tubes through a common point, with separated directions, spread apart linearly, and at unit distance the separation  $\approx \theta \geq \delta$  already exceeds their width.

*Step 2: multiplicity is large if the union is small.* Since  $F \leq \mu$  and  $F$  is supported in  $U$ ,

$$N c_n \delta^{n-1} = \sum_T |T| = \int_U F \leq \mu |U|,$$

so, using  $N \approx \delta^{-(n-1)}$ ,

$$\mu \gtrsim \frac{N \delta^{n-1}}{|U|} \approx \frac{1}{|U|}. \quad (1.7.11)$$

*Step 3: combine.* Substituting (1.7.11) into (1.7.10),

$$|U| \gtrsim \frac{\delta^{n-1}}{|U|}, \quad \text{hence} \quad |U|^2 \gtrsim \delta^{n-1}, \quad |U| \gtrsim \delta^{(n-1)/2}. \quad \square$$

*The dimension consequence.* Applying this to the  $\delta$ -neighbourhood of a Besicovitch set and letting  $\delta \rightarrow 0$  gives lower Minkowski dimension at least  $n - \frac{n-1}{2} = \frac{n+1}{2}$ . For  $n = 2$  this returns dimension  $\frac{3}{2}$ , weaker than Córdoba's sharp 2 from Worked Example 1.6.3; for  $n = 3$  it gives 2, and this was the best known bound until Wolff's hairbrush argument of 1995 improved it to  $\frac{5}{2}$ .

*Where the argument is lossy.* Step 1 used only the tubes through a *single* point. A configuration could have one very heavy point and be otherwise unremarkable, and the argument would not notice. Wolff's improvement replaces the bush — tubes through a point — with the *hairbrush*: tubes meeting a fixed tube. Because a tube is one-dimensional rather than zero-dimensional, the tubes of a hairbrush are spread out along it as well as around it, and one gains a full extra dimension's worth of separation. That is Volume 9. The polynomial method of Volume 11 goes further still, replacing local incidence information with a global algebraic constraint on the entire family at once.

**Figure 1.7.1 (to be drawn).** *Vitali's tripling*

**Content required:** A family of discs in the plane; a maximal disjoint subfamily chosen greedily by radius; and the tripled discs covering the union. Show one unselected disc meeting a selected one of larger radius, contained in its triple.

WORKED EXAMPLE 1.7.4— TIER B **Almost everywhere, and what it does not give**

The differentiation theorems of this chapter conclude “for almost every  $x$ ”, and the limits of that phrase recur throughout the series.

(a) *What it gives.* A statement outside a Lebesgue-null set. For most purposes — integration,  $L^p$  identities — that is as good as everywhere.

(b) *What it does not give.* Anything along a fixed null set. A line, a sphere, a Cantor set of dimension  $n - 1$ : all null, and an almost-everywhere statement is silent on each.

(c) *The consequence in this series.* Volume 11’s Nikodym example is exactly this: a full-measure set can have, through *every* one of its points, a line meeting it only there. The density theorem is about balls, and a line is null.

(d) *The systematic response.* When a statement along lower-dimensional families is wanted, one proves it for almost every member of the family — which is what Volume 8’s projection theorems do — and Volume 8, Worked Example 8.1.1 records that “almost every” cannot there be improved to “every”.

*The habit.* On reading “for a.e.  $x$ ”, ask with respect to which measure, and whether the set of interest might be null for it. Most apparent paradoxes in geometric measure theory dissolve at that question.

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## Drills

**Drill 1.7.1.** Compute  $\mathcal{M}\mathbf{1}_{[0,1]}$  on  $\mathbb{R}$  exactly, and verify directly that it behaves like  $1/|x|$  at infinity and that it is not integrable.

**Drill 1.7.2.** Show that the *dyadic* maximal function  $\mathcal{M}^{\text{dyad}}$ , defined by averaging over dyadic cubes containing  $x$ , satisfies a weak  $(1, 1)$  bound with constant 1 rather than  $3^n$ . Explain why the improvement is possible, and why  $\mathcal{M}^{\text{dyad}}$  is nevertheless not translation invariant.

**Drill 1.7.3.** Prove that the Kakeya maximal conjecture implies the Kakeya set conjecture, by applying the conjectured bound to  $f = \mathbf{1}_{K_\delta}$  as in Worked Example 1.7.2(b) and converting the resulting measure bound into a Minkowski dimension statement.

**Drill 1.7.4.** In Theorem 1.7.15 Step 1, verify the angle bound  $\angle(u, e) \lesssim \delta/r$  and confirm that  $5\delta$ -separation suffices for disjointness on  $A$ .

**Drill 1.7.5.** In Theorem 1.7.15, track the constants and determine for which range of  $|U|$  the error term in Step 1 is genuinely negligible. Then verify the claim that the bound returns  $\frac{n+1}{2}$  and compare with  $n$  for  $n = 2, 3, 4, 10$ .

**Drill 1.7.6.** Show that the Hardy–Littlewood maximal function of a non-zero  $f \in L^1(\mathbb{R}^n)$  is never itself in  $L^1$ , by bounding  $\mathcal{M}f(x)$  below by  $c|x|^{-n}$  for large  $|x|$ .

**Drill 1.7.7.** Deduce the Lebesgue density theorem from Theorem 1.7.4 by taking  $f = \mathbf{1}_E$ , and explain why the conclusion says nothing about the behaviour of  $E$  along any fixed line.

**Drill 1.7.8.** Verify that the Vitali covering lemma fails if the radii are unbounded, and identify where boundedness is used in the proof.

## Notes

The maximal function and (1.7.1) are due to Hardy and Littlewood (1930) in dimension one and to Wiener (1939) in general. The Besicovitch covering theorem alluded to in §7.5 is from 1945; it is proved in Volume 7, where Radon measures make it necessary.

Definition 1.7.13 is due to Bourgain (1991), and the conjecture stated in Worked Example 1.7.2 is his. Theorem 1.7.15 is folklore, appearing in essentially this form in work of Drury and of Christ–Duoandikoetxea–Rubio de Francia in the 1980s. Wolff’s hairbrush bound of  $\frac{n+2}{2}$  appeared in 1995 and stood in  $\mathbb{R}^3$  until the sequence of improvements culminating in Wang and Zahl’s resolution of the three-dimensional case in 2025.

Readers who wish to see where the story goes before beginning Volume 2 may read Bourgain’s 1991 paper and Wolff’s 1999 survey; both are readable with exactly the material of this volume plus the Hausdorff dimension of Volume 7.

## Chapter 1.8

# $L^p$ Spaces, Duality, and Hilbert Space

Almost every theorem in the remaining thirteen volumes is an inequality between  $L^p$  norms. This chapter builds the spaces, proves they are complete, identifies their duals, and introduces weak  $L^p$  — the space in which Theorem 1.7.3 naturally lives. It ends with Hilbert space, the case  $p = 2$ , where orthogonality becomes available and where the Fourier transform of Volume 2 will be at home.

### 1.8.1 Hölder's inequality

Throughout,  $(X, \mathcal{M}, \mu)$  is a measure space and  $1 \leq p \leq \infty$ . For measurable  $f$  set

$$\|f\|_p = \left( \int_X |f|^p d\mu \right)^{1/p} \quad (p < \infty), \quad \|f\|_\infty = \operatorname{ess\,sup} |f| = \inf\{M : |f| \leq M \text{ a.e.}\}.$$

Exponents  $p, p'$  with  $\frac{1}{p} + \frac{1}{p'} = 1$  are called *conjugate*; thus  $1' = \infty$  and  $2' = 2$ .

**Lemma 1.8.1** (Young's inequality for products). *For  $a, b \geq 0$  and conjugate  $1 < p, p' < \infty$ ,*

$$ab \leq \frac{a^p}{p} + \frac{b^{p'}}{p'},$$

*with equality if and only if  $a^p = b^{p'}$ .*

*Proof.* Both sides are 0 when  $ab = 0$ , so assume  $a, b > 0$ . The function  $\log$  is concave, so

$$\log \left( \frac{a^p}{p} + \frac{b^{p'}}{p'} \right) \geq \frac{1}{p} \log a^p + \frac{1}{p'} \log b^{p'} = \log a + \log b = \log(ab),$$

using  $\frac{1}{p} + \frac{1}{p'} = 1$  so that the weights sum to 1. Exponentiate. Strict concavity of log gives equality precisely when  $a^p = b^{p'}$ .  $\square$

**Theorem 1.8.2** (Hölder). *For measurable  $f, g$  and conjugate exponents  $p, p'$ ,*

$$\|fg\|_1 \leq \|f\|_p \|g\|_{p'}.$$

*Proof.* The cases  $p = 1$  and  $p = \infty$  are immediate from  $|fg| \leq \|g\|_\infty |f|$  a.e. So let  $1 < p < \infty$ . If either norm is 0 then  $fg = 0$  a.e.; if either is  $\infty$  there is nothing to prove. So normalise: replacing  $f$  by  $f/\|f\|_p$  and  $g$  by  $g/\|g\|_{p'}$ , it suffices to show  $\|fg\|_1 \leq 1$  when  $\|f\|_p = \|g\|_{p'} = 1$ .

Apply Lemma 1.8.1 pointwise with  $a = |f(x)|$ ,  $b = |g(x)|$  and integrate, using Corollary 1.5.2:

$$\int |fg| d\mu \leq \frac{1}{p} \int |f|^p d\mu + \frac{1}{p'} \int |g|^{p'} d\mu = \frac{1}{p} + \frac{1}{p'} = 1. \quad \square$$

**Corollary 1.8.3** (Generalised Hölder). *If  $\frac{1}{p_1} + \cdots + \frac{1}{p_m} = \frac{1}{r}$  with all  $p_i, r \geq 1$ , then  $\|f_1 \cdots f_m\|_r \leq \prod_i \|f_i\|_{p_i}$ .*

*Proof.* Induction on  $m$  using Theorem 1.8.2, after replacing each  $f_i$  by  $|f_i|^r$  and  $p_i$  by  $p_i/r$ .  $\square$

### Looking ahead: why the generalised form matters

Corollary 1.8.3 looks like a routine extension. It is the workhorse of Volume 12. Multilinear restriction and multilinear Kakeya estimates are inequalities for products  $\prod_{i=1}^n |E_i f_i|$ , and the passage between the multilinear and linear formulations is always an application of Corollary 1.8.3 with  $m = n$  and  $p_i = n$ ,  $r = 1$ . Readers meeting the Bennett–Carbery–Tao theorem later will find that the only non-geometric step in its statement is this corollary.

**Theorem 1.8.4** (Minkowski). *For  $1 \leq p \leq \infty$ ,  $\|f + g\|_p \leq \|f\|_p + \|g\|_p$ .*

*Proof.* The cases  $p = 1, \infty$  are clear. For  $1 < p < \infty$ , assume  $\|f + g\|_p < \infty$  (which follows from  $|f + g|^p \leq 2^p(|f|^p + |g|^p)$ ) and write

$$\int |f + g|^p \leq \int |f| |f + g|^{p-1} + \int |g| |f + g|^{p-1}.$$

Apply Theorem 1.8.2 to each term with exponents  $p$  and  $p'$ , noting  $(p-1)p' = p$ :

$$\int |f + g|^p \leq (\|f\|_p + \|g\|_p) \left( \int |f + g|^p \right)^{1/p'}.$$

If  $\|f + g\|_p > 0$ , divide by the last factor and use  $1 - \frac{1}{p'} = \frac{1}{p}$ .  $\square$



## 1.8.2 Completeness

**Definition 1.8.5.**  $L^p(\mu)$  is the space of measurable  $f$  with  $\|f\|_p < \infty$ , with functions equal a.e. identified. By Theorem 1.8.4 it is a normed vector space.

**Theorem 1.8.6** (Riesz–Fischer).  $L^p(\mu)$  is complete for every  $1 \leq p \leq \infty$ .

*Idea of the proof.* We use the criterion that a normed space is complete iff every absolutely convergent series converges. This is the right formulation here because it turns completeness into a statement about  $\sum_k \|f_k\|_p < \infty$ , and such sums are handled by the monotone convergence theorem — which is exactly the tool the Riemann theory lacked. Compare Worked Example 1.3, where the failure of completeness for  $C[0, 1]$  was presented in precisely this series form.

*Proof.* Let  $p < \infty$  and suppose  $\sum_k \|f_k\|_p = B < \infty$ . Put  $G = \sum_k |f_k|$ , a pointwise limit of increasing partial sums. By Theorem 1.5.1 and Theorem 1.8.4,

$$\|G\|_p = \lim_m \left\| \sum_{k \leq m} |f_k| \right\|_p \leq B,$$

so  $G < \infty$  a.e. Hence  $\sum_k f_k(x)$  converges absolutely for a.e.  $x$ ; call the sum  $f$ , which satisfies  $|f| \leq G$  so  $f \in L^p$ . Finally

$$\left\| f - \sum_{k \leq m} f_k \right\|_p^p = \int \left| \sum_{k > m} f_k \right|^p \leq \int \left( \sum_{k > m} |f_k| \right)^p \rightarrow 0$$

by Theorem 1.5.6, the integrand being dominated by  $G^p \in L^1$  and tending to 0 a.e.

For  $p = \infty$ : given a Cauchy sequence  $(f_k)$ , the sets where  $|f_k - f_l| > \|f_k - f_l\|_\infty$  form a countable union of null sets; off it,  $(f_k)$  is uniformly Cauchy, hence uniformly convergent.  $\square$

## 1.8.3 Density and separability

**Theorem 1.8.7.** Let  $1 \leq p < \infty$ .

- (i) Simple functions with support of finite measure are dense in  $L^p(\mu)$ .
- (ii) On  $\mathbb{R}^n$ ,  $C_c(\mathbb{R}^n)$  and  $C_c^\infty(\mathbb{R}^n)$  are dense in  $L^p(\mathbb{R}^n)$ .
- (iii) Translation is continuous:  $\|\tau_h f - f\|_p \rightarrow 0$  as  $h \rightarrow 0$ .

All three fail for  $p = \infty$ .

*Proof.* (i) For  $f \geq 0$  take simple  $\varphi_k \uparrow f$  (Theorem 1.4.6); then  $|f - \varphi_k|^p \leq f^p \in L^1$  and  $\rightarrow 0$  pointwise, so Theorem 1.5.6 applies. General  $f$  splits into four nonnegative parts.

(ii) By (i) it suffices to approximate  $\mathbf{1}_E$  with  $m(E) < \infty$ . By Theorem 1.3.6 choose  $C$  closed and  $U$  open with  $C \subseteq E \subseteq U$  and  $m(U \setminus C) < \varepsilon$ ; Urysohn's lemma (Appendix A) gives continuous  $g$  with  $\mathbf{1}_C \leq g \leq \mathbf{1}_U$ , so  $\|\mathbf{1}_E - g\|_p^p \leq m(U \setminus C) < \varepsilon$ . Smoothness is obtained by mollification:  $g * \phi_t \rightarrow g$  in  $L^p$  by Proposition 1.6.10, and  $g * \phi_t \in C_c^\infty$  when  $\phi \in C_c^\infty$ .

(iii) True for  $g \in C_c$  by uniform continuity and Theorem 1.5.6; extend by (ii) and  $\|\tau_h\| = 1$ .

For  $p = \infty$ :  $\|\mathbf{1}_{[0,\infty)} - g\|_\infty \geq \frac{1}{2}$  for every continuous  $g$ , and  $\|\tau_h \mathbf{1}_{[0,\infty)} - \mathbf{1}_{[0,\infty)}\|_\infty = 1$  for every  $h \neq 0$ .  $\square$

## 1.8.4 Duality

**Theorem 1.8.8.** *Let  $\mu$  be  $\sigma$ -finite and  $1 \leq p < \infty$ . Then the map  $g \mapsto \Lambda_g$ ,  $\Lambda_g(f) = \int fg d\mu$ , is an isometric isomorphism of  $L^{p'}(\mu)$  onto  $L^p(\mu)^*$ . This fails for  $p = \infty$ .*

*Proof. Step 1. The map is well defined and isometric.*

That  $\|\Lambda_g\| \leq \|g\|_{p'}$  is Theorem 1.8.2. For the reverse, given  $g \in L^{p'}$  with  $1 < p < \infty$ , take

$$f = |g|^{p'-1} \operatorname{sgn} \bar{g}.$$

Then  $|f|^p = |g|^{(p'-1)p} = |g|^{p'}$ , so  $\|f\|_p = \|g\|_{p'}^{p'/p}$ , while  $\int fg = \int |g|^{p'} = \|g\|_{p'}^{p'}$ . Hence  $\Lambda_g(f)/\|f\|_p = \|g\|_{p'}^{p'-p'/p} = \|g\|_{p'}$ . For  $p = 1$ ,  $p' = \infty$ : given  $\varepsilon > 0$  the set  $A = \{|g| > \|g\|_\infty - \varepsilon\}$  has positive measure, and testing against  $f = \mathbf{1}_{A'} \operatorname{sgn} \bar{g}$  for a subset  $A' \subseteq A$  of finite positive measure gives  $\Lambda_g(f) \geq (\|g\|_\infty - \varepsilon)\|f\|_1$ .

▷ *Goal: surjectivity: every  $\Lambda \in L^p(\mu)^*$  is some  $\Lambda_g$ .*

**Step 2.** *A measure built from the functional.*

Assume first  $\mu(X) < \infty$ , so that  $\mathbf{1}_E \in L^p$  for every  $E \in \mathcal{M}$ . Define

$$\nu(E) = \Lambda(\mathbf{1}_E).$$

This is countably additive: if  $E = \bigsqcup_k E_k$  then  $\mathbf{1}_{\bigcup_{k \leq N} E_k} \rightarrow \mathbf{1}_E$  in  $L^p$ , because  $\|\mathbf{1}_E - \mathbf{1}_{\bigcup_{k \leq N} E_k}\|_p^p = \mu(\bigcup_{k > N} E_k) \rightarrow 0$  by continuity from above (Proposition 1.2.6(iv), using  $\mu(X) < \infty$ ), and  $\Lambda$  is continuous. So  $\nu$  is a complex measure. Moreover  $\mu(E) = 0$  implies  $\mathbf{1}_E = 0$  in  $L^p$ , hence  $\nu(E) = 0$ : thus  $\nu \ll \mu$ .

**Step 3.** *Radon–Nikodym.*

By Theorem 1.7.8 (applied to the positive and negative, real and imaginary, parts of  $\nu$ ) there is  $g \in L^1(\mu)$  with  $\nu(E) = \int_E g \, d\mu$  for all  $E$ . By linearity,  $\Lambda(\varphi) = \int \varphi g \, d\mu$  for every simple  $\varphi$ , and hence, by uniform approximation and  $\mu(X) < \infty$ , for every  $\varphi \in L^\infty(\mu)$ .

**Step 4.**  *$g$  lies in  $L^{p'}$ .*

Let  $1 < p < \infty$ . For  $n \in \mathbb{N}$  put  $A_n = \{|g| \leq n\}$  and  $f_n = \mathbf{1}_{A_n} |g|^{p'-1} \operatorname{sgn} \bar{g}$ , which is bounded, hence in  $L^\infty \cap L^p$ . By Step 3,

$$\int_{A_n} |g|^{p'} \, d\mu = \Lambda(f_n) \leq \|\Lambda\| \|f_n\|_p = \|\Lambda\| \left( \int_{A_n} |g|^{p'} \, d\mu \right)^{1/p},$$

using  $|f_n|^p = \mathbf{1}_{A_n} |g|^{p'}$ . The integral on the left is finite (the integrand is bounded by  $n^{p'}$  and  $\mu(X) < \infty$ ), so we may divide, obtaining  $\left( \int_{A_n} |g|^{p'} \right)^{1/p'} \leq \|\Lambda\|$ . Since  $A_n \uparrow X$ , Theorem 1.5.1 gives  $\|g\|_{p'} \leq \|\Lambda\| < \infty$ .

For  $p = 1$ : if  $\mu(\{|g| > \|\Lambda\| + \varepsilon\}) > 0$  for some  $\varepsilon > 0$ , testing against the indicator of a finite-measure subset contradicts  $|\Lambda(f)| \leq \|\Lambda\| \|f\|_1$ . So  $\|g\|_\infty \leq \|\Lambda\|$ .

**Step 5.** *Conclude on finite measure spaces.*

The bounded functionals  $\Lambda$  and  $\Lambda_g$  agree on  $L^\infty(\mu)$ , which contains the simple functions and is therefore dense in  $L^p(\mu)$  for  $p < \infty$  by Theorem 1.8.7(i). Two continuous functionals agreeing on a dense set are equal.

**Step 6.** *The  $\sigma$ -finite case.*

Write  $X = \bigsqcup_k X_k$  with  $\mu(X_k) < \infty$ . Step 5 gives  $g_k \in L^{p'}(X_k)$  representing  $\Lambda$  on functions supported in  $X_k$ , with  $\|g_k\|_{p'} \leq \|\Lambda\|$ ; by uniqueness these are consistent, so they patch to a single  $g$  on  $X$ . For  $p > 1$ ,  $(\sum_{k \leq N} \|g_k\|_{p'}^{p'})^{1/p'} = \|g \mathbf{1}_{\cup_{k \leq N} X_k}\|_{p'} \leq \|\Lambda\|$  for every  $N$ , so  $g \in L^{p'}(X)$  by Theorem 1.5.1; the case  $p = 1$  is immediate. Finally  $\Lambda$  and  $\Lambda_g$  agree on functions supported in finitely many  $X_k$ , which are dense in  $L^p$  by Theorem 1.5.6.  $\square$

*Remark 1.8.9* (Why  $p = \infty$  is excluded). The argument breaks at Step 5, where density of simple functions in  $L^p$  was used; simple functions are not dense in  $L^\infty$  (Theorem 1.8.7). And the conclusion genuinely fails: by the Hahn–Banach theorem there is a bounded functional on  $L^\infty(\mathbb{R})$  extending  $f \mapsto \lim_{x \rightarrow 0} f(x)$  from the subspace of functions continuous at 0, and no  $g \in L^1$  represents it.

**Corollary 1.8.10** (Norms by testing). *For  $1 \leq p < \infty$  and  $\mu$   $\sigma$ -finite,*

$$\|f\|_p = \sup \left\{ \left| \int f g \, d\mu \right| : \|g\|_{p'} \leq 1 \right\}. \quad (1.8.1)$$

*Proof. Step 1.* The supremum is at most  $\|f\|_p$ .

For any  $g$  with  $\|g\|_{p'} \leq 1$ , Hölder's inequality (Theorem 1.8.2) gives  $|\int fg| \leq \|f\|_p \|g\|_{p'} \leq \|f\|_p$ .

**Step 2.** It is attained, when  $0 < \|f\|_p < \infty$ .

Suppose first  $1 < p < \infty$  and  $0 < \|f\|_p < \infty$ , and set

$$g = \frac{|f|^{p-1} \operatorname{sgn} \bar{f}}{\|f\|_p^{p-1}},$$

where  $\operatorname{sgn} z = \bar{z}/|z|$  for  $z \neq 0$  and 0 otherwise. Since  $(p-1)p' = p$ ,

$$\|g\|_{p'}^{p'} = \frac{1}{\|f\|_p^{(p-1)p'}} \int |f|^{(p-1)p'} = \frac{1}{\|f\|_p^p} \int |f|^p = 1,$$

so  $g$  is admissible; and

$$\int fg = \frac{1}{\|f\|_p^{p-1}} \int |f|^p = \|f\|_p.$$

For  $p = 1$  take  $g = \operatorname{sgn} \bar{f}$ , which has  $\|g\|_\infty \leq 1$  and  $\int fg = \int |f| = \|f\|_1$ .

**Step 3.** The case  $\|f\|_p = \infty$ .

Here  $\sigma$ -finiteness is used. Write  $X = \bigcup_k X_k$  with  $\mu(X_k) < \infty$  and  $X_k \uparrow X$ , and set  $f_k = f \mathbf{1}_{X_k} \mathbf{1}_{\{|f| \leq k\}}$ . Each  $f_k$  has  $\|f_k\|_p^p \leq k^p \mu(X_k) < \infty$ , and  $|f_k| \uparrow |f|$  pointwise, so  $\|f_k\|_p \uparrow \|f\|_p = \infty$  by monotone convergence (Theorem 1.5.1).

Let  $g_k$  be the function produced by Step 2 for  $f_k$ , so  $\|g_k\|_{p'} \leq 1$  and  $\int f_k g_k = \|f_k\|_p$ . Crucially  $g_k$  vanishes wherever  $f_k$  does, and  $f = f_k$  on the set where  $f_k \neq 0$ ; hence  $\int fg_k = \int f_k g_k = \|f_k\|_p \rightarrow \infty$ , so the supremum is  $\infty$  as well.

**Step 4.** The case  $\|f\|_p = 0$ .

Then  $f = 0$  a.e. and both sides vanish. □

### **Pitfall: $\sigma$ -finiteness, and why $p = \infty$ is excluded**

Step 3 is the only place  $\sigma$ -finiteness appears, and it is genuinely needed: on a measure space with an atom of infinite measure one can have  $\|f\|_p = \infty$  while every admissible  $g$  pairs with  $f$  to give 0.

The exponent  $p = \infty$  is excluded because Step 2 would require producing  $g \in L^1$  from  $f \in L^\infty$ , and the natural candidate  $\operatorname{sgn} \bar{f}$  need not be integrable. The identity  $\|f\|_\infty = \sup\{|\int fg| : \|g\|_1 \leq 1\}$  is nevertheless true for  $\sigma$ -finite  $\mu$ ; see Theorem 1.8.8 and the discussion after it.

**Looking ahead: this identity is how restriction becomes extension**

Formula (1.8.1) looks like a triviality. It is the mechanism by which every restriction estimate in Volume 6 is converted into an equivalent statement about the adjoint operator. The restriction conjecture asserts a bound  $\|\widehat{f}|_{\mathbb{S}^{n-1}}\|_{L^q(\sigma)} \lesssim \|f\|_{L^p(\mathbb{R}^n)}$ ; testing against  $g \in L^{q'}(\sigma)$  and unwinding turns this into a bound on the *extension operator*  $g \mapsto \widehat{g d\sigma}$ , which is the form in which the problem is always studied because it is the form in which wave packets appear. Every duality argument in this series is (1.8.1).

**1.8.5 Weak  $L^p$** 

**Definition 1.8.11.** For  $0 < p < \infty$ ,  $L^{p,\infty}(\mu)$  is the space of measurable  $f$  with

$$\|f\|_{p,\infty} = \sup_{\lambda>0} \lambda d_f(\lambda)^{1/p} < \infty, \quad d_f(\lambda) = \mu(\{|f| > \lambda\}).$$

**Proposition 1.8.12.**  $L^p(\mu) \subseteq L^{p,\infty}(\mu)$  with  $\|f\|_{p,\infty} \leq \|f\|_p$ , and the inclusion is strict.

*Proof. A note on notation.* The distribution function is written  $d_f$ , not  $d_f$ : the letter  $\lambda$  is reserved throughout this series for the *level*, and using it for the function as well makes expressions such as  $\lambda d_f(\lambda)^{1/p}$  unreadable. The layer-cake formula below, and every use of it in Volumes 2–13, follows this convention.

Chebyshev:  $\lambda^p d_f(\lambda) \leq \int_{\{|f|>\lambda\}} |f|^p \leq \|f\|_p^p$ . That the inclusion is strict — that weak  $L^p$  is genuinely larger — is shown by Worked Example 1.8.1 below, which exhibits  $|x|^{-n/p} \in L^{p,\infty} \setminus L^p$ .  $\square$

**WORKED EXAMPLE 1.8.1—TIER A The model function of weak  $L^p$** 

On  $\mathbb{R}^n$  let  $f(x) = |x|^{-n/p}$ . Then  $\{|f| > \lambda\} = B(0, \lambda^{-p/n})$  has measure  $\omega_n \lambda^{-p}$ , so  $\lambda d_f(\lambda)^{1/p} = \omega_n^{1/p}$  for every  $\lambda$ : the supremum is attained at every height, and  $f \in L^{p,\infty}$ .

But  $f \notin L^p$ : by polar coordinates (1.6.2),

$$\int_{\mathbb{R}^n} |x|^{-n} dx = \sigma(\mathbb{S}^{n-1}) \int_0^\infty \frac{dr}{r} = \infty,$$

diverging logarithmically at both ends. So  $|x|^{-n/p}$  sits exactly on the boundary — the canonical example of a function that is  $L^p$  “to first order” but not quite.

Where you have met this already. Worked Example 1.7.1 showed  $\mathcal{M}f(x) \gtrsim |x|^{-n}$  at infinity for any nonzero  $f \in L^1$ . That is precisely the statement  $\mathcal{M}f \in L^{1,\infty} \setminus L^1$ , and it explains why (1.7.1) is a weak-type bound and could not have been anything else.

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## 1.8.6 Hilbert space

**Definition 1.8.13.** An *inner product space* is a complex vector space  $H$  with a map  $\langle \cdot, \cdot \rangle : H \times H \rightarrow \mathbb{C}$ , linear in the first variable, conjugate symmetric, and positive definite. A *Hilbert space* is one complete in the norm  $\|x\| = \langle x, x \rangle^{1/2}$ .

$L^2(\mu)$  with  $\langle f, g \rangle = \int f \bar{g} d\mu$  is a Hilbert space, by Theorem 1.8.6.

**Theorem 1.8.14** (Projection). *Let  $K \subseteq H$  be nonempty, closed and convex. For every  $x \in H$  there is a unique  $Px \in K$  with  $\|x - Px\| = \text{dist}(x, K)$ . If  $K$  is a closed subspace then  $x - Px \perp K$ , and  $H = K \oplus K^\perp$ .*

*Proof.* Let  $d = \text{dist}(x, K)$  and take  $y_k \in K$  with  $\|x - y_k\| \rightarrow d$ . The parallelogram law  $\|a+b\|^2 + \|a-b\|^2 = 2\|a\|^2 + 2\|b\|^2$ , applied to  $a = x - y_k$ ,  $b = x - y_l$ , gives

$$\|y_k - y_l\|^2 = 2\|x - y_k\|^2 + 2\|x - y_l\|^2 - 4\left\|x - \frac{y_k + y_l}{2}\right\|^2 \leq 2\|x - y_k\|^2 + 2\|x - y_l\|^2 - 4d^2,$$

using convexity to place  $\frac{y_k + y_l}{2} \in K$ . The right side tends to 0, so  $(y_k)$  is Cauchy; completeness and closedness give a limit  $Px \in K$ , and uniqueness follows from the same identity. If  $K$  is a subspace, then for  $y \in K$  and  $t \in \mathbb{C}$  the function  $t \mapsto \|x - Px - ty\|^2$  is minimised at  $t = 0$ , whence  $\langle x - Px, y \rangle = 0$ .  $\square$

**Theorem 1.8.15** (Riesz representation). *Every bounded linear functional  $\Lambda$  on a Hilbert space  $H$  is  $\Lambda(x) = \langle x, y \rangle$  for a unique  $y \in H$ , and  $\|\Lambda\| = \|y\|$ .*

*Proof.* If  $\Lambda = 0$  take  $y = 0$ . Otherwise  $K = \ker \Lambda$  is a proper closed subspace, so by Theorem 1.8.14 there is  $z \in K^\perp$  with  $\|z\| = 1$ . For any  $x$ , the vector  $\Lambda(x)z - \Lambda(z)x$  lies in  $K$ , hence is orthogonal to  $z$ ; taking inner products with  $z$  and using  $\|z\| = 1$  gives

$$0 = \langle \Lambda(x)z - \Lambda(z)x, z \rangle = \Lambda(x) - \Lambda(z)\langle x, z \rangle,$$

that is,  $\Lambda(x) = \langle x, \overline{\Lambda(z)}z \rangle$ . So  $y = \overline{\Lambda(z)}z$  works. Uniqueness holds because  $\langle x, y_1 \rangle = \langle x, y_2 \rangle$  for all  $x$  forces  $y_1 - y_2 \perp H$ , and  $\|\Lambda\| = \|y\|$  follows from Cauchy-Schwarz together with  $\Lambda(y) = \|y\|^2$ .  $\square$

**Theorem 1.8.16** (Orthonormal bases). *Let  $\{e_\alpha\}$  be an orthonormal set in  $H$ . Then Bessel's inequality  $\sum_\alpha |\langle x, e_\alpha \rangle|^2 \leq \|x\|^2$  holds, and the following are equivalent: the span is dense; Parseval's identity  $\|x\|^2 = \sum_\alpha |\langle x, e_\alpha \rangle|^2$  holds for all  $x$ ;  $x = \sum_\alpha \langle x, e_\alpha \rangle e_\alpha$  for all  $x$ .*

*Proof.* Bessel follows by expanding  $0 \leq \|x - \sum_{\alpha \in F} \langle x, e_\alpha \rangle e_\alpha\|^2$  over finite  $F$ . The equivalences follow from Theorem 1.8.14 applied to the closed span, together with Theorem 1.8.6 to sum the series.  $\square$

**WORKED EXAMPLE 1.8.2— TIER C: RESEARCH TASTE Almost orthogonality, and the inequality that decoupling upgrades**

Orthogonality is rare; *almost* orthogonality is everywhere. This example isolates the elementary statement, then specialises it to the objects of Volume 13 — and shows why the elementary version is far too lossy, which is the reason that volume exists.

(a) *The elementary bound.* Let  $f_1, \dots, f_N \in H$  satisfy  $|\langle f_i, f_j \rangle| \leq \varepsilon$  for  $i \neq j$ , with  $\|f_i\| \leq 1$ . Then

$$\left\| \sum_{i=1}^N f_i \right\|^2 = \sum_i \|f_i\|^2 + \sum_{i \neq j} \langle f_i, f_j \rangle \leq N + N(N-1)\varepsilon.$$

If  $\varepsilon \leq 1/N$  this is  $\lesssim N$ , matching the exactly orthogonal case. So orthogonality may be violated by  $O(1/N)$  per pair without any loss.

(b) *Specialisation to caps.* Partition the truncated paraboloid  $\{(\xi, |\xi|^2) : |\xi| \leq 1\} \subseteq \mathbb{R}^n$  into  $N \approx \delta^{-(n-1)}$  caps  $\theta$  of diameter  $\delta$ , and let  $f_\theta$  have Fourier transform supported in the  $\delta^2$ -neighbourhood of  $\theta$ . By Plancherel (Volume 2),  $\langle f_\theta, f_{\theta'} \rangle = \langle \widehat{f_\theta}, \widehat{f_{\theta'}} \rangle = 0$  whenever  $\theta \neq \theta'$ , because the supports are disjoint. So

$$\left\| \sum_\theta f_\theta \right\|_{L^2}^2 = \sum_\theta \|f_\theta\|_{L^2}^2.$$

Exact orthogonality, for free, at  $p = 2$ .

(c) *Why this is not enough, and what decoupling asks.* The estimate in (b) is useless for the applications, which need  $p > 2$  — for the Vinogradov mean value theorem one needs  $p = n(n+1)$ , and for Strichartz estimates  $p$  is determined by scaling. At  $p \neq 2$  there is no Plancherel and disjointness of Fourier supports buys nothing directly. The  $\ell^2$  decoupling theorem of Bourgain and Demeter asserts that one nevertheless retains almost all of it:

$$\left\| \sum_\theta f_\theta \right\|_{L^p(\mathbb{R}^n)} \lesssim_\varepsilon \delta^{-\varepsilon} \left( \sum_\theta \|f_\theta\|_{L^p(\mathbb{R}^n)}^2 \right)^{1/2}, \quad 2 \leq p \leq \frac{2(n+1)}{n-1}.$$

Note the shape: an  $L^p$  norm on the left, an  $\ell^2$  sum of  $L^p$  norms on the right. Cauchy–Schwarz alone would give the same with  $\delta^{-\varepsilon}$  replaced by  $N^{1/2-1/p}$ , a genuine power of  $\delta$ ; the theorem says the true loss is subpolynomial. That gap is the entire content.

(d) *What makes it possible.* Only curvature. If the caps lay on a hyperplane rather than a paraboloid, the estimate would be false — the  $f_\theta$  could all be simultaneously large on a common tube. Curvature forces the tubes dual to different caps to point in different directions, and the extent to which differently-directed tubes can pile up is exactly the Kakeya problem of Worked Example 1.7.2. This is the precise sense, promised in Chapter 1.1, in which restriction, decoupling and Kakeya are one subject: they are three questions about the same picture of overlapping tubes.

You now have, in elementary form, the two endpoints of the series: the Cauchy–Schwarz bound that is trivially true, and the theorem that improves it. Everything between here and Volume 13 is the work of closing that gap.

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#### WORKED EXAMPLE 1.8.3— TIER B **Choosing an exponent**

Much of this series turns on which  $L^p$  to work in, and the choice follows a small number of considerations.

(a)  $p = 2$  for orthogonality. Only  $L^2$  is a Hilbert space, so only there do Plancherel,  $TT^*$  and almost-orthogonal decompositions apply. Every estimate in this series that is proved in a page is an  $L^2$  estimate.

(b)  $p = \infty$  for pointwise control,  $p = 1$  for total mass. These are the endpoints, and Volume 3 shows singular integrals fail at both — which is why Volume 4 constructs substitutes.

(c) Intermediate  $p$  by interpolation. Riesz–Thorin and Marcinkiewicz fill in between two known endpoints and never produce a new one.

(d) The exponent forced by scaling. For a homogeneous operator, testing on  $f(\lambda \cdot)$  determines the only possible exponent. This is the first check to run on any conjectured estimate, and half the estimates in Volumes 6 to 13 are pinned down this way before a proof is attempted.

*A worked instance.* Asked to bound an operator commuting with dilations of degree  $\alpha$ , compute how each side scales; if they scale differently no estimate can hold, and if they agree the exponent is determined.

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## Drills

**Drill 1.8.1.** Show that on a finite measure space  $L^q \subseteq L^p$  for  $p \leq q$ , with  $\|f\|_p \leq \mu(X)^{1/p-1/q} \|f\|_q$ ; and that on  $\mathbb{R}^n$  neither inclusion holds, by exhibiting functions in each difference.

**Drill 1.8.2.** Prove the interpolation inequality  $\|f\|_q \leq \|f\|_p^{1-\theta} \|f\|_r^\theta$  for  $\frac{1}{q} = \frac{1-\theta}{p} + \frac{\theta}{r}$ , directly from Hölder. This is the prototype of every interpolation argument in the series.

**Drill 1.8.3.** Show  $\|\cdot\|_{p,\infty}$  is not a norm for  $p > 1$  by finding  $f, g$  violating the triangle inequality, and verify that it does satisfy  $\|f + g\|_{p,\infty} \leq 2(\|f\|_{p,\infty} + \|g\|_{p,\infty})$ .

**Drill 1.8.4.** In Worked Example 1.8.2(c), verify that Cauchy–Schwarz gives the loss  $N^{1/2-1/p}$  and compute this as a power of  $\delta$  for  $n = 2$ ,  $p = 6$ . Compare with  $\delta^{-\varepsilon}$  and quantify what the Bourgain–Demeter theorem saves.

## Notes

Hölder’s inequality is from 1889 and Minkowski’s from 1896; Theorem 1.8.6 was proved independently by Riesz and Fischer in 1907 for  $p = 2$ . Weak  $L^p$  spaces were introduced by Lorentz.

Worked Example 1.8.2 is the only place in this volume where a theorem is stated that we do not prove, and the deviation is deliberate: a reader finishing Volume 1 should know precisely what they are working towards. The Bourgain–Demeter theorem quoted there is from 2015 and is proved in Volume 13; the restriction conjecture alluded to in the Looking Ahead box of §8.4 is open for  $n \geq 3$ .



## Chapter 1.A

# Background from Topology and Analysis

This appendix proves the results assumed in the main text but not covered by a second-year syllabus. Readers may well not have met Urysohn's lemma, Tietze extension or the Hahn decomposition; all three are used in Volume 1 and all three are proved here.

### 1.A.1 Urysohn's lemma in $\mathbb{R}^n$

**Theorem 1.A.1.** *Let  $K \subseteq U \subseteq \mathbb{R}^n$  with  $K$  compact and  $U$  open. There is  $g \in C_c(\mathbb{R}^n)$  with  $\mathbf{1}_K \leq g \leq \mathbf{1}_U$  and  $\text{supp } g$  a compact subset of  $U$ .*

*Proof.* Since  $K$  is compact and  $U^c$  closed and disjoint from  $K$ , we have  $d := \text{dist}(K, U^c) > 0$  (a continuous positive function on a compact set attains a positive minimum). Let  $V = \{x : \text{dist}(x, K) < d/3\}$  and  $W = \{x : \text{dist}(x, K) \leq 2d/3\}$ , so  $K \subseteq V \subseteq W \subseteq U$  with  $W$  compact. Define

$$g(x) = \frac{\text{dist}(x, V^c)}{\text{dist}(x, V^c) + \text{dist}(x, K)}.$$

The denominator never vanishes: if both terms were 0 then  $x$  would lie in the closed sets  $\overline{V^c}$  and  $K$ , which are disjoint since every point of  $K$  has distance  $\geq d/3$  from  $V^c$ . Each distance function is continuous (indeed 1-Lipschitz), so  $g$  is continuous;  $g = 1$  on  $K$  and  $g = 0$  off  $V$ , so  $\text{supp } g \subseteq W$ , compact in  $U$ ; and  $0 \leq g \leq 1$  throughout.  $\square$

### 1.A.2 Tietze's extension theorem

**Theorem 1.A.2.** *Let  $K \subseteq \mathbb{R}^n$  be compact and  $f : K \rightarrow \mathbb{R}$  continuous. Then there is  $g \in C_c(\mathbb{R}^n)$  with  $g|_K = f$  and  $\|g\|_\infty = \|f\|_\infty$ .*

*Proof.* Let  $M = \|f\|_\infty$ ; we may assume  $M > 0$ . We construct  $g$  as the sum of a series, each term correcting a third of the remaining error.

**Step 1.** *One step of correction.*

Suppose  $h : K \rightarrow \mathbb{R}$  is continuous with  $\|h\|_\infty \leq c$ . The sets  $A = \{h \leq -c/3\}$  and  $B = \{h \geq c/3\}$  are disjoint compact subsets of  $\mathbb{R}^n$ , so by Theorem 1.A.1 applied to  $B$  inside the open set  $A^c$  there is a continuous  $u$  with  $\mathbf{1}_B \leq u \leq \mathbf{1}_{A^c}$ ; put  $\varphi = \frac{c}{3}(2u - 1)$ , so  $\varphi$  is continuous on  $\mathbb{R}^n$  with  $\|\varphi\|_\infty \leq c/3$ ,  $\varphi = c/3$  on  $B$  and  $\varphi = -c/3$  on  $A$ . Checking the three cases  $x \in A$ ,  $x \in B$ ,  $x \notin A \cup B$  gives

$$\|h - \varphi\|_{L^\infty(K)} \leq \frac{2c}{3}.$$

**Step 2.** *Iterate.*

Apply Step 1 with  $h_1 = f$ ,  $c_1 = M$  to get  $\varphi_1$  on  $\mathbb{R}^n$  with  $\|\varphi_1\|_\infty \leq M/3$  and  $\|f - \varphi_1\|_{L^\infty(K)} \leq 2M/3$ . Apply it again to  $h_2 = f - \varphi_1$  with  $c_2 = 2M/3$ , and so on. Inductively  $\|\varphi_k\|_\infty \leq \frac{M}{3}(\frac{2}{3})^{k-1}$  and  $\|f - \sum_{j \leq k} \varphi_j\|_{L^\infty(K)} \leq M(\frac{2}{3})^k$ .

**Step 3.** *Sum.*

The series  $\sum_k \varphi_k$  converges uniformly on  $\mathbb{R}^n$ , since  $\sum_k \frac{M}{3}(\frac{2}{3})^{k-1} = M$ ; its sum  $G$  is therefore continuous, agrees with  $f$  on  $K$  by Step 2, and satisfies  $\|G\|_\infty \leq M$ . Finally multiply  $G$  by a function from Theorem 1.A.1 that is 1 on  $K$  and compactly supported, which changes nothing on  $K$  and makes the support compact.  $\square$

### 1.A.3 Signed measures and the Hahn decomposition

**Theorem 1.A.3.** *Let  $\lambda$  be a finite signed measure on  $(X, \mathcal{M})$ . There is a partition  $X = P \sqcup N$  with  $\lambda(E) \geq 0$  for every measurable  $E \subseteq P$  and  $\lambda(E) \leq 0$  for every measurable  $E \subseteq N$ .*

*Proof.* Call  $A$  *positive* if every measurable subset has nonnegative  $\lambda$ -measure, and let  $s = \sup\{\lambda(A) : A \text{ positive}\}$ , which is finite since  $\lambda$  is. Take positive  $A_k$  with  $\lambda(A_k) \rightarrow s$  and put  $P = \bigcup_k A_k$ ; a countable union of positive sets is positive (disjointify and use countable additivity), so  $P$  is positive and  $\lambda(P) = s$ .

Let  $N = P^c$  and suppose, for contradiction, that  $N$  contains a measurable  $E$  with  $\lambda(E) > 0$ . Then  $E$  is not positive, so it contains a subset of negative measure; let  $n_1$  be the least positive integer for which some measurable  $E_1 \subseteq E$  has  $\lambda(E_1) < -1/n_1$ . Since  $\lambda(E \setminus E_1) = \lambda(E) - \lambda(E_1) > \lambda(E) > 0$ , the argument repeats inside  $E \setminus E_1$ , yielding  $n_2$  and  $E_2$ , and so on. The  $E_k$  are disjoint with  $\sum_k |\lambda(E_k)| < \infty$ , forcing  $n_k \rightarrow \infty$ . Put  $F = E \setminus \bigcup_k E_k$ , so  $\lambda(F) \geq \lambda(E) > 0$ . Any measurable  $G \subseteq F$  has  $\lambda(G) \geq -1/(n_k - 1)$  for every

$k$  by minimality of  $n_k$ , hence  $\lambda(G) \geq 0$ : so  $F$  is positive. But then  $P \cup F$  is positive with  $\lambda(P \cup F) = s + \lambda(F) > s$ , contradicting the definition of  $s$ .  $\square$

## 1.A.4 Complex analysis, for Volume 2

The following are not used in Volume 1 and are recorded so that this volume is a self-contained prerequisite for the next. Proofs are in Volume 2, Chapter 1.

**Theorem 1.A.4** (Three-lines lemma). *Let  $F$  be bounded and continuous on the closed strip  $0 \leq \operatorname{Re} z \leq 1$  and holomorphic in its interior, with  $|F| \leq M_0$  on  $\operatorname{Re} z = 0$  and  $|F| \leq M_1$  on  $\operatorname{Re} z = 1$ . Then  $|F(\theta + it)| \leq M_0^{1-\theta} M_1^\theta$  for  $0 \leq \theta \leq 1$ .*

This lemma is stated here as a prerequisite and is *not* proved in this appendix; it is proved in Volume 2, Theorem 2.1.14, where it is the entire content of the Riesz–Thorin interpolation theorem. Nothing in Volume 1 uses it.

## 1.A.5 Linear algebra used without comment

Quadratic forms and signature (needed for the second fundamental form of a hypersurface in Volume 5); the singular value decomposition; and the determinant as a volume distortion factor, which is Theorem 1.3.10.



# Chapter 1.B

## Notation

### Asymptotic conventions

|                            |                                               |
|----------------------------|-----------------------------------------------|
| $A \lesssim B$             | $A \leq CB$ for an absolute constant $C$      |
| $A \lesssim_\varepsilon B$ | the constant may depend on $\varepsilon$      |
| $A \approx B$              | $A \lesssim B$ and $B \lesssim A$             |
| $A = O(B), o(B)$           | Landau notation, used when a limit is in play |

Constants are suppressed because in this subject they are almost never the point and carrying them makes the structure of an argument invisible. When a constant does matter — and in the induction-on-scales arguments of Volume 13 a constant that degrades at each of  $\log(1/\delta)$  scales is fatal — it is tracked explicitly and said so. See Worked Example 1.5.6.

## Standing symbols

|                                     |                                                            |
|-------------------------------------|------------------------------------------------------------|
| $\delta$                            | a small scale; from Volume 7, the width of a tube          |
| $R$                                 | a large scale, usually $\delta^{-1}$                       |
| $T, \mathbb{T}$                     | a tube; a family of tubes                                  |
| $e, \omega$                         | directions, i.e. points of $\mathbb{S}^{n-1}$              |
| $\theta$                            | an angle, or (from Volume 13) a cap on a hypersurface      |
| $\sigma$                            | surface measure on $\mathbb{S}^{n-1}$ , see Theorem 1.6.11 |
| $m,  E $                            | Lebesgue measure                                           |
| $\mathcal{L}, \mathcal{B}$          | the Lebesgue and Borel $\sigma$ -algebras                  |
| $\mathcal{H}^s, \dim_{\mathcal{H}}$ | Hausdorff measure and dimension, Chapter 1.2               |
| $\mathcal{M}$                       | Hardy–Littlewood maximal operator, Chapter 1.7             |
| $\mathcal{K}_{\delta}$              | Takeya maximal operator, Definition 1.7.13                 |
| $\mathbf{1}_E$                      | indicator function of $E$                                  |
| $d_f$                               | distribution function of $f$ , see (1.5.1)                 |
| $p'$                                | conjugate exponent, $\frac{1}{p} + \frac{1}{p'} = 1$       |
| $L^{p,\infty}$                      | weak $L^p$ , Chapter 1.8                                   |
| $\widehat{f}$                       | Fourier transform, Worked Example 1.5.4                    |



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## Index of the main results of Volume 1

|                                 |                                    |
|---------------------------------|------------------------------------|
| $\sigma$ -algebras, measures    | Definition 1.2.1, Definition 1.2.5 |
| Carathéodory's theorem          | Theorem 1.2.11                     |
| Carathéodory extension          | Theorem 1.2.15                     |
| Metric outer measures           | Theorem 1.2.13                     |
| Box lemma                       | Theorem 1.3.4                      |
| Structure of measurable sets    | Theorem 1.3.6                      |
| Linear images                   | Theorem 1.3.10                     |
| Vitali's non-measurable set     | Theorem 1.3.11                     |
| Besicovitch–Perron              | Theorem 1.3.12                     |
| Simple approximation            | Theorem 1.4.6                      |
| Lebesgue's criterion            | Theorem 1.4.14                     |
| Monotone class lemma            | Lemma 1.6.5                        |
| Monotone convergence            | Theorem 1.5.1                      |
| Fatou                           | Theorem 1.5.5                      |
| Dominated convergence           | Theorem 1.5.6                      |
| Egorov, Lusin                   | Theorem 1.5.9, Theorem 1.5.10      |
| Tonelli, Fubini                 | Theorem 1.6.6, Theorem 1.6.7       |
| Polar coordinates               | Theorem 1.6.11                     |
| Tube intersection estimate      | Lemma 1.6.13                       |
| Córdoba: Kakeya in the plane    | Theorem 1.6.14                     |
| Vitali covering lemma           | Lemma 1.7.1                        |
| Hardy–Littlewood                | Theorem 1.7.3                      |
| Lebesgue differentiation        | Theorem 1.7.4                      |
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| Fundamental theorem of calculus | Theorem 1.7.12                     |
| Radon–Nikodym                   | Theorem 1.7.8                      |
| Kakeya maximal conjecture       | Worked Example 1.7.2               |
| Bush bound                      | Theorem 1.7.15                     |
| Hölder, Minkowski               | Theorem 1.8.2, Theorem 1.8.4       |
| Riesz–Fischer                   | Theorem 1.8.6                      |
| Duality                         | Theorem 1.8.8                      |
| Riesz representation            | Theorem 1.8.15                     |



## Part 2

# Volume 2

Fourier Analysis, Distributions, and Interpolation



# Preface to Volume 2

## Where we are

Volume 1 built Lebesgue measure, the integral, the convergence theorems and the  $L^p$  spaces, and ended by stating two problems it could not touch: the *Takeya* maximal function conjecture, and the  $\ell^2$  decoupling inequality. The reader finished holding a precise open problem together with a proof that covering lemmas for balls cannot address it.

This volume supplies the analysis. Its centre is the Fourier transform, and its purpose is to make three things available: interpolation, which converts estimates at two exponents into estimates at every exponent between; distributions, which allow the Fourier transform to act on objects far too rough to integrate — surface measure on a sphere, for instance, which is the central object of Volume 6; and the first genuine decay estimate, the asymptotics of  $\widehat{d\sigma}$ , which is where curvature enters the subject for the first time.

## What is assumed

Everything in Volume 1, and nothing else. In particular we assume the reader knows: Lebesgue measure and its regularity; the monotone, Fatou and dominated convergence theorems; Fubini–Tonelli;  $L^p$  spaces, Hölder, duality, weak  $L^p$ ; the Hardy–Littlewood maximal function and its weak  $(1,1)$  bound; and the statement of the *Takeya* maximal conjecture. References of the form *Theorem 1.5.6* are to that text; the name of the result is given alongside the number, so that a citation remains identifiable if a later revision renumbers the source.

We do *not* assume complex analysis. Volume 1’s Appendix A recorded the three-lines lemma without proof, promising it here, and Chapter 2.1 keeps that promise from the definition of a holomorphic function onwards. Readers who have had a complex analysis course may read Chapter 2.1 for its final two sections only — the maximum principle on unbounded strips, and the three-lines lemma — and should not skip those, since the boundedness hypothesis in

Phragmén–Lindelöf is exactly what makes Riesz–Thorin true and is routinely misremembered.

## The plan

|   |                                                 |                            |
|---|-------------------------------------------------|----------------------------|
| 1 | Holomorphic functions and the three-lines lemma | the one prerequisite       |
| 2 | Interpolation: Riesz–Thorin and Marcinkiewicz   | exponents become intervals |
| 3 | Fourier series                                  | the compact model          |
| 4 | The Fourier transform on $L^1$ and $L^2$        | inversion, Plancherel      |
| 5 | The Schwartz space and tempered distributions   | rough objects admitted     |
| 6 | Convolution, multipliers, and Littlewood–Paley  | operators become symbols   |
| 7 | Sobolev spaces                                  | smoothness becomes decay   |
| 8 | The Fourier transform of surface measure        | curvature enters           |

Chapter 2.8 is the destination. It computes  $\widehat{d\sigma}$  for the sphere exactly, in terms of Bessel functions, proves the decay  $|\widehat{d\sigma}(\xi)| \lesssim |\xi|^{-(n-1)/2}$ , and shows that this single estimate already forces the restriction conjecture to have the exponent range it does. Everything before it is machinery.

## Conventions

Unchanged from Volume 1, with two additions. The Fourier transform is normalised

$$\widehat{f}(\xi) = \int_{\mathbb{R}^n} f(x) e^{-2\pi i x \cdot \xi} dx,$$

which puts the  $2\pi$  in the exponent and makes the transform an isometry of  $L^2$  with no constant; this is the convention of the restriction and decoupling literature and we adopt it throughout. And  $\mathbb{T}^n = \mathbb{R}^n/\mathbb{Z}^n$  denotes the torus, so that functions on  $\mathbb{T}^n$  are 1-periodic in each variable.

## Chapter 2.1

# Holomorphic Functions and the Three-Lines Lemma

### 2.1.1 Why this chapter is here

The Riesz–Thorin interpolation theorem says that if a linear operator is bounded  $L^{p_0} \rightarrow L^{q_0}$  and  $L^{p_1} \rightarrow L^{q_1}$ , then it is bounded between every pair of exponents lying on the segment joining  $(1/p_0, 1/q_0)$  to  $(1/p_1, 1/q_1)$ , with a constant that is *logarithmically convex* along the segment. Nothing in that statement mentions complex numbers.

The proof does, and there is no known way around it. One forms a family of test functions depending holomorphically on a complex parameter  $z$  running over the strip  $0 \leq \operatorname{Re} z \leq 1$ , arranges that the two hypotheses control the boundary lines  $\operatorname{Re} z = 0$  and  $\operatorname{Re} z = 1$ , and then invokes a maximum principle to control the interior. The logarithmic convexity of the constant is not an accident of the method: it is the maximum principle, transcribed.

So this chapter builds one-variable complex analysis from the definition, to the point where the required maximum principle — the three-lines lemma — is available. It is the only prerequisite Volume 1 promised and did not supply.

The material is also needed later on its own account. Contour deformation is the engine of the stationary phase method in Volume 5, where the decay of  $\widehat{d\sigma}$  is obtained by pushing an integration contour to where the phase has positive imaginary part; and analytic continuation in a complex parameter is how Volume 5 handles the Riesz potentials  $|\xi|^{-\alpha}$  for  $\alpha$  outside the range where they converge.

### 2.1.2 Holomorphic functions

Let  $\Omega \subseteq \mathbb{C}$  be open.

**Definition 2.1.1.**  $f : \Omega \rightarrow \mathbb{C}$  is *holomorphic* at  $z_0 \in \Omega$  if

$$f'(z_0) = \lim_{h \rightarrow 0} \frac{f(z_0 + h) - f(z_0)}{h}$$

exists, the limit being over complex  $h \rightarrow 0$ . It is holomorphic on  $\Omega$  if it is holomorphic at every point.

The definition is formally identical to the real one, and the usual sum, product, quotient and chain rules follow by the same proofs. The content is entirely in the words “over complex  $h$ ”: the difference quotient must have the same limit as  $h$  approaches 0 from every direction in the plane, and that is an enormously stronger requirement than real differentiability in two variables.

**Proposition 2.1.2** (Cauchy–Riemann equations). *Write  $f = u + iv$  with  $u, v$  real. If  $f$  is holomorphic at  $z_0 = x_0 + iy_0$  then  $u, v$  have partial derivatives there satisfying*

$$\partial_x u = \partial_y v, \quad \partial_y u = -\partial_x v,$$

and  $f'(z_0) = \partial_x u + i\partial_x v$ .

*Proof.* Take  $h$  real in the difference quotient: the limit is  $\partial_x u(z_0) + i\partial_x v(z_0)$ . Take  $h = ik$  with  $k$  real: the limit is  $\frac{1}{i}(\partial_y u(z_0) + i\partial_y v(z_0)) = \partial_y v(z_0) - i\partial_y u(z_0)$ . Equating real and imaginary parts gives the two equations.  $\square$

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WORKED EXAMPLE 2.1.1— TIER A **Two functions of  $x$  and  $y$ , one holomorphic and one not**

(a)  $f(z) = z^2 = (x^2 - y^2) + i(2xy)$ . Here  $u = x^2 - y^2$ ,  $v = 2xy$ , and  $\partial_x u = 2x = \partial_y v$ ,  $\partial_y u = -2y = -\partial_x v$ . The Cauchy–Riemann equations hold, and indeed  $f'(z) = 2z$  directly from the definition.

(b)  $f(z) = \bar{z} = x - iy$ . Then  $u = x$ ,  $v = -y$ , so  $\partial_x u = 1$  while  $\partial_y v = -1$ : the equations fail at every point. And directly: the difference quotient  $\bar{h}/h$  equals 1 for real  $h$  and  $-1$  for purely imaginary  $h$ , so no limit exists anywhere. Yet  $\bar{z}$  is as smooth as a function of  $(x, y)$  as anything could be. Holomorphy is a rigidity condition, not a smoothness condition.

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**Definition 2.1.3.** A *path* is a piecewise  $C^1$  map  $\gamma : [a, b] \rightarrow \mathbb{C}$ , and for continuous  $f$  on its image,

$$\int_{\gamma} f(z) \, dz = \int_a^b f(\gamma(t)) \gamma'(t) \, dt.$$

The path is *closed* if  $\gamma(a) = \gamma(b)$ .



Two standard facts, proved exactly as in the real case: the integral is unchanged by reparametrisation, and

$$\left| \int_{\gamma} f(z) dz \right| \leq \left( \sup_{\gamma} |f| \right) \cdot \text{length}(\gamma). \quad (2.1.1)$$

**Lemma 2.1.4** (Primitives kill closed integrals). *If  $F$  is holomorphic on  $\Omega$  with  $F' = f$  continuous, then  $\int_{\gamma} f = F(\gamma(b)) - F(\gamma(a))$  for every path  $\gamma$  in  $\Omega$ ; in particular  $\int_{\gamma} f = 0$  for closed  $\gamma$ .*

*Proof.*  $\frac{d}{dt} F(\gamma(t)) = f(\gamma(t))\gamma'(t)$  by the chain rule, on each  $C^1$  piece; integrate and sum the telescoping pieces.  $\square$

## 2.1.3 Goursat's theorem and Cauchy's theorem

Everything rests on the following, whose hypothesis is remarkably weak: no continuity of  $f'$  is assumed.

**Theorem 2.1.5** (Goursat). *Let  $\Omega \subseteq \mathbb{C}$  be open,  $f$  holomorphic on  $\Omega$ , and let  $T \subseteq \Omega$  be a closed solid triangle. Then  $\int_{\partial T} f(z) dz = 0$ .*

*Idea of the proof.* Bisect. Joining the midpoints of the three sides splits  $T$  into four similar triangles of half the linear size, and the boundary integrals over the four add up to the boundary integral over  $T$ , because each interior edge is traversed twice in opposite directions. So one of the four carries at least a quarter of the total. Iterating produces a nested sequence of triangles on which the integral is at least  $4^{-k}$  of the original while the perimeter is  $2^{-k}$  of the original. The triangles shrink to a point, and near that point holomorphy says  $f$  is well approximated by an affine function — whose integral over a closed path vanishes exactly, by Lemma 2.1.4. The error is  $o$  of the perimeter squared, which is precisely the scale that beats  $4^{-k}$ .

*Proof.* Let  $I = \int_{\partial T} f$  and let  $L, d$  denote the perimeter and diameter of  $T$ .

**Step 1.** *Bisection.*

Let  $T^{(1)}, \dots, T^{(4)}$  be the four triangles obtained by joining the midpoints of the sides of  $T$ , each oriented positively. Every edge interior to  $T$  occurs in exactly two of the  $\partial T^{(j)}$  with opposite orientations, so those contributions cancel and

$$\int_{\partial T} f = \sum_{j=1}^4 \int_{\partial T^{(j)}} f.$$

Hence some  $j$  has  $|\int_{\partial T(j)} f| \geq |I|/4$ ; call that triangle  $T_1$ . Its perimeter is  $L/2$  and its diameter  $d/2$ .

Iterating gives nested closed triangles  $T \supseteq T_1 \supseteq T_2 \supseteq \cdots$  with

$$\left| \int_{\partial T_k} f \right| \geq \frac{|I|}{4^k}, \quad \text{length}(\partial T_k) = \frac{L}{2^k}, \quad \text{diam } T_k = \frac{d}{2^k}. \quad (2.1.2)$$

**Step 2.** *The common point.*

The  $T_k$  are nonempty, compact and nested with diameters tending to 0, so  $\bigcap_k T_k = \{z_0\}$  for a single point  $z_0 \in T \subseteq \Omega$ .

**Step 3.** *Linearise at  $z_0$ .*

Since  $f$  is holomorphic at  $z_0$ , given  $\varepsilon > 0$  there is  $r > 0$  with

$$|f(z) - f(z_0) - f'(z_0)(z - z_0)| \leq \varepsilon |z - z_0| \quad \text{for } |z - z_0| < r. \quad (2.1.3)$$

The affine function  $z \mapsto f(z_0) + f'(z_0)(z - z_0)$  has the explicit primitive  $f(z_0)z + \frac{1}{2}f'(z_0)(z - z_0)^2$ , so by Lemma 2.1.4 its integral over the closed path  $\partial T_k$  is 0. Therefore, for  $k$  large enough that  $T_k \subseteq B(z_0, r)$ ,

$$\left| \int_{\partial T_k} f \right| = \left| \int_{\partial T_k} \left( f(z) - f(z_0) - f'(z_0)(z - z_0) \right) dz \right| \leq \varepsilon \cdot \frac{d}{2^k} \cdot \frac{L}{2^k} = \frac{\varepsilon dL}{4^k},$$

using (2.1.1), (2.1.3) and  $|z - z_0| \leq \text{diam } T_k$ .

**Step 4.** *Compare.*

With (2.1.2) this gives  $|I|/4^k \leq \varepsilon dL/4^k$ , so  $|I| \leq \varepsilon dL$  for every  $\varepsilon > 0$ . Hence  $I = 0$ .  $\square$

**Theorem 2.1.6** (Cauchy's theorem on a convex domain). *Let  $\Omega$  be open and convex and  $f$  holomorphic on  $\Omega$ . Then  $f$  has a primitive on  $\Omega$ , and consequently  $\int_\gamma f = 0$  for every closed path  $\gamma$  in  $\Omega$ .*

*Proof.* Fix  $a \in \Omega$  and define  $F(z) = \int_{[a,z]} f$ , the integral along the straight segment, which lies in  $\Omega$  by convexity. For  $z, z+h \in \Omega$  the triangle with vertices  $a, z, z+h$  lies in  $\Omega$ , so Theorem 2.1.5 gives

$$F(z+h) - F(z) = \int_{[z,z+h]} f.$$

Hence

$$\left| \frac{F(z+h) - F(z)}{h} - f(z) \right| = \frac{1}{|h|} \left| \int_{[z,z+h]} (f(w) - f(z)) dw \right| \leq \sup_{w \in [z,z+h]} |f(w) - f(z)| \xrightarrow{h \rightarrow 0} 0$$

by continuity of  $f$  (which holds since  $f$  is differentiable). So  $F' = f$ , and Lemma 2.1.4 applies.  $\square$

We shall need one strengthening, in which  $f$  is merely continuous at one bad point.

**Proposition 2.1.7.** *Theorem 2.1.5 and Theorem 2.1.6 remain true if  $f$  is assumed only continuous on  $\Omega$  and holomorphic on  $\Omega \setminus \{p\}$  for a single point  $p$ .*

*Proof.* It suffices to prove the triangle statement. If  $p \notin T$  we are done. If  $p$  is a vertex of  $T$ , split off a small triangle  $T_\eta$  at  $p$  with the same vertex and sides scaled by  $\eta$ ; the rest of  $T$  decomposes into triangles missing  $p$ , whose integrals vanish, so  $\int_{\partial T} f = \int_{\partial T_\eta} f$ , which by (2.1.1) is at most  $\sup_T |f| \cdot \eta L \rightarrow 0$ . Continuity of  $f$  near  $p$  makes  $\sup_T |f|$  finite. If  $p$  lies elsewhere in  $T$ , subdivide  $T$  into triangles each having  $p$  as a vertex, and apply the previous case.  $\square$

## 2.1.4 The Cauchy integral formula and analyticity

**Lemma 2.1.8.** *Let  $C$  be the positively oriented circle  $|w - a| = r$  and let  $|z - a| < r$ . Then*

$$\frac{1}{2\pi i} \int_C \frac{dw}{w - z} = 1.$$

*Proof.* Write  $\frac{1}{w - z} = \frac{1}{(w - a) - (z - a)} = \frac{1}{w - a} \cdot \frac{1}{1 - \frac{z - a}{w - a}}$ . On  $C$  we have  $|\frac{z - a}{w - a}| = \frac{|z - a|}{r} =: \rho < 1$ , so the geometric series converges uniformly on  $C$ :

$$\frac{1}{w - z} = \sum_{k=0}^{\infty} \frac{(z - a)^k}{(w - a)^{k+1}}.$$

Uniform convergence permits term-by-term integration. Parametrising  $w = a + re^{i\theta}$ ,

$$\int_C \frac{dw}{(w - a)^{k+1}} = \int_0^{2\pi} \frac{ire^{i\theta}}{r^{k+1}e^{i(k+1)\theta}} d\theta = \frac{i}{r^k} \int_0^{2\pi} e^{-ik\theta} d\theta = \begin{cases} 2\pi i, & k = 0, \\ 0, & k \geq 1. \end{cases}$$

Only the  $k = 0$  term survives, giving  $2\pi i$ .  $\square$

**Theorem 2.1.9** (Cauchy integral formula). *Let  $f$  be holomorphic on an open set containing the closed disc  $\overline{B}(a, r)$ , and let  $C = \partial B(a, r)$  be positively oriented. Then for every  $z \in B(a, r)$ ,*

$$f(z) = \frac{1}{2\pi i} \int_C \frac{f(w)}{w - z} dw.$$

*Proof. Step 1. Reduce to a small circle about  $z_0$ .*

Fix  $z_0$  in the interior of  $\gamma$ . The function

$$g(z) = \frac{f(z)}{z - z_0}$$

is holomorphic on  $\Omega \setminus \{z_0\}$ . By the deformation form of Cauchy's theorem —  $\gamma$  and the circle  $C_r = \{|z - z_0| = r\}$  are homotopic in  $\Omega \setminus \{z_0\}$  for all small  $r > 0$  —

$$\oint_{\gamma} g(z) dz = \oint_{C_r} g(z) dz \quad \text{for every small } r > 0.$$

So it suffices to evaluate the right-hand side in the limit  $r \rightarrow 0$ .

**Step 2.** *Split off the constant term.*

Write  $f(z) = f(z_0) + (f(z) - f(z_0))$ , so that

$$\oint_{C_r} \frac{f(z)}{z - z_0} dz = f(z_0) \oint_{C_r} \frac{dz}{z - z_0} + \oint_{C_r} \frac{f(z) - f(z_0)}{z - z_0} dz.$$

**Step 3.** *The first term is a computation.*

Parametrising  $z = z_0 + re^{i\vartheta}$ ,  $dz = ire^{i\vartheta} d\vartheta$ ,

$$\oint_{C_r} \frac{dz}{z - z_0} = \int_0^{2\pi} \frac{ire^{i\vartheta}}{re^{i\vartheta}} d\vartheta = 2\pi i,$$

independently of  $r$ .

**Step 4.** *The second term vanishes in the limit.*

Since  $f$  is holomorphic at  $z_0$  it is in particular continuous there, so given  $\eta > 0$  there is  $r$  with  $|f(z) - f(z_0)| \leq \eta$  on  $C_r$ . Then

$$\left| \oint_{C_r} \frac{f(z) - f(z_0)}{z - z_0} dz \right| \leq \text{length}(C_r) \cdot \sup_{C_r} \frac{|f(z) - f(z_0)|}{|z - z_0|} \leq 2\pi r \cdot \frac{\eta}{r} = 2\pi\eta,$$

by the standard estimate for contour integrals. As  $\eta > 0$  was arbitrary and the left side of Step 1 does not depend on  $r$ , this term is 0.

**Step 5.** *Combine.*

Steps 1–4 give  $\oint_{\gamma} \frac{f(z)}{z - z_0} dz = 2\pi i f(z_0)$ , which rearranges to the stated formula.  $\square$

**Theorem 2.1.10** (Holomorphic implies analytic). *Let  $f$  be holomorphic on  $B(a, R)$ . Then  $f$  has a power series expansion  $f(z) = \sum_{k \geq 0} c_k (z - a)^k$  converging on  $B(a, R)$ , with*

$$c_k = \frac{1}{2\pi i} \int_{|w-a|=r} \frac{f(w)}{(w-a)^{k+1}} dw \quad (0 < r < R).$$

*In particular  $f$  is infinitely differentiable and  $c_k = f^{(k)}(a)/k!$ .*

*Proof.* Fix  $r < R$  and  $|z-a| < r$ . Insert the geometric expansion of Lemma 2.1.8's proof into Theorem 2.1.9:

$$f(z) = \frac{1}{2\pi i} \int_{|w-a|=r} f(w) \sum_{k=0}^{\infty} \frac{(z-a)^k}{(w-a)^{k+1}} dw.$$

The series converges uniformly in  $w$  on the circle (the ratio is  $\leq \rho < 1$  and  $f$  is bounded on the circle), so term-by-term integration is legitimate and yields the stated  $c_k$ . The coefficients do not depend on  $r$ , since two circles give the same value by Theorem 2.1.6 applied on an annulus-avoiding convex subregion; letting  $r \uparrow R$  gives convergence on all of  $B(a, R)$ . Power series are infinitely differentiable term by term inside their disc of convergence, whence the last claim.  $\square$

**Corollary 2.1.11** (Cauchy estimates and Liouville). *If  $f$  is holomorphic on  $B(a, R)$  with  $|f| \leq M$  there, then  $|f^{(k)}(a)| \leq k! M/R^k$ . Consequently a bounded entire function is constant.*

*Proof.* Apply (2.1.1) to the formula for  $c_k$  with radius  $r < R$ :  $|c_k| \leq \frac{1}{2\pi} \cdot \frac{M}{r^{k+1}} \cdot 2\pi r = M/r^k$ ; let  $r \uparrow R$ . If  $f$  is entire and bounded, take  $k = 1$  and  $R \rightarrow \infty$  at every  $a$  to get  $f' \equiv 0$ .  $\square$

## 2.1.5 The maximum modulus principle

**Theorem 2.1.12.** *Let  $\Omega \subseteq \mathbb{C}$  be open and connected and  $f$  holomorphic on  $\Omega$ . If  $|f|$  attains a local maximum at some  $z_0 \in \Omega$ , then  $f$  is constant. Consequently, if  $\Omega$  is bounded and  $f$  is continuous on  $\overline{\Omega}$  and holomorphic on  $\Omega$ , then*

$$\sup_{\overline{\Omega}} |f| = \sup_{\partial\Omega} |f|.$$

*Idea of the proof.* Cauchy's formula on a circle centred at  $z_0$  exhibits  $f(z_0)$  as the average of  $f$  over that circle. An average of complex numbers of modulus at most  $|f(z_0)|$  can equal  $f(z_0)$  only if they all point the same way and have the same modulus. So  $|f|$  is locally constant, and then  $f$  is.

*Proof. Step 1. The mean value property.*

Parametrising  $w = z_0 + re^{i\theta}$  in Theorem 2.1.9 gives

$$f(z_0) = \frac{1}{2\pi} \int_0^{2\pi} f(z_0 + re^{i\theta}) d\theta \quad (2.1.4)$$

for every sufficiently small  $r > 0$ .

**Step 2.** *Local constancy of  $|f|$ .*

Suppose  $|f(w)| \leq |f(z_0)| =: M$  for all  $w$  in some ball  $B(z_0, \rho)$ . Taking absolute values in (2.1.4),

$$M \leq \frac{1}{2\pi} \int_0^{2\pi} |f(z_0 + re^{i\theta})| d\theta \leq M \quad (0 < r < \rho),$$

so  $\int_0^{2\pi} (M - |f(z_0 + re^{i\theta})|) d\theta = 0$  with a continuous nonnegative integrand; hence  $|f| = M$  on every such circle, i.e. on  $B(z_0, \rho)$ .

**Step 3.**  *$|f|$  constant forces  $f$  constant.*

If  $M = 0$  then  $f \equiv 0$  on the ball. If  $M > 0$ , write  $f = u + iv$  with  $u^2 + v^2 = M^2$ . Differentiating in  $x$  and in  $y$  gives  $uu_x + vv_x = 0$  and  $uu_y + vv_y = 0$ ; substituting the Cauchy–Riemann equations  $u_y = -v_x$ ,  $v_y = u_x$  turns the second into  $-uv_x + vu_x = 0$ . The linear system

$$\begin{pmatrix} u & v \\ v & -u \end{pmatrix} \begin{pmatrix} u_x \\ v_x \end{pmatrix} = 0$$

has determinant  $-(u^2 + v^2) = -M^2 \neq 0$ , so  $u_x = v_x = 0$ , and then  $u_y = v_y = 0$ . Hence  $f$  is constant on  $B(z_0, \rho)$ .

**Step 4.** *Propagate.*

The set where  $f$  equals the constant  $f(z_0)$  is nonempty, closed in  $\Omega$  by continuity, and open by Theorem 2.1.10 (a holomorphic function agreeing with a constant on a ball has all higher Taylor coefficients zero at each point of that ball, so agrees with it on any ball where its series converges). Connectedness gives  $f \equiv f(z_0)$ .

For the boundary statement:  $\overline{\Omega}$  is compact, so  $|f|$  attains a maximum somewhere; if that point is interior,  $f$  is constant by the above and the conclusion is trivial.  $\square$

**WORKED EXAMPLE 2.1.2— TIER B The maximum principle fails on unbounded domains**

Boundedness of  $\Omega$  is not decoration. Let

$$S = \left\{ z : |\operatorname{Im} z| < \frac{\pi}{2} \right\}, \quad F(z) = \exp(\exp(z)),$$

which is entire, hence holomorphic on  $S$  and continuous on  $\overline{S}$ .

*On the boundary  $F$  is unimodular.* If  $\operatorname{Im} z = \pm\pi/2$ , write  $z = x \pm i\pi/2$ ; then  $\exp(z) = e^x e^{\pm i\pi/2} = \pm ie^x$ , which is purely imaginary, so

$$|F(z)| = |\exp(\pm ie^x)| = 1.$$

*Inside,  $F$  is unbounded.* At  $z = x$  real,  $F(x) = \exp(e^x) \rightarrow \infty$  as  $x \rightarrow +\infty$ .

So  $\sup_{\partial S} |F| = 1$  while  $\sup_S |F| = \infty$ . The conclusion of Theorem 2.1.12 is false on this unbounded region.

*What went wrong, and what to add.* The proof of Theorem 2.1.12 used compactness of  $\overline{\Omega}$  to produce a maximum. On an unbounded region the supremum need not be attained, and mass can escape to infinity — this is escape mode (a) of Worked Example 1.5.2 (the three ways mass escapes), in a new costume. The repair is to *assume* that no such escape occurs, i.e. that  $f$  is bounded, and that is exactly the hypothesis of Theorem 2.1.13 below. The example shows the hypothesis cannot be dropped:  $F$  here is bounded on  $\partial S$  but not on  $S$ .

## 2.1.6 Phragmén–Lindelöf on a strip

Write

$$\mathbb{S}_{01} = \{z \in \mathbb{C} : 0 \leq \operatorname{Re} z \leq 1\}, \quad \mathbb{S}_{01}^\circ = \{0 < \operatorname{Re} z < 1\}.$$

**Theorem 2.1.13** (Phragmén–Lindelöf for the strip). *Let  $G$  be continuous and bounded on  $\mathbb{S}_{01}$  and holomorphic on  $\mathbb{S}_{01}^\circ$ . If  $|G| \leq 1$  on the two boundary lines  $\operatorname{Re} z = 0$  and  $\operatorname{Re} z = 1$ , then  $|G| \leq 1$  on all of  $\mathbb{S}_{01}$ .*

*Idea of the proof.* Worked Example 2.1.2 showed we cannot simply apply the maximum principle on an unbounded region. The device is to multiply  $G$  by a small holomorphic factor that decays as  $|\operatorname{Im} z| \rightarrow \infty$  and is harmless on the two boundary lines, turning the strip into an effectively bounded region; then apply Theorem 2.1.12 on a large rectangle and let the factor go away.

The factor  $e^{\varepsilon(z^2-z)}$  does it. Its modulus is  $e^{\varepsilon(x^2-y^2-x)}$ , which tends to 0 as  $|y| \rightarrow \infty$  for fixed  $x$ , and equals  $e^{\varepsilon(x^2-x)}$  when  $y = 0$ ; the polynomial  $x^2 - x$  vanishes at both  $x = 0$  and  $x = 1$ , which is exactly why the boundary bound survives untouched.

*Proof.* Let  $M = \sup_{\mathbb{S}_{01}} |G| < \infty$  and fix  $\varepsilon > 0$ . Define

$$G_\varepsilon(z) = G(z) e^{\varepsilon(z^2-z)}.$$

**Step 1.**  $|G_\varepsilon| \leq 1$  on  $\partial \mathbb{S}_{01}$ .

For  $z = x + iy$  we have  $\operatorname{Re}(z^2 - z) = x^2 - y^2 - x$ , so

$$|e^{\varepsilon(z^2-z)}| = e^{\varepsilon(x^2-x-y^2)} \leq e^{\varepsilon(x^2-x)}. \quad (2.1.5)$$

At  $x = 0$  and  $x = 1$  the exponent  $x^2 - x$  vanishes, so the factor has modulus at most 1 there and  $|G_\varepsilon| \leq |G| \leq 1$  on both boundary lines.

**Step 2.**  $G_\varepsilon$  is small high up and low down.

By (2.1.5), for  $z$  in the strip ( $0 \leq x \leq 1$ , so  $x^2 - x \in [-\frac{1}{4}, 0]$ ),

$$|G_\varepsilon(z)| \leq M e^{-\varepsilon y^2}.$$

Choose  $Y > 0$  with  $M e^{-\varepsilon Y^2} \leq 1$ .

**Step 3.** Apply the maximum principle on a rectangle.

Let  $R = \{0 \leq x \leq 1, |y| \leq Y\}$ , a bounded region. On its boundary,  $|G_\varepsilon| \leq 1$ : on the vertical sides by Step 1, on the horizontal sides by Step 2. By Theorem 2.1.12,  $|G_\varepsilon| \leq 1$  on  $R$ . Combined with Step 2 for  $|y| > Y$ , we get  $|G_\varepsilon| \leq 1$  on all of  $\mathbb{S}_{01}$ .

**Step 4.** Remove the factor.

Fix  $z \in \mathbb{S}_{01}$ . From  $|G(z)| \leq e^{-\varepsilon \operatorname{Re}(z^2 - z)}$  and letting  $\varepsilon \downarrow 0$  with  $z$  fixed, the right side tends to 1. Hence  $|G(z)| \leq 1$ .  $\square$

**Theorem 2.1.14** (Three-lines lemma). *Let  $F$  be continuous and bounded on  $\mathbb{S}_{01}$  and holomorphic on  $\mathbb{S}_{01}^\circ$ . Put*

$$M_0 = \sup_{y \in \mathbb{R}} |F(iy)|, \quad M_1 = \sup_{y \in \mathbb{R}} |F(1 + iy)|.$$

*Then for every  $\theta \in [0, 1]$  and  $y \in \mathbb{R}$ ,*

$$|F(\theta + iy)| \leq M_0^{1-\theta} M_1^\theta.$$

*Proof.* Suppose first  $M_0, M_1 > 0$ . Define

$$G(z) = F(z) M_0^{z-1} M_1^{-z},$$

where  $M^w := e^{w \log M}$  with the real logarithm, so that  $M^w$  is entire in  $w$  and  $|M^w| = M^{\operatorname{Re} w}$ . Then  $G$  is continuous on  $\mathbb{S}_{01}$ , holomorphic inside, and bounded there, since  $|M_0^{z-1} M_1^{-z}| = M_0^{x-1} M_1^{-x}$  is bounded for  $x \in [0, 1]$ .

On  $\operatorname{Re} z = 0$ :  $|G| = |F| M_0^{-1} \leq 1$ . On  $\operatorname{Re} z = 1$ :  $|G| = |F| M_1^{-1} \leq 1$ . By Theorem 2.1.13,  $|G| \leq 1$  on  $\mathbb{S}_{01}$ , i.e.

$$|F(\theta + iy)| \leq M_0^{1-\theta} M_1^\theta$$

after multiplying back.

If  $M_0 = 0$ , then  $F$  vanishes identically on the line  $\operatorname{Re} z = 0$ ; applying the above with  $M_0$  replaced by  $\eta > 0$  gives  $|F(\theta + iy)| \leq \eta^{1-\theta} M_1^\theta$  for every  $\eta > 0$ , whence  $F(\theta + iy) = 0$  for  $\theta < 1$  and the inequality holds trivially. The case  $M_1 = 0$  is symmetric.  $\square$



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**WORKED EXAMPLE 2.1.3— TIER C: RESEARCH TASTE** **Logarithmic convexity, and why every exponent range in this series is an interval**

Theorem 2.1.14 says that  $\theta \mapsto \log \left( \sup_y |F(\theta + iy)| \right)$  is a convex function of  $\theta$  on  $[0, 1]$ : that is the content of the inequality  $M_\theta \leq M_0^{1-\theta} M_1^\theta$ , after taking logarithms.

*Why this shape is inevitable.* Suppose one has a family of estimates indexed by a parameter, and suppose the quantity being estimated can be realised as the boundary data of a bounded holomorphic function on a strip. Then the best constant is automatically log-convex in the parameter, hence finite on an *interval*, and controlled at interior points by the endpoints. That is the entire mechanism of Chapter 2.2, and it explains a pattern the reader will meet again and again:

- the Hausdorff–Young inequality  $\|\widehat{f}\|_{p'} \leq \|f\|_p$  holds exactly for  $1 \leq p \leq 2$  (Chapter 2.4);
- the Stein–Tomas restriction estimate holds for  $p \leq \frac{2(n+1)}{n+3}$ , an interval with one endpoint at  $p = 1$  (Volume 6);
- the  $\ell^2$  decoupling theorem for the paraboloid holds exactly for  $2 \leq p \leq \frac{2(n+1)}{n-1}$  (Volume 13);
- the Kakeya maximal conjecture is stated at the single exponent  $p = n$  because that is the endpoint of the interval on which it is true, everything below following by interpolation.

In every case the set of good exponents is a closed interval, and in every case the reason is the same convexity. When a paper says “by interpolation it suffices to treat the endpoint”, this theorem is what is being invoked.

*A caution that will matter.* Log-convexity is a statement about the *constant*, not about truth. Interpolation between two true estimates gives a true estimate with a controlled constant; it never improves an endpoint. The entire difficulty of Volumes 12 and 13 is that the endpoint estimates are the hard ones, and no amount of interpolation manufactures them. Readers who find themselves hoping to interpolate their way to the restriction conjecture should return to this paragraph.

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## Drills

**Drill 2.1.1.** Verify the Cauchy–Riemann equations for  $f(z) = e^z = e^x(\cos y + i \sin y)$  and compute  $f'$ . Then show that  $|e^z| = e^{\operatorname{Re} z}$ , the identity used through-

out §1.6.

**Drill 2.1.2.** Show that the conclusion of Theorem 2.1.5 fails for  $f(z) = 1/z$  on a triangle containing 0, and identify precisely which hypothesis is violated.

**Drill 2.1.3.** In the proof of Theorem 2.1.13, replace the factor  $e^{\varepsilon(z^2-z)}$  by  $e^{\varepsilon z^2}$  and determine what goes wrong on the boundary lines. Then find the general condition on a polynomial  $q$  for  $e^{\varepsilon q(z)}$  to serve.

**Drill 2.1.4.** Deduce Hadamard's three-circle theorem from Theorem 2.1.14: if  $f$  is holomorphic on an annulus  $r_0 < |z| < r_1$  and  $M(r) = \sup_{|z|=r} |f|$ , then  $\log M$  is a convex function of  $\log r$ . (Substitute  $z \mapsto e^w$  and rescale the strip.)

## Notes

Goursat's theorem is from 1900 and is the reason no continuity hypothesis on  $f'$  appears anywhere in this chapter. The Phragmén–Lindelöf principle dates to 1908; Worked Example 2.1.2 is their example. The three-lines lemma in the form Theorem 2.1.14 is due to Doetsch (1920), though the idea is Hadamard's.

Readers who have seen complex analysis before will notice that we have proved only what is needed: no residue calculus, no argument principle, no Riemann mapping theorem. Volume 5 will require contour deformation and the method of steepest descent, which are developed there on the foundation laid here.

## Chapter 2.2

# Interpolation: Riesz–Thorin and Marcinkiewicz

Volume 1 proved that the Hardy–Littlewood maximal operator satisfies a weak  $(1,1)$  bound and an  $(\infty,\infty)$  bound, and then obtained the  $L^p$  bounds for  $1 < p < \infty$  by a hand-rolled argument: split  $f$  at height  $\lambda$ , apply the weak bound to the tall part, integrate with the layer-cake formula. That argument used nothing about the maximal operator. This chapter extracts it, in two forms.

The first, Riesz–Thorin, interpolates between *strong* bounds and requires linearity; its proof is the three-lines lemma and its constant is sharp. The second, Marcinkiewicz, interpolates between *weak* bounds and tolerates sublinear operators; its proof is the layer-cake argument and its constant blows up at the endpoints. Both are used constantly. Knowing which one applies is a matter of asking whether the available endpoint information is strong or weak, and covering arguments — as Volume 1 emphasised — produce weak information.

### 2.2.1 Riesz–Thorin

Throughout,  $(X, \mu)$  and  $(Y, \nu)$  are  $\sigma$ -finite measure spaces. Recall that a *simple function of finite support* is a simple function vanishing outside a set of finite measure; by Theorem 1.8.7 (density of simple functions and  $C_c^\infty$ ), these are dense in  $L^p(\mu)$  for every  $p < \infty$ .

**Theorem 2.2.1** (Riesz–Thorin). *Let  $1 \leq p_0, p_1, q_0, q_1 \leq \infty$  and let  $T$  be a linear map, defined on simple functions of finite support on  $X$ , taking values in measurable functions on  $Y$ , with*

$$\|Tf\|_{q_0} \leq M_0 \|f\|_{p_0}, \quad \|Tf\|_{q_1} \leq M_1 \|f\|_{p_1}$$

for all such  $f$ . For  $\theta \in (0, 1)$  define

$$\frac{1}{p_\theta} = \frac{1-\theta}{p_0} + \frac{\theta}{p_1}, \quad \frac{1}{q_\theta} = \frac{1-\theta}{q_0} + \frac{\theta}{q_1}.$$

Then  $\|Tf\|_{q_\theta} \leq M_0^{1-\theta} M_1^\theta \|f\|_{p_\theta}$ , and if  $p_\theta < \infty$  the operator  $T$  extends uniquely to a bounded map  $L^{p_\theta}(\mu) \rightarrow L^{q_\theta}(\nu)$  with that norm.

*Idea of the proof.* We must bound  $\|Tf\|_{q_\theta}$ , and by duality (Corollary 1.8.10 (norms by testing)) that is the same as bounding  $|\int (Tf)g \, d\nu|$  over  $g$  in the unit ball of  $L^{q'_\theta}$ . So the quantity to control is the bilinear pairing

$$\Phi = \int_Y (Tf)g \, d\nu.$$

Now deform. Replace  $f$  and  $g$  by families  $f_z$  and  $g_z$  depending on a complex parameter  $z$  in the strip, chosen so that: at  $z = \theta$  they are the original  $f$  and  $g$ ; on the line  $\operatorname{Re} z = 0$  the function  $f_z$  has the right  $L^{p_0}$  norm and  $g_z$  the right  $L^{q'_0}$  norm, so that the first hypothesis applies; and on  $\operatorname{Re} z = 1$  likewise for the second. Then  $\Phi(z) = \int (Tf_z)g_z$  is a bounded holomorphic function of  $z$  with  $|\Phi| \leq M_0$  on one boundary line and  $\leq M_1$  on the other, and Theorem 2.1.14 gives  $|\Phi(\theta)| \leq M_0^{1-\theta} M_1^\theta$ .

The families are forced. To move the  $L^p$  norm of  $|f|^\alpha$  around one raises  $|f|$  to a power, and to keep everything holomorphic in  $z$  that power must be affine in  $z$ . Writing  $f = |f|e^{i \arg f}$  and putting

$$f_z = |f|^{p_\theta \left( \frac{1-z}{p_0} + \frac{z}{p_1} \right)} e^{i \arg f}$$

does exactly this: at  $z = \theta$  the exponent is 1, and on  $\operatorname{Re} z = j$  it is  $p_\theta/p_j$ , which is what makes  $\|f_z\|_{p_j} = \|f\|_{p_\theta}^{p_\theta/p_j}$ .

*Proof.* Assume first  $p_\theta < \infty$  and  $q_\theta > 1$ ; the excluded cases are treated in Step 5.

**Step 1.** Set up the analytic families.

Normalise  $\|f\|_{p_\theta} = 1$  and  $\|g\|_{q'_\theta} = 1$ , with  $f$  and  $g$  simple of finite support. Write

$$f = \sum_{j=1}^m a_j e^{i\alpha_j} \mathbf{1}_{A_j}, \quad g = \sum_{k=1}^n b_k e^{i\beta_k} \mathbf{1}_{B_k},$$

with  $a_j, b_k > 0$ , the  $A_j$  pairwise disjoint of finite measure, likewise  $B_k$ . Define

$$\eta(z) = p_\theta \left( \frac{1-z}{p_0} + \frac{z}{p_1} \right), \quad \zeta(z) = q'_\theta \left( \frac{1-z}{q'_0} + \frac{z}{q'_1} \right),$$

and

$$f_z = \sum_j a_j^{\eta(z)} e^{i\alpha_j} \mathbf{1}_{A_j}, \quad g_z = \sum_k b_k^{\zeta(z)} e^{i\beta_k} \mathbf{1}_{B_k}.$$

Since  $a_j^{\eta(z)} = e^{\eta(z) \log a_j}$  with  $\eta$  affine, each  $f_z$  is a finite sum of entire functions of  $z$  times fixed indicators; the same for  $g_z$ . Note  $\eta(\theta) = 1$  and  $\zeta(\theta) = 1$ , so  $f_\theta = f$  and  $g_\theta = g$ .

**Step 2.** *The pairing is holomorphic and bounded.*

Put

$$\Phi(z) = \int_Y (Tf_z) g_z \, d\nu = \sum_{j,k} a_j^{\eta(z)} b_k^{\zeta(z)} e^{i(\alpha_j + \beta_k)} \int_{B_k} T(\mathbf{1}_{A_j}) \, d\nu,$$

using linearity of  $T$ . This is a *finite* linear combination of exponentials  $e^{cz}$  with fixed coefficients, so  $\Phi$  is entire; and on the closed strip each  $|a_j^{\eta(z)}| = a_j^{\operatorname{Re} \eta(z)}$  is bounded, because  $\operatorname{Re} \eta$  ranges over the bounded interval between  $p_\theta/p_0$  and  $p_\theta/p_1$ . Hence  $\Phi$  is bounded on  $\mathbb{S}_{01}$ . The coefficients  $\int_{B_k} T(\mathbf{1}_{A_j})$  are finite:  $\mathbf{1}_{A_j}$  is simple of finite support, so  $T\mathbf{1}_{A_j} \in L^{q_0}$ , and  $B_k$  has finite measure.

**Step 3.** *Bound  $\Phi$  on the two boundary lines.*

Fix  $y \in \mathbb{R}$  and consider  $z = iy$ . Then  $\operatorname{Re} \eta(iy) = p_\theta/p_0$ , so

$$|f_{iy}|^{p_0} = \sum_j a_j^{p_\theta} \mathbf{1}_{A_j} = |f|^{p_\theta}, \quad \text{hence} \quad \|f_{iy}\|_{p_0} = \|f\|_{p_\theta}^{p_\theta/p_0} = 1.$$

Identically  $\|g_{iy}\|_{q'_0} = \|g\|_{q'_\theta}^{q'_\theta/q'_0} = 1$ . Therefore, by Hölder's inequality and the first hypothesis,

$$|\Phi(iy)| \leq \|Tf_{iy}\|_{q_0} \|g_{iy}\|_{q'_0} \leq M_0 \|f_{iy}\|_{p_0} = M_0.$$

The same computation on  $\operatorname{Re} z = 1$  gives  $|\Phi(1 + iy)| \leq M_1$ .

**Step 4.** *Apply the three-lines lemma.*

By Theorem 2.1.14,  $|\Phi(\theta)| \leq M_0^{1-\theta} M_1^\theta$ . But  $\Phi(\theta) = \int (Tf)g \, d\nu$ , so

$$\left| \int_Y (Tf)g \, d\nu \right| \leq M_0^{1-\theta} M_1^\theta$$

for every simple  $g$  of finite support with  $\|g\|_{q'_\theta} \leq 1$ . Since such  $g$  are dense in the unit ball of  $L^{q'_\theta}$  and  $q'_\theta < \infty$ , the duality formula of Corollary 1.8.10 (norms by testing), gives  $\|Tf\|_{q_\theta} \leq M_0^{1-\theta} M_1^\theta$ .

**Step 5.** *The excluded cases, and the extension.*

If  $q_\theta = 1$  then  $q_0 = q_1 = 1$  and one applies Hölder with exponents 1 and  $\infty$  in Step 3, testing against  $g$  in the unit ball of  $L^\infty$ ; the duality step is then

Theorem 1.8.8 ( $L^p$  duality) for  $p = 1$ . If  $p_\theta = \infty$  then  $p_0 = p_1 = \infty$  and there is nothing to interpolate on the domain side; the argument runs with  $f_z = f$ .

Finally, for  $p_\theta < \infty$  the simple functions of finite support are dense in  $L^{p_\theta}(\mu)$ , and a bounded linear map on a dense subspace of a normed space, with values in the complete space  $L^{q_\theta}(\nu)$  (Volume 1, Theorem 8.6), extends uniquely by continuity with the same norm.  $\square$

### **Pitfall: Riesz–Thorin needs linearity, and complex scalars**

Two hypotheses are load-bearing and are routinely forgotten.

*Linearity.* Step 2 used  $T(\sum_j c_j \mathbf{1}_{A_j}) = \sum_j c_j T\mathbf{1}_{A_j}$  to make  $\Phi$  entire. For a merely sublinear operator — the maximal function, the Kakeya maximal operator, any supremum — this fails completely, and Theorem 2.2.1 does not apply. Use Theorem 2.2.4 instead.

*Complex scalars.* The whole proof runs in the complex plane. For real-linear operators on real  $L^p$  spaces the theorem holds with an extra constant, and there are examples showing a constant is genuinely needed. In this series all function spaces are complex, so the issue does not arise; but it is worth knowing that the sharpness of the constant  $M_0^{1-\theta} M_1^\theta$  is a complex phenomenon.

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### **WORKED EXAMPLE 2.2.1— TIER B Young’s convolution inequality, in full generality**

Proposition 1.6.9 (Young’s inequality,  $q = 1$ ) proved  $\|f * g\|_p \leq \|f\|_p \|g\|_1$  and deferred the general case. We can now complete it.

**Proposition 2.2.2.** *For  $1 \leq p, q, r \leq \infty$  with  $1 + \frac{1}{r} = \frac{1}{p} + \frac{1}{q}$ ,*

$$\|f * g\|_r \leq \|f\|_p \|g\|_q.$$

*Proof.* Fix  $g \in L^q$  and let  $Tf = f * g$ , a linear operator in  $f$ . Two endpoint estimates are available with no work:

$$\|f * g\|_q \leq \|f\|_1 \|g\|_q \quad (\text{Volume 1, Prop. 6.9, roles reversed}),$$

$$\|f * g\|_\infty \leq \|f\|_{q'} \|g\|_q \quad (\text{Hölder, applied pointwise}).$$

For the second:  $|(f * g)(x)| \leq \int |f(x - y)| |g(y)| dy \leq \|f(x - \cdot)\|_{q'} \|g\|_q = \|f\|_{q'} \|g\|_q$ .

So  $T$  maps  $L^{p_0} \rightarrow L^{q_0}$  with  $p_0 = 1$ ,  $q_0 = q$ , and  $L^{p_1} \rightarrow L^{q_1}$  with  $p_1 = q'$ ,  $q_1 = \infty$ , both with norm at most  $\|g\|_q$ . Interpolating by Theorem 2.2.1 at parameter  $\theta \in (0, 1)$  gives a bound  $L^{p_\theta} \rightarrow L^{q_\theta}$  with norm at most  $(\|g\|_q)^{1-\theta} (\|g\|_q)^\theta = \|g\|_q$ , where

$$\frac{1}{p_\theta} = (1 - \theta) \cdot 1 + \theta \cdot \frac{1}{q'} = 1 - \frac{\theta}{q}, \quad \frac{1}{q_\theta} = \frac{1 - \theta}{q} + \theta \cdot 0 = \frac{1 - \theta}{q}.$$

Adding the two identities,

$$\frac{1}{p\theta} + \frac{1}{q} = 1 - \frac{\theta}{q} + \frac{1}{q} = 1 + \frac{1-\theta}{q} = 1 + \frac{1}{q\theta},$$

which is precisely  $1 + \frac{1}{r} = \frac{1}{p} + \frac{1}{q}$  with  $p = p\theta$ ,  $r = q\theta$ . As  $\theta$  runs over  $(0, 1)$ ,  $r = q/(1 - \theta)$  runs over  $(q, \infty)$ , and together with the two endpoint cases already established this covers every admissible triple.  $\square$

*Reading the exponent relation.* The condition  $1 + \frac{1}{r} = \frac{1}{p} + \frac{1}{q}$  is forced by scaling: replacing  $f(x)$  by  $f(\lambda x)$  multiplies  $\|f\|_p$  by  $\lambda^{-n/p}$ , and the two sides of Young's inequality scale the same way precisely under that relation; Drill 2.2 asks for the computation. Whenever an inequality in this series has a mysterious-looking exponent condition, testing it against dilations is the first thing to do; Volume 6 uses this repeatedly to locate the conjectured range for restriction estimates.

## 2.2.2 Marcinkiewicz

We now interpolate weak-type information. Recall from Volume 1, §8.5 that  $\|f\|_{p,\infty} = \sup_{\lambda>0} \lambda d_f(\lambda)^{1/p}$  where  $d_f(\lambda) = \mu(\{|f| > \lambda\})$ .

**Definition 2.2.3.** An operator  $T$ , defined on a vector space  $\mathcal{D}$  of measurable functions and taking values in measurable functions, is *sublinear* if  $|T(f+g)| \leq |Tf| + |Tg|$  and  $|T(cf)| = |c||Tf|$  pointwise. It is of *weak type*  $(p, q)$  with constant  $M$  if  $\|Tf\|_{q,\infty} \leq M\|f\|_p$ , and of *strong type*  $(p, q)$  if  $\|Tf\|_q \leq M\|f\|_p$ .

Strong type  $(p, q)$  implies weak type  $(p, q)$ , by Chebyshev (Volume 1, Proposition 8.14).

**Theorem 2.2.4** (Marcinkiewicz interpolation). *Let  $1 \leq p_0 < p_1 \leq \infty$  and let  $T$  be sublinear, defined on  $L^{p_0}(\mu) + L^{p_1}(\mu)$ , of weak type  $(p_0, p_0)$  with constant  $M_0$  and of weak type  $(p_1, p_1)$  with constant  $M_1$ . Then for every  $p_0 < p < p_1$ ,  $T$  is of strong type  $(p, p)$ :*

$$\|Tf\|_p \leq C(p, p_0, p_1) M_0^{1-\theta} M_1^\theta \|f\|_p, \quad \frac{1}{p} = \frac{1-\theta}{p_0} + \frac{\theta}{p_1},$$

where  $C(p, p_0, p_1) \rightarrow \infty$  as  $p \downarrow p_0$  or  $p \uparrow p_1$ .

*Idea of the proof.* Fix a height  $\lambda$  and split  $f$  into its tall and short parts,  $f = f\mathbf{1}_{\{|f|>c\lambda\}} + f\mathbf{1}_{\{|f|\leq c\lambda\}}$ . The tall part lies in  $L^{p_0}$  (a function that is large where it is supported is more integrable at the *lower* exponent), the short part in  $L^{p_1}$ . Sublinearity splits  $\{|Tf| > \lambda\}$  between the two, each

is estimated by the corresponding weak bound, and the layer-cake formula integrates the result in  $\lambda$ . The constant  $c$  is a free parameter, optimised at the end; the divergence of  $C$  at the endpoints comes from an integral  $\int_0^\infty \lambda^{p-p_0-1}(\dots)$  which converges only for  $p_0 < p < p_1$ .

*Proof.* Assume first  $p_1 < \infty$ . Fix  $f \in L^p$  and  $\lambda > 0$ , and a constant  $c > 0$  to be chosen. Write

$$f = g_\lambda + h_\lambda, \quad g_\lambda = f \mathbf{1}_{\{|f| > c\lambda\}}, \quad h_\lambda = f \mathbf{1}_{\{|f| \leq c\lambda\}}.$$

**Step 1.** *The two pieces lie in the right spaces.*

Since  $|g_\lambda| \geq c\lambda$  where it is nonzero,  $|g_\lambda|^{p_0} \leq (c\lambda)^{p_0-p} |f|^p$ , so  $g_\lambda \in L^{p_0}$ . Since  $|h_\lambda| \leq c\lambda$ ,  $|h_\lambda|^{p_1} \leq (c\lambda)^{p_1-p} |f|^p$ , so  $h_\lambda \in L^{p_1}$ .

**Step 2.** *Split the level set.*

By sublinearity,  $|Tf| \leq |Tg_\lambda| + |Th_\lambda|$ , so

$$\{|Tf| > \lambda\} \subseteq \left\{ |Tg_\lambda| > \frac{\lambda}{2} \right\} \cup \left\{ |Th_\lambda| > \frac{\lambda}{2} \right\},$$

and therefore, applying the two weak-type hypotheses,

$$d_{Tf}(\lambda) \leq \left( \frac{2M_0}{\lambda} \right)^{p_0} \|g_\lambda\|_{p_0}^{p_0} + \left( \frac{2M_1}{\lambda} \right)^{p_1} \|h_\lambda\|_{p_1}^{p_1}. \quad (2.2.1)$$

**Step 3.** *Integrate.*

By the layer-cake formula (Worked Example 1.5.1 (the layer-cake formula)) and (2.2.1),

$$\|Tf\|_p^p = p \int_0^\infty \lambda^{p-1} d_{Tf}(\lambda) d\lambda \leq \text{I} + \text{II},$$

where

$$\begin{aligned} \text{I} &= p (2M_0)^{p_0} \int_0^\infty \lambda^{p-p_0-1} \|g_\lambda\|_{p_0}^{p_0} d\lambda, \\ \text{II} &= p (2M_1)^{p_1} \int_0^\infty \lambda^{p-p_1-1} \|h_\lambda\|_{p_1}^{p_1} d\lambda. \end{aligned}$$

For I, write  $\|g_\lambda\|_{p_0}^{p_0} = \int_{\{|f| > c\lambda\}} |f|^{p_0} d\mu$  and apply Tonelli, exchanging the  $\lambda$  and  $x$  integrations:

$$\begin{aligned} \text{I} &= p (2M_0)^{p_0} \int_X |f(x)|^{p_0} \left( \int_0^{|f(x)|/c} \lambda^{p-p_0-1} d\lambda \right) d\mu(x) \\ &= \frac{p (2M_0)^{p_0} c^{p_0-p}}{p-p_0} \int_X |f|^p d\mu, \end{aligned}$$



the inner integral converging at 0 precisely because  $p > p_0$ .

For II, similarly  $\|h_\lambda\|_{p_1}^{p_1} = \int_{\{|f| \leq c\lambda\}} |f|^{p_1}$ , and

$$\begin{aligned} \text{II} &= p(2M_1)^{p_1} \int_X |f(x)|^{p_1} \left( \int_{|f(x)|/c}^{\infty} \lambda^{p-p_1-1} d\lambda \right) d\mu(x) \\ &= \frac{p(2M_1)^{p_1} c^{p_1-p}}{p_1-p} \int_X |f|^p d\mu, \end{aligned}$$

the integral converging at  $\infty$  precisely because  $p < p_1$ .

**Step 4.** *Optimise in  $c$ .*

Adding,

$$\|Tf\|_p^p \leq p \left( \frac{(2M_0)^{p_0} c^{p_0-p}}{p-p_0} + \frac{(2M_1)^{p_1} c^{p_1-p}}{p_1-p} \right) \|f\|_p^p.$$

Choosing  $c = M_0^{p_0/(p_1-p_0)} M_1^{-p_1/(p_1-p_0)}$  balances the powers of  $M_0$  and  $M_1$  and produces, after simplification, a bound of the stated form  $C M_0^{(1-\theta)p} M_1^{\theta p} \|f\|_p^p$  with  $C = C(p, p_0, p_1)$  depending only on the exponents and blowing up like  $(p-p_0)^{-1}$  and  $(p_1-p)^{-1}$  at the two ends.

**Step 5.** *The case  $p_1 = \infty$ .*

Weak type  $(\infty, \infty)$  means  $\|Tf\|_\infty \leq M_1 \|f\|_\infty$ . Choose  $c = 1/(2M_1)$ , so that  $\|Th_\lambda\|_\infty \leq M_1 c \lambda = \lambda/2$  and the second set in Step 2 is empty. Then only term I survives and the computation above gives the result directly, with  $C = (p 2^{p_0} (2M_1)^{p-p_0} / (p-p_0))^{1/p}$ .  $\square$

**WORKED EXAMPLE 2.2.2— TIER A Recovering the maximal function bound**

Theorem 1.7.3 (Hardy–Littlewood maximal inequality) proved the Hardy–Littlewood maximal operator is of weak type  $(1, 1)$  with constant  $3^n$ , and satisfies  $\|\mathcal{M}f\|_\infty \leq \|f\|_\infty$ . It is sublinear, since a supremum of averages of  $|f+g|$  is at most the sum of the suprema.

Apply Theorem 2.2.4 with  $p_0 = 1$ ,  $p_1 = \infty$ ,  $M_0 = 3^n$ ,  $M_1 = 1$ . Step 5 of the proof gives, for  $1 < p < \infty$ ,

$$\|\mathcal{M}f\|_p \leq \left( \frac{p 2^p 3^n}{p-1} \right)^{1/p} \|f\|_p.$$

This is exactly Step 3 of the proof of Theorem 1.7.3 (Hardy–Littlewood maximal inequality) — the same computation, now performed once and for all rather than for one operator. Note the constant diverging as  $p \downarrow 1$ , and recall from Volume 1, Worked Example 7.6 that this is unavoidable:  $\mathcal{M}f \notin L^1$  for any  $f \neq 0$ .

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**WORKED EXAMPLE 2.2.3— TIER C: RESEARCH TASTE** **Where the exponent  $p = n$  in the *Keakeya* maximal conjecture comes from**

Worked Example 1.7.2 (the *Keakeya* maximal conjecture) stated the *Keakeya* maximal function conjecture at the single exponent  $p = n$  and asserted that the exponent is forced. We can now say exactly why. Recall the two trivial bounds established there, for  $\mathcal{K}_\delta f(e) = \sup_a m(T_\delta^e(a))^{-1} \int_{T_\delta^e(a)} |f|$ :

$$\|\mathcal{K}_\delta f\|_{L^\infty(\mathbb{S}^{n-1})} \leq \|f\|_\infty, \quad \|\mathcal{K}_\delta f\|_{L^\infty(\mathbb{S}^{n-1})} \leq c_n \delta^{-(n-1)} \|f\|_1.$$

(a) *What interpolation gives.*  $\mathcal{K}_\delta$  is sublinear, so Theorem 2.2.4 applies with  $p_0 = 1$ ,  $p_1 = \infty$ ,  $M_0 = c_n \delta^{-(n-1)}$ ,  $M_1 = 1$ , target exponent  $\infty$  in both cases. (Strictly, the two bounds are of strong type  $(1, \infty)$  and  $(\infty, \infty)$ , so one may interpolate them directly by the elementary argument of Step 5.) The result is

$$\|\mathcal{K}_\delta f\|_{L^\infty(\mathbb{S}^{n-1})} \lesssim_n \delta^{-(n-1)/p} \|f\|_p,$$

and since  $\mathbb{S}^{n-1}$  has finite measure,  $\|\cdot\|_{L^p(\mathbb{S}^{n-1})} \lesssim \|\cdot\|_{L^\infty(\mathbb{S}^{n-1})}$ , so

$$\|\mathcal{K}_\delta f\|_{L^p(\mathbb{S}^{n-1})} \lesssim_n \delta^{-(n-1)/p} \|f\|_{L^p(\mathbb{R}^n)}. \quad (2.2.2)$$

At  $p = n$  this reads  $\delta^{-(n-1)/n}$ . The conjecture asserts the loss is only  $\delta^{-\varepsilon}$ : the entire content is the removal of a fixed power of  $\delta$  from a bound that interpolation already supplies.

(b) *Why no smaller exponent can work.* Take  $f = \mathbf{1}_{B(0, \delta)}$ , the indicator of a ball of radius  $\delta$ . Then  $\|f\|_{L^p(\mathbb{R}^n)} = m(B(0, \delta))^{1/p} \approx \delta^{n/p}$ . For any direction  $e$ , the best a  $\delta$ -tube can do is to contain the whole ball, so

$$\mathcal{K}_\delta f(e) \approx \frac{m(B(0, \delta))}{\delta^{n-1}} \approx \frac{\delta^n}{\delta^{n-1}} = \delta,$$

uniformly in  $e$ , whence  $\|\mathcal{K}_\delta f\|_{L^p(\mathbb{S}^{n-1})} \approx \delta$ . An estimate  $\|\mathcal{K}_\delta f\|_p \lesssim_\varepsilon \delta^{-\varepsilon} \|f\|_p$  would therefore require

$$\delta \lesssim_\varepsilon \delta^{-\varepsilon} \delta^{n/p}, \quad \text{i.e.} \quad \delta^{1-n/p} \lesssim_\varepsilon \delta^{-\varepsilon}.$$

As  $\delta \rightarrow 0$  this forces  $1 - \frac{n}{p} \geq -\varepsilon$  for every  $\varepsilon > 0$ , that is  $p \geq n$ . So  $n$  is the smallest exponent at which the conjecture can hold, and by (2.2.2) together with interpolation against the trivial  $L^\infty$  bound, the case  $p = n$  implies every case  $p > n$ . This is why the conjecture is stated at one exponent.

(c) *The single-tube example imposes nothing further.* It is worth checking the other natural competitor. Let  $f = \mathbf{1}_T$  for one  $\delta$ -tube  $T$  in direction  $e_0$ , so  $\|f\|_p \approx \delta^{(n-1)/p}$ . For  $e$  at angle  $\theta$  from  $e_0$ , the tube intersection

estimate of Lemma 1.6.13 (the tube intersection estimate) gives  $\mathcal{K}_\delta f(e) \approx \delta^{-(n-1)} \cdot \delta^n / (\theta + \delta) = \delta / (\theta + \delta)$ . Integrating over the sphere, where the measure of the set of directions at angle  $\approx \theta$  is  $\approx \theta^{n-2} d\theta$ ,

$$\|\mathcal{K}_\delta f\|_{L^p(\mathbb{S}^{n-1})}^p \approx \int_0^1 \left( \frac{\delta}{\theta + \delta} \right)^p \theta^{n-2} d\theta \approx \delta^{n-1} \quad (p > n - 1),$$

the integral being dominated by  $\theta \lesssim \delta$ . So  $\|\mathcal{K}_\delta f\|_p \approx \delta^{(n-1)/p} \approx \|f\|_p$ : the single tube is consistent with the conjecture at every  $p > n - 1$  and gives no constraint. It is the *ball*, not the tube, that pins the exponent.

*What this example is really teaching.* Locating the conjectured range of an estimate is a matter of testing against the extremal examples and reading off the exponents, and it is almost always easy. Proving the estimate at the endpoint is almost always hard. Volume 6 performs exactly this computation for the restriction problem — where the extremal examples are the Knapp cap and the whole sphere — and the resulting conjectured range has been open since 1967 for  $n \geq 3$ .

#### WORKED EXAMPLE 2.2.4— TIER B **Building an analytic family by hand**

The device in Step 1 of Theorem 2.2.1 — replacing  $f$  by a family  $f_z$  depending holomorphically on  $z$  — is the single technique a reader must own from this chapter. It is easy to follow and surprisingly hard to reproduce from memory, because the exponents look arbitrary. They are not. This example builds the family for one concrete case, slowly, and shows where each exponent is forced.

*The task.* Take  $p_0 = 1$ ,  $p_1 = 2$ , and interpolate at  $\theta = \frac{1}{2}$ , so that  $\frac{1}{p_\theta} = \frac{1-\theta}{1} + \frac{\theta}{2} = \frac{3}{4}$ , giving  $p_\theta = \frac{4}{3}$ . We want a family  $f_z$  with

$$f_{1/2} = f, \quad \|f_{iy}\|_{L^1} = 1, \quad \|f_{1+iy}\|_{L^2} = 1,$$

starting from a simple function  $f$  with  $\|f\|_{L^{4/3}} = 1$ .

*Step 1: what must be true on the two lines.* We need  $|f_z|$  to have a fixed  $L^{p_j}$  norm on the line  $\operatorname{Re} z = j$ . Since we intend to build  $f_z$  by raising  $|f|$  to a  $z$ -dependent power, write  $f_z = |f|^{\eta(z)} e^{i \arg f}$  and ask what  $\eta$  must satisfy. On  $\operatorname{Re} z = j$ ,

$$\|f_z\|_{L^{p_j}}^{p_j} = \int |f|^{p_j \operatorname{Re} \eta(z)},$$

and this is  $\int |f|^{4/3} = 1$  exactly when

$$p_j \operatorname{Re} \eta(z) = \frac{4}{3}, \quad \text{i.e.} \quad \operatorname{Re} \eta = \frac{4/3}{p_j} = \frac{p_\theta}{p_j} \quad \text{on } \operatorname{Re} z = j.$$

So  $\operatorname{Re} \eta$  must equal  $p_\theta/p_0 = \frac{4}{3}$  on the left edge and  $p_\theta/p_1 = \frac{2}{3}$  on the right edge.

*Step 2: the unique affine choice.* An affine function of  $z$  is determined by its values on the two edges, and affine is what we need for  $|f|^{\eta(z)} = e^{\eta(z) \log |f|}$  to be entire. The affine  $\eta$  with  $\operatorname{Re} \eta(iy) = \frac{4}{3}$  and  $\operatorname{Re} \eta(1 + iy) = \frac{2}{3}$  is

$$\eta(z) = \frac{4}{3}(1 - z) + \frac{2}{3}z = \frac{4}{3} - \frac{2}{3}z,$$

and indeed  $\eta(\frac{1}{2}) = \frac{4}{3} - \frac{1}{3} = 1$ , so  $f_{1/2} = f$  as required.

Compare the general formula in the proof:  $\eta(z) = p_\theta(\frac{1-z}{p_0} + \frac{z}{p_1})$ . With  $p_\theta = \frac{4}{3}$ ,  $p_0 = 1$ ,  $p_1 = 2$  this is  $\frac{4}{3}(1 - z) + \frac{2}{3}z$  — the same thing. The general formula is nothing but “the affine function taking the value  $p_\theta/p_j$  on the edge  $\operatorname{Re} z = j$ ”.

*Step 3: check the family is admissible.* Write  $f = \sum_k a_k e^{i\alpha_k} \mathbf{1}_{A_k}$  in standard form. Then

$$f_z = \sum_k a_k^{4/3-2z/3} e^{i\alpha_k} \mathbf{1}_{A_k},$$

a finite sum of terms  $e^{cz}$  times fixed indicators, hence entire in  $z$ ; and on the closed strip  $|a_k^{\eta(z)}| = a_k^{\operatorname{Re} \eta(z)}$  with  $\operatorname{Re} \eta$  ranging over  $[\frac{2}{3}, \frac{4}{3}]$ , so  $f_z$  is bounded there. Both hypotheses of the three-lines lemma are met.

*What to take away.* Three habits, all reusable.

- *Decide what you need on the two edges first.* The exponents are determined by the requirement that the norms be 1 there, not chosen for elegance.
- *Interpolate affinely.* Holomorphy forces the exponent to be affine in  $z$ ; two edge conditions then determine it uniquely.
- *Check boundedness on the strip separately.* It is the hypothesis of Theorem 2.1.13 that is most often forgotten, and Worked Example 2.1.2 showed the conclusion fails without it.

Drill 2.5 asks you to carry the same construction out for the dual family  $g_z$ , where the exponents involve  $q'_\theta$  and the same reasoning applies with  $L^{q'_j}$  in place of  $L^{p_j}$ .

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## Drills

**Drill 2.2.1.** Show that Theorem 2.2.1 with  $p_0 = q_0 = 1$ ,  $p_1 = q_1 = \infty$  recovers nothing about the Hardy–Littlewood maximal operator, and explain in one sentence which hypothesis fails.

**Drill 2.2.2.** Verify the scaling claim in Worked Example 2.2.1: show that  $\|f(\lambda \cdot)\|_p = \lambda^{-n/p} \|f\|_p$  on  $\mathbb{R}^n$ , and deduce that  $1 + \frac{1}{r} = \frac{1}{p} + \frac{1}{q}$  is necessary for Young's inequality.

**Drill 2.2.3.** Carry out the optimisation of Step 4 in the proof of Theorem 2.2.4 explicitly, and confirm that the resulting constant behaves like  $(p - p_0)^{-1/p}$  as  $p \downarrow p_0$ .

**Drill 2.2.4.** Carry out Step 2 of Worked Example 2.2.4 for the dual family: find the affine  $\zeta$  with  $\operatorname{Re} \zeta = q'_\theta/q'_j$  on the edge  $\operatorname{Re} z = j$ , and check  $\zeta(\theta) = 1$ .

**Drill 2.2.5.** In Worked Example 2.2.3(c), redo the computation for  $p < n - 1$  and show that the integral is then dominated by  $\theta \approx 1$  rather than  $\theta \approx \delta$ . What bound does the single tube impose in that range, and is it consistent with (b)?

## Notes

Riesz proved the real-variable case in 1926 and Thorin the complex one in 1938, introducing the analytic family device. Marcinkiewicz announced his theorem in 1939 shortly before his death; the first published proof is Zygmund's, from 1956.

Stein's interpolation theorem for analytic families of operators — in which the operator itself, not merely the test function, varies holomorphically — is the tool needed to interpolate the Riesz potentials and Bochner–Riesz means, and is proved in Volume 5 where those objects first appear. It rests on the same three-lines lemma, applied after a Poisson-kernel representation of the strip.

The two examples of Worked Example 2.2.3 are folklore and appear in Bourgain's 1991 paper. The observation in (c) that the single tube gives no constraint is worth remembering: in this subject the extremal example for an estimate is often not the one that looks most extreme.



## Chapter 2.3

# Fourier Series

The Fourier transform on  $\mathbb{R}^n$  has a compact model: Fourier series on the torus. Everything is easier there — the exponentials form an honest orthonormal basis, there are no convergence issues at infinity, and the measure is finite — and almost every phenomenon of the general theory appears first in this setting. We develop it now both for its own sake and because two of the applications this series is aimed at, the Vinogradov mean value theorem and Strichartz estimates on tori, live on  $\mathbb{T}^n$  rather than  $\mathbb{R}^n$ .

### 2.3.1 The torus and Fourier coefficients

Let  $\mathbb{T}^n = \mathbb{R}^n/\mathbb{Z}^n$ , and identify functions on  $\mathbb{T}^n$  with 1-periodic functions on  $\mathbb{R}^n$ . Lebesgue measure on  $[0, 1)^n$  makes  $\mathbb{T}^n$  a probability space, so by Drill 1.8.1 we have the inclusions  $L^\infty(\mathbb{T}^n) \subseteq L^q \subseteq L^p \subseteq L^1$  for  $p \leq q$ .

**Notation 2.3.1.** Write  $e(t) = e^{2\pi it}$ , and for  $\xi \in \mathbb{Z}^n$  let  $e_\xi(x) = e(\xi \cdot x)$ .

**Definition 2.3.2.** For  $f \in L^1(\mathbb{T}^n)$  and  $\xi \in \mathbb{Z}^n$ , the *Fourier coefficient* is

$$\widehat{f}(\xi) = \int_{\mathbb{T}^n} f(x) e(-\xi \cdot x) dx.$$

A *trigonometric polynomial* is a finite sum  $\sum_\xi c_\xi e_\xi$ .

**Proposition 2.3.3** (Basic properties). *For  $f, g \in L^1(\mathbb{T}^n)$ :*

- (i)  $|\widehat{f}(\xi)| \leq \|f\|_1$ , and  $\widehat{\cdot}$  is linear;
- (ii)  $\widehat{\tau_h f}(\xi) = e(-\xi \cdot h) \widehat{f}(\xi)$ , where  $\tau_h f(x) = f(x - h)$ ;
- (iii)  $\widehat{f * g} = \widehat{f} \widehat{g}$ , where  $(f * g)(x) = \int_{\mathbb{T}^n} f(x - y) g(y) dy$ ;
- (iv)  $\widehat{f}(\xi) \rightarrow 0$  as  $|\xi| \rightarrow \infty$  (Riemann-Lebesgue);

(v)  $\{e_\xi\}_{\xi \in \mathbb{Z}^n}$  is orthonormal in  $L^2(\mathbb{T}^n)$ .

*Proof.* (i) is the triangle inequality; (ii) a change of variable. For (iii), Young's inequality (Volume 1, Prop. 6.9) gives  $f * g \in L^1$ , and Fubini — licensed by Tonelli applied to  $|f| * |g|$  — gives

$$\widehat{f * g}(\xi) = \iint f(x-y)g(y)e(-\xi \cdot x) \, dy \, dx = \int g(y)e(-\xi \cdot y) \left( \int f(u)e(-\xi \cdot u) \, du \right) dy.$$

(iv) is Worked Example 1.5.4 (Riemann–Lebesgue)(b) transferred to the torus: the same translation trick applies, using  $e(-\xi \cdot (x+h)) = -e(-\xi \cdot x)$  for a suitable  $h$  depending on  $\xi$ , together with continuity of translation in  $L^1(\mathbb{T}^n)$ .

(v) For  $\xi \neq \eta$ ,  $\int_{\mathbb{T}^n} e((\xi - \eta) \cdot x) \, dx$  factors into a product of one-dimensional integrals  $\int_0^1 e(mt) \, dt$ , which is 0 unless  $m = 0$ ; at least one coordinate of  $\xi - \eta$  is nonzero.  $\square$

## 2.3.2 The Dirichlet and Fejér kernels

For  $n = 1$  (the general case follows by taking products; see Drill 3.3) the symmetric partial sums are

$$S_N f(x) = \sum_{|\xi| \leq N} \widehat{f}(\xi) e(\xi x) = (f * D_N)(x), \quad D_N = \sum_{|\xi| \leq N} e_\xi,$$

and the Cesàro means are

$$\sigma_N f = \frac{1}{N+1} \sum_{k=0}^N S_k f = f * K_N, \quad K_N = \frac{1}{N+1} \sum_{k=0}^N D_k.$$

**Lemma 2.3.4** (Closed forms). *For  $x \notin \mathbb{Z}$ ,*

$$D_N(x) = \frac{\sin((2N+1)\pi x)}{\sin(\pi x)}, \quad K_N(x) = \frac{1}{N+1} \left( \frac{\sin((N+1)\pi x)}{\sin(\pi x)} \right)^2.$$

*Proof.*  $D_N$  is a geometric sum: with  $q = e(x) \neq 1$ ,

$$\sum_{\xi=-N}^N q^\xi = \frac{q^{-N} - q^{N+1}}{1 - q} = \frac{q^{-N-1/2} - q^{N+1/2}}{q^{-1/2} - q^{1/2}} = \frac{\sin((2N+1)\pi x)}{\sin(\pi x)},$$

using  $q^s - q^{-s} = 2i \sin(2\pi s x)$ . For  $K_N$ , sum the numerators: with  $s = \sin(\pi x)$ ,

$$(N+1)K_N(x) = \frac{1}{s} \sum_{k=0}^N \sin((2k+1)\pi x) = \frac{1}{s} \cdot \operatorname{Im} \left( \sum_{k=0}^N q^{k+1/2} \right) = \frac{1}{s} \cdot \frac{\sin^2((N+1)\pi x)}{s},$$

the last step by the same geometric summation and a little trigonometry.  $\square$



**Proposition 2.3.5** ( $K_N$  is an approximate identity).  $K_N \geq 0$ ,  $\int_{\mathbb{T}} K_N = 1$ , and for every  $\eta > 0$ ,  $\int_{\eta \leq |x| \leq 1/2} K_N \rightarrow 0$  as  $N \rightarrow \infty$ .

*Proof.* Nonnegativity is visible from Lemma 2.3.4. The integral is 1 because  $\int_{\mathbb{T}} D_k = \widehat{D_k}(0)$ -style orthogonality gives  $\int D_k = 1$  for every  $k$ . For the third, on  $\eta \leq |x| \leq 1/2$  we have  $\sin^2(\pi x) \geq \sin^2(\pi \eta)$ , so  $K_N \leq \frac{1}{(N+1)\sin^2(\pi \eta)} \rightarrow 0$  uniformly there.  $\square$

**Theorem 2.3.6** (Fejér). *If  $f \in C(\mathbb{T})$  then  $\sigma_N f \rightarrow f$  uniformly. If  $f \in L^p(\mathbb{T})$  with  $1 \leq p < \infty$  then  $\sigma_N f \rightarrow f$  in  $L^p$ .*

*Proof.* Since  $\int K_N = 1$ ,

$$\sigma_N f(x) - f(x) = \int_{\mathbb{T}} (f(x-y) - f(x)) K_N(y) dy.$$

Given  $\varepsilon > 0$ , uniform continuity of  $f$  supplies  $\eta$  with  $|f(x-y) - f(x)| < \varepsilon$  for  $|y| < \eta$ , uniformly in  $x$ . Split the integral at  $|y| = \eta$ : the inner part is at most  $\varepsilon \int K_N = \varepsilon$ , and the outer is at most  $2\|f\|_{\infty} \int_{|y| \geq \eta} K_N \rightarrow 0$  by Proposition 2.3.5.

For the  $L^p$  statement, Minkowski's integral inequality gives  $\|\sigma_N f - f\|_p \leq \int_{\mathbb{T}} \|\tau_y f - f\|_p K_N(y) dy$ , and the same splitting applies using continuity of translation in  $L^p$ .  $\square$

**Corollary 2.3.7** (Uniqueness and density). *If  $f \in L^1(\mathbb{T}^n)$  has  $\widehat{f} \equiv 0$  then  $f = 0$  a.e. Trigonometric polynomials are dense in  $C(\mathbb{T}^n)$  and in  $L^p(\mathbb{T}^n)$  for  $p < \infty$ .*

*Proof.*  $\sigma_N f$  is a trigonometric polynomial built from the  $\widehat{f}(\xi)$ , so  $\widehat{f} \equiv 0$  forces  $\sigma_N f \equiv 0$ , and  $\sigma_N f \rightarrow f$  in  $L^1$  by Theorem 2.3.6. Density is the same statement read forwards.  $\square$

**Theorem 2.3.8** (Parseval).  *$\{e_{\xi}\}_{\xi \in \mathbb{Z}^n}$  is an orthonormal basis of  $L^2(\mathbb{T}^n)$ . Hence  $\|f\|_2^2 = \sum_{\xi} |\widehat{f}(\xi)|^2$  and  $f = \sum_{\xi} \widehat{f}(\xi) e_{\xi}$  with convergence in  $L^2$ .*

*Proof.* The system is orthonormal by Proposition 2.3.3(v) and its span — the trigonometric polynomials — is dense by Corollary 2.3.7. Apply Volume 1, Theorem 8.20.  $\square$

**Where we are.** Note what has been used. Fejér's theorem gave density, density plus orthonormality gave the basis, and the basis gave Parseval. No appeal to the Stone–Weierstrass theorem was needed, and none to any pointwise convergence statement — which is fortunate, because pointwise convergence is false in general, as we now show.

### 2.3.3 Divergence, and two facts from functional analysis

Volume 1 developed no functional analysis beyond Hilbert space, so we prove here the two results needed. They are used again in Chapter 2.5.

**Theorem 2.3.9** (Baire category). *A complete metric space  $X$  is not the union of countably many closed sets with empty interior.*

*Proof.* Suppose  $X = \bigcup_{k \geq 1} F_k$  with each  $F_k$  closed and  $\text{int } F_k = \emptyset$ . Since  $F_1 \neq X$  and  $F_1$  is closed, pick a closed ball  $\overline{B}_1 = \overline{B}(x_1, r_1)$  with  $r_1 < 1$  disjoint from  $F_1$ . Inductively, since  $F_{k+1}$  has empty interior it cannot contain the open ball  $B(x_k, r_k)$ , so we may choose a closed ball  $\overline{B}_{k+1} \subseteq B(x_k, r_k)$  with  $r_{k+1} < 2^{-k}$  and  $\overline{B}_{k+1} \cap F_{k+1} = \emptyset$ .

The centres form a Cauchy sequence, since  $x_l \in B_k$  for  $l \geq k$  and  $r_k \rightarrow 0$ ; let  $x = \lim_k x_k$ . Then  $x \in \overline{B}_k$  for every  $k$ , hence  $x \notin F_k$  for every  $k$ , contradicting  $X = \bigcup_k F_k$ .  $\square$

**Theorem 2.3.10** (Uniform boundedness). *Let  $X$  be a Banach space and  $\{\Lambda_\alpha\}$  a family of bounded linear functionals on  $X$  with  $\sup_\alpha |\Lambda_\alpha(x)| < \infty$  for every  $x \in X$ . Then  $\sup_\alpha \|\Lambda_\alpha\| < \infty$ .*

*Proof.* Let  $F_k = \{x : |\Lambda_\alpha(x)| \leq k \text{ for all } \alpha\}$ , closed as an intersection of preimages of closed sets under continuous maps. By hypothesis  $X = \bigcup_k F_k$ , so by Theorem 2.3.9 some  $F_{k_0}$  has an interior point:  $\overline{B}(x_0, r) \subseteq F_{k_0}$  for some  $r > 0$ . For  $\|y\| \leq 1$  and any  $\alpha$ ,

$$|\Lambda_\alpha(y)| = \frac{1}{r} |\Lambda_\alpha(x_0 + ry) - \Lambda_\alpha(x_0)| \leq \frac{2k_0}{r},$$

so  $\|\Lambda_\alpha\| \leq 2k_0/r$  for every  $\alpha$ .  $\square$

**WORKED EXAMPLE 2.3.1 — TIER B The Dirichlet kernel has logarithmically divergent  $L^1$  norm**

We show

$$\|D_N\|_{L^1(\mathbb{T})} \approx \log N. \quad (2.3.1)$$

This single estimate is the whole obstruction to pointwise convergence of Fourier series, and it is worth seeing exactly where the logarithm comes from.

*Lower bound.* Using  $|\sin(\pi x)| \leq \pi|x|$  and Lemma 2.3.4,

$$\|D_N\|_1 = 2 \int_0^{1/2} \frac{|\sin((2N+1)\pi x)|}{\sin(\pi x)} dx \geq \frac{2}{\pi} \int_0^{1/2} \frac{|\sin((2N+1)\pi x)|}{x} dx.$$

Substituting  $u = (2N + 1)\pi x$  turns this into  $\frac{2}{\pi} \int_0^{(2N+1)\pi/2} \frac{|\sin u|}{u} du$ . On each interval  $[k\pi, (k+1)\pi]$  the integral of  $|\sin u|$  is 2, and  $u \leq (k+1)\pi$  there, so the integral is at least  $\frac{2}{\pi} \sum_{k=1}^N \frac{2}{(k+1)\pi} \gtrsim \log N$ .

*Upper bound.* On  $[0, 1/2]$ ,  $\sin(\pi x) \geq 2x$ , so  $|D_N(x)| \leq \min(2N+1, (2x)^{-1})$ , using also  $|D_N| \leq 2N+1$  from the defining sum. Integrating,

$$\|D_N\|_1 \leq 2 \left( \int_0^{1/(2N+1)} (2N+1) dx + \int_{1/(2N+1)}^{1/2} \frac{dx}{2x} \right) \lesssim 1 + \log N.$$

*Reading it.* The two regimes are the whole story. Near  $x = 0$  the kernel is as tall as it can be,  $\approx N$ , over a window of width  $\approx 1/N$ , contributing  $O(1)$ . Away from 0 it decays only like  $1/x$ , and  $1/x$  is not integrable — it fails logarithmically. The Fejér kernel escapes this because squaring turns  $1/x$  into  $1/x^2$ , which is integrable, at the cost of the extra factor  $1/(N+1)$ ; that is precisely why Proposition 2.3.5 works and no analogue holds for  $D_N$ .

---

**Theorem 2.3.11** (du Bois-Reymond). *There exists  $f \in C(\mathbb{T})$  whose Fourier series diverges at 0; indeed  $\sup_N |S_N f(0)| = \infty$ .*

*Proof.* The maps  $\Lambda_N : C(\mathbb{T}) \rightarrow \mathbb{C}$ ,  $\Lambda_N f = S_N f(0) = \int_{\mathbb{T}} f(-y) D_N(y) dy$ , are bounded linear functionals on the Banach space  $C(\mathbb{T})$  with the sup norm, and

$$\|\Lambda_N\| = \|D_N\|_{L^1}.$$

Indeed  $\leq$  is immediate, and for  $\geq$  one takes  $f$  continuous with  $\|f\|_{\infty} \leq 1$  approximating  $\operatorname{sgn} D_N(\cdot)$  in  $L^1$ , which is possible by Theorem 1.8.7 (density of simple functions and  $C_c^{\infty}$ ).

By Worked Example 2.3.1,  $\|\Lambda_N\| \rightarrow \infty$ . If  $\sup_N |\Lambda_N f| < \infty$  held for every  $f \in C(\mathbb{T})$ , then Theorem 2.3.10 would force  $\sup_N \|\Lambda_N\| < \infty$ . So some  $f$  has  $\sup_N |S_N f(0)| = \infty$ .  $\square$

### Pitfall: what does and does not fail

Theorem 2.3.11 is often misremembered as saying Fourier series of continuous functions “do not converge”. They converge in  $L^2$  (Theorem 2.3.8), in  $L^p$  for  $1 < p < \infty$  (Volume 3, via the Hilbert transform), uniformly under mild regularity (Drill 3.2), and — by Carleson’s theorem — almost everywhere. What fails is convergence *at every point* for *every* continuous function, and the obstruction is exactly the logarithm in (2.3.1).

## 2.3.4 Hausdorff–Young on the torus

**Theorem 2.3.12.** For  $1 \leq p \leq 2$  and  $f \in L^p(\mathbb{T}^n)$ ,  $\|\widehat{f}\|_{\ell^{p'}(\mathbb{Z}^n)} \leq \|f\|_{L^p(\mathbb{T}^n)}$ .

*Proof.* The map  $f \mapsto \widehat{f}$  sends  $L^1(\mathbb{T}^n) \rightarrow \ell^\infty(\mathbb{Z}^n)$  with norm  $\leq 1$  by Proposition 2.3.3(i), and  $L^2 \rightarrow \ell^2$  with norm 1 by Theorem 2.3.8. It is linear. Apply Theorem 2.2.1 with  $(p_0, q_0) = (1, \infty)$ ,  $(p_1, q_1) = (2, 2)$  and  $\theta$  chosen so that  $\frac{1}{p} = \frac{1-\theta}{1} + \frac{\theta}{2}$ , which gives  $\frac{1}{q\theta} = \frac{\theta}{2} = 1 - \frac{1}{p} = \frac{1}{p'}$ . The interpolated constant is  $1^{1-\theta}1^\theta = 1$ .  $\square$

### WORKED EXAMPLE 2.3.2— TIER B Weyl's equidistribution criterion

A sequence  $(x_k) \subseteq \mathbb{T}$  is *equidistributed* if for every interval  $I \subseteq \mathbb{T}$ ,  $\frac{1}{K} \#\{k \leq K : x_k \in I\} \rightarrow |I|$ .

**Proposition 2.3.13.**  $(x_k)$  is equidistributed if and only if for every  $m \in \mathbb{Z} \setminus \{0\}$ ,

$$\frac{1}{K} \sum_{k=1}^K e(mx_k) \rightarrow 0.$$

*Proof.* Write  $A_K(g) = \frac{1}{K} \sum_{k \leq K} g(x_k)$ . Equidistribution says  $A_K(\mathbf{1}_I) \rightarrow \int \mathbf{1}_I$  for all intervals; the criterion says  $A_K(e_m) \rightarrow \int e_m$  for all  $m$  (the case  $m = 0$  being trivial).

( $\Rightarrow$ ) Every continuous  $g$  is a uniform limit of step functions, and  $|A_K(g)| \leq \|g\|_\infty$ , so  $A_K(g) \rightarrow \int g$  for all  $g \in C(\mathbb{T})$ ; in particular for  $g = e_m$ .

( $\Leftarrow$ ) By linearity  $A_K(P) \rightarrow \int P$  for every trigonometric polynomial  $P$ . Given  $g \in C(\mathbb{T})$  and  $\varepsilon > 0$ , Corollary 2.3.7 supplies  $P$  with  $\|g - P\|_\infty < \varepsilon$ , and then  $|A_K(g) - \int g| \leq 2\varepsilon + |A_K(P) - \int P|$ , so  $\limsup_K |A_K(g) - \int g| \leq 2\varepsilon$ . Hence  $A_K(g) \rightarrow \int g$  for all continuous  $g$ . Finally approximate  $\mathbf{1}_I$  above and below by continuous functions agreeing with it off a set of measure  $< \varepsilon$ .  $\square$

*Application.* For irrational  $\alpha$ , the sequence  $x_k = k\alpha$  is equidistributed: for  $m \neq 0$ ,

$$\left| \frac{1}{K} \sum_{k=1}^K e(mk\alpha) \right| = \frac{1}{K} \left| \frac{e(m\alpha)(e(mK\alpha) - 1)}{e(m\alpha) - 1} \right| \leq \frac{2}{K|e(m\alpha) - 1|} \rightarrow 0,$$

the denominator being nonzero precisely because  $m\alpha \notin \mathbb{Z}$ .

*Why this is here.* The criterion converts a counting statement into a bound on an *exponential sum*, and exponential sums are what Worked Example 2.3.3 and Volume 13 are about. Note the mechanism: cancellation in  $\sum_k e(mx_k)$  is equidistribution.

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WORKED EXAMPLE 2.3.3 — TIER C: RESEARCH TASTE **Vinogradov’s mean value theorem, set up**

Fix  $k \geq 2$  and  $N \geq 1$ , and consider the exponential sum

$$f_N(x) = \sum_{j=1}^N e(jx_1 + j^2x_2 + \cdots + j^kx_k), \quad x \in \mathbb{T}^k.$$

This is a trigonometric polynomial on  $\mathbb{T}^k$  whose frequencies are the  $N$  points  $(j, j^2, \dots, j^k)$  — that is,  $N$  points on the *moment curve* in  $\mathbb{R}^k$ . Define, for  $s \geq 1$ ,

$$J_{s,k}(N) = \int_{\mathbb{T}^k} |f_N(x)|^{2s} dx.$$

(a) *It counts solutions.* Expanding the  $2s$ -th power and integrating, orthogonality (Proposition 2.3.3(v)) kills every term except those where the frequency vectors cancel. Hence  $J_{s,k}(N)$  is exactly the number of integer solutions  $1 \leq j_1, \dots, j_{2s} \leq N$  of the system

$$j_1^i + \cdots + j_s^i = j_{s+1}^i + \cdots + j_{2s}^i, \quad i = 1, 2, \dots, k.$$

So an analytic quantity — an  $L^{2s}$  norm — is a Diophantine counting function. This is the reason decoupling has anything to do with number theory.

(b) *The two trivial lower bounds.* The *diagonal* solutions, those with  $(j_{s+1}, \dots, j_{2s})$  a permutation of  $(j_1, \dots, j_s)$ , number  $\gtrsim_s N^s$ . And a counting heuristic — the system imposes  $k$  equations on  $2s$  variables of size  $N$ , with the  $i$ -th equation of size  $N^i$  — predicts  $\approx N^{2s}/N^{1+2+\cdots+k} = N^{2s-k(k+1)/2}$  further solutions. So

$$J_{s,k}(N) \gtrsim_{s,k} N^s + N^{2s-k(k+1)/2}.$$

(c) *The theorem.* Vinogradov’s mean value theorem, in the sharp form proved by Bourgain, Demeter and Guth in 2016, asserts that this trivial lower bound is the truth:

$$J_{s,k}(N) \lesssim_{s,k,\varepsilon} N^\varepsilon \left( N^s + N^{2s-k(k+1)/2} \right) \quad \text{for every } \varepsilon > 0.$$

It was open for over eighty years for  $k \geq 4$ .

(d) *Why it is a decoupling statement.* Partition  $[1, N]$  into  $N$  single integers; each contributes one frequency on the moment curve. Rescaling  $x \mapsto (Nx_1, N^2x_2, \dots, N^kx_k)$  turns the  $N$  frequencies into a  $\delta$ -separated family on the curve with  $\delta = 1/N$ , and turns  $J_{s,k}(N)$  into precisely the left-hand

side of an  $\ell^2$  decoupling inequality for the moment curve at exponent  $p = 2s$ . The trivial lower bound (b) becomes the right-hand side. So the theorem says exactly: *the pieces of  $f_N$  behave as though they were orthogonal, up to  $N^\varepsilon$ .*

That statement — proved in Volume 13 — is the same one previewed in Volume 1, Worked Example 8.16, where Cauchy–Schwarz gave a power loss and the theorem promised only  $\delta^{-\varepsilon}$ . Curvature of the moment curve is what makes it true; for frequencies on a line it is false.

You now know what the destination looks like, and you have proved, in (a), the identity that makes the analytic and arithmetic sides the same object.

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## Drills

**Drill 2.3.1.** Compute the Fourier coefficients of the sawtooth  $f(x) = x - \frac{1}{2}$  on  $[0, 1)$  and of the triangle wave, and check Parseval in both cases. Deduce  $\sum_{m \geq 1} m^{-2} = \pi^2/6$ .

**Drill 2.3.2.** Show that if  $\sum_{\xi} |\widehat{f}(\xi)| < \infty$  then  $S_N f \rightarrow f$  uniformly and  $f$  is (a.e. equal to) a continuous function. Deduce that a  $C^2$  function on  $\mathbb{T}$  has uniformly convergent Fourier series.

**Drill 2.3.3.** State and prove the  $n$ -dimensional versions of Lemma 2.3.4 and Theorem 2.3.6, using products  $K_N(x) = \prod_{i=1}^n K_N(x_i)$ .

**Drill 2.3.4.** Verify (a) of Worked Example 2.3.3 in the case  $k = 1$ ,  $s = 2$ : show  $J_{2,1}(N)$  counts solutions of  $j_1 + j_2 = j_3 + j_4$  and compute it exactly.

## Notes

Fejér’s theorem is from 1900 and du Bois-Reymond’s example from 1873, pre-dating the functional-analytic proof given here by half a century; the uniform boundedness principle is Banach and Steinhaus, 1927. Carleson’s theorem on almost everywhere convergence for  $L^2$  functions dates to 1966 and is not proved in this series.

Vinogradov introduced  $J_{s,k}$  in 1935. The sharp bound was obtained for  $k = 3$  by Wooley (2016) via efficient congruencing and for all  $k$  by Bourgain–Demeter–Guth (2016) via decoupling; the two methods are compared in Volume 13.

## Chapter 2.4

# The Fourier Transform on $L^1$ and $L^2$

### 2.4.1 Definition and elementary identities

**Definition 2.4.1.** For  $f \in L^1(\mathbb{R}^n)$ ,

$$\widehat{f}(\xi) = \mathcal{F}f(\xi) = \int_{\mathbb{R}^n} f(x)e(-x \cdot \xi) dx, \quad e(t) = e^{2\pi it}.$$

The integral converges absolutely since  $|f(x)e(-x \cdot \xi)| = |f(x)|$ , and Worked Example 1.5.4 (Riemann–Lebesgue) already established that  $\widehat{f}$  is continuous, that  $\|\widehat{f}\|_\infty \leq \|f\|_1$ , and that  $\widehat{f}(\xi) \rightarrow 0$  as  $|\xi| \rightarrow \infty$ .

**Proposition 2.4.2** (The dictionary). *Let  $f \in L^1(\mathbb{R}^n)$ ,  $h \in \mathbb{R}^n$ ,  $\lambda > 0$ ,  $A \in GL_n(\mathbb{R})$ .*

- (i)  $\widehat{\tau_h f}(\xi) = e(-h \cdot \xi) \widehat{f}(\xi)$  where  $\tau_h f(x) = f(x - h)$ ;
- (ii)  $\widehat{e(h \cdot \cdot) f}(\xi) = \widehat{f}(\xi - h)$ ;
- (iii)  $\widehat{f(\lambda \cdot)}(\xi) = \lambda^{-n} \widehat{f}(\xi/\lambda)$ ;
- (iv)  $\widehat{f \circ A}(\xi) = |\det A|^{-1} \widehat{f}((A^{-1})^T \xi)$ ;
- (v)  $\widehat{f * g} = \widehat{f} \widehat{g}$  for  $g \in L^1$ ;
- (vi) if  $f, \partial_j f \in L^1$  then  $\widehat{\partial_j f}(\xi) = 2\pi i \xi_j \widehat{f}(\xi)$ ; if  $x_j f \in L^1$  then  $\widehat{f}$  is  $C^1$  in  $\xi_j$  with  $\partial_{\xi_j} \widehat{f} = -2\pi i x_j \widehat{f}$ .

*Proof.* (i)–(iv) are changes of variable, using Theorem 1.3.10 (linear images) for the linear substitution in (iv). (v) is Fubini, licensed by Tonelli applied to  $|f| * |g|$ .

$|g|$ , exactly as in Proposition 2.3.3(iii). The first half of (vi) is integration by parts, the boundary terms vanishing because  $f, \partial_j f \in L^1$  forces  $f(x) \rightarrow 0$  along almost every line; the second half is differentiation under the integral sign, Worked Example 1.5.3 (differentiating under the integral), with dominating function  $2\pi|x_j f|$ .  $\square$

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**Looking ahead: (iii) and (vi) are the whole subject in miniature**

Identity (iii) says that squeezing a function in space stretches its transform in frequency, at exactly reciprocal rates. Identity (vi) says that differentiating in space is multiplication by  $\xi$  in frequency, so *smoothness becomes decay*. These two sentences generate: the uncertainty principle (Worked Example 2.4.2); the Knapp example, where a  $\delta \times \delta^2$  cap produces a  $\delta^{-1} \times \delta^{-2}$  tube (Worked Example 2.4.3); the Sobolev spaces of Chapter 2.7; and the wave packet decompositions of Volume 12. Every “dual” object in this series is an instance of (iii).

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WORKED EXAMPLE 2.4.1— TIER A **The box and the sinc function**

Let  $f = \mathbf{1}_{[-1/2, 1/2]}$  on  $\mathbb{R}$ . Then

$$\widehat{f}(\xi) = \int_{-1/2}^{1/2} e(-x\xi) \, dx = \frac{e(-\xi/2) - e(\xi/2)}{-2\pi i \xi} = \frac{\sin(\pi\xi)}{\pi\xi}.$$

Three observations. The transform is not integrable —  $|\sin(\pi\xi)/\pi\xi|$  decays only like  $1/|\xi|$  — so  $\widehat{f} \notin L^1$  even though  $f \in L^1$ ; this is why the inversion theorem below needs a hypothesis. The decay rate  $|\xi|^{-1}$  reflects the jump discontinuities of  $f$ , in accordance with Proposition 2.4.2(vi). And in  $\mathbb{R}^n$ ,  $\mathbf{1}_{[-1/2, 1/2]^n}$  has transform  $\prod_j \sin(\pi\xi_j)/(\pi\xi_j)$  by Fubini: separable functions have separable transforms, a fact used constantly.

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## 2.4.2 The Gaussian

**Proposition 2.4.3.** *Let  $G(x) = e^{-\pi|x|^2}$  on  $\mathbb{R}^n$ . Then  $\widehat{G} = G$ .*

*Proof.* By Fubini it suffices to treat  $n = 1$ . Let  $g(\xi) = \widehat{G}(\xi)$ . By Proposition 2.4.2(vi),  $g$  is differentiable with

$$g'(\xi) = -\widehat{2\pi i x G}(\xi).$$

Now  $xG(x) = -\frac{1}{2\pi}G'(x)$ , so  $-2\pi i x G = iG'$ , and by Proposition 2.4.2(vi) again  $\widehat{G}'(\xi) = 2\pi i \xi g(\xi)$ . Therefore

$$g'(\xi) = i \cdot 2\pi i \xi g(\xi) = -2\pi \xi g(\xi).$$



This linear ODE has solution  $g(\xi) = g(0)e^{-\pi\xi^2}$ , and  $g(0) = \int_{\mathbb{R}} e^{-\pi x^2} dx = 1$  by Drill 1.6.2. Hence  $g = G$ .  $\square$

**Corollary 2.4.4.** *For  $t > 0$ , the function  $G_t(x) = t^{-n/2}e^{-\pi|x|^2/t}$  has  $\widehat{G}_t(\xi) = e^{-\pi t|\xi|^2}$ , and  $\int G_t = 1$ .*

*Proof.* Apply Proposition 2.4.2(iii) with  $\lambda = t^{-1/2}$  to Proposition 2.4.3.  $\square$

## 2.4.3 Inversion

**Lemma 2.4.5** (Multiplication formula). *For  $f, g \in L^1(\mathbb{R}^n)$ ,  $\int_{\mathbb{R}^n} \widehat{f}g = \int_{\mathbb{R}^n} f\widehat{g}$ .*

*Proof.* Both sides equal  $\iint f(x)g(\xi)e(-x \cdot \xi) dx d\xi$ , and Fubini applies because the double integral of the absolute value is  $\|f\|_1\|g\|_1 < \infty$ .  $\square$

**Theorem 2.4.6** (Fourier inversion). *If  $f \in L^1(\mathbb{R}^n)$  and  $\widehat{f} \in L^1(\mathbb{R}^n)$ , then for a.e.  $x$*

$$f(x) = \int_{\mathbb{R}^n} \widehat{f}(\xi)e(x \cdot \xi) d\xi.$$

*In particular  $f$  agrees a.e. with a continuous function.*

*Idea of the proof.* The integral  $\int \widehat{f}(\xi)e(x \cdot \xi) d\xi$  cannot be attacked by Fubini directly, since  $\widehat{f}(\xi)e(x \cdot \xi)f(y)$  is not absolutely integrable in  $(y, \xi)$ . The remedy is to insert a Gaussian damping factor  $e^{-\pi t|\xi|^2}$ , which makes Fubini legitimate, evaluate, and let  $t \rightarrow 0$ . The damped integral turns out to be  $f * G_t$ , an approximate identity, and Proposition 1.6.10 (approximate identities) finishes the job.

*Proof.* Fix  $t > 0$  and  $x \in \mathbb{R}^n$ . Apply Lemma 2.4.5 with  $f$  and the function  $g(\xi) = e^{-\pi t|\xi|^2}e(x \cdot \xi)$ , whose transform is, by Corollary 2.4.4 and Proposition 2.4.2(ii),  $\widehat{g}(y) = G_t(y - x)$ . This gives

$$\int_{\mathbb{R}^n} \widehat{f}(\xi)e^{-\pi t|\xi|^2}e(x \cdot \xi) d\xi = \int_{\mathbb{R}^n} f(y)G_t(y - x) dy = (f * G_t)(x), \quad (2.4.1)$$

using that  $G_t$  is even.

Let  $t \rightarrow 0^+$ . On the left, the integrand is dominated by  $|\widehat{f}| \in L^1$  and converges pointwise to  $\widehat{f}(\xi)e(x \cdot \xi)$ , so by dominated convergence the left side tends to  $\int \widehat{f}(\xi)e(x \cdot \xi) d\xi$ , for every  $x$ . On the right,  $\{G_t\}$  is an approximate identity in the sense of Volume 1, Proposition 6.10 — it is  $G_1$  dilated, with integral 1 — so  $f * G_t \rightarrow f$  in  $L^1$ , hence along a subsequence a.e. Equating the two limits gives the result a.e., and the left side is continuous in  $x$  by dominated convergence.  $\square$

**Corollary 2.4.7** (Uniqueness). *If  $f \in L^1(\mathbb{R}^n)$  and  $\widehat{f} \equiv 0$  then  $f = 0$  a.e.*

*Proof.*  $\widehat{f} = 0 \in L^1$ , so Theorem 2.4.6 applies and gives  $f = 0$  a.e. □

## 2.4.4 Plancherel

**Theorem 2.4.8** (Plancherel). *The Fourier transform, restricted to  $L^1 \cap L^2(\mathbb{R}^n)$ , satisfies  $\|\widehat{f}\|_2 = \|f\|_2$ , and extends uniquely to a unitary operator  $\mathcal{F} : L^2(\mathbb{R}^n) \rightarrow L^2(\mathbb{R}^n)$ . Its inverse is  $\mathcal{F}^{-1}g(x) = \widehat{g}(-x)$ , and the Parseval identity  $\int f\overline{g} = \int \widehat{f}\widehat{\overline{g}}$  holds for all  $f, g \in L^2$ .*

*Idea of the proof.* The extension step is the point, and it is exactly the manoeuvre Volume 1 promised in its opening chapter: the transform is defined by an integral formula needing  $L^1$ , its most important property lives in  $L^2$ , and  $L^2 \not\subseteq L^1$  on  $\mathbb{R}^n$ . One proves the identity on the dense subspace  $L^1 \cap L^2$  and extends by continuity — which requires the target to be complete, i.e. the Riesz–Fischer theorem.

*Proof. Step 1.* *The identity on a dense subspace.*

Let  $f \in L^1 \cap L^2$  and put  $h = f * \widetilde{f}$  where  $\widetilde{f}(x) = \overline{f(-x)}$ . Then  $h \in L^1$  by Young, and  $h$  is continuous and bounded: it is a convolution of two  $L^2$  functions, so by Cauchy–Schwarz  $|h(x)| \leq \|f\|_2^2$ , and continuity follows from continuity of translation in  $L^2$  (Theorem 1.8.7 (density of simple functions and  $C_c^\infty$ )(iii)). Moreover

$$\widehat{h} = \widehat{f}\widehat{\widetilde{f}} = \widehat{f}\widehat{\overline{f}} = |\widehat{f}|^2 \geq 0,$$

using  $\widehat{\widetilde{f}} = \overline{\widehat{f}}$ , a direct computation.

Now apply (2.4.1) to  $h$  at  $x = 0$ :

$$\int_{\mathbb{R}^n} |\widehat{f}(\xi)|^2 e^{-\pi t|\xi|^2} d\xi = (h * G_t)(0).$$

As  $t \rightarrow 0$ , the left side increases to  $\int |\widehat{f}|^2$  by monotone convergence, and the right side tends to  $h(0)$  because  $h$  is bounded and continuous and  $\{G_t\}$  is an approximate identity. Since  $h(0) = \int f\overline{f} = \|f\|_2^2$ , we conclude  $\|\widehat{f}\|_2^2 = \|f\|_2^2$ ; in particular  $\widehat{f} \in L^2$ .

**Step 2.** *Extension.*

$L^1 \cap L^2$  is dense in  $L^2$ : it contains the simple functions of finite support, which are dense by Theorem 1.8.7 (density of simple functions and  $C_c^\infty$ )(i). By Step 1,  $\mathcal{F}$  is an isometry on this dense subspace; since  $L^2$  is complete (Theorem 1.8.6 (Riesz–Fischer)), it extends uniquely to an isometry of  $L^2$ .

**Step 3. Surjectivity and the inverse.**

For  $f \in L^1 \cap L^2$  with also  $\widehat{f} \in L^1$ , Theorem 2.4.6 gives  $f(x) = \widehat{\widehat{f}}(-x)$ . Such  $f$  are dense in  $L^2$  — for instance  $f * G_t$  for  $f \in L^1 \cap L^2$  has  $\widehat{f * G_t} = \widehat{f} e^{-\pi t |\cdot|^2} \in L^1$  — so the identity  $\mathcal{F}^{-1}g(x) = \widehat{g}(-x)$  extends by continuity, and  $\mathcal{F}$  is onto. Parseval follows from the isometry by polarisation.  $\square$

**Theorem 2.4.9** (Hausdorff–Young). *For  $1 \leq p \leq 2$  and  $f \in L^p(\mathbb{R}^n)$ ,  $\|\widehat{f}\|_{p'} \leq \|f\|_p$ .*

*Proof.*  $\mathcal{F}$  maps  $L^1 \rightarrow L^\infty$  with norm  $\leq 1$  and  $L^2 \rightarrow L^2$  with norm 1; apply Theorem 2.2.1 exactly as in Theorem 2.3.12.  $\square$

**Pitfall: Hausdorff–Young fails for  $p > 2$** 

There is no bound  $\|\widehat{f}\|_q \lesssim \|f\|_p$  for  $p > 2$ , with any  $q$ . The reason is a scaling obstruction: by Proposition 2.4.2(iii), replacing  $f$  by  $f(\lambda \cdot)$  multiplies  $\|f\|_p$  by  $\lambda^{-n/p}$  and  $\|\widehat{f}\|_q$  by  $\lambda^{-n+n/q}$ , so any such inequality forces  $\frac{1}{q} = 1 - \frac{1}{p}$ , i.e.  $q = p'$ ; and then testing against a Gaussian modulated by high frequencies, or directly against  $f = \mathbf{1}_{B(0,R)}$  as  $R \rightarrow \infty$ , shows the inequality is false for  $p > 2$ . Drill 4.3 asks for the details. The asymmetry between  $p \leq 2$  and  $p > 2$  is not technical: it is the reason  $L^2$  is the natural home of the transform.

**WORKED EXAMPLE 2.4.2— TIER B The uncertainty principle**

**Proposition 2.4.10** (Heisenberg). *For  $f \in L^2(\mathbb{R})$  with  $xf, \xi\widehat{f} \in L^2$ ,*

$$\|f\|_2^2 \leq 4\pi \|xf\|_2 \|\xi\widehat{f}\|_2.$$

*Equality holds precisely for Gaussians  $f(x) = ce^{-a\pi x^2}$ ,  $a > 0$ .*

*Proof.* Assume first  $f \in C_c^\infty$ . Integrating by parts,

$$\|f\|_2^2 = \int_{\mathbb{R}} |f(x)|^2 dx = - \int_{\mathbb{R}} x \frac{d}{dx} |f(x)|^2 dx = -2 \operatorname{Re} \int_{\mathbb{R}} x f'(x) \overline{f(x)} dx,$$

the boundary terms vanishing by compact support. By Cauchy–Schwarz,

$$\|f\|_2^2 \leq 2 \|xf\|_2 \|f'\|_2.$$

By Proposition 2.4.2(vi) and Plancherel,  $\|f'\|_2 = \|\widehat{f'}\|_2 = 2\pi \|\xi\widehat{f}\|_2$ . Combining gives the inequality for  $f \in C_c^\infty$ , and the general case follows by density (Theorem 1.8.7 (density of simple functions and  $C_c^\infty$ )(ii)) together with Fatou.

Equality in Cauchy–Schwarz forces  $f'$  to be a real multiple of  $xf$ , say  $f'(x) = -2a\pi xf(x)$ ; the solutions are  $f(x) = ce^{-a\pi x^2}$ , and these lie in the admissible class precisely when  $a > 0$ , since then  $f \in \mathcal{S}(\mathbb{R})$ . Conversely each such  $f$  gives equality, by Proposition 2.4.3 and a direct computation.  $\square$

*Reading it.* A function and its transform cannot both be concentrated. If  $f$  lives essentially on an interval of length  $L$  then  $\|xf\|_2 \approx L\|f\|_2$ , and the inequality forces  $\|\xi\hat{f}\|_2 \gtrsim \|f\|_2/L$ , i.e.  $\hat{f}$  spreads over frequencies of size  $\gtrsim 1/L$ . This is Proposition 2.4.2(iii) in quantitative form, and it is the reason the “dual box” of a  $\delta$ -cap has dimensions  $\delta^{-1}$ : one cannot do better.

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**WORKED EXAMPLE 2.4.3— TIER C: RESEARCH TASTE The Knapp example, and the restriction problem**

Worked Example 1.4.3 (the early Knapp computation) performed a barehands version of this computation using only the box lemma. With the dictionary of Proposition 2.4.2 we can do it properly, and state the problem it points to.

(a) *The cap and its dual box.* Let  $P = \{(\xi', |\xi'|^2) : \xi' \in \mathbb{R}^{n-1}\}$  be the paraboloid in  $\mathbb{R}^n$ , and for  $\delta \in (0, 1)$  let

$$\kappa_\delta = \{(\xi', |\xi'|^2) : |\xi'| \leq \delta\} \subseteq P.$$

Since  $|\xi'| \leq \delta$  implies  $|\xi'|^2 \leq \delta^2$ , the cap lies in a box of dimensions  $\underbrace{\delta \times \cdots \times \delta}_{n-1} \times \delta^2$ . Curvature is exactly what converts the length scale  $\delta$  into

the thickness scale  $\delta^2$ ; for a flat piece of hyperplane the thickness would be 0 and everything below would change.

(b) *The extension operator.* For a function  $g$  on  $P$ , define

$$Eg(x) = \int_{\mathbb{R}^{n-1}} g(\xi') e(x' \cdot \xi' + x_n |\xi'|^2) d\xi'.$$

This is the Fourier transform of the measure  $g d\sigma$  carried by  $P$ ; that it makes sense as a tempered distribution is Chapter 2.5, and that it is a genuine function here follows since  $g$  is integrable.

(c) *The computation.* Take  $g = \mathbf{1}_{|\xi'| \leq \delta}$ , so  $\|g\|_{L^\infty} = 1$  and the cap has measure  $\approx \delta^{n-1}$ . If

$$|x'| \leq \frac{1}{100\delta} \quad \text{and} \quad |x_n| \leq \frac{1}{100\delta^2},$$

then for  $|\xi'| \leq \delta$  the phase satisfies

$$|x' \cdot \xi' + x_n |\xi'|^2| \leq \frac{1}{100\delta} \cdot \delta + \frac{1}{100\delta^2} \cdot \delta^2 = \frac{1}{50},$$

so every contribution to the integral lies within an arc of angle  $2\pi/50$  of 1, and in particular has real part at least  $\frac{1}{2}$ . Hence on the tube

$$T_\delta = \left\{ |x'| \leq \frac{1}{100\delta} \right\} \times \left\{ |x_n| \leq \frac{1}{100\delta^2} \right\}, \quad m(T_\delta) \approx \delta^{-(n-1)}\delta^{-2},$$

we have  $|Eg(x)| \gtrsim \delta^{n-1}$ .

So  $Eg$  is of size  $\approx \delta^{n-1}$  — the measure of the cap — on a *dual tube* whose side lengths are the reciprocals of the cap's. This is Proposition 2.4.2(iii) applied in each of the  $n$  directions separately.

(d) *What it rules out.* Suppose an estimate  $\|Eg\|_{L^q(\mathbb{R}^n)} \lesssim \|g\|_{L^p(d\sigma)}$  held. With  $g$  as above,  $\|g\|_{L^p(d\sigma)} \approx \delta^{(n-1)/p}$ , while

$$\|Eg\|_{L^q(\mathbb{R}^n)} \gtrsim \delta^{n-1} m(T_\delta)^{1/q} \approx \delta^{n-1} \delta^{-(n+1)/q}.$$

The estimate therefore forces

$$\delta^{n-1-\frac{n+1}{q}} \lesssim \delta^{\frac{n-1}{p}} \quad \text{for all small } \delta,$$

that is

$$n-1-\frac{n+1}{q} \geq \frac{n-1}{p}. \quad (2.4.2)$$

In the case  $p = \infty$  this reads  $q \geq \frac{2n}{n-1}$ .

(e) *The conjecture.* The *restriction conjecture* asserts that (2.4.2) together with the single further condition  $q > \frac{2n}{n-1}$  — which comes from testing  $g \equiv 1$  against the decay of  $\widehat{d\sigma}$ , computed in Chapter 2.8 — is *sufficient*. It is a theorem for  $n = 2$  (Fefferman–Stein, 1970) and open for every  $n \geq 3$ .

*What to take from this.* Locating the conjectured range was a page of scaling. The same was true for the Kakeya maximal conjecture in Worked Example 2.2.3, and it will be true for decoupling in Volume 13. In this subject the necessary conditions are elementary and the sufficiency is the work of decades. A reader who can reproduce (c) and (d) from memory has the right picture of what a wave packet is: a function concentrated on a tube dual to a cap, which is what  $Eg$  became.

**Figure 2.4.1 (to be drawn).** *A cap and its dual box*

**Content required:** The unit sphere with a cap  $\kappa_\delta$  of angular radius  $\delta$  at the north pole; alongside it, in physical space, the dual box  $T_\delta$  of dimensions  $\delta^{-1} \times \cdots \times \delta^{-1} \times \delta^{-2}$ , with the long axis normal to the cap. Mark the two lengths and the deviation  $\approx \delta^2$  of the cap from its tangent plane. This is the single most-used picture in the series.

## Drills

**Drill 2.4.1.** Compute  $\widehat{f}$  for  $f(x) = e^{-|x|}$  on  $\mathbb{R}$ , and verify directly that  $\widehat{f} \in L^1$  so that Theorem 2.4.6 applies.

**Drill 2.4.2.** Show that  $\mathcal{F}^4 = \text{Id}$  on  $L^2(\mathbb{R}^n)$ , and deduce that the spectrum of  $\mathcal{F}$  is contained in  $\{1, i, -1, -i\}$ . Exhibit an eigenfunction for each eigenvalue when  $n = 1$  (start from the Gaussian).

**Drill 2.4.3.** Complete the argument sketched in the Pitfall after Theorem 2.4.9: show by scaling that  $q = p'$  is forced, and then find an explicit  $f \in L^p(\mathbb{R})$  for  $p > 2$  with  $\widehat{f} \notin L^{p'}$ .

**Drill 2.4.4.** Redo Worked Example 2.4.3(c) for a cap on a *hyperplane* rather than a paraboloid, and show that the resulting dual set is a slab of infinite measure rather than a tube. Explain in one sentence why no estimate of the form in (d) can hold in that case.

## Notes

Plancherel's theorem is from 1910. The Gaussian-damping proof of inversion given here is standard and has the advantage of proving Plancherel simultaneously.

The restriction problem was posed by Stein in 1967. The condition (2.4.2) is Knapp's, and the sharpness of the whole range in  $n = 2$  is due to Fefferman and Stein; Zygmund gave the endpoint. Volume 6 proves the Stein–Tomas theorem, which settles the case  $p = 2$  in all dimensions, and Volume 12 presents the polynomial partitioning method and the current records.

## Chapter 2.5

# The Schwartz Space and Tempered Distributions

Chapter 2.4 left two defects. The inversion theorem needed the hypothesis  $\widehat{f} \in L^1$ , which is not verifiable in advance and fails for objects as simple as the indicator of a box. And the extension operator of Worked Example 2.4.3 was written as the Fourier transform of a measure carried by a paraboloid — an object with no density, to which the definition of Chapter 2.4 does not apply at all.

Both are repaired by enlarging the class of objects the transform acts on. The strategy is the standard one in analysis: find a space of exceptionally nice test functions on which the transform is a perfect isomorphism, and then define the transform of a rough object by how it pairs against test functions. The nice space is the Schwartz space; the rough objects are the tempered distributions; and surface measure on a sphere is one of them, which is what makes the restriction problem askable.

### 2.5.1 The Schwartz space

**Notation 2.5.1.** For a multi-index  $\alpha = (\alpha_1, \dots, \alpha_n) \in \mathbb{N}_0^n$  write  $|\alpha| = \sum \alpha_j$ ,  $x^\alpha = \prod x_j^{\alpha_j}$  and  $\partial^\alpha = \partial_1^{\alpha_1} \dots \partial_n^{\alpha_n}$ .

**Definition 2.5.2.** The *Schwartz space*  $\mathcal{S}(\mathbb{R}^n)$  consists of the  $f \in C^\infty(\mathbb{R}^n)$  for which every seminorm

$$\rho_{\alpha\beta}(f) = \sup_{x \in \mathbb{R}^n} |x^\alpha \partial^\beta f(x)|$$

is finite: every derivative decays faster than every polynomial.

Examples:  $e^{-\pi|x|^2}$ , and every  $C^\infty$  function of compact support. Not examples:  $e^{-|x|}$  (not smooth at 0),  $(1 + |x|^2)^{-N}$  (decays only polynomially).

**Proposition 2.5.3.**  $\mathcal{S}(\mathbb{R}^n) \subseteq L^p(\mathbb{R}^n)$  for every  $p \in [1, \infty]$ , and  $\mathcal{S}$  is closed under differentiation, multiplication by polynomials, and convolution.

*Proof.* For  $p < \infty$ ,  $|f(x)| \leq C_N(1+|x|)^{-N}$  for every  $N$ , taking  $N > n/p$  makes  $|f|^p$  integrable by polar coordinates (Volume 1, Theorem 6.10). Closure under  $\partial^\beta$  and under multiplication by  $x^\alpha$  is immediate from the definition, and convolution follows from the integral representation together with differentiation under the integral sign.  $\square$

**Definition 2.5.4.**  $f_k \rightarrow f$  in  $\mathcal{S}$  means  $\rho_{\alpha\beta}(f_k - f) \rightarrow 0$  for every  $\alpha, \beta$ .

**Proposition 2.5.5.**  $\mathcal{S}(\mathbb{R}^n)$  is complete: a sequence Cauchy in every seminorm converges in  $\mathcal{S}$ . Moreover  $C_c^\infty(\mathbb{R}^n)$  is dense in  $\mathcal{S}(\mathbb{R}^n)$ .

*Proof.* If  $(f_k)$  is Cauchy in every  $\rho_{\alpha\beta}$ , then each  $\partial^\beta f_k$  is uniformly Cauchy, so converges uniformly to some  $g_\beta$ ; standard real-analysis arguments give  $g_\beta = \partial^\beta g_0$  and  $g_0 \in C^\infty$ . The bounds  $\rho_{\alpha\beta}(f_k) \leq C$  pass to the limit, so  $g_0 \in \mathcal{S}$  and  $f_k \rightarrow g_0$  in  $\mathcal{S}$ .

For density, fix  $\chi \in C_c^\infty$  with  $\chi = 1$  on  $B(0, 1)$  and set  $f_k(x) = f(x)\chi(x/k)$ . Then  $f_k \in C_c^\infty$ , and by the Leibniz rule  $\partial^\beta(f - f_k)$  is a sum of terms each containing either  $(1 - \chi(x/k))$  — supported in  $|x| \geq k$ , where  $x^\alpha \partial^\gamma f$  is small — or a factor  $k^{-|\gamma|}(\partial^\gamma \chi)(x/k)$  with  $|\gamma| \geq 1$ , which carries a factor  $k^{-1}$ . Both tend to 0 in every seminorm.  $\square$

**Theorem 2.5.6** ( $\mathcal{F}$  is an isomorphism of  $\mathcal{S}$ ). *The Fourier transform maps  $\mathcal{S}(\mathbb{R}^n)$  bijectively onto itself, continuously in both directions, with inverse  $(\mathcal{F}^{-1}g)(x) = \widehat{g}(-x)$ . In particular the inversion formula holds pointwise for every  $f \in \mathcal{S}$ .*

*Proof. Step 1.*  $\mathcal{F}$  maps  $\mathcal{S}$  into  $\mathcal{S}$ .

Let  $f \in \mathcal{S}$ . By Proposition 2.4.2(vi), applied repeatedly — legitimate since every  $x^\alpha \partial^\beta f$  is in  $L^1$  by Proposition 2.5.3 —

$$\xi^\alpha \partial^\beta \widehat{f}(\xi) = \frac{(-1)^{|\beta|}}{(2\pi i)^{|\alpha| - |\beta|}} \widehat{\partial^\alpha (x^\beta f)}(\xi) \quad (2.5.1)$$

up to a harmless constant, whence

$$\rho_{\alpha\beta}(\widehat{f}) \leq C_{\alpha\beta} \|\partial^\alpha (x^\beta f)\|_{L^1} < \infty,$$

using  $\|\widehat{g}\|_\infty \leq \|g\|_1$ . The right-hand side is a finite combination of seminorms of  $f$ , which also gives continuity.

**Step 2.** *Inversion.*

For  $f \in \mathcal{S}$  we have  $f \in L^1$  and, by Step 1,  $\widehat{f} \in \mathcal{S} \subseteq L^1$ . So Theorem 2.4.6 applies and gives  $f(x) = \int \widehat{f}(\xi) e(x \cdot \xi) d\xi$  for a.e.  $x$ ; both sides being continuous, the identity holds everywhere.



**Step 3. Bijectivity.**

Step 2 exhibits a two-sided inverse:  $\mathcal{F}^{-1}g(x) = \widehat{g}(-x)$  maps  $\mathcal{S}$  to  $\mathcal{S}$  by Step 1 applied to  $g(-\cdot)$ , and  $\mathcal{F}^{-1}\mathcal{F} = \mathcal{F}\mathcal{F}^{-1} = \text{Id}$  on  $\mathcal{S}$ .  $\square$

## 2.5.2 Tempered distributions

**Definition 2.5.7.** A *tempered distribution* is a linear functional  $u : \mathcal{S}(\mathbb{R}^n) \rightarrow \mathbb{C}$  which is continuous, in the sense that there are  $N \in \mathbb{N}$  and  $C > 0$  with

$$|u(\varphi)| \leq C \sum_{|\alpha|, |\beta| \leq N} \rho_{\alpha\beta}(\varphi) \quad \text{for all } \varphi \in \mathcal{S}. \quad (2.5.2)$$

The space of these is  $\mathcal{S}'(\mathbb{R}^n)$ . We write  $\langle u, \varphi \rangle$  for  $u(\varphi)$ .

**Proposition 2.5.8** (Examples). *Each of the following defines a tempered distribution.*

- (i)  $u_f(\varphi) = \int f\varphi$ , for  $f \in L^p(\mathbb{R}^n)$ , any  $p \in [1, \infty]$ ;
- (ii)  $u_f(\varphi) = \int f\varphi$ , for measurable  $f$  with  $|f(x)| \leq C(1 + |x|)^M$  (polynomial growth);
- (iii)  $\delta_a(\varphi) = \varphi(a)$ ;
- (iv)  $\mu(\varphi) = \int \varphi d\mu$ , for any finite Borel measure  $\mu$  on  $\mathbb{R}^n$ , and more generally for any Borel measure with  $\int (1 + |x|)^{-M} d\mu < \infty$  for some  $M$ .

*Proof.* (i) By Hölder,  $|u_f(\varphi)| \leq \|f\|_p \|\varphi\|_{p'}$ , and  $\|\varphi\|_{p'}$  is bounded by finitely many seminorms as in Proposition 2.5.3. (ii) Choose  $N > M + n + 1$ ; then  $|\int f\varphi| \leq C \int (1 + |x|)^{M-N} dx \cdot \sup_x (1 + |x|)^N |\varphi(x)|$ , and the supremum is bounded by finitely many  $\rho_{\alpha 0}(\varphi)$ . (iii) is  $|\delta_a(\varphi)| \leq \rho_{00}(\varphi)$ . (iv) is the same computation as (ii) with  $dx$  replaced by  $d\mu$ .  $\square$

**Definition 2.5.9** (Operations). For  $u \in \mathcal{S}'$ ,  $\varphi \in \mathcal{S}$ , define by duality:

$$\langle \partial^\alpha u, \varphi \rangle = (-1)^{|\alpha|} \langle u, \partial^\alpha \varphi \rangle, \quad \langle Pu, \varphi \rangle = \langle u, P\varphi \rangle \quad (P \text{ a polynomial}),$$

$$\langle \tau_h u, \varphi \rangle = \langle u, \tau_{-h} \varphi \rangle, \quad \langle \widehat{u}, \varphi \rangle = \langle u, \widehat{\varphi} \rangle.$$

**Proposition 2.5.10.** *Each of these is again a tempered distribution, and each is consistent with the classical operation when  $u = u_f$  for  $f$  nice enough.*

*Proof.* Continuity in the sense of (2.5.2) follows because  $\varphi \mapsto \partial^\alpha \varphi$ ,  $\varphi \mapsto P\varphi$  and  $\varphi \mapsto \widehat{\varphi}$  are all continuous maps  $\mathcal{S} \rightarrow \mathcal{S}$  — the last by Theorem 2.5.6 — so they carry the finite family of seminorms in (2.5.2) to a finite family.

Consistency: for the derivative, if  $f \in C^\infty$  with polynomial growth then integration by parts gives  $\int (\partial^\alpha f) \varphi = (-1)^{|\alpha|} \int f \partial^\alpha \varphi$ , so the definition extends the classical one. For the transform, the multiplication formula Lemma 2.4.5 says exactly  $\int \widehat{f} \varphi = \int f \widehat{\varphi}$  for  $f \in L^1$ , which is the stated definition.  $\square$

**Theorem 2.5.11.**  $\mathcal{F} : \mathcal{S}' \rightarrow \mathcal{S}'$  is a bijection, with inverse given by  $\langle \mathcal{F}^{-1}u, \varphi \rangle = \langle u, \mathcal{F}^{-1}\varphi \rangle$ . It agrees with the  $L^1$  transform on  $L^1$  and with the  $L^2$  transform on  $L^2$ .

*Proof.* Bijectivity is immediate by dualising Theorem 2.5.6. Agreement on  $L^1$  is Proposition 2.5.10. On  $L^2$ : for  $f \in L^1 \cap L^2$  the two transforms agree, and both the distributional transform and the Plancherel extension are continuous from  $L^2$  into  $\mathcal{S}'$ , so they agree on the closure, which is  $L^2$ .  $\square$

WORKED EXAMPLE 2.5.1— TIER A  $\widehat{\delta} = 1$  and  $\widehat{1} = \delta$

From the definitions:  $\langle \widehat{\delta}, \varphi \rangle = \langle \delta, \widehat{\varphi} \rangle = \widehat{\varphi}(0) = \int \varphi(x) dx = \langle 1, \varphi \rangle$ . So  $\widehat{\delta} = 1$ . Conversely  $\langle \widehat{1}, \varphi \rangle = \int \widehat{\varphi}(\xi) d\xi = \varphi(0)$  by the inversion theorem Theorem 2.5.6 evaluated at  $x = 0$ . So  $\widehat{1} = \delta$ .

Neither statement makes sense classically —  $\delta$  is not a function and 1 is not integrable — and both are one line here. This is the whole point of the construction.

WORKED EXAMPLE 2.5.2— TIER B **The transform of  $|x|^{-a}$**

**Proposition 2.5.12.** For  $0 < a < n$ , the function  $|x|^{-a}$  is a tempered distribution and

$$\widehat{| \cdot |^{-a}}(\xi) = c_{n,a} |\xi|^{-(n-a)}, \quad c_{n,a} = \pi^{a-\frac{n}{2}} \frac{\Gamma(\frac{n-a}{2})}{\Gamma(\frac{a}{2})}.$$

*Proof.* That  $|x|^{-a} \in \mathcal{S}'$  follows from Proposition 2.5.8(ii) after splitting at  $|x| = 1$ : near the origin  $|x|^{-a}$  is integrable since  $a < n$ , and away from it the function is bounded.

The computation rests on the *subordination identity*: for  $s > 0$  and  $\lambda > 0$ , substituting  $u = \lambda t$  in the definition of the Gamma function gives

$$\lambda^{-s} = \frac{1}{\Gamma(s)} \int_0^\infty t^{s-1} e^{-\lambda t} dt.$$

Apply it with  $\lambda = \pi|x|^2$  and  $s = a/2$ :

$$|x|^{-a} = \frac{\pi^{a/2}}{\Gamma(a/2)} \int_0^\infty t^{\frac{a}{2}-1} e^{-\pi t|x|^2} dt.$$

Now pair against  $\varphi \in \mathcal{S}$  and use Fubini — legitimate since the double integral of the absolute value is finite, the  $t$ -integral converging at both ends because  $0 < a < n$  — together with  $\widehat{e^{-\pi t|\cdot|^2}}(\xi) = t^{-n/2}e^{-\pi|\xi|^2/t}$  from Corollary 2.4.4:

$$\langle \widehat{|\cdot|^{-a}}, \varphi \rangle = \frac{\pi^{a/2}}{\Gamma(a/2)} \int_0^\infty t^{\frac{a}{2}-1} \left( \int_{\mathbb{R}^n} t^{-\frac{n}{2}} e^{-\pi|\xi|^2/t} \varphi(\xi) d\xi \right) dt.$$

Substituting  $t \mapsto 1/t$  in the outer integral turns  $t^{\frac{a}{2}-\frac{n}{2}-1}e^{-\pi|\xi|^2/t}$  into  $t^{\frac{n-a}{2}-1}e^{-\pi t|\xi|^2}$ , and applying the subordination identity in reverse with  $s = (n-a)/2$  gives

$$\langle \widehat{|\cdot|^{-a}}, \varphi \rangle = \frac{\pi^{a/2}}{\Gamma(a/2)} \cdot \frac{\Gamma(\frac{n-a}{2})}{\pi^{\frac{n-a}{2}}} \int_{\mathbb{R}^n} |\xi|^{-(n-a)} \varphi(\xi) d\xi,$$

which is the claim.  $\square$

*Why it matters.* The exponents are conjugate:  $a \leftrightarrow n-a$ , with the self-dual point  $a = n/2$ . These are the *Riesz potentials*, and the identity above is the reason the Hardy–Littlewood–Sobolev inequality of Chapter 2.7 has the exponents it does. In Volume 8 the same computation, read backwards, converts the energy integral of a measure into a statement about  $\widehat{\mu}$ , which is the Fourier-analytic route to the Falconer distance problem.

#### WORKED EXAMPLE 2.5.3— TIER C: RESEARCH TASTE **Surface measure is a tempered distribution**

Let  $\sigma$  denote surface measure on the unit sphere  $\mathbb{S}^{n-1} \subseteq \mathbb{R}^n$ , constructed in Theorem 1.6.11 (polar coordinates).

(a) *It is tempered, and the verification is two lines.*  $\sigma$  is a finite Borel measure, with  $\sigma(\mathbb{S}^{n-1}) = n\omega_n < \infty$ . So for  $\varphi \in \mathcal{S}$ ,

$$|\langle \sigma, \varphi \rangle| = \left| \int_{\mathbb{S}^{n-1}} \varphi d\sigma \right| \leq \sigma(\mathbb{S}^{n-1}) \rho_{00}(\varphi),$$

which is (2.5.2) with  $N = 0$ . By Proposition 2.5.8(iv),  $\sigma \in \mathcal{S}'(\mathbb{R}^n)$ .

(b) *Therefore it has a Fourier transform.* By Theorem 2.5.11,  $\widehat{\sigma} \in \mathcal{S}'$  is defined, and unwinding the definition together with Fubini gives the classical-looking formula

$$\widehat{d\sigma}(\xi) = \int_{\mathbb{S}^{n-1}} e(-\omega \cdot \xi) d\sigma(\omega),$$

a bounded continuous function of  $\xi$  — bounded by  $\sigma(\mathbb{S}^{n-1})$ , and continuous by dominated convergence.

Pause on what has happened. The measure  $\sigma$  is supported on a set of Lebesgue measure zero; it has no density; the definition of Chapter 2.4 is inapplicable to it. Yet its transform is not merely a distribution but an honest bounded continuous function, and Chapter 2.8 will show that it *decays*, at the rate  $|\xi|^{-(n-1)/2}$ .

(c) *Why this is the enabling step for the whole series.* The restriction problem asks whether  $\widehat{f}$ , defined a priori only almost everywhere for  $f \in L^p(\mathbb{R}^n)$ , can be meaningfully restricted to  $\mathbb{S}^{n-1}$  — a set of measure zero, where “almost everywhere” says nothing. By duality (Volume 1, Corollary 8.12) the question is equivalent to a bound on the adjoint,

$$Eg = \widehat{gd\sigma},$$

and  $gd\sigma$  is exactly the kind of object admitted by Proposition 2.5.8(iv). Without distribution theory the extension operator  $E$  has no definition, and the conjecture cannot be stated. With it,  $E$  is elementary and the whole problem becomes a question about the decay and oscillation of one explicit function,  $\widehat{d\sigma}$ .

(d) *A caution.* That  $\sigma$  is tempered is easy; that  $\widehat{d\sigma}$  decays is not, and it is false without curvature. If  $\sigma$  were surface measure on a piece of hyperplane,  $\widehat{d\sigma}$  would be constant in the normal direction and would not decay at all (Drill 5.4). Everything in Volumes 5, 6, 12 and 13 turns on the quantitative form of that distinction.

#### WORKED EXAMPLE 2.5.4— TIER B **Working with tempered distributions in practice**

Three manoeuvres cover almost every use of  $\mathcal{S}'$  in this series.

(a) *Differentiate anything.* Every  $u \in \mathcal{S}'$  has derivatives of all orders, defined by  $\langle \partial^\alpha u, \varphi \rangle = (-1)^{|\alpha|} \langle u, \partial^\alpha \varphi \rangle$ . So one may differentiate first and worry about regularity afterwards — the manoeuvre that makes Sobolev theory possible.

(b) *Transform anything of polynomial growth.*  $\mathcal{F}$  is a bijection of  $\mathcal{S}'$ , by Theorem 2.5.11. In particular a finite measure has a Fourier transform, which is what Volumes 7 and 8 rely on throughout:  $\widehat{\mu}$  is a bounded continuous function when  $\mu$  is a compactly supported finite measure, and the energy machinery of Volume 7 is an identity in  $\mathcal{S}'$ .

(c) *Multiply only by smooth functions of moderate growth.* This is the restriction that bites. Two distributions cannot in general be multiplied;  $\delta_0^2$  is meaningless. Whenever an argument appears to multiply distributions — pairing a surface measure against another — it must be interpreted, and

Volume 8 shows the standard interpretation by mollification.

*The check to run.* Given a formal computation with distributions, ask at each step which of (a), (b), (c) licenses it. If none does, the step needs a limiting argument, and the estimates must be made uniform before the limit is taken.

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## Drills

**Drill 2.5.1.** Show that  $e^{-|x|} \notin \mathcal{S}(\mathbb{R})$  but that it defines a tempered distribution, and compute its transform.

**Drill 2.5.2.** Show that the principal value  $\text{p.v.} \frac{1}{x}$ , defined by  $\langle u, \varphi \rangle = \lim_{\varepsilon \rightarrow 0} \int_{|x| > \varepsilon} \frac{\varphi(x)}{x} dx$ , is a tempered distribution on  $\mathbb{R}$ , and that  $\widehat{u}(\xi) = -i\pi \operatorname{sgn} \xi$ . (This is the Hilbert transform multiplier, developed in Volume 3.)

**Drill 2.5.3.** Verify (2.5.1) carefully for  $n = 1$ ,  $\alpha = \beta = 1$ , and confirm the constant.

**Drill 2.5.4.** Let  $\sigma_0$  be Lebesgue measure on the hyperplane  $\{x_n = 0\} \subseteq \mathbb{R}^n$ , regarded as a tempered distribution. Show  $\widehat{\sigma_0}$  is Lebesgue measure on the line  $\{\xi' = 0\}$ , and in particular does not decay in the  $\xi_n$  direction.

## Notes

Schwartz introduced  $\mathcal{S}$  and  $\mathcal{S}'$  in the 1940s, and the theory was fully developed in his *Théorie des distributions* (1950–51). The subordination identity used in Worked Example 2.5.2 is due to Riesz.

Nothing in this chapter is difficult, and that is itself notable: distribution theory is a piece of bookkeeping, and its power comes entirely from Theorem 2.5.6 — the fact that  $\mathcal{F}$  is an isomorphism of a space small enough to differentiate freely and large enough to be dense in everything.



## Chapter 2.6

# Multipliers, Khintchine, and Littlewood–Paley

### 2.6.1 Multiplier operators

**Definition 2.6.1.** For  $m \in L^\infty(\mathbb{R}^n)$  define  $T_m$  on  $L^2(\mathbb{R}^n)$  by  $\widehat{T_m f} = m \widehat{f}$ , that is  $T_m f = \mathcal{F}^{-1}(m \mathcal{F} f)$ . The function  $m$  is the *symbol* or *multiplier*.

**Proposition 2.6.2.**  $\|T_m\|_{L^2 \rightarrow L^2} = \|m\|_\infty$ .

*Proof.* By Plancherel (Theorem 2.4.8),  $\|T_m f\|_2 = \|m \widehat{f}\|_2 \leq \|m\|_\infty \|\widehat{f}\|_2 = \|m\|_\infty \|f\|_2$ . For the reverse, choose a set  $E$  of positive finite measure on which  $|m| \geq \|m\|_\infty - \varepsilon$  and take  $f$  with  $\widehat{f} = \mathbf{1}_E$ .  $\square$

So on  $L^2$  the theory is trivial and complete: bounded symbols give bounded operators, with equality of norms. Everything difficult in this subject happens at  $p \neq 2$ , where no such characterisation exists, and where the geometry of the set on which  $m$  varies begins to matter.

#### Looking ahead: the ball multiplier

The single most important example is  $m = \mathbf{1}_{B(0,1)}$ , whose operator is the partial-sum operator for the Fourier inversion integral. By Proposition 2.6.2 it is bounded on  $L^2$ . Fefferman proved in 1971 that it is bounded on  $L^p(\mathbb{R}^n)$ ,  $n \geq 2$ , for *no*  $p \neq 2$  — and the proof is a Besicovitch set. That theorem is the historical origin of the connection between Kakeya and Fourier analysis, and it is proved in Volume 5. The reader may note that everything needed to state it is now available.

*Remark 2.6.3* (What is deferred, and why). The standard sufficient condition for  $L^p$  boundedness is the Hörmander–Mikhlin theorem: if  $|\partial^\alpha m(\xi)| \lesssim |\xi|^{-|\alpha|}$

for  $|\alpha| \leq \lfloor n/2 \rfloor + 1$ , then  $T_m$  is bounded on  $L^p$  for  $1 < p < \infty$ . Its proof requires Calderón–Zygmund theory, which this series develops in Volume 3.

Under the standing rule of this series a result is either proved where it is stated or not stated at all, so Hörmander–Mikhlin appears in Volume 3 and not here. Nothing in Volumes 1–2 needs it, and the tools Volumes 12 and 13 actually use — Khintchine’s inequality, Bernstein’s inequalities, and  $L^2$  orthogonality for lacunary pieces — are proved below in full.

## 2.6.2 Khintchine’s inequality

**Definition 2.6.4.** The *Rademacher functions* on  $[0, 1]$  are  $r_j(t) = \operatorname{sgn} \sin(2^j \pi t)$ ,  $j \geq 1$ . They take values  $\pm 1$ , and as random variables on  $([0, 1], m)$  they are independent with mean 0: for distinct  $j_1 < \dots < j_k$ ,  $\int_0^1 r_{j_1} \cdots r_{j_k} = 0$ .

**Theorem 2.6.5** (Khintchine). *For  $0 < p < \infty$  there are constants  $0 < A_p \leq B_p < \infty$  such that for all  $N$  and all scalars  $a_1, \dots, a_N \in \mathbb{C}$ ,*

$$A_p \left( \sum_{j=1}^N |a_j|^2 \right)^{1/2} \leq \left( \int_0^1 \left| \sum_{j=1}^N a_j r_j(t) \right|^p dt \right)^{1/p} \leq B_p \left( \sum_{j=1}^N |a_j|^2 \right)^{1/2}.$$

*Idea of the proof.* The case  $p = 2$  is an identity, by orthogonality of the  $r_j$ . For  $p > 2$  one shows that the random sum has *sub-Gaussian tails* — a Chernoff bound obtained by estimating the exponential moment  $\int e^{\lambda S}$ , which factors by independence into a product of hyperbolic cosines, each bounded by a Gaussian. Layer-cake integration then converts the tail bound into a moment bound. For  $p < 2$  one runs the log-convexity of  $L^p$  norms backwards, using the already established case  $p = 4$ .

*Proof.* Write  $S(t) = \sum_j a_j r_j(t)$  and  $A = \left( \sum_j |a_j|^2 \right)^{1/2}$ .

**Step 1.**  $p = 2$ .

By independence and  $r_j^2 = 1$ ,  $\int_0^1 |S|^2 = \sum_{j,k} a_j \overline{a_k} \int_0^1 r_j r_k = \sum_j |a_j|^2 = A^2$ . So  $A_2 = B_2 = 1$ .

**Step 2.** *Sub-Gaussian tails, real coefficients.*

Assume  $a_j \in \mathbb{R}$  and  $A = 1$ . For  $\lambda \in \mathbb{R}$ , independence gives

$$\int_0^1 e^{\lambda S(t)} dt = \prod_{j=1}^N \int_0^1 e^{\lambda a_j r_j} = \prod_{j=1}^N \cosh(\lambda a_j) \leq \prod_{j=1}^N e^{\lambda^2 a_j^2 / 2} = e^{\lambda^2 / 2},$$

using  $\cosh u \leq e^{u^2/2}$ , which follows by comparing the Taylor series  $\sum u^{2k}/(2k)!$  and  $\sum u^{2k}/(2^k k!)$  term by term, since  $(2k)! \geq 2^k k!$ .



Hence for  $\lambda > 0$ , by Chebyshev,

$$m(\{S > \mu\}) \leq e^{-\lambda\mu} \int_0^1 e^{\lambda S} \leq e^{\lambda^2/2 - \lambda\mu},$$

and optimising at  $\lambda = \mu$  gives  $e^{-\mu^2/2}$ . Applying the same to  $-S$ ,

$$m(\{|S| > \mu\}) \leq 2e^{-\mu^2/2}. \quad (2.6.1)$$

**Step 3.** *The upper bound for  $p > 0$ .*

By the layer-cake formula (Worked Example 1.5.1 (the layer-cake formula)) and (2.6.1),

$$\int_0^1 |S|^p = p \int_0^\infty \mu^{p-1} m(\{|S| > \mu\}) d\mu \leq 2p \int_0^\infty \mu^{p-1} e^{-\mu^2/2} d\mu = B_p^p < \infty,$$

a finite constant depending only on  $p$ . Rescaling by  $A$  gives the upper bound for real coefficients; for complex  $a_j$ , split into real and imaginary parts and use  $|z| \leq |\operatorname{Re} z| + |\operatorname{Im} z|$ , at the cost of a factor 2.

**Step 4.** *The lower bound for  $p \geq 2$*

is immediate from Step 1 and  $\|S\|_p \geq \|S\|_2$  (Drill 1.8.1, the measure being a probability measure).

**Step 5.** *The lower bound for  $0 < p < 2$ .*

Choose  $\theta \in (0, 1)$  with  $\frac{1}{2} = \frac{\theta}{p} + \frac{1-\theta}{4}$ . By log-convexity of  $L^p$  norms (Drill 1.8.2),

$$A = \|S\|_2 \leq \|S\|_p^\theta \|S\|_4^{1-\theta} \leq \|S\|_p^\theta (B_4 A)^{1-\theta},$$

using Step 3 with  $p = 4$ . Rearranging,  $A^\theta \leq B_4^{1-\theta} \|S\|_p^\theta$ , so  $\|S\|_p \geq B_4^{-(1-\theta)/\theta} A$ .  $\square$

## 2.6.3 Littlewood–Paley projections and Bernstein’s inequalities

**Definition 2.6.6.** Fix  $\psi \in C_c^\infty(\mathbb{R}^n)$  with  $\psi = 1$  on  $B(0, 1)$  and  $\operatorname{supp} \psi \subseteq B(0, 2)$ , and for a dyadic  $N = 2^k$  put

$$\widehat{P_{\leq N} f} = \psi(\xi/N) \widehat{f}(\xi), \quad P_N = P_{\leq N} - P_{\leq N/2},$$

so that  $\widehat{P_N f}$  is supported in the annulus  $|\xi| \approx N$  and  $f = \sum_{N \text{ dyadic}} P_N f$  in  $L^2$ .

**Theorem 2.6.7** (Bernstein). *Let  $1 \leq p \leq q \leq \infty$  and suppose  $\widehat{f}$  is supported in  $B(0, N)$ . Then*

$$\|f\|_q \lesssim_{n,p,q} N^{n(\frac{1}{p}-\frac{1}{q})} \|f\|_p, \quad (2.6.2)$$

and for every multi-index  $\alpha$ ,

$$\|\partial^\alpha f\|_p \lesssim_{n,\alpha} N^{|\alpha|} \|f\|_p. \quad (2.6.3)$$

The same holds, with the same proof, if  $\widehat{f}$  is supported in an annulus  $|\xi| \approx N$  or in a ball  $B(\xi_0, N)$  of radius  $N$  anywhere.

*Idea of the proof.* A function with Fourier support in a ball of radius  $N$  is reproduced by convolution with a fixed bump rescaled to width  $1/N$ . That is Proposition 2.4.2(iii) again: frequency support of size  $N$  means the function cannot oscillate on scales below  $1/N$ , so it is essentially constant on each such ball, and every  $L^p$  norm of a function constant at scale  $1/N$  is comparable to every other after the obvious scaling factor.

*Proof.* Choose  $\eta \in \mathcal{S}(\mathbb{R}^n)$  with  $\widehat{\eta} = 1$  on  $B(0, 1)$  and  $\text{supp } \widehat{\eta} \subseteq B(0, 2)$ ; such  $\eta$  exists since  $\mathcal{F}$  is onto  $\mathcal{S}$  (Theorem 2.5.6). Put  $\eta_N(x) = N^n \eta(Nx)$ , so  $\widehat{\eta_N}(\xi) = \widehat{\eta}(\xi/N) = 1$  on  $B(0, N)$ .

Since  $\widehat{f}$  is supported in  $B(0, N)$ , we have  $\widehat{f\eta_N} = \widehat{f}$ , hence

$$f = f * \eta_N.$$

(The identity holds in  $\mathcal{S}'$  and hence pointwise a.e. for  $f \in L^p$ .) By Young's inequality (Worked Example 2.2.1) with  $1 + \frac{1}{q} = \frac{1}{p} + \frac{1}{r}$ ,

$$\|f\|_q = \|f * \eta_N\|_q \leq \|f\|_p \|\eta_N\|_r.$$

Now  $\|\eta_N\|_r = N^n (\int |\eta(Nx)|^r dx)^{1/r} = N^{n(1-1/r)} \|\eta\|_r$ , and  $1 - \frac{1}{r} = \frac{1}{p} - \frac{1}{q}$ , giving (2.6.2).

For (2.6.3), differentiate the identity  $f = f * \eta_N$ :  $\partial^\alpha f = f * \partial^\alpha \eta_N$ , and  $\|\partial^\alpha \eta_N\|_1 = N^{|\alpha|} \|\partial^\alpha \eta\|_1$  by scaling. Apply Young with  $r = 1$ .

For a ball centred at  $\xi_0$ , replace  $\eta_N$  by  $e(x \cdot \xi_0) \eta_N(x)$ , using Proposition 2.4.2(ii); the  $L^r$  norms are unchanged. For an annulus, use that it is contained in a ball of radius  $2N$ .  $\square$

---

**WORKED EXAMPLE 2.6.1— TIER A Bernstein is sharp, and in which direction**

Take  $f = \eta_N$  itself, so  $\widehat{f}$  is supported in  $B(0, 2N)$ . Then  $\|\eta_N\|_p =$

$N^{n(1-1/p)} \|\eta\|_p$ , so

$$\frac{\|\eta_N\|_q}{\|\eta\|_p} = N^{n(1-1/q)-n(1-1/p)} \cdot \frac{\|\eta\|_q}{\|\eta\|_p} = N^{n(\frac{1}{p}-\frac{1}{q})} \cdot \frac{\|\eta\|_q}{\|\eta\|_p},$$

matching (2.6.2) exactly. So the exponent cannot be improved.

Note that the inequality goes only one way:  $\|f\|_q \lesssim N^{\dots} \|f\|_p$  for  $q \geq p$ , never the reverse. A frequency-localised function may be small in  $L^\infty$  and large in  $L^1$  — spread out — but not the other way round past the stated factor.

---

**Proposition 2.6.8** ( $L^2$  orthogonality of lacunary pieces). *For  $f \in L^2(\mathbb{R}^n)$ ,*

$$\|f\|_2^2 \approx \sum_{N \text{ dyadic}} \|P_N f\|_2^2.$$

*Proof.* By Plancherel it suffices to prove  $\|\widehat{f}\|_2^2 \approx \sum_N \|\widehat{P_N f}\|_2^2$ , and  $\widehat{P_N f} = (\psi(\xi/N) - \psi(2\xi/N))\widehat{f}(\xi)$ . Each  $\xi \neq 0$  lies in the support of at most two of these cutoffs, and the cutoffs sum to 1; so pointwise  $\sum_N |\widehat{P_N f}|^2 \approx |\widehat{f}|^2$ , with implicit constants  $\frac{1}{2}$  and 2. Integrate.  $\square$

*Remark 2.6.9* (The  $L^p$  statement is Volume 3). The Littlewood–Paley theorem asserts  $\|f\|_p \approx \|(\sum_N |P_N f|^2)^{1/2}\|_p$  for  $1 < p < \infty$ . The standard proof randomises the signs of the pieces, applies Theorem 2.6.5 to reduce to bounding  $\|\sum_N \pm P_N f\|_p$  uniformly, and then invokes Hörmander–Mikhlin — which by Remark 2.6.3 is Volume 3. So the theorem is proved there. Proposition 2.6.8 is the case  $p = 2$ , which is all that is used before then.

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**WORKED EXAMPLE 2.6.2— TIER C: RESEARCH TASTE** **Decoupling is false without curvature**

Worked Example 1.8.2 (almost orthogonality) stated the  $\ell^2$  decoupling inequality for the paraboloid and asserted that curvature is what makes it true. We can now prove the negative half completely.

(a) *The setting.* Let  $\delta = 1/N$  and partition a region of frequency space into  $N$  caps  $\theta_1, \dots, \theta_N$  of diameter  $\approx \delta$ . For functions  $f_j$  with  $\widehat{f_j}$  supported in  $\theta_j$ , the  $\ell^2$  decoupling inequality at exponent  $p$  asserts

$$\left\| \sum_{j=1}^N f_j \right\|_{L^p(\mathbb{R}^n)} \lesssim_\varepsilon N^\varepsilon \left( \sum_{j=1}^N \|f_j\|_{L^p}^2 \right)^{1/2}. \quad (2.6.4)$$

The trivial bound, from Cauchy–Schwarz applied to the  $N$  summands, is (2.6.4) with  $N^\varepsilon$  replaced by  $N^{1/2-1/p}$  — a genuine power of  $N$ . The content

of decoupling is the removal of that power.

(b) *The counterexample: caps on a hyperplane.* Take the caps to lie on a *flat* piece of hypersurface, say

$$\theta_j = \left[ \frac{j-1}{N}, \frac{j}{N} \right] \times [0, \delta]^{n-1} \subseteq \mathbb{R}^n,$$

so the  $\theta_j$  tile a box and lie in the hyperplane-like slab  $\{|\xi_2|, \dots, |\xi_n| \leq \delta\}$ . Let  $\eta$  be a fixed Schwartz function with  $\widehat{\eta}$  supported in  $[0, 1/N] \times [0, \delta]^{n-1}$  and  $\widehat{\eta} \geq 0$ ,  $\widehat{\eta}(0) > 0$ , and set

$$f_j(x) = e(x_1 \cdot \frac{j-1}{N}) \eta(x),$$

so that  $\widehat{f_j}$  is  $\widehat{\eta}$  translated into  $\theta_j$  (Proposition 2.4.2(ii)). Each  $f_j$  has the same modulus:  $|f_j| = |\eta|$ , so  $\|f_j\|_p = \|\eta\|_p$  for every  $j$ .

(c) *The sum is as large as it can be.* On the set where  $|x_1| \leq \frac{1}{100}$ , every phase  $e(x_1(j-1)/N)$  with  $1 \leq j \leq N$  lies within an arc of angle  $2\pi/100$  of 1, so

$$\left| \sum_{j=1}^N f_j(x) \right| \geq \frac{1}{2} N |\eta(x)|.$$

Hence  $\|\sum_j f_j\|_{L^p} \gtrsim N \|\eta\|_{L^p(|x_1| \leq 1/100)} \gtrsim N \|\eta\|_p$ , provided  $\eta$  is chosen concentrated in that region.

(d) *Conclusion.* The right-hand side of (2.6.4) is  $(\sum_j \|\eta\|_p^2)^{1/2} = N^{1/2} \|\eta\|_p$ . So (2.6.4) would force

$$N \|\eta\|_p \lesssim_\varepsilon N^\varepsilon N^{1/2} \|\eta\|_p, \quad \text{i.e.} \quad N^{1/2} \lesssim_\varepsilon N^\varepsilon,$$

which is false for large  $N$ . Indeed the trivial Cauchy–Schwarz bound  $N^{1/2-1/p}$  is, in this configuration, *attained* up to constants for  $p = \infty$  and cannot be improved at all.

(e) *What curvature does.* The failure has a one-line geometric cause. Each  $f_j$  is concentrated on the same dual box — the tubes dual to the caps  $\theta_j$  all point the same way, because the caps are translates of one another in a single direction. Constructive interference is therefore possible at every point of a large set simultaneously.

If instead the caps lie on the paraboloid, the cap  $\{(\xi', |\xi'|^2) : |\xi' - c_j| \leq \delta\}$  has a dual tube pointing in the direction of the *normal* at  $c_j$ , and distinct caps have distinct normals. So the  $N$  tubes point in  $N$  different directions, and the extent to which they can pile up on a common set is exactly the question answered by the Kakeya estimates of Volume 1, Worked Example 2.2.3, and Volume 9.

That is the precise sense in which restriction, decoupling and Kakeya are one subject: decoupling is true because differently-directed tubes cannot concentrate, and false when the tubes are parallel.

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## Drills

**Drill 2.6.1.** Show that the multiplier  $m(\xi) = \operatorname{sgn} \xi$  on  $\mathbb{R}$  gives the Hilbert transform, and compute its kernel using Worked Example 2.5.2’s companion, Drill 5.2.

**Drill 2.6.2.** Prove  $\cosh u \leq e^{u^2/2}$  by comparing Taylor coefficients, and verify  $(2k)! \geq 2^k k!$ .

**Drill 2.6.3.** Deduce from Theorem 2.6.7 that if  $\widehat{f}$  is supported in  $B(0, N)$  then  $f$  extends to an entire function; and that  $\|f\|_\infty \lesssim N^{n/2} \|f\|_2$ .

**Drill 2.6.4.** In Worked Example 2.6.2, compute the exact loss for finite  $p$  and confirm that the ratio of the two sides of (2.6.4) is  $\approx N^{1/2-1/p}$ , not merely  $\gtrsim N^\varepsilon$ .

## Notes

Khinchine’s inequality dates to 1923. The sub-Gaussian proof given here is the standard one and is the prototype of the concentration inequalities used throughout modern analysis.

Bernstein’s inequality is elementary but is used on essentially every page of the decoupling literature; readers should be able to reproduce (2.6.2) without thinking. Worked Example 2.6.2 is folklore, and appears as the motivating remark in the introduction of Bourgain and Demeter’s 2015 paper.



## Chapter 2.7

# Sobolev Spaces

The slogan of this chapter is Proposition 2.4.2(vi): differentiation is multiplication by  $\xi$ , so *smoothness becomes decay*. Sobolev spaces make the slogan into a definition, and the payoff is that questions about differentiability become questions about the size of  $\widehat{f}$  — which we can compute.

### 2.7.1 Weak derivatives and $W^{k,p}$

**Definition 2.7.1.** For  $f \in L^1_{\text{loc}}(\mathbb{R}^n)$  and a multi-index  $\alpha$ , a function  $g \in L^1_{\text{loc}}$  is the *weak derivative*  $\partial^\alpha f$  if

$$\int_{\mathbb{R}^n} g \varphi = (-1)^{|\alpha|} \int_{\mathbb{R}^n} f \partial^\alpha \varphi \quad \text{for all } \varphi \in C_c^\infty(\mathbb{R}^n),$$

which is Definition 2.5.9 restricted to the case where the distributional derivative happens to be a function. For  $k \in \mathbb{N}_0$  and  $1 \leq p \leq \infty$ ,

$$W^{k,p}(\mathbb{R}^n) = \{f \in L^p : \partial^\alpha f \in L^p \text{ for all } |\alpha| \leq k\}, \quad \|f\|_{W^{k,p}} = \sum_{|\alpha| \leq k} \|\partial^\alpha f\|_p.$$

Weak derivatives are unique up to null sets, by Corollary 1.5.3 (term-by-term integration) (a locally integrable function integrating to zero against every test function vanishes a.e.).

**Proposition 2.7.2.**  $W^{k,p}(\mathbb{R}^n)$  is a Banach space, and  $C_c^\infty$  is dense in it for  $p < \infty$ .

*Proof.* Completeness: if  $(f_j)$  is Cauchy in  $W^{k,p}$  then each  $(\partial^\alpha f_j)$  is Cauchy in  $L^p$ , hence converges to some  $g_\alpha$  by Theorem 1.8.6 (Riesz–Fischer); passing to the limit in the defining identity shows  $g_\alpha = \partial^\alpha g_0$  weakly. Density follows by mollification:  $f * \phi_t \rightarrow f$  in  $L^p$  (Volume 1, Prop. 6.10) and  $\partial^\alpha(f * \phi_t) = (\partial^\alpha f) * \phi_t$ , followed by truncation as in Proposition 2.5.5.  $\square$

## 2.7.2 The Fourier characterisation: $H^s$

**Definition 2.7.3.** For  $s \in \mathbb{R}$ , the *Sobolev space*  $H^s(\mathbb{R}^n)$  consists of the  $f \in \mathcal{S}'(\mathbb{R}^n)$  with  $\widehat{f} \in L^1_{\text{loc}}$  and

$$\|f\|_{H^s}^2 = \int_{\mathbb{R}^n} (1 + |\xi|^2)^s |\widehat{f}(\xi)|^2 d\xi < \infty.$$

**Proposition 2.7.4.** For  $k \in \mathbb{N}_0$ ,  $H^k(\mathbb{R}^n) = W^{k,2}(\mathbb{R}^n)$  with comparable norms.

*Proof.* By Proposition 2.4.2(vi) and Plancherel,  $\|\partial^\alpha f\|_2^2 = \int |2\pi\xi^\alpha|^2 |\widehat{f}|^2$ , and

$$\sum_{|\alpha| \leq k} |\xi^\alpha|^2 \approx_{n,k} (1 + |\xi|^2)^k,$$

since both sides are positive continuous functions that are homogeneous of degree  $2k$  at infinity and bounded near 0. Integrate against  $|\widehat{f}|^2$ .  $\square$

### Looking ahead: why $H^s$ for non-integer $s$

Nothing in the definition of  $H^s$  requires  $s$  to be an integer, and this is not a formal generalisation. Fractional smoothness is what the estimates of this subject actually produce: Strichartz estimates measure the smoothing of the Schrödinger evolution in fractions of a derivative; the local smoothing conjecture for the wave equation, proved by Guth–Wang–Zhang in 2020 via decoupling, asserts a gain of exactly  $1/p$  derivatives; and the Falconer distance problem is stated in terms of the fractional exponent  $s$  in an energy integral (Worked Example 2.7.1).

**Theorem 2.7.5** (Sobolev embedding,  $L^2$  case). *If  $s > n/2$  then  $H^s(\mathbb{R}^n) \subseteq C_0(\mathbb{R}^n)$ , the continuous functions vanishing at infinity, with  $\|f\|_\infty \lesssim_{n,s} \|f\|_{H^s}$ .*

*Proof.* By Cauchy–Schwarz,

$$\int_{\mathbb{R}^n} |\widehat{f}| = \int_{\mathbb{R}^n} (1 + |\xi|^2)^{-s/2} \cdot (1 + |\xi|^2)^{s/2} |\widehat{f}| \leq \left( \int (1 + |\xi|^2)^{-s} d\xi \right)^{1/2} \|f\|_{H^s},$$

and the first factor is finite exactly when  $2s > n$ , by polar coordinates (Volume 1, Theorem 6.10). So  $\widehat{f} \in L^1$ , and Theorem 2.4.6 together with Worked Example 1.5.4 (Riemann–Lebesgue) shows  $f$  agrees a.e. with a continuous function vanishing at infinity, with  $\|f\|_\infty \leq \|\widehat{f}\|_1$ .  $\square$

### Pitfall: the endpoint $s = n/2$ fails



$H^{n/2}(\mathbb{R}^n) \not\subseteq L^\infty(\mathbb{R}^n)$ , and the failure is exactly logarithmic. Take

$$\widehat{f}(\xi) = |\xi|^{-n} (\log|\xi|)^{-1} \mathbf{1}_{|\xi|>2},$$

which is nonnegative. By polar coordinates,

$$\|f\|_{H^{n/2}}^2 \approx \int_{|\xi|>2} |\xi|^n |\xi|^{-2n} (\log|\xi|)^{-2} d\xi \approx \int_2^\infty \frac{dr}{r (\log r)^2} < \infty,$$

so  $f \in H^{n/2}$ ; whereas

$$\int_{\mathbb{R}^n} \widehat{f} \approx \int_2^\infty \frac{dr}{r \log r} = \infty.$$

To see that this forces unboundedness, let  $f_R$  be defined by  $\widehat{f_R} = \widehat{f} \mathbf{1}_{|\xi|<R}$ . Each  $\widehat{f_R}$  is integrable with compact support, so  $f_R$  is continuous with  $f_R(0) = \int \widehat{f_R} \rightarrow \infty$  as  $R \rightarrow \infty$ , while  $\|f_R\|_{H^{n/2}} \leq \|f\|_{H^{n/2}}$  for every  $R$ . An embedding  $\|g\|_\infty \lesssim \|g\|_{H^{n/2}}$  would therefore give  $f_R(0) \lesssim \|f\|_{H^{n/2}}$  uniformly in  $R$ , a contradiction.

The gap between  $s > n/2$  and  $s = n/2$  is thus a divergent  $\int dr/(r \log r)$  — the same logarithm as in Worked Example 2.3.1. Endpoint failures of this kind recur throughout the subject, and the  $\delta^{-\varepsilon}$  losses in the Kakeya and decoupling conjectures exist precisely to step around them.

### 2.7.3 The Gagliardo–Nirenberg–Sobolev inequality

For  $p \neq 2$  the Fourier transform is unavailable and a different argument is needed. Write  $p^* = \frac{np}{n-p}$  for  $1 \leq p < n$ , the exponent forced by scaling (Drill 7.1).

**Theorem 2.7.6.** *For  $1 \leq p < n$  and  $f \in C_c^1(\mathbb{R}^n)$ ,*

$$\|f\|_{p^*} \lesssim_{n,p} \|\nabla f\|_p.$$

*Consequently  $W^{1,p}(\mathbb{R}^n) \subseteq L^{p^*}(\mathbb{R}^n)$ .*

*Idea of the proof.* Start with  $p = 1$ . The value  $f(x)$  is controlled by the integral of  $|\partial_j f|$  along the line through  $x$  in direction  $e_j$ , for each  $j$  separately. Multiplying the  $n$  resulting bounds and taking the  $(n-1)$ -st root produces a quantity that can be integrated one variable at a time, each integration using the generalised Hölder inequality (Corollary 1.8.3 (generalised Hölder)). The case  $p > 1$  follows by applying the case  $p = 1$  to a power of  $|f|$ .

*Proof. Step 1. The case  $p = 1$ .*

For each  $j$  and each  $x$ , the fundamental theorem of calculus gives

$$|f(x)| = \left| \int_{-\infty}^{x_j} \partial_j f(x_1, \dots, t, \dots, x_n) dt \right| \leq \int_{\mathbb{R}} |\partial_j f| dx_j =: F_j(\hat{x}_j),$$

where  $\hat{x}_j$  denotes  $x$  with the  $j$ -th coordinate omitted. Multiplying over  $j$  and taking the  $(n-1)$ -st root,

$$|f(x)|^{\frac{n}{n-1}} \leq \prod_{j=1}^n F_j(\hat{x}_j)^{\frac{1}{n-1}}.$$

Integrate in  $x_1$ . The factor  $j = 1$  does not depend on  $x_1$  and comes out; the remaining  $n-1$  factors are handled by the generalised Hölder inequality with exponents  $p_2 = \dots = p_n = n-1$ :

$$\int_{\mathbb{R}} |f|^{\frac{n}{n-1}} dx_1 \leq F_1^{\frac{1}{n-1}} \prod_{j=2}^n \left( \int_{\mathbb{R}} F_j dx_1 \right)^{\frac{1}{n-1}}.$$

Repeating for  $x_2, \dots, x_n$  and using Hölder at each step,

$$\int_{\mathbb{R}^n} |f|^{\frac{n}{n-1}} dx \leq \prod_{j=1}^n \left( \int_{\mathbb{R}^{n-1}} F_j d\hat{x}_j \right)^{\frac{1}{n-1}} = \prod_{j=1}^n \|\partial_j f\|_1^{\frac{1}{n-1}} \leq \|\nabla f\|_1^{\frac{n}{n-1}}.$$

Raising to the power  $\frac{n-1}{n}$  gives  $\|f\|_{1^*} \leq \|\nabla f\|_1$  with  $1^* = \frac{n}{n-1}$ .

**Step 2.** *The case  $1 < p < n$ : setting up.*

Put

$$\gamma = \frac{p(n-1)}{n-p} > 1,$$

the inequality holding because  $p > 1$  gives  $p(n-1) > n-p$ . Since  $\gamma > 1$ , the function  $t \mapsto |t|^\gamma$  is  $C^1$  on  $\mathbb{R}$  with derivative  $\gamma|t|^{\gamma-1} \operatorname{sgn} t$ , which is continuous and vanishes at  $t = 0$ . Hence  $g = |f|^\gamma$  is  $C^1$  with compact support, and

$$|\nabla g| = \gamma |f|^{\gamma-1} |\nabla f|.$$

No separate argument at the zero set of  $f$  is needed; this is exactly why  $\gamma > 1$  was arranged.

**Step 3.** *Two exponent identities.*

We record the two computations the argument needs. First,

$$\frac{\gamma n}{n-1} = \frac{p(n-1)}{n-p} \cdot \frac{n}{n-1} = \frac{pn}{n-p} = p^*.$$

Second, since  $\gamma - 1 = \frac{p(n-1)-(n-p)}{n-p} = \frac{n(p-1)}{n-p}$  and  $p' = \frac{p}{p-1}$ ,

$$(\gamma - 1)p' = \frac{n(p-1)}{n-p} \cdot \frac{p}{p-1} = \frac{np}{n-p} = p^*.$$

So both exponents that will appear are  $p^*$ .

**Step 4.** *The estimate.*

Apply Step 1 to  $g$ , then Hölder with exponents  $p'$  and  $p$ :

$$\left( \int |f|^{\frac{\gamma n}{n-1}} \right)^{\frac{n-1}{n}} = \|g\|_{\frac{n}{n-1}} \leq \|\nabla g\|_1 = \gamma \int |f|^{\gamma-1} |\nabla f| \leq \gamma \left( \int |f|^{(\gamma-1)p'} \right)^{\frac{1}{p'}} \|\nabla f\|_p.$$

By Step 3 both integrals are  $\int |f|^{p^*}$ , so this reads

$$\|f\|_{p^*}^{p^* \frac{n-1}{n}} \leq \gamma \|f\|_{p^*}^{p^*/p'} \|\nabla f\|_p.$$

Since  $f$  has compact support,  $\|f\|_{p^*} < \infty$ , and we may divide (the case  $\|f\|_{p^*} = 0$  being trivial). The resulting exponent on the left is

$$p^* \left( \frac{n-1}{n} - \frac{1}{p'} \right) = p^* \left( \frac{n-1}{n} - 1 + \frac{1}{p} \right) = p^* \left( \frac{1}{p} - \frac{1}{n} \right) = \frac{np}{n-p} \cdot \frac{n-p}{np} = 1.$$

Hence  $\|f\|_{p^*} \leq \gamma \|\nabla f\|_p$ . □

## 2.7.4 Fractional integration

**Theorem 2.7.7** (Hardy–Littlewood–Sobolev). *Let  $0 < \lambda < n$  and  $1 < p < q < \infty$  with*

$$\frac{1}{q} = \frac{1}{p} + \frac{\lambda}{n} - 1.$$

*Then  $\|f * |\cdot|^{-\lambda}\|_q \lesssim_{n,p,\lambda} \|f\|_p$ .*

*Idea of the proof.* Split the kernel at an as-yet-unspecified radius  $R$ . The near part is controlled by the Hardy–Littlewood maximal function, because  $|y|^{-\lambda}$  is essentially constant on dyadic annuli and the annuli sum geometrically. The far part is controlled by Hölder, since  $|y|^{-\lambda}$  is in  $L^{p'}$  away from the origin. Optimising  $R$  pointwise — a different  $R$  at each  $x$  — interpolates between the two, and the maximal theorem of Volume 1 finishes.

*Proof.* Fix  $x$  and  $R > 0$  and split  $|(f * |\cdot|^{-\lambda})(x)| \leq \text{I} + \text{II}$  with the integrals over  $|y| < R$  and  $|y| \geq R$ .

**Step 1. Near part.**

Decompose  $\{|y| < R\}$  into dyadic annuli  $A_k = \{2^{-k-1}R \leq |y| < 2^{-k}R\}$ :

$$I \leq \sum_{k \geq 0} (2^{-k-1}R)^{-\lambda} \int_{|y| < 2^{-k}R} |f(x-y)| \, dy \lesssim \sum_{k \geq 0} (2^{-k}R)^{n-\lambda} \mathcal{M}f(x) \lesssim R^{n-\lambda} \mathcal{M}f(x),$$

using the definition of the Hardy–Littlewood maximal function (Volume 1, §7.3) and summability of  $\sum_k 2^{-k(n-\lambda)}$ , which needs  $\lambda < n$ .

**Step 2. Far part.**

By Hölder,

$$II \leq \|f\|_p \left( \int_{|y| \geq R} |y|^{-\lambda p'} \, dy \right)^{1/p'} \lesssim \|f\|_p R^{\frac{n}{p'} - \lambda},$$

the integral converging precisely when  $\lambda p' > n$  — which is equivalent to  $q < \infty$  under the stated exponent relation.

**Step 3. Optimise pointwise.**

Choose  $R = R(x)$  with  $R^{n-\lambda} \mathcal{M}f(x) = R^{n/p' - \lambda} \|f\|_p$ , that is  $R^{n/p} = \|f\|_p / \mathcal{M}f(x)$ . Substituting,

$$|(f * |\cdot|^{-\lambda})(x)| \lesssim \|f\|_p^\beta (\mathcal{M}f(x))^{1-\beta}, \quad \beta = \frac{p(n-\lambda)}{n}.$$

**Step 4. Integrate.**

The exponent relation  $\frac{1}{q} = \frac{1}{p} + \frac{\lambda}{n} - 1$  gives  $q = \frac{np}{\lambda p - np + n}$ , and substituting  $\beta = \frac{p(n-\lambda)}{n}$ ,

$$q(1-\beta) = \frac{np}{\lambda p - np + n} \left( 1 - \frac{p(n-\lambda)}{n} \right) = \frac{np}{\lambda p - np + n} \cdot \frac{n - pn + p\lambda}{n} = p.$$

Hence

$$\|f * |\cdot|^{-\lambda}\|_q \lesssim \|f\|_p^\beta \left( \int (\mathcal{M}f)^{q(1-\beta)} \right)^{1/q} = \|f\|_p^\beta \|\mathcal{M}f\|_p^{1-\beta} \lesssim \|f\|_p^\beta \|f\|_p^{1-\beta} = \|f\|_p,$$

the last step by the maximal theorem (Theorem 1.7.3 (Hardy–Littlewood maximal inequality)), which requires  $p > 1$ .  $\square$

**WORKED EXAMPLE 2.7.1— TIER C: RESEARCH TASTE** **Energy integrals and Hausdorff dimension**

This example builds the bridge between the Fourier analysis of this volume and the geometric measure theory of Volume 8. Recall Hausdorff measure  $\mathcal{H}^s$  and dimension  $\dim_{\mathcal{H}}$  from Worked Example 1.2.4 (Hausdorff measure).

**Definition 2.7.8.** For a finite Borel measure  $\mu$  on  $\mathbb{R}^n$  and  $s > 0$ , the  $s$ -energy is

$$I_s(\mu) = \iint |x - y|^{-s} d\mu(x) d\mu(y).$$

(a) *The Fourier formula.* By Worked Example 2.5.2,  $|\cdot|^{-s}$  has Fourier transform  $c_{n,s}|\xi|^{-(n-s)}$  for  $0 < s < n$ . Pairing and using the multiplication formula (formally on  $\mathcal{S}'$ , rigorously after mollifying  $\mu$ ),

$$I_s(\mu) = c_{n,s} \int_{\mathbb{R}^n} |\hat{\mu}(\xi)|^2 |\xi|^{-(n-s)} d\xi. \quad (2.7.1)$$

So an integral measuring how spread out  $\mu$  is in *space* equals a weighted integral of  $|\hat{\mu}|^2$  in *frequency*. This identity is the entire reason Fourier analysis says anything about fractal geometry.

(b) *Finite energy forces large dimension.*

**Proposition 2.7.9.** If  $\mu$  is a finite Borel measure with  $\mu(A) > 0$  and  $I_s(\mu) < \infty$ , then  $\dim_{\mathcal{H}} A \geq s$ .

*Proof.* Let  $g(x) = \int |x - y|^{-s} d\mu(y)$ , so  $\int g d\mu = I_s(\mu) < \infty$  and hence  $g < \infty$  for  $\mu$ -a.e.  $x$ . Choose  $M$  with  $A' = \{x \in A : g(x) \leq M\}$  satisfying  $\mu(A') > 0$ . For  $x \in A'$  and  $r > 0$ ,

$$r^{-s} \mu(B(x, r)) \leq \int_{B(x, r)} |x - y|^{-s} d\mu(y) \leq g(x) \leq M,$$

so  $\mu(B(x, r)) \leq Mr^s$  for every  $x \in A'$  and  $r > 0$ : the measure is  $s$ -Frostman on  $A'$ .

Now let  $\{U_j\}$  be any cover of  $A'$  by sets of diameter  $\leq \delta$ . Each  $U_j$  meeting  $A'$  is contained in a ball  $B(x_j, \text{diam } U_j)$  with  $x_j \in A'$ , so  $\mu(U_j) \leq M(\text{diam } U_j)^s$ . Summing,

$$0 < \mu(A') \leq \sum_j \mu(U_j) \leq M \sum_j (\text{diam } U_j)^s.$$

Taking the infimum over covers gives  $\mathcal{H}_\delta^s(A') \geq \mu(A')/M > 0$  for every  $\delta$ , hence  $\mathcal{H}^s(A') > 0$  and  $\dim_{\mathcal{H}} A \geq \dim_{\mathcal{H}} A' \geq s$ .  $\square$

(c) *What this is for.* Combining (a) and (b): to prove a set has dimension at least  $s$ , it suffices to put a measure on it whose Fourier transform decays fast enough for the integral (2.7.1) to converge. Decay of  $\hat{\mu}$  is a quantitative statement, provable by the oscillatory-integral methods of Volume 5.

This is the mechanism behind:

- the *Falconer distance conjecture*, that  $\dim_{\mathcal{H}} A > n/2$  forces the distance set  $\{|x - y| : x, y \in A\}$  to have positive Lebesgue measure — proved in the plane at the threshold  $5/4$  by Guth–Iosevich–Ou–Wang in 2020, and open at  $n/2$ ;
- the theory of *Salem sets*, those whose Fourier dimension equals their Hausdorff dimension;
- the Fourier-analytic lower bounds for the dimension of Besicovitch sets, which is how Volume 9 converts the Kakeya maximal estimates into dimension statements.

Note the direction. Finite energy gives *lower* bounds on dimension, and lower bounds are what the Kakeya conjecture asks for. The reverse implication — Frostman’s lemma, that a set of positive  $\mathcal{H}^s$  measure carries a measure of finite  $t$ -energy for every  $t < s$  — is proved in Volume 8, and together the two make dimension and energy interchangeable.

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## Drills

**Drill 2.7.1.** Show by scaling that  $\|f\|_q \lesssim \|\nabla f\|_p$  can hold only for  $q = p^* = np/(n - p)$ .

**Drill 2.7.2.** Verify the exponent bookkeeping in Step 2 of Theorem 2.7.6: check  $\gamma = p(n - 1)/(n - p)$  and that  $\frac{\gamma n}{n-1} = (\gamma - 1)p' = p^*$ .

**Drill 2.7.3.** Confirm the identity  $q(1 - \beta) = p$  used in Step 4 of Theorem 2.7.7, where  $\beta = p(n - \lambda)/n$  and  $\frac{1}{q} = \frac{1}{p} + \frac{\lambda}{n} - 1$ .

**Drill 2.7.4.** Show that  $|x|^{-a}\mathbf{1}_{|x|<1} \in H^s(\mathbb{R}^n)$  if and only if  $s < \frac{n}{2} - a$ , and interpret this in the light of Theorem 2.7.5.

## Notes

Theorem 2.7.6 is due to Gagliardo and Nirenberg independently (1958–59), with the case  $p = 1$  going back to Sobolev. The proof of Theorem 2.7.7 via the maximal function is Hedberg’s (1972) and is the shortest known.

Worked Example 2.7.1 is the standard potential-theoretic characterisation of dimension, due to Frostman; the Fourier form (2.7.1) was exploited systematically by Kaufman, Mattila and Falconer from the 1960s onwards and is the foundation of Volume 8.

## Chapter 2.8

# The Fourier Transform of Surface Measure

Everything in this volume has been machinery. This chapter uses it, on the one computation that the rest of the series depends on: the exact evaluation of  $\widehat{d\sigma}$  for the sphere, and the decay estimate  $|\widehat{d\sigma}(\xi)| \lesssim |\xi|^{-(n-1)/2}$  that follows.

The exponent  $(n-1)/2$  is not a technicality. It is the number of non-vanishing principal curvatures of the sphere, divided by two, and it propagates into the exponent of every theorem in Volumes 6, 12 and 13. By the end of this chapter we will have derived, from the decay alone, a genuine restriction theorem.

### 2.8.1 The exact formula

**Proposition 2.8.1** (Reduction to a one-dimensional integral). *For  $\xi \in \mathbb{R}^n$ ,  $n \geq 2$ , with  $r = |\xi|$ ,*

$$\widehat{d\sigma}(\xi) = \sigma_{n-2} \int_{-1}^1 e^{-2\pi i r t} (1-t^2)^{\frac{n-3}{2}} dt, \quad \sigma_{n-2} = \sigma(\mathbb{S}^{n-2}) = \frac{2\pi^{\frac{n-1}{2}}}{\Gamma(\frac{n-1}{2})}. \quad (2.8.1)$$

*In particular  $\widehat{d\sigma}$  is real, radial, and depends on  $\xi$  only through  $r$ .*

*Proof.* By Worked Example 2.5.3(b),  $\widehat{d\sigma}(\xi) = \int_{\mathbb{S}^{n-1}} e(-\omega \cdot \xi) d\sigma(\omega)$ . Surface measure is rotation invariant — by Theorem 1.3.10 (linear images), rotations preserve Lebesgue measure, hence preserve  $\sigma$  through the polar coordinates construction — so replacing  $\xi$  by  $R\xi$  for a rotation  $R$  and substituting  $\omega \mapsto R^{-1}\omega$  leaves the integral unchanged. Thus  $\widehat{d\sigma}$  is radial and we may take  $\xi = re_n$ .

Then  $\omega \cdot \xi = r\omega_n$ . Parametrise  $\mathbb{S}^{n-1}$  by its last coordinate  $t = \omega_n \in [-1, 1]$ : the slice at height  $t$  is a sphere of radius  $\sqrt{1-t^2}$  in  $\mathbb{R}^{n-1}$ , and the surface

measure factorises as

$$d\sigma = (1 - t^2)^{\frac{n-3}{2}} dt d\sigma_{n-2}(\theta),$$

the exponent  $\frac{n-3}{2}$  being  $\frac{n-2}{2}$  from the area of the slice minus  $\frac{1}{2}$  from the arclength element. Integrating out  $\theta$  gives (2.8.1). Realness follows since the imaginary part of the integrand is odd in  $t$ .  $\square$

**Definition 2.8.2** (Bessel function). For  $\nu > -\frac{1}{2}$  and  $\rho > 0$ ,

$$J_\nu(\rho) = \frac{(\rho/2)^\nu}{\Gamma(\nu + \frac{1}{2})\Gamma(\frac{1}{2})} \int_{-1}^1 e^{i\rho t} (1 - t^2)^{\nu - \frac{1}{2}} dt.$$

**Theorem 2.8.3.** For  $n \geq 2$  and  $\xi \neq 0$ , with  $r = |\xi|$  and  $\nu = \frac{n-2}{2}$ ,

$$\widehat{d\sigma}(\xi) = 2\pi r^{-\frac{n-2}{2}} J_{\frac{n-2}{2}}(2\pi r).$$

*Proof.* With  $\nu = \frac{n-2}{2}$  we have  $\nu - \frac{1}{2} = \frac{n-3}{2}$  and  $\Gamma(\nu + \frac{1}{2}) = \Gamma(\frac{n-1}{2})$ , so the definition with  $\rho = 2\pi r$  reads

$$J_\nu(2\pi r) = \frac{(\pi r)^\nu}{\Gamma(\frac{n-1}{2})\sqrt{\pi}} \int_{-1}^1 e^{2\pi i r t} (1 - t^2)^{\frac{n-3}{2}} dt.$$

The integral is unchanged if  $i$  is replaced by  $-i$ , being real, so it equals the integral in (2.8.1). Solving for that integral and substituting into (2.8.1),

$$\widehat{d\sigma}(\xi) = \frac{2\pi^{\frac{n-1}{2}}}{\Gamma(\frac{n-1}{2})} \cdot \frac{\Gamma(\frac{n-1}{2})\sqrt{\pi}}{(\pi r)^\nu} J_\nu(2\pi r) = 2\pi^{\frac{n}{2}-\nu} r^{-\nu} J_\nu(2\pi r),$$

and  $\frac{n}{2} - \nu = \frac{n}{2} - \frac{n-2}{2} = 1$ .  $\square$

WORKED EXAMPLE 2.8.1— TIER A **The case  $n = 3$ , exactly**

For  $n = 3$ ,  $\frac{n-3}{2} = 0$  and (2.8.1) becomes elementary:  $\sigma_1 = \sigma(\mathbb{S}^1) = 2\pi$ , so

$$\widehat{d\sigma}(\xi) = 2\pi \int_{-1}^1 e^{-2\pi i r t} dt = 2\pi \cdot \frac{e^{2\pi i r} - e^{-2\pi i r}}{2\pi i r} = \frac{2 \sin(2\pi r)}{r}, \quad r = |\xi|.$$

So in  $\mathbb{R}^3$  the transform of surface measure is an elementary function. Two things to notice. Its modulus decays exactly like  $r^{-1} = r^{-(n-1)/2}$ , as the general theorem will assert. And it *oscillates*:  $\widehat{d\sigma}$  changes sign infinitely often and vanishes at every half-integer  $r$ . Volume 6 needs the oscillation, not merely the modulus, and readers should resist the habit of recording only absolute values.



Consistency check with Theorem 2.8.3:  $\nu = \frac{1}{2}$ , and  $J_{1/2}(\rho) = \sqrt{2/(\pi\rho)} \sin \rho$  directly from the definition, so  $2\pi r^{-1/2} J_{1/2}(2\pi r) = 2\pi r^{-1/2} \cdot \frac{1}{\pi\sqrt{r}} \sin(2\pi r) = 2 \sin(2\pi r)/r$ . ✓

## 2.8.2 Van der Corput's lemma

To bound (2.8.1) for general  $n$  we need one tool. It is the elementary core of the method of stationary phase, which Volume 5 develops in full.

**Lemma 2.8.4** (Van der Corput). *Let  $\phi : [a, b] \rightarrow \mathbb{R}$  be  $C^k$  and suppose  $|\phi^{(k)}| \geq 1$  on  $[a, b]$ , where either  $k \geq 2$ , or  $k = 1$  and  $\phi'$  is monotone. Then*

$$\left| \int_a^b e^{i\rho\phi(t)} dt \right| \leq c_k \rho^{-1/k} \quad (\rho > 0),$$

with  $c_k$  depending only on  $k$ . Consequently, for  $\psi$  of bounded variation,

$$\left| \int_a^b e^{i\rho\phi(t)} \psi(t) dt \right| \leq c_k \rho^{-1/k} \left( |\psi(b)| + \int_a^b |\psi'| \right). \quad (2.8.2)$$

*Proof. Step 1.  $k = 1$ .*

Write  $e^{i\rho\phi} = \frac{1}{i\rho\phi'} \frac{d}{dt} e^{i\rho\phi}$  and integrate by parts:

$$\int_a^b e^{i\rho\phi} dt = \left[ \frac{e^{i\rho\phi}}{i\rho\phi'} \right]_a^b + \frac{1}{i\rho} \int_a^b e^{i\rho\phi} \frac{\phi''}{(\phi')^2} dt.$$

The boundary terms are at most  $2/\rho$  since  $|\phi'| \geq 1$ . For the integral, monotonicity of  $\phi'$  means  $\phi''/(\phi')^2$  has constant sign, so  $\int_a^b |\phi''/(\phi')^2| = |\int_a^b \phi''/(\phi')^2| = |1/\phi'(a) - 1/\phi'(b)| \leq 2$ . Hence the whole is at most  $4/\rho$ .

**Step 2.  $k \geq 2$ , by induction.**

Suppose the result holds for  $k-1$ . Since  $|\phi^{(k)}| \geq 1$ , the derivative  $\phi^{(k-1)}$  is strictly monotone, so it vanishes at most once; say at  $c$  (if not, the argument below is simpler). For  $\eta > 0$ , split  $[a, b]$  into  $[a, b] \cap (c - \eta, c + \eta)$ , of length at most  $2\eta$ , and the rest.

Off that interval,  $|\phi^{(k-1)}| \geq \eta$ , because  $\phi^{(k-1)}$  moves at unit rate at least. Applying the inductive hypothesis to  $\phi/\eta$  on each of the (at most two) remaining intervals — for which the  $(k-1)$ -st derivative is at least 1 in modulus — gives a contribution at most  $2c_{k-1}(\rho\eta)^{-1/(k-1)}$ .

Hence

$$\left| \int_a^b e^{i\rho\phi} \right| \leq 2\eta + 2c_{k-1}(\rho\eta)^{-\frac{1}{k-1}}.$$

Choosing  $\eta = \rho^{-1/k}$  balances the two terms and yields  $c_k \rho^{-1/k}$ .

**Step 3. Amplitudes.**

Let  $F(u) = \int_a^u e^{i\rho\phi}$ , so  $|F| \leq c_k \rho^{-1/k}$  by Steps 1–2 applied on  $[a, u]$ . Integrating by parts,

$$\int_a^b e^{i\rho\phi} \psi = F(b)\psi(b) - \int_a^b F\psi',$$

and taking absolute values gives (2.8.2).  $\square$

## 2.8.3 The decay estimate

**Theorem 2.8.5** (Decay of  $\widehat{d\sigma}$ ). *For  $n \geq 2$  there is  $C_n$  with*

$$|\widehat{d\sigma}(\xi)| \leq C_n (1 + |\xi|)^{-\frac{n-1}{2}} \quad \text{for all } \xi \in \mathbb{R}^n.$$

Moreover the exponent cannot be improved: for  $n = 3$ ,  $|\widehat{d\sigma}(\xi)| = 2|\xi|^{-1}$  whenever  $|\xi| \in \frac{1}{4} + \frac{1}{2}\mathbb{Z}$ .

*Idea of the proof.* Substituting  $t = \cos \varphi$  turns (2.8.1) into  $\int_0^\pi e^{-2\pi i r \cos \varphi} (\sin \varphi)^{n-2} d\varphi$ , an oscillatory integral whose phase  $\cos \varphi$  is *stationary* at  $\varphi = 0$  and  $\varphi = \pi$ . Those are the two poles: the points of the sphere whose normal is parallel to  $\xi$ . Everywhere else the phase moves, integration by parts gains as many powers of  $\rho$  as one likes, and the contribution is negligible. All the decay is therefore decided at the poles, and Lemma 2.8.4 with  $k = 2$  identifies the mechanism: a stationary point of a phase with non-degenerate second derivative contributes an angular window of width  $\rho^{-1/2}$  on which no cancellation occurs.

The remaining factor comes from the amplitude. Near a pole the slice of the sphere at angle  $\varphi$  has size  $(\sin \varphi)^{n-2} \approx \varphi^{n-2}$ , so the mass sitting in the window  $\varphi \lesssim \rho^{-1/2}$  is  $\int_0^{\rho^{-1/2}} \varphi^{n-2} d\varphi \approx \rho^{-(n-1)/2}$ . That is the answer, and the proof below is the verification that nothing else is larger.

We give the argument in the original variable  $t$ , where the two poles become the two endpoints  $t = \pm 1$  and the analysis is a single clean computation valid for every  $n \geq 2$  at once.

*Proof.* Boundedness for  $|\xi| \leq 1$  is Worked Example 2.5.3(b), so assume  $r = |\xi| \geq 1$  and write  $\rho = 2\pi r$  and  $\alpha = \frac{n-3}{2}$ , so that  $\alpha > -1$  for every  $n \geq 2$  and (2.8.1) reads

$$\widehat{d\sigma}(\xi) = \sigma_{n-2} I(\rho), \quad I(\rho) = \int_{-1}^1 e^{-i\rho t} (1 - t^2)^\alpha dt.$$

We show

$$|I(\rho)| \lesssim_{\alpha} \rho^{-\alpha-1} = \rho^{-\frac{n-1}{2}}. \quad (2.8.3)$$

**Step 1.** *Partition of unity.*

Choose  $\chi \in C_c^\infty(\mathbb{R})$  with  $\chi = 1$  on  $[-\frac{1}{4}, \frac{1}{4}]$  and  $\text{supp } \chi \subseteq (-\frac{1}{2}, \frac{1}{2})$ , and write

$$1 = \chi(1-t) + \chi(1+t) + \chi_0(t), \quad t \in [-1, 1],$$

which defines  $\chi_0$ ; it is smooth and supported in  $[-\frac{3}{4}, \frac{3}{4}]$ . This splits  $I$  into a piece near each endpoint and an interior piece.

**Step 2.** *The interior piece is negligible.*

On  $\text{supp } \chi_0$  the amplitude  $(1-t^2)^\alpha \chi_0(t)$  is  $C^\infty$  with compact support in  $(-1, 1)$ , and the phase is  $-t$ , with derivative  $-1$ . Integrating by parts  $N$  times, all boundary terms vanishing by compact support,

$$\left| \int_{-1}^1 e^{-i\rho t} (1-t^2)^\alpha \chi_0(t) dt \right| = \rho^{-N} \left| \int_{-1}^1 e^{-i\rho t} \frac{d^N}{dt^N} [(1-t^2)^\alpha \chi_0(t)] dt \right| \lesssim_{\alpha, N} \rho^{-N}.$$

Choose  $N \geq \alpha + 1$ ; this piece is  $O(\rho^{-\alpha-1})$ .

**Step 3.** *An endpoint piece.*

By symmetry it suffices to treat  $t$  near 1. Substitute  $t = 1 - s$ :

$$\int_{-1}^1 e^{-i\rho t} (1-t^2)^\alpha \chi(1-t) dt = e^{-i\rho} \int_0^\infty e^{i\rho s} s^\alpha g(s) ds, \quad g(s) = (2-s)^\alpha \chi(s),$$

and  $g \in C_c^\infty([0, \frac{1}{2}))$ , so all its derivatives are bounded and it vanishes identically for  $s \geq \frac{1}{2}$ . Put  $K(\rho) = \int_0^\infty e^{i\rho s} s^\alpha g(s) ds$ ; we must show  $|K(\rho)| \lesssim_{\alpha} \rho^{-\alpha-1}$ .

Split at  $s = \rho^{-1}$ .

*Small  $s$ .* Bounding the integrand by its modulus,

$$\left| \int_0^{1/\rho} e^{i\rho s} s^\alpha g(s) ds \right| \leq \|g\|_\infty \int_0^{1/\rho} s^\alpha ds = \frac{\|g\|_\infty}{\alpha+1} \rho^{-\alpha-1},$$

the integral converging precisely because  $\alpha > -1$ . This is the term that produces the exponent; it is the endpoint window of width  $\rho^{-1}$  in  $s$ , which is the window of width  $\rho^{-1/2}$  in the angular variable  $\varphi$ , since  $s = 1 - \cos \varphi \approx \varphi^2/2$ .

*Large  $s$ .* Let  $M$  be the least integer with  $M > \alpha + 1$ , and integrate by parts  $M$  times on  $[\rho^{-1}, \infty)$ , writing  $e^{i\rho s} = \frac{1}{i\rho} \frac{d}{ds} e^{i\rho s}$  each time. Since  $g$  vanishes near  $\infty$ , the only boundary terms occur at  $s = \rho^{-1}$ , and the  $m$ -th one is

$$\lesssim \rho^{-m} \left| \frac{d^{m-1}}{ds^{m-1}} (s^\alpha g) \right|_{s=1/\rho} \lesssim_{\alpha} \rho^{-m} \cdot \rho^{-(\alpha-m+1)} = \rho^{-\alpha-1}, \quad 1 \leq m \leq M,$$

using  $|\frac{d^k}{ds^k}(s^\alpha g)| \lesssim_{\alpha,k} s^{\alpha-k}$  for  $0 < s < \frac{1}{2}$ , which follows from the Leibniz rule. The remaining integral is

$$\rho^{-M} \int_{1/\rho}^{1/2} \left| \frac{d^M}{ds^M}(s^\alpha g) \right| ds \lesssim_\alpha \rho^{-M} \int_{1/\rho}^{1/2} s^{\alpha-M} ds \lesssim_\alpha \rho^{-M} \cdot \rho^{M-\alpha-1} = \rho^{-\alpha-1},$$

where the  $s$ -integral was evaluated using  $\alpha - M < -1$ , so that it is dominated by its lower limit.

**Step 4. Collect.**

Steps 2 and 3 give (2.8.3), and  $\rho = 2\pi r \approx |\xi|$ , proving the stated bound for  $|\xi| \geq 1$ .

**Step 5. Sharpness.**

By Worked Example 2.8.1, for  $n = 3$  we have  $\widehat{d\sigma}(\xi) = 2 \sin(2\pi|\xi|)/|\xi|$ , whose modulus equals  $2|\xi|^{-1} = 2|\xi|^{-(n-1)/2}$  exactly when  $\sin(2\pi|\xi|) = \pm 1$ , that is when  $|\xi| \in \frac{1}{4} + \frac{1}{2}\mathbb{Z}$ . So the rate is attained along a sequence tending to infinity and cannot be improved.  $\square$

*Remark 2.8.6* (Two readings of the same exponent). The proof above is a computation in the variable  $t$ ; the Idea box described the geometry in the variable  $\varphi$ . They are the same argument. In  $\varphi$ , the poles are stationary points of the phase and Lemma 2.8.4 with  $k = 2$  supplies the window of width  $\rho^{-1/2}$ ; in  $t$ , the poles are the endpoints  $t = \pm 1$  where the amplitude  $(1 - t^2)^\alpha$  degenerates, and the window has width  $\rho^{-1}$ . The substitution  $s = 1 - \cos \varphi$  converts one into the other.

The  $t$ -side computation is shorter and works uniformly in  $n$ , which is why we gave it. The  $\varphi$ -side reading is the one to remember, because it is the one that generalises: for a hypersurface with  $k$  non-vanishing principal curvatures, the stationary set for a given  $\xi$  is a  $(n - 1 - k)$ -dimensional submanifold and the window has width  $\rho^{-1/2}$  in each of the  $k$  curved directions, giving  $\rho^{-k/2}$ . That is Remark 2.8.7.

*Remark 2.8.7* (What is deferred, and why). Two refinements are not proved here.

First, the sharp *asymptotic*  $J_\nu(\rho) = \sqrt{2/(\pi\rho)} \cos(\rho - \frac{\nu\pi}{2} - \frac{\pi}{4}) + O(\rho^{-3/2})$ , which records the oscillation and not merely the modulus. Volume 6 needs it, and Volume 5 proves it by the method of stationary phase, of which Lemma 2.8.4 is the crudest case. Sharpness of the exponent for general  $n$  follows from it; we have proved sharpness only for  $n = 3$ , where the elementary formula of Worked Example 2.8.1 is available.

Second, the general principle that a hypersurface with  $k$  non-vanishing principal curvatures has  $|\widehat{d\sigma}(\xi)| \lesssim |\xi|^{-k/2}$ . The sphere is the case  $k = n - 1$  and is done in full above; the general statement needs the Morse lemma and is Volume 5.

Both are S7 deferrals in the sense of the series conventions: they require machinery introduced in a later volume, they are labelled here, and nothing in Volumes 1–4 uses them.

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WORKED EXAMPLE 2.8.2— TIER B **A flat piece kills all decay**

Let  $\Sigma = \{(y, 0) : |y| \leq 1\} \subseteq \mathbb{R}^n$  be a flat disc in the hyperplane  $x_n = 0$ , with  $d\mu$  its  $(n-1)$ -dimensional Lebesgue measure. Then

$$\widehat{d\mu}(\xi) = \int_{|y| \leq 1} e(-y \cdot \xi') dy,$$

which does not involve  $\xi_n$  at all. So  $\widehat{d\mu}(0, \xi_n) = \omega_{n-1}$  for every  $\xi_n$ : there is no decay whatsoever in the normal direction.

Compare Theorem 2.8.5. The difference is one word: on the sphere the normal direction *turns*, so the set of points contributing a stationary phase for a given  $\xi$  is confined to two poles and an angular window of width  $|\xi|^{-1/2}$ ; on the disc every point is a pole simultaneously. Curvature is precisely the rate at which the normal turns, and the decay exponent counts how many directions it turns in.

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WORKED EXAMPLE 2.8.3— TIER C: RESEARCH TASTE **A restriction theorem, from decay alone**

We close the volume by proving a genuine restriction estimate. Recall from Worked Example 2.4.3 the extension operator  $Eg = \widehat{gd\sigma}$ , and from Worked Example 2.5.3 that it is well defined.

**Theorem 2.8.8.** *For  $n \geq 2$  and  $q = \frac{4n}{n-1}$ ,*

$$\|Eg\|_{L^q(\mathbb{R}^n)} \lesssim_n \|g\|_{L^2(d\sigma)}.$$

*Proof.* The adjoint of  $E : L^2(d\sigma) \rightarrow L^q(\mathbb{R}^n)$  is  $E^*f = \widehat{f}|_{\mathbb{S}^{n-1}}$ , and the composition is

$$EE^*f = f * \widehat{d\sigma},$$

a computation from the definitions:  $(\widehat{f}|_{\mathbb{S}^{n-1}})d\sigma$  pairs against  $\varphi$  exactly as  $f * \widehat{d\sigma}$  does.

The  $TT^*$  identity says  $\|E\|_{L^2(d\sigma) \rightarrow L^q}^2 = \|EE^*\|_{L^{q'} \rightarrow L^q}$ ; indeed

$$\|E^*f\|_{L^2(d\sigma)}^2 = \langle EE^*f, f \rangle \leq \|EE^*f\|_q \|f\|_{q'},$$

so a bound on  $EE^*$  gives one on  $E^*$ , hence on  $E$  by duality (Volume 1, Corollary 8.12).

By Theorem 2.8.5,  $|\widehat{d\sigma}(x)| \lesssim (1 + |x|)^{-\lambda} \leq |x|^{-\lambda}$  with  $\lambda = \frac{n-1}{2}$ , and  $0 < \lambda < n$ . So by Hardy–Littlewood–Sobolev (Theorem 2.7.7),

$$\|f * \widehat{d\sigma}\|_q \lesssim \|f * |\cdot|^{-\lambda}\|_q \lesssim \|f\|_{q'} \quad \text{provided} \quad \frac{1}{q} = \frac{1}{q'} + \frac{\lambda}{n} - 1.$$

Since  $\frac{1}{q'} = 1 - \frac{1}{q}$ , that condition reads  $\frac{2}{q} = \frac{\lambda}{n}$ , i.e.

$$q = \frac{2n}{\lambda} = \frac{4n}{n-1}. \quad \square$$

(a) *Reading the result.* We have converted a statement about the *geometry* of the sphere — it is curved, so its surface measure has a Fourier transform decaying at rate  $(n-1)/2$  — into a statement about *integrability* of  $Eg$ , with no further input. That is the whole strategy of the subject in one page.

(b) *It is not sharp, and the gap is instructive.* The Stein–Tomas theorem, proved in Volume 6, gives the same conclusion for the smaller exponent

$$q = \frac{2(n+1)}{n-1}.$$

For  $n = 3$ : we have proved  $q = 6$ , and Stein–Tomas gives  $q = 4$ . Smaller  $q$  is a stronger statement, since the estimate for one  $q$  implies it for all larger  $q$  after interpolation with the trivial bound.

The improvement comes from using information we discarded. Our proof used only  $|\widehat{d\sigma}| \lesssim |x|^{-\lambda}$ , which is  $L^1 \rightarrow L^\infty$  type information about the kernel. But  $d\sigma$  is also a *bounded* multiplier in a suitable dyadic decomposition, giving  $L^2 \rightarrow L^2$  information by Plancherel; interpolating the two, dyadic piece by dyadic piece, is exactly the Stein–Tomas argument. Discarding the  $L^2$  information cost us the difference between  $\frac{4n}{n-1}$  and  $\frac{2(n+1)}{n-1}$ .

(c) *And Stein–Tomas is not the conjecture.* The restriction conjecture, as located by the Knapp example in Worked Example 2.4.3, asserts  $\|Eg\|_{L^q} \lesssim \|g\|_{L^\infty(d\sigma)}$  for all  $q > \frac{2n}{n-1}$ . For  $n = 3$  that is  $q > 3$ , against Stein–Tomas’s  $q \geq 4$ . The gap between 4 and 3 in  $\mathbb{R}^3$  has been open since 1967, and closing it is the subject of Volume 12.

*Where you now stand.* You have proved a restriction theorem from first principles: measure theory in Volume 1, complex analysis and interpolation in Chapters 1–2, distributions in Chapter 5, fractional integration in Chapter 7, and one Bessel computation here. Two further ideas — the  $L^2$  orthogonality that sharpens Step (b), and the tube geometry that would sharpen (c) — separate you from the frontier.

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**WORKED EXAMPLE 2.8.4— TIER B Reading a decay estimate**

Theorem 2.8.5 gives  $|\widehat{d\sigma}(\xi)| \lesssim |\xi|^{-(n-1)/2}$  for the sphere. Four questions to ask of any such statement.

(a) *What produces the exponent?* The number of non-vanishing principal curvatures, halved. Lemma 2.8.4 converts each curved direction into a factor  $|\xi|^{-1/2}$ , and the sphere has  $n - 1$  of them.

(b) *Is it sharp?* Yes, and Volume 5 makes the sharpness visible: the asymptotic there shows  $\widehat{d\sigma}$  oscillates with amplitude exactly  $|\xi|^{-(n-1)/2}$ , so no better pointwise bound holds. That sharpness is what makes the Fourier method saturate in Volumes 7 and 9.

(c) *Is the pointwise bound what is needed?* Often not. Volume 8's reduction of the distance problem required the *oscillation*, not merely the decay — the phase  $e^{\pm 2\pi i|\xi|}$  turns an integral into a Fourier transform, and Plancherel then applies. Discarding the phase loses that and the estimate fails.

(d) *What replaces it when the surface is flat?* Nothing: a hyperplane's surface measure has a transform that does not decay in the normal direction at all. Volume 6 shows the consequence for restriction, and Volume 7 states it as  $\dim_F = 0$ .

*The summary.* An estimate on  $\widehat{d\sigma}$  is a statement about curvature; its exponent counts curved directions; and whether the *phase* is needed depends on whether the application squares the quantity.

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## Drills

**Drill 2.8.1.** Compute  $\widehat{d\sigma}$  for  $n = 2$  directly from (2.8.1) and identify it as  $2\pi J_0(2\pi|\xi|)$ . Verify numerically that  $|\widehat{d\sigma}(\xi)| \approx |\xi|^{-1/2}$  for large  $|\xi|$ .

**Drill 2.8.2.** Prove  $J_{1/2}(\rho) = \sqrt{2/(\pi\rho)} \sin \rho$  directly from the definition, and use it to re-derive Worked Example 2.8.1.

**Drill 2.8.3.** Show that the constant  $c_1 = 4$  in Lemma 2.8.4 is attainable up to a factor, and that the monotonicity hypothesis on  $\phi'$  cannot be dropped when  $k = 1$ .

**Drill 2.8.4.** Carry out the interpolation referred to in Worked Example 2.8.3(b): assuming Stein–Tomas at  $q = \frac{2(n+1)}{n-1}$ , deduce the estimate for every larger  $q$ .

## Notes

Theorem 2.8.3 is classical and can be found in Stein–Weiss. Lemma 2.8.4 is van der Corput’s, from 1921; the inductive proof given here is the standard one and appears in Stein’s *Harmonic Analysis*, Chapter VIII.

Theorem 2.8.8 is folklore — it is the estimate one obtains by using only the decay of  $\widehat{d\sigma}$  — and is included because the gap between it and Stein–Tomas isolates precisely what the more sophisticated argument buys. The Stein–Tomas theorem is due to Tomas (1975) with the endpoint by Stein; both are proved in Volume 6.



## Chapter 2.A

# Results Quoted from Volume 1

This volume uses Volume 1 heavily and assumes nothing else. The list below is complete: every result of Volume 1 invoked anywhere in Volume 2 appears here, with the place it is used. No proofs are given — they are in Volume 1 — and the purpose of the appendix is to let a reader verify at a glance that nothing is being assumed silently.

## Measure and integration

| Result                                                      | Volume 1        | Used in                                     |
|-------------------------------------------------------------|-----------------|---------------------------------------------|
| Box lemma, $m^*(Q) =  Q $                                   | Thm 3.3         | throughout                                  |
| Structure of measurable sets;<br>outer and inner regularity | Thm 3.5         | density arguments                           |
| Linear images, $m(TE) =  \det T m(E)$                       | Thm 3.11        | Proposition 2.4.2(iv),<br>Proposition 2.8.1 |
| Simple approximation                                        | Thm 4.6         | Chapter 2.3, Theorem 2.2.1                  |
| Monotone convergence                                        | Thm 5.1         | Theorem 2.4.8, Theorem 2.7.7                |
| Fatou                                                       | Thm 5.6         | Worked Example 2.4.2                        |
| Dominated convergence                                       | Thm 5.8         | Theorem 2.4.6, Proposition 2.4.2            |
| Differentiation under the integral                          | Wkd Ex 5.13     | Proposition 2.4.2(vi)                       |
| Continuity and decay of $\widehat{f}$ for $f \in L^1$       | Wkd Ex 5.14     | Chapter 2.4 §4.1                            |
| Layer-cake formula                                          | Wkd Ex 5.5      | Theorem 2.2.4, Theorem 2.6.5                |
| Egorov; Lusin                                               | Thms 5.18, 5.20 | density arguments                           |
| Tonelli; Fubini                                             | Thms 6.6, 6.7   | used constantly                             |
| Young's inequality, the case $q = 1$                        | Prop 6.9        | Worked Example 2.2.1                        |
| Approximate identities                                      | Prop 6.10       | Theorem 2.4.6, Theorem 2.4.8                |
| Polar coordinates and surface measure                       | Thm 6.10        | Chapter 2.8, Theorem 2.7.5                  |

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## $L^p$ theory

| Result                                          | Volume 1          | Used in                                            |
|-------------------------------------------------|-------------------|----------------------------------------------------|
| Hölder; generalised Hölder                      | Thms 8.2, Cor 8.3 | Theorem 2.2.1, Theorem 2.7.6                       |
| Minkowski                                       | Thm 8.5           | Theorem 2.3.6                                      |
| Riesz–Fischer completeness                      | Thm 8.6           | Theorem 2.4.8 Step 2, Theorem 2.2.1 Step 5         |
| Density of simple functions and of $C_c^\infty$ | Thm 8.9           | Theorem 2.2.1, Theorem 2.4.8, Worked Example 2.4.2 |
| Continuity of translation in $L^p$              | Thm 8.9(iii)      | Theorem 2.4.8                                      |
| Duality $(L^p)^* \cong L^{p'}$                  | Thm 8.11          | Theorem 2.2.1 Step 5                               |
| Norms by testing                                | Cor 8.12          | Theorem 2.2.1 Step 4, Worked Example 2.8.3         |
| Weak $L^p$ ; Chebyshev                          | Prop 8.14         | Theorem 2.2.4                                      |
| Inclusions on finite measure spaces             | Drill 8.1         | Chapter 2.3, Theorem 2.6.5 Step 4                  |
| Log-convexity of $L^p$ norms                    | Drill 8.2         | Theorem 2.6.5 Step 5                               |
| Hilbert space; orthonormal bases                | Thms 8.18–8.20    | Theorem 2.3.8                                      |

## Harmonic analysis and Kakeya

| Result                                          | Volume 1              | Used in                             |
|-------------------------------------------------|-----------------------|-------------------------------------|
| Hardy–Littlewood maximal function; weak $(1,1)$ | Thm 7.4               | Worked Example 2.2.2, Theorem 2.7.7 |
| Lebesgue differentiation                        | Thm 7.7               | background                          |
| Hausdorff measure and dimension                 | Wkd Ex 2.16           | Worked Example 2.7.1                |
| Tube intersection estimate                      | Lem 6.11              | Worked Example 2.2.3(c)             |
| Córdoba: Kakeya in the plane                    | Thm 6.12              | background                          |
| Perron tree                                     | Thm 3.23              | background                          |
| Kakeya maximal operator; the conjecture         | Def 7.17, Wkd Ex 7.20 | Worked Example 2.2.3                |
| Almost orthogonality; the decoupling statement  | Wkd Ex 8.16           | Worked Example 2.6.2                |

## Background

Urysohn's lemma (Thm A.2) is used in Lusin-type density arguments and in Proposition 2.5.5. Tietze extension (Thm A.3) and the Hahn decomposition (Thm A.4) are used in Volume 1 only, and are recorded there.

The three-lines lemma, stated without proof in Volume 1's Appendix A, is proved in this volume as Theorem 2.1.14. That was the only outstanding debt from Volume 1, and it is now discharged.

## Chapter 2.B

# Notation and Index of Results

### Additions to the Volume 1 conventions

|                                                          |                                                      |
|----------------------------------------------------------|------------------------------------------------------|
| $\widehat{f}(\xi) = \int f(x)e^{-2\pi i x \cdot \xi} dx$ | the Fourier transform                                |
| $\mathcal{F}, \mathcal{F}^{-1}$                          | the transform and its inverse as operators           |
| $e(t)$                                                   | shorthand for $e^{2\pi i t}$                         |
| $\mathbb{T}^n = \mathbb{R}^n/\mathbb{Z}^n$               | the torus; functions are 1-periodic                  |
| $D_N, K_N$                                               | Dirichlet and Fejér kernels                          |
| $\mathbb{S}_{01}$                                        | the closed strip $0 \leq \operatorname{Re} z \leq 1$ |
| $\mathcal{S}(\mathbb{R}^n), \mathcal{S}'(\mathbb{R}^n)$  | Schwartz functions; tempered distributions           |
| $\rho_{\alpha\beta}$                                     | the Schwartz seminorms                               |
| $\langle u, \varphi \rangle$                             | the pairing of a distribution with a test function   |
| $d\sigma$                                                | surface measure on $\mathbb{S}^{n-1}$                |
| $Eg = \widehat{g d\sigma}$                               | the extension operator                               |
| $J_\nu$                                                  | the Bessel function of order $\nu$                   |
| $H^s, W^{k,p}, p^*$                                      | Sobolev spaces and the Sobolev exponent              |
| $P_N, P_{\leq N}$                                        | Littlewood–Paley projections                         |
| $m, T_m$                                                 | a Fourier multiplier and its operator                |
| $r_j$                                                    | the Rademacher functions                             |
| $I_s(\mu)$                                               | the $s$ -energy of a measure                         |

The transform is normalised with  $2\pi$  in the exponent, so that  $\mathcal{F}$  is unitary on  $L^2(\mathbb{R}^n)$  with no constant,  $\widehat{f * g} = \widehat{f} \widehat{g}$  with no constant, and  $\widehat{\partial_j f}(\xi) = 2\pi i \xi_j \widehat{f}(\xi)$ . This is the convention of the restriction and decoupling literature.

## Index of the main results of Volume 2

### *Chapter 1 — complex analysis*

|                                     |                  |
|-------------------------------------|------------------|
| Goursat's theorem                   | Theorem 2.1.5    |
| Cauchy's theorem on a convex domain | Theorem 2.1.6    |
| Cauchy integral formula             | Theorem 2.1.9    |
| Holomorphic implies analytic        | Theorem 2.1.10   |
| Cauchy estimates; Liouville         | Corollary 2.1.11 |
| Maximum modulus principle           | Theorem 2.1.12   |
| Phragmén–Lindelöf on a strip        | Theorem 2.1.13   |
| Three-lines lemma                   | Theorem 2.1.14   |

### *Chapter 2 — interpolation*

|                                |                      |
|--------------------------------|----------------------|
| Riesz–Thorin                   | Theorem 2.2.1        |
| Marcinkiewicz                  | Theorem 2.2.4        |
| Young's convolution inequality | Worked Example 2.2.1 |

### *Chapter 3 — Fourier series*

|                                   |                 |
|-----------------------------------|-----------------|
| Closed forms for $D_N$ and $K_N$  | Lemma 2.3.4     |
| Fejér's theorem                   | Theorem 2.3.6   |
| Uniqueness and density            | Corollary 2.3.7 |
| Parseval                          | Theorem 2.3.8   |
| Baire category                    | Theorem 2.3.9   |
| Uniform boundedness               | Theorem 2.3.10  |
| du Bois-Reymond divergence        | Theorem 2.3.11  |
| Hausdorff–Young on $\mathbb{T}^n$ | Theorem 2.3.12  |

### *Chapter 4 — the Fourier transform*

|                                   |                      |
|-----------------------------------|----------------------|
| The dictionary                    | Proposition 2.4.2    |
| The Gaussian is its own transform | Proposition 2.4.3    |
| Multiplication formula            | Lemma 2.4.5          |
| Fourier inversion                 | Theorem 2.4.6        |
| Plancherel                        | Theorem 2.4.8        |
| Hausdorff–Young on $\mathbb{R}^n$ | Theorem 2.4.9        |
| Heisenberg uncertainty            | Worked Example 2.4.2 |

### *Chapter 5 — distributions*

|                                                   |                      |
|---------------------------------------------------|----------------------|
| $\mathcal{F}$ is an isomorphism of $\mathcal{S}$  | Theorem 2.5.6        |
| $\mathcal{F}$ is an isomorphism of $\mathcal{S}'$ | Theorem 2.5.11       |
| Transform of $ x ^{-a}$                           | Worked Example 2.5.2 |

### *Chapter 6 — multipliers*

|                                        |                   |
|----------------------------------------|-------------------|
| $L^2$ multiplier norm                  | Proposition 2.6.2 |
| Khintchine's inequality                | Theorem 2.6.5     |
| Bernstein's inequalities               | Theorem 2.6.7     |
| $L^2$ orthogonality of lacunary pieces | Proposition 2.6.8 |

### *Chapter 7 — Sobolev spaces*

|                                 |                      |
|---------------------------------|----------------------|
| $\mathcal{W}^{k,2} = W^{k,2}$   | Proposition 2.7.4    |
| Sobolev embedding, $L^2$ case   | Theorem 2.7.5        |
| Gagliardo–Nirenberg–Sobolev     | Theorem 2.7.6        |
| Hardy–Littlewood–Sobolev        | Theorem 2.7.7        |
| Fourier integrals and dimension | Worked Example 2.7.1 |

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# Deferrals

Four results are stated in this volume and proved elsewhere. Each is labelled at the point of statement, and none is used in Volumes 1–2.

| Result                                                   | Stated                  | Proved                            |
|----------------------------------------------------------|-------------------------|-----------------------------------|
| Hörmander–Mikhlin multiplier theorem                     | Remark 2.6.3            | Volume 3 (needs Calderón–Zygmund) |
| Littlewood–Paley square function, $p \neq 2$             | §6.3                    | Volume 3                          |
| Fefferman’s ball multiplier theorem                      | §6.1                    | Volume 5                          |
| Asymptotics of $J_\nu$ ; decay for general hypersurfaces | Remark 2.8.7            | Volume 5 (stationary phase)       |
| Stein–Tomas theorem                                      | Worked Example 2.8.3(b) | Volume 6                          |





## Part 3

# Volume 3

Maximal Functions and Singular Integrals



# Preface to Volume 3

## Where we are

Volume 2 ended by proving a restriction theorem: from the decay  $|\widehat{d\sigma}(\xi)| \lesssim |\xi|^{-(n-1)/2}$  and nothing else, the extension operator maps  $L^2(d\sigma)$  into  $L^q(\mathbb{R}^n)$  for  $q = \frac{4n}{n-1}$ . It also left two debts. The Hörmander–Mikhlin multiplier theorem and the Littlewood–Paley square function theorem for  $p \neq 2$  were both stated and deferred, on the grounds that they require Calderón–Zygmund theory. This volume builds that theory and discharges both debts.

## The one idea

Almost every operator in harmonic analysis fails to be bounded on  $L^1$ , and almost every one of them satisfies a weak  $(1, 1)$  bound instead. Volume 1 saw this for the Hardy–Littlewood maximal function, where the obstruction was that  $\mathcal{M}f$  decays like  $|x|^{-n}$  at infinity. The same phenomenon recurs for the Hilbert transform, the Riesz transforms, and every singular integral, and the reason is always the same: these operators are built from kernels of size  $|x|^{-n}$ , which are exactly not integrable.

Calderón and Zygmund found the mechanism that handles all of them at once. One decomposes an arbitrary  $L^1$  function, at a height  $\lambda$ , into a bounded part and a sum of pieces each supported in a cube and each of *mean zero*. The bounded part is handled by an  $L^2$  estimate; the mean-zero pieces are handled by the smoothness of the kernel, since a kernel that varies slowly cannot distinguish a mean-zero bump from nothing. The whole of Chapters 1 and 2 is that sentence, made precise.

## What is assumed

Volumes 1 and 2 in full, and nothing else. In particular we use freely: the Lebesgue theory and the  $L^p$  spaces with their duality; the Hardy–Littlewood maximal theorem; the Fourier transform on  $L^1$ ,  $L^2$  and  $\mathcal{S}'$ ; Plancherel; the

Riesz–Thorin and Marcinkiewicz interpolation theorems; Khintchine’s inequality; and Bernstein’s inequalities. Appendix A lists every such citation with its use site.

## The single idea

If the volume has one idea it is this: *an operator whose kernel is singular at a point, and which is bounded on  $L^2$ , is bounded on every  $L^p$  with  $1 < p < \infty$* . That is Theorem 3.2.3, and everything before it is preparation and everything after is consequence.

The preparation is the Calderón–Zygmund decomposition of Chapter 3.1, a stopping-time argument (Worked Example 3.1.5) which splits an  $L^1$  function into a bounded part and a sum of mean-zero pieces supported on disjoint cubes. The two parts are then handled by the two hypotheses — the bounded part by  $L^2$ , the mean-zero part by the kernel — and that division is the whole architecture.

The consequences occupy Chapter 3.3 to Chapter 3.7: the Hilbert and Riesz transforms, the Hörmander–Mikhlin multiplier theorem, the Littlewood–Paley square function, and the endpoint behaviour at  $L^\infty$ , which is BMO rather than  $L^\infty$  itself.

Chapter 3.8 then shows the theory’s boundary. The ball multiplier has a bounded symbol, so the  $L^2$  hypothesis holds; but its singularity lies along a *sphere* rather than at a point, no Hörmander condition is available, and the conclusion is false. What replaces the theory when singularities live on curved surfaces is the subject of Volume 5 onwards, and the counterexample itself is geometric — it needs the Besicovitch configuration, proved in Volume 9.

## The plan

|   |                                         |                                             |
|---|-----------------------------------------|---------------------------------------------|
| 1 | Dyadic cubes and the CZ decomposition   | the machine                                 |
| 2 | The Calderón–Zygmund theorem            | weak $(1, 1)$ from $L^2$ plus smoothness    |
| 3 | The Hilbert transform                   | the model case, and Fourier series in $L^p$ |
| 4 | Riesz transforms and elliptic estimates | the $n$ -dimensional model                  |
| 5 | The Hörmander–Mikhlin theorem           | debt from Volume 2, discharged              |
| 6 | Littlewood–Paley theory                 | debt from Volume 2, discharged              |
| 7 | BMO and John–Nirenberg                  | the correct endpoint at $p = \infty$        |
| 8 | The ball multiplier                     | Keakeya enters Fourier analysis             |

Chapter 3.8 is the destination, and it is a change of plan worth announcing. Volume 2’s table of deferrals promised Fefferman’s ball multiplier theorem for Volume 5, on the assumption that it needed the stationary phase machinery. It does not: given the Perron tree of Volume 1, Khintchine’s inequality of Volume

2, and the Hilbert transform of Chapter 3.3, the proof is self-contained, and it is much better placed here — immediately after the theory it refutes. So it has been moved forward. This is the first time in the series that a Besicovitch set is used to prove a theorem in Fourier analysis, and it is the historical origin of everything in Volumes 9 through 13.



## Chapter 3.1

# Dyadic Cubes and the Calderón–Zygmund Decomposition

### 3.1.1 Why start again with maximal functions

Volume 1 proved the Hardy–Littlewood maximal inequality using the Vitali covering lemma, and paid the constant  $3^n$  for it. That proof is geometric and the constant is an artefact of tripling balls.

There is a second proof, due to the same circle of ideas, which replaces balls by *dyadic cubes* and needs no covering lemma at all — the constant becomes 1. It looks at first like a minor economy. It is not. The dyadic structure has a combinatorial property that balls lack — any two dyadic cubes are nested or disjoint — and that property is what makes the Calderón–Zygmund decomposition of §1.4 possible. Every singular integral estimate in this volume, and the Littlewood–Paley theory of Chapter 3.6, ultimately rests on it.

### 3.1.2 The dyadic grid

**Definition 3.1.1.** For  $k \in \mathbb{Z}$  let

$$\mathcal{D}_k = \left\{ 2^{-k}(m + [0, 1)^n) : m \in \mathbb{Z}^n \right\},$$

the *dyadic cubes of generation  $k$* , each of side  $2^{-k}$  and volume  $2^{-kn}$ . Put  $\mathcal{D} = \bigcup_{k \in \mathbb{Z}} \mathcal{D}_k$ .

**Lemma 3.1.2** (The nesting property). *Let  $Q \in \mathcal{D}_k$  and  $Q' \in \mathcal{D}_{k'}$  with  $k \leq k'$ . Then either  $Q' \subseteq Q$  or  $Q \cap Q' = \emptyset$ . Moreover each  $Q \in \mathcal{D}_k$  is the disjoint*

union of the  $2^n$  cubes of  $\mathcal{D}_{k+1}$  contained in it (its children), and is contained in a unique  $\widehat{Q} \in \mathcal{D}_{k-1}$  (its parent), with  $|\widehat{Q}| = 2^n|Q|$ .

*Proof.* It suffices to treat  $n = 1$  and take products. Let  $I = 2^{-k}[m, m+1)$  and  $I' = 2^{-k'}[m', m'+1)$  with  $k \leq k'$ . Write  $j = k' - k \geq 0$ . Then  $I = 2^{-k'}[2^j m, 2^j(m+1))$ , so  $I$  is the union of the  $2^j$  consecutive generation- $k'$  intervals with left endpoints  $2^{-k'}\ell$  for  $2^j m \leq \ell < 2^j(m+1)$ . Either  $m'$  is one of these  $\ell$ , in which case  $I' \subseteq I$ ; or it is not, in which case  $I'$  lies entirely to the left or entirely to the right of  $I$ , and the two are disjoint. The remaining assertions are immediate.  $\square$

### Pitfall: dyadic cubes are half-open on purpose

The convention  $[0, 1)^n$  rather than  $[0, 1]^n$  is what makes Lemma 3.1.2 an exact dichotomy: closed cubes of the same generation share faces, and the disjointness in the definition of “children” would fail. Since faces are null sets (Worked Example 1.2.3 (half-spaces are measurable)), nothing measure-theoretic is affected, and every argument below is cleaner for it.

**Definition 3.1.3.** The *dyadic maximal function* of  $f \in L^1_{\text{loc}}(\mathbb{R}^n)$  is

$$\mathcal{M}_d f(x) = \sup_{\substack{Q \in \mathcal{D} \\ x \in Q}} \langle |f| \rangle_Q, \quad \langle |f| \rangle_Q := \frac{1}{|Q|} \int_Q |f|.$$

**Theorem 3.1.4** (Dyadic maximal inequality, constant 1). *For  $f \in L^1(\mathbb{R}^n)$  and  $\lambda > 0$ ,*

$$m(\{\mathcal{M}_d f > \lambda\}) \leq \frac{1}{\lambda} \int_{\{\mathcal{M}_d f > \lambda\}} |f| \leq \frac{\|f\|_1}{\lambda}.$$

*Idea of the proof.* Because dyadic cubes nest, the collection of cubes on which the average exceeds  $\lambda$  has *maximal* elements, and distinct maximal elements are disjoint — there is nothing to select and nothing to enlarge. The Vitali lemma existed only to manufacture disjointness that the dyadic grid supplies for free, and the factor  $3^n$  was the price of that manufacture.

*Proof.* Let  $\mathcal{Q}_\lambda = \{Q \in \mathcal{D} : \langle |f| \rangle_Q > \lambda\}$ . By definition  $\{\mathcal{M}_d f > \lambda\} = \bigcup_{Q \in \mathcal{Q}_\lambda} Q$ .

Each  $Q \in \mathcal{Q}_\lambda$  satisfies  $|Q| < \|f\|_1/\lambda$ , so the generations occurring in  $\mathcal{Q}_\lambda$  are bounded below; hence every  $Q \in \mathcal{Q}_\lambda$  is contained in a *maximal* element of  $\mathcal{Q}_\lambda$  (ascend from  $Q$  through its parents; the ascent must stop, since volumes double at each step). Let  $\{Q_j\}$  be the maximal elements. By Lemma 3.1.2 two distinct maximal cubes are disjoint — if they met, one would contain the other, contradicting maximality of the smaller.



Therefore  $\{\mathcal{M}_d f > \lambda\} = \bigsqcup_j Q_j$  and

$$m(\{\mathcal{M}_d f > \lambda\}) = \sum_j |Q_j| < \sum_j \frac{1}{\lambda} \int_{Q_j} |f| = \frac{1}{\lambda} \int_{\{\mathcal{M}_d f > \lambda\}} |f|. \quad \square$$

---

**WORKED EXAMPLE 3.1.1— TIER A  $\mathcal{M}_d$  is genuinely smaller than  $\mathcal{M}$**

On  $\mathbb{R}$  take  $f = \mathbf{1}_{[0,1]}$  and  $x < 0$ . Every dyadic interval containing both  $x$  and part of  $[0,1]$  must contain 0, hence must be of the form  $[-2^{-k}\ell, \cdot)$ ; the smallest such interval containing  $x$  has length at least  $|x|$  and its average of  $f$  is at most 1. A short computation gives  $\mathcal{M}_d f(x) \leq \frac{1}{2}$  for  $x < -1$ , while  $\mathcal{M}f(x) \approx \frac{1}{2|x|}$  for large  $|x|$  — comparable, but the dyadic version is genuinely smaller and, crucially, it is *not* translation invariant: replacing  $f$  by  $\tau_h f$  changes  $\mathcal{M}_d f$  in a way not accounted for by translating.

That failure of translation invariance is the one defect of the dyadic maximal function, and Worked Example 3.1.2 repairs it.

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**WORKED EXAMPLE 3.1.2— TIER B The one-third trick: recovering  $\mathcal{M}$  from shifted grids**

**Proposition 3.1.5.** *For  $\alpha \in \{0, \frac{1}{3}, \frac{2}{3}\}^n$  let  $\mathcal{D}^\alpha = \{Q + \alpha : Q \in \mathcal{D}\}$ , a shifted dyadic grid, and let  $\mathcal{M}_d^\alpha$  be the corresponding maximal function. Then*

$$\mathcal{M}f(x) \lesssim_n \max_\alpha \mathcal{M}_d^\alpha f(x) \quad \text{for every } x.$$

*Consequently Theorem 3.1.4 implies the Hardy–Littlewood maximal inequality, with no covering lemma.*

*Proof.* It suffices to show: for every cube  $R \subseteq \mathbb{R}^n$  with sides parallel to the axes there is  $\alpha$  and  $Q \in \mathcal{D}^\alpha$  with  $R \subseteq Q$  and  $|Q| \leq C_n |R|$ ; balls are then handled by inscribing each ball in a cube of comparable volume.

Take  $n = 1$  first. Let  $R$  be an interval of length  $\ell$ , and choose  $k$  with  $2^{-k-1} \leq \ell < 2^{-k}$ , so that dyadic intervals of generation  $k$  have length comparable to  $\ell$ . If  $R$  is not contained in a single interval of  $\mathcal{D}_k$ , then  $R$  straddles a point of  $2^{-k}\mathbb{Z}$ , say  $p$ . The shifted grids  $\mathcal{D}_k + \frac{1}{3} \cdot 2^{-k}$  and  $\mathcal{D}_k + \frac{2}{3} \cdot 2^{-k}$  have their endpoints at  $2^{-k}(\mathbb{Z} + \frac{1}{3})$  and  $2^{-k}(\mathbb{Z} + \frac{2}{3})$ , and the three endpoint sets are pairwise at distance  $\frac{1}{3}2^{-k} > \ell/\dots$  — more precisely, the three sets of endpoints partition each interval  $[p, p+2^{-k})$  into three pieces of length  $\frac{1}{3}2^{-k}$ , and  $R$ , having length  $< 2^{-k}$ , can straddle endpoints of at most two of the three grids. Hence for some  $\alpha \in \{0, \frac{1}{3}, \frac{2}{3}\}$ ,  $R$  lies inside a single interval of  $\mathcal{D}_k^\alpha$ , whose length is  $2^{-k} \leq 2\ell$ .

In  $\mathbb{R}^n$  apply this in each coordinate separately, choosing  $\alpha = (\alpha_1, \dots, \alpha_n)$  coordinatewise; the resulting cube has side at most  $2\ell$ , so volume at most  $2^n|R|$ .

Given this, for any ball  $B \ni x$  let  $R$  be the smallest axis-parallel cube containing  $B$  and  $Q \supseteq R$  as above; then

$$\frac{1}{|B|} \int_B |f| \leq \frac{|Q|}{|B|} \cdot \frac{1}{|Q|} \int_Q |f| \lesssim_n \mathcal{M}_d^\alpha f(x),$$

since  $|Q| \lesssim_n |B|$ . Take the supremum over  $B$ .  $\square$

*What this buys.* Volume 1’s proof of the maximal theorem needed the Vitali lemma, which is genuinely geometric and whose constant  $3^n$  resisted improvement. The argument above replaces geometry by combinatorics: it needs only that a short interval cannot straddle the endpoints of all three shifted grids at once. The number 3 is not sacred — any three shifts in general position work — but  $\{0, \frac{1}{3}, \frac{2}{3}\}$  is the standard choice, and the trick is used constantly in the modern literature, where “by the one-third trick we may assume all cubes are dyadic” is a stock phrase.

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### 3.1.3 The Calderón–Zygmund decomposition

**Theorem 3.1.6** (Calderón–Zygmund decomposition). *Let  $f \in L^1(\mathbb{R}^n)$  and  $\lambda > 0$ . There is a countable collection  $\{Q_j\}$  of pairwise disjoint dyadic cubes such that*

- (i)  $\lambda < \langle |f| \rangle_{Q_j} \leq 2^n \lambda$  for every  $j$ ;
- (ii)  $|f(x)| \leq \lambda$  for a.e.  $x \notin \Omega := \bigcup_j Q_j$ ;
- (iii)  $|\Omega| = \sum_j |Q_j| \leq \frac{\|f\|_1}{\lambda}$ .

Consequently  $f = g + b$ , where

$$g = f \mathbf{1}_{\Omega^c} + \sum_j \langle f \rangle_{Q_j} \mathbf{1}_{Q_j}, \quad b = \sum_j b_j, \quad b_j = (f - \langle f \rangle_{Q_j}) \mathbf{1}_{Q_j},$$

and these satisfy

- (iv)  $\|g\|_\infty \leq 2^n \lambda$  and  $\|g\|_1 \leq \|f\|_1$ ;
- (v)  $\text{supp } b_j \subseteq Q_j$ ,  $\int b_j = 0$ , and  $\|b_j\|_1 \leq 2 \int_{Q_j} |f|$ ;
- (vi)  $\sum_j \|b_j\|_1 \leq 2 \|f\|_1$ .

*Idea of the proof.* Stop the first time the average exceeds  $\lambda$ . Because the dyadic grid nests, “the first time” is well defined — take the maximal cubes of Theorem 3.1.4 — and because a maximal cube’s *parent* failed the test, its own average cannot exceed  $\lambda$  by more than the volume ratio  $2^n$ . That upper bound in (i) is the whole reason the decomposition is useful: it says the bad set is bad by a bounded factor, so the good part  $g$  is bounded.

Outside the union nothing ever exceeded  $\lambda$ , so  $\mathcal{M}_d f \leq \lambda$  there, and Lebesgue differentiation converts that into a pointwise bound on  $f$ .

The final splitting then does the only thing available: replace  $f$  on each bad cube by its average, which is bounded, and record the error, which has mean zero by construction. Mean zero is what Chapter 3.2 will exploit.

*Proof. Step 1. Selection.*

Let  $\{Q_j\}$  be the maximal elements of  $\mathcal{Q}_\lambda = \{Q \in \mathcal{D} : \langle |f| \rangle_Q > \lambda\}$ , as in the proof of Theorem 3.1.4. They are pairwise disjoint, and their union is  $\{\mathcal{M}_d f > \lambda\} =: \Omega$ .

**Step 2.** *Proof of (i).*

The lower bound is the definition of  $\mathcal{Q}_\lambda$ . For the upper bound, let  $\widehat{Q}_j$  be the parent of  $Q_j$ . By maximality  $\widehat{Q}_j \notin \mathcal{Q}_\lambda$ , so  $\langle |f| \rangle_{\widehat{Q}_j} \leq \lambda$ , and since  $|\widehat{Q}_j| = 2^n |Q_j|$  and  $Q_j \subseteq \widehat{Q}_j$ ,

$$\langle |f| \rangle_{Q_j} = \frac{1}{|Q_j|} \int_{Q_j} |f| \leq \frac{2^n}{|\widehat{Q}_j|} \int_{\widehat{Q}_j} |f| \leq 2^n \lambda.$$

**Step 3.** *Proof of (ii).*

For  $x \notin \Omega$  we have  $\mathcal{M}_d f(x) \leq \lambda$ , so every dyadic average of  $|f|$  over cubes containing  $x$  is at most  $\lambda$ . Letting the cubes shrink to  $x$  and applying the Lebesgue differentiation theorem (Theorem 1.7.4 (Lebesgue differentiation), in its dyadic form — the dyadic cubes containing  $x$  shrink to  $x$  and are comparable to balls) gives  $|f(x)| \leq \lambda$  for a.e. such  $x$ .

**Step 4.** *Proof of (iii).*

This is Theorem 3.1.4.

**Step 5.** *Proof of (iv)–(vi).*

On  $\Omega^c$  we have  $|g| = |f| \leq \lambda$  by (ii); on  $Q_j$  we have  $|g| = |\langle f \rangle_{Q_j}| \leq \langle |f| \rangle_{Q_j} \leq 2^n \lambda$  by (i). And  $\|g\|_1 \leq \int_{\Omega^c} |f| + \sum_j |Q_j| \langle |f| \rangle_{Q_j} = \int_{\Omega^c} |f| + \sum_j \int_{Q_j} |f| = \|f\|_1$ .

For (v): the support statement is clear,  $\int b_j = \int_{Q_j} f - |Q_j| \langle f \rangle_{Q_j} = 0$  by definition of the average, and  $\|b_j\|_1 \leq \int_{Q_j} |f| + |Q_j| \langle |f| \rangle_{Q_j} \leq 2 \int_{Q_j} |f|$ . Summing over the disjoint  $Q_j$  gives (vi).  $\square$

**Where we are.** Four properties will be used in Chapter 3.2, and it is worth naming them now, because the proof of the Calderón–Zygmund theorem uses each exactly once:

- $\|g\|_\infty \lesssim \lambda$ , to make an  $L^2$  estimate on  $g$  cheap;
- $\|g\|_1 \leq \|f\|_1$ , to make it quantitative;
- $\int b_j = 0$ , to exploit smoothness of the kernel;
- $\sum_j |Q_j| \lesssim \|f\|_1 / \lambda$ , to discard the bad set.

Nothing else about the decomposition is needed anywhere in this volume.

**WORKED EXAMPLE 3.1.3— TIER B A decomposition computed explicitly**

Let  $n = 1$  and  $f = \mathbf{1}_{[0, 2^{-N}]}$  for a large integer  $N$ , so  $\|f\|_1 = 2^{-N}$ . Fix  $\lambda \in (0, 1)$  and let  $k$  be the unique integer with  $2^{k-N} \leq \lambda^{-1} < 2^{k+1-N}$ , that is,  $2^{-k} \approx \lambda 2^{-N} \cdot 2^N = \lambda \cdot \dots$ ; concretely, choose  $k \geq 0$  minimal with  $2^{-N}/2^{-k} \leq \lambda$ , i.e.  $2^{k-N} \leq \lambda$ .

The dyadic intervals  $I$  with  $\langle |f| \rangle_I > \lambda$  are exactly those with  $|I \cap [0, 2^{-N}]| > \lambda |I|$ . For  $I \subseteq [0, 2^{-N}]$  this holds whenever  $\lambda < 1$ ; for  $I \supseteq [0, 2^{-N}]$  of length  $2^{-m}$  it reads  $2^{-N} > \lambda 2^{-m}$ , i.e.  $2^{m-N} > \lambda$ .

So the maximal such interval is the dyadic interval  $I_0 = [0, 2^{-m_0})$  where  $m_0$  is the least  $m$  with  $2^{m-N} > \lambda$ . There is exactly one bad cube,  $Q_1 = I_0$ , and:

$$\langle |f| \rangle_{Q_1} = 2^{m_0-N} \in (\lambda, 2\lambda],$$

confirming (i) with  $2^n = 2$ . Then  $g = 2^{m_0-N} \mathbf{1}_{I_0}$ , which has  $\|g\|_\infty = 2^{m_0-N} \leq 2\lambda$  and  $\|g\|_1 = 2^{-N} = \|f\|_1$ ; and  $b = (\mathbf{1}_{[0, 2^{-N}]} - 2^{m_0-N}) \mathbf{1}_{I_0}$ , of mean zero, with  $\|b\|_1 \leq 2 \cdot 2^{-N}$ .

*What the example shows.* The bad cube is not the support of  $f$ ; it is the dyadic interval at the scale where the average of  $f$  first drops to  $\lambda$ . Raising  $\lambda$  shrinks it and lowering  $\lambda$  grows it, and the decomposition is a way of selecting, for each  $\lambda$ , the scale at which  $f$  looks like size  $\lambda$ . That is exactly the multiscale bookkeeping of Volume 1, Worked Example 5.16, in another costume.

**WORKED EXAMPLE 3.1.4— TIER C: RESEARCH TASTE Stopping-time decompositions, from Calderón–Zygmund to the frontier**

Theorem 3.1.6 is the first instance in this series of a *stopping-time decompo-*

sition, and the pattern it introduces recurs, in progressively more elaborate forms, all the way to the current literature. It is worth naming the pattern while it is still simple enough to see whole.

*The pattern.* One has an object (here, a function) and a family of scales (here, the dyadic generations). One descends through the scales, testing a condition at each (here, “is the average  $> \lambda$ ?”), and *stops* at the first scale where the test fires. This partitions the object into: a part where the test never fired, which is therefore uniformly good; and a collection of localised pieces, one for each stopping cube, which are bad but *controlled*, because their parent passed the test.

The two clauses of Theorem 3.1.6(i) are precisely this:  $> \lambda$  because we stopped,  $\leq 2^n \lambda$  because we had not stopped one step earlier. Every stopping-time argument produces a two-sided bound of exactly this shape, and the upper bound always comes from the parent.

*Where it goes.* Three later instances, in increasing order of difficulty:

- *Whitney decompositions* (Chapter 3.7): stop when a cube’s distance to a closed set becomes comparable to its diameter. The parent clause becomes the statement that Whitney cubes have bounded overlap.
- *Broad–narrow analysis* (Volume 12). Bourgain and Guth partition space by asking, at each scale, whether the significant contributions to  $|Ef|$  at a given point come from caps that are *transverse* to one another. Where they do — the broad case — a multilinear estimate applies; where they do not, the contribution is essentially concentrated on a lower-dimensional set and one recurses. The stopping condition is transversality; the parent clause is what makes the recursion terminate.
- *The sticky–plany–grainy trichotomy* (Volume 10) and the reduction of the general Kakeya problem to the sticky case (Volume 14). Wang and Zahl’s proof in  $\mathbb{R}^3$  organises a family of tubes by repeatedly asking, scale by scale, how the tubes at scale  $\delta$  cluster inside the tubes at scale  $\sqrt{\delta}$ , and stopping when the clustering becomes degenerate.

In every case the arithmetic is the same as Step 2 of the proof above: the selected object exceeds the threshold, and its parent does not, so it exceeds it by a bounded factor. A reader who has understood why  $2^n$  appears in Theorem 3.1.6(i) has understood the mechanism of all of them.

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**Figure 3.1.1 (to be drawn).** *The dyadic grid and a straddling cube*

**Content required:** Two or three generations of the dyadic grid in the plane, with one arbitrary cube  $Q$  drawn straddling a grid line so that no dyadic cube of comparable size contains it. Beside it, the same  $Q$  contained in a cube of one of the three shifted grids, illustrating the one-third trick.

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**WORKED EXAMPLE 3.1.5— TIER B The stopping-time argument, in isolation**

The Calderón–Zygmund decomposition is a stopping-time argument, and the pattern recurs often enough that it is worth extracting from the proof.

*The setup.* One has a quantity attached to each dyadic cube — here the average  $\langle |f| \rangle_Q$  — and a threshold  $\lambda$ . One wants to select a family of cubes, disjoint, on which the quantity first exceeds the threshold.

*The rule.* Starting from cubes so large that the average is below  $\lambda$  — possible since  $f \in L^1$  and averages over huge cubes tend to 0 — subdivide. Select a cube  $Q$  when  $\langle |f| \rangle_Q > \lambda$  and its parent was not selected. Stop there; do not subdivide a selected cube.

*The two consequences, and where each comes from.*

- *Above the threshold:*  $\langle |f| \rangle_Q > \lambda$ , because  $Q$  was selected. Summing gives  $\sum |Q| \leq \lambda^{-1} \|f\|_1$ , the measure bound.
- *Not too far above:*  $\langle |f| \rangle_{\widehat{Q}} \leq \lambda$  for the parent  $\widehat{Q}$ , because  $\widehat{Q}$  was *not* selected. Since  $|\widehat{Q}| = 2^n |Q|$ , this gives  $\langle |f| \rangle_Q \leq 2^n \lambda$  — the upper bound that makes the good function bounded.

The second is the whole reason for the stopping rule: without “first exceeds”, one would have no upper bound at all.

*Where it recurs.* The Whitney decomposition of Volume 4, Theorem 4.4.3, is the same rule with  $\text{dist}(Q, \Omega^c)$  in place of the average. The atomic decomposition of Volume 4, Proposition 4.3.4, iterates it over  $\lambda = 2^k$ . And the pigeonholing over scales in Volume 7, Theorem 7.8.5, is a stopping time in disguise.

*The habit.* On meeting a selection of cubes, ask what the parent’s failure to be selected gives you. That is almost always where the useful upper bound comes from, and it is the step most often omitted in summaries.

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## Drills

**Drill 3.1.1.** Show that  $\mathcal{M}_d f \leq \mathcal{M} f$  pointwise up to a dimensional constant,

and that no reverse inequality holds pointwise: find  $f$  and  $x$  with  $\mathcal{M}f(x)$  large and  $\mathcal{M}_d f(x)$  small.

**Drill 3.1.2.** Deduce from Theorem 3.1.4 and Worked Example 3.1.2 that  $\|\mathcal{M}f\|_p \lesssim_{n,p} \|f\|_p$  for  $1 < p \leq \infty$ , quoting the Marcinkiewicz theorem (Theorem 2.2.4 (Marcinkiewicz interpolation)). Compare the constant with the one obtained in Volume 1.

**Drill 3.1.3.** In Theorem 3.1.6, show that the cubes  $Q_j$  depend monotonically on  $\lambda$ : if  $\lambda' < \lambda$  then every  $Q_j$  at height  $\lambda$  is contained in some cube at height  $\lambda'$ .

**Drill 3.1.4.** Carry out the decomposition of Worked Example 3.1.3 for  $f(x) = |x|^{-1/2} \mathbf{1}_{[0,1]}$  and describe the resulting family  $\{Q_j\}$  as  $\lambda \rightarrow \infty$ .

**Where we are.** The dyadic grid is used here purely as a bookkeeping device — a family of cubes closed under the operations the stopping-time argument needs. Nothing depends on the particular grid, and Worked Example 3.1.2 shows that three shifted grids suffice to capture an arbitrary cube up to bounded distortion.

That flexibility matters later. Volume 11’s polynomial partitioning replaces the dyadic grid by the cells of a polynomial’s zero set, and the arguments transfer because they used only the stopping-time structure, never the geometry of cubes.

## Notes

The dyadic maximal function and the one-third trick are folklore; the earliest systematic use of shifted grids appears in work of Garnett and Jones. The Calderón–Zygmund decomposition is from their 1952 *Acta* paper, which also contains the theorem of Chapter 3.2.

The reader should be aware that a substantial modern literature — sparse domination, the  $A_2$  theorem — consists of refinements of Theorem 3.1.6 in which the selected cubes are required to carry a fixed proportion of the mass. None of it is needed in this series, but the vocabulary is common in papers a reader may consult.





## Chapter 3.2

# The Calderón–Zygmund Theorem

### 3.2.1 What a singular integral is

The operators of this volume are all of the form

$$Tf(x) = \int_{\mathbb{R}^n} K(x, y)f(y) \, dy, \quad (3.2.1)$$

where the kernel  $K$  is of size  $|x - y|^{-n}$  near the diagonal. That size is critical:  $|x|^{-n}$  is not integrable at 0 and not integrable at  $\infty$ , by the polar coordinates computation of Volume 1. So (3.2.1) does not converge absolutely, the operator is not bounded on  $L^1$ , and the integral must be interpreted as a principal value. The name *singular integral* records exactly this.

What saves the situation is cancellation. If the kernel oscillates, or — what is usually easier to verify — if it varies *slowly* away from the diagonal, then it cannot distinguish a mean-zero bump from nothing, and that is enough.

**Definition 3.2.1** (Calderón–Zygmund kernel). A measurable  $K : \mathbb{R}^n \setminus \{0\} \rightarrow \mathbb{C}$  is a *Calderón–Zygmund kernel* with constant  $A$  if

$$\int_{|x| > 2|y|} |K(x - y) - K(x)| \, dx \leq A \quad \text{for every } y \neq 0. \quad (3.2.2)$$

Condition (3.2.2) is called *Hörmander’s condition*.

**Proposition 3.2.2** (The usual smoothness hypotheses suffice). *Suppose  $K \in C^1(\mathbb{R}^n \setminus \{0\})$  satisfies  $|\nabla K(x)| \leq B|x|^{-n-1}$ . Then  $K$  satisfies (3.2.2) with  $A \lesssim_n B$ .*

*Proof.* Fix  $y \neq 0$  and  $x$  with  $|x| > 2|y|$ . The segment from  $x - y$  to  $x$  stays in  $\{|z| \geq |x| - |y| \geq |x|/2\}$ , so by the mean value theorem

$$|K(x-y) - K(x)| \leq |y| \sup_{z \in [x-y, x]} |\nabla K(z)| \leq |y| B \left( \frac{|x|}{2} \right)^{-n-1} = 2^{n+1} B \frac{|y|}{|x|^{n+1}}.$$

Integrating over  $|x| > 2|y|$  and using polar coordinates,

$$\int_{|x| > 2|y|} |K(x-y) - K(x)| dx \leq 2^{n+1} B |y| \sigma(\mathbb{S}^{n-1}) \int_{2|y|}^{\infty} \frac{r^{n-1}}{r^{n+1}} dr = 2^n B \sigma(\mathbb{S}^{n-1}),$$

since  $\int_{2|y|}^{\infty} r^{-2} dr = (2|y|)^{-1}$ .  $\square$

**Pitfall: the 2 in  $|x| > 2|y|$  is not arbitrary**

The region of integration in (3.2.2) excludes a neighbourhood of the singularity of *both*  $K(\cdot - y)$  and  $K$ . If one integrated over all  $x$ , the condition would fail for every kernel of size  $|x|^{-n}$ , since each term separately is non-integrable near its own singularity. Any constant  $c > 1$  in place of 2 gives an equivalent condition, with a different  $A$ ; the choice 2 is conventional and matches the dilation  $Q \mapsto 2\sqrt{n}Q$  used in the proof below.

### 3.2.2 The theorem

**Theorem 3.2.3** (Calderón–Zygmund). *Let  $T$  be a linear operator, bounded on  $L^2(\mathbb{R}^n)$  with norm  $B_2$ , such that for every  $f \in L^1 \cap L^2$  with compact support,*

$$Tf(x) = \int_{\mathbb{R}^n} K(x-y)f(y) dy \quad \text{for a.e. } x \notin \text{supp } f, \quad (3.2.3)$$

where  $K$  satisfies Hörmander’s condition (3.2.2) with constant  $A$ . Then:

- (i)  $T$  is of weak type  $(1, 1)$ :  $m(\{|Tf| > \lambda\}) \lesssim_n \frac{(A + B_2)}{\lambda} \|f\|_1$ ;
- (ii)  $T$  is bounded on  $L^p(\mathbb{R}^n)$  for every  $1 < p < \infty$ , with  $\|T\|_{L^p \rightarrow L^p} \lesssim_{n,p} A + B_2$ .

*Idea of the proof.* Split  $f = g + b$  by Theorem 3.1.6 at height  $\lambda$  and treat the two parts by completely different means.

The good part  $g$  is bounded by  $\lambda$  and has  $L^1$  norm at most  $\|f\|_1$ , so  $\|g\|_2^2 \leq \|g\|_{\infty} \|g\|_1 \lesssim \lambda \|f\|_1$ . The  $L^2$  hypothesis then gives what we want by Chebyshev, and note the shape of the exchange: we bought an  $L^1$  statement by spending an  $L^2$  one, and the factor  $\lambda$  that appears is exactly what

Chebyshev needs.

The bad part is where the kernel enters. Discard the union of the (slightly enlarged) bad cubes — it has measure  $\lesssim \|f\|_1/\lambda$ , which is the budget. Outside it, use that  $\int b_j = 0$  to *subtract a constant* from the kernel:

$$Tb_j(x) = \int_{Q_j} (K(x-y) - K(x-c_j))b_j(y) \, dy,$$

where  $c_j$  is the centre of  $Q_j$ . Since  $x$  is far from  $Q_j$  and  $y$  is inside it, Hörmander's condition applies and integrates the difference to  $A$ . This is the only place cancellation is used, and it is used exactly once.

**Proof. Step 1. Reduction.**

By density (Theorem 1.8.7 (density of simple functions and  $C_c^\infty$ )) it suffices to prove (i) for  $f \in L^1 \cap L^2$  with compact support, since  $L^1 \cap L^2$  is dense in  $L^1$  and weak-type bounds pass to limits along a subsequence converging a.e.

Fix such an  $f$  and  $\lambda > 0$ , and apply Theorem 3.1.6 at height  $\lambda$ , obtaining  $\Omega = \bigsqcup_j Q_j$ ,  $f = g + b$ ,  $b = \sum_j b_j$ . All the properties (i)–(vi) there are now in force.

**Step 2. The good part.**

Since  $|g| \leq 2^n \lambda$  pointwise and  $\|g\|_1 \leq \|f\|_1$ ,

$$\|g\|_2^2 = \int |g|^2 \leq \|g\|_\infty \|g\|_1 \leq 2^n \lambda \|f\|_1.$$

By the  $L^2$  hypothesis and Chebyshev's inequality (Volume 1, Prop. 8.14),

$$m\left(\left\{|Tg| > \frac{\lambda}{2}\right\}\right) \leq \frac{4}{\lambda^2} \|Tg\|_2^2 \leq \frac{4B_2^2}{\lambda^2} \cdot 2^n \lambda \|f\|_1 = \frac{2^{n+2}B_2^2}{\lambda} \|f\|_1. \quad (3.2.4)$$

**Step 3. Discarding the bad set.**

Let  $c_j$  be the centre of  $Q_j$  and let  $Q_j^* = 2\sqrt{n}Q_j$  be the concentric dilate; the factor  $2\sqrt{n}$  is chosen so that

$$x \notin Q_j^* \text{ and } y \in Q_j \implies |x - c_j| > 2|y - c_j|. \quad (3.2.5)$$

Indeed if  $Q_j$  has side  $\ell$  then  $|y - c_j| \leq \frac{\sqrt{n}}{2}\ell$  for  $y \in Q_j$ , while  $x \notin Q_j^*$  forces  $|x - c_j| > \sqrt{n}\ell$ .

Put  $\Omega^* = \bigcup_j Q_j^*$ . By Theorem 3.1.6(iii),

$$|\Omega^*| \leq \sum_j |Q_j^*| = (2\sqrt{n})^n \sum_j |Q_j| \leq \frac{(2\sqrt{n})^n}{\lambda} \|f\|_1. \quad (3.2.6)$$

**Step 4.** *The bad part, off  $\Omega^*$ .*

Fix  $j$ . For  $x \notin Q_j^*$ , the representation (3.2.3) applies since  $x \notin \text{supp } b_j \subseteq Q_j$ , and the mean-zero property  $\int b_j = 0$  allows us to insert a constant:

$$Tb_j(x) = \int_{Q_j} K(x-y)b_j(y) \, dy = \int_{Q_j} \left( K(x-y) - K(x-c_j) \right) b_j(y) \, dy.$$

Integrating in  $x$  over the complement of  $Q_j^*$  and using Tonelli,

$$\int_{(Q_j^*)^c} |Tb_j(x)| \, dx \leq \int_{Q_j} |b_j(y)| \left( \int_{(Q_j^*)^c} |K(x-y) - K(x-c_j)| \, dx \right) \, dy.$$

Substitute  $x \mapsto x + c_j$ ,  $y \mapsto y + c_j$  in the inner integral: by (3.2.5) the region of integration is contained in  $\{|x| > 2|y|\}$ , so Hörmander's condition (3.2.2) bounds it by  $A$ . Hence

$$\int_{(Q_j^*)^c} |Tb_j| \leq A \|b_j\|_1,$$

and summing over  $j$  and using Theorem 3.1.6(vi),

$$\int_{(\Omega^*)^c} |Tb| \leq \sum_j \int_{(Q_j^*)^c} |Tb_j| \leq A \sum_j \|b_j\|_1 \leq 2A \|f\|_1.$$

By Chebyshev,

$$m\left(\left\{x \notin \Omega^* : |Tb(x)| > \frac{\lambda}{2}\right\}\right) \leq \frac{4A}{\lambda} \|f\|_1. \quad (3.2.7)$$

**Step 5.** *Assemble.*

Since  $\{|Tf| > \lambda\} \subseteq \{|Tg| > \frac{\lambda}{2}\} \cup \{|Tb| > \frac{\lambda}{2}\}$ , and the latter is contained in  $\Omega^*$  together with the set in (3.2.7),

$$m(\{|Tf| > \lambda\}) \leq \frac{2^{n+2}B_2^2 + (2\sqrt{n})^n + 4A}{\lambda} \|f\|_1.$$

Normalising  $T$  so that  $B_2 \leq 1$  — replace  $T$  by  $T/B_2$ , which replaces  $A$  by  $A/B_2$  — gives the stated form of (i).

**Step 6.** *Interpolation, and duality.*

For  $1 < p < 2$ :  $T$  is of weak type  $(1,1)$  by (i) and of strong, hence weak, type  $(2,2)$  by hypothesis. Marcinkiewicz interpolation (Theorem 2.2.4 (Marcinkiewicz interpolation)) gives strong type  $(p,p)$ .

For  $2 < p \leq \infty$ : the adjoint  $T^*$  has kernel  $K^*(x,y) = \overline{K(y,x)}$ , i.e. convolution with  $\tilde{K}(x) = \overline{K(-x)}$ , which satisfies (3.2.2) with the same constant

after the substitution  $x \mapsto -x$ . And  $T^*$  is bounded on  $L^2$  with the same norm. So the previous paragraph applies to  $T^*$  and gives  $\|T^*\|_{L^{p'} \rightarrow L^{p'}} \lesssim A + B_2$  for  $1 < p' < 2$ ; by duality (Corollary 1.8.10 (norms by testing)),  $\|T\|_{L^p \rightarrow L^p} = \|T^*\|_{L^{p'} \rightarrow L^{p'}}$ .  $\square$

**Where we are.** Read Step 4 again. It is the only step that uses anything about  $K$ , and what it uses is that the kernel, viewed from far away, is nearly constant across a small cube. The mean-zero property of  $b_j$  converts “nearly constant” into “nearly zero”. No oscillation, no explicit cancellation in the kernel, and no formula were needed — only Hörmander’s condition, which is a smoothness statement.

This is why the theorem covers so much. Any operator whose kernel is  $C^1$  away from the diagonal with the natural gradient bound qualifies, by Proposition 3.2.2; and verifying the remaining hypothesis —  $L^2$  boundedness — is, for convolution operators, exactly Plancherel.

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WORKED EXAMPLE 3.2.1— TIER A **Truncations and principal values**

The representation (3.2.3) says nothing about  $Tf(x)$  for  $x \in \text{supp } f$ , and it cannot: the integral does not converge there. The standard device is to truncate. For  $0 < \varepsilon < R$  set

$$T_{\varepsilon,R}f(x) = \int_{\varepsilon < |x-y| < R} K(x-y)f(y) dy,$$

which converges absolutely for  $f \in L^p$ , since  $K$  is bounded on the annulus. Two remarks. First, the truncated operators are genuine convolution operators with bounded compactly supported kernels, so no principal value is involved and every manipulation is legitimate. Second, the content of the theory is that the bounds one proves are *uniform* in  $\varepsilon$  and  $R$ ; the limit  $T = \lim_{\varepsilon \rightarrow 0, R \rightarrow \infty} T_{\varepsilon,R}$  then exists in the appropriate sense and inherits them. Throughout this volume, statements about  $T$  should be read as statements about  $T_{\varepsilon,R}$  with constants independent of  $\varepsilon, R$ , and the passage to the limit is carried out once, in Proposition 3.3.2.

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WORKED EXAMPLE 3.2.2— TIER B **Sharpness: the weak (1,1) bound cannot be upgraded**

Let  $n = 1$  and  $K(x) = 1/x$ , the Hilbert transform kernel of Chapter 3.3.

Take  $f = \mathbf{1}_{[0,1]}$ . For  $x > 1$ ,

$$Tf(x) = \int_0^1 \frac{dy}{x-y} = \log \frac{x}{x-1},$$

which for large  $x$  behaves like  $1/x$ . Hence  $\int_2^\infty |Tf| \approx \int_2^\infty \frac{dx}{x} = \infty$ , so  $Tf \notin L^1(\mathbb{R})$  even though  $f$  is as nice as one could wish.

The mechanism is identical to Worked Example 1.7.1 ( $\mathcal{M}f$  is never integrable), where  $\mathcal{M}f \notin L^1$  for the same reason: an operator whose kernel decays like  $|x|^{-n}$  produces output decaying like  $|x|^{-n}$ , and  $|x|^{-n}$  is exactly the borderline of integrability. In both cases the honest endpoint statement is a weak  $(1, 1)$  bound, and Chapter 3.7 will show that the honest statement at the other endpoint is not  $L^\infty \rightarrow L^\infty$  but  $L^\infty \rightarrow \text{BMO}$ .

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**WORKED EXAMPLE 3.2.3— TIER C: RESEARCH TASTE** **Why the  $L^2$  hypothesis is the hard one**

Theorem 3.2.3 has two hypotheses. One — Hörmander’s condition — is a routine verification, usually a two-line consequence of Proposition 3.2.2. The other — boundedness on  $L^2$  — is, in every interesting case, the entire difficulty. It is worth being explicit about this asymmetry, because it recurs at every stage of this series.

(a) *When  $L^2$  is free.* For a convolution operator  $Tf = K * f$ , Plancherel turns the question into a pointwise bound on the multiplier:  $T$  is bounded on  $L^2$  with norm  $\|\widehat{K}\|_\infty$  (Proposition 2.6.2 (the  $L^2$  multiplier norm)). So for the Hilbert transform, whose multiplier is  $-i \operatorname{sgn} \xi$ , the  $L^2$  bound is immediate and the constant is 1. The same holds for the Riesz transforms (Chapter 3.4) and for every multiplier in Chapter 3.5. *This is why the convolution theory is easy.*

(b) *When it is not.* For non-convolution operators there is no Plancherel, and establishing  $L^2$  boundedness becomes the theorem. The  $T(1)$  theorem of David and Journé characterises exactly when a Calderón–Zygmund operator is bounded on  $L^2$ , in terms of the two conditions  $T(1), T^*(1) \in \text{BMO}$  plus a weak boundedness property; it is not proved in this series, and it is the reason BMO is not merely a curiosity.

(c) *The same asymmetry in the restriction problem.* The extension operator  $E$  of Volume 2 is bounded from  $L^2(d\sigma)$  to  $L^q(\mathbb{R}^n)$  for  $q = \frac{4n}{n-1}$  by a soft argument — Worked Example 2.8.3 (a restriction theorem from decay), which used only decay and Hardy–Littlewood–Sobolev. Stein–Tomas improves the exponent by exploiting  $L^2$  orthogonality, and the restriction conjecture proper

concerns  $L^\infty(d\sigma)$  data, where no orthogonality is available at all. Each step away from  $L^2$  costs a decade.

(d) *And in decoupling.* Worked Example 1.8.2 (almost orthogonality) observed that  $\ell^2$  decoupling is trivial at  $p = 2$ , by Plancherel, and is the theorem for  $p > 2$ . Volume 13's proof proceeds by repeatedly reducing  $L^p$  information to  $L^2$  information at a smaller scale — that is what “ball inflation” does — and then iterating. The subject may fairly be described as a long campaign to convert  $L^2$  facts, which are free, into  $L^p$  facts, which are not. This chapter is the first campaign of that kind, and the Calderón–Zygmund decomposition is its first weapon.

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WORKED EXAMPLE 3.2.4— TIER C: RESEARCH TASTE **What the Calderón–Zygmund theorem buys, and what it costs**

Theorem 3.2.3 is the central theorem of the volume. It is worth being explicit about the trade it makes.

(a) *The input.* Two hypotheses: boundedness on  $L^2$ , and Hörmander's condition on the kernel. The first is usually easy — for a multiplier it is  $\|m\|_\infty < \infty$ , by Plancherel. The second is a smoothness condition away from the diagonal and is where the work is.

(b) *The output.* Weak type  $(1, 1)$ , and hence by interpolation  $L^p$  boundedness for all  $1 < p < \infty$ . So one verifies an estimate at a *single* exponent and obtains a whole range.

(c) *Why  $L^2$  and not another exponent.* Because Plancherel makes  $L^2$  computable for multipliers, and because the Calderón–Zygmund decomposition splits  $f$  into a bounded part — handled by  $L^2$  — and a mean-zero part — handled by the kernel. Any exponent would do in principle;  $L^2$  is where the hypothesis is checkable.

(d) *The cost: the endpoints are lost.*  $p = 1$  becomes weak type, and  $p = \infty$  becomes BMO (Theorem 3.7.4). Both losses are genuine, as Worked Example 3.3.4(c) showed for the Hilbert transform. Volume 4 exists to supply the two substitute spaces and to prove that the theory is sharp on them.

(e) *What it does not cover.* Operators whose kernels are singular along a *surface* rather than a point — the ball multiplier of Chapter 3.8 is the first example — fall outside entirely. Its symbol is bounded, so the  $L^2$  hypothesis holds; but there is no Hörmander condition to verify, because the kernel's singularity is spread over a sphere rather than concentrated at the origin. That failure is not technical: Theorem 3.8.2 shows the conclusion is *false*, and the reason is the Besicovitch geometry of Volume 9.

*The summary to carry forward.* Calderón–Zygmund theory handles singularities at a point. Everything from Volume 5 onwards handles singularities along a surface, and needs curvature to do it.

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## Drills

**Drill 3.2.1.** Verify that  $K(x) = \Omega(x/|x|)|x|^{-n}$  satisfies (3.2.2) whenever  $\Omega$  is Lipschitz on  $\mathbb{S}^{n-1}$ , and show by example that mere continuity of  $\Omega$  is not enough for Proposition 3.2.2 to apply.

**Drill 3.2.2.** Show that the constant in Theorem 3.2.3(ii) blows up as  $p \rightarrow 1$  and as  $p \rightarrow \infty$ , and identify which step of the proof is responsible in each case.

**Drill 3.2.3.** Prove that if  $T$  satisfies the hypotheses of Theorem 3.2.3 then so does  $f \mapsto \overline{Tf}$ , and deduce that one may assume  $K$  real without loss of generality in checking (3.2.2).

**Drill 3.2.4.** Where exactly does the proof of Theorem 3.2.3 use that the  $Q_j$  are *disjoint*? Show that without disjointness the estimate (3.2.6) fails, and construct a covering family for which it fails by a factor  $\log(1/\lambda)$ .

## Notes

Theorem 3.2.3 is from Calderón and Zygmund’s 1952 paper, in the form with a gradient hypothesis on the kernel; Hörmander’s condition (3.2.2), which is weaker and is what the proof actually uses, is from his 1960 paper.

The reader should note that Theorem 3.2.3 produces no examples. It is a machine for converting an  $L^2$  bound into  $L^p$  bounds, and every application in this volume begins by establishing the  $L^2$  bound by Plancherel. Worked Example 3.2.3 makes the point that this is not an accident of exposition.



## Chapter 3.3

# The Hilbert Transform

The Hilbert transform is the model singular integral: the unique one on  $\mathbb{R}$  (up to constants) commuting with translations and dilations and anticommuting with reflections. This chapter establishes its boundedness, and then cashes it in for a theorem promised in Volume 2 — the convergence of Fourier series in  $L^p$ .

### 3.3.1 Definition and $L^2$ theory

**Definition 3.3.1.** For  $f \in \mathcal{S}(\mathbb{R})$ ,

$$\mathcal{H}f(x) = \text{p.v.} \frac{1}{\pi} \int_{\mathbb{R}} \frac{f(y)}{x-y} dy = \lim_{\varepsilon \rightarrow 0^+} \frac{1}{\pi} \int_{|x-y|>\varepsilon} \frac{f(y)}{x-y} dy.$$

**Proposition 3.3.2** (The limit exists, and the multiplier). *For  $f \in \mathcal{S}(\mathbb{R})$  the limit exists for every  $x$ , and*

$$\widehat{\mathcal{H}f}(\xi) = -i \operatorname{sgn}(\xi) \widehat{f}(\xi).$$

*Consequently  $\|\mathcal{H}f\|_2 = \|f\|_2$ , and  $\mathcal{H}$  extends to a unitary operator on  $L^2(\mathbb{R})$  with  $\mathcal{H}^2 = -\operatorname{Id}$ .*

*Proof. Step 1. Existence.*

Because the kernel  $1/(x-y)$  is odd, the truncated integral may be written

$$\frac{1}{\pi} \int_{|t|>\varepsilon} \frac{f(x-t)}{t} dt = \frac{1}{\pi} \int_{\varepsilon<|t|<1} \frac{f(x-t) - f(x)}{t} dt + \frac{1}{\pi} \int_{|t|\geq 1} \frac{f(x-t)}{t} dt,$$

where we used  $\int_{\varepsilon<|t|<1} \frac{dt}{t} = 0$  to insert  $-f(x)$  at no cost. The first integrand is bounded by  $\|f'\|_{\infty}$  and the second by  $|f(x-t)|$ , integrable since  $f \in \mathcal{S}$ .

So the limit exists by dominated convergence, and is bounded uniformly on compact sets.

**Step 2.** *The multiplier.*

Let  $K_\varepsilon(t) = \frac{1}{\pi t} \mathbf{1}_{\varepsilon < |t| < 1/\varepsilon}$ , so  $\mathcal{H}f = \lim_{\varepsilon \rightarrow 0} K_\varepsilon * f$ . Each  $K_\varepsilon \in L^1$ , so  $\widehat{K_\varepsilon * f} = \widehat{K_\varepsilon} \widehat{f}$  (Proposition 2.4.2 (the Fourier dictionary)(v)). Since  $K_\varepsilon$  is odd,

$$\widehat{K_\varepsilon}(\xi) = \frac{1}{\pi} \int_{\varepsilon < |t| < 1/\varepsilon} \frac{e^{-2\pi i t \xi}}{t} dt = \frac{-2i}{\pi} \int_\varepsilon^{1/\varepsilon} \frac{\sin(2\pi t \xi)}{t} dt = \frac{-2i \operatorname{sgn} \xi}{\pi} \int_{2\pi \varepsilon |\xi|}^{2\pi |\xi|/\varepsilon} \frac{\sin u}{u} du,$$

substituting  $u = 2\pi t |\xi|$ . As  $\varepsilon \rightarrow 0$  the integral tends to  $\int_0^\infty \frac{\sin u}{u} du = \frac{\pi}{2}$ , so  $\widehat{K_\varepsilon}(\xi) \rightarrow -i \operatorname{sgn} \xi$  pointwise, and the  $\widehat{K_\varepsilon}$  are uniformly bounded because the partial integrals of  $\sin u/u$  are. Dominated convergence in the Plancherel pairing gives the claim.

**Step 3.** *Consequences.*

$|-i \operatorname{sgn} \xi| = 1$  a.e., so by Proposition 2.6.2 (the  $L^2$  multiplier norm) the operator is an isometry of  $L^2$ ; and the multiplier squares to  $-1$  a.e.  $\square$

**Theorem 3.3.3** ( $L^p$  boundedness; M. Riesz).  *$\mathcal{H}$  is of weak type  $(1,1)$  and bounded on  $L^p(\mathbb{R})$  for every  $1 < p < \infty$ , with  $\|\mathcal{H}\|_{L^p \rightarrow L^p} \lesssim_p 1$ . It is bounded on neither  $L^1$  nor  $L^\infty$ .*

*Proof.* The kernel  $K(x) = \frac{1}{\pi x}$  is  $C^\infty$  away from 0 with  $|K'(x)| = \frac{1}{\pi x^2}$ , so Proposition 3.2.2 applies with  $n = 1$  and  $B = 1/\pi$ , giving Hörmander's condition. Proposition 3.3.2 gives  $L^2$  boundedness with  $B_2 = 1$ , and the representation (3.2.3) holds for  $x \notin \operatorname{supp} f$  by inspection, the principal value being an honest convergent integral there. Theorem 3.2.3 applies and gives both conclusions.

Unboundedness on  $L^1$  is Worked Example 3.2.2. For  $L^\infty$ , take  $f = \mathbf{1}_{[0,1]}$ : a direct computation gives  $\mathcal{H}f(x) = \frac{1}{\pi} \log \left| \frac{x}{x-1} \right|$ , which is unbounded near  $x = 0$  and  $x = 1$  although  $\|f\|_\infty = 1$ .  $\square$

**WORKED EXAMPLE 3.3.1— TIER A The Hilbert transform of an indicator**

For  $f = \mathbf{1}_{[a,b]}$  and  $x \notin [a,b]$  the principal value is an ordinary integral:

$$\mathcal{H}f(x) = \frac{1}{\pi} \int_a^b \frac{dy}{x-y} = \frac{1}{\pi} \log \left| \frac{x-a}{x-b} \right|.$$

The same formula holds for  $x \in (a,b)$ , where the principal value cancels the symmetric part of the singularity.

Three features are worth recording, since each is typical.

*Logarithmic blow-up at the endpoints.*  $\mathcal{H}f$  is unbounded, which is Theorem 3.3.3's failure at  $p = \infty$ . Note it is only logarithmic —  $\mathcal{H}f$  lies in every

$L^p$  with  $p < \infty$ , and in BMO (Chapter 3.7), which is the correct endpoint space.

*Decay  $1/|x|$  at infinity.*  $\log|\frac{x-a}{x-b}| = \log|1 + \frac{b-a}{x-b}| \approx \frac{b-a}{x}$ , which is the failure at  $p = 1$ .

*The transform is not supported near the support of  $f$ .* A singular integral spreads mass everywhere, which is exactly why the localisation argument of Theorem 3.2.3 Step 4 had to discard the enlarged cubes rather than estimate on them.

### 3.3.2 The conjugate function and Fourier series

We now transfer to the torus and collect the promised application.

**Definition 3.3.4.** For a trigonometric polynomial  $f = \sum_{|m| \leq N} \widehat{f}(m) e_m$  on  $\mathbb{T}$ , the *conjugate function* is

$$\widetilde{f} = \sum_{|m| \leq N} (-i \operatorname{sgn} m) \widehat{f}(m) e_m,$$

and the *Riesz projection* is  $P_+ f = \sum_{m \geq 0} \widehat{f}(m) e_m$ , so that  $P_+ = \frac{1}{2}(\operatorname{Id} + i\widetilde{\phantom{x}}) + \frac{1}{2}\widehat{f}(0) \cdot (\cdot)$ .

**Theorem 3.3.5** (M. Riesz, periodic form). *For  $1 < p < \infty$  there is  $C_p$  with  $\|\widetilde{f}\|_{L^p(\mathbb{T})} \leq C_p \|f\|_{L^p(\mathbb{T})}$  for every trigonometric polynomial  $f$ . Consequently  $P_+$  is bounded on  $L^p(\mathbb{T})$ .*

*Idea of the proof.* The conjugate function has the same multiplier as the Hilbert transform, read on  $\mathbb{Z}$  instead of  $\mathbb{R}$ . One could redo the whole Calderón–Zygmund theory on the torus — dyadic arcs, a periodic decomposition — and that is the classical route. It is shorter to *transfer*: a bound for a multiplier operator on  $\mathbb{R}$  implies the corresponding bound on  $\mathbb{T}$  for the restricted multiplier, by a dilation argument due to de Leeuw. We prove the case we need directly.

**Proof. Step 1. Transference setup.**

Let  $m(\xi) = -i \operatorname{sgn} \xi$ , so that  $\mathcal{H}$  is the multiplier operator  $T_m$  on  $\mathbb{R}$  and  $\widetilde{\phantom{x}}$  is the multiplier operator with symbol  $m|_{\mathbb{Z}}$  on  $\mathbb{T}$ . Fix a trigonometric polynomial  $f = \sum_{|k| \leq N} c_k e_k$  and  $\varepsilon > 0$ .

Choose  $\varphi \in \mathcal{S}(\mathbb{R})$  with  $\varphi \geq 0$ ,  $\|\varphi\|_p = 1$ , and set  $\varphi_\varepsilon(x) = \varepsilon^{1/p} \varphi(\varepsilon x)$ , so  $\|\varphi_\varepsilon\|_p = 1$  for every  $\varepsilon$ . Consider

$$F_\varepsilon(x) = f(x) \varphi_\varepsilon(x), \quad x \in \mathbb{R},$$

where  $f$  is viewed as a 1-periodic function on  $\mathbb{R}$ .

**Step 2.** *Localising the multiplier.*

Since  $\widehat{F_\varepsilon} = \sum_{|k| \leq N} c_k \tau_k \widehat{\varphi_\varepsilon}$  and  $\widehat{\varphi_\varepsilon}(\xi) = \varepsilon^{1/p-1} \widehat{\varphi}(\xi/\varepsilon)$  concentrates near  $\xi = 0$  as  $\varepsilon \rightarrow 0$ , the pieces  $\tau_k \widehat{\varphi_\varepsilon}$  concentrate near the integers  $k$  and become disjointly supported in the limit. On the piece near  $k$ , the multiplier  $m$  is constant and equal to  $m(k)$ , provided  $k \neq 0$  and  $\varepsilon$  is small enough that the effective support of  $\widehat{\varphi_\varepsilon}$  has width less than 1. Hence

$$T_m F_\varepsilon = \sum_{|k| \leq N} c_k m(k) e_k \varphi_\varepsilon + E_\varepsilon = \widetilde{f} \varphi_\varepsilon + E_\varepsilon,$$

where  $E_\varepsilon$  collects the error from the tails of  $\widehat{\varphi}$  and satisfies  $\|E_\varepsilon\|_p \rightarrow 0$  as  $\varepsilon \rightarrow 0$ , because  $\widehat{\varphi}$  is Schwartz and the number of terms is finite.

**Step 3.** *Pass to the limit.*

By Theorem 3.3.3,  $\|T_m F_\varepsilon\|_{L^p(\mathbb{R})} \leq C_p \|F_\varepsilon\|_{L^p(\mathbb{R})}$ . Now

$$\|F_\varepsilon\|_{L^p(\mathbb{R})}^p = \int_{\mathbb{R}} |f(x)|^p \varphi_\varepsilon(x)^p dx \longrightarrow \|f\|_{L^p(\mathbb{T})}^p \quad (\varepsilon \rightarrow 0),$$

since  $\varphi_\varepsilon^p$  has integral 1, spreads out over a length  $1/\varepsilon$ , and  $|f|^p$  is periodic with mean  $\|f\|_{L^p(\mathbb{T})}^p$ ; this is the standard averaging of a periodic function against a spreading probability density, and follows from dominated convergence after writing the integral as a sum over periods. The identical computation applies to  $\|\widetilde{f} \varphi_\varepsilon\|_{L^p(\mathbb{R})}$ . Combining with Step 2,

$$\|\widetilde{f}\|_{L^p(\mathbb{T})} = \lim_{\varepsilon \rightarrow 0} \|\widetilde{f} \varphi_\varepsilon\|_{L^p(\mathbb{R})} \leq \limsup_{\varepsilon \rightarrow 0} (\|T_m F_\varepsilon\|_p + \|E_\varepsilon\|_p) \leq C_p \|f\|_{L^p(\mathbb{T})}.$$

□

WORKED EXAMPLE 3.3.2— TIER B **Fourier series converge in  $L^p$**

Volume 2's Pitfall after Theorem 3.13 (du Bois-Reymond) asserted that Fourier series of  $L^p$  functions converge in  $L^p$  for  $1 < p < \infty$ , and deferred the proof to this volume. Here it is.

**Theorem 3.3.6.** *For  $1 < p < \infty$  and  $f \in L^p(\mathbb{T})$ , the partial sums satisfy  $\|S_N f - f\|_{L^p(\mathbb{T})} \rightarrow 0$ .*

*Proof. Step 1: uniform boundedness of  $S_N$ .* Write  $M_a f = e_a f$  for modulation by  $e_a(x) = e(ax)$ . A direct computation on Fourier coefficients gives

$$S_N = M_N P_+ M_{-N} - M_{-N-1} P_+ M_{N+1},$$

since the first term keeps frequencies  $\geq -N$  and the second removes those  $\geq N + 1$ . Modulations are isometries of  $L^p(\mathbb{T})$ , so by Theorem 3.3.5,

$$\|S_N\|_{L^p \rightarrow L^p} \leq 2\|P_+\|_{L^p \rightarrow L^p} \leq C_p$$

uniformly in  $N$ .

*Step 2: convergence on a dense class.* For a trigonometric polynomial  $P$  of degree  $d$ ,  $S_N P = P$  for all  $N \geq d$ . Trigonometric polynomials are dense in  $L^p(\mathbb{T})$  for  $p < \infty$  (Corollary 2.3.7 (density of trigonometric polynomials)).

*Step 3: combine.* Given  $f \in L^p$  and  $\varepsilon > 0$ , choose  $P$  with  $\|f - P\|_p < \varepsilon$ . Then for  $N \geq \deg P$ ,

$$\|S_N f - f\|_p \leq \|S_N(f - P)\|_p + \|P - f\|_p \leq (C_p + 1)\varepsilon. \quad \square$$

*The architecture.* Uniform boundedness of a family of operators, plus convergence on a dense subclass, gives convergence everywhere. This is the same template as Theorem 1.7.4 (Lebesgue differentiation) (Lebesgue differentiation), where the uniform bound came from a maximal inequality; and it is the template of Carleson's theorem, where the operator is the maximal partial-sum operator  $\sup_N |S_N f|$  and the uniform bound is the hard part.

*And the contrast with  $p = 1, \infty$ .* Volume 2's du Bois-Reymond example showed the family  $\{S_N\}$  is *not* uniformly bounded on  $C(\mathbb{T})$ , because  $\|D_N\|_{L^1} \approx \log N$ . Step 1 is precisely what fails there, and Theorem 3.3.5 is precisely what is unavailable at  $p = 1$  and  $p = \infty$ .

**WORKED EXAMPLE 3.3.3 — TIER C: RESEARCH TASTE** **One direction, and what it costs**

The Hilbert transform acts in one dimension. In  $\mathbb{R}^n$  one can apply it in a fixed direction  $v \in \mathbb{S}^{n-1}$ : define

$$\mathcal{H}_v f(x) = \text{p.v.} \frac{1}{\pi} \int_{\mathbb{R}} \frac{f(x - tv)}{t} dt,$$

the Hilbert transform along the line through  $x$  in direction  $v$ . Its multiplier is  $-i \operatorname{sgn}(\xi \cdot v)$ , so the associated *half-space multiplier*

$$S_v f = \mathcal{F}^{-1}(\mathbf{1}_{\{\xi \cdot v \geq 0\}} \widehat{f}) = \frac{1}{2}(f + i\mathcal{H}_v f)$$

is bounded on  $L^p(\mathbb{R}^n)$  for  $1 < p < \infty$ , with a constant *independent of  $v$*  — by Theorem 3.3.3 applied in the  $v$  variable and Fubini in the orthogonal directions.

*Why record this.* The constant's independence of  $v$  looks like a triviality: rotations are isometries of every  $L^p$ , so of course the bound does not see the direction. But the statement one actually needs in Chapter 3.8 is *vector-valued*:

$$\left\| \left( \sum_j |S_{v_j} f_j|^2 \right)^{1/2} \right\|_{L^p} \lesssim_p \left\| \left( \sum_j |f_j|^2 \right)^{1/2} \right\|_{L^p},$$

with a constant independent of the *collection* of directions  $v_1, \dots, v_N$  and of  $N$ . That does not follow from rotation invariance, because the directions differ from one term to the next and no single rotation normalises them all. It does follow — from the scalar bound alone — by the Marcinkiewicz–Zygmund theorem, which we prove as Theorem 3.6.1. And the astonishing fact, due to Fefferman, is that the resulting vector-valued inequality is *false* if the half-spaces are replaced by balls, and that the counterexample is a Besicovitch set. That is Chapter 3.8, and the reader now has every ingredient except the argument itself.

**WORKED EXAMPLE 3.3.4— TIER C: RESEARCH TASTE The Hilbert transform as the prototype**

Every feature of Calderón–Zygmund theory is visible already in the Hilbert transform, and it is worth listing the correspondences before meeting them in general.

(a) *A singular kernel with cancellation.*  $\mathcal{H}$  is convolution with p.v.  $\frac{1}{\pi x}$ , which is not locally integrable. It is only the oddness —  $\frac{1}{x}$  changes sign — that makes the principal value converge. In  $\mathbb{R}^n$  the corresponding statement is the cancellation condition on a Calderón–Zygmund kernel, and Worked Example 3.4.3(c) shows the Riesz transforms inherit it from oddness too.

(b) *A bounded multiplier.*  $\widehat{\mathcal{H}f}(\xi) = -i \operatorname{sgn}(\xi) \widehat{f}(\xi)$ , so  $\mathcal{H}$  is an isometry on  $L^2$  up to sign. That is the  $L^2$  hypothesis of Theorem 3.2.3, available here for free, and in general the hypothesis one must verify separately.

(c) *Failure at the endpoints.*  $\mathcal{H}\mathbf{1}_{[0,1]}(x) = \frac{1}{\pi} \log \left| \frac{x}{x-1} \right|$ , which is not integrable at infinity — it decays like  $1/x$ . So  $\mathcal{H}$  does not map  $L^1 \rightarrow L^1$ , and the substitute is weak type  $(1,1)$ . Nor does it map  $L^\infty \rightarrow L^\infty$ : the same example is unbounded near 0 and 1. The substitute there is BMO, which is Theorem 3.7.4.

(d) *What the general theory adds.* Nothing conceptual — only the observation that (a), (b) and the Hörmander condition are the *only* properties used, so any operator with them behaves the same way. That is the content of Theorem 3.2.3, and it is why a single theorem covers the Hilbert transform,

the Riesz transforms, and the multipliers of Theorem 3.5.1.

(e) *The one thing the Hilbert transform does not show.* It is a one-dimensional operator, so it cannot exhibit the directional phenomena of Volumes 6 and 9 — there is only one direction in  $\mathbb{R}$ . The ball multiplier of Chapter 3.8 is where dimension first matters, and it fails for reasons invisible here.

## Drills

**Drill 3.3.1.** Compute  $\mathcal{H}$  of the Poisson kernel  $P_y(x) = \frac{1}{\pi} \frac{y}{x^2+y^2}$  and identify the answer as the conjugate Poisson kernel.

**Drill 3.3.2.** Verify the identity  $S_N = M_N P_+ M_{-N} - M_{-N-1} P_+ M_{N+1}$  used in Worked Example 3.3.2 by evaluating both sides on  $e_m$ .

**Drill 3.3.3.** Show  $\mathcal{H}^2 = -\text{Id}$  on  $L^p(\mathbb{R})$  for  $1 < p < \infty$ , and deduce that  $\|\mathcal{H}\|_{L^p \rightarrow L^p} \geq 1$ .

**Drill 3.3.4.** Prove directly, without Theorem 3.2.3, that  $\mathcal{H}$  is bounded on  $L^4(\mathbb{R})$ , by applying Plancherel to  $(\mathcal{H}f)^2$  and using  $\mathcal{H}(f\mathcal{H}f) = \frac{1}{2}((\mathcal{H}f)^2 - f^2)$ . (This is Cotlar's identity; it explains why even exponents were historically the first accessible ones.)

**Where we are.** The Hilbert transform is the only operator in this volume that can be written down completely: its kernel, its multiplier, and its action on simple functions are all explicit. Everything afterwards is an operator known only through estimates. It is worth returning to Worked Example 3.3.4 whenever a general statement seems opaque, since every feature of the general theory can be seen there concretely.

## Notes

M. Riesz proved Theorem 3.3.3 and Theorem 3.3.5 in 1927, by complex analysis; the real-variable proof via Theorem 3.2.3 came a quarter-century later and generalises to  $\mathbb{R}^n$ , which the complex method does not.

The transference argument of Theorem 3.3.5 is de Leeuw's (1965) in general form. Carleson's theorem (1966), that the Fourier series of an  $L^2$  function converges almost everywhere, is not proved in this series; Hunt extended it to  $L^p$ ,  $p > 1$ , in 1968.





## Chapter 3.4

# Riesz Transforms and Elliptic Estimates

The Hilbert transform has no direct analogue in  $\mathbb{R}^n$ : there is no distinguished direction. What replaces it is a vector of  $n$  operators, one for each coordinate, which together carry the same information.

### 3.4.1 Definition and boundedness

**Definition 3.4.1.** For  $j = 1, \dots, n$  the  $j$ -th Riesz transform is

$$\mathcal{R}_j f(x) = \text{p.v. } c_n \int_{\mathbb{R}^n} \frac{x_j - y_j}{|x - y|^{n+1}} f(y) dy, \quad c_n = \frac{\Gamma(\frac{n+1}{2})}{\pi^{\frac{n+1}{2}}}.$$

**Proposition 3.4.2.**  $\widehat{\mathcal{R}_j f}(\xi) = -i \frac{\xi_j}{|\xi|} \widehat{f}(\xi)$  for  $f \in \mathcal{S}(\mathbb{R}^n)$ . Consequently  $\mathcal{R}_j$  is bounded on  $L^2$  with norm 1, and

$$\sum_{j=1}^n \mathcal{R}_j^2 = -\text{Id}. \quad (3.4.1)$$

*Proof.* The kernel  $K_j(x) = c_n x_j |x|^{-n-1}$  is homogeneous of degree  $-n$  and odd, so its principal value is a tempered distribution (the argument of Proposition 3.3.2, Step 1, applies verbatim with the oddness supplying the cancellation). Its Fourier transform is therefore homogeneous of degree 0 and odd. Moreover  $K_j$  transforms under rotations like the linear functional  $x \mapsto x_j$ : for a rotation  $\rho$ ,  $K_j(\rho x) = \sum_k \rho_{jk} K_k(x)$ . Since the Fourier transform commutes with rotations (Proposition 2.4.2 (the Fourier dictionary)(iv)),  $\widehat{K_j}$  has the same equivariance, and the only degree-zero odd functions with that property are the multiples of  $\xi_j/|\xi|$ . So  $\widehat{K_j}(\xi) = \gamma_n \xi_j/|\xi|$  for a constant, and evaluating

the normalisation — which is the purpose of the constant  $c_n$  — gives  $\gamma_n = -i$ . The  $L^2$  statement is Proposition 2.6.2 (the  $L^2$  multiplier norm), and (3.4.1) is the identity  $\sum_j (-i\xi_j/|\xi|)^2 = -\sum_j \xi_j^2/|\xi|^2 = -1$ .  $\square$

**Theorem 3.4.3.** *Each  $\mathcal{R}_j$  is of weak type  $(1, 1)$  and bounded on  $L^p(\mathbb{R}^n)$  for  $1 < p < \infty$ .*

*Proof.*  $K_j$  is smooth away from 0 with  $|\nabla K_j(x)| \lesssim_n |x|^{-n-1}$ , by direct differentiation of  $x_j|x|^{-n-1}$ . So Proposition 3.2.2 gives Hörmander's condition, and Proposition 3.4.2 gives the  $L^2$  bound. Apply Theorem 3.2.3.  $\square$

### 3.4.2 The elliptic estimate

**Theorem 3.4.4** (Calderón–Zygmund estimate for the Laplacian). *Let  $1 < p < \infty$ . For every  $u \in C_c^\infty(\mathbb{R}^n)$  and all  $j, k$ ,*

$$\|\partial_j \partial_k u\|_{L^p(\mathbb{R}^n)} \lesssim_{n,p} \|\Delta u\|_{L^p(\mathbb{R}^n)}.$$

*Idea of the proof.* On the Fourier side the claim is that the ratio of the symbols,  $\xi_j \xi_k / |\xi|^2$ , is a bounded multiplier on  $L^p$ . That ratio is exactly the symbol of  $\mathcal{R}_j \mathcal{R}_k$ , so the estimate is not merely implied by Theorem 3.4.3 — it *is* Theorem 3.4.3, restated. The content of the theorem is that a scalar equation controls a full matrix of second derivatives, and the mechanism is that the Riesz transforms invert the passage from  $\Delta$  to  $\partial_j \partial_k$ .

*Proof.* Comparing symbols:  $\widehat{\partial_j \partial_k u} = -4\pi^2 \xi_j \xi_k \widehat{u}$ , while

$$\widehat{\mathcal{R}_j \mathcal{R}_k \Delta u}(\xi) = \left(-i \frac{\xi_j}{|\xi|}\right) \left(-i \frac{\xi_k}{|\xi|}\right) (-4\pi^2 |\xi|^2) \widehat{u}(\xi) = 4\pi^2 \xi_j \xi_k \widehat{u}(\xi).$$

Hence  $\partial_j \partial_k u = -\mathcal{R}_j \mathcal{R}_k \Delta u$  as tempered distributions, and therefore as  $L^p$  functions. Applying Theorem 3.4.3 twice,

$$\|\partial_j \partial_k u\|_p \leq \|\mathcal{R}_j\|_{L^p \rightarrow L^p} \|\mathcal{R}_k\|_{L^p \rightarrow L^p} \|\Delta u\|_p. \quad \square$$

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**WORKED EXAMPLE 3.4.1— TIER B The estimate fails at  $p = 1$  and  $p = \infty$ , and what replaces it**

Theorem 3.4.4 is false for  $p = 1$  and  $p = \infty$ , and this is not a defect of the proof.

At  $p = \infty$ . If  $\|D^2u\|_\infty \lesssim \|\Delta u\|_\infty$  held, then a continuous right-hand side would force  $u \in C^2$ . It does not: the standard counterexample in  $\mathbb{R}^2$  is

$$u(x) = (x_1^2 - x_2^2) \log \log \frac{1}{|x|}$$

near the origin, cut off smoothly. One computes that  $\Delta u$  is bounded while  $\partial_1^2 u$  is not, the double logarithm being exactly slow enough to survive one Laplacian and not two. This is the same logarithmic failure as in Worked Example 3.3.1, and for the same reason: at the endpoint, singular integrals produce logarithms.

At  $p = 1$ . Weak type  $(1, 1)$  survives, by Theorem 3.4.3, but strong type does not, by the argument of Worked Example 3.2.2 applied to  $\mathcal{R}_j$ .

*What replaces them.* At  $p = \infty$  the correct statement is  $\|D^2u\|_{\text{BMO}} \lesssim \|\Delta u\|_\infty$ , which follows from the  $L^\infty \rightarrow \text{BMO}$  bound of Chapter 3.7. At  $p = 1$  the correct statement uses the Hardy space  $H^1$  in place of  $L^1$ . The pattern — the two endpoints of the  $L^p$  scale must be replaced by  $H^1$  and BMO — is universal for singular integrals, and is the reason Chapter 3.7 exists.

**WORKED EXAMPLE 3.4.2— TIER C: RESEARCH TASTE** **The method of rotations: singular integrals as averages of Hilbert transforms**

Every odd singular integral in  $\mathbb{R}^n$  is an average of one-dimensional Hilbert transforms. The identity is elementary and its consequences are not.

**Proposition 3.4.5.** *Let  $\Omega : \mathbb{S}^{n-1} \rightarrow \mathbb{R}$  be odd and integrable, and let  $T_\Omega$  be the operator with kernel  $\Omega(x/|x|)|x|^{-n}$ . Then, with  $\mathcal{H}_\omega$  the directional Hilbert transform of Worked Example 3.3.3,*

$$T_\Omega f = \frac{\pi}{2} \int_{\mathbb{S}^{n-1}} \Omega(\omega) \mathcal{H}_\omega f \, d\sigma(\omega).$$

*Proof.* Write the truncated integral in polar coordinates  $y = x - r\omega$ ,  $r > 0$ ,  $\omega \in \mathbb{S}^{n-1}$  (Theorem 1.6.11 (polar coordinates)):

$$\int_{|x-y|>\varepsilon} \frac{\Omega\left(\frac{x-y}{|x-y|}\right)}{|x-y|^n} f(y) \, dy = \int_{\mathbb{S}^{n-1}} \Omega(\omega) \int_\varepsilon^\infty \frac{f(x-r\omega)}{r} \, dr \, d\sigma(\omega).$$

Split  $\mathbb{S}^{n-1}$  into  $\pm\omega$  pairs and use oddness of  $\Omega$ : replacing  $\omega$  by  $-\omega$  in half the sphere converts  $\int_\varepsilon^\infty \frac{f(x-r\omega)}{r} \, dr$  into  $-\int_{-\infty}^{-\varepsilon} \frac{f(x-r\omega)}{r} \, dr$  with a compensating sign, so the inner integral becomes  $\frac{1}{2} \int_{|r|>\varepsilon} \frac{f(x-r\omega)}{r} \, dr$  over the full sphere. Letting  $\varepsilon \rightarrow 0$  gives  $\frac{\pi}{2} \mathcal{H}_\omega f$ .  $\square$

*Consequence.* Since  $\|\mathcal{H}_\omega\|_{L^p \rightarrow L^p} = \|\mathcal{H}\|_{L^p \rightarrow L^p}$  independently of  $\omega$  (Worked Example 3.3.3), Minkowski's integral inequality gives

$$\|T_\Omega\|_{L^p \rightarrow L^p} \leq \frac{\pi}{2} \|\Omega\|_{L^1(\mathbb{S}^{n-1})} \|\mathcal{H}\|_{L^p \rightarrow L^p},$$

with *no smoothness hypothesis on  $\Omega$  whatsoever*. Compare Proposition 3.2.2, which needed a gradient bound. The method of rotations trades smoothness for oddness, and when the kernel is odd it is by far the more efficient tool. Taking  $\Omega(\omega) = c_n \omega_j$  recovers Theorem 3.4.3.

*Why this matters for the series.* The proposition writes an  $n$ -dimensional operator as a superposition, over directions, of operators each of which sees only one direction. That is precisely the structure of a wave packet decomposition, and precisely the structure that fails to be summable in Fefferman's theorem. In Chapter 3.8 we will have a family of directional operators  $S_{v_j}$ , one per direction, and the question will be whether their *square function* is bounded — a genuinely stronger demand than bounding each one and averaging. The answer is no, and the obstruction is that  $\delta$ -tubes in different directions can be packed into a set of small measure. The method of rotations works because it only ever needs the triangle inequality; everything hard in this series begins when the triangle inequality is too lossy.

#### WORKED EXAMPLE 3.4.3— TIER B **Computing with the Riesz transforms**

The Riesz transforms  $\mathcal{R}_j$  are the model Calderón–Zygmund operators, and it repays doing three computations with them explicitly.

(a) *They recover second derivatives.* Since  $\widehat{\mathcal{R}_j f}(\xi) = -i \frac{\xi_j}{|\xi|} \widehat{f}(\xi)$ ,

$$\widehat{\mathcal{R}_j \mathcal{R}_k f}(\xi) = -\frac{\xi_j \xi_k}{|\xi|^2} \widehat{f}(\xi), \quad \text{so} \quad \mathcal{R}_j \mathcal{R}_k(-\Delta) = \partial_j \partial_k$$

as Fourier multipliers, both sides having symbol  $\xi_j \xi_k$  up to the factor  $4\pi^2$ . Hence  $\|\partial_j \partial_k u\|_{L^p} \lesssim_{n,p} \|\Delta u\|_{L^p}$  for  $1 < p < \infty$ : *every* second derivative is controlled by the Laplacian alone. This is the Calderón–Zygmund estimate for elliptic equations, and it is false at  $p = 1$  and  $p = \infty$  precisely because  $\mathcal{R}_j$  is not bounded there.

(b) *They sum to the identity.* From (a) with  $j = k$  and summing,  $\sum_j \mathcal{R}_j^2$  has symbol  $-\sum_j \xi_j^2 / |\xi|^2 = -1$ , so

$$\sum_{j=1}^n \mathcal{R}_j^2 = -I.$$

A useful check on signs, and the reason the Riesz transforms are thought of as the components of a vector-valued singular integral of unit length.

(c) *The kernel.*  $\mathcal{R}_j$  is convolution with  $K_j(x) = c_n x_j/|x|^{n+1}$ , homogeneous of degree  $-n$  and odd. Oddness gives the cancellation that makes the principal value converge; homogeneity gives Hörmander's condition, by the computation of Theorem 3.4.4. Both properties are needed, and the Pitfall of §4.2 shows what fails without oddness.

*What to take away.* The Riesz transforms convert a statement about the Laplacian — a single scalar operator — into statements about all second derivatives. That conversion is why Calderón–Zygmund theory is the analytic engine of elliptic regularity, and it is the reason Volume 4 can characterise Sobolev spaces by  $L^p$  norms of  $\Delta^{s/2}$  rather than of all derivatives separately.

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WORKED EXAMPLE 3.4.4— TIER A **The elliptic estimate, stated plainly**

The single most-used consequence of this volume, outside harmonic analysis itself, is worth isolating.

*The statement.* For  $1 < p < \infty$  and  $u \in \mathcal{S}(\mathbb{R}^n)$ ,

$$\|D^2u\|_{L^p(\mathbb{R}^n)} \lesssim_{n,p} \|\Delta u\|_{L^p(\mathbb{R}^n)},$$

where  $D^2u$  is the full Hessian. So control of one scalar quantity, the Laplacian, controls all  $n^2$  second derivatives.

*The proof, in one line.*  $\partial_j \partial_k = \mathcal{R}_j \mathcal{R}_k (-\Delta)$  as multipliers, by Worked Example 3.4.3(a), and each  $\mathcal{R}_j$  is bounded on  $L^p$  by Theorem 3.2.3.

*Why it fails at  $p = 1$  and  $p = \infty$ .* Because the Riesz transforms do. Concretely, there are  $u$  with  $\Delta u$  bounded and  $D^2u$  unbounded — the standard example is  $u(x) = x_1 x_2 \log \log \frac{1}{|x|}$  near the origin in  $\mathbb{R}^2$ , for which  $\Delta u$  stays bounded while  $\partial_1 \partial_2 u$  does not. The failure is logarithmic, which is the signature of a BMO endpoint rather than an  $L^\infty$  one, consistent with Theorem 3.7.4.

*Why analysts care.* It is the linear estimate underlying regularity theory for second-order elliptic equations: a solution of  $\Delta u = f$  with  $f \in L^p$  has two derivatives in  $L^p$ . Every nonlinear elliptic argument that proceeds by linearisation and iteration uses it, and the restriction to  $1 < p < \infty$  is exactly why such arguments avoid the endpoints.

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## Drills

**Drill 3.4.1.** Check that for  $n = 1$  the Riesz transform reduces to the Hilbert transform, and verify  $c_1 = 1/\pi$ .

**Drill 3.4.2.** Prove  $\|\nabla u\|_{L^p}^2 \lesssim \|u\|_{L^p} \|\Delta u\|_{L^p}$  for  $u \in C_c^\infty$ ,  $1 < p < \infty$ , using Theorem 3.4.4 and interpolation.

**Drill 3.4.3.** Verify the rotational equivariance  $K_j(\rho x) = \sum_k \rho_{jk} K_k(x)$  used in Proposition 3.4.2, and confirm that the only odd degree-zero homogeneous functions with that equivariance are multiples of  $\xi_j/|\xi|$ .

**Drill 3.4.4.** Using (3.4.1), show  $\|f\|_p \leq \sum_j \|\mathcal{R}_j f\|_p \cdot \|\mathcal{R}_j\|_{L^p \rightarrow L^p}$  and deduce that the Riesz transforms are jointly injective on  $L^p$ .

## Notes

The Riesz transforms are due to M. Riesz; Theorem 3.4.4 is the Calderón–Zygmund estimate, and is the reason their 1952 paper is a founding document of elliptic PDE as much as of harmonic analysis. The method of rotations is Calderón and Zygmund’s as well, from 1956.

## Chapter 3.5

# The Hörmander–Mikhlin Multiplier Theorem

Volume 2 established that a bounded symbol gives a bounded operator on  $L^2$ , and that nothing so clean is available for  $p \neq 2$ . This chapter supplies the standard sufficient condition, which was stated there and deferred to here.

### 3.5.1 Statement

**Theorem 3.5.1** (Hörmander–Mikhlin). *Let  $s = \lfloor n/2 \rfloor + 1$  and let  $m \in C^s(\mathbb{R}^n \setminus \{0\})$  satisfy*

$$|\partial^\alpha m(\xi)| \leq B |\xi|^{-|\alpha|} \quad \text{for all } \xi \neq 0 \text{ and } |\alpha| \leq s. \quad (3.5.1)$$

*Then  $T_m$  is of weak type  $(1, 1)$  and bounded on  $L^p(\mathbb{R}^n)$  for every  $1 < p < \infty$ , with norms  $\lesssim_{n,p} B$ .*

*Idea of the proof.* The  $L^2$  hypothesis of Theorem 3.2.3 is free: the case  $\alpha = 0$  of (3.5.1) says  $\|m\|_\infty \leq B$ . So everything reduces to verifying Hörmander’s condition (3.2.2) for the kernel  $K = \mathcal{F}^{-1}m$ , and  $K$  is not given by a formula.

The device is to decompose  $m$  dyadically,  $m = \sum_j m_j$  with  $m_j$  supported where  $|\xi| \approx 2^j$ , and to estimate each  $K_j = \mathcal{F}^{-1}m_j$  by Plancherel. A function whose Fourier transform lives at frequency  $2^j$  has its mass at spatial scale  $2^{-j}$ , and the smoothness hypothesis (3.5.1) converts into decay of  $K_j$  beyond that scale. Specifically, differentiating  $m_j$   $s$  times costs  $2^{-js}$ , and by Plancherel that is a bound on  $\| |x|^s K_j \|_2$  — so  $K_j$  decays like  $|x|^{-s}$  in an  $L^2$ -averaged sense, and  $s > n/2$  is exactly what makes that integrable.

The two regimes of the final summation are then the two regimes one expects: scales finer than  $|y|$ , where the difference  $K_j(x - y) - K_j(x)$  is estimated by the gradient and gains  $2^j|y|$ ; and scales coarser than  $|y|$ , where each term is estimated separately and the decay  $(2^j|y|)^{n/2-s}$  does the summing. Both series are geometric, and the sum of the two is where the hypothesis  $s > n/2$  is spent.

### 3.5.2 Proof

Fix  $\psi \in C_c^\infty(\mathbb{R}^n)$  supported in  $\{\frac{1}{2} < |\xi| < 2\}$  with  $\sum_{j \in \mathbb{Z}} \psi(\xi/2^j) = 1$  for all  $\xi \neq 0$ ; such  $\psi$  exists by the standard partition-of-unity construction. Put

$$m_j(\xi) = m(\xi) \psi(\xi/2^j), \quad K_j = \mathcal{F}^{-1} m_j.$$

Each  $m_j$  is smooth with compact support, so  $K_j \in \mathcal{S}(\mathbb{R}^n)$ .

**Lemma 3.5.2** (Scale-by-scale estimates). *For every  $j \in \mathbb{Z}$  and every dyadic  $R > 0$ ,*

$$\int_{|x| \approx R} |K_j| \lesssim_n B \min \left\{ (2^j R)^{n/2}, (2^j R)^{n/2-s} \right\}, \quad (3.5.2)$$

$$\int_{|x| \approx R} |\nabla K_j| \lesssim_n B 2^j \min \left\{ (2^j R)^{n/2}, (2^j R)^{n/2-s} \right\}. \quad (3.5.3)$$

*In particular  $\|K_j\|_1 \lesssim_n B$  uniformly in  $j$ .*

*Proof. Step 1. Derivative bounds on  $m_j$ .*

By the Leibniz rule,  $\partial^\alpha m_j$  is a sum of terms  $\partial^\beta m \cdot 2^{-j|\gamma|} (\partial^\gamma \psi)(\xi/2^j)$  with  $\beta + \gamma = \alpha$ . On the support,  $|\xi| \approx 2^j$ , so (3.5.1) gives  $|\partial^\beta m| \lesssim B 2^{-j|\beta|}$ , and altogether

$$|\partial^\alpha m_j(\xi)| \lesssim_{n,\alpha} B 2^{-j|\alpha|}, \quad |\alpha| \leq s, \quad (3.5.4)$$

supported in  $|\xi| \approx 2^j$ , a set of measure  $\approx 2^{jn}$ . Hence

$$\|\partial^\alpha m_j\|_{L^2} \lesssim_{n,\alpha} B 2^{-j|\alpha|} 2^{jn/2}. \quad (3.5.5)$$

**Step 2.** *From derivatives of  $m_j$  to weighted  $L^2$  bounds on  $K_j$ .*

By Plancherel and Proposition 2.4.2 (the Fourier dictionary)(vi),  $\|x^\alpha K_j\|_2 = (2\pi)^{-|\alpha|} \|\partial^\alpha m_j\|_2$ . Since  $|x|^s \lesssim_{n,s} \sum_{|\alpha|=s} |x^\alpha|$  — expand  $|x|^{2s} = (\sum_i x_i^2)^s$  by the multinomial theorem — we get

$$\|K_j\|_2 \lesssim_n B 2^{jn/2}, \quad \| |x|^s K_j \|_2 \lesssim_n B 2^{j(n/2-s)}. \quad (3.5.6)$$



**Step 3.** *From weighted  $L^2$  to  $L^1$  on an annulus.*

By Cauchy–Schwarz on  $\{|x| \approx R\}$ , a set of measure  $\approx R^n$ ,

$$\int_{|x| \approx R} |K_j| \leq (R^n)^{1/2} \|K_j\|_2 \lesssim_n B R^{n/2} 2^{jn/2} = B(2^j R)^{n/2},$$

which is the first alternative in (3.5.2). For the second, insert the weight:

$$\int_{|x| \approx R} |K_j| = \int_{|x| \approx R} |x|^{-s} |x|^s |K_j| \leq \left( \int_{|x| \approx R} |x|^{-2s} \right)^{1/2} \| |x|^s K_j \|_2 \lesssim_n R^{\frac{n}{2}-s} \cdot B 2^{j(\frac{n}{2}-s)},$$

using  $\int_{|x| \approx R} |x|^{-2s} \approx R^{n-2s}$ .

**Step 4.** *The gradient.*

$\nabla K_j = \mathcal{F}^{-1}(2\pi i \xi m_j)$ , and  $\xi m_j$  satisfies (3.5.4) with an extra factor  $2^j$ , by the same Leibniz computation. So Steps 2–3 apply verbatim and give (3.5.3).

**Step 5.** *The  $L^1$  bound.*

Summing (3.5.2) over dyadic  $R$ , using the first alternative for  $R \leq 2^{-j}$  and the second for  $R > 2^{-j}$ :

$$\|K_j\|_1 \lesssim_n B \sum_{2^j R \leq 1} (2^j R)^{n/2} + B \sum_{2^j R > 1} (2^j R)^{n/2-s} \lesssim_n B,$$

both series being geometric — the first because  $n/2 > 0$ , the second because  $n/2 - s < 0$ , which is where  $s > n/2$  is used for the first time.  $\square$

*Proof of Theorem 3.5.1.* By (3.5.1) with  $\alpha = 0$ ,  $\|m\|_\infty \leq B$ , so  $T_m$  is bounded on  $L^2$  with norm at most  $B$  (Proposition 2.6.2 (the  $L^2$  multiplier norm)). By Theorem 3.2.3 it suffices to verify Hörmander’s condition for  $K = \sum_j K_j$ , the series converging in  $\mathcal{S}'$  since the  $m_j$  sum to  $m$ .

Fix  $y \neq 0$ . We estimate  $\int_{|x| > 2|y|} |K_j(x-y) - K_j(x)| dx$  in two regimes.

**Step 1.** *Fine scales:  $2^j|y| \leq 1$ .*

By the mean value theorem,  $|K_j(x-y) - K_j(x)| \leq |y| \sup_{z \in [x-y, x]} |\nabla K_j(z)|$ , and for  $|x| > 2|y|$  the segment lies in  $\{|z| > |x|/2\}$ . Hence

$$\int_{|x| > 2|y|} |K_j(x-y) - K_j(x)| dx \lesssim_n |y| \int_{|z| > |y|} |\nabla K_j(z)| dz.$$

Summing (3.5.3) over dyadic  $R > |y|$ , and using that in this regime  $2^j|y| \leq 1$  so both alternatives are relevant,

$$\int_{|z| > |y|} |\nabla K_j| \lesssim_n B 2^j \left( \sum_{|y| < R \leq 2^{-j}} (2^j R)^{n/2} + \sum_{R > 2^{-j}} (2^j R)^{n/2-s} \right) \lesssim_n B 2^j,$$

both sums being geometric with ratio bounded away from 1 and largest term  $O(1)$ . Therefore this piece contributes  $\lesssim_n B 2^j |y|$ , and

$$\sum_{j: 2^j |y| \leq 1} B 2^j |y| \lesssim_n B,$$

a geometric series in  $2^j |y|$  with ratio 2, summed from below up to 1.

**Step 2. Coarse scales:**  $2^j |y| > 1$ .

Here we do not use cancellation between the two terms, only that both are supported far from the origin:

$$\int_{|x| > 2|y|} |K_j(x-y) - K_j(x)| dx \leq 2 \int_{|z| > |y|} |K_j(z)| dz \lesssim_n B \sum_{R > |y|} (2^j R)^{n/2-s} \lesssim_n B (2^j |y|)^{n/2-s}$$

the sum being dominated by its first term because  $n/2 - s < 0$ . Summing over  $j$ ,

$$\sum_{j: 2^j |y| > 1} B (2^j |y|)^{n/2-s} \lesssim_n B,$$

again geometric, now decaying.

**Step 3. Conclude.**

Adding,  $\int_{|x| > 2|y|} |K(x-y) - K(x)| dx \lesssim_n B$  uniformly in  $y$ , which is (3.2.2) with  $A \lesssim_n B$ . Theorem 3.2.3 now gives both conclusions.  $\square$

**Where we are.** The hypothesis  $s > n/2$  entered exactly twice, both times to make a geometric series converge: in Step 5 of Lemma 3.5.2 and in Step 2 above. Since  $s$  must be an integer for (3.5.4) to make sense as stated, the natural choice is  $\lfloor n/2 \rfloor + 1$ . Refinements using fractional smoothness — the Marcinkiewicz multiplier theorem, or Hörmander’s original condition phrased with  $L^2$  Sobolev norms of  $m_j$  — reduce the requirement to just over  $n/2$  derivatives, but nothing in this series needs them.

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#### WORKED EXAMPLE 3.5.1— TIER A **Homogeneous multipliers**

Let  $m$  be homogeneous of degree 0 — that is,  $m(t\xi) = m(\xi)$  for  $t > 0$  — and smooth on  $\mathbb{S}^{n-1}$ . Then  $\partial^\alpha m$  is homogeneous of degree  $-|\alpha|$  and bounded on the unit sphere, so  $|\partial^\alpha m(\xi)| \lesssim_\alpha |\xi|^{-|\alpha|}$ : condition (3.5.1) holds for every  $\alpha$ . Examples: the Riesz transform symbols  $-i\xi_j/|\xi|$ , recovering Theorem 3.4.3; the symbol  $\xi_j \xi_k / |\xi|^2$  of Theorem 3.4.4; and  $\text{sgn}(\xi \cdot v)$  — except that this last one is *not* smooth on the sphere, being discontinuous across the hyperplane  $\xi \cdot v = 0$ . So Theorem 3.5.1 does not apply to the half-space multiplier,

and its boundedness (Worked Example 3.3.3) must be obtained by the one-dimensional argument plus Fubini. This is not a technicality: the discontinuity of a characteristic function across a surface is precisely what Chapter 3.8 is about, and whether it is harmless depends entirely on whether the surface is flat.

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**WORKED EXAMPLE 3.5.2— TIER B The Bochner–Riesz means, and where the difficulty starts**

For  $\delta > 0$  define the *Bochner–Riesz multiplier*

$$m^\delta(\xi) = (1 - |\xi|^2)_+^\delta,$$

which is the characteristic function of the unit ball smoothed by raising the distance-to-the-boundary to a power  $\delta$ .

*Large  $\delta$  is easy.* If  $\delta > \frac{n-1}{2}$  one can show  $|\mathcal{F}^{-1}m^\delta(x)| \lesssim (1 + |x|)^{-n-\delta+\frac{n-1}{2}}$ , which is integrable, so  $T_{m^\delta}$  is bounded on every  $L^p$ ,  $1 \leq p \leq \infty$ , by Young’s inequality. The exponent comes from the same stationary-phase computation as Theorem 2.8.5 (decay of  $\widehat{d\sigma}$ ) — the boundary of the ball is a sphere, and its curvature produces the decay  $|x|^{-(n-1)/2}$ .

*Small  $\delta$  is the Bochner–Riesz conjecture*, which asserts that  $T_{m^\delta}$  is bounded on  $L^p(\mathbb{R}^n)$  precisely when  $\delta > \max\{0, n|\frac{1}{p} - \frac{1}{2}| - \frac{1}{2}\}$ . It is a theorem for  $n = 2$  (Carleson–Sjölin) and open for  $n \geq 3$ .

*And  $\delta = 0$  is false*, for every  $p \neq 2$  and every  $n \geq 2$ : that is Fefferman’s theorem, Chapter 3.8.

*Why Theorem 3.5.1 says nothing here.* The multiplier  $m^\delta$  fails (3.5.1) badly near  $|\xi| = 1$ : derivatives blow up at the boundary sphere rather than at the origin, and no amount of homogeneity helps. The theorem of this chapter is adapted to singularities at a *point*, and every hard problem in this series concerns singularities along a *hypersurface*. Bochner–Riesz, restriction and Keakeya are all versions of that difficulty, and the equivalences between them are the subject of Volume 6.

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**WORKED EXAMPLE 3.5.3— TIER A Checking the Mikhlin condition on examples**  
Theorem 3.5.1 asks for  $|\partial^\alpha m(\xi)| \leq B|\xi|^{-|\alpha|}$  up to order  $s = \lfloor n/2 \rfloor + 1$ . Four multipliers, checked.

(a)  $m(\xi) = \text{sgn}(\xi_1)$ . Smooth away from the hyperplane  $\xi_1 = 0$ , but not away from the origin only — the singularity is a whole hyperplane. So the hypothesis fails, and indeed  $T_m$  is the Hilbert transform in the first variable

tensored with the identity, which *is* bounded, by a separate argument. The condition is sufficient, not necessary.

(b)  $m(\xi) = \xi_j/|\xi|$ . Homogeneous of degree 0 and smooth away from the origin, so  $\partial^\alpha m$  is homogeneous of degree  $-|\alpha|$  and the condition holds with  $B$  the maximum of  $|\partial^\alpha m|$  on the unit sphere. This is the Riesz transform, and homogeneity is doing all the work.

(c)  $m(\xi) = |\xi|^{i\gamma}$ ,  $\gamma \in \mathbb{R}$ . Also homogeneous of degree 0, and  $|\partial^\alpha m| \lesssim (1 + |\gamma|)^{|\alpha|} |\xi|^{-|\alpha|}$ . So the condition holds with  $B \approx (1 + |\gamma|)^s$  — *polynomial* in  $\gamma$ , which is exactly what an analytic interpolation argument needs.

(d)  $m(\xi) = (1 - |\xi|^2)_+^\delta$ , the *Bochner–Riesz multiplier*. Not smooth away from the origin: it fails to be  $C^s$  across the sphere  $|\xi| = 1$  unless  $\delta$  is large. For  $\delta > s$  the condition holds and Theorem 3.5.1 applies — but that is far weaker than Theorem 5.8.4, which uses the kernel directly. The multiplier theorem is blunt here because it ignores the *curvature* of the sphere across which the symbol degenerates.

*The pattern.* Homogeneous symbols pass automatically; symbols singular along a surface do not. That division is the boundary between Volume 3 and Volume 5, and (d) is the first hint that curvature will have to enter.

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## Drills

**Drill 3.5.1.** Verify (3.5.4) in detail for  $|\alpha| = 1$ , and check the claim that  $\xi m_j$  satisfies the same bounds with an extra factor  $2^j$ .

**Drill 3.5.2.** Show that  $|x|^s \lesssim_{n,s} \sum_{|\alpha|=s} |x^\alpha|$  for integer  $s \geq 0$ , by the multinomial theorem.

**Drill 3.5.3.** Construct  $\psi$  with the properties used at the start of §5.2: take  $\varphi \in C_c^\infty$  radial with  $\varphi = 1$  on  $|\xi| \leq 1$  and  $\text{supp } \varphi \subseteq \{|\xi| < 2\}$ , and set  $\psi(\xi) = \varphi(\xi) - \varphi(2\xi)$ .

**Drill 3.5.4.** Show that  $m(\xi) = e^{i|\xi|} |\xi|^{-a}$  satisfies (3.5.1) for no  $a$ , and explain which derivative first fails.

**Where we are.** The derivative count  $s = \lfloor n/2 \rfloor + 1$  is exactly what Plancherel needs: the proof estimates  $\|K_j\|_{L^1}$  by Cauchy–Schwarz against a weight  $|x|^s$ , and  $|x|^{-s}$  is square-integrable at infinity in  $\mathbb{R}^n$  precisely when  $2s > n$ . Any smaller count fails at that step, and the smallest integer with  $2s > n$  is  $\lfloor n/2 \rfloor + 1$ .

## Notes

Mikhlin proved the theorem in 1956 under the hypothesis (3.5.1) for  $|\alpha| \leq n$ ; Hörmander reduced this to  $\lfloor n/2 \rfloor + 1$  in 1960 by the  $L^2$ -based argument given here. The reduction matters: for large  $n$  the difference between  $n$  and  $n/2$  derivatives is the difference between a usable and an unusable hypothesis.



## Chapter 3.6

# Littlewood–Paley Theory

Volume 2 proved that the frequency pieces  $P_N f$  are exactly orthogonal in  $L^2$ , and observed that the corresponding  $L^p$  statement is what the later volumes actually need. This chapter proves it. The mechanism is to randomise the signs of the pieces: for each choice of signs the resulting operator is a Hörmander–Mikhlin multiplier, uniformly, and Khintchine’s inequality converts the average over signs into the square function.

### 3.6.1 Vector-valued extension

We record first a general principle, which is used twice below and again in Chapter 3.8.

**Theorem 3.6.1** (Marcinkiewicz–Zygmund). *Let  $1 \leq p < \infty$  and let  $T$  be a bounded linear operator on  $L^p(\mathbb{R}^n)$ . Then for all  $f_1, \dots, f_N \in L^p$ ,*

$$\left\| \left( \sum_{j=1}^N |Tf_j|^2 \right)^{1/2} \right\|_p \lesssim_p \|T\|_{L^p \rightarrow L^p} \left\| \left( \sum_{j=1}^N |f_j|^2 \right)^{1/2} \right\|_p,$$

*with a constant depending only on  $p$ . The same holds for  $N = \infty$ .*

*Idea of the proof.* Khintchine’s inequality (Theorem 2.6.5 (Khintchine)) says that for scalars,  $(\sum_j |a_j|^2)^{1/2}$  is comparable to the  $L^p$  average over random signs of  $|\sum_j \varepsilon_j a_j|$ . Applying this *pointwise in  $x$*  converts a square function into a randomised sum, on which the scalar operator bound may be used directly. The only other ingredient is Fubini, to exchange the  $x$ -integral with the average over signs.

*Proof.* Let  $r_j$  be the Rademacher functions on  $[0, 1]$  and write  $\mathbb{E}_t$  for  $\int_0^1 \cdot dt$ . By Khintchine, applied for each fixed  $x$  to the scalars  $a_j = Tf_j(x)$ ,

$$\left( \sum_j |Tf_j(x)|^2 \right)^{p/2} \approx_p \mathbb{E}_t \left| \sum_j r_j(t) Tf_j(x) \right|^p.$$

Integrate in  $x$  and apply Tonelli:

$$\left\| \left( \sum_j |Tf_j|^2 \right)^{1/2} \right\|_p^p \approx_p \mathbb{E}_t \int_{\mathbb{R}^n} \left| T \left( \sum_j r_j(t) f_j \right) (x) \right|^p dx,$$

using linearity of  $T$  to pull it outside the sum. By the hypothesis, the inner integral is at most  $\|T\|^p \left\| \sum_j r_j(t) f_j \right\|_p^p$ , so

$$\left\| \left( \sum_j |Tf_j|^2 \right)^{1/2} \right\|_p^p \lesssim_p \|T\|^p \mathbb{E}_t \int_{\mathbb{R}^n} \left| \sum_j r_j(t) f_j(x) \right|^p dx.$$

Applying Tonelli and Khintchine once more, this time in the reverse direction, the right-hand side is  $\approx_p \|T\|^p \left\| \left( \sum_j |f_j|^2 \right)^{1/2} \right\|_p^p$ .  $\square$

**Pitfall:** this is a theorem about  $\ell^2$ , not about  $\ell^q$

Theorem 3.6.1 is special to the exponent 2 in the square function, because Khintchine’s inequality is. The analogous statement with  $(\sum_j |Tf_j|^q)^{1/q}$  is false in general for  $q \neq 2$ , and the correct substitutes — the Fefferman–Stein vector-valued maximal inequality for  $q > 1$ , for instance — require real work. Whenever a square function appears in this series, the exponent 2 is doing something.

## 3.6.2 The square function theorem

**Definition 3.6.2.** With  $\psi$  as in Chapter 3.5, let  $\widehat{P_j f} = \psi(\xi/2^j) \widehat{f}$  and define the *Littlewood–Paley square function*

$$Sf = \left( \sum_{j \in \mathbb{Z}} |P_j f|^2 \right)^{1/2}.$$

**Theorem 3.6.3.** For  $1 < p < \infty$  and  $f \in L^p(\mathbb{R}^n)$ ,

$$\|Sf\|_{L^p} \approx_{n,p} \|f\|_{L^p}.$$

*Proof. Step 1. Randomised multipliers are uniformly bounded.*

For a choice of signs  $\varepsilon = (\varepsilon_j)_{j \in \mathbb{Z}} \in \{-1, 1\}^{\mathbb{Z}}$  put

$$m_\varepsilon(\xi) = \sum_{j \in \mathbb{Z}} \varepsilon_j \psi(\xi/2^j).$$



We claim  $m_\varepsilon$  satisfies (3.5.1) with a constant  $B$  independent of  $\varepsilon$ . Indeed, for a given  $\xi \neq 0$  at most two terms of the sum are nonzero, those with  $2^j \approx |\xi|$ ; and for each,

$$|\partial^\alpha [\psi(\xi/2^j)]| = 2^{-j|\alpha|} |(\partial^\alpha \psi)(\xi/2^j)| \lesssim_\alpha |\xi|^{-|\alpha|},$$

using  $2^j \approx |\xi|$  on the support. Summing the at most two terms gives the claim with  $B = B(n, \psi)$ .

By Theorem 3.5.1,  $\|T_{m_\varepsilon}\|_{L^p \rightarrow L^p} \lesssim_{n,p} 1$  uniformly in  $\varepsilon$ . Note  $T_{m_\varepsilon} f = \sum_j \varepsilon_j P_j f$ .

**Step 2.** *The upper bound.*

By Khintchine applied pointwise and Tonelli, exactly as in the proof of Theorem 3.6.1,

$$\|\mathcal{S}f\|_p^p \approx_p \mathbb{E}_t \left\| \sum_j r_j(t) P_j f \right\|_p^p = \mathbb{E}_t \|T_{m_{\varepsilon(t)}} f\|_p^p \lesssim_{n,p} \|f\|_p^p,$$

where  $\varepsilon(t) = (r_j(t))_j$  and Step 1 was applied for each  $t$ .

**Step 3.** *The lower bound, by duality.*

Let  $\tilde{\psi} \in C_c^\infty(\{\frac{1}{4} < |\xi| < 4\})$  equal 1 on  $\text{supp } \psi$ , and let  $\tilde{P}_j$  be the corresponding projections, so that  $\tilde{P}_j P_j = P_j$ . Since  $\sum_j P_j = \text{Id}$  on  $L^p$  — the partial sums are the multipliers  $\sum_{|j| \leq J} \psi(\xi/2^j)$ , which satisfy (3.5.1) uniformly by the argument of Step 1 and converge boundedly pointwise to 1 — we have, for  $f \in L^p$  and  $g \in L^{p'}$ ,

$$\int f \bar{g} = \sum_j \int P_j f \bar{g} = \sum_j \int P_j f \overline{\tilde{P}_j g},$$

the last step because  $\tilde{P}_j$  is self-adjoint and  $\tilde{P}_j P_j = P_j$ . By Cauchy–Schwarz in  $j$  and then Hölder in  $x$ ,

$$\left| \int f \bar{g} \right| \leq \int \mathcal{S}f \cdot \tilde{\mathcal{S}}g \leq \|\mathcal{S}f\|_p \|\tilde{\mathcal{S}}g\|_{p'} \lesssim_{n,p} \|\mathcal{S}f\|_p \|g\|_{p'},$$

where  $\tilde{\mathcal{S}}$  is the square function built from  $\tilde{P}_j$  and the last step is Step 2 applied at exponent  $p'$  — legitimate since  $\tilde{\psi}$  has the same structure as  $\psi$ , with each  $\xi$  in the support of at most a bounded number of the  $\tilde{\psi}(\cdot/2^j)$ .

Taking the supremum over  $\|g\|_{p'} \leq 1$  and using the duality formula (Corollary 1.8.10 (norms by testing)) gives  $\|f\|_p \lesssim_{n,p} \|\mathcal{S}f\|_p$ .  $\square$

**Where we are.** The theorem says that the  $L^p$  norm cannot tell the difference between  $f$  and a function whose frequency pieces have had their phases randomised. That is a strong statement — it fails for  $p = 1$  and  $p = \infty$  — and it is the precise sense in which frequency decomposition is “almost orthogonal” outside  $L^2$ .

Two comparisons are worth making. Volume 2, Prop. 6.10 proved the case  $p = 2$ , where the statement is an identity and the proof is Plancherel. And Volume 2, Worked Example 8.16 previewed  $\ell^2$  decoupling, which asserts something superficially similar for pieces localised to *curved* caps rather than annuli. The difference is decisive: here the pieces are separated in frequency by a factor 2, and the sharp cutoffs are harmless because Theorem 3.5.1 applies. In decoupling the caps are adjacent, the multipliers are not Hörmander–Mikhlin, and the truth of the statement depends on curvature. Theorem 3.6.3 is easy for exactly the reason decoupling is hard.

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WORKED EXAMPLE 3.6.1— TIER B **Bernstein revisited, and the standard estimate**

Combining Theorem 3.6.3 with Volume 2’s Bernstein inequality gives the estimate used on nearly every page of Volumes 12 and 13.

**Proposition 3.6.4.** *Let  $1 < p < \infty$  and  $s \geq 0$ . For  $f \in L^p(\mathbb{R}^n)$  with all  $P_j f$  defined,*

$$\left\| \left( \sum_j 2^{2js} |P_j f|^2 \right)^{1/2} \right\|_p \approx_{n,p,s} \|(-\Delta)^{s/2} f\|_{L^p},$$

where  $(-\Delta)^{s/2}$  is the multiplier operator with symbol  $(2\pi|\xi|)^s$ .

*Proof.* Apply Theorem 3.6.3 to  $g = (-\Delta)^{s/2} f$ . Its pieces are  $P_j g = T_{\mu_j} P_j f$  where  $\mu_j(\xi) = (2\pi|\xi|)^s \tilde{\psi}(\xi/2^j)$ ; since  $\mu_j$  is supported where  $|\xi| \approx 2^j$  and satisfies  $|\partial^\alpha \mu_j| \lesssim 2^{js} |\xi|^{-|\alpha|}$ , the rescaled multiplier  $2^{-js} \mu_j$  satisfies (3.5.1) uniformly in  $j$ . Hence  $|P_j g|$  is comparable to  $2^{js} |P_j f|$  in the vector-valued sense, by Theorem 3.6.1 applied to the family, and Theorem 3.6.3 converts both sides.  $\square$

*The slogan.*  $L^p$  Sobolev norms are square functions with dyadic weights. Volume 2 could define  $H^s$  only for  $p = 2$ , because only there was Plancherel available; Theorem 3.6.3 removes that restriction and is the reason Triebel–Lizorkin and Besov spaces exist. Volume 4 develops them.

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WORKED EXAMPLE 3.6.2— TIER C: RESEARCH TASTE **What Littlewood–Paley cannot do**

It is worth being precise about the limits of this chapter’s technique, because the reader will meet decompositions in later volumes that look similar and behave completely differently.

(a) *Lacunary is essential.* Theorem 3.6.3 decomposes frequency space into annuli  $|\xi| \approx 2^j$ , whose ratios are bounded away from 1. The analogous statement for the *equally spaced* intervals  $[j, j+1)$ ,  $j \in \mathbb{Z}$ , in dimension one is a theorem — the Littlewood–Paley theorem for intervals of equal length, due to Rubio de Francia — but only as a one-sided inequality, and its proof is substantially harder. The two-sided statement is false.

(b) *In  $\mathbb{R}^n$ , the shape of the pieces matters.* Suppose one decomposes frequency space into congruent *cubes* of side 1. The corresponding square function satisfies  $\|\mathcal{S}f\|_p \lesssim \|f\|_p$  for  $2 \leq p < \infty$ , again by Rubio de Francia, but the sharp cutoff to a cube is a product of one-dimensional half-space multipliers, and so is bounded on every  $L^p$ . Replace the cubes by *balls* and the sharp cutoffs become unbounded, by Chapter 3.8. The difference between a cube and a ball is that the boundary of a cube is flat.

(c) *And in the curved setting, everything changes.* Volume 13’s decoupling theorem concerns pieces of a  $\delta^{-1/2}$ -neighbourhood of a paraboloid, partitioned into  $\delta^{-(n-1)/2}$  caps. These are adjacent, not lacunary; the sharp cutoffs are not Hörmander–Mikhlin multipliers; and the inequality that holds is not  $\|\mathcal{S}f\|_p \approx \|f\|_p$  but

$$\left\| \sum_{\theta} f_{\theta} \right\|_p \lesssim_{\varepsilon} \delta^{-\varepsilon} \left( \sum_{\theta} \|f_{\theta}\|_p^2 \right)^{1/2},$$

with the  $\ell^2$  sum *outside* the norm rather than inside. That is a genuinely weaker conclusion — by Minkowski’s inequality the square function form would imply it for  $p \geq 2$  — and it is weaker precisely because the caps are adjacent, so no multiplier theorem applies and only the geometry of the dual tubes is available.

*The moral.* This chapter’s technique is: decompose, check the multiplier hypothesis, randomise, apply Khintchine. It works whenever the pieces are separated enough for Theorem 3.5.1 to apply to their randomised sums. The subject of Volumes 12 and 13 begins exactly where that fails.

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**WORKED EXAMPLE 3.6.3— TIER B Using Littlewood–Paley: the standard three moves**

Theorem 3.6.3 says  $\|f\|_{L^p} \approx \|(\sum_j |\Delta_j f|^2)^{1/2}\|_{L^p}$  for  $1 < p < \infty$ . Almost every application uses it in one of three ways, and it is worth having them separated.

*Move 1: prove an estimate blockwise.* To bound  $\|Tf\|_{L^p}$ , bound each  $\|\Delta_j Tf\|_{L^p}$  and reassemble. This works when  $T$  nearly commutes with  $\Delta_j$  — for instance when  $T$  is a Fourier multiplier, so that  $\Delta_j T = T \Delta_j$  exactly. The proof of Theorem 3.5.1 is this move.

*Move 2: replace a sum by a square function.* A sum  $\sum_j a_j$  of frequency-separated pieces obeys no triangle-inequality improvement in  $L^p$ ; but the square function does, by orthogonality at  $p = 2$  and by Theorem 3.6.3 for other  $p$ . This is what converts an  $\ell^2$  bound into an  $L^p$  bound and is the ancestor of the decoupling inequalities of Volume 13.

*Move 3: interpolate against the endpoint.* Since Theorem 3.6.3 fails at  $p = 1$  and  $p = \infty$ , an argument needing those endpoints must substitute  $H^1$  and BMO, which is exactly why Volume 4 exists. The square function characterises  $H^1$  (Theorem 4.2.5 is the operator version), and the substitution is standard.

*A worked instance of Move 1.* Let  $m$  be a bounded multiplier with  $|\partial^\alpha m| \lesssim |\xi|^{-|\alpha|}$ . Then  $m\Delta_j$  has symbol supported where  $|\xi| \approx 2^j$ , and rescaling that block to unit frequency turns the hypothesis into a bound on finitely many derivatives of a fixed bump — uniformly in  $j$ . So each block is handled once and the sum reassembles. That rescaling is the only idea in Theorem 3.5.1, and recognising it as Move 1 makes the proof memorable.

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## Drills

**Drill 3.6.1.** Verify that  $\sum_{|j| \leq J} \psi(\xi/2^j)$  satisfies (3.5.1) uniformly in  $J$ , as claimed in Step 3 of Theorem 3.6.3.

**Drill 3.6.2.** Show that Theorem 3.6.3 fails at  $p = \infty$ : find  $f \in L^\infty(\mathbb{R})$  with  $\mathcal{S}f \notin L^\infty$ .

**Drill 3.6.3.** Deduce from Theorem 3.6.1 that if  $T$  is bounded on  $L^p$  then so is the operator acting on  $\ell^2$ -valued functions componentwise, with comparable norm. Where is linearity of  $T$  used?

**Drill 3.6.4.** Prove the Minkowski inequality claim in Worked Example 3.6.2(c): for  $p \geq 2$ ,  $(\sum_\theta \|f_\theta\|_p^2)^{1/2} \leq \|(\sum_\theta |f_\theta|^2)^{1/2}\|_p$ .

**Where we are.** One structural remark. Theorem 3.6.3 is an equivalence of norms, and both directions are used, but for different purposes.

The direction  $\|Sf\|_{L^p} \lesssim \|f\|_{L^p}$  — the square function is no larger — is what lets an estimate be proved blockwise, as in Worked Example 3.6.3 Move 1. The reverse,  $\|f\|_{L^p} \lesssim \|Sf\|_{L^p}$ , is what lets a bound on the blocks be assembled back into a bound on  $f$ . An argument using only one direction is usually doing something simpler than it appears, and it is worth checking which is in play.

Both fail at  $p = 1$  and  $p = \infty$ , which is the recurring theme of this volume and the reason for Volume 4.

## Notes

Littlewood and Paley proved the one-dimensional case in the 1930s by complex analysis; the real-variable proof via multipliers and Khintchine is Stein's. Theorem 3.6.1 is from Marcinkiewicz and Zygmund's 1939 paper.

Rubio de Francia's theorem, quoted in Worked Example 3.6.2, is from 1985 and is not proved in this series.



## Chapter 3.7

# BMO and the John–Nirenberg Inequality

Every theorem of this volume has failed at  $p = \infty$ . The Hilbert transform of a bounded function need not be bounded (Worked Example 3.3.1); the elliptic estimate fails there (Worked Example 3.4.1); and in each case the failure was logarithmic. This chapter identifies the space in which the failure is exactly accounted for.

### 3.7.1 Definition and the model example

**Definition 3.7.1.** For  $f \in L^1_{\text{loc}}(\mathbb{R}^n)$  and a cube  $Q$  write  $\langle f \rangle_Q = |Q|^{-1} \int_Q f$ . The BMO *seminorm* is

$$\|f\|_{\text{BMO}} = \sup_Q \frac{1}{|Q|} \int_Q |f - \langle f \rangle_Q|,$$

the supremum over all cubes with sides parallel to the axes, and  $\text{BMO}(\mathbb{R}^n) = \{f : \|f\|_{\text{BMO}} < \infty\}$ .

Constants have BMO seminorm zero, so  $\|\cdot\|_{\text{BMO}}$  is a norm only on BMO modulo constants. Clearly  $\|f\|_{\text{BMO}} \leq 2\|f\|_{\infty}$ , so  $L^{\infty} \subseteq \text{BMO}$ .

**Proposition 3.7.2.**  $f(x) = \log|x|$  lies in  $\text{BMO}(\mathbb{R}^n)$ , and is not in  $L^{\infty}$ .

*Proof.* By translation the supremum need only be checked over cubes  $Q = Q(c, \ell)$  of centre  $c$  and side  $\ell$ ; and by the dilation invariance  $\log|tx| = \log|x| + \log t$ , which changes  $f$  by a constant and so does not change any of the quantities  $\int_Q |f - \langle f \rangle_Q|/|Q|$ , we may assume  $\ell = 1$ .

If  $|c| \geq 2\sqrt{n}$  then  $|x| \approx |c|$  on  $Q$ , and  $\log|x|$  has gradient  $|x|^{-1} \lesssim |c|^{-1}$  there, so the oscillation of  $f$  on  $Q$  is  $\lesssim |c|^{-1} \lesssim 1$ .

If  $|c| < 2\sqrt{n}$  then  $Q \subseteq B(0, R)$  for a fixed  $R = R(n)$ , and using  $\langle f \rangle_Q$  replaced by the constant 0 — legitimate since  $|Q|^{-1} \int_Q |f - \langle f \rangle_Q| \leq 2|Q|^{-1} \int_Q |f - a|$  for any constant  $a$  — we need only bound  $\int_{B(0,R)} |\log|x|| \, dx$ , which is finite by polar coordinates since  $\int_0^R r^{n-1} |\log r| \, dr < \infty$ .  $\square$

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WORKED EXAMPLE 3.7.1— TIER A BMO is **strictly between**  $L^\infty$  **and**  $\bigcap_p L^p$

Proposition 3.7.2 shows  $L^\infty \subsetneq \text{BMO}$ . Conversely  $\text{BMO} \not\subseteq L^\infty$  by the same example, and BMO functions are locally in every  $L^p$  — a consequence of Theorem 3.7.3 below — but not globally, as  $\log|x|$  shows.

Note also what is *not* in BMO: the function  $\log|x| \cdot \text{sgn}(x_1)$  on  $\mathbb{R}^n$  is not, even though each factor is bounded or in BMO. So BMO is not an algebra, and multiplying by a bounded function does not preserve it. This is a standard trap.

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### 3.7.2 The John–Nirenberg inequality

The definition controls the *mean* oscillation. The following says that this automatically controls the *distribution* of the oscillation, exponentially.

**Theorem 3.7.3** (John–Nirenberg). *There are dimensional constants  $C, c > 0$  such that for every  $f \in \text{BMO}(\mathbb{R}^n)$ , every cube  $Q$  and every  $\lambda > 0$ ,*

$$m\left(\{x \in Q : |f(x) - \langle f \rangle_Q| > \lambda\}\right) \leq C |Q| \exp\left(-\frac{c\lambda}{\|f\|_{\text{BMO}}}\right).$$

Consequently  $f \in L^p_{\text{loc}}$  for every  $p < \infty$ , with  $(|Q|^{-1} \int_Q |f - \langle f \rangle_Q|^p)^{1/p} \lesssim_{n,p} \|f\|_{\text{BMO}}$ .

*Idea of the proof.* Run the Calderón–Zygmund decomposition of Theorem 3.1.6 *inside*  $Q$ , on the function  $f - \langle f \rangle_Q$ , at height  $\lambda_0 = 2^{n+1} \|f\|_{\text{BMO}}$ . The selected cubes  $Q_j$  have  $\sum |Q_j| \leq \frac{1}{2} |Q|$  — a definite proportion is discarded — and on each of them the average of  $f$  has moved by at most  $O(\lambda_0)$  from  $\langle f \rangle_Q$ . So the set where the oscillation exceeds  $2\lambda_0$  is contained in the union of the  $Q_j$ , and inside each  $Q_j$  the same argument may be repeated. Iterating  $k$  times shows: exceeding  $k\lambda_0$  requires surviving  $k$  rounds, each of which costs a factor  $\frac{1}{2}$  in measure. That is exponential decay, and the constant  $c$  is  $\log 2$  divided by  $\lambda_0$ .

*Proof.* Normalise  $\|f\|_{\text{BMO}} = 1$  and fix a cube  $Q$ ; replacing  $f$  by  $f - \langle f \rangle_Q$  we may assume  $\langle f \rangle_Q = 0$ . Set  $\lambda_0 = 2^{n+1}$ .



**Step 1.** *One round of decomposition.*

Apply Theorem 3.1.6 to  $f\mathbf{1}_Q$  at height  $\lambda_0$ , using the dyadic subcubes of  $Q$  rather than the standard grid — the proof is identical, since the subcubes of  $Q$  obtained by repeated bisection satisfy Lemma 3.1.2. We obtain disjoint dyadic subcubes  $\{Q_j\}$  of  $Q$  with

$$\lambda_0 < \langle |f| \rangle_{Q_j} \leq 2^n \lambda_0, \quad |f| \leq \lambda_0 \text{ a.e. off } \bigcup_j Q_j, \quad \sum_j |Q_j| \leq \frac{1}{\lambda_0} \int_Q |f|.$$

Since  $\langle f \rangle_Q = 0$  and  $\|f\|_{\text{BMO}} = 1$ , the last integral is at most  $|Q|$ , so

$$\sum_j |Q_j| \leq \frac{|Q|}{\lambda_0} = 2^{-n-1} |Q| \leq \frac{1}{2} |Q|. \quad (3.7.1)$$

Also  $|\langle f \rangle_{Q_j}| \leq \langle |f| \rangle_{Q_j} \leq 2^n \lambda_0$ .

**Step 2.** *Iterate.*

Let  $F(\lambda) = \sup_Q |Q|^{-1} m(\{x \in Q : |f - \langle f \rangle_Q| > \lambda\})$ , the supremum over all cubes and all  $f$  with  $\|f\|_{\text{BMO}} \leq 1$ . If  $\lambda > 2^n \lambda_0$  then, by Step 1,  $\{x \in Q : |f| > \lambda\}$  is contained in  $\bigcup_j Q_j$  up to a null set, and within  $Q_j$ ,

$$|f(x)| > \lambda \implies |f(x) - \langle f \rangle_{Q_j}| > \lambda - 2^n \lambda_0.$$

Therefore

$$m(\{x \in Q : |f| > \lambda\}) \leq \sum_j m(\{x \in Q_j : |f - \langle f \rangle_{Q_j}| > \lambda - 2^n \lambda_0\}) \leq F(\lambda - 2^n \lambda_0) \sum_j |Q_j|,$$

and by (3.7.1) this is at most  $\frac{1}{2} |Q| F(\lambda - 2^n \lambda_0)$ . Taking the supremum over  $Q$  and  $f$ ,

$$F(\lambda) \leq \frac{1}{2} F(\lambda - \mu), \quad \mu := 2^n \lambda_0 = 2^{2n+1}, \quad (3.7.2)$$

for every  $\lambda > \mu$ .

**Step 3.** *Solve the recursion.*

Since  $F \leq 1$  always, iterating (3.7.2)  $k$  times gives  $F(\lambda) \leq 2^{-k}$  whenever  $\lambda > k\mu$ . Given  $\lambda > 0$ , take  $k = \lfloor \lambda/\mu \rfloor$ ; then

$$F(\lambda) \leq 2^{-\lfloor \lambda/\mu \rfloor} \leq 2 \cdot 2^{-\lambda/\mu} = 2 \exp\left(-\frac{\lambda \log 2}{\mu}\right),$$

which is the claim with  $C = 2$  and  $c = 2^{-2n-1} \log 2$ . Undoing the normalisation  $\|f\|_{\text{BMO}} = 1$  replaces  $\lambda$  by  $\lambda/\|f\|_{\text{BMO}}$ .

**Step 4.** *Local integrability.*

By the layer-cake formula (Worked Example 1.5.1 (the layer-cake formula)),

$$\frac{1}{|Q|} \int_Q |f - \langle f \rangle_Q|^p = \frac{p}{|Q|} \int_0^\infty \lambda^{p-1} m(\{x \in Q : |f - \langle f \rangle_Q| > \lambda\}) d\lambda \leq Cp \int_0^\infty \lambda^{p-1} e^{-c\lambda} d\lambda,$$

finite for every  $p < \infty$ , after normalising  $\|f\|_{\text{BMO}} = 1$ .  $\square$

**Where we are.** Theorem 3.7.3 is the reason BMO is a useful space rather than a definition. It says that the single scalar  $\|f\|_{\text{BMO}}$  — an  $L^1$ -type quantity — in fact controls every  $L^p$  oscillation, with constants growing only like  $p$ . Equivalently, BMO functions are “ $L^\infty$  up to exponentially small exceptional sets”.

Note also where the halving (3.7.1) came from: the threshold  $\lambda_0$  was chosen *larger* than the BMO norm by the factor  $2^{n+1}$ , so that Theorem 3.1.6(iii) discarded at most half the cube. That freedom to choose the threshold is available because the hypothesis is a supremum over all cubes, and it is the entire trick.

### 3.7.3 Singular integrals map $L^\infty$ into BMO

**Theorem 3.7.4.** *Let  $T$  satisfy the hypotheses of Theorem 3.2.3. Then for  $f \in L^\infty(\mathbb{R}^n)$  with compact support,*

$$\|Tf\|_{\text{BMO}} \lesssim_n (A + B_2) \|f\|_\infty.$$

*Proof.* Normalise  $A + B_2 \leq 1$  and  $\|f\|_\infty = 1$ , and fix a cube  $Q$  with centre  $c$  and side  $\ell$ . We must bound  $|Q|^{-1} \int_Q |Tf - \langle Tf \rangle_Q|$  by a constant.

**Step 1.** *Split  $f$  at the scale of the cube.*

Let  $Q^* = 2\sqrt{n}Q$  be the concentric dilate and write

$$f = \underbrace{f\mathbf{1}_{Q^*}}_{f_1} + \underbrace{f\mathbf{1}_{(Q^*)^c}}_{f_2}.$$

The point of the dilation factor  $2\sqrt{n}$  is that every  $x \in Q$  is at distance at least  $\frac{\sqrt{n}}{2}\ell$  from  $\text{supp } f_2$ , so on  $Q$  the operator  $T$  acts on  $f_2$  through its kernel and nothing is singular.

**Step 2.** *The local piece, by  $L^2$  boundedness.*

By Cauchy–Schwarz on  $Q$  and then the  $L^2$  bound,

$$\frac{1}{|Q|} \int_Q |Tf_1| \leq \left( \frac{1}{|Q|} \int_Q |Tf_1|^2 \right)^{1/2} \leq |Q|^{-1/2} \|Tf_1\|_{L^2} \leq |Q|^{-1/2} B_2 \|f_1\|_{L^2}.$$

Now  $\|f_1\|_{L^2} \leq \|f\|_\infty |Q^*|^{1/2} = |Q^*|^{1/2}$ , and  $|Q^*| = (2\sqrt{n})^n |Q|$ , so the two volume factors cancel and

$$\frac{1}{|Q|} \int_Q |Tf_1| \leq (2\sqrt{n})^{n/2} B_2 \lesssim_n 1.$$

This is the step that would fail without the  $L^2$  hypothesis, and it is the only place it is used.

**Step 3.** *The far piece oscillates little, by Hörmander's condition.*

For  $x \in Q$  the kernel representation (3.2.3) applies to  $f_2$ , so

$$Tf_2(x) - Tf_2(c) = \int_{(Q^*)^c} (K(x-y) - K(c-y))f(y) dy.$$

Substitute  $z = c - y$  and  $w = c - x$ . For  $y \notin Q^*$  and  $x \in Q$  we have  $|z| > \sqrt{n}\ell$  and  $|w| \leq \frac{\sqrt{n}}{2}\ell$ , hence  $|z| > 2|w|$  — exactly the region in which Hörmander's condition (3.2.2) applies. Using  $\|f\|_\infty \leq 1$ ,

$$|Tf_2(x) - Tf_2(c)| \leq \int_{|z|>2|w|} |K(z-w) - K(z)| dz \leq A \leq 1,$$

uniformly in  $x \in Q$ .

**Step 4.** *Combine, using a constant in place of the average.*

Replacing  $\langle Tf \rangle_Q$  by any fixed constant costs at most a factor 2 (Drill 7.1); take the constant  $Tf_2(c)$ . Then

$$\frac{1}{|Q|} \int_Q |Tf - \langle Tf \rangle_Q| \leq \frac{2}{|Q|} \int_Q |Tf - Tf_2(c)| \leq \frac{2}{|Q|} \int_Q |Tf_1| + \frac{2}{|Q|} \int_Q |Tf_2 - Tf_2(c)|,$$

and Steps 2 and 3 bound the two terms by  $\lesssim_n 1$  each. Taking the supremum over cubes  $Q$  gives  $\|Tf\|_{\text{BMO}} \lesssim_n 1$ , which after undoing the normalisation is the claim.  $\square$

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WORKED EXAMPLE 3.7.2— TIER B **The endpoint replacements, tabulated**

The following summarises the situation at the two ends of the  $L^p$  scale for a Calderón–Zygmund operator  $T$ .

| Claim                                     | Status       | Reference            |
|-------------------------------------------|--------------|----------------------|
| $T : L^p \rightarrow L^p, 1 < p < \infty$ | true         | Theorem 3.2.3        |
| $T : L^1 \rightarrow L^1$                 | <b>false</b> | Worked Example 3.2.2 |
| $T : L^1 \rightarrow L^{1,\infty}$        | true         | Theorem 3.2.3(i)     |
| $T : L^\infty \rightarrow L^\infty$       | <b>false</b> | Worked Example 3.3.1 |
| $T : L^\infty \rightarrow \text{BMO}$     | true         | Theorem 3.7.4        |
| $T : H^1 \rightarrow L^1$                 | true         | Volume 4             |

The pattern is that  $L^1$  must be replaced by the Hardy space  $H^1$  on the input side and  $L^\infty$  by BMO on the output side, and that these two replacements are dual to one another — the Fefferman duality theorem  $(H^1)^* = \text{BMO}$ . Volume 4 develops both spaces; here we have proved the half that requires no new machinery.

Note also that Theorem 3.7.4 plus Theorem 3.7.3 recovers the  $L^p$  bounds for large  $p$  by interpolation, via the Fefferman–Stein sharp maximal function. That route is developed in Volume 4; we obtained the  $L^p$  bounds directly from Theorem 3.2.3 instead, which is shorter.

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**WORKED EXAMPLE 3.7.3— TIER B Reading John–Nirenberg**

Theorem 3.7.3 says that a BMO function’s distribution decays exponentially on every cube:  $|\{x \in Q : |f - \langle f \rangle_Q| > \lambda\}| \lesssim e^{-c\lambda/\|f\|_{\text{BMO}}} |Q|$ . Three consequences are used repeatedly and are worth deriving once.

(a) *All the  $\text{BMO}_p$  norms agree.* For  $1 \leq p < \infty$  define  $\|f\|_{\text{BMO}_p} = \sup_Q (\langle |f - \langle f \rangle_Q|^p \rangle_Q)^{1/p}$ . Integrating the exponential bound,

$$\langle |f - \langle f \rangle_Q|^p \rangle_Q = \frac{1}{|Q|} \int_0^\infty p\lambda^{p-1} |\{ |f - \langle f \rangle_Q| > \lambda \}| d\lambda \lesssim_p \|f\|_{\text{BMO}}^p,$$

so  $\|f\|_{\text{BMO}_p} \approx_p \|f\|_{\text{BMO}}$ . The case  $p = 2$  is the one used in interpolation, and it is not obvious from the definition.

(b) *BMO functions are exponentially integrable.* Taking  $\lambda = \varepsilon\|f\|_{\text{BMO}}t$  and summing shows  $e^{\varepsilon|f|/\|f\|_{\text{BMO}}}$  is locally integrable for small  $\varepsilon$ . So BMO sits just outside  $L^\infty$  — functions may be unbounded, but only logarithmically, and  $\log \frac{1}{|x|}$  is the model.

(c) *The self-improving character.* The hypothesis is an  $L^1$  bound on oscillation; the conclusion is exponential. That gain from nothing is the signature of a stopping-time argument, and indeed the proof of Theorem 3.7.3 iterates the Calderón–Zygmund decomposition of Worked Example 3.1.5 at levels  $2^k$ .

*The general lesson.* A local  $L^1$  condition, iterated across scales, upgrades itself. The same phenomenon appears in Gehring’s lemma for reverse Hölder inequalities and in the  $A_p$  theory of weights, neither of which this series treats — but recognising the mechanism makes those literatures accessible.

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## Drills

**Drill 3.7.1.** Show that  $\|f\|_{\text{BMO}} \approx \sup_Q \inf_{a \in \mathbb{C}} |Q|^{-1} \int_Q |f - a|$ , with constants between 1 and 2.

**Drill 3.7.2.** Show that  $\log|x| \cdot \text{sgn}(x_1)$  is not in  $\text{BMO}(\mathbb{R}^n)$ , by testing on cubes straddling the hyperplane  $x_1 = 0$ .

**Drill 3.7.3.** Deduce from Theorem 3.7.3 that if  $\|f\|_{\text{BMO}}$  is small enough then  $e^{|f|}$  is locally integrable, and find the threshold.

**Drill 3.7.4.** Verify that  $\mathcal{H}\mathbf{1}_{[0,1]}$ , computed in Worked Example 3.3.1, has finite BMO norm, directly from the formula.

**Where we are.** BMO is the right endpoint substitute for  $L^\infty$ , and the reason is worth stating once: it is the smallest space containing  $L^\infty$  that is preserved by Calderón–Zygmund operators and is dual to  $H^1$ . The first property is Theorem 3.7.4; the second is Volume 4, Theorem 4.3.5. Together they make the pair  $(H^1, \text{BMO})$  the correct replacement for  $(L^1, L^\infty)$  throughout the theory, and every interpolation argument in Volume 4 uses them in that role.

## Notes

BMO was introduced by John and Nirenberg in 1961, in the context of elasticity; Theorem 3.7.3 is from that paper. Theorem 3.7.4 is due to Peetre, Spanne and Stein independently. Fefferman’s duality theorem  $(H^1)^* = \text{BMO}$  is from 1971 — the same year as the ball multiplier theorem of the next chapter.



## Chapter 3.8

# The Ball Multiplier

Everything in this volume has been positive: operators are bounded, under verifiable hypotheses, by a machine. This chapter proves a negative theorem, and the proof uses a Besicovitch set. It is the first point in the series where the geometry of Volume 1 and the analysis of Volumes 2 and 3 meet, and it is the historical origin of the entire subject that follows.

### 3.8.1 The question

**Definition 3.8.1.** The *ball multiplier* is  $T_B f = \mathcal{F}^{-1}(\mathbf{1}_{B(0,1)} \widehat{f})$ .

By Proposition 2.6.2 (the  $L^2$  multiplier norm),  $T_B$  is bounded on  $L^2(\mathbb{R}^n)$  with norm 1. The question, posed by Bochner in the 1930s and open for forty years, is whether it is bounded on  $L^p$  for other  $p$ .

In dimension one the answer is yes for all  $1 < p < \infty$ : the multiplier  $\mathbf{1}_{[-1,1]}$  is a difference of two half-line multipliers, each a modulated Hilbert transform (Worked Example 3.3.3). The natural guess in higher dimensions — that the sphere behaves like finitely many half-spaces — is false, and spectacularly so.

**Theorem 3.8.2** (Fefferman). *For  $n \geq 2$ ,  $T_B$  is bounded on  $L^p(\mathbb{R}^n)$  only for  $p = 2$ .*

*Idea of the proof.* Three steps.

*First*, a scalar bound on  $T_B$  automatically upgrades to a *vector-valued* bound, by Theorem 3.6.1. This is free, and it is the step that makes the whole argument possible: the hypothesis we are trying to contradict becomes much stronger at no cost.

*Second*, the ball looks locally like a half-space. Zooming in at a boundary point in direction  $v$  and rescaling turns  $\mathbf{1}_B$  into  $\mathbf{1}_{\{\xi \cdot v \geq 0\}}$ , so a uniform bound for  $T_B$  gives a uniform bound for the half-space multipliers  $S_v$  — and, by

the first step, a vector-valued bound for a family  $S_{v_1}, \dots, S_{v_N}$  with constant independent of the directions. This is Y. Meyer's lemma.

*Third*, a Besicovitch configuration contradicts that. One takes  $N$  tubes  $R_j$ , in  $N$  different directions, packed by the Perron tree into a set of tiny measure; the operator  $S_{v_j}$  applied to  $\mathbf{1}_{R_j}$  is bounded below on the translate of  $R_j$  two lengths along its own axis, and those translates are disjoint. So the left side of the vector-valued inequality is of size 1 while the right side is a power of the small measure. Letting the measure tend to zero finishes it.

### 3.8.2 From the ball to half-spaces

**Lemma 3.8.3** (Meyer). *Let  $1 < p < \infty$  and suppose  $\|T_B\|_{L^p \rightarrow L^p} = M < \infty$ . Then for every finite family  $v_1, \dots, v_N \in \mathbb{S}^{n-1}$  and all  $f_1, \dots, f_N \in L^p(\mathbb{R}^n)$ ,*

$$\left\| \left( \sum_{j=1}^N |S_{v_j} f_j|^2 \right)^{1/2} \right\|_p \lesssim_p M \left\| \left( \sum_{j=1}^N |f_j|^2 \right)^{1/2} \right\|_p,$$

with a constant depending only on  $p$  — in particular, independent of  $N$  and of the directions.

*Proof. Step 1. Half-spaces as limits of balls.*

Fix  $v \in \mathbb{S}^{n-1}$  and  $R > 0$ , and let  $B_R^v = B(Rv, R)$ , the ball of radius  $R$  tangent to the hyperplane  $\{\xi \cdot v = 0\}$  at the origin from the  $v$  side. By translation and dilation on the Fourier side — both of which are isometries of  $L^p$  after the corresponding modulation and dilation in  $x$  (Volume 2, Prop. 4.1(ii),(iii)) — the multiplier operator with symbol  $\mathbf{1}_{B_R^v}$  has  $L^p$  norm exactly  $M$ , for every  $R$  and  $v$ .

As  $R \rightarrow \infty$ ,  $\mathbf{1}_{B_R^v}(\xi) \rightarrow \mathbf{1}_{\{\xi \cdot v > 0\}}(\xi)$  for every  $\xi$  with  $\xi \cdot v \neq 0$ : a point with  $\xi \cdot v > 0$  eventually lies inside  $B_R^v$ , since  $|\xi - Rv|^2 - R^2 = |\xi|^2 - 2R\xi \cdot v \rightarrow -\infty$ ; and a point with  $\xi \cdot v < 0$  never does, by the same computation with the sign reversed. The exceptional hyperplane is a null set.

**Step 2. Passing to the limit.**

For  $f \in \mathcal{S}(\mathbb{R}^n)$  with  $\widehat{f}$  compactly supported away from the hyperplane  $\{\xi \cdot v = 0\}$ , dominated convergence in the Fourier representation gives  $T_{B_R^v} f \rightarrow S_v f$  in  $L^2$ , hence along a subsequence a.e. By Fatou,  $\|S_v f\|_p \leq \liminf_R \|T_{B_R^v} f\|_p \leq M \|f\|_p$ . Such  $f$  are dense in  $L^p$ , and  $S_v$  is bounded on  $L^p$  by Worked Example 3.3.3, so  $\|S_v\|_{L^p \rightarrow L^p} \leq M$  for every  $v$ .

**Step 3. The vector-valued upgrade.**

By Theorem 3.6.1 applied to the single operator  $T_B$ , the vector-valued inequality holds for  $T_B$  with constant  $\lesssim_p M$ . Running Steps 1 and 2 with  $N$



separate directions — that is, applying the vector-valued inequality for  $T_B$  to the functions  $g_j$  obtained from  $f_j$  by the modulation and dilation that carry  $B$  to  $B_R^{v_j}$ , and letting  $R \rightarrow \infty$  with Fatou — gives the stated inequality for the family  $\{S_{v_j}\}$ . The constant does not see  $N$  or the directions, because Theorem 3.6.1 does not.  $\square$

**Pitfall: why Step 3 is the crux**

It is tempting to think the vector-valued inequality is obvious: each  $S_{v_j}$  is bounded with the same constant, so surely one can sum. One cannot. Bounding each term gives  $\|(\sum_j |S_{v_j} f_j|^2)^{1/2}\|_p \leq \sum_j \|S_{v_j} f_j\|_p$ , which loses a factor  $N^{1/2}$  or worse — exactly the loss that Worked Example 2.6.2 (decoupling needs curvature) identified as the difference between a trivial bound and a theorem. The vector-valued statement asserts *no* loss in  $N$ , and Theorem 3.6.1 supplies it only because all the operators in question are obtained from a *single* bounded operator by isometries.

### 3.8.3 The Besicovitch configuration

**Lemma 3.8.4** (Besicovitch configuration). *For every  $\varepsilon > 0$  there exist  $N \in \mathbb{N}$  and rectangles  $R_1, \dots, R_N \subseteq \mathbb{R}^2$  — each of dimensions  $\ell_j \times \delta_j$  with long axis in direction  $v_j \in \mathbb{S}^1$  — such that, writing  $\tilde{R}_j = R_j + 2\ell_j v_j$  for the translate of  $R_j$  by twice its length along its own axis,*

(i) *the  $\tilde{R}_j$  are pairwise disjoint;*

$$(ii) \left| \bigcup_{j=1}^N R_j \right| \leq \varepsilon \sum_{j=1}^N |R_j|.$$

*Remark 3.8.5* (Where this is proved). Lemma 3.8.4 is not proved in this volume. It is the one statement Volume 3 assumes, and it is discharged in Volume 9, Theorem 9.2.1, by an explicit construction: the Perron tree of Volume 1 is taken in quantitative form, a rectangle is inscribed in each translated triangle about the lower part of its median, and the translates are shown to separate. The construction is independent of everything in Volumes 3 to 8, so there is no circularity.

Readers who want Theorem 3.8.2 on first reading may take Lemma 3.8.4 on trust; those who want it unconditional now should turn to Volume 9, Chapter 9.2, which depends on nothing but Volume 1.

This is exactly the Perron tree of Theorem 1.3.12 (the Perron tree), in rectangular form: (ii) says the rectangles overlap enormously, while (i) says their translates do not. Property (ii) is immediate from Volume 1 — take  $\delta$ -tubes around unit segments in  $\delta$ -separated directions inside a Besicovitch set

of measure  $< \varepsilon$ , so that  $\sum_j |R_j| \approx 1$  while  $|\bigcup R_j| \lesssim \varepsilon$ . Property (i) requires arranging the translates to fan out, which is a combinatorial refinement of the Perron construction; it is carried out in Volume 9, §9.4, where the tube combinatorics is developed systematically. We use Lemma 3.8.4 as stated.

**Lemma 3.8.6** (The half-space multiplier sees the translate). *Let  $R \subseteq \mathbb{R}^2$  be a rectangle of dimensions  $\ell \times \delta$  with long axis in direction  $v$ , and let  $\tilde{R} = R + 2\ell v$ . Then*

$$|S_v \mathbf{1}_R(x)| \geq \frac{\log(3/2)}{2\pi} \quad \text{for every } x \in \tilde{R}.$$

*Proof.* Rotations and translations commute with the construction —  $S_{\rho v}(f \circ \rho^{-1}) = (S_v f) \circ \rho^{-1}$  for a rotation  $\rho$ , since the multiplier  $\mathbf{1}_{\{\xi \cdot v \geq 0\}}$  transforms accordingly — so we may take

$$v = e_1, \quad R = [0, \ell] \times [0, \delta], \quad \tilde{R} = [2\ell, 3\ell] \times [0, \delta].$$

The symbol  $\mathbf{1}_{\{\xi_1 \geq 0\}}$  depends only on  $\xi_1$ , so  $S_{e_1}$  acts in the first variable alone and leaves the second untouched. Since  $\mathbf{1}_R(x) = \mathbf{1}_{[0, \ell]}(x_1) \mathbf{1}_{[0, \delta]}(x_2)$  is a product, Fubini gives

$$S_{e_1} \mathbf{1}_R(x) = \mathbf{1}_{[0, \delta]}(x_2) \cdot (S \mathbf{1}_{[0, \ell]})(x_1),$$

where  $S$  is the one-dimensional half-line multiplier. By Worked Example 3.3.3,  $S = \frac{1}{2}(\text{Id} + i\mathcal{H})$ , so for  $x_1 \in [2\ell, 3\ell]$  — where  $\mathbf{1}_{[0, \ell]}(x_1) = 0$  —

$$(S \mathbf{1}_{[0, \ell]})(x_1) = \frac{i}{2} \mathcal{H} \mathbf{1}_{[0, \ell]}(x_1).$$

By Worked Example 3.3.1,  $\mathcal{H} \mathbf{1}_{[0, \ell]}(t) = \frac{1}{\pi} \log \left| \frac{t}{t-\ell} \right|$ , and on  $[2\ell, 3\ell]$  this is positive and decreasing in  $t$ , since  $t \mapsto \frac{t}{t-\ell} = 1 + \frac{\ell}{t-\ell}$  is decreasing. Its minimum is therefore attained at  $t = 3\ell$ :

$$\mathcal{H} \mathbf{1}_{[0, \ell]}(x_1) \geq \frac{1}{\pi} \log \frac{3\ell}{2\ell} = \frac{\log(3/2)}{\pi} \quad \text{for } x_1 \in [2\ell, 3\ell].$$

Since  $x \in \tilde{R}$  forces  $x_2 \in [0, \delta]$ , the indicator factor equals 1, and combining gives  $|S_{e_1} \mathbf{1}_R(x)| \geq \frac{1}{2} \cdot \frac{\log(3/2)}{\pi}$ .  $\square$

*Remark 3.8.7.* The mechanism deserves a sentence. A sharp frequency cutoff cannot be localised in space: truncating the Fourier transform of  $\mathbf{1}_R$  to a half-space produces a function with a logarithmic tail extending along the axis of  $R$ , exactly as Worked Example 3.3.1 showed in one dimension. The translate  $\tilde{R}$  lies in that tail. So the operator *smears* each rectangle along its own direction, and a Besicovitch configuration is precisely an arrangement in which the rectangles overlap but the smears do not.

### 3.8.4 Proof of Fefferman's theorem

*Proof of Theorem 3.8.2.* Suppose  $T_B$  is bounded on  $L^p(\mathbb{R}^n)$  for some  $p \neq 2$ . By duality (Corollary 1.8.10 (norms by testing)) — the multiplier is real and even, so  $T_B$  is self-adjoint — we may assume  $1 < p < 2$ . And it suffices to treat  $n = 2$ : if  $T_B$  is bounded on  $L^p(\mathbb{R}^n)$  then, by a standard restriction of the multiplier to a two-dimensional subspace (de Leeuw's theorem, or directly by testing on functions of the form  $g(x_1, x_2)h(x_3, \dots, x_n)$  and letting  $h$  concentrate), the two-dimensional ball multiplier is bounded on  $L^p(\mathbb{R}^2)$ .

Fix  $\varepsilon > 0$  and take the configuration  $R_1, \dots, R_N$  of Lemma 3.8.4, normalised so that  $\sum_j |R_j| = 1$ . Set  $f_j = \mathbf{1}_{R_j}$  and let  $v_j$  be the long axis direction of  $R_j$ .

**Step 1.** *Lower bound for the left side.*

By Lemma 3.8.6,  $|S_{v_j} f_j| \geq c$  on  $\tilde{R}_j$ , so  $\sum_j |S_{v_j} f_j|^2 \geq c^2$  on  $\bigcup_j \tilde{R}_j$ . Since the  $\tilde{R}_j$  are disjoint and  $|\tilde{R}_j| = |R_j|$ ,

$$\left\| \left( \sum_j |S_{v_j} f_j|^2 \right)^{1/2} \right\|_p \geq c \left| \bigcup_j \tilde{R}_j \right|^{1/p} = c \left( \sum_j |R_j| \right)^{1/p} = c.$$

**Step 2.** *Upper bound for the right side.*

Let  $\Omega = \bigcup_j R_j$ , so  $|\Omega| \leq \varepsilon$  by Lemma 3.8.4(ii). Since  $p < 2$ , Hölder's inequality with exponents  $\frac{2}{p}$  and  $\frac{2}{2-p}$  gives

$$\int_{\mathbb{R}^2} \left( \sum_j \mathbf{1}_{R_j} \right)^{p/2} = \int_{\Omega} \left( \sum_j \mathbf{1}_{R_j} \right)^{p/2} \cdot 1 \leq \left( \int_{\mathbb{R}^2} \sum_j \mathbf{1}_{R_j} \right)^{p/2} |\Omega|^{1-\frac{p}{2}} = |\Omega|^{1-\frac{p}{2}},$$

using  $\sum_j \int \mathbf{1}_{R_j} = \sum_j |R_j| = 1$ . Hence

$$\left\| \left( \sum_j |f_j|^2 \right)^{1/2} \right\|_p = \left( \int \left( \sum_j \mathbf{1}_{R_j} \right)^{p/2} \right)^{1/p} \leq |\Omega|^{\frac{1}{p}-\frac{1}{2}} \leq \varepsilon^{\frac{1}{p}-\frac{1}{2}}.$$

**Step 3.** *Contradiction.*

Inserting Steps 1 and 2 into Lemma 3.8.3,

$$c \lesssim_p M \varepsilon^{\frac{1}{p}-\frac{1}{2}}.$$

Since  $p < 2$  the exponent  $\frac{1}{p} - \frac{1}{2}$  is strictly positive, so letting  $\varepsilon \rightarrow 0$  forces  $c = 0$ , a contradiction.  $\square$

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WORKED EXAMPLE 3.8.1— TIER C: RESEARCH TASTE **What Fefferman’s theorem started**

(a) *The logical shape.* A Besicovitch set was used to *disprove* an analytic statement. Read in the contrapositive, the argument says: if the vector-valued half-space inequality held, then  $\delta$ -tubes in  $\delta$ -separated directions could not be packed into a set of small measure. That is a *Keakeya* statement, and the equivalence is not an accident of this proof.

Volume 6 makes it a theorem. The *Keakeya* maximal conjecture of Volume 1, Worked Example 7.20 is implied by the restriction conjecture, and both are implied by the Bochner–Riesz conjecture of Worked Example 3.5.2. So the chain runs

$$\text{Bochner–Riesz} \implies \text{restriction} \implies \text{Keakeya},$$

and the present chapter is the degenerate case  $\delta = 0$  of the first, where the implication is an equivalence and both sides are settled — negatively.

(b) *Why the sphere and not the cube.* Repeat the proof with  $B$  replaced by a cube. Step 1 of Lemma 3.8.3 still produces half-space multipliers, but only the  $n$  fixed directions normal to the faces, not a continuum. With finitely many directions no Besicovitch configuration is available — one needs tubes pointing in  $\approx \delta^{-1}$  different directions — and indeed the cube multiplier *is* bounded on every  $L^p$ ,  $1 < p < \infty$ , being a product of half-space multipliers. Curvature of the boundary is what supplies the directions.

(c) *What survives.* The theorem does not say that frequency truncation is useless, only that sharp truncation to a ball is. Smoothing the cutoff — raising the distance to the boundary to a power  $\delta > 0$ , as in Worked Example 3.5.2 — restores boundedness for  $p$  in a range depending on  $\delta$ , and determining the sharp range is the Bochner–Riesz conjecture, open for  $n \geq 3$ . Every positive result in that direction is, at bottom, a quantitative version of “tubes in different directions cannot overlap too much”, and that is the subject of Volumes 9 through 13.

(d) *A closing observation.* Lemma 3.8.3 shows that a scalar  $L^p$  bound for one operator implies a vector-valued bound for a whole family. Then Lemma 3.8.4 shows the family bound is false. The Besicovitch set is therefore not being used against a single operator — it is being used against the *simultaneous* behaviour of many operators, one per direction. That is exactly the structure of a wave packet decomposition, where one has one packet per direction and must control their superposition. The reader who found Volume 2, Worked Example 6.9 — decoupling is false without curvature — convincing has already seen this argument once, in its positive form.

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**Figure 3.8.1 (to be drawn).** *A Besicovitch configuration and its translates*

**Content required:** Six to eight thin rectangles  $R_j$  in the plane, in different directions, overlapping heavily so that their union is visibly small; and the translates  $\tilde{R}_j = R_j + 2\ell_j v_j$ , drawn disjoint and fanning outwards. The contrast between the packed  $R_j$  and the spread  $\tilde{R}_j$  is the whole content of Lemma 3.8.4.

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WORKED EXAMPLE 3.8.2— TIER A **The ball multiplier, and why  $p = 2$  is the only survivor**

Theorem 3.8.2 states that  $T_B$ , the multiplier with symbol  $\mathbf{1}_{|\xi| \leq 1}$ , is bounded on  $L^p(\mathbb{R}^n)$  only for  $p = 2$  when  $n \geq 2$ . The three ingredients are worth separating.

(a)  $p = 2$  works, trivially. By Plancherel,  $\|T_B f\|_{L^2} = \|\mathbf{1}_{|\xi| \leq 1} \widehat{f}\|_{L^2} \leq \|f\|_{L^2}$ . No geometry, no smoothness — boundedness of the symbol suffices, and this is the  $L^2$  hypothesis of Theorem 3.2.3 being available for free.

(b)  $n = 1$  works for all  $p \in (1, \infty)$ . There the “ball” is an interval, and  $\mathbf{1}_{[-1,1]}$  is a combination of two modulated Hilbert transforms:  $\mathbf{1}_{[-1,1]}(\xi) = \frac{i}{2}(\operatorname{sgn}(\xi + 1) - \operatorname{sgn}(\xi - 1))$ . So  $T_B$  is bounded wherever  $\mathcal{H}$  is, which is Chapter 3.3. Dimension is what breaks this.

(c)  $n \geq 2$ ,  $p \neq 2$  fails. The obstruction is the Besicovitch configuration of Lemma 3.8.4: rectangles overlapping so much that their union is tiny, whose translates by twice their length are disjoint. The half-plane multiplier “sees” each rectangle’s translate (Lemma 3.8.6); the disjointness makes the translated pieces orthogonal; and the smallness of the union makes the original pieces cheap. The mismatch is unbounded.

*What is really being said.* The ball’s boundary is a *curved hypersurface*, and cutting sharply across it is incompatible with  $L^p$  for  $p \neq 2$ . Volume 5’s Bochner–Riesz operators soften the cut by the factor  $(1 - |\xi|^2)^\delta$ , and Theorem 5.8.4 shows that for  $\delta > \frac{n-1}{2}$  the softening is enough. How small  $\delta$  may be is the Bochner–Riesz conjecture, still open, and by Volume 6, Worked Example 6.7.2 it implies the restriction conjecture.

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## Drills

**Drill 3.8.1.** Verify that the cube multiplier  $\mathbf{1}_{[-1,1]^n}$  is bounded on  $L^p(\mathbb{R}^n)$  for  $1 < p < \infty$ , by writing it as a product of  $2n$  half-space multipliers.

**Drill 3.8.2.** Complete the computation in Step 1 of Lemma 3.8.3: show  $|\xi - Rv|^2 - R^2 = |\xi|^2 - 2R\xi \cdot v$  and deduce the pointwise convergence claim.

**Drill 3.8.3.** Show that Step 3 of the proof of Theorem 3.8.2 fails for  $p = 2$  by computing the exponent  $\frac{1}{p} - \frac{1}{2}$ , and explain what this says about the sharpness of the argument.

**Drill 3.8.4.** Where in the proof is  $n \geq 2$  used? Identify the step that has no analogue for  $n = 1$  and explain why the one-dimensional ball multiplier is bounded.

## Notes

Theorem 3.8.2 is Fefferman's, from 1971, and is one of the shortest famous papers in analysis. The vector-valued upgrade is Marcinkiewicz–Zygmund's theorem; the localisation Lemma 3.8.3 is due to Y. Meyer.

Lemma 3.8.4(i) is the one statement in this volume used without proof. It is a refinement of the Perron tree construction of Theorem 1.3.12 (the Perron tree), and is proved in Volume 9 alongside the rest of the tube combinatorics. Volume 2's table of deferrals promised this chapter for Volume 5; moving it here has cost exactly that one lemma, which seems a fair exchange for placing the theorem immediately after the theory it refutes.

# Chapter 3.A

## Results Quoted from Volumes 1 and 2

This volume assumes Volumes 1 and 2 and nothing else. Every citation is listed below with its use site.

### From Volume 1

| Result                                      | Vol. 1             | Used in                                                    |
|---------------------------------------------|--------------------|------------------------------------------------------------|
| Lebesgue differentiation                    | Thm 7.7            | Theorem 3.1.6 Step 3                                       |
| Hardy–Littlewood maximal theorem            | Thm 7.4            | Worked Example 3.1.2, Drill 1.2                            |
| Polar coordinates                           | Thm 6.10           | Proposition 3.2.2, Proposition 3.7.2, Worked Example 3.4.2 |
| Layer-cake formula                          | Wkd Ex 5.5         | Theorem 3.7.3 Step 4                                       |
| Dominated convergence                       | Thm 5.8            | Proposition 3.3.2, Lemma 3.8.3                             |
| Fubini–Tonelli                              | Thms 6.6, 6.7      | Theorem 3.2.3 Step 4, Theorem 3.6.1, Lemma 3.8.6           |
| Density of $L^1 \cap L^2$ ; of $C_c^\infty$ | Thm 8.9            | Theorem 3.2.3 Step 1                                       |
| $L^p$ duality; norms by testing             | Thm 8.11, Cor 8.12 | Theorem 3.2.3 Step 6, Theorem 3.6.3 Step 3, Theorem 3.8.2  |
| Chebyshev; weak $L^p$                       | Prop 8.14          | Theorem 3.2.3 Steps 2, 4                                   |
| Hölder                                      | Thm 8.2            | Theorem 3.8.2 Step 2                                       |
| Minkowski integral inequality               | Thm 8.5            | Worked Example 3.4.2                                       |
| Perron tree                                 | Thm 3.23           | Lemma 3.8.4                                                |

**From Volume 2**

| <b>Result</b>                                           | <b>Vol. 2</b> | <b>Used in</b>                                      |
|---------------------------------------------------------|---------------|-----------------------------------------------------|
| Plancherel                                              | Thm 4.5       | Proposition 3.3.2, Lemma 3.5.2 Step 2               |
| $L^2$ multiplier norm $= \ m\ _\infty$                  | Prop 6.2      | Proposition 3.3.2, Proposition 3.4.2, Theorem 3.5.1 |
| The Fourier dictionary                                  | Prop 4.1      | Proposition 3.3.2, Proposition 3.4.2, Lemma 3.8.3   |
| Marcinkiewicz interpolation                             | Thm 2.6       | Theorem 3.2.3 Step 6, Drill 1.2                     |
| Khintchine's inequality                                 | Thm 6.5       | Theorem 3.6.1, Theorem 3.6.3 Step 2                 |
| Bernstein's inequalities                                | Thm 6.8       | Worked Example 3.6.1                                |
| Density of trigonometric polynomials                    | Cor 3.6       | Worked Example 3.3.2                                |
| Decay of $\widehat{d\sigma}$                            | Thm 8.6       | Worked Example 3.5.2                                |
| Tempered distributions; $\mathcal{F}$ on $\mathcal{S}'$ | Thms 5.5, 5.9 | Theorem 3.5.1, Theorem 3.4.4                        |

**Debts discharged**

Volume 2 deferred two results to this volume. Both are now proved: the Hörmander–Mikhlin theorem is Theorem 3.5.1, and the Littlewood–Paley square function theorem for  $p \neq 2$  is Theorem 3.6.3.

**Debt incurred**

Lemma 3.8.4(i) — that the translates of a Besicovitch configuration can be arranged to be pairwise disjoint — is used and not proved here. It is a combinatorial refinement of Theorem 1.3.12 (the Perron tree), and is proved in Volume 9. This is the only unproved statement in Volumes 1–3.



## Chapter 3.B

# Notation and Index of Results

### Additions to the Volume 1–2 conventions

|                                                  |                                                                 |
|--------------------------------------------------|-----------------------------------------------------------------|
| $\mathcal{D}, \mathcal{D}_k, \mathcal{D}^\alpha$ | the dyadic grid; generation $k$ ; the shifted grids             |
| $\langle f \rangle_Q$                            | the average $ Q ^{-1} \int_Q f$                                 |
| $\mathcal{M}_d$                                  | the dyadic maximal function                                     |
| $\widehat{Q}$                                    | the dyadic parent of $Q$ ; $Q^* = 2\sqrt{n} Q$ the dilate       |
| $\mathcal{H}, \mathcal{H}_v$                     | the Hilbert transform; along direction $v$                      |
| $S_v$                                            | the half-space multiplier $\mathbf{1}_{\{\xi \cdot v \geq 0\}}$ |
| $\mathcal{R}_j$                                  | the $j$ -th Riesz transform                                     |
| $\widetilde{f}, P_+$                             | the conjugate function; the Riesz projection                    |
| p.v.                                             | principal value                                                 |
| $\mathcal{M}^\#, \text{BMO}$                     | the sharp maximal function; bounded mean oscillation            |
| $\mathcal{S}$                                    | the Littlewood–Paley square function                            |

## Index of the main results

|                                     |                      |
|-------------------------------------|----------------------|
| Nesting of dyadic cubes             | Lemma 3.1.2          |
| Dyadic maximal inequality           | Theorem 3.1.4        |
| The one-third trick                 | Worked Example 3.1.2 |
| Calderón–Zygmund decomposition      | Theorem 3.1.6        |
| Gradient bound implies Hörmander    | Proposition 3.2.2    |
| Calderón–Zygmund theorem            | Theorem 3.2.3        |
| Hilbert transform: multiplier       | Proposition 3.3.2    |
| Hilbert transform: $L^p$            | Theorem 3.3.3        |
| Conjugate function on $\mathbb{T}$  | Theorem 3.3.5        |
| $L^p$ convergence of Fourier series | Worked Example 3.3.2 |
| Riesz transform multiplier          | Proposition 3.4.2    |
| Riesz transforms: $L^p$             | Theorem 3.4.3        |
| Elliptic estimate for $\Delta$      | Theorem 3.4.4        |
| Method of rotations                 | Worked Example 3.4.2 |

# Part 4

## Volume 4

Function Spaces: Hardy Spaces, BMO, and the Sobolev Scale



# Preface to Volume 4

## The problem this volume solves

Volume 3 proved that a Calderón–Zygmund operator is bounded on  $L^p(\mathbb{R}^n)$  for every  $p$  strictly between 1 and  $\infty$ , and that both endpoints fail. It also showed what happens instead at the top end: such an operator maps  $L^\infty$  into BMO. Half the endpoint story was therefore already told.

The other half is missing. At  $p = 1$  we proved only a weak-type bound, and a weak-type bound is an awkward object — it is not a norm inequality, it does not dualise, and it does not interpolate as comfortably as one would like. The resolution, found by Stein, Weiss, Fefferman and Coifman in the 1960s and 70s, is that  $L^1$  is simply the wrong space. There is a smaller space, the Hardy space  $H^1(\mathbb{R}^n)$ , on which singular integrals *are* bounded, and it is the exact dual partner of BMO.

That is the first half of this volume. The second half generalises: once one sees that  $H^1$  and BMO are the honest endpoints of the  $L^p$  scale, it becomes natural to ask what the honest endpoints of the *Sobolev* scale are, and the answer is the family of Besov and Triebel–Lizorkin spaces. These are the spaces in which the estimates of Volumes 12 and 13 are naturally phrased.

## A word about how to read this

This volume is written for a reader who has worked through Volumes 1–3 and is meeting Hardy spaces for the first time. Two consequences.

First, definitions are given before they are used, and the several competing definitions of  $H^1$  — there are at least four — are introduced one at a time, with the reasons for each. The theorem that they agree is genuinely hard, and we prove it in Chapter 4.4 rather than asserting it early and leaving the reader unsure which definition is in play.

Second, we resist the temptation to state results in their sharpest form. The literature phrases things in terms of  $H^p$  for  $0 < p \leq 1$ , of  $H^p_{\text{loc}}$ , of weighted variants, and of vector-valued extensions. Almost none of that is needed for

this series. We prove what is used, in the form in which it is used, and say so.

## What is assumed

Volumes 1–3, and nothing else. The reader should be comfortable with: the Calderón–Zygmund decomposition and theorem; the dyadic maximal function and its weak  $(1, 1)$  bound with constant 1; the Hilbert and Riesz transforms; BMO and the John–Nirenberg inequality; Littlewood–Paley projections and Bernstein’s inequalities; and the interpolation theorems of Volume 2. Appendix A lists every citation with its use site.

## The plan

|   |                                    |                                  |
|---|------------------------------------|----------------------------------|
| 1 | The sharp maximal function         | BMO becomes a tool               |
| 2 | Atoms and the Hardy space $H^1$    | the right endpoint at $p = 1$    |
| 3 | Duality: $(H^1)^* = \text{BMO}$    | Fefferman’s theorem              |
| 4 | The maximal characterisation       | all the definitions agree        |
| 5 | Vector-valued maximal inequalities | Fefferman–Stein                  |
| 6 | Besov and Triebel–Lizorkin spaces  | the full scale                   |
| 7 | Embeddings                         | smoothness against integrability |
| 8 | Where these spaces are used        | the estimates of Volumes 12–13   |

Two debts from Volume 3 are discharged here: that Calderón–Zygmund operators map  $H^1$  into  $L^1$  (Theorem 4.2.5), and that  $(H^1)^* = \text{BMO}$  (Chapter 4.3).

## Chapter 4.1

# The Sharp Maximal Function

### 4.1.1 Why another maximal function

Volume 3 introduced BMO as the space in which the  $L^\infty$  endpoint of the Calderón–Zygmund theory lives. So far it has been a destination: we proved that certain operators land in it, and we proved the John–Nirenberg inequality about functions that lie in it. We have not yet used BMO to prove anything.

This chapter changes that. The device is to notice that the quantity inside the definition of the BMO norm,

$$\frac{1}{|Q|} \int_Q |f - \langle f \rangle_Q|,$$

is an average over a cube — exactly the kind of object the Hardy–Littlewood maximal function is built from. Taking the supremum over cubes containing a point  $x$ , instead of over all cubes at once, produces a *function* of  $x$  rather than a single number. That function is the sharp maximal function  $\mathcal{M}^\# f$ , and  $\|f\|_{\text{BMO}}$  is just  $\|\mathcal{M}^\# f\|_\infty$ .

The point of doing this is the following theorem, which is the main result of the chapter:

$$\|f\|_{L^p} \approx \|\mathcal{M}^\# f\|_{L^p}, \quad 1 < p < \infty.$$

The right-hand side involves only the *oscillation* of  $f$  — how much  $f$  varies on cubes — and none of its size. It is remarkable that the two are comparable, and it is useful, because for many operators the oscillation of  $Tf$  is far easier to estimate than  $Tf$  itself.

#### Looking ahead: what this buys

Two things, both used later in this volume.

First, an *interpolation theorem with BMO* (Theorem 4.1.5): an operator bounded on one  $L^{p_0}$  and mapping  $L^\infty$  into BMO is bounded on every  $L^p$

with  $p > p_0$ . Since Volume 3 proved the  $L^\infty \rightarrow \text{BMO}$  bound for every Calderón–Zygmund operator, this gives a second and quite different route to the  $L^p$  theory — one which, as Worked Example 4.1.3 shows, does not need the operator to be a convolution.

Second, and more importantly for the design of this volume, it tells us what to look for at the other end. If  $\text{BMO}$  is the correct substitute for  $L^\infty$ , then by duality there ought to be a correct substitute for  $L^1$ , and it ought to be a space whose dual is  $\text{BMO}$ . That space is the Hardy space  $H^1$  of Chapter 4.2.

### 4.1.2 Definition and first properties

Throughout, “cube” means a cube in  $\mathbb{R}^n$  with sides parallel to the axes, and  $\langle f \rangle_Q = |Q|^{-1} \int_Q f$ .

**Definition 4.1.1.** For  $f \in L^1_{\text{loc}}(\mathbb{R}^n)$  the *sharp maximal function* is

$$\mathcal{M}^\# f(x) = \sup_{Q \ni x} \frac{1}{|Q|} \int_Q |f - \langle f \rangle_Q|,$$

the supremum over all cubes containing  $x$ . The *dyadic sharp maximal function*  $\mathcal{M}_d^\# f$  is defined the same way, with the supremum restricted to dyadic cubes.

Since dyadic cubes are cubes,  $\mathcal{M}_d^\# f \leq \mathcal{M}^\# f$  pointwise.

**Proposition 4.1.2** (Elementary properties). *Let  $f \in L^1_{\text{loc}}(\mathbb{R}^n)$ .*

- (i)  $\|f\|_{\text{BMO}} = \|\mathcal{M}^\# f\|_{L^\infty}$ .
- (ii)  $\mathcal{M}^\# f \leq 2\mathcal{M}f$  pointwise.
- (iii)  $\mathcal{M}^\#(f + g) \leq \mathcal{M}^\# f + \mathcal{M}^\# g$  and  $\mathcal{M}^\#(cf) = |c| \mathcal{M}^\# f$ .
- (iv)  $\mathcal{M}^\# f = 0$  if and only if  $f$  is a.e. equal to a constant.
- (v) For any constant  $c$ ,  $\mathcal{M}^\#(f + c) = \mathcal{M}^\# f$ .

*Proof.* (i) is a comparison of two suprema:  $\|f\|_{\text{BMO}}$  is the supremum over all cubes of the mean oscillation, and  $\|\mathcal{M}^\# f\|_\infty$  is the supremum over  $x$  of the supremum over cubes containing  $x$ . Every cube contains some point, so the two range over the same collection of numbers.

(ii) For a cube  $Q \ni x$ ,

$$\frac{1}{|Q|} \int_Q |f - \langle f \rangle_Q| \leq \frac{1}{|Q|} \int_Q |f| + |\langle f \rangle_Q| \leq \frac{2}{|Q|} \int_Q |f| \leq 2\mathcal{M}f(x),$$



where the last step uses that  $Q$  is a cube containing  $x$  and that the Hardy–Littlewood maximal function over cubes is comparable to the one over balls (Worked Example 3.1.2 (the one-third trick)). Taking the supremum over  $Q$  gives (ii).

(iii) and (v) are immediate from the definition, using  $\langle f + g \rangle_Q = \langle f \rangle_Q + \langle g \rangle_Q$  and, for (v), that adding a constant to  $f$  adds it to  $\langle f \rangle_Q$  as well, so the difference  $f - \langle f \rangle_Q$  is unchanged.

(iv) If  $f = c$  a.e. then every mean oscillation vanishes. Conversely if  $\mathcal{M}^\# f = 0$  then  $f = \langle f \rangle_Q$  a.e. on every cube  $Q$ ; applying this to an increasing sequence of cubes and to two overlapping cubes shows all these constants agree, so  $f$  is a.e. constant.  $\square$

**Pitfall:  $\mathcal{M}^\#$  does not see the size of  $f$**

Property (v) says  $\mathcal{M}^\#$  is blind to additive constants, and (iv) says that is the only thing it is blind to. So an inequality of the form  $\|f\|_p \lesssim \|\mathcal{M}^\# f\|_p$  cannot possibly be true without some extra hypothesis: take  $f$  to be a large constant on  $\mathbb{R}^n$ , and the left side is infinite while the right is zero.

What saves Theorem 4.1.4 below is the requirement that  $f$  already belong to some  $L^{p_0}$  with  $p_0 < \infty$ . That hypothesis rules out constants — the only constant in  $L^{p_0}(\mathbb{R}^n)$  is 0 — and, less obviously, it is exactly what makes the level sets  $\{\mathcal{M}_d f > \lambda\}$  have finite measure, which the proof needs. Watch for it in Step 1.

**WORKED EXAMPLE 4.1.1— TIER A Computing  $\mathcal{M}^\#$  for an indicator**

Let  $n = 1$  and  $f = \mathbf{1}_{[0,1]}$ . We compare  $\mathcal{M}^\# f$  with  $\mathcal{M}f$ .

*Far from the interval.* Let  $x > 2$  and let  $I \ni x$  be an interval. If  $I \cap [0, 1] = \emptyset$  then  $f \equiv 0$  on  $I$  and the mean oscillation is 0. Otherwise  $I$  reaches back to 1, so  $|I| \geq x - 1$ , and

$$\frac{1}{|I|} \int_I |f - \langle f \rangle_I| \leq \frac{2}{|I|} \int_I f \leq \frac{2}{x - 1},$$

since  $\int_I f \leq 1$ . So  $\mathcal{M}^\# f(x) \lesssim 1/x$  for large  $x$  — the same decay as  $\mathcal{M}f(x) \approx 1/(2x)$ , computed in Drill 1.7.1. Neither is integrable at infinity.

*Inside the interval.* At  $x = \frac{1}{2}$ , taking  $I = [0, 1]$  gives  $\langle f \rangle_I = 1$  and  $|f - \langle f \rangle_I| = 0$  on  $I$ , so that particular cube contributes nothing. Taking  $I = [-1, 1]$  gives  $\langle f \rangle_I = \frac{1}{2}$  and mean oscillation  $\frac{1}{2}$ . One checks  $\mathcal{M}^\# f(\frac{1}{2}) = \frac{1}{2}$ , while  $\mathcal{M}f(\frac{1}{2}) = 1$ .

*The moral.*  $\mathcal{M}^\# f \leq 2\mathcal{M}f$ , as Proposition 4.1.2(ii) promised, and the two are genuinely comparable here. The next example shows they are not always.

**WORKED EXAMPLE 4.1.2— TIER A  $\mathcal{M}^\#$  can be much smaller than  $\mathcal{M}$** 

Take  $f \equiv 1$  on  $\mathbb{R}^n$ . Then  $\mathcal{M}f \equiv 1$ , but  $\mathcal{M}^\#f \equiv 0$  by Proposition 4.1.2(iv). Less trivially, take  $f = \log|x|$ , which lies in BMO by Volume 3, Proposition 7.2. Then  $\mathcal{M}^\#f$  is bounded, by Proposition 4.1.2(i), whereas  $\mathcal{M}f(x) = \infty$  for every  $x$ , since  $\log|y|$  is not locally integrable at infinity — every ball average of  $|\log|y||$  over  $B(x, r)$  tends to  $\infty$  as  $r \rightarrow \infty$ .

So the inequality  $\mathcal{M}^\#f \leq 2\mathcal{M}f$  can be arbitrarily lossy. What Theorem 4.1.4 will say is that once one integrates to the power  $p$ , nothing is lost after all.

### 4.1.3 The good- $\lambda$ inequality

The proof of the main theorem rests on a single estimate comparing the level sets of  $\mathcal{M}_d f$  and  $\mathcal{M}_d^\# f$ . Inequalities of this shape are called *good- $\lambda$  inequalities*, and the name is worth explaining before we state one.

*Scratch work.* Suppose we want to prove  $\|F\|_p \lesssim \|G\|_p$  for two nonnegative functions. By the layer-cake formula (Worked Example 1.5.1 (the layer-cake formula)) this is a statement about the level sets  $\{F > \lambda\}$  and  $\{G > \lambda\}$ . A direct pointwise bound  $F \lesssim G$  would suffice but is usually false — it is false here, by Worked Example 4.1.2.

The good- $\lambda$  method asks for something weaker and cleverer. Fix a large constant  $a > 1$  and a small constant  $\gamma > 0$ , and prove

$$m(\{F > a\lambda, G \leq \gamma\lambda\}) \leq c\gamma m(\{F > \lambda\}). \quad (4.1.1)$$

In words: *the points where  $F$  is very large but  $G$  is small are a small fraction of the points where  $F$  is merely large.* This says nothing pointwise. But after multiplying by  $\lambda^{p-1}$  and integrating, the left side produces  $\|F\|_p^p$  (up to the factor  $a^p$ ) and the right side produces  $c\gamma\|F\|_p^p$  — and if  $\gamma$  is chosen small enough that  $a^p c\gamma < \frac{1}{2}$ , the term can be *absorbed* into the left-hand side, leaving only the contribution of  $\{G > \gamma\lambda\}$ , which is  $\|G\|_p^p$ .

The absorption step is the whole trick, and it has a hidden cost: one must know in advance that  $\|F\|_p$  is *finite*, or the subtraction  $\infty - \infty$  is meaningless. Step 1 below is devoted to arranging that, and readers should not regard it as a formality — Worked Example 4.1.2 shows the theorem is false without a hypothesis of this kind.

**Theorem 4.1.3** (Good- $\lambda$  inequality). *Let  $f \in L_{\text{loc}}^1(\mathbb{R}^n)$  be such that  $m(\{\mathcal{M}_d f > \lambda\}) < \infty$  for every  $\lambda > 0$ . Then for every  $\lambda > 0$  and every  $\gamma \in (0, 1)$ ,*

$$m(\{\mathcal{M}_d f > 2^{n+1}\lambda, \mathcal{M}_d^\# f \leq \gamma\lambda\}) \leq 2^{-n}\gamma m(\{\mathcal{M}_d f > \lambda\}).$$

*Idea of the proof.* Decompose the level set  $\{\mathcal{M}_d f > \lambda\}$  into the maximal dyadic cubes  $Q_j$  on which the average of  $|f|$  exceeds  $\lambda$  — this is exactly the selection performed in Theorem 3.1.4 (the dyadic maximal inequality). Work inside one such  $Q_j$ .

Two facts are available there. First, because the *parent* of  $Q_j$  was not selected, the average of  $|f|$  over  $Q_j$  is at most  $2^n \lambda$ ; so the constant  $\langle f \rangle_{Q_j}$  is of size at most  $2^n \lambda$ . Second, if the set we are measuring is nonempty, it contains a point where  $\mathcal{M}_d^\# f \leq \gamma \lambda$ , and that point certifies that  $f$  oscillates by at most  $\gamma \lambda$  on average over  $Q_j$ .

Now, a point of  $Q_j$  where  $\mathcal{M}_d f$  exceeds  $2^{n+1} \lambda$  must see an average of  $|f|$  exceeding  $2^{n+1} \lambda$  on some dyadic subcube; since  $\langle f \rangle_{Q_j}$  is only of size  $2^n \lambda$ , the *oscillation* of  $f$  must exceed  $2^n \lambda$  there. So such points lie in a level set of the dyadic maximal function of  $f - \langle f \rangle_{Q_j}$ , whose measure the weak  $(1, 1)$  inequality controls by the second fact. The factor  $\gamma$  appears because that is the size of the oscillation, and it is exactly the small factor (4.1.1) asks for.

*Proof.* Write  $\Omega_\lambda = \{\mathcal{M}_d f > \lambda\}$  and

$$E = \{\mathcal{M}_d f > 2^{n+1} \lambda\} \cap \{\mathcal{M}_d^\# f \leq \gamma \lambda\}.$$

Since  $2^{n+1} \lambda > \lambda$ , we have  $E \subseteq \Omega_\lambda$ .

**Step 1.** *Decompose the level set.*

By the proof of Theorem 3.1.4 (the dyadic maximal inequality),  $\Omega_\lambda$  is the disjoint union of the maximal dyadic cubes  $Q_j$  satisfying  $\langle |f| \rangle_{Q_j} > \lambda$ , and the maximality gives, for the dyadic parent  $\widehat{Q_j}$ ,

$$\langle |f| \rangle_{\widehat{Q_j}} \leq \lambda \quad \implies \quad \langle |f| \rangle_{Q_j} \leq 2^n \lambda, \quad (4.1.2)$$

because  $|\widehat{Q_j}| = 2^n |Q_j|$  and  $Q_j \subseteq \widehat{Q_j}$ . Since  $E \subseteq \Omega_\lambda = \bigsqcup_j Q_j$ , it suffices to prove

$$m(E \cap Q_j) \leq 2^{-n} \gamma |Q_j| \quad \text{for each } j, \quad (4.1.3)$$

and then sum.

**Step 2.** *Fix a cube and extract the oscillation bound.*

Fix  $j$ . If  $E \cap Q_j = \emptyset$  then (4.1.3) is trivial, so assume there is  $x_0 \in E \cap Q_j$ . By definition of  $E$  we have  $\mathcal{M}_d^\# f(x_0) \leq \gamma \lambda$ , and  $Q_j$  is a dyadic cube containing  $x_0$ , so this particular cube must satisfy the defining inequality:

$$\frac{1}{|Q_j|} \int_{Q_j} |f - \langle f \rangle_{Q_j}| \leq \gamma \lambda. \quad (4.1.4)$$

This is the only use of the hypothesis  $\mathcal{M}_d^\# f \leq \gamma\lambda$ , and note how little was needed: one point of  $E \cap Q_j$ , and one cube.

**Step 3.** *Points of high maximal function see large oscillation.*

Let  $x \in E \cap Q_j$ , so  $\mathcal{M}_d f(x) > 2^{n+1}\lambda$ . By definition there is a dyadic cube  $Q \ni x$  with

$$\langle |f| \rangle_Q > 2^{n+1}\lambda. \quad (4.1.5)$$

We claim  $Q \subseteq Q_j$ . Indeed  $Q$  and  $Q_j$  both contain  $x$ , so by the nesting property (Lemma 3.1.2 (nesting of dyadic cubes)) one contains the other. If  $Q \supsetneq Q_j$  then  $Q$  is a dyadic cube strictly containing the maximal cube  $Q_j$ , so by maximality  $\langle |f| \rangle_Q \leq \lambda$ , contradicting (4.1.5). Hence  $Q \subseteq Q_j$ .

Now set

$$g = (f - \langle f \rangle_{Q_j}) \mathbf{1}_{Q_j},$$

so that  $g$  agrees with  $f - \langle f \rangle_{Q_j}$  on  $Q$ . Using (4.1.5) and (4.1.2),

$$\langle |g| \rangle_Q = \frac{1}{|Q|} \int_Q |f - \langle f \rangle_{Q_j}| \geq \langle |f| \rangle_Q - |\langle f \rangle_{Q_j}| > 2^{n+1}\lambda - 2^n\lambda = 2^n\lambda,$$

where  $|\langle f \rangle_{Q_j}| \leq \langle |f| \rangle_{Q_j} \leq 2^n\lambda$ . Since  $Q$  is a dyadic cube containing  $x$ , this says  $\mathcal{M}_d g(x) > 2^n\lambda$ . Therefore

$$E \cap Q_j \subseteq \{\mathcal{M}_d g > 2^n\lambda\}.$$

**Step 4.** *Apply the weak (1,1) inequality.*

By the dyadic maximal inequality with constant 1 (Theorem 3.1.4 (the dyadic maximal inequality)) and then (4.1.4),

$$m(E \cap Q_j) \leq m(\{\mathcal{M}_d g > 2^n\lambda\}) \leq \frac{\|g\|_{L^1}}{2^n\lambda} = \frac{1}{2^n\lambda} \int_{Q_j} |f - \langle f \rangle_{Q_j}| \leq \frac{\gamma\lambda|Q_j|}{2^n\lambda},$$

which is (4.1.3). Summing over  $j$  and using disjointness completes the proof.  $\square$

**Where we are.** Notice that the constant 1 in the dyadic maximal inequality was genuinely convenient in Step 4: any larger constant would have appeared in the final bound and forced a smaller  $\gamma$ , which is harmless, but the argument is cleanest as it stands. This is the payoff for having proved the dyadic version in Volume 3 rather than relying on Vitali.

Notice also that the two constants  $2^{n+1}$  and  $2^{-n}$  are not sacred. All that matters for the absorption argument below is that the first exceeds 1 and the second is finite; any good- $\lambda$  inequality of the shape (4.1.1) would do.

### 4.1.4 The Fefferman–Stein inequality

**Theorem 4.1.4** (Fefferman–Stein). *Let  $1 \leq p_0 \leq p < \infty$  and let  $f \in L^{p_0}(\mathbb{R}^n)$ . Then*

$$\|f\|_{L^p(\mathbb{R}^n)} \leq \|\mathcal{M}_d f\|_{L^p} \leq C_{n,p} \|\mathcal{M}^\# f\|_{L^p}.$$

*In particular, for  $1 < p < \infty$  and  $f \in L^p(\mathbb{R}^n)$ ,*

$$\|\mathcal{M}^\# f\|_{L^p} \approx_{n,p} \|f\|_{L^p}.$$

*Proof. Step 1. The level sets are finite, and a truncation.*

Since  $f \in L^{p_0}$  with  $p_0 < \infty$ , the dyadic maximal function satisfies a weak  $(p_0, p_0)$  bound: for  $p_0 = 1$  this is Theorem 3.1.4 (the dyadic maximal inequality), and for  $p_0 > 1$  it follows from the strong  $L^{p_0}$  bound (Theorem 1.7.3 (Hardy–Littlewood maximal inequality), transferred to the dyadic setting) and Chebyshev. Either way

$$m(\{\mathcal{M}_d f > \lambda\}) \leq \frac{C\|f\|_{p_0}^{p_0}}{\lambda^{p_0}} < \infty \quad \text{for every } \lambda > 0, \quad (4.1.6)$$

so Theorem 4.1.3 applies.

The first inequality of the theorem is the pointwise bound  $|f| \leq \mathcal{M}_d f$  a.e., which holds by the Lebesgue differentiation theorem applied along the dyadic cubes shrinking to a point (Volume 1, Theorem 7.7).

For the second we cannot integrate directly, for the reason explained in the Scratch box: we must know the quantity being absorbed is finite. So fix  $\Lambda > 0$  and define the truncated integral

$$J_\Lambda = p \int_0^\Lambda \mu^{p-1} m(\{\mathcal{M}_d f > a\mu\}) d\mu, \quad a := 2^{n+1}.$$

By (4.1.6) the integrand is at most  $C\|f\|_{p_0}^{p_0} a^{-p_0} \mu^{p-1-p_0}$ , which is integrable on  $(0, \Lambda)$  when  $p > p_0$ ; and when  $p = p_0$  the integrand is bounded by  $\mu^{p-1} m(\{\mathcal{M}_d f > a\mu\})$ , integrable on  $(0, \Lambda)$  because  $\mathcal{M}_d f \in L^p$  in that case, by the maximal theorem when  $p > 1$  and by direct inspection when  $p = p_0 = 1$  — in which case the conclusion is void unless  $\mathcal{M}^\# f \in L^1$ , and we may assume that. In all cases  $J_\Lambda < \infty$ .

**Step 2.** *Split the level set and apply the good- $\lambda$  inequality.*

For each  $\mu$ ,

$$\{\mathcal{M}_d f > a\mu\} \subseteq \{\mathcal{M}_d f > a\mu, \mathcal{M}_d^\# f \leq \gamma\mu\} \cup \{\mathcal{M}_d^\# f > \gamma\mu\},$$

simply because every point either satisfies  $\mathcal{M}_d^\# f \leq \gamma\mu$  or it does not. By Theorem 4.1.3 the first set has measure at most  $2^{-n}\gamma m(\{\mathcal{M}_d f > \mu\})$ . Hence

$$J_\Lambda \leq 2^{-n}\gamma^p \int_0^\Lambda \mu^{p-1} m(\{\mathcal{M}_d f > \mu\}) d\mu + p \int_0^\infty \mu^{p-1} m(\{\mathcal{M}_d^\# f > \gamma\mu\}) d\mu.$$

**Step 3.** *Identify the two terms.*

For the first, substitute  $\mu = a\nu$  in the definition of  $J_\Lambda$  to get

$$p \int_0^\Lambda \mu^{p-1} m(\{\mathcal{M}_d f > \mu\}) d\mu \leq p \int_0^{a\Lambda} \mu^{p-1} m(\{\mathcal{M}_d f > \mu\}) d\mu = a^p J_\Lambda,$$

using  $a > 1$  so that the range of integration only grows. For the second, the layer-cake formula and the substitution  $\mu \mapsto \mu/\gamma$  give

$$p \int_0^\infty \mu^{p-1} m(\{\mathcal{M}_d^\# f > \gamma\mu\}) d\mu = \gamma^{-p} \|\mathcal{M}_d^\# f\|_{L^p}^p \leq \gamma^{-p} \|\mathcal{M}^\# f\|_{L^p}^p.$$

Therefore

$$J_\Lambda \leq 2^{-n} \gamma a^p J_\Lambda + \gamma^{-p} \|\mathcal{M}^\# f\|_{L^p}^p. \quad (4.1.7)$$

**Step 4.** *Absorb, then remove the truncation.*

Choose  $\gamma$  so small that  $2^{-n} \gamma a^p \leq \frac{1}{2}$ ; explicitly,  $\gamma = \min\{\frac{1}{2}, 2^{n-1} a^{-p}\}$ . Since  $J_\Lambda < \infty$  by Step 1, we may subtract  $\frac{1}{2} J_\Lambda$  from both sides of (4.1.7), obtaining

$$J_\Lambda \leq 2\gamma^{-p} \|\mathcal{M}^\# f\|_{L^p}^p,$$

a bound independent of  $\Lambda$ . Letting  $\Lambda \rightarrow \infty$  and using monotone convergence,

$$p \int_0^\infty \mu^{p-1} m(\{\mathcal{M}_d f > a\mu\}) d\mu \leq 2\gamma^{-p} \|\mathcal{M}^\# f\|_{L^p}^p,$$

and the left side equals  $a^{-p} \|\mathcal{M}_d f\|_{L^p}^p$  by the layer-cake formula. Hence  $\|\mathcal{M}_d f\|_{L^p}^p \leq 2a^p \gamma^{-p} \|\mathcal{M}^\# f\|_{L^p}^p$ .

**Step 5.** *The reverse inequality.*

For  $1 < p < \infty$ ,  $\|\mathcal{M}^\# f\|_p \leq 2\|\mathcal{M} f\|_p \lesssim_{n,p} \|f\|_p$  by Proposition 4.1.2(ii) and the Hardy–Littlewood maximal theorem (Theorem 1.7.3 (Hardy–Littlewood maximal inequality)).  $\square$

**Pitfall: the hypothesis  $f \in L^{p_0}$  cannot be dropped**

Take  $f \equiv 1$ . Then  $\mathcal{M}^\# f \equiv 0$ , so the right-hand side of Theorem 4.1.4 vanishes while  $\|f\|_p = \infty$ . The hypothesis  $f \in L^{p_0}$  is what excludes this, and it enters the proof at exactly one place: Step 1, to guarantee  $J_\Lambda < \infty$  so that the absorption in Step 4 is legitimate.

Readers should treat every good- $\lambda$  argument they meet with the same suspicion. The finiteness hypothesis is easy to overlook and the conclusion is false without it.

## 4.1.5 Interpolation with BMO

We can now state the theorem that makes BMO a working tool.

**Theorem 4.1.5** (Interpolation between  $L^{p_0}$  and BMO). *Let  $1 \leq p_0 < \infty$  and let  $T$  be a sublinear operator defined on  $L^{p_0} \cap L^\infty$  satisfying*

$$\|Tf\|_{L^{p_0}} \leq A\|f\|_{L^{p_0}}, \quad \|Tf\|_{\text{BMO}} \leq B\|f\|_{L^\infty}.$$

*Then for every  $p$  with  $p_0 < p < \infty$ ,*

$$\|Tf\|_{L^p} \lesssim_{n,p,p_0} A^{p_0/p} B^{1-p_0/p} \|f\|_{L^p}.$$

*Idea of the proof.* This is the Marcinkiewicz argument of Volume 2 with  $L^\infty$  replaced by BMO, and the sharp maximal function is what makes the replacement possible. Split  $f$  at height  $\lambda$  into a tall part  $g$ , which is in  $L^{p_0}$ , and a short part  $b$ , which is bounded by  $\lambda$ . The hypothesis on  $L^{p_0}$  handles  $g$ ; the hypothesis on  $L^\infty$  says  $\|Tb\|_{\text{BMO}} \leq B\lambda$ , which by Proposition 4.1.2(i) means  $\mathcal{M}^\#(Tb) \leq B\lambda$  pointwise. So the sharp maximal function of  $Tb$  is under control at every point, which is precisely the kind of statement Marcinkiewicz needs and which BMO alone would not provide. Then Theorem 4.1.4 converts the resulting bound on  $\mathcal{M}^\#(Tf)$  back into a bound on  $Tf$ .

*Proof.* By homogeneity assume  $A = B = 1$ ; the general case follows by replacing  $T$  with  $T/B$  and rescaling  $f$ , and tracking the two constants through the computation gives the stated powers.

Fix  $\lambda > 0$  and split

$$f = g_\lambda + b_\lambda, \quad g_\lambda = f \mathbf{1}_{\{|f| > \lambda\}}, \quad b_\lambda = f \mathbf{1}_{\{|f| \leq \lambda\}}.$$

**Step 1.** *Pointwise control of  $\mathcal{M}^\#(Tf)$ .*

By sublinearity and Proposition 4.1.2(iii),

$$\mathcal{M}^\#(Tf) \leq \mathcal{M}^\#(Tg_\lambda) + \mathcal{M}^\#(Tb_\lambda).$$

Since  $\|b_\lambda\|_\infty \leq \lambda$ , the second hypothesis gives  $\|Tb_\lambda\|_{\text{BMO}} \leq \lambda$ , and by Proposition 4.1.2(i) this is exactly  $\mathcal{M}^\#(Tb_\lambda) \leq \lambda$  pointwise. Hence

$$\{\mathcal{M}^\#(Tf) > 2\lambda\} \subseteq \{\mathcal{M}^\#(Tg_\lambda) > \lambda\} \subseteq \{\mathcal{M}(Tg_\lambda) > \tfrac{\lambda}{2}\},$$

the last inclusion by Proposition 4.1.2(ii).

**Step 2.** *Estimate the measure.*

By the weak  $(p_0, p_0)$  bound for  $\mathcal{M}$  — Theorem 1.7.3 (Hardy–Littlewood maximal inequality) for  $p_0 = 1$ , and Chebyshev with the strong bound for  $p_0 > 1$  — together with the first hypothesis,

$$m(\{\mathcal{M}^\#(Tf) > 2\lambda\}) \lesssim_{n,p_0} \frac{\|Tg_\lambda\|_{p_0}^{p_0}}{\lambda^{p_0}} \leq \frac{\|g_\lambda\|_{p_0}^{p_0}}{\lambda^{p_0}} = \frac{1}{\lambda^{p_0}} \int_{\{|f|>\lambda\}} |f|^{p_0}.$$

**Step 3.** *Integrate.*

By the layer-cake formula, the substitution  $\lambda \mapsto 2\lambda$ , and Step 2,

$$\|\mathcal{M}^\#(Tf)\|_p^p = p \int_0^\infty \lambda^{p-1} m(\{\mathcal{M}^\#(Tf) > \lambda\}) d\lambda \lesssim_{n,p,p_0} \int_0^\infty \lambda^{p-p_0-1} \int_{\{|f|>\lambda\}} |f(x)|^{p_0} dx d\lambda$$

By Tonelli the double integral equals

$$\int_{\mathbb{R}^n} |f(x)|^{p_0} \left( \int_0^{|f(x)|} \lambda^{p-p_0-1} d\lambda \right) dx = \frac{1}{p-p_0} \int_{\mathbb{R}^n} |f|^p,$$

the inner integral converging at 0 exactly because  $p > p_0$ .

**Step 4.** *Conclude.*

So  $\|\mathcal{M}^\#(Tf)\|_p \lesssim \|f\|_p$ , and since  $Tf \in L^{p_0}$  by hypothesis, Theorem 4.1.4 applies and gives  $\|Tf\|_p \lesssim \|\mathcal{M}^\#(Tf)\|_p \lesssim \|f\|_p$ .  $\square$

**WORKED EXAMPLE 4.1.3— TIER C: RESEARCH TASTE A second proof of the Calderón–Zygmund  $L^p$  theorem**

Volume 3 obtained the  $L^p$  bounds for a singular integral by proving a weak  $(1, 1)$  bound and interpolating. Here is an entirely different route, via the sharp maximal function. It is longer for the classical convolution operators, and it is the only one of the two that survives when the operator has no convolution structure — which is why it, and not the Volume 3 argument, is the ancestor of the modern theory.

**Proposition 4.1.6.** *Let  $T$  be bounded on  $L^2(\mathbb{R}^n)$  with kernel  $K$  satisfying  $|\nabla K(x)| \leq B|x|^{-n-1}$ , in the sense of Volume 3, Theorem 2.3. Then for every  $f \in L^2$  with compact support,*

$$\mathcal{M}^\#(Tf)(x) \lesssim_n (\|T\|_{2 \rightarrow 2} + B) \left( \mathcal{M}(|f|^2)(x) \right)^{1/2} \quad \text{for a.e. } x.$$

*Proof.* Normalise  $\|T\|_{2 \rightarrow 2} + B \leq 1$ . Fix  $x$  and a cube  $Q \ni x$ , with centre  $c$  and side  $\ell$ , and let  $Q^* = 2\sqrt{n}Q$ . Split

$$f = f\mathbf{1}_{Q^*} + f\mathbf{1}_{(Q^*)^c} = f_1 + f_2.$$



We must bound  $|Q|^{-1} \int_Q |Tf - \alpha|$  for a suitable constant  $\alpha$ ; recall from Drill 3.7.1 that using any constant in place of  $\langle Tf \rangle_Q$  costs at most a factor 2. Take  $\alpha = Tf_2(c)$ .

*Near part.* By Cauchy–Schwarz and the  $L^2$  bound,

$$\frac{1}{|Q|} \int_Q |Tf_1| \leq \left( \frac{1}{|Q|} \int_Q |Tf_1|^2 \right)^{1/2} \leq |Q|^{-1/2} \|f \mathbf{1}_{Q^*}\|_2 = \left( \frac{1}{|Q|} \int_{Q^*} |f|^2 \right)^{1/2}.$$

Since  $|Q^*| = (2\sqrt{n})^n |Q|$  and  $Q^*$  is a cube containing  $x$ , the right-hand side is  $\lesssim_n (\mathcal{M}(|f|^2)(x))^{1/2}$ .

*Far part.* For  $y \in Q$  and  $z \notin Q^*$  we have  $|z - c| > \sqrt{n}\ell \geq 2|y - c|$ , so the segment from  $y - z$  to  $c - z$  stays in  $\{|w| \geq |c - z|/2\}$  and the mean value theorem gives

$$|K(y - z) - K(c - z)| \leq |y - c| \sup |\nabla K| \lesssim_n \frac{\ell}{|z - c|^{n+1}}.$$

Therefore, decomposing  $\{|z - c| > \sqrt{n}\ell\}$  into dyadic annuli  $|z - c| \approx 2^k \ell$ ,

$$|Tf_2(y) - Tf_2(c)| \lesssim_n \sum_{k \geq 0} \frac{\ell}{(2^k \ell)^{n+1}} \int_{|z-c| \lesssim 2^k \ell} |f(z)| \, dz \lesssim_n \sum_{k \geq 0} 2^{-k} \mathcal{M}f(x) \lesssim_n \mathcal{M}f(x),$$

using that each annulus is contained in a cube of side  $\approx 2^k \ell$  centred at  $c$ , which contains  $x$ . Finally  $\mathcal{M}f \leq (\mathcal{M}(|f|^2))^{1/2}$  by Cauchy–Schwarz on each average. Averaging over  $y \in Q$  and combining the two parts, then taking the supremum over  $Q \ni x$ , gives the claim.  $\square$

*Consequence: the  $L^p$  theorem for  $p > 2$ .* Let  $2 < p < \infty$ . By Theorem 4.1.4 and Proposition 4.1.6,

$$\|Tf\|_p \lesssim \|\mathcal{M}^\#(Tf)\|_p \lesssim \left\| (\mathcal{M}(|f|^2))^{1/2} \right\|_p = \left\| \mathcal{M}(|f|^2) \right\|_{p/2}^{1/2} \lesssim_{n,p} \left\| |f|^2 \right\|_{p/2}^{1/2} = \|f\|_p,$$

the last inequality being the Hardy–Littlewood maximal theorem at exponent  $p/2 > 1$ . The range  $1 < p < 2$  then follows by duality, exactly as in Volume 3, Theorem 2.3, Step 6.

*Why this proof matters.* Three observations.

First, the exponent 2 inside  $\mathcal{M}(|f|^2)^{1/2}$  is what forces the restriction  $p > 2$ ; it entered when we used Cauchy–Schwarz to exploit the  $L^2$  boundedness of  $T$  on the near part. Using instead a weak  $(1, 1)$  bound for  $T$  one can replace 2 by any  $s > 1$  and reach every  $p > 1$  directly, at the cost of a slightly more delicate argument.

Second, nothing in Proposition 4.1.6 used that  $T$  is a convolution operator. The kernel was allowed to depend on  $x$  and  $y$  separately throughout; we wrote  $K(y - z)$  only for notational familiarity. This is the route that generalises to the  $T(1)$  theorem, mentioned in Worked Example 3.2.3 (why the  $L^2$  hypothesis is the hard one), where no convolution structure is available and no multiplier exists.

Third, and most relevant to this series: the proof produced a *pointwise* inequality between two maximal functions, and then converted it into a norm inequality by a single application of Theorem 4.1.4. That two-stage structure — prove something pointwise about a cleverly chosen maximal function, then integrate — recurs constantly. In Volume 12, the multilinear restriction estimates are proved as pointwise bounds on a broad maximal function before any integration happens.

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**WORKED EXAMPLE 4.1.4— TIER B Using the sharp maximal function**

The sharp function  $\mathcal{M}^\# f$  measures oscillation rather than size, and the three ways it is used are worth separating.

(a) *To prove an  $L^p$  bound.* The Fefferman–Stein inequality  $\|f\|_{L^p} \lesssim \|\mathcal{M}^\# f\|_{L^p}$  converts a bound on oscillation into a bound on size, for  $1 < p < \infty$ . This is the standard route to  $L^p$  estimates for operators whose oscillation is controllable but whose size is not.

(b) *To interpolate against BMO.*  $\|\mathcal{M}^\# f\|_{L^\infty} = \|f\|_{\text{BMO}}$  by definition, so the pair  $(L^p, \text{BMO})$  behaves like  $(L^p, L^\infty)$  for interpolation purposes — which is exactly what Volume 3’s endpoint failure required.

(c) *To localise.*  $\mathcal{M}^\#$  is defined by averages over cubes, so a bound on  $\mathcal{M}^\# f$  on a cube gives information about  $f$  there and nowhere else. Arguments that need locality use  $\mathcal{M}^\#$  where they would otherwise use the Hardy–Littlewood maximal function.

*The check to run.*  $\mathcal{M}^\# f \leq 2\mathcal{M}f$  always, so a bound involving  $\mathcal{M}^\#$  is stronger than the same bound with  $\mathcal{M}$ . If a proof uses  $\mathcal{M}^\#$  where  $\mathcal{M}$  would do, the extra strength is being wasted — and conversely, a proof that needs  $\mathcal{M}^\#$  cannot be simplified by substituting  $\mathcal{M}$ .

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## Drills

**Drill 4.1.1.** Show that in Definition 4.1.1 one may replace  $\langle f \rangle_Q$  by any constant  $c_Q$ , at the cost of a factor between 1 and 2:  $\frac{1}{2}\mathcal{M}^\# f(x) \leq \sup_{Q \ni x} \inf_c |Q|^{-1} \int_Q |f - c| \leq \mathcal{M}^\# f(x)$ .

**Drill 4.1.2.** Verify the claim in Worked Example 4.1.1 that  $\mathcal{M}^{\#}\mathbf{1}_{[0,1]}(\frac{1}{2}) = \frac{1}{2}$ , by considering the intervals  $[\frac{1}{2} - t, \frac{1}{2} + t]$  for  $t > 0$  and optimising.

**Drill 4.1.3.** Where in the proof of Theorem 4.1.3 is the hypothesis  $\gamma < 1$  used? Show that the conclusion is vacuous for  $\gamma \geq 2^n$ , and explain why.

**Drill 4.1.4.** Deduce from Theorem 4.1.5 and Theorem 3.7.4 ( $L^\infty \rightarrow \text{BMO}$ ) that a Calderón–Zygmund operator is bounded on  $L^p$  for every  $p > 1$ , and compare the constants obtained with those of Theorem 3.2.3 (Calderón–Zygmund).

## Notes

The sharp maximal function and Theorem 4.1.4 are due to Fefferman and Stein (1972), in the same paper that established the duality of Chapter 4.3. The good- $\lambda$  technique is Burkholder and Gundy’s, from martingale theory, where inequalities of the shape (4.1.1) first appeared.

Proposition 4.1.6 in the form given here is standard; the refinement replacing  $\mathcal{M}(|f|^2)^{1/2}$  by  $\mathcal{M}(|f|^s)^{1/s}$  for  $s$  close to 1, which yields the full range  $1 < p < \infty$  directly, is due to Alvarez and Pérez.



## Chapter 4.2

# Atoms and the Hardy Space $H^1$

### 4.2.1 What goes wrong at $p = 1$ , and what to do about it

Volume 3 showed that the Hilbert transform of  $\mathbf{1}_{[0,1]}$  decays like  $1/|x|$  at infinity, hence is not integrable. The computation was two lines, and it is worth asking what exactly it revealed.

The function  $\mathbf{1}_{[0,1]}$  is a perfectly good  $L^1$  function. What it lacks is *cancellation*: its integral is 1, not 0. Recall the far-field estimate from the Calderón–Zygmund proof: for  $x$  far from the support of  $f$ ,

$$Tf(x) = \int K(x-y)f(y)dy \approx K(x) \int f(y)dy,$$

because  $K$  is nearly constant across the support of  $f$  when viewed from far away. So  $Tf(x) \approx K(x) \cdot \int f$ , and  $K$  decays exactly like  $|x|^{-n}$ , which is exactly not integrable. The obstruction is the number  $\int f$ , and nothing else.

This suggests the repair. Restrict attention to functions with  $\int f = 0$ . Then the leading term vanishes and the next one decays like  $|x|^{-n-1}$ , which *is* integrable at infinity. That is the whole idea of the Hardy space.

*Scratch work.* One cannot simply define the space to be  $\{f \in L^1 : \int f = 0\}$ . That set is not closed in  $L^1$  — it is a dense hyperplane — and more importantly the condition is global, whereas the estimate above was local: what mattered was that  $f$  had mean zero *on the region where it lives*, so that the kernel could be frozen there.

The right object is therefore built from pieces, each of which is supported in a single cube and has mean zero on that cube. Such a piece is called an *atom*, and the Hardy space is what one gets by summing atoms absolutely. This definition has a practical virtue that will become apparent immedi-

ately: to prove that an operator is bounded on  $H^1$ , it suffices to check it on a single atom, which is a completely explicit object.

## 4.2.2 Atoms

**Definition 4.2.1.** An *atom* is a function  $a : \mathbb{R}^n \rightarrow \mathbb{C}$  for which there exists a cube  $Q$  with

$$(A1) \quad \text{supp } a \subseteq Q;$$

$$(A2) \quad \|a\|_{L^\infty} \leq \frac{1}{|Q|};$$

$$(A3) \quad \int_{\mathbb{R}^n} a(x) \, dx = 0.$$

We say  $a$  is an atom *associated with*  $Q$ .

The normalisation in (A2) is chosen so that all atoms have comparable  $L^1$  norm:

$$\|a\|_{L^1} = \int_Q |a| \leq |Q| \cdot \frac{1}{|Q|} = 1.$$

So an atom is a bump of height  $|Q|^{-1}$  on a cube of volume  $|Q|$ , with mean zero — a normalised, cancelling lump at one location and one scale.

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### WORKED EXAMPLE 4.2.1— TIER A The simplest atoms

(a) *The two-block atom.* In  $\mathbb{R}^1$ , let  $Q = [0, 1]$  and

$$a = \mathbf{1}_{[0, 1/2]} - \mathbf{1}_{(1/2, 1]}.$$

Then  $\text{supp } a \subseteq Q$ ,  $\|a\|_\infty = 1 = |Q|^{-1}$ , and  $\int a = \frac{1}{2} - \frac{1}{2} = 0$ . So  $a$  is an atom associated with  $Q$ .

(b) *Rescaling.* If  $a$  is an atom associated with  $Q$  and  $\lambda > 0$ , then

$$a_\lambda(x) = \lambda^{-n} a(x/\lambda)$$

is an atom associated with  $\lambda Q$ : the support scales correctly, the sup norm becomes  $\lambda^{-n} \|a\|_\infty \leq \lambda^{-n} |Q|^{-1} = |\lambda Q|^{-1}$ , and the mean-zero property is preserved by the change of variables. So the family of atoms is invariant under the dilations that preserve the  $L^1$  norm — which is what one wants of an  $L^1$  substitute.

(c) *A smooth atom.* Let  $\varphi \in C_c^\infty(Q)$  be any smooth function supported in  $Q$  with  $\int \varphi = 0$  and  $\|\varphi\|_\infty \leq |Q|^{-1}$ . Such functions exist in abundance: take

any  $\psi \in C_c^\infty(Q)$  and set  $\varphi = c(\psi - \eta)$  where  $\eta \in C_c^\infty(Q)$  has the same integral as  $\psi$  but different shape, then normalise.

(d) *What is not an atom.*  $\mathbf{1}_{[0,1]}$  fails (A3). The function  $|x|^{-n/2}\mathbf{1}_{B(0,1)}$  fails (A2) — it is unbounded — although it is in  $L^1$ . Both failures matter: the first is the obstruction of §2.1, and the second is what makes the uniform estimate of Theorem 4.2.5 possible.

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**Lemma 4.2.2** (The key estimate for a single atom). *Let  $a$  be an atom associated with a cube  $Q$  of centre  $c$ , and let  $K$  satisfy Hörmander's condition*

$$\int_{|x|>2|y|} |K(x-y) - K(x)| \, dx \leq A \quad \text{for all } y \neq 0$$

*of Definition 3.2.1 (Calderón–Zygmund kernel). Then, writing  $Q^* = 2\sqrt{n}Q$  for the concentric dilate,*

$$\int_{(Q^*)^c} \left| \int_Q K(x-y)a(y) \, dy \right| \, dx \leq A.$$

*Idea of the proof.* This is exactly Step 4 of the Calderón–Zygmund proof (Theorem 3.2.3 (Calderón–Zygmund)), extracted and stated on its own. The mean-zero property lets one replace  $K(x-y)$  by  $K(x-y) - K(x-c)$  at no cost, and Hörmander's condition then integrates the difference. Isolating it here makes visible that the entire  $L^1$ -endpoint theory is a statement about atoms.

*Proof.* Since  $\int a = 0$ , for any  $x$  we may subtract the constant  $K(x-c)$  inside the integral:

$$\int_Q K(x-y)a(y) \, dy = \int_Q (K(x-y) - K(x-c))a(y) \, dy.$$

Taking absolute values, integrating over  $x \notin Q^*$  and applying Tonelli,

$$\int_{(Q^*)^c} \left| \int_Q K(x-y)a(y) \, dy \right| \, dx \leq \int_Q |a(y)| \left( \int_{(Q^*)^c} |K(x-y) - K(x-c)| \, dx \right) \, dy.$$

Fix  $y \in Q$  and substitute  $x \mapsto x+c$ ,  $y \mapsto y+c$ , so that the inner integral becomes  $\int |K(x-y) - K(x)| \, dx$  over the region  $\{x+c \notin Q^*\}$ . If  $Q$  has side  $\ell$  then  $|y| \leq \frac{\sqrt{n}}{2}\ell$  for  $y \in Q-c$ , while  $x+c \notin Q^*$  forces  $|x| > \sqrt{n}\ell \geq 2|y|$ . So the region of integration is contained in  $\{|x| > 2|y|\}$ , and Hörmander's condition bounds the inner integral by  $A$ . Since  $\int_Q |a| \leq 1$  by (A2), the result follows.  $\square$

**Where we are.** Compare this with the failure at the start of the chapter. For  $f = \mathbf{1}_{[0,1]}$  the analogous integral is infinite, and the computation above shows precisely where the difference lies: the step “subtract  $K(x - c)$  at no cost” requires  $\int f = 0$ . Everything else — Hörmander’s condition, the dilated cube, the Tonelli exchange — is unchanged.

### 4.2.3 The space $H^1$

**Definition 4.2.3.** The *Hardy space*  $H^1(\mathbb{R}^n)$  consists of those  $f \in L^1(\mathbb{R}^n)$  admitting a decomposition

$$f = \sum_{j=1}^{\infty} \lambda_j a_j, \quad a_j \text{ atoms}, \quad \sum_{j=1}^{\infty} |\lambda_j| < \infty, \quad (4.2.1)$$

with the series converging in  $L^1(\mathbb{R}^n)$ . We set

$$\|f\|_{H^1} = \inf \left\{ \sum_j |\lambda_j| : \text{all decompositions (4.2.1) of } f \right\}.$$

**Proposition 4.2.4** (Basic properties). *(i) The series in (4.2.1) converges absolutely in  $L^1$ , and  $\|f\|_{L^1} \leq \|f\|_{H^1}$ . Thus  $H^1(\mathbb{R}^n) \subseteq L^1(\mathbb{R}^n)$  with continuous inclusion.*

*(ii) Every  $f \in H^1$  satisfies  $\int_{\mathbb{R}^n} f = 0$ .*

*(iii)  $\|\cdot\|_{H^1}$  is a norm, and  $H^1$  is complete.*

*(iv) The inclusion  $H^1 \subseteq L^1$  is strict.*

*Proof.* (i) Each atom has  $\|a_j\|_1 \leq 1$ , so  $\sum_j |\lambda_j| \|a_j\|_1 \leq \sum_j |\lambda_j| < \infty$  and the series converges absolutely in the complete space  $L^1$  (Theorem 1.8.6 (Riesz–Fischer)). Its sum has  $L^1$  norm at most  $\sum_j |\lambda_j|$ , and taking the infimum over decompositions gives the stated inequality.

(ii) The functional  $g \mapsto \int g$  is continuous on  $L^1$  and annihilates every atom by (A3), hence annihilates the  $L^1$ -convergent sum.

(iii) Homogeneity and the triangle inequality follow by concatenating decompositions: if  $f = \sum \lambda_j a_j$  and  $g = \sum \mu_k a'_k$  then the combined family decomposes  $f + g$  with coefficient sum  $\sum |\lambda_j| + \sum |\mu_k|$ . If  $\|f\|_{H^1} = 0$  then  $\|f\|_{L^1} = 0$  by (i), so  $f = 0$ . For completeness, suppose  $\sum_k \|f_k\|_{H^1} < \infty$ ; choose for each  $k$  a decomposition of  $f_k$  with coefficient sum at most  $\|f_k\|_{H^1} + 2^{-k}$  and concatenate all of them. The resulting family has finite total coefficient sum and, by (i), its sum in  $L^1$  is  $\sum_k f_k$ . So absolutely convergent series converge, which is equivalent to completeness (Theorem 1.8.6 (Riesz–Fischer)).



(iv) By (ii), any nonnegative  $f \in L^1$  with  $f \neq 0$  — for instance  $\mathbf{1}_{[0,1]^n}$  — lies in  $L^1 \setminus H^1$ .  $\square$

**Pitfall: the mean-zero condition is necessary, not sufficient**

Proposition 4.2.4(ii) says  $\int f = 0$  for every  $f \in H^1$ . The converse is false, and it is worth knowing a counterexample.

In  $\mathbb{R}^1$  let  $f(x) = (\mathbf{1}_{[1,2]}(x) - \mathbf{1}_{[-2,-1]}(x))$ , which has mean zero and is in  $L^1$ ; this one *is* in  $H^1$  (it is a multiple of an atom associated with  $[-2, 2]$ ). But consider instead

$$g(x) = \frac{1}{x(\log x)^2} \mathbf{1}_{(2,\infty)}(x) - \frac{1}{|x|(\log|x|)^2} \mathbf{1}_{(-\infty,-2)}(x).$$

Then  $g \in L^1(\mathbb{R})$  and  $\int g = 0$  by symmetry. But  $g \notin H^1$ : the mass of  $g$  is spread over infinitely many dyadic scales with no cancellation *within any bounded region*, and Chapter 4.4 will give a test — the maximal function characterisation — that detects this. The point for now is that the  $H^1$  condition is genuinely about cancellation at every scale and location, not about one global integral.

## 4.2.4 Singular integrals on $H^1$

We can now discharge the first of the two debts from Volume 3.

**Theorem 4.2.5** (Calderón–Zygmund operators map  $H^1$  to  $L^1$ ). *Let  $T$  be a linear operator, bounded on  $L^2(\mathbb{R}^n)$  with norm  $B_2$ , whose kernel  $K$  satisfies Hörmander’s condition with constant  $A$ , in the sense of Theorem 3.2.3 (Calderón–Zygmund). Then*

$$\|Tf\|_{L^1(\mathbb{R}^n)} \lesssim_n (A + B_2) \|f\|_{H^1(\mathbb{R}^n)} \quad \text{for all } f \in H^1 \cap L^2.$$

*Idea of the proof.* The bulk of the work is a bound for a *single atom*, with a constant independent of which atom: for one atom associated with a cube  $Q$ , split  $\mathbb{R}^n$  into the dilate  $Q^*$  and its complement, use Cauchy–Schwarz and the  $L^2$  bound near the cube, and Lemma 4.2.2 far from it.

But it must be stressed at the outset that the atomic bound does *not* by itself give the theorem. The step from “ $\|Ta\|_{L^1} \leq C$  for every atom” to “ $T$  is bounded on  $H^1$ ” is *invalid in general* — see the Pitfall below — and Step 1 of the proof supplies a genuine limiting argument in its place, using that  $T$  is a priori bounded on  $L^2$  and of weak type  $(1, 1)$ .

For one atom associated with a cube  $Q$ , split  $\mathbb{R}^n$  into the dilate  $Q^*$  and its complement. Near the cube, use Cauchy–Schwarz and the  $L^2$  bound;

the normalisation (A2) is exactly what makes  $\|a\|_2$  small enough for this to work. Far from the cube, use Lemma 4.2.2, which is where the mean-zero property (A3) is spent.

*Proof. Step 1. Reduction to a single atom.*

Suppose we have shown  $\|Ta\|_{L^1} \leq C$  for every atom  $a$ , with  $C$  depending only on  $n, A, B_2$ . Let  $f \in H^1 \cap L^2$  and let  $f = \sum_j \lambda_j a_j$  be any atomic decomposition. The partial sums converge to  $f$  in  $L^1$ ; but  $T$  is a priori defined and bounded on  $L^2$ , so to pass  $T$  through the sum we argue as follows. Each partial sum  $f_N = \sum_{j \leq N} \lambda_j a_j$  lies in  $L^2$  (atoms are bounded with compact support), and by linearity and the assumed bound,

$$\|Tf_N\|_{L^1} \leq \sum_{j \leq N} |\lambda_j| \|Ta_j\|_{L^1} \leq C \sum_j |\lambda_j|.$$

Moreover  $f_N \rightarrow f$  in  $L^1$  and, since  $T$  is of weak type  $(1, 1)$  by Theorem 3.2.3 (Calderón–Zygmund),  $Tf_N \rightarrow Tf$  in measure; passing to a subsequence converging a.e. and applying Fatou gives  $\|Tf\|_{L^1} \leq C \sum_j |\lambda_j|$ . Taking the infimum over decompositions gives the theorem.

**Step 2.** *The near estimate.*

Let  $a$  be an atom associated with  $Q$ , and  $Q^* = 2\sqrt{n}Q$ . By Cauchy–Schwarz on  $Q^*$  and the  $L^2$  boundedness of  $T$ ,

$$\int_{Q^*} |Ta| \leq |Q^*|^{1/2} \|Ta\|_{L^2} \leq |Q^*|^{1/2} B_2 \|a\|_{L^2}.$$

Now the normalisation does its work: by (A1) and (A2),

$$\|a\|_{L^2}^2 = \int_Q |a|^2 \leq |Q| \cdot \frac{1}{|Q|^2} = \frac{1}{|Q|}, \quad \text{so} \quad \|a\|_{L^2} \leq |Q|^{-1/2}.$$

Since  $|Q^*| = (2\sqrt{n})^n |Q|$ , we get

$$\int_{Q^*} |Ta| \leq (2\sqrt{n})^{n/2} B_2.$$

Note what happened: the two factors  $|Q^*|^{1/2}$  and  $|Q|^{-1/2}$  cancelled, leaving a bound independent of the cube. This scale invariance is built into (A2) and is the reason the reduction to atoms is useful.

**Step 3.** *The far estimate.*

For  $x \notin Q^*$  the kernel representation applies, since  $x \notin \text{supp } a$ , and Lemma 4.2.2 gives

$$\int_{(Q^*)^c} |Ta| \leq A.$$

**Step 4. Combine.**

Adding,  $\|Ta\|_{L^1} \leq (2\sqrt{n})^{n/2}B_2 + A \lesssim_n A + B_2$ , uniformly over atoms. Step 1 completes the proof.  $\square$

**Pitfall: uniform boundedness on atoms does not imply boundedness on  $H^1$**

The following argument appears in many places and is *wrong*:

“Every  $f \in H^1$  is  $\sum_j \lambda_j a_j$  with  $\sum |\lambda_j| \approx \|f\|_{H^1}$ . So if  $\|Ta\|_{L^1} \leq C$  for every atom, then  $\|Tf\|_{L^1} \leq C \sum |\lambda_j| \approx C\|f\|_{H^1}$ .”

The gap is at “so”:  $T$  has not been shown to commute with the infinite sum, and for a general linear  $T$  defined only on finite combinations of atoms it need not. Bownik (2005), refining an example of Meyer, constructed a linear functional on finite combinations of  $(1, \infty)$ -atoms which is uniformly bounded on atoms and admits *no* bounded extension to  $H^1$ ; composing it with a fixed  $L^1$  function gives an operator uniformly bounded on atoms which is not bounded from  $H^1$  to  $L^1$ .

*Why Theorem 4.2.5 is nonetheless true.* Because  $T$  is not merely defined on atoms: it is given as bounded on  $L^2$ , and by Theorem 3.2.3 (Calderón–Zygmund) it is of weak type  $(1, 1)$ . Those two facts let Step 1 pass to the limit honestly —  $f_N \rightarrow f$  in  $L^1$  forces  $Tf_N \rightarrow Tf$  in measure, and Fatou does the rest. The theorem is stated for  $f \in H^1 \cap L^2$  for the same reason, and the extension to all of  $H^1$  is then by density.

*What to check when reading.* Whenever an argument establishes a bound on atoms and concludes a bound on  $H^1$ , look for the additional hypothesis that licenses the limit. For Calderón–Zygmund operators it is the  $L^2$  boundedness; for other operators one may instead use  $(1, q)$ -atoms with  $q < \infty$ , for which Meda, Sjögren and Vallarino (2008) showed the naive argument *is* valid. With  $(1, \infty)$ -atoms and no further hypothesis, it is not.

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**WORKED EXAMPLE 4.2.2— TIER B The Hilbert transform of an atom, computed**

It is worth seeing Theorem 4.2.5 in a case where everything is explicit.

Let  $n = 1$ ,  $Q = [0, 1]$  and  $a = \mathbf{1}_{[0, 1/2]} - \mathbf{1}_{(1/2, 1]}$ , the atom of Worked Example 4.2.1(a). By Worked Example 3.3.1 (the Hilbert transform of an indicator),  $\mathcal{H}\mathbf{1}_{[u, v]}(x) = \frac{1}{\pi} \log \left| \frac{x-u}{x-v} \right|$ , so by linearity

$$\mathcal{H}a(x) = \frac{1}{\pi} \log \left| \frac{x}{x - \frac{1}{2}} \right| - \frac{1}{\pi} \log \left| \frac{x - \frac{1}{2}}{x - 1} \right| = \frac{1}{\pi} \log \left| \frac{x(x-1)}{(x - \frac{1}{2})^2} \right|.$$

*Decay at infinity.* For large  $|x|$ , expand:

$$\frac{x(x-1)}{(x-\frac{1}{2})^2} = \frac{x^2-x}{x^2-x+\frac{1}{4}} = 1 - \frac{1/4}{x^2-x+\frac{1}{4}},$$

so log of it is  $\approx -\frac{1}{4x^2}$ , giving  $|\mathcal{H}a(x)| \approx \frac{1}{4\pi x^2}$  as  $|x| \rightarrow \infty$ . This *is* integrable. *Compare with the non-atom.* For  $f = \mathbf{1}_{[0,1]}$ , Volume 3 computed  $\mathcal{H}f(x) = \frac{1}{\pi} \log|\frac{x}{x-1}| \approx \frac{1}{\pi x}$ , which is not.

*Reading the difference.* The mean-zero condition improved the decay from  $|x|^{-1}$  to  $|x|^{-2}$  — from  $|x|^{-n}$  to  $|x|^{-n-1}$  in general — and  $|x|^{-n-1}$  is integrable at infinity while  $|x|^{-n}$  sits exactly on the boundary. That single power is the whole content of the  $H^1$  theory, and Lemma 4.2.2 is the general version of the cancellation just performed by hand.

*A caution.*  $\mathcal{H}a$  also has logarithmic singularities at  $x = 0, \frac{1}{2}, 1$ , which are locally integrable, so they are harmless. Cancellation fixes the behaviour at infinity; it does nothing about local singularities, and it does not need to.

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**WORKED EXAMPLE 4.2.3— TIER C: RESEARCH TASTE    Why one checks operators on atoms, and when that is not enough**

Step 1 of Theorem 4.2.5 is the standard method: to bound an operator on  $H^1$ , bound it on atoms. The method is so convenient that it is worth being precise about what it does and does not establish, because the point is frequently got wrong in the literature.

(a) *What the argument actually proved.* We showed  $\|Ta\|_{L^1} \leq C$  for atoms, and concluded  $\|Tf\|_{L^1} \leq C\|f\|_{H^1}$  — but the passage from one to the other used more than linearity. It used that  $T$  was already known to be defined on  $f$  (through the  $L^2$  theory) and that  $Tf_N \rightarrow Tf$  in some usable sense (through the weak  $(1,1)$  bound). Without such an a priori statement, the inequality on atoms says nothing about  $f$ : a general  $f \in H^1$  has infinitely many atomic decompositions, and there is no reason for a merely finitely-additive operator to give the same answer on each.

(b) *The genuine counterexample.* Bownik (2005) constructed a linear operator defined on all finite linear combinations of atoms, uniformly bounded on atoms into  $L^1$ , which does *not* extend to a bounded operator  $H^1 \rightarrow L^1$ . So the naive statement — “uniformly bounded on atoms implies bounded on  $H^1$ ” — is false as stated. What is true, and what we used, is that the implication holds when the operator is already known to be continuous into some space in which the atomic series converges. For every operator in this series that hypothesis is available for free, from the  $L^2$  theory.

(c) *Why the difficulty exists.* The infimum in Definition 4.2.3 is over decompositions, and a given  $f$  has many. Two decompositions with nearly optimal coefficient sums can be completely different collections of atoms, and there is no canonical choice. Chapter 4.4 removes this discomfort by proving that  $H^1$  has an equivalent, *decomposition-free* description in terms of a maximal function — and that theorem, not the atomic definition, is what makes  $H^1$  a natural object rather than a construction.

(d) *The same issue elsewhere in the series.* A wave packet decomposition (Volume 12) writes a function as a sum of pieces adapted to tubes, and the decomposition is again far from unique. Estimates are proved piece by piece and summed, exactly as here, and the same care is required: one must know in advance that the object being estimated is well defined independently of the decomposition. In the restriction literature this is usually arranged by working with a fixed partition of unity and proving all estimates uniform in the choice. The lesson transfers: *a decomposition is a tool for estimating a function that already exists, not a definition of one.*

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## Drills

**Drill 4.2.1.** Show that if  $a$  is an atom associated with  $Q$  and  $\tilde{Q} \supseteq Q$  is a larger cube, then  $a$  is  $|\tilde{Q}|/|Q|$  times an atom associated with  $\tilde{Q}$ . Deduce that in Definition 4.2.3 one may restrict to atoms associated with dyadic cubes, at the cost of a dimensional constant.

**Drill 4.2.2.** Verify that  $\|a\|_{L^p} \leq |Q|^{1/p-1}$  for every atom associated with  $Q$  and every  $1 \leq p \leq \infty$ , and identify where this was used in Theorem 4.2.5.

**Drill 4.2.3.** Compute  $\mathcal{R}_1 a$  explicitly in  $\mathbb{R}^2$  for an atom of the form  $a = |Q|^{-1}(\mathbf{1}_{Q_+} - \mathbf{1}_{Q_-})$ , where  $Q_{\pm}$  are the left and right halves of a square  $Q$ , and confirm the decay rate  $|x|^{-3}$ .

**Drill 4.2.4.** Show that  $H^1(\mathbb{R}^n)$  is not closed under multiplication by bounded functions, by multiplying an atom by  $\text{sgn}$  of a coordinate. Contrast with the fact that  $L^1$  is.

## Notes

The atomic decomposition originates with Coifman (1974) in dimension one and Latter (1978) in general. The definition of  $H^1$  historically came first in a different form — via boundary values of holomorphic functions on the

upper half plane, whence the name — and the equivalence with the atomic description is Chapter 4.4.

Worked Example 4.2.3(b) refers to M. Bownik, *Boundedness of operators defined on atoms*, Proc. Amer. Math. Soc. 133 (2005). The subtlety it exposes had been overlooked in a number of published arguments.

## Chapter 4.3

# Duality: $(H^1)^* = \text{BMO}$

### 4.3.1 The statement, and why it is plausible

Volume 3 proved that a singular integral maps  $L^\infty$  into BMO; Chapter 4.2 proved that it maps  $H^1$  into  $L^1$ . The two statements look like mirror images, and they are. This chapter proves that BMO is *exactly* the dual of  $H^1$ , which explains the symmetry: the second statement is the adjoint of the first.

Before the proof, a plausibility check. If  $g$  is to define a functional on  $H^1$  by  $f \mapsto \int fg$ , the natural first test is an atom. Let  $a$  be an atom associated with a cube  $Q$ . Because  $\int a = 0$ , we may subtract any constant from  $g$  without changing the pairing, and the natural constant is  $\langle g \rangle_Q$ :

$$\left| \int a g \right| = \left| \int_Q a (g - \langle g \rangle_Q) \right| \leq \|a\|_{L^\infty} \int_Q |g - \langle g \rangle_Q| \leq \frac{1}{|Q|} \int_Q |g - \langle g \rangle_Q|.$$

The right-hand side is at most  $\|g\|_{\text{BMO}}$ , by definition. So the pairing of  $g$  against *any* atom is bounded by  $\|g\|_{\text{BMO}}$  — with no dependence on the cube. The two normalisations were made for each other: the atom's height  $|Q|^{-1}$  cancels the cube's volume, and the atom's mean-zero property is what licenses the subtraction of  $\langle g \rangle_Q$ .

That is the whole of the easy half, and §3.2 does no more than make it precise. The converse — that every bounded functional arises this way — is genuinely harder, and occupies §3.3 and §3.4.

**Notation 4.3.1.** Write  $H_{\text{fin}}^1$  for the set of *finite* linear combinations of atoms. By Definition 4.2.3 it is dense in  $H^1$ : truncating an atomic decomposition after  $N$  terms changes the  $H^1$  norm by at most  $\sum_{j>N} |\lambda_j| \rightarrow 0$ .

### 4.3.2 The easy half

**Theorem 4.3.2.** *Let  $g \in \text{BMO}(\mathbb{R}^n)$ . Then the map*

$$\Lambda_g(f) = \int_{\mathbb{R}^n} f g \quad (f \in H_{\text{fin}}^1)$$

*is well defined and extends uniquely to a bounded linear functional on  $H^1(\mathbb{R}^n)$ , with*

$$\|\Lambda_g\|_{(H^1)^*} \leq 2\|g\|_{\text{BMO}}.$$

*Moreover  $\Lambda_g = \Lambda_{g'}$  if and only if  $g - g'$  is constant.*

*Proof. Step 1. The integral converges.*

Let  $f \in H_{\text{fin}}^1$ , so  $f$  is bounded with compact support, say  $\text{supp } f \subseteq Q_0$ . By John–Nirenberg (Theorem 3.7.3 (John–Nirenberg)),  $g$  is in  $L^2(Q_0)$ , hence in  $L^1(Q_0)$ , so  $\int f g$  converges absolutely.

**Step 2.** *The bound on atoms.*

Let  $a$  be an atom associated with  $Q$ . Since  $\int a = 0$  and  $a$  is supported in  $Q$ ,

$$\int a g = \int_Q a (g - \langle g \rangle_Q),$$

and therefore, by (A2),

$$\left| \int a g \right| \leq \frac{1}{|Q|} \int_Q |g - \langle g \rangle_Q| \leq \|g\|_{\text{BMO}}. \quad (4.3.1)$$

**Step 3.** *Extend by linearity and density.*

If  $f = \sum_{j \leq N} \lambda_j a_j$  then  $|\Lambda_g(f)| \leq \sum_{j \leq N} |\lambda_j| \|g\|_{\text{BMO}}$  by (4.3.1). Taking the infimum over all finite atomic representations of  $f$  — note this infimum may exceed  $\|f\|_{H^1}$ , since the latter allows infinite decompositions — gives  $|\Lambda_g(f)| \leq \|g\|_{\text{BMO}} \|f\|_{H_{\text{fin}}^1}$ .

To pass to  $\|f\|_{H^1}$ : given any decomposition  $f = \sum_{j=1}^{\infty} \lambda_j a_j$  with  $\sum_j |\lambda_j| \leq \|f\|_{H^1} + \varepsilon$ , the truncations  $f_N$  lie in  $H_{\text{fin}}^1$ , satisfy  $|\Lambda_g(f_N)| \leq \|g\|_{\text{BMO}} \sum_j |\lambda_j|$ , and converge to  $f$  in  $L^1$ , hence —  $f$  and all  $f_N$  being supported in a common compact set only when the decomposition is finite — one argues instead on  $H^1$  directly: the functional  $\Lambda_g$  is bounded on the dense subspace  $H_{\text{fin}}^1$  with constant  $\|g\|_{\text{BMO}}$ , so it extends uniquely and boundedly to  $H^1$ , with the same constant. The factor 2 in the statement is slack absorbed for the complex case, where one splits into real and imaginary parts.



**Step 4.** *The kernel of the map  $g \mapsto \Lambda_g$ .*

If  $g$  is constant then  $\int ag = g \int a = 0$  for every atom, so  $\Lambda_g = 0$ . Conversely suppose  $\Lambda_g = 0$ . Fix a cube  $Q$  and let  $h \in L^\infty(Q)$  with  $\int_Q h = 0$ ; then  $h/(|Q|\|h\|_\infty)$  is an atom, so  $\int_Q hg = 0$ . A locally integrable function annihilating every bounded mean-zero function supported in  $Q$  is a.e. constant on  $Q$ : apply this with  $h = \mathbf{1}_A - \frac{|A|}{|B|}\mathbf{1}_B$  for measurable  $A, B \subseteq Q$  to see that  $\langle g \rangle_A = \langle g \rangle_B$  for all such  $A, B$ , and let  $A$  shrink to a Lebesgue point. Since  $Q$  was arbitrary and cubes overlap, the constants agree.  $\square$

**Pitfall: the pairing is an extension, not always an integral**

For general  $f \in H^1$  and  $g \in \text{BMO}$ , the integral  $\int fg$  need not converge absolutely:  $f$  is only in  $L^1$  and  $g$  need only be locally in every  $L^p$ , with no global bound. Theorem 4.3.2 defines  $\Lambda_g$  on a dense subspace and *extends* it, which is a different and weaker assertion.

This is the same situation as the Fourier transform on  $L^2$  (Volume 2, Theorem 4.5), which is also defined by extension from a dense subspace where an integral formula happens to converge. The reader should not expect to compute  $\Lambda_g(f)$  by integrating, except when  $f$  has compact support.

### 4.3.3 $L^2$ -atoms

For the converse we need to know that the atomic space is not too small — more precisely, that a mean-zero  $L^2$  function supported in a cube already lies in  $H^1$ , with the norm one would guess by scaling. This requires an argument, and the argument is where the exponent 2 earns its place.

**Definition 4.3.3.** An  $L^2$ -atom associated with a cube  $Q$  is a function  $h$  with  $\text{supp } h \subseteq Q$ ,  $\int h = 0$ , and  $\|h\|_{L^2} \leq |Q|^{-1/2}$ .

Every atom is an  $L^2$ -atom: by (A1) and (A2),  $\|a\|_{L^2}^2 \leq |Q| \cdot |Q|^{-2} = |Q|^{-1}$ . The content of the next result is the converse inclusion, at the level of norms.

**Proposition 4.3.4** ( $L^2$ -atoms belong to  $H^1$ ). *There is  $C_n$  such that every  $L^2$ -atom  $h$  satisfies  $h \in H^1(\mathbb{R}^n)$  with  $\|h\|_{H^1} \leq C_n$ . Equivalently, for every  $h \in L^2$  supported in a cube  $Q$  with  $\int h = 0$ ,*

$$\|h\|_{H^1} \leq C_n |Q|^{1/2} \|h\|_{L^2}.$$

*Scratch work.* Why should this be true, and why should  $L^2$  matter?

We must manufacture atoms out of  $h$ , and the only tool for manufacturing mean-zero pieces is the Calderón–Zygmund decomposition of Volume 3. So run it — but at which height? Running it once, at a single height  $\lambda$ ,

produces a good part and a bad part; the bad part is a sum of mean-zero pieces supported in cubes, which is promising, but the good part is merely bounded, not mean-zero, and so is not made of atoms.

The way out is to run the decomposition at *every* dyadic height  $2^k$  simultaneously and telescope. The good parts  $g_k$  at successive heights differ by something supported in the level set at height  $2^k$  and having mean zero on each of its cubes — an atom, up to normalisation. Since  $g_k \rightarrow h$ , the telescoping series  $\sum_k (g_{k+1} - g_k)$  reconstructs  $h$  out of atoms.

Now the arithmetic. The piece at height  $2^k$  has size  $\approx 2^k$  and lives on the level set  $\Omega_k$ , so it contributes  $\approx 2^k |\Omega_k|$  to the total coefficient sum. If we knew only  $h \in L^1$ , Chebyshev would give  $|\Omega_k| \lesssim 2^{-k} \|h\|_1$  and the contribution would be  $\approx \|h\|_1$  at *every*  $k$  — a divergent sum. With  $h \in L^2$  we get  $|\Omega_k| \lesssim 4^{-k} \|h\|_2^2$  instead, and the contribution is  $\approx 2^{-k} \|h\|_2^2$ , which is summable.

That is the entire role of the exponent 2: it converts a borderline divergent series into a geometric one. Any exponent  $q > 1$  would do, and the corresponding statement for  $L^q$ -atoms is true for every  $q > 1$  and false for  $q = 1$ .

*Proof.* By translating and dilating we may assume  $Q = [0, 1]^n$ , so  $|Q| = 1$  and  $\|h\|_{L^2} \leq 1$ ; the general case follows because both sides of the stated inequality transform the same way under  $x \mapsto \lambda x + x_0$ . Note  $\|h\|_{L^1} \leq \|h\|_{L^2} |Q|^{1/2} \leq 1$  by Cauchy–Schwarz.

Let  $\mathcal{D}(Q)$  be the dyadic subcubes of  $Q$  and let

$$\mathcal{M}_Q h(x) = \sup_{\substack{R \in \mathcal{D}(Q) \\ x \in R}} \langle |h| \rangle_R, \quad x \in Q,$$

the dyadic maximal function relative to  $Q$ .

**Step 1.** *The level sets and their measures.*

For  $k \geq 0$  set  $\Omega_k = \{x \in Q : \mathcal{M}_Q h(x) > 2^k\}$ , and  $\Omega_{-1} = Q$ . These decrease in  $k$ . By the dyadic maximal theorem on  $L^2$  — Theorem 3.1.4 (the dyadic maximal inequality) combined with Marcinkiewicz interpolation, giving  $\|\mathcal{M}_Q h\|_{L^2} \leq 2 \|h\|_{L^2}$  — and Chebyshev,

$$|\Omega_k| \leq \frac{\|\mathcal{M}_Q h\|_{L^2}^2}{4^k} \leq \frac{4}{4^k}. \quad (4.3.2)$$

This is the estimate the Scratch box identified as decisive.

For  $k \geq 0$ ,  $\Omega_k$  is the disjoint union of the maximal cubes  $Q_j^k \in \mathcal{D}(Q)$  with  $\langle |h| \rangle_{Q_j^k} > 2^k$ , and as in Theorem 3.1.6 (the Calderón–Zygmund decomposition) the parent bound gives

$$\langle |h| \rangle_{Q_j^k} \leq 2^{n+k}. \quad (4.3.3)$$

(For  $k = 0$  the maximal cubes may include  $Q$  itself, in which case  $\Omega_0 = Q$ ; nothing below is affected.)

**Step 2.** *The good functions.*

For  $k \geq 0$  define

$$g_k = h \mathbf{1}_{Q \setminus \Omega_k} + \sum_j \langle h \rangle_{Q_j^k} \mathbf{1}_{Q_j^k},$$

and set  $g_{-1} = \langle h \rangle_Q \mathbf{1}_Q = 0$ , using  $\int h = 0$ . Two properties:

$$\|g_k\|_{L^\infty} \leq 2^{n+k} \quad (4.3.4)$$

— off  $\Omega_k$  we have  $|h| \leq 2^k$  a.e. by Lebesgue differentiation, and on  $Q_j^k$  we have  $|\langle h \rangle_{Q_j^k}| \leq \langle |h| \rangle_{Q_j^k} \leq 2^{n+k}$  by (4.3.3) — and

$$\|h - g_k\|_{L^1} \leq 2 \int_{\Omega_k} |h| \longrightarrow 0 \quad (k \rightarrow \infty), \quad (4.3.5)$$

since  $h - g_k$  is supported in  $\Omega_k$  and is there the difference of  $h$  and its averages, and  $|\Omega_k| \rightarrow 0$  by (4.3.2) while  $h$  is integrable (absolute continuity of the integral, Proposition 1.7.10 (absolute continuity of the integral)).

**Step 3.** *The telescoping series.*

By (4.3.5) and  $g_{-1} = 0$ ,

$$h = \sum_{k \geq 0} (g_k - g_{k-1}) \quad \text{in } L^1(\mathbb{R}^n).$$

Fix  $k \geq 0$ . Since  $\Omega_k \subseteq \Omega_{k-1}$  and  $g_k = g_{k-1} = h$  off  $\Omega_{k-1}$ , the difference  $g_k - g_{k-1}$  is supported in  $\Omega_{k-1}$ , so it splits as

$$g_k - g_{k-1} = \sum_i b_i^k, \quad b_i^k := (g_k - g_{k-1}) \mathbf{1}_{Q_i^{k-1}},$$

a sum over the maximal cubes constituting  $\Omega_{k-1}$ . (For  $k = 0$  read  $\Omega_{-1} = Q$  with the single cube  $Q$ .)

**Step 4.** *Each piece is an atom, up to a factor.*

Fix  $i$  and  $k$ . Support:  $b_i^k$  is supported in  $Q_i^{k-1}$  by construction. Size: by (4.3.4),

$$\|b_i^k\|_{L^\infty} \leq \|g_k\|_\infty + \|g_{k-1}\|_\infty \leq 2^{n+k} + 2^{n+k-1} \leq 2^{n+k+1}.$$

Mean zero: on  $Q_i^{k-1}$  we have  $g_{k-1} = \langle h \rangle_{Q_i^{k-1}}$ , so  $\int_{Q_i^{k-1}} g_{k-1} = \int_{Q_i^{k-1}} h$ . And

$$\int_{Q_i^{k-1}} g_k = \int_{Q_i^{k-1} \setminus \Omega_k} h + \sum_{j: Q_j^k \subseteq Q_i^{k-1}} \int_{Q_j^k} h = \int_{Q_i^{k-1}} h,$$

because every  $Q_j^k$  meeting  $Q_i^{k-1}$  is contained in it — both are dyadic, and  $Q_j^k$  cannot strictly contain the maximal cube  $Q_i^{k-1}$  since  $\langle |h| \rangle_{Q_j^k} > 2^k > 2^{k-1}$  would contradict maximality. Subtracting,  $\int b_i^k = 0$ .

Therefore

$$b_i^k = \mu_i^k a_i^k, \quad \mu_i^k := 2^{n+k+1} |Q_i^{k-1}|,$$

with  $a_i^k$  an atom associated with  $Q_i^{k-1}$ .

**Step 5.** *Sum the coefficients.*

Using (4.3.2) and  $\sum_i |Q_i^{k-1}| = |\Omega_{k-1}|$ ,

$$\sum_{k \geq 0} \sum_i \mu_i^k = \sum_{k \geq 0} 2^{n+k+1} |\Omega_{k-1}| \leq 2^{n+1} |Q| + \sum_{k \geq 1} 2^{n+k+1} \cdot \frac{4}{4^{k-1}} \leq 2^{n+1} + 2^{n+5} \sum_{k \geq 1} 2^{-k} \lesssim_n 1.$$

So  $h = \sum_{k,i} \mu_i^k a_i^k$  with summable coefficients, and  $\|h\|_{H^1} \leq C_n$ .  $\square$

**Where we are.** Look again at Step 5. The factor  $2^k$  came from the height of the piece and the factor  $4^{-k}$  from (4.3.2); their product  $2^{-k}$  is summable. Had we only assumed  $h \in L^1$ , (4.3.2) would read  $|\Omega_k| \lesssim 2^{-k}$  and the product would be  $2^0 = 1$  at every  $k$  — divergent. The proposition is false for  $L^1$ : by Proposition 4.2.4(iv) there are mean-zero  $L^1$  functions not in  $H^1$ .

### 4.3.4 The hard half

**Theorem 4.3.5** (Fefferman). *The map  $g \mapsto \Lambda_g$  of Theorem 4.3.2 is an isomorphism of  $\text{BMO}(\mathbb{R}^n)/\{\text{constants}\}$  onto  $(H^1(\mathbb{R}^n))^*$ . That is, every bounded linear functional  $\Lambda$  on  $H^1$  is of the form  $\Lambda = \Lambda_g$  for some  $g \in \text{BMO}$ , unique up to an additive constant, and*

$$\frac{1}{2} \|\Lambda_g\| \leq \|g\|_{\text{BMO}} \leq C_n \|\Lambda_g\|.$$

*Idea of the proof.* Given  $\Lambda$ , we must produce  $g$ . The only general device for producing a function that represents a functional is the Riesz representation theorem in a Hilbert space — so we look for a Hilbert space on which  $\Lambda$  is bounded.

Proposition 4.3.4 supplies exactly that. On the space of mean-zero  $L^2$  functions supported in a fixed cube  $Q$ , the  $H^1$  norm is dominated by  $|Q|^{1/2} \|\cdot\|_{L^2}$ , so  $\Lambda$  is a bounded functional for the  $L^2$  norm, and Riesz representation gives a  $g^Q \in L^2(Q)$  representing it there.

Two things then remain. The functions  $g^Q$  for different cubes must be patched into a single  $g$  — they agree up to constants, which is exactly the

ambiguity the theorem allows. And  $g$  must be shown to lie in BMO, which is done by feeding  $\Lambda$  the one test function designed to extract the mean oscillation: a normalised multiple of  $\text{sgn}(g - \langle g \rangle_Q)$ , adjusted to have mean zero.

*Proof.* Let  $\Lambda \in (H^1)^*$  with  $\|\Lambda\| = 1$ .

**Step 1.**  $\Lambda$  is bounded on a Hilbert space, for each cube.

Fix a cube  $Q$  and let

$$L_0^2(Q) = \left\{ h \in L^2(\mathbb{R}^n) : \text{supp } h \subseteq Q, \int h = 0 \right\},$$

a closed subspace of  $L^2(Q)$  — closed because both defining conditions are preserved under  $L^2$  limits, the second by Cauchy–Schwarz on the finite-measure set  $Q$ . Hence  $L_0^2(Q)$  is a Hilbert space.

By Proposition 4.3.4, every  $h \in L_0^2(Q)$  lies in  $H^1$  with  $\|h\|_{H^1} \leq C_n |Q|^{1/2} \|h\|_{L^2}$ . Therefore

$$|\Lambda(h)| \leq \|h\|_{H^1} \leq C_n |Q|^{1/2} \|h\|_{L^2} \quad (h \in L_0^2(Q)), \quad (4.3.6)$$

so  $\Lambda$  restricts to a bounded functional on  $L_0^2(Q)$  of norm at most  $C_n |Q|^{1/2}$ .

**Step 2.** *Represent it.*

By the Riesz representation theorem (Theorem 1.8.15 (Riesz representation)) there is a unique  $G^Q \in L_0^2(Q)$  with

$$\Lambda(h) = \int_{\mathbb{R}^n} h \overline{G^Q} \quad \text{for all } h \in L_0^2(Q),$$

and  $\|G^Q\|_{L^2} \leq C_n |Q|^{1/2}$ . Write  $g^Q = \overline{G^Q}$ .

**Step 3.** *Consistency across cubes.*

Let  $Q \subseteq Q'$ . For  $h \in L_0^2(Q) \subseteq L_0^2(Q')$  both representations apply, so  $\int h(g^Q - g^{Q'}) = 0$  for every  $h \in L_0^2(Q)$ . A function in  $L^2(Q)$  orthogonal to every mean-zero element of  $L^2(Q)$  lies in the orthogonal complement of  $L_0^2(Q)$  inside  $L^2(Q)$ , which is precisely the constants. Hence

$$g^{Q'}|_Q = g^Q + c_{Q,Q'} \quad \text{for a constant } c_{Q,Q'}.$$

**Step 4.** *Patch.*

Let  $Q_1 \subseteq Q_2 \subseteq \dots$  be cubes with  $\bigcup_m Q_m = \mathbb{R}^n$ . Define  $g$  on  $\mathbb{R}^n$  by setting  $g = g^{Q_1}$  on  $Q_1$  and, inductively,  $g = g^{Q_{m+1}} - c_m$  on  $Q_{m+1}$ , where  $c_m$  is

chosen by Step 3 so that the new definition agrees with the old on  $Q_m$ . This is consistent, and produces  $g \in L^2_{\text{loc}}(\mathbb{R}^n) \subseteq L^1_{\text{loc}}$  with

$$\Lambda(h) = \int h g \quad \text{for every } h \in L^2_0(Q_m) \text{ and every } m, \quad (4.3.7)$$

since adding a constant to  $g$  does not change the pairing against a mean-zero  $h$ .

**Step 5.**  $g$  lies in BMO.

Fix a cube  $Q$ ; by enlarging we may assume  $Q \subseteq Q_m$  for some  $m$ . Put

$$s = \overline{\text{sgn}(g - \langle g \rangle_Q)} \mathbf{1}_Q, \quad h = \frac{1}{|Q|} (s - \langle s \rangle_Q \mathbf{1}_Q),$$

where  $\text{sgn } z = z/|z|$  for  $z \neq 0$  and 0 otherwise. Then  $h$  is supported in  $Q$ , has  $\int h = 0$  by construction, and satisfies  $\|h\|_{L^\infty} \leq 2|Q|^{-1}$ , hence  $\|h\|_{L^2} \leq 2|Q|^{-1/2}$ ; so  $h \in L^2_0(Q)$ .

Now compute the pairing. Since  $\int h = 0$  we may replace  $g$  by  $g - \langle g \rangle_Q$ , and since  $\int_Q (s - \langle s \rangle_Q) \bar{c} = 0$  for constants  $c$ ,

$$\int h g = \frac{1}{|Q|} \int_Q (s - \langle s \rangle_Q)(g - \langle g \rangle_Q) = \frac{1}{|Q|} \int_Q s (g - \langle g \rangle_Q) = \frac{1}{|Q|} \int_Q |g - \langle g \rangle_Q|,$$

where the middle equality drops the term  $\langle s \rangle_Q \int_Q (g - \langle g \rangle_Q) = 0$ , and the last uses  $s(g - \langle g \rangle_Q) = |g - \langle g \rangle_Q|$  pointwise.

Combining with (4.3.7) and (4.3.6),

$$\frac{1}{|Q|} \int_Q |g - \langle g \rangle_Q| = |\Lambda(h)| \leq C_n |Q|^{1/2} \|h\|_{L^2} \leq 2C_n.$$

Taking the supremum over cubes,  $\|g\|_{\text{BMO}} \leq 2C_n = 2C_n \|\Lambda\|$ .

**Step 6.**  $\Lambda = \Lambda_g$ .

Both are bounded functionals on  $H^1$ . They agree on every atom: an atom  $a$  associated with  $Q$  lies in  $L^2_0(Q)$ , so (4.3.7) gives  $\Lambda(a) = \int a g = \Lambda_g(a)$ . Hence they agree on  $H^1_{\text{fin}}$  by linearity, and on  $H^1$  by density and continuity.

Uniqueness up to constants is Theorem 4.3.2, Step 4, and the two-sided norm comparison collects Steps 5 and Theorem 4.3.2.  $\square$

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**WORKED EXAMPLE 4.3.1— TIER A Reading John–Nirenberg through the duality**

The John–Nirenberg inequality (Theorem 3.7.3 (John–Nirenberg)) says a BMO function has exponentially small oscillation tails. Theorem 4.3.5 gives it a second meaning.

By duality,  $\|g\|_{\text{BMO}} \approx \sup\{|\int ag| : a \text{ an atom}\}$ . Now  $L^2$ -atoms are, by Proposition 4.3.4, atoms up to a bounded factor, so the same supremum over  $L^2$ -atoms is comparable. Unwinding, for every cube  $Q$  and every  $h \in L^2$  supported in  $Q$  with  $\int h = 0$ ,

$$\left| \int_Q h g \right| \lesssim_n |Q|^{1/2} \|h\|_{L^2} \|g\|_{\text{BMO}},$$

which by duality in  $L^2(Q)$  says exactly

$$\left( \frac{1}{|Q|} \int_Q |g - \langle g \rangle_Q|^2 \right)^{1/2} \lesssim_n \|g\|_{\text{BMO}}.$$

So the  $L^2$  mean oscillation is controlled by the  $L^1$  mean oscillation — a special case of John–Nirenberg, obtained here with no stopping-time argument at all, purely from Proposition 4.3.4.

The general  $L^p$  case follows the same way from the  $L^p$ -atom version of Proposition 4.3.4, whose proof is identical with 2 replaced by  $p$ . What duality does *not* give is the exponential bound itself, which genuinely needs the iteration of Volume 3.

WORKED EXAMPLE 4.3.2— TIER C: RESEARCH TASTE **What the duality is good for**

(a) *Interpolation.*  $H^1$  and BMO are the correct endpoints of the  $L^p$  scale in a precise sense: complex interpolation between  $H^1$  and  $L^\infty$  gives  $L^p$  for  $1 < p < \infty$ , as does interpolation between  $L^1$  and BMO. The Fefferman–Stein inequality of Chapter 4.1 is one concrete consequence, and Theorem 4.1.5 another. Neither statement is available with  $L^1$  and  $L^\infty$  in place of  $H^1$  and BMO, because the Calderón–Zygmund operators are unbounded on both.

(b) *Adjointns become free.* Volume 3 proved  $T : L^\infty \rightarrow \text{BMO}$  and Chapter 4.2 proved  $T : H^1 \rightarrow L^1$ . Given Theorem 4.3.5, these are adjoint statements: the second applied to  $T^*$  dualises to the first for  $T$ . So half the endpoint theory was a consequence of the other half all along, and the duality is what makes that visible.

(c) *Carleson measures, and the connection to later volumes.* Fefferman’s original formulation of Theorem 4.3.5 came with a companion characterisation:  $g \in \text{BMO}$  if and only if the measure

$$d\mu_g(x, t) = |\nabla u(x, t)|^2 t \, dx \, dt$$

is a *Carleson measure* on the upper half space, where  $u$  is the Poisson extension of  $g$ ; that is,  $\mu_g(Q \times (0, \ell(Q))) \lesssim |Q|$  for every cube. The condition says:

the energy of  $g$  at scales below  $\ell(Q)$  and locations inside  $Q$  is at most  $|Q|$  — a bound on the total amount of oscillation available across all scales at once. That is a multiscale counting statement of exactly the type this series is about. In Volume 13 the analogous bookkeeping appears as the constraint that a wave packet decomposition cannot have too much mass at too many scales simultaneously, and the summation over dyadic scales in the decoupling proofs is the same computation with tubes in place of Carleson boxes. We do not prove the Carleson characterisation — it needs the Littlewood–Paley theory of Chapter 4.6 together with a stopping-time argument — but it is worth knowing that the multiscale accounting of the later volumes has an ancestor here.

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WORKED EXAMPLE 4.3.3— TIER B **Using**  $(H^1)^* = \text{BMO}$

Theorem 4.3.5 is used in one of two ways, and it is worth separating them.

*Move 1: prove a bound on  $H^1$  by testing against BMO.* To show  $\|f\|_{H^1} \lesssim A$ , it suffices to show  $|\int fg| \lesssim A\|g\|_{\text{BMO}}$  for all  $g$  in a dense subclass. This converts a statement about atomic decompositions — an infimum over representations — into a statement about pairings, which is usually easier to verify.

*Move 2: prove a bound on BMO by testing against atoms.* To show  $\|g\|_{\text{BMO}} \lesssim A$ , show  $|\int ag| \lesssim A$  for every atom  $a$ . Since atoms are explicit — bounded, supported on a cube, mean zero — this is often a direct computation.

*A worked instance of Move 2.* Let  $T$  be a Calderón–Zygmund operator and  $g \in L^\infty$ . For an atom  $a$  on a cube  $Q$ ,

$$\int a(Tg) = \int (T^*a)g,$$

and  $T^*$  is again Calderón–Zygmund, so  $\|T^*a\|_{L^1} \lesssim 1$  by Theorem 4.2.5. Hence  $|\int a(Tg)| \lesssim \|g\|_{L^\infty}$ , and by Move 2,  $\|Tg\|_{\text{BMO}} \lesssim \|g\|_{L^\infty}$  — which is Theorem 3.7.4, obtained here by duality rather than directly.

*The caution.* Move 1 requires knowing the pairing extends, and the Pitfall of Chapter 4.2 shows that atomic arguments need care at exactly this point. Testing against a dense subclass is safe; testing against atoms alone is the step that requires the operator to be a priori bounded somewhere.

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**WORKED EXAMPLE 4.3.4— TIER B Interpolating with  $H^1$  and BMO**

The pair  $(H^1, \text{BMO})$  substitutes for  $(L^1, L^\infty)$ , and the mechanics of using it are worth stating.

(a) *The substitution.* An operator bounded  $H^1 \rightarrow L^1$  and  $L^\infty \rightarrow \text{BMO}$  is bounded on  $L^p$  for  $1 < p < \infty$ , by interpolation. This is how Volume 3's Calderón–Zygmund theorem is recovered from its two endpoint statements.

(b) *Why the endpoints must be replaced.*  $L^1 \rightarrow L^1$  and  $L^\infty \rightarrow L^\infty$  are both false for singular integrals, as Worked Example 3.3.4(c) showed concretely for the Hilbert transform.

(c) *The interpolation theorem needed.* Not Riesz–Thorin, which is for  $L^p$  pairs, but the complex interpolation of the pair  $(H^1, \text{BMO})$  — a genuinely different theorem, and one this series states rather than proves.

(d) *A worked use.* To show a Calderón–Zygmund operator is bounded on  $L^p$ , one may either run the decomposition directly (Volume 3, Theorem 3.2.3) or prove the two endpoint statements and interpolate. The second route is longer but generalises to operators where the direct decomposition fails.

*The caution.* Duality (Theorem 4.3.5) and interpolation are different tools. Duality converts a statement about  $H^1$  into one about BMO; interpolation converts two endpoint statements into a range. Using one where the other is meant is a common confusion.

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## Drills

**Drill 4.3.1.** Show directly from Theorem 4.3.2 that BMO is complete, using that the dual of a normed space is always complete.

**Drill 4.3.2.** Verify the claim in Step 3 of Theorem 4.3.5 that the orthogonal complement of  $L_0^2(Q)$  inside  $L^2(Q)$  is the space of constants.

**Drill 4.3.3.** Carry out the proof of Proposition 4.3.4 with  $L^2$  replaced by  $L^q$ ,  $1 < q < \infty$ , and identify where the argument breaks at  $q = 1$ .

**Drill 4.3.4.** Let  $g \in \text{BMO}$  and  $a$  an atom associated with  $Q$ . Show  $|\int ag| \leq (|Q|^{-1} \int_Q |g - \langle g \rangle_Q|^2)^{1/2}$  and compare with (4.3.1). Which is sharper, and why does it not matter?

## Notes

Theorem 4.3.5 is Fefferman's, announced in 1971 and published with Stein in 1972; it is the theorem that made BMO a central object. The proof given

here, via  $L^2$ -atoms and Riesz representation, is the standard modern one and is due in this form to Coifman and Weiss.

Proposition 4.3.4 is where the atomic theory is genuinely used rather than merely convenient, and readers should note that it is the only place in this volume where a Calderón–Zygmund decomposition is run at infinitely many heights at once. The technique — telescope over dyadic thresholds and sum a geometric series — recurs in the proof of the maximal characterisation in Chapter 4.4.

## Chapter 4.4

# Maximal Functions and the Whitney Decomposition

### 4.4.1 The discomfort with the atomic definition

Definition 4.2.3 defines  $H^1$  by a decomposition, and Worked Example 2.11 recorded the discomfort this causes: a given  $f$  has infinitely many atomic decompositions, none canonical, and the norm is an infimum one can rarely compute. Worse, the definition gives no way to *recognise* membership. Faced with a concrete  $f$ , one has no test to apply — only the hope of finding a decomposition.

This chapter addresses the discomfort. We introduce a maximal function  $\mathcal{M}_\varphi f$ , built from  $f$  by a completely explicit recipe, and prove that

$$f \in H^1 \quad \implies \quad \mathcal{M}_\varphi f \in L^1,$$

with  $\|\mathcal{M}_\varphi f\|_1 \lesssim \|f\|_{H^1}$ . The converse is also true — it is the Fefferman–Stein characterisation, and it is the statement that makes  $H^1$  a natural object rather than a construction — but its proof is long and this series does not give it. Remark 4.4.5 explains that decision and what depends on it, which is nothing.

We also prove the Whitney decomposition (Theorem 4.4.3), a geometric tool used in the converse and reused independently in Volume 11.

#### Looking ahead: a test we can already use

The Pitfall in §2.3 exhibited a mean-zero  $L^1$  function

$$g(x) = \frac{\operatorname{sgn}(x)}{|x|(\log|x|)^2} \mathbf{1}_{|x|>2}$$

and asserted, without proof, that it is not in  $H^1$ . With the atomic definition there was nothing to be done: one cannot prove a decomposition does not

exist by failing to find one. Once Theorem 4.4.4 is available the verification is a computation — see Drill 4.4.

## 4.4.2 Maximal functions attached to a function

**Definition 4.4.1.** Fix  $\varphi \in \mathcal{S}(\mathbb{R}^n)$  with  $\int_{\mathbb{R}^n} \varphi = 1$ , and write  $\varphi_t(x) = t^{-n} \varphi(x/t)$  for  $t > 0$ . For  $f \in L^1(\mathbb{R}^n)$  define the *radial maximal function*

$$\mathcal{M}_\varphi f(x) = \sup_{t>0} |(\varphi_t * f)(x)|.$$

The recipe is: smooth  $f$  at every scale  $t$ , and record the largest value seen at  $x$ . Two remarks orient the definition.

First, this is the Hardy–Littlewood maximal function with the sharp cutoff replaced by a smooth bump. Indeed if one took  $\varphi = |B(0, 1)|^{-1} \mathbf{1}_{B(0, 1)}$  — not Schwartz, but morally the same — then  $\varphi_t * |f|$  is exactly the average of  $|f|$  over  $B(x, t)$ . The difference is that we do *not* take absolute values inside:  $\varphi_t * f$ , not  $\varphi_t * |f|$ . So  $\mathcal{M}_\varphi f$  can be much smaller than  $\mathcal{M}f$  when  $f$  oscillates, and that is the entire point.

Second,  $\mathcal{M}_\varphi f \geq |f|$  a.e., since  $\varphi_t * f \rightarrow f$  a.e. as  $t \rightarrow 0$  (Proposition 1.6.10 (approximate identities), together with the Lebesgue differentiation theorem). So  $\|\mathcal{M}_\varphi f\|_{L^1} \geq \|f\|_{L^1}$ , and the condition  $\mathcal{M}_\varphi f \in L^1$  is strictly stronger than  $f \in L^1$ .

**Pitfall:**  $\mathcal{M}_\varphi f \in L^1$  is a real restriction

For a general  $f \in L^1$  one should expect  $\mathcal{M}_\varphi f \notin L^1$ , for the same reason  $\mathcal{M}f \notin L^1$  (Worked Example 1.7.1 ( $\mathcal{M}f$  is never integrable)): if  $\int f \neq 0$  then for large  $|x|$  and  $t \approx |x|$ ,

$$(\varphi_t * f)(x) \approx t^{-n} \varphi(x/t) \int f \approx \frac{c}{|x|^n} \int f,$$

which is not integrable. So the condition  $\mathcal{M}_\varphi f \in L^1$  already forces  $\int f = 0$  — consistent with Proposition 4.2.4(ii), and a first sign that the right space is being detected.

**Theorem 4.4.2** (The easy direction). *Let  $\varphi \in \mathcal{S}(\mathbb{R}^n)$  with  $\int \varphi = 1$ . There is  $C_{n,\varphi}$  such that*

$$\|\mathcal{M}_\varphi f\|_{L^1(\mathbb{R}^n)} \leq C_{n,\varphi} \|f\|_{H^1(\mathbb{R}^n)} \quad \text{for all } f \in H^1(\mathbb{R}^n).$$

*Idea of the proof.* By sublinearity of  $f \mapsto \mathcal{M}_\varphi f$  it is enough to prove  $\|\mathcal{M}_\varphi a\|_{L^1} \leq C$  for a single atom, uniformly. This is the same reduction as in Theorem 4.2.5, and the same two-region split works: near the cube, use that  $\mathcal{M}_\varphi$  is trivially bounded by  $\|a\|_\infty$  and that the cube is small; far from the cube, use the mean-zero property to replace  $\varphi_t(x - y)$  by its difference from  $\varphi_t(x - c)$ .

The one new feature is that we must optimise over  $t$ . The far-field bound depends on  $t$ , and the supremum defining  $\mathcal{M}_\varphi$  ranges over all  $t$ ; the computation in Step 3 shows the worst case is  $t \approx |x - c|$  and that the resulting bound decays like  $|x - c|^{-n-1}$  — one power better than integrability requires, exactly as in Worked Example 4.2.2.

**Proof. Step 1. Reduction to an atom.**

If  $f = \sum_j \lambda_j a_j$  in  $L^1$ , then  $\varphi_t * f = \sum_j \lambda_j (\varphi_t * a_j)$  pointwise for each  $t$ , so  $\mathcal{M}_\varphi f \leq \sum_j |\lambda_j| \mathcal{M}_\varphi a_j$  pointwise, and therefore  $\|\mathcal{M}_\varphi f\|_1 \leq \sum_j |\lambda_j| \|\mathcal{M}_\varphi a_j\|_1$ . Taking the infimum over decompositions, it suffices to bound  $\|\mathcal{M}_\varphi a\|_1$  by a constant independent of the atom.

Let  $a$  be an atom associated with a cube  $Q$  of centre  $c$  and side  $\ell$ , and let  $Q^* = 2\sqrt{n}Q$ .

**Step 2. Near the cube.**

For any  $t$  and any  $x$ , Young's inequality gives

$$|(\varphi_t * a)(x)| \leq \|\varphi_t\|_{L^1} \|a\|_{L^\infty} \leq \frac{\|\varphi\|_{L^1}}{|Q|},$$

using  $\|\varphi_t\|_1 = \|\varphi\|_1$  for every  $t$ . Hence  $\mathcal{M}_\varphi a \leq \|\varphi\|_1 |Q|^{-1}$  everywhere, and

$$\int_{Q^*} \mathcal{M}_\varphi a \leq |Q^*| \cdot \frac{\|\varphi\|_1}{|Q|} = (2\sqrt{n})^n \|\varphi\|_1.$$

As in Theorem 4.2.5, the volume factors cancel and the bound does not see the cube.

**Step 3. Far from the cube.**

Let  $x \notin Q^*$ , and write  $r = |x - c|$ , so  $r > \sqrt{n}\ell$ . Using  $\int a = 0$ ,

$$(\varphi_t * a)(x) = \int_Q \left( \varphi_t(x - y) - \varphi_t(x - c) \right) a(y) \, dy.$$

For  $y \in Q$  we have  $|y - c| \leq \frac{\sqrt{n}}{2}\ell < \frac{r}{2}$ , so the segment from  $x - y$  to  $x - c$  lies in  $\{|z| \geq r/2\}$ . By the mean value theorem and the scaling  $\nabla(\varphi_t)(z) = t^{-n-1}(\nabla\varphi)(z/t)$ ,

$$|\varphi_t(x - y) - \varphi_t(x - c)| \leq |y - c| t^{-n-1} \sup_{|z| \geq r/2} |(\nabla\varphi)(z/t)|.$$

Since  $\varphi \in \mathcal{S}$ , for any  $N$  there is  $C_N$  with  $|\nabla \varphi(w)| \leq C_N(1 + |w|)^{-N}$ ; take  $N = n + 2$ . Then

$$|(\varphi_t * a)(x)| \leq \frac{\sqrt{n}\ell}{2} C_{n+2} t^{-n-1} \left(1 + \frac{r}{2t}\right)^{-(n+2)} \int_Q |a| \leq C \ell t^{-n-1} \left(1 + \frac{r}{2t}\right)^{-(n+2)},$$

using  $\int_Q |a| \leq 1$ .

It remains to maximise the right-hand side over  $t > 0$ . If  $t \geq r$  the last factor is between  $2^{-(n+2)}$  and 1, and  $t^{-n-1} \leq r^{-n-1}$ . If  $t < r$  then  $1 + \frac{r}{2t} \geq \frac{r}{2t}$ , so

$$t^{-n-1} \left(\frac{r}{2t}\right)^{-(n+2)} = 2^{n+2} \frac{t}{r^{n+2}} \leq \frac{2^{n+2}}{r^{n+1}}.$$

In both regimes the bound is  $\lesssim_n r^{-n-1}$ , so

$$\mathcal{M}_{\varphi} a(x) \lesssim_{n,\varphi} \frac{\ell}{|x - c|^{n+1}} \quad (x \notin Q^*). \quad (4.4.1)$$

**Step 4.** *Integrate the far part.*

By polar coordinates (Theorem 1.6.11 (polar coordinates)) and  $r > \sqrt{n}\ell$  on the region of integration,

$$\int_{(Q^*)^c} \mathcal{M}_{\varphi} a \lesssim_{n,\varphi} \ell \int_{|x-c| > \sqrt{n}\ell} \frac{dx}{|x - c|^{n+1}} = \ell \sigma(\mathbb{S}^{n-1}) \int_{\sqrt{n}\ell}^{\infty} \frac{dr}{r^2} = \frac{\sigma(\mathbb{S}^{n-1})}{\sqrt{n}},$$

again independent of  $\ell$ . Adding Steps 2 and 4 completes the proof.  $\square$

**Where we are.** Notice that (4.4.1) is the same estimate as Worked Example 4.2.2: an atom is seen from far away with decay  $|x|^{-n-1}$ , one power better than the  $|x|^{-n}$  that a general  $L^1$  function produces, and that one power is exactly the difference between integrable and not. The Hilbert transform and the smoothing  $\varphi_t$  are quite different operations, and the mechanism is identical — freeze the kernel at the centre of the cube, and let the mean-zero property kill the frozen term.

### 4.4.3 The Whitney decomposition

The converse direction requires cutting an open set into cubes whose sizes are comparable to their distance from the boundary. This is a purely geometric construction, useful far beyond the present chapter, and we prove it here.

**Theorem 4.4.3** (Whitney). *Let  $\Omega \subsetneq \mathbb{R}^n$  be open and nonempty. There is a countable family  $\{Q_j\}$  of dyadic cubes with pairwise disjoint interiors such that*

$$\Omega = \bigcup_j Q_j \quad \text{and} \quad \text{diam } Q_j \leq \text{dist}(Q_j, \Omega^c) \leq 4 \text{diam } Q_j \quad \text{for every } j.$$

*Idea of the proof.* The two inequalities pull in opposite directions: the left says the cube is small enough to sit comfortably inside  $\Omega$ , the right says it is not wastefully small. A cube of side comparable to  $\frac{1}{3} \text{dist}(x, \Omega^c)$  satisfies both, and such a cube exists at every  $x \in \Omega$  because the dyadic side lengths  $2^{-k}$  pass within a factor 2 of any target. Selecting the maximal cubes with this property makes the family disjoint, by the nesting lemma.

**Proof. Step 1.** *The admissible family is nonempty at every point.*

Let

$$\mathcal{S} = \left\{ Q \in \mathcal{D} : Q \subseteq \Omega, \text{diam } Q \leq \text{dist}(Q, \Omega^c) \leq 4 \text{diam } Q \right\}.$$

Fix  $x \in \Omega$  and put  $d = \text{dist}(x, \Omega^c)$ , which is positive since  $\Omega$  is open, and finite since  $\Omega^c \neq \emptyset$ . For  $k \in \mathbb{Z}$  let  $Q^{(k)}$  be the unique cube of  $\mathcal{D}_k$  containing  $x$ , so  $\text{diam } Q^{(k)} = \sqrt{n} 2^{-k}$ . Choose  $k$  with

$$\frac{d}{6} < \sqrt{n} 2^{-k} \leq \frac{d}{3},$$

possible since the numbers  $\sqrt{n} 2^{-k}$  decrease by factors of 2. Then for any  $z \in Q^{(k)}$  we have  $|z - x| \leq \text{diam } Q^{(k)} \leq d/3$ , so

$$\text{dist}(Q^{(k)}, \Omega^c) \geq d - \frac{d}{3} = \frac{2d}{3} > \frac{d}{3} \geq \text{diam } Q^{(k)},$$

which also shows  $Q^{(k)} \subseteq \Omega$ ; and

$$\text{dist}(Q^{(k)}, \Omega^c) \leq d < 6 \cdot \sqrt{n} 2^{-k} = 6 \text{diam } Q^{(k)}.$$

This gives the left inequality and the right one with 6 in place of 4; running the same computation with the window  $\frac{d}{8} < \sqrt{n} 2^{-k} \leq \frac{d}{4}$  gives  $\text{dist} \geq \frac{3d}{4}$  and  $\text{dist} \leq d < 4 \cdot \sqrt{n} 2^{-k}$ , hence  $Q^{(k)} \in \mathcal{S}$ . So every  $x \in \Omega$  lies in some member of  $\mathcal{S}$ , and  $\bigcup_{Q \in \mathcal{S}} Q = \Omega$ .

**Step 2.** *Maximal elements exist.*

If  $Q \in \mathcal{S}$  then  $\text{diam } Q \leq \text{dist}(Q, \Omega^c) \leq \text{dist}(x_0, \Omega^c)$  for any fixed  $x_0 \in Q$ , so the side lengths of members of  $\mathcal{S}$  containing a given point are bounded above. Hence every  $Q \in \mathcal{S}$  is contained in a maximal element of  $\mathcal{S}$ : ascend through dyadic parents, and the ascent terminates because side lengths double.

**Step 3.** *Maximal elements are disjoint and cover.*

Let  $\{Q_j\}$  enumerate the maximal elements of  $\mathcal{S}$ ; the family is countable because  $\mathcal{D}$  is. If two distinct maximal cubes met, then by the nesting property (Lemma 3.1.2 (nesting of dyadic cubes)) one would contain the other, contradicting maximality of the smaller. Their union is  $\Omega$  by Step 1 and Step 2.  $\square$

**WORKED EXAMPLE 4.4.1— TIER A What a Whitney decomposition looks like**

Take  $n = 1$  and  $\Omega = (0, 1)$ . The Whitney cubes are the dyadic intervals  $[2^{-k-1}, 2^{-k})$  and their reflections  $(1 - 2^{-k}, 1 - 2^{-k-1}]$ , for  $k \geq 1$ , together with the middle interval  $[\frac{1}{4}, \frac{3}{4}]$  suitably subdivided. Each has length comparable to its distance from  $\{0, 1\}$ : the interval  $[2^{-k-1}, 2^{-k})$  has length  $2^{-k-1}$  and distance  $2^{-k-1}$  from 0.

Two features are typical. The cubes accumulate at the boundary, with sizes shrinking geometrically — there are infinitely many of them. And the number of cubes of any given size is bounded: in  $\mathbb{R}^n$  one shows that at most  $C_n$  Whitney cubes of a given generation can touch any fixed one, which is why sums over Whitney cubes converge in practice.

#### 4.4.4 The converse, and the limits of what we prove

**Theorem 4.4.4** (Fefferman–Stein maximal characterisation). *Let  $\varphi \in \mathcal{S}(\mathbb{R}^n)$  with  $\int \varphi = 1$ , and let  $f \in L^1(\mathbb{R}^n)$ . Then  $f \in H^1(\mathbb{R}^n)$  if and only if  $\mathcal{M}_\varphi f \in L^1(\mathbb{R}^n)$ , and in that case  $\|f\|_{H^1} \approx_{n,\varphi} \|\mathcal{M}_\varphi f\|_{L^1}$ .*

Theorem 4.4.2 is the implication  $\Rightarrow$ . We do not prove the converse in this series, and this section explains what the obstruction is, why the omission is safe, and what the reader should do about it.

*Scratch work.* The architecture of the converse is clear enough. Set  $\Omega_k = \{\mathcal{M}_\varphi f > 2^k\}$ , an open set — openness is why a smooth  $\varphi$  is used rather than a sharp cutoff — with  $|\Omega_k| \leq 2^{-k} \|\mathcal{M}_\varphi f\|_1$ . Decompose each  $\Omega_k$  into Whitney cubes by Theorem 4.4.3, take a smooth partition of unity subordinate to them, and split  $f$  into localised pieces. Subtracting a weighted average on each piece makes it mean zero. Telescoping over  $k$  then assembles the pieces into a series, whose coefficient sum is controlled by

$$\sum_k 2^k |\Omega_k| \approx \|\mathcal{M}_\varphi f\|_{L^1}$$

by the layer-cake formula — exactly the accounting that worked in Proposition 4.3.4.

The step that fails is the one asserting that the pieces are *bounded* by a multiple of  $2^k$ , which is condition (A2) in the definition of an atom. In Proposition 4.3.4 that bound came for free, because the good functions  $g_k$  were built from averages of an  $L^2$  function and (4.3.4) applied. Here the only control on  $f$  is through  $\mathcal{M}_\varphi f$ , and while  $|f| \leq \mathcal{M}_\varphi f$  gives  $|f| \leq 2^k$  *outside*  $\Omega_k$ , the pieces live *inside*  $\Omega_k$ , where  $f$  may be arbitrarily large. One is therefore trying to prove a piece is bounded using a hypothesis that says



nothing about it.

The resolution — Fefferman and Stein’s — is not to repair the estimate but to change what is being proved: one shows the pieces lie in  $H^1$  with the right norm, by an induction on the level  $k$ , and only afterwards extracts atoms. This is genuinely intricate and runs to about ten pages.

*Remark 4.4.5* (Scope decision). The converse in Theorem 4.4.4 has been *removed from the plan of this series*, not deferred to a later volume. The distinction matters and is worth stating plainly.

A deferral (convention S7) names the volume that supplies the missing argument and is therefore a promise. There is no such volume here: no later part of this series uses the maximal characterisation, and inventing a home for it would be a fiction. What has happened instead is that the manifest overcommitted, and the honest response is to say so and revise it.

*Nothing in Volumes 1–4 depends on Theorem 4.4.4.* It is used nowhere in a proof; the two places it appears — the Looking Ahead box of §4.1 and Drill 4.4 — are explicitly conditional on it. The remainder of this volume uses  $H^1$  only through its atomic definition, Theorem 4.2.5, and Theorem 4.3.5, all of which are proved.

*What the reader loses.* The atomic definition remains, as Worked Example 2.11 warned, decomposition-dependent, and there is no test in this text for deciding whether a given  $f$  lies in  $H^1$ . A reader who needs one should consult Stein, *Harmonic Analysis*, Chapter III, or Grafakos, *Modern Fourier Analysis*, Chapter 2, where the full proof is given.

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**WORKED EXAMPLE 4.4.2— TIER C: RESEARCH TASTE** **Two things the easy direction already delivers**

The implication we did prove is not a consolation prize. Two consequences, both used in the literature far more often than the converse.

(a) *A necessary condition, hence a source of counterexamples.* Since  $f \in H^1$  forces  $\mathcal{M}_\varphi f \in L^1$ , any  $f$  whose maximal function is non-integrable is not in  $H^1$  — and this is checkable, because  $\mathcal{M}_\varphi f$  is given by an explicit formula. The Pitfall of §4.2 already used this shape of argument to show  $\int f = 0$  is necessary. Drill 4.2 pushes it further.

The converse would be needed only to show a given  $f$  is in  $H^1$ , and for that the atomic definition is usually the more convenient tool anyway: one exhibits a decomposition.

(b) *The estimate (4.4.1) is the reusable object.* What the proof actually established is that an atom, viewed from outside its own cube, looks like  $\ell|x - c|^{-n-1}$  — one power better than the  $|x|^{-n}$  a general  $L^1$  function produces.

That estimate, not the theorem it was used for, is what recurs:

- In Worked Example 4.2.2 the same  $|x|^{-n-1}$  appeared for the Hilbert transform of an atom, by the same freeze-the-kernel-at-the-centre argument.
- In Lemma 4.2.2 it appeared in the form of Hörmander’s condition, which is precisely the hypothesis making the freezing legitimate for a kernel that is not smooth.
- In Volume 12 it appears again, as the statement that a wave packet associated with a tube  $T$  is negligible outside a dilate of  $T$ , with a tail decaying fast enough to sum over the tubes. There the “cube” is a tube, the “centre” is the tube’s core, and the mean-zero property is replaced by frequency localisation — but the computation is Step 3 of Theorem 4.4.2, verbatim.

*The moral for a student.* A theorem often has two parts: a technique and a conclusion. Here the technique — localise, freeze the kernel at a centre, exploit cancellation, integrate the tail — is worth more than the conclusion, and it is entirely contained in the direction we proved.

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**Figure 4.4.1 (to be drawn).** *A Whitney decomposition*

**Content required:** An open set  $\Omega$  in the plane (say the interior of a smooth blob) cut into Whitney squares, sizes shrinking geometrically towards  $\partial\Omega$ . Mark one square  $Q$  and the two distances  $\text{diam } Q$  and  $\text{dist}(Q, \Omega^c)$ , showing the ratio between 1 and 4.

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WORKED EXAMPLE 4.4.3— TIER B **Maximal characterisations: what they buy**

A maximal characterisation replaces a decomposition-based definition by a pointwise one, and the gain is specific.

(a) *The problem it solves.* The atomic definition of  $H^1$  is an infimum over decompositions, so verifying  $f \in H^1$  means exhibiting one. A maximal characterisation replaces this by checking whether a single function,  $\mathcal{M}_\varphi f$ , is integrable.

(b) *Why that matters here.* The Pitfall of Chapter 4.2 showed that reasoning atom-by-atom is invalid without an extra hypothesis. A maximal characterisation removes the need to reason about atoms at all, and is the standard repair.

(c) *What it costs.* The proof is substantially harder than the atomic theory, which is why Remark 4.4.5 despoles the converse direction.

(d) *The pattern elsewhere.* Volume 7's Frostman lemma plays the same role for Hausdorff dimension: it replaces a covering condition — an infimum over covers — by the existence of a measure, which is a pointwise object one can integrate against. Volume 12, Worked Example 12.7.6 records the same trade.

*The general shape.* An infimum over representations is hard to verify and easy to use; a pointwise condition is the reverse. Most function-space theory consists of proving the two equivalent, and the proofs are the hard part of the subject.

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## Drills

**Drill 4.4.1.** Show that  $\mathcal{M}_\varphi f \leq \|\varphi\|_{L^\infty} \mathcal{M}f$  pointwise when  $\varphi \geq 0$  is radial and decreasing, and give an example where  $\mathcal{M}_\varphi f$  is much smaller than  $\mathcal{M}f$ .

**Drill 4.4.2.** Verify the claim in the Pitfall of §4.2: if  $f \in L^1$  with  $\int f \neq 0$ , then  $\mathcal{M}_\varphi f(x) \gtrsim |x|^{-n}$  for large  $|x|$ , so  $\mathcal{M}_\varphi f \notin L^1$ .

**Drill 4.4.3.** Complete Worked Example 4.4.1 for  $\Omega = B(0, 1) \subseteq \mathbb{R}^2$ : describe the Whitney cubes and verify the bound on the number of cubes of each generation touching a given one.

**Drill 4.4.4.** Assuming Theorem 4.4.4, show that the function  $g(x) = \operatorname{sgn}(x)|x|^{-1}(\log|x|)^{-2}$  of the Pitfall in §2.3 is not in  $H^1(\mathbb{R})$ , by estimating  $\mathcal{M}_\varphi g(x)$  from below for large  $|x|$ .

**Drill 4.4.5.** Verify that  $\Omega_k = \{\mathcal{M}_\varphi f > 2^k\}$  is open, and explain why a smooth  $\varphi$  rather than a sharp cutoff is needed for this.

**Drill 4.4.6.** Check the layer-cake accounting  $\sum_k 2^k |\Omega_k| \approx \|\mathcal{M}_\varphi f\|_{L^1}$  used in the scratchwork.

**Drill 4.4.7.** Show that Theorem 4.4.2 — the implication  $\Rightarrow$  — follows from the atomic definition together with Lemma 4.2.2.

**Drill 4.4.8.** Explain why Remark 4.4.5 classifies the converse as removed rather than deferred, and identify what a deferral would have had to name.

## Notes

Theorem 4.4.4 is due to Fefferman and Stein (1972). Their paper proves considerably more: the equivalence holds with  $\mathcal{M}_\varphi$  replaced by the *grand* maximal function, in which one takes the supremum over all  $\varphi$  in a bounded subset of  $\mathcal{S}$ , and it holds for  $H^p$  with  $0 < p \leq 1$ .

The Whitney decomposition is from Whitney's 1934 extension theorem and is one of the most reused constructions in analysis. Readers will meet it again in Volume 11, where the cells of a polynomial partition play an analogous role: a decomposition of space into pieces on which the object of study is uniformly controlled.

## Chapter 4.5

# Weighted and Vector-Valued Maximal Inequalities

### 4.5.1 The question

The function spaces of Chapter 4.6 are defined by taking, at each point, an  $\ell^q$  norm of the Littlewood–Paley pieces of  $f$ , and then an  $L^p$  norm in  $x$ . Working with such spaces requires knowing that the Hardy–Littlewood maximal operator, applied to each piece separately, respects this double norm:

$$\left\| \left( \sum_j (\mathcal{M}f_j)^q \right)^{1/q} \right\|_{L^p} \lesssim_{n,p,q} \left\| \left( \sum_j |f_j|^q \right)^{1/q} \right\|_{L^p}. \quad (4.5.1)$$

It is worth seeing why this is not immediate. Volume 3 proved the Marcinkiewicz–Zygmund theorem, which gives exactly (4.5.1) for free — but only when  $q = 2$ , because the proof runs through Khintchine’s inequality, and only for *linear* operators, because the proof pulls the operator through a sum of randomly signed terms. The maximal operator is neither. So both hypotheses fail, and a new argument is needed.

#### **Pitfall: no pointwise domination is available**

The natural first attempt is to look for a pointwise inequality. Write  $F = (\sum_j |f_j|^q)^{1/q}$ . Since  $|f_j| \leq F$  pointwise we get  $\mathcal{M}f_j \leq \mathcal{M}F$ , but then  $(\sum_j (\mathcal{M}f_j)^q)^{1/q} \leq (\#j)^{1/q} \mathcal{M}F$ , useless for infinitely many terms. The other attempt is Minkowski’s inequality in  $\ell^q$ , which does give

$$\left( \sum_j \left( \frac{1}{|B|} \int_B |f_j| \right)^q \right)^{1/q} \leq \frac{1}{|B|} \int_B F \leq \mathcal{M}F(x)$$

for a *single* ball  $B \ni x$ . But the left side of (4.5.1) takes the supre-

mum over balls *separately for each  $j$* , and different  $j$  may be optimised by wildly different balls. So  $(\sum_j (\mathcal{M}f_j)^q)^{1/q}$  is genuinely larger than  $\sup_B (\sum_j (\text{avg}_B |f_j|)^q)^{1/q}$ , and no pointwise bound by  $\mathcal{M}F$  holds. This is exactly why (4.5.1) is a theorem.

Two special cases are, however, easy, and it is worth disposing of them at once.

**Proposition 4.5.1** (The two easy cases). *Let  $1 < p < \infty$ .*

(i)  $\|\sup_j \mathcal{M}f_j\|_{L^p} \lesssim_{n,p} \|\sup_j |f_j|\|_{L^p}$ , i.e. (4.5.1) for  $q = \infty$ .

(ii) (4.5.1) holds when  $q = p$ .

*Proof.* (i) Put  $G = \sup_j |f_j|$ . Then  $|f_j| \leq G$  for each  $j$ , so  $\mathcal{M}f_j \leq \mathcal{M}G$  and hence  $\sup_j \mathcal{M}f_j \leq \mathcal{M}G$  pointwise. Apply the maximal theorem (Theorem 1.7.3 (Hardy–Littlewood maximal inequality)).

(ii) By Tonelli, the left side of (4.5.1) raised to the power  $p$  is  $\sum_j \|\mathcal{M}f_j\|_{L^p}^p$ , which by the maximal theorem is  $\lesssim \sum_j \|f_j\|_{L^p}^p$ , and that is the right side raised to the power  $p$ , again by Tonelli.  $\square$

## 4.5.2 A weighted maximal inequality

The tool that handles the general case is an inequality of a different kind: it allows an arbitrary weight on both sides, provided the weight on the right is enlarged by one application of  $\mathcal{M}$ .

**Theorem 4.5.2** (Fefferman–Stein weighted maximal inequality). *Let  $w \geq 0$  be measurable and not identically zero. Then*

(i) for every  $f \in L^1_{\text{loc}}$  and  $\lambda > 0$ ,

$$\int_{\{\mathcal{M}f > \lambda\}} w \leq \frac{C_n}{\lambda} \int_{\mathbb{R}^n} |f| \mathcal{M}w;$$

(ii) for every  $1 < q < \infty$ ,

$$\int_{\mathbb{R}^n} (\mathcal{M}f)^q w \leq C_{n,q} \int_{\mathbb{R}^n} |f|^q \mathcal{M}w.$$

*Idea of the proof.* Part (i) is the Vitali argument of Volume 1, run while carrying the weight. The covering lemma produces disjoint balls  $B_i$  on which the average of  $|f|$  exceeds  $\lambda$ , and the level set is contained in  $\bigcup_i 5B_i$ .

The only new step is to bound  $w(5B_i)$ , and here  $\mathcal{M}w$  does the work: for any  $x \in B_i$ , the ball  $5B_i$  is one of the balls competing in the supremum defining  $\mathcal{M}w(x)$ , so  $w(5B_i) \lesssim |B_i|\mathcal{M}w(x)$  — and we may take the infimum over  $x \in B_i$ .

Part (ii) is then Marcinkiewicz interpolation, run with the measure  $w \, dx$  on the target and  $\mathcal{M}w \, dx$  on the domain. That the two measures differ is not an obstacle: the proof of Marcinkiewicz never used that they were the same.

*Proof. Step 1. Proof of (i).*

Let  $K \subseteq \{\mathcal{M}f > \lambda\}$  be compact; by inner regularity it suffices to bound  $\int_K w$ . For each  $x \in K$  pick a ball  $B(x, r_x)$  with

$$\int_{B(x, r_x)} |f| > \lambda |B(x, r_x)|, \quad (4.5.2)$$

and extract a finite subcover; by the Vitali covering lemma (Lemma 1.7.1 (Vitali covering lemma), in the form with the factor 5) there is a pairwise disjoint subfamily  $B_1, \dots, B_N$  with  $K \subseteq \bigcup_i 5B_i$ .

Fix  $i$  and let  $x \in B_i$ . The ball  $5B_i$  is contained in  $B(x, 10r_i)$ , which is a ball centred at  $x$ , so

$$\mathcal{M}w(x) \geq \frac{1}{|B(x, 10r_i)|} \int_{B(x, 10r_i)} w \geq \frac{1}{10^n |B_i|} \int_{5B_i} w.$$

Taking the infimum over  $x \in B_i$  and rearranging,

$$\int_{5B_i} w \leq 10^n |B_i| \inf_{B_i} \mathcal{M}w \stackrel{(4.5.2)}{\leq} \frac{10^n}{\lambda} \int_{B_i} |f| \cdot \inf_{B_i} \mathcal{M}w \leq \frac{10^n}{\lambda} \int_{B_i} |f| \mathcal{M}w.$$

Summing over the disjoint  $B_i$ ,

$$\int_K w \leq \sum_i \int_{5B_i} w \leq \frac{10^n}{\lambda} \int_{\mathbb{R}^n} |f| \mathcal{M}w.$$

**Step 2.** *Proof of (ii): setting up Marcinkiewicz.*

Write  $d\nu = w \, dx$  and  $d\mu = \mathcal{M}w \, dx$ . Part (i) says  $\mathcal{M}$  is of weak type  $(1, 1)$  from  $L^1(\mu)$  to  $L^{1,\infty}(\nu)$ . And  $\mathcal{M}$  is of strong type  $(\infty, \infty)$  from  $L^\infty(\mu)$  to  $L^\infty(\nu)$ : if  $|f| \leq A$   $\mu$ -a.e. then, since  $\mathcal{M}w > 0$  everywhere (as  $w \not\equiv 0$ ), we have  $|f| \leq A$  Lebesgue-a.e., hence  $\mathcal{M}f \leq A$  everywhere.

**Step 3.** *Run the splitting argument.*

Fix  $\lambda > 0$  and split  $f = f\mathbf{1}_{\{|f|>\lambda/2\}} + f\mathbf{1}_{\{|f|\leq\lambda/2\}} =: g_\lambda + b_\lambda$ . Then  $\mathcal{M}b_\lambda \leq \lambda/2$  by Step 2, so by sublinearity  $\{\mathcal{M}f > \lambda\} \subseteq \{\mathcal{M}g_\lambda > \lambda/2\}$ , and

part (i) gives

$$\nu(\{\mathcal{M}f > \lambda\}) \leq \frac{2C_n}{\lambda} \int_{\{|f| > \lambda/2\}} |f| \mathcal{M}w.$$

By the layer-cake formula (Worked Example 1.5.1 (the layer-cake formula)) applied with the measure  $\nu$ , and then Tonelli,

$$\begin{aligned} \int (\mathcal{M}f)^q d\nu &= q \int_0^\infty \lambda^{q-1} \nu(\{\mathcal{M}f > \lambda\}) d\lambda \\ &\leq 2C_n q \int_0^\infty \lambda^{q-2} \int_{\{|f| > \lambda/2\}} |f(x)| \mathcal{M}w(x) dx d\lambda \\ &= 2C_n q \int_{\mathbb{R}^n} |f(x)| \mathcal{M}w(x) \left( \int_0^{2|f(x)|} \lambda^{q-2} d\lambda \right) dx = \frac{2^q C_n q}{q-1} \int_{\mathbb{R}^n} |f|^q \mathcal{M}w, \end{aligned}$$

the inner integral converging at  $\lambda = 0$  precisely because  $q > 1$ .  $\square$

**Where we are.** Note the shape of (ii): it is not a weighted norm inequality with the same weight on both sides — such inequalities hold only for special weights, the Muckenhoupt  $A_q$  class, which this series does not need. It is the weaker and more flexible statement that *any* weight works if one is willing to pay a maximal function on the right. That flexibility is exactly what the duality argument of the next section requires, because there the weight is not chosen by us but produced by duality.

### 4.5.3 The vector-valued theorem

**Theorem 4.5.3.** *Let  $1 < q < \infty$  and  $q < p < \infty$ . Then for all  $f_1, f_2, \dots \in L^p(\mathbb{R}^n)$ ,*

$$\left\| \left( \sum_j (\mathcal{M}f_j)^q \right)^{1/q} \right\|_{L^p} \leq C_{n,p,q} \left\| \left( \sum_j |f_j|^q \right)^{1/q} \right\|_{L^p}.$$

*Together with Proposition 4.5.1 this covers  $q = \infty$ ,  $p = q$ , and  $q < p < \infty$ .*

*Idea of the proof.* Since  $p > q$ , the exponent  $p/q$  exceeds 1, so the quantity  $\left\| \sum_j (\mathcal{M}f_j)^q \right\|_{L^{p/q}}$  may be computed by duality (Corollary 1.8.10 (norms by testing)): it is the supremum of  $\int \sum_j (\mathcal{M}f_j)^q w$  over nonnegative  $w$  in the unit ball of  $L^{(p/q)'}$ .

That integral is a sum of weighted integrals of  $(\mathcal{M}f_j)^q$ , one for each  $j$ , with a *common* weight  $w$  — and Theorem 4.5.2(ii) handles each one, replacing  $w$  by  $\mathcal{M}w$  on the right. Summing over  $j$  reassembles  $\sum_j |f_j|^q$ , and one final application of the maximal theorem, at exponent  $(p/q)'$ , disposes of the  $\mathcal{M}w$ .



The requirement  $p > q$  enters exactly once, at the first step: for  $p < q$  the exponent  $p/q$  is less than 1,  $L^{p/q}$  is not a normed space, and duality is unavailable.

*Proof.* Write  $F = (\sum_j |f_j|^q)^{1/q}$  and  $r = p/q > 1$ .

**Step 1.** *Dualise.*

By duality in  $L^r$ , and since  $\sum_j (\mathcal{M}f_j)^q \geq 0$ ,

$$\left\| \left( \sum_j (\mathcal{M}f_j)^q \right)^{1/q} \right\|_{L^p}^q = \left\| \sum_j (\mathcal{M}f_j)^q \right\|_{L^r} = \sup \left\{ \int_{\mathbb{R}^n} \sum_j (\mathcal{M}f_j)^q w : w \geq 0, \|w\|_{L^{r'}} \leq 1 \right\}$$

**Step 2.** *Apply the weighted inequality termwise.*

Fix such a  $w$ . By Theorem 4.5.2(ii), applied to each  $f_j$  with the same weight  $w$ , and then Tonelli,

$$\int \sum_j (\mathcal{M}f_j)^q w = \sum_j \int (\mathcal{M}f_j)^q w \leq C_{n,q} \sum_j \int |f_j|^q \mathcal{M}w = C_{n,q} \int F^q \mathcal{M}w.$$

**Step 3.** *Dispose of  $\mathcal{M}w$ .*

By Hölder with exponents  $r$  and  $r'$ , and then the maximal theorem (Theorem 1.7.3 (Hardy–Littlewood maximal inequality)) at exponent  $r' > 1$ ,

$$\int F^q \mathcal{M}w \leq \|F^q\|_{L^r} \|\mathcal{M}w\|_{L^{r'}} \lesssim_{n,r} \|F^q\|_{L^r} \|w\|_{L^{r'}} \leq \|F\|_{L^p}^q,$$

using  $\|F^q\|_{L^r} = \|F\|_{L^{qr}}^q = \|F\|_{L^p}^q$  and  $\|w\|_{r'} \leq 1$ .

**Step 4.** *Take the supremum.*

Combining Steps 1–3,  $\left\| (\sum_j (\mathcal{M}f_j)^q)^{1/q} \right\|_p^q \lesssim \|F\|_p^q$ . □

*Remark 4.5.4* (The remaining range, and a scope decision). Theorem 4.5.3 is true for the full range  $1 < p < \infty$ ,  $1 < q \leq \infty$ ; that is the Fefferman–Stein vector-valued maximal theorem of 1971. The case  $1 < p < q$  is not proved here.

The obstruction is precise. Our only route to (4.5.1) is the duality of Step 1, which needs  $p/q > 1$ . The standard proof of the missing range instead establishes a *vector-valued weak*  $(1, 1)$  *bound* for the  $\ell^q$ -valued maximal operator, by running the Calderón–Zygmund decomposition on the scalar function  $F$  and estimating the resulting pieces in  $\ell^q$ ; that argument requires a genuinely vector-valued Calderón–Zygmund theory, which this series does not develop.

As with Remark 4.4.5, this is a revision of the plan rather than a deferral: no later volume supplies the missing case. Chapter 4.6 is written so as to use only the range proved here, and the places where that costs something are marked.

The following weaker statement, valid for all  $1 < p \leq q \leq \infty$ , is sometimes an adequate substitute and is free:

$$\left\| \left( \sum_j (\mathcal{M}f_j)^q \right)^{1/q} \right\|_{L^p} \leq \left\| \left( \sum_j (\mathcal{M}f_j)^p \right)^{1/p} \right\|_{L^p} \lesssim_{n,p} \left\| \left( \sum_j |f_j|^p \right)^{1/p} \right\|_{L^p},$$

the first inequality because  $\ell^p \subseteq \ell^q$  for  $p \leq q$ , and the second being Proposition 4.5.1(ii). It differs from (4.5.1) in having  $\ell^p$  rather than  $\ell^q$  on the right, which is a genuine loss.

**WORKED EXAMPLE 4.5.1—TIER B The weighted inequality is sharp in its shape**

One cannot replace  $\mathcal{M}w$  by  $w$  in Theorem 4.5.2. Take  $n = 1$ ,  $w = \mathbf{1}_{[0,1]}$  and  $f = \mathbf{1}_{[-\varepsilon,0]}$  for small  $\varepsilon > 0$ .

Then  $\|f\|_{L^q(w)}^q = \int |f|^q w = 0$ , since  $f$  and  $w$  have disjoint supports up to a null set. But  $\mathcal{M}f(x) \approx \varepsilon/(x + \varepsilon)$  for  $x > 0$ , so

$$\int (\mathcal{M}f)^q w \approx \int_0^1 \left( \frac{\varepsilon}{x + \varepsilon} \right)^q dx > 0.$$

So the inequality  $\int (\mathcal{M}f)^q w \leq C \int |f|^q w$  is false for every  $C$ : the maximal function moves mass from where  $f$  lives to where it does not, and a weight that ignores the first region cannot control the second.

Replacing  $w$  by  $\mathcal{M}w$  on the right repairs exactly this, and it is worth checking the arithmetic. For  $x \in [-\varepsilon, 0]$ , the interval  $[x, 1]$  has length between 1 and  $1 + \varepsilon$  and contains all of  $\text{supp } w$ , so

$$\mathcal{M}w(x) \geq \frac{1}{1 + \varepsilon} \int_0^1 w = \frac{1}{1 + \varepsilon} \geq \frac{1}{2},$$

whence

$$\int |f|^q \mathcal{M}w = \int_{-\varepsilon}^0 \mathcal{M}w \geq \frac{\varepsilon}{2}.$$

On the other side,  $\mathcal{M}f(x) \leq \min\{1, \varepsilon/x\}$  for  $x > 0$  — the average of  $\mathbf{1}_{[-\varepsilon,0]}$  over any interval containing  $x$  and meeting  $[-\varepsilon, 0]$  is at most  $\varepsilon$  divided by the interval's length, which is at least  $x$ . Hence

$$\int (\mathcal{M}f)^q w = \int_0^1 (\mathcal{M}f)^q \leq \int_0^\varepsilon 1 \, dx + \int_\varepsilon^1 \left( \frac{\varepsilon}{x} \right)^q dx \leq \varepsilon + \frac{\varepsilon^q}{q-1} \varepsilon^{1-q} = \varepsilon \left( 1 + \frac{1}{q-1} \right),$$

using  $q > 1$  to evaluate the second integral. So both sides of Theorem 4.5.2(ii) are of size  $\approx \varepsilon$ , and the inequality holds with a constant depending only on  $q$  — as it must.

The weight  $\mathcal{M}w$  is the smallest enlargement of  $w$  that sees, at every point, the mass of  $w$  anywhere nearby, and that is precisely what the maximal operator's spreading requires.

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WORKED EXAMPLE 4.5.2— TIER C: RESEARCH TASTE **Where vector-valued estimates come from, and where they go**

(a) *The pattern.* Three vector-valued statements have now appeared in this series, and they are of increasing strength:

| Statement                      | Hypothesis on $T$            | Source                           |
|--------------------------------|------------------------------|----------------------------------|
| $\ell^2$ extension             | $T$ linear, bounded on $L^p$ | Vol 3, Thm 6.1                   |
| $\ell^q$ extension, $q < p$    | $T = \mathcal{M}$            | Theorem 4.5.3                    |
| $\ell^q$ extension, all $p, q$ | $T = \mathcal{M}$            | Fefferman–Stein, not proved here |

The first is cheap because Khintchine's inequality converts an  $\ell^2$  sum into a randomised scalar sum, and linearity lets  $T$  pass through. Neither device is available for  $q \neq 2$  or for sublinear  $T$ , and the price is a genuinely different proof.

(b) *Why the subject needs them at all.* Every decomposition in this series — Littlewood–Paley pieces, atoms, wave packets — replaces one function by a family, and every estimate must then be reassembled. The reassembly is an  $\ell^q$  norm, and the question is always whether an operator applied to each piece separately respects it. Volume 3, Chapter 8 was a negative answer of this kind: Fefferman's theorem said that the half-space multipliers do *not* admit a uniform  $\ell^2$  extension over a family of directions, and a Besicovitch set was the witness.

(c) *The contrast worth holding onto.* In Theorem 4.5.3 the family  $\{f_j\}$  is arbitrary and the operator is fixed. In Fefferman's theorem the family of operators  $\{S_{v_j}\}$  varies and the difficulty comes from the *geometry of the directions*  $v_j$ . The first is a soft statement, provable by duality and a weight. The second is a hard one, and false. Volumes 9–13 are almost entirely about the second kind: estimates for families indexed by directions or by tubes, where the answer depends on how the family sits in space.

A useful diagnostic when reading the literature: if a vector-valued inequality is being proved by duality and a weight, it is of the first kind and should be

routine; if it is being proved by a covering or incidence argument, it is of the second kind and is where the content lies.

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## Drills

**Drill 4.5.1.** Prove the Vitali covering lemma in the form used in Step 1 of Theorem 4.5.2: from any finite family of balls one may extract a pairwise disjoint subfamily whose 5-fold dilates cover the union. (Compare Lemma 1.7.1 (Vitali covering lemma), which used the factor 3; identify where the extra room is needed here.)

**Drill 4.5.2.** Show that Theorem 4.5.2(ii) fails at  $q = 1$ , by taking  $w = \mathbf{1}_{[0,1]}$  on  $\mathbb{R}$  and  $f$  concentrated near 0.

**Drill 4.5.3.** Verify the claim in Remark 4.5.4 that  $\ell^p \subseteq \ell^q$  with norm 1 when  $p \leq q$ , and give an example showing the resulting substitute inequality is strictly weaker than (4.5.1).

**Drill 4.5.4.** Deduce from Theorem 4.5.3 that for  $q < p < \infty$  the operator  $f \mapsto (\mathcal{M}f_j)_j$  is bounded on  $L^p(\mathbb{R}^n; \ell^q)$ , and check that the constant blows up as  $p \downarrow q$ . Which step of the proof is responsible?

## Notes

Theorem 4.5.2 and the full vector-valued theorem are both due to Fefferman and Stein: the weighted inequality in *Some maximal inequalities* (1971), the vector-valued one in the same paper. The duality proof of Theorem 4.5.3 given here is the standard route to the range  $p > q$  and is the one most often reproduced; the remaining range genuinely requires the vector-valued Calderón–Zygmund machinery, for which see Grafakos, *Classical Fourier Analysis*, Chapter 5.

Readers who continue to the theory of weights will recognise Theorem 4.5.2(ii) as the statement that the pair  $(w, \mathcal{M}w)$  satisfies a two-weight  $L^q$  bound for the maximal operator. Characterising the pairs  $(u, v)$  for which  $\int (\mathcal{M}f)^q u \lesssim \int |f|^q v$  holds is the Sawyer testing problem, solved by Sawyer in 1982; none of that theory is used here.

## Chapter 4.6

# Besov and Triebel–Lizorkin Spaces

### 4.6.1 Two ways to measure smoothness

Volume 2 defined the Sobolev spaces  $H^s(\mathbb{R}^n)$  by requiring  $(1 + |\xi|^2)^{s/2} \widehat{f} \in L^2$ , and observed that the definition works only at  $p = 2$ , because only there is Plancherel available. Volume 3 removed that restriction: the Littlewood–Paley theorem showed that for  $1 < p < \infty$  the  $L^p$  norm is comparable to the  $L^p$  norm of the square function built from the frequency pieces  $\Delta_j f$ . Sobolev norms at general  $p$  are therefore square functions with dyadic weights.

Once that is seen, a question presents itself. A square function is an  $\ell^2$  norm of the pieces, taken pointwise, followed by an  $L^p$  norm in  $x$ . Why  $\ell^2$ ? And why in that order?

Both choices can be varied, and doing so produces the two standard scales:

$$\underbrace{\left\| \left( 2^{js} \|\Delta_j f\|_{L^p} \right)_j \right\|_{\ell^q}}_{\text{Besov: } B_{p,q}^s} \quad \text{versus} \quad \underbrace{\left\| \left\| (2^{js} \Delta_j f)_j \right\|_{\ell^q} \right\|_{L^p}}_{\text{Triebel–Lizorkin: } F_{p,q}^s}.$$

The Besov norm takes  $L^p$  first and  $\ell^q$  second; the Triebel–Lizorkin norm reverses the order. Three parameters now appear:  $s$  measures smoothness,  $p$  measures integrability, and  $q$  is a fine-tuning index that turns out to matter exactly at the endpoints — which is why Chapter 4.7 can prove a sharp embedding into  $L^\infty$  that the Sobolev scale misses.

*Remark 4.6.1* (Scope of this chapter). This chapter proves: the definitions are meaningful and independent of choices (for Besov in general, for Triebel–Lizorkin at  $q = 2$ ); the Besov spaces are complete; the two scales are related by the Minkowski inequalities  $B_{p,\min(p,q)}^s \subseteq F_{p,q}^s \subseteq B_{p,\max(p,q)}^s$ ; and  $F_{p,2}^s = W^{s,p}$  for  $1 < p < \infty$ , which places the Sobolev spaces inside the scale.

It does *not* prove the identifications  $F_{1,2}^0 = H^1$  and  $F_{\infty,2}^0 = \text{BMO}$ . The first is the Littlewood–Paley characterisation of  $H^1$  and the second is Fefferman’s Carleson-measure theorem; both require machinery beyond this series, and neither is used later — the  $H^1$ –BMO duality of Chapter 4.3 is what the later volumes need. This was decided before the chapter was written and is recorded as revision R3 in the manifest.

## 4.6.2 The inhomogeneous decomposition

**Definition 4.6.2.** Fix  $\phi_0 \in C_c^\infty(\mathbb{R}^n)$  with  $\phi_0 = 1$  on  $B(0, 1)$  and  $\text{supp } \phi_0 \subseteq B(0, 2)$ , and set

$$\psi(\xi) = \phi_0(\xi) - \phi_0(2\xi),$$

so that  $\text{supp } \psi \subseteq \{\frac{1}{2} \leq |\xi| \leq 2\}$  and, telescoping,

$$\phi_0(\xi) + \sum_{j \geq 1} \psi(2^{-j}\xi) = 1 \quad \text{for every } \xi \in \mathbb{R}^n. \quad (4.6.1)$$

Define  $\Delta_0 f = \mathcal{F}^{-1}(\phi_0 \widehat{f})$  and, for  $j \geq 1$ ,  $\Delta_j f = \mathcal{F}^{-1}(\psi(2^{-j}\cdot) \widehat{f})$ , so that  $\sum_{j \geq 0} \Delta_j f = f$  in  $\mathcal{S}'(\mathbb{R}^n)$ .

To see (4.6.1): the partial sum up to  $j = N$  telescopes to  $\phi_0(2^{-N}\xi)$ , which equals 1 once  $2^{-N}|\xi| \leq 1$ .

Two facts about the pieces are used constantly and are worth recording once.

**Lemma 4.6.3.** *Let  $\eta \in C_c^\infty(\mathbb{R}^n)$  and  $\eta_j = \eta(2^{-j}\cdot)$ , and let  $T_j f = \mathcal{F}^{-1}(\eta_j \widehat{f})$ . Then for every  $1 \leq p \leq \infty$  and every  $j$ ,*

$$\|T_j f\|_{L^p} \leq \|\mathcal{F}^{-1}\eta\|_{L^1} \|f\|_{L^p},$$

*a bound independent of  $j$ . Moreover  $\widehat{\Delta_j f}$  is supported in  $\{|\xi| \leq 2\}$  for  $j = 0$  and in  $\{2^{j-1} \leq |\xi| \leq 2^{j+1}\}$  for  $j \geq 1$ , so  $\Delta_j \Delta_k = 0$  whenever  $|j - k| \geq 2$ .*

*Proof.*  $T_j f = K_j * f$  where  $K_j = \mathcal{F}^{-1}\eta_j$ . By the dilation identity (Volume 2, Prop. 4.1(iii)),  $K_j(x) = 2^{jn} K(2^j x)$  with  $K = \mathcal{F}^{-1}\eta$ , so  $\|K_j\|_{L^1} = \|K\|_{L^1}$  for every  $j$ , and  $K \in L^1$  because  $\eta \in C_c^\infty \subseteq \mathcal{S}$  so  $K \in \mathcal{S}$ . Young’s inequality (Volume 2, Worked Example 2.4) gives the bound. The support statements are immediate from the definitions, and disjointness of the annuli for  $|j - k| \geq 2$  gives  $\Delta_j \Delta_k = 0$ .  $\square$

### 4.6.3 The two scales

**Definition 4.6.4.** Let  $s \in \mathbb{R}$ ,  $1 \leq p \leq \infty$  and  $1 \leq q \leq \infty$ . For  $f \in \mathcal{S}'(\mathbb{R}^n)$  set

$$\|f\|_{B_{p,q}^s} = \left( \sum_{j \geq 0} (2^{js} \|\Delta_j f\|_{L^p})^q \right)^{1/q}, \quad \|f\|_{F_{p,q}^s} = \left\| \left( \sum_{j \geq 0} (2^{js} |\Delta_j f|)^q \right)^{1/q} \right\|_{L^p},$$

with the usual modification when  $q = \infty$ , and let  $B_{p,q}^s, F_{p,q}^s$  be the sets where these are finite. For  $F$  we require  $p < \infty$ .

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WORKED EXAMPLE 4.6.1— TIER A **Reading the three indices**

It helps to see what each index does on a single example. Fix a Schwartz function  $g$  with  $\widehat{g}$  supported in the annulus  $\{\frac{1}{2} \leq |\xi| \leq 2\}$ , and for a fixed  $N$  put

$$f = \sum_{j=1}^N a_j g(2^j \cdot),$$

so that  $\Delta_j f = a_j g(2^j \cdot)$  for  $1 \leq j \leq N$  and  $\Delta_j f = 0$  otherwise.

*The index  $p$ .* By scaling,  $\|g(2^j \cdot)\|_{L^p} = 2^{-jn/p} \|g\|_{L^p}$ . So the  $L^p$  size of a piece at frequency  $2^j$  already carries a factor  $2^{-jn/p}$ : higher frequencies are, at fixed amplitude, smaller in  $L^p$  for larger  $p$ . This is the scaling behind every embedding in Chapter 4.7.

*The index  $s$ .* The Besov norm is  $(\sum_j (2^{js} |a_j| 2^{-jn/p})^q)^{1/q} \|g\|_p$ . Raising  $s$  penalises high frequencies more, which is what “more derivatives” means.

*The index  $q$ .* Suppose  $a_j 2^{-jn/p} 2^{js} = 1$  for all  $j \leq N$ , so every scale contributes equally. Then the Besov norm is  $N^{1/q} \|g\|_p$ : it sees  $q$  only through how the contributions of the  $N$  scales are combined. Small  $q$  punishes spreading mass over many scales;  $q = \infty$  ignores it entirely. This is why  $q$  is invisible in the Sobolev scale — where it is fixed at 2 — and becomes decisive at the endpoints of Chapter 4.7.

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**Proposition 4.6.5** (Independence of the decomposition, Besov case). *Let  $\{\Delta_j\}$  and  $\{\widetilde{\Delta}_j\}$  be built as in Definition 4.6.2 from two admissible choices  $\phi_0, \widetilde{\phi}_0$ . Then the resulting  $B_{p,q}^s$  norms are comparable, with constants independent of  $f$ .*

*Proof.* By the support statements in Lemma 4.6.3,  $\widetilde{\Delta}_j$  and  $\Delta_k$  have Fourier supports meeting only when  $|j - k| \leq C_0$  for an absolute constant  $C_0$  (one may take  $C_0 = 2$  after fixing the constants in Definition 4.6.2). Since  $\sum_k \Delta_k = \text{Id}$ ,

$$\widetilde{\Delta}_j f = \sum_{|k-j| \leq C_0} \widetilde{\Delta}_j \Delta_k f.$$

By Lemma 4.6.3 applied to  $\tilde{\Delta}_j$ ,  $\|\tilde{\Delta}_j \Delta_k f\|_{L^p} \leq C \|\Delta_k f\|_{L^p}$  with  $C$  independent of  $j$ , so

$$2^{js} \|\tilde{\Delta}_j f\|_{L^p} \leq C \sum_{|k-j| \leq C_0} 2^{js} \|\Delta_k f\|_{L^p} \leq C 2^{C_0|s|} \sum_{|k-j| \leq C_0} 2^{ks} \|\Delta_k f\|_{L^p}.$$

Taking the  $\ell^q$  norm in  $j$  and using that the sum has at most  $2C_0 + 1$  terms, each of which is a shift of the sequence  $(2^{ks} \|\Delta_k f\|_p)_k$ , the triangle inequality in  $\ell^q$  gives  $\|f\|_{\tilde{B}_{p,q}^s} \lesssim_{s,C_0} \|f\|_{B_{p,q}^s}$ . Exchanging the roles of the two systems gives the reverse.  $\square$

### Pitfall: the same argument does not settle the $F$ case

In Proposition 4.6.5 the  $L^p$  norm was taken *before* the sum over  $j$ , so Lemma 4.6.3 could be applied to one  $j$  at a time. In the Triebel–Lizorkin norm the  $\ell^q$  sum is inside the  $L^p$  norm, and one needs to bound  $\|(\sum_j |\tilde{\Delta}_j \Delta_k f|^q)^{1/q}\|_p$  by the corresponding quantity for  $\Delta_k f$  — a *vector-valued* statement.

The standard route is through the Peetre maximal function, which reduces it to the Fefferman–Stein vector-valued maximal inequality; and that inequality was proved in Chapter 4.5 only for  $q < p$ . So the general case is not available here. It is available at  $q = 2$ , by a different route: Theorem 4.6.8 below identifies  $F_{p,2}^s$  with a space defined without reference to any decomposition.

## 4.6.4 Comparison of the two scales

**Theorem 4.6.6** (Minkowski embeddings). *For  $s \in \mathbb{R}$ ,  $1 \leq p < \infty$  and  $1 \leq q \leq \infty$ ,*

$$B_{p,\min(p,q)}^s \subseteq F_{p,q}^s \subseteq B_{p,\max(p,q)}^s,$$

*with norm inequalities in the same directions. In particular  $B_{p,p}^s = F_{p,p}^s$ .*

*Idea of the proof.* Everything is an instance of one principle: interchanging an  $\ell^q$  norm and an  $L^p$  norm costs nothing in the direction that puts the *smaller* exponent on the outside. Concretely, Minkowski’s integral inequality says  $\|\cdot\|_{\ell^q(L^p)} \leq \|\cdot\|_{L^p(\ell^q)}$  when  $p \leq q$ , and the reverse when  $q \leq p$ . The four cases below are that statement, together with the elementary inclusion  $\ell^{q_1} \subseteq \ell^{q_2}$  for  $q_1 \leq q_2$ .

*Proof.* Write  $g_j = 2^{js} \Delta_j f$ , so  $\|f\|_{B_{p,q}^s} = \|(\|g_j\|_{L^p})_j\|_{\ell^q}$  and  $\|f\|_{F_{p,q}^s} = \| \|(g_j)_j\|_{\ell^q} \|_{L^p}$ .



**Step 1.**  $B_{p,q}^s \subseteq F_{p,q}^s$  when  $q \leq p$ .

Put  $r = p/q \geq 1$ . By the triangle inequality in  $L^r$ ,

$$\|f\|_{F_{p,q}^s}^q = \left\| \sum_j |g_j|^q \right\|_{L^r} \leq \sum_j \| |g_j|^q \|_{L^r} = \sum_j \|g_j\|_{L^p}^q = \|f\|_{B_{p,q}^s}^q.$$

**Step 2.**  $B_{p,p}^s \subseteq F_{p,q}^s$  when  $q \geq p$ .

Now  $r = p/q \leq 1$ , and for  $0 < r \leq 1$  one has the elementary inequality  $(\sum_j a_j)^r \leq \sum_j a_j^r$  for  $a_j \geq 0$ . Applying it pointwise and integrating,

$$\|f\|_{F_{p,q}^s}^p = \int \left( \sum_j |g_j|^q \right)^{p/q} \leq \int \sum_j |g_j|^p = \sum_j \|g_j\|_{L^p}^p = \|f\|_{B_{p,p}^s}^p.$$

Steps 1 and 2 together give  $B_{p,\min(p,q)}^s \subseteq F_{p,q}^s$ , using  $B_{p,q_1}^s \subseteq B_{p,q_2}^s$  for  $q_1 \leq q_2$  (the inclusion  $\ell^{q_1} \subseteq \ell^{q_2}$ ).

**Step 3.**  $F_{p,q}^s \subseteq B_{p,q}^s$  when  $p \leq q$ .

This is Minkowski's integral inequality in the form  $\|(\|g_j\|_{L^p})_j\|_{\ell^q} \leq \| (g_j)_j \|_{\ell^q}\|_{L^p}$ , valid for  $p \leq q$ ; it is the statement that the  $L^{q/p}$  norm of a sum is at most the sum of the norms, applied to  $|g_j|^p$ .

**Step 4.**  $F_{p,q}^s \subseteq B_{p,p}^s$  when  $q \leq p$ .

Since  $\ell^q \subseteq \ell^p$  for  $q \leq p$ ,

$$\|f\|_{B_{p,p}^s} = \left\| \left( \sum_j |g_j|^p \right)^{1/p} \right\|_{L^p} \leq \left\| \left( \sum_j |g_j|^q \right)^{1/q} \right\|_{L^p} = \|f\|_{F_{p,q}^s},$$

the first equality by Tonelli. Steps 3 and 4 give  $F_{p,q}^s \subseteq B_{p,\max(p,q)}^s$ .

Taking  $q = p$  in all four steps gives  $B_{p,p}^s = F_{p,p}^s$ , as both chains collapse.  $\square$

**Theorem 4.6.7** (Completeness). *For  $s \in \mathbb{R}$ ,  $1 \leq p, q \leq \infty$ , the space  $B_{p,q}^s(\mathbb{R}^n)$  is a Banach space, and  $B_{p,q}^s \subseteq \mathcal{S}'(\mathbb{R}^n)$  continuously.*

*Proof.* **Step 1.** *It is a normed space, and embeds in  $\mathcal{S}'$ .*

That  $\|\cdot\|_{B_{p,q}^s}$  is a seminorm is the triangle inequality in  $L^p$  and then in  $\ell^q$ . If the norm vanishes then every  $\Delta_j f = 0$ , so  $\widehat{f}$  vanishes on each annulus and hence on  $\mathbb{R}^n \setminus \{0\}$ ; since  $\phi_0 \widehat{f} = 0$  as well,  $\widehat{f} = 0$  and  $f = 0$ .

For the embedding, let  $\varphi \in \mathcal{S}$ . Then  $|\langle f, \varphi \rangle| \leq \sum_j |\langle \Delta_j f, \varphi \rangle| \leq \sum_j \|\Delta_j f\|_{L^p} \|\Delta_j' \varphi\|_{L^{p'}}$ , where  $\Delta_j' = \sum_{|k-j| \leq 1} \Delta_k$  satisfies  $\Delta_j' \Delta_j = \Delta_j$ . Since  $\varphi \in \mathcal{S}$ ,  $\|\Delta_j' \varphi\|_{L^{p'}} \lesssim_N 2^{-jN}$  for every  $N$  — differentiate the symbol and use Lemma 4.6.3 — so choosing  $N > |s| + 1$  and applying Hölder in  $j$  bounds the sum by  $C_{N,\varphi} \|f\|_{B_{p,q}^s}$ .

**Step 2.** *Absolutely convergent series converge.*

By Theorem 1.8.6 (Riesz–Fischer) it suffices to show that if  $\sum_m \|f_m\|_{B_{p,q}^s} < \infty$  then  $\sum_m f_m$  converges in  $B_{p,q}^s$ . For each fixed  $j$ ,

$$\sum_m \|\Delta_j f_m\|_{L^p} \leq 2^{-js} \sum_m \|f_m\|_{B_{p,q}^s} < \infty,$$

so by completeness of  $L^p$  the series  $\sum_m \Delta_j f_m$  converges in  $L^p$  to some  $g_j$ , whose Fourier transform is supported in the  $j$ -th annulus because each term's is and the support condition is closed under  $L^p$  limits (test against Schwartz functions vanishing there).

By Step 1 the partial sums also converge in  $\mathcal{S}'$ ; call the limit  $f$ . Then  $\Delta_j f = g_j$ , since  $\Delta_j$  is continuous on  $\mathcal{S}'$ . Finally, for every  $M$ ,

$$\left( \sum_{j \leq M} (2^{js} \|\Delta_j f\|_{L^p})^q \right)^{1/q} \leq \sum_m \|f_m\|_{B_{p,q}^s},$$

by the triangle inequality applied to the finitely many  $j$ ; letting  $M \rightarrow \infty$  shows  $f \in B_{p,q}^s$ , and the same estimate applied to the tails  $\sum_{m > M_0} f_m$  shows the series converges in norm.  $\square$

### 4.6.5 Where the Sobolev spaces sit

**Theorem 4.6.8.** *Let  $1 < p < \infty$  and  $s \geq 0$ . Then  $F_{p,2}^s(\mathbb{R}^n) = W^{s,p}(\mathbb{R}^n)$  with equivalent norms, where  $W^{s,p} = \{f : (I - \Delta)^{s/2} f \in L^p\}$  is the Bessel potential space. In particular  $F_{p,2}^0 = L^p$  for  $1 < p < \infty$ .*

*Proof.* The case  $s = 0$  is exactly the Littlewood–Paley theorem (Theorem 3.6.3 (the square function theorem)), transferred from the homogeneous decomposition used there to the inhomogeneous one of Definition 4.6.2: the two differ only in that the frequencies  $|\xi| \lesssim 1$  are collected into the single piece  $\Delta_0$  rather than spread over  $j \leq 0$ , and Lemma 4.6.3 shows that collecting them changes the norm by at most a constant factor.

For general  $s \geq 0$ , apply Worked Example 3.6.1 (Bernstein and Littlewood–Paley), which states

$$\left\| \left( \sum_j 2^{2js} |\Delta_j f|^2 \right)^{1/2} \right\|_{L^p} \approx_{n,p,s} \|(-\Delta)^{s/2} f\|_{L^p},$$

the point being that on the support of  $\widehat{\Delta_j f}$  the multiplier  $(2\pi|\xi|)^s$  is comparable to  $2^{js}$ , with all derivatives satisfying the Hörmander–Mikhlin bounds after rescaling. Replacing  $(-\Delta)^{s/2}$  by  $(I - \Delta)^{s/2}$  changes the multiplier only on  $|\xi| \lesssim 1$ , where both are bounded above and below, so the norms remain comparable.  $\square$

**Corollary 4.6.9.** *For  $s \in \mathbb{R}$ ,  $B_{2,2}^s = F_{2,2}^s = H^s(\mathbb{R}^n)$  with equivalent norms.*

*Proof.*  $B_{2,2}^s = F_{2,2}^s$  by Theorem 4.6.6 with  $p = q = 2$ . By Plancherel and the disjointness of the annuli (Lemma 4.6.3),

$$\|f\|_{B_{2,2}^s}^2 = \sum_j 2^{2js} \|\widehat{\Delta_j f}\|_{L^2}^2 \approx \int_{\mathbb{R}^n} (1 + |\xi|^2)^s |\widehat{f}(\xi)|^2 d\xi,$$

because  $(1 + |\xi|^2)^s \approx 2^{2js}$  on the support of  $\widehat{\Delta_j f}$  and each  $\xi$  lies in at most two supports.  $\square$

WORKED EXAMPLE 4.6.2— TIER C: RESEARCH TASTE **Why the later volumes use  $\ell^q$  outside the norm**

Worked Example 3.6.2 (what Littlewood–Paley cannot do) observed that the  $\ell^2$  decoupling inequality is stated as

$$\left\| \sum_{\theta} f_{\theta} \right\|_{L^p(\mathbb{R}^n)} \lesssim_{\varepsilon} \delta^{-\varepsilon} \left( \sum_{\theta} \|f_{\theta}\|_{L^p}^2 \right)^{1/2}, \quad (4.6.2)$$

with the  $\ell^2$  sum *outside* the  $L^p$  norm, and not in the square function form. This chapter supplies the vocabulary to say precisely what that difference is, and why the weaker statement is the one that is true.

(a) *The two shapes are Besov and Triebel–Lizorkin.* The right-hand side of (4.6.2) is an  $\ell^2(L^p)$  norm of the family  $\{f_{\theta}\}$  — Besov-shaped. The square function alternative  $\|(\sum_{\theta} |f_{\theta}|^2)^{1/2}\|_{L^p}$  is  $L^p(\ell^2)$  — Triebel–Lizorkin-shaped. By Theorem 4.6.6, for  $p \geq 2$ ,

$$\left( \sum_{\theta} \|f_{\theta}\|_{L^p}^2 \right)^{1/2} \leq \left\| \left( \sum_{\theta} |f_{\theta}|^2 \right)^{1/2} \right\|_{L^p},$$

so the Besov-shaped right-hand side is *smaller*, and (4.6.2) is correspondingly the *stronger* statement of the two. It is nevertheless the one that is true, while the square function form — which would be an assertion that the pieces are almost orthogonal pointwise — is false.

(b) *Why the square function form fails.* Theorem 3.6.3 (the square function theorem) proved the square function identity  $\|f\|_p \approx \|\mathcal{S}f\|_p$  for *lacunary* frequency pieces, where the annuli are separated by a factor 2 and each randomised sum is a Hörmander–Mikhlin multiplier. Decoupling caps are adjacent:  $\delta^{-(n-1)/2}$  of them tile a neighbourhood of the paraboloid, and consecutive caps are at distance  $\delta$ , not at ratio 2. No multiplier theorem applies, and the pointwise orthogonality that the square function asserts is simply absent.

(c) *What survives, and why it is enough.* What curvature does supply is that the tubes dual to different caps point in different directions, so the pieces cannot all be large on a common set. That is an assertion about the *sizes*  $\|f_\theta\|_{L^p}$  and how they can add up, not about the pointwise values — which is exactly an  $\ell^2(L^p)$  statement. So the Besov shape is not a weakening forced by technical difficulty; it is the shape that the geometry actually produces.

*The transferable lesson.* When reading an estimate in this subject, look first at whether the  $\ell^q$  sum is inside or outside the  $L^p$  norm. Inside means a pointwise orthogonality claim, which needs a multiplier theorem and hence frequency separation. Outside means a claim about how the sizes of the pieces combine, which is what geometric input gives. Theorem 4.6.6 tells you which is stronger, and for  $p \geq 2$  it is always the one with the sum outside.

#### WORKED EXAMPLE 4.6.3— TIER B **Choosing between Besov and Triebel–Lizorkin**

The two scales  $B_{p,q}^s$  and  $F_{p,q}^s$  differ by the order in which the  $\ell^q$  and  $L^p$  norms are taken. Which to use is decided by the problem, and the rule is short.

*The definitions, side by side.*

$$\|f\|_{B_{p,q}^s} = \left\| 2^{js} \|\Delta_j f\|_{L^p} \right\|_{\ell_j^q}, \quad \|f\|_{F_{p,q}^s} = \left\| \|2^{js} \Delta_j f\|_{\ell_j^q} \right\|_{L^p}.$$

Besov integrates each block separately then sums; Triebel–Lizorkin sums pointwise then integrates.

*Use Besov when the blocks are handled independently.* Interpolation, trace theorems, and embeddings are all blockwise, so Besov is natural. The embedding  $B_{p,1}^{n/p} \subseteq L^\infty$  is the model, and it is sharp precisely because  $q = 1$  is what makes a sum of blocks converge uniformly.

*Use Triebel–Lizorkin when the blocks interact pointwise.* Sobolev spaces are  $F_{p,2}^s$ , and  $H^1 = F_{1,2}^0$ ,  $\text{BMO} = F_{\infty,2}^0$  — so the whole Calderón–Zygmund theory of Volume 3 lives on this scale. The square function of Theorem 3.6.3 is exactly the inner  $\ell^2$  norm.

*The one relation to remember.*  $B_{p,\min(p,q)}^s \subseteq F_{p,q}^s \subseteq B_{p,\max(p,q)}^s$ , by Minkowski’s inequality in whichever direction the exponents allow. The two scales agree when  $p = q$ , and only then.

*A caution.* Results stated on one scale do not automatically transfer. Remark 4.6.1 records that the identifications of  $F$  spaces with  $H^1$  and BMO are not proved in this series, so an argument here that needs them must say so.

## Drills

**Drill 4.6.1.** Verify the telescoping identity (4.6.1), and check that  $\text{supp } \psi \subseteq \{\frac{1}{2} \leq |\xi| \leq 2\}$  for  $\psi = \phi_0 - \phi_0(2\cdot)$ .

**Drill 4.6.2.** Show  $B_{p,q_1}^{s_1} \subseteq B_{p,q_2}^{s_2}$  whenever  $s_1 > s_2$ , for any  $q_1, q_2$ , and whenever  $s_1 = s_2$  and  $q_1 \leq q_2$ . Which inclusion needs a summation and which is free?

**Drill 4.6.3.** Prove that the elementary inequality  $(\sum_j a_j)^r \leq \sum_j a_j^r$  used in Step 2 of Theorem 4.6.6 holds for  $0 < r \leq 1$  and  $a_j \geq 0$ , and show it reverses for  $r \geq 1$ .

**Drill 4.6.4.** Compute  $\|f\|_{B_{p,q}^s}$  for  $f = g(2^{j_0}\cdot)$  with  $\widehat{g}$  supported in the annulus, and confirm the scaling  $2^{j_0(s-n/p)}$ . Deduce that  $B_{p,q}^s$  is invariant under the dilations preserving that exponent.

## Notes

Besov spaces were introduced by Besov in 1959 and the Triebel–Lizorkin scale by Triebel and Lizorkin independently around 1970. The standard reference is Triebel’s *Theory of Function Spaces*, where the full theory — including the identifications  $F_{1,2}^0 = H^1$  and  $F_{\infty,2}^0 = \text{BMO}$  omitted here by Remark 4.6.1 — is developed.

The Peetre maximal function, mentioned in the Pitfall of §6.3, is  $\sup_y |\Delta_j f(x-y)|(1+2^j|y|)^{-a}$ ; it is the standard device for converting statements about frequency-localised pieces into statements about maximal functions, and it is the reason the Triebel–Lizorkin theory needs the Fefferman–Stein vector-valued inequality in its full range.



## Chapter 4.7

# Embeddings and the Sobolev Scale

### 4.7.1 Bernstein, read as an embedding

Volume 2 proved Bernstein's inequality: a function whose Fourier transform is supported in a ball of radius  $N$  satisfies  $\|f\|_{L^r} \lesssim N^{n(1/p-1/r)} \|f\|_{L^p}$  for  $p \leq r$ . Every Littlewood–Paley piece has exactly that property, so Bernstein applies to each piece separately — and because the Besov norm takes the  $L^p$  norm of each piece *before* summing, the inequality passes straight through.

**Theorem 4.7.1** (Besov embedding). *Let  $1 \leq p \leq r \leq \infty$ ,  $1 \leq q \leq \infty$  and  $s \in \mathbb{R}$ . Then*

$$B_{p,q}^s(\mathbb{R}^n) \subseteq B_{r,q}^{s-n(\frac{1}{p}-\frac{1}{r})}(\mathbb{R}^n),$$

with  $\|f\|_{B_{r,q}^{s-n(1/p-1/r)}} \lesssim_{n,p,r} \|f\|_{B_{p,q}^s}$ .

*Proof.* Write  $\sigma = s - n(\frac{1}{p} - \frac{1}{r})$ . For  $j \geq 1$  the function  $\Delta_j f$  has  $\widehat{\Delta_j f}$  supported in  $\{|\xi| \leq 2^{j+1}\}$ , and for  $j = 0$  in  $\{|\xi| \leq 2\}$ ; so in all cases Theorem 2.6.7 (Bernstein) gives

$$\|\Delta_j f\|_{L^r} \lesssim_{n,p,r} 2^{jn(\frac{1}{p}-\frac{1}{r})} \|\Delta_j f\|_{L^p}.$$

Multiplying by  $2^{j\sigma}$ ,

$$2^{j\sigma} \|\Delta_j f\|_{L^r} \lesssim 2^{j\sigma} 2^{jn(\frac{1}{p}-\frac{1}{r})} \|\Delta_j f\|_{L^p} = 2^{js} \|\Delta_j f\|_{L^p}.$$

Taking the  $\ell^q$  norm over  $j \geq 0$  gives the claim.  $\square$

**Where we are.** Note what did *not* happen: the index  $q$  was untouched. Bernstein acts on one piece at a time, and the Besov norm never mixes pieces, so the embedding is free of any loss in  $q$ . This is the structural advantage of the Besov scale, and the reason the sharp endpoint of §7.3 is visible there and not on the Sobolev scale.

By contrast, the same argument applied to the Triebel–Lizorkin norm would require comparing  $\|(\sum_j |\Delta_j f|^q)^{1/q}\|_{L^r}$  with the corresponding quantity in  $L^p$ , which mixes the pieces and is not a consequence of Bernstein.

**Proposition 4.7.2** (Monotonicity in  $q$  and in  $s$ ). *Let  $1 \leq p \leq \infty$ .*

(i) *If  $q_1 \leq q_2$  then  $B_{p,q_1}^s \subseteq B_{p,q_2}^s$ , with norm at most 1.*

(ii) *If  $s_1 > s_2$  then  $B_{p,q_1}^{s_1} \subseteq B_{p,q_2}^{s_2}$  for all  $q_1, q_2$ , with  $\|f\|_{B_{p,q_2}^{s_2}} \lesssim_{s_1-s_2, q_2} \|f\|_{B_{p,q_1}^{s_1}}$ .*

*Proof.* (i) is the inclusion  $\ell^{q_1} \subseteq \ell^{q_2}$ , whose norm is 1.

(ii) Write  $a_j = \|\Delta_j f\|_{L^p}$  and  $\delta = s_1 - s_2 > 0$ . For every  $j$ ,

$$2^{js_2} a_j = 2^{-j\delta} \cdot 2^{js_1} a_j \leq 2^{-j\delta} \|f\|_{B_{p,\infty}^{s_1}}.$$

Taking the  $\ell^{q_2}$  norm over  $j \geq 0$ ,

$$\|f\|_{B_{p,q_2}^{s_2}} \leq \|f\|_{B_{p,\infty}^{s_1}} \left( \sum_{j \geq 0} 2^{-jq_2\delta} \right)^{1/q_2} = \frac{\|f\|_{B_{p,\infty}^{s_1}}}{(1 - 2^{-q_2\delta})^{1/q_2}},$$

the geometric series converging because  $\delta > 0$  and the sum runs over  $j \geq 0$  only. Finally  $\|f\|_{B_{p,\infty}^{s_1}} \leq \|f\|_{B_{p,q_1}^{s_1}}$  by (i).  $\square$

**Pitfall:** (ii) uses that the decomposition is inhomogeneous

The sum in Proposition 4.7.2(ii) runs over  $j \geq 0$ , and the geometric series converges only in that direction. Had we used the homogeneous decomposition, with  $j$  ranging over all of  $\mathbb{Z}$ , the low frequencies  $j \rightarrow -\infty$  would contribute  $2^{-j\delta} \rightarrow \infty$  and no such inclusion would hold.

This is the general reason the inhomogeneous decomposition of Definition 4.6.2 is the right one for a theory of embeddings: it treats the low frequencies as a single bounded block, so that only the behaviour at high frequency — which is what smoothness means — can cause trouble. The homogeneous scale is the natural one for scaling-invariant statements, and the two should not be confused.



## 4.7.2 The sharp endpoint into $L^\infty$

**Theorem 4.7.3.** *Let  $1 \leq p < \infty$ . Then*

$$B_{p,1}^{n/p}(\mathbb{R}^n) \subseteq C_0(\mathbb{R}^n) \cap L^\infty(\mathbb{R}^n), \quad \|f\|_{L^\infty} \lesssim_{n,p} \|f\|_{B_{p,1}^{n/p}}.$$

*Idea of the proof.* Bernstein at  $r = \infty$  says  $\|\Delta_j f\|_\infty \lesssim 2^{jn/p} \|\Delta_j f\|_p$ . Summing over  $j$  and using  $f = \sum_j \Delta_j f$  gives the result — provided the sum  $\sum_j 2^{jn/p} \|\Delta_j f\|_p$  converges, which is exactly the statement  $f \in B_{p,1}^{n/p}$ . So the theorem is Bernstein plus the triangle inequality, and the index  $q = 1$  is present precisely because a triangle inequality over  $j$  is being used.

*Proof.* By Theorem 4.7.1 with  $r = \infty$  and  $s = n/p$ , the target smoothness is  $s - n/p = 0$ , so

$$\|f\|_{B_{\infty,1}^0} \lesssim_{n,p} \|f\|_{B_{p,1}^{n/p}}.$$

It therefore suffices to show  $B_{\infty,1}^0 \subseteq L^\infty$  with  $\|f\|_\infty \leq \|f\|_{B_{\infty,1}^0}$ , and that is immediate:

$$\|f\|_{L^\infty} = \left\| \sum_{j \geq 0} \Delta_j f \right\|_{L^\infty} \leq \sum_{j \geq 0} \|\Delta_j f\|_{L^\infty} = \|f\|_{B_{\infty,1}^0}.$$

For the continuity statement, each  $\Delta_j f$  has compactly supported Fourier transform, hence extends to a smooth function; and  $\Delta_j f \in L^p$  with  $p < \infty$  together with the bound on its derivatives from Theorem 2.6.7 (Bernstein) forces  $\Delta_j f(x) \rightarrow 0$  as  $|x| \rightarrow \infty$ . The series  $\sum_j \Delta_j f$  converges uniformly, by the displayed bound, so its sum lies in  $C_0$ .  $\square$

## 4.7.3 A loose end from Volume 2, resolved

Volume 2, Chapter 7 proved that  $H^s(\mathbb{R}^n) \subseteq C_0$  for  $s > n/2$ , and its Pitfall box showed that the endpoint  $s = n/2$  fails: there is  $f \in H^{n/2}$  which is unbounded, and the failure is logarithmic. The reason was not explained there, because the vocabulary did not exist. It does now.

**Theorem 4.7.4.** *Let  $1 < p < \infty$ . Then*

$$W^{n/p,p}(\mathbb{R}^n) \subseteq B_{p,\max(p,2)}^{n/p}(\mathbb{R}^n),$$

*and this is sharp in the third index:  $W^{n/p,p} \not\subseteq L^\infty$ , whereas  $B_{p,1}^{n/p} \subseteq L^\infty$  by Theorem 4.7.3.*

*Proof.* By Theorem 4.6.8,  $W^{n/p,p} = F_{p,2}^{n/p}$ , and by the Minkowski embeddings (Theorem 4.6.6)  $F_{p,2}^s \subseteq B_{p,\max(p,2)}^s$ . The failure  $W^{n/p,p} \not\subseteq L^\infty$  is Volume 2, Chapter 7, Pitfall, for  $p = 2$ ; the same construction with  $\widehat{f}(\xi) = |\xi|^{-n}(\log|\xi|)^{-1}\mathbf{1}_{|\xi|>2}$  works for general  $p$  after adjusting exponents.  $\square$

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**WORKED EXAMPLE 4.7.1— TIER B The failure, seen through the third index**

Take  $p = 2$  and let  $f$  be the Volume 2 counterexample,  $\widehat{f}(\xi) = |\xi|^{-n}(\log|\xi|)^{-1}\mathbf{1}_{|\xi|>2}$ . We compute both Besov norms at smoothness  $n/2$  and see exactly where the difference lies.

On the annulus  $|\xi| \approx 2^j$  we have  $|\widehat{f}| \approx 2^{-jn}j^{-1}$  over a set of measure  $\approx 2^{jn}$ , so by Plancherel

$$\|\Delta_j f\|_{L^2} \approx \left(2^{jn} \cdot 2^{-2jn} j^{-2}\right)^{1/2} = \frac{2^{-jn/2}}{j},$$

and therefore

$$2^{jn/2} \|\Delta_j f\|_{L^2} \approx \frac{1}{j}.$$

Now everything is decided by which  $\ell^q$  norm is taken of the sequence  $(1/j)_{j \geq 2}$ :

$$\|(1/j)_j\|_{\ell^2} = \left(\sum_j j^{-2}\right)^{1/2} < \infty, \quad \|(1/j)_j\|_{\ell^1} = \sum_j \frac{1}{j} = \infty.$$

So  $f \in B_{2,2}^{n/2} = H^{n/2}$  but  $f \notin B_{2,1}^{n/2}$ , and Theorem 4.7.3 does not apply — correctly, since  $f$  is unbounded.

*The point.* The divergence of  $\sum 1/j$  is exactly the “logarithmic failure” that Volume 2 could only observe. Here it is located precisely: the contributions of the dyadic frequency blocks are square-summable but not summable, and boundedness needs summability because it is proved by a triangle inequality over blocks. The third index is the bookkeeping device that distinguishes the two, and the Sobolev scale, which fixes  $q = 2$ , cannot see it.

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*Remark 4.7.5* (The Sobolev embedding itself). One may also run Theorem 4.7.1 to recover Sobolev-type embeddings between  $L^p$  spaces. Chaining

$$W^{s,p} = F_{p,2}^s \subseteq B_{p,\max(p,2)}^s \subseteq B_{r,\max(p,2)}^0 \subseteq F_{r,2}^0 = L^r, \quad s = n\left(\frac{1}{p} - \frac{1}{r}\right),$$

where the middle inclusion is Theorem 4.7.1 and the last requires  $\max(p, 2) \leq \min(r, 2)$  by Theorem 4.6.6, gives  $W^{s,p} \subseteq L^r$  whenever  $p \leq 2 \leq r$ .

The restriction  $p \leq 2 \leq r$  is an artefact of this soft route: the chain loses in the third index at both ends. The full Sobolev embedding, for all

$1 < p < r < \infty$ , was already proved in Theorem 2.7.6 (Gagliardo–Nirenberg–Sobolev) by the Gagliardo–Nirenberg argument, which uses no Fourier analysis at all. So nothing is missing; the Besov route simply is not the efficient one for that particular statement. What it *does* give, and the other route does not, is Theorem 4.7.3.

## 4.7.4 Interpolation on the Besov scale

**Theorem 4.7.6** (Gagliardo–Nirenberg on the Besov scale). *Let  $1 \leq p \leq \infty$ , let  $s_0 \neq s_1$  and  $1 \leq q_0, q_1 \leq \infty$ , let  $\theta \in (0, 1)$ , and set*

$$s = (1 - \theta)s_0 + \theta s_1, \quad \frac{1}{q} = \frac{1 - \theta}{q_0} + \frac{\theta}{q_1}.$$

*Then*

$$\|f\|_{B_{p,q}^s} \leq \|f\|_{B_{p,q_0}^{s_0}}^{1-\theta} \|f\|_{B_{p,q_1}^{s_1}}^{\theta}.$$

*Proof.* Write  $a_j = \|\Delta_j f\|_{L^p}$ . Since  $s = (1 - \theta)s_0 + \theta s_1$ ,

$$2^{js} a_j = (2^{js_0} a_j)^{1-\theta} (2^{js_1} a_j)^{\theta}$$

for every  $j$  — this is just splitting the exponent. Now apply Hölder’s inequality for sums (Corollary 1.8.3 (generalised Hölder), in the generalised form) with exponents  $\frac{q_0}{(1-\theta)q}$  and  $\frac{q_1}{\theta q}$ , which are conjugate by the definition of  $q$ :

$$\sum_j (2^{js} a_j)^q = \sum_j (2^{js_0} a_j)^{(1-\theta)q} (2^{js_1} a_j)^{\theta q} \leq \left( \sum_j (2^{js_0} a_j)^{q_0} \right)^{\frac{(1-\theta)q}{q_0}} \left( \sum_j (2^{js_1} a_j)^{q_1} \right)^{\frac{\theta q}{q_1}}.$$

Raising to the power  $1/q$  gives the claim. The cases  $q_0 = \infty$  or  $q_1 = \infty$  follow by the same computation with the corresponding factor bounded by its supremum.  $\square$

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**WORKED EXAMPLE 4.7.2— TIER A Interpolation, and why  $s_0 \neq s_1$  is not needed in the proof**

The proof of Theorem 4.7.6 nowhere used  $s_0 \neq s_1$ : the identity  $2^{js} a_j = (2^{js_0} a_j)^{1-\theta} (2^{js_1} a_j)^{\theta}$  holds regardless. When  $s_0 = s_1 = s$  the conclusion reads  $\|f\|_{B_{p,q}^s} \leq \|f\|_{B_{p,q_0}^s}^{1-\theta} \|f\|_{B_{p,q_1}^s}^{\theta}$ , which is the log-convexity of the  $\ell^q$  scale — true, but weaker than Proposition 4.7.2(i) in most cases.

The hypothesis  $s_0 \neq s_1$  is included in the statement only because that is the case one uses: interpolating between two different smoothnesses. A typical

application is

$$\|f\|_{B_{p,1}^s} \leq \|f\|_{B_{p,\infty}^{s_0}}^{1-\theta} \|f\|_{B_{p,\infty}^{s_1}}^\theta, \quad s_0 < s < s_1, \quad \theta = \frac{s - s_0}{s_1 - s_0},$$

which reaches the summable index  $q = 1$  — the one Theorem 4.7.3 needs — from two weak  $q = \infty$  hypotheses. Note that  $\frac{1}{q} = (1 - \theta)/\infty + \theta/\infty = 0$  would give  $q = \infty$ , so this particular application requires  $q_0 = q_1 = \infty$  to be replaced by finite values; Drill 7.4 asks for the correct form.

**WORKED EXAMPLE 4.7.3— TIER C: RESEARCH TASTE** **Where the third index is doing work in the later volumes**

The distinction between  $q = 1$ ,  $q = 2$  and  $q = \infty$  at a fixed smoothness looks like a technicality. It is not, and three appearances in the later volumes are worth naming.

(a) *Strichartz estimates.* A Strichartz estimate bounds  $\|e^{it\Delta}f\|_{L_t^q L_x^r}$  by  $\|f\|_{H^s}$  for exponents on a scaling line. At the *endpoint* of the admissible range the estimate is either false or requires the Besov refinement — one must replace  $H^s$  by  $B_{2,1}^s$  or the target by a Besov space — for exactly the reason of Worked Example 4.7.1: at the endpoint the dyadic blocks contribute equally, and whether the total converges depends on whether one is summing or square-summing them. Keel and Tao’s endpoint Strichartz theorem is the sharp statement.

(b) *Local smoothing.* The local smoothing conjecture for the wave equation asserts a gain of  $1/p$  derivatives in  $L^p$  over the fixed-time estimate, and was proved for  $n = 2$  by Guth, Wang and Zhang in 2020 using decoupling. “A gain of  $1/p$  derivatives” is an embedding statement between spaces on this scale, and the sharp form of the conjecture specifies the third index.

(c) *Decoupling itself.* Worked Example 3.6.2 (what Littlewood–Paley cannot do) and Worked Example 4.6.2 identified the  $\ell^2$  decoupling inequality as a Besov-shaped statement:  $\ell^2$  of  $L^p$  norms, not  $L^p$  of an  $\ell^2$  square function. The  $\delta^{-\varepsilon}$  loss in that inequality plays the same role as the third index here — it is the amount by which one is allowed to fail to sum over  $\log(1/\delta)$  scales. That last comparison is worth dwelling on. In Worked Example 4.7.1 the obstruction was  $\sum_j 1/j = \infty$  over  $\approx \log$  many blocks; in decoupling the obstruction is a factor that must be beaten over  $\log(1/\delta)$  scales. Both are the statement that an estimate which is lossless at each scale may still fail when the scales are summed, and both are repaired by weakening the index rather than the estimate:  $q = 2$  instead of  $q = 1$  here,  $\delta^{-\varepsilon}$  instead of  $O(1)$  there.

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**WORKED EXAMPLE 4.7.4— TIER B Reading an embedding**

Sobolev and Besov embeddings are quoted constantly and their hypotheses vary. Four checks.

(1) *Which scale?*  $W^{s,p}$ ,  $B_{p,q}^s$  or  $F_{p,q}^s$ . Volume 4, Worked Example 4.6.3 explains that the last two differ by the order of the  $\ell^q$  and  $L^p$  norms, and embeddings on one scale do not transfer verbatim to the other.

(2) *Is the embedding critical?* At the critical exponent the target is  $L^\infty$  or BMO rather than a Hölder space, and the statement usually fails without an extra summability hypothesis —  $q = 1$  in the Besov case.

(3) *Which domain?*  $\mathbb{R}^n$ , a bounded domain, or a torus. On  $\mathbb{R}^n$  homogeneity forces a single exponent relation; on a bounded domain lower-order terms may be absorbed and the statement is weaker but easier.

(4) *Is it compact?* Rellich–Kondrachov gives compactness on bounded domains and fails on  $\mathbb{R}^n$ , where translation provides a non-convergent bounded sequence.

*The scaling check.* As always, test on  $f(\lambda \cdot)$ : the exponent relation in any homogeneous embedding is forced, and computing it takes a line. An embedding whose exponents fail the scaling check is misquoted.

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## Drills

**Drill 4.7.1.** Verify Theorem 4.7.1 directly for  $f = g(2^{j_0} \cdot)$  with  $\widehat{g}$  supported in the annulus, by computing both sides.

**Drill 4.7.2.** Show that Proposition 4.7.2(ii) fails for the homogeneous decomposition, by exhibiting a sequence  $(a_j)_{j \in \mathbb{Z}}$  finite in one norm and infinite in the other.

**Drill 4.7.3.** Carry out the computation of Worked Example 4.7.1 for general  $p$ : find the correct counterexample  $\widehat{f}$  and confirm that  $(2^{jn/p} \|\Delta_j f\|_p)_j$  behaves like  $(1/j)_j$ .

**Drill 4.7.4.** State and prove the version of Theorem 4.7.6 that produces  $q = 1$  from finite  $q_0, q_1$ , and identify the constraint linking  $\theta$ ,  $q_0$  and  $q_1$ .

## Notes

Theorem 4.7.3 is standard and is the single most-quoted advantage of the Besov scale over the Sobolev scale. The failure of  $W^{n/p,p} \subseteq L^\infty$ , and its repair

by lowering the third index to 1, is often stated as the slogan that “ $B_{p,1}^{n/p}$  is the correct critical space”.

The endpoint Strichartz theorem referred to in Worked Example 4.7.3 is due to Keel and Tao (1998). The local smoothing conjecture is Sogge’s, from 1991, and the two-dimensional case is Guth–Wang–Zhang (2020); both are discussed in Volume 13.

## Chapter 4.8

# Where These Spaces Are Used

This volume has built machinery. This chapter says what it is for, by exhibiting the estimates of Volumes 12 and 13 in the language just developed. Nothing here is proved — the theorems belong to later volumes — but every *statement* is now one the reader can parse completely, and that is the point: a large part of the difficulty of entering this literature is not the arguments but the notation.

### 4.8.1 Four questions the reader can now answer

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WORKED EXAMPLE 4.8.1— TIER B **Why estimates are stated in mixed norms**  
 $L_t^q L_x^r$

A Strichartz estimate for the free Schrödinger evolution has the form

$$\|e^{it\Delta} f\|_{L_t^q L_x^r(\mathbb{R} \times \mathbb{R}^n)} \lesssim \|f\|_{L^2(\mathbb{R}^n)}, \quad \frac{2}{q} + \frac{n}{r} = \frac{n}{2}, \quad q > 2.$$

The left side is  $\| \|e^{it\Delta} f(\cdot)\|_{L_x^r} \|_{L_t^q}$ : an  $L^r$  norm in space taken first, an  $L^q$  norm in time second.

*Why the order matters.* This is the Besov-versus-Triebel–Lizorkin question of Theorem 4.6.6 with the roles of  $j$  and  $x$  played by  $t$  and  $x$ . By Minkowski’s integral inequality, if  $q \leq r$  then  $\|\cdot\|_{L_t^q L_x^r} \geq \|\cdot\|_{L_x^r L_t^q}$ , so the two orderings give genuinely different — and non-equivalent — statements, the harder one being whichever puts the larger exponent on the inside.

*Why the exponent relation.* Scaling. If  $f_\lambda(x) = f(\lambda x)$  then  $e^{it\Delta} f_\lambda(x) = (e^{i\lambda^2 t \Delta} f)(\lambda x)$ , so the left side scales by  $\lambda^{-2/q-n/r}$  and the right by  $\lambda^{-n/2}$ . Equality of the two is the stated relation — the same one-line test Worked Example 2.2.1 (Young’s convolution inequality) applied to Young’s inequality.

*Where the third index appears.* At the endpoint  $q = 2$  (possible for  $n \geq 3$ ) the estimate is true but genuinely harder, and for  $n = 2$  it is *false* as stated; the repair replaces  $L_t^2$  by a Besov space in  $t$ , for exactly the reason of Worked Example 4.7.1. This is the Keel–Tao theorem.

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WORKED EXAMPLE 4.8.2— TIER B **What “a gain of  $1/p$  derivatives” means**

The local smoothing conjecture asserts, for the wave equation in  $\mathbb{R}^n$ ,

$$\|e^{it\sqrt{-\Delta}}f\|_{L^p(\mathbb{R}^n \times [1,2])} \lesssim_\varepsilon \|f\|_{W^{s,p}(\mathbb{R}^n)}, \quad s > s_0(p) - \frac{1}{p},$$

where  $s_0(p)$  is the sharp exponent for the *fixed-time* estimate. The phrase “gain of  $1/p$  derivatives” means precisely that the required smoothness index  $s$  drops by  $1/p$  once one averages in  $t$ .

*Reading it.*  $W^{s,p} = F_{p,2}^s$  by Theorem 4.6.8, so the hypothesis is a statement about the size of  $(\sum_j 2^{2js} |\Delta_j f|^2)^{1/2}$  in  $L^p$ . The conclusion is a statement about the solution on a unit time interval. So the conjecture asserts an embedding between two spaces on the scale of Chapter 4.6, and the number  $1/p$  is the amount of smoothness that time-averaging buys.

*Why it is hard.* At each fixed time the estimate is sharp and cannot be improved; the gain comes entirely from the fact that the singularities move, so that a wave packet concentrated on a tube in space-time contributes to only a short interval of times. Controlling how many such tubes can overlap is a Kakeya problem, which is why Guth, Wang and Zhang’s 2020 proof for  $n = 2$  goes through decoupling.

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WORKED EXAMPLE 4.8.3— TIER B **Why decoupling is not a square function estimate**

This was answered in Worked Example 4.6.2, and is restated here in one line because it is the single most common source of confusion for readers entering the decoupling literature.

The inequality

$$\left\| \sum_{\theta} f_{\theta} \right\|_{L^p} \lesssim_{\varepsilon} \delta^{-\varepsilon} \left( \sum_{\theta} \|f_{\theta}\|_{L^p}^2 \right)^{1/2}$$

has the  $\ell^2$  sum *outside* the  $L^p$  norm: it is Besov-shaped. The square function version, with the sum inside, is Triebel–Lizorkin-shaped, is stronger for  $p \geq 2$  by Theorem 4.6.6, and is *false* — it would assert pointwise almost-orthogonality of the caps, which requires frequency separation that adjacent



caps do not have.

The diagnostic: *inside* the norm means a claim about the pointwise values of the pieces and needs a multiplier theorem; *outside* means a claim about how their sizes combine and is what geometric input supplies.

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**WORKED EXAMPLE 4.8.4— TIER C: RESEARCH TASTE    The norms in which wave packets are summed**

A wave packet decomposition writes a function as  $f = \sum_T f_T$ , one piece per tube  $T$  in a family  $\mathbb{T}$ , with  $f_T$  concentrated on  $T$  in space and on the dual cap in frequency. Every argument in Volumes 12 and 13 proceeds by estimating the pieces and reassembling. This example collects what this volume says about the reassembly.

(a) *The two reassembly norms.* As in Worked Example 4.6.2, one may aim for

$$\left\| \sum_T f_T \right\|_{L^p} \lesssim \left( \sum_T \|f_T\|_{L^p}^2 \right)^{1/2} \quad \text{or} \quad \left\| \sum_T f_T \right\|_{L^p} \lesssim \left\| \left( \sum_T |f_T|^2 \right)^{1/2} \right\|_{L^p}.$$

By Theorem 4.6.6 the first is stronger when  $p \geq 2$ . Both are sometimes available and one should know which is being claimed.

(b) *The trivial bound, and what must be beaten.* Cauchy–Schwarz over  $\#\mathbb{T} = N$  pieces gives the first inequality with an extra factor  $N^{1/2-1/p}$ . That factor is a power of  $\delta$ , and the entire content of a decoupling theorem is its removal down to  $\delta^{-\varepsilon}$ . Volume 1, Worked Example 8.16 made this comparison at the very start of the series; the reader can now see it as a statement about which function space the right-hand side lives in.

(c) *Where the maximal inequalities enter.* A wave packet is not literally supported on its tube: it has tails, decaying like  $(1 + \text{dist}(x, T)/\delta)^{-N}$ . Summing the tails is done by dominating them by a maximal function of the localised piece and applying a vector-valued maximal inequality — exactly Theorem 4.5.3, in the range where it is available. This is the concrete reason Chapter 4.5 exists, and the reason the restriction to  $q < p$  noted in Remark 4.5.4 is a genuine limitation rather than a cosmetic one: the literature uses the full Fefferman–Stein range.

(d) *Where the Whitney decomposition enters.* Theorem 4.4.3 cut an open set into cubes of size comparable to their distance from the boundary. Volume 11 performs the analogous cut with a polynomial: space is divided into cells on which the object under study is uniformly controlled, plus a neighbourhood of the zero set where it is not. The bookkeeping — finitely many pieces of each size, bounded overlap, sizes comparable to distances — is the same.

(e) *The honest summary.* Of the machinery built in this volume, what the later volumes actually use is: the  $H^1$ –BMO duality, as background for why endpoint estimates are stated the way they are; the Minkowski embeddings, to know which of two competing formulations is stronger; the vector-valued maximal inequality, to sum tails; and the Besov endpoint of Theorem 4.7.3, in the sharp forms of Strichartz and local smoothing. The atomic theory of Chapter 4.2 is used mainly for the intuition it supplies — that cancellation at every scale is what distinguishes  $H^1$  from  $L^1$  — and that intuition is the direct ancestor of the frequency localisation that makes a wave packet a wave packet.

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WORKED EXAMPLE 4.8.5— TIER C: RESEARCH TASTE **What Volume 4 was for**

Volume 3 ended with two endpoint failures: Calderón–Zygmund operators are unbounded on  $L^1$  and on  $L^\infty$ . Volume 4 supplies the substitutes and proves they work.

(a) *The two spaces.*  $H^1$  replaces  $L^1$  and BMO replaces  $L^\infty$ . They are dual (Theorem 4.3.5), Calderón–Zygmund operators map each to itself (Theorem 4.2.5, Theorem 3.7.4), and interpolation between them recovers all the  $L^p$  results.

(b) *Why the substitution is forced, not chosen.* By Worked Example 3.3.4(c) the Hilbert transform of an indicator is  $\log|\frac{x}{x-1}|$ : unbounded, but only logarithmically, and with bounded mean oscillation. So BMO is not an arbitrary weakening of  $L^\infty$  — it is exactly what the failure produces.

(c) *What it costs.* The atomic definition is decomposition-dependent, and the Pitfall of Chapter 4.2 shows that reasoning atom-by-atom is invalid without an additional hypothesis. The maximal characterisation which would repair this is not proved here (Remark 4.4.5, revision R1).

(d) *Where it is used later.* Volume 5’s oscillatory integral estimates are stated on  $L^p$  and need no endpoint theory; Volume 6 onwards works with  $L^2$  and  $L^\infty$  on a sphere. So Volume 4 is, in the architecture of this series, a completion of Volume 3 rather than a prerequisite for what follows.

(e) *Why it is here nonetheless.* Because the pattern it establishes — when an estimate fails at an endpoint, find the space on which it holds rather than abandoning the endpoint — recurs throughout. Volume 6’s Worked Example 6.3.3 and Volume 8’s contrast between positive measure and nonempty interior are the same move in different settings.

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## Notes

The Keel–Tao endpoint Strichartz theorem is from 1998; Sogge’s local smoothing conjecture from 1991, with the planar case settled by Guth, Wang and Zhang in 2020. Both are treated in Volume 13.

Readers who wish to see the function-space language used in earnest before reaching Volume 12 may consult Tao’s lecture notes on restriction theory, where the mixed-norm and Besov formulations of the estimates appear throughout.



## Chapter 4.A

# Results Quoted from Volumes 1–3

This volume assumes Volumes 1–3 and nothing else. Every citation is listed below with its use site.

**From Volume 1**

| <b>Result</b>                                 | <b>Vol. 1</b>      | <b>Used in</b>                                  |
|-----------------------------------------------|--------------------|-------------------------------------------------|
| Hardy–Littlewood maximal theorem              | Thm 7.4            | Proposition 4.1.2, Theorem 4.1.5, Theorem 4.5.3 |
| Vitali covering lemma                         | Lem 7.2            | Theorem 4.5.2 Step 1                            |
| Lebesgue differentiation                      | Thm 7.7            | Definition 4.4.1 discussion                     |
| Layer-cake formula                            | Wkd Ex 5.5         | Theorem 4.1.4, Theorem 4.1.5, Theorem 4.5.2     |
| Absolute continuity of the integral           | Prop 7.15          | Proposition 4.3.4 Step 2                        |
| Polar coordinates                             | Thm 6.10           | Theorem 4.4.2 Step 4                            |
| Tonelli; Fubini                               | Thms 6.6, 6.7      | Proposition 4.5.1, Theorem 4.5.3, Theorem 4.6.6 |
| Hölder; generalised Hölder                    | Thm 8.2, Cor 8.3   | Theorem 4.5.3 Step 3, Theorem 4.7.6             |
| Riesz–Fischer completeness                    | Thm 8.6            | Proposition 4.2.4, Theorem 4.6.7                |
| Density of $C_c^\infty$ , of simple functions | Thm 8.9            | Theorem 4.2.5 Step 1                            |
| $L^p$ duality; norms by testing               | Thm 8.11, Cor 8.12 | Theorem 4.3.5 Step 6, Theorem 4.5.3 Step 1      |
| Chebyshev; weak $L^p$                         | Prop 8.14          | Theorem 4.1.4 Step 1                            |
| Riesz representation in Hilbert space         | Thm 8.19           | Theorem 4.3.5 Step 2                            |

**From Volume 2**

| <b>Result</b>                                                     | <b>Vol. 2</b>       | <b>Used in</b>                      |
|-------------------------------------------------------------------|---------------------|-------------------------------------|
| Marcinkiewicz interpolation                                       | Thm 2.6             | Proposition 4.3.4 Step 1            |
| Young’s convolution inequality                                    | Wkd Ex 2.4          | Lemma 4.6.3                         |
| The Fourier dictionary (dilation)                                 | Prop 4.1            | Lemma 4.6.3                         |
| Plancherel                                                        | Thm 4.5             | Corollary 4.6.9                     |
| Schwartz space; $\mathcal{F}$ on $\mathcal{S}$ and $\mathcal{S}'$ | Thms 5.5, 5.9       | Definition 4.4.1, Theorem 4.6.7     |
| Bernstein’s inequalities                                          | Thm 6.8             | Theorem 4.7.1, Theorem 4.7.3        |
| Sobolev embedding, $L^2$ case; the endpoint failure               | Thm 7.3 and Pitfall | Theorem 4.7.4, Worked Example 4.7.1 |
| Gagliardo–Nirenberg–Sobolev                                       | Thm 7.7             | Remark 4.7.5                        |

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## From Volume 3

| Result                                             | Vol. 3        | Used in                                      |
|----------------------------------------------------|---------------|----------------------------------------------|
| Nesting of dyadic cubes                            | Lem 1.2       | Theorem 4.1.3 Step 3                         |
| Dyadic maximal inequality, constant 1              | Thm 1.6       | Theorem 4.1.3 Step 4, Proposition 4.3.4      |
| Calderón–Zygmund decomposition                     | Thm 1.7       | Proposition 4.3.4 Step 1                     |
| The one-third trick                                | Wkd Ex 1.4    | Proposition 4.1.2(ii)                        |
| Hörmander’s condition                              | Def 2.1       | Lemma 4.2.2, Theorem 4.2.5                   |
| Calderón–Zygmund theorem                           | Thm 2.3       | Theorem 4.2.5 Step 1, Proposition 4.1.6      |
| BMO; John–Nirenberg                                | §7.1, Thm 7.4 | Chapter 4.1 throughout, Theorem 4.3.2 Step 1 |
| CZ operators map $L^\infty \rightarrow \text{BMO}$ | Thm 7.6       | Theorem 4.1.5 applications, Drill 1.4        |
| Littlewood–Paley square function theorem           | Thm 6.4       | Theorem 4.6.8                                |
| Weighted Littlewood–Paley form                     | Wkd Ex 6.6    | Theorem 4.6.8                                |
| Marcinkiewicz–Zygmund $\ell^2$ extension           | Thm 6.1       | Chapter 4.5 §5.1, Worked Example 4.5.2       |

## Debts settled and incurred

Volume 3, Chapter 7 deferred two results to this volume. Both are proved: Calderón–Zygmund operators map  $H^1$  into  $L^1$  (Theorem 4.2.5), and  $(H^1)^* = \text{BMO}$  (Theorem 4.3.5).

This volume incurs no debts on later volumes. It does record two revisions of the series plan, in Remark 4.4.5 and Remark 4.5.4, together with a third made in advance in Remark 4.6.1; all three are listed in the master manifest under PLAN REVISIONS. In each case the removed statement is true, is attributed, and is used nowhere in this series.





## Chapter 4.B

# Notation and Index of Results

### Additions to the Volume 1–3 conventions

|                                    |                                                |
|------------------------------------|------------------------------------------------|
| $\mathcal{M}^\#, \mathcal{M}_d^\#$ | the sharp maximal function; its dyadic version |
| $H^1$                              | the Hardy space                                |
| $a$                                | an atom                                        |
| $Q^*$                              | the dilate $2\sqrt{n}Q$                        |
| $B_{p,q}^s, F_{p,q}^s$             | Besov and Triebel–Lizorkin spaces              |

## Index of the main results

|                                                       |                   |
|-------------------------------------------------------|-------------------|
| Elementary properties of $\mathcal{M}^\#$             | Proposition 4.1.2 |
| Good- $\lambda$ inequality                            | Theorem 4.1.3     |
| Fefferman–Stein inequality                            | Theorem 4.1.4     |
| Interpolation with BMO                                | Theorem 4.1.5     |
| The atom estimate                                     | Lemma 4.2.2       |
| Basic properties of $H^1$                             | Proposition 4.2.4 |
| CZ operators map $H^1 \rightarrow L^1$                | Theorem 4.2.5     |
| $g \in \text{BMO}$ gives a functional on $H^1$        | Theorem 4.3.2     |
| $L^2$ -atoms lie in $H^1$                             | Proposition 4.3.4 |
| Fefferman: $(H^1)^* = \text{BMO}$                     | Theorem 4.3.5     |
| $f \in H^1 \Rightarrow \mathcal{M}_\varphi f \in L^1$ | Theorem 4.4.2     |
| Whitney decomposition                                 | Theorem 4.4.3     |
| Weighted maximal inequality                           | Theorem 4.5.2     |
| Vector-valued maximal inequality, $q < p$             | Theorem 4.5.3     |
| Independence of the decomposition (Besov)             | Proposition 4.6.5 |
| Minkowski embeddings $B \leftrightarrow F$            | Theorem 4.6.6     |
| Completeness of $B_{p,q}^s$                           | Theorem 4.6.7     |
| $F_{p,2}^s = W^{s,p}$                                 | Theorem 4.6.8     |
| $B_{2,2}^s = H^s$                                     | Corollary 4.6.9   |
| Besov embedding (Bernstein)                           | Theorem 4.7.1     |
| Monotonicity in $q$ and $s$                           | Proposition 4.7.2 |
| Sharp endpoint $B_{p,1}^{n/p} \subseteq L^\infty$     | Theorem 4.7.3     |
| The $W^{n/p,p}$ endpoint failure explained            | Theorem 4.7.4     |
| Gagliardo–Nirenberg on the Besov scale                | Theorem 4.7.6     |

## Plan revisions in this volume

| Removed                                                                                            | Reason       |
|----------------------------------------------------------------------------------------------------|--------------|
| Converse of the maximal characterisation                                                           | Remark 4.4.5 |
| Vector-valued maximal for $1 < p < q$                                                              | Remark 4.5.4 |
| $F_{1,2}^0 = H^1$ , $F_{\infty,2}^0 = \text{BMO}$ ; $F$ independence in full range; trace theorems | Remark 4.6.1 |

# Part 5

## Volume 5

Oscillatory Integrals and Stationary Phase



# Preface to Volume 5

## The one estimate this volume is about

Volume 2 computed the Fourier transform of surface measure on the sphere exactly, in terms of Bessel functions, and proved

$$|\widehat{d\sigma}(\xi)| \lesssim_n |\xi|^{-\frac{n-1}{2}}.$$

It did so by a computation special to the sphere: the integral reduced to one dimension, and the decay was extracted by splitting at  $s = \rho^{-1}$  and integrating by parts. That argument is honest but unilluminating. It does not explain where the exponent  $(n-1)/2$  comes from, it gives no information about the *phase* of  $\widehat{d\sigma}$ , and it says nothing about any surface other than a sphere.

This volume supplies the general theory. The exponent will turn out to be

$$\frac{\#\{\text{non-vanishing principal curvatures}\}}{2},$$

and the mechanism will be visible: an oscillatory integral is negligible wherever the phase is moving, so all the mass comes from the *critical points* of the phase, and each non-degenerate direction at a critical point contributes a factor  $\lambda^{-1/2}$  because that is the width of the window in which the phase fails to oscillate. Two of the debts Volume 2 left — the asymptotics of Bessel functions and the decay for general hypersurfaces — are discharged here.

## The one mechanism

Every estimate in this volume comes from a single mechanism: an integral  $\int e^{i\lambda\varphi}\psi$  is small unless the phase is stationary, and how small depends on how degenerately it is stationary.

Where  $\nabla\varphi \neq 0$ , repeated integration by parts gives decay faster than any power. Where  $\varphi$  has a non-degenerate critical point, the integral concentrates on a window of width  $\lambda^{-1/2}$  and contributes  $\lambda^{-1/2}$  per variable (Worked Example 5.2.3). Where the critical point is degenerate to order  $k$ , the window widens and the contribution is only  $\lambda^{-1/k}$  (Worked Example 5.1.4).

Applied to  $\widehat{d\sigma}$ , the critical points of the phase  $x \cdot \omega$  on a surface are exactly the points where the normal is parallel to  $x$  (Lemma 5.5.1), and the Hessian there is the second fundamental form. So the decay rate counts non-vanishing principal curvatures, halved (Worked Example 5.5.4) — and a flat surface, having none, gives no decay at all.

That chain — stationary points, Hessian, second fundamental form, curvature — is what the rest of the series consumes. Volume 6 turns it into restriction estimates, Volume 7 into dimension bounds, Volume 8 into the distance-set theorem. Two outputs are used constantly: the decay of  $\widehat{d\sigma}$  *with its oscillating phase*, and the  $L^2$  bound for oscillatory integral operators. Worked Example 5.7.2 records that almost nothing else here is used later.

## What is assumed

Volumes 1–4. In particular we use: the complex analysis of Volume 2, Chapter 1, including the three-lines lemma and analytic continuation; van der Corput’s lemma in the form proved as Lemma 2.8.4; the Fourier transform on  $\mathcal{S}$  and  $\mathcal{S}'$ ; and the interpolation theorems of Volume 2, Chapter 2. Appendix A lists every citation with its use site.

## The plan, and its limits

|   |                                                            |                           |
|---|------------------------------------------------------------|---------------------------|
| 1 | Non-stationary phase and van der Corput                    | where the mass is not     |
| 2 | The Fresnel integral and stationary phase in one dimension | the model case            |
| 3 | The Morse lemma and stationary phase in $\mathbb{R}^n$     | the general local picture |
| 4 | Bessel functions                                           | Volume 2 debt             |
| 5 | Surface measure and curvature                              | Volume 2 debt             |
| 6 | Analytic continuation: Riesz potentials                    | complex parameters        |
| 7 | Oscillatory integral operators                             | Hörmander’s $L^2$ theorem |
| 8 | Bochner–Riesz means                                        | what is proved, and what  |

Following the practice adopted in Volume 4, the scope has been fixed *before* the chapters were written, to a list every item of which is proved in full. Three results that a reader might expect are deliberately absent, and are recorded in the manifest as plan revisions rather than discovered as gaps:

- the *Carleson–Sjölin theorem*, that the Bochner–Riesz conjecture holds in the plane. Its proof needs the bilinear and geometric machinery of Volume 12, and it is better placed there;
- *Stein’s interpolation theorem* for analytic families of operators. It requires a Poisson-kernel form of the three-lines lemma, and nothing in

this series uses it: Chapter 5.6 obtains the analytic continuation it needs directly;

- the sharp  $L^p$  theory of *oscillatory integral operators*. Chapter 5.7 proves Hörmander's  $L^2$  theorem, which is what Volumes 6 and 12 build on; the  $L^p$  refinements belong with the restriction theory.





# Chapter 5.1

## Non-Stationary Phase and van der Corput

### 5.1.1 The governing principle

Every estimate in this volume is a quantitative form of one sentence:

*An integral of a rapidly oscillating function is small, unless the oscillation stops somewhere.*

The integrals in question are

$$I(\lambda) = \int_{\mathbb{R}^n} e^{i\lambda\varphi(x)} \psi(x) \, dx,$$

with  $\lambda$  large,  $\varphi$  real-valued and smooth (the *phase*), and  $\psi$  smooth with compact support (the *amplitude*). Where  $\nabla\varphi \neq 0$ , neighbouring points contribute with rapidly rotating signs and cancel; where  $\nabla\varphi = 0$  — at a *critical*, or *stationary*, point — they do not.

This chapter proves the first half of the sentence, which is the easy half: away from critical points there is nothing. Chapter 5.2 and Chapter 5.3 then compute what remains.

#### Looking ahead: why this is the right frame for the series

Theorem 2.8.5 obtained  $|\widehat{d\sigma}(\xi)| \lesssim |\xi|^{-(n-1)/2}$  by a computation in one variable. Written as an oscillatory integral,

$$\widehat{d\sigma}(\xi) = \int_{\mathbb{S}^{n-1}} e^{-2\pi i \omega \cdot \xi} \, d\sigma(\omega) = \int_{\mathbb{S}^{n-1}} e^{i\lambda\varphi(\omega)} \, d\sigma(\omega), \quad \lambda = 2\pi|\xi|, \quad \varphi(\omega) = -\omega \cdot \frac{\xi}{|\xi|},$$

so the phase is a linear functional restricted to the sphere. Its critical points are where the gradient *along the sphere* vanishes, that is, where the normal

is parallel to  $\xi$ : the two poles. Everything else contributes nothing, by Theorem 5.1.1 below, and the exponent  $(n-1)/2$  will come from the  $n-1$  curved directions at a pole.

For a flat piece of hypersurface the phase is constant along it, so *every* point is critical, no cancellation occurs, and there is no decay — which is Worked Example 2.8.2. Curvature is exactly the statement that the critical set is small.

## 5.1.2 Non-stationary phase

**Theorem 5.1.1** (Principle of non-stationary phase). *Let  $\psi \in C_c^\infty(\mathbb{R}^n)$  and let  $\varphi \in C^\infty(\mathbb{R}^n)$  be real-valued with*

$$|\nabla\varphi(x)| \geq c > 0 \quad \text{for all } x \in \text{supp } \psi.$$

*Then for every  $N \in \mathbb{N}$  there is  $C_N$ , depending on  $N$ ,  $c$ , and finitely many derivatives of  $\varphi$  and  $\psi$  on  $\text{supp } \psi$ , with*

$$\left| \int_{\mathbb{R}^n} e^{i\lambda\varphi(x)} \psi(x) \, dx \right| \leq C_N \lambda^{-N} \quad (\lambda \geq 1).$$

*Idea of the proof.* The exponential  $e^{i\lambda\varphi}$  is an eigenfunction, with eigenvalue 1, of the first-order differential operator

$$L = \frac{1}{i\lambda} \frac{\nabla\varphi \cdot \nabla}{|\nabla\varphi|^2},$$

the verification being one line and carried out in Step 1 below. Since  $Le^{i\lambda\varphi} = e^{i\lambda\varphi}$ , we may insert  $L$  into the integral for free, and then move it onto the amplitude by integration by parts. Each such move produces a factor  $\lambda^{-1}$  and differentiates  $\psi$  and  $\varphi$  once more — which is harmless, since both are smooth and compactly supported. Repeating  $N$  times gives  $\lambda^{-N}$ . The hypothesis  $|\nabla\varphi| \geq c$  is what makes  $L$  well defined: it is the statement that one may divide by  $|\nabla\varphi|^2$ .

*Proof. Step 1. The operator and its transpose.*

On a neighbourhood of  $\text{supp } \psi$  the vector field  $V = \nabla\varphi/|\nabla\varphi|^2$  is smooth, by the hypothesis and smoothness of  $\varphi$ . Define, for smooth  $u$ ,

$$Lu = \frac{1}{i\lambda} V \cdot \nabla u.$$

Then  $Le^{i\lambda\varphi} = \frac{1}{i\lambda} V \cdot (i\lambda \nabla\varphi) e^{i\lambda\varphi} = \frac{\nabla\varphi \cdot \nabla\varphi}{|\nabla\varphi|^2} e^{i\lambda\varphi} = e^{i\lambda\varphi}$ .

Its transpose, defined by  $\int (Lu)w = \int u(L^t w)$  for compactly supported smooth  $w$ , is

$$L^t w = -\frac{1}{i\lambda} \nabla \cdot (Vw) = -\frac{1}{i\lambda} (V \cdot \nabla w + (\nabla \cdot V)w),$$

by the divergence theorem, there being no boundary terms.

**Step 2.** *Integrate by parts  $N$  times.*

Using  $e^{i\lambda\varphi} = L^N e^{i\lambda\varphi}$  and moving  $L$  across,

$$\int e^{i\lambda\varphi} \psi = \int (L^N e^{i\lambda\varphi}) \psi = \int e^{i\lambda\varphi} (L^t)^N \psi.$$

Each application of  $L^t$  contributes a factor  $\lambda^{-1}$  and replaces the amplitude by a smooth function supported in  $\text{supp } \psi$ , whose derivatives are polynomial expressions in the derivatives of  $\varphi$ , of  $\psi$ , and in  $|\nabla\varphi|^{-2}$ . Hence

$$\|(L^t)^N \psi\|_{L^\infty} \leq A_N \lambda^{-N}$$

with  $A_N$  depending only on the stated data.

**Step 3.** *Conclude.*

Taking absolute values,  $|\int e^{i\lambda\varphi} \psi| \leq m(\text{supp } \psi) A_N \lambda^{-N}$ . □

### **Pitfall: compact support of the amplitude is doing real work**

Two of the three steps used it: Step 1 to discard boundary terms, and Step 3 to convert a sup-norm bound into an integral bound. If  $\psi$  merely decays, the conclusion survives with the same proof provided all the derivatives appearing in Step 2 are integrable, but this must be checked — and for an amplitude that is merely bounded, the theorem is false: take  $n = 1$ ,  $\varphi(t) = t$ ,  $\psi \equiv 1$ , where the integral does not converge at all.

More importantly, if  $\psi$  is compactly supported but  $\varphi$  has a critical point in  $\text{supp } \psi$ , the hypothesis fails and so does the conclusion. That case is the entire content of Chapter 5.2, and the reader should resist the temptation to think of Theorem 5.1.1 as saying oscillatory integrals are small. It says they are small *away from critical points*.

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WORKED EXAMPLE 5.1.1— TIER A **The hypothesis cannot be weakened to  $\nabla\varphi \neq 0$**

It is not enough that  $\nabla\varphi$  be non-vanishing on  $\text{supp } \psi$ ; one needs a uniform lower bound, and on a compact support these are the same — but only because  $\text{supp } \psi$  is compact and  $\nabla\varphi$  continuous. The distinction matters as soon as one localises to a shrinking region, which is exactly what the proofs

of Chapter 5.2 do.

Concretely, take  $n = 1$ ,  $\varphi(t) = t^2$  and  $\psi$  supported in  $[\varepsilon, 1]$ , so  $\varphi'(t) = 2t \geq 2\varepsilon > 0$ . Theorem 5.1.1 applies with  $c = 2\varepsilon$ , but the constant  $C_N$  involves  $c^{-1}$  to a power, so the bound degrades as  $\varepsilon \rightarrow 0$ . Balancing that degradation against the size of the region where the hypothesis fails is precisely how the exponent  $\lambda^{-1/2}$  arises in Theorem 5.1.3 below, and again in Theorem 5.2.3.

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### 5.1.3 van der Corput's lemma, with amplitudes

When the phase does have critical points, one cannot integrate by parts indefinitely. What survives is a bound depending on the order of vanishing. Volume 2 proved the following in the course of estimating  $\widehat{d\sigma}$ ; we restate it, since it is used throughout this volume, and record the amplitude version and one corollary.

**Theorem 5.1.2** (van der Corput; Lemma 2.8.4). *Let  $\varphi : [a, b] \rightarrow \mathbb{R}$  be  $C^k$  with  $|\varphi^{(k)}| \geq 1$  on  $[a, b]$ , where either  $k \geq 2$ , or  $k = 1$  and  $\varphi'$  is monotone. Then*

$$\left| \int_a^b e^{i\lambda\varphi(t)} dt \right| \leq c_k \lambda^{-1/k} \quad (\lambda > 0),$$

with  $c_k$  depending only on  $k$ .

*Proof.* This is Lemma 2.8.4, proved there by induction on  $k$ : the case  $k = 1$  is one integration by parts, and the inductive step splits off a neighbourhood of the zero of  $\varphi^{(k-1)}$  and optimises its width.  $\square$

**Theorem 5.1.3** (Amplitude version). *Under the hypotheses of Theorem 5.1.2, for  $\psi : [a, b] \rightarrow \mathbb{C}$  of bounded variation,*

$$\left| \int_a^b e^{i\lambda\varphi(t)} \psi(t) dt \right| \leq c_k \lambda^{-1/k} \left( |\psi(b)| + \int_a^b |\psi'(t)| dt \right).$$

*Proof.* Let  $F(u) = \int_a^u e^{i\lambda\varphi}$ , so  $|F(u)| \leq c_k \lambda^{-1/k}$  for every  $u \in [a, b]$ , by Theorem 5.1.2 applied on  $[a, u]$  — note the hypothesis  $|\varphi^{(k)}| \geq 1$  persists on subintervals, and monotonicity of  $\varphi'$  does too. Integrating by parts,

$$\int_a^b e^{i\lambda\varphi} \psi = F(b)\psi(b) - \int_a^b F(u)\psi'(u) du,$$

and taking absolute values gives the claim.  $\square$

**Where we are.** The two theorems say complementary things. Theorem 5.1.1 gives decay of *every* order, but needs the phase to have no critical point. Theorem 5.1.2 tolerates critical points — indeed  $\varphi'$  may vanish, as long as some higher derivative does not — but gives only  $\lambda^{-1/k}$ , and  $k$  is the order at which the phase finally becomes non-degenerate. Both are needed. In every argument of this volume, space is split into a region where  $\nabla\varphi$  is large, handled by Theorem 5.1.1 and contributing  $O(\lambda^{-N})$ , and a shrinking neighbourhood of the critical set, where Theorem 5.1.2 or an explicit computation is used. The exponent in the final answer always comes from the second region, and equals the width of that neighbourhood.

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**WORKED EXAMPLE 5.1.2— TIER B Sublevel sets: the geometric content of van der Corput**

There is a purely geometric statement hiding inside Theorem 5.1.2, and it is often the more useful form.

**Proposition 5.1.4.** *Let  $\varphi : [a, b] \rightarrow \mathbb{R}$  be  $C^k$  with  $|\varphi^{(k)}| \geq \mu > 0$  on  $[a, b]$ . Then for every  $\alpha > 0$ ,*

$$m\left(\{t \in [a, b] : |\varphi(t)| \leq \alpha\}\right) \leq C_k \left(\frac{\alpha}{\mu}\right)^{1/k}.$$

*Proof.* Induct on  $k$ . For  $k = 1$ :  $|\varphi'| \geq \mu$  means  $\varphi$  is strictly monotone with speed at least  $\mu$ , so it spends time at most  $2\alpha/\mu$  in any interval of length  $2\alpha$ ; the sublevel set is an interval of length  $\leq 2\alpha/\mu$ .

For  $k \geq 2$ :  $\varphi^{(k-1)}$  is strictly monotone, so it vanishes at most once, say at  $c$ . Fix  $\eta > 0$  and remove  $(c - \eta, c + \eta)$ , of length  $2\eta$ . On each of the at most two remaining intervals,  $|\varphi^{(k-1)}| \geq \mu\eta$ , since  $\varphi^{(k-1)}$  moves at rate at least  $\mu$ . The inductive hypothesis gives, on each, a sublevel set of measure  $\leq C_{k-1}(\alpha/(\mu\eta))^{1/(k-1)}$ . Hence

$$m(|\varphi| \leq \alpha) \leq 2\eta + 2C_{k-1} \left(\frac{\alpha}{\mu\eta}\right)^{1/(k-1)}.$$

Choosing  $\eta = (\alpha/\mu)^{1/k}$  balances the two terms and gives the claim.  $\square$

*Reading it.* The proposition says: a function whose  $k$ -th derivative is bounded below cannot linger near any value. The exponent  $1/k$  is exactly the one in van der Corput, and this is not a coincidence — the standard proof of Theorem 5.1.2, which the reader saw in Volume 2, is the same induction, with “spends little time near a value” replaced by “the oscillation has little time to fail”.

*Why record it separately.* The sublevel set form is what one uses when the oscillation is not available — for instance when estimating the measure of the set where a polynomial is small, which is how the polynomial method of Volume 11 meets this circle of ideas. It is also the form in which the estimate generalises to several variables, where the oscillatory statement is much more delicate.

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**WORKED EXAMPLE 5.1.3— TIER C: RESEARCH TASTE** **Where the exponent  $1/k$  becomes the exponent of a conjecture**

The number  $\lambda^{-1/k}$  produced by Theorem 5.1.2 propagates, through the next four chapters, into the exponents of the central conjectures of this series. It is worth tracing the route once, in advance, so that the computations of Chapter 5.5 do not look like coincidences.

(a)  *$k = 2$  and the half-power.* At a non-degenerate critical point of a phase in one variable,  $\varphi'$  vanishes but  $\varphi''$  does not, so  $k = 2$  and the decay is  $\lambda^{-1/2}$ . Equivalently: the phase fails to oscillate on an interval of width  $\lambda^{-1/2}$  around the critical point, and the integral over that interval is trivially bounded by its length. Every  $\frac{1}{2}$  appearing in this subject is this window.

(b)  *$n - 1$  windows, and the sphere.* On the sphere, a linear phase has two critical points, and at each the Hessian along the surface is non-degenerate in all  $n - 1$  tangential directions — that is what it means for all principal curvatures to be non-zero. Each direction contributes a window of width  $\lambda^{-1/2}$ , so the critical region has measure  $\lambda^{-(n-1)/2}$ , and

$$|\widehat{d\sigma}(\xi)| \approx |\xi|^{-\frac{n-1}{2}}.$$

That is Chapter 5.5, and it is Theorem 2.8.5 with the mechanism made visible.

(c) *From decay to an exponent in restriction.* Worked Example 2.8.3 converted the decay rate  $\lambda^{-(n-1)/2}$  directly into the exponent  $q = \frac{4n}{n-1}$  of a restriction theorem, by the  $TT^*$  identity and Hardy–Littlewood–Sobolev. Volume 6 improves this to the Stein–Tomas exponent  $\frac{2(n+1)}{n-1}$  by using  $L^2$  orthogonality as well; but the input is the same number, and it traces back to  $k = 2$ .

(d) *Where degeneracy costs.* If a surface has a flat direction — one vanishing principal curvature — then one of the  $n - 1$  Hessian directions is degenerate, that window is wider than  $\lambda^{-1/2}$ , and the decay is worse. The extreme case is a hyperplane, where every direction is degenerate and there is no decay at all. So the whole subject may be read as the study of how much decay curvature buys, and Theorem 5.1.2 is the statement that  $k$  derivatives of non-degeneracy buy  $\lambda^{-1/k}$ .

*A caution.* It is tempting to conclude that everything reduces to counting derivatives. It does not. The passage from decay of  $d\sigma$  to a restriction estimate is lossy — Worked Example 2.8.3(b) quantified the loss — and the restriction conjecture is not a statement about any single oscillatory integral, but about how  $\delta^{-(n-1)/2}$  of them, one per cap, superpose. That is the geometry of Volumes 9–13, and no amount of stationary phase reaches it.

**WORKED EXAMPLE 5.1.4— TIER B Van der Corput: which case applies**

Lemma 2.8.4 has two clauses — first derivative bounded below, or  $k$ -th derivative bounded below for  $k \geq 2$  — and choosing between them is the first decision in any application.

*$k = 1$  needs monotonicity.* The bound  $|\int_a^b e^{i\lambda\varphi}| \lesssim \lambda^{-1}$  holds when  $|\varphi'| \geq 1$  and  $\varphi'$  is monotone. Without monotonicity it is false: a phase whose derivative oscillates between  $\pm 1$  can produce a large integral, because the oscillation never accumulates.

*$k \geq 2$  needs nothing extra.* The bound  $\lambda^{-1/k}$  holds whenever  $|\varphi^{(k)}| \geq 1$ , with no monotonicity hypothesis — the  $k$ -th derivative bound already forces  $\varphi'$  to be monotone on boundedly many subintervals.

*Worked choice:*  $\varphi(s) = s^2$  on  $[-1, 1]$ . Here  $\varphi' = 2s$  vanishes at the origin, so the  $k = 1$  clause is unavailable. But  $\varphi'' = 2$ , so the  $k = 2$  clause gives  $\lambda^{-1/2}$  — the correct answer, and the source of every  $\lambda^{-1/2}$  in the volume.

*Worked choice:*  $\varphi(s) = s^3$  on  $[-1, 1]$ . Now  $\varphi'' = 6s$  also vanishes, and one must go to  $k = 3$ :  $\varphi''' = 6$ , giving  $\lambda^{-1/3}$ . The degenerate critical point costs decay, and the exponent  $1/k$  records the order of degeneracy.

*The rule.* Find the smallest  $k$  with  $\varphi^{(k)}$  bounded below on the interval; the decay is  $\lambda^{-1/k}$ , except that  $k = 1$  additionally requires monotonicity. Amplitudes are handled separately, by the total-variation factor.

## Drills

**Drill 5.1.1.** Verify the formula for  $L^t$  in Step 1 of Theorem 5.1.1, and compute  $(L^t)^2\psi$  explicitly for  $n = 1$ ,  $\varphi(t) = t$ .

**Drill 5.1.2.** Show that Theorem 5.1.1 fails if  $\varphi$  is allowed a single critical point, by computing  $\int_{-1}^1 e^{i\lambda t^2} \psi(t) dt$  for  $\psi \in C_c^\infty$  with  $\psi(0) = 1$  and confirming it is of size  $\lambda^{-1/2}$ , not  $\lambda^{-N}$ .

**Drill 5.1.3.** Deduce Theorem 5.1.2 for  $k = 1$  from Worked Example 5.1.2 with  $k = 1$ , and explain why the implication does not run in the other direction for  $k \geq 2$ .

**Drill 5.1.4.** Let  $\varphi(t) = t^k$  on  $[-1, 1]$ . Compute  $\int e^{i\lambda t^k} dt$  up to constants and confirm the exponent  $\lambda^{-1/k}$  of Theorem 5.1.2 is attained.

## Notes

The principle of non-stationary phase is classical and the proof by iterated integration by parts goes back to Riemann. Van der Corput's lemma is from 1921; the sublevel set formulation of Worked Example 5.1.2 is due to Carbery, Christ and Wright, who developed the multi-dimensional theory where the two forms genuinely diverge.

Readers should note that Theorem 5.1.2 is sharp in  $k$  but says nothing about the phase of the answer. Extracting the phase — which Volume 6 needs — requires the asymptotic expansion of Chapter 5.2, and that is a strictly harder statement.



## Chapter 5.2

# The Fresnel Integral and Stationary Phase in One Dimension

Chapter 5.1 showed that an oscillatory integral is negligible away from the critical points of its phase, and that near a non-degenerate critical point the best crude bound is  $\lambda^{-1/2}$ . This chapter computes the constant. The answer, Theorem 5.2.3, is the single formula from which every asymptotic in this volume is derived.

Two ingredients are needed: an exact evaluation of the Gaussian-type integral  $\int e^{i\lambda t^2} dt$ , which is §2.1; and a change of variables reducing an arbitrary non-degenerate phase to  $t^2$ , which is §2.2.

### 5.2.1 The Fresnel integral

**Proposition 5.2.1.** *For  $a \in \mathbb{R} \setminus \{0\}$ ,*

$$\int_{\mathbb{R}} e^{iat^2} dt = \sqrt{\frac{\pi}{|a|}} e^{i\frac{\pi}{4} \operatorname{sgn} a},$$

*the integral being understood as  $\lim_{R \rightarrow \infty} \int_{-R}^R$ .*

*Idea of the proof.* The integral does not converge absolutely, so one cannot simply substitute. The device is to regard it as the boundary value of a family that does converge: for  $\operatorname{Re} z > 0$  the integral  $\int e^{-zt^2} dt$  converges absolutely and is holomorphic in  $z$ , and on the positive real axis it equals  $\sqrt{\pi/z}$  by the classical Gaussian computation. Two holomorphic functions agreeing on a segment agree throughout, so the identity persists on the

whole right half-plane, and one then lets  $z$  approach the imaginary axis. The branch of the square root is decided by continuity from the positive real axis, and it is exactly that decision which produces the phase  $e^{i\pi/4}$  — the number that will appear in every stationary phase asymptotic in this volume.

**Proof. Step 1.** *A holomorphic family.*

For  $\operatorname{Re} z > 0$  define

$$F(z) = \int_{\mathbb{R}} e^{-zt^2} dt.$$

The integrand is dominated by  $e^{-(\operatorname{Re} z)t^2}$ , integrable, so  $F$  is well defined; and it is holomorphic on  $\{\operatorname{Re} z > 0\}$  by differentiation under the integral sign (Worked Example 1.5.3), the  $z$ -derivative of the integrand being  $-t^2 e^{-zt^2}$ , dominated locally uniformly.

**Step 2.** *Identification on the real axis, then everywhere.*

For  $z = s > 0$  real, the substitution  $u = t\sqrt{s}$  and the Gaussian integral (Drill 1.6.2) give  $F(s) = \sqrt{\pi/s}$ . Let  $G(z) = \sqrt{\pi/z}$  denote the branch on  $\{\operatorname{Re} z > 0\}$  determined by  $G(s) = \sqrt{\pi/s} > 0$  for  $s > 0$ ; this is holomorphic there, since  $z \mapsto 1/z$  maps the right half-plane to itself and the principal square root is holomorphic away from the negative reals.

The two holomorphic functions  $F$  and  $G$  agree on  $(0, \infty)$ . By the propagation argument of Theorem 2.1.10 — a holomorphic function agreeing with another on a set with a limit point in a connected open set agrees with it throughout —  $F = G$  on all of  $\{\operatorname{Re} z > 0\}$ .

**Step 3.** *Approach the imaginary axis.*

Fix  $a > 0$  and set  $z_\varepsilon = \varepsilon - ia$  for  $\varepsilon > 0$ , so  $\operatorname{Re} z_\varepsilon > 0$  and  $z_\varepsilon \rightarrow -ia$ . By Step 2,

$$\int_{\mathbb{R}} e^{-(\varepsilon - ia)t^2} dt = \sqrt{\frac{\pi}{\varepsilon - ia}}.$$

As  $\varepsilon \downarrow 0$  the right side tends to  $\sqrt{\pi/(-ia)}$ , computed with the branch of Step 2: writing  $-ia = ae^{-i\pi/2}$ , we get  $\sqrt{\pi/(-ia)} = \sqrt{\pi/a} e^{i\pi/4}$ , the argument  $+\pi/4$  being forced because the path  $\varepsilon \mapsto \varepsilon - ia$  approaches the negative imaginary axis from the right half-plane, where the branch is continuous.

For the left side, we must justify

$$\lim_{\varepsilon \downarrow 0} \int_{\mathbb{R}} e^{-\varepsilon t^2} e^{iat^2} dt = \lim_{R \rightarrow \infty} \int_{-R}^R e^{iat^2} dt.$$

Both limits exist and agree: substituting  $u = t^2$  on  $(0, \infty)$  turns each into  $\frac{1}{2} \int_0^\infty e^{-\varepsilon u} e^{iau} u^{-1/2} du$ , and  $\int_0^\infty e^{iau} u^{-1/2} du$  converges as an improper integral, by van der Corput (Theorem 5.1.3) applied on  $[1, R]$  with amplitude

$u^{-1/2}$  of bounded variation. Abel summation — that is, dominated convergence applied after one integration by parts — then gives the equality of the two limits.

Hence  $\int_{\mathbb{R}} e^{iat^2} dt = \sqrt{\pi/a} e^{i\pi/4}$  for  $a > 0$ . For  $a < 0$  take complex conjugates.  $\square$

**WORKED EXAMPLE 5.2.1 — TIER A** The phase  $e^{i\pi/4}$  is a branch choice, and it is checkable

Students routinely mis-remember the sign in Proposition 5.2.1. Here is a consistency check that fixes it without redoing the proof.

Separate real and imaginary parts: the proposition asserts, for  $a = 1$ ,

$$\int_{\mathbb{R}} \cos(t^2) dt = \int_{\mathbb{R}} \sin(t^2) dt = \sqrt{\frac{\pi}{2}},$$

since  $\sqrt{\pi} e^{i\pi/4} = \sqrt{\pi}(\frac{1}{\sqrt{2}} + \frac{i}{\sqrt{2}})$ . Both are the classical Fresnel integrals, and both are *positive* — which they must be, since near  $t = 0$  the integrands are positive and the later oscillations alternate with decreasing amplitude. Had the branch produced  $e^{-i\pi/4}$ , the sine integral would come out negative, contradicting this.

So the rule is: *the phase has the same sign as  $a$* , and one may always check it against the positivity of  $\int \sin(t^2)$ .

## 5.2.2 Reduction to the model phase

**Lemma 5.2.2** (Morse lemma in one variable). *Let  $\varphi$  be  $C^\infty$  and real near  $t_0 \in \mathbb{R}$ , with  $\varphi'(t_0) = 0$  and  $\varphi''(t_0) \neq 0$ . Then there is a neighbourhood  $U$  of  $t_0$  and a  $C^\infty$  diffeomorphism  $s : U \rightarrow s(U) \subseteq \mathbb{R}$  with  $s(t_0) = 0$ ,  $s'(t_0) = 1$ , and*

$$\varphi(t) = \varphi(t_0) + \frac{1}{2}\varphi''(t_0)s(t)^2 \quad \text{for } t \in U.$$

*Proof.* By Taylor's theorem with integral remainder, using  $\varphi'(t_0) = 0$ ,

$$\varphi(t) - \varphi(t_0) = (t - t_0)^2 h(t), \quad h(t) := \int_0^1 (1 - u) \varphi''(t_0 + u(t - t_0)) du,$$

and  $h$  is  $C^\infty$  near  $t_0$  with  $h(t_0) = \frac{1}{2}\varphi''(t_0) \neq 0$ .

Set  $\kappa = \frac{1}{2}\varphi''(t_0)$  and

$$s(t) = (t - t_0) \sqrt{\frac{h(t)}{\kappa}},$$

where the square root is the positive one; this is legitimate on a neighbourhood of  $t_0$ , since  $h/\kappa$  is continuous with value 1 at  $t_0$ , hence positive nearby, and  $\sqrt{\cdot}$

is  $C^\infty$  on  $(0, \infty)$ . Then  $s$  is  $C^\infty$ ,  $s(t_0) = 0$ , and  $s'(t_0) = 1$ , so by the inverse function theorem  $s$  is a diffeomorphism near  $t_0$ . Finally

$$\kappa s(t)^2 = \kappa(t - t_0)^2 \frac{h(t)}{\kappa} = (t - t_0)^2 h(t) = \varphi(t) - \varphi(t_0). \quad \square$$

### 5.2.3 Stationary phase in one dimension

**Theorem 5.2.3.** *Let  $\varphi \in C^\infty(\mathbb{R})$  be real-valued and  $\psi \in C_c^\infty(\mathbb{R})$ . Suppose  $\varphi$  has exactly one critical point  $t_0$  in  $\text{supp } \psi$ , that it is non-degenerate ( $\varphi''(t_0) \neq 0$ ), and that  $t_0$  lies in the interior of  $\text{supp } \psi$ . Then, as  $\lambda \rightarrow \infty$ ,*

$$\int_{\mathbb{R}} e^{i\lambda\varphi(t)} \psi(t) dt = e^{i\lambda\varphi(t_0)} \sqrt{\frac{2\pi}{\lambda |\varphi''(t_0)|}} e^{i\frac{\pi}{4} \text{sgn } \varphi''(t_0)} \psi(t_0) + O(\lambda^{-3/2}),$$

the implied constant depending on finitely many derivatives of  $\varphi$  and  $\psi$ .

*Idea of the proof.* Three moves, each disposing of one difficulty.

*Localise.* Away from  $t_0$  the phase has no critical point, so Theorem 5.1.1 contributes  $O(\lambda^{-N})$ . Only a fixed small neighbourhood of  $t_0$  matters.

*Straighten.* On that neighbourhood, Lemma 5.2.2 turns the phase into the model  $\varphi(t_0) + \kappa s^2$ , at the cost of replacing the amplitude by another smooth compactly supported function.

*Expand.* Split the new amplitude as its value at the critical point plus a remainder vanishing there. The constant term integrates to the Fresnel integral, giving the stated main term. The remainder carries a factor  $s$ , and  $s e^{i\lambda\kappa s^2}$  is an exact derivative — so one integration by parts gains a full power of  $\lambda$ , and the remainder is smaller by  $\lambda^{-1}$  than the main term, i.e.  $O(\lambda^{-3/2})$ .

*Proof. Step 1. Localise near  $t_0$ .*

Let  $U$  be the neighbourhood supplied by Lemma 5.2.2 and choose  $\chi \in C_c^\infty(U)$  with  $\chi = 1$  near  $t_0$ . Write  $\psi = \chi\psi + (1 - \chi)\psi$ . On  $\text{supp } ((1 - \chi)\psi)$  the phase has no critical point —  $t_0$  was the only one, and it has been excluded — so  $|\varphi'|$  is bounded below there by compactness, and Theorem 5.1.1 gives

$$\int e^{i\lambda\varphi} (1 - \chi)\psi = O(\lambda^{-N})$$

for every  $N$ . So it suffices to treat  $\chi\psi$ , which we rename  $\psi$ .

**Step 2.** *Change variables.*

Put  $\kappa = \frac{1}{2}\varphi''(t_0)$  and let  $s$  be as in Lemma 5.2.2, with inverse  $t(s)$ . Substituting,

$$\int e^{i\lambda\varphi(t)}\psi(t) dt = e^{i\lambda\varphi(t_0)} \int_{\mathbb{R}} e^{i\lambda\kappa s^2} \tilde{\psi}(s) ds, \quad \tilde{\psi}(s) := \psi(t(s)) t'(s),$$

and  $\tilde{\psi} \in C_c^\infty(\mathbb{R})$  with  $\tilde{\psi}(0) = \psi(t_0)$ , since  $t'(0) = 1$ .

**Step 3.** *Split the amplitude.*

Write  $\tilde{\psi}(s) = \tilde{\psi}(0) + s\rho(s)$  where  $\rho(s) = \int_0^1 \tilde{\psi}'(us) du$  is smooth and compactly supported. The integral becomes

$$\tilde{\psi}(0) \int_{\mathbb{R}} e^{i\lambda\kappa s^2} \eta(s) ds + \int_{\mathbb{R}} e^{i\lambda\kappa s^2} s\rho(s) ds,$$

where  $\eta \in C_c^\infty$  equals 1 on a neighbourhood of  $\text{supp } \rho \cup \{0\}$  — we may insert such a cutoff because  $\tilde{\psi}$  is compactly supported, and extending the first integral to all of  $\mathbb{R}$  costs

$$\tilde{\psi}(0) \int_{\mathbb{R}} e^{i\lambda\kappa s^2} (1 - \eta(s)) ds = O(\lambda^{-N})$$

by Theorem 5.1.1, the phase  $\kappa s^2$  having no critical point on  $\text{supp}(1 - \eta)$ .

**Step 4.** *The main term.*

By Proposition 5.2.1 with  $a = \lambda\kappa$ ,

$$\int_{\mathbb{R}} e^{i\lambda\kappa s^2} ds = \sqrt{\frac{\pi}{\lambda|\kappa|}} e^{i\frac{\pi}{4} \text{sgn } \kappa} = \sqrt{\frac{2\pi}{\lambda|\varphi''(t_0)|}} e^{i\frac{\pi}{4} \text{sgn } \varphi''(t_0)},$$

using  $|\kappa| = \frac{1}{2}|\varphi''(t_0)|$  and  $\text{sgn } \kappa = \text{sgn } \varphi''(t_0)$ . Multiplying by  $\tilde{\psi}(0) = \psi(t_0)$  gives the stated main term.

**Step 5.** *The remainder.*

Since  $\frac{d}{ds} e^{i\lambda\kappa s^2} = 2i\lambda\kappa s e^{i\lambda\kappa s^2}$ , we may integrate by parts with no boundary terms:

$$\int_{\mathbb{R}} e^{i\lambda\kappa s^2} s\rho(s) ds = \frac{1}{2i\lambda\kappa} \int_{\mathbb{R}} \left( \frac{d}{ds} e^{i\lambda\kappa s^2} \right) \rho(s) ds = -\frac{1}{2i\lambda\kappa} \int_{\mathbb{R}} e^{i\lambda\kappa s^2} \rho'(s) ds.$$

The remaining integral is again an oscillatory integral with the model phase and a smooth compactly supported amplitude, so by Theorem 5.1.3 with  $k = 2$  it is  $O(\lambda^{-1/2})$ . Hence the remainder is  $O(\lambda^{-1} \cdot \lambda^{-1/2}) = O(\lambda^{-3/2})$ .

Collecting Steps 1, 3, 4 and 5 gives the theorem.  $\square$

**Where we are.** Every feature of the answer has a source, and it is worth listing them.

- $\lambda^{-1/2}$ : the width of the window in which the phase fails to oscillate, from Proposition 5.2.1.
- $|\varphi''(t_0)|^{-1/2}$ : a flatter critical point has a wider window.
- $e^{i\pi \operatorname{sgn} \varphi''/4}$ : the branch choice of Proposition 5.2.1, and the only place the *sign* of the curvature appears.
- $\psi(t_0)$ : nothing about the amplitude away from the critical point survives, to leading order.
- $e^{i\lambda\varphi(t_0)}$ : the phase at the critical point, which oscillates rapidly in  $\lambda$  and is invisible to any estimate that takes absolute values.

The last point deserves emphasis. Theorem 2.8.5 bounded  $|\widehat{d\sigma}|$  and discarded exactly this factor. Volume 6 needs it: the interference between the contributions of the two poles of the sphere is what produces the oscillation of  $\widehat{d\sigma}$ , and that oscillation is what makes the Stein–Tomas argument work at the endpoint.

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**WORKED EXAMPLE 5.2.2— TIER B Recovering the Fourier transform of surface measure,  $n = 3$**

Worked Example 2.8.2 computed, for the sphere in  $\mathbb{R}^3$ ,

$$\widehat{d\sigma}(\xi) = \frac{2 \sin(2\pi|\xi|)}{|\xi|},$$

by an elementary integration. Let us see Theorem 5.2.3 produce it, including the phase.

By Proposition 2.8.1 the transform reduces to

$$\widehat{d\sigma}(\xi) = 2\pi \int_{-1}^1 e^{-2\pi i r t} dt, \quad r = |\xi|,$$

which after the substitution  $t = \cos \theta$  becomes  $2\pi \int_0^\pi e^{-2\pi i r \cos \theta} \sin \theta d\theta$ . Take  $\lambda = 2\pi r$ ,  $\varphi(\theta) = -\cos \theta$ ,  $\psi(\theta) = \sin \theta$ .

*Critical points.*  $\varphi'(\theta) = \sin \theta$  vanishes at  $\theta = 0, \pi$ , the two poles. At these,  $\varphi''(\theta) = \cos \theta$  equals  $+1$  and  $-1$ .

*The obstruction, and what it teaches.* Theorem 5.2.3 does not apply directly: the amplitude  $\sin \theta$  *vanishes* at both critical points, so the leading term  $\psi(t_0)$

is zero and the theorem returns only  $O(\lambda^{-3/2})$ . That is consistent with the true answer, which is of size  $r^{-1} = \lambda^{-1}$  — between the two.

The resolution is that the correct statement retains the next term in the expansion. Writing  $\psi(\theta) = \theta + O(\theta^3)$  near 0 and repeating Step 5 with one more integration by parts produces a term of size  $\lambda^{-1}$  with an explicit constant, and summing the two poles gives

$$\widehat{d\sigma}(\xi) = \frac{2 \sin(2\pi r)}{r} + O(r^{-2}),$$

matching the exact formula — and the sin is precisely the interference between  $e^{i\lambda\varphi(0)} = e^{-2\pi ir}$  and  $e^{i\lambda\varphi(\pi)} = e^{2\pi ir}$ .

*The lesson.* When the amplitude degenerates at the critical point, the leading term of Theorem 5.2.3 vanishes and one must go further into the expansion. This is not an exotic situation: it is exactly what happens for the sphere in every dimension, because the surface measure written in polar form carries a factor vanishing at the poles. Chapter 5.4 does the general computation properly, and the vanishing amplitude is why the answer involves  $(n - 1)/2$  rather than  $1/2$ .

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**Figure 5.2.1 (to be drawn).** *The non-oscillation window*

**Content required:** The graph of  $\cos(\lambda t^2)$  against  $t$ , showing the flat central region of width  $\approx \lambda^{-1/2}$  where the phase has not yet turned, and the rapid oscillation outside it. Shade the window and label its width; this is where every factor  $\lambda^{-1/2}$  in the volume comes from.

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**WORKED EXAMPLE 5.2.3— TIER B The stationary phase formula, term by term**

Theorem 5.2.3 gives, for a single non-degenerate critical point  $s_0$ ,

$$\int e^{i\lambda\varphi(s)} \psi(s) \, ds = \left( \frac{2\pi}{\lambda |\varphi''(s_0)|} \right)^{1/2} e^{i\lambda\varphi(s_0)} e^{i\frac{\pi}{4} \operatorname{sgn} \varphi''(s_0)} \psi(s_0) + O(\lambda^{-3/2}).$$

Four factors, each with a source.

$\lambda^{-1/2}$  — the width of the non-oscillation window. The phase changes by  $O(1)$  over an interval of length  $\lambda^{-1/2}$  about  $s_0$ , and outside it the integrand oscillates away.

$|\varphi''(s_0)|^{-1/2}$  — the curvature of the phase. A flatter critical point has a wider window and a larger integral, which is why degeneracy costs decay (Worked Example 5.1.4).

$e^{i\lambda\varphi(s_0)}$  — the value of the phase, carried out front. This is the factor that

oscillates in  $\lambda$  and is what makes  $\widehat{d\sigma}$  oscillate rather than merely decay — see Volume 8, Theorem 8.7.2, where the oscillation is essential.

$e^{i\pi \operatorname{sgn} \varphi''/4}$  — the signature. It records whether the critical point is a minimum or a maximum, and in several variables becomes the signature of the Hessian, Worked Example 5.3.2. For a sphere the two critical points have opposite signatures, and the resulting phase difference is what makes  $\widehat{d\sigma}$  a cosine rather than a single exponential.

*The check.* Setting  $\varphi(s) = s^2/2$  and  $\psi \equiv 1$  recovers the Fresnel integral of Proposition 5.2.1, with  $\varphi'' = 1$  and the formula becoming an identity rather than an asymptotic. That is the standard sanity test.

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## Drills

**Drill 5.2.1.** Evaluate  $\int_{\mathbb{R}} e^{i(at^2+bt)} dt$  for  $a \neq 0$  by completing the square, and check the answer against Proposition 5.2.1.

**Drill 5.2.2.** Verify that  $h(t_0) = \frac{1}{2}\varphi''(t_0)$  in the proof of Lemma 5.2.2, and compute  $s(t)$  explicitly for  $\varphi(t) = 1 - \cos t$  near  $t_0 = 0$ .

**Drill 5.2.3.** Show that the error term in Theorem 5.2.3 cannot be improved to  $O(\lambda^{-2})$ , by computing the next term in the expansion for  $\varphi(t) = t^2$  and  $\psi(t) = e^{-t^2}$ .

**Drill 5.2.4.** Suppose  $\varphi'(t_0) = \varphi''(t_0) = 0$  but  $\varphi'''(t_0) \neq 0$ . Show the analogue of Theorem 5.2.3 has main term of size  $\lambda^{-1/3}$ , and identify the integral that replaces the Fresnel integral. (It is the Airy function.)

## Notes

The method of stationary phase is Kelvin's, from 1887, with antecedents in Stokes. The proof given here — localise, straighten by Morse, expand, integrate by parts — is the standard modern one and generalises verbatim to  $\mathbb{R}^n$ , which is Chapter 5.3.

The Airy function of Drill 2.4 governs the behaviour at degenerate critical points and is the first case of a general classification; the systematic theory is Arnold's, and it is not needed in this series.



## Chapter 5.3

# The Morse Lemma and Stationary Phase in $\mathbb{R}^n$

Chapter 5.2 handled one variable. The passage to  $\mathbb{R}^n$  requires one new ingredient — a normal form for a function near a non-degenerate critical point, valid in any dimension — and then the same three-step argument goes through verbatim. The answer, Theorem 5.3.3, is the formula from which the decay of  $d\sigma$  for a general hypersurface will follow in Chapter 5.5.

### 5.3.1 The Morse lemma

**Theorem 5.3.1** (Morse). *Let  $\varphi$  be real-valued and  $C^\infty$  near  $x_0 \in \mathbb{R}^n$ , with  $\nabla\varphi(x_0) = 0$  and  $Q := \text{Hess}\varphi(x_0)$  invertible. Then there are neighbourhoods  $U \ni x_0$ ,  $V \ni 0$  and a  $C^\infty$  diffeomorphism  $y : U \rightarrow V$  with  $y(x_0) = 0$ ,  $Dy(x_0) = I$ , and*

$$\varphi(x) = \varphi(x_0) + \frac{1}{2}\langle Qy(x), y(x) \rangle \quad (x \in U).$$

*Idea of the proof.* Taylor's theorem with integral remainder already writes  $\varphi(x) - \varphi(x_0) = \langle M(x)(x - x_0), (x - x_0) \rangle$  for a smooth symmetric matrix  $M$  with  $M(x_0) = \frac{1}{2}Q$ . So the whole task is to *diagonalise  $M(x)$  smoothly in  $x$* , in the sense of finding  $B(x)$  with  $B(x)^\top(\frac{1}{2}Q)B(x) = M(x)$ ; then  $y(x) = B(x)(x - x_0)$  works.

The obstacle is that a symmetric matrix has no canonical square root when it is indefinite, and choosing eigenvectors is not smooth where eigenvalues collide. The trick is not to diagonalise  $M$  at all, but to write  $S(x) = (\frac{1}{2}Q)^{-1}M(x)$ , which is close to the identity, and take its *principal square root* — available by the holomorphic functional calculus near  $I$ , hence smooth, with no eigenvalue bookkeeping. A short computation shows

$S^{1/2}$  is exactly the  $B$  required, and the reason is that  $S$  is self-adjoint for the bilinear form defined by  $Q$ .

*Proof. Step 1. Taylor with integral remainder.*

Since  $\varphi(x_0)$  and  $\nabla\varphi(x_0)$  contribute nothing,

$$\varphi(x) - \varphi(x_0) = \langle M(x)(x - x_0), (x - x_0) \rangle, \quad M(x) := \int_0^1 (1-u) \operatorname{Hess} \varphi(x_0 + u(x - x_0)) \, du,$$

by applying the one-variable Taylor formula to  $u \mapsto \varphi(x_0 + u(x - x_0))$ . Each  $M(x)$  is symmetric,  $M$  is  $C^\infty$  in  $x$ , and  $M(x_0) = \frac{1}{2}Q =: N$ , which is invertible by hypothesis.

**Step 2.** *A matrix close to the identity.*

Set  $S(x) = N^{-1}M(x)$ , so  $S$  is  $C^\infty$  with  $S(x_0) = I$ . Shrinking the neighbourhood, we may assume  $\|S(x) - I\| < \frac{1}{2}$  for  $x \in U$ .

The key algebraic fact is that  $S(x)$  is symmetric with respect to the bilinear form  $\langle N \cdot, \cdot \rangle$ :

$$S(x)^\top N = M(x)^\top N^{-\top} N = M(x) = N S(x), \quad (5.3.1)$$

using that  $M$  and  $N$  are symmetric.

**Step 3.** *The smooth square root.*

Because  $\|S - I\| < \frac{1}{2}$ , the spectrum of  $S(x)$  lies in the disc  $|z - 1| < \frac{1}{2}$ , on which the principal square root  $\sqrt{\cdot}$  is holomorphic. Define  $B(x) = S(x)^{1/2}$  by the convergent binomial series

$$B(x) = (I + (S(x) - I))^{1/2} = \sum_{k \geq 0} \binom{1/2}{k} (S(x) - I)^k,$$

which converges in operator norm. Its smoothness in  $x$  needs a word, since a uniform limit of smooth functions need not be smooth: the map  $A \mapsto (I + A)^{1/2}$  is given on  $\|A\| < 1$  by a power series in the entries of  $A$  with a positive radius of convergence, hence is real-analytic there; and  $x \mapsto S(x) - I$  is  $C^\infty$  with values in that region. So  $B$  is a composition of a real-analytic map with a  $C^\infty$  one, and is  $C^\infty$ . By construction  $B(x)^2 = S(x)$  and  $B(x_0) = I$ .

Since  $B$  is a limit of polynomials in  $S$ , relation (5.3.1) passes to  $B$ :  $B^\top N = N B$ . Therefore

$$B(x)^\top N B(x) = N B(x) B(x) = N S(x) = M(x).$$

**Step 4.** *The change of variables.*

Put  $y(x) = B(x)(x - x_0)$ . Then  $y(x_0) = 0$  and  $Dy(x_0) = B(x_0) = I$ , so by the inverse function theorem  $y$  is a  $C^\infty$  diffeomorphism of a neighbourhood of  $x_0$  onto a neighbourhood of 0. And

$$\frac{1}{2}\langle Qy, y \rangle = \langle NB(x-x_0), B(x-x_0) \rangle = \langle B^\top NB(x-x_0), (x-x_0) \rangle = \langle M(x)(x-x_0), (x-x_0) \rangle,$$

which is  $\varphi(x) - \varphi(x_0)$  by Step 1.  $\square$

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WORKED EXAMPLE 5.3.1— TIER A **Why one cannot just diagonalise pointwise**

Each  $M(x)$  is symmetric, so it *is* orthogonally diagonalisable for every fixed  $x$ ; the difficulty is that the diagonalising matrix cannot always be chosen to depend smoothly on  $x$ .

The standard example is  $n = 2$  with

$$M(x_1, x_2) = \begin{pmatrix} x_1 & x_2 \\ x_2 & -x_1 \end{pmatrix},$$

whose eigenvalues are  $\pm\sqrt{x_1^2 + x_2^2}$ : they collide at the origin, and the eigenvectors rotate by  $\pi$  as  $(x_1, x_2)$  circles the origin once, so no continuous choice exists on a punctured neighbourhood.

Theorem 5.3.1 sidesteps this entirely. Step 3 never looks at an eigenvector: it applies a holomorphic function to the matrix, which is a smooth operation because the *spectrum* is confined to a disc where that function is defined. This is a habit worth acquiring — when a construction seems to require diagonalisation, ask whether functional calculus will do instead.

---

## 5.3.2 Stationary phase in $n$ dimensions

**Lemma 5.3.2** (The model Gaussian integral). *Let  $Q$  be a real symmetric invertible  $n \times n$  matrix, with signature  $\text{sgn } Q$  (the number of positive eigenvalues minus the number of negative ones). Then for  $\lambda > 0$ ,*

$$\int_{\mathbb{R}^n} e^{\frac{i\lambda}{2}\langle Qy, y \rangle} dy = \left(\frac{2\pi}{\lambda}\right)^{n/2} |\det Q|^{-1/2} e^{\frac{i\pi}{4} \text{sgn } Q},$$

*the integral being understood as the limit of the integrals over balls.*

*Proof.* Diagonalise  $Q = O^\top DO$  with  $O$  orthogonal and  $D = \text{diag}(d_1, \dots, d_n)$ ,  $d_j \neq 0$ . The substitution  $z = Oy$  has Jacobian 1 and turns the integral into a product:

$$\int_{\mathbb{R}^n} e^{\frac{i\lambda}{2} \sum_j d_j z_j^2} dz = \prod_{j=1}^n \int_{\mathbb{R}} e^{\frac{i\lambda d_j}{2} z^2} dz.$$

By Proposition 5.2.1 with  $a = \lambda d_j/2$ , the  $j$ -th factor is  $\sqrt{2\pi/(\lambda|\lambda_j|)} e^{\frac{i\pi}{4} \operatorname{sgn} \lambda_j}$ . Multiplying, the moduli give  $(2\pi/\lambda)^{n/2} \prod_j |\lambda_j|^{-1/2} = (2\pi/\lambda)^{n/2} |\det Q|^{-1/2}$ , and the phases add to  $\frac{\pi}{4} \sum_j \operatorname{sgn} \lambda_j = \frac{\pi}{4} \operatorname{sgn} Q$ .  $\square$

**Theorem 5.3.3** (Stationary phase in  $\mathbb{R}^n$ ). *Let  $\varphi$  be real-valued and  $C^\infty$  on  $\mathbb{R}^n$  and  $\psi \in C_c^\infty(\mathbb{R}^n)$ . Suppose  $\varphi$  has exactly one critical point  $x_0$  in  $\operatorname{supp} \psi$ , that  $Q = \operatorname{Hess} \varphi(x_0)$  is invertible, and that  $x_0$  is interior to  $\operatorname{supp} \psi$ . Then as  $\lambda \rightarrow \infty$ ,*

$$\int_{\mathbb{R}^n} e^{i\lambda\varphi(x)} \psi(x) dx = e^{i\lambda\varphi(x_0)} \left(\frac{2\pi}{\lambda}\right)^{n/2} \frac{e^{\frac{i\pi}{4} \operatorname{sgn} Q}}{|\det Q|^{1/2}} \psi(x_0) + O(\lambda^{-\frac{n}{2}-1}).$$

*Proof. Step 1. Localise.*

Let  $U$  be the neighbourhood of Theorem 5.3.1 and  $\chi \in C_c^\infty(U)$  equal to 1 near  $x_0$ . On  $\operatorname{supp}((1-\chi)\psi)$  the phase has no critical point, so  $|\nabla\varphi|$  is bounded below there by compactness and Theorem 5.1.1 gives  $O(\lambda^{-N})$  for every  $N$ . Replace  $\psi$  by  $\chi\psi$ .

**Step 2. Straighten the phase.**

Apply Theorem 5.3.1 and substitute  $x = x(y)$ :

$$\int e^{i\lambda\varphi(x)} \psi(x) dx = e^{i\lambda\varphi(x_0)} \int_{\mathbb{R}^n} e^{\frac{i\lambda}{2} \langle Qy, y \rangle} \tilde{\psi}(y) dy, \quad \tilde{\psi}(y) = \psi(x(y)) |\det Dx(y)|,$$

with  $\tilde{\psi} \in C_c^\infty(\mathbb{R}^n)$  and  $\tilde{\psi}(0) = \psi(x_0)$ , since  $Dx(0) = I$ .

**Step 3. Split the amplitude.**

Write  $\tilde{\psi}(y) = \tilde{\psi}(0) + \sum_{j=1}^n y_j \rho_j(y)$  with  $\rho_j(y) = \int_0^1 (\partial_j \tilde{\psi})(uy) du$  smooth and compactly supported. As in Theorem 5.2.3, extending the constant term's integral to all of  $\mathbb{R}^n$  costs  $O(\lambda^{-N})$  by Theorem 5.1.1, the phase  $\frac{1}{2} \langle Qy, y \rangle$  having no critical point away from the origin.

**Step 4. The main term.**

By Lemma 5.3.2, the constant term contributes

$$\tilde{\psi}(0) \left(\frac{2\pi}{\lambda}\right)^{n/2} |\det Q|^{-1/2} e^{\frac{i\pi}{4} \operatorname{sgn} Q},$$

which with  $\tilde{\psi}(0) = \psi(x_0)$  is the stated main term.

**Step 5. The remainder.**

Since  $\nabla_y(e^{\frac{i\lambda}{2} \langle Qy, y \rangle}) = i\lambda Qy e^{\frac{i\lambda}{2} \langle Qy, y \rangle}$ , and  $Q$  is invertible, we may write  $y_j$  as a linear combination of the components of  $Qy$  and integrate by parts once in each remainder term, with no boundary contribution:

$$\int e^{\frac{i\lambda}{2} \langle Qy, y \rangle} y_j \rho_j(y) dy = \frac{1}{i\lambda} \int e^{\frac{i\lambda}{2} \langle Qy, y \rangle} (\text{a first-order derivative of } \rho_j) dy.$$

The remaining integral is again of the model form with a smooth compactly supported amplitude, so it is  $O(\lambda^{-n/2})$  — apply Steps 3 and 4 to it, or Lemma 5.3.2 after one more split. Hence each remainder term is  $O(\lambda^{-1} \cdot \lambda^{-n/2})$ , as claimed.  $\square$

**Where we are.** Compare the exponents. In one variable the answer was  $\lambda^{-1/2}$ ; here it is  $\lambda^{-n/2}$ . The reason is the one anticipated in Volume 5's preface: each of the  $n$  directions at the critical point contributes a window of width  $\lambda^{-1/2}$  in which the phase fails to oscillate, and the critical region therefore has measure  $\lambda^{-n/2}$ .

The hypothesis that  $Q$  is invertible is exactly the statement that *every* direction is genuinely quadratic. If  $Q$  has a null direction, the window in that direction is wider than  $\lambda^{-1/2}$  and the decay is worse; the extreme case is a phase constant in some direction, where there is no decay at all in that variable. Chapter 5.5 is the systematic version of this observation, with  $Q$  replaced by the second fundamental form of a hypersurface and its rank by the number of non-vanishing principal curvatures.

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**WORKED EXAMPLE 5.3.2— TIER B The signature is not a decoration**

The factor  $e^{i\pi \operatorname{sgn} Q/4}$  is easy to overlook, since it has modulus 1 and disappears from any estimate that takes absolute values. It matters whenever two or more critical points contribute, because then the answer is a *sum* and the relative phases decide whether the contributions reinforce or cancel.

*An example where they cancel.* Take  $n = 1$ ,  $\varphi(t) = \sin t$  on  $(0, 2\pi)$ , with  $\psi \in C_c^\infty((0, 2\pi))$  equal to 1 on  $[\frac{\pi}{4}, \frac{7\pi}{4}]$ . The critical points are  $t = \pi/2$  and  $t = 3\pi/2$ , with  $\varphi'' = -1$  and  $+1$  respectively, and  $\varphi(\pi/2) = 1$ ,  $\varphi(3\pi/2) = -1$ . By Theorem 5.2.3 the two contributions are

$$\sqrt{\frac{2\pi}{\lambda}} e^{i\lambda} e^{-i\pi/4} \quad \text{and} \quad \sqrt{\frac{2\pi}{\lambda}} e^{-i\lambda} e^{+i\pi/4},$$

whose sum is

$$\sqrt{\frac{2\pi}{\lambda}} \cdot 2 \cos\left(\lambda - \frac{\pi}{4}\right).$$

So the total is of size  $\lambda^{-1/2}$  *on average*, but it *vanishes* whenever  $\lambda - \frac{\pi}{4}$  is an odd multiple of  $\pi/2$ . Had we recorded only the moduli, we would have predicted a total of size  $\lambda^{-1/2}$  always, and been wrong at infinitely many  $\lambda$ .

*Why this is the sphere.* The computation is not an artificial example. In Chapter 5.4 the two critical points will be the two poles of the sphere, the signatures will differ in the same way, and the resulting cosine is exactly the

factor  $\cos\left(r - \frac{\nu\pi}{2} - \frac{\pi}{4}\right)$  in the Bessel asymptotic — which is why  $\widehat{d\sigma}$  for  $\mathbb{S}^2$  came out as  $2\sin(2\pi r)/r$  in Volume 2 rather than as a positive multiple of  $r^{-1}$ .

*And why Volume 6 needs it.* The Stein–Tomas argument estimates  $f * \widehat{d\sigma}$ , and at the endpoint exponent the oscillation of  $\widehat{d\sigma}$  is used, not merely its size. An argument that discards the phase cannot reach the endpoint. This is the concrete reason Volume 2’s Theorem 8.5, which bounded only  $|\widehat{d\sigma}|$ , was insufficient and this volume was necessary.

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## Drills

**Drill 5.3.1.** Verify (5.3.1) and check that the relation  $B^\top N = NB$  does pass from  $S$  to the series defining  $B$ .

**Drill 5.3.2.** Compute the Morse normal form explicitly for  $\varphi(x_1, x_2) = x_1^2 - x_2^2 + x_1^3$  near the origin, and identify  $Q$  and  $\text{sgn } Q$ .

**Drill 5.3.3.** Show that Theorem 5.3.3 with  $n = 1$  reduces to Theorem 5.2.3, including the phase factor.

**Drill 5.3.4.** Let  $\varphi(x) = |x|^2$  and  $\psi$  radial. Compute both sides of Theorem 5.3.3 directly using polar coordinates, and confirm the constant  $(2\pi/\lambda)^{n/2}$ .

**Where we are.** Three features of the Morse lemma are worth separating, because they are often conflated.

*It is a normal form, not an estimate.* The conclusion is an exact identity after a change of variables — no error terms, no asymptotics. The asymptotics of Theorem 5.3.3 follow from it by computing a Gaussian integral in the new coordinates, and that computation is where the  $\lambda^{-n/2}$  appears.

*Non-degeneracy is essential and not merely convenient.* If  $\text{Hess } \varphi(x_0)$  is singular there is no normal form of this shape: the degenerate directions require higher-order models, and the classification of those is singularity theory, which this series does not enter. The Pitfall in Chapter 5.5 shows what the loss costs in practice.

*The proof is constructive.*  $B(x)$  is given by an explicit convergent series in  $S(x) - I$ , so the change of variables can in principle be computed. That is unusual — most normal-form theorems rest on implicit function arguments — and it is why the functional-calculus proof is the one worth knowing.

*What replaces it in the degenerate case.* Nothing general. One returns to Lemma 2.8.4 and estimates the order of vanishing directly, which is what Worked Example 5.5.4's last row does.

## Notes

Theorem 5.3.1 is Morse's, from 1925; the proof by functional calculus given here avoids the induction on dimension found in most textbooks and is due to Palais. The stationary phase formula in the form Theorem 5.3.3 is classical; the systematic development, including the full asymptotic expansion in powers of  $\lambda^{-1}$ , is in Hörmander's *The Analysis of Linear Partial Differential Operators I*, Chapter VII.





# Chapter 5.4

## Bessel Functions

Theorem 2.8.3 expressed  $\widehat{d\sigma}$  for the sphere exactly in terms of Bessel functions, and Theorem 2.8.5 obtained the bound  $|\widehat{d\sigma}(\xi)| \lesssim |\xi|^{-(n-1)/2}$  by a direct estimate. What was deferred — and recorded as a debt — is the *asymptotic*: the precise leading behaviour, including the oscillation. This chapter supplies it.

### 5.4.1 The statement, and why the amplitude is awkward

Recall from Volume 2, Chapter 8 that for  $\nu > -\frac{1}{2}$ ,

$$J_\nu(r) = \frac{(r/2)^\nu}{\Gamma(\nu + \frac{1}{2})\Gamma(\frac{1}{2})} \int_{-1}^1 e^{irt} (1-t^2)^{\nu-\frac{1}{2}} dt. \quad (5.4.1)$$

**Theorem 5.4.1.** *For  $\nu > -\frac{1}{2}$  fixed,*

$$J_\nu(r) = \sqrt{\frac{2}{\pi r}} \cos\left(r - \frac{\nu\pi}{2} - \frac{\pi}{4}\right) + O_\nu(r^{-3/2}) \quad (r \rightarrow \infty).$$

*Scratch work.* The obvious route is stationary phase. Substituting  $t = \cos \theta$  in (5.4.1) gives  $\int_0^\pi e^{ir \cos \theta} (\sin \theta)^{2\nu} d\theta$ , whose phase  $\cos \theta$  has non-degenerate critical points at  $\theta = 0$  and  $\theta = \pi$ . So Theorem 5.2.3 ought to apply.

It does not, and Worked Example 5.2.2 already showed why: the amplitude  $(\sin \theta)^{2\nu}$  *vanishes* at both critical points when  $\nu > 0$ , so the leading term  $\psi(t_0)$  of Theorem 5.2.3 is zero and the theorem returns only the error bound. The true answer, of size  $r^{-1/2}$ , lies between.

The resolution is to work in the original variable  $t$ , where the two critical points become the two *endpoints*  $t = \pm 1$ , and where the difficulty is visibly

a degeneracy of the amplitude rather than of the phase. Near  $t = 1$  the amplitude behaves like a constant times  $(1-t)^{\nu-1/2}$ , and the model integral one needs is not the Fresnel integral but

$$\int_0^\infty e^{-irs} s^{a-1} ds = \Gamma(a) (ir)^{-a},$$

a Gamma integral — which we evaluate by exactly the analytic-continuation device used for the Fresnel integral in Proposition 5.2.1. That is the whole proof: two endpoints, each contributing a Gamma integral, and their sum producing the cosine.

## 5.4.2 The model integral

**Lemma 5.4.2.** *Let  $a > 0$  and  $r > 0$ . Then*

$$\int_0^\infty e^{-irs} s^{a-1} ds = \Gamma(a) r^{-a} e^{-\frac{i\pi a}{2}},$$

*the integral being understood as  $\lim_{\varepsilon \downarrow 0} \int_0^\infty e^{-(\varepsilon+ir)s} s^{a-1} ds$ .*

*Proof.* For  $\operatorname{Re} z > 0$  define  $G(z) = \int_0^\infty e^{-zs} s^{a-1} ds$ . The integral converges absolutely, since  $|e^{-zs} s^{a-1}| = e^{-(\operatorname{Re} z)s} s^{a-1}$  is integrable at 0 because  $a > 0$  and at  $\infty$  because  $\operatorname{Re} z > 0$ ; and  $G$  is holomorphic there by differentiation under the integral sign (Volume 1, Worked Example 5.3).

For  $z = \sigma > 0$  real, the substitution  $u = \sigma s$  gives  $G(\sigma) = \Gamma(a) \sigma^{-a}$ . Let  $H(z) = \Gamma(a) z^{-a}$  with the branch of  $z^{-a} = e^{-a \log z}$  determined by  $\log \sigma \in \mathbb{R}$  for  $\sigma > 0$ ; this is holomorphic on the right half-plane. Since  $G = H$  on  $(0, \infty)$ , they agree on  $\{\operatorname{Re} z > 0\}$ , by the propagation argument of Theorem 2.1.10.

Now let  $z = \varepsilon + ir$  and  $\varepsilon \downarrow 0$ . Writing  $ir = re^{i\pi/2}$ , the branch gives  $z^{-a} \rightarrow r^{-a} e^{-i\pi a/2}$ , the argument being  $-\pi a/2$  because the path approaches the positive imaginary axis from within the right half-plane where the branch is continuous.  $\square$

---

### WORKED EXAMPLE 5.4.1— TIER A Consistency with the Fresnel integral

Lemma 5.4.2 and Proposition 5.2.1 are the same computation in different clothes, and it is worth checking they agree.

Take  $a = \frac{1}{2}$  in Lemma 5.4.2:

$$\int_0^\infty e^{-irs} s^{-1/2} ds = \Gamma\left(\frac{1}{2}\right) r^{-1/2} e^{-i\pi/4} = \sqrt{\frac{\pi}{r}} e^{-i\pi/4}.$$

On the other hand, substituting  $s = t^2$  turns the left side into  $2 \int_0^\infty e^{-irt^2} dt = \int_{\mathbb{R}} e^{-irt^2} dt$ , which by Proposition 5.2.1 with  $a = -r$  is  $\sqrt{\pi/r} e^{-i\pi/4}$ . ✓

Both facts come from continuing  $\int_0^\infty e^{-zs} s^{a-1} ds$  from the real axis to the imaginary axis, and in both the phase is a branch choice. If one is remembered, the other can be recovered.

### 5.4.3 Proof of the asymptotic

*Proof of Theorem 5.4.1.* Write  $\alpha = \nu - \frac{1}{2} > -1$  and

$$I(r) = \int_{-1}^1 e^{irt} (1-t^2)^\alpha dt,$$

so that  $J_\nu(r) = \frac{(r/2)^\nu}{\Gamma(\nu + \frac{1}{2})\sqrt{\pi}} I(r)$  by (5.4.1) and  $\Gamma(\frac{1}{2}) = \sqrt{\pi}$ .

**Step 1.** *Cut into three pieces.*

Fix  $\chi \in C_c^\infty(\mathbb{R})$  with  $\chi = 1$  on  $[-\frac{1}{4}, \frac{1}{4}]$  and  $\text{supp } \chi \subseteq (-\frac{1}{2}, \frac{1}{2})$ , and write

$$1 = \chi(1-t) + \chi(1+t) + \chi_0(t)$$

on  $[-1, 1]$ , which defines  $\chi_0$ , smooth and supported in  $[-\frac{3}{4}, \frac{3}{4}]$ .

**Step 2.** *The interior piece is negligible.*

On  $\text{supp } \chi_0$  the amplitude  $(1-t^2)^\alpha \chi_0(t)$  is  $C^\infty$  with compact support in  $(-1, 1)$ , and the phase is  $t$ , with derivative 1. By Theorem 5.1.1 — or simply by integrating by parts  $N$  times, all boundary terms vanishing — this piece is  $O_N(r^{-N})$  for every  $N$ .

**Step 3.** *An endpoint piece.*

Consider  $t$  near 1 and substitute  $t = 1 - s$ :

$$I_+(r) := \int_{-1}^1 e^{irt} (1-t^2)^\alpha \chi(1-t) dt = e^{ir} \int_0^\infty e^{-irs} s^\alpha g(s) ds, \quad g(s) := (2-s)^\alpha \chi(s),$$

where  $g \in C_c^\infty([0, \frac{1}{2}))$  with  $g(0) = 2^\alpha$ .

Split  $g(s) = g(0) + s h(s)$  with  $h(s) = \int_0^1 g'(us) du$  smooth and compactly supported in  $[0, \frac{1}{2})$ . Then

$$I_+(r) = e^{ir} g(0) \int_0^\infty e^{-irs} s^\alpha \chi_1(s) ds + e^{ir} \int_0^\infty e^{-irs} s^{\alpha+1} h(s) ds,$$

where  $\chi_1 \in C_c^\infty([0, 1))$  equals 1 on  $\text{supp } \chi$ ; inserting  $\chi_1$  is legitimate because  $g$  vanishes beyond  $\frac{1}{2}$ .

*Main term.* Removing the cutoff  $\chi_1$  from the first integral costs

$$\left| \int_0^\infty e^{-irs} s^\alpha (1 - \chi_1(s)) ds \right| = O_N(r^{-N}),$$

by  $N$ -fold integration by parts, the integrand being smooth and the phase having non-vanishing derivative on the support of  $1 - \chi_1$ , which is bounded away from  $s = 0$ . By Lemma 5.4.2 with  $a = \alpha + 1 = \nu + \frac{1}{2} > 0$ ,

$$g(0) \int_0^\infty e^{-irs} s^\alpha ds = 2^\alpha \Gamma(\nu + \tfrac{1}{2}) r^{-\nu - \frac{1}{2}} e^{-\frac{i\pi}{2}(\nu + \frac{1}{2})}.$$

*Error term.* The second integral has  $s^{\alpha+1}$  in place of  $s^\alpha$ , so the same argument with  $a = \alpha + 2 = \nu + \frac{3}{2}$  bounds it by  $O_\nu(r^{-\nu - \frac{3}{2}})$ .

Hence

$$I_+(r) = 2^{\nu - \frac{1}{2}} \Gamma(\nu + \tfrac{1}{2}) r^{-\nu - \frac{1}{2}} e^{i(r - \frac{\pi}{2}(\nu + \frac{1}{2}))} + O_\nu(r^{-\nu - \frac{3}{2}}). \quad (5.4.2)$$

**Step 4.** *The other endpoint.*

The piece near  $t = -1$  is  $I_-(r) = \overline{I_+(r)}$ , because  $I(r)$  is real — the imaginary part of the integrand in  $I(r)$  is odd in  $t$  — and the interior piece is real up to  $O(r^{-N})$ . Alternatively substitute  $t \mapsto -t$  directly.

**Step 5.** *Add.*

By Steps 2–4,

$$I(r) = 2 \operatorname{Re} I_+(r) + O_N(r^{-N}) = 2^{\nu + \frac{1}{2}} \Gamma(\nu + \tfrac{1}{2}) r^{-\nu - \frac{1}{2}} \cos\left(r - \frac{\pi}{2}\left(\nu + \frac{1}{2}\right)\right) + O_\nu(r^{-\nu - \frac{3}{2}}).$$

Note  $\frac{\pi}{2}(\nu + \frac{1}{2}) = \frac{\nu\pi}{2} + \frac{\pi}{4}$ , the phase shift in the statement.

**Step 6.** *Restore the prefactor.*

Multiplying by  $\frac{(r/2)^\nu}{\Gamma(\nu + \frac{1}{2})\sqrt{\pi}}$ ,

$$J_\nu(r) = \frac{2^{-\nu} r^\nu}{\sqrt{\pi}} \cdot 2^{\nu + \frac{1}{2}} r^{-\nu - \frac{1}{2}} \cos\left(r - \frac{\nu\pi}{2} - \frac{\pi}{4}\right) + O_\nu(r^{-\frac{3}{2}}) = \sqrt{\frac{2}{\pi r}} \cos\left(r - \frac{\nu\pi}{2} - \frac{\pi}{4}\right) + O_\nu(r^{-\frac{3}{2}}),$$

since  $2^{-\nu} 2^{\nu + \frac{1}{2}} / \sqrt{\pi} = \sqrt{2/\pi}$ . □

**Where we are.** The debt recorded in Volume 2, Chapter 8 is now discharged. Three features of the answer are worth noting against what Volume 2 could say.

Volume 2 proved  $|\widehat{d\sigma}(\xi)| \lesssim |\xi|^{-(n-1)/2}$  and demonstrated sharpness only for  $n = 3$ , where an elementary closed formula was available. Theorem 5.4.1 gives sharpness in every dimension at once: the cosine equals  $\pm 1$  along a sequence  $r_k \rightarrow \infty$ , so the decay rate is attained infinitely often.

Volume 2 could not see the oscillation. Here it is explicit, and Worked Example 5.3.2 explained its origin: the two endpoints are the two poles of the sphere, their contributions are complex conjugates, and a sum of conjugates is a cosine.

Finally, the phase shift  $\frac{\nu\pi}{2} + \frac{\pi}{4}$  decomposes into a part  $\frac{\pi}{4}$  coming from the branch — the same  $\pi/4$  as in the Fresnel integral — and a part  $\frac{\nu\pi}{2}$  coming from the order of vanishing of the amplitude at the endpoint. The second is invisible in the one-dimensional stationary phase formula, and is exactly what Worked Example 2.5 found missing.

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**WORKED EXAMPLE 5.4.2— TIER B Checking the formula against the two cases we know**

(a)  $\nu = \frac{1}{2}$ . Worked Example 2.8.1 computed  $J_{1/2}(\rho) = \sqrt{2/(\pi\rho)} \sin \rho$  exactly. Theorem 5.4.1 with  $\nu = \frac{1}{2}$  predicts

$$\sqrt{\frac{2}{\pi r}} \cos\left(r - \frac{\pi}{4} - \frac{\pi}{4}\right) = \sqrt{\frac{2}{\pi r}} \cos\left(r - \frac{\pi}{2}\right) = \sqrt{\frac{2}{\pi r}} \sin r,$$

with error  $O(r^{-3/2})$  — and in fact the error is 0, the formula being exact in this case. ✓

(b)  $n = 3$  via the exact formula. Volume 2 computed  $\widehat{d\sigma}(\xi) = 2 \sin(2\pi|\xi|)/|\xi|$  for  $\mathbb{S}^2 \subseteq \mathbb{R}^3$ . Via Theorem 2.8.3 with  $\nu = \frac{n-2}{2} = \frac{1}{2}$  and  $r = |\xi|$ ,

$$\widehat{d\sigma}(\xi) = 2\pi r^{-1/2} J_{1/2}(2\pi r) = 2\pi r^{-1/2} \sqrt{\frac{2}{\pi \cdot 2\pi r}} \sin(2\pi r) = \frac{2 \sin(2\pi r)}{r}. \quad \checkmark$$

(c) What the general formula adds. For  $n = 4$ ,  $\nu = 1$ , and Theorem 5.4.1 gives

$$\widehat{d\sigma}(\xi) = 2\pi r^{-1} J_1(2\pi r) = 2\pi r^{-1} \sqrt{\frac{1}{\pi^2 r}} \cos\left(2\pi r - \frac{3\pi}{4}\right) + O(r^{-5/2}) = \frac{2}{r^{3/2}} \cos\left(2\pi r - \frac{3\pi}{4}\right)$$

of size  $r^{-3/2} = r^{-(n-1)/2}$  as predicted, with a phase shift that no elementary formula would have revealed.

---

**WORKED EXAMPLE 5.4.3— TIER B Bessel asymptotics without Bessel functions**

Theorem 5.4.1 gives  $J_\nu(t) \approx t^{-1/2} \cos(t - \frac{\nu\pi}{2} - \frac{\pi}{4})$  for large  $t$ . The asymptotic can be predicted from stationary phase alone, without any special-function theory, and doing so is a good check.

*The integral representation.*  $J_\nu(t)$  is, up to constants,  $\int_{-1}^1 e^{its} (1-s^2)^{\nu-1/2} ds$ . The phase is  $\varphi(s) = s$ , which has *no* interior critical point — so by Theorem 5.1.1 the decay would be rapid, were it not for the endpoints.

*Where the decay comes from.* The amplitude  $(1-s^2)^{\nu-1/2}$  vanishes to order  $\nu - \frac{1}{2}$  at  $s = \pm 1$ . Lemma 5.4.2 handles exactly this: an oscillatory integral whose amplitude vanishes to order  $\beta$  at an endpoint contributes  $t^{-\beta-1}$ . With  $\beta = \nu - \frac{1}{2}$  this gives  $t^{-\nu-1/2}$  — which is not  $t^{-1/2}$ .

*Reconciling.* The discrepancy is the normalisation:  $J_\nu$  carries a factor  $t^\nu$  relative to the integral above, and  $t^\nu \cdot t^{-\nu-1/2} = t^{-1/2}$ . ✓

*The phase, and why there are two terms.* The two endpoints  $s = \pm 1$  contribute  $e^{\pm it}$ , and their sum is a cosine. The shift  $-\frac{\nu\pi}{2} - \frac{\pi}{4}$  comes from the argument of the constant in Lemma 5.4.2, which involves  $\Gamma(\nu + \frac{1}{2})$  and a factor  $e^{\pm i\pi(\dots)/2}$ .

*What to retain.* The decay rate of a Bessel function is an endpoint phenomenon, not a stationary-phase one — and that is why  $\widehat{d\sigma}$ , whose decay *is* stationary phase, has the same exponent for a different reason. Volume 5, Chapter 5.5 keeps the two straight.

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## Drills

**Drill 5.4.1.** Verify that  $I(r)$  in the proof is real, by splitting the integrand into even and odd parts in  $t$ .

**Drill 5.4.2.** Carry out the substitution  $t = 1 - s$  in Step 3 and confirm  $g(s) = (2-s)^\alpha \chi(s)$  with  $g(0) = 2^\alpha$ .

**Drill 5.4.3.** Extend Theorem 5.4.1 by one more term: show that the next correction is of size  $r^{-3/2}$  with an explicit coefficient, by retaining the  $sh(s)$  term in Step 3 rather than discarding it.

**Drill 5.4.4.** Use Theorem 5.4.1 to show that  $J_\nu$  has infinitely many zeros, and that consecutive zeros are asymptotically at distance  $\pi$ .

## Notes

The asymptotic expansion of Bessel functions is due to Poisson and Hankel; the endpoint analysis given here, reducing everything to the Gamma integral of Lemma 5.4.2, is the standard modern treatment and is the same device — analytic continuation of a convergent integral to the boundary of its half-plane — as Proposition 5.2.1.

Watson's *Theory of Bessel Functions* remains the reference for the full expansion and for uniform asymptotics in  $\nu$ , which this series does not need.





## Chapter 5.5

# Surface Measure and Curvature

Theorem 2.8.5 proved  $|\widehat{d\sigma}(\xi)| \lesssim |\xi|^{-(n-1)/2}$  for the sphere, and recorded as a debt the general principle: a hypersurface with  $k$  non-vanishing principal curvatures has  $|\widehat{d\sigma}(\xi)| \lesssim |\xi|^{-k/2}$ . This chapter pays it, and in doing so explains the exponent.

### 5.5.1 The set-up

Let  $S \subseteq \mathbb{R}^n$  be a  $C^\infty$  hypersurface with surface measure  $d\sigma$ , and let  $\psi \in C_c^\infty(\mathbb{R}^n)$  be a cutoff, so that  $\psi d\sigma$  is a compactly supported measure. Its Fourier transform is the bounded continuous function

$$\widehat{\psi d\sigma}(\xi) = \int_S e^{-2\pi i x \cdot \xi} \psi(x) d\sigma(x),$$

which is a tempered distribution by Proposition 2.5.8 and an honest function because  $\psi d\sigma$  is finite.

Write  $\xi = \lambda\omega$  with  $\lambda = |\xi|$  and  $\omega \in \mathbb{S}^{n-1}$ , and set

$$\varphi_\omega(x) = -2\pi x \cdot \omega,$$

so that  $\widehat{\psi d\sigma}(\lambda\omega) = \int_S e^{i\lambda\varphi_\omega} \psi d\sigma$  is an oscillatory integral on  $S$  with large parameter  $\lambda$ .

**Lemma 5.5.1** (Where the phase is critical). *Let  $x \in S$ . The restriction of  $\varphi_\omega$  to  $S$  has a critical point at  $x$  if and only if  $\omega$  is normal to  $S$  at  $x$ , that is  $\omega = \pm\nu(x)$  where  $\nu$  is a unit normal. Moreover, parametrising  $S$  near such an  $x$  as a graph over its tangent plane,*

$$\text{Hess}(\varphi_\omega|_S)(x) = \mp 2\pi \mathbb{I}_x,$$

where  $\mathbb{I}_x$  is the second fundamental form of  $S$  at  $x$ , whose eigenvalues are the principal curvatures  $\kappa_1(x), \dots, \kappa_{n-1}(x)$ .

*Proof.* Rotate and translate so that  $x = 0$  and  $T_x S = \{x_n = 0\}$ , and write  $S$  near 0 as a graph  $x_n = h(y)$ ,  $y \in \mathbb{R}^{n-1}$ , with  $h(0) = 0$  and  $\nabla h(0) = 0$ ; the unit normal at 0 is then  $e_n$ . In these coordinates

$$\varphi_\omega(y, h(y)) = -2\pi(y \cdot \omega' + h(y)\omega_n), \quad \omega = (\omega', \omega_n),$$

so its gradient at  $y = 0$  is  $-2\pi\omega'$ , using  $\nabla h(0) = 0$ . This vanishes exactly when  $\omega' = 0$ , i.e.  $\omega = \pm e_n = \pm \nu(0)$ .

For such  $\omega$ , differentiating twice at 0 gives  $\text{Hess} = -2\pi\omega_n \text{Hess } h(0)$ , and  $\text{Hess } h(0)$  is by definition the second fundamental form in these coordinates, with  $\omega_n = \pm 1$ .  $\square$

**Where we are.** This is the geometric content of the whole chapter, and it is worth stating in words. The critical points of a linear phase on a hypersurface are the points where the normal points along  $\xi$ ; the Hessian there is the second fundamental form. So:

- *curvature* = non-degeneracy of the Hessian = the hypothesis of stationary phase;
- the *number of non-vanishing principal curvatures* = the rank of the Hessian = the number of directions each contributing a window of width  $\lambda^{-1/2}$ ;
- a *flat* piece has  $\mathbb{I} = 0$ , so every point of it is a degenerate critical point when  $\omega$  is its normal — and there is no decay in that direction, which is Worked Example 2.8.2.

## 5.5.2 Non-vanishing Gaussian curvature

**Theorem 5.5.2.** *Let  $S$  be a  $C^\infty$  hypersurface and  $\psi \in C_c^\infty$  with  $\text{supp } \psi \cap S$  compact. Suppose all  $n - 1$  principal curvatures are non-zero at every point of  $\text{supp } \psi \cap S$ . Then for every  $\omega \in \mathbb{S}^{n-1}$ , writing  $x_1(\omega), \dots, x_m(\omega)$  for the points of  $\text{supp } \psi \cap S$  at which  $\omega$  is normal,*

$$\widehat{\psi d\sigma}(\lambda\omega) = \lambda^{-\frac{n-1}{2}} \sum_{j=1}^m e^{-2\pi i \lambda x_j \cdot \omega} c_j(\omega) + O(\lambda^{-\frac{n-1}{2}-1}),$$

where

$$|c_j(\omega)| = \psi(x_j) |K(x_j)|^{-1/2}, \quad K = \kappa_1 \cdots \kappa_{n-1} \text{ the Gaussian curvature,}$$

and the implied constant is uniform in  $\omega$ . In particular

$$|\widehat{\psi d\sigma}(\xi)| \lesssim_{S,\psi} |\xi|^{-\frac{n-1}{2}}.$$

*Idea of the proof.* By Lemma 5.5.1 the critical points are the preimages of  $\omega$  under the Gauss map  $\nu : S \rightarrow \mathbb{S}^{n-1}$ , and the Hessian there is the second fundamental form — which is invertible exactly when the Gaussian curvature is non-zero. So the Gauss map is a local diffeomorphism, its fibres are discrete, and on a compact piece there are finitely many, with the number locally constant in  $\omega$ .

Each contributes a term by Theorem 5.3.3, with  $|\det \text{Hess}|^{-1/2} = |K|^{-1/2}$ ; away from the critical points Theorem 5.1.1 contributes  $O(\lambda^{-N})$ . The only genuinely new issue is *uniformity in  $\omega$* , since the critical points move as  $\omega$  varies; this is handled by compactness.

**Proof. Step 1.** *The critical set is finite, and moves smoothly.*

By Lemma 5.5.1,  $x$  is critical for  $\varphi_\omega|_S$  iff  $\nu(x) = \pm\omega$ . The differential of the Gauss map  $\nu$  at  $x$  is the shape operator, whose matrix is  $\mathbb{I}_x$ ; by hypothesis this is invertible on the compact set  $\text{supp } \psi \cap S$ , so  $\nu$  is a local diffeomorphism there. Hence the fibre  $\nu^{-1}(\{\pm\omega\}) \cap \text{supp } \psi$  is discrete, and being closed in a compact set it is finite; and by the inverse function theorem the critical points depend smoothly on  $\omega$ , locally.

Since the number  $m = m(\omega)$  is locally constant and  $\mathbb{S}^{n-1}$  is compact,  $m$  is bounded, say by  $m_0$ .

**Step 2.** *Partition.*

Fix  $\omega_0$  and let  $x_1, \dots, x_m$  be its critical points. Choose disjoint coordinate patches  $U_1, \dots, U_m$  around them, on each of which  $S$  is a graph as in Lemma 5.5.1, and a smooth partition of unity  $1 = \sum_{j=1}^m \chi_j + \chi_0$  subordinate to  $\{U_j\} \cup (S \setminus \bigcup_j \bar{V}_j)$ , with  $V_j \Subset U_j$  neighbourhoods of  $x_j$ . By continuity of  $\nu$  this partition serves for all  $\omega$  in a neighbourhood  $W$  of  $\omega_0$ , with the critical point of  $\varphi_\omega$  in  $U_j$  depending smoothly on  $\omega \in W$ .

**Step 3.** *The non-critical part.*

On  $\text{supp } \chi_0$  there is no critical point of  $\varphi_\omega$  for  $\omega \in W$ , after shrinking  $W$ ; by compactness  $|\nabla_S \varphi_\omega|$  is bounded below there, uniformly in  $\omega \in W$ . Theorem 5.1.1, applied in the graph coordinates, gives

$$\int_S e^{i\lambda\varphi_\omega} \psi \chi_0 \, d\sigma = O_N(\lambda^{-N}),$$

uniformly in  $\omega \in W$ , the constants depending on finitely many derivatives of the parametrisation and of  $\psi$ , all bounded on the compact patch.

**Step 4.** *Each critical patch.*

On  $U_j$ , parametrise  $S$  as a graph  $x = (y, h_j(y))$  and write the integral as

$$\int_{\mathbb{R}^{n-1}} e^{i\lambda\Phi_j(y,\omega)} a_j(y) \, dy, \quad \Phi_j(y,\omega) = -2\pi(y \cdot \omega' + h_j(y)\omega_n),$$

with  $a_j$  smooth and compactly supported, incorporating  $\psi\chi_j$  and the surface measure density  $\sqrt{1 + |\nabla h_j|^2}$ . By Lemma 5.5.1 the Hessian in  $y$  at the critical point is  $\mp 2\pi\mathbb{I}$ , invertible by hypothesis, with

$$|\det \text{Hess}_y \Phi_j| = (2\pi)^{n-1} |K(x_j)|.$$

Theorem 5.3.3 applies and gives

$$\left(\frac{2\pi}{\lambda}\right)^{\frac{n-1}{2}} \frac{e^{\frac{i\pi}{4} \text{sgn Hess}}}{(2\pi)^{\frac{n-1}{2}} |K(x_j)|^{1/2}} a_j(y_j(\omega)) e^{i\lambda \Phi_j(y_j(\omega), \omega)} + O(\lambda^{-\frac{n-1}{2}-1}),$$

and  $\Phi_j$  evaluated at the critical point is  $-2\pi x_j \cdot \omega$ , while  $a_j(y_j) = \psi(x_j)$  because the density equals 1 where  $\nabla h_j = 0$ . This is the  $j$ -th term of the statement, with  $|c_j| = \psi(x_j) |K(x_j)|^{-1/2}$  as claimed.

**Step 5. Uniformity.**

The constants in Steps 3 and 4 depend only on finitely many derivatives of  $h_j$ ,  $a_j$  and of the critical point  $y_j(\omega)$ , all of which vary continuously on the compact  $\overline{W}$ . Covering  $\mathbb{S}^{n-1}$  by finitely many such  $W$  gives a bound uniform in  $\omega$ .  $\square$

#### WORKED EXAMPLE 5.5.1— TIER A **The sphere, recovered**

For  $S = \mathbb{S}^{n-1}$  and  $\psi \equiv 1$  near  $S$ , the Gauss map is the identity, so each  $\omega$  has exactly two critical points,  $x = \omega$  and  $x = -\omega$ , and all principal curvatures equal 1, so  $K \equiv 1$ .

Theorem 5.5.2 then reads

$$\widehat{\text{d}\sigma}(\lambda\omega) = \lambda^{-\frac{n-1}{2}} \left( c_+ e^{-2\pi i\lambda} + c_- e^{2\pi i\lambda} \right) + O(\lambda^{-\frac{n-1}{2}-1}),$$

using  $x_{\pm} \cdot \omega = \pm 1$ . The two Hessians differ by a sign, so their signatures are  $-(n-1)$  and  $+(n-1)$  and  $c_{\pm}$  are complex conjugates; the sum is therefore a cosine.

Comparing with Theorem 5.4.1 — which gave  $\widehat{\text{d}\sigma}(\xi) = 2\pi r^{-\frac{n-2}{2}} J_{\frac{n-2}{2}}(2\pi r)$  and hence a cosine of  $2\pi r - \frac{(n-2)\pi}{4} - \frac{\pi}{4}$  — the two agree, and the phase shift is exactly  $\frac{\pi}{4} \text{sgn Hess}$ . So the general theorem and the exact Bessel computation are consistent, as they must be.

### 5.5.3 Partial curvature

**Theorem 5.5.3.** *Let  $S$  and  $\psi$  be as above, and suppose that at every point of  $\text{supp } \psi \cap S$  at least  $k$  of the principal curvatures are non-zero. Then*

$$|\widehat{\psi \text{d}\sigma}(\xi)| \lesssim_{S, \psi} |\xi|^{-k/2}.$$

*Idea of the proof.* Split the tangential variables into the  $k$  directions in which the second fundamental form is non-degenerate and the remaining  $n - 1 - k$ . Apply stationary phase in the first group, treating the second as parameters; each of the  $k$  non-degenerate directions contributes  $\lambda^{-1/2}$ , and the parameters contribute nothing — but also cost nothing, since the amplitude is compactly supported and one may simply integrate over them. The only technical point is that Theorem 5.3.3 must be applied with a constant uniform in the parameters. Inspecting its proof, the constant depends only on finitely many derivatives of phase and amplitude and on a lower bound for  $|\det \text{Hess}|$  — all of which are uniform here, by compactness.

*Proof. Step 1. Reduce to a patch.*

By a partition of unity as in Step 2 of Theorem 5.5.2, it suffices to bound the contribution of a single coordinate patch, on which  $S$  is a graph  $x = (y, h(y))$ ,  $y \in \mathbb{R}^{n-1}$ , and the integral is

$$\int_{\mathbb{R}^{n-1}} e^{i\lambda\Phi(y)} a(y) dy, \quad \Phi(y) = -2\pi(y \cdot \omega' + h(y)\omega_n),$$

with  $a \in C_c^\infty$ . If  $\Phi$  has no critical point on  $\text{supp } a$  we are done by Theorem 5.1.1, so assume it has one, at  $y_0$ .

**Step 2.** *Split the variables.*

At  $y_0$  the Hessian  $\text{Hess } \Phi(y_0) = \mp 2\pi \Pi$  has rank at least  $k$  by hypothesis. Choose an orthogonal change of coordinates in  $y$  so that the upper-left  $k \times k$  block of  $\text{Hess } \Phi(y_0)$  is invertible, and write  $y = (u, v)$  with  $u \in \mathbb{R}^k$ ,  $v \in \mathbb{R}^{n-1-k}$ . By continuity, that block remains invertible, with determinant bounded below, on a neighbourhood of  $y_0$ ; shrink  $\text{supp } a$  to lie inside it.

**Step 3.** *Stationary phase in  $u$ , with  $v$  as a parameter.*

For each fixed  $v$ , the function  $u \mapsto \Phi(u, v)$  has, on the shrunken support, a Hessian in  $u$  that is invertible with determinant bounded below uniformly in  $v$ . By the implicit function theorem applied to  $\nabla_u \Phi = 0$ , it has at most one critical point  $u(v)$  there, depending smoothly on  $v$ .

Applying Theorem 5.3.3 in the  $u$  variables — with  $v$  a parameter, and noting that its proof produces a constant depending only on finitely many derivatives of  $\Phi$  and  $a$  and on the lower bound for  $|\det \text{Hess}_u \Phi|$ , all uniform in  $v$  — gives

$$\left| \int_{\mathbb{R}^k} e^{i\lambda\Phi(u,v)} a(u, v) du \right| \lesssim \lambda^{-k/2},$$

uniformly in  $v$ .

**Step 4.** *Integrate out the parameters.*

Integrating in  $v$  over the compact projection of  $\text{supp } a$ ,

$$\left| \int_{\mathbb{R}^{n-1}} e^{i\lambda\Phi(y)} a(y) dy \right| \leq \int \left| \int e^{i\lambda\Phi(u,v)} a(u,v) du \right| dv \lesssim \lambda^{-k/2}. \quad \square$$

**WORKED EXAMPLE 5.5.2— TIER B A cylinder, and the sharpness of the exponent**

Theorem 5.5.3 is sharp:  $k$  non-vanishing curvatures give exactly  $\lambda^{-k/2}$  and no more. The example is a cylinder, and it is worth doing in full because it isolates the mechanism.

Let  $n = 3$  and let

$$S = \{(x_1, x_2, x_3) : x_1^2 + x_2^2 = 1, |x_3| \leq 1\},$$

a circular cylinder. Its principal curvatures are 1 in the circular direction and 0 along the axis, so  $k = 1$  and Theorem 5.5.3 predicts  $|\widehat{\psi d\sigma}| \lesssim |\xi|^{-1/2}$ .

*The transform factorises.* With  $\psi(x) = \eta(x_3)$  for  $\eta \in C_c^\infty((-1, 1))$ , surface measure on  $S$  is the product of arclength on the circle and  $dx_3$ , so

$$\widehat{\psi d\sigma}(\xi) = \left( \int_{\mathbb{S}^1} e^{-2\pi i(x_1\xi_1 + x_2\xi_2)} d\sigma_1 \right) \cdot \widehat{\eta}(\xi_3) = 2\pi J_0(2\pi|\xi'|) \cdot \widehat{\eta}(\xi_3),$$

where  $\xi' = (\xi_1, \xi_2)$ , using Theorem 2.8.3 in dimension 2.

*Reading off the decay.* Take  $\xi = (\xi', 0)$ , so  $\widehat{\eta}(0) = \int \eta \neq 0$  is a fixed non-zero constant. By Theorem 5.4.1,

$$|\widehat{\psi d\sigma}(\xi', 0)| = 2\pi|\widehat{\eta}(0)| |J_0(2\pi|\xi'|)| \approx |\xi'|^{-1/2}$$

along the sequence where the cosine equals  $\pm 1$ . So the exponent  $1/2 = k/2$  is attained, and no better bound holds.

*Where the flat direction shows.* Now take  $\xi = (0, 0, \xi_3)$ , along the axis. Then  $|\xi'| = 0$  and  $J_0(0) = 1$ , so  $\widehat{\psi d\sigma}(0, 0, \xi_3) = 2\pi\widehat{\eta}(\xi_3)$ , which decays rapidly — but only because we *cut off* in  $x_3$ . For an infinite cylinder the transform in that direction is a delta, with no decay at all: the flat direction contributes nothing, exactly as Lemma 5.5.1 predicts, since the whole line  $\{x' = \text{const}\}$  is a critical set for the phase.

*The moral.* Decay in a given direction  $\omega$  is governed by the curvature of  $S$  at the points where  $\omega$  is normal, and by nothing else. A surface may be highly curved somewhere and flat elsewhere; the transform will decay quickly in the directions normal to the curved part and slowly in those normal to the flat part.

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**Pitfall:**  $k/2$  is sharp only when the degenerate directions are flat

Worked Example 5.5.2 showed the exponent  $k/2$  is attained, and it is easy to read that as saying Theorem 5.5.3 is sharp in general. It is not. The example was a cylinder, which is *affine* in its degenerate direction, and that is exactly the case in which nothing better is true.

Suppose instead the degeneracy is of finite order. Take  $n = 3$  and the graph  $x_3 = x_1^2 + x_2^4$ , whose second fundamental form at the origin has rank 1: one principal curvature vanishes. Theorem 5.5.3 gives  $|\widehat{\psi d\sigma}(\xi)| \lesssim |\xi|^{-1/2}$ . But in the normal direction the phase is  $x_1^2 + x_2^4$ , and the two variables separate:

$$\left| \int e^{i\lambda(x_1^2 + x_2^4)} a(x) dx \right| \lesssim \lambda^{-1/2} \cdot \lambda^{-1/4} = \lambda^{-3/4},$$

the first factor by Theorem 5.2.3 and the second by van der Corput with  $k = 4$  (Theorem 5.1.2). So the true decay is  $|\xi|^{-3/4}$ , better than the theorem's  $|\xi|^{-1/2}$ .

The reason Theorem 5.5.3 misses this is structural: its proof discards the degenerate variables, integrating them out in Step 4 and extracting nothing from them, so any oscillation they carry is thrown away. Determining the sharp exponent for a surface with finite-order degeneracies means reading it off the *Newton polyhedron* of the phase — Varchenko's theory — which this series does not develop and does not need: every surface appearing in Volumes 6 to 13 has non-vanishing Gaussian curvature, where Theorem 5.5.2 applies and is sharp.

WORKED EXAMPLE 5.5.3— TIER C: RESEARCH TASTE **From decay to the restriction conjecture's exponents**

The debt from Volume 2 is now paid, and it is worth recording precisely what has been bought and what has not.

(a) *What is now available.* For the sphere and, more generally, for any hypersurface with non-vanishing Gaussian curvature:

$$\widehat{d\sigma}(\xi) = |\xi|^{-\frac{n-1}{2}} \sum_{\pm} a_{\pm}(\omega) e^{\mp 2\pi i |\xi|} + O(|\xi|^{-\frac{n+1}{2}}),$$

with the *phase* explicit. Volume 2 had only the modulus, and only for the sphere.

(b) *What that yields immediately.* Worked Example 2.8.3 converted the bound  $|\widehat{d\sigma}| \lesssim |\xi|^{-(n-1)/2}$  into the restriction estimate  $\|Eg\|_{L^q} \lesssim \|g\|_{L^2(d\sigma)}$  at  $q = \frac{4n}{n-1}$ , by  $TT^*$  and Hardy–Littlewood–Sobolev. That argument now applies

verbatim to every hypersurface with non-vanishing curvature, and with  $k$  curvatures gives the corresponding exponent  $q = \frac{4n}{k}$ .

(c) *What the phase buys.* Volume 6 proves the Stein–Tomas theorem, with the better exponent  $\frac{2(n+1)}{n-1}$ . Its proof decomposes  $\widehat{d\sigma}$  dyadically and interpolates between an  $L^1 \rightarrow L^\infty$  bound coming from the decay and an  $L^2 \rightarrow L^2$  bound coming from Plancherel. The second uses that  $\widehat{d\sigma}$  *oscillates*: the dyadic pieces have Fourier transforms supported near the sphere, which is what makes the  $L^2$  estimate available. An argument with only  $|\widehat{d\sigma}|$  cannot see this, which is why Volume 2’s Theorem 8.5 was insufficient and Theorem 5.5.2 was needed.

(d) *What no amount of stationary phase will buy.* The restriction conjecture concerns  $L^\infty(d\sigma)$  data, and asserts an estimate for  $\|Eg\|_{L^q}$  down to  $q > \frac{2n}{n-1}$ . That is not a statement about the decay of a single oscillatory integral. Writing  $g = \sum_\theta g_\theta$  over  $\delta$ -caps,  $Eg$  is a superposition of  $\delta^{-(n-1)/2}$  wave packets, each concentrated on a tube dual to its cap, and the question is how much those tubes can overlap. Stationary phase describes one packet; the conjecture is about  $\delta^{-(n-1)/2}$  of them at once.

That is the boundary of this volume. Chapter 5.7 pushes the oscillatory integral method as far as it goes — to an  $L^2$  bound for operators, which is what Stein–Tomas needs — and everything past it is the geometry of Volumes 9–13.

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**Figure 5.5.1 (to be drawn).** *Critical points are normal directions*

**Content required:** A closed convex curve in the plane with a direction  $\omega$  marked; the two points where the normal is parallel to  $\omega$ , and the tangent lines there. Beside it, the same picture for a curve with a flat segment, where the whole segment is critical. Together these illustrate Lemma 5.5.1 and why flatness destroys decay.

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WORKED EXAMPLE 5.5.4— TIER A **Decay exponents for surfaces one meets**

Theorem 5.5.3 gives decay  $|\xi|^{-k/2}$  where  $k$  is the number of non-vanishing principal curvatures. A table, with the value of  $k$  in each case.



| Surface in $\mathbb{R}^n$                     | $k$     | decay              | used in               |
|-----------------------------------------------|---------|--------------------|-----------------------|
| sphere $\mathbb{S}^{n-1}$                     | $n - 1$ | $ \xi ^{-(n-1)/2}$ | everywhere            |
| paraboloid, truncated                         | $n - 1$ | $ \xi ^{-(n-1)/2}$ | Vol 6, Schrödinger    |
| cone, truncated                               | $n - 2$ | $ \xi ^{-(n-2)/2}$ | Vol 6, wave equation  |
| cylinder $\mathbb{S}^{n-2} \times \mathbb{R}$ | $n - 2$ | $ \xi ^{-(n-2)/2}$ | Worked Example 5.5.2  |
| hyperplane                                    | 0       | none               | Vol 6, no restriction |
| graph of $x_3 = x_1^2 + x_2^4$ at 0           | 1       | $ \xi ^{-1/2}$     | the Pitfall of §5.4   |

*Reading the last row.* The theorem gives  $k = 1$  and hence  $|\xi|^{-1/2}$ , because one principal curvature vanishes at the origin. But the *true* decay in the degenerate direction is  $|\xi|^{-3/4}$ , since the vanishing is of finite order and Lemma 2.8.4 with  $k = 4$  applies there. So Theorem 5.5.3 is not sharp for degenerate surfaces — it is sharp only when the vanishing curvatures vanish identically, as for a cylinder.

*The practical rule.* Count non-vanishing principal curvatures for a lower bound on the decay; if some curvature vanishes at isolated points, the true decay may be better and requires a separate van der Corput computation at the order of vanishing.

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## Drills

**Drill 5.5.1.** Verify the claim in Step 4 of Theorem 5.5.2 that the surface measure density  $\sqrt{1 + |\nabla h|^2}$  equals 1 at a critical point.

**Drill 5.5.2.** Compute the Gaussian curvature of the paraboloid  $x_n = |x'|^2$  and confirm that Theorem 5.5.2 applies on any compact piece. What happens as the piece recedes to infinity?

**Drill 5.5.3.** For the cylinder of Worked Example 5.5.2, compute  $\widehat{\psi d\sigma}(\xi)$  for  $\xi = (\xi', \xi_3)$  with both parts non-zero, and identify the decay rate in terms of the angle between  $\xi$  and the axis.

**Drill 5.5.4.** Let  $S$  be the graph of  $x_3 = x_1^2$  in  $\mathbb{R}^3$ , a parabolic cylinder. Determine  $k$  at each point and state the decay rate that Theorem 5.5.3 gives.

**Where we are.** This chapter is the hinge of the series, and it is worth stating its output in the form the later volumes consume.

*What goes in.* A compact hypersurface  $S$  with its surface measure, and the stationary phase machinery of Chapter 5.2 and Chapter 5.3.

*What comes out.* Two statements. First,  $|\widehat{d\sigma}(\xi)| \lesssim |\xi|^{-k/2}$  where  $k$  is the curvature rank (Theorem 5.5.3). Second, and used less often but essentially, the full asymptotic with its oscillating phase (Theorem 5.5.2):  $\widehat{d\sigma}$  does not merely decay, it decays *while oscillating like a cosine in  $|\xi|$* .

*Why the second matters.* Volume 8's reduction of the distance-set problem turns an integral in the radial variable into a Fourier transform, and Plancherel then applies exactly. With the pointwise bound alone the argument fails — the Pitfall there records that no choice of weight repairs it. So the phase is not decoration; it is the content.

*Where the geometry enters, once.* In Lemma 5.5.1: the phase  $x \cdot \omega$  is stationary on  $S$  exactly at the points where the normal is parallel to  $x$ , and its Hessian there is the second fundamental form. Every appearance of curvature in Volumes 6 through 9 traces back to that single lemma.

## Notes

Theorem 5.5.2 is classical, going back to Herz (1962) for the sphere and to Littman (1963) in general; Theorem 5.5.3 is Littman's. The formulation in terms of the Gauss map, which makes the finiteness of the critical set transparent, is standard in the modern literature.

Readers should be aware that the uniformity in  $\omega$  obtained in Step 5 is genuinely needed for the applications: Volume 6 integrates the estimate over  $\omega$ , and a bound with an  $\omega$ -dependent constant would be useless there.

## Chapter 5.6

# Analytic Continuation: Riesz Potentials

Twice already this volume has evaluated a divergent-looking integral by continuing a convergent one: the Fresnel integral in Proposition 5.2.1 and the Gamma integral in Lemma 5.4.2. Both times the device was to regard the quantity as the boundary value of a holomorphic function of a complex parameter. This chapter turns the device into a systematic tool, applied to the family  $|x|^{-a}$ , and produces a family of distributions on which the Fourier transform acts as the reflection  $a \mapsto n - a$ .

### 5.6.1 The problem

Worked Example 2.5.2 proved that for  $0 < a < n$  the function  $|x|^{-a}$  is a tempered distribution with

$$\widehat{|\cdot|^{-a}}(\xi) = c_{n,a} |\xi|^{-(n-a)}, \quad c_{n,a} = \pi^{a-\frac{n}{2}} \frac{\Gamma(\frac{n-a}{2})}{\Gamma(\frac{a}{2})}. \quad (5.6.1)$$

The restriction  $0 < a < n$  was needed twice: for  $a < n$  so that  $|x|^{-a}$  is locally integrable at the origin, and for  $a > 0$  so that the same holds for  $|\xi|^{-(n-a)}$ .

For  $a \geq n$  the function  $|x|^{-a}$  is not locally integrable, so  $\int |x|^{-a} \varphi$  diverges and the pairing that defines a distribution is meaningless. This is not a technicality to be waved away: the exponent  $a = n$  is exactly the one at which the Riesz potentials degenerate, and the objects on the far side of it — most importantly the delta function — are the ones one wants.

*Scratch work.* The remedy is the one used twice already. Regard

$$F_\varphi(a) = \int_{\mathbb{R}^n} |x|^{-a} \varphi(x) \, dx$$

as a function of the complex variable  $a$ . On the strip  $0 < \operatorname{Re} a < n$  it converges and is holomorphic. If it extends meromorphically to a larger region, then the extension *defines*  $|x|^{-a}$  as a distribution there, and any identity proved on the strip — such as (5.6.1) — automatically persists, because two meromorphic functions agreeing on an open set agree wherever both are defined.

The continuation is obtained by subtracting the Taylor polynomial of  $\varphi$  at the origin, which is where the divergence lives. Each subtracted monomial can be integrated explicitly and contributes one simple pole. So the poles are computable, and their locations turn out to be  $a = n, n+2, n+4, \dots$  — with the odd values absent, for a reason of symmetry that is worth seeing.

## 5.6.2 Meromorphic continuation

**Theorem 5.6.1.** *For  $\varphi \in \mathcal{S}(\mathbb{R}^n)$  the function*

$$F_\varphi(a) = \int_{\mathbb{R}^n} |x|^{-a} \varphi(x) \, dx, \quad 0 < \operatorname{Re} a < n,$$

*extends to a meromorphic function on  $\mathbb{C}$  whose only singularities are simple poles at*

$$a = n, n+2, n+4, \dots$$

*with*

$$\operatorname{Res}_{a=n+2k} F_\varphi = - \frac{\sigma(\mathbb{S}^{n-1})}{2^k k! n(n+2) \cdots (n+2k-2)} (\Delta^k \varphi)(0) \quad \text{for } k \geq 0,$$

*where for  $k = 0$  the empty product is 1. Moreover, for each fixed  $a$  outside the poles,  $\varphi \mapsto F_\varphi(a)$  is a tempered distribution.*

**Proof. Step 1.** *Split at the unit ball.*

Write  $F_\varphi(a) = A(a) + B(a)$  with

$$A(a) = \int_{|x| < 1} |x|^{-a} \varphi(x) \, dx, \quad B(a) = \int_{|x| \geq 1} |x|^{-a} \varphi(x) \, dx.$$

The integral  $B$  converges absolutely for *every*  $a \in \mathbb{C}$ : on  $|x| \geq 1$  we have  $||x|^{-a}| = |x|^{-\operatorname{Re} a} \leq |x|^{|\operatorname{Re} a|}$ , which is polynomially bounded, and  $\varphi$  decays

faster than any polynomial. By differentiation under the integral sign (Worked Example 1.5.3),  $B$  is entire.

**Step 2.** *Subtract the Taylor polynomial.*

Fix  $N \in \mathbb{N}$  and write, for  $|x| < 1$ ,

$$\varphi(x) = \sum_{|\alpha| \leq N} \frac{\partial^\alpha \varphi(0)}{\alpha!} x^\alpha + R_N(x), \quad |R_N(x)| \leq C_N |x|^{N+1},$$

by Taylor's theorem. The remainder term  $\int_{|x| < 1} |x|^{-a} R_N(x) dx$  converges absolutely whenever  $\operatorname{Re} a < n + N + 1$ , and is holomorphic there by the same argument as in Step 1.

**Step 3.** *The explicit monomial integrals.*

For a multi-index  $\alpha$ , polar coordinates (Theorem 1.6.11) give

$$\int_{|x| < 1} |x|^{-a} x^\alpha dx = \left( \int_{\mathbb{S}^{n-1}} \omega^\alpha d\sigma(\omega) \right) \int_0^1 r^{-a+|\alpha|+n-1} dr = \frac{m_\alpha}{n + |\alpha| - a},$$

valid for  $\operatorname{Re} a < n + |\alpha|$ , where  $m_\alpha = \int_{\mathbb{S}^{n-1}} \omega^\alpha d\sigma$ .

Now  $m_\alpha = 0$  unless every  $\alpha_j$  is even: replacing  $\omega_j$  by  $-\omega_j$  is a measure-preserving symmetry of the sphere under which  $\omega^\alpha$  changes sign when  $\alpha_j$  is odd. *This is why only even  $|\alpha|$  produce poles*, and hence why the poles occur at  $a = n + 2k$  and not at  $a = n + 1, n + 3, \dots$

**Step 4.** *Assemble.*

Combining,

$$F_\varphi(a) = B(a) + \int_{|x| < 1} |x|^{-a} R_N(x) dx + \sum_{\substack{|\alpha| \leq N \\ \alpha \text{ even}}} \frac{\partial^\alpha \varphi(0)}{\alpha!} \frac{m_\alpha}{n + |\alpha| - a}.$$

The first two terms are holomorphic on  $\operatorname{Re} a < n + N + 1$  and the third is a finite sum of simple poles at  $a = n + |\alpha|$  with  $|\alpha|$  even. Since  $N$  was arbitrary, this defines the meromorphic continuation to all of  $\mathbb{C}$ , and the poles are exactly at  $a = n + 2k$ .

**Step 5.** *The residues.*

The pole at  $a = n + 2k$  comes from the terms with  $|\alpha| = 2k$ , and  $\operatorname{Res}_{a=n+2k} \frac{1}{n+2k-a} = -1$ . So

$$\operatorname{Res}_{a=n+2k} F_\varphi = - \sum_{|\alpha|=2k} \frac{\partial^\alpha \varphi(0)}{\alpha!} m_\alpha.$$

It remains to identify this sum as a multiple of  $\Delta^k \varphi(0)$ . Both sides are linear functionals of the degree- $2k$  part of the Taylor expansion of  $\varphi$ , invariant under

rotations — the left because  $d\sigma$  is, the right because  $\Delta$  is — and the space of rotation-invariant such functionals is one-dimensional. Comparing on  $\varphi(x) = |x|^{2k}$ , for which  $\Delta^k |x|^{2k} = 2^k k! n(n+2) \cdots (n+2k-2)$  and  $\sum_{|\alpha|=2k} \frac{m_\alpha}{\alpha!} \cdot$  (coefficient)  $= \sigma(\mathbb{S}^{n-1})$ , gives the stated constant.

**Step 6. *Temperedness.***

For fixed  $a$  off the poles, each of the three terms in Step 4 is bounded by a finite sum of Schwartz seminorms of  $\varphi$ :  $B$  by the decay of  $\varphi$ , the remainder integral by  $\sup_{|\alpha| \leq N+1} \|\partial^\alpha \varphi\|_\infty$  through the Taylor bound, and the finite sum by the derivatives at 0. So  $\varphi \mapsto F_\varphi(a)$  satisfies the continuity estimate of Volume 2, Definition 5.7.  $\square$

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**WORKED EXAMPLE 5.6.1— TIER A The residue at  $a = n$  is a delta function**

Take  $k = 0$  in Theorem 5.6.1. The residue is

$$\operatorname{Res}_{a=n} F_\varphi = -\sigma(\mathbb{S}^{n-1}) \varphi(0),$$

that is, as distributions,

$$\operatorname{Res}_{a=n} |x|^{-a} = -\sigma(\mathbb{S}^{n-1}) \delta_0.$$

This is worth pausing on. The family  $|x|^{-a}$  consists of honest functions for  $a < n$ , and at  $a = n$  it blows up — but the blow-up is not formless. The residue is a delta: the mass that was spread over  $\mathbb{R}^n$  concentrates entirely at the origin as  $a \uparrow n$ , and the rate of concentration is measured by  $\sigma(\mathbb{S}^{n-1})$ .

The computation behind it is the divergence of  $\int_0^1 r^{-a+n-1} dr$  at  $a = n$ , which is the statement that  $|x|^{-n}$  just fails to be integrable at the origin — the same borderline that made  $\mathcal{M}f \notin L^1$  in Volume 1 and made singular integrals fail on  $L^1$  in Volume 3. Here the failure is not an obstruction but the source of the delta.

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### 5.6.3 The Riesz family and the reflection formula

The poles of Theorem 5.6.1 can be removed by dividing by a Gamma function with matching poles. The resulting family is entire, and the Fourier transform acts on it as simply as possible.

**Definition 5.6.2.** For  $a \in \mathbb{C}$  define the *Riesz distribution*

$$Z_a = \frac{\pi^{-a/2}}{\Gamma(\frac{n-a}{2})} |x|^{-a},$$

where  $|x|^{-a}$  is understood via Theorem 5.6.1.

**Proposition 5.6.3.**  $a \mapsto Z_a$  is an entire family of tempered distributions: for every  $\varphi \in \mathcal{S}$ , the function  $a \mapsto \langle Z_a, \varphi \rangle$  is entire.

*Proof.* By Theorem 5.6.1 the only singularities of  $\langle |x|^{-a}, \varphi \rangle$  are simple poles at  $a = n + 2k$ ,  $k \geq 0$ . The function  $1/\Gamma(\frac{n-a}{2})$  is entire —  $1/\Gamma$  is — and vanishes exactly where  $\Gamma(\frac{n-a}{2})$  has poles, namely where  $\frac{n-a}{2} \in \{0, -1, -2, \dots\}$ , that is  $a = n + 2k$ ; and those zeros are simple. So each simple pole is cancelled by a simple zero, and the product is holomorphic at  $a = n + 2k$ ; elsewhere both factors are holomorphic.  $\square$

**Theorem 5.6.4** (Reflection formula). For every  $a \in \mathbb{C}$ ,

$$\widehat{Z}_a = Z_{n-a}$$

as tempered distributions.

*Idea of the proof.* On the strip  $0 < \operatorname{Re} a < n$  this is Volume 2's computation (5.6.1), rearranged — the normalisation in Definition 5.6.2 was chosen for precisely this purpose. Both sides are entire in  $a$  by Proposition 5.6.3 and the continuity of the Fourier transform on  $\mathcal{S}'$ , so agreement on a strip forces agreement everywhere. That is the entire proof: the analytic continuation does the work that would otherwise require a separate argument at every exponent.

*Proof. Step 1.* Verify the identity on the strip.

Let  $0 < a < n$  be real. By (5.6.1),

$$\widehat{Z}_a = \frac{\pi^{-a/2}}{\Gamma(\frac{n-a}{2})} c_{n,a} |\xi|^{-(n-a)} = \frac{\pi^{-a/2}}{\Gamma(\frac{n-a}{2})} \cdot \pi^{a-\frac{n}{2}} \frac{\Gamma(\frac{n-a}{2})}{\Gamma(\frac{a}{2})} |\xi|^{-(n-a)} = \frac{\pi^{\frac{a-n}{2}}}{\Gamma(\frac{a}{2})} |\xi|^{-(n-a)}.$$

On the other hand, substituting  $a \mapsto n - a$  in Definition 5.6.2,

$$Z_{n-a} = \frac{\pi^{-(n-a)/2}}{\Gamma(\frac{a}{2})} |\xi|^{-(n-a)} = \frac{\pi^{\frac{a-n}{2}}}{\Gamma(\frac{a}{2})} |\xi|^{-(n-a)}.$$

The two agree.

**Step 2.** Continue.

Fix  $\varphi \in \mathcal{S}$ . The functions

$$a \mapsto \langle \widehat{Z}_a, \varphi \rangle = \langle Z_a, \widehat{\varphi} \rangle \quad \text{and} \quad a \mapsto \langle Z_{n-a}, \varphi \rangle$$

are both entire: the first because  $a \mapsto Z_a$  is entire (Proposition 5.6.3) and  $\widehat{\varphi} \in \mathcal{S}$  by Theorem 2.5.6; the second because  $a \mapsto n - a$  is entire and  $Z$  is. By Step 1 they agree for  $a \in (0, n)$ , a set with a limit point, so by the propagation argument of Theorem 2.1.10 they agree for all  $a \in \mathbb{C}$ . As  $\varphi$  was arbitrary,  $\widehat{Z}_a = Z_{n-a}$ .  $\square$

**WORKED EXAMPLE 5.6.2— TIER B Reading off the special values**

The reflection formula turns several familiar identities into instances of one statement, and it is a good exercise to extract them.

(a)  $a = 0$ . Here  $Z_0 = \frac{1}{\Gamma(n/2)}|x|^0 = \frac{1}{\Gamma(n/2)}$ , the constant function. Theorem 5.6.4 gives

$$Z_n = \widehat{Z_0} = \frac{1}{\Gamma(n/2)}\widehat{1} = \frac{\delta_0}{\Gamma(n/2)},$$

using Worked Example 2.5.2. So the Riesz distribution at the critical exponent  $a = n$  is a multiple of the delta function — consistent with Worked Example 5.6.1, where the pole of  $|x|^{-a}$  had residue proportional to  $\delta$ , and the factor  $1/\Gamma(\frac{n-a}{2})$  turns that residue into a finite value.

(b)  $a = n/2$ . The family is self-dual there:  $\widehat{Z_{n/2}} = Z_{n/2}$ , so  $\pi^{-n/4}|x|^{-n/2}/\Gamma(n/4)$  is its own Fourier transform. Compare the Gaussian, Proposition 2.4.3, which is the other standard self-dual object; the two are quite different functions, and the coincidence of being fixed by  $\mathcal{F}$  is a statement about the exponent  $n/2$ , not about the shape.

(c) *Negative  $a$ .* For  $a < 0$ ,  $|x|^{-a} = |x|^{|a|}$  is a polynomially growing function — a perfectly good tempered distribution with no continuation required — and Theorem 5.6.4 computes its transform as  $Z_{n-a}$  with  $n-a > n$ , i.e. a distribution supported at the origin when  $n-a \in \{n, n+2, \dots\}$ . For instance  $a = -2$  gives  $|x|^2$  on the left, and on the right  $Z_{n+2}$ , which by the zero of  $1/\Gamma$  is a multiple of  $\Delta\delta_0$ . This is the familiar fact that the transform of a polynomial is a derivative of a delta, obtained here as a special value of an analytic family.

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**WORKED EXAMPLE 5.6.3— TIER C: RESEARCH TASTE Analytic continuation as a proof technique**

This chapter used a single move three times, and it is worth abstracting, because it is one of the standard tools of the subject and appears in the research literature under several names.

(a) *The move.* To prove an identity or an estimate at a parameter value where the objects do not converge, embed the problem in a family depending holomorphically on a complex parameter, prove it where things do converge, and continue. The identity theorem then transports the conclusion.

Instances so far: Proposition 5.2.1, continuing  $\int e^{-zt^2}$  from  $\operatorname{Re} z > 0$  to the imaginary axis; Lemma 5.4.2, doing the same for  $\int e^{-zs}s^{a-1}$ ; and Theorem 5.6.4, continuing in the exponent rather than in a variable of integration.



(b) *Where it is used later.* Chapter 5.8 defines the Bochner–Riesz means  $S^\delta$  for  $\operatorname{Re} \delta > 0$  and continues in  $\delta$ ; the resulting family is exactly of the type above, and the kernel estimates one needs at real  $\delta$  are obtained from the range where the integrals converge absolutely.

(c) *The version we do not prove.* Continuing an *estimate* rather than an identity requires more: one has a family of operators  $T_z$  and bounds at  $\operatorname{Re} z = 0$  and  $\operatorname{Re} z = 1$ , and wants a bound in between. That is Stein’s interpolation theorem for analytic families, which strengthens Riesz–Thorin (Theorem 2.2.1) by allowing the operator, not only the test function, to vary. Its proof needs a Poisson-kernel form of the three-lines lemma, and it is recorded as revision R4: nothing in this series requires it, because Chapter 5.8 works with a single operator at a time.

Readers going on to the Bochner–Riesz literature will meet Stein interpolation constantly, and should know that it is the natural completion of the technique of this chapter rather than a separate idea.

(d) *A caution about what continuation cannot do.* It transports identities and, with Stein’s theorem, bounds — but only within the region where the family is holomorphic. It cannot cross a pole, and it cannot manufacture information that was absent on the initial domain. In particular, no analytic continuation will prove the restriction conjecture: the conjecture is not the boundary value of a family of estimates that hold on an open set of parameters, it is a statement about a fixed operator whose difficulty is geometric. Recognising which problems yield to continuation and which do not is a large part of knowing the subject.

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WORKED EXAMPLE 5.6.4— TIER B **Analytic continuation as a bookkeeping device**

Several constants in this volume are defined first for parameters where an integral converges, and then extended. The device is worth isolating.

*The pattern.* A quantity  $I(z)$  is given by an integral converging for  $\operatorname{Re} z$  in a half-plane, is shown holomorphic there, and is then continued by an identity valid on the overlap — typically integration by parts, which trades a power of the variable for a derivative and enlarges the domain.

*The instance.*  $\Gamma(z) = \int_0^\infty t^{z-1} e^{-t} dt$  converges for  $\operatorname{Re} z > 0$ ; the identity  $\Gamma(z+1) = z\Gamma(z)$  continues it to  $\operatorname{Re} z > -1$ , then  $-2$ , and so on, with simple poles at the non-positive integers. Lemma 5.4.2 is the oscillatory analogue.

*Why it is more than convenience.* The continued function satisfies the same identities as the original, by the identity theorem — two holomorphic functions agreeing on a set with a limit point agree everywhere. So a formula

proved where the integral converges holds where it does not, *provided both sides continue*. That proviso is the only place care is needed.

*Where it is used.* The constant in Proposition 5.2.1, whose branch is fixed by continuation from the real axis; the Riesz potentials of Volume 7, Proposition 7.6.5, where  $\widehat{|x|^{-t}} = c_{n,t}|\xi|^{t-n}$  is first proved for  $0 < t < n$  and then extended; and the Schrödinger propagator of Volume 6, Proposition 6.4.4, whose Gaussian kernel is the heat kernel continued to imaginary time.

*The caution.* Continuation preserves identities, not inequalities. A bound  $|I(z)| \leq M$  on a half-plane says nothing about the continued function elsewhere — for that one needs a Phragmén–Lindelöf argument, which is what Theorem 2.1.14 in Volume 2 supplies.

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## Drills

**Drill 5.6.1.** Verify that  $m_\alpha = \int_{S^{n-1}} \omega^\alpha d\sigma$  vanishes when some  $\alpha_j$  is odd, and compute  $m_\alpha$  for  $\alpha = (2, 0, \dots, 0)$ .

**Drill 5.6.2.** Check the constant in Theorem 5.6.1's residue formula for  $k = 1$  by computing both sides on  $\varphi(x) = e^{-\pi|x|^2}$ .

**Drill 5.6.3.** Show directly from Definition 5.6.2 that  $Z_a$  is homogeneous of degree  $-a$ , and deduce that  $\widehat{Z}_a$  is homogeneous of degree  $a - n$  — consistent with Theorem 5.6.4.

**Drill 5.6.4.** Use Theorem 5.6.4 to compute the Fourier transform of  $|x|^{-n-2}$ , interpreted via Theorem 5.6.1, and identify the answer as a multiple of  $\Delta\delta_0$  plus a correction.

**Where we are.** A remark on why this chapter exists at all, since analytic continuation is not an estimate and proves nothing about oscillatory integrals directly.

The constants produced by stationary phase are contour integrals, and contour integrals are most easily evaluated where they converge absolutely — which is usually not where they are needed. Proposition 5.2.1 is the model: the Fresnel integral  $\int e^{is^2} ds$  does not converge absolutely at all, and its value  $\sqrt{\pi}e^{i\pi/4}$  is obtained by continuing the Gaussian  $\int e^{-as^2} ds = \sqrt{\pi/a}$  from  $a > 0$  to  $a = -i$ .

That single continuation supplies the factor  $e^{i\pi/4}$  in Theorem 5.2.3, hence the signature in Theorem 5.3.3, hence the phase in Theorem 5.5.2 — which Volume 8 then requires. So the chapter's apparent detour into complex

analysis is the source of a factor that later turns out to be indispensable. The general shape is worth naming: *compute where it converges, continue to where it is needed, and check that both sides continue*. The last clause is the only place errors occur.

## Notes

The meromorphic continuation of  $|x|^{-a}$  and the reflection formula are due to M. Riesz (1949); the systematic treatment of families of distributions depending holomorphically on a parameter is Gelfand and Shilov's.

The normalisation in Definition 5.6.2 is chosen so that Theorem 5.6.4 has no constant. Readers will meet several other normalisations in the literature, differing by factors of  $\pi$  and 2; the invariant statement is that there is a choice making the Fourier transform act as  $a \mapsto n - a$ , and any two such choices differ by a factor  $c(a)$  with  $c(a)c(n - a) = 1$ .



## Chapter 5.7

# Oscillatory Integral Operators

So far the oscillation has been integrated out to a number. This chapter treats oscillatory integrals as *operators*: the phase depends on two variables, one is integrated and one is left free, and the question is boundedness on  $L^2$ . The answer, Theorem 5.7.3, is what Volume 6 builds the Stein–Tomas theorem on.

### 5.7.1 The set-up and the theorem

**Definition 5.7.1.** Let  $\Phi \in C^\infty(\mathbb{R}^n \times \mathbb{R}^n)$  be real-valued and  $a \in C_c^\infty(\mathbb{R}^n \times \mathbb{R}^n)$ . For  $\lambda > 0$  define

$$(I_\lambda f)(x) = \int_{\mathbb{R}^n} e^{i\lambda\Phi(x,y)} a(x,y) f(y) \, dy.$$

Since  $a$  is compactly supported,  $I_\lambda$  is bounded on every  $L^p$  with norm  $O(1)$ , uniformly in  $\lambda$ , simply by the triangle inequality. The content of the theory is that oscillation improves this to a power of  $\lambda$ , and the improvement is governed by a non-degeneracy condition on the phase.

**Definition 5.7.2.** The *mixed Hessian* of  $\Phi$  is the  $n \times n$  matrix

$$\Phi''_{xy}(x,y) = \left( \frac{\partial^2 \Phi}{\partial x_i \partial y_j}(x,y) \right)_{i,j=1}^n.$$

The phase is *non-degenerate* on a set if  $\det \Phi''_{xy} \neq 0$  there.

**Theorem 5.7.3** (Hörmander). *Suppose  $\det \Phi''_{xy} \neq 0$  at every point of  $\text{supp } a$ . Then*

$$\|I_\lambda f\|_{L^2(\mathbb{R}^n)} \lesssim \lambda^{-n/2} \|f\|_{L^2(\mathbb{R}^n)} \quad (\lambda \geq 1),$$

*with the constant depending on finitely many derivatives of  $\Phi$  and  $a$  and on a lower bound for  $|\det \Phi''_{xy}|$  on  $\text{supp } a$ .*

*Idea of the proof.* Estimate  $I_\lambda I_\lambda^*$  rather than  $I_\lambda$ . Its kernel is

$$K_\lambda(x, z) = \int_{\mathbb{R}^n} e^{i\lambda(\Phi(x, y) - \Phi(z, y))} a(x, y) \overline{a(z, y)} dy,$$

an oscillatory integral in  $y$  with phase  $\Psi(y) = \Phi(x, y) - \Phi(z, y)$ . The crucial observation is that

$$\nabla_y \Psi(y) = \left( \int_0^1 \Phi''_{xy}(z + t(x - z), y) dt \right) (x - z),$$

so the non-degeneracy hypothesis makes  $|\nabla_y \Psi| \gtrsim |x - z|$ . The phase therefore moves at rate  $\lambda|x - z|$ , and non-stationary phase gives  $|K_\lambda(x, z)| \lesssim_N (1 + \lambda|x - z|)^{-N}$ .

A kernel of that shape has  $\int |K_\lambda(x, z)| dz \lesssim \lambda^{-n}$ , so the Schur test bounds  $I_\lambda I_\lambda^*$  by  $\lambda^{-n}$  on  $L^2$ , and taking square roots gives  $\lambda^{-n/2}$ .

Note where the exponent comes from: it is the volume of the set  $\{|x - z| \lesssim \lambda^{-1}\}$ , that is, the region in which the two copies of the phase have not yet separated. Each of the  $n$  variables contributes a factor  $\lambda^{-1}$ , and the square root of  $\lambda^{-n}$  is  $\lambda^{-n/2}$ .

**Lemma 5.7.4** (Schur test). *Let  $K$  be measurable on  $\mathbb{R}^n \times \mathbb{R}^n$  with*

$$A = \sup_x \int |K(x, z)| dz < \infty, \quad B = \sup_z \int |K(x, z)| dx < \infty.$$

*Then  $Tf(x) = \int K(x, z)f(z) dz$  satisfies  $\|Tf\|_{L^2} \leq \sqrt{AB} \|f\|_{L^2}$ .*

*Proof.* By Cauchy–Schwarz, splitting  $|K| = |K|^{1/2} \cdot |K|^{1/2}$ ,

$$|Tf(x)|^2 \leq \left( \int |K(x, z)| dz \right) \left( \int |K(x, z)| |f(z)|^2 dz \right) \leq A \int |K(x, z)| |f(z)|^2 dz.$$

Integrating in  $x$  and applying Tonelli,

$$\|Tf\|_{L^2}^2 \leq A \int |f(z)|^2 \left( \int |K(x, z)| dx \right) dz \leq AB \|f\|_{L^2}^2. \quad \square$$

*Proof of Theorem 5.7.3. Step 1. Localise.*

By compactness of  $\text{supp } a$  and continuity of  $\det \Phi''_{xy}$ , there are  $c > 0$  and a finite cover of  $\text{supp } a$  by open sets  $U_1 \times V_1, \dots, U_M \times V_M$  of diameter less than  $\rho$ , on each of which  $|\det \Phi''_{xy}| \geq c$  and  $\|(\Phi''_{xy})^{-1}\| \leq c^{-1}$ . Take a subordinate partition of unity and write  $a = \sum_{j \leq M} a_j$ ; then  $I_\lambda = \sum_j I_\lambda^{(j)}$  and it suffices to bound each summand,  $M$  being finite. So assume  $\text{supp } a \subseteq U \times V$  with  $U$  convex of diameter  $< \rho$ .

**Step 2.** *The kernel of  $I_\lambda I_\lambda^*$ .*

A direct computation gives

$$(I_\lambda I_\lambda^* g)(x) = \int K_\lambda(x, z) g(z) dz, \quad K_\lambda(x, z) = \int e^{i\lambda \Psi_{x,z}(y)} a(x, y) \overline{a(z, y)} dy,$$

with  $\Psi_{x,z}(y) = \Phi(x, y) - \Phi(z, y)$ .

**Step 3.** *The gradient of the phase is bounded below.*

For  $x, z \in U$ , the segment from  $z$  to  $x$  lies in  $U$  by convexity, so the fundamental theorem of calculus applied to  $t \mapsto \nabla_y \Phi(z + t(x - z), y)$  gives

$$\nabla_y \Psi_{x,z}(y) = M(x, z, y)(x - z), \quad M = \int_0^1 \Phi''_{xy}(z + t(x - z), y) dt.$$

Each  $\Phi''_{xy}$  in the integrand has inverse of norm at most  $c^{-1}$ , and the integrand varies by at most  $\varepsilon(\rho) \rightarrow 0$  across  $U$ ; choosing  $\rho$  small enough that  $\varepsilon(\rho) < c/2$ , the average  $M$  is invertible with  $\|M^{-1}\| \leq 2c^{-1}$ . Hence

$$|\nabla_y \Psi_{x,z}(y)| \geq \frac{c}{2} |x - z| \quad \text{for } x, z \in U, y \in V. \quad (5.7.1)$$

**Step 4.** *Kernel decay.*

Fix  $x \neq z$ . Apply Theorem 5.1.1 to the  $y$ -integral with large parameter  $\lambda|x - z|$  — more precisely, write  $\lambda \Psi_{x,z} = (\lambda|x - z|) \tilde{\Psi}$  with  $\tilde{\Psi} = \Psi_{x,z}/|x - z|$ , which by (5.7.1) satisfies  $|\nabla_y \tilde{\Psi}| \geq c/2$  and has derivatives bounded uniformly in  $x, z$ . Then

$$|K_\lambda(x, z)| \lesssim_N (\lambda|x - z|)^{-N}.$$

Combining with the trivial bound  $|K_\lambda| \lesssim 1$ ,

$$|K_\lambda(x, z)| \lesssim_N (1 + \lambda|x - z|)^{-N}, \quad (5.7.2)$$

and  $K_\lambda(x, z) = 0$  unless  $x, z \in U$ .

**Step 5.** *Schur.*

Taking  $N = n + 1$  in (5.7.2) and integrating,

$$\int |K_\lambda(x, z)| dz \lesssim \int_{\mathbb{R}^n} \frac{dz}{(1 + \lambda|x - z|)^{n+1}} = \lambda^{-n} \int_{\mathbb{R}^n} \frac{dw}{(1 + |w|)^{n+1}} \lesssim \lambda^{-n},$$

by the substitution  $w = \lambda(z - x)$ , the last integral converging by polar coordinates. The same bound holds with the roles of  $x$  and  $z$  exchanged, since (5.7.2) is symmetric. By Lemma 5.7.4,  $\|I_\lambda I_\lambda^*\|_{L^2 \rightarrow L^2} \lesssim \lambda^{-n}$ .

**Step 6.** *Take the square root.*

For any bounded operator,  $\|I_\lambda\|^2 = \|I_\lambda I_\lambda^*\|$ , since  $\|I_\lambda f\|_2^2 = \langle I_\lambda^* I_\lambda f, f \rangle$  and the operator norms of  $I^* I$  and  $I I^*$  agree. Hence  $\|I_\lambda\|_{L^2 \rightarrow L^2} \lesssim \lambda^{-n/2}$ .  $\square$

**Pitfall:  $TT^*$  is not a trick, it is the only available move**

One cannot estimate  $I_\lambda$  directly, because there is nothing to integrate by parts against: the phase  $\Phi(x, y)$  oscillates in  $y$  for fixed  $x$ , but the  $L^2$  norm in  $x$  sees no oscillation at all.

Forming  $I_\lambda I_\lambda^*$  creates the needed structure: the kernel involves the *difference*  $\Phi(x, y) - \Phi(z, y)$ , whose  $y$ -gradient is controlled by  $|x - z|$  through the mixed Hessian. So the operator's smoothing is converted into decay of a kernel off the diagonal, which is a form one can estimate.

The same move appears in Worked Example 2.8.3, where  $EE^*$  was convolution with  $\widehat{d\sigma}$  and the decay of  $\widehat{d\sigma}$  became the whole input; and it will appear again in Volume 6 for the Stein–Tomas theorem. Whenever an operator resists direct estimation, the first thing to try is  $TT^*$ .

**WORKED EXAMPLE 5.7.1 — TIER B The extension operator is an oscillatory integral operator**

Worked Example 2.4.3 introduced the extension operator for the paraboloid,

$$Eg(x) = \int_{\mathbb{R}^{n-1}} g(\xi') e^{2\pi i(x' \cdot \xi' + x_n |\xi'|^2)} d\xi'.$$

This is exactly of the form Definition 5.7.1, and identifying the parameters explains where the scaling in the restriction problem comes from.

*Rescaling to unit frequency.* Fix  $R \geq 1$  and consider  $Eg$  on the ball  $|x| \leq R$ . Substituting  $x = R\tilde{x}$  gives

$$Eg(R\tilde{x}) = \int_{\mathbb{R}^{n-1}} g(\xi') e^{2\pi i R(\tilde{x}' \cdot \xi' + \tilde{x}_n |\xi'|^2)} d\xi',$$

which is  $I_\lambda$  with  $\lambda = 2\pi R$ , phase  $\Phi(\tilde{x}, \xi') = \tilde{x}' \cdot \xi' + \tilde{x}_n |\xi'|^2$ , and amplitude a cutoff to the cap.

*The mixed Hessian.* Here  $x \in \mathbb{R}^n$  and  $\xi' \in \mathbb{R}^{n-1}$ , so the phase is not square and Theorem 5.7.3 does not literally apply — this is the first sign that restriction is not merely an oscillatory-integral-operator problem. Computing anyway,

$$\frac{\partial^2 \Phi}{\partial \tilde{x}_i \partial \xi'_j} = \begin{cases} \delta_{ij}, & i \leq n-1, \\ 2\xi'_j, & i = n, \end{cases}$$

an  $n \times (n-1)$  matrix of full rank  $n-1$  everywhere. So the phase is non-degenerate in the appropriate rectangular sense.

*What the theorem gives, and what it does not.* Applying Theorem 5.7.3 in the  $n-1$  variables — treating  $\tilde{x}_n$  as a parameter, as in Theorem 5.5.3 — gives  $\|Eg\|_{L^2(|x| \leq R)} \lesssim R^{1/2} \|g\|_{L^2}$ , which after unwinding is the trivial  $L^2$  bound



and no more. The restriction conjecture concerns  $L^q$  for  $q < \infty$  and  $L^\infty$  data, and no  $L^2$  theorem reaches it.

*Why this is worth knowing.* The identification is standard in the literature: papers on restriction routinely write  $E$  as an oscillatory integral operator with parameter  $R$ , and speak of “the operator at scale  $R$ ”. A reader who has seen the translation will recognise the phrase; one who has not will find the rescaling mysterious. The genuine content of Volumes 12 and 13 begins where Theorem 5.7.3 stops, and knowing exactly where that is — at the passage from  $L^2$  to  $L^q$  — is worth more than any single estimate.

**WORKED EXAMPLE 5.7.2— TIER C: RESEARCH TASTE** **Oscillatory integral operators, and where this volume stops**

Chapter 5.7 treats operators  $T_\lambda f(x) = \int e^{i\lambda\varphi(x,y)}\psi(x,y)f(y) \, dy$ , and proves the  $L^2$  theory. It is worth saying what is proved and what is not.

(a) *What is proved.* Theorem 5.7.3: if the mixed Hessian  $\partial^2\varphi/\partial x\partial y$  is non-degenerate, then  $\|T_\lambda\|_{L^2 \rightarrow L^2} \lesssim \lambda^{-n/2}$ . The proof is  $TT^*$  followed by non-stationary phase, and it is sharp.

(b) *What is not.* The  $L^p$  theory for  $p \neq 2$ , which requires the Carleson–Sjölin condition and is descoped as revision R4. Nothing in this series uses it: Volume 6 works with  $L^2$  data on the sphere, and the extension operator estimates there are proved directly.

(c) *Why  $L^2$  suffices for what follows.* The Stein–Tomas theorem of Volume 6 is an  $L^2(d\sigma)$  statement, and its proof uses only the decay of  $\widehat{d\sigma}$  and interpolation — not the  $L^p$  theory of oscillatory operators. The restriction conjecture concerns  $L^\infty(d\sigma)$ , which the oscillatory-operator machinery does not reach either.

(d) *The connection worth knowing.* An extension operator *is* an oscillatory integral operator with  $\lambda = |\xi|$  and phase  $x \cdot \omega$ ; so the two subjects are the same one, viewed at fixed and at varying frequency. The Carleson–Sjölin theory is what a full treatment would use, and readers heading for the literature on local smoothing will meet it there.

(e) *The summary.* This volume supplies exactly two things to the rest of the series: the decay of  $\widehat{d\sigma}$  with its phase, and the  $L^2$  bound above. Both are used constantly; nothing else here is.

## Drills

**Drill 5.7.1.** Verify the identity  $\|T\|^2 = \|TT^*\|$  used in Step 6, for a bounded

operator on a Hilbert space.

**Drill 5.7.2.** Show that Theorem 5.7.3 fails without the non-degeneracy hypothesis, by taking  $\Phi(x, y) = x_1 y_1$  in  $\mathbb{R}^2$  — so that  $\Phi$  does not depend on  $x_2$  or  $y_2$  — and computing  $\|I_\lambda\|$ .

**Drill 5.7.3.** Carry out Step 5 with general  $N > n$  and determine how the constant depends on  $N$ .

**Drill 5.7.4.** Compute the mixed Hessian for  $\Phi(x, y) = |x - y|$  on a region where  $x \neq y$ , and determine its rank. (This is the phase of the wave propagator; the rank deficiency is why the wave equation is harder than the Schrödinger equation.)

**Where we are.** The  $TT^*$  device used in Theorem 5.7.3 is worth recording separately, since Volume 6 opens with it again in a different setting.

The point is always the same: an operator  $T$  may be hard to estimate directly, but  $TT^*$  is a *convolution-like* object whose kernel can be written down. Here  $T_\lambda T_\lambda^*$  has kernel  $\int e^{i\lambda(\varphi(x, y) - \varphi(x', y))} \psi \bar{\psi} dy$ , and the non-degeneracy of the mixed Hessian makes the phase non-stationary in  $y$  for  $x \neq x'$  — so Theorem 5.1.1 applies and the kernel decays. Schur's test then finishes it.

Two features generalise. The identity  $\|T\|_{L^2 \rightarrow L^2}^2 = \|TT^*\|_{L^2 \rightarrow L^2}$  is exact, so nothing is lost in passing to  $TT^*$ ; and the hypothesis one needs on  $\varphi$  is a statement about *differences*  $\varphi(x, y) - \varphi(x', y)$ , which is why non-degeneracy of the mixed Hessian — and not of  $\varphi$  itself — is the right condition.

Volume 6, Proposition 6.1.5 is the same identity with  $T = \mathcal{E}$ , and there the kernel of  $\mathcal{E}\mathcal{E}^*$  is  $\widehat{d\sigma}$  itself. So the two volumes' central computations are one computation.

## Notes

Theorem 5.7.3 is Hörmander's, from 1973. The  $L^p$  theory — which requires curvature conditions on the level sets of the phase rather than merely non-degeneracy of the mixed Hessian — is due to Stein and to Carleson–Sjölin in the plane; it is recorded as revision R4 and not proved here.

The rectangular case appearing in Worked Example 5.7.1, where  $x$  and  $y$  have different dimensions, is the setting of the restriction problem and is treated systematically in Volume 6.

## Chapter 5.8

# Bochner–Riesz Means

Volume 3 proved that the sharp cutoff to the unit ball is a bounded Fourier multiplier on  $L^p(\mathbb{R}^n)$  only for  $p = 2$ , and that the obstruction is a Besicovitch set. This chapter shows what happens when the cutoff is smoothed: raising the distance to the boundary to a positive power  $\delta$  restores boundedness, and for  $\delta$  large enough the proof is a computation with the tools of this volume.

The chapter is the payoff of Volume 5, in that its proof uses Lemma 5.4.2, Theorem 5.4.1 and Theorem 5.5.2 together.

### 5.8.1 The operators

**Definition 5.8.1.** For  $\delta > 0$  the *Bochner–Riesz multiplier* of order  $\delta$  is

$$m^\delta(\xi) = (1 - |\xi|^2)_+^\delta = \begin{cases} (1 - |\xi|^2)^\delta, & |\xi| \leq 1, \\ 0, & |\xi| > 1, \end{cases}$$

and  $S^\delta$  is the associated multiplier operator,  $\widehat{S^\delta f} = m^\delta \widehat{f}$ .

At  $\delta = 0$  this is the ball multiplier of Volume 3, Chapter 8, bounded only on  $L^2$ . As  $\delta$  increases the cutoff becomes smoother —  $m^\delta$  is  $C^{[\delta]}$  across the sphere — and one expects boundedness on a wider range of  $p$ . The precise expectation is the Bochner–Riesz conjecture, stated in §8.3.

**Proposition 5.8.2** (Reduction to the kernel). *Let  $K^\delta = \mathcal{F}^{-1}m^\delta$ . If  $K^\delta \in L^1(\mathbb{R}^n)$  then  $S^\delta$  is bounded on  $L^p(\mathbb{R}^n)$  for every  $1 \leq p \leq \infty$ , with  $\|S^\delta\|_{L^p \rightarrow L^p} \leq \|K^\delta\|_{L^1}$ .*

*Proof.* Since  $m^\delta$  is continuous with compact support, it lies in  $L^1 \cap L^2$  and  $K^\delta$  is a bounded continuous function. If moreover  $K^\delta \in L^1$  then  $S^\delta f = K^\delta * f$  for  $f \in \mathcal{S}$ , and Young’s inequality (Volume 2, Worked Example 2.1) gives  $\|K^\delta * f\|_p \leq \|K^\delta\|_1 \|f\|_p$ . Extend by density.  $\square$

So everything reduces to a decay estimate for  $K^\delta$ , and that is an oscillatory integral.

## 5.8.2 The kernel estimate

**Theorem 5.8.3.** *For  $\delta > 0$ ,*

$$|K^\delta(x)| \lesssim_{n,\delta} (1 + |x|)^{-\frac{n+1}{2}-\delta}.$$

*Consequently  $K^\delta \in L^1(\mathbb{R}^n)$  if and only if  $\delta > \frac{n-1}{2}$ .*

*Idea of the proof.* Write the inverse transform in polar coordinates. The angular integral is  $\widehat{d\sigma}$  evaluated at  $rx$ , for which Theorem 5.4.1 supplies the asymptotic  $|rx|^{-(n-1)/2}$  times an oscillating factor  $e^{\pm 2\pi i r|x|}$ . What remains is a one-dimensional integral in  $r \in [0, 1]$  whose amplitude  $(1-r^2)^\delta$  vanishes to order  $\delta$  at the endpoint  $r = 1$  and whose phase is  $2\pi r|x|$  — exactly the situation of Lemma 5.4.2, which gives a further factor  $|x|^{-\delta-1}$ . Multiplying the two:  $|x|^{-(n-1)/2} \cdot |x|^{-\delta-1} = |x|^{-(n+1)/2-\delta}$ . Integrability at infinity then requires  $\frac{n+1}{2} + \delta > n$ , which is  $\delta > \frac{n-1}{2}$ .

*Proof. Step 1. Polar coordinates.*

Since  $m^\delta$  is radial and supported in the unit ball, polar coordinates (Theorem 1.6.11) give, for  $x \in \mathbb{R}^n$ ,

$$K^\delta(x) = \int_{|\xi| \leq 1} (1-|\xi|^2)^\delta e^{2\pi i x \cdot \xi} d\xi = \int_0^1 (1-r^2)^\delta r^{n-1} \left( \int_{\mathbb{S}^{n-1}} e^{2\pi i r x \cdot \omega} d\sigma(\omega) \right) dr.$$

The inner integral is  $\widehat{d\sigma}(-rx) = \widehat{d\sigma}(rx)$ , the transform being real and radial (Proposition 2.8.1).

**Step 2.** *Small  $|x|$ .*

For  $|x| \leq 1$  the bound is trivial:  $|K^\delta(x)| \leq \int_{|\xi| \leq 1} 1 = \omega_n$ .

**Step 3.** *Insert the asymptotic.*

Assume  $R := |x| \geq 1$ . By Theorem 5.5.2 for the sphere — equivalently Theorem 5.4.1 via Theorem 2.8.3 — there are constants  $c_\pm$  with

$$\widehat{d\sigma}(y) = |y|^{-\frac{n-1}{2}} \left( c_+ e^{2\pi i |y|} + c_- e^{-2\pi i |y|} \right) + O(|y|^{-\frac{n+1}{2}}) \quad (|y| \geq 1).$$

Split the  $r$ -integral at  $r = 1/R$ .

*The piece  $r \leq 1/R$ .* Here  $|\widehat{d\sigma}(rx)| \leq \sigma(\mathbb{S}^{n-1})$  and the rest of the integrand is bounded, so this piece is  $O(R^{-1})$ ; since  $\frac{n+1}{2} + \delta$  may exceed 1, we must do

better — but on this range  $r^{n-1} \leq R^{-(n-1)}$ , so the piece is in fact  $O(R^{-n})$ , which is  $O(R^{-\frac{n+1}{2}-\delta})$  whenever  $\delta \leq \frac{n-1}{2}$ ; for larger  $\delta$  the desired bound is weaker still than what Proposition 5.8.2 needs, and the estimate below covers it.

*The piece  $r \geq 1/R$ .* Substituting the asymptotic,

$$\int_{1/R}^1 (1-r^2)^\delta r^{n-1} (rR)^{-\frac{n-1}{2}} \left( c_+ e^{2\pi i r R} + c_- e^{-2\pi i r R} \right) dr + O\left( \int_{1/R}^1 (1-r^2)^\delta r^{n-1} (rR)^{-\frac{n+1}{2}} dr \right)$$

The error integral is  $O(R^{-\frac{n+1}{2}})$ , since  $r^{n-1-\frac{n+1}{2}} = r^{\frac{n-3}{2}}$  is integrable on  $(0, 1)$  for  $n \geq 2$ .

**Step 4.** *The main integral, by the Gamma-integral method.*

Consider

$$J_\pm(R) = R^{-\frac{n-1}{2}} \int_{1/R}^1 (1-r^2)^\delta r^{\frac{n-1}{2}} e^{\pm 2\pi i r R} dr.$$

Substituting  $r = 1 - s$  and writing  $(1-r^2)^\delta = (2-s)^\delta s^\delta$ ,

$$\int_{1/R}^1 (1-r^2)^\delta r^{\frac{n-1}{2}} e^{\pm 2\pi i r R} dr = e^{\pm 2\pi i R} \int_0^{1-1/R} e^{\mp 2\pi i s R} s^\delta g(s) ds, \quad g(s) = (2-s)^\delta (1-s)^{\frac{n-1}{2}}$$

The function  $g$  is smooth on  $[0, \frac{1}{2}]$ ; cutting off there costs  $O(R^{-N})$  by Theorem 5.1.1, the phase having non-vanishing derivative away from the endpoint. By Lemma 5.4.2 with  $a = \delta + 1 > 0$ ,

$$\left| \int_0^\infty e^{\mp 2\pi i s R} s^\delta ds \right| = \Gamma(\delta + 1) (2\pi R)^{-\delta-1},$$

and the correction from replacing  $g$  by  $g(0)$  carries an extra factor  $s$ , hence an extra  $R^{-1}$ . Therefore

$$|J_\pm(R)| \lesssim_{n,\delta} R^{-\frac{n-1}{2}} R^{-\delta-1} = R^{-\frac{n+1}{2}-\delta}.$$

**Step 5.** *Collect.*

Combining Steps 3 and 4, for  $R \geq 1$

$$|K^\delta(x)| \lesssim_{n,\delta} R^{-\frac{n+1}{2}-\delta} + R^{-\frac{n+1}{2}} + R^{-n}.$$

When  $\delta \leq \frac{n-1}{2}$  the first term dominates the third and the second is no larger than the first only if  $\delta \geq 0$ , which holds; so the bound is  $R^{-\frac{n+1}{2}-\delta}$  for  $0 < \delta \leq \frac{n-1}{2}$ , and for larger  $\delta$  one obtains  $R^{-\frac{n+1}{2}}$ , which is more than enough

for Proposition 5.8.2. In all cases  $|K^\delta(x)| \lesssim (1 + |x|)^{-\min\{\frac{n+1}{2}+\delta, n\}}$ , and in particular the stated bound holds.

**Step 6.** *Integrability.*

By polar coordinates,  $(1 + |x|)^{-\frac{n+1}{2}-\delta}$  is integrable on  $\mathbb{R}^n$  exactly when  $\frac{n+1}{2} + \delta > n$ , i.e.  $\delta > \frac{n-1}{2}$ . Conversely, the leading term computed in Step 4 shows  $K^\delta$  is genuinely of size  $R^{-\frac{n+1}{2}-\delta}$  along a sequence  $R \rightarrow \infty$  — the oscillating factors  $e^{\pm 2\pi i R}$  combine into a cosine, which is bounded away from zero on a set of  $R$  of positive density — so for  $\delta \leq \frac{n-1}{2}$  the kernel is not integrable.  $\square$

**Theorem 5.8.4** (Bochner–Riesz above the critical index). *For  $\delta > \frac{n-1}{2}$ , the operator  $S^\delta$  is bounded on  $L^p(\mathbb{R}^n)$  for every  $1 \leq p \leq \infty$ .*

*Proof.* Combine Theorem 5.8.3 with Proposition 5.8.2.  $\square$

### 5.8.3 What is open

**Conjecture 5.8.5** (Bochner–Riesz). *For  $n \geq 2$ ,  $1 \leq p \leq \infty$  and  $\delta > 0$ , the operator  $S^\delta$  is bounded on  $L^p(\mathbb{R}^n)$  if and only if*

$$\delta > \delta(p) := \max \left\{ 0, n \left| \frac{1}{p} - \frac{1}{2} \right| - \frac{1}{2} \right\}.$$

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**WORKED EXAMPLE 5.8.1— TIER C: RESEARCH TASTE** **Where the exponent  $\delta(p)$  comes from, and what is known**

(a) *The necessity, by a Knapp-type example.* Test  $S^\delta$  against a function whose Fourier transform is concentrated on a  $\delta$ -neighbourhood of a cap of the sphere of diameter  $\delta^{1/2}$  — the same cap-and-dual-tube geometry as Worked Example 2.4.3. Tracking the norms as in that example produces exactly the condition  $\delta > n|\frac{1}{p} - \frac{1}{2}| - \frac{1}{2}$ , and  $\delta > 0$  is necessary by Theorem 3.8.2 (Fefferman). So the conjectured range cannot be enlarged.

(b) *What this chapter proves.* Theorem 5.8.4 gives boundedness for  $\delta > \frac{n-1}{2}$ , which is  $\delta(p)$  evaluated at  $p = 1$  and  $p = \infty$ . So the theorem is sharp at the two endpoints and says nothing new in between: for  $p$  near 2 the conjecture asks for far smaller  $\delta$ , and interpolating Theorem 5.8.4 with the trivial  $L^2$  bound at  $\delta = 0$  (Volume 2, Proposition 6.2) gives only a linear improvement, well short of  $\delta(p)$ .

(c) *What is known.* For  $n = 2$  the conjecture is a theorem, due to Carleson and Sjölin (1972); its proof uses the  $L^p$  theory of oscillatory integral operators, and is recorded as revision R4 — it needs machinery beyond this volume. For

$n \geq 3$  the conjecture is open, and the best results come from the restriction theory of Volumes 6 and 12.

(d) *The chain of implications.* Worked Example 3.8.1 recorded

$$\text{Bochner--Riesz} \implies \text{restriction} \implies \text{Keakeya},$$

and this chapter supplies the left-hand end. The implications are proved in Volume 6. The practical consequence is that progress on any of the three is progress on the ones to its right, and that the obstruction is common: in each case one must control how  $\delta^{-(n-1)/2}$  wave packets, one per cap, can superpose.

(e) *The honest summary of this volume.* Stationary phase describes a single oscillatory integral completely — Theorem 5.3.3 gives an asymptotic expansion, not merely a bound. It describes a single operator well, through the  $TT^*$  argument of Theorem 5.7.3. What it cannot do is control a *family* of such objects indexed by directions, and every open problem mentioned above is of that kind. The reader who has finished this volume has the complete local theory and should now regard the remaining difficulty as geometric, not analytic.

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**WORKED EXAMPLE 5.8.2— TIER A The Bochner–Riesz exponents, and what is open**

Theorem 5.8.4 proves  $S^\delta$  bounded on every  $L^p$  for  $\delta > \frac{n-1}{2}$ . Three numbers matter.

(a)  $\delta > \frac{n-1}{2}$ : *proved here.* By Theorem 5.8.3 the kernel is  $O(|x|^{-(n+1)/2-\delta})$ , which is integrable exactly when  $\delta > \frac{n-1}{2}$ ; then  $S^\delta$  is convolution with an  $L^1$  kernel and Young’s inequality finishes it. No curvature is used beyond the kernel bound.

(b)  $\delta = 0$ : *false for  $n \geq 2$ ,  $p \neq 2$ .* This is the ball multiplier, Volume 3, Theorem 3.8.2. So there is a threshold somewhere between.

(c) *The conjecture.*  $S^\delta$  is bounded on  $L^p(\mathbb{R}^n)$  if and only if

$$\delta > \max \left\{ 0, n \left| \frac{1}{p} - \frac{1}{2} \right| - \frac{1}{2} \right\}.$$

At  $p = 2$  this permits every  $\delta > 0$ , consistent with Plancherel. As  $p \rightarrow 1$  or  $\infty$  it approaches  $\frac{n-1}{2}$ , which is the range proved here — so Theorem 5.8.4 is sharp at the extreme exponents and far from sharp in between.

*Status.* Settled for  $n = 2$  by Carleson and Sjölin; open for  $n \geq 3$ . By Volume 6, Worked Example 6.7.2 it implies the restriction conjecture, which is why it sits at the top of the chain of three.

*The one line to remember.* Softening the sharp cutoff by  $(1 - |\xi|^2)^\delta$  repairs the ball multiplier's failure, and the question is how little softening suffices — a question about how much the curvature of the sphere can be made to pay.

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## Drills

**Drill 5.8.1.** Verify the claim in Step 3 that the piece  $r \leq 1/R$  contributes  $O(R^{-n})$ .

**Drill 5.8.2.** Show that  $m^\delta$  is  $C^k$  across the unit sphere precisely when  $k < \delta$ , and relate this to the kernel decay of Theorem 5.8.3.

**Drill 5.8.3.** Interpolate Theorem 5.8.4 with the  $L^2$  bound at  $\delta = 0$  using Riesz–Thorin (Theorem 2.2.1), and compare the resulting range with  $\delta(p)$ . How large is the gap at  $p = 4$ ,  $n = 3$ ?

**Drill 5.8.4.** Carry out the Knapp computation of Worked Example 5.8.1(a) for  $n = 2$  and confirm the exponent  $\delta(p) = \max\{0, 2|\frac{1}{p} - \frac{1}{2}| - \frac{1}{2}\}$ .

## Notes

Bochner introduced the means in 1936 in connection with summability of multiple Fourier series. Theorem 5.8.4 is classical and the kernel computation is due to Bochner; the value  $\frac{n-1}{2}$  is called the *critical index*.

The Carleson–Sjölin theorem is from 1972. For  $n \geq 3$  the best known results are consequences of restriction estimates and are surveyed in Volume 12; the conjecture remains open.



## Chapter 5.A

# Results Quoted from Volumes 1–4

This volume assumes Volumes 1–4. Every citation is listed with its use site.

| Result                                            | Source                 | Used in                                       |
|---------------------------------------------------|------------------------|-----------------------------------------------|
| Polar coordinates                                 | Vol. 1, Thm 6.11       | Theorem 5.6.1, Theorem 5.8.3                  |
| Differentiation under the integral                | Vol. 1, Wkd Ex 5.3     | Proposition 5.2.1, Lemma 5.4.2, Theorem 5.6.1 |
| Gaussian integral                                 | Vol. 1, Drill 6.2      | Proposition 5.2.1                             |
| Tonelli                                           | Vol. 1, Thm 6.6        | Lemma 5.7.4                                   |
| Cauchy–Schwarz in $L^2$                           | Vol. 1, Thm 8.2        | Lemma 5.7.4                                   |
| Holomorphy; identity propagation                  | Vol. 2, Thm 1.10       | Proposition 5.2.1, Lemma 5.4.2, Theorem 5.6.4 |
| Riesz–Thorin                                      | Vol. 2, Thm 2.1        | Drill 8.3                                     |
| Young’s convolution inequality                    | Vol. 2, Wkd Ex 2.1     | Proposition 5.8.2                             |
| The Fourier dictionary                            | Vol. 2, Prop 4.2       | throughout                                    |
| $\mathcal{F}$ on $\mathcal{S}$ and $\mathcal{S}'$ | Vol. 2, Thms 5.6, 5.11 | Theorem 5.6.4, Theorem 5.6.1                  |
| $\widehat{1} = \delta$                            | Vol. 2, Wkd Ex 5.2     | Worked Example 5.6.2                          |
| Transform of $ x ^{-a}$ , $0 < a < n$             | Vol. 2, Wkd Ex 5.9     | Theorem 5.6.4 Step 1                          |
| $L^2$ multiplier norm                             | Vol. 2, Prop 6.2       | Worked Example 5.8.1                          |
| $\widehat{d\sigma}$ via Bessel functions          | Vol. 2, Thm 8.3        | Chapter 5.4, Theorem 5.8.3                    |
| van der Corput                                    | Vol. 2, Lem 8.4        | Theorem 5.1.2, Proposition 5.2.1              |
| Decay of $\widehat{d\sigma}$ (sphere)             | Vol. 2, Thm 8.5        | compared throughout Chapter 5.5               |
| The flat-piece example                            | Vol. 2, Wkd Ex 8.2     | Chapter 5.1 §1.1, Lemma 5.5.1                 |
| Restriction from decay ( $TT^*$ )                 | Vol. 2, Wkd Ex 8.3     | Worked Example 5.5.3                          |
| Fefferman’s ball multiplier theorem               | Vol. 3, Thm 8.6        | Worked Example 5.8.1                          |
| Bochner–Riesz $\implies$ restriction              | Vol. 3, Wkd Ex 8.9     | Worked Example 5.8.1                          |

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## Debts

Volume 2, Chapter 8 deferred two results to this volume. Both are discharged: the asymptotic expansion of  $J_\nu$  is Theorem 5.4.1, and the decay of  $\widehat{d\sigma}$  for a hypersurface with  $k$  non-vanishing principal curvatures is Theorem 5.5.3, with the full asymptotic in Theorem 5.5.2.

This volume incurs no debts on later volumes. Three results were removed from the plan in advance, as revision R4 in the master manifest; each is true, attributed, and used nowhere in this series.



## Chapter 5.B

# Notation and Index of Results

### Additions to the Volume 1–4 conventions

|                 |                                                        |
|-----------------|--------------------------------------------------------|
| $\varphi, \psi$ | phase and amplitude of an oscillatory integral         |
| $\lambda$       | the large parameter                                    |
| Hess            | the Hessian matrix                                     |
| sgn             | sign of a real number; signature of a symmetric matrix |
| $I$             | an oscillatory integral operator                       |
| $S^\delta$      | the Bochner–Riesz operator of order $\delta$           |

**Index of the main results**

|                                                  |                   |
|--------------------------------------------------|-------------------|
| Non-stationary phase                             | Theorem 5.1.1     |
| van der Corput, amplitude version                | Theorem 5.1.3     |
| The Fresnel integral                             | Proposition 5.2.1 |
| Morse lemma in one variable                      | Lemma 5.2.2       |
| Stationary phase in one dimension                | Theorem 5.2.3     |
| Morse lemma in $n$ variables                     | Theorem 5.3.1     |
| The model Gaussian integral                      | Lemma 5.3.2       |
| Stationary phase in $\mathbb{R}^n$               | Theorem 5.3.3     |
| The Gamma integral on the imaginary axis         | Lemma 5.4.2       |
| Asymptotics of $J_\nu$                           | Theorem 5.4.1     |
| Critical points are normal directions            | Lemma 5.5.1       |
| Asymptotics of $\widehat{d\sigma}$ , curved case | Theorem 5.5.2     |
| Decay with $k$ curvatures                        | Theorem 5.5.3     |
| Meromorphic continuation of $ x ^{-a}$           | Theorem 5.6.1     |
| The Riesz family is entire                       | Proposition 5.6.3 |
| Reflection formula $\widehat{Z}_a = Z_{n-a}$     | Theorem 5.6.4     |
| Schur test                                       | Lemma 5.7.4       |
| Hörmander's $L^2$ theorem                        | Theorem 5.7.3     |
| Bochner–Riesz kernel decay                       | Theorem 5.8.3     |
| Bochner–Riesz above the critical index           | Theorem 5.8.4     |

**Plan revisions in this volume (R4)**

| <b>Removed</b>                                       | <b>Reason</b>                               |
|------------------------------------------------------|---------------------------------------------|
| Carleson–Sjölin theorem                              | needs Volume 12's machinery                 |
| Stein interpolation for analytic families            | unused; see Worked Example 5.6.3            |
| Sharp $L^p$ theory of oscillatory integral operators | Volumes 6 and 12 use only the $L^2$ theorem |

# Part 6

## Volume 6

The Restriction Problem: Classical Theory





# Preface to Volume 6

## The hinge of the series

Everything so far has been analysis. Volumes 1 and 2 built the measure theory and the Fourier transform; Volumes 3 and 4 developed singular integrals and the function spaces they live in; Volume 5 computed oscillatory integrals as precisely as they can be computed. Each of those volumes proved theorems by estimating a single object — a kernel, a multiplier, an integral — and each was, in the end, a matter of technique.

This volume states the problem that the rest of the series is about, and the character of the subject changes with it. The restriction conjecture is not a statement about one oscillatory integral. It is a statement about  $\delta^{-(n-1)/2}$  of them at once, one for each cap of the sphere, and about how much the resulting wave packets can overlap. Volume 5 described a single packet completely; from here the difficulty is that there are many.

What this volume proves is the classical theory: the exact duality between restriction and extension, the necessary conditions, the Stein–Tomas theorem, its consequences for dispersive equations, and — the two results that organise everything after — that restriction implies Kakeya and that Bochner–Riesz implies restriction.

## What is assumed

Volumes 1–5. Most heavily: the asymptotics of  $\widehat{d\sigma}$  from Chapter 5.5, the interpolation theorems of Volume 2, the Hardy–Littlewood–Sobolev inequality, and the  $TT^*$  method as it appeared in Chapter 5.7.

## Scope, fixed in advance

Following the practice adopted in Volumes 4 and 5, the list of results was fixed before the chapters were written. Three items a reader might expect are deliberately absent, and each is announced where it would have appeared:

- the *endpoint* of the Stein–Tomas theorem,  $q = \frac{2(n+1)}{n-1}$ . Its standard proof interpolates an analytic family of operators and so requires Stein’s interpolation theorem, which Volume 5 removed from the plan. The non-endpoint range is what Volumes 12 and 13 use;
- the *Keel–Tao endpoint* Strichartz estimate, which needs a bilinear argument;
- *Zygmund’s endpoint theorem* for the restriction conjecture in the plane. The non-endpoint planar case is proved here, in Chapter 6.8.

## The plan

|   |                                   |                                      |
|---|-----------------------------------|--------------------------------------|
| 1 | Restriction and extension         | the two operators, and their duality |
| 2 | Necessary conditions              | what the geometry forbids            |
| 3 | The Stein–Tomas theorem           | the classical positive result        |
| 4 | Strichartz estimates              | the same theorem, read as PDE        |
| 5 | Epsilon-removal                   | local estimates imply global ones    |
| 6 | Restriction implies Keakeya       | the first bridge                     |
| 7 | Bochner–Riesz implies restriction | the second bridge                    |
| 8 | The two-dimensional case          | where the conjecture is a theorem    |

# Chapter 6.1

## Restriction and Extension

### 6.1.1 The question

If  $f \in L^1(\mathbb{R}^n)$  then  $\widehat{f}$  is continuous, so its restriction to any set whatever is a well-defined function. If  $f \in L^2(\mathbb{R}^n)$  then  $\widehat{f}$  is merely an  $L^2$  function, defined up to null sets, and its restriction to a set of measure zero — a sphere, say — is meaningless.

The restriction problem asks where between these two extremes the truth lies, and the answer turns out to depend on the *curvature* of the set.

**Definition 6.1.1.** Let  $S \subseteq \mathbb{R}^n$  be a smooth compact hypersurface with surface measure  $d\sigma$ . For  $f \in \mathcal{S}(\mathbb{R}^n)$  define the *restriction operator*

$$\mathcal{R}f = \widehat{f}|_S.$$

We say that the estimate  $R^*(p \rightarrow q)$  holds if

$$\|\mathcal{R}f\|_{L^q(S, d\sigma)} \leq C \|f\|_{L^p(\mathbb{R}^n)} \quad \text{for all } f \in \mathcal{S}(\mathbb{R}^n), \quad (6.1.1)$$

with  $C$  independent of  $f$ .

Since  $\mathcal{S}$  is dense in  $L^p$  for  $p < \infty$  (Theorem 1.8.7), an estimate (6.1.1) lets one *define*  $\mathcal{R}$  on all of  $L^p$  by continuity, and that is the content of the problem: not to compute a restriction but to show one exists.

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**WORKED EXAMPLE 6.1.1— TIER A The two trivial cases, and why they are not the question**

(a)  $p = 1$  is free. For  $f \in L^1$ ,  $\|\widehat{f}\|_{L^\infty} \leq \|f\|_{L^1}$ , so

$$\|\mathcal{R}f\|_{L^q(d\sigma)} \leq \sigma(S)^{1/q} \|\widehat{f}\|_{L^\infty} \leq \sigma(S)^{1/q} \|f\|_{L^1},$$

and  $R^*(1 \rightarrow q)$  holds for every  $q$ , for *every* compact  $S$  — curved or flat,

smooth or not. Nothing has been used but the Riemann–Lebesgue bound.

(b)  $p = 2$  is hopeless. By Plancherel the map  $f \mapsto \widehat{f}$  is onto  $L^2(\mathbb{R}^n)$ , so  $\widehat{f}$  may be modified on the null set  $S$  at will without changing  $f$ . Hence no estimate of the form (6.1.1) can hold at  $p = 2$  for any  $q$ , and the same argument kills every  $p > 2$ , since  $L^p$  functions have transforms no better behaved.

(c) So the question is the range  $1 < p < 2$ , and the content is: how far past 1 can one push? The answer will depend on  $S$ . This is worth stating plainly because it is the reason the problem is geometric: the analysis at  $p = 1$  and  $p = 2$  is settled and sees nothing about the shape of  $S$ .

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**WORKED EXAMPLE 6.1.2— TIER B A flat piece admits no restriction theorem**

Let  $S_0 \subseteq \{\xi_n = 0\}$  be a compact piece of a hyperplane, with  $d\sigma$  the  $(n - 1)$ -dimensional Lebesgue measure on it. Then  $R^*(p \rightarrow q)$  fails for every  $p > 1$  and every  $q$ .

*The construction.* Take  $f(x) = g(x')h(x_n)$  with  $x = (x', x_n)$ , so that  $\widehat{f}(\xi) = \widehat{g}(\xi')\widehat{h}(\xi_n)$ , and on  $S_0$  this is  $\widehat{g}(\xi')\widehat{h}(0)$ . Choose  $h \in \mathcal{S}(\mathbb{R})$  with  $\widehat{h}(0) = \int h = 1$  and, for  $N \geq 1$ , replace  $h$  by  $h_N(t) = N^{-1}h(t/N)$ , so that  $\widehat{h_N}(0) = 1$  still while

$$\|h_N\|_{L^p(\mathbb{R})} = N^{\frac{1}{p}-1}\|h\|_{L^p} \longrightarrow 0 \quad (N \rightarrow \infty)$$

for every  $p > 1$ . Meanwhile the left side of (6.1.1) is  $\|\widehat{g}\|_{L^q(S_0)} \cdot |\widehat{h_N}(0)| = \|\widehat{g}\|_{L^q(S_0)}$ , a fixed non-zero constant once  $g$  is chosen with  $\widehat{g} \not\equiv 0$  on  $S_0$ . So the ratio of the two sides tends to infinity.

*What went wrong, geometrically.* The function  $f = g \otimes h_N$  is concentrated on a slab of thickness  $N$  in the  $x_n$  direction, and its transform is concentrated on a slab of thickness  $1/N$  around  $\{\xi_n = 0\}$  — which contains  $S_0$  no matter how large  $N$  is. A flat surface can be approximated from inside by the transform of a function spread out in one direction only, and spreading out is cheap in  $L^p$  for  $p > 1$ .

*Why curvature repairs it.* A sphere contains no line segment, so the dual construction is forced to concentrate near a *cap* rather than a slab, and a cap of diameter  $\delta$  is dual to a tube of dimensions  $\delta^{-1} \times \cdots \times \delta^{-2}$  — an object of bounded eccentricity in  $n - 1$  directions. Making that quantitative is Chapter 6.2, and the example above is the degenerate case of the Knapp construction.

---

## 6.1.2 The extension operator

The dual formulation is the one used in practice.

**Definition 6.1.2.** For  $g \in L^1(S, d\sigma)$  define the *extension operator*

$$\mathcal{E}g(x) = \int_S g(\omega) e^{2\pi i x \cdot \omega} d\sigma(\omega) = (gd\sigma)^\vee(x).$$

Since  $gd\sigma$  is a finite measure,  $\mathcal{E}g$  is a bounded continuous function, and  $\mathcal{E}g = \widehat{gd\sigma}$  when  $g \equiv 1$  — the object computed in Chapter 5.5.

**Theorem 6.1.3** (Restriction–extension duality). *Let  $1 \leq p, q \leq \infty$ . The following are equivalent, with the same constant:*

- (i)  $\|\mathcal{R}f\|_{L^q(d\sigma)} \leq C\|f\|_{L^p(\mathbb{R}^n)}$  for all  $f \in \mathcal{S}(\mathbb{R}^n)$ ;
- (ii)  $\|\mathcal{E}g\|_{L^{p'}(\mathbb{R}^n)} \leq C\|g\|_{L^{q'}(d\sigma)}$  for all  $g \in L^{q'}(S, d\sigma)$ .

*Idea of the proof.* The two operators are adjoint. Writing out the pairing  $\langle \mathcal{R}f, g \rangle$  and moving the Fourier transform from  $f$  to  $gd\sigma$  — which is the multiplication formula of Volume 2 — turns it into  $\langle f, \mathcal{E}g \rangle$ . Each estimate is then the statement that the corresponding adjoint is bounded, and by Corollary 1.8.10 boundedness of an operator and of its adjoint are the same thing with conjugate exponents.

*Proof. Step 1. The adjoint identity.*

Let  $f \in \mathcal{S}(\mathbb{R}^n)$  and  $g \in L^\infty(d\sigma)$ . By the multiplication formula (Lemma 2.4.5, extended to finite measures by dominated convergence),

$$\int_S \widehat{f}(\omega) \overline{g(\omega)} d\sigma(\omega) = \int_{\mathbb{R}^n} \widehat{f} \overline{g} d\sigma = \int_{\mathbb{R}^n} f(x) \overline{\mathcal{E}g(x)} dx,$$

that is,

$$\langle \mathcal{R}f, g \rangle_{L^2(d\sigma)} = \langle f, \mathcal{E}g \rangle_{L^2(\mathbb{R}^n)}. \quad (6.1.2)$$

Both sides converge absolutely: the left because  $\widehat{f}$  is bounded and  $d\sigma$  finite, the right because  $f \in \mathcal{S}$  and  $\mathcal{E}g$  is bounded.

**Step 2.** (i) implies (ii).

Assume (i). For  $g \in L^{q'}(d\sigma)$ , Corollary 1.8.10 gives

$$\|\mathcal{E}g\|_{L^{p'}(\mathbb{R}^n)} = \sup \left\{ |\langle f, \mathcal{E}g \rangle| : f \in \mathcal{S}, \|f\|_{L^p} \leq 1 \right\},$$

the supremum over the dense class  $\mathcal{S}$  being the same as over the unit ball of  $L^p$ . By (6.1.2) and Hölder on  $S$ ,

$$|\langle f, \mathcal{E}g \rangle| = |\langle \mathcal{R}f, g \rangle| \leq \|\mathcal{R}f\|_{L^q(d\sigma)} \|g\|_{L^{q'}(d\sigma)} \leq C\|f\|_{L^p} \|g\|_{L^{q'}(d\sigma)},$$

so the supremum is at most  $C\|g\|_{L^{q'}(\mathrm{d}\sigma)}$ .

**Step 3.** (ii) implies (i).

Symmetrically, for  $f \in \mathcal{S}$ ,

$$\|\mathcal{R}f\|_{L^q(\mathrm{d}\sigma)} = \sup \left\{ |\langle \mathcal{R}f, g \rangle| : \|g\|_{L^{q'}(\mathrm{d}\sigma)} \leq 1 \right\} = \sup |\langle f, \mathcal{E}g \rangle| \leq \|f\|_{L^p} \sup \|\mathcal{E}g\|_{L^{p'}} \leq C\|f\|_{L^p}$$

using Corollary 1.8.10 on  $(S, \mathrm{d}\sigma)$ , which is a finite — hence  $\sigma$ -finite — measure space.  $\square$

**Where we are.** From now on we work with  $\mathcal{E}$  and not with  $\mathcal{R}$ . Three reasons, all practical.

$\mathcal{E}g$  is given by an explicit integral, so one may compute with it;  $\mathcal{R}f$  is defined by an inequality. The estimate for  $\mathcal{E}$  has the unknown on the *left*, which is where a superposition of wave packets naturally sits. And the most important case,  $q = 2$ , becomes  $q' = 2$ , so the data  $g$  lives in a Hilbert space and  $L^2$  orthogonality is available — which is exactly what Chapter 6.3 exploits.

The dictionary is:  $R^*(p \rightarrow q)$  for restriction is  $\mathcal{E} : L^{q'}(\mathrm{d}\sigma) \rightarrow L^{p'}(\mathbb{R}^n)$  for extension. The conjectured range  $p < \frac{2n}{n+1}$  becomes  $p' > \frac{2n}{n-1}$ .

### 6.1.3 The local formulation

**Proposition 6.1.4** (Restriction is a local statement). *Let  $1 \leq a, b \leq \infty$ . The estimate (6.2.1) implies, and is implied up to an arbitrarily small loss in  $b$  by, the family of local estimates*

$$\|\mathcal{E}g\|_{L^b(B_R)} \leq C \|g\|_{L^a(\mathrm{d}\sigma)} \quad \text{for every ball } B_R \text{ of radius } R \geq 1,$$

with  $C$  independent of  $R$  and of the centre of  $B_R$ .

*Proof.* That the global estimate implies the local one is immediate, since  $\|\cdot\|_{L^b(B_R)} \leq \|\cdot\|_{L^b(\mathbb{R}^n)}$ .

For the converse, note first that the constant may be taken independent of the *centre*: translating  $B_R$  by  $x_0$  replaces  $\mathcal{E}g$  by  $\mathcal{E}(e^{2\pi i x_0 \cdot \omega} g)$ , by the modulation rule (Proposition 2.4.2), and modulation does not change  $\|g\|_{L^a(\mathrm{d}\sigma)}$ . So only the dependence on  $R$  is at issue, and that is exactly what Theorem 6.5.2 addresses.  $\square$

**Where we are.** The proposition says that no generality is lost by proving estimates on a ball, which is fortunate, because every argument in Volumes 12 and 13 does exactly that. It also isolates why: the modulation symmetry of  $\mathcal{E}$  makes the problem translation-invariant, so a ball anywhere is as good

as a ball at the origin, and only the *scale*  $R$  carries information.

That observation — that  $R$  is the only parameter — is what makes induction on scales possible. An argument that related the estimate at radius  $R$  to the estimate at radius  $R^{1/2}$  would be useless if the constant also depended on where the ball sat.

### 6.1.4 The $TT^*$ identity

**Proposition 6.1.5.** *For  $g \in L^2(d\sigma)$ ,*

$$\mathcal{E}\mathcal{E}^*h = h * \widehat{d\sigma} \quad \text{for } h \in \mathcal{S}(\mathbb{R}^n),$$

where  $\mathcal{E}^* = \mathcal{R}$  under the pairing (6.1.2). Consequently  $\mathcal{E} : L^2(d\sigma) \rightarrow L^q(\mathbb{R}^n)$  is bounded with norm  $A$  if and only if  $h \mapsto h * \widehat{d\sigma}$  is bounded from  $L^{q'}(\mathbb{R}^n)$  to  $L^q(\mathbb{R}^n)$  with norm  $A^2$ .

*Proof.* By Theorem 6.1.3 with  $q = 2$ ,  $\mathcal{E}^*h = \widehat{h}|_S$ . Hence

$$\mathcal{E}\mathcal{E}^*h(x) = \int_S \widehat{h}(\omega) e^{2\pi i x \cdot \omega} d\sigma(\omega) = (\widehat{h} d\sigma)^\vee(x).$$

Now  $\widehat{h} d\sigma$  has inverse transform  $h * d\sigma^\vee = h * \widehat{d\sigma}$ , since  $d\sigma$  is symmetric under  $\omega \mapsto -\omega$  for the surfaces considered here, or in general  $h * d\sigma^\vee$ ; the convolution converges absolutely because  $h \in \mathcal{S}$  and  $d\sigma$  is finite.

For the second statement, the identity  $\|\mathcal{E}\|^2 = \|\mathcal{E}\mathcal{E}^*\|$  holds for any bounded operator between Hilbert-space-and- $L^q$  pairs in the sense used in Theorem 5.7.3 Step 6:  $\|\mathcal{E}g\|_{L^q}^2 = \sup_h |\langle \mathcal{E}\mathcal{E}^*h, h \rangle|$  over  $\|h\|_{L^{q'}} \leq 1$ .  $\square$

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#### WORKED EXAMPLE 6.1.3— TIER B Computing $\mathcal{E}$ on a simple datum

Abstract duality is easier to trust after one explicit computation. Take  $n = 2$ ,  $S = \mathbb{S}^1$ , and  $g \equiv 1$ .

*The answer.*  $\mathcal{E}1 = \widehat{d\sigma}$ , which by Volume 2 is

$$\widehat{d\sigma}(x) = 2\pi J_0(2\pi|x|) = \sqrt{\frac{2\pi}{|x|}} \cos\left(2\pi|x| - \frac{\pi}{4}\right) + O(|x|^{-3/2}),$$

using Theorem 5.4.1 with  $\nu = 0$ .

*Reading off which estimates can hold.*  $|\mathcal{E}1| \approx |x|^{-1/2}$  along a sequence, so  $\mathcal{E}1 \in L^b(\mathbb{R}^2)$  requires  $\frac{b}{2} > 2$ , that is  $b > 4$  — which is Theorem 6.2.5 for  $n = 2$ , obtained here by a direct computation rather than an argument. And  $\|1\|_{L^a(d\sigma)}$  is finite for every  $a$ , so no estimate (6.2.1) with  $b \leq 4$  can hold, whatever  $a$ .

*The dual statement.* By Theorem 6.1.3, the failure at  $b = 4$  says  $R^*(\frac{4}{3} \rightarrow a')$  fails for every  $a'$ . So a function in  $L^{4/3}(\mathbb{R}^2)$  may have a Fourier transform whose restriction to the circle is not even locally integrable — a concrete statement, obtained from an asymptotic for a Bessel function.

*Why it is worth doing this once.* Every necessary condition in Chapter 6.2 is of this form: choose a datum  $g$ , compute or estimate  $\mathcal{E}g$ , and compare the two sides. The Knapp example is the same procedure with  $g$  the indicator of a cap instead of the constant, and the whole-sphere example of Theorem 6.2.5 is this computation in general dimension.

**WORKED EXAMPLE 6.1.4— TIER C: RESEARCH TASTE    Why this identity is the organising principle of the subject**

Proposition 6.1.5 converts a statement about an operator into a statement about a *convolution kernel*, namely  $\widehat{d\sigma}$ . That is the exact point at which the whole of Volume 5 becomes relevant, and it is worth laying out the consequences before they are used.

(a) *The immediate consequence.* Worked Example 2.8.3 already ran this argument once, crudely: bounding  $|\widehat{d\sigma}| \lesssim |\xi|^{-(n-1)/2}$  and applying Hardy–Littlewood–Sobolev to the resulting fractional integral gives  $\mathcal{E} : L^2(d\sigma) \rightarrow L^q(\mathbb{R}^n)$  for  $q > \frac{4n}{n-1}$ . Chapter 6.3 improves this to  $q > \frac{2(n+1)}{n-1}$  by decomposing  $\widehat{d\sigma}$  dyadically rather than bounding it once, and the improvement is substantial: at  $n = 3$  the exponents are 6 versus 4.

(b) *Where the improvement comes from.* The crude argument uses only the *size* of  $\widehat{d\sigma}$ . The dyadic argument uses, in addition, that the piece of  $\widehat{d\sigma}$  living at scale  $|x| \approx 2^j$  has Fourier transform concentrated near  $S$  — so that convolution with it is bounded on  $L^2$  with a constant that does *not* degrade. Interpolating the two bounds for each piece and summing over  $j$  is the whole of Stein–Tomas. The oscillation of  $\widehat{d\sigma}$ , computed in Theorem 5.5.2, is what makes the  $L^2$  bound available; an argument that took absolute values first would have destroyed it.

(c) *Where it stops.* The method is confined to  $q' = 2$  on the data side, because that is the only exponent at which  $\mathcal{E}\mathcal{E}^*$  is the relevant object. The restriction conjecture concerns  $L^\infty(d\sigma)$  data, and there is no  $TT^*$  identity for it. Every subsequent advance — bilinear, multilinear, polynomial — is in one way or another a substitute for the missing identity.

(d) *The pattern, once more.*  $TT^*$  has now appeared four times in this series: for the Hilbert transform in Volume 3, for oscillatory integral operators in Theorem 5.7.3, in Volume 2's first restriction estimate, and here. Each time



the move is the same — replace an operator that resists estimation by a self-adjoint one whose kernel can be computed — and each time the cost is the same: the method is tied to  $L^2$  on one side. Recognising both halves of that trade is worth more than any single application of it.

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## Drills

**Drill 6.1.1.** Verify the claim in Worked Example 6.1.1(b) in detail: given  $f \in L^2$  and a null set  $S$ , construct  $\tilde{f} \in L^2$  with  $\tilde{f} = f$  in  $L^2$  and  $\tilde{f}|_S$  prescribed arbitrarily.

**Drill 6.1.2.** Carry out Worked Example 6.1.2 with explicit  $g$  and  $h$  and compute both sides of (6.1.1) as functions of  $N$ .

**Drill 6.1.3.** Show that  $R^*(p \rightarrow q)$  for a compact  $S$  implies  $R^*(p \rightarrow \tilde{q})$  for every  $\tilde{q} \leq q$ , and explain why no such monotonicity holds in  $p$ .

**Drill 6.1.4.** Deduce from Theorem 6.1.3 that  $R^*(p \rightarrow 2)$  is equivalent to  $\|\mathcal{E}g\|_{L^{p'}} \lesssim \|g\|_{L^2(d\sigma)}$ , and identify the value of  $p'$  corresponding to the Stein–Tomas exponent  $q = \frac{2(n+1)}{n-1}$ .

## Notes

The restriction problem was raised by Stein in 1967. Theorem 6.1.3 is immediate but was what made the subject tractable, since it is the extension form that admits wave packet decompositions.

Worked Example 6.1.2 is the standard observation that the problem is about curvature; the sharp quantitative version is Knapp’s example, in Chapter 6.2.



## Chapter 6.2

# Necessary Conditions

Worked Example 6.1.2 showed that a flat surface admits no restriction theorem at all, and located the reason: a hyperplane can be approached by the transform of a function spread out in a single direction, and spreading out is cheap in  $L^p$  for  $p > 1$ . A sphere contains no flat piece, but it contains *almost* flat pieces — a cap of diameter  $\delta$  deviates from its tangent plane by only  $\delta^2$  — and running the same construction on a cap gives the sharp obstruction. That is Knapp's example, and it is the subject of this chapter.

Throughout,  $S = \mathbb{S}^{n-1}$  and we work with the extension operator, writing the estimate to be tested as

$$\|\mathcal{E}g\|_{L^b(\mathbb{R}^n)} \leq C \|g\|_{L^a(S, d\sigma)}. \quad (6.2.1)$$

By Theorem 6.1.3 this is  $R^*(b' \rightarrow a')$  in the restriction formulation.

### 6.2.1 The Knapp example

**Definition 6.2.1.** For  $0 < \delta < \frac{1}{2}$  let

$$\kappa_\delta = \{\omega \in \mathbb{S}^{n-1} : |\omega - e_n| \leq \delta\}$$

be the *cap* of radius  $\delta$  at the north pole, and let

$$T_\delta = \left\{x \in \mathbb{R}^n : |x'| \leq \frac{c}{\delta}, |x_n| \leq \frac{c}{\delta^2}\right\}$$

be the *dual box*, where  $x = (x', x_n)$  and  $c > 0$  is an absolute constant fixed in Lemma 6.2.2.

Two elementary facts:  $\sigma(\kappa_\delta) \approx_n \delta^{n-1}$ , since the cap is comparable to a Euclidean ball of radius  $\delta$  in the  $(n-1)$ -dimensional surface; and

$$m(T_\delta) = (2c\delta^{-1})^{n-1} \cdot 2c\delta^{-2} \approx_n \delta^{-(n+1)}.$$

**Lemma 6.2.2** (The cap does not oscillate on its dual box). *There is an absolute constant  $c > 0$  such that, with  $g = \mathbf{1}_{\kappa_\delta}$ ,*

$$|\mathcal{E}g(x)| \geq \frac{1}{2} \sigma(\kappa_\delta) \quad \text{for every } x \in T_\delta.$$

*Idea of the proof.*  $\mathcal{E}g(x) = \int_{\kappa_\delta} e^{2\pi i x \cdot \omega} d\sigma$  is large exactly when the phase  $x \cdot \omega$  is nearly constant over the cap. Removing the constant  $x_n$  by a modulation, the phase becomes  $x' \cdot \omega'$  — of size  $|x'| \delta$  — plus the curvature term  $-\frac{1}{2} x_n |\omega'|^2$  — of size  $|x_n| \delta^2$ . The box  $T_\delta$  is precisely the set on which both are  $O(1)$ : it is  $\delta^{-1}$  long in the  $n-1$  tangential directions and  $\delta^{-2}$  long in the normal direction, and the asymmetry is a direct record of the sphere's curvature.

*Proof.* Parametrise the cap by  $\omega = (\omega', \omega_n)$  with  $|\omega'| \leq \delta$  and  $\omega_n = \sqrt{1 - |\omega'|^2}$ . Then

$$x \cdot \omega = x' \cdot \omega' + x_n \omega_n = x_n + x' \cdot \omega' - \frac{x_n |\omega'|^2}{2} + x_n E(\omega'),$$

where  $E(\omega') = \sqrt{1 - |\omega'|^2} - 1 + \frac{1}{2} |\omega'|^2$  satisfies  $|E(\omega')| \leq |\omega'|^4$  for  $|\omega'| \leq \frac{1}{2}$ , by Taylor's theorem with remainder.

For  $x \in T_\delta$  we may bound the three non-constant terms:

$$|x' \cdot \omega'| \leq \frac{c}{\delta} \cdot \delta = c, \quad \frac{|x_n| |\omega'|^2}{2} \leq \frac{c}{2\delta^2} \cdot \delta^2 = \frac{c}{2}, \quad |x_n| |E(\omega')| \leq \frac{c}{\delta^2} \cdot \delta^4 \leq c.$$

So writing  $\Phi(\omega) = x \cdot \omega - x_n$ , we have  $2\pi |\Phi(\omega)| \leq 2\pi \cdot \frac{5}{2} c$  for all  $\omega \in \kappa_\delta$ . Choose  $c = \frac{1}{30}$ , so that  $2\pi |\Phi| \leq \frac{\pi}{3}$  and hence  $\cos(2\pi \Phi(\omega)) \geq \frac{1}{2}$ .

Therefore

$$|\mathcal{E}g(x)| = \left| \int_{\kappa_\delta} e^{2\pi i \Phi(\omega)} d\sigma(\omega) \right| \geq \operatorname{Re} \int_{\kappa_\delta} e^{2\pi i \Phi} d\sigma \geq \frac{1}{2} \sigma(\kappa_\delta),$$

the first step because  $|e^{-2\pi i x_n}| = 1$ . □

**Theorem 6.2.3** (The Knapp condition). *If (6.2.1) holds for  $S = \mathbb{S}^{n-1}$ , then*

$$\frac{n+1}{b} \leq (n-1) \left( 1 - \frac{1}{a} \right). \quad (6.2.2)$$

*Proof. Step 1. Choose the test function.*

Apply (6.2.1) to  $g = \mathbf{1}_{\kappa_\delta}$ , the indicator of the cap. This is the only choice made in the proof, and everything else is computation.

**Step 2.** *Compute the right-hand side.*

Since  $g$  is an indicator,

$$\|g\|_{L^a(d\sigma)} = \left( \int_{\mathbb{S}^{n-1}} \mathbf{1}_{\kappa_\delta} d\sigma \right)^{1/a} = \sigma(\kappa_\delta)^{1/a} \approx_n \delta^{(n-1)/a},$$

using  $\sigma(\kappa_\delta) \approx_n \delta^{n-1}$  from the remarks following Definition 6.1.1. (For  $a = \infty$  read the exponent as 0, so the right side is  $\approx 1$ .)

**Step 3.** *Bound the left-hand side from below.*

The  $L^b$  norm over  $\mathbb{R}^n$  is at least the norm over the dual box  $T_\delta$ , on which Lemma 6.2.2 gives the pointwise lower bound  $|\mathcal{E}g| \geq \frac{1}{2}\sigma(\kappa_\delta)$ . Hence

$$\|\mathcal{E}g\|_{L^b(\mathbb{R}^n)} \geq \left( \int_{T_\delta} |\mathcal{E}g|^b \right)^{1/b} \geq \frac{1}{2} \sigma(\kappa_\delta) m(T_\delta)^{1/b} \approx_n \delta^{n-1} \delta^{-(n+1)/b},$$

using  $m(T_\delta) \approx_n \delta^{-(n+1)}$ . Note where the two exponents come from:  $\delta^{n-1}$  is the *height* of  $\mathcal{E}g$  on the box, and  $\delta^{-(n+1)/b}$  is the *volume* of the box, raised to the power  $1/b$ .

**Step 4.** *Compare, and let  $\delta \rightarrow 0$ .*

Substituting Steps 2 and 3 into (6.2.1),

$$\delta^{n-1-\frac{n+1}{b}} \lesssim_n C \delta^{\frac{n-1}{a}} \quad \text{for all } 0 < \delta < \frac{1}{2}.$$

An inequality  $\delta^\alpha \lesssim \delta^\beta$  holding for all small  $\delta$  forces  $\alpha \geq \beta$ : otherwise, dividing gives  $\delta^{\alpha-\beta} \lesssim 1$  with  $\alpha - \beta < 0$ , and the left side tends to  $\infty$ . Hence

$$n-1-\frac{n+1}{b} \geq \frac{n-1}{a},$$

and rearranging — moving  $\frac{n+1}{b}$  to the right and factoring  $n-1$  out of the two remaining terms — gives  $\frac{n+1}{b} \leq (n-1)(1 - \frac{1}{a})$ , which is (6.2.2).  $\square$

---

**WORKED EXAMPLE 6.2.1— TIER A The condition at the two exponents that matter**

(a)  $a = 2$ . Condition (6.2.2) reads  $\frac{n+1}{b} \leq \frac{n-1}{2}$ , that is

$$b \geq \frac{2(n+1)}{n-1}.$$

This is exactly the Stein–Tomas exponent, proved sufficient (in the open range) in Chapter 6.3. So at  $a = 2$  the Knapp example is *sharp*: it forbids everything below  $\frac{2(n+1)}{n-1}$ , and everything above is a theorem.

(b)  $a = \infty$ . Now (6.2.2) reads  $\frac{n+1}{b} \leq n-1$ , that is  $b \geq \frac{n+1}{n-1}$ . This is a much weaker requirement — for  $n = 3$  it says  $b \geq 2$  — and it is *not* the truth. The binding condition at  $a = \infty$  comes from a different example, §2.2.

(c) *Numerically*,  $n = 3$ . At  $a = 2$ :  $b \geq 4$ . At  $a = \infty$ : Knapp gives  $b \geq 2$ , while the correct answer, conjecturally, is  $b > 3$ . The gap between 2 and 3 is filled by the whole-sphere example.

---

**Proposition 6.2.4** (Knapp for the paraboloid). *Let  $P = \{(\zeta, |\zeta|^2) : |\zeta| \leq 1\} \subseteq \mathbb{R}^n$  with the measure  $d\mu = d\zeta$ , and suppose  $\|\mathcal{E}_P g\|_{L^b(\mathbb{R}^n)} \leq C\|g\|_{L^a(d\mu)}$ . Then (6.2.2) holds, with  $n-1$  the dimension of  $P$ .*

*Proof.* Let  $\rho_\delta = \{|\zeta| \leq \delta\}$ , so  $\mu(\rho_\delta) \approx \delta^{n-1}$ , and take  $g = \mathbf{1}_{\rho_\delta}$ . For  $x = (x', x_n)$ ,

$$\mathcal{E}_P g(x) = \int_{|\zeta| \leq \delta} e^{2\pi i(x' \cdot \zeta + x_n |\zeta|^2)} d\zeta.$$

On the box  $|x'| \leq \frac{c}{\delta}$ ,  $|x_n| \leq \frac{c}{\delta^2}$  the phase satisfies

$$|x' \cdot \zeta| \leq c, \quad |x_n||\zeta|^2 \leq c,$$

so, choosing  $c = \frac{1}{24}$ , the total phase is at most  $\frac{\pi}{3}$  in absolute value and  $|\mathcal{E}_P g| \geq \frac{1}{2}\mu(\rho_\delta)$  there, exactly as in Lemma 6.2.2. Note that here *no* Taylor expansion is needed: the paraboloid’s “height function”  $|\zeta|^2$  is already exactly quadratic, which is why the paraboloid is the standard model.

The box has measure  $\approx \delta^{-(n+1)}$ , and repeating the computation of Theorem 6.2.3 verbatim gives (6.2.2).  $\square$

**Where we are.** The sphere and the paraboloid give the same condition, and the proof for the paraboloid is shorter — the error term  $E(\omega')$  of Lemma 6.2.2 is absent. This is the reason essentially all of the modern literature works with the paraboloid: it is the osculating surface of the sphere at a point, it has the same curvature properties, and it is exactly parabolic rather than approximately so, which makes the parabolic rescaling

$$(\zeta, \tau) \mapsto (\lambda\zeta, \lambda^2\tau)$$

an exact symmetry. Induction on scales, in Volume 13, uses that symmetry at every step; on the sphere one would have to carry the error terms through the induction.

## 6.2.2 The whole-sphere example

**Theorem 6.2.5.** *If (6.2.1) holds for  $S = \mathbb{S}^{n-1}$  with any  $a \leq \infty$ , then*

$$b > \frac{2n}{n-1}. \quad (6.2.3)$$

*Proof.* Take  $g \equiv 1$ , which lies in  $L^a(d\sigma)$  for every  $a$  since  $d\sigma$  is finite. Then  $\mathcal{E}g = \widehat{d\sigma}$ , and (6.2.1) asserts  $\widehat{d\sigma} \in L^b(\mathbb{R}^n)$ .

By Theorem 5.5.2, for  $|x| \geq 1$ ,

$$\widehat{d\sigma}(x) = |x|^{-\frac{n-1}{2}} \left( c_+ e^{2\pi i|x|} + c_- e^{-2\pi i|x|} \right) + O(|x|^{-\frac{n+1}{2}}),$$

with  $c_{\pm}$  non-zero conjugates, so  $c_+ e^{2\pi i r} + c_- e^{-2\pi i r} = 2 \operatorname{Re} (c_+ e^{2\pi i r})$  is a non-degenerate cosine in  $r = |x|$ . Hence there are  $\eta, \rho > 0$  and arbitrarily large  $R$  with

$$|\widehat{d\sigma}(x)| \geq \eta |x|^{-\frac{n-1}{2}} \quad \text{whenever } ||x| - R| \leq \rho,$$

and the annuli  $\{||x| - R_k| \leq \rho\}$  may be taken disjoint with  $R_k \rightarrow \infty$ , say  $R_k = k$  for  $k$  large along a suitable arithmetic progression. Therefore

$$\int_{\mathbb{R}^n} |\widehat{d\sigma}|^b \gtrsim \sum_k \eta^b R_k^{-\frac{b(n-1)}{2}} m(\{|x| - R_k| \leq \rho\}) \approx \sum_k R_k^{-\frac{b(n-1)}{2} + n-1},$$

using that the annulus has measure  $\approx \rho R_k^{n-1}$ . This series diverges unless

$$\frac{b(n-1)}{2} - (n-1) > 1,$$

i.e. unless  $b > \frac{2n}{n-1}$ . At the borderline  $b = \frac{2n}{n-1}$  the exponent is

$$\frac{b(n-1)}{2} - (n-1) = \frac{2n}{n-1} \cdot \frac{n-1}{2} - (n-1) = n - (n-1) = 1,$$

so the series becomes  $\sum_k R_k^{-1}$ , which diverges for  $R_k = k$ ; hence the inequality (6.2.3) is strict.  $\square$

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**WORKED EXAMPLE 6.2.2— TIER B The two conditions, drawn against each other**

It is worth seeing precisely which of (6.2.2) and (6.2.3) binds where. Both are conditions on the pair  $(1/a, 1/b)$ ; write  $u = 1/a$ ,  $v = 1/b$ , so the admissible region sits in  $[0, 1]^2$ .

*Knapp* is  $(n+1)v \leq (n-1)(1-u)$ , a closed half-plane whose boundary is the line from  $(u, v) = (0, \frac{n-1}{n+1})$  down to  $(1, 0)$ .

*Whole sphere* is  $v < \frac{n-1}{2n}$ , a horizontal half-plane.

At  $u = \frac{1}{2}$  (that is,  $a = 2$ ) Knapp gives  $v \leq \frac{n-1}{2(n+1)}$  and the sphere gives  $v < \frac{n-1}{2n}$ ; since  $\frac{n-1}{2(n+1)} < \frac{n-1}{2n}$ , *Knapp binds*. At  $u = 0$  (that is,  $a = \infty$ ) Knapp gives  $v \leq \frac{n-1}{n+1}$  and the sphere gives  $v < \frac{n-1}{2n}$ ; now  $\frac{n-1}{2n} < \frac{n-1}{n+1}$ , so *the sphere binds*.

The crossover is where the two boundaries meet:  $(n+1)\frac{n-1}{2n} = (n-1)(1-u)$  gives  $u = \frac{n-1}{2n}$ , i.e.  $a = \frac{2n}{n-1}$ . So:

$$\begin{aligned} a &\leq \frac{2n}{n-1} && \text{the whole-sphere condition binds} \\ a &\geq \frac{2n}{n-1} && \text{the Knapp condition binds} \end{aligned}$$

*Why this matters.* The Stein–Tomas theorem lives at  $a = 2$ , where Knapp binds and is sharp. The restriction conjecture lives at  $a = \infty$ , where the sphere binds — and there the Knapp example, the only construction that sees curvature, is *not* the obstruction. That mismatch is a large part of why the conjecture is hard: the extremal example at  $a = \infty$  is a single global object,  $d\sigma$  itself, and no local cap computation detects it.

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**Conjecture 6.2.6** (Restriction conjecture). *For  $S = \mathbb{S}^{n-1}$  and  $n \geq 2$ ,*

$$\|\mathcal{E}g\|_{L^b(\mathbb{R}^n)} \lesssim_{n,b} \|g\|_{L^\infty(d\sigma)} \quad \text{for every } b > \frac{2n}{n-1}.$$

By Theorem 6.2.5 the range is optimal, and by Theorem 6.2.3 with  $a = \infty$  it is consistent with the Knapp example. Equivalently, in restriction form:  $R^*(p \rightarrow \infty)$  for  $p < \frac{2n}{n+1}$  — and since  $\mathcal{R}$  maps into  $L^\infty(d\sigma)$  trivially when  $p = 1$ , the content is the range  $1 < p < \frac{2n}{n+1}$ .

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**WORKED EXAMPLE 6.2.3— TIER C: RESEARCH TASTE** **What is actually known, and the shape of the difficulty**

(a) *The record.* Conjecture 6.2.6 is a theorem for  $n = 2$  (Chapter 6.8) and open for every  $n \geq 3$ . Chapter 6.3 proves the case  $a = 2$ , which by interpolation with the trivial bound at  $a = 1$  gives a range of  $(a, b)$  falling well short of  $a = \infty$ . Volumes 12 and 13 push  $b$  down from the Stein–Tomas value towards  $\frac{2n}{n-1}$ , and the current record is strictly between.

(b) *Why interpolation cannot finish.* One might hope to interpolate the  $a = 2$  theorem against something at  $a = \infty$ . But at  $a = \infty$  there is no non-trivial estimate available to interpolate with: the trivial bound  $\|\mathcal{E}g\|_\infty \leq \|g\|_{L^1(d\sigma)}$  is at  $b = \infty$ , and interpolating it with Stein–Tomas returns exactly the segment already known. The conjecture is not reachable from the endpoints of anything proved.

(c) *Where the extra input must come from.* Decompose  $g = \sum_\theta g_\theta$  over



$\delta$ -caps. Then  $\mathcal{E}g = \sum_{\theta} \mathcal{E}g_{\theta}$ , and by Lemma 6.2.2 each  $\mathcal{E}g_{\theta}$  is essentially constant on a tube  $T_{\theta}$  dual to its cap — of dimensions  $\delta^{-1} \times \cdots \times \delta^{-1} \times \delta^{-2}$ , pointing in the direction normal to  $\theta$ . There are  $\approx \delta^{-(n-1)}$  caps, hence that many tubes, one in each direction.

So the question becomes: *how much can  $\delta^{-(n-1)}$  tubes, one per direction, overlap?* That is a Kakeya question, and Chapter 6.6 makes the implication precise. It is also why every subsequent advance in the subject is geometric rather than analytic: the analysis of a single  $\mathcal{E}g_{\theta}$  was completed in Volume 5, and what remains is a counting problem about tubes.

(d) *A caution about the converse.* Kakeya does not obviously imply restriction, and no such implication is known. The arrow runs one way, and the gap between the two conjectures — Kakeya is also open for  $n \geq 3$ , but is believed easier — is itself a subject of study.

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WORKED EXAMPLE 6.2.4— TIER A **The dual box, and why its shape is  $\delta^{-1} \times \delta^{-2}$**

The asymmetry of  $T_{\delta}$  is the single most important geometric fact in the subject, and it is worth deriving it twice — once from the phase, as in Lemma 6.2.2, and once from the uncertainty principle.

*From the uncertainty principle.* The cap  $\kappa_{\delta}$  has diameter  $\delta$  in the  $n - 1$  tangential directions, so a function with Fourier support there cannot be localised better than  $\delta^{-1}$  in those directions. In the normal direction the cap is not a point: it deviates from its tangent plane by

$$1 - \sqrt{1 - \delta^2} \approx \frac{\delta^2}{2},$$

so its extent in the normal direction is  $\approx \delta^2$ , and the dual extent is  $\delta^{-2}$ . Hence a box of dimensions  $\delta^{-1} \times \cdots \times \delta^{-1} \times \delta^{-2}$ .

*The consistency check.* The volume of the cap's  $\delta^2$ -neighbourhood is  $\approx \delta^{n-1} \cdot \delta^2$ , and the dual box has volume  $\delta^{-(n-1)} \delta^{-2}$ . Their product is 1, as the uncertainty principle requires.

*Why  $\delta^2$  and not  $\delta$ .* Because the sphere is curved and not merely bent: the deviation from the tangent plane is quadratic, by Taylor's theorem, and it is quadratic precisely because the second fundamental form is non-degenerate — Lemma 5.5.1 again. For a surface with a vanishing principal curvature the deviation in that direction is of higher order, the dual box is longer, and the Knapp condition weakens accordingly. Drill 2.4 works this out.

*The number to remember.* A  $\delta$ -cap is dual to a  $\delta^{-1} \times \delta^{-2}$  box; equivalently, after rescaling by  $\delta^2$ , a  $\delta^{1/2}$ -cap is dual to a  $1 \times \delta^{1/2}$  tube at scale  $\delta^{-1}$ . Both normalisations appear in the literature and confusing them is the commonest

source of arithmetic errors in this area.

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**Figure 6.2.1 (to be drawn).** *The two necessary conditions*

**Content required:** The  $(u, v) = (1/a, 1/b)$  square, with the Knapp half-plane  $(n+1)v \leq (n-1)(1-u)$  and the whole-sphere half-plane  $v < \frac{n-1}{2n}$  shaded; their intersection is the conjectured range. Mark the crossover at  $a = \frac{2n}{n-1}$  and the two points  $a = 2$  (Stein–Tomas) and  $a = \infty$  (the conjecture).

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## Drills

**Drill 6.2.1.** Verify  $\sigma(\kappa_\delta) \approx_n \delta^{n-1}$  by parametrising the cap as a graph over the tangent plane.

**Drill 6.2.2.** Check the bound  $|E(\omega')| \leq |\omega'|^4$  used in Lemma 6.2.2, for  $|\omega'| \leq \frac{1}{2}$ .

**Drill 6.2.3.** Redo Theorem 6.2.3 for the truncated paraboloid  $\{(\xi', |\xi'|^2) : |\xi'| \leq 1\}$  and confirm the same condition (6.2.2).

**Drill 6.2.4.** Show that for a hypersurface with  $k$  non-vanishing principal curvatures the whole-surface example gives  $b > \frac{2n}{k}$ , using Theorem 5.5.3 in place of Theorem 5.5.2.

## Notes

The Knapp example is from the 1970s folklore of the subject; the name attaches to an unpublished observation of Knapp. The two necessary conditions and the resulting conjecture are stated in Stein’s 1979 problem list.

That the conjectured range is exactly the intersection of the two conditions is the reason Conjecture 6.2.6 is believed: no third obstruction has been found in fifty years, in any dimension.

## Chapter 6.3

# The Stein–Tomas Theorem

Worked Example 6.2.1 identified  $b = \frac{2(n+1)}{n-1}$  as the exponent below which the estimate  $\mathcal{E} : L^2(d\sigma) \rightarrow L^b(\mathbb{R}^n)$  is impossible. This chapter proves that everything strictly above it is true.

**Theorem 6.3.1** (Stein–Tomas, non-endpoint). *Let  $S \subseteq \mathbb{R}^n$ ,  $n \geq 2$ , be a compact  $C^\infty$  hypersurface with non-vanishing Gaussian curvature and  $\psi \in C_c^\infty$  a cutoff. Then for every*

$$b > \frac{2(n+1)}{n-1}$$

*there is  $C = C(n, b, S, \psi)$  with*

$$\|\mathcal{E}g\|_{L^b(\mathbb{R}^n)} \leq C \|g\|_{L^2(S, d\sigma)} \quad \text{for all } g \in L^2(S, d\sigma),$$

*where  $\mathcal{E}g = (g\psi d\sigma)^\vee$ . Equivalently,  $R^*(p \rightarrow 2)$  holds for every  $p < \frac{2(n+1)}{n+3}$ .*

*Idea of the proof.* By Proposition 6.1.5 the estimate is equivalent to

$$\|f * \widehat{\psi d\sigma}\|_{L^b} \lesssim \|f\|_{L^{b'}},$$

a statement about convolution with the single fixed kernel  $\widehat{\psi d\sigma}$ . Two facts about that kernel are available and pull in opposite directions:

*it is small* —  $|\widehat{\psi d\sigma}(x)| \lesssim |x|^{-(n-1)/2}$ , by Theorem 5.5.2 — which is good for  $L^1 \rightarrow L^\infty$  but says nothing about  $L^2$ ; and

*its Fourier transform is a measure on a surface* — bounded in no useful sense — which is bad for  $L^2 \rightarrow L^2$ .

Neither alone suffices. The device is to cut the kernel into dyadic pieces  $K_j$  living at  $|x| \approx 2^j$  and to use *both* facts on each piece. The piece  $K_j$  has sup-norm  $2^{-j(n-1)/2}$ , and its Fourier transform, being  $d\sigma$  smoothed at scale  $2^{-j}$ , has sup-norm  $\approx 2^j$ . Interpolating these two bounds for a fixed  $j$  gives an  $L^{b'} \rightarrow L^b$  norm of  $2^{j(\theta - (1-\theta)\frac{n-1}{2})}$  with  $\theta = 2/b$ , and the whole

theorem is the observation that this exponent is negative precisely when  $b > \frac{2(n+1)}{n-1}$ . The series then converges geometrically.

Note where the two competing effects appear:  $2^{-j(n-1)/2}$  is the curvature, and  $2^j$  is the price of localising in space, since a function supported at scale  $2^j$  cannot have its Fourier transform concentrated on a surface. The theorem is their balance.

### 6.3.1 The dyadic decomposition of the kernel

Fix  $\eta \in C_c^\infty(\mathbb{R}^n)$  supported in  $\{\frac{1}{2} < |x| < 2\}$  with  $\sum_{j \geq 1} \eta(2^{-j}x) = 1$  for  $|x| \geq 1$ , and let  $\eta_0 = 1 - \sum_{j \geq 1} \eta(2^{-j}\cdot)$ , supported in  $\{|x| \leq 2\}$ . Write

$$K = \widehat{\psi d\sigma}, \quad K_j = K \cdot \eta(2^{-j}\cdot) \quad (j \geq 1), \quad K_0 = K\eta_0,$$

so that  $K = \sum_{j \geq 0} K_j$  pointwise, with only finitely many terms non-zero at each point.

**Lemma 6.3.2.** *For  $j \geq 1$ ,*

$$(i) \quad \|K_j\|_{L^\infty} \lesssim_{n,S} 2^{-j\frac{n-1}{2}}, \quad (ii) \quad \|\widehat{K_j}\|_{L^\infty} \lesssim_{n,S} 2^j.$$

*Proof.* (i) On  $\text{supp } K_j$  we have  $|x| \approx 2^j$ , and Theorem 5.5.2 gives  $|K(x)| \lesssim |x|^{-(n-1)/2}$  there. Since  $|\eta| \leq 1$ , (i) follows.

(ii) Write  $\eta_j = \eta(2^{-j}\cdot)$ , so  $K_j = K\eta_j$  and hence

$$\widehat{K_j} = \widehat{K} * \widehat{\eta_j} = (\psi d\sigma) * \widehat{\eta_j},$$

using  $\widehat{K} = \psi d\sigma$  by Fourier inversion (Theorem 2.5.11, the measure being tempered). Now  $\widehat{\eta_j}(\xi) = 2^{jn}\widehat{\eta}(2^j\xi)$ , so  $\|\widehat{\eta_j}\|_{L^1} = \|\widehat{\eta}\|_{L^1}$  and, since  $\widehat{\eta} \in \mathcal{S}$ , for every  $N$

$$|\widehat{\eta_j}(\xi)| \lesssim_N \frac{2^{jn}}{(1 + 2^j|\xi|)^N}.$$

Therefore, for any  $\xi$ ,

$$|\widehat{K_j}(\xi)| \leq \int_S |\widehat{\eta_j}(\xi - \omega)| |\psi(\omega)| d\sigma(\omega) \lesssim_N 2^{jn} \int_S \frac{d\sigma(\omega)}{(1 + 2^j|\xi - \omega|)^N}.$$

Decompose  $S$  into the pieces where  $|\xi - \omega| \approx 2^{-j}2^k$ ,  $k \geq 0$ . Since  $S$  is a hypersurface,  $\sigma(S \cap B(\xi, r)) \lesssim_S r^{n-1}$  for  $r \leq 1$ , and  $\sigma(S) < \infty$ ; so the  $k$ -th piece has measure  $\lesssim \min\{(2^{k-j})^{n-1}, 1\}$  and contributes  $\lesssim 2^{jn}2^{-kN} \min\{(2^{k-j})^{n-1}, 1\}$ . Taking  $N = n$  and summing over  $k \geq 0$ ,

$$|\widehat{K_j}(\xi)| \lesssim 2^{jn} 2^{-j(n-1)} \sum_{k \geq 0} 2^{-kn} 2^{k(n-1)} \lesssim 2^j,$$

the series  $\sum_k 2^{-k}$  converging. □

**Where we are.** The exponent  $2^j$  in (ii) is worth understanding, because it is the only place the cost of localisation enters. Smoothing  $d\sigma$  at scale  $2^{-j}$  spreads its mass over a slab of thickness  $2^{-j}$  around  $S$ ; the density of that slab is its total mass,  $O(1)$ , divided by its volume,  $\approx 2^{-j}$ . Hence  $2^j$ . The two exponents in Lemma 6.3.2 therefore have quite different origins:  $2^{-j(n-1)/2}$  comes from curvature and would be  $2^0$  for a flat piece, while  $2^j$  comes only from measure theory and is the same for every hypersurface. The theorem is the statement that curvature wins when  $b$  is large enough.

## 6.3.2 Proof of the theorem

*Proof of Theorem 6.3.1. Step 1. Reduce to the kernel.*

By Proposition 6.1.5 it suffices to prove

$$\|f * K\|_{L^b(\mathbb{R}^n)} \lesssim \|f\|_{L^{b'}(\mathbb{R}^n)}, \quad K = \widehat{\psi d\sigma}, \quad (6.3.1)$$

for  $f \in \mathcal{S}$ ; the general case follows by density.

**Step 2.** *The two endpoint bounds for a single piece.*

Let  $T_j f = f * K_j$ . By Lemma 6.3.2(i) and Young's inequality (Worked Example 2.2.1),

$$\|T_j f\|_{L^\infty} \leq \|K_j\|_{L^\infty} \|f\|_{L^1} \lesssim 2^{-j \frac{n-1}{2}} \|f\|_{L^1}. \quad (6.3.2)$$

By Lemma 6.3.2(ii) and Plancherel —  $T_j$  is the multiplier operator with symbol  $\widehat{K_j}$ , so Proposition 2.6.2 applies —

$$\|T_j f\|_{L^2} \leq \|\widehat{K_j}\|_{L^\infty} \|f\|_{L^2} \lesssim 2^j \|f\|_{L^2}. \quad (6.3.3)$$

**Step 3.** *Interpolate.*

Apply the Riesz–Thorin theorem (Theorem 2.2.1) to  $T_j$  between the pair  $(L^1, L^\infty)$  of (6.3.2) and the pair  $(L^2, L^2)$  of (6.3.3), with interpolation parameter  $\theta \in (0, 1)$ . The exponents are

$$\frac{1}{p_\theta} = \frac{1-\theta}{1} + \frac{\theta}{2} = 1 - \frac{\theta}{2}, \quad \frac{1}{q_\theta} = \frac{1-\theta}{\infty} + \frac{\theta}{2} = \frac{\theta}{2},$$

so  $q_\theta = 2/\theta$  and  $p_\theta = q'_\theta$ . Setting  $\theta = 2/b$  gives  $q_\theta = b$  and  $p_\theta = b'$ , and the interpolated norm is

$$\|T_j\|_{L^{b'} \rightarrow L^b} \lesssim \left(2^{-j \frac{n-1}{2}}\right)^{1-\theta} (2^j)^\theta = 2^{j(\theta - (1-\theta) \frac{n-1}{2})}. \quad (6.3.4)$$

**Step 4.** *The exponent is negative exactly in the stated range.*

Write  $\beta(\theta) = \theta - (1 - \theta)\frac{n-1}{2}$ . Then

$$\beta(\theta) < 0 \iff 2\theta < (1 - \theta)(n - 1) \iff \theta(n + 1) < n - 1 \iff \theta < \frac{n - 1}{n + 1}.$$

Since  $\theta = 2/b$ , this is  $\frac{2}{b} < \frac{n-1}{n+1}$ , that is

$$b > \frac{2(n + 1)}{n - 1},$$

which is the hypothesis. So  $\beta := \beta(2/b) < 0$  and, by (6.3.4),

$$\sum_{j \geq 1} \|T_j\|_{L^{b'} \rightarrow L^b} \lesssim \sum_{j \geq 1} 2^{j\beta} < \infty.$$

**Step 5.** *The bounded piece.*

$K_0$  is bounded —  $\widehat{\psi d\sigma}$  is a bounded continuous function — and compactly supported, hence lies in  $L^r(\mathbb{R}^n)$  for every  $r$ . Taking  $r$  with  $\frac{1}{b} = \frac{1}{b'} + \frac{1}{r} - 1$ , which is possible since  $b > b'$ , Young's inequality gives  $\|T_0 f\|_{L^b} \lesssim \|K_0\|_{L^r} \|f\|_{L^{b'}}$ .

**Step 6.** *Sum.*

By Steps 4 and 5 and the triangle inequality in  $L^b$ ,

$$\|f * K\|_{L^b} \leq \sum_{j \geq 0} \|T_j f\|_{L^b} \lesssim \|f\|_{L^{b'}},$$

which is (6.3.1). The equivalent restriction statement follows from Theorem 6.1.3:  $b = p'$  gives  $p = b' < \left(\frac{2(n+1)}{n-1}\right)' = \frac{2(n+1)}{n+3}$ .  $\square$

**Corollary 6.3.3** (The  $L^2$  trace estimate). *Let  $S$  be as in Theorem 6.3.1. Then for every  $b > \frac{2(n+1)}{n-1}$  and every  $f \in \mathcal{S}(\mathbb{R}^n)$ ,*

$$\|\widehat{f}\|_{L^2(S, d\sigma)} \lesssim_{n,b} \|f\|_{L^{b'}(\mathbb{R}^n)}, \quad b' = \frac{b}{b-1}.$$

*In particular the restriction of  $\widehat{f}$  to  $S$  is a well-defined element of  $L^2(d\sigma)$  for every  $f \in L^p(\mathbb{R}^n)$  with  $p < \frac{2(n+1)}{n+3}$ .*

*Proof.* Theorem 6.3.1 gives  $\mathcal{E} : L^2(d\sigma) \rightarrow L^b(\mathbb{R}^n)$ . By Theorem 6.1.3 with  $a = q' = 2$  — so that  $q = 2$  as well — and  $p' = b$ , this is equivalent to  $\mathcal{R} : L^{b'}(\mathbb{R}^n) \rightarrow L^2(d\sigma)$  with the same constant. The final sentence is the case  $p = b'$ , together with the density of  $\mathcal{S}$  in  $L^p$  (Theorem 1.8.7), which lets the estimate define  $\mathcal{R}$  on all of  $L^p$  by continuity; and  $b > \frac{2(n+1)}{n-1}$  translates to  $b' < \frac{2(n+1)}{n+3}$  by conjugation.  $\square$

**Where we are.** This corollary is the form in which Stein–Tomas is usually applied, and it is worth stating separately because it answers the question Chapter 6.1 began with: for which  $p$  does the restriction of  $\hat{f}$  to a sphere exist? The answer is  $p < \frac{2(n+1)}{n+3}$ , with values in  $L^2(d\sigma)$ .

At  $n = 2$  this is  $p < \frac{6}{5}$ ; at  $n = 3$ ,  $p < \frac{4}{3}$ . Compare Worked Example 6.1.1: the trivial range was  $p = 1$  only, so Stein–Tomas has genuinely enlarged it, and Conjecture 6.2.6 asserts one can go to  $p < \frac{2n}{n+1}$  — that is  $\frac{4}{3}$  at  $n = 2$  and  $\frac{3}{2}$  at  $n = 3$  — at the cost of weakening the target to  $L^\infty(d\sigma)$ .

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WORKED EXAMPLE 6.3.1— TIER A **The exponent in low dimensions**

$n = 2$ .  $\frac{2(n+1)}{n-1} = 6$ . So  $\mathcal{E} : L^2(d\sigma) \rightarrow L^b(\mathbb{R}^2)$  for  $b > 6$ , and in restriction form  $R^*(p \rightarrow 2)$  for  $p < \frac{6}{5}$ . The restriction conjecture in the plane asks for  $b > 4$ , so Stein–Tomas is genuinely weaker — and Chapter 6.8 closes the gap.

$n = 3$ .  $\frac{2(n+1)}{n-1} = 4$ , and the conjecture asks for  $b > 3$ . The current record lies strictly between; see Worked Example 6.2.3.

*Large  $n$ .*  $\frac{2(n+1)}{n-1} \rightarrow 2$  and  $\frac{2n}{n-1} \rightarrow 2$ , so both the theorem and the conjecture approach the trivial exponent  $b = 2$ . The *difference*  $\frac{2(n+1)}{n-1} - \frac{2n}{n-1} = \frac{2}{n-1}$  also tends to 0: in high dimensions Stein–Tomas is close to sharp in absolute terms. This is misleading. The natural measure of progress is not  $b$  but the exponent in the associated Kakeya bound, and there the gap does not close.

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WORKED EXAMPLE 6.3.2— TIER B **Where each hypothesis is used**

It is worth tracing which feature of  $S$  each step needed, since the theorem is often quoted for surfaces where one of them fails.

*Non-vanishing Gaussian curvature* entered once, in Lemma 6.3.2(i), to give the decay  $|x|^{-(n-1)/2}$ . If only  $k$  principal curvatures are non-zero, Theorem 5.5.3 gives  $|x|^{-k/2}$  instead, and repeating Steps 3 and 4 with  $\frac{n-1}{2}$  replaced by  $\frac{k}{2}$  yields

$$\beta(\theta) < 0 \iff \theta < \frac{k}{k+2} \iff b > \frac{2(k+2)}{k},$$

the Stein–Tomas theorem for surfaces of curvature rank  $k$ . For the cylinder of Worked Example 5.5.2,  $k = 1$  and  $n = 3$ : the exponent is 6 rather than 4. *Compactness* entered in Step 5 and in the finiteness of  $\sigma(S)$  used in Lemma 6.3.2(ii). For a non-compact surface such as the full paraboloid one localises first and sums, at the cost of an  $\varepsilon$ .

*Smoothness* entered only through Theorem 5.5.2, which needs enough derivatives for stationary phase;  $C^\infty$  is convenient and far more than necessary.

*What was never used* is any relation between  $S$  and the sphere. The proof is insensitive to the global shape of  $S$ : only the local curvature at the points where each direction is normal matters, which is exactly what Lemma 5.5.1 said.

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**WORKED EXAMPLE 6.3.3— TIER C: RESEARCH TASTE    The endpoint, and why we stop short of it**

Theorem 6.3.1 proves the open range  $b > \frac{2(n+1)}{n-1}$ . The endpoint  $b = \frac{2(n+1)}{n-1}$  is also true — that is Tomas’s theorem, completed by Stein — and this series does not prove it. The reason is worth being precise about, because it is a structural limitation rather than an oversight.

(a) *Exactly where the proof fails.* At the endpoint,  $\theta = \frac{2}{b}$  equals  $\frac{n-1}{n+1}$  and so  $\beta(\theta) = 0$ . Then (6.3.4) gives  $\|T_j\|_{L^{b'} \rightarrow L^b} \lesssim 1$  for every  $j$ , and Step 6 sums a constant over  $j$  — divergent. The loss is only logarithmic, but a logarithm is a logarithm.

(b) *What repairs it.* One embeds the operators  $T_j$  in an analytic family  $T_z$ , holomorphic in a complex parameter, arranging that the  $L^1 \rightarrow L^\infty$  bound holds on one edge of a strip and the  $L^2 \rightarrow L^2$  bound on the other, and then interpolates the *family* rather than each member. Because the interpolation is applied once to the whole sum rather than term by term, the divergent series never appears. This is Stein’s interpolation theorem for analytic families — and it was removed from the plan of this series as revision R4 in Volume 5, on the grounds that nothing here needs it.

(c) *Whether the endpoint is needed later.* It is not. Volumes 12 and 13 use Stein–Tomas as an input to induction-on-scales arguments, which already carry  $\delta^{-\varepsilon}$  losses; an  $\varepsilon$  in the exponent is invisible against  $\delta^{-\varepsilon}$ . A reader who wants the endpoint will find it in Stein’s *Harmonic Analysis*, Chapter IX.

(d) *The general lesson.* Twice now a proof has come down to whether a geometric series has ratio less than one — here, and in the Hörmander–Mikhlin theorem of Volume 3, Chapter 5. In both cases the open range falls out immediately and the endpoint requires a genuinely different technique. That is the normal state of affairs in this subject, and it is worth recognising the pattern: *when an argument sums a geometric series over scales, the endpoint is a separate theorem.*

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**WORKED EXAMPLE 6.3.4— TIER B    Why the two bounds on  $K_j$  must be combined, and not used singly**



Lemma 6.3.2 gives two estimates on each dyadic piece. Using either alone gives nothing, and seeing exactly how each fails is the quickest route to understanding what interpolation buys.

*Using only  $L^1 \rightarrow L^\infty$ .* Summing (6.3.2) over  $j$  gives

$$\|f * K\|_{L^\infty} \lesssim \left( \sum_{j \geq 1} 2^{-j \frac{n-1}{2}} \right) \|f\|_{L^1} \lesssim \|f\|_{L^1},$$

a valid but useless conclusion: it is the statement  $\mathcal{E} : L^2(d\sigma) \rightarrow L^\infty$ , which holds trivially by Cauchy–Schwarz and gives  $b = \infty$ .

*Using only  $L^2 \rightarrow L^2$ .* Summing (6.3.3) gives  $\sum_j 2^j = \infty$ . The series diverges, and this is not an artefact:  $\widehat{d\sigma}$  is not a bounded multiplier of  $L^2$  — indeed  $\widehat{\widehat{d\sigma}} = d\sigma$  is a measure, not a function — so no  $L^2$  bound for the full convolution exists.

*What interpolation does.* It produces, for each  $j$ , a bound  $2^{j\beta(\theta)}$  that *interpolates between a convergent and a divergent series*, and asks for which  $\theta$  the result still converges. The answer,  $\theta < \frac{n-1}{n+1}$ , is exactly the trade-off between the decay  $2^{-j(n-1)/2}$  that curvature supplies and the growth  $2^j$  that localisation costs.

*The general shape.* This is the standard architecture of an estimate proved by decomposition: one bound improves with  $j$ , another worsens, and the theorem is the range of exponents where the geometric mean decays. Chapter 3.5’s proof of the Hörmander–Mikhlin theorem has exactly this form, with  $\lambda^{-N}$  against  $2^{|j|^\alpha}$ ; so does the proof of Theorem 5.8.3. Once recognised, such proofs can be read by locating the two competing bounds and computing where their product is summable, without following the details.

## Drills

**Drill 6.3.1.** Verify the exponent identity  $p_\theta = q'_\theta$  in Step 3, and confirm that  $\theta = 2/b$  gives  $q_\theta = b$ .

**Drill 6.3.2.** Carry out the summation in Lemma 6.3.2(ii) with general  $N$  and determine the smallest  $N$  for which the argument closes.

**Drill 6.3.3.** Deduce from Theorem 6.3.1 and Theorem 6.1.3 that  $\|\widehat{f}|_S\|_{L^2(d\sigma)} \lesssim \|f\|_{L^p}$  for  $p < \frac{2(n+1)}{n+3}$ , and check this exponent at  $n = 2, 3$ .

**Drill 6.3.4.** Show that Theorem 6.3.1 is false for  $b < \frac{2(n+1)}{n-1}$  by combining it with Theorem 6.2.3, and identify the extremal  $g$ .

## Notes

Tomas proved the non-endpoint case in 1975 by the argument given here; Stein obtained the endpoint shortly after, by analytic interpolation. The theorem is the first non-trivial restriction estimate and remains, for  $a = 2$ , the sharp one.

The dyadic decomposition of  $\widehat{d\sigma}$  used in Lemma 6.3.2 is the prototype for the wave packet decompositions of Volume 12: in both cases a single object is cut into pieces at different scales, each piece is estimated by two incompatible methods, and the results are interpolated.

## Chapter 6.4

# Strichartz Estimates

The Stein–Tomas theorem is a statement about a sphere. Applied instead to a paraboloid or a cone, the very same theorem becomes a statement about the Schrödinger or wave equation, and it is in that guise that it is most used. This chapter performs the translation.

### 6.4.1 The paraboloid is the Schrödinger evolution

**Definition 6.4.1.** For  $u_0 \in \mathcal{S}(\mathbb{R}^d)$  the free Schrödinger evolution is

$$(e^{it\Delta}u_0)(y) = \int_{\mathbb{R}^d} \widehat{u_0}(\zeta) e^{2\pi i(y \cdot \zeta - 2\pi t|\zeta|^2)} d\zeta,$$

the unique Schwartz-class solution of  $i\partial_t u = -\Delta u$ ,  $u(0) = u_0$ .

**Lemma 6.4.2.** Let  $n = d + 1$  and let

$$P = \{(\zeta, |\zeta|^2) : |\zeta| \leq 1\} \subseteq \mathbb{R}^n$$

be the truncated paraboloid, with measure  $d\mu = d\zeta$  in the graph parametrisation. Then for  $g$  supported in  $|\zeta| \leq 1$ ,

$$\mathcal{E}_P g(y, s) = \int_{|\zeta| \leq 1} g(\zeta) e^{2\pi i(y \cdot \zeta + s|\zeta|^2)} d\zeta = (e^{it\Delta}u_0)(y) \Big|_{t=-s/2\pi},$$

where  $u_0$  is defined by  $\widehat{u_0} = g$ .

*Proof.* Substituting  $t = -s/2\pi$  in the definition gives phase  $2\pi(y \cdot \zeta - 2\pi t|\zeta|^2) = 2\pi(y \cdot \zeta + s|\zeta|^2)$ , which is the phase in  $\mathcal{E}_P$ . The two integrals then agree term by term.  $\square$

So an estimate for  $\mathcal{E}_P$  on  $\mathbb{R}^{d+1} = \mathbb{R}_y^d \times \mathbb{R}_s$  is a space-time estimate for the Schrödinger equation, with  $\|g\|_{L^2(d\mu)} = \|\widehat{u_0}\|_{L^2} = \|u_0\|_{L^2}$  by Plancherel.

**Theorem 6.4.3** (Strichartz, non-endpoint, frequency-localised). *Let  $d \geq 1$  and  $n = d + 1$ . For every*

$$q > \frac{2(d+2)}{d}$$

*there is  $C = C(d, q)$  such that, for all  $u_0 \in L^2(\mathbb{R}^d)$  with  $\widehat{u_0}$  supported in  $\{|\zeta| \leq 1\}$ ,*

$$\|e^{it\Delta}u_0\|_{L^q(\mathbb{R}_y^d \times \mathbb{R}_t)} \leq C \|u_0\|_{L^2(\mathbb{R}^d)}.$$

*Proof.* The truncated paraboloid  $P \subseteq \mathbb{R}^n$  has non-vanishing Gaussian curvature: in the graph parametrisation  $\zeta \mapsto (\zeta, |\zeta|^2)$  the second fundamental form at every point is a positive multiple of the identity, since  $\text{Hess}(|\zeta|^2) = 2I$  (Lemma 5.5.1). It is compact. So Theorem 6.3.1 applies with  $n = d + 1$  and gives

$$\|\mathcal{E}_P g\|_{L^q(\mathbb{R}^{d+1})} \lesssim \|g\|_{L^2(d\mu)} \quad \text{for } q > \frac{2(n+1)}{n-1} = \frac{2(d+2)}{d}.$$

By Lemma 6.4.2 the left-hand side is  $\|e^{it\Delta}u_0\|_{L^q(dy \, ds)}$ , and the change of variable  $s = -2\pi t$  multiplies it by the constant  $(2\pi)^{1/q}$ . By Plancherel the right-hand side is  $\|u_0\|_{L^2}$ .  $\square$

**WORKED EXAMPLE 6.4.1— TIER A Reading the exponent, and checking it by scaling**

The theorem as stated has the same exponent  $q$  in space and in time, and is restricted to frequencies  $|\zeta| \leq 1$ . Both features deserve comment.

(a) *Scaling determines the admissible pairs.* Drop the frequency restriction and ask for which  $(q, r)$  one may hope

$$\|e^{it\Delta}u_0\|_{L_t^q L_y^r} \lesssim \|u_0\|_{L^2}.$$

If  $u_0^\lambda(y) = u_0(\lambda y)$  then  $e^{it\Delta}u_0^\lambda(y) = (e^{i\lambda^2 t \Delta}u_0)(\lambda y)$ , so the left side scales by  $\lambda^{-2/q-d/r}$  and the right by  $\lambda^{-d/2}$ . Equality forces

$$\frac{2}{q} + \frac{d}{r} = \frac{d}{2}, \tag{6.4.1}$$

the *admissibility* relation. Theorem 6.4.3 is the diagonal case  $q = r$ , for which (6.4.1) gives  $\frac{2}{q} + \frac{d}{q} = \frac{d}{2}$ , i.e.  $q = \frac{2(d+2)}{d}$  — exactly the endpoint of the theorem's range. So the theorem is sharp up to the endpoint on the diagonal, and the frequency localisation is what allows the strict inequality  $q > \frac{2(d+2)}{d}$  without violating scaling.

(b) *Why localisation is needed for  $q > \text{the diagonal value}$ .* Without it, (6.4.1) must hold exactly; a strict inequality would be inconsistent with the scaling

above. Localising to  $|\zeta| \leq 1$  breaks the scale invariance and buys the open range.

(c)  $d = 1$ . The exponent is  $q > 6$ , and (6.4.1) on the diagonal gives  $q = 6$ : the classical  $L^6_{t,y}$  Strichartz estimate for the one-dimensional Schrödinger equation.

**Proposition 6.4.4** (The dispersive estimate). *For  $u_0 \in L^1(\mathbb{R}^d) \cap L^2(\mathbb{R}^d)$  and  $t \neq 0$ ,*

$$\|e^{it\Delta}u_0\|_{L^\infty(\mathbb{R}^d)} \leq (4\pi|t|)^{-d/2}\|u_0\|_{L^1(\mathbb{R}^d)}.$$

*Proof.* By the Gaussian computation of Volume 2 — Proposition 2.4.3 together with the analytic continuation of Lemma 5.4.2, applied to the purely imaginary parameter — the Schrödinger propagator is convolution with

$$K_t(y) = (4\pi it)^{-d/2} e^{i|y|^2/4t},$$

where  $(4\pi it)^{-d/2}$  is computed with the branch continued from the positive real axis, exactly as in Proposition 5.2.1. Hence  $|K_t(y)| = (4\pi|t|)^{-d/2}$  for every  $y$ , and Young's inequality gives

$$\|e^{it\Delta}u_0\|_{L^\infty} = \|K_t * u_0\|_{L^\infty} \leq \|K_t\|_{L^\infty}\|u_0\|_{L^1} = (4\pi|t|)^{-d/2}\|u_0\|_{L^1}. \quad \square$$

**Where we are.** Proposition 6.4.4 is the other classical route to Strichartz estimates: one interpolates the  $L^1 \rightarrow L^\infty$  decay above against the  $L^2 \rightarrow L^2$  conservation  $\|e^{it\Delta}u_0\|_{L^2} = \|u_0\|_{L^2}$ , and then applies a fractional integration in  $t$ . That argument gives the same admissible pairs and is the one usually presented in PDE courses.

The two routes are genuinely different in what they reveal. The dispersive route shows the estimates come from the *decay of the propagator*; the Stein–Tomas route of Theorem 6.4.3 shows they come from the *curvature of the characteristic surface*. Both are correct, and the second is the one that generalises to the problems of this series, since it is curvature, not decay, that the Kakeya obstruction limits.

## 6.4.2 The cone is the wave evolution

**Proposition 6.4.5.** *Let  $C = \{(\zeta, |\zeta|) : \frac{1}{2} \leq |\zeta| \leq 1\} \subseteq \mathbb{R}^{d+1}$  be a truncated cone,  $d \geq 2$ . Then  $C$  has  $d - 1$  non-vanishing principal curvatures and one vanishing, and consequently*

$$\|\mathcal{E}_C g\|_{L^q(\mathbb{R}^{d+1})} \lesssim \|g\|_{L^2(d\sigma)} \quad \text{for } q > \frac{2(d+1)}{d-1}.$$

*Proof.* The cone is ruled by the lines  $\{(r\theta, r) : r > 0\}$  for  $\theta \in \mathbb{S}^{d-1}$ , and is affine along each, so one principal curvature vanishes. In the transverse directions it is a dilate of the sphere  $\mathbb{S}^{d-1}$  and the remaining  $d-1$  principal curvatures are non-zero on the compact piece  $\frac{1}{2} \leq |\zeta| \leq 1$ . So the curvature rank is  $k = d-1$ .

By Worked Example 6.3.2, the Stein–Tomas argument with curvature rank  $k$  gives the range  $q > \frac{2(k+2)}{k}$ ; with  $k = d-1$  this is  $q > \frac{2(d+1)}{d-1}$ .  $\square$

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**WORKED EXAMPLE 6.4.2— TIER B Why the wave equation is harder than the Schrödinger equation**

Compare the two exponents just obtained, both in  $\mathbb{R}^{d+1}$ :

$$\text{Schrödinger (paraboloid): } q > \frac{2(d+2)}{d}, \quad \text{wave (cone): } q > \frac{2(d+1)}{d-1}.$$

The second is larger — at  $d = 3$ ,  $q > 4$  against  $q > \frac{10}{3}$  — so the wave estimate is weaker.

*The reason is one missing curvature.* The paraboloid curves in all  $d$  tangential directions; the cone is flat along its rulings and curves in only  $d-1$ . By Lemma 5.5.1 the rank of the second fundamental form is exactly what stationary phase converts into decay, so the cone's  $\widehat{d\sigma}$  decays like  $|x|^{-(d-1)/2}$  where the paraboloid's decays like  $|x|^{-d/2}$ , and Lemma 6.3.2(i) carries the difference straight into the exponent.

*Where the flat direction comes from, physically.* A solution of the wave equation propagates at unit speed in a fixed direction; the ruling of the cone is the record of that. The Schrödinger evolution has no finite propagation speed and its characteristic surface, the paraboloid, is curved everywhere. So the analytic difference traces back to a difference in the physics.

*A caution.* The cone estimate above is not the sharpest known — the wave equation admits estimates in mixed norms  $L_t^q L_y^r$  that the diagonal Stein–Tomas argument does not see, and the local smoothing conjecture asserts a further gain after averaging in time. Worked Example 4.8.2 in Volume 4 described that gain as an embedding between function spaces, and Volume 13 proves the planar case.

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**WORKED EXAMPLE 6.4.3— TIER B The admissible pairs, computed**

Theorem 6.4.3 is the diagonal case  $q = r$ . The estimates actually used in practice are off-diagonal, and it is worth deriving the full family, since the derivation is short and the table is otherwise memorised without reason.

*The two constraints.* A pair  $(q, r)$  is *admissible* for the Schrödinger equation

in  $\mathbb{R}^d$  if

$$\|e^{it\Delta}u_0\|_{L_t^q L_y^r} \lesssim \|u_0\|_{L^2}.$$

Scaling forces  $\frac{2}{q} + \frac{d}{r} = \frac{d}{2}$ , by Worked Example 6.4.1. A second constraint comes from requiring  $q \geq 2$ : the Knapp example, transported through Lemma 6.4.2, produces a solution concentrated on a slab and shows the estimate fails for  $q < 2$ .

*The endpoints of the family.* Setting  $q = \infty$  gives  $r = 2$ , which is just conservation of the  $L^2$  norm — true and trivial. Setting  $q = 2$  gives  $\frac{d}{r} = \frac{d}{2} - 1$ , that is  $r = \frac{2d}{d-2}$ , which requires  $d \geq 3$ ; this is the Keel–Tao endpoint, true but not proved here.

*Worked cases.*

| $d$ | diagonal $q = r$ | endpoint $(q, r)$                 | a typical off-diagonal pair   |
|-----|------------------|-----------------------------------|-------------------------------|
| 1   | 6                | none ( $q = 2$ needs $d \geq 3$ ) | $(\infty, 2), (8, 4)$         |
| 2   | 4                | none                              | $(\infty, 2), (4, 4), (6, 3)$ |
| 3   | $\frac{10}{3}$   | $(2, 6)$                          | $(\infty, 2), (4, 3)$         |

Each entry is checked by substituting into  $\frac{2}{q} + \frac{d}{r} = \frac{d}{2}$ : for  $d = 3$ ,  $(q, r) = (4, 3)$  gives  $\frac{2}{4} + \frac{3}{3} = \frac{1}{2} + 1 = \frac{3}{2} = \frac{d}{2}$ . ✓

*What Theorem 6.3.1 delivers, and what it does not.* The Stein–Tomas route gives exactly the diagonal pairs, since it produces an  $L^b$  bound in all  $n = d + 1$  space-time variables at once. Off-diagonal pairs require interpolating the diagonal estimate against the trivial pair  $(\infty, 2)$  — which recovers the segment between them, and that is the whole admissible family in the range  $q >$  diagonal. The endpoint is what no interpolation reaches.

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**WORKED EXAMPLE 6.4.4— TIER C: RESEARCH TASTE** **What Strichartz estimates are for**

The translation of this chapter is not a formal exercise: it is the reason the restriction problem is studied outside harmonic analysis, and it is worth knowing what the estimates do.

(a) *Well-posedness of nonlinear equations.* Consider the cubic nonlinear Schrödinger equation  $i\partial_t u + \Delta u = |u|^2 u$ . Solving it by Picard iteration requires a space in which the Duhamel term  $\int_0^t e^{i(t-s)\Delta}(|u|^2 u)(s) ds$  is controlled by the data, and Theorem 6.4.3 supplies exactly that: the space-time norm on the left is what the nonlinearity is estimated in, and the  $L^2$  norm on the right is the conserved quantity. The admissible pairs (6.4.1) are precisely

the norms available for the iteration, which is why they are tabulated in every paper on the subject.

*(b) The endpoint matters here, and we do not have it.* For some applications one needs the endpoint of the admissible range,  $q = 2$ , available for  $d \geq 3$ . That is the Keel–Tao theorem, whose proof is bilinear and which this series does not prove — announced in the preface to this volume. The non-endpoint estimates suffice for local well-posedness; the endpoint is used for scattering and for critical problems.

*(c) The direction of the implication.* Restriction estimates imply Strichartz estimates, as this chapter shows, and the implication is essentially an identity — Lemma 6.4.2 is a change of variables. It does not run backwards in any useful sense: a Strichartz estimate is a statement about  $L^2$  data, which is the case  $a = 2$ , and the restriction conjecture concerns  $a = \infty$ . So progress on the conjecture does not follow from progress on dispersive equations.

*(d) What this says about the shape of the subject.* Everything in this chapter was a corollary: one theorem about surfaces, applied to two particular surfaces, gave the classical estimates for the two classical dispersive equations. The mathematical content sits entirely in Theorem 6.3.1 and hence, through Lemma 6.3.2, in the curvature of the surface. That is the sense in which the restriction problem is the fundamental one and the PDE estimates are consequences.

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## Drills

**Drill 6.4.1.** Verify Lemma 6.4.2 by direct substitution, tracking the constant introduced by  $t = -s/2\pi$ .

**Drill 6.4.2.** Confirm that the second fundamental form of the paraboloid  $\zeta \mapsto (\zeta, |\zeta|^2)$  is  $2I$  at the origin, and compute it at a general point.

**Drill 6.4.3.** Derive the admissibility relation (6.4.1) for the wave equation, where the scaling is  $u_0^\lambda(y) = u_0(\lambda y)$  with evolution  $e^{it\sqrt{-\Delta}}$ , and compare with Proposition 6.4.5.

**Drill 6.4.4.** Show that Theorem 6.4.3 fails without the frequency localisation, by scaling a fixed non-zero  $u_0$ .

**Where we are.** It is worth recording what this chapter did *not* require. No property of the Schrödinger equation was used beyond the shape



of its characteristic surface; the evolution never appeared except through Lemma 6.4.2, which is a change of variables. In particular nothing about conservation laws, dispersion in the physical sense, or the structure of the equation entered.

That is a general feature. A linear constant-coefficient dispersive equation  $i\partial_t u = P(D)u$  has characteristic surface  $\{(\zeta, P(\zeta))\}$ , and its space-time estimates are the restriction estimates for that surface. The Schrödinger equation gives a paraboloid, the wave equation a cone, the Airy equation  $\partial_t u + \partial_x^3 u = 0$  the curve  $(\zeta, \zeta^3)$  — which has an inflection at the origin, where the curvature vanishes, and correspondingly worse estimates near that frequency.

So the classification of dispersive estimates *is* the classification of surfaces by their curvature, and the results of Volume 5 are what performs it. A reader who wants the estimates for a new equation should compute the second fundamental form of its characteristic surface, find the rank, and read off the exponent from Worked Example 6.3.2.

## Notes

Strichartz proved the estimate that bears his name in 1977, deducing it from Stein–Tomas exactly as here. The systematic theory of admissible pairs is due to Ginibre–Velo and Keel–Tao; the latter also settled the endpoint.

The observation that a dispersive equation’s characteristic surface carries the relevant curvature, and that the flat directions of that surface are the directions of propagation, is the organising idea of the field.



## Chapter 6.5

# Epsilon-Removal

Every argument in Volumes 12 and 13 proves a *local* estimate: a bound for  $\mathcal{E}g$  on a ball of radius  $R$ , with a constant growing like  $R^\varepsilon$ . The conjecture, by contrast, is global and has no loss. This short chapter proves that the two are equivalent up to shrinking the exponent, so that the  $R^\varepsilon$  losses carried by induction-on-scales arguments are harmless.

### 6.5.1 Local estimates

**Definition 6.5.1.** Let  $S \subseteq \mathbb{R}^n$  be a compact hypersurface and  $1 \leq b \leq \infty$ . Say that the *local estimate*  $\mathcal{E}(2 \rightarrow b; \varepsilon)$  holds if for every  $R \geq 1$  and every  $g \in L^2(d\sigma)$ ,

$$\|\mathcal{E}g\|_{L^b(B_R)} \leq C_\varepsilon R^\varepsilon \|g\|_{L^2(d\sigma)},$$

where  $B_R$  is any ball of radius  $R$  and  $C_\varepsilon$  does not depend on  $R$  or  $g$ .

The global estimate is the case  $\varepsilon = 0$  with  $B_R$  replaced by  $\mathbb{R}^n$ ; it plainly implies the local one. The content is the converse.

**Theorem 6.5.2** (Epsilon-removal). *Suppose the local estimate  $\mathcal{E}(2 \rightarrow b_0; \varepsilon)$  holds for every  $\varepsilon > 0$ . Then the global estimate*

$$\|\mathcal{E}g\|_{L^b(\mathbb{R}^n)} \lesssim_b \|g\|_{L^2(d\sigma)}$$

*holds for every  $b > b_0$ .*

*Idea of the proof.* Two ingredients. First, an  $L^b$  norm over  $\mathbb{R}^n$  can be assembled from norms over balls of radius  $R$ : cover  $\mathbb{R}^n$  by such balls with bounded overlap and sum. That alone loses a factor equal to the number of balls, which is infinite.

The repair is the second ingredient: interpolation with a bound that is *better*

than needed at a larger exponent. The global estimate at  $b = \infty$  is free —  $\|\mathcal{E}g\|_\infty \leq \|g\|_{L^1(d\sigma)} \lesssim \|g\|_{L^2(d\sigma)}$  since  $d\sigma$  is finite — and interpolating the local estimate at  $b_0$  against it at  $\infty$  converts the  $R^\varepsilon$  loss into a *gain*  $R^{-\eta}$  at any exponent  $b$  strictly between, which is summable over a dyadic family of scales.

*Proof.* Fix  $b > b_0$  and choose  $\theta \in (0, 1)$  with

$$\frac{1}{b} = \frac{1-\theta}{b_0} + \frac{\theta}{\infty} = \frac{1-\theta}{b_0}, \quad \text{that is} \quad 1-\theta = \frac{b_0}{b} \in (0, 1).$$

**Step 1.** *The two bounds on a ball.*

Let  $B_R$  be a ball of radius  $R \geq 1$ . The hypothesis with parameter  $\varepsilon$  gives

$$\|\mathcal{E}g\|_{L^{b_0}(B_R)} \leq C_\varepsilon R^\varepsilon \|g\|_{L^2(d\sigma)},$$

and the trivial bound gives

$$\|\mathcal{E}g\|_{L^\infty(B_R)} \leq \|g\|_{L^1(d\sigma)} \leq \sigma(S)^{1/2} \|g\|_{L^2(d\sigma)},$$

by Cauchy–Schwarz on the finite measure space  $(S, d\sigma)$ .

**Step 2.** *Interpolate on the ball.*

For any function  $F$  and any exponents, Hölder’s inequality gives  $\|F\|_{L^b} \leq \|F\|_{L^{b_0}}^{1-\theta} \|F\|_{L^\infty}^\theta$  when  $\frac{1}{b} = \frac{1-\theta}{b_0}$ . Applying this on  $B_R$  with  $F = \mathcal{E}g$  and using Step 1,

$$\|\mathcal{E}g\|_{L^b(B_R)} \lesssim (R^\varepsilon)^{1-\theta} \|g\|_{L^2(d\sigma)} = R^{\varepsilon b_0/b} \|g\|_{L^2(d\sigma)}. \quad (6.5.1)$$

This is still a loss, and no better than the hypothesis; the gain comes in Step 3, from how the balls are assembled.

**Step 3.** *Sum over dyadic annuli.*

Decompose  $\mathbb{R}^n = B_1 \cup \bigcup_{k \geq 1} A_k$  with  $A_k = \{2^{k-1} \leq |x| < 2^k\}$ . On  $A_k$  apply the trivial bound and (6.5.1) at scale  $R = 2^k$  once more, but now exploiting that  $A_k$  has finite measure: by Hölder on  $A_k$ ,

$$\|\mathcal{E}g\|_{L^b(A_k)} \leq \|\mathcal{E}g\|_{L^{b_0}(A_k)}^{1-\theta} \|\mathcal{E}g\|_{L^\infty(A_k)}^\theta.$$

For the second factor we use a *decay* estimate rather than the trivial one. Since  $\mathcal{E}g = (gd\sigma)^\vee$  and  $d\sigma$  has non-vanishing curvature, the same computation as Lemma 6.3.2(i) applied to  $\mathcal{E}\mathcal{E}^*$  shows

$$\|\mathcal{E}g\|_{L^\infty(A_k)} \lesssim \|g\|_{L^2(d\sigma)}$$

uniformly, which is all that is required; and for the first factor (6.5.1) with  $R = 2^k$  gives  $2^{k\varepsilon} \|g\|_{L^2(d\sigma)}$ .

To obtain a summable bound, we instead run the interpolation with the exponent  $b_1 \in (b_0, b)$  and use that the hypothesis holds for *every*  $\varepsilon$ : choosing  $\varepsilon$  so small that

$$\eta := \frac{n}{b_1} - \frac{n}{b} - \varepsilon > 0$$

— possible since  $b_1 < b$  makes  $\frac{n}{b_1} - \frac{n}{b} > 0$  — and applying Hölder on  $A_k$  between  $L^{b_1}$  and  $L^b$ ,

$$\|\mathcal{E}g\|_{L^b(A_k)} \leq m(A_k)^{\frac{1}{b} - \frac{1}{b_1}} \|\mathcal{E}g\|_{L^{b_1}(A_k)} \lesssim 2^{kn(\frac{1}{b} - \frac{1}{b_1})} 2^{k\varepsilon} \|g\|_{L^2(d\sigma)} = 2^{-k\eta} \|g\|_{L^2(d\sigma)}.$$

**Step 4. Conclude.**

Summing the geometric series,

$$\|\mathcal{E}g\|_{L^b(\mathbb{R}^n)}^b = \|\mathcal{E}g\|_{L^b(B_1)}^b + \sum_{k \geq 1} \|\mathcal{E}g\|_{L^b(A_k)}^b \lesssim \|g\|_{L^2(d\sigma)}^b \left(1 + \sum_{k \geq 1} 2^{-kb\eta}\right) \lesssim \|g\|_{L^2(d\sigma)}^b.$$

□

**Where we are.** The mechanism is worth isolating, because it recurs. An  $R^\varepsilon$  loss is harmless whenever it can be played off against a *power* gain coming from somewhere else — here, from Hölder on an annulus of measure  $2^{kn}$  at two nearby exponents, which supplies  $2^{-kn(1/b_1 - 1/b)}$ . Since  $\varepsilon$  may be taken as small as one likes and the power gain is fixed once  $b > b_1$ , the gain always wins.

This is why the literature is content to prove local estimates with  $\delta^{-\varepsilon}$  losses, and why the phrase “by epsilon-removal” appears without comment in papers on decoupling. It is also why the *endpoint* is never obtained this way: at  $b = b_1$  the power gain vanishes and nothing is left to beat the loss.

**Proposition 6.5.3** (Local and global are equivalent, up to the exponent). *For a compact hypersurface  $S$  with non-vanishing curvature and  $1 \leq b_0 < \infty$ , the following are equivalent:*

- (i) the local estimate  $\mathcal{E}(2 \rightarrow b_0; \varepsilon)$  holds for every  $\varepsilon > 0$ ;
- (ii) the global estimate  $\|\mathcal{E}g\|_{L^b} \lesssim_b \|g\|_{L^2(d\sigma)}$  holds for every  $b > b_0$ .

*Proof.* That (i) implies (ii) is Theorem 6.5.2. Conversely, assume (ii) and fix  $\varepsilon > 0$ . Choose  $b > b_0$  with

$$\frac{n}{b_0} - \frac{n}{b} < \varepsilon,$$

possible since the left side tends to 0 as  $b \downarrow b_0$ . By Hölder on a ball of radius  $R$ , whose measure is  $\approx R^n$ ,

$$\|\mathcal{E}g\|_{L^{b_0}(B_R)} \leq m(B_R)^{\frac{1}{b_0}-\frac{1}{b}} \|\mathcal{E}g\|_{L^b(B_R)} \lesssim R^{\frac{n}{b_0}-\frac{n}{b}} \|g\|_{L^2(d\sigma)} \leq R^\varepsilon \|g\|_{L^2(d\sigma)},$$

using (ii) for the middle step. This is  $\mathcal{E}(2 \rightarrow b_0; \varepsilon)$ .  $\square$

**Where we are.** So nothing is lost by working locally, provided one is willing to state conclusions as open ranges. That is the convention throughout Volumes 12 and 13, and Proposition 6.5.3 is the licence for it.

The proposition also explains a phrasing the reader will meet constantly. A paper proves “the estimate holds for  $b_0$ ” meaning the local estimate with  $R^\varepsilon$ ; a survey then reports “the estimate holds for  $b > b_0$ ” meaning the global one. Both are correct and they say the same thing, but only because of this equivalence.

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WORKED EXAMPLE 6.5.1— TIER A **Why one cannot simply cover space by balls**

The naive approach is to cover  $\mathbb{R}^n$  by  $\approx (M/R)^n$  balls of radius  $R$  filling  $B_M$ , apply the local estimate on each, and sum. Doing so gives

$$\|\mathcal{E}g\|_{L^b(B_M)}^b \leq \sum_{\text{balls}} \|\mathcal{E}g\|_{L^b(B_R)}^b \lesssim \left(\frac{M}{R}\right)^n (R^\varepsilon)^b \|g\|_{L^2(d\sigma)}^b,$$

which diverges as  $M \rightarrow \infty$  for any fixed  $R$ . Nothing about the local estimate prevents  $\mathcal{E}g$  from being equally large on every one of the balls.

What rescues the argument is that  $\mathcal{E}g$  *cannot* in fact be equally large everywhere: it is the transform of a measure on a curved surface and so decays. The proof above encodes this by working with annuli, whose measure grows, and by paying for that growth with the difference of two exponents. Curvature enters only through the requirement  $b > b_0$ , i.e. that some non-trivial estimate holds at all.

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WORKED EXAMPLE 6.5.2— TIER B **The same device, in the form used in Volume 13**

Decoupling estimates are stated with an unavoidable  $\delta^{-\varepsilon}$ :

$$\left\| \sum_{\theta} f_{\theta} \right\|_{L^p(B_{\delta^{-1}})} \lesssim_{\varepsilon} \delta^{-\varepsilon} \left( \sum_{\theta} \|f_{\theta}\|_{L^p}^2 \right)^{1/2}.$$

The  $\varepsilon$  is not an artefact: at the critical exponent it is known that no  $\delta$ -free constant is possible. So decoupling is inherently a local, lossy statement, and any global conclusion drawn from it must remove the loss.

The removal is always of the type proved here — trade the  $\delta^{-\varepsilon}$  against a power gain obtained by moving slightly off the critical exponent. The practical consequence, and the reason to have seen the argument once: *a theorem proved by decoupling is never sharp at its endpoint*. Every application in Volume 13 states its exponent as an open condition, and Theorem 6.5.2 is why that is the natural form.

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WORKED EXAMPLE 6.5.3— TIER B **Tracking an  $\varepsilon$  through an induction on scales**

Induction-on-scales arguments produce constants of the form  $C_\varepsilon R^\varepsilon$ , and a reader meeting them for the first time reasonably worries that the  $\varepsilon$ 's accumulate. They do not, and seeing why once removes the worry permanently.

*The typical structure.* One proves a statement  $\mathcal{S}(R)$  at scale  $R$  assuming  $\mathcal{S}(R^{1/2})$  at the square root of that scale, with a bound of the shape

$$C(R) \leq K \cdot C(R^{1/2})^\alpha$$

for constants  $K$  and  $\alpha$  depending only on the dimension. The question is what  $C(R)$  grows like.

*The case  $\alpha = 1$ .* Iterating  $m$  times reaches scale  $R^{2^{-m}}$ , and

$$C(R) \leq K^m C(R^{2^{-m}}).$$

Stopping when  $R^{2^{-m}} \approx 2$ , that is  $m \approx \log \log R$ , and using that  $C$  is bounded at bounded scales,

$$C(R) \lesssim K^{\log \log R} = (\log R)^{\log K}.$$

So the induction costs a *power of a logarithm* — which is  $\lesssim_\varepsilon R^\varepsilon$  for every  $\varepsilon > 0$ , and therefore removable by Theorem 6.5.2. This is the mechanism behind essentially every  $\delta^{-\varepsilon}$  in the literature.

*Why one does not simply keep the logarithm.* Because the exponent  $\log K$  is not usually computed, and because a second induction would square it. Carrying “ $\lesssim_\varepsilon R^\varepsilon$ ” is both weaker and more robust: it is stable under composition, since  $R^{\varepsilon_1} R^{\varepsilon_2} = R^{\varepsilon_1 + \varepsilon_2}$  and the  $\varepsilon$ 's may be chosen in advance to sum to the target.

*The one thing to watch.* The constant  $C_\varepsilon$  does blow up as  $\varepsilon \rightarrow 0$  — typically like  $\exp(\varepsilon^{-1})$  or worse — so an argument that needs  $\varepsilon$  to depend on  $R$  is invalid. In every correct application,  $\varepsilon$  is fixed first and  $R$  tends to infinity afterwards. Checking the order of quantifiers is the standard way to find an error in this kind of proof, and it is worth doing whenever a bound looks too strong.

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**WORKED EXAMPLE 6.5.4— TIER A The bookkeeping, on a concrete pair of exponents**

Take  $n = 3$  and  $b_0 = 4$ , the Stein–Tomas exponent, and suppose we have the local estimate at  $b_0 = 4$  with loss  $R^\varepsilon$ . What does Theorem 6.5.2 give at  $b = 5$ ?

*Choosing the intermediate exponent.* Take  $b_1 = \frac{b_0+b}{2} = \frac{9}{2}$ . Then

$$\frac{n}{b_1} - \frac{n}{b} = \frac{3}{9/2} - \frac{3}{5} = \frac{2}{3} - \frac{3}{5} = \frac{1}{15}.$$

So any  $\varepsilon < \frac{1}{15}$  works, and the resulting decay rate is  $\eta = \frac{1}{15} - \varepsilon$ .

*The sum.* With  $\varepsilon = \frac{1}{30}$ ,  $\eta = \frac{1}{30}$ , and Step 4 of the proof sums  $\sum_k 2^{-5k/30} = \sum_k 2^{-k/6}$ , a geometric series with ratio  $2^{-1/6} \approx 0.891$ . Convergent, though not fast: about 40 terms are needed for two-digit accuracy, which is to say scales from 1 up to  $2^{40}$ .

*What this shows.* The convergence is genuinely slow near the endpoint, and the constant in Theorem 6.5.2 degenerates as  $b \downarrow b_0$ : from Step 4 the constant is  $\approx (1 - 2^{-b\eta})^{-1/b}$ , which blows up as  $\eta \rightarrow 0$ . So epsilon-removal is not free — it costs a constant that is large when the exponent is close to critical, and no amount of care improves this, since the endpoint statement is a different theorem.

*A useful habit.* When reading a claimed estimate at an exponent, ask whether it is open or closed. If open, expect a removable loss and a constant that blows up at the endpoint; if closed, expect a genuinely harder argument. The two cases are rarely distinguished in abstracts and almost always in the theorem statements.

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## Drills

**Drill 6.5.1.** Verify the Hölder inequality  $\|F\|_{L^b} \leq \|F\|_{L^{b_0}}^{1-\theta} \|F\|_{L^\infty}^\theta$  used in Step 2, and identify the relation between  $b, b_0, \theta$ .

**Drill 6.5.2.** Carry out Step 3 with explicit  $b_1 = \frac{b_0+b}{2}$  and determine how small  $\varepsilon$  must be taken.

**Drill 6.5.3.** Show that Theorem 6.5.2 gives nothing at  $b = b_0$ , by tracking  $\eta$  as  $b \downarrow b_0$ .

**Drill 6.5.4.** State and prove the analogue of Theorem 6.5.2 for  $L^a(d\sigma)$  data with  $a \neq 2$ , and check that the proof uses nothing about  $a$ .



**Where we are.** A last word on why this chapter is short and why it nevertheless has to exist.

The mathematics is two applications of Hölder. What it buys is a change of *register*: after Proposition 6.5.3, one may prove estimates on a ball of radius  $R$ , with any constant that is  $O(R^\varepsilon)$ , and still be proving a theorem about  $\mathbb{R}^n$ . Every technique in the volumes ahead — induction on scales, broad–narrow decomposition, polynomial partitioning, decoupling — produces exactly that kind of statement and nothing better, because each of them subdivides a ball and recurses.

Without this chapter those techniques would prove only local statements, and the conjectures they are aimed at are global. With it, the two are the same problem. That is why the result is quoted so often and proved so rarely: it is the bridge between the way the subject is practised and the way its questions are posed.

## Notes

Epsilon-removal in this context is due to Tao (1999), in a sharper form than the one proved here: his argument removes the loss without moving the exponent, at the cost of a more delicate decomposition. The version above is the one that suffices for this series and is the standard textbook treatment.



## Chapter 6.6

# Restriction Implies Kakeya

Worked Example 6.2.3 argued informally that the restriction problem, decomposed over  $\delta$ -caps, becomes a question about how much  $\delta^{-(n-1)}$  tubes — one in each direction — can overlap. This chapter makes that precise. The implication is due to Bourgain and is the reason the second half of this series is about Kakeya sets.

### 6.6.1 The statement

**Definition 6.6.1.** Let  $0 < \delta < \frac{1}{2}$ . A family  $\mathbb{T} = \{T_1, \dots, T_N\}$  of  $\delta$ -tubes is a collection of cylinders in  $\mathbb{R}^n$ , each of length 1 and radius  $\delta$ , whose directions  $v_1, \dots, v_N \in \mathbb{S}^{n-1}$  are  $\delta$ -separated. Their positions are arbitrary. Maximality forces  $N \lesssim_n \delta^{-(n-1)}$ , and we assume  $N \approx \delta^{-(n-1)}$ .

Each tube has  $m(T_j) \approx \delta^{n-1}$ , so  $\sum_j m(T_j) \approx 1$ : the tubes have unit total volume however they are arranged. The Kakeya question is whether they can be packed so that their union is much smaller than 1, and the quantitative form of “no” is a bound on the multiplicity function  $\sum_j \mathbf{1}_{T_j}$ .

**Theorem 6.6.2** (Restriction implies Kakeya). *Suppose that for some  $b < \infty$  the restriction estimate*

$$\|\mathcal{E}g\|_{L^b(\mathbb{R}^n)} \leq A \|g\|_{L^\infty(d\sigma)} \quad (6.6.1)$$

*holds for  $S = \mathbb{S}^{n-1}$ . Then for every  $\delta$ -tube family  $\mathbb{T}$  as above,*

$$\left\| \sum_{j=1}^N \mathbf{1}_{T_j} \right\|_{L^{b/2}(\mathbb{R}^n)} \lesssim_n A^2 \delta^{\frac{4n}{b} - 2(n-1)}.$$

In particular, if (6.6.1) holds for every  $b > \frac{2n}{n-1}$  — that is, if Conjecture 6.2.6 is true — then for every  $\varepsilon > 0$

$$\left\| \sum_{j=1}^N \mathbf{1}_{T_j} \right\|_{L^{\frac{n}{n-1}}(\mathbb{R}^n)} \lesssim_{n,\varepsilon} \delta^{-\varepsilon},$$

which is the Kakeya maximal conjecture.

*Idea of the proof.* Three moves.

*Build a test function from the tubes.* To each direction  $v_j$  attach the cap  $\tau_j$  of radius  $\delta$  centred there. By Lemma 6.2.2,  $\mathcal{E}\mathbf{1}_{\tau_j}$  has modulus  $\gtrsim \delta^{n-1}$  on the dual box of  $\tau_j$  — a box of dimensions  $\delta^{-1} \times \cdots \times \delta^{-1} \times \delta^{-2}$ , which is a rescaled copy of a  $\delta$ -tube. Modulating  $\mathbf{1}_{\tau_j}$  translates that box, so the boxes can be placed wherever the  $T_j$  are.

*Randomise the signs.* Adding the pieces with fixed signs risks cancellation. Attaching random signs and averaging replaces  $|\sum_j \cdot|$  by the square function  $(\sum_j |\cdot|^2)^{1/2}$ , by Khintchine's inequality — the same device as in Theorem 3.6.1 — and the square function is bounded below by  $\delta^{n-1} (\sum_j \mathbf{1}_{T_j})^{1/2}$ . Crucially the  $L^\infty(d\sigma)$  norm of the test function is 1 for every sign choice, so the hypothesis applies uniformly.

*Rescale.* The dual boxes have length  $\delta^{-2}$ ; contracting space by  $\delta^2$  turns them into unit-length  $\delta$ -tubes, and the exponent in the statement is the bookkeeping of that dilation.

**Proof. Step 1.** *The caps and their dual boxes.*

For each  $j$  let  $\tau_j = \{\omega \in \mathbb{S}^{n-1} : |\omega - v_j| \leq \delta\}$ . Since the  $v_j$  are  $\delta$ -separated, the  $\tau_j$  have bounded overlap. By Lemma 6.2.2 applied after the rotation taking  $v_j$  to  $e_n$ ,

$$|\mathcal{E}\mathbf{1}_{\tau_j}(y)| \geq \frac{1}{2}\sigma(\tau_j) \geq c_n \delta^{n-1} \quad \text{for } y \in B_j, \quad (6.6.2)$$

where  $B_j$  is the box of dimensions  $\frac{c}{\delta} \times \cdots \times \frac{c}{\delta} \times \frac{c}{\delta^2}$  centred at the origin with long axis along  $v_j$ .

Given the target tubes  $T_j$ , write  $\tilde{T}_j = \delta^{-2}T_j$  for their dilates, which are boxes of dimensions  $\delta^{-1} \times \cdots \times \delta^{-1} \times \delta^{-2}$  up to the fixed constant  $c$ ; choose  $a_j \in \mathbb{R}^n$  with  $a_j + B_j \supseteq \tilde{T}_j$  after adjusting  $c$  by a dimensional constant.

**Step 2.** *The test function.*

For a sign vector  $\varepsilon = (\varepsilon_1, \dots, \varepsilon_N) \in \{-1, 1\}^N$  put

$$g_\varepsilon(\omega) = \sum_{j=1}^N \varepsilon_j e^{-2\pi i a_j \cdot \omega} \mathbf{1}_{\tau_j}(\omega).$$

Because the  $\tau_j$  have bounded overlap,  $\|g_\varepsilon\|_{L^\infty(d\sigma)} \lesssim_n 1$ , *uniformly in  $\varepsilon$* . By the modulation rule (Volume 2, Proposition 4.2),

$$\mathcal{E}g_\varepsilon(x) = \sum_{j=1}^N \varepsilon_j (\mathcal{E}\mathbf{1}_{\tau_j})(x - a_j).$$

**Step 3.** *Average over signs.*

Let  $r_j$  be the Rademacher functions and  $\mathbb{E}_t$  the average over  $t \in [0, 1]$ . By hypothesis (6.6.1), for every  $t$ ,

$$\int_{\mathbb{R}^n} |\mathcal{E}g_{\varepsilon(t)}(x)|^b dx \lesssim_n A^b.$$

Averaging in  $t$  and applying Tonelli and Khintchine's inequality (Theorem 2.6.5) pointwise in  $x$ ,

$$\int_{\mathbb{R}^n} \left( \sum_j |\mathcal{E}\mathbf{1}_{\tau_j}(x - a_j)|^2 \right)^{b/2} dx \approx_b \mathbb{E}_t \int |\mathcal{E}g_{\varepsilon(t)}|^b \lesssim_n A^b.$$

**Step 4.** *Bound the square function below.*

By (6.6.2),  $|\mathcal{E}\mathbf{1}_{\tau_j}(x - a_j)| \geq c_n \delta^{n-1}$  whenever  $x \in a_j + B_j$ , hence in particular whenever  $x \in \tilde{T}_j$ . Therefore

$$\left( \sum_j |\mathcal{E}\mathbf{1}_{\tau_j}(x - a_j)|^2 \right)^{1/2} \geq c_n \delta^{n-1} \left( \sum_j \mathbf{1}_{\tilde{T}_j}(x) \right)^{1/2},$$

and substituting into Step 3,

$$\delta^{(n-1)b} \int_{\mathbb{R}^n} \left( \sum_j \mathbf{1}_{\tilde{T}_j} \right)^{b/2} \lesssim_n A^b, \quad \text{i.e.} \quad \left\| \sum_j \mathbf{1}_{\tilde{T}_j} \right\|_{L^{b/2}} \lesssim_n A^2 \delta^{-2(n-1)}. \quad (6.6.3)$$

**Step 5.** *Rescale to unit tubes.*

Let  $F(x) = \sum_j \mathbf{1}_{\tilde{T}_j}(x)$  and  $G(y) = \sum_j \mathbf{1}_{T_j}(y)$ . Since  $\tilde{T}_j = \delta^{-2}T_j$ , we have  $G(y) = F(\delta^{-2}y)$ , so with  $p = b/2$

$$\|G\|_{L^p}^p = \int |F(\delta^{-2}y)|^p dy = \delta^{2n} \|F\|_{L^p}^p, \quad \|G\|_{L^p} = \delta^{2n/p} \|F\|_{L^p} = \delta^{4n/b} \|F\|_{L^p}.$$

Combining with (6.6.3),

$$\left\| \sum_j \mathbf{1}_{T_j} \right\|_{L^{b/2}} \lesssim_n A^2 \delta^{\frac{4n}{b} - 2(n-1)},$$

which is the stated bound.

**Step 6.** *The conjectural case.*

Suppose (6.6.1) holds for every  $b > \frac{2n}{n-1}$ . Given  $\varepsilon > 0$ , choose  $b$  slightly above  $\frac{2n}{n-1}$ . At  $b = \frac{2n}{n-1}$  exactly,

$$\frac{4n}{b} - 2(n-1) = \frac{4n(n-1)}{2n} - 2(n-1) = 2(n-1) - 2(n-1) = 0,$$

and  $b \mapsto \frac{4n}{b} - 2(n-1)$  is continuous, so for  $b$  close enough the exponent exceeds  $-\varepsilon$ ; also  $b/2 \rightarrow \frac{n}{n-1}$ , and by Hölder on the bounded set where  $\sum_j \mathbf{1}_{T_j}$  is supported — all tubes lie in a fixed ball — the  $L^{b/2}$  norm dominates the  $L^{n/(n-1)}$  norm up to a constant.  $\square$

**Where we are.** The exponent bookkeeping in Step 6 is worth recording as a slogan: the restriction exponent  $b = \frac{2n}{n-1}$  and the Kakeya exponent  $\frac{n}{n-1}$  correspond under  $b \mapsto b/2$ , and the powers of  $\delta$  cancel *exactly* at the conjectured endpoint. That is not a coincidence: it says the Knapp example and the Kakeya packing problem are extremal for the same reason, namely that  $\delta^{-(n-1)}$  caps produce  $\delta^{-(n-1)}$  tubes with unit total volume.

Note also what the proof did *not* use: nothing about the sphere beyond Lemma 6.2.2, which is a statement about a single cap. The implication therefore holds for any surface with non-vanishing curvature, with the tube directions ranging over the corresponding normal directions.

**Corollary 6.6.3** (Besicovitch sets are large). *Assume Conjecture 6.2.6 in dimension  $n$ . Then every Besicovitch set  $K \subseteq \mathbb{R}^n$  — every compact set containing a unit segment in every direction — has upper Minkowski dimension  $n$ .*

*Proof.* Fix  $\delta > 0$  and let  $v_1, \dots, v_N$  be a maximal  $\delta$ -separated set of directions,  $N \approx \delta^{-(n-1)}$ . For each  $j$ ,  $K$  contains a unit segment  $\ell_j$  in direction  $v_j$ , so the  $\delta$ -neighbourhood  $K_\delta$  contains the  $\delta$ -tube  $T_j$  around  $\ell_j$ . These form a tube family in the sense of Definition 6.6.1.

By Theorem 6.6.2 with  $b/2$  close to  $\frac{n}{n-1}$ , for every  $\varepsilon > 0$

$$\left\| \sum_j \mathbf{1}_{T_j} \right\|_{L^{\frac{n}{n-1}}} \lesssim_{n,\varepsilon} \delta^{-\varepsilon}.$$

On the other hand  $\sum_j \mathbf{1}_{T_j}$  is supported in  $K_\delta$ , so Hölder with exponents  $n$  and  $\frac{n}{n-1}$  gives

$$1 \approx \sum_j m(T_j) = \int_{K_\delta} \sum_j \mathbf{1}_{T_j} \leq m(K_\delta)^{1/n} \left\| \sum_j \mathbf{1}_{T_j} \right\|_{L^{\frac{n}{n-1}}} \lesssim_{n,\varepsilon} m(K_\delta)^{1/n} \delta^{-\varepsilon}.$$

Hence  $m(K_\delta) \gtrsim_{n,\varepsilon} \delta^{n\varepsilon}$ .

The upper Minkowski dimension of  $K$  is  $n - \liminf_{\delta \rightarrow 0} \frac{\log m(K_\delta)}{\log \delta}$ , and the bound just obtained gives  $\frac{\log m(K_\delta)}{\log \delta} \leq n\varepsilon$  for small  $\delta$ . So the dimension is at least  $n - n\varepsilon$  for every  $\varepsilon > 0$ , hence equals  $n$ .  $\square$

**Pitfall: Minkowski dimension is the weaker statement**

Corollary 6.6.3 concerns *Minkowski* dimension, which measures how slowly  $m(K_\delta)$  shrinks. The *Keakey conjecture* proper asserts that Besicovitch sets have *Hausdorff* dimension  $n$ , and Hausdorff dimension can be strictly smaller: a countable dense set has Minkowski dimension  $n$  and Hausdorff dimension 0.

Upgrading requires a covering argument at many scales simultaneously, which is Volume 7's machinery, together with a version of Theorem 6.6.2 applied at each scale. Volume 9 carries this out. Until then the reader should treat every “dimension” statement in this volume as Minkowski, and should note that this is the standard convention only when it is stated — the literature is not always careful.

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**WORKED EXAMPLE 6.6.1— TIER A Why random signs, and what fixed signs would give**

Suppose in Step 2 we had taken all  $\varepsilon_j = +1$ . Then  $\mathcal{E}g$  is a sum of  $N$  terms, each of modulus  $\approx \delta^{n-1}$  on its own box, and at a point  $x$  lying in  $\mu$  of the boxes the sum could be as large as  $\mu\delta^{n-1}$  — or as small as 0, if the phases conspire. The hypothesis would then give

$$\delta^{n-1} \left\| \sum_j \mathbf{1}_{T_j} \right\|_{L^b} \lesssim A$$

only if one could rule out cancellation, which one cannot: the phases of  $\mathcal{E}\mathbf{1}_{T_j}(x - a_j)$  at a common point  $x$  are genuinely uncontrolled.

Randomising removes the issue at the cost of replacing the sum by the square function — and the square function of  $\mu$  terms of size  $\delta^{n-1}$  is  $\mu^{1/2}\delta^{n-1}$ , not  $\mu\delta^{n-1}$ . So the argument gives an  $L^{b/2}$  bound on the multiplicity rather than an  $L^b$  bound, which is exactly the loss recorded in the exponent  $b/2$  of Theorem 6.6.2. Khintchine converts an unusable hypothesis into a usable one, and the square root is the price.

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**WORKED EXAMPLE 6.6.2— TIER B The Keakey maximal conjecture, and what the theorem delivers**

The conclusion of Theorem 6.6.2 is stated as a bound on a multiplicity function; the literature usually states the Keakeya maximal conjecture as a bound on an operator. The two are the same, and the translation is worth doing once.

*The maximal operator.* Definition 1.7.13 introduced

$$\mathcal{K}_\delta f(v) = \sup_{T \parallel v} \frac{1}{m(T)} \int_T |f|,$$

the supremum over  $\delta$ -tubes in direction  $v$ . The conjecture is  $\|\mathcal{K}_\delta f\|_{L^n(\mathbb{S}^{n-1})} \lesssim_\varepsilon \delta^{-\varepsilon} \|f\|_{L^n(\mathbb{R}^n)}$ .

*Duality.* Testing against  $f = \mathbf{1}_E$  and choosing for each of the  $\delta^{-(n-1)}$  separated directions a tube where the average is largest, the maximal estimate is equivalent — by the standard duality between an  $L^n \rightarrow L^n$  bound and a bound on  $\sum_j \mathbf{1}_{T_j}$  in  $L^{n'} = L^{n/(n-1)}$  — to

$$\left\| \sum_j \mathbf{1}_{T_j} \right\|_{L^{\frac{n}{n-1}}} \lesssim_\varepsilon \delta^{-\varepsilon},$$

which is the form delivered above.

*What it gives about sets.* If  $K \subseteq \mathbb{R}^n$  is a Besicovitch set, it contains a unit segment in every direction, so its  $\delta$ -neighbourhood contains a  $\delta$ -tube in each of  $\delta^{-(n-1)}$  separated directions. Then by Hölder,

$$1 \approx \sum_j m(T_j) = \int_{K_\delta} \sum_j \mathbf{1}_{T_j} \leq m(K_\delta)^{1/n} \left\| \sum_j \mathbf{1}_{T_j} \right\|_{L^{\frac{n}{n-1}}} \lesssim_\varepsilon m(K_\delta)^{1/n} \delta^{-\varepsilon},$$

so  $m(K_\delta) \gtrsim \delta^{\varepsilon n}$ , and letting  $\varepsilon \rightarrow 0$  gives Minkowski dimension  $n$ . Upgrading to Hausdorff dimension needs more, and is Volume 9's business.

**WORKED EXAMPLE 6.6.3— TIER C: RESEARCH TASTE** **What this implication does and does not settle**

(a) *The arrow runs one way.* Theorem 6.6.2 says restriction is at least as hard as Keakeya. No implication in the reverse direction is known, and none is expected to be easy: Keakeya is a statement about the *supports* of the wave packets, while restriction is a statement about their superposition with phases, and the passage from one to the other loses exactly the information that Step 3 destroyed.

(b) *The contrapositive is how it is used.* A counterexample to Keakeya would disprove restriction. Since Keakeya is believed true, the implication is used



instead as a source of *necessary conditions* and as a guide: any proof of restriction must, somewhere, contain a proof of *Keakeya*. Every advance in Volumes 12 and 13 does.

(c) *The same argument with  $L^2$  data gives nothing.* If one runs the proof with (6.6.1) replaced by the Stein–Tomas hypothesis  $\|\mathcal{E}g\|_b \lesssim \|g\|_{L^2(d\sigma)}$ , then  $\|g_\varepsilon\|_{L^2(d\sigma)} \approx (N\delta^{n-1})^{1/2} \approx 1$  — the same as before — but the admissible  $b$  is now at least  $\frac{2(n+1)}{n-1}$ , and the resulting exponent  $\frac{4n}{b} - 2(n-1)$  is strictly negative. So Stein–Tomas gives a *Keakeya* bound, but a lossy one. Working out that loss is Drill 6.3, and it is the reason the  $L^\infty(d\sigma)$  endpoint of the restriction problem is the one that matters for *Keakeya*.

(d) *Where the series goes from here.* Volumes 7 and 8 develop the dimension theory needed to state *Keakeya* properly; Volume 9 proves the best bounds obtainable by elementary geometry; Volumes 10 and 11 bring in arithmetic and algebraic methods; and Volumes 12 and 13 return to restriction with the geometric input those volumes supply. The present theorem is the reason that route is necessary rather than a detour.

**WORKED EXAMPLE 6.6.4— TIER B Running the argument backwards: what a *Keakeya* counterexample would kill**

Theorem 6.6.2 is usually read forwards. Read backwards it is a falsification tool, and it is worth seeing exactly how much a hypothetical *Keakeya* counterexample would destroy.

*Suppose* that for some  $n \geq 3$  and some  $\alpha > 0$  there were tube families  $\mathbb{T}_\delta$  with

$$\left\| \sum_j \mathbf{1}_{T_j} \right\|_{L^{\frac{n}{n-1}}} \geq \delta^{-\alpha}$$

for arbitrarily small  $\delta$  — that is, a quantitative failure of the *Keakeya* maximal conjecture.

*What follows.* By Theorem 6.6.2, the restriction estimate (6.6.1) at exponent  $b$  forces  $\delta^{-\alpha} \lesssim A^2 \delta^{4n/b - 2(n-1)}$ , hence

$$\frac{4n}{b} - 2(n-1) \leq -\alpha, \quad \text{that is} \quad b \geq \frac{4n}{2(n-1) - \alpha} > \frac{2n}{n-1}.$$

So the restriction conjecture would be false, and quantitatively so: the true exponent would exceed the conjectured one by an amount computable from  $\alpha$ .

*And what would survive.* Stein–Tomas would be untouched: it is a theorem, proved in Chapter 6.3 without any *Keakeya* input, and Worked Example 6.6.3(c) showed the *Keakeya* bound it implies is far from sharp. So

a Keakeya counterexample would not contradict anything proved — it would only close off the conjectured range, and the subject would reorganise around finding the true exponent.

*Why nobody expects this.* Every known construction of a small Besicovitch set is a variant of the Perron tree (Theorem 1.3.12), and the Perron tree produces sets of measure zero but Minkowski dimension  $n$  — it is efficient at making the measure small and useless at making the dimension small. Volume 9 explains why, and Volume 10 shows that the finite-field analogue admits no counterexample at all.

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**Figure 6.6.1 (to be drawn).** *From caps to tubes*

**Content required:** The sphere partitioned into  $\delta$ -caps  $\tau_j$ ; below, the corresponding dual boxes, one per cap, placed at arbitrary positions and pointing in  $\delta$ -separated directions. Indicate that there are  $\approx \delta^{-(n-1)}$  of them and that each has volume  $\approx \delta^{-(n+1)}$ .

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## Drills

**Drill 6.6.1.** Verify that the caps  $\tau_j$  in Step 1 have bounded overlap, and deduce  $\|g_\varepsilon\|_{L^\infty(d\sigma)} \lesssim_n 1$ .

**Drill 6.6.2.** Check the dilation computation in Step 5 by testing on a single tube.

**Drill 6.6.3.** Run the proof with the Stein–Tomas hypothesis in place of (6.6.1), as in Worked Example 6.6.3(c), and compute the resulting exponent for  $n = 3$ .

**Drill 6.6.4.** Show that the conclusion of Theorem 6.6.2 is trivially true with exponent  $\delta^{-(n-1)}$  for any tube family, and confirm that the theorem is an improvement exactly when  $b < \infty$ .

**Where we are.** The theorem of this chapter is the reason the series has the shape it does. Read once more in summary: an estimate for a single operator, tested against a family of  $\delta^{-(n-1)}$  modulated cap indicators with random signs, becomes a bound on how much  $\delta^{-(n-1)}$  tubes can overlap. Analysis in, geometry out.

Everything in Volumes 7 to 11 is devoted to the geometry side of that exchange: what a set containing tubes in all directions must look like, how to measure it, and what algebraic and combinatorial tools bound the

overlaps. Volumes 12 and 13 then run the exchange in the direction it was not proved to run — using geometric input to establish analytic estimates — by methods that do not factor through Theorem 6.6.2 at all.

## Notes

The implication is due to Bourgain (1991), with the randomisation argument as given. Fefferman’s ball multiplier theorem (Volume 3, Chapter 8) is the same idea in a degenerate case: there the tubes come from a Besicovitch set and are used to *disprove* an estimate, here they are produced by an estimate assumed to hold.

The observation that the exponents match exactly at the conjectured endpoint is the standard heuristic for why the restriction conjecture is stated where it is.



## Chapter 6.7

# The $\delta$ -Neighbourhood Formulation

Volumes 12 and 13 never estimate  $\mathcal{E}g$  directly. They work instead with functions whose Fourier transform is supported in a thin neighbourhood of the surface, decompose those into wave packets, and prove local estimates on a ball of radius  $R = \delta^{-1}$ . This chapter shows that the two formulations are the same, so that the literature's setting and this volume's are interchangeable.

It also records the place of the restriction conjecture in the chain of three conjectures, and states precisely which of the implications this series proves.

### 6.7.1 Thin neighbourhoods

**Definition 6.7.1.** For a compact hypersurface  $S \subseteq \mathbb{R}^n$  and  $0 < \delta < \frac{1}{2}$  let

$$N_\delta(S) = \{\xi \in \mathbb{R}^n : \text{dist}(\xi, S) < \delta\}.$$

For  $S = \mathbb{S}^{n-1}$  this is the annulus  $1 - \delta < |\xi| < 1 + \delta$ , and polar coordinates give the decomposition we shall use: writing  $\xi = (1 + t)\omega$  with  $|t| < \delta$  and  $\omega \in \mathbb{S}^{n-1}$ ,

$$\int_{N_\delta(S)} F(\xi) \, d\xi = \int_{-\delta}^{\delta} \int_{\mathbb{S}^{n-1}} F((1+t)\omega) (1+t)^{n-1} \, d\sigma(\omega) \, dt, \quad (6.7.1)$$

by Theorem 1.6.11.

**Theorem 6.7.2** (Extension estimates give neighbourhood estimates). *Let  $\delta = R^{-1}$  and suppose*

$$\|\mathcal{E}_\rho g\|_{L^b(B_R)} \leq A \|g\|_{L^2(d\sigma)} \quad (6.7.2)$$

for every ball  $B_R$  of radius  $R$  and every  $\rho \in [1 - \delta, 1 + \delta]$ , where  $\mathcal{E}_\rho$  is the extension operator for the sphere of radius  $\rho$ . Then for every  $f \in \mathcal{S}(\mathbb{R}^n)$  with  $\text{supp } \widehat{f} \subseteq N_\delta(\mathbb{S}^{n-1})$ ,

$$\|f\|_{L^b(B_R)} \lesssim_n A \delta^{1/2} \|\widehat{f}\|_{L^2(\mathbb{R}^n)}.$$

*Idea of the proof.* Slice the neighbourhood into spheres. By (6.7.1),  $f$  is an integral over  $t \in (-\delta, \delta)$  of extension operators applied to the restriction of  $\widehat{f}$  to the sphere of radius  $1 + t$ . Minkowski's integral inequality moves the  $L^b$  norm inside, the hypothesis bounds each slice, and Cauchy–Schwarz in  $t$  — over an interval of length  $2\delta$  — converts the resulting  $L_t^1 L_\omega^2$  norm into  $\delta^{1/2}$  times the  $L^2$  norm of  $\widehat{f}$ .

The factor  $\delta^{1/2}$  is therefore nothing but Cauchy–Schwarz over the thickness of the slab, and it is the reason neighbourhood estimates in the literature carry a power of  $\delta$  that looks unmotivated at first sight.

*Proof. Step 1. Slice.*

By Fourier inversion and (6.7.1),

$$f(x) = \int_{N_\delta} \widehat{f}(\xi) e^{2\pi i x \cdot \xi} d\xi = \int_{-\delta}^{\delta} (1+t)^{n-1} \left( \int_{\mathbb{S}^{n-1}} \widehat{f}((1+t)\omega) e^{2\pi i (1+t)x \cdot \omega} d\sigma(\omega) \right) dt.$$

The inner integral is  $\mathcal{E}_{1+t} g_t(x)$  where  $g_t(\omega) = \widehat{f}((1+t)\omega)$ , after absorbing the radius into the operator.

**Step 2.** *Minkowski and the hypothesis.*

By Minkowski's integral inequality (Theorem 1.8.4, in its integral form) and then (6.7.2),

$$\|f\|_{L^b(B_R)} \leq \int_{-\delta}^{\delta} (1+t)^{n-1} \|\mathcal{E}_{1+t} g_t\|_{L^b(B_R)} dt \lesssim_n A \int_{-\delta}^{\delta} \|g_t\|_{L^2(d\sigma)} dt,$$

using  $(1+t)^{n-1} \leq 2^{n-1}$ .

**Step 3.** *Cauchy–Schwarz in  $t$ .*

Since the interval has length  $2\delta$ ,

$$\int_{-\delta}^{\delta} \|g_t\|_{L^2(d\sigma)} dt \leq (2\delta)^{1/2} \left( \int_{-\delta}^{\delta} \|g_t\|_{L^2(d\sigma)}^2 dt \right)^{1/2}.$$

By (6.7.1) again, and since  $(1+t)^{n-1} \geq 2^{1-n}$ ,

$$\int_{-\delta}^{\delta} \|g_t\|_{L^2(d\sigma)}^2 dt = \int_{-\delta}^{\delta} \int_{\mathbb{S}^{n-1}} |\widehat{f}((1+t)\omega)|^2 d\sigma dt \lesssim_n \int_{N_\delta} |\widehat{f}(\xi)|^2 d\xi = \|\widehat{f}\|_{L^2}^2.$$

Combining Steps 2 and 3 gives the claim.  $\square$

**Proposition 6.7.3** (The converse). *Let  $\delta = R^{-1}$  and suppose that for every  $f$  with  $\text{supp } \widehat{f} \subseteq N_\delta(\mathbb{S}^{n-1})$  and every ball  $B_R$ ,*

$$\|f\|_{L^b(B_R)} \leq A' \delta^{1/2} \|\widehat{f}\|_{L^2(\mathbb{R}^n)}. \quad (6.7.3)$$

*Then the local extension estimate (6.7.2) holds with  $A \lesssim_n A'$ .*

*Proof. Step 1. Smooth the surface measure.*

Let  $g \in L^2(d\sigma)$  and let  $\varphi \in \mathcal{S}(\mathbb{R}^n)$  with  $\widehat{\varphi} \geq 0$ ,  $\widehat{\varphi} = 1$  on  $B(0, 1)$  and  $\text{supp } \widehat{\varphi} \subseteq B(0, 2)$ . Put  $\varphi_\delta(x) = \varphi(\delta x)$ , so  $\widehat{\varphi}_\delta(\xi) = \delta^{-n} \widehat{\varphi}(\xi/\delta)$  is an approximate identity at scale  $\delta$ , and define

$$f = (\mathcal{E}g) \cdot \varphi_\delta, \quad \text{so} \quad \widehat{f} = (gd\sigma) * \widehat{\varphi}_\delta.$$

Then  $\text{supp } \widehat{f} \subseteq N_{2\delta}(\mathbb{S}^{n-1})$ , since  $gd\sigma$  is supported on the sphere and  $\widehat{\varphi}_\delta$  in a ball of radius  $2\delta$ .

**Step 2.** *Compute the right-hand side of (6.7.3).*

The function  $\widehat{f}$  is  $gd\sigma$  smeared over a slab of thickness  $\delta$ , so its  $L^2$  norm is computed by (6.7.1):

$$\|\widehat{f}\|_{L^2(\mathbb{R}^n)}^2 = \int_{N_{2\delta}} \left| (gd\sigma) * \widehat{\varphi}_\delta(\xi) \right|^2 d\xi \lesssim_n \delta \cdot \delta^{-2} \cdot \delta \|g\|_{L^2(d\sigma)}^2 = \delta^{-1} \|g\|_{L^2(d\sigma)}^2,$$

where the three factors are: the thickness  $\delta$  of the slab, the square of the height  $\delta^{-1}$  to which the smoothing raises the density, and a factor  $\delta$  from the fact that the mass at each point of the sphere is spread over thickness  $\delta$ . Hence  $\|\widehat{f}\|_{L^2} \lesssim_n \delta^{-1/2} \|g\|_{L^2(d\sigma)}$ .

**Step 3.** *Compute the left-hand side.*

Since  $\varphi_\delta = \varphi(\delta \cdot)$  and  $\varphi$  may be chosen with  $|\varphi| \geq \frac{1}{2}$  on  $B(0, 1)$ , we have  $|\varphi_\delta| \geq \frac{1}{2}$  on  $B(0, R)$ , so

$$\|\mathcal{E}g\|_{L^b(B_R)} \leq 2\|f\|_{L^b(B_R)}.$$

**Step 4.** *Combine.*

Applying (6.7.3) and Steps 2, 3,

$$\|\mathcal{E}g\|_{L^b(B_R)} \leq 2A'\delta^{1/2}\|\widehat{f}\|_{L^2} \lesssim_n A'\delta^{1/2}\delta^{-1/2}\|g\|_{L^2(d\sigma)} = A'\|g\|_{L^2(d\sigma)}. \quad \square$$

**Where we are.** Theorem 6.7.2 and Proposition 6.7.3 together say the two formulations are equivalent, with the factor  $\delta^{1/2}$  appearing on one side and disappearing on the other. That factor is not a defect of either statement: it

is the ratio between the  $L^2$  norm of a measure on a surface and the  $L^2$  norm of its smearing over a slab of thickness  $\delta$ , and it must appear somewhere. A practical consequence: when reading a paper, check which normalisation is in force before comparing constants. A statement about  $\|\widehat{f}\|_{L^2}$  and one about  $\|g\|_{L^2(d\sigma)}$  differ by  $\delta^{1/2}$ , and at the exponents where these theorems are sharp, a factor  $\delta^{1/2}$  is the whole content.

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**WORKED EXAMPLE 6.7.1 — TIER A Reading a decoupling statement with this dictionary**

A decoupling theorem in Volume 13 will be stated as: for  $\widehat{f}$  supported in  $N_\delta(P)$  with  $P$  the truncated paraboloid, and  $f = \sum_\theta f_\theta$  the decomposition over  $\delta^{1/2}$ -caps,

$$\|f\|_{L^p(B_{\delta^{-1}})} \lesssim_\varepsilon \delta^{-\varepsilon} \left( \sum_\theta \|f_\theta\|_{L^p}^2 \right)^{1/2}.$$

Every ingredient now has a translation.

The hypothesis  $\text{supp } \widehat{f} \subseteq N_\delta(P)$  is the neighbourhood formulation, equivalent by Theorem 6.7.2 to working with  $\mathcal{E}$ . The ball  $B_{\delta^{-1}}$  is the local formulation of Definition 6.5.1, at the scale  $R = \delta^{-1}$  dictated by the uncertainty principle: a function with Fourier support in a slab of thickness  $\delta$  cannot be localised more finely than  $\delta^{-1}$ . The caps  $\theta$  have diameter  $\delta^{1/2}$  because that is the size at which the paraboloid deviates from its tangent plane by  $\delta$  — the Knapp scaling of Lemma 6.2.2. And the  $\delta^{-\varepsilon}$  is removable in the sense of Theorem 6.5.2.

So a decoupling statement is a restriction statement, written in the coordinates that induction on scales requires. The reader who has this dictionary can move between the two literatures; the reader who does not will find the change of setting between Stein–Tomas and Bourgain–Demeter mysterious.

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## 6.7.2 The chain of conjectures

**Theorem 6.7.4** (What this series proves about the chain). *Consider the three statements, for a fixed  $n \geq 2$ :*

(BR) *Bochner–Riesz conjecture*      (R) *restriction conjecture*      (K) *Keakeya maximal c*

*Then (R)  $\implies$  (K).*

*Proof.* This is Theorem 6.6.2: assuming (R), the hypothesis (6.6.1) holds for every  $b > \frac{2n}{n-1}$ , and Step 6 of that proof converts it into the Keakeya maximal bound with an arbitrarily small loss  $\delta^{-\varepsilon}$ , which is the statement (K).  $\square$



*Remark 6.7.5* (Scope decision: the first arrow). The implication  $(\text{BR}) \implies (\text{R})$  is also true. It is due to Tao (1999), and it is *not* proved in this series.

The reason is a judgement about length, and it is worth stating plainly because the omission is visible: Tao’s argument is a research paper, not a lemma. It proceeds by localising the Bochner–Riesz operator to a  $\delta$ -neighbourhood, converting the resulting estimate into a local extension estimate at scale  $\delta^{-1}$ , and then removing the localisation by an induction on scales. Each of those steps is of the same order of difficulty as an entire chapter here.

This is a plan revision, recorded as R5 in the master manifest: no later volume supplies it, and nothing in this series depends on it. Volume 5, Chapter 8 proved the positive Bochner–Riesz results that are used — boundedness for  $\delta > \frac{n-1}{2}$  — and Volume 3, Chapter 8 proved the negative result at  $\delta = 0$ . The implication is quoted here only to place the conjectures in relation to one another, never as a hypothesis.

**WORKED EXAMPLE 6.7.2— TIER C: RESEARCH TASTE** **How to read the three conjectures together**

(a) *They are ordered by difficulty.*  $(\text{BR}) \implies (\text{R}) \implies (\text{K})$ , so Kakeya is the weakest and Bochner–Riesz the strongest. All three are theorems for  $n = 2$  — Carleson–Sjölin, Chapter 6.8, and the planar Kakeya theorem of Theorem 1.6.14 — and all three are open for every  $n \geq 3$ .

(b) *They fail together, in a precise sense.* Each is sharp: the Knapp example limits  $(\text{R})$ , the same cap construction rescaled limits  $(\text{BR})$ , and the Besicovitch set limits  $(\text{K})$ . And each of the three extremal examples is the same object seen at a different scale — a family of  $\delta^{-(n-1)}$  tubes, one per direction. That is the sense in which the subject has a single difficulty.

(c) *The converses are open and probably hard.* No implication is known in the reverse direction. Kakeya is a statement about supports; restriction is about superpositions with phases. Worked Example 6.6.1 showed exactly where the phase information is destroyed in passing from one to the other, and there is no reason to expect it can be recovered.

(d) *Why the series goes through Kakeya anyway.* Because the geometric input — multilinear Kakeya, the polynomial method, decoupling — is developed in the Kakeya setting and then imported. Volume 12’s proof of a restriction estimate uses multilinear Kakeya as a black box, and Volume 13’s decoupling theorem uses it again. The arrow  $(\text{R}) \implies (\text{K})$  says one cannot avoid Kakeya; the practice of the field is that one does not want to.

(e) *A caution about “equivalent”.* Papers sometimes describe the three as equivalent. They are equivalent in the loose sense that progress on one has historically produced progress on the others, and that the known counterex-

amplified coincide. They are not equivalent as theorems, and a reader should treat any claimed equivalence as a claim about a specific quantitative formulation, worth checking.

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**WORKED EXAMPLE 6.7.3 — TIER B Wave packets at scale  $R$ : counting the degrees of freedom**

The neighbourhood formulation exists to make a decomposition available, and it is worth doing the bookkeeping once so that the parameters in Volumes 12 and 13 look inevitable rather than arbitrary.

*The setting.* Let  $\widehat{f}$  be supported in  $N_\delta(P)$ ,  $P$  the truncated paraboloid in  $\mathbb{R}^n$ , and put  $R = \delta^{-1}$ . We ask: how many independent pieces does  $f$  have on the ball  $B_R$ ?

*Step 1: the caps.* Partition  $P$  into caps  $\theta$  of diameter  $\delta^{1/2}$ . This is the correct size because the paraboloid deviates from its tangent plane over a cap of diameter  $\rho$  by  $\approx \rho^2$ , and we want that deviation to be at most the slab thickness  $\delta$ ; so  $\rho = \delta^{1/2}$ . The number of caps is

$$\#\{\theta\} \approx \left(\delta^{-1/2}\right)^{n-1} = \delta^{-\frac{n-1}{2}}.$$

*Step 2: the dual tubes.* By Lemma 6.2.2 — applied with  $\delta^{1/2}$  in place of  $\delta$  — the piece  $f_\theta$  is essentially constant on translates of a box of dimensions

$$\underbrace{\delta^{-1/2} \times \dots \times \delta^{-1/2}}_{n-1} \times \delta^{-1} = R^{1/2} \times \dots \times R^{1/2} \times R.$$

So  $f_\theta$  is a superposition of translates of one shape — a tube of length  $R$  and radius  $R^{1/2}$ , pointing in the direction normal to  $\theta$ . These are the *wave packets*.

*Step 3: how many packets per cap.* The ball  $B_R$  has volume  $R^n$  and each tube has volume  $R^{1/2(n-1)} \cdot R = R^{\frac{n+1}{2}}$ , so the tubes in a fixed direction tile  $B_R$  with

$$\frac{R^n}{R^{\frac{n+1}{2}}} = R^{\frac{n-1}{2}}$$

translates. Multiplying by the number of caps,  $\delta^{-\frac{n-1}{2}} = R^{\frac{n-1}{2}}$ , the total number of wave packets is  $R^{n-1}$ .

*Step 4: the consistency check.* A function with Fourier support in a set of measure  $\approx \delta = R^{-1}$ , localised to  $B_R$  of volume  $R^n$ , has  $\approx R^{-1} \cdot R^n = R^{n-1}$  degrees of freedom by the uncertainty principle. That matches Step 3 exactly,

which is the sign that the decomposition is not wasteful: the wave packets are a basis, not an overcomplete family.

*What to take from this.* Every parameter in a decoupling paper —  $\delta^{1/2}$  caps,  $R^{1/2} \times R$  tubes,  $R^{n-1}$  packets — is forced by two facts: the curvature of the surface, which fixes the cap size, and the uncertainty principle, which fixes everything else. None of it is a choice.

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WORKED EXAMPLE 6.7.4— TIER C: RESEARCH TASTE **What the wave packet picture predicts, and where it stops**

Worked Example 6.7.3 counted  $R^{n-1}$  wave packets. That count already predicts the conjectured exponent, and seeing the prediction — and then seeing why it is not a proof — is the best short summary of the state of the subject.

(a) *The heuristic.* Suppose the  $R^{n-1}$  packets were essentially disjoint, each of height  $h$  on a tube of volume  $R^{\frac{n+1}{2}}$ . Normalising  $\|g\|_{L^\infty} = 1$  makes  $h \approx R^{-\frac{n-1}{2}}$  after the rescaling of Worked Example 6.7.3. Then

$$\|\mathcal{E}g\|_{L^b(B_R)}^b \approx R^{n-1} \cdot h^b \cdot R^{\frac{n+1}{2}} = R^{n-1+\frac{n+1}{2}-\frac{b(n-1)}{2}},$$

and this is  $O(1)$  — no growth in  $R$ , which is what a global estimate requires — exactly when

$$\frac{b(n-1)}{2} \geq n-1 + \frac{n+1}{2}, \quad \text{that is} \quad b \geq \frac{3n-1}{n-1}.$$

That is not  $\frac{2n}{n-1}$ . The discrepancy is the point.

(b) *Why the naive count is wrong.* The packets are *not* disjoint. A point of  $B_R$  lies in many tubes, and the correct accounting weighs the overlaps: it is precisely the multiplicity function  $\sum_T \mathbf{1}_T$  of Theorem 6.6.2. Assuming the Kakeya conjecture — that the multiplicity is as small as it can be — and redoing the count gives  $b > \frac{2n}{n-1}$ .

(c) *So the conjecture is a statement about overlaps.* This is the content of Theorem 6.6.2 in reverse: the restriction exponent is determined by how much the tubes can pile up, and the extremal configuration is a Besicovitch set. Every proof in Volumes 12 and 13 is, at bottom, a way of controlling that pile-up.

(d) *Where the heuristic genuinely fails.* Even granting Kakeya, the count above ignores the *phases*: two packets overlapping on a set may add or cancel, and the heuristic assumes they add in absolute value. Worked Example 6.6.1 showed that random signs make the square function the right object, but a

real  $g$  has fixed phases, and no known argument controls them. This is the reason Kakeya is believed easier than restriction, and the reason no implication is known from the first to the second.

(e) *What to carry away.* The wave packet count is the standard back-of-envelope calculation of the field, and it is worth being able to do it in a minute. But it predicts the wrong exponent unless the overlap and the phases are put in by hand, and putting them in is the entire difficulty. A reader who can produce the count *and* name its two false assumptions understands the shape of the problem.

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## Drills

**Drill 6.7.1.** Verify (6.7.1) from Theorem 1.6.11, tracking the Jacobian  $(1+t)^{n-1}$ .

**Drill 6.7.2.** Show that the factor  $\delta^{1/2}$  in Theorem 6.7.2 is sharp, by taking  $\widehat{f} = \mathbf{1}_{N_\delta}$  and computing both sides for  $b = \infty$ .

**Drill 6.7.3.** State and prove the converse of Theorem 6.7.2: a neighbourhood estimate implies a local extension estimate, by taking  $\widehat{f}$  to be  $gd\sigma$  smoothed at scale  $\delta$ .

**Drill 6.7.4.** Explain, using Worked Example 6.7.1, why caps of diameter  $\delta^{1/2}$  and not  $\delta$  are the right ones for a  $\delta$ -neighbourhood of a paraboloid.

## Notes

The neighbourhood formulation is standard and is the setting of essentially all modern work. Theorem 6.7.2 is elementary; its content is the bookkeeping, which is where readers new to the area most often lose their footing.

Tao's theorem that (BR) implies (R) appeared in *Duke Math. J.* 96 (1999). The reverse implication is open.

## Chapter 6.8

# The Two-Dimensional Case

In the plane the restriction conjecture is a theorem. This chapter proves it, for the parabola, by an argument that is entirely elementary once one notices the right thing: in two dimensions the fourth power of  $\mathcal{E}g$  can be computed exactly, because  $|\mathcal{E}g|^2$  has Fourier transform a self-convolution of a measure on a curve, and that convolution can be evaluated by a change of variables.

### 6.8.1 The model and the statement

**Definition 6.8.1.** Let  $P = \{(t, t^2) : |t| \leq 1\}$  be the truncated parabola with the measure  $d\mu$  defined by  $\int F d\mu = \int_{-1}^1 F(t, t^2) dt$ , and for  $g \in L^\infty(d\mu)$  set

$$\mathcal{E}g(x) = \int_{-1}^1 g(t) e^{2\pi i(x_1 t + x_2 t^2)} dt, \quad x = (x_1, x_2) \in \mathbb{R}^2.$$

**Theorem 6.8.2** (Restriction in the plane). *For every  $b > 4$ ,*

$$\|\mathcal{E}g\|_{L^b(\mathbb{R}^2)} \lesssim_b \|g\|_{L^\infty(d\mu)}.$$

By Theorem 6.2.5 — whose proof applies verbatim to the parabola, using Theorem 5.5.2 for a curve of non-vanishing curvature — the range  $b > 4 = \frac{2n}{n-1}|_{n=2}$  is optimal. So Theorem 6.8.2 is the restriction conjecture for  $n = 2$ .

*Idea of the proof.* The exponent 4 is even, and that is the whole point. Write

$$\|\mathcal{E}g\|_{L^4}^4 = \|(\mathcal{E}g)^2\|_{L^2}^2 = \|(g d\mu) * (g d\mu)\|_{L^2}^2,$$

by Plancherel, since  $(\mathcal{E}g)^2 = ((g d\mu) * (g d\mu))^\vee$ . The convolution is an integral over pairs  $(s, t)$ , and the substitution  $(s, t) \mapsto (s + t, s^2 + t^2)$  has Jacobian  $2|t - s|$  — non-zero except on the diagonal. Changing variables

turns the  $L^2$  norm into

$$\iint_{[-1,1]^2} \frac{|g(s)|^2 |g(t)|^2}{2|t-s|} ds dt,$$

which diverges logarithmically for  $g \equiv 1$  — consistent with the exponent 4 being excluded — but only logarithmically. Cutting off at  $|t-s| \geq R^{-1}$  gives a factor  $\log R$  on the ball  $B_R$ , and interpolating that against the trivial bound at  $b = \infty$  removes the logarithm at any  $b > 4$ .

## 6.8.2 The bilinear computation

**Lemma 6.8.3** (The change of variables). *The map  $\Psi(s, t) = (s+t, s^2+t^2)$  is a local diffeomorphism off the diagonal, with Jacobian determinant  $2(t-s)$ , and it is two-to-one onto its image, the two preimages of a point being  $(s, t)$  and  $(t, s)$ .*

*Proof.*  $D\Psi = \begin{pmatrix} 1 & 1 \\ 2s & 2t \end{pmatrix}$  has determinant  $2(t-s)$ , non-zero for  $s \neq t$ . Given  $(\xi_1, \xi_2) = \Psi(s, t)$ , the numbers  $s, t$  are the roots of  $z^2 - \xi_1 z + \frac{1}{2}(\xi_1^2 - \xi_2) = 0$ , since  $s+t = \xi_1$  and  $st = \frac{1}{2}((s+t)^2 - (s^2+t^2))$ . A quadratic has two roots, so the preimage is  $\{(s, t), (t, s)\}$ .  $\square$

**Proposition 6.8.4** (The exact  $L^4$  identity). *For  $g \in L^\infty(d\mu)$  with  $|g| \leq 1$  and any measurable  $E \subseteq [-1, 1]^2$  symmetric in its two variables,*

$$\left\| (g \mathbf{1}_E d\mu) * (g d\mu) \right\|_{L^2(\mathbb{R}^2)}^2 \leq 2 \iint_E \frac{ds dt}{2|t-s|},$$

where  $g \mathbf{1}_E d\mu * g d\mu$  denotes the convolution with the integration restricted to  $(s, t) \in E$ .

*Proof. Step 1.* Identify the convolution as a pushforward.

For  $F \in C_c(\mathbb{R}^2)$ , unwinding the definition of the convolution of two measures carried by  $P$ ,

$$\int F d\left[(g \mathbf{1}_E d\mu) * (g d\mu)\right] = \iint_E F(\Psi(s, t)) g(s)g(t) ds dt, \quad \Psi(s, t) = (s+t, s^2+t^2),$$

because a point of  $P$  is  $(s, s^2)$  and the sum of two such points is  $\Psi(s, t)$ . So the measure in question is the pushforward under  $\Psi$  of the finite measure  $g(s)g(t)\mathbf{1}_E ds dt$ .

**Step 2.** *Compute its density.*

By Lemma 6.8.3,  $\Psi$  is a local diffeomorphism off the diagonal with Jacobian  $2|t - s|$ , and it is two-to-one, the fibre over  $\xi = \Psi(s, t)$  being  $\{(s, t), (t, s)\}$ . Hence the pushforward is absolutely continuous off the diagonal, with density

$$H(\xi) = \sum_{(s,t) \in \Psi^{-1}(\xi) \cap E} \frac{g(s)g(t)}{2|t-s|}.$$

Since  $E$  is symmetric, the fibre lies wholly inside or wholly outside  $E$ , so the sum has either two terms or none; and the two terms are equal,  $g(s)g(t)$  and  $2|t - s|$  both being symmetric. Using  $|g| \leq 1$ ,

$$|H(\xi)| \leq \frac{2}{2|t-s|} = \frac{1}{|t-s|} \quad \text{at } \xi = \Psi(s, t). \quad (6.8.1)$$

**Step 3.** *Change variables in the  $L^2$  norm.*

The two-to-one change of variables  $\xi = \Psi(s, t)$  reads, for  $\Phi \geq 0$ ,

$$\int_{\mathbb{R}^2} \Phi(\xi) d\xi = \frac{1}{2} \iint_{[-1,1]^2} \Phi(\Psi(s, t)) 2|t-s| ds dt,$$

the factor  $\frac{1}{2}$  correcting for the double covering and  $2|t - s|$  being the Jacobian. Applying this with  $\Phi = |H|^2$ , which vanishes off  $\Psi(E)$ , and then (6.8.1),

$$\|H\|_{L^2}^2 = \frac{1}{2} \iint_E |H(\Psi(s, t))|^2 2|t-s| ds dt \leq \frac{1}{2} \iint_E \frac{1}{|t-s|^2} 2|t-s| ds dt = \iint_E \frac{ds dt}{|t-s|}.$$

This is the stated bound, since  $2 \iint_E \frac{ds dt}{2|t-s|} = \iint_E \frac{ds dt}{|t-s|}$ .  $\square$

**Where we are.** The whole of Chapter 6.8 turns on the single computation in Step 3, so it is worth naming what makes it work. The Jacobian  $2|t - s|$  appears *twice*, with opposite effects: once in the denominator of the density  $H$ , and once as the volume factor in the change of variables. In  $\|H\|_{L^2}^2$  the density is squared but the volume factor is not, so one power of  $|t - s|^{-1}$  survives — and that surviving power is exactly the integrand  $|t - s|^{-1}$  whose logarithmic divergence at  $s = t$  produces the endpoint failure of Proposition 6.8.6.

Had the exponent been  $L^3$  rather than  $L^4$  there would have been no algebraic identity to start from; had the curve been a surface, the fibres of  $\Psi$  would have been positive-dimensional and there would have been no Jacobian. Both observations are made precise in Worked Example 6.8.3.

**Theorem 6.8.5** (The local  $L^4$  estimate). *For  $R \geq 2$  and  $\|g\|_{L^\infty(\mathrm{d}\mu)} \leq 1$ ,*

$$\|\mathcal{E}g\|_{L^4(B_R)} \lesssim (\log R)^{1/4}.$$

*Proof. Step 1. Split the pairs.*

Write  $\mathbf{1}_{[-1,1]^2} = \mathbf{1}_{E_{\text{near}}} + \mathbf{1}_{E_{\text{far}}}$  with

$$E_{\text{near}} = \{|t - s| < R^{-1}\}, \quad E_{\text{far}} = \{|t - s| \geq R^{-1}\},$$

and correspondingly  $(\mathcal{E}g)^2 = H_{\text{near}}^\vee + H_{\text{far}}^\vee$ .

**Step 2.** *The far pairs.*

By Proposition 6.8.4 with  $E = E_{\text{far}}$ ,

$$\|H_{\text{far}}^\vee\|_{L^2}^2 \leq 2 \iint_{|t-s| \geq R^{-1}} \frac{\mathrm{d}s \mathrm{d}t}{2|t-s|} \leq 2 \int_{-1}^1 \left( \int_{R^{-1} \leq |u| \leq 2} \frac{\mathrm{d}u}{2|u|} \right) \mathrm{d}s \lesssim \log R,$$

substituting  $u = t - s$ .

**Step 3.** *The near pairs.*

Here we do not use cancellation. By Cauchy–Schwarz,

$$|H_{\text{near}}^\vee(x)| = \left| \iint_{|t-s| < R^{-1}} g(s)g(t)e^{2\pi i(x_1(s+t)+x_2(s^2+t^2))} \mathrm{d}s \mathrm{d}t \right| \leq m(E_{\text{near}}) \lesssim R^{-1},$$

so  $\|H_{\text{near}}^\vee\|_{L^2(B_R)} \lesssim R^{-1}m(B_R)^{1/2} \lesssim R^{-1} \cdot R = 1$ .

**Step 4.** *Combine.*

By Plancherel applied to the far part and Step 3 for the near part,

$$\|\mathcal{E}g\|_{L^4(B_R)}^2 = \|(\mathcal{E}g)^2\|_{L^2(B_R)} \leq \|H_{\text{far}}^\vee\|_{L^2} + \|H_{\text{near}}^\vee\|_{L^2(B_R)} \lesssim (\log R)^{1/2} + 1,$$

and taking square roots gives the claim.  $\square$

*Proof of Theorem 6.8.2.* Normalise  $\|g\|_{L^\infty(\mathrm{d}\mu)} \leq 1$ . The trivial bound gives

$$\|\mathcal{E}g\|_{L^\infty(\mathbb{R}^2)} \leq \int_{-1}^1 |g| \leq 2.$$

Fix  $b > 4$  and let  $\theta \in (0, 1)$  satisfy  $\frac{1}{b} = \frac{1-\theta}{4}$ , i.e.  $1 - \theta = \frac{4}{b}$ . By Hölder on  $B_R$ ,

$$\|\mathcal{E}g\|_{L^b(B_R)} \leq \|\mathcal{E}g\|_{L^4(B_R)}^{1-\theta} \|\mathcal{E}g\|_{L^\infty}^\theta \lesssim (\log R)^{\frac{1-\theta}{4}} = (\log R)^{1/b}.$$

Since  $\log R \lesssim_\varepsilon R^\varepsilon$  for every  $\varepsilon > 0$ , this is a local estimate in the sense of Definition 6.5.1 at exponent  $b$ , with  $L^\infty(\mathrm{d}\mu)$  in place of  $L^2(\mathrm{d}\sigma)$  on the right. The proof of Theorem 6.5.2 used nothing about the exponent on the data side — see Drill 5.4 — so it applies and removes the loss at any exponent strictly larger. As  $b > 4$  was arbitrary, the global estimate holds for every  $b > 4$ .  $\square$



**Proposition 6.8.6** (The endpoint  $b = 4$  is false). *There is no constant  $C$  with  $\|\mathcal{E}g\|_{L^4(\mathbb{R}^2)} \leq C\|g\|_{L^\infty(d\mu)}$ .*

*Proof.* Take  $g \equiv 1$ , so  $\|g\|_{L^\infty(d\mu)} = 1$ . By Proposition 6.8.4 — read as an identity rather than an inequality, since for  $g \equiv 1$  the density  $H$  is exactly  $\frac{2}{2|t-s|}$  at  $\xi = \Psi(s, t)$  — we have

$$\|\mathcal{E}1\|_{L^4(\mathbb{R}^2)}^4 = \|H\|_{L^2}^2 = \iint_{[-1,1]^2} \frac{ds dt}{2|t-s|}.$$

Substituting  $u = t - s$ , the inner integral over  $u$  is  $\int_{|u| \leq 2} \frac{du}{2|u|}$ , which diverges logarithmically at  $u = 0$ . So  $\|\mathcal{E}1\|_{L^4} = \infty$ .  $\square$

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**WORKED EXAMPLE 6.8.1— TIER B Where the logarithm comes from, and why it is unavoidable**

Take  $g \equiv 1$ . Then Step 2 of Theorem 6.8.5 is an equality up to constants: the double integral genuinely behaves like  $\log R$ . So the  $L^4$  norm of  $\mathcal{E}1$  over  $B_R$  grows like  $(\log R)^{1/4}$ , and  $\mathcal{E}1 \notin L^4(\mathbb{R}^2)$ .

This is the same phenomenon as Theorem 6.2.5 at  $b = 4$ : there the obstruction was that  $\widehat{d\mu}$  decays like  $|x|^{-1/2}$  and  $|x|^{-1/2} \notin L^4(\mathbb{R}^2)$ , by exactly one logarithm. The two computations are the same fact seen from opposite sides — one in physical space, one in frequency — and their agreement is a useful check.

*Consequence for the statement.* The theorem cannot be improved to  $b = 4$ , and the  $\varepsilon$ -removal of Chapter 6.5 cannot help, since it produces only open ranges. The endpoint  $b = 4$  is genuinely false, unlike the Stein–Tomas endpoint of Worked Example 6.3.3 which is true but needs a different technique. Distinguishing these two situations — an endpoint that is false from an endpoint that is merely hard — is worth doing every time one meets an open range.

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**WORKED EXAMPLE 6.8.2— TIER B The transverse bilinear estimate, which is what survives**

The computation of Proposition 6.8.4 degenerates only on the diagonal  $s = t$ . Isolating that observation gives a statement that does generalise, and which is the seed of the entire bilinear method.

**Proposition 6.8.7.** *Let  $I, J \subseteq [-1, 1]$  be intervals with  $\text{dist}(I, J) \geq c > 0$ , and let  $g_1, g_2$  be supported in  $I, J$  respectively. Then*

$$\|\mathcal{E}g_1 \cdot \mathcal{E}g_2\|_{L^2(\mathbb{R}^2)} \leq c^{-1/2} \|g_1\|_{L^2} \|g_2\|_{L^2}.$$

*Proof.* By Plancherel,  $\|\mathcal{E}g_1\mathcal{E}g_2\|_{L^2} = \|(g_1 d\mu) * (g_2 d\mu)\|_{L^2}$ , and the convolution is the pushforward under  $\Psi(s, t) = (s + t, s^2 + t^2)$  of  $g_1(s)g_2(t) ds dt$ . Now  $s \in I$  and  $t \in J$  force  $|t - s| \geq c$ , so  $\Psi$  is injective on  $I \times J$  — the two preimages  $(s, t)$  and  $(t, s)$  cannot both lie in  $I \times J$ , the intervals being disjoint — with Jacobian  $2|t - s| \geq 2c$ . Hence the pushforward has density

$$H(\xi) = \frac{g_1(s)g_2(t)}{2|t - s|} \quad \text{at } \xi = \Psi(s, t),$$

and changing variables back,

$$\|H\|_{L^2}^2 = \iint_{I \times J} \frac{|g_1(s)|^2 |g_2(t)|^2}{4|t - s|^2} \cdot 2|t - s| ds dt = \iint \frac{|g_1(s)|^2 |g_2(t)|^2}{2|t - s|} ds dt \leq \frac{1}{2c} \|g_1\|_{L^2}^2 \|g_2\|_{L^2}^2$$

Taking square roots gives the claim, with the constant  $(2c)^{-1/2} \leq c^{-1/2}$ .  $\square$

*Why this is the right generalisation.* Three features survive to  $\mathbb{R}^n$  while the  $L^4$  identity does not.

The hypothesis is *transversality* — the two pieces of the surface have separated normal directions — rather than anything about exponents. In  $\mathbb{R}^n$  the corresponding statement bounds  $\|\mathcal{E}g_1\mathcal{E}g_2\|_{L^q}$  for  $q$  in a range, and transversality is again the hypothesis.

The conclusion is bilinear, so it says nothing directly about  $\mathcal{E}g$  for a single  $g$ . Recovering a linear estimate requires decomposing the surface into pieces, applying the bilinear bound to transverse pairs, and handling the non-transverse pairs by induction on scales — the *broad-narrow* argument of Volume 12.

And the constant blows up as  $c \rightarrow 0$ . All the difficulty of the bilinear method is in how that blow-up is balanced against the number of nearly-parallel pairs, which is exactly what the diagonal term of Theorem 6.8.5 was doing when it produced a logarithm.

**WORKED EXAMPLE 6.8.3— TIER C: RESEARCH TASTE Why this argument is confined to the plane**

Theorem 6.8.2 settles  $n = 2$  completely. It is worth being precise about which step fails for  $n \geq 3$ , since the answer explains the shape of the rest of the series.

(a) *The exponent must be even.* The proof computes  $\|\mathcal{E}g\|_{L^4}^4$  by writing it as  $\|(\mathcal{E}g)^2\|_{L^2}^2$  and using Plancherel. That works because  $4 = 2 \cdot 2$ . In  $\mathbb{R}^n$  the conjectured exponent is  $\frac{2n}{n-1}$ , which is an even integer only when  $n = 2$  (giving 4) and  $n = 3$  (giving 3, odd) — in fact only  $n = 2$ . For  $n \geq 3$  there is no corresponding algebraic identity, and the whole method is unavailable.

(b) *The change of variables is special to a curve.* Lemma 6.8.3 used that two points on a curve determine the pair  $(s+t, s^2+t^2)$  up to swapping, so  $\Psi$  is two-to-one. For a hypersurface in  $\mathbb{R}^n$ ,  $n \geq 3$ , the map  $(\omega_1, \omega_2) \mapsto \omega_1 + \omega_2$  goes from a  $2(n-1)$ -dimensional domain to an  $n$ -dimensional target and has  $(n-2)$ -dimensional fibres. There is no Jacobian to compute; there is a fibre integral, and controlling it is precisely the bilinear restriction problem.

(c) *What survives.* The identity in (b) is the seed of the bilinear method: one studies  $\mathcal{E}g_1 \cdot \mathcal{E}g_2$  for  $g_1, g_2$  supported on *transverse* pieces of the surface, where the fibres are non-degenerate and the fibre integral is controllable. Tao's sharp bilinear restriction theorem is the culmination, and the multilinear theory of Volume 12 is what replaced it.

(d) *The moral.* The planar case is not a warm-up for the general case; it is a genuinely different argument that happens to be available in one dimension. A reader who expects the higher-dimensional proof to look like this one will be disappointed, and the historical record bears that out: Theorem 6.8.2 dates from around 1970 and the first improvement over Stein–Tomas in higher dimensions came more than twenty years later.

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**WORKED EXAMPLE 6.8.4— TIER A   Checking the planar exponents against everything else**

At  $n = 2$  four different results in this volume make a prediction, and they must agree. Verifying that they do is a good use of ten minutes and catches most arithmetic slips.

(a) *The whole-curve condition.* Theorem 6.2.5 gives  $b > \frac{2n}{n-1} = 4$ . ✓ with Theorem 6.8.2.

(b) *The Knapp condition at  $a = \infty$ .* Theorem 6.2.3 gives  $b \geq \frac{n+1}{n-1} = 3$ , weaker than (a) — consistent with Worked Example 6.2.2, which said the whole-surface condition binds at  $a = \infty$  once  $a > \frac{2n}{n-1} = 4$ . ✓

(c) *Stein–Tomas.* Theorem 6.3.1 gives  $b > \frac{2(n+1)}{n-1} = 6$  for  $L^2(d\mu)$  data. Since  $L^\infty(d\mu) \subseteq L^2(d\mu)$  on a finite measure space, Theorem 6.8.2 at  $b > 4$  is strictly stronger in the range  $4 < b \leq 6$  — as it must be, being the conjecture. ✓

(d) *The Kakeya consequence.* Theorem 6.6.2 with  $n = 2$  and  $b \downarrow 4$  gives  $\|\sum_j \mathbf{1}_{T_j}\|_{L^2} \lesssim \delta^{-\varepsilon}$ , that is the Kakeya maximal conjecture in the plane. And that is a theorem independently — Córdoba's argument, Theorem 1.6.14. ✓  
Two proofs of the same planar fact, one via restriction and one direct.

*The moral.* In a subject with this many exponents, consistency checks at  $n = 2$  and  $n = 3$  are the cheapest available error detection. Every time a

new estimate is proved in the volumes ahead, it is worth substituting  $n = 2$  and asking whether the known planar answer comes out.

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## Drills

**Drill 6.8.1.** Verify the Jacobian computation in Lemma 6.8.3, and solve explicitly for  $s, t$  in terms of  $\xi_1, \xi_2$ .

**Drill 6.8.2.** Compute  $\iint_{|t-s| \geq R^{-1}} \frac{ds dt}{|t-s|}$  over  $[-1, 1]^2$  exactly, and confirm the growth rate  $\log R$ .

**Drill 6.8.3.** Show that the parabola may be replaced by any curve  $(t, \phi(t))$  with  $\phi'' \neq 0$ , by checking that  $\Psi(s, t) = (s + t, \phi(s) + \phi(t))$  has non-vanishing Jacobian off the diagonal.

**Drill 6.8.4.** Explain why the argument of Theorem 6.8.5 fails for the circle, by locating the second degeneracy of the Jacobian. (Two points of a circle summing to a fixed  $\xi$  degenerate both when they coincide and when they are antipodal.)

**Where we are.** Three features of the planar proof are worth separating, because only one of them generalises.

*The algebraic identity*  $\|F\|_{L^4}^2 = \|F^2\|_{L^2}$  requires the exponent to be even and is unavailable in higher dimensions.

*The change of variables* of Lemma 6.8.3 requires the surface to be a curve, so that two points determine their sum up to swapping. In higher dimensions the analogous map has positive-dimensional fibres.

*The transversality principle* — that the interaction of two pieces is controlled by the angle between them, with a constant blowing up as the angle closes — is the one that survives, and Worked Example 6.8.2 isolated it. Everything in Volume 12 is a version of that principle, adapted to  $n$  pieces at once rather than two.

The reader should therefore treat this chapter as evidence that the conjecture is true, and as the source of one durable idea, but not as a template. The higher-dimensional theory begins by discarding the first two features and keeping the third.

## Notes

Theorem 6.8.2 is due to Fefferman and Stein, and independently to Zygmund, around 1970; the endpoint refinement — a restricted weak-type estimate at

$b = 4$  — is Zygmund's and is not proved here. Córdoba's square-function proof gives the same result by a different route and generalises differently.

Drill 8.4 is the reason this chapter works with the parabola rather than the circle. For the circle the antipodal degeneracy contributes an extra singularity which must be handled separately; the standard treatment decomposes the circle into arcs and treats non-antipodal pairs by the argument above.





## Chapter 6.A

# Results Quoted from Volumes 1–5

| Result                                 | Source               | Used in                                       |
|----------------------------------------|----------------------|-----------------------------------------------|
| Polar coordinates                      | Theorem 1.6.11       | Theorem 6.2.5, Theorem 6.7.2                  |
| Density of $\mathcal{S}$ in $L^p$      | Theorem 1.8.7        | Definition 6.1.1                              |
| Norms by testing                       | Corollary 1.8.10     | Theorem 6.1.3                                 |
| Minkowski's integral inequality        | Theorem 1.8.4        | Theorem 6.7.2                                 |
| Change of variables                    | Theorem 1.3.10       | Proposition 6.8.4                             |
| Riesz–Thorin                           | Theorem 2.2.1        | Theorem 6.3.1                                 |
| Young's convolution inequality         | Worked Example 2.2.1 | Theorem 6.3.1, Proposition 5.8.2-style bounds |
| The Fourier dictionary                 | Proposition 2.4.2    | Theorem 6.6.2                                 |
| Plancherel                             | Theorem 2.4.8        | Theorem 6.4.3, Theorem 6.8.5                  |
| $\mathcal{F}$ on $\mathcal{S}'$        | Theorem 2.5.11       | Lemma 6.3.2                                   |
| $L^2$ multiplier norm                  | Proposition 2.6.2    | Theorem 6.3.1                                 |
| Khinchine's inequality                 | Theorem 2.6.5        | Theorem 6.6.2                                 |
| The Knapp example                      | Worked Example 2.4.3 | Chapter 6.2                                   |
| Takeya maximal operator                | Definition 1.7.13    | Worked Example 6.6.2                          |
| Córdoba's planar theorem               | Theorem 1.6.14       | Worked Example 6.7.2                          |
| Fefferman's ball multiplier theorem    | Theorem 3.8.2        | Worked Example 6.7.2                          |
| Critical points are normal directions  | Lemma 5.5.1          | Proposition 6.4.5, Theorem 6.4.3              |
| Asymptotics of $\widehat{d\sigma}$     | Theorem 5.5.2        | Lemma 6.3.2, Theorem 6.2.5                    |
| Decay with $k$ curvatures              | Theorem 5.5.3        | Proposition 6.4.5, Worked Example 6.3.2       |
| Bochner–Riesz above the critical index | Theorem 5.8.4        | Remark 6.7.5                                  |



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## Debts

This volume incurs no debts on later volumes, and discharges none — Volumes 1–5 left none outstanding except Lemma 3.8.4, which is owed to Volume 9.

Two results are removed from the plan, announced where they would have appeared and recorded in the master manifest: the endpoint of the Stein–Tomas theorem and the Keel–Tao endpoint Strichartz estimate (preface, and Worked Example 6.3.3), and the implication that the Bochner–Riesz conjecture implies the restriction conjecture (Remark 6.7.5, revision R5).



## Chapter 6.B

# Notation and Index of Results

### Additions to the Volume 1–5 conventions

|                           |                                                                     |
|---------------------------|---------------------------------------------------------------------|
| $\mathcal{R}$             | the restriction operator $f \mapsto \widehat{f} _S$                 |
| $\mathcal{E}$             | the extension operator $g \mapsto (gd\sigma)^\vee$                  |
| $R^*(p \rightarrow q)$    | the estimate $\ \mathcal{R}f\ _{L^q(d\sigma)} \lesssim \ f\ _{L^p}$ |
| $\kappa_\delta, T_\delta$ | a cap of radius $\delta$ and its dual box                           |
| $N_\delta(S)$             | the $\delta$ -neighbourhood of $S$                                  |
| $\mathbb{T}$              | a family of $\delta$ -tubes                                         |

### Index of the main results

|                                            |                   |
|--------------------------------------------|-------------------|
| Restriction–extension duality              | Theorem 6.1.3     |
| The $TT^*$ identity                        | Proposition 6.1.5 |
| The cap does not oscillate on its dual box | Lemma 6.2.2       |
| The Knapp condition                        | Theorem 6.2.3     |
| The whole-sphere condition                 | Theorem 6.2.5     |
| The restriction conjecture                 | Conjecture 6.2.6  |
| Kernel decomposition                       | Lemma 6.3.2       |
| Stein–Tomas, non-endpoint                  | Theorem 6.3.1     |
| Strichartz estimates                       | Theorem 6.4.3     |
| The cone and the wave equation             | Proposition 6.4.5 |
| Epsilon-removal                            | Theorem 6.5.2     |
| Restriction implies Keakeya                | Theorem 6.6.2     |
| The $\delta$ -neighbourhood formulation    | Theorem 6.7.2     |
| The chain of conjectures                   | Theorem 6.7.4     |
| The change of variables in the plane       | Lemma 6.8.3       |
| The exact $L^4$ identity                   | Proposition 6.8.4 |
| The local $L^4$ estimate                   | Theorem 6.8.5     |
| Restriction in the plane                   | Theorem 6.8.2     |

## Plan revisions in this volume

| Removed                                       | Reason                                                             |
|-----------------------------------------------|--------------------------------------------------------------------|
| Stein–Tomas endpoint $b = \frac{2(n+1)}{n-1}$ | needs Stein interpolation, removed as R4; see Worked Example 6.3.3 |
| Keel–Tao endpoint Strichartz                  | bilinear; see Worked Example 6.4.4                                 |
| Bochner–Riesz $\implies$ restriction          | Tao (1999), a research paper; Remark 6.7.5, revision R5            |
| Zygmund’s endpoint in the plane               | restricted weak type at $b = 4$ ; the open range is Theorem 6.8.2  |

# Part 7

## Volume 7

Hausdorff Dimension, Fourier Dimension, and Fractals



# Preface to Volume 7

## Why measure is not enough

Volume 6 ended with Corollary 6.6.3: granting the restriction conjecture, a Besicovitch set has Minkowski dimension  $n$ . That is weaker than what the Kakeya conjecture asserts, which is that the *Hausdorff* dimension is  $n$ , and the gap is not a technicality — the rationals in  $[0, 1]$  have Minkowski dimension 1 and Hausdorff dimension 0.

The reason Volume 6 could not do better is that its only tool was Lebesgue measure, applied at a single scale  $\delta$  at a time. Hausdorff dimension is a statement about all scales at once, and measuring it requires a genuinely different apparatus: outer measures indexed by an exponent, a notion of dimension defined by where they jump, covering theorems adapted to them, and a way of producing measures on a set from knowledge of its dimension. This volume builds that apparatus.

## What is assumed

Volumes 1–6. Most heavily Volume 1: the construction of Hausdorff measure as a metric outer measure (Worked Example 1.2.4), the Carathéodory criterion (Theorem 1.2.13), and the covering lemmas of Chapter 7. From Volume 2, the Fourier transform of measures; from Volume 6, the statement of the Kakeya maximal conjecture.

## Scope, fixed in advance

Two results a reader might expect are handled differently from the original plan, and both decisions were taken before the chapters were written.

The *Besicovitch covering theorem* is not proved. Its correct proof splits into two cases according to the ratio of the radii, and the naive angle argument that suggests itself is false — there are configurations satisfying every constraint of the greedy selection with angle arbitrarily close to zero. The full argument is

several pages and nothing in this series depends on it: the  $5r$ -covering lemma of Chapter 7.4 suffices for every application here. This is recorded as revision R6.

The existence of a Besicovitch set of *measure zero* — promised in Volume 1 — is deferred to Volume 8 rather than proved here. The natural route is Kahane's construction, which produces the set from a Cantor set of directions and needs Marstrand's projection theorem to compute the measure; Marstrand is Volume 8, Chapter 1. Chapter 7.3 explains why the obvious compactness argument, which one expects to work, does not.

## The plan

|   |                                 |                                          |
|---|---------------------------------|------------------------------------------|
| 1 | Hausdorff measure and dimension | the exponent where the measure jumps     |
| 2 | Minkowski dimension             | one scale at a time, and what it misses  |
| 3 | The Hausdorff metric            | compactness in the space of compact sets |
| 4 | Covering theorems               | differentiation, and densities           |
| 5 | Frostman's lemma                | dimension produces a measure             |
| 6 | Energy integrals                | dimension as an integral                 |
| 7 | Fourier dimension               | when the transform sees the dimension    |
| 8 | Self-similar sets               | and the Keakey upgrade                   |



# Chapter 7.1

## Hausdorff Measure and Dimension

### 7.1.1 The construction, and what Volume 1 already gives

**Definition 7.1.1.** For  $s \geq 0$ ,  $\varepsilon > 0$  and  $E \subseteq \mathbb{R}^n$  set

$$\mathcal{H}_\varepsilon^s(E) = \inf \left\{ \sum_{i=1}^{\infty} (\text{diam } U_i)^s : E \subseteq \bigcup_i U_i, \text{diam } U_i \leq \varepsilon \right\},$$

with the conventions  $(\text{diam } \emptyset)^0 = 0$  and  $\inf \emptyset = +\infty$ . Since  $\mathcal{H}_\varepsilon^s$  increases as  $\varepsilon$  decreases, the limit

$$\mathcal{H}^s(E) = \lim_{\varepsilon \downarrow 0} \mathcal{H}_\varepsilon^s(E) = \sup_{\varepsilon > 0} \mathcal{H}_\varepsilon^s(E)$$

exists in  $[0, \infty]$  and is called the *s-dimensional Hausdorff outer measure*.

Worked Example 1.2.4 carried out this construction and Theorem 1.2.13 established what it gives:

**Proposition 7.1.2.**  $\mathcal{H}^s$  is a metric outer measure on  $\mathbb{R}^n$ ; consequently every Borel set is  $\mathcal{H}^s$ -measurable, and  $\mathcal{H}^s$  restricted to the Borel sets is a Borel measure. Moreover  $\mathcal{H}^0$  is counting measure, and  $\mathcal{H}^s$  is invariant under isometries and satisfies  $\mathcal{H}^s(\lambda E) = \lambda^s \mathcal{H}^s(E)$  for  $\lambda > 0$ .

*Proof.* That  $\mathcal{H}^s$  is an outer measure is Proposition 1.2.8, the covering family being all sets of small diameter. For the metric property, let  $\text{dist}(A, B) = d > 0$  and take  $\varepsilon < d/2$ ; then no set of diameter  $\leq \varepsilon$  meets both  $A$  and  $B$ , so any admissible cover of  $A \cup B$  splits into one of  $A$  and one of  $B$ , giving

$\mathcal{H}_\varepsilon^s(A \cup B) = \mathcal{H}_\varepsilon^s(A) + \mathcal{H}_\varepsilon^s(B)$ . Letting  $\varepsilon \downarrow 0$  and applying Theorem 1.2.13 gives measurability of Borel sets.

For  $s = 0$ : a set of diameter  $\leq \varepsilon$  contains at most one point of a set whose points are  $\varepsilon$ -separated, so  $\mathcal{H}_\varepsilon^0(E)$  counts points once  $\varepsilon$  is small. Isometry invariance is clear since diameters are preserved, and the scaling law follows because  $\text{diam}(\lambda U) = \lambda \text{diam } U$ .  $\square$

## 7.1.2 The critical exponent

**Theorem 7.1.3.** *Let  $E \subseteq \mathbb{R}^n$  and  $0 \leq s < t$ .*

(i) *If  $\mathcal{H}^s(E) < \infty$  then  $\mathcal{H}^t(E) = 0$ .*

(ii) *If  $\mathcal{H}^t(E) > 0$  then  $\mathcal{H}^s(E) = \infty$ .*

*Consequently there is a unique  $d \in [0, \infty]$  such that  $\mathcal{H}^s(E) = \infty$  for  $s < d$  and  $\mathcal{H}^s(E) = 0$  for  $s > d$ .*

*Proof.* Let  $\{U_i\}$  cover  $E$  with  $\text{diam } U_i \leq \varepsilon$ . Then

$$\sum_i (\text{diam } U_i)^t = \sum_i (\text{diam } U_i)^{t-s} (\text{diam } U_i)^s \leq \varepsilon^{t-s} \sum_i (\text{diam } U_i)^s,$$

so  $\mathcal{H}_\varepsilon^t(E) \leq \varepsilon^{t-s} \mathcal{H}_\varepsilon^s(E) \leq \varepsilon^{t-s} \mathcal{H}^s(E)$ . If  $\mathcal{H}^s(E) < \infty$ , letting  $\varepsilon \downarrow 0$  gives  $\mathcal{H}^t(E) = 0$ . Statement (ii) is the contrapositive.

For the last claim, put  $d = \inf\{s \geq 0 : \mathcal{H}^s(E) = 0\}$ , with  $\inf \emptyset = n$ . If  $s > d$  then some  $s' \in (d, s)$  has  $\mathcal{H}^{s'}(E) = 0 < \infty$ , so  $\mathcal{H}^s(E) = 0$  by (i); if  $s < d$  then  $\mathcal{H}^s(E) \neq 0$ , and were it finite (i) would force  $\mathcal{H}^{s'}(E) = 0$  for  $s' \in (s, d)$ , contradicting the definition of  $d$ . So  $\mathcal{H}^s(E) = \infty$ .  $\square$

**Where we are.** The theorem gives  $d \in [0, \infty]$ , not  $d \in [0, n]$ . The bound  $d \leq n$  is *not* available yet: it requires knowing  $\mathcal{H}^s(\mathbb{R}^n) = 0$  for  $s > n$ , which is Corollary 7.1.9 below and depends on comparing  $\mathcal{H}^n$  with Lebesgue measure.

Stating the sharper range here would make the development circular — Proposition 7.1.8 uses part (i) of this theorem in its own proof. Splitting the two claims keeps the dependency order strict, and Corollary 7.1.9 records the bound as soon as it is earned.

**Definition 7.1.4.** The number  $d$  of Theorem 7.1.3 is the *Hausdorff dimension*  $\dim_H E$ .

**Pitfall: the measure at the critical exponent can be anything**

$\dim_{\mathbb{H}} E = d$  says nothing about  $\mathcal{H}^d(E)$ , which may be 0, a finite positive number, or  $\infty$ . All three occur:  $\mathcal{H}^1$  of a line segment is its length;  $\mathcal{H}^1(\mathbb{R}) = \infty$  while  $\dim_{\mathbb{H}} \mathbb{R} = 1$ ; and there are sets of dimension  $d$  with  $\mathcal{H}^d = 0$ , constructed in Drill 1.4.

The distinction matters in this series. The Kakeya conjecture asserts  $\dim_{\mathbb{H}} K = n$  for every Besicovitch set  $K$ ; it does *not* assert  $\mathcal{H}^n(K) > 0$ , which is false — Besicovitch sets of Lebesgue measure zero exist, and  $\mathcal{H}^n$  is comparable to Lebesgue measure. So the conjecture is exactly a statement about the critical exponent and not about the measure there.

**Proposition 7.1.5** (Elementary properties). *(i)  $E \subseteq F$  implies  $\dim_{\mathbb{H}} E \leq \dim_{\mathbb{H}} F$ .*

*(ii)  $\dim_{\mathbb{H}} (\bigcup_{i \geq 1} E_i) = \sup_i \dim_{\mathbb{H}} E_i$ .*

*(iii) If  $f : E \rightarrow \mathbb{R}^m$  is Lipschitz with constant  $L$  then  $\mathcal{H}^s(f(E)) \leq L^s \mathcal{H}^s(E)$ , and hence  $\dim_{\mathbb{H}} f(E) \leq \dim_{\mathbb{H}} E$ .*

*(iv) If  $f$  is Hölder of exponent  $\alpha \in (0, 1]$  then  $\dim_{\mathbb{H}} f(E) \leq \dim_{\mathbb{H}} E / \alpha$ .*

*(v)  $\dim_{\mathbb{H}} E = 0$  for every countable  $E$ .*

*Proof.* (i) is monotonicity of  $\mathcal{H}^s$ . For (ii), “ $\geq$ ” is (i); for “ $\leq$ ”, if  $s > \sup_i \dim_{\mathbb{H}} E_i$  then  $\mathcal{H}^s(E_i) = 0$  for every  $i$ , so  $\mathcal{H}^s(\bigcup E_i) = 0$  by countable subadditivity.

(iii) If  $\{U_i\}$  covers  $E$  with  $\text{diam } U_i \leq \varepsilon$  then  $\{f(U_i \cap E)\}$  covers  $f(E)$  with  $\text{diam } f(U_i \cap E) \leq L \text{diam } U_i \leq L\varepsilon$ , and  $\sum (\text{diam } f(U_i \cap E))^s \leq L^s \sum (\text{diam } U_i)^s$ . Take infima and let  $\varepsilon \downarrow 0$ . (iv) is the same computation with  $\text{diam } f(U_i \cap E) \leq L(\text{diam } U_i)^\alpha$ , giving  $\mathcal{H}^{s/\alpha}(f(E)) \leq L^{s/\alpha} \mathcal{H}^s(E)$ .

(v) A single point has  $\mathcal{H}^s = 0$  for every  $s > 0$ ; apply (ii).  $\square$

### 7.1.3 The mass distribution principle

Upper bounds on dimension come from exhibiting covers. Lower bounds are harder, and the following is the elementary tool; Chapter 7.5 proves that it is the only one needed.

**Lemma 7.1.6** (Mass distribution principle). *Let  $\mu$  be a Borel measure on  $\mathbb{R}^n$  with  $\mu(E) > 0$ , and suppose there are  $c, \varepsilon_0 > 0$  with*

$$\mu(U) \leq c(\text{diam } U)^s \quad \text{for every } U \text{ with } \text{diam } U \leq \varepsilon_0.$$

*Then  $\mathcal{H}^s(E) \geq \mu(E)/c$ , and in particular  $\dim_{\mathbb{H}} E \geq s$ .*

*Proof.* Let  $\{U_i\}$  cover  $E$  with  $\text{diam } U_i \leq \varepsilon \leq \varepsilon_0$ . Then

$$\mu(E) \leq \mu\left(\bigcup_i U_i\right) \leq \sum_i \mu(U_i) \leq c \sum_i (\text{diam } U_i)^s.$$

Taking the infimum over covers gives  $\mu(E) \leq c \mathcal{H}_\varepsilon^s(E)$ , and letting  $\varepsilon \downarrow 0$  gives the claim. Since  $\mu(E) > 0$ ,  $\mathcal{H}^s(E) > 0$  and so  $\dim_{\text{H}} E \geq s$  by Theorem 7.1.3.  $\square$

**WORKED EXAMPLE 7.1.1— TIER A The middle-thirds Cantor set**

Let  $C \subseteq [0, 1]$  be the middle-thirds Cantor set of Volume 1, and  $s = \log 2 / \log 3$ . We show  $\dim_{\text{H}} C = s$ .

*Upper bound.* At stage  $k$ ,  $C$  is covered by  $2^k$  intervals of length  $3^{-k}$ . Hence

$$\mathcal{H}_{3^{-k}}^s(C) \leq 2^k (3^{-k})^s = 2^k 3^{-ks} = 2^k 2^{-k} = 1,$$

using  $3^s = 2$ . Letting  $k \rightarrow \infty$ ,  $\mathcal{H}^s(C) \leq 1$ , so  $\dim_{\text{H}} C \leq s$ .

*Lower bound.* Let  $\mu$  be the natural measure on  $C$ : the unique Borel measure giving mass  $2^{-k}$  to each of the  $2^k$  stage- $k$  intervals. (It exists by Theorem 1.2.15 applied to the algebra generated by those intervals.) Let  $U$  have  $\text{diam } U \leq 1$  and choose  $k$  with  $3^{-(k+1)} \leq \text{diam } U < 3^{-k}$ . An interval of length  $< 3^{-k}$  meets at most 2 of the stage- $k$  intervals, since consecutive ones are separated by a gap of length  $3^{-k}$ . Hence

$$\mu(U) \leq 2 \cdot 2^{-k} = 2 \cdot 3^{-ks} \leq 2 \cdot (3 \text{diam } U)^s = 2 \cdot 3^s (\text{diam } U)^s = 6 (\text{diam } U)^s.$$

By Lemma 7.1.6 with  $c = 6$ ,  $\mathcal{H}^s(C) \geq \mu(C)/6 = 1/6 > 0$ , so  $\dim_{\text{H}} C \geq s$ .

*Conclusion.*  $\dim_{\text{H}} C = \log 2 / \log 3 \approx 0.6309$ , and  $\frac{1}{6} \leq \mathcal{H}^s(C) \leq 1$ . In fact  $\mathcal{H}^s(C) = 1$ , but the sharp constant needs a density argument and is not needed here.

**WORKED EXAMPLE 7.1.2— TIER A A set of every dimension**

For  $s \in (0, 1)$  let  $C_\lambda$  be the Cantor set obtained by removing, at each stage, the middle proportion  $1 - 2\lambda$  of every interval, so that stage  $k$  consists of  $2^k$  intervals of length  $\lambda^k$ , with  $0 < \lambda < \frac{1}{2}$ .

Repeating Worked Example 7.1.1 verbatim with 3 replaced by  $\lambda^{-1}$  gives

$$\dim_{\text{H}} C_\lambda = \frac{\log 2}{\log(1/\lambda)},$$

which runs over all of  $(0, 1)$  as  $\lambda$  runs over  $(0, \frac{1}{2})$ . Taking products  $C_\lambda \times [0, 1]^k$  realises every value in  $(0, n)$  in  $\mathbb{R}^n$ , using the product formula of Drill 1.3.

So the Hausdorff dimension takes every value in  $[0, n]$ , and in particular it is not restricted to integers. That is the whole point of the notion, and the contrast with topological dimension — which is an integer — is what makes it the right tool for Kakeya, where the conjectured answer  $n$  is an integer but every known lower bound is not.

**Proposition 7.1.7** (Dimension of a countable union and of a graph of a measure). *Let  $E \subseteq \mathbb{R}^n$  be Borel with  $\mathcal{H}^s(E) < \infty$ , and let  $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$  be Lipschitz. Then for  $\mathcal{H}^s$ -almost every value the fibre is small in the following sense: if  $s > m$  then*

$$\int_{\mathbb{R}^m} \mathcal{H}^{s-m}(E \cap f^{-1}(y)) \, dy \lesssim_{n,m,s} L^m \mathcal{H}^s(E),$$

where  $L$  is the Lipschitz constant. In particular  $\mathcal{H}^{s-m}(E \cap f^{-1}(y)) < \infty$  for Lebesgue-almost every  $y$ .

*Proof.* Let  $\{U_i\}$  cover  $E$  with  $\text{diam } U_i \leq \varepsilon$  and  $\sum_i (\text{diam } U_i)^s \leq \mathcal{H}^s(E) + \eta$ . For each  $i$ ,  $f(U_i)$  has diameter at most  $L \text{diam } U_i$ , so it is contained in a ball of that radius and

$$m(f(U_i)) \leq \omega_m (L \text{diam } U_i)^m.$$

For a fixed  $y$ , the sets  $U_i$  with  $y \in f(U_i)$  cover  $E \cap f^{-1}(y)$  and have diameter at most  $\varepsilon$ , so

$$\mathcal{H}_\varepsilon^{s-m}(E \cap f^{-1}(y)) \leq \sum_i (\text{diam } U_i)^{s-m} \mathbf{1}_{f(U_i)}(y).$$

Integrating in  $y$  and using the display above,

$$\int \mathcal{H}_\varepsilon^{s-m}(E \cap f^{-1}(y)) \, dy \leq \sum_i (\text{diam } U_i)^{s-m} m(f(U_i)) \leq \omega_m L^m \sum_i (\text{diam } U_i)^s.$$

Letting  $\varepsilon \downarrow 0$ , with Fatou on the left and the choice of cover on the right, gives the bound with  $\mathcal{H}^s(E) + \eta$ ; let  $\eta \downarrow 0$ .  $\square$

**Where we are.** Proposition 7.1.7 is a Fubini-type statement for Hausdorff measure, and it is the reason dimension behaves predictably under slicing: a set of dimension  $s$  meets almost every fibre of a Lipschitz map to  $\mathbb{R}^m$  in dimension at most  $s - m$ . The reverse inequality is false in general — fibres can be smaller than the count suggests, since  $f$  may be far from surjective — which is why the statement is one-sided.

It will be used in Volume 8 in the contrapositive: to prove a set is large, one shows many of its slices are large. That is exactly the structure of

Marstrand's projection theorem, where the slices are the fibres of  $\pi_\theta$  and the averaging over  $\theta$  is what Worked Example 7.6.2(a) performs.

## 7.1.4 Comparison with Lebesgue measure

**Proposition 7.1.8.** *There are constants  $0 < c_n \leq C_n < \infty$  with*

$$c_n m(E) \leq \mathcal{H}^n(E) \leq C_n m(E) \quad \text{for every } E \subseteq \mathbb{R}^n.$$

*In particular  $\mathcal{H}^s(\mathbb{R}^n) = 0$  for  $s > n$ , and  $\dim_{\mathbb{H}} E \leq n$  always.*

*Proof. Step 1.*  $m \lesssim \mathcal{H}^n$ .

Any set  $U$  of diameter  $d$  is contained in a closed ball of radius  $d$ , so  $m(U) \leq \omega_n d^n$  where  $\omega_n = m(B(0, 1))$ . Given a cover  $\{U_i\}$  of  $E$ ,  $m(E) \leq \sum_i m(U_i) \leq \omega_n \sum_i (\text{diam } U_i)^n$ . Taking the infimum,  $m(E) \leq \omega_n \mathcal{H}_\varepsilon^n(E) \leq \omega_n \mathcal{H}^n(E)$ .

**Step 2.**  $\mathcal{H}^n \lesssim m$ .

It suffices to treat  $E$  of finite Lebesgue measure. Given  $\eta > 0$ , choose an open  $V \supseteq E$  with  $m(V) \leq m(E) + \eta$  (outer regularity, Theorem 1.3.6). Decompose  $V$  into dyadic cubes  $Q_i$  with disjoint interiors and  $\text{diam } Q_i \leq \varepsilon$ , which is possible by subdividing the standard dyadic decomposition of an open set. Then  $\{Q_i\}$  covers  $E$  and, writing  $\ell_i$  for the side length,  $\text{diam } Q_i = \sqrt{n} \ell_i$ , so

$$\sum_i (\text{diam } Q_i)^n = n^{n/2} \sum_i \ell_i^n = n^{n/2} \sum_i m(Q_i) = n^{n/2} m(V) \leq n^{n/2} (m(E) + \eta).$$

Hence  $\mathcal{H}_\varepsilon^n(E) \leq n^{n/2} (m(E) + \eta)$ ; let  $\varepsilon \downarrow 0$  and  $\eta \downarrow 0$ .

**Step 3.** *Consequences.*

By Step 2,  $\mathcal{H}^n$  is finite on bounded sets, so Theorem 7.1.3(i) gives  $\mathcal{H}^s = 0$  on bounded sets for  $s > n$ , and countable subadditivity extends this to  $\mathbb{R}^n$ .  $\square$

**Corollary 7.1.9** (The dimension is at most  $n$ ). *For every  $E \subseteq \mathbb{R}^n$ ,  $\dim_{\mathbb{H}} E \leq n$ ; that is, the exponent  $d$  of Theorem 7.1.3 lies in  $[0, n]$ .*

*Proof.* By Proposition 7.1.8,  $\mathcal{H}^n$  is finite on bounded sets. Fix a bounded  $E$ ; then  $\mathcal{H}^n(E) < \infty$ , so Theorem 7.1.3(i) gives  $\mathcal{H}^s(E) = 0$  for every  $s > n$ , whence  $d \leq n$ . A general  $E$  is a countable union of bounded sets and  $\dim_{\mathbb{H}}$  is countably stable (Proposition 7.1.5(ii)).  $\square$

*Remark 7.1.10.* The sharp constant is  $\mathcal{H}^n = 2^n \omega_n^{-1} m$  with the normalisation of Definition 7.1.1, and proving it requires the isodiametric inequality — that among sets of given diameter the ball has largest volume. We do not prove it, and nothing here needs it: every use of Proposition 7.1.8 is qualitative, being a statement about which sets are null.

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**WORKED EXAMPLE 7.1.3— TIER B Dimension of a graph, and how much regularity buys**

Let  $f : [0, 1] \rightarrow \mathbb{R}$  and  $\Gamma_f = \{(x, f(x)) : x \in [0, 1]\}$  its graph. The dimension of  $\Gamma_f$  measures how badly  $f$  oscillates, and the computation is a good exercise in both bounds.

*If  $f$  is Lipschitz.* The map  $x \mapsto (x, f(x))$  is Lipschitz with constant  $\sqrt{1 + L^2}$ , so by Proposition 7.1.5(iii),  $\dim_{\mathbb{H}} \Gamma_f \leq \dim_{\mathbb{H}} [0, 1] = 1$ ; and the projection  $(x, y) \mapsto x$  is Lipschitz and maps  $\Gamma_f$  onto  $[0, 1]$ , so  $\dim_{\mathbb{H}} \Gamma_f \geq 1$ . Hence  $\dim_{\mathbb{H}} \Gamma_f = 1$ : a Lipschitz graph is one-dimensional, as it should be.

*If  $f$  is only Hölder of exponent  $\alpha$ .* Now  $x \mapsto (x, f(x))$  is Hölder of exponent  $\alpha$ , and Proposition 7.1.5(iv) gives  $\dim_{\mathbb{H}} \Gamma_f \leq 1/\alpha$ . That is a genuine loss, and it is attained: the Weierstrass function  $W(x) = \sum_k \lambda^{-k\alpha} \cos(\lambda^k x)$  is Hölder- $\alpha$  and its graph has dimension  $2 - \alpha$  for  $\lambda$  large — a theorem of Shen (2018), which we do not prove.

*The lesson.* Between 1 and  $1/\alpha$  there is room, and the upper bound from Hölder continuity alone is not sharp:  $2 - \alpha < 1/\alpha$  for  $\alpha \in (0, 1)$ . Dimension is not determined by a modulus of continuity, and computing it always requires information about the specific set.

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**WORKED EXAMPLE 7.1.4— TIER C: RESEARCH TASTE The Kakeya conjecture, stated properly at last**

Volume 1 introduced Besicovitch sets and Volume 6 could only reach Minkowski dimension. We can now state the conjecture the series is aimed at.

**Conjecture 7.1.11** (Kakeya). *For  $n \geq 2$ , every Besicovitch set  $K \subseteq \mathbb{R}^n$  — every compact set containing a unit line segment in every direction — satisfies  $\dim_{\mathbb{H}} K = n$ .*

(a) *Why  $\dim_{\mathbb{H}}$  and not  $m$ .* Because the measure statement is *false*: there are Besicovitch sets of Lebesgue measure zero, and by Proposition 7.1.8 therefore with  $\mathcal{H}^n(K) = 0$ . So the strongest true statement must be about the critical exponent, and by the Pitfall in §1.2 a set can have  $\dim_{\mathbb{H}} K = n$  with  $\mathcal{H}^n(K) = 0$ . The conjecture asks for exactly that.

(b) *Why  $\dim_{\mathbb{H}}$  and not  $\dim_{\mathbb{M}}$ .* Because Minkowski dimension is the weaker assertion: Chapter 7.2 proves  $\dim_{\mathbb{H}} \leq \dim_{\mathbb{M}}$ , so a Minkowski bound follows from a Hausdorff bound and not conversely. The known results reflect this — Minkowski-dimension bounds for Besicovitch sets were obtained decades before the corresponding Hausdorff bounds, and in some dimensions the gap remains.

(c) *What is known.*  $n = 2$  is a theorem:  $\dim_{\mathbb{H}} K = 2$  for every planar Besicovitch set, by an argument in the spirit of Theorem 1.6.14, carried out in Volume 9. For  $n = 3$  the conjecture was settled by Wang and Zahl in 2025 and is the subject of Volume 14. For  $n \geq 4$  it is open, with the best bounds coming from the polynomial and multilinear methods of Volumes 11 and 12.

(d) *What this volume contributes.* Nothing towards the conjecture directly. It builds the language in which the conjecture is stated and the tools — Frostman’s lemma, energy integrals — by which dimension is estimated in practice. Chapter 7.8 closes the loop by showing how a Kakeya maximal function bound of the kind Volume 6 produced upgrades to a Hausdorff dimension bound, which is the step Volume 6 could not take.

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**WORKED EXAMPLE 7.1.5— TIER B Proving a lower bound on dimension: the standard workflow**

Upper bounds on  $\dim_{\mathbb{H}}$  come from exhibiting a cover, which is usually easy. Lower bounds are the work, and Lemma 7.1.6 reduces them to a fixed routine. Here it is, with the three places errors are made.

*The routine.* To show  $\dim_{\mathbb{H}} E \geq s$ : (1) construct a measure  $\mu$  on  $E$  with  $\mu(E) > 0$ ; (2) verify  $\mu(U) \lesssim (\text{diam } U)^s$  for all small  $U$ ; (3) apply Lemma 7.1.6.

*Step 1 in practice.* For a set built by a nested construction —  $E = \bigcap_k E_k$  with  $E_k$  a union of  $N_k$  pieces — define  $\mu$  by giving each stage- $k$  piece mass  $1/N_k$  and extending by Theorem 1.2.15. Consistency requires each stage- $k$  piece to contain the same number  $N_{k+1}/N_k$  of stage- $(k+1)$  pieces; when it does not, one weights proportionally instead.

*Step 2 in practice.* Given  $U$  with  $\text{diam } U$  small, find the stage  $k$  with  $\text{diam } U$  comparable to the stage- $k$  piece size  $\rho_k$ , and count how many stage- $k$  pieces  $U$  can meet. *This is where the separation between pieces is used:* if consecutive pieces are separated by a gap  $\gtrsim \rho_k$ , then  $U$  meets  $O(1)$  of them and  $\mu(U) \lesssim 1/N_k \approx \rho_k^s$ .

*The three errors.*

- *Forgetting the gaps.* Without separation,  $U$  may meet many pieces and the bound fails. A construction with pieces touching — for instance removing nothing — gives  $\mu(U) \approx \text{diam } U$ , and only  $s \leq 1$  follows.
- *Checking only  $U$  from the construction.* The hypothesis of Lemma 7.1.6 quantifies over *all* sets  $U$ , not just the pieces. It is the arbitrary  $U$  that forces the choice of  $k$  in Step 2.



- *Losing*  $\mu(E) > 0$ . If the mass is normalised at each stage the total is 1 automatically; if instead one assigns absolute masses, they may tend to zero and the conclusion is vacuous. Theorem 7.5.1 deals with exactly this difficulty in general.

*A check on the routine.* Applying it to  $E = [0, 1]$  with the pieces being dyadic intervals gives  $\mu = \text{Lebesgue}$ ,  $\mu(U) \leq \text{diam } U$ , and  $s = 1$  — the correct answer, and a reassurance that no step is losing anything.

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## Drills

**Drill 7.1.1.** Show that in Definition 7.1.1 one may restrict the covering sets to be open, or to be closed, or to be convex, without changing  $\mathcal{H}^s$ .

**Drill 7.1.2.** Prove that  $\mathcal{H}^1$  on  $\mathbb{R}$  equals Lebesgue measure exactly, with no constant.

**Drill 7.1.3.** Show  $\dim_{\text{H}}(E \times F) \geq \dim_{\text{H}} E + \dim_{\text{H}} F$  for Borel  $E, F$ , using Lemma 7.1.6 and a product measure. (Equality can fail; the example is due to Besicovitch and Moran.)

**Drill 7.1.4.** Construct a set  $E \subseteq [0, 1]$  with  $\dim_{\text{H}} E = s$  and  $\mathcal{H}^s(E) = 0$ , by taking a Cantor set whose ratios vary with the stage.

## Notes

Hausdorff introduced the measure in 1918, extending Carathéodory's construction of 1914. The mass distribution principle is folklore; its converse, that a measure with the required decay always exists, is Frostman's lemma (Chapter 7.5) and is much deeper.

The value  $\mathcal{H}^s(C) = 1$  for the middle-thirds Cantor set is due to Marion and to Ayer–Strichartz; the sharp constant in Proposition 7.1.8 is classical and follows from the isodiametric inequality.



## Chapter 7.2

# Minkowski Dimension

Hausdorff dimension is defined by a covering problem in which the covering sets may have wildly different sizes. Minkowski dimension asks the same question with all sets of the *same* size, and is correspondingly cruder and much easier to estimate. This chapter develops it and quantifies exactly what is lost — which is what Volume 6 ran into.

### 7.2.1 Box counting

**Definition 7.2.1.** For a bounded  $E \subseteq \mathbb{R}^n$  and  $\delta > 0$  let  $N(E, \delta)$  be the least number of sets of diameter at most  $\delta$  needed to cover  $E$ . The *lower* and *upper Minkowski dimensions* are

$$\underline{\dim}_M E = \liminf_{\delta \downarrow 0} \frac{\log N(E, \delta)}{\log(1/\delta)}, \quad \overline{\dim}_M E = \limsup_{\delta \downarrow 0} \frac{\log N(E, \delta)}{\log(1/\delta)}.$$

When they agree the common value is written  $\dim_M E$ .

**Proposition 7.2.2** (The definitions agree). *In Definition 7.2.1 one may replace  $N(E, \delta)$  by any of*

- (i) *the least number of closed balls of radius  $\delta$  covering  $E$ ;*
- (ii) *the number of  $\delta$ -grid cubes meeting  $E$ ;*
- (iii) *the largest number of disjoint balls of radius  $\delta$  centred in  $E$ ;*
- (iv)  *$m(E_\delta)\delta^{-n}$ , where  $E_\delta$  is the  $\delta$ -neighbourhood,*

*without changing either limit.*

*Proof.* Write  $N_1, \dots, N_4$  for the four quantities. Each is comparable to  $N = N(E, \delta)$  up to a constant factor and a bounded change of scale, and since

$$\frac{\log(CN(E, c\delta))}{\log(1/\delta)} = \frac{\log N(E, c\delta)}{\log(1/(c\delta))} \cdot \frac{\log(1/(c\delta))}{\log(1/\delta)} + \frac{\log C}{\log(1/\delta)}$$

and both correction factors tend to 1 and 0, such changes do not affect the limits.

The comparisons: a set of diameter  $\delta$  lies in a ball of radius  $\delta$ , so  $N_1 \leq N$ ; a ball of radius  $\delta$  has diameter  $2\delta$ , so  $N \leq N_1(E, \delta/2)$ . A  $\delta$ -grid cube has diameter  $\sqrt{n}\delta$  and any set of diameter  $\delta$  meets at most  $2^n$  grid cubes, giving  $N_2 \leq 2^n N$  and  $N \leq N_2$  after rescaling. For (iii), if  $B_1, \dots, B_{N_3}$  are disjoint balls of radius  $\delta$  centred in  $E$  then no set of diameter  $\delta$  meets two of them, so  $N_3 \leq N$ ; conversely a maximal such disjoint family has the property that the doubled balls cover  $E$ , so  $N_1(E, 2\delta) \leq N_3$ . For (iv), if  $N_3$  disjoint balls of radius  $\delta$  are centred in  $E$  then  $E_\delta$  contains them, so  $m(E_\delta) \geq N_3 \omega_n \delta^n$ ; and  $E_\delta$  is covered by the  $N_1(E, \delta)$  balls of radius  $2\delta$ , giving  $m(E_\delta) \leq N_1 \omega_n (2\delta)^n$ .  $\square$

**Proposition 7.2.3** (Minkowski content). *Let  $E \subseteq \mathbb{R}^n$  be bounded and  $s \geq 0$ . Define the upper and lower  $s$ -dimensional Minkowski contents*

$$\overline{\mathcal{M}}^s(E) = \limsup_{\delta \downarrow 0} \frac{m(E_\delta)}{\delta^{n-s}}, \quad \underline{\mathcal{M}}^s(E) = \liminf_{\delta \downarrow 0} \frac{m(E_\delta)}{\delta^{n-s}}.$$

*Then  $\overline{\dim}_M E = \inf\{s : \overline{\mathcal{M}}^s(E) = 0\}$  and  $\underline{\dim}_M E = \inf\{s : \underline{\mathcal{M}}^s(E) = 0\}$ , and*

$$\mathcal{H}^s(E) \lesssim_n \underline{\mathcal{M}}^s(E).$$

*Proof.* By Proposition 7.2.2(iv),  $m(E_\delta) \approx_n N(E, \delta)\delta^n$ , so  $m(E_\delta)\delta^{s-n} \approx_n N(E, \delta)\delta^s$ . The quantity  $N(E, \delta)\delta^s$  tends to 0 precisely when  $\liminf_{\delta} \frac{\log N}{\log(1/\delta)} < s$ , which gives the two identities.

For the inequality, cover  $E$  by  $N(E, \delta)$  sets of diameter  $\delta$ ; then  $\mathcal{H}_\delta^s(E) \leq N(E, \delta)\delta^s \approx_n m(E_\delta)\delta^{s-n}$ . Taking  $\liminf$  along a sequence realising  $\underline{\mathcal{M}}^s$  and using  $\mathcal{H}^s = \lim_{\delta} \mathcal{H}_\delta^s$  gives the claim.  $\square$

**Where we are.** Proposition 7.2.3 makes precise the sense in which Minkowski dimension is a statement about volumes of neighbourhoods, which is the form in which it appeared in Corollary 6.6.3. It also gives the inequality  $\mathcal{H}^s \lesssim \underline{\mathcal{M}}^s$ , a quantitative version of  $\dim_H \leq \underline{\dim}_M$ , and shows the loss is genuinely in the direction one would guess: a neighbourhood bound controls the Hausdorff measure from above, never from below.

There is no reverse inequality. The rationals in  $[0, 1]$  have  $\underline{\mathcal{M}}^1 = 1$  and  $\mathcal{H}^1 = 0$ , so the gap can be total.

## 7.2.2 The comparison with Hausdorff dimension

**Theorem 7.2.4.** *For every bounded  $E \subseteq \mathbb{R}^n$ ,*

$$\dim_{\text{H}} E \leq \underline{\dim}_{\text{M}} E \leq \overline{\dim}_{\text{M}} E.$$

*Proof.* Only the first inequality needs proof. Let  $s < \dim_{\text{H}} E$ , so  $\mathcal{H}^s(E) = \infty$  by Theorem 7.1.3; in particular  $\mathcal{H}_{\delta}^s(E) \geq 1$  for all small  $\delta$ . Covering  $E$  by  $N(E, \delta)$  sets of diameter at most  $\delta$ ,

$$1 \leq \mathcal{H}_{\delta}^s(E) \leq N(E, \delta) \delta^s,$$

so  $N(E, \delta) \geq \delta^{-s}$  and  $\frac{\log N(E, \delta)}{\log(1/\delta)} \geq s$  for all small  $\delta$ . Hence  $\underline{\dim}_{\text{M}} E \geq s$ , and letting  $s \uparrow \dim_{\text{H}} E$  gives the claim.  $\square$

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WORKED EXAMPLE 7.2.1— TIER A **The rationals: the inequality is strict**

Let  $E = \mathbb{Q} \cap [0, 1]$ .

*Hausdorff.*  $E$  is countable, so  $\dim_{\text{H}} E = 0$  by Proposition 7.1.5(v).

*Minkowski.*  $E$  is dense in  $[0, 1]$ , so any cover of  $E$  by sets of diameter  $\delta$  also covers  $[0, 1]$  after taking closures — the closure of a set of diameter  $\delta$  has diameter  $\delta$ . Hence  $N(E, \delta) \geq N([0, 1], \delta) \geq \delta^{-1}$ , and  $\underline{\dim}_{\text{M}} E = 1$ .

*What the example shows.* The two dimensions differ by the whole range. Minkowski dimension cannot see countability because it uses covers by sets of a single size, and a countable dense set looks, at every fixed scale, exactly like the interval containing it. Hausdorff dimension sees it because it may use covers whose sets shrink at different rates — assigning to the  $k$ -th rational a set of diameter  $\eta 2^{-k}$  gives  $\sum (\text{diam } U_i)^s \leq \eta^s \sum 2^{-ks} < \infty$  for every  $s > 0$ .

*The moral for this series.* Whenever a bound is proved by covering at a single scale  $\delta$  — which is what every argument in Volume 6 did — what is bounded is the Minkowski dimension. Upgrading requires an argument that survives passing to covers of varying size, and that is Chapter 7.5.

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**Proposition 7.2.5** (Stability).  $\overline{\dim}_{\text{M}}$  is finitely stable:  $\overline{\dim}_{\text{M}}(E \cup F) = \max\{\overline{\dim}_{\text{M}} E, \overline{\dim}_{\text{M}} F\}$ . It is not countably stable, and  $\underline{\dim}_{\text{M}}$  is not even finitely stable.

*Proof.* Finite stability:  $N(E \cup F, \delta) \leq N(E, \delta) + N(F, \delta) \leq 2 \max\{N(E, \delta), N(F, \delta)\}$ , and the factor 2 disappears in the limit; the reverse inequality is monotonicity.

Countable stability fails by Worked Example 7.2.1:  $\mathbb{Q} \cap [0, 1]$  is a countable union of points, each of Minkowski dimension 0, but has Minkowski dimension 1.

For  $\underline{\dim}_{\text{M}}$ , the standard counterexample is a pair of sets whose counting functions oscillate out of phase: choose sequences  $\delta_k \rightarrow 0$  and build  $E$  with  $N(E, \delta) \approx \delta^{-1}$  for  $\delta$  near  $\delta_{2k}$  and  $\approx \delta^{-1/2}$  near  $\delta_{2k+1}$ , and  $F$  the reverse.

Then  $\underline{\dim}_M E = \underline{\dim}_M F = \frac{1}{2}$  while  $\underline{\dim}_M(E \cup F) = 1$ . Drill 2.3 asks for the construction.  $\square$

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**WORKED EXAMPLE 7.2.2— TIER B Minkowski content, and when the two dimensions agree**

By Proposition 7.2.2(iv),  $\overline{\dim}_M E = n - \liminf_{\delta \rightarrow 0} \frac{\log m(E_\delta)}{\log \delta}$ , so Minkowski dimension is exactly a statement about how fast the volume of the  $\delta$ -neighbourhood shrinks. This is the form in which it appears in Volume 6 — Corollary 6.6.3 bounded  $m(K_\delta)$  from below — and it is worth recording when the cruder notion happens to be the sharp one.

*They agree for self-similar sets.* Chapter 7.8 proves that if  $E$  is the attractor of an iterated function system satisfying the open set condition then  $\dim_H E = \dim_M E$ , given by the similarity dimension. The Cantor set of Worked Example 7.1.1 is the model case:  $N(C, 3^{-k}) = 2^k$  exactly, so  $\dim_M C = \log 2 / \log 3 = \dim_H C$ .

*They agree for sets with a regularity property.* If  $E$  carries a measure  $\mu$  with  $\mu(B(x, r)) \approx r^s$  for all  $x \in E$  and  $r \leq 1$  — an *Ahlfors–David regular set* — then both dimensions equal  $s$ . The lower bound is Lemma 7.1.6 and the upper bound is a packing argument.

*They differ exactly when the set is inhomogeneous across scales.* The rationals are the extreme case. A more interesting one: the set  $\{0\} \cup \{1/k : k \geq 1\}$  has  $\dim_H = 0$  but  $\dim_M = \frac{1}{2}$ , computed in Drill 2.1. Here nothing is countable-dense; the discrepancy comes from the points accumulating at a polynomial rate, which a single-scale cover cannot exploit and a variable-scale cover can.

*Practical consequence.* When reading a dimension bound in this subject, always check which dimension is meant. Bounds for Besicovitch sets in the literature are stated for both, the Minkowski versions are typically older and easier, and a paper proving a Hausdorff bound will usually say so in its title.

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**WORKED EXAMPLE 7.2.3— TIER C: RESEARCH TASTE Why Volume 6 stopped at Minkowski, and what it would take to go further**

Corollary 6.6.3 derived, from the restriction conjecture, that  $m(K_\delta) \gtrsim_\epsilon \delta^\epsilon$  for a Besicovitch set  $K$ . By Proposition 7.2.2(iv) that is exactly a Minkowski statement, and by Worked Example 7.2.1 it is strictly weaker than the Kakeya conjecture.

(a) *Where the single scale is built in.* The proof went through a family of  $\delta$ -tubes with  $\delta$ -separated directions. That family exists at each fixed  $\delta$ , and the estimate obtained bounds the multiplicity of tubes at that scale. Nothing in the argument relates the scales to one another.

(b) *What a Hausdorff bound would require.* By Lemma 7.1.6, to prove  $\dim_{\text{H}} K \geq s$  it suffices to produce a measure  $\mu$  on  $K$  with  $\mu(B(x, r)) \lesssim r^s$  for all  $r$  simultaneously. So one needs an estimate that is uniform across scales, not one estimate per scale.

(c) *The standard route.* Suppose, for contradiction, that  $\dim_{\text{H}} K < s$ . Then  $\mathcal{H}^s(K) = 0$ , so there is a cover of  $K$  by balls  $B(x_i, r_i)$  with  $\sum r_i^s$  small and the  $r_i$  of different sizes. Grouping the balls by dyadic size and applying a single-scale estimate to each group produces, after summing, a contradiction — provided the single-scale estimate carries a factor that beats the number of groups, which is  $\approx \log(1/\delta)$ . Since Volume 6's estimates carry  $\delta^{-\varepsilon}$  and  $\delta^{-\varepsilon}$  beats any power of a logarithm, this works.

(d) *Where it is done.* Chapter 7.8 carries out (c) in detail. The conclusion is that the *Keakeya maximal* conjecture — a family of single-scale statements — implies the *Keakeya* conjecture for Hausdorff dimension, so the two formulations are equivalent after all. That equivalence is why the literature moves between them freely, and it is the last thing this volume owes the reader.

(e) *A caution.* The implication in (d) requires the single-scale estimate with a  $\delta^{-\varepsilon}$  loss for every  $\varepsilon$ . A single-scale bound with a fixed power loss  $\delta^{-c}$  gives only  $\dim_{\text{H}} \geq n - c'$  for a corresponding  $c'$ , and the relation between  $c$  and  $c'$  is not the identity. Tracking that relation correctly is where dimension bounds in the literature are most often misquoted.

#### WORKED EXAMPLE 7.2.4— TIER B **Computing a box dimension in practice**

Minkowski dimension is the one that can actually be computed, and the computation follows a fixed recipe. Here it is on three examples of increasing difficulty.

(a) *A self-similar set: the Cantor set.* At scale  $\delta = 3^{-k}$  the natural cover is by the  $2^k$  construction intervals, so  $N(C, 3^{-k}) \leq 2^k$ . For the lower bound, the  $2^k$  intervals are separated by gaps of length  $3^{-k}$ , so no set of diameter  $3^{-k}$  meets two of them and  $N(C, 3^{-k}) \geq 2^k$ . Hence  $N = 2^k$  exactly and

$$\dim_{\text{M}} C = \lim_k \frac{k \log 2}{k \log 3} = \frac{\log 2}{\log 3}.$$

Intermediate  $\delta$  are handled by monotonicity:  $N(E, \delta)$  is decreasing in  $\delta$ , so it is enough to compute along a geometric sequence.

(b) *A set with a rate:*  $E = \{0\} \cup \{k^{-1} : k \geq 1\}$ . Split at the point where consecutive gaps equal  $\delta$ . The gap between  $1/k$  and  $1/(k+1)$  is  $\approx k^{-2}$ , so the gaps exceed  $\delta$  when  $k < \delta^{-1/2}$ . For those  $k$  each point needs its own interval:

$\approx \delta^{-1/2}$  intervals. The remaining points lie in  $[0, \delta^{-1/2-1}] = [0, \delta^{1/2}]$ , an interval of length  $\delta^{1/2}$ , needing  $\approx \delta^{-1/2}$  intervals. Total  $N(E, \delta) \approx \delta^{-1/2}$ , so  $\dim_{\mathbf{M}} E = \frac{1}{2}$ .

*The recipe visible in (b).* Find the scale at which the set's structure changes — here, where the gaps drop below  $\delta$  — and count separately on each side. Nearly every box-counting computation has this shape.

(c) *A set where the limit does not exist.* Let  $E = \{0\} \cup \bigcup_k [a_{k+1}, a_k]$  with  $a_k$  decreasing very fast, say  $a_k = 2^{-k!}$ . At  $\delta = a_k$  the set looks essentially like the single interval  $[a_{k+1}, a_k]$  of length  $\approx a_k$ , so  $N \approx 1$ ; at  $\delta = a_{k+1}$  it looks like a union of intervals of total length  $\approx a_k \gg a_{k+1}$ , so  $N \approx a_k/a_{k+1}$ , which is enormous. The ratio  $\log N / \log(1/\delta)$  therefore oscillates, and  $\underline{\dim}_{\mathbf{M}} E < \overline{\dim}_{\mathbf{M}} E$ .

*What (c) shows.* The two Minkowski dimensions genuinely differ for sets built at rapidly separating scales, and such sets are not pathological — the Besicovitch sets produced by iterated constructions have exactly this character. This is a second reason, beyond Worked Example 7.2.1, to prefer  $\dim_{\mathbf{H}}$ : it is a single number, always.

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## Drills

**Drill 7.2.1.** Show that  $E = \{0\} \cup \{1/k : k \geq 1\}$  has  $\dim_{\mathbf{M}} E = \frac{1}{2}$ , by counting how many  $\delta$ -intervals are needed near 0 and away from it.

**Drill 7.2.2.** Prove that  $\overline{\dim}_{\mathbf{M}} \overline{E} = \overline{\dim}_{\mathbf{M}} E$  for every bounded  $E$ , and deduce that Minkowski dimension cannot distinguish a set from its closure.

**Drill 7.2.3.** Carry out the construction sketched in Proposition 7.2.5 showing that  $\underline{\dim}_{\mathbf{M}}$  is not finitely stable.

**Drill 7.2.4.** Show that for a compact set  $E$  with  $\dim_{\mathbf{H}} E < s < \overline{\dim}_{\mathbf{M}} E$  there is a cover of  $E$  by balls of varying radii with  $\sum r_i^s$  arbitrarily small, but none by balls of a single radius.

**Where we are.** The two dimensions answer different questions, and it is worth having the distinction in one sentence.

$\dim_{\mathbf{M}}$  asks: *at scale  $\delta$ , how many pieces does  $E$  occupy?* It is a question about a snapshot, and it is answered by a count.

$\dim_{\mathbf{H}}$  asks: *how efficiently can  $E$  be covered if one is allowed to use different scales in different places?* It is a question about all scales at once, and it is answered by an optimisation.



Everything in this chapter follows from that difference. Minkowski dimension is computable, finitely stable, blind to countability, and insensitive to closure. Hausdorff dimension is none of those things, and is the one the Kakeya conjecture is about. Theorem 7.8.5 will show that the gap can nonetheless be bridged when the single-scale estimates come with arbitrarily small power losses, which is the situation in this subject — and that is why the literature is able to work almost entirely with snapshots.

## Notes

Box dimension appears in work of Bouligand and of Pontrjagin–Schnirelmann in the 1930s; the name “Minkowski dimension” records Minkowski’s use of the  $\delta$ -neighbourhood. The equivalences of Proposition 7.2.2 are standard and worth doing once.

The failure of countable stability is the reason Hausdorff dimension is the notion of choice in geometric measure theory, despite being harder to estimate. For the sets arising in this series — Besicovitch sets, and the Furstenberg sets of Volume 14 — the two often coincide, but proving that they do is generally as hard as computing either.



## Chapter 7.3

# The Hausdorff Metric and Blaschke Selection

Several constructions in this volume and the next produce a set as a limit of approximations: the attractor of an iterated function system in Chapter 7.8, and the Besicovitch set of measure zero promised in Volume 1. Both need a topology on the space of compact sets in which such limits exist. This chapter supplies it, and then explains — with some care, because the point is easy to get wrong — exactly what it does *not* give.

### 7.3.1 The metric

**Definition 7.3.1.** Let  $\mathcal{K}(\mathbb{R}^n)$  denote the collection of nonempty compact subsets of  $\mathbb{R}^n$ . For  $A, B \in \mathcal{K}$  set

$$d_H(A, B) = \inf \{ \varepsilon > 0 : A \subseteq B_\varepsilon \text{ and } B \subseteq A_\varepsilon \},$$

where  $A_\varepsilon = \{x : \text{dist}(x, A) \leq \varepsilon\}$ .

**Proposition 7.3.2.**  $d_H$  is a metric on  $\mathcal{K}(\mathbb{R}^n)$ , and

$$d_H(A, B) = \max \left\{ \sup_{a \in A} \text{dist}(a, B), \sup_{b \in B} \text{dist}(b, A) \right\}.$$

*Proof.* The displayed identity is a restatement of the definition:  $A \subseteq B_\varepsilon$  says exactly  $\sup_{a \in A} \text{dist}(a, B) \leq \varepsilon$ , and the infimum of such  $\varepsilon$  is attained because  $A$  is compact and  $\text{dist}(\cdot, B)$  continuous.

Symmetry is clear and  $d_H(A, A) = 0$ . If  $d_H(A, B) = 0$  then every  $a \in A$  has  $\text{dist}(a, B) = 0$ , hence  $a \in B$  as  $B$  is closed; and symmetrically, so  $A = B$ . For the triangle inequality, if  $A \subseteq B_\varepsilon$  and  $B \subseteq C_\eta$  then  $A \subseteq (C_\eta)_\varepsilon = C_{\eta+\varepsilon}$ , and symmetrically.  $\square$

**Theorem 7.3.3** (Completeness).  $(\mathcal{K}(\mathbb{R}^n), d_H)$  is a complete metric space.

*Proof.* Let  $(A_k)$  be Cauchy. Define

$$A = \left\{ x \in \mathbb{R}^n : \text{there are } x_k \in A_k \text{ with } x_k \rightarrow x \right\}.$$

**Step 1.**  $A$  is nonempty and compact.

Choose  $k_1 < k_2 < \dots$  with  $d_H(A_j, A_{k_i}) \leq 2^{-i}$  for  $j \geq k_i$ . Pick  $y_1 \in A_{k_1}$ ; inductively pick  $y_{i+1} \in A_{k_{i+1}}$  with  $|y_{i+1} - y_i| \leq 2^{-i}$ , possible since  $A_{k_i} \subseteq (A_{k_{i+1}})_{2^{-i}}$ . Then  $(y_i)$  is Cauchy in  $\mathbb{R}^n$ ; its limit lies in  $A$  by construction, so  $A \neq \emptyset$ . Boundedness of  $A$  follows since a Cauchy sequence in  $\mathcal{K}$  is uniformly bounded, and  $A$  is closed by a diagonal argument.

**Step 2.**  $A_k \rightarrow A$ .

Given  $\varepsilon > 0$  pick  $K$  with  $d_H(A_j, A_k) \leq \varepsilon/2$  for  $j, k \geq K$ .

$A \subseteq (A_k)_\varepsilon$  for  $k \geq K$ : if  $x \in A$ , take  $x_j \in A_j$  with  $x_j \rightarrow x$ ; for large  $j \geq K$ ,  $|x - x_j| \leq \varepsilon/2$  and  $\text{dist}(x_j, A_k) \leq \varepsilon/2$ , so  $\text{dist}(x, A_k) \leq \varepsilon$ .

$A_k \subseteq A_\varepsilon$  for  $k \geq K$ : let  $x \in A_k$ . Build  $z_j \in A_j$  for  $j \geq k$  with  $|z_{j+1} - z_j|$  summable and  $z_k = x$ , exactly as in Step 1 using the Cauchy property along a rapidly converging subsequence; the limit lies in  $A$  and is within  $\varepsilon$  of  $x$ .  $\square$

**Theorem 7.3.4** (Blaschke selection). Let  $B \subseteq \mathbb{R}^n$  be compact. Then  $\mathcal{K}(B) = \{A \in \mathcal{K}(\mathbb{R}^n) : A \subseteq B\}$  is compact in  $d_H$ . In particular every sequence of nonempty compact subsets of  $B$  has a subsequence converging in  $d_H$  to a nonempty compact subset of  $B$ .

*Proof.* By Theorem 7.3.3 and the fact that  $\mathcal{K}(B)$  is closed in  $\mathcal{K}(\mathbb{R}^n)$  — a Hausdorff limit of subsets of  $B$  lies in  $B$ , as  $B$  is closed — it suffices to prove total boundedness.

Fix  $\varepsilon > 0$  and let  $F \subseteq B$  be a finite  $\varepsilon$ -net, so  $B \subseteq \bigcup_{y \in F} B(y, \varepsilon)$ . For  $A \in \mathcal{K}(B)$  put  $A' = \{y \in F : \text{dist}(y, A) \leq \varepsilon\}$ , a nonempty subset of  $F$ . Then  $d_H(A, A') \leq 2\varepsilon$ : every  $a \in A$  lies within  $\varepsilon$  of some  $y \in F$ , and that  $y$  lies in  $A'$ ; conversely every  $y \in A'$  is within  $\varepsilon$  of  $A$ .

Since  $F$  has finitely many subsets, the finitely many sets  $A'$  form a  $2\varepsilon$ -net for  $\mathcal{K}(B)$ .  $\square$

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**WORKED EXAMPLE 7.3.1 — TIER A** Hausdorff limits can be much smaller, or much larger

(a) *Smaller.* Let  $A_k = \{0, \frac{1}{k}, \frac{2}{k}, \dots, 1\} \subseteq [0, 1]$ . Then  $A_k \rightarrow [0, 1]$ , and the limit is much larger than any  $A_k$ : finite sets converge to an interval.

(b) *Larger.* Let  $A_k = [0, 1]$  for all  $k$ ; trivially  $A_k \rightarrow [0, 1]$ . Now let  $A_k = [0, 1] \cup \{2\}$  for even  $k$  and  $[0, 1]$  for odd  $k$ . This is not Cauchy:  $d_H$  between

consecutive terms is 1. Hausdorff convergence is sensitive to isolated far-away pieces, which is why compactness in Theorem 7.3.4 requires a common bounded container.

(c) *Dimension is not continuous.* In (a) each  $A_k$  is finite, so  $\dim_{\mathbb{H}} A_k = 0$ , while  $\dim_{\mathbb{H}}[0, 1] = 1$ . So  $\dim_{\mathbb{H}}$  is not continuous, and not even upper semicontinuous, along Hausdorff-convergent sequences. Drill 3.2 shows it is not lower semicontinuous either.

**Proposition 7.3.5** (Supports converge, measures need not). *Let  $\mu_k$  be probability measures on a fixed compact  $B \subseteq \mathbb{R}^n$  with  $\mu_k \rightarrow \mu$  weak-\*. Then*

$$\text{spt } \mu \subseteq \liminf_k \text{spt } \mu_k := \left\{ x : \text{dist}(x, \text{spt } \mu_k) \rightarrow 0 \right\},$$

and if moreover  $\text{spt } \mu_k \rightarrow K$  in  $d_{\mathbb{H}}$  then  $\text{spt } \mu \subseteq K$ . The inclusion may be strict.

*Proof.* Let  $x \in \text{spt } \mu$  and  $\varepsilon > 0$ , so  $\mu(B(x, \varepsilon)) > 0$ . Choose  $\varphi \in C_c(B(x, \varepsilon))$  with  $0 \leq \varphi \leq 1$  and  $\int \varphi d\mu > 0$ . By weak-\* convergence,  $\int \varphi d\mu_k \rightarrow \int \varphi d\mu > 0$ , so for large  $k$  we have  $\mu_k(B(x, \varepsilon)) > 0$  and hence  $\text{dist}(x, \text{spt } \mu_k) \leq \varepsilon$ . As  $\varepsilon > 0$  was arbitrary,  $x \in \liminf_k \text{spt } \mu_k$ .

If  $\text{spt } \mu_k \rightarrow K$  in  $d_{\mathbb{H}}$  then  $\liminf_k \text{spt } \mu_k \subseteq K$ , since  $K$  is closed and every point of the liminf is a limit of points of the  $\text{spt } \mu_k$ .

For strictness, let  $\mu_k = (1 - \frac{1}{k})\delta_0 + \frac{1}{k}\delta_1$  on  $B = [0, 1]$ . Then  $\text{spt } \mu_k = \{0, 1\}$  for every  $k$ , so  $\text{spt } \mu_k \rightarrow \{0, 1\}$ , while  $\mu_k \rightarrow \delta_0$  weak-\* and  $\text{spt } \delta_0 = \{0\} \subsetneq \{0, 1\}$ .  $\square$

**Where we are.** Proposition 7.3.5 is the exact statement needed in Step 5 of Theorem 7.5.1: a weak-\* limit of measures supported in shrinking neighbourhoods of  $E$  is supported in  $E$ . It also records the asymmetry that runs through this chapter — *supports can only shrink in the limit, never grow* — which is the same one-sidedness as Proposition 7.3.6 for measure, and the same reason compactness never proves a set is small.

## 7.3.2 Measure along Hausdorff limits

**Proposition 7.3.6.** *Let  $A_k, A \in \mathcal{K}(B)$  with  $A_k \rightarrow A$  in  $d_{\mathbb{H}}$ . Then*

$$\limsup_{k \rightarrow \infty} m(A_k) \leq m(A).$$

*That is, Lebesgue measure is upper semicontinuous on  $\mathcal{K}(B)$ .*

*Proof.* Fix  $\varepsilon > 0$ . For large  $k$ ,  $A_k \subseteq A_\varepsilon$ , so  $m(A_k) \leq m(A_\varepsilon)$  and hence  $\limsup_k m(A_k) \leq m(A_\varepsilon)$ . Since  $A$  is compact,  $A_\varepsilon \downarrow A$  as  $\varepsilon \downarrow 0$  and the sets have finite measure, so  $m(A_\varepsilon) \rightarrow m(A)$  by continuity from above (Proposition 1.2.6). Letting  $\varepsilon \downarrow 0$  gives the claim.  $\square$

**Pitfall: why compactness does *not* produce a measure-zero Besicovitch set**

Here is an argument that looks conclusive and is not. Let

$$\mathcal{B} = \{A \in \mathcal{K}(\overline{B}(0, 2)) : A \text{ contains a unit segment in every direction}\}.$$

One checks  $\mathcal{B}$  is closed — Drill 3.3 — hence compact by Theorem 7.3.4. Volume 1, Theorem 1.3.12 gives elements of  $\mathcal{B}$  of arbitrarily small measure, so  $\inf_{\mathcal{B}} m = 0$ . Take  $A_k \in \mathcal{B}$  with  $m(A_k) \rightarrow 0$  and a convergent subsequence  $A_k \rightarrow A \in \mathcal{B}$ . Is  $m(A) = 0$ ?

*No conclusion follows.* Proposition 7.3.6 gives  $\limsup_k m(A_k) \leq m(A)$ , that is  $0 \leq m(A)$  — vacuous. Upper semicontinuity bounds the limit of the measures *by* the measure of the limit, which is the wrong direction. A u.s.c. function on a compact set attains its maximum, not its minimum, and the infimum of  $m$  over  $\mathcal{B}$  need not be attained.

*And the failure is real, not an artefact.* Worked Example 7.3.1(a) shows sets of measure zero converging to a set of measure one. So the Hausdorff limit of small Besicovitch sets can perfectly well be large, and indeed the limit of a Perron tree construction typically is: the triangles are translated further and further, and their limit fills out a region.

*What is actually needed.* A construction producing a *decreasing* sequence, so that the limit is an intersection and continuity from above applies. The Perron tree does not decrease. The standard route instead builds the set from a Cantor set of directions and computes its measure by Fubini, which requires knowing that certain sums of Cantor sets are null — a projection theorem. That is Marstrand's theorem, proved in Volume 8, and the construction is Kahane's; accordingly the measure-zero Besicovitch set is deferred to Volume 8 rather than proved here.

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**WORKED EXAMPLE 7.3.2— TIER B Contraction mappings on  $\mathcal{K}$ , and why attractors exist**

The positive use of Theorem 7.3.3 in this volume is the existence of attractors, which Chapter 7.8 needs.

*The setup.* Let  $f_1, \dots, f_m : \mathbb{R}^n \rightarrow \mathbb{R}^n$  be contractions, with  $|f_i(x) - f_i(y)| \leq r_i|x - y|$  and  $r_i < 1$ . Define  $F : \mathcal{K} \rightarrow \mathcal{K}$  by

$$F(A) = \bigcup_{i=1}^m f_i(A),$$

a compact set, being a finite union of continuous images of a compact set.

*F is a contraction on  $\mathcal{K}$ .* Put  $r = \max_i r_i < 1$ . If  $A \subseteq B_\varepsilon$  then  $f_i(A) \subseteq f_i(B)_{r_i\varepsilon} \subseteq F(B)_{r\varepsilon}$ , so  $F(A) \subseteq F(B)_{r\varepsilon}$ ; symmetrically. Hence

$$d_H(F(A), F(B)) \leq r d_H(A, B).$$

*Conclusion.* By Theorem 7.3.3 and the Banach fixed point theorem,  $F$  has a unique fixed point  $E \in \mathcal{K}$ , and  $F^k(A) \rightarrow E$  for every starting  $A$ . This  $E$  is the *attractor* of the system, and satisfies  $E = \bigcup_i f_i(E)$ .

*Why this is worth isolating.* The existence of the Cantor set, the Sierpiński gasket and the von Koch curve are all instances, obtained without any construction: one writes down the maps and the set exists. The work then shifts entirely to computing its dimension, which is Chapter 7.8.

**WORKED EXAMPLE 7.3.3— TIER C: RESEARCH TASTE Compactness arguments in this subject: when they work**

The Pitfall above is a case where a compactness argument fails for a structural reason, and it is worth extracting the general principle, since compactness is attempted often in this area and succeeds only sometimes.

(a) *Compactness gives existence of a limit, never a property of it.* Theorem 7.3.4 produces a limit set; whether the limit inherits a property depends on whether that property is *closed* under Hausdorff convergence. Containing a segment in every direction is closed (Drill 3.3). Having small measure is not. Having small dimension is not (Worked Example 7.3.1(c)).

(b) *The properties that are closed are the “containment” ones.* If  $P$  is of the form “ $A$  contains a member of each of a compact family of compact sets”, then  $P$  is closed, by the argument of Drill 3.3.akeya-type hypotheses are all of this form, which is why compactness reductions are standard: one may always assume a Besicovitch set is a limit of nice ones.

(c) *The quantities that are semicontinuous have the wrong sign.* Measure and dimension are the quantities one wants to bound *above*, and Proposition 7.3.6 bounds them below. This asymmetry is not accidental: it reflects that a limit can be larger than everything approaching it, never smaller. So compactness

never proves a set is small.

(d) *Where it is nonetheless used.* In Volume 14, the reduction of the Kakeya conjecture to the *sticky* case is a compactness argument: one extracts a limiting configuration of tubes with additional structure, and the structure is closed under the limit even though the measure is not. The technique is legitimate; what must be checked, every time, is that the property being inherited is of the type in (b).

**WORKED EXAMPLE 7.3.4— TIER B Verifying that a property is closed: the standard argument**

Worked Example 7.3.3(b) claimed that “containing a member of each of a compact family” is closed under Hausdorff convergence. The verification is short and is worth doing once, since every compactness reduction in this subject uses it.

*The claim.* Let  $\mathcal{S}$  be a family of nonempty compact subsets of  $\mathbb{R}^n$  which is itself compact in  $d_H$ . Then

$$\mathcal{P} = \{A \in \mathcal{K}(B) : \exists S \in \mathcal{S}, S \subseteq A\}$$

is closed in  $\mathcal{K}(B)$ .

*Proof.* Let  $A_k \in \mathcal{P}$  with  $A_k \rightarrow A$ , and pick  $S_k \in \mathcal{S}$  with  $S_k \subseteq A_k$ . By compactness of  $\mathcal{S}$ , a subsequence  $S_{k_j} \rightarrow S \in \mathcal{S}$ . We claim  $S \subseteq A$ .

Let  $x \in S$ . Since  $S_{k_j} \rightarrow S$ , there are  $x_j \in S_{k_j}$  with  $x_j \rightarrow x$  — by Proposition 7.3.2,  $\text{dist}(x, S_{k_j}) \leq d_H(S, S_{k_j}) \rightarrow 0$ . Each  $x_j \in S_{k_j} \subseteq A_{k_j}$ , and  $A_{k_j} \rightarrow A$ , so  $\text{dist}(x_j, A) \leq d_H(A_{k_j}, A) \rightarrow 0$ . Therefore  $\text{dist}(x, A) \leq |x - x_j| + \text{dist}(x_j, A) \rightarrow 0$ , and  $x \in A$  as  $A$  is closed.  $\square$

*Application to Besicovitch sets.* Take  $\mathcal{S}$  to be the family of unit segments contained in  $\overline{B}(0, 2)$ . It is compact: a segment is determined by its midpoint and direction, both ranging over compact sets, and the map from (midpoint, direction) to the segment is continuous into  $d_H$ . Hence “contains a unit segment” is closed.

For “contains a segment in *every* direction” one applies the argument for each fixed direction  $\omega$  —  $\mathcal{S}_\omega$ , the segments in direction  $\omega$ , is compact — and intersects over  $\omega$ , an intersection of closed sets. This is the assertion of Drill 3.3.

*The contrast.* Try the same argument for “ $m(A) \leq \varepsilon$ ”. Given  $A_k \rightarrow A$  with  $m(A_k) \leq \varepsilon$ , one wants  $m(A) \leq \varepsilon$ ; but Proposition 7.3.6 gives only  $m(A) \geq \limsup m(A_k)$ , and Worked Example 7.3.1(a) shows the conclusion is false. The difference is that containment is preserved by limits and measure is not.



## Drills

**Drill 7.3.1.** Show  $\mathcal{K}(\mathbb{R}^n)$  is separable, and that  $d_H(A, B)$  may be computed using only the extreme points when  $A, B$  are convex.

**Drill 7.3.2.** Exhibit compact  $A_k \rightarrow A$  with  $\dim_H A_k = 1$  for all  $k$  and  $\dim_H A = 0$ , showing  $\dim_H$  is not lower semicontinuous either.

**Drill 7.3.3.** Prove that the family  $\mathcal{B}$  in the Pitfall is closed in  $d_H$ . (Use that the unit segments contained in  $\overline{B}(0, 2)$  form a compact family.)

**Drill 7.3.4.** Verify the contraction estimate in Worked Example 7.3.2 in detail, and compute the attractor of  $f_1(x) = x/3$ ,  $f_2(x) = x/3 + 2/3$  on  $\mathbb{R}$ .

## Notes

The Hausdorff metric dates to Hausdorff's 1914 book; Blaschke's selection theorem is from 1916 and was proved for convex bodies, where it is usually stated. The formulation for general compact sets is the one used here and is the same proof.

The fixed point argument of Worked Example 7.3.2 is Hutchinson's (1981), and is the standard modern foundation for self-similar sets.



## Chapter 7.4

# Covering Theorems and Densities

Volume 1 proved the Vitali covering lemma for Lebesgue measure and used it for the Hardy–Littlewood maximal function. Hausdorff measures are not translation invariant and need not be doubling, so that argument does not transfer directly. This chapter isolates what does: a covering lemma valid in any metric space, and the differentiation and density theorems it supports.

### 7.4.1 The $5r$ -covering lemma

**Theorem 7.4.1** ( $5r$ -covering lemma). *Let  $\mathcal{F}$  be a family of closed balls in  $\mathbb{R}^n$  with  $\sup\{r(B) : B \in \mathcal{F}\} < \infty$ . Then there is a countable pairwise disjoint subfamily  $\mathcal{G} \subseteq \mathcal{F}$  with*

$$\bigcup_{B \in \mathcal{F}} B \subseteq \bigcup_{B \in \mathcal{G}} 5B,$$

where  $5B$  denotes the ball concentric with  $B$  of five times the radius.

*Idea of the proof.* Select greedily by size, always taking a ball of nearly maximal radius among those disjoint from what has been chosen. Any unselected ball must meet a selected one of comparable or larger radius — otherwise it would have been available at a stage where it was nearly maximal — and a ball meeting another of at least two-thirds its radius is contained in the fivefold dilate of that one.

The constant 5 is  $1 + 2 \cdot 2$ : a factor 2 for the radius ratio and a factor 2 for crossing the selected ball. It is not optimal, but nothing here depends on the constant.

*Proof. Step 1. Stratify by radius.*

Let  $R = \sup_{\mathcal{F}} r(B)$  and for  $j \geq 0$  put

$$\mathcal{F}_j = \left\{ B \in \mathcal{F} : \frac{R}{2^{j+1}} < r(B) \leq \frac{R}{2^j} \right\}.$$

**Step 2. Select.**

Let  $\mathcal{G}_0$  be a maximal pairwise disjoint subfamily of  $\mathcal{F}_0$ ; such a family exists by Zorn's lemma, and is countable since the balls are disjoint with radii bounded below. Inductively, let  $\mathcal{G}_j$  be a maximal pairwise disjoint subfamily of

$$\left\{ B \in \mathcal{F}_j : B \cap B' = \emptyset \text{ for all } B' \in \mathcal{G}_0 \cup \cdots \cup \mathcal{G}_{j-1} \right\},$$

and set  $\mathcal{G} = \bigcup_j \mathcal{G}_j$ , a countable pairwise disjoint family.

**Step 3. Cover.**

Let  $B = B(x, r) \in \mathcal{F}$ , say  $B \in \mathcal{F}_j$ . By maximality of  $\mathcal{G}_j$ ,  $B$  meets some  $B' = B(x', r') \in \mathcal{G}_0 \cup \cdots \cup \mathcal{G}_j$  — either  $B \in \mathcal{G}_j$  itself, or  $B$  was excluded, meaning it meets an earlier selected ball, or it could have been added to  $\mathcal{G}_j$ , contradicting maximality.

Since  $B' \in \mathcal{F}_i$  with  $i \leq j$ ,

$$r' > \frac{R}{2^{i+1}} \geq \frac{R}{2^{j+1}} \geq \frac{r}{2}.$$

Now for  $y \in B$ , using that  $B \cap B' \neq \emptyset$ ,

$$|y - x'| \leq |y - x| + |x - x'| \leq r + (r + r') = 2r + r' \leq 4r' + r' = 5r',$$

so  $B \subseteq 5B'$ . □

**Pitfall: the disjoint family need not cover anything**

Theorem 7.4.1 produces a disjoint family whose *dilates* cover. The disjoint family itself may cover very little: if  $\mathcal{F}$  consists of all balls  $B(x, 1)$  with  $x \in \mathbb{R}^n$ , a maximal disjoint subfamily covers a fixed proportion of space and no more.

This is why the lemma is used to bound measures from above — one estimates  $\mu(\bigcup \mathcal{F}) \leq \sum \mu(5B)$  — and is useless for bounds from below. When a lower bound is wanted, one needs a covering theorem producing a family that covers with bounded overlap, and that is the Besicovitch covering theorem, discussed in Remark 7.4.2.

*Remark 7.4.2* (Scope decision: the Besicovitch covering theorem). The Besicovitch covering theorem asserts that, given a family of balls centred at every

point of a set  $A$ , one may extract  $N(n)$  subfamilies, each disjoint, which together cover  $A$  — so the selected balls cover  $A$  with overlap at most  $N(n)$ , and no dilation is needed. It is not proved in this series.

The obstruction is worth naming, because the proof that suggests itself is false. One selects balls greedily with radii within a factor  $\frac{3}{4}$  of maximal, so that for  $i < j$  one has  $r_j < \frac{4}{3}r_i$ , and centres are never inside previously selected balls, so  $|x_i - x_j| > r_i$ . One then hopes that if  $z$  lies in both  $B_i$  and  $B_j$ , the angle subtended at  $z$  is bounded below, which would bound the overlap by a sphere-packing count. *It is not.* A direct search over the constraints produces configurations with angle less than 0.011 radians — for instance  $r_i = 1$ ,  $r_j = 1.283$ ,  $|x_i - z| = 0.068$ ,  $|x_j - z| = 1.144$ ,  $|x_i - x_j| = 1.076$  — all of which satisfy every condition above. A correct proof must split according to the ratio of the radii and treat comparable and very different radii by different arguments.

That argument runs to several pages, and nothing in this series needs it: every application here is an upper bound on a measure, for which Theorem 7.4.1 suffices. Recorded as revision R6 in the master manifest.

## 7.4.2 Differentiation and density

**Theorem 7.4.3** (Vitali covering theorem for a Radon measure). *Let  $\mu$  be a finite Borel measure on  $\mathbb{R}^n$ ,  $A \subseteq \mathbb{R}^n$  Borel, and  $\mathcal{F}$  a family of closed balls such that every point of  $A$  is the centre of balls in  $\mathcal{F}$  of arbitrarily small radius. Then for every  $\varepsilon > 0$  there is a countable disjoint  $\mathcal{G} \subseteq \mathcal{F}$  with*

$$\mu\left(A \setminus \bigcup_{B \in \mathcal{G}} B\right) = 0 \quad \text{provided } \mu \text{ is doubling,}$$

*and in general  $\mu(A \setminus \bigcup_{\mathcal{G}} B) \leq \varepsilon$  whenever  $\mathcal{G}$  is chosen so that  $\sum_{\mathcal{G}} \mu(5B) \geq \mu(\bigcup \mathcal{F}) - \varepsilon$ .*

*Proof.* Apply Theorem 7.4.1 to  $\mathcal{F}$ , obtaining a disjoint  $\mathcal{G}$  with  $\bigcup \mathcal{F} \subseteq \bigcup_{\mathcal{G}} 5B$ . Then

$$\mu\left(\bigcup \mathcal{F}\right) \leq \sum_{B \in \mathcal{G}} \mu(5B).$$

If  $\mu$  is doubling,  $\mu(5B) \leq C\mu(B)$ , so  $\mu(\bigcup \mathcal{F}) \leq C \sum_{\mathcal{G}} \mu(B) = C\mu(\bigcup \mathcal{G})$ ; applying this to the residual family covering  $A \setminus \bigcup \mathcal{G}$  and iterating gives a disjoint family exhausting  $A$  up to  $\mu$ -measure zero, by the standard geometric-decay argument of Volume 1, Theorem 1.7.7.

Without doubling, the displayed inequality is still available and is the form used below.  $\square$

**Theorem 7.4.4** (Density of  $\mathcal{H}^s$ ). *Let  $E \subseteq \mathbb{R}^n$  be  $\mathcal{H}^s$ -measurable with  $\mathcal{H}^s(E) < \infty$ . Then for  $\mathcal{H}^s$ -almost every  $x \notin E$ ,*

$$\lim_{r \downarrow 0} \frac{\mathcal{H}^s(E \cap B(x, r))}{(2r)^s} = 0,$$

*and for  $\mathcal{H}^s$ -almost every  $x \in E$ ,*

$$\limsup_{r \downarrow 0} \frac{\mathcal{H}^s(E \cap B(x, r))}{(2r)^s} \geq 2^{-s}.$$

*Proof. Step 1. The upper bound off  $E$ .*

Let  $F = \{x \notin E : \limsup_r \mathcal{H}^s(E \cap B(x, r))(2r)^{-s} > \lambda\}$  for  $\lambda > 0$ ; we show  $\mathcal{H}^s(F) = 0$ , which suffices on letting  $\lambda \downarrow 0$  along a sequence.

Since  $\mathcal{H}^s \upharpoonright E$  is a finite Borel measure, it is outer regular (Theorem 1.3.6 applies to finite Borel measures on  $\mathbb{R}^n$ ), so given  $\eta > 0$  there is an open  $V \supseteq E$  with  $\mathcal{H}^s(E \cap V) \leq \mathcal{H}^s(E) + \eta$  — and,  $F$  being disjoint from  $E$ , we may arrange  $\mathcal{H}^s(E \cap V') \leq \eta$  for an open  $V' \supseteq F$  by choosing  $V'$  inside the complement of a compact subset of  $E$  carrying most of its measure.

Every  $x \in F$  is the centre of arbitrarily small balls  $B \subseteq V'$  with  $\mathcal{H}^s(E \cap B) > \lambda(\text{diam } B)^s$ . By Theorem 7.4.1 extract a disjoint subfamily  $\mathcal{G}$  whose fivefold dilates cover  $F$ . Then

$$\mathcal{H}_{10r_0}^s(F) \leq \sum_{\mathcal{G}} (\text{diam } 5B)^s = 5^s \sum_{\mathcal{G}} (\text{diam } B)^s \leq \frac{5^s}{\lambda} \sum_{\mathcal{G}} \mathcal{H}^s(E \cap B) \leq \frac{5^s}{\lambda} \mathcal{H}^s(E \cap V') \leq \frac{5^s}{\lambda}$$

the balls being disjoint and contained in  $V'$ , and  $r_0$  a bound for their radii. Letting  $r_0 \downarrow 0$  and then  $\eta \downarrow 0$  gives  $\mathcal{H}^s(F) = 0$ .

**Step 2. The lower bound on  $E$ .**

Let  $G = \{x \in E : \limsup_r \mathcal{H}^s(E \cap B(x, r))(2r)^{-s} < 2^{-s}\lambda\}$  with  $\lambda < 1$ ; we show  $\mathcal{H}^s(G) = 0$ . Every  $x \in G$  is the centre of arbitrarily small balls with  $\mathcal{H}^s(E \cap B) < \lambda(\text{diam } B)^s$ . Choose an open  $V \supseteq G$  with  $\mathcal{H}^s(E \cap V) \leq \mathcal{H}^s(G) + \eta$  and restrict to balls inside  $V$ ; a  $5r$ -selection as in Step 1 gives a cover of  $G$  by sets  $\{B\}$  with

$$\sum_B (\text{diam } B)^s > \frac{1}{\lambda} \sum_B \mathcal{H}^s(E \cap B) \geq \frac{\mathcal{H}^s(G)}{\lambda},$$

where the last step uses that the chosen balls, after a Vitali-type exhaustion, cover  $G$  up to  $\mathcal{H}^s$ -null sets. Since also  $\sum_B (\text{diam } B)^s \geq \mathcal{H}_{r_0}^s(G)$ , letting  $r_0 \downarrow 0$  yields  $\mathcal{H}^s(G) \geq \mathcal{H}^s(G)/\lambda$ , forcing  $\mathcal{H}^s(G) = 0$  as  $\lambda < 1$ .  $\square$

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**WORKED EXAMPLE 7.4.1— TIER A Densities detect dimension**

Let  $E \subseteq \mathbb{R}^n$  with  $0 < \mathcal{H}^s(E) < \infty$ . Theorem 7.4.4 says that at  $\mathcal{H}^s$ -almost every point of  $E$  the ratio  $\mathcal{H}^s(E \cap B(x, r))/(2r)^s$  does not tend to 0, while at almost every point off  $E$  it does.

*Consequence: a set cannot have finite positive measure in two dimensions.* If  $0 < \mathcal{H}^s(E) < \infty$  and  $t > s$ , then for a.e.  $x \in E$

$$\limsup_{r \rightarrow 0} \frac{\mathcal{H}^t(E \cap B(x, r))}{(2r)^t} \leq \limsup_{r \rightarrow 0} (2r)^{s-t} \cdot \frac{\mathcal{H}^s(E \cap B(x, r))}{(2r)^s} \cdot \frac{\mathcal{H}^t}{\mathcal{H}^s}$$

— and the factor  $(2r)^{s-t} \rightarrow \infty$ , consistent with  $\mathcal{H}^t(E) = 0$  from Theorem 7.1.3. The densities give a local, and therefore more flexible, version of the critical exponent.

*Where this is used.* The density theorem is the tool for showing a set is *not* rectifiable, and in Volume 8 it underlies the proof that a set of finite  $\mathcal{H}^1$  measure with positive lower density is contained in countably many curves. In this series its role is more modest: it certifies that the Frostman measure of Chapter 7.5 is genuinely spread over the set rather than concentrated on a small part.

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**WORKED EXAMPLE 7.4.2— TIER B Why the 5 can be replaced by any  $\lambda > 1$ , and why nobody bothers**

The constant in Theorem 7.4.1 can be improved. Selecting balls with radii within a factor  $\theta < 1$  of maximal at each stage, the computation in Step 3 becomes

$$|y - x'| \leq 2r + r' \leq \left(\frac{2}{\theta} + 1\right)r',$$

so the dilation factor is  $1 + 2/\theta$ , which tends to 3 as  $\theta \uparrow 1$ . With a more careful selection one can reach any  $\lambda > 3$ , and in a general metric space 3 is optimal.

*Why it does not matter here.* Every use of Theorem 7.4.1 in this volume ends with a sum  $\sum (\text{diam } 5B)^s = 5^s \sum (\text{diam } B)^s$ , and the factor  $5^s$  is absorbed into a constant that is then sent to zero by a separate limit. The statement being proved — that some measure vanishes — is insensitive to it.

*When constants do matter.* They matter when the covering lemma is applied  $\log(1/\delta)$  times, once per scale, since then a factor  $C$  per application becomes  $C^{\log(1/\delta)} = \delta^{-\log C}$ , a power loss. This happens in multiscale arguments, and it is the reason such arguments are stated with  $\delta^{-\varepsilon}$  rather than with explicit constants — see Worked Example 6.5.3. A reader who finds a covering constant being tracked carefully should suspect a multiscale induction is

coming.

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**WORKED EXAMPLE 7.4.3— TIER C: RESEARCH TASTE** **What covering theorems are, structurally**

Three covering results have now appeared: Vitali in Volume 1, the  $5r$ -lemma here, and Besicovitch, which is stated and not proved. They differ along two axes, and the classification is worth having.

(a) *What is assumed about the balls.* Vitali and the  $5r$ -lemma assume nothing: any family. Besicovitch assumes the balls are centred at the points of the set being covered. That extra hypothesis is what buys the conclusion without dilation.

(b) *What is concluded.* Vitali and  $5r$  give a disjoint family whose dilates cover; Besicovitch gives a bounded-overlap family that covers. The first is an upper bound tool, the second a lower bound tool, as the Pitfall in §4.1 explained.

(c) *What is assumed about the measure.* The  $5r$ -lemma is purely geometric — no measure appears. Vitali's covering theorem for a measure needs doubling, or else the weaker conclusion of Theorem 7.4.3. Besicovitch needs nothing about the measure, which is exactly why it is the tool for differentiating arbitrary Radon measures, and why its omission here would matter if this series needed that.

(d) *What this series actually needs.* Only (i) upper bounds on Hausdorff measures, from the  $5r$ -lemma; and (ii) the density theorem, proved above from the same lemma. The measures appearing later — Frostman measures on Besicovitch sets, and the natural measures on self-similar sets — are constructed with whatever regularity is required, so the general theory is not needed. This is a recurring pattern in the subject: the general machinery exists, and specific problems are usually solved by building measures with the properties one wants rather than by invoking it.

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**WORKED EXAMPLE 7.4.4— TIER A** **The  $5r$ -lemma applied by hand**

The lemma is used so often that it is worth performing once explicitly.

*The task.* Let  $E \subseteq \mathbb{R}$  be the set  $\{1/k : k = 1, \dots, 100\}$  and  $\mathcal{F}$  the family of intervals  $B(1/k, r_k)$  with  $r_k = 1/(4k^2)$ . Extract a disjoint subfamily whose 5-fold dilates cover.

*Selection.* Radii decrease in  $k$ , so the greedy rule takes  $k = 1$  first:  $B(1, \frac{1}{4}) = (\frac{3}{4}, \frac{5}{4})$ . It contains no other  $1/k$ . Next available is  $k = 2$ :  $B(\frac{1}{2}, \frac{1}{16})$ , disjoint



from the first. Continuing, one checks that  $B(1/k, r_k)$  and  $B(1/(k+1), r_{k+1})$  are disjoint, since

$$\frac{1}{k} - \frac{1}{k+1} = \frac{1}{k(k+1)} \quad \text{while} \quad r_k + r_{k+1} = \frac{1}{4k^2} + \frac{1}{4(k+1)^2} < \frac{1}{2k^2} < \frac{1}{k(k+1)}$$

for  $k \geq 2$ . So *every* ball is selected,  $\mathcal{G} = \mathcal{F}$ , and the covering conclusion is trivial.

*A case where selection bites.* Enlarge the radii to  $r_k = 1/(2k)$ . Now  $B(1, \frac{1}{2}) = (\frac{1}{2}, \frac{3}{2})$  contains  $1/k$  for  $k = 1$  only, but  $B(\frac{1}{2}, \frac{1}{4})$  meets it. The greedy rule takes  $k = 1$ , discards  $k = 2$ , and proceeds; the discarded ball  $B(\frac{1}{2}, \frac{1}{4})$  satisfies  $B(\frac{1}{2}, \frac{1}{4}) \subseteq 5B(1, \frac{1}{2}) = (-\frac{3}{2}, \frac{7}{2})$  ✓, as the lemma promises.

*What the example illustrates.* The dilation factor is needed only for the *discarded* balls, and how much is needed depends on the ratio of radii — which is why the proof's Step 3 turns on  $r' > r/2$ . When all radii are comparable no dilation is needed at all, and the lemma reduces to a packing count.

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**Figure 7.4.1 (to be drawn).** *The 5r-covering lemma*

**Content required:** A cluttered family of discs of various radii; the greedily selected disjoint subfamily in heavy outline; and their 5-fold dilates, dashed, covering everything. Show one discarded disc meeting a selected one of at least half its radius, which is the configuration Step 3 of Theorem 7.4.1 analyses.

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## Drills

**Drill 7.4.1.** Verify the claim in Worked Example 7.4.2 that selecting with ratio  $\theta$  gives dilation factor  $1 + 2/\theta$ .

**Drill 7.4.2.** Show that Theorem 7.4.1 fails if the radii are unbounded, by exhibiting a family of balls in  $\mathbb{R}$  for which no disjoint subfamily has dilates covering the union.

**Drill 7.4.3.** Deduce from Theorem 7.4.4 that a set  $E$  with  $0 < \mathcal{H}^s(E) < \infty$  satisfies  $\mathcal{H}^s(E \cap B(x, r)) \lesssim r^s$  for  $\mathcal{H}^s$ -a.e.  $x \in E$  and all small  $r$  — the conclusion Lemma 7.1.6 takes as a hypothesis.

**Drill 7.4.4.** Confirm the numerical configuration in Remark 7.4.2: check that  $r_i = 1$ ,  $r_j = 1.283$ ,  $|x_i - z| = 0.068$ ,  $|x_j - z| = 1.144$ ,  $|x_i - x_j| = 1.076$  satisfies  $r_j < \frac{4}{3}r_i$  and  $|x_i - x_j| > r_i$ , and compute the angle at  $z$ .

## Notes

The  $5r$ -lemma in this form is due to Vitali and Wiener; the sharp constant 3 in general metric spaces is standard. The Besicovitch covering theorem is from 1945 and its constant  $N(n)$  grows exponentially in  $n$ ; the sharp value is known only for  $n = 1, 2$ .

Theorem 7.4.4 is Besicovitch's; the sharp constant in the lower bound — that the upper density is exactly 1 for  $\mathcal{H}^s$ -a.e. point when  $0 < \mathcal{H}^s(E) < \infty$  and  $s$  is an integer — is much harder and is part of the theory of rectifiability, treated in Volume 8.

## Chapter 7.5

# Frostman's Lemma

Lemma 7.1.6 said that a measure on  $E$  with  $\mu(B(x, r)) \lesssim r^s$  proves  $\dim_{\text{H}} E \geq s$ . Frostman's lemma is the converse: whenever  $\dim_{\text{H}} E > s$ , such a measure exists. It converts the definition of dimension — an infimum over covers, which is awkward to use — into the existence of an object one can compute with, and it is the reason dimension is tractable at all.

**Theorem 7.5.1** (Frostman). *Let  $E \subseteq \mathbb{R}^n$  be compact and  $s \geq 0$ . Then  $\mathcal{H}^s(E) > 0$  if and only if there is a nonzero finite Borel measure  $\mu$  supported in  $E$  with*

$$\mu(B(x, r)) \leq r^s \quad \text{for every } x \in \mathbb{R}^n, r > 0. \quad (7.5.1)$$

Consequently

$$\dim_{\text{H}} E = \sup \left\{ s \geq 0 : \exists \mu \neq 0 \text{ on } E \text{ with } \mu(B(x, r)) \lesssim r^s \right\}.$$

*Idea of the proof.* One direction is Lemma 7.1.6. For the other, the measure is built by a dyadic construction. Start by placing mass  $\ell(Q)^s$  on every dyadic cube  $Q$  of some small generation  $m$  meeting  $E$  — far too much mass, in general. Then sweep upwards through the generations, and whenever a cube carries more than  $\ell(Q)^s$ , scale down the mass inside it uniformly so that it carries exactly  $\ell(Q)^s$ .

After the sweep, every dyadic cube carries at most  $\ell(Q)^s$ , which is (7.5.1) up to a dimensional constant. The work is to show the total mass has not collapsed to zero, and that is where  $\mathcal{H}^s(E) > 0$  enters: each cube that was scaled down is *saturated*, carrying exactly  $\ell(Q)^s$ , and the saturated cubes at the top of the construction form a cover of  $E$  — so their total  $\sum \ell(Q)^s$  is bounded below by  $\mathcal{H}_{\infty}^s(E) > 0$ , which is precisely the surviving mass.

*Proof.* The “if” direction is Lemma 7.1.6 with  $c = 1$  — note that a set  $U$  of diameter  $d$  is contained in a ball of radius  $d$ , so (7.5.1) gives  $\mu(U) \leq (\text{diam } U)^s$ .

For “only if”, assume  $\mathcal{H}^s(E) > 0$ . We use the *dyadic content*

$$\mathcal{H}_\infty^s(E) = \inf \left\{ \sum_i (\text{diam } U_i)^s : E \subseteq \bigcup U_i \right\},$$

the case  $\varepsilon = \infty$  of Definition 7.1.1. Since  $\mathcal{H}_\infty^s \leq \mathcal{H}^s$  and, conversely,  $\mathcal{H}_\infty^s(E) = 0$  forces  $\mathcal{H}^s(E) = 0$  — given a cover with  $\sum (\text{diam } U_i)^s < \eta$ , each  $U_i$  has  $\text{diam } U_i < \eta^{1/s}$ , so the same cover is admissible for  $\mathcal{H}_{\eta^{1/s}}^s$  — we have  $\mathcal{H}_\infty^s(E) > 0$ . Normalise so that  $E \subseteq [0, 1]^n$ .

**Step 1.** *Initialise.*

Fix  $m \geq 1$ . Define a measure  $\mu_m$  by giving each dyadic cube  $Q$  of generation  $m$  that meets  $E$  a uniformly distributed mass  $\ell(Q)^s = 2^{-ms}$ , and mass 0 to the others.

**Step 2.** *Sweep upwards.*

For  $k = m - 1, m - 2, \dots, 0$  in turn, define  $\mu^{(k)}$  from  $\mu^{(k+1)}$  (starting from  $\mu^{(m)} = \mu_m$ ) by: for each dyadic  $Q$  of generation  $k$ , if  $\mu^{(k+1)}(Q) > \ell(Q)^s = 2^{-ks}$ , replace  $\mu^{(k+1)}$  on  $Q$  by

$$\frac{2^{-ks}}{\mu^{(k+1)}(Q)} \mu^{(k+1)} \upharpoonright Q,$$

and otherwise leave it unchanged. Set  $\mu = \mu^{(0)}$ .

Each step only decreases mass and only inside a single cube, so after the sweep

$$\mu(Q) \leq \ell(Q)^s \quad \text{for every dyadic } Q \text{ of generation } 0 \leq k \leq m, \quad (7.5.2)$$

because the value at generation  $k$  was capped at that stage and later stages only rescale  $\mu$  on larger cubes by factors  $\leq 1$ .

**Step 3.** *Mass does not vanish.*

Call a dyadic cube  $Q$  *saturated* if  $\mu(Q) = \ell(Q)^s$ . Every point of  $E$  lies in a saturated cube: the generation- $m$  cube containing it started with mass  $2^{-ms}$ , and if no cube containing it was ever capped then it retains that mass and is itself saturated; otherwise the largest capped cube containing it is saturated, since capping sets its mass exactly to  $\ell^s$  and later stages rescale only by factors applied to still larger cubes — and if such a larger cube were rescaled, it would itself be the saturated one.

Let  $\{Q_i\}$  be the maximal saturated cubes; they are disjoint and cover  $E$ . By definition of  $\mathcal{H}_\infty^s$ , and since  $\text{diam } Q_i = \sqrt{n} \ell(Q_i)$ ,

$$\mathcal{H}_\infty^s(E) \leq \sum_i (\text{diam } Q_i)^s = n^{s/2} \sum_i \ell(Q_i)^s = n^{s/2} \sum_i \mu(Q_i) = n^{s/2} \mu(\mathbb{R}^n),$$

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so  $\mu(\mathbb{R}^n) \geq n^{-s/2} \mathcal{H}_\infty^s(E) > 0$ , a bound independent of  $m$ .

**Step 4.** *From dyadic cubes to balls.*

Let  $B = B(x, r)$  with  $r < 1$  and choose  $k$  with  $2^{-k-1} \leq r < 2^{-k}$ . The ball  $B$  meets at most  $2^n$  dyadic cubes of generation  $k$ , so by (7.5.2)

$$\mu(B) \leq 2^n 2^{-ks} \leq 2^n (2r)^s = 2^{n+s} r^s.$$

For  $r \geq 1$ ,  $\mu(B) \leq \mu(\mathbb{R}^n) \leq 1 \cdot r^s$  after the normalisation below.

**Step 5.** *Pass to the limit and normalise.*

The measures  $\mu = \mu_{(m)}$  constructed for each  $m$  are supported in the compact set  $E_{\sqrt{n}2^{-m}}$  and have total mass in  $[n^{-s/2} \mathcal{H}_\infty^s(E), 1]$ . By weak-\* compactness of bounded sets of measures on a compact set — the Banach–Alaoglu theorem, or Theorem 1.8.15 applied on  $C(K)$  — a subsequence converges weak-\* to a measure  $\mu$  supported in  $E$ , with  $\mu(\mathbb{R}^n) \geq n^{-s/2} \mathcal{H}_\infty^s(E) > 0$ , and the bound of Step 4 passes to the limit since it holds for closed balls and weak-\* limits satisfy  $\mu(B) \leq \liminf \mu_{(m)}(B')$  for slightly larger  $B'$ . Dividing  $\mu$  by  $2^{n+s}$  gives (7.5.1).

The final display of the theorem now follows: “ $\geq$ ” from Lemma 7.1.6, and “ $\leq$ ” because for  $s < \dim_{\mathbb{H}} E$  we have  $\mathcal{H}^s(E) = \infty > 0$  and the construction applies.  $\square$

**Where we are.** The construction is worth summarising in one sentence: *put down too much mass, then cap every cube at its allowance, and observe that the cubes where the cap bites form a cover, so the surviving mass is bounded below by the Hausdorff content.*

Two features are essential. The sweep must go *upwards*, from small cubes to large, so that capping a cube is never undone by a later step. And the lower bound must be against  $\mathcal{H}_\infty^s$ , the content with no restriction on the diameters, because the maximal saturated cubes can be large. That  $\mathcal{H}_\infty^s(E) > 0$  is equivalent to  $\mathcal{H}^s(E) > 0$  is the small lemma at the start of the proof, and it is the only place the equivalence is used.

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#### WORKED EXAMPLE 7.5.1— TIER A The Frostman measure on the Cantor set

For  $C$  the middle-thirds Cantor set and  $s = \log 2 / \log 3$ , the construction of Theorem 7.5.1 recovers the natural measure of Worked Example 7.1.1 — which is a useful check, since that measure was written down by hand.

*Why the sweep does nothing.* Work with base-3 intervals rather than dyadic cubes. At generation  $k$ ,  $C$  is covered by  $2^k$  intervals of length  $3^{-k}$ , and the initialisation gives each mass  $3^{-ks} = 2^{-k}$ . A generation- $(k-1)$  interval meeting  $C$  contains exactly two generation- $k$  intervals meeting  $C$ , so it carries

mass  $2 \cdot 2^{-k} = 2^{-(k-1)}$ , which is exactly its allowance  $3^{-(k-1)s}$ . Nothing is ever capped.

*Consequence.*  $\mu$  is the natural measure, every interval meeting  $C$  is saturated, and  $\mu(C) = 1$ . The construction is sharp here because  $C$  is *Ahlfors–David regular*: mass and allowance agree at every scale and location.

*Where capping does bite.* Take  $E = C \cup [2, 3]$  and  $s = \log 2 / \log 3$ . Now the interval  $[2, 3]$  has  $\mathcal{H}^s = \infty$ , so the initialisation gives it enormous mass, and every containing cube is capped. The Frostman measure that emerges is supported essentially on  $[2, 3]$  and gives  $C$  almost nothing — correctly, since  $\dim_{\mathbb{H}} E = 1 > s$  and the dimension is carried by the interval. Frostman's lemma finds a measure witnessing the dimension, not a canonical one.

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#### WORKED EXAMPLE 7.5.2— TIER B **Frostman as a max-flow min-cut theorem**

There is a second proof of Theorem 7.5.1, due to Howroyd and to Kahane–Salem in the classical case, which explains the construction rather than merely performing it.

*The network.* Build a graph whose vertices are the dyadic cubes of generations 0 through  $m$  meeting  $E$ , with an edge from each cube to each of its children, a source above generation 0 and a sink below generation  $m$ . Give the edge into a cube  $Q$  capacity  $\ell(Q)^s$ .

*Flows are Frostman measures.* A flow from source to sink assigns to each cube the amount passing through it, and the capacity constraint is exactly  $\mu(Q) \leq \ell(Q)^s$  — condition (7.5.2). So admissible flows *are* the measures we are trying to construct, and the maximum flow is the largest total mass available.

*Cuts are covers.* A cut is a set of edges whose removal disconnects source from sink, that is, a collection of cubes meeting every root-to-leaf path — a *cover* of  $E$  by dyadic cubes. Its capacity is  $\sum \ell(Q)^s$ , which is comparable to  $\sum (\text{diam } Q)^s$ .

*The theorem.* Max-flow min-cut says the maximum mass equals the minimum cover cost, that is

$$\max \{ \mu(E) : \mu \text{ Frostman} \} \approx_n \mathcal{H}_{\infty}^s(E).$$

Frostman's lemma is the direction “content positive implies flow positive”, and the capping construction of Theorem 7.5.1 is precisely the standard augmenting-path algorithm run once.

*Why this is worth knowing.* It explains why the constant is dimensional and unavoidable, why the maximal saturated cubes appear (they are the min cut), and it generalises: the same argument gives Frostman-type lemmas for

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measures satisfying other growth conditions, simply by changing the capacities.

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**Where we are.** The lemma is an equivalence between two very different-looking statements:

$$\mathcal{H}^s(E) > 0 \quad \Longleftrightarrow \quad \exists \mu \neq 0 \text{ on } E \text{ with } \mu(B(x, r)) \leq r^s.$$

The left side quantifies over *all covers* of  $E$  — an infimum over an enormous family. The right side asserts the existence of a *single object*. Converting a universally quantified statement into an existentially quantified one is what makes the hypothesis usable, and it is the reason Frostman’s lemma is the most-cited result of this volume.

The same conversion happens elsewhere in the series and is worth recognising as a pattern: Theorem 6.1.3 turned a restriction estimate (a bound for all  $f$ ) into an extension estimate (a bound for all  $g$ ), which is not obviously progress until one observes that  $\mathcal{E}g$  can be written down and  $\mathcal{R}f$  cannot. Frostman does the same for dimension.

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WORKED EXAMPLE 7.5.3 — TIER C: RESEARCH TASTE **What Frostman’s lemma is for**

Every subsequent lower bound on a dimension in this series goes through Theorem 7.5.1, and it is worth listing the pattern before the applications arrive.

(a) *The pattern.* To prove  $\dim_{\mathrm{H}} E \geq s$ : produce a measure on  $E$  satisfying (7.5.1) and apply Lemma 7.1.6. To *use*  $\dim_{\mathrm{H}} E > s$ : apply Theorem 7.5.1 to obtain such a measure, then compute with it. The second direction is the one that makes the hypothesis usable, and it is why the lemma is quoted far more often than it is proved.

(b) *Energy.* Chapter 7.6 converts (7.5.1) into the finiteness of an integral,  $I_t(\mu) = \iint |x - y|^{-t} d\mu d\mu$ . That is a smoother condition, it is stable under the operations one wants to perform — projections, convolutions — and it has a Fourier expression. Almost every computation in Volume 8 is an energy computation, and Frostman is what supplies the measure to compute with.

(c) *Keakeya.* Chapter 7.8 proves that a Keakeya maximal function bound implies a Hausdorff dimension bound. The argument assumes  $\dim_{\mathrm{H}} K < s$ , extracts from the definition a cover with  $\sum r_i^s$  small, groups it by scale, and applies the single-scale estimate to each group. Frostman is not needed for that direction; but the reverse question — what a Besicovitch set of small

dimension would look like — is studied through the Frostman measures it would carry, and that is how the arithmetic methods of Volume 10 proceed.

(d) *A caution about “the” Frostman measure.* As Worked Example 7.5.1 showed, the measure is far from unique and need not be spread over the whole set. Arguments that require the measure to see every part of  $E$  must say so and arrange it, usually by applying the lemma to a subset chosen in advance. Errors of this kind are common in the literature and are worth watching for.

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**WORKED EXAMPLE 7.5.4— TIER B Using Frostman: the standard three-line opening**

Almost every argument in Volumes 8 and 14 begins the same way, and it is worth rehearsing the moves in isolation so that they are recognised later.

*The situation.* One is given a compact  $E$  with  $\dim_{\mathrm{H}} E > s$  and must prove some quantitative statement about  $E$ .

*Move 1: get a measure.* Choose  $t$  with  $s < t < \dim_{\mathrm{H}} E$ . By Theorem 7.5.1 there is a nonzero finite  $\mu$  on  $E$  with  $\mu(B(x, r)) \leq r^t$  for all  $x, r$ . Normalise  $\mu(E) = 1$ .

*Move 2: convert to energy.* By Step 2 of Theorem 7.6.2,  $I_{t'}(\mu) < \infty$  for every  $t' < t$ ; fix such a  $t' > s$ . One now has a finite number  $I_{t'}(\mu)$  to work with instead of a supremum condition.

*Move 3: go to the Fourier side if the problem is two-point.* By Proposition 7.6.5,  $\int |\widehat{\mu}(\xi)|^2 |\xi|^{t'-n} d\xi < \infty$ , so  $\widehat{\mu}$  is small in an averaged sense. Any bilinear or convolution quantity built from  $E$  can now be estimated by Plancherel.

*A worked instance: the distance set has positive measure when  $\dim_{\mathrm{H}} E > \frac{n+1}{2}$ .* Sketching the shape only, since Volume 8 gives the details: let  $\nu$  be the pushforward of  $\mu \times \mu$  under  $(x, y) \mapsto |x - y|$ . One shows  $\nu$  has an  $L^2$  density by writing

$$\int |\widehat{\nu}(\tau)|^2 d\tau \approx \int_0^\infty \left( \int_{\mathbb{S}^{n-1}} |\widehat{\mu}(\tau\omega)|^2 d\sigma(\omega) \right)^2 \tau^{n-1} d\tau,$$

and the inner spherical average is controlled by the decay of  $\widehat{d\sigma}$  from Theorem 5.5.2 together with Move 3. The exponent  $\frac{n+1}{2}$  emerges from matching the two, exactly as in Worked Example 7.7.2.

*What to watch.* Moves 1 and 2 each lose an epsilon in the exponent, which is why conclusions are always stated for  $\dim_{\mathrm{H}} E > s$  strictly and never at equality. And the measure from Move 1 is not canonical (Worked Example 7.5.1),



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so any step requiring  $\mu$  to charge a particular part of  $E$  must arrange it by applying Frostman to a suitable subset first.

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**Figure 7.5.1 (to be drawn).** *The Frostman sweep*

**Content required:** A dyadic tree of four or five generations over an interval, with mass  $\ell(Q)^s$  placed at the bottom; arrows showing the upward sweep; and the capped (saturated) cubes marked, forming an antichain that covers the set. This antichain is the min cut of Worked Example 7.5.2.

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## Drills

**Drill 7.5.1.** Prove carefully that  $\mathcal{H}_\infty^s(E) = 0$  implies  $\mathcal{H}^s(E) = 0$ , the small lemma used at the start of the proof.

**Drill 7.5.2.** Verify (7.5.2): that capping at generation  $k$  is not undone by later stages of the sweep.

**Drill 7.5.3.** Carry out Worked Example 7.5.1 for the Cantor set  $C_\lambda$  of Worked Example 7.1.2 and confirm that nothing is capped there either.

**Drill 7.5.4.** Show that the conclusion of Theorem 7.5.1 fails for non-compact  $E$  by taking  $E = \mathbb{Q} \cap [0, 1]$  and  $s = 0$ , and identify which step of the proof breaks.

**Where we are.** One structural remark before leaving the chapter. Theorem 7.5.1 is stated for *compact*  $E$ , and compactness is used exactly once — in Step 5, to extract a weak-\* convergent subsequence. Everything before that step is a finite construction at a fixed depth  $m$ .

That localisation of the hypothesis is worth noticing, because it tells one what to do when compactness is unavailable. For a Borel set one applies the lemma to compact subsets and takes a supremum, which works because Hausdorff measure is inner regular on Borel sets of finite measure. For a general analytic set the required inner regularity is Choquet's capacitability theorem. And in a metric space with no local compactness the weak-\* argument fails entirely, which is why Howroyd's max-flow proof (Worked Example 7.5.2) — which never passes to a limit — is the one that generalises.

## Notes

Frostman proved the lemma in his 1935 thesis. The dyadic construction given here is the standard modern one; the max-flow interpretation of Worked Example 7.5.2 is due to Howroyd (1995), who used it to extend the lemma to general metric spaces, where the compactness argument of Step 5 is unavailable.

The result is sometimes stated for analytic rather than compact sets; that version requires a capacitability theorem of Choquet and is not needed here.

## Chapter 7.6

# Energy Integrals

Frostman's lemma produces a measure with a bound at every scale and location. That is a supremum condition, and suprema are awkward to combine. The energy integral replaces it by an integral condition, which behaves well under the operations Volume 8 needs — projections, convolutions, and Fourier transforms.

### 7.6.1 The energy of a measure

**Definition 7.6.1.** For a finite Borel measure  $\mu$  on  $\mathbb{R}^n$  and  $t > 0$ , the  $t$ -energy is

$$I_t(\mu) = \iint \frac{d\mu(x) d\mu(y)}{|x - y|^t} = \int (k_t * \mu) d\mu, \quad k_t(x) = |x|^{-t}.$$

**Theorem 7.6.2.** Let  $E \subseteq \mathbb{R}^n$  be compact.

- (i) If  $\mu$  is a nonzero finite measure on  $E$  with  $I_t(\mu) < \infty$ , then  $\dim_{\text{H}} E \geq t$ .
- (ii) If  $\dim_{\text{H}} E > t$ , there is a nonzero finite measure  $\mu$  on  $E$  with  $I_t(\mu) < \infty$ .

Consequently

$$\dim_{\text{H}} E = \sup \{t \geq 0 : \exists \mu \neq 0 \text{ on } E, I_t(\mu) < \infty\}.$$

*Proof. Step 1.* (i): finite energy forces a Frostman bound on a large subset.

Let  $\mu$  be as stated, normalised to  $\mu(E) = 1$ . Put

$$F = \left\{ x \in E : \limsup_{r \downarrow 0} \frac{\mu(B(x, r))}{r^t} > 0 \right\}.$$

We claim  $\mu(F) = 0$ . Indeed, for  $x \in E$ ,

$$(k_t * \mu)(x) = \int \frac{d\mu(y)}{|x - y|^t} \geq \int_{B(x, r)} \frac{d\mu(y)}{|x - y|^t} \geq \frac{\mu(B(x, r))}{r^t},$$

so if  $\limsup_r \mu(B(x, r))r^{-t} = c > 0$  then  $(k_t * \mu)(x) \geq c$ ; but more is true: decomposing  $B(x, r)$  into dyadic annuli one finds  $(k_t * \mu)(x) \geq \sum_j 2^{jt} \mu(B(x, 2^{-j}) \setminus B(x, 2^{-j-1}))$ , and if  $\mu(B(x, 2^{-j})) \geq c2^{-jt}$  for infinitely many  $j$  the sum diverges. Since  $I_t(\mu) = \int (k_t * \mu) d\mu < \infty$ , we have  $(k_t * \mu) < \infty$   $\mu$ -a.e., so  $\mu(F) = 0$ .

Hence  $\mu(E \setminus F) = 1$ , and on  $E \setminus F$  we have  $\mu(B(x, r)) \leq r^t$  for all small  $r$ , uniformly after restricting to a subset  $E'$  of positive measure by Egorov's theorem (Theorem 1.5.9) applied to the functions  $x \mapsto \sup_{r \leq 1/k} \mu(B(x, r))r^{-t}$ . Then  $\mu \upharpoonright E'$  satisfies the hypothesis of Lemma 7.1.6, giving  $\dim_{\mathbb{H}} E \geq \dim_{\mathbb{H}} E' \geq t$ .

**Step 2.** (ii): *a Frostman measure at a higher exponent has finite energy.*

Let  $t < s < \dim_{\mathbb{H}} E$ . By Theorem 7.5.1 there is a nonzero  $\mu$  on  $E$  with  $\mu(B(x, r)) \leq r^s$  for all  $x, r$ . We show  $I_t(\mu) < \infty$ .

Fix  $x \in \text{spt } \mu$  and decompose by dyadic annuli  $A_j = \{y : 2^{-j-1} < |x - y| \leq 2^{-j}\}$  for  $j \geq 0$ , together with  $\{|x - y| > 1\}$ :

$$(k_t * \mu)(x) \leq \mu(\mathbb{R}^n) + \sum_{j \geq 0} 2^{(j+1)t} \mu(A_j) \leq \mu(\mathbb{R}^n) + 2^t \sum_{j \geq 0} 2^{jt} \mu(B(x, 2^{-j})).$$

By the Frostman bound  $\mu(B(x, 2^{-j})) \leq 2^{-js}$ , so the sum is at most  $\sum_j 2^{j(t-s)} < \infty$  since  $t < s$ . Hence  $k_t * \mu$  is bounded, and

$$I_t(\mu) = \int (k_t * \mu) d\mu \leq \|k_t * \mu\|_{L^\infty} \mu(\mathbb{R}^n) < \infty. \quad \square$$

**Where we are.** Step 2 is the direction used constantly, and it is worth extracting: a Frostman measure of exponent  $s$  has finite  $t$ -energy for every  $t < s$ . The loss of the endpoint is real — a measure with  $\mu(B(x, r)) \approx r^s$  exactly has  $I_s(\mu) = \infty$ , since the sum  $\sum_j 2^{j(s-s)}$  diverges logarithmically. This is the same borderline that appeared in Proposition 6.8.6 and in the failure of  $\mathcal{M}f \in L^1$ : a quantity that is exactly critical fails by a logarithm. In practice one always works at  $t < s$  and lets  $t \uparrow s$  at the end, which is why dimension statements in this subject are almost never attained at the critical exponent.

**Proposition 7.6.3** (Energy under maps and products). *Let  $\mu$  be a finite Borel measure on  $\mathbb{R}^n$  and  $t > 0$ .*

(i) *If  $f$  is bi-Lipschitz with constants  $L_1 \leq L_2$ , then  $I_t(f_{\#}\mu) \leq L_1^{-t} I_t(\mu)$ .*

(ii) *If  $\nu$  is a finite Borel measure on  $\mathbb{R}^m$  with  $I_u(\nu) < \infty$ , then  $I_{t+u}(\mu \times \nu) < \infty$  whenever  $I_t(\mu) < \infty$  and one of the two measures is bounded.*

*Proof.* (i) By the change of variables formula for pushforwards,

$$I_t(f_{\#}\mu) = \iint \frac{d\mu(x) d\mu(y)}{|f(x) - f(y)|^t} \leq \iint \frac{d\mu(x) d\mu(y)}{(L_1|x - y|)^t} = L_1^{-t} I_t(\mu),$$

using  $|f(x) - f(y)| \geq L_1|x - y|$ .

(ii) For  $(x, y), (x', y') \in \mathbb{R}^n \times \mathbb{R}^m$  we have  $|(x, y) - (x', y')| \geq \max\{|x - x'|, |y - y'|\}$ , so

$$\frac{1}{|(x, y) - (x', y')|^{t+u}} \leq \frac{1}{|x - x'|^t |y - y'|^u},$$

since  $\max\{a, b\}^{t+u} \geq a^t b^u$  for  $a, b > 0$ . Integrating against  $\mu \times \nu$  twice and applying Tonelli factorises the double integral into  $I_t(\mu) I_u(\nu)$ .  $\square$

**Corollary 7.6.4.**  $\dim_{\mathbb{H}}(E \times F) \geq \dim_{\mathbb{H}} E + \dim_{\mathbb{H}} F$  for compact  $E \subseteq \mathbb{R}^n$ ,  $F \subseteq \mathbb{R}^m$ .

*Proof.* Let  $t < \dim_{\mathbb{H}} E$  and  $u < \dim_{\mathbb{H}} F$ . By Theorem 7.6.2(ii) there are measures  $\mu$  on  $E$  and  $\nu$  on  $F$  with  $I_t(\mu), I_u(\nu) < \infty$ . By Proposition 7.6.3(ii),  $I_{t+u}(\mu \times \nu) < \infty$ , and  $\mu \times \nu$  is supported in  $E \times F$ . By Theorem 7.6.2(i),  $\dim_{\mathbb{H}}(E \times F) \geq t + u$ ; let  $t \uparrow \dim_{\mathbb{H}} E$  and  $u \uparrow \dim_{\mathbb{H}} F$ .  $\square$

**Where we are.** Corollary 7.6.4 is one inequality, and it is the only one that holds in general: there are compact  $E, F \subseteq \mathbb{R}$  with  $\dim_{\mathbb{H}} E = \dim_{\mathbb{H}} F = 0$  and  $\dim_{\mathbb{H}}(E \times F) = 1$ . The construction alternates the scales at which  $E$  and  $F$  are “active”, so that at every scale one factor is rich while the other is poor, and the product is rich throughout — the same mechanism as the failure of finite stability for  $\underline{\dim}_{\mathbb{M}}$  in Proposition 7.2.5.

The example is due to Besicovitch and Moran. Its moral is the one the whole volume keeps returning to: Hausdorff dimension is an infimum over covers with freely varying scales, and any quantity built by combining single-scale data can behave unexpectedly.

## 7.6.2 The Fourier expression

**Proposition 7.6.5.** Let  $0 < t < n$  and let  $\mu$  be a finite Borel measure on  $\mathbb{R}^n$  with compact support. Then

$$I_t(\mu) = c_{n,t} \int_{\mathbb{R}^n} |\widehat{\mu}(\xi)|^2 |\xi|^{t-n} d\xi,$$

where  $c_{n,t} > 0$  is the constant of Theorem 5.6.4, and both sides may be infinite simultaneously.

*Proof.* By Volume 2, the distributional identity  $\widehat{k}_t = c_{n,t}|\xi|^{t-n}$  holds for  $0 < t < n$  — this is Worked Example 2.5.2, restated by Theorem 5.6.4 as  $\widehat{Z}_t = Z_{n-t}$ . Since  $\mu$  is a compactly supported finite measure,  $\widehat{\mu}$  is a bounded continuous function, and Parseval's identity for tempered distributions (Theorem 2.5.11) gives

$$I_t(\mu) = \langle k_t * \mu, \mu \rangle = \langle \widehat{k}_t \widehat{\mu}, \widehat{\mu} \rangle = c_{n,t} \int |\widehat{\mu}(\xi)|^2 |\xi|^{t-n} d\xi,$$

the middle step because the Fourier transform turns convolution into multiplication and is unitary in the appropriate pairing. Both sides are integrals of nonnegative quantities, so the identity holds in  $[0, \infty]$ .  $\square$

**Corollary 7.6.6.** *If  $\mu$  is a nonzero compactly supported measure with  $|\widehat{\mu}(\xi)| \lesssim |\xi|^{-\beta}$  for some  $\beta > 0$ , then  $\dim_{\mathbb{H}}(\text{spt } \mu) \geq \min\{2\beta, n\}$ .*

*Proof.* By Proposition 7.6.5,  $I_t(\mu) \lesssim \int |\xi|^{-2\beta+t-n} d\xi$  over  $|\xi| \geq 1$ , plus a finite contribution from  $|\xi| \leq 1$  since  $\widehat{\mu}$  is bounded. By polar coordinates the integral converges when  $-2\beta + t - n < -n$ , that is  $t < 2\beta$ . So  $I_t(\mu) < \infty$  for every  $t < \min\{2\beta, n\}$ , and Theorem 7.6.2(i) applies.  $\square$

**WORKED EXAMPLE 7.6.1— TIER A The energy of the Cantor measure**

Let  $\mu$  be the natural measure on the middle-thirds Cantor set  $C$  and  $s = \log 2 / \log 3$ . Worked Example 7.1.1 gave  $\mu(I) \leq 6(\text{diam } I)^s$ , so by Step 2 of Theorem 7.6.2,  $I_t(\mu) < \infty$  for every  $t < s$ .

*Direct verification.* For  $x \in C$ ,  $\mu(B(x, 3^{-k})) = 2^{-k} = 3^{-ks}$  exactly, so

$$(k_t * \mu)(x) \approx \sum_{k \geq 0} 3^{kt} 3^{-ks} = \sum_{k \geq 0} 3^{k(t-s)},$$

convergent precisely for  $t < s$  and divergent at  $t = s$ . So the Cantor measure has finite  $t$ -energy exactly for  $t < s$ , and  $I_s(\mu) = \infty$  — the endpoint loss of the Checkpoint above, made explicit.

*Conclusion.*  $\dim_{\mathbb{H}} C \geq t$  for every  $t < s$ , hence  $\dim_{\mathbb{H}} C \geq s$ , recovering Worked Example 7.1.1 by a different route. The energy computation is longer here, but unlike the covering argument it will survive being pushed through a projection in Volume 8.

**WORKED EXAMPLE 7.6.2— TIER B Energy is stable under the operations covering is not**

The reason to prefer energy over the Frostman condition is that it passes through maps and convolutions. Three instances, all used in Volume 8.

(a) *Projections.* Let  $\pi_\theta(x) = x \cdot \theta$  be the orthogonal projection of  $\mathbb{R}^2$  onto the line in direction  $\theta$ . Then

$$\int_0^\pi I_t(\pi_{\theta\#}\mu) d\theta = \iint \left( \int_0^\pi \frac{d\theta}{|(x-y) \cdot \theta|^t} \right) d\mu d\mu \approx_t I_t(\mu) \quad (0 < t < 1),$$

because the inner integral is  $\approx |x-y|^{-t}$  — substituting  $u = (x-y)/|x-y|$  reduces it to a fixed convergent integral over  $\theta$ . So finite  $t$ -energy is inherited by *almost every* projection. That single computation is Marstrand's projection theorem, and it is impossible to see through the covering definition of dimension.

(b) *Convolutions and distance sets.* The distance set  $\Delta(E) = \{|x-y| : x, y \in E\}$  is studied through the pushforward of  $\mu \times \mu$  under  $(x, y) \mapsto |x-y|$ , whose energy is again computable from  $I_t(\mu)$ . Proposition 7.6.5 turns the computation into one about  $\widehat{\mu}$ , and the decay of  $\widehat{d\sigma}$  from Volume 5 enters. That is the Mattila integral of Volume 8.

(c) *The Fourier side.* Proposition 7.6.5 says the energy is a weighted  $L^2$  norm of  $\widehat{\mu}$ . Any estimate for  $\widehat{\mu}$  — for instance one coming from a restriction theorem — therefore becomes a dimension bound, by Corollary 7.6.6. This is the bridge between Volumes 6 and 8, and it does not exist at the level of coverings.

*The general point.* Hausdorff dimension is defined by an infimum over covers and is therefore hard to compute and harder to transport. Frostman converts it to the existence of a measure; energy converts that to the finiteness of an integral; and integrals can be estimated, exchanged, and Fourier transformed. Each step trades exactness for flexibility, and the flexibility is what the applications need.

**WORKED EXAMPLE 7.6.3— TIER C: RESEARCH TASTE    Where the energy method stops**

Corollary 7.6.6 bounds dimension below by twice the Fourier decay rate, and it is worth understanding why the factor 2 is there and why the method saturates.

(a) *The factor 2 is Plancherel.* The energy is quadratic in  $\mu$ , so a decay rate  $\beta$  for  $\widehat{\mu}$  contributes  $2\beta$  to the exponent. There is no way to improve this within the method: the energy is the only quantity for which Proposition 7.6.5 holds.

(b) *The bound is often far from sharp.* For the Cantor set,  $\widehat{\mu}$  does not decay at all —  $\dim_F C = 0$ , proved in Chapter 7.7 — so Corollary 7.6.6 gives nothing, while the true dimension is  $\log 2 / \log 3$ . The Fourier method sees only the

part of the dimension that is visible to the transform.

(c) *The saturation point is  $n/2$ .* Since  $\dim_{\mathbb{H}} \leq n$ , the bound  $2\beta$  is useful only for  $\beta < n/2$ ; and a measure on a set of dimension  $s$  cannot have  $\hat{\mu}$  decaying faster than  $|\xi|^{-s/2}$ , by Proposition 7.6.5 run backwards. So the method is sharp exactly for *Salem sets*, where  $\dim_{\mathbb{F}} = \dim_{\mathbb{H}}$ , and lossy otherwise.

(d) *What this predicts about Kakeya.* A Besicovitch set has  $\dim_{\mathbb{H}} = n$  conjecturally, so a measure on it with  $\hat{\mu}$  decaying like  $|\xi|^{-n/2}$  would prove the conjecture by Corollary 7.6.6. No such measure is known to exist, and the Fourier-analytic bound obtainable this way — Chapter 7.7 carries it out — gives only  $\dim_{\mathbb{H}} \geq (n+1)/2$ , the bush bound of Volume 1. So the energy method reproduces the easiest known bound and nothing beyond it. Every improvement in Volumes 9 to 14 is geometric or combinatorial, not Fourier-analytic, and this is the precise sense in which the Fourier method is exhausted.

**WORKED EXAMPLE 7.6.4— TIER B Estimating an energy: the dyadic annulus method**

Every energy computation in this series reduces to the same manoeuvre, used twice already in Theorem 7.6.2. It is worth isolating.

*The method.* To estimate  $(k_t * \mu)(x) = \int |x-y|^{-t} d\mu(y)$ , split the domain into the annuli  $A_j = \{2^{-j-1} < |x-y| \leq 2^{-j}\}$ ,  $j \geq 0$ , plus the region  $|x-y| > 1$ . On  $A_j$  the kernel is  $\approx 2^{jt}$  and the mass is  $\mu(A_j) \leq \mu(B(x, 2^{-j}))$ , giving

$$(k_t * \mu)(x) \lesssim \mu(\mathbb{R}^n) + \sum_{j \geq 0} 2^{jt} \mu(B(x, 2^{-j})).$$

Everything then depends on how  $\mu(B(x, r))$  behaves.

*Case 1: a Frostman measure of exponent  $s$ .* Then  $\mu(B(x, 2^{-j})) \leq 2^{-js}$  and the sum is  $\sum_j 2^{j(t-s)}$ : convergent for  $t < s$ , divergent for  $t \geq s$ . This is Step 2 of Theorem 7.6.2, and it gives finite energy with a bound *uniform in  $x$* , hence  $I_t(\mu) < \infty$ .

*Case 2: an Ahlfors–David regular measure,  $\mu(B(x, r)) \approx r^s$ .* Now the sum is  $\approx \sum_j 2^{j(t-s)}$  from both sides, so  $I_t(\mu) < \infty$  exactly for  $t < s$  and  $= \infty$  for  $t \geq s$ . The Cantor measure of Worked Example 7.6.1 is the model.

*Case 3: a measure with a heavier point.* If  $\mu$  has an atom, every  $\mu(B(x, r))$  is bounded below near that point and  $I_t(\mu) = \infty$  for every  $t > 0$ . So finite energy already forces  $\mu$  to be non-atomic — a small observation, but it is why one never has to exclude atoms by hand.



*The general shape.* A convergent geometric series appears exactly when the kernel's growth rate  $t$  is strictly less than the measure's decay rate  $s$ , and the endpoint always fails logarithmically. Recognising this pattern lets one read off the answer to most energy questions without computing: *compare the exponent in the kernel with the exponent in the measure, and the smaller wins, strictly.*

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## Drills

**Drill 7.6.1.** Verify the dyadic annulus estimate in Step 1 of Theorem 7.6.2 and check that the divergence is genuine.

**Drill 7.6.2.** Compute  $I_t(\mu)$  for  $\mu$  Lebesgue measure on the unit ball, and confirm it is finite exactly for  $t < n$ .

**Drill 7.6.3.** Carry out the inner-integral computation in Worked Example 7.6.2(a) and determine the constant's dependence on  $t$  as  $t \uparrow 1$ .

**Drill 7.6.4.** Show that Corollary 7.6.6 is sharp for the measure  $d\sigma$  on  $\mathbb{S}^{n-1}$ : compute  $\beta$  from Theorem 5.5.2 and compare  $2\beta$  with  $\dim_{\mathbb{H}} \mathbb{S}^{n-1} = n - 1$ .

## Notes

The energy characterisation of dimension is due to Frostman; the Fourier expression Proposition 7.6.5 is standard and is the starting point of Mattila's book, which is the reference for this chapter and for Volume 8.

The observation in Worked Example 7.6.3(d) — that the Fourier method gives exactly  $(n+1)/2$  for Kakeya — is due to Drury and to Christ–Duoandikoetxea–Rubio de Francia, and was the best known bound for over a decade.



# Chapter 7.7

## Fourier Dimension

Corollary 7.6.6 bounds Hausdorff dimension below by twice the decay rate of  $\widehat{\mu}$ . That suggests defining a second dimension from the decay alone, and comparing. The comparison is instructive: the two agree for some sets and differ maximally for others, and which case one is in decides whether Fourier analysis is any use.

### 7.7.1 Definition and the basic inequality

**Definition 7.7.1.** For a compact  $E \subseteq \mathbb{R}^n$ , the *Fourier dimension* is

$$\dim_{\mathbf{F}} E = \sup \left\{ \beta \in [0, n] : \exists \mu \neq 0 \text{ on } E \text{ with } |\widehat{\mu}(\xi)| \lesssim |\xi|^{-\beta/2} \right\},$$

the supremum of the empty set being 0.

The normalisation  $\beta/2$  is chosen so that Corollary 7.6.6 reads:

**Theorem 7.7.2.** For every compact  $E \subseteq \mathbb{R}^n$ ,

$$\dim_{\mathbf{F}} E \leq \dim_{\mathbf{H}} E.$$

*Proof.* Let  $\beta < \dim_{\mathbf{F}} E$  and take  $\mu \neq 0$  on  $E$  with  $|\widehat{\mu}(\xi)| \lesssim |\xi|^{-\beta/2}$ . By Corollary 7.6.6 with the decay exponent  $\beta/2$  in place of  $\beta$ ,  $\dim_{\mathbf{H}} E \geq \min\{\beta, n\} = \beta$ . Let  $\beta \uparrow \dim_{\mathbf{F}} E$ .  $\square$

**Definition 7.7.3.**  $E$  is a *Salem set* if  $\dim_{\mathbf{F}} E = \dim_{\mathbf{H}} E$ .

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WORKED EXAMPLE 7.7.1— TIER A **The sphere is Salem; a smooth surface piece need not be**

*The sphere.* Take  $E = \mathbb{S}^{n-1}$  and  $\mu = d\sigma$ . By Theorem 5.5.2,  $|\widehat{d\sigma}(\xi)| \lesssim |\xi|^{-(n-1)/2}$ , so  $\beta = n - 1$  is admissible and  $\dim_{\mathbf{F}} \mathbb{S}^{n-1} \geq n - 1$ . Since

$\dim_{\mathbb{H}} \mathbb{S}^{n-1} = n - 1$ , Theorem 7.7.2 gives equality: the sphere is Salem.

*A piece of a hyperplane is not.* Let  $E = [0, 1]^{n-1} \times \{0\}$ , so  $\dim_{\mathbb{H}} E = n - 1$ . Any measure on  $E$  has  $\widehat{\mu}(\xi)$  independent of  $\xi_n$  —  $\mu$  is a product with  $\delta_0$  in the last coordinate — so  $\widehat{\mu}$  does not decay along the  $\xi_n$  axis at all, and  $\dim_{\mathbb{F}} E = 0$ .

*The moral.* Fourier dimension measures how *spread out in direction* the set is, not merely how large. A flat piece is invisible to the transform in the normal direction, whatever its Hausdorff dimension. This is the same distinction as in Volume 5: curvature produces decay, flatness does not, and  $\dim_{\mathbb{F}}$  is the systematic version of that statement.

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**Theorem 7.7.4.** *The middle-thirds Cantor set  $C$  has  $\dim_{\mathbb{F}} C = 0$ , while  $\dim_{\mathbb{H}} C = \log 2 / \log 3$ .*

*Proof.* Let  $\mu$  be *any* nonzero finite measure supported in  $C$ . We show  $\widehat{\mu}(\xi) \not\rightarrow 0$  along a sequence, which gives  $\dim_{\mathbb{F}} C = 0$ .

The key is that  $C$  is invariant under  $x \mapsto 3x$  modulo the construction: more precisely,  $C \subseteq \{x : \{3^k x\} \in [0, \frac{1}{3}] \cup [\frac{2}{3}, 1] \ \forall k \geq 0\}$ , where  $\{\cdot\}$  is the fractional part, since the base-3 digits of points of  $C$  are all 0 or 2.

Consider  $\xi_k = 3^k$ . For  $x \in C$  write  $x = \sum_{j \geq 1} a_j 3^{-j}$  with  $a_j \in \{0, 2\}$ . Then

$$3^k x = \underbrace{\sum_{j \leq k} a_j 3^{k-j}}_{\text{an integer}} + \sum_{j > k} a_j 3^{k-j},$$

and the tail lies in  $[0, 1]$  with  $\sum_{j > k} a_j 3^{k-j} \leq 2 \sum_{i \geq 1} 3^{-i} = 1$ . So  $e^{2\pi i 3^k x} = e^{2\pi i \theta_k(x)}$  with  $\theta_k(x) = \sum_{j > k} a_j 3^{k-j} \in [0, 1]$ , and in fact  $\theta_k(x) \in [0, \frac{1}{3}] \cup [\frac{2}{3}, 1]$  since  $a_{k+1} \in \{0, 2\}$  contributes 0 or  $\frac{2}{3}$  and the remainder is at most  $\frac{1}{3}$ .

Therefore  $\operatorname{Re} e^{2\pi i 3^k x} = \cos(2\pi \theta_k(x)) \geq \cos(2\pi/3) = -\frac{1}{2}$ , and on the part where  $\theta_k \in [0, \frac{1}{3}]$  it is  $\geq -\frac{1}{2}$  as well, with  $\cos \geq \frac{1}{2}$  when  $\theta_k \in [0, \frac{1}{6}] \cup [\frac{5}{6}, 1]$ . A more careful count is unnecessary: the point is that  $e^{2\pi i 3^k x}$  takes values in a fixed arc of the circle of length less than the whole, uniformly in  $k$ , so

$$|\widehat{\mu}(3^k)| = \left| \int_C e^{-2\pi i 3^k x} d\mu(x) \right| \geq \left| \operatorname{Re} \int_C e^{-2\pi i 3^k x} d\mu \right| \geq c \mu(C)$$

for an absolute  $c > 0$ , since the integrand has real part bounded below by a positive constant on a fixed proportion of  $\mu$ -mass, and the estimate is uniform in  $k$  because the digit structure is.

Hence  $\widehat{\mu}(3^k) \not\rightarrow 0$ , so no decay exponent  $\beta > 0$  is admissible and  $\dim_{\mathbb{F}} C = 0$ . □

**Pitfall: Fourier dimension is not a dimension in the usual sense**

$\dim_F$  fails properties one expects. It is not countably stable, nor even finitely stable:  $\dim_F(E \cup F)$  can be 0 while  $\dim_F E$  is large, because a measure on the union may be forced to charge the bad part. It is not monotone in the naive way either — a subset can have larger Fourier dimension than the set, since fewer constraints bind.

It is also *not* determined by the set's geometry in any local sense: it is a statement about which measures the set carries, and Theorem 7.7.4 shows a set can be geometrically substantial and Fourier-analytically invisible.

The reader should treat  $\dim_F$  as a convenient name for “the best decay exponent available on this set” rather than as a dimension. Its only role in this series is Theorem 7.7.2, used in the direction  $\dim_H \geq \dim_F$ .

## 7.7.2 The Fourier bound for Besicovitch sets

*Remark 7.7.5* (Scope decision: Drury's bound is not reproved here). The energy method yields, for every Besicovitch set  $K \subseteq \mathbb{R}^n$ ,

$$\dim_H K \geq \frac{n+1}{2},$$

a theorem of Drury (1983) and, independently, Christ–Duoandikoetxea–Rubio de Francia. *We do not prove it in this series.*

The reason is that we have it already, by a shorter and entirely different argument: Volume 1, Theorem 1.7.15 proved exactly this bound by the bush argument, counting tube intersections, with no Fourier analysis and no Frostman measure. Proving it here would add a second proof of a known result at the cost of several pages of careful bookkeeping with dual slabs.

What is worth extracting from the Fourier route is not the bound but the *explanation of why  $(n+1)/2$  is where it stops*, and Worked Example 7.7.2 gives that without needing the full argument. Readers wanting the complete computation will find it in Mattila's book, Chapter 22.

Recorded as revision R7 in the master manifest.

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### WORKED EXAMPLE 7.7.2—TIER B Why $(n+1)/2$ , and why the Fourier method cannot pass it

The exponent can be predicted by matching two dimension counts, and the count also shows why no refinement of the method will improve it.

*The heuristic.* A measure  $\mu$  of dimension  $t$  has, by Proposition 7.6.5, finite  $I_t$ , and the borderline case corresponds to  $|\widehat{\mu}(\xi)| \approx |\xi|^{-t/2}$ . A Besicovitch set contains a unit segment in every direction, so it must in particular support the

directional spreading measured by  $d\sigma$ , whose transform decays like  $|\xi|^{-(n-1)/2}$  (Theorem 5.5.2).

*The dimension count.* A segment is 1-dimensional; the directions form an  $(n-1)$ -dimensional sphere. If the segments were entirely independent of their directions the union would have dimension  $1 + (n-1) = n$ . If instead each segment is determined by its direction — the opposite extreme — one gets only  $\max\{1, n-1\}$ . The Fourier method sees the average of these two behaviours, because  $I_t$  is a two-point average, and delivers the midpoint

$$1 + \frac{n-1}{2} = \frac{n+1}{2}.$$

*The bush picture, which makes it exact.* Volume 1’s proof of Theorem 1.7.15 is the same statement without averaging: a bush of  $\mu$  tubes through a common point occupies volume  $\gtrsim \mu\delta^{n-1}$ , because at unit distance from the centre the tubes have separated. Balancing that against the trivial bound gives  $\frac{n+1}{2}$  exactly.

*Why the method saturates there.* The bush argument uses only that many tubes pass through *one* point — a two-point condition, matching what  $I_t$  can see. Wolff’s hairbrush argument, in Volume 9, uses that many tubes pass through a *line*, which is a three-point condition, and gains  $\frac{n+2}{2}$ . No amount of care with the Fourier transform reaches it, for the reason set out in Worked Example 7.7.3(a): the transform of a measure encodes pair correlations and nothing more.

**Where we are.** Three sets, three verdicts, and they organise the chapter.

*The sphere* has  $\dim_F = \dim_H = n-1$ : curvature makes  $\widehat{d\sigma}$  decay at the maximal rate compatible with the dimension. Fourier methods are sharp.

*A hyperplane piece* has  $\dim_F = 0$  and  $\dim_H = n-1$ : no decay in the normal direction. Fourier methods see nothing.

*The Cantor set* has  $\dim_F = 0$  and  $\dim_H = \log 2 / \log 3$ : arithmetic self-similarity obstructs decay even though the set is curved in no direction at all — it is totally disconnected. Fourier methods again see nothing, for a different reason.

The pattern is that  $\dim_F$  measures a form of *directional and arithmetic spreading*, and  $\dim_H$  measures size. They agree only when the set is “generic” in the appropriate sense, and every deterministic construction in this series — Perron trees, Cantor sets, self-similar attractors — is arithmetic and therefore not generic.

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**WORKED EXAMPLE 7.7.3— TIER C: RESEARCH TASTE** **When is Fourier analysis the right tool for a dimension problem?**

This chapter and the last give a criterion, worth stating because it settles the question for the rest of the series.

(a) *The method sees only  $I_t$ , a two-point quantity.* By Proposition 7.6.5, everything the Fourier transform contributes is encoded in  $\iint |x - y|^{-t} d\mu d\mu$  — an average over *pairs* of points. Any feature of a set requiring three or more points to detect is invisible.

(b) *Consequently it is sharp exactly for Salem sets.* If  $\dim_F E = \dim_H E$  the method loses nothing; otherwise it loses exactly the gap. Random sets are typically Salem — Brownian images, random Cantor sets — and deterministic self-similar sets typically are not, by the mechanism of Theorem 7.7.4: exact self-similarity creates arithmetic structure, and arithmetic structure obstructs decay.

(c) *Besicovitch sets are not Salem, and this is why Volumes 9–14 exist.* The Fourier bound stops at  $\frac{n+1}{2}$  while the truth is conjectured to be  $n$ . The obstruction is (a): a Besicovitch set’s smallness is a statement about how *many tubes meet*, which is a three-point or higher condition, and the two-point function cannot see it.

(d) *Where Fourier analysis does remain essential.* Not forakeya, but for the distance set problem of Volume 8, where the quantity of interest — the pushforward of  $\mu \times \mu$  under  $(x, y) \mapsto |x - y|$  — is genuinely a two-point object. There the Mattila integral is the sharp tool and the combinatorial methods are the lossy ones. Knowing which regime a problem is in before choosing a method is most of the skill.

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**WORKED EXAMPLE 7.7.4— TIER B** **Detecting arithmetic structure through  $\widehat{\mu}$**

Theorem 7.7.4 showed  $\dim_F C = 0$  because  $\widehat{\mu}(3^k)$  does not tend to zero. The mechanism is general and worth naming, since it decides in practice whether a set is Salem.

*The principle.* If  $\text{spt } \mu$  is invariant — even approximately — under a map  $x \mapsto \lambda x$  with  $\lambda$  an integer, then the frequencies  $\lambda^k$  all “see” the same structure, and  $\widehat{\mu}(\lambda^k)$  cannot decay. Exact self-similarity with an integer ratio is the extreme case.

*Worked instance: the Cantor set with ratio  $1/4$ .* Let  $C_{1/4}$  be the attractor of  $f_1(x) = x/4$ ,  $f_2(x) = x/4 + 3/4$ , with  $\dim_H = \frac{1}{2}$  by Theorem 7.8.3. Points

have base-4 digits in  $\{0, 3\}$ , so for the natural measure

$$\widehat{\mu}(4^k) = \int e^{-2\pi i 4^k x} d\mu(x)$$

and  $4^k x$  has fractional part determined by the digits after position  $k$ , which again lie in  $\{0, 3\}$ . The same argument as Theorem 7.7.4 gives  $|\widehat{\mu}(4^k)| \geq c > 0$ , so  $\dim_F C_{1/4} = 0$  although  $\dim_H C_{1/4} = \frac{1}{2}$ .

*Contrast: a random Cantor set.* Build a Cantor set by keeping, at each stage and independently, a random one of the four quarters. The resulting set has  $\dim_H = \frac{1}{2}$  almost surely, and now the digits are independent random variables, so  $\widehat{\mu}(\xi)$  is a product of independent factors each of modulus  $< 1$ ; a second-moment computation gives  $|\widehat{\mu}(\xi)| \lesssim |\xi|^{-1/4+\varepsilon}$  almost surely, so the set is Salem. The computation is Salem's and is not carried out here (Chapter 7.7 scope note).

*The diagnostic.* Deterministic and arithmetically structured  $\Rightarrow$  probably  $\dim_F = 0$ , and Fourier methods will fail. Random or curved  $\Rightarrow$  probably Salem, and Fourier methods will be sharp. Spheres are curved and Salem (Worked Example 7.7.1); Cantor sets are arithmetic and are not. Besicovitch sets are neither, which is why the situation there (Worked Example 7.7.3) is genuinely intermediate and hard.

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## Drills

**Drill 7.7.1.** Show  $\dim_F E = 0$  for every  $E$  contained in a hyperplane, and more generally for every  $E$  contained in a smooth surface of dimension  $< n - 1$ .

**Drill 7.7.2.** Show that  $\dim_F$  is not finitely stable, by taking  $E$  a Salem set of dimension  $\frac{1}{2}$  in  $\mathbb{R}$  and  $F$  a point.

**Drill 7.7.3.** For a  $\delta$ -tube in direction  $\omega$ , show that the dual slab has dimensions  $\delta^{-1} \times \cdots \times \delta^{-1} \times 1$ , and that the measure of  $\{\omega : \xi \text{ lies in that slab}\} \approx \min\{1, (\delta|\xi|)^{-(n-1)}\}$ .

**Drill 7.7.4.** Using Worked Example 7.7.2, predict the exponent the same heuristic would give for a set containing a  $k$ -plane in every direction, and compare with the bush bound for such sets.



## Notes

Salem constructed sets with  $\dim_F = \dim_H$  in 1951, using random Cantor sets; Kahane gave the Brownian construction. That the middle-thirds Cantor set has  $\dim_F = 0$  is classical and is the model instance of arithmetic structure obstructing decay.

Drury's bound  $\frac{n+1}{2}$  (1983), obtained independently by Christ–Duong and Koetxea–Rubio de Francia, stood as the best for general  $n$  until Bourgain's and Wolff's work in the 1990s. Remark 7.7.5 explains why it is not reproved here.



## Chapter 7.8

# Self-Similar Sets and the Keakeya Upgrade

Two loose ends remain. The first is a class of sets whose dimension can be computed exactly, which is what makes the theory testable. The second is the promise of Worked Example 7.2.3: that a Keakeya *maximal* bound, which is a family of single-scale statements of the kind Volume 6 produces, upgrades to a bound on Hausdorff dimension.

### 7.8.1 Iterated function systems

**Definition 7.8.1.** A *similarity* of ratio  $r > 0$  is a map  $f : \mathbb{R}^n \rightarrow \mathbb{R}^n$  with  $|f(x) - f(y)| = r|x - y|$ . An *iterated function system* is a finite family  $f_1, \dots, f_m$  of similarities of ratios  $r_1, \dots, r_m \in (0, 1)$ ; by Worked Example 7.3.2 it has a unique nonempty compact *attractor*  $E$  with  $E = \bigcup_i f_i(E)$ .

**Definition 7.8.2.** The system satisfies the *open set condition* if there is a nonempty bounded open  $V$  with

$$\bigcup_{i=1}^m f_i(V) \subseteq V \quad \text{and} \quad f_i(V) \cap f_j(V) = \emptyset \text{ for } i \neq j.$$

The *similarity dimension* is the unique  $s \geq 0$  with  $\sum_{i=1}^m r_i^s = 1$ .

**Theorem 7.8.3** (Hutchinson). *If an iterated function system of similarities satisfies the open set condition with attractor  $E$  and similarity dimension  $s$ , then*

$$\dim_{\text{H}} E = \dim_{\text{M}} E = s, \quad 0 < \mathcal{H}^s(E) < \infty.$$

*Proof.* Write  $\mathcal{I}_k = \{1, \dots, m\}^k$  and, for  $\mathbf{i} = (i_1, \dots, i_k)$ , put  $f_{\mathbf{i}} = f_{i_1} \circ \dots \circ f_{i_k}$  and  $r_{\mathbf{i}} = r_{i_1} \cdots r_{i_k}$ . Iterating  $E = \bigcup_i f_i(E)$  gives  $E = \bigcup_{\mathbf{i} \in \mathcal{I}_k} f_{\mathbf{i}}(E)$  for every  $k$ .

**Step 1.** *Upper bound.*

Since  $f_{\mathbf{i}}$  is a similarity of ratio  $r_{\mathbf{i}}$ ,  $\text{diam } f_{\mathbf{i}}(E) = r_{\mathbf{i}} \text{diam } E$ . So

$$\sum_{\mathbf{i} \in \mathcal{I}_k} (\text{diam } f_{\mathbf{i}}(E))^s = (\text{diam } E)^s \sum_{\mathbf{i} \in \mathcal{I}_k} r_{\mathbf{i}}^s = (\text{diam } E)^s \left( \sum_i r_i^s \right)^k = (\text{diam } E)^s,$$

using the defining property of  $s$  and that the sum over  $\mathcal{I}_k$  factorises. Since  $\max_{\mathbf{i}} r_{\mathbf{i}} \leq (\max_i r_i)^k \rightarrow 0$ , these covers have diameters tending to 0, so  $\mathcal{H}^s(E) \leq (\text{diam } E)^s < \infty$  and  $\dim_{\text{H}} E \leq s$ . The same covers, refined so that all  $r_{\mathbf{i}}$  are comparable, give  $\overline{\dim}_{\text{M}} E \leq s$  by Drill 8.2.

**Step 2.** *A natural measure.*

Define  $\mu$  as the unique Borel probability measure with  $\mu = \sum_i r_i^s \mu \circ f_i^{-1}$ ; it exists by the same contraction argument as Worked Example 7.3.2, applied on the complete metric space of probability measures with the Wasserstein metric, or equivalently as the distribution of the random point  $\lim_k f_{i_1} \circ \cdots \circ f_{i_k}(x_0)$  with the  $i_j$  chosen independently with probabilities  $r_i^s$ . Iterating,

$$\mu(f_{\mathbf{i}}(E)) = r_{\mathbf{i}}^s \quad \text{for } \mathbf{i} \in \mathcal{I}_k. \quad (7.8.1)$$

**Step 3.** *The open set condition controls overlaps.*

Let  $V$  be as in the definition; we may assume  $\bar{V} \supseteq E$ , replacing  $V$  by a large ball if necessary and noting the conditions persist. The sets  $f_{\mathbf{i}}(V)$ ,  $\mathbf{i} \in \mathcal{I}_k$ , are pairwise disjoint, open, and of diameter  $r_{\mathbf{i}} \text{diam } V$ .

Fix  $x$  and  $0 < \rho < \text{diam } V$ . Let  $\mathcal{J}$  be the set of  $\mathbf{i}$  — of varying lengths — that are minimal subject to  $r_{\mathbf{i}} \leq \rho / \text{diam } V$ ; these index a disjoint family whose union covers  $E$ , and  $r_{\mathbf{i}} \geq r_{\min} \rho / \text{diam } V$  where  $r_{\min} = \min_i r_i$ .

The sets  $f_{\mathbf{i}}(V)$  for  $\mathbf{i} \in \mathcal{J}$  are disjoint, open, of diameter  $\approx \rho$ , and each contains a ball of radius  $c\rho$  where  $c$  depends on  $V$  and  $r_{\min}$  — because  $V$  contains a ball and similarities map balls to balls. A volume count therefore bounds the number of  $\mathbf{i} \in \mathcal{J}$  with  $f_{\mathbf{i}}(V) \cap B(x, \rho) \neq \emptyset$  by a constant  $N = N(n, V, r_{\min})$ .

**Step 4.** *The Frostman bound, and the lower bound.*

By Step 3 and (7.8.1),

$$\mu(B(x, \rho)) \leq \sum_{\substack{\mathbf{i} \in \mathcal{J} \\ f_{\mathbf{i}}(V) \cap B(x, \rho) \neq \emptyset}} \mu(f_{\mathbf{i}}(E)) = \sum r_{\mathbf{i}}^s \leq N \left( \frac{\rho}{\text{diam } V} \right)^s \lesssim \rho^s.$$

By Lemma 7.1.6,  $\mathcal{H}^s(E) \geq \mu(E)/C > 0$  and  $\dim_{\text{H}} E \geq s$ .

**Step 5.** *Collect.*

Steps 1 and 4 give  $\dim_{\text{H}} E = s$  and  $0 < \mathcal{H}^s(E) < \infty$ . Since  $\dim_{\text{H}} E \leq \overline{\dim}_{\text{M}} E \leq \dim_{\text{M}} E \leq s$  by Theorem 7.2.4 and Step 1, all three coincide.  $\square$

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 WORKED EXAMPLE 7.8.1— TIER A **The classical examples**

*Cantor set.*  $f_1(x) = x/3$ ,  $f_2(x) = x/3 + 2/3$  on  $\mathbb{R}$ , with  $V = (0, 1)$ . Then  $2 \cdot 3^{-s} = 1$  gives  $s = \log 2 / \log 3$ , recovering Worked Example 7.1.1 — and now with  $0 < \mathcal{H}^s(C) < \infty$ , which the covering argument alone did not give.

*Sierpiński gasket.* Three maps of ratio  $\frac{1}{2}$  fixing the vertices of a triangle, with  $V$  its interior. Then  $3 \cdot 2^{-s} = 1$ , so  $s = \log 3 / \log 2 \approx 1.585$ .

*von Koch curve.* Four maps of ratio  $\frac{1}{3}$ ;  $4 \cdot 3^{-s} = 1$  gives  $s = \log 4 / \log 3 \approx 1.262$ . The curve has infinite length ( $\mathcal{H}^1 = \infty$ ) and zero area, which the dimension records precisely.

*A non-example.* Take  $f_1(x) = x/2$ ,  $f_2(x) = x/2 + 1/4$  on  $\mathbb{R}$ . The similarity dimension solves  $2 \cdot 2^{-s} = 1$ , giving  $s = 1$ ; but the attractor is  $[0, 1/2]$ , which indeed has dimension 1. Here the open set condition fails —  $f_1((0, 1/2))$  and  $f_2((0, 1/2))$  overlap — and the formula happens to remain correct. Drill 8.3 gives a system where it fails.

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## 7.8.2 From maximal estimates to Hausdorff dimension

**Definition 7.8.4.** For  $0 < \delta < 1$  and  $s > 0$ , say that the *Keakeya maximal bound*  $K(s)$  holds if for every family  $\mathbb{T}$  of  $\delta$ -tubes with  $\delta$ -separated directions and every  $\varepsilon > 0$ ,

$$\left\| \sum_{T \in \mathbb{T}} \mathbf{1}_T \right\|_{L^p(\mathbb{R}^n)} \lesssim_{n, \varepsilon} \delta^{-\varepsilon}, \quad p = \frac{s}{s-1},$$

uniformly in  $\delta$ .

For  $s = n$  this is the Keakeya maximal conjecture in the form of Worked Example 6.6.2.

**Theorem 7.8.5.** *Suppose  $K(s)$  holds for some  $1 < s \leq n$ . Then every Besicovitch set  $K \subseteq \mathbb{R}^n$  has  $\dim_{\mathbb{H}} K \geq s$ .*

*Idea of the proof.* Suppose not, so  $\mathcal{H}^{s'}(K) = 0$  for some  $s' < s$  and there is a cover of  $K$  by balls with  $\sum r_i^{s'}$  arbitrarily small. The balls have many different radii; group them by dyadic size. For each direction, the unit segment of  $K$  in that direction is covered, so a definite proportion of its length is covered by balls of *some* single dyadic size — by pigeonholing over the  $\approx \log(1/\delta)$  available sizes. Apply the single-scale hypothesis at that size.

The loss from pigeonholing is a factor  $\log(1/\delta)$ , and the hypothesis carries  $\delta^{-\varepsilon}$  for every  $\varepsilon$ ; since  $\delta^{-\varepsilon}$  beats any power of a logarithm, the argument

closes. This is the mechanism promised in Worked Example 7.2.3(c).

*Proof.* Let  $s' < s$  and suppose  $\mathcal{H}^{s'}(K) = 0$ ; we derive a contradiction, which gives  $\dim_{\mathbb{H}} K \geq s'$  for every  $s' < s$  and hence the claim.

**Step 1.** *A cover, grouped by scale.*

Choose a cover  $\{B_i = B(x_i, r_i)\}$  of  $K$  with  $\sum_i r_i^{s'} < \eta$ , where  $\eta > 0$  will be chosen at the end; we may take all  $r_i < \frac{1}{100}$ . For  $k \geq 1$  let  $\mathcal{B}_k = \{i : 2^{-k-1} < r_i \leq 2^{-k}\}$  and  $N_k = \#\mathcal{B}_k$ . Then

$$\sum_{k \geq 1} N_k 2^{-(k+1)s'} \leq \sum_i r_i^{s'} < \eta. \quad (7.8.2)$$

**Step 2.** *Pigeonhole over scales for each direction.*

Let  $\omega \in \mathbb{S}^{n-1}$  and  $\ell_\omega \subseteq K$  a unit segment in direction  $\omega$ . The balls cover  $\ell_\omega$ , so  $\sum_k \mathcal{H}^1(\ell_\omega \cap \bigcup_{i \in \mathcal{B}_k} B_i) \geq 1$ . Choose weights  $c_k = c/k^2$  with  $\sum_k c_k = 1$ ; then there is  $k = k(\omega)$  with

$$\mathcal{H}^1\left(\ell_\omega \cap \bigcup_{i \in \mathcal{B}_k} B_i\right) \geq \frac{c}{k^2}. \quad (7.8.3)$$

Partition  $\mathbb{S}^{n-1}$  into the sets  $\Omega_k = \{\omega : k(\omega) = k\}$ ; some  $\Omega_k$  has  $\sigma(\Omega_k) \geq c_k \sigma(\mathbb{S}^{n-1})$ , again by weighting. Fix that  $k$  and write  $\delta = 2^{-k}$ .

**Step 3.** *Build a tube family.*

Choose a maximal  $\delta$ -separated set of directions in  $\Omega_k$ ; since  $\sigma(\Omega_k) \gtrsim k^{-2}$ , there are  $M \gtrsim k^{-2} \delta^{-(n-1)}$  of them. For each such  $\omega$ , let  $T_\omega$  be the  $\delta$ -tube around  $\ell_\omega$ . By (7.8.3), the balls of  $\mathcal{B}_k$  — each of radius  $\approx \delta$  — cover a proportion  $\gtrsim k^{-2}$  of  $\ell_\omega$ , hence

$$m\left(T_\omega \cap \bigcup_{i \in \mathcal{B}_k} 2B_i\right) \gtrsim \frac{1}{k^2} m(T_\omega) \approx \frac{\delta^{n-1}}{k^2}.$$

**Step 4.** *Apply the hypothesis.*

Let  $U = \bigcup_{i \in \mathcal{B}_k} 2B_i$ , so  $m(U) \leq N_k C_n \delta^n$ . With  $F = \sum_\omega \mathbf{1}_{T_\omega}$  over the  $M$  chosen directions, Step 3 gives

$$\int_U F = \sum_\omega m(T_\omega \cap U) \gtrsim M \frac{\delta^{n-1}}{k^2} \gtrsim \frac{\delta^{-(n-1)}}{k^2} \cdot \frac{\delta^{n-1}}{k^2} = \frac{1}{k^4}.$$

On the other hand, by Hölder with exponents  $p = \frac{s}{s-1}$  and  $p' = s$ , and the hypothesis  $\mathbf{K}(s)$ ,

$$\int_U F \leq m(U)^{1/s} \|F\|_{L^p} \lesssim_\varepsilon (N_k \delta^n)^{1/s} \delta^{-\varepsilon}.$$

**Step 5.** *Compare, and use the budget.*

Combining,

$$\frac{1}{k^4} \lesssim_\varepsilon (N_k \delta^n)^{1/s} \delta^{-\varepsilon}, \quad \text{so} \quad N_k \gtrsim_\varepsilon \frac{\delta^{-n} \delta^{s\varepsilon}}{k^{4s}}.$$

Insert this into (7.8.2), using  $\delta = 2^{-k}$  and keeping only the  $k$ -th term:

$$\eta > N_k 2^{-(k+1)s'} \gtrsim_\varepsilon \frac{2^{kn} 2^{-ks\varepsilon}}{k^{4s}} 2^{-(k+1)s'} \gtrsim_\varepsilon \frac{2^{k(n-s'-s\varepsilon)}}{k^{4s}}.$$

Choose  $\varepsilon$  with  $s\varepsilon < n - s'$ , which is possible since  $s' < s \leq n$  gives  $n - s' > 0$ . Then the exponent  $n - s' - s\varepsilon$  is positive, so the right-hand side tends to  $\infty$  with  $k$ ; and  $k \rightarrow \infty$  as  $\eta \rightarrow 0$ , since a cover with  $\sum r_i^{s'} < \eta$  forces all  $r_i$  small. Taking  $\eta$  small enough gives a contradiction.  $\square$

**Where we are.** The upgrade is complete: a family of single-scale estimates, each with an arbitrarily small power loss, yields a Hausdorff dimension bound. Combined with Theorem 6.6.2, the chain now runs

restriction conjecture  $\implies$  Kakeya maximal conjecture  $\implies \dim_{\mathbb{H}} K = n$  for every Besicovitch set  $K$ .

the first implication from Volume 6 and the second from Theorem 7.8.5 with  $s = n$ .

Two features of the proof deserve emphasis. The  $\delta^{-\varepsilon}$  in the hypothesis is essential: Step 5 needs  $s\varepsilon < n - s'$ , and with a fixed power loss  $\delta^{-c}$  one obtains only  $\dim_{\mathbb{H}} K \geq n - sc$ . And the two pigeonholings — over scales, and over directions — each cost a factor  $k^{-2}$ , that is a power of  $\log(1/\delta)$ , which the  $\delta^{-\varepsilon}$  absorbs. This is exactly the bookkeeping of Worked Example 6.5.3.

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**WORKED EXAMPLE 7.8.2— TIER B** **Besicovitch sets are not self-similar, and what replaces the formula**

Theorem 7.8.3 computes dimensions exactly. It is worth being clear why it says nothing about Kakeya.

*No Besicovitch set is the attractor of a similarity IFS with the open set condition.* If it were, Theorem 7.8.3 would give  $0 < \mathcal{H}^s(K) < \infty$  with  $s = \dim_{\mathbb{H}} K$ . But a Besicovitch set can have  $m(K) = 0$  while  $\dim_{\mathbb{H}} K = n$  is conjectured, and by Proposition 7.1.8 that forces  $\mathcal{H}^n(K) = 0$  — incompatible with  $\mathcal{H}^s(K) > 0$  at  $s = n$ . So either the conjecture fails or such a set is not self-similar, and the conjecture is believed.

*More directly: self-similar sets are homogeneous, Besicovitch sets are not.* The attractor looks the same at every point and scale, up to the finitely

many maps. A Besicovitch set is built by translating triangles at each stage of a Perron construction, and the translations differ from stage to stage; the set has no exact scaling symmetry.

*What replaces the dimension formula.* Nothing exact. Every bound on  $\dim_{\mathbb{H}} K$  in this series is an inequality, obtained by a counting argument at a single scale and upgraded by Theorem 7.8.5. The absence of a formula is not a gap in the theory — it is the reason the problem is open.

*A useful contrast.* Over a finite field the analogous problem *does* have an exact answer: Dvir’s theorem, in Volume 10, shows aakeya set in  $\mathbb{F}_q^n$  has  $\gtrsim q^n$  points, with no loss at all. The polynomial method supplies there what self-similarity supplies for fractals: exact algebraic structure. Volume 10, Chapter 8 explains why the transfer to  $\mathbb{R}^n$  fails.

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**WORKED EXAMPLE 7.8.3— TIER C: RESEARCH TASTE** **What Volume 7 has and has not delivered**

(a) *The language.* Hausdorff dimension is now defined, its elementary theory established, and theakeya conjecture stated in its proper form (Conjecture 7.1.11). The comparison with Minkowski dimension (Theorem 7.2.4) quantifies exactly what Volume 6’s single-scale arguments were missing.

(b) *The tools.* Frostman’s lemma (Theorem 7.5.1) converts a dimension hypothesis into a measure; energy (Theorem 7.6.2) converts that measure into an integral; Proposition 7.6.5 converts the integral into a statement about  $\hat{\mu}$ . These three steps are used in every dimension estimate of Volume 8.

(c) *The upgrade.* Theorem 7.8.5 closes the gap Volume 6 left, so that all subsequent work may be done at a single scale.

(d) *What is deliberately absent.* The Besicovitch covering theorem (Remark 7.4.2); the measure-zero Besicovitch set, deferred to Volume 8 where Marstrand’s theorem makes Kahane’s construction available (Chapter 7.3); Salem set constructions (Chapter 7.7); and Drury’s Fourier proof of the  $(n + 1)/2$  bound, which Volume 1 already established by the bush (Remark 7.7.5).

(e) *What this volume does not do.* It proves no new bound on the dimension of a Besicovitch set. Everything here is machinery, and the machinery is two-point — as Worked Example 7.7.3 made precise. The improvements begin in Volume 9, and every one of them is a statement about configurations of three or more tubes.

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**WORKED EXAMPLE 7.8.4— TIER B Reading a dimension bound in the literature**

Papers on *Keakeya* state bounds in several inequivalent ways, and the differences matter. This volume now supplies the dictionary; here it is, applied to the four formulations one meets.

(a) “*Every Besicovitch set in  $\mathbb{R}^n$  has  $\dim_{\mathrm{H}} \geq s$ .*” The Hausdorff statement, the strongest. By Theorem 7.2.4 it implies the Minkowski version.

(b) “*Every Besicovitch set has  $\dim_{\mathrm{M}} \geq s$ .*” Weaker. Historically these came first, often by several years, because a single-scale argument suffices — compare Corollary 6.6.3.

(c) “*The Keakeya maximal operator satisfies  $\|\mathcal{K}_{\delta}f\|_{L^q(\mathbb{S}^{n-1})} \lesssim_{\varepsilon} \delta^{-\varepsilon}\|f\|_{L^p}$ .*” A single-scale family. By Theorem 7.8.5 it implies (a) with the corresponding  $s$ , *provided* the loss is  $\delta^{-\varepsilon}$  for every  $\varepsilon$ . With a fixed loss  $\delta^{-c}$  it implies only a weaker  $s$ , by Drill 8.4 — a distinction routinely elided in surveys.

(d) “ *$m(K_{\delta}) \gtrsim \delta^{n-s}$  for every Besicovitch set.*” Equivalent to (b), by Proposition 7.2.3.

*Worked instance: the bush bound.* Volume 1, Theorem 1.7.15 proves the maximal-operator form (c) with  $s = \frac{n+1}{2}$ . Via Theorem 7.8.5 this gives (a): every Besicovitch set has  $\dim_{\mathrm{H}} \geq \frac{n+1}{2}$ . That is the same conclusion Drury obtained by Fourier analysis (Remark 7.7.5), and the two routes are genuinely independent.

*What to check when reading.* Which dimension; whether the loss is  $\delta^{-\varepsilon}$  or a fixed power; and whether the statement is for all Besicovitch sets or for a restricted class such as the sticky ones of Volume 14. Three questions, and the answers change the strength of the result substantially.

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## Drills

**Drill 7.8.1.** Verify that the sum over  $\mathcal{I}_k$  factorises as  $(\sum_i r_i^s)^k$ , as used in Step 1 of Theorem 7.8.3.

**Drill 7.8.2.** Show that refining the covers in Step 1 so that all  $r_i$  lie in  $[\rho r_{\min}, \rho]$  gives  $\overline{\dim}_{\mathrm{M}} E \leq s$ .

**Drill 7.8.3.** Exhibit an IFS of similarities for which the open set condition fails and  $\dim_{\mathrm{H}} E$  is strictly less than the similarity dimension.

**Drill 7.8.4.** Track the constants in Theorem 7.8.5 to show that  $K(s)$  with a fixed loss  $\delta^{-c}$  in place of  $\delta^{-\varepsilon}$  gives only  $\dim_{\mathrm{H}} K \geq n - sc$ .

**Where we are.** Volume 7 is machinery, and it is worth stating exactly what has been added to the series' capability.

Before this volume, a dimension statement could be made only about Minkowski dimension, and only at a single scale. After it:  $\dim_{\text{H}}$  is defined and its elementary theory available (Chapter 7.1); the gap to  $\dim_{\text{M}}$  is quantified (Theorem 7.2.4); a dimension hypothesis can be converted into a measure (Theorem 7.5.1), the measure into an integral (Theorem 7.6.2), and the integral into a Fourier statement (Proposition 7.6.5); and a family of single-scale bounds upgrades to a Hausdorff bound (Theorem 7.8.5).

That last theorem is what licenses the practice of the rest of the series. Volumes 9 through 14 prove estimates for  $\delta$ -tube families at a fixed  $\delta$ , with losses  $\delta^{-\varepsilon}$ , and never mention Hausdorff measure again. They are entitled to do so because of Theorem 7.8.5, and a reader who wants to know what a tube-counting estimate says about dimension should return here.

## Notes

Theorem 7.8.3 is Hutchinson's (1981), building on Moran (1946). The open set condition is Moran's; that it can be replaced by the weaker *strong separation* or by conditions permitting some overlap is a substantial literature not needed here.

Theorem 7.8.5 is standard and appears in essentially this form in Bourgain's and Wolff's papers; the pigeonholing over scales is the step that makes the  $\delta^{-\varepsilon}$  hypothesis indispensable, and is the reason all Keakeya maximal statements in the literature carry it.



# Chapter 7.A

## Results Quoted from Volumes 1–6

| Result                               | Source               | Used in                                    |
|--------------------------------------|----------------------|--------------------------------------------|
| Outer measure from a covering family | Proposition 1.2.3    | Proposition 7.1.2                          |
| Metric outer measures                | Theorem 1.2.13       | Proposition 7.1.2                          |
| Hausdorff measure, constructed       | Worked Example 1.2.4 | Definition 7.1.1                           |
| Continuity of measures               | Proposition 1.2.6    | Proposition 7.3.6                          |
| Outer regularity                     | Theorem 1.3.6        | Proposition 7.1.8, Theorem 7.4.4           |
| Carathéodory extension               | Theorem 1.2.15       | Worked Example 7.1.1                       |
| Egorov                               | Theorem 1.5.9        | Theorem 7.6.2                              |
| Vitali covering theorem              | Theorem 1.7.7        | Theorem 7.4.3                              |
| Riesz representation                 | Theorem 1.8.15       | Theorem 7.5.1                              |
| The Perron tree                      | Theorem 1.3.12       | Chapter 7.3                                |
| The bush bound                       | Theorem 1.7.15       | Remark 7.7.5, Worked Example 7.7.2         |
| $\mathcal{F}$ on $\mathcal{S}'$      | Theorem 2.5.11       | Proposition 7.6.5                          |
| Transform of $ x ^{-a}$              | Worked Example 2.5.2 | Proposition 7.6.5                          |
| The Riesz reflection formula         | Theorem 5.6.4        | Proposition 7.6.5                          |
| Asymptotics of $\widehat{d\sigma}$   | Theorem 5.5.2        | Worked Example 7.7.1, Worked Example 7.7.2 |
| Takeya maximal, tube form            | Worked Example 6.6.2 | Theorem 7.8.5                              |
| Restriction implies Takeya           | Theorem 6.6.2        | Chapter 7.8 Checkpoint                     |
| Besicovitch sets, Minkowski bound    | Corollary 6.6.3      | Worked Example 7.2.3                       |
| Epsilon bookkeeping                  | Worked Example 6.5.3 | Theorem 7.8.5                              |

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## Debts

Volume 1 left two results to this volume. Neither is proved here, and the dispositions differ:

*The Besicovitch covering theorem* is removed from the series (Remark 7.4.2, revision R6). Nothing depends on it; the  $5r$ -lemma (Theorem 7.4.1) serves every purpose here.

*A Besicovitch set of measure zero* is deferred to Volume 8, where Marstrand's projection theorem makes Kahane's construction available. The Pitfall in Chapter 7.3 explains why the compactness argument one expects to work does not.

This volume incurs one further omission of its own: Drury's Fourier-analytic proof of  $\dim_{\mathbb{H}} K \geq \frac{n+1}{2}$  (Remark 7.7.5, revision R7), the same bound having been proved in Volume 1 by the bush argument.



## Chapter 7.B

# Notation and Index of Results

### Additions to the Volume 1–6 conventions

|                                                                  |                                                                      |
|------------------------------------------------------------------|----------------------------------------------------------------------|
| $\mathcal{H}^s, \mathcal{H}_\varepsilon^s, \mathcal{H}_\infty^s$ | Hausdorff measure, its $\varepsilon$ -approximation, and the content |
| $\dim_H, \dim_M, \dim_F$                                         | Hausdorff, Minkowski and Fourier dimension                           |
| $N(E, \delta)$                                                   | covering number at scale $\delta$                                    |
| $d_H, \mathcal{K}(B)$                                            | Hausdorff metric; compact subsets of $B$                             |
| $I_t(\mu)$                                                       | the $t$ -energy of $\mu$                                             |
| $\text{spt } \mu$                                                | the support of $\mu$                                                 |
| $K(s)$                                                           | the Kakeya maximal bound at exponent $s$                             |

## Index of the main results

|                                                                                          |                   |
|------------------------------------------------------------------------------------------|-------------------|
| $\mathcal{H}^s$ is a metric outer measure                                                | Proposition 7.1.2 |
| The critical exponent                                                                    | Theorem 7.1.3     |
| Elementary properties of $\dim_{\mathbf{H}}$                                             | Proposition 7.1.5 |
| Mass distribution principle                                                              | Lemma 7.1.6       |
| $\mathcal{H}^n \approx$ Lebesgue                                                         | Proposition 7.1.8 |
| Box-counting definitions agree                                                           | Proposition 7.2.2 |
| $\dim_{\mathbf{H}} \leq \underline{\dim}_{\mathbf{M}} \leq \overline{\dim}_{\mathbf{M}}$ | Theorem 7.2.4     |
| Completeness of the Hausdorff metric                                                     | Theorem 7.3.3     |
| Blaschke selection                                                                       | Theorem 7.3.4     |
| Upper semicontinuity of measure                                                          | Proposition 7.3.6 |
| The $5r$ -covering lemma                                                                 | Theorem 7.4.1     |
| Vitali for a Radon measure                                                               | Theorem 7.4.3     |
| Density theorem for $\mathcal{H}^s$                                                      | Theorem 7.4.4     |
| Frostman's lemma                                                                         | Theorem 7.5.1     |
| Energy characterises dimension                                                           | Theorem 7.6.2     |
| The Fourier expression of energy                                                         | Proposition 7.6.5 |
| Decay implies dimension                                                                  | Corollary 7.6.6   |
| $\dim_{\mathbf{F}} \leq \dim_{\mathbf{H}}$                                               | Theorem 7.7.2     |
| $\dim_{\mathbf{F}}$ of the Cantor set is 0                                               | Theorem 7.7.4     |
| Hutchinson's dimension formula                                                           | Theorem 7.8.3     |
| Maximal bounds give Hausdorff bounds                                                     | Theorem 7.8.5     |

## Plan revisions in this volume

| Removed                               | Reason                                                                                                                                        |
|---------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| Besicovitch covering theorem          | the naive angle argument is false; the correct proof splits by radius ratio and is several pages. Nothing needs it. Remark 7.4.2, revision R6 |
| Drury's Fourier bound $\frac{n+1}{2}$ | the same bound is proved in Volume 1 by the bush argument. Remark 7.7.5, revision R7                                                          |
| Salem set constructions               | need random measures; unused                                                                                                                  |
| Measure-zero Besicovitch set          | <i>deferred</i> , not removed: Volume 8, via Marstrand and Kahane                                                                             |



# Part 8

## Volume 8

Projections, Distance Sets, and Besicovitch Sets



# Preface to Volume 8

## Dimension under maps

Volume 7 built the apparatus for measuring a set: Hausdorff dimension, Frostman measures, energy integrals, and the Fourier expression of energy. This volume uses it, on two questions that have driven the subject and that are the natural companions of the Kakeya problem.

The first is what happens to dimension under *projection*. If  $E$  lies in the plane and one shadows it onto a line, the shadow can only be smaller — but by how much, and for how many directions? Marstrand’s answer, that almost every shadow is as large as it can be, is the model for every “generic” statement in the subject.

The second is the *distance set*  $\Delta(E) = \{|x - y| : x, y \in E\}$ . Falconer asked how large  $E$  must be for  $\Delta(E)$  to have positive measure, conjectured the threshold  $\dim_{\mathbb{H}} E > n/2$ , and proved  $\frac{n+1}{2}$ . That proof is the centrepiece of this volume, and it is a direct application of two things already built: the decay of  $\widehat{d\sigma}$  from Volume 5, and the Fourier expression of energy from Volume 7.

## A note on the route to Falconer’s theorem

Readers who consult the literature will find two approaches. Falconer’s original argument is an  $L^\infty$  estimate: one shows the distance measure has a *bounded density*. Mattila’s later approach is an  $L^2$  estimate, reducing the problem to spherical averages of  $\widehat{\mu}$ , and it is the framework in which every subsequent improvement has been made.

This volume proves Falconer’s theorem by the first route, in Chapter 8.5, and introduces the second in Chapter 8.7 as the tool for the improvements. The distinction is not cosmetic: *the Mattila integral does not converge at the exponent  $\frac{n+1}{2}$*  — the pointwise decay of  $\widehat{d\sigma}$  alone never makes it converge, in any dimension — so a reader who tries to prove Falconer’s theorem by that route will fail, and should know in advance that the failure is structural rather

than a missing idea.

## What is assumed

Volumes 1–7. Most heavily: Frostman’s lemma and the energy machinery of Volume 7, Chapters 5 and 6; the asymptotics of  $\widehat{d\sigma}$  from Volume 5, Chapter 5; and the Fourier transform of measures from Volume 2.

## Scope, fixed in advance

Following the practice of Volumes 4 through 7, the list of results was fixed before writing. Three items are handled by announcement rather than proof, each recorded in the manifest:

- the existence of a Besicovitch set of *measure zero*, deferred here from Volume 7. Chapter 8.3 explains why the natural construction needs the Besicovitch projection theorem, which is outside the scope of this series, and records the decision;
- the improvements on Falconer’s exponent — Wolff’s  $\frac{4}{3}$  in the plane, Erdoğan’s  $\frac{n}{2} + \frac{1}{3}$ , and the current record  $\frac{5}{4}$  of Guth, Iosevich, Ou and Wang. Chapter 8.7 states them, explains what each contributes, and proves the reduction they all rest on;
- distance sets over finite fields, which need a separate Fourier analysis on  $\mathbb{F}_q^d$  and are not used later here.

## How the two halves of the volume fit

The volume divides cleanly. Chapter 8.1 to Chapter 8.3 concern *projections*: what a shadow of a set can look like, how many directions can be exceptional, and what the answers say about product sets and Besicovitch sets. Chapter 8.4 to Chapter 8.8 concern *distances*.

The two halves share a method and differ in what they can achieve with it. Both proceed by producing a Frostman measure, converting a dimension hypothesis into an energy bound, and passing to the Fourier side. In the projection problem that suffices completely: Theorem 8.1.2 is sharp, and Theorem 8.2.1 bounds the exceptional set optimally in its own terms. In the distance problem it does not: Theorem 8.5.1 falls short of the conjectured exponent by exactly  $\frac{1}{2}$ , and closing the gap has required, over forty years, the import of successively stronger restriction estimates.

The reason for the difference is worth stating at the outset. A projection is a *linear* map, and the transversality that drives Lemma 8.1.3 is exact: a linear functional vanishes to first order and no better. A distance is a *quadratic* function, its level sets are spheres, and the relevant transversality is the curvature of those spheres — which enters through  $\widehat{d\sigma}$  and is quantitatively weaker than one would like. So the projection theory is complete and the distance theory is not, and the discrepancy traces back to the difference between a hyperplane and a sphere.

## The plan

|   |                                   |                                            |
|---|-----------------------------------|--------------------------------------------|
| 1 | Projections                       | Marstrand's theorem, by Kaufman's argument |
| 2 | Exceptional sets                  | how many directions can be bad             |
| 3 | Product sets and Besicovitch sets | and a scope decision                       |
| 4 | The distance measure              | its Fourier representation                 |
| 5 | Falconer's theorem                | the $L^\infty$ route                       |
| 6 | Interiors                         | Mattila–Sjölin                             |
| 7 | The Mattila integral              | where the improvements come from           |
| 8 | The $n/2$ barrier                 | what would be needed                       |



# Chapter 8.1

## Projections

### 8.1.1 The question

**Definition 8.1.1.** For  $\theta \in [0, \pi)$  let  $e_\theta = (\cos \theta, \sin \theta)$  and let  $\pi_\theta : \mathbb{R}^2 \rightarrow \mathbb{R}$  be the orthogonal projection  $\pi_\theta(x) = x \cdot e_\theta$ .

Since  $\pi_\theta$  is Lipschitz with constant 1, Proposition 7.1.5(iii) gives

$$\dim_{\text{H}} \pi_\theta(E) \leq \min \{ \dim_{\text{H}} E, 1 \}$$

for every  $E \subseteq \mathbb{R}^2$  and every  $\theta$ , the second bound because the image lies in a line. The question is whether equality holds, and the answer is that it holds for almost every  $\theta$  but not for all: the projection of a line segment onto the perpendicular direction is a point.

**Theorem 8.1.2** (Marstrand). *Let  $E \subseteq \mathbb{R}^2$  be compact.*

(i) *For Lebesgue-almost every  $\theta \in [0, \pi)$ ,*

$$\dim_{\text{H}} \pi_\theta(E) = \min \{ \dim_{\text{H}} E, 1 \}.$$

(ii) *If  $\dim_{\text{H}} E > 1$  then  $m(\pi_\theta(E)) > 0$  for almost every  $\theta$ .*

*Idea of the proof.* Both halves are one computation, performed twice with different exponents.

Take a measure  $\mu$  on  $E$  of finite  $t$ -energy, supplied by Theorem 7.6.2. Its projection  $\pi_{\theta\#}\mu$  is a measure on the line, and its energy is

$$I_t(\pi_{\theta\#}\mu) = \iint \frac{d\mu(x) d\mu(y)}{|(x - y) \cdot e_\theta|^t}.$$

Averaging over  $\theta$  and exchanging the order of integration — which is legit-

imate, everything being nonnegative — the inner integral becomes

$$\int_0^\pi \frac{d\theta}{|(x-y) \cdot e_\theta|^t} = \frac{1}{|x-y|^t} \int_0^\pi \frac{d\theta}{|u \cdot e_\theta|^t}, \quad u = \frac{x-y}{|x-y|},$$

and the remaining integral is a *constant*, finite exactly when  $t < 1$ , because  $u \cdot e_\theta$  vanishes to first order in  $\theta$ . So the average of  $I_t(\pi_{\theta\#}\mu)$  over  $\theta$  is a constant times  $I_t(\mu)$ , and if the latter is finite then so is the former for almost every  $\theta$ .

Part (ii) is the same with the Fourier expression instead of the energy: one computes  $\int_0^\pi \int_{\mathbb{R}} |\widehat{\pi_{\theta\#}\mu}(r)|^2 dr d\theta$  and recognises it, after polar coordinates, as  $I_1(\mu)$ .

**Lemma 8.1.3** (The angular integral). *For  $0 < t < 1$  and any unit vector  $u \in \mathbb{S}^1$ ,*

$$c_t := \int_0^\pi \frac{d\theta}{|u \cdot e_\theta|^t} = \int_0^\pi \frac{d\varphi}{|\cos \varphi|^t} < \infty,$$

*independent of  $u$ ; and  $c_t = \infty$  for  $t \geq 1$ .*

*Proof.* Write  $u = e_\alpha$ ; then  $u \cdot e_\theta = \cos(\theta - \alpha)$ , and the substitution  $\varphi = \theta - \alpha$  together with  $\pi$ -periodicity of  $|\cos|$  gives the stated identity, independent of  $\alpha$  and hence of  $u$ .

For convergence,  $|\cos \varphi|$  vanishes only at  $\varphi = \pi/2$ , where  $|\cos \varphi| = |\sin(\varphi - \pi/2)| \approx |\varphi - \pi/2|$ . So near that point the integrand is  $\approx |\varphi - \pi/2|^{-t}$ , integrable exactly for  $t < 1$ .  $\square$

*Proof of Theorem 8.1.2. Step 1. Part (i), lower bound.*

Let  $t < \min\{\dim_{\mathbb{H}} E, 1\}$ . By Theorem 7.6.2(ii) there is a nonzero finite  $\mu$  on  $E$  with  $I_t(\mu) < \infty$ . For each  $\theta$ , the projected measure  $\pi_{\theta\#}\mu$  is a nonzero finite measure on  $\pi_\theta(E)$ , and by the change-of-variables formula for pushforwards,

$$I_t(\pi_{\theta\#}\mu) = \iint \frac{d\mu(x) d\mu(y)}{|(x-y) \cdot e_\theta|^t}.$$

Integrating in  $\theta$  and applying Tonelli — the integrand is nonnegative and jointly measurable — and then Lemma 8.1.3,

$$\int_0^\pi I_t(\pi_{\theta\#}\mu) d\theta = \iint \left( \int_0^\pi \frac{d\theta}{|(x-y) \cdot e_\theta|^t} \right) d\mu d\mu = c_t I_t(\mu) < \infty. \quad (8.1.1)$$

A nonnegative function with finite integral is finite almost everywhere, so  $I_t(\pi_{\theta\#}\mu) < \infty$  for a.e.  $\theta$ , and Theorem 7.6.2(i) gives  $\dim_{\mathbb{H}} \pi_\theta(E) \geq t$  for those  $\theta$ .



Taking a sequence  $t_k \uparrow \min\{\dim_{\mathbb{H}} E, 1\}$  and intersecting the countably many full-measure sets gives the lower bound for a.e.  $\theta$ ; the upper bound was noted above.

**Step 2. Part (ii).**

Assume  $\dim_{\mathbb{H}} E > 1$  and choose  $\mu$  on  $E$  with  $I_1(\mu) < \infty$ , again by Theorem 7.6.2(ii). The projected measure has Fourier transform

$$\widehat{\pi_{\theta\#}\mu}(r) = \int_{\mathbb{R}} e^{-2\pi i r v} d(\pi_{\theta\#}\mu)(v) = \int_{\mathbb{R}^2} e^{-2\pi i r x \cdot e_{\theta}} d\mu(x) = \widehat{\mu}(re_{\theta}).$$

Therefore, by polar coordinates in  $\mathbb{R}^2$  (Theorem 1.6.11), writing  $\xi = re_{\theta}$  with  $r \in \mathbb{R}$  and  $\theta \in [0, \pi)$  so that  $d\xi = |r| dr d\theta$ ,

$$\int_0^{\pi} \int_{\mathbb{R}} |\widehat{\pi_{\theta\#}\mu}(r)|^2 dr d\theta = \int_0^{\pi} \int_{\mathbb{R}} |\widehat{\mu}(re_{\theta})|^2 dr d\theta = \int_{\mathbb{R}^2} \frac{|\widehat{\mu}(\xi)|^2}{|\xi|} d\xi \approx I_1(\mu),$$

the last step by Proposition 7.6.5 with  $n = 2$ ,  $t = 1$ , since  $|\xi|^{t-n} = |\xi|^{-1}$ .

So the left side is finite, and hence  $\widehat{\pi_{\theta\#}\mu} \in L^2(\mathbb{R})$  for a.e.  $\theta$ . By Plancherel (Theorem 2.4.8) the measure  $\pi_{\theta\#}\mu$  is then absolutely continuous with an  $L^2$  density, so it cannot be supported on a Lebesgue-null set. Since it is supported in  $\pi_{\theta}(E)$ , that set has positive measure.  $\square$

**Where we are.** The two parts used the two forms of the energy from Volume 7: the integral form  $I_t(\mu)$  for part (i), and the Fourier form  $\int |\widehat{\mu}|^2 |\xi|^{t-n}$  for part (ii). Neither computation is more than an exchange of integrals; the content is entirely in Lemma 8.1.3 — that a linear functional vanishes to first order, so its reciprocal power is integrable exactly below exponent 1.

That single fact is why the threshold in part (i) is 1 and not something else, and it is why the theorem has no analogue for projections onto lines in  $\mathbb{R}^3$  with the same exponent: there the relevant vanishing is still of first order, but the sphere of directions is two-dimensional, and the bookkeeping changes. Drill 1.4 works out the general case.

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**WORKED EXAMPLE 8.1.1— TIER A** Why “almost every” cannot be improved to “every”

Let  $E$  be a line segment in direction  $v$ . Then  $\dim_{\mathbb{H}} E = 1$ , and  $\pi_{\theta}(E)$  is a segment of length  $|v \cdot e_{\theta}|$  times the length of  $E$  — which is a single point when  $e_{\theta} \perp v$ . So  $\dim_{\mathbb{H}} \pi_{\theta}(E) = 0$  for that one  $\theta$ , and 1 for every other.

The exceptional set here is a single direction, of dimension 0. Chapter 8.2 shows this is typical: the exceptional set is always small, with a bound de-

pending on the dimension one is asking for.

*A richer example.* Let  $E = \ell_1 \cup \ell_2$  be two segments in directions  $v_1 \neq v_2$ . Now two directions are exceptional, and the exceptional set still has dimension 0. Taking  $E$  a countable union of segments in a dense set of directions makes the exceptional set countable and dense — still of dimension 0, but no longer closed. So the exceptional set need not be closed, and statements about it must be measure- or dimension-theoretic.

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**WORKED EXAMPLE 8.1.2— TIER B Running the argument for a concrete set**

Take  $E = C \times C$  where  $C$  is the middle-thirds Cantor set. By Corollary 7.6.4 and Theorem 7.8.3,

$$\dim_{\mathrm{H}}(C \times C) = 2 \frac{\log 2}{\log 3} \approx 1.2619 > 1.$$

So Theorem 8.1.2(ii) applies: for almost every  $\theta$ , the projection  $\pi_{\theta}(C \times C)$  has positive Lebesgue measure.

*What the measure  $\mu$  is.* Take  $\mu = \mu_C \times \mu_C$  with  $\mu_C$  the natural Cantor measure of Worked Example 7.1.1. By Proposition 7.6.3(ii),  $I_t(\mu) < \infty$  for every  $t < 2 \log 2 / \log 3$ , in particular for  $t = 1$ .

*What the theorem does not say.* It gives no information about any *particular*  $\theta$ . And indeed the exceptional set here is genuinely interesting: for  $\theta$  with  $\tan \theta$  rational of small height the projection can be much smaller than expected, and the set of  $\theta$  for which  $\pi_{\theta}(C \times C)$  has measure zero is uncountable. Determining it exactly is open.

*A caution about products.*  $\pi_{\theta}(C \times C)$  is, up to scaling, the sumset  $\cos \theta \cdot C + \sin \theta \cdot C$ . So Marstrand's theorem is a statement about sums of Cantor sets, and the arithmetic of such sumsets is the subject of a large literature — the same arithmetic that made  $\dim_{\mathrm{F}} C = 0$  in Theorem 7.7.4. Projections and sumsets are the same question in different notation, and it is worth carrying both readings.

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**WORKED EXAMPLE 8.1.3— TIER C: RESEARCH TASTE What Marstrand's theorem is a model of**

The theorem is the prototype for a family of statements that recur throughout the rest of this series, and the shape is worth isolating.

(a) *The shape.* One has a family of maps  $\{T_{\theta}\}$  indexed by a parameter, and wants a lower bound on the size of  $T_{\theta}(E)$ . Proving it for a fixed  $\theta$  is impossible — there are always bad parameters. Instead one averages a quantity over

$\theta$ , exchanges the order of integration, and finds the inner integral is a finite constant. Then the quantity is finite for almost every  $\theta$ .

(b) *Where the finiteness comes from.* Always from a transversality statement: the map  $\theta \mapsto T_\theta(x) - T_\theta(y)$  vanishes to a controlled order. Here it is Lemma 8.1.3; in Volume 12 it is the transversality hypothesis of the multilinear restriction theorem; in Volume 14 it is the non-degeneracy in the Furstenberg set problem.

(c) *What the method never gives.* A statement for every  $\theta$ , and a statement for a specified  $\theta$ . Both are usually much harder. *Keakeya* is the extreme case: a Besicovitch set is exactly a configuration where the bad directions have been arranged to be *all* of them, and no averaging argument can see it.

(d) *The connection to Keakeya, made precise.* Chapter 8.3 explains that a Besicovitch set of measure zero would follow from knowing that  $m(\pi_\theta(C \times C)) = 0$  for a.e.  $\theta$  for a suitable Cantor set — a statement in the *opposite* direction to Theorem 8.1.2(ii), requiring  $\dim_{\mathbb{H}}(C \times C) \leq 1$  with the measure still vanishing. That is the Besicovitch projection theorem, and it is genuinely harder than Marstrand.

#### WORKED EXAMPLE 8.1.4— TIER B **Computing a projection dimension from scratch**

Theorem 8.1.2 is an existence statement about almost every  $\theta$ . To see the machinery working, here is the whole computation carried out for one set, with every constant tracked.

*The set.* Let  $C_\lambda \subseteq [0, 1]$  be the Cantor set of ratio  $\lambda < \frac{1}{2}$  from Worked Example 7.1.2, so  $d := \dim_{\mathbb{H}} C_\lambda = \log 2 / \log(1/\lambda)$ , and put  $E = C_\lambda \times C_\lambda$  in the plane, so  $\dim_{\mathbb{H}} E = 2d$  by Corollary 7.6.4 and Theorem 7.8.3.

*Step 1: the measure.* Let  $\mu_\lambda$  be the natural measure on  $C_\lambda$ , so  $\mu_\lambda(I) \leq c(\text{diam } I)^d$  as in Worked Example 7.1.1, and set  $\mu = \mu_\lambda \times \mu_\lambda$ . By Proposition 7.6.3(ii),  $I_t(\mu) < \infty$  for every  $t < 2d$ .

*Step 2: the average.* By (8.1.1), for  $t < \min\{2d, 1\}$ ,

$$\int_0^\pi I_t(\pi_{\theta\#}\mu) \, d\theta = c_t I_t(\mu), \quad c_t = \int_0^\pi |\cos \varphi|^{-t} \, d\varphi.$$

Both factors are finite, so the left side is finite.

*Step 3: the conclusion.*  $I_t(\pi_{\theta\#}\mu) < \infty$  for a.e.  $\theta$ , hence  $\dim_{\mathbb{H}} \pi_\theta(E) \geq t$  for a.e.  $\theta$ ; letting  $t \uparrow \min\{2d, 1\}$  through a sequence,

$$\dim_{\mathbb{H}} \pi_\theta(C_\lambda \times C_\lambda) = \min\{2d, 1\} \quad \text{for a.e. } \theta.$$

*Reading the answer.* If  $\lambda < \frac{1}{4}$  then  $d < \frac{1}{2}$ , so  $2d < 1$  and almost every projection has dimension exactly  $2d$  — the projection loses nothing. If  $\lambda > \frac{1}{4}$  then  $2d > 1$  and almost every projection has dimension 1; by Theorem 8.1.2(ii) it has positive measure as well. The transition is at  $\lambda = \frac{1}{4}$ , where  $C_\lambda$  has dimension exactly  $\frac{1}{2}$  and  $E$  has dimension exactly 1 — and there the theorem says nothing about measure, which is precisely the borderline of Worked Example 8.3.1.

*What to practise.* The three steps are always the same: produce a measure of finite  $t$ -energy for every  $t$  below the dimension; average the projected energy over the parameter; read off “finite for almost every parameter”. Only Step 2 involves any thought, and only through the angular integral. Drill 1.4 asks for the same computation in  $\mathbb{R}^n$ .

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**Proposition 8.1.4** (Projections in  $\mathbb{R}^n$ ). *Let  $E \subseteq \mathbb{R}^n$  be compact,  $1 \leq m < n$ , and let  $\pi_V$  denote orthogonal projection onto an  $m$ -plane  $V$ , with  $\gamma_{n,m}$  the natural rotation-invariant probability measure on the Grassmannian  $G(n, m)$ . Then for  $\gamma_{n,m}$ -almost every  $V$ ,*

$$\dim_H \pi_V(E) = \min\{\dim_H E, m\}.$$

*Proof.* The upper bound is Proposition 7.1.5(iii),  $\pi_V$  being 1-Lipschitz, and  $\dim_H \pi_V(E) \leq m$  since the image lies in an  $m$ -plane.

For the lower bound, let  $t < \min\{\dim_H E, m\}$  and take  $\mu$  on  $E$  with  $I_t(\mu) < \infty$  (Theorem 7.6.2). Exactly as in (8.1.1),

$$\int_{G(n,m)} I_t(\pi_V \# \mu) d\gamma_{n,m}(V) = \iint \left( \int_{G(n,m)} \frac{d\gamma_{n,m}(V)}{|\pi_V(x-y)|^t} \right) d\mu d\mu,$$

and the inner integral equals  $|x-y|^{-t} c_{n,m,t}$  with

$$c_{n,m,t} = \int_{G(n,m)} |\pi_V u|^{-t} d\gamma_{n,m}(V), \quad u \in \mathbb{S}^{n-1} \text{ arbitrary,}$$

independent of  $u$  by rotation invariance of  $\gamma_{n,m}$ .

It remains to see  $c_{n,m,t} < \infty$  for  $t < m$ . The quantity  $|\pi_V u|$  is small exactly when  $u$  is nearly orthogonal to  $V$ , and

$$\gamma_{n,m}\{V : |\pi_V u| < \lambda\} \lesssim_{n,m} \lambda^m$$

— the set of  $m$ -planes nearly orthogonal to a fixed direction has measure  $\lambda^m$ , since  $m$  independent conditions each of probability  $\lambda$  must hold. By the layer-cake formula (Worked Example 1.5.1),

$$c_{n,m,t} = \int_0^\infty \gamma_{n,m}\{|\pi_V u|^{-t} > \lambda\} d\lambda \lesssim 1 + \int_1^\infty \lambda^{-m/t} d\lambda < \infty$$

precisely when  $m/t > 1$ . The argument then concludes as in Theorem 8.1.2, letting  $t \uparrow \min\{\dim_{\mathbb{H}} E, m\}$ .  $\square$

**Where we are.** The only thing that changed from the planar case is the exponent in the sublevel estimate:  $\gamma_{n,m}\{|\pi_V u| < \lambda\} \lesssim \lambda^m$  rather than  $\lambda^1$ . That is why the threshold is  $m$  rather than 1, and it is the general principle of Worked Example 8.2.4: *the exponent in the conclusion is the exponent in the sublevel estimate*.

The same substitution runs through Theorem 8.2.1: the exceptional set of  $m$ -planes for dimension  $t$  has  $\gamma_{n,m}$ -dimension at most  $t$ , by replacing Lemma 8.2.2 with its Grassmannian analogue. Drill 1.4 asks for the details.

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**Figure 8.1.1 (to be drawn).** *Projections of a set onto lines*

**Content required:** A compact set  $E$  in the plane (say a Cantor-like dust) with its shadows  $\pi_\theta(E)$  drawn on three lines: a generic  $\theta$ , where the shadow is as large as the dimension permits; and an exceptional  $\theta$ , where it collapses. Mark the angle and indicate that the exceptional directions form a Lebesgue-null set, of dimension bounded by Theorem 8.2.1.

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## Drills

**Drill 8.1.1.** Verify that  $c_t$  in Lemma 8.1.3 satisfies  $c_t \rightarrow \infty$  as  $t \uparrow 1$ , and compute its rate.

**Drill 8.1.2.** Show that Theorem 8.1.2(i) fails if “almost every” is replaced by “every”, using Worked Example 8.1.1, and that the exceptional set can be dense.

**Drill 8.1.3.** Check the polar-coordinate computation in Step 2, being careful that  $r$  ranges over all of  $\mathbb{R}$  while  $\theta$  ranges over  $[0, \pi)$ .

**Drill 8.1.4.** State and prove the analogue of Theorem 8.1.2(i) for projections  $\mathbb{R}^n \rightarrow \mathbb{R}^m$ ,  $m < n$ , onto  $m$ -planes, identifying the exponent at which the angular integral diverges.

**Where we are.** Before leaving the chapter, note what Theorem 8.1.2 does *not* require: no regularity of  $E$ , no structure beyond compactness, and no relation between  $E$  and the family of projections. That generality is what makes the theorem a model, and it is also what limits it — a statement true for every compact set of a given dimension cannot distinguish between sets, and the exceptional set is where all the individuality lives.

Chapter 8.2 therefore does the more informative work, and the modern literature on projections is almost entirely about exceptional sets rather than about the almost-everywhere statement.

## Notes

Marstrand proved the theorem in 1954 by a geometric argument. The proof given here is Kaufman's (1968), and its brevity is the reason the potential-theoretic formulation of dimension became standard: the entire content is Lemma 8.1.3.

Mattila's book is the reference for this chapter and the next. The problem of determining the exceptional set exactly for a specific  $E$ , such as  $C \times C$ , is open and is the subject of the projection theorems mentioned in Volume 14.

## Chapter 8.2

# Exceptional Sets

Theorem 8.1.2 says the set of bad directions is Lebesgue-null. Worked Example 8.1.1 showed it can be nonempty, and even dense. This chapter bounds its *dimension*, by running Kaufman’s computation against a Frostman measure on the exceptional set rather than against Lebesgue measure — a move worth learning, since it is the standard way to upgrade an “almost every” statement.

**Theorem 8.2.1** (Kaufman). *Let  $E \subseteq \mathbb{R}^2$  be compact and  $0 < t \leq 1$ . Then*

$$\dim_{\mathrm{H}} \left\{ \theta \in [0, \pi) : \dim_{\mathrm{H}} \pi_{\theta}(E) < t \right\} \leq t.$$

*Idea of the proof.* Suppose the exceptional set  $F_t$  had dimension exceeding  $s$  for some  $s > t$ . Frostman then supplies a measure  $\nu$  on  $F_t$  with  $\nu(B(\theta, r)) \leq r^s$ . Run the computation of Theorem 8.1.2 with  $d\theta$  replaced by  $d\nu(\theta)$ .

The only step that changes is the angular integral: instead of  $\int_0^\pi |u \cdot e_\theta|^{-t} d\theta$ , one must bound  $\int |u \cdot e_\theta|^{-t} d\nu(\theta)$ . The set where  $|u \cdot e_\theta|$  is small is an interval of comparable length around the direction perpendicular to  $u$ , and  $\nu$  gives it mass at most  $r^s$  rather than  $r$ . So the integral converges when  $t < s$  — one power better than Lebesgue allows.

That gives  $I_t(\pi_{\theta\#}\mu) < \infty$  for  $\nu$ -almost every  $\theta$ , so  $\dim_{\mathrm{H}} \pi_{\theta}(E) \geq t$  for  $\nu$ -almost every  $\theta$ ; but every  $\theta \in F_t$  violates that, and  $\nu(F_t) > 0$ . Contradiction.

**Lemma 8.2.2** (The weighted angular integral). *Let  $\nu$  be a finite Borel measure on  $[0, \pi)$  with  $\nu(I) \leq |I|^s$  for every interval  $I$ , where  $0 < s \leq 1$ . Then for  $0 < t < s$  and every unit vector  $u$ ,*

$$\int_0^\pi \frac{d\nu(\theta)}{|u \cdot e_\theta|^t} \leq C_{s,t} < \infty,$$

with  $C_{s,t}$  independent of  $u$ .

*Proof. Step 1. Locate where the integrand is large.*

Write  $u = e_\alpha$ , so that  $u \cdot e_\theta = \cos(\theta - \alpha)$ . The integrand  $|u \cdot e_\theta|^{-t}$  is large exactly where  $\cos(\theta - \alpha)$  is small, that is near  $\theta = \alpha \pm \frac{\pi}{2}$  — the directions perpendicular to  $u$ . Quantitatively,  $|\cos \varphi| < \lambda$  forces  $\varphi$  to lie within  $\arcsin \lambda$  of  $\pm \frac{\pi}{2}$ , and  $\arcsin \lambda \leq \frac{\pi}{2} \lambda$  for  $\lambda \in [0, 1]$ . So

$$\{\theta \in [0, \pi) : |u \cdot e_\theta| < \lambda\}$$

is contained in an interval of length at most  $\pi\lambda$ , modulo  $\pi$ .

**Step 2.** *Apply the Frostman hypothesis.*

By assumption  $\nu(I) \leq |I|^s$  for intervals, so

$$\nu\{\theta : |u \cdot e_\theta| < \lambda\} \leq (\pi\lambda)^s. \quad (8.2.1)$$

This is the sublevel set estimate discussed in Worked Example 8.2.4, and it is the only place the hypothesis on  $\nu$  is used. Note the gain over Lebesgue measure: there the same set has measure  $\approx \lambda$ , and here it is  $\lambda^s$  with  $s$  possibly much smaller — which is what will allow a larger exponent  $t$ .

**Step 3.** *Integrate by layers.*

Put  $f(\theta) = |u \cdot e_\theta|^{-t}$ . By the layer-cake formula (Worked Example 1.5.1),

$$\int f \, d\nu = \int_0^\infty \nu\{f > \lambda\} \, d\lambda = \int_0^1 \nu\{f > \lambda\} \, d\lambda + \int_1^\infty \nu\{f > \lambda\} \, d\lambda.$$

The first integral is at most  $\nu([0, \pi)) \leq \pi^s$ . For the second, note  $f(\theta) > \lambda$  is equivalent to  $|u \cdot e_\theta| < \lambda^{-1/t}$ , so by (8.2.1),

$$\int_1^\infty \nu\{f > \lambda\} \, d\lambda \leq \pi^s \int_1^\infty \lambda^{-s/t} \, d\lambda.$$

**Step 4.** *Check convergence.*

The last integral converges if and only if  $s/t > 1$ , that is  $t < s$ , with value  $\frac{t}{s-t}$ . Hence

$$\int f \, d\nu \leq \pi^s \left(1 + \frac{t}{s-t}\right) = \pi^s \frac{s}{s-t} =: C_{s,t},$$

which is finite, independent of  $u$  — no step used anything about  $u$  beyond  $|u| = 1$  — and blows up as  $t \uparrow s$ .  $\square$

*Proof of Theorem 8.2.1.* Write  $F_t = \{\theta : \dim_{\mathbb{H}} \pi_\theta(E) < t\}$  and suppose, for contradiction, that  $\dim_{\mathbb{H}} F_t > t$ . Fix  $s$  with  $t < s < \dim_{\mathbb{H}} F_t$ .



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**Step 1.** *Two measures.*

Since  $\dim_{\mathbb{H}} F_t > s$ , Theorem 7.5.1 — applied to a compact subset of  $F_t$  of dimension exceeding  $s$ , which exists by inner regularity — gives a nonzero finite  $\nu$  carried by  $F_t$  with  $\nu(B(\theta, r)) \leq r^s$ , hence  $\nu(I) \leq |I|^s$  for intervals.

Since  $t \leq 1$  and  $F_t \neq \emptyset$  forces  $\dim_{\mathbb{H}} E \geq t$  — otherwise every projection would trivially have dimension less than  $t$  and  $F_t$  would be all of  $[0, \pi)$ , of dimension  $1 \leq$  nothing useful; we treat that degenerate case separately in Step 4 — Theorem 7.6.2(ii) gives a nonzero finite  $\mu$  on  $E$  with  $I_{t'}(\mu) < \infty$  for any fixed  $t' < t$ .

**Step 2.** *Average against  $\nu$ .*

Exactly as in (8.1.1), but integrating  $d\nu(\theta)$ ,

$$\int_0^\pi I_{t'}(\pi_{\theta\#}\mu) d\nu(\theta) = \iint \left( \int_0^\pi \frac{d\nu(\theta)}{|(x-y) \cdot e_\theta|^{t'}} \right) d\mu d\mu \leq C_{s,t'} I_{t'}(\mu) < \infty,$$

by Lemma 8.2.2, applicable since  $t' < t < s$ .

**Step 3.** *Conclude.*

The left-hand side is finite, so  $I_{t'}(\pi_{\theta\#}\mu) < \infty$  for  $\nu$ -almost every  $\theta$ , whence  $\dim_{\mathbb{H}} \pi_\theta(E) \geq t'$  for  $\nu$ -almost every  $\theta$  by Theorem 7.6.2(i).

Letting  $t' \uparrow t$  along a sequence and intersecting countably many  $\nu$ -full sets,  $\dim_{\mathbb{H}} \pi_\theta(E) \geq t$  for  $\nu$ -a.e.  $\theta$ . But  $\nu$  is carried by  $F_t$ , on which  $\dim_{\mathbb{H}} \pi_\theta(E) < t$  by definition, and  $\nu \neq 0$  — a contradiction.

**Step 4.** *The degenerate case.*

If  $\dim_{\mathbb{H}} E < t$  then  $F_t = [0, \pi)$  and the conclusion  $\dim_{\mathbb{H}} F_t \leq t$  reads  $1 \leq t$ , which holds since we may assume  $t \geq 1$ ; and for  $t < 1$  with  $\dim_{\mathbb{H}} E < t$  the statement is vacuous once one notes  $\dim_{\mathbb{H}} \pi_\theta E \leq \dim_{\mathbb{H}} E < t$  for all  $\theta$ , so the theorem is asserting  $\dim_{\mathbb{H}} [0, \pi) = 1 \leq t$  — which requires  $t = 1$ . We therefore read Theorem 8.2.1 as asserting content only when  $t \leq \dim_{\mathbb{H}} E$ , which is the case of interest.  $\square$

**Corollary 8.2.3.** *For compact  $E \subseteq \mathbb{R}^2$  with  $\dim_{\mathbb{H}} E \leq 1$ ,*

$$\dim_{\mathbb{H}} \left\{ \theta : \dim_{\mathbb{H}} \pi_\theta(E) < \dim_{\mathbb{H}} E \right\} \leq \dim_{\mathbb{H}} E.$$

*Proof.* Apply Theorem 8.2.1 with  $t = \dim_{\mathbb{H}} E$  and note that the set in question is the increasing union over  $t' < \dim_{\mathbb{H}} E$  of the sets  $\{\theta : \dim_{\mathbb{H}} \pi_\theta E < t'\}$ , each of dimension at most  $t'$ ; countable stability (Proposition 7.1.5(ii)) gives the bound.  $\square$

**WORKED EXAMPLE 8.2.1— TIER A The bound is attained**

Theorem 8.2.1 is sharp. Let  $0 < t \leq 1$  and let  $L$  be a compact set of directions with  $\dim_{\mathbb{H}} L = t$ . Build

$$E = \bigcup_{\theta \in L} \ell_{\theta},$$

a union of unit segments, one in each direction of  $L$ , arranged so that  $E$  is compact — for instance all through a common point, a bush.

For  $\theta \in L$ , the projection  $\pi_{\theta^{\perp}}$  collapses  $\ell_{\theta}$  to a point. Arranging the bush so that  $\dim_{\mathbb{H}} E > t$  — which holds since  $E$  contains a segment, giving  $\dim_{\mathbb{H}} E \geq 1 \geq t$ , with strict inequality when  $t < 1$  — the projection  $\pi_{\theta^{\perp}}(E)$  consists of the images of the other segments, whose directions range over  $L$ , and one computes  $\dim_{\mathbb{H}} \pi_{\theta^{\perp}}(E) \leq \dim_{\mathbb{H}} L = t < \dim_{\mathbb{H}} E$ . So every  $\theta^{\perp}$  with  $\theta \in L$  is exceptional, and the exceptional set contains an isometric copy of  $L$ , of dimension  $t$ .

*What this shows.* No bound better than  $t$  is possible, so the theorem is optimal in its own terms. What is *not* known is the sharp statement for a *fixed*  $E$  of given dimension — the modern projection theorems of Orponen, Shmerkin and Wang, surveyed in Volume 14, give improvements under extra hypotheses on  $E$ .

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**WORKED EXAMPLE 8.2.2— TIER B Upgrading “almost every” to a dimension bound: the general recipe**

The passage from Theorem 8.1.2 to Theorem 8.2.1 is a manoeuvre worth having in hand, since it applies whenever an averaging argument gives an almost-everywhere statement.

*The recipe.* Suppose one has proved

$$\int_{\Theta} Q(\theta) \, d\theta < \infty$$

for some nonnegative quantity  $Q$ , concluding  $Q(\theta) < \infty$  for a.e.  $\theta$ . To bound the dimension of  $\{Q = \infty\}$ :

- (1) suppose  $\dim_{\mathbb{H}}\{Q = \infty\} > s$  and take a Frostman measure  $\nu$  of exponent  $s$  on it;
- (2) repeat the original computation with  $d\theta$  replaced by  $d\nu(\theta)$ ;
- (3) the only step that changes is the inner integral, which must now be estimated against  $\nu$  rather than Lebesgue;
- (4) if it still converges,  $Q < \infty$   $\nu$ -a.e., contradicting  $\nu \neq 0$  on  $\{Q = \infty\}$ .

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Where the work is. Entirely in step (3). Here the inner integral was  $\int |u \cdot e_\theta|^{-t} d\nu$ , and Lemma 8.2.2 showed the Frostman condition buys exactly one extra power: convergence for  $t < s$  rather than  $t < 1$ . In general one needs a bound of the form

$$\nu\{\theta : |\Phi(\theta)| < \lambda\} \lesssim \lambda^s,$$

where  $\Phi$  is the transversality function — a *sublevel set estimate* for  $\nu$ , of exactly the kind proved in Volume 5, Worked Example 5.1.2.

Where it is used again. Volume 14’s projection theorems are all of this form, with  $\Phi$  more complicated and the sublevel estimate correspondingly harder. Recognising the recipe makes those statements readable even when their proofs are not.

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**WORKED EXAMPLE 8.2.3— TIER C: RESEARCH TASTE** **What is not known, and why the gap is interesting**

(a) *The dimension of the exceptional set for a fixed  $E$ .* Theorem 8.2.1 bounds it by  $t$ , and Worked Example 8.2.1 shows that is attained — but only by a set  $E$  built for the purpose. For a specific  $E$ , such as  $C \times C$ , nothing like a complete answer is known.

(b) *Falconer’s improvement, and its limits.* Falconer showed that for  $t > \frac{1}{2}$  the exceptional set for  $\dim_{\mathbb{H}} \pi_\theta E < t$  has dimension at most  $2t - \dim_{\mathbb{H}} E$  — better than Kaufman when  $\dim_{\mathbb{H}} E > t$ . Combining the two gives the best elementary bound, and the truth is conjectured to be  $\max\{0, \dim_{\mathbb{H}} E - t\}$ , still open in general.

(c) *The connection to Kakeya.* A Besicovitch set is, by Worked Example 8.1.3(c), a configuration in which *every* direction is exceptional for a related problem. So the projection theorems and the Kakeya problem are two ends of the same question: how large can the set of bad parameters be? Projection theory says “small”; Besicovitch’s construction says “in the Kakeya setting, everything”. Understanding the transition is the content of the Furstenberg set problem, which Volume 14 discusses and which Ren and Wang settled in 2023.

(d) *A methodological point.* Every result in this chapter came from an averaging argument plus a sublevel-set estimate. Neither ingredient is geometric; both are potential-theoretic. That is why this material is comparatively complete while the Kakeya problem is not: the moment the bad parameters can be *all* of them, averaging is useless, and one must count tubes.

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**WORKED EXAMPLE 8.2.4— TIER B Sublevel set estimates: the ingredient to check**

Worked Example 8.2.2 said that the whole content of an upgrade is a bound of the form  $\nu\{\theta : |\Phi(\theta)| < \lambda\} \lesssim \lambda^s$ . Such bounds are exactly the sublevel set estimates of Volume 5, Worked Example 5.1.2, and it is worth seeing how they are produced.

*The generic principle.* If  $\Phi$  is  $C^1$  with  $|\Phi'| \geq c$  on the relevant interval, then  $\{|\Phi| < \lambda\}$  is contained in an interval of length  $2\lambda/c$ ; if  $\nu(I) \leq |I|^s$  this gives  $\nu\{|\Phi| < \lambda\} \leq (2\lambda/c)^s$ . Every estimate in this chapter is that computation.

*Instance 1: Lemma 8.2.2.* Here  $\Phi(\theta) = \cos(\theta - \alpha)$  and  $|\Phi'| = |\sin(\theta - \alpha)|$ , which is *not* bounded below — it vanishes where  $\Phi$  is extremal. But one does not need it there: near the zeros of  $\Phi$ , which are where  $\theta - \alpha = \pm\frac{\pi}{2}$ , the derivative is  $\pm 1$ . So the hypothesis holds exactly where it is needed, and that is the content of the estimate  $\{|\cos(\theta - \alpha)| < \lambda\} \subseteq$  an interval of length  $\pi\lambda$ .

*Instance 2: higher-order vanishing.* If instead  $\Phi$  vanished to order  $k$  at its zeros —  $\Phi(\theta) \approx (\theta - \theta_0)^k$  — the sublevel set would be an interval of length  $\approx \lambda^{1/k}$ , and the bound would become  $\nu\{|\Phi| < \lambda\} \lesssim \lambda^{s/k}$ . Repeating Lemma 8.2.2 then gives convergence for  $t < s/k$  instead of  $t < s$ , and the exceptional-set bound weakens accordingly.

*Instance 3: where this bites in the literature.* The modern projection theorems replace the family  $\{\pi_\theta\}$  by a family of nonlinear maps, and the transversality function  $\Phi$  is then a genuine object of study. The Ren–Wang solution of the Furstenberg problem, in Volume 14, turns on exactly such an estimate. A reader who can identify  $\Phi$  and its order of vanishing can predict the exponent in any such theorem before reading the proof.

*The exercise to do once.* Take  $\Phi(\theta) = \cos(\theta - \alpha)$ , verify the order of vanishing is 1, and confirm that the exponent in Theorem 8.2.1 is  $s$  and not  $s/2$  or  $2s$ . Then repeat with  $\Phi(\theta) = \cos^2(\theta - \alpha)$  and check the exponent halves.

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**Proposition 8.2.4** (Falconer’s complementary bound). *Let  $E \subseteq \mathbb{R}^2$  be compact with  $\dim_{\mathbb{H}} E > t$  and  $\frac{1}{2} < t \leq 1$ . Then*

$$\dim_{\mathbb{H}} \left\{ \theta : \dim_{\mathbb{H}} \pi_\theta(E) < t \right\} \leq 2t - \dim_{\mathbb{H}} E.$$

*Proof.* Write  $d = \dim_{\mathbb{H}} E$  and suppose the exceptional set  $F_t$  has  $\dim_{\mathbb{H}} F_t > s$  for some  $s > 2t - d$ ; we derive a contradiction. Fix  $t' < t$  and  $d' < d$  with  $s > 2t' - d'$ , possible by continuity.

By Theorem 7.5.1 take  $\nu \neq 0$  on  $F_t$  with  $\nu(I) \leq |I|^s$ , and  $\mu \neq 0$  on  $E$  with  $\mu(B(x, r)) \leq r^{d'}$ ; by Theorem 7.6.2 we may also assume  $I_{t'}(\mu) < \infty$ .

The proof of Theorem 8.2.1 bounded the inner integral by  $|x - y|^{-t'}$  times a constant. Here we use the Frostman property of  $\mu$  as well. Splitting the pair integral according to  $|x - y|$  and applying Lemma 8.2.2 on each piece,

$$\int_0^\pi I_{t'}(\pi_{\theta\#}\mu) \, d\nu(\theta) \lesssim \iint \frac{d\mu(x) \, d\mu(y)}{|x - y|^{t'}} \cdot \sup_u \int_0^\pi \frac{d\nu(\theta)}{|u \cdot e_\theta|^{t'}},$$

which is finite provided  $t' < s$ . When  $s \geq t$  this already gives the contradiction, as in Theorem 8.2.1.

When  $s < t$  the estimate above fails, and one uses instead that  $\mu$  has dimension  $d' > 2t' - s$ : the mass of  $\mu \times \mu$  in the set where  $|(x - y) \cdot e_\theta| < \lambda|x - y|$  is at most  $\lambda^{d' - t'}$  by the Frostman bound applied to slabs, and combining with  $\nu\{|u \cdot e_\theta| < \lambda\} \leq \lambda^s$  gives convergence exactly when  $t' < \frac{s + d'}{2}$ , that is  $s > 2t' - d'$  — which is the hypothesis. So  $I_{t'}(\pi_{\theta\#}\mu) < \infty$  for  $\nu$ -a.e.  $\theta$ , contradicting  $\nu(F_t) > 0$  as before.  $\square$

**Where we are.** Theorem 8.2.1 and Proposition 8.2.4 are complementary: Kaufman's bound  $t$  is better when  $\dim_{\mathbb{H}} E < t$ , Falconer's bound  $2t - \dim_{\mathbb{H}} E$  is better when  $\dim_{\mathbb{H}} E > t$ , and they agree at  $\dim_{\mathbb{H}} E = t$ . Taking the minimum gives the best elementary estimate,

$$\dim_{\mathbb{H}} \{\theta : \dim_{\mathbb{H}} \pi_\theta E < t\} \leq \min\{t, 2t - \dim_{\mathbb{H}} E\}.$$

The conjectured truth is  $\max\{0, \dim_{\mathbb{H}} E - t\}$  — rather different, and open. Modern work of Orponen, Shmerkin and Wang, surveyed in Volume 14, improves on both under additional hypotheses.

## Drills

**Drill 8.2.1.** Verify the containment  $\{\theta : |\cos(\theta - \alpha)| < \lambda\} \subseteq$  an interval of length  $\pi\lambda$ , used in Lemma 8.2.2.

**Drill 8.2.2.** Compute  $C_{s,t}$  in Lemma 8.2.2 explicitly, and check that it blows up as  $t \uparrow s$ .

**Drill 8.2.3.** Deduce from Theorem 8.2.1 that if  $\dim_{\mathbb{H}} E = 1$  then  $\dim_{\mathbb{H}} \pi_\theta(E) = 1$  for all  $\theta$  outside a set of dimension at most 1 — and explain why this is vacuous, so that Corollary 8.2.3 is the useful statement.

**Drill 8.2.4.** Carry out Worked Example 8.2.2 for the statement “ $\widehat{\mu}(re_\theta)$  is square-integrable in  $r$  for a.e.  $\theta$ ” from Theorem 8.1.2(ii), and find the sublevel estimate needed.

## Notes

Theorem 8.2.1 is Kaufman's (1968), proved in the same paper as the short proof of Marstrand's theorem. Falconer's complementary bound is from 1982; the sharp conjecture is due to him and remains open.

The recipe of Worked Example 8.2.2 — replace Lebesgue measure by a Frostman measure on the exceptional set — is one of the most transferable techniques in the subject, and is worth trying whenever an argument produces an almost-everywhere conclusion.

## Chapter 8.3

# Product Sets and Besicovitch Sets

Volume 1 constructed Besicovitch sets of arbitrarily small measure and promised one of measure *zero*; Volume 7 explained why the compactness argument one expects to work does not, and deferred the construction here. This chapter settles the matter. The natural route runs through the projection theory just developed, and it is worth following far enough to see exactly where it stops.

### 8.3.1 Projections of product sets

**Proposition 8.3.1.** *Let  $A, B \subseteq \mathbb{R}$  be compact and  $\theta \in (0, \pi/2)$ . Then, up to a bi-Lipschitz change of coordinates on the target line,*

$$\pi_\theta(A \times B) = \cos \theta \cdot A + \sin \theta \cdot B.$$

*In particular  $\pi_{\pi/4}(A \times B)$  is a rescaled copy of the sumset  $A + B$ , and the projection in direction  $\theta$  is the sumset  $A + \lambda B$  with  $\lambda = \tan \theta$ , rescaled.*

*Proof.* By definition  $\pi_\theta(a, b) = a \cos \theta + b \sin \theta$ , and as  $(a, b)$  ranges over  $A \times B$  this ranges over  $\cos \theta A + \sin \theta B$ . Dividing by  $\cos \theta \neq 0$  is a bi-Lipschitz map of the line.  $\square$

**Where we are.** So the projection theorems of Chapter 8.1 and Chapter 8.2 are, when applied to product sets, statements about sumsets  $A + \lambda B$ : Marstrand says that for almost every  $\lambda$  the sumset is as large as the dimensions allow, and Kaufman bounds the set of bad  $\lambda$ .

This is the reason projection theory and additive combinatorics keep meeting. Volume 10 will develop sumset inequalities for their own sake and use them on Kakeya; the present chapter is the geometric face of the same

material.

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WORKED EXAMPLE 8.3.1— TIER A **The four-corner Cantor set**

Let  $K$  be the attractor of the four similarities of ratio  $\frac{1}{4}$  taking the unit square to its four corner squares of side  $\frac{1}{4}$ . By Theorem 7.8.3, since  $4 \cdot 4^{-s} = 1$  gives  $s = 1$ ,

$$\dim_{\mathbb{H}} K = 1, \quad 0 < \mathcal{H}^1(K) < \infty.$$

So  $K$  is a one-dimensional set of finite positive length, and it is totally disconnected — it contains no rectifiable curve.

*Its projections.*  $\pi_{\theta}(K)$  has dimension 1 for almost every  $\theta$ , by Theorem 8.1.2(i), which is the most Theorem 8.1.2 can say since  $\dim_{\mathbb{H}} K = 1$  is not  $> 1$ . But a stronger and quite different statement is true:  $m(\pi_{\theta}(K)) = 0$  for almost every  $\theta$ . The projection in direction  $\theta$  has dimension 1 and measure zero, simultaneously.

*Why this does not follow from anything proved here.* Theorem 8.1.2 gives a lower bound on dimension and, when  $\dim_{\mathbb{H}} E > 1$ , positive measure. It says nothing when  $\dim_{\mathbb{H}} E = 1$  exactly, which is precisely the borderline the four-corner set sits on. The vanishing of the measure is the Besicovitch projection theorem, discussed in Remark 8.3.3.

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### 8.3.2 The construction that would work, and why we stop

**Proposition 8.3.2** (Kahane's reduction). *Let  $A \subseteq [0, 1]$  be compact with  $A - A \supseteq [-1, 1]$ , and set*

$$E = \bigcup_{a, b \in A} [(a, 0), (b, 1)] \subseteq \mathbb{R}^2,$$

*the union of the segments joining  $(a, 0)$  to  $(b, 1)$ . Then*

(i)  *$E$  is compact and contains a segment in every direction of slope at least 1 in absolute value;*

(ii)  $m(E) = \int_0^1 m((1-y)A + yA) dy;$

(iii) *consequently  $m(E) = 0$  if and only if  $m((1-y)A + yA) = 0$  for almost every  $y \in (0, 1)$ .*

*Proof.* (i)  $E$  is the image of the compact set  $A \times A \times [0, 1]$  under the continuous map  $(a, b, y) \mapsto ((1-y)a + yb, y)$ , hence compact. The segment from  $(a, 0)$  to



$(b, 1)$  has direction vector  $(b - a, 1)$ ; as  $(a, b)$  ranges over  $A \times A$ , the number  $b - a$  ranges over  $A - A \supseteq [-1, 1]$ , so every direction  $(v, 1)$  with  $|v| \leq 1$  occurs.

(ii) The slice of  $E$  at height  $y \in [0, 1]$  is  $E_y = \{(1 - y)a + yb : a, b \in A\} = (1 - y)A + yA$ , so Fubini (Theorem 1.6.7) gives  $m(E) = \int_0^1 m(E_y) dy$ .

(iii) Immediate from (ii), the integrand being nonnegative.  $\square$

*Remark 8.3.3* (Scope decision: the measure-zero Besicovitch set). Proposition 8.3.2 reduces the construction to a single question: is there a compact  $A \subseteq [0, 1]$  with  $A - A \supseteq [-1, 1]$  and

$$m((1 - y)A + yA) = 0 \quad \text{for a.e. } y \in (0, 1) ? \quad (8.3.1)$$

We do not answer it in this series, and the existence of a Besicovitch set of measure zero is accordingly removed from the plan. This is revision R8.

The obstruction is precise and worth recording, because the route looks as though the machinery of this volume should close it.

By Proposition 8.3.1,  $(1 - y)A + yA$  is a projection of  $A \times A$ . So (8.3.1) asks for a set whose almost-every projection is Lebesgue-null. Theorem 8.1.2(ii) says this forces  $\dim_H(A \times A) \leq 1$ , that is  $\dim_H A \leq \frac{1}{2}$  by Corollary 7.6.4 and Theorem 7.8.3. But the requirement  $A - A \supseteq [-1, 1]$  pushes the other way: for a self-similar  $A$  with two maps of ratio  $r$ , the difference set contains an interval only when  $r \geq \frac{1}{3}$ , and then

$$\dim_H A = \frac{\log 2}{\log(1/r)} \geq \frac{\log 2}{\log 3} \approx 0.631 > \frac{1}{2}.$$

So  $\dim_H(A \times A) > 1$ , and Theorem 8.1.2(ii) gives the *opposite* of what is wanted: almost every projection has *positive* measure. The two constraints are in direct conflict for self-similar  $A$ .

The classical resolution uses a set  $A$  that is not self-similar, sitting exactly on the borderline  $\dim_H(A \times A) = 1$  — as the four-corner set of Worked Example 8.3.1 does — where Theorem 8.1.2 is silent and the vanishing of the measure is the content of the *Besicovitch projection theorem*: a purely unrectifiable set of finite  $\mathcal{H}^1$  measure has Lebesgue-null projection in almost every direction. That theorem is a substantial piece of geometric measure theory, requiring the density and rectifiability theory this series does not develop.

*What is lost.* Nothing that is used. The Kakeya conjecture concerns Hausdorff dimension (Conjecture 7.1.11), not measure; Volume 3's ball multiplier theorem uses the configuration lemma, not measure zero; and Volume 1's Theorem 1.3.12, giving measure less than every  $\varepsilon$ , is what all the applications in this series invoke. The measure-zero refinement is a celebrated fact and, for the purposes of these volumes, an ornament.

*A note on the record.* This is the third disposition of the item: promised in Volume 1, deferred from Volume 7 to Volume 8, and now removed. The

pattern is worth naming — an attractive result kept being postponed because each time the tools to hand *nearly* sufficed. Naming the obstruction, as above, is what should have happened at the first deferral.

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WORKED EXAMPLE 8.3.2— TIER B **What the reduction does deliver**

Proposition 8.3.2 is not wasted even though (8.3.1) is left open here. It converts a two-dimensional geometric question into a one-dimensional arithmetic one, and that conversion is used repeatedly.

(a) *Small measure, cheaply.* Take  $A$  self-similar with ratio  $r = \frac{1}{3}$ , so  $A - A = [-1, 1]$  and  $\dim_{\mathcal{H}} A = \log 2 / \log 3$ . Then  $A$  is Lebesgue null, and although  $m((1 - y)A + yA)$  need not vanish, one can estimate it: at resolution  $\delta$ ,  $A$  is covered by  $\approx \delta^{-\log 2 / \log 3}$  intervals, so the sumset is covered by  $\approx \delta^{-2 \log 2 / \log 3}$  intervals of length  $\delta$ , giving  $m(((1 - y)A + yA)_{\delta}) \lesssim \delta^{1 - 2 \log 2 / \log 3}$ . Since  $2 \log 2 / \log 3 \approx 1.262 > 1$ , this is not small — consistent with Remark 8.3.3.

(b) *The correct reading.* The computation in (a) says the construction is *efficient*: it produces a set of dimension 2 from a set  $A$  of dimension well below 1. That efficiency is what makes Kakeya sets small, and it is the same phenomenon as the Perron tree, seen arithmetically instead of geometrically.

(c) *The general principle.* A Kakeya-type set built from a Cantor set of parameters has its measure computed by Fubini as an average of sumset measures. Volume 10's arithmetic method for Kakeya is this observation systematised: one studies the additive structure of the parameter set instead of the geometry of the tubes. Bourgain's bound, and the Katz–Łaba–Tao improvement, both proceed this way.

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WORKED EXAMPLE 8.3.3— TIER C: RESEARCH TASTE **Rectifiability, and the theory this series omits**

Remark 8.3.3 appealed to the Besicovitch projection theorem. It is worth saying what that theorem belongs to, since it is the one substantial branch of geometric measure theory this series does without.

(a) *The dichotomy.* A set  $E$  with  $\mathcal{H}^1(E) < \infty$  in the plane splits into a *rectifiable* part, contained in countably many Lipschitz curves, and a *purely unrectifiable* part, meeting every Lipschitz curve in  $\mathcal{H}^1$ -measure zero. Every such  $E$  decomposes uniquely this way.

(b) *The theorem.* Projections detect the dichotomy exactly: a rectifiable set has projections of positive measure in almost every direction, and a purely unrectifiable set has projections of measure zero in almost every direction. The four-corner set of Worked Example 8.3.1 is the standard purely unrectifiable

example.

(c) *Why it is hard.* The proof of the unrectifiable half uses the density theory of Theorem 7.4.4 pushed much further — one shows that a set with projections of positive measure has tangent-like behaviour at almost every point, and then that this forces rectifiability. The full argument is the subject of several chapters in Mattila’s book.

(d) *Why this series omits it.* Because Kakeya and restriction are about sets of dimension close to  $n$ , where the rectifiable/unrectifiable dichotomy carries no information: every set of dimension  $> 1$  in the plane is unrectifiable in the relevant sense, and the theory says nothing further. The dichotomy is sharp exactly at integer dimension with finite measure, which is a regime this subject does not visit.

(e) *The exception worth knowing.* It does visit that regime once. The Furstenberg set problem, in Volume 14, concerns sets meeting each line of a family in dimension  $s$ , and at  $s = 1$  the rectifiability theory becomes relevant. Ren and Wang’s solution does not use it, but the connection is why the problem was long regarded as belonging to geometric measure theory rather than harmonic analysis.

#### WORKED EXAMPLE 8.3.4— TIER B **Translating between projections and sumsets**

Proposition 8.3.1 identifies  $\pi_\theta(A \times B)$  with  $\cos \theta A + \sin \theta B$ . The dictionary is used in both directions, and it is worth rehearsing each.

*Projection facts read as sumset facts.*

- Theorem 8.1.2(i) becomes: for almost every  $\lambda$ ,  $\dim_{\mathbb{H}}(A + \lambda B) = \min\{\dim_{\mathbb{H}} A + \dim_{\mathbb{H}} B, 1\}$ . So a sumset is as large as the dimensions permit, for almost every dilation ratio.
- Theorem 8.1.2(ii) becomes: if  $\dim_{\mathbb{H}} A + \dim_{\mathbb{H}} B > 1$  then  $A + \lambda B$  has positive measure for almost every  $\lambda$ .
- Theorem 8.2.1 becomes: the set of bad ratios  $\lambda$  has dimension at most  $t$ , where  $t$  is the dimension one is asking for.

*Sumset facts read as projection facts.* Conversely, an arithmetic statement about  $A + \lambda B$  transfers. For instance if  $A$  is the middle-thirds Cantor set then  $A + A = [0, 2]$  — every number in  $[0, 2]$  is a sum of two base-3 numbers with digits in  $\{0, 2\}$  — so  $\pi_{\pi/4}(A \times A)$  is a full interval, even though  $\dim_{\mathbb{H}}(A \times A) \approx 1.26$  gives no such guarantee for a specific  $\theta$ .

*Where the dictionary is essential.* The measure-zero Besicovitch question of Remark 8.3.3 is a sumset question, (8.3.1), and it was only by reading it as a projection question that the obstruction became visible. Conversely, Volume 10 will prove sumset inequalities — Plünnecke–Ruzsa, Balog–Szemerédi–Gowers — and apply them to tube configurations; that application is the same translation running the other way.

*A caution.* The dictionary is exact only for *product* sets. A general  $E \subseteq \mathbb{R}^2$  has projections that are not sumsets of anything, and the arithmetic intuition can mislead. When a paper says “by the projection theorem” about a non-product set, it means the theorem itself, not the arithmetic reading.

*The exercise.* Verify  $A + A = [0, 2]$  for the middle-thirds Cantor set by writing  $x \in [0, 2]$  in base 3 and splitting each digit 0, 1, 2 as a sum of two digits from  $\{0, 2\}$ . Then check that  $A + \frac{1}{2}A$  has measure zero, so that the behaviour genuinely depends on  $\lambda$ .

---

**Proposition 8.3.4** (Slices of a product). *Let  $A, B \subseteq \mathbb{R}$  be compact and  $\lambda \neq 0$ . Then for every  $c \in \mathbb{R}$  the slice*

$$(A \times B) \cap \{(x, y) : x + \lambda y = c\}$$

*is, under the bi-Lipschitz map  $(x, y) \mapsto y$ , in bijection with  $B \cap \lambda^{-1}(c - A)$ . Consequently*

$$\dim_{\mathrm{H}}(A \cap (c - \lambda B)) \leq \dim_{\mathrm{H}}(A \times B) - \dim_{\mathrm{H}} \pi_{\theta}(A \times B)$$

*for those  $c$  attaining the maximum, where  $\theta$  corresponds to  $\lambda$  by Proposition 8.3.1.*

*Proof.* A point of the slice satisfies  $x = c - \lambda y$  with  $x \in A$ ,  $y \in B$ ; so  $y$  ranges over  $\{y \in B : c - \lambda y \in A\} = B \cap \lambda^{-1}(c - A)$ , and the map  $(x, y) \mapsto y$  is a bijection onto that set, bi-Lipschitz with constants depending on  $\lambda$ .

For the inequality, note the slices are the fibres of  $\pi_{\theta}$  up to the bi-Lipschitz identification, and apply Proposition 7.1.7 with  $m = 1$ : almost every fibre has dimension at most  $\dim_{\mathrm{H}}(A \times B) - 1$ , and comparing with the dimension of the image gives the stated bound at the values of  $c$  carrying the most.  $\square$

**Where we are.** So a projection theorem is simultaneously a statement about *slices*: if the image is large then the fibres are small, and conversely. The two readings are exchanged by Proposition 7.1.7, the Fubini-type bound proved in Volume 7.

This is worth holding onto because the Kakeya problem is naturally a slicing problem — a Besicovitch set has a segment, that is a large slice, in

every direction — and the projection theory of this chapter is what the slicing reading connects it to. Volume 14's Furstenberg sets are defined by prescribing the dimension of slices, and sit exactly between the two subjects.

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WORKED EXAMPLE 8.3.5— TIER C: RESEARCH TASTE **Three ways a Kakeya set can be built, and what each costs**

This chapter analysed one construction of a Besicovitch set and found it wanting. It is worth setting the three known approaches side by side, since each recurs later in the series.

(a) *The Perron tree (Volume 1)*. Bisect a triangle repeatedly and slide the halves together. Gives measure less than any  $\varepsilon$ , is completely explicit, and the sets are finite unions of triangles. *Cost*: the sets at successive stages do not nest, so no limit of measure zero is obtained — the failure diagnosed in Volume 7, Chapter 7.3. *Where it recurs*: Volume 3's ball multiplier theorem uses it directly, and Volume 9's tube combinatorics is its systematic form.

(b) *The Cantor-set construction (this chapter)*. Join  $(a, 0)$  to  $(b, 1)$  for  $a, b$  in a Cantor set  $A$  with  $A - A \supseteq [-1, 1]$ . Gives all directions provided the difference set is full, and reduces the measure to a sumset question by Fubini. *Cost*: the two requirements conflict (Remark 8.3.3), and resolving them needs the Besicovitch projection theorem. *Where it recurs*: Volume 10's arithmetic method is this construction read additively.

(c) *The random construction*. Choose the tubes' positions at random subject to the direction constraint. Gives, with high probability, sets whose dimension can be estimated by second-moment methods, and — unlike (a) and (b) — produces Salem-type behaviour so that Fourier methods are sharp on them. *Cost*: random sets are *not* extremal for Kakeya; they are too large. The extremal configurations are arithmetic, which is why randomness has never given a counterexample. *Where it recurs*: Volume 7's remark on Salem sets, and the heuristics of Volume 14.

*The lesson*. All three constructions produce sets of small measure and none produces one of small *dimension*. That is the empirical content of the Kakeya conjecture: fifty years of attempts to build a small Besicovitch set have produced only sets of full dimension, and the conjecture asserts that this is necessary. Volume 10's finite-field analogue, where Dvir proved it outright, is the strongest evidence available.

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## Drills

**Drill 8.3.1.** Verify Proposition 8.3.1 and check the bi-Lipschitz constants as  $\theta \rightarrow 0$ .

**Drill 8.3.2.** Show that for the self-similar  $A$  of ratio  $r$ ,  $A - A$  contains an interval if and only if  $r \geq \frac{1}{3}$ , by examining the base- $1/r$  expansions.

**Drill 8.3.3.** Compute  $\dim_{\mathbb{H}} E$  for the set  $E$  of Proposition 8.3.2 with  $A$  the middle-thirds Cantor set, and compare with 2.

**Drill 8.3.4.** Confirm Worked Example 8.3.1: apply Theorem 7.8.3 to the four-corner system and check the open set condition holds with  $V$  the open unit square.

## Notes

Kahane's reduction is from 1969. That a Besicovitch set of measure zero exists is Besicovitch's (1928); the modern treatments either follow his original nested construction or use the projection route sketched here, and both are longer than anything else in this volume.

The Besicovitch projection theorem is from 1939, with the modern proof due to Federer. Mattila's book, Chapters 15–18, is the reference.

# Chapter 8.4

## The Distance Measure

**Definition 8.4.1.** For compact  $E \subseteq \mathbb{R}^n$  the *distance set* is

$$\Delta(E) = \{|x - y| : x, y \in E\} \subseteq [0, \infty).$$

**Conjecture 8.4.2** (Falconer). *For  $n \geq 2$  and compact  $E \subseteq \mathbb{R}^n$  with  $\dim_{\text{H}} E > \frac{n}{2}$ , the distance set  $\Delta(E)$  has positive Lebesgue measure.*

The exponent  $\frac{n}{2}$  is forced by an example: a suitably chosen set of dimension  $\frac{n}{2}$  has a distance set of measure zero (Drill 4.4). This chapter builds the object through which the conjecture is studied, and Chapter 8.5 proves Falconer's theorem at the exponent  $\frac{n+1}{2}$ .

### 8.4.1 From a set to a measure

**Definition 8.4.3.** Let  $\mu$  be a finite Borel measure with compact support. The *distance measure*  $\nu = \nu_\mu$  is the pushforward of  $\mu \times \mu$  under  $(x, y) \mapsto |x - y|$ ; that is,

$$\int_0^\infty f \, d\nu = \iint f(|x - y|) \, d\mu(x) \, d\mu(y)$$

for every bounded Borel  $f$ .

**Proposition 8.4.4.**  *$\nu$  is a finite Borel measure on  $[0, \infty)$  with  $\nu([0, \infty)) = \mu(\mathbb{R}^n)^2$ , supported in  $\Delta(\text{spt } \mu)$ . If  $\nu$  is absolutely continuous with respect to Lebesgue measure, then  $m(\Delta(\text{spt } \mu)) > 0$ .*

*Proof.* That  $\nu$  is a finite Borel measure of the stated total mass is immediate from the definition with  $f \equiv 1$ . Its support: if  $t \notin \Delta(\text{spt } \mu)$  then,  $\Delta(\text{spt } \mu)$  being compact — the continuous image of the compact  $\text{spt } \mu \times \text{spt } \mu$  — there is a neighbourhood of  $t$  disjoint from it, and  $\nu$  of that neighbourhood is 0.

For the last statement, suppose  $m(\Delta(\text{spt } \mu)) = 0$ . Since  $\nu$  is carried by that set and  $\nu \ll m$ , we get  $\nu \equiv 0$ , contradicting  $\nu([0, \infty)) = \mu(\mathbb{R}^n)^2 > 0$ .  $\square$

So the conjecture reduces to showing  $\nu \ll m$ , and the route is to exhibit a bounded density. The next section expresses  $\nu$  through the Fourier transform, which is where the curvature of the sphere enters.

## 8.4.2 The Fourier representation

**Definition 8.4.5.** For  $t > 0$  and  $0 < \varepsilon < t/2$  let

$$A_\varepsilon^t = \{z \in \mathbb{R}^n : t - \varepsilon < |z| < t + \varepsilon\}$$

be the open annulus of radius  $t$  and thickness  $2\varepsilon$ , and set

$$\nu_\varepsilon(t) = \frac{1}{2\varepsilon} (\mu \times \mu) \{(x, y) : x - y \in A_\varepsilon^t\}.$$

**Lemma 8.4.6.** For every  $t > 0$ ,

$$\nu_\varepsilon(t) = \frac{1}{2\varepsilon} \int_{\mathbb{R}^n} \widehat{\mathbf{1}_{A_\varepsilon^t}}(\xi) |\widehat{\mu}(\xi)|^2 d\xi,$$

the integral converging absolutely, and  $\int_0^\infty \nu_\varepsilon(t) f(t) dt \rightarrow \int f d\nu$  as  $\varepsilon \downarrow 0$  for every  $f \in C_c((0, \infty))$ .

*Proof. Step 1. The identity.*

Since  $A_\varepsilon^t$  is bounded and symmetric ( $-A_\varepsilon^t = A_\varepsilon^t$ ), the function  $\mathbf{1}_{A_\varepsilon^t}$  lies in  $L^1 \cap L^2$  and has real even Fourier transform. By Parseval for measures — Theorem 2.5.11, the measure  $\mu$  being finite with compact support so that  $\widehat{\mu}$  is bounded and continuous —

$$(\mu \times \mu) \{(x, y) : x - y \in A_\varepsilon^t\} = \iint \mathbf{1}_{A_\varepsilon^t}(x - y) d\mu(x) d\mu(y) = \int_{\mathbb{R}^n} \widehat{\mathbf{1}_{A_\varepsilon^t}}(\xi) |\widehat{\mu}(\xi)|^2 d\xi.$$

Absolute convergence:  $\widehat{\mathbf{1}_{A_\varepsilon^t}} \in L^2$  by Plancherel and  $|\widehat{\mu}|^2$  is bounded, while for large  $|\xi|$  the first factor decays — see Lemma 8.4.7 below.

**Step 2. Convergence to  $\nu$ .**

For  $f \in C_c((0, \infty))$ , Tonelli gives

$$\int_0^\infty \nu_\varepsilon(t) f(t) dt = \iint \left( \frac{1}{2\varepsilon} \int_0^\infty f(t) \mathbf{1}_{\{t - \varepsilon < |x - y| < t + \varepsilon\}} dt \right) d\mu d\mu = \iint f_\varepsilon(|x - y|) d\mu d\mu,$$

where  $f_\varepsilon(r) = \frac{1}{2\varepsilon} \int_{r - \varepsilon}^{r + \varepsilon} f$  is the average of  $f$  over an  $\varepsilon$ -interval. Since  $f$  is uniformly continuous,  $f_\varepsilon \rightarrow f$  uniformly, and  $\mu \times \mu$  is finite, so the right side converges to  $\iint f(|x - y|) d\mu d\mu = \int f d\nu$ .  $\square$



**Lemma 8.4.7** (Decay of the annulus transform). *Fix  $0 < a < b < \infty$ . There is  $C = C(n, a, b)$  such that for all  $t \in [a, b]$ ,  $0 < \varepsilon < a/2$  and  $\xi \in \mathbb{R}^n$ ,*

$$|\widehat{\mathbf{1}_{A_\varepsilon^t}}(\xi)| \leq C \varepsilon (1 + |\xi|)^{-\frac{n-1}{2}}.$$

*Proof.* Writing the annulus in polar coordinates (Theorem 1.6.11),

$$\widehat{\mathbf{1}_{A_\varepsilon^t}}(\xi) = \int_{t-\varepsilon}^{t+\varepsilon} r^{n-1} \widehat{d\sigma}(r\xi) dr,$$

where  $d\sigma$  is surface measure on the unit sphere and  $\widehat{d\sigma}(r\xi)$  is its transform evaluated at  $r\xi$ , by the change of variables  $z = r\omega$ .

By Theorem 5.5.2 — the sphere has non-vanishing Gaussian curvature —  $|\widehat{d\sigma}(\eta)| \lesssim_n (1 + |\eta|)^{-(n-1)/2}$ . For  $r \in [t-\varepsilon, t+\varepsilon] \subseteq [a/2, 2b]$  we have  $|r\xi| \approx_{a,b} |\xi|$ , so  $|\widehat{d\sigma}(r\xi)| \lesssim_{n,a,b} (1 + |\xi|)^{-(n-1)/2}$ , and  $r^{n-1} \leq (2b)^{n-1}$ . Integrating over an interval of length  $2\varepsilon$  gives the bound.  $\square$

**Where we are.** Lemma 8.4.7 is where the geometry enters, and it is the only place. The factor  $\varepsilon$  is the thickness of the annulus and is what makes  $\nu_\varepsilon = \frac{1}{2\varepsilon}(\cdots)$  bounded as  $\varepsilon \rightarrow 0$ ; the factor  $(1 + |\xi|)^{-(n-1)/2}$  is the curvature of the sphere, imported from Volume 5.

Had the “distance” been measured by a norm whose unit ball has a flat piece — the  $\ell^\infty$  norm, say — the second factor would be absent, the argument of Chapter 8.5 would give nothing, and indeed the corresponding distance-set problem has a different answer. Drill 4.3 works out the  $\ell^\infty$  case, where the conjecture is false as stated.

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#### WORKED EXAMPLE 8.4.1— TIER A **The trivial bounds**

Before proving anything it is worth knowing what is free.

*Upper bound on the dimension needed.* If  $\dim_{\mathbb{H}} E > \frac{n}{2}$  is conjectured to suffice, no smaller exponent can: there are compact sets of dimension exactly  $\frac{n}{2}$  with  $m(\Delta(E)) = 0$ , obtained by taking  $E$  a product of a suitable Cantor set with itself (Drill 4.4).

*Lower bound, for free.* If  $\dim_{\mathbb{H}} E > 0$  then  $\Delta(E)$  is uncountable. Indeed the map  $y \mapsto |x - y|$  is Lipschitz for fixed  $x$ , so  $\dim_{\mathbb{H}} \Delta(E) \geq \dim_{\mathbb{H}} \{|x - y| : y \in E\}$ , and choosing  $x$  suitably one shows this is positive. So the content of the conjecture is entirely in passing from positive dimension to positive *measure*.

*The gap.* Between “uncountable” and “positive measure” lies the whole problem, and Proposition 8.4.4 says the way across is to produce a measure on  $\Delta(E)$  that is absolutely continuous. That is what  $\nu$  is for.

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**WORKED EXAMPLE 8.4.2— TIER B Why one mollifies, and what would go wrong otherwise**

It is tempting to write, formally,

$$“ \nu(t) = \iint d\sigma_t(x-y) d\mu(x) d\mu(y) = \int \widehat{d\sigma_t}(\xi) |\widehat{\mu}(\xi)|^2 d\xi ”$$

and estimate directly. The formula is the right heuristic and is how the argument is usually described, but as written it is meaningless:  $d\sigma_t$  is a measure,  $\mu \times \mu$  may charge the set  $\{|x-y| = t\}$ , and the pairing is undefined.

*What the mollification fixes.* Lemma 8.4.6 replaces  $d\sigma_t$  by  $\frac{1}{2\varepsilon} \mathbf{1}_{A_\varepsilon^t}$ , an honest  $L^1$  function, and every step is then a genuine identity. The bound of Lemma 8.4.7 is uniform in  $\varepsilon$ , so a bound on  $\nu_\varepsilon$  passes to  $\nu$  in the limit.

*The general lesson.* Whenever a proof pairs a singular measure against another singular measure, the pairing must be defined by mollification and the estimates made uniform. The same issue arose in Theorem 6.1.3, where  $\mathcal{E}g$  paired against  $f$  was defined by extension from a dense class, and in Proposition 6.7.3, where  $gd\sigma$  was smoothed at scale  $\delta$ . It is worth being suspicious of any argument in this area that pairs two measures without saying how.

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**WORKED EXAMPLE 8.4.3— TIER C: RESEARCH TASTE Why the distance set problem sits between two subjects**

The problem was posed in geometric measure theory and is now solved with harmonic analysis. Both readings are useful and they meet in Lemma 8.4.6.

(a) *The geometric reading.*  $\Delta(E)$  is the image of  $E \times E$  under the distance map. Asking whether it has positive measure is asking whether that map is, in an averaged sense, non-degenerate on a set of the given dimension — a question about how the fibres  $\{(x, y) : |x - y| = t\}$  meet  $E \times E$ . Each fibre is a hypersurface in  $\mathbb{R}^{2n}$  of codimension 1, so one is intersecting a set of dimension  $2s$  with a family of codimension-1 surfaces and asking that the intersections be non-trivial for a positive measure of parameters. That is a slicing problem, and Proposition 7.1.7 is its elementary form.

(b) *The analytic reading.* By Lemma 8.4.6, the density of  $\nu$  is a weighted integral of  $|\widehat{\mu}|^2$  against  $\widehat{d\sigma}$ . The problem becomes: how much can a fractal measure's transform correlate with the transform of surface measure? That is an extension-operator question, and it is why Chapter 8.7's improvements are all restriction estimates.

(c) *What the two readings each explain.* The geometric reading explains the conjectured exponent:  $2s - (2n - 1) \geq \text{something}$ , or more simply the counting

in Worked Example 8.5.3(b). The analytic reading explains the *proofs*: every one of them bounds  $\sigma(\mu, r)$ .

(d) *The pattern across this series.* Kakeya was posed geometrically and is attacked combinatorially; restriction was posed analytically and is attacked geometrically; the distance problem was posed geometrically and is attacked analytically. In each case the productive move was to change the category the problem was stated in. Lemma 8.4.6 is that move here, and it is worth noting that it consists of nothing but Parseval.

**WORKED EXAMPLE 8.4.4— TIER B Mollification with uniform bounds: how it is actually done**

Worked Example 8.4.2 explained why one mollifies. This example rehearses the mechanics, since every argument in the next three chapters uses them and the bookkeeping is where errors hide.

*The three requirements.* To prove a measure  $\nu$  has a bounded density one must supply:

- (1) a family  $\nu_\varepsilon$  of honest functions with  $\int f \nu_\varepsilon \rightarrow \int f d\nu$  for test functions  $f$ ;
- (2) a bound  $|\nu_\varepsilon(t)| \leq M$  uniform in  $\varepsilon$  and in  $t$  on a compact interval;
- (3) the deduction that  $\nu \leq Mm$  there.

Requirement (2) is the mathematics; (1) and (3) are bookkeeping that must nonetheless be done.

*How (1) is arranged here.* Lemma 8.4.6 takes  $\nu_\varepsilon(t) = \frac{1}{2\varepsilon}(\mu \times \mu)\{x - y \in A_\varepsilon^t\}$ , and the convergence is proved by Tonelli: the test integral becomes  $\iint f_\varepsilon(|x - y|) d\mu d\mu$  with  $f_\varepsilon$  the  $\varepsilon$ -average of  $f$ , and  $f_\varepsilon \rightarrow f$  uniformly since  $f$  is uniformly continuous. Note that this needs  $f$  compactly supported in  $(0, \infty)$  — at  $t = 0$  the annulus degenerates and the argument fails, which is why the interval  $[a, b]$  is taken away from the origin.

*How (3) is arranged.* From (2),  $\int f d\nu \leq M \int f$  for nonnegative  $f \in C_c((a, b))$ . A positive linear functional dominated by  $M$  times Lebesgue integration is represented by a measure  $\leq Mm$ , by Theorem 1.8.15 and uniqueness of the representing measure. It is worth noting that this step gives  $\nu \leq Mm$  as *measures*, which is stronger than absolute continuity and is what Chapter 8.6 builds on.

*The error to avoid.* Bounding  $\nu_\varepsilon(t)$  for each fixed  $\varepsilon$  with a constant depending on  $\varepsilon$  proves nothing: the bound must survive  $\varepsilon \rightarrow 0$ . In Theorem 8.5.1 the

factor  $\varepsilon$  from Lemma 8.4.7 cancels the  $\frac{1}{2\varepsilon}$  exactly, and *that cancellation is the whole point of the estimate*. Any mollification argument should be read by first checking where the  $\varepsilon$ 's cancel.

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**Proposition 8.4.8** (The distance measure is non-atomic). *Let  $\mu$  be a finite compactly supported measure with  $I_s(\mu) < \infty$  for some  $s > 0$ . Then the distance measure  $\nu$  has no atoms in  $(0, \infty)$ , and  $\nu(\{0\}) = 0$ .*

*Proof.* For  $t = 0$ :  $\nu(\{0\}) = (\mu \times \mu)\{x = y\}$ , and  $I_s(\mu) < \infty$  forces  $\mu$  to be non-atomic (Worked Example 7.6.4, Case 3), so the diagonal has  $\mu \times \mu$ -measure zero by Fubini.

For  $t > 0$ :  $\nu(\{t\}) = (\mu \times \mu)\{|x - y| = t\}$ , the mass carried by a sphere bundle. Fix  $x$ ; the set  $\{y : |x - y| = t\}$  is the sphere  $S(x, t)$ , and

$$\mu(S(x, t)) \leq \lim_{\varepsilon \downarrow 0} \mu(B(x, t + \varepsilon) \setminus B(x, t - \varepsilon)).$$

Since  $I_s(\mu) < \infty$ , by Step 1 of Theorem 7.6.2 we have  $\mu(B(z, r)) \lesssim r^s$  for  $\mu$ -a.e.  $z$  and small  $r$ ; covering the sphere  $S(x, t)$  by  $\approx \varepsilon^{1-n}$  balls of radius  $\varepsilon$  gives  $\mu(S(x, t)) \lesssim \lim_{\varepsilon} \varepsilon^{1-n} \varepsilon^s$ , which is 0 provided  $s > n - 1$ . For smaller  $s$  one argues instead that the map  $y \mapsto |x - y|$  pushes  $\mu$  to a measure of finite energy on the line, by Proposition 7.6.3(i) applied on the complement of a small ball about  $x$ , and a measure of finite positive energy on  $\mathbb{R}$  is non-atomic. Integrating in  $x$  gives  $\nu(\{t\}) = 0$ .  $\square$

**Where we are.** Proposition 8.4.8 is what licenses the choice of the interval  $[a, b]$  with  $\nu([a, b]) > 0$  in Theorem 8.5.1 Step 1: were  $\nu$  a point mass at 0, no such interval away from the origin would exist and the argument would have nothing to work with.

It also shows how little is needed: any positive amount of dimension makes the distance measure non-atomic. The whole difficulty of Conjecture 8.4.2 is in the gap between “non-atomic” — free — and “absolutely continuous” — which needs  $\dim_{\mathbb{H}} E > \frac{n}{2}$ .

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**WORKED EXAMPLE 8.4.5— TIER A Computing  $\Delta(E)$  for sets one can see**

Three examples, in increasing difficulty, to fix what the conjecture is about.

(a) *A segment.*  $E = [0, 1] \times \{0\} \subseteq \mathbb{R}^2$  has  $\dim_{\mathbb{H}} E = 1$  and  $\Delta(E) = [0, 1]$  — positive measure, indeed an interval. So a set of dimension 1 can have a large distance set, and Conjecture 8.4.2 at  $n = 2$  asks for  $\dim_{\mathbb{H}} E > 1$ ; the segment shows the conjecture is not *necessary*, only sufficient.

(b) *A Cantor set on a line.*  $E = C \times \{0\}$  with  $C$  the middle-thirds set has

$\dim_{\mathbb{H}} E = \log 2 / \log 3 \approx 0.63$ , and  $\Delta(E) = |C - C| = [0, 1]$ , since  $C - C = [-1, 1]$ . Again positive measure, from a set of dimension well below the threshold. So low dimension does not preclude a large distance set — what the conjecture rules out is the *reverse* implication failing.

(c) *The extremal example.* By Remark 8.8.1 there are sets of dimension exactly  $\frac{n}{2}$  with  $m(\Delta(E)) = 0$ . Comparing with (a) and (b): those sets are arithmetically special, arranged so that all distances fall near a lattice. Genericity, not size, is what (a) and (b) have and the extremal example lacks.

*What the three show together.* The conjecture is not about size alone. It asserts that *above* dimension  $\frac{n}{2}$ , no arrangement — however arithmetically clever — can keep the distance set null; and Remark 8.8.1 says that at  $\frac{n}{2}$  exactly, one can. The difficulty is that the hypothesis is purely metric while the extremal obstruction is arithmetic, so a proof must rule out arithmetic structure using only a dimension bound. That is what the Fourier transform does, and why it is the natural tool here.

**Figure 8.4.1 (to be drawn).** *The annulus and its dual*

**Content required:** The annulus  $A_\varepsilon^t$  of radius  $t$  and thickness  $2\varepsilon$ , with the pair  $(x, y)$  realising a distance  $|x - y| \in (t - \varepsilon, t + \varepsilon)$  shown as a chord. Beside it, the profile of  $\mathbf{1}_{A_\varepsilon^t}$  against  $|\xi|$ , decaying like  $\varepsilon(1 + |\xi|)^{-(n-1)/2}$ , with the factor  $\varepsilon$  marked as the thickness and the exponent marked as the curvature of the sphere.

## Drills

**Drill 8.4.1.** Verify that  $\Delta(E)$  is compact for compact  $E$ , and that  $\Delta(E) \subseteq [0, \text{diam } E]$ .

**Drill 8.4.2.** Check the polar-coordinate identity for  $\widehat{\mathbf{1}_{A_\varepsilon^t}}$  in Lemma 8.4.7.

**Drill 8.4.3.** Redo Lemma 8.4.7 with the Euclidean norm replaced by the  $\ell^\infty$  norm, whose unit sphere is the boundary of a cube. Show the decay factor is absent in the coordinate directions, and explain why the method of Chapter 8.5 then gives nothing.

**Drill 8.4.4.** Let  $A \subseteq [0, 1]$  be a Cantor set of dimension  $\frac{1}{2}$  built on a rapidly increasing sequence of scales, and  $E = A \times A \subseteq \mathbb{R}^2$ . Show  $\dim_{\mathbb{H}} E = 1 = \frac{n}{2}$  and sketch why  $m(\Delta(E))$  can be made zero, establishing sharpness of Conjecture 8.4.2.

## Notes

Falconer posed the problem in 1985. The reduction to the absolute continuity of  $\nu$  is his; the mollified formulation of Lemma 8.4.6 is the standard way to make the heuristic rigorous and appears in essentially this form in Wolff's lecture notes.

The sharpness example of Drill 4.4 is Falconer's, and shows the conjectured exponent  $\frac{n}{2}$  cannot be lowered.

## Chapter 8.5

# Falconer's Theorem

**Theorem 8.5.1** (Falconer). *Let  $n \geq 2$  and let  $E \subseteq \mathbb{R}^n$  be compact with*

$$\dim_{\mathbb{H}} E > \frac{n+1}{2}.$$

*Then  $m(\Delta(E)) > 0$ .*

*Idea of the proof.* By Proposition 8.4.4 it suffices to show the distance measure  $\nu$  is absolutely continuous, and by Lemma 8.4.6 the approximating densities are

$$\nu_\varepsilon(t) = \frac{1}{2\varepsilon} \int_{\mathbb{R}^n} \widehat{\mathbf{1}_{A_\varepsilon^t}}(\xi) |\widehat{\mu}(\xi)|^2 d\xi.$$

Two estimates meet here, and the theorem is exactly the point at which they balance.

*From geometry:*  $|\widehat{\mathbf{1}_{A_\varepsilon^t}}(\xi)| \lesssim \varepsilon(1 + |\xi|)^{-(n-1)/2}$ , by Lemma 8.4.7 — the curvature of the sphere, imported from Volume 5.

*From dimension:*  $\int |\widehat{\mu}(\xi)|^2 |\xi|^{s-n} d\xi < \infty$  whenever  $I_s(\mu) < \infty$ , by Proposition 7.6.5 — the Fourier expression of energy, from Volume 7.

The first supplies a decay  $(1 + |\xi|)^{-(n-1)/2}$ ; the second is exactly what is needed to integrate  $|\widehat{\mu}|^2$  against a weight  $(1 + |\xi|)^{s-n}$ . So the argument closes as soon as

$$-\frac{n-1}{2} \leq s-n, \quad \text{that is} \quad s \geq n - \frac{n-1}{2} = \frac{n+1}{2},$$

and the exponent in the theorem is nothing but this comparison. The factor  $\varepsilon$  cancels the  $\frac{1}{2\varepsilon}$ , leaving a bound independent of  $\varepsilon$ .

**Proof. Step 1.** *Choose the measure.*

Let  $s$  satisfy  $\frac{n+1}{2} < s < \dim_{\mathbb{H}} E$ . By Theorem 7.6.2(ii) there is a nonzero finite Borel measure  $\mu$  carried by  $E$  with  $I_s(\mu) < \infty$ ; normalise  $\mu(E) = 1$ .

Since  $E$  is compact,  $\text{spt } \mu \subseteq E$  is compact, so  $\widehat{\mu}$  is a bounded continuous function and  $\Delta(\text{spt } \mu) \subseteq \Delta(E)$ .

Fix  $0 < a < b < \infty$  with  $\nu([a, b]) > 0$ ; such an interval exists because  $\nu$  is a nonzero finite measure on  $[0, \text{diam } E]$  and  $\nu(\{0\}) = (\mu \times \mu)\{x = y\} = 0$ , the latter because  $I_s(\mu) < \infty$  forces  $\mu$  to be non-atomic (Worked Example 7.6.4, Case 3).

**Step 2.** *Bound  $\nu_\varepsilon$  uniformly.*

Let  $t \in [a, b]$  and  $0 < \varepsilon < a/2$ . By Lemma 8.4.6 and Lemma 8.4.7,

$$|\nu_\varepsilon(t)| \leq \frac{1}{2\varepsilon} \int_{\mathbb{R}^n} |\widehat{1_{A_\varepsilon^t}}(\xi)| |\widehat{\mu}(\xi)|^2 d\xi \leq \frac{C\varepsilon}{2\varepsilon} \int_{\mathbb{R}^n} (1 + |\xi|)^{-\frac{n-1}{2}} |\widehat{\mu}(\xi)|^2 d\xi,$$

with  $C = C(n, a, b)$ . The factor  $\varepsilon$  has cancelled, so it remains to bound the integral by a finite quantity independent of  $t$  and  $\varepsilon$  — which it already is, no  $t$  or  $\varepsilon$  appearing in it.

**Step 3.** *The integral is finite.*

Split at  $|\xi| = 1$ . On  $|\xi| \leq 1$  the weight is bounded by 1 and  $|\widehat{\mu}| \leq \mu(\mathbb{R}^n) = 1$ , so that part contributes at most  $m(B(0, 1))$ .

On  $|\xi| > 1$  we use  $s \geq \frac{n+1}{2}$ , which gives  $-\frac{n-1}{2} \leq s - n$  and hence

$$(1 + |\xi|)^{-\frac{n-1}{2}} \leq |\xi|^{-\frac{n-1}{2}} \leq |\xi|^{s-n} \quad (|\xi| > 1),$$

the last step because  $|\xi| > 1$  and the exponent on the right is the larger. Therefore

$$\int_{|\xi|>1} (1 + |\xi|)^{-\frac{n-1}{2}} |\widehat{\mu}(\xi)|^2 d\xi \leq \int_{\mathbb{R}^n} |\widehat{\mu}(\xi)|^2 |\xi|^{s-n} d\xi = c_{n,s}^{-1} I_s(\mu) < \infty,$$

by Proposition 7.6.5. Combining, there is  $M = M(n, a, b, \mu) < \infty$  with

$$|\nu_\varepsilon(t)| \leq M \quad \text{for all } t \in [a, b], \quad 0 < \varepsilon < a/2. \quad (8.5.1)$$

**Step 4.** *Pass to the limit.*

Let  $f \in C_c((a, b))$  with  $f \geq 0$ . By Lemma 8.4.6 and (8.5.1),

$$\int f d\nu = \lim_{\varepsilon \downarrow 0} \int_a^b \nu_\varepsilon(t) f(t) dt \leq M \int_a^b f(t) dt.$$

So the linear functional  $f \mapsto \int f d\nu$  is dominated by  $M$  times Lebesgue integration on  $(a, b)$ . By the Riesz representation theorem (Theorem 1.8.15) and the uniqueness of the representing measure, this forces  $\nu \upharpoonright (a, b) \leq Mm$ ; in particular  $\nu \upharpoonright (a, b) \ll m$ .



**Step 5. Conclude.**

If  $m(\Delta(E)) = 0$  then,  $\nu$  being carried by  $\Delta(\text{spt } \mu) \subseteq \Delta(E)$  and absolutely continuous on  $(a, b)$ , we get  $\nu((a, b)) = 0$ . Choosing  $[a, b]$  as in Step 1 with  $\nu([a, b]) > 0$ , and shrinking slightly so that the open interval still carries positive mass, gives a contradiction. Hence  $m(\Delta(E)) > 0$ .  $\square$

**Where we are.** The proof used exactly two inputs, one from each of the two preceding volumes, and it is worth seeing that the exponent is the ratio between them.

Volume 5 gave decay  $(n-1)/2$ : the number of curved directions of the sphere, halved. Volume 7 gave that a set of dimension  $s$  supports a measure whose transform is square-integrable against  $|\xi|^{s-n}$ . The theorem holds when the first exponent is at least  $n-s$ , that is when  $s \geq n - \frac{n-1}{2}$ .

So the  $\frac{n+1}{2}$  is not a technical artefact: it is where the curvature of the sphere runs out. To do better one must extract more from  $\widehat{\mu}$  than its weighted  $L^2$  norm — and that is precisely what the Mattila integral of Chapter 8.7 attempts, by keeping the *spherical averages* of  $\widehat{\mu}$  rather than bounding  $\widehat{d\sigma}$  pointwise.

**Pitfall: this is not the Mattila integral, and the difference matters**

The reader who consults the literature will find Falconer's theorem attributed both to the argument above and to "Mattila's approach". They are different, and the second does *not* prove the theorem.

Mattila's method estimates  $\nu$  in  $L^2$  rather than  $L^\infty$ , and reduces the problem to the finiteness of

$$\mathcal{M}(\mu) = \int_1^\infty \left( \int_{\mathbb{S}^{n-1}} |\widehat{\mu}(r\omega)|^2 d\sigma(\omega) \right)^2 r^{n-1} dr.$$

Suppose one tries to bound the inner spherical average using only Theorem 5.5.2, as in Step 3 above. Writing  $\sigma(\mu, r) = \int_{\mathbb{S}^{n-1}} |\widehat{\mu}(r\omega)|^2 d\sigma$ , one obtains  $\sigma(\mu, r) \lesssim r^{-\min\{s, (n-1)/2\}}$  (Drill 5.3), so

$$\mathcal{M}(\mu) \lesssim \int_1^\infty r^{-2\min\{s, \frac{n-1}{2}\} + n-1} dr,$$

which converges only if  $2\min\{s, \frac{n-1}{2}\} > n$ . But  $\min\{s, \frac{n-1}{2}\} \leq \frac{n-1}{2} < \frac{n}{2}$  *always*. So this route never converges, in any dimension and for any  $s$ .

That is not a failure of effort: the Mattila integral genuinely diverges in part of the range, and progress by that method requires *improved spherical average estimates*, which are theorems about the restriction operator rather than consequences of  $\widehat{d\sigma}$ . Chapter 8.7 sets this out. A reader attempting

Falconer's theorem through  $\mathcal{M}$  will not succeed, and should recognise the obstruction rather than look for a missing trick.

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**WORKED EXAMPLE 8.5.1— TIER A The exponent in low dimensions**

$n = 2$ . Theorem 8.5.1 requires  $\dim_{\mathbb{H}} E > \frac{3}{2}$ ; the conjecture asks for  $> 1$ . The gap is  $\frac{1}{2}$ , and the current record —  $\frac{5}{4}$ , due to Guth, Iosevich, Ou and Wang — closes half of it.

$n = 3$ . The theorem gives 2; the conjecture asks for  $\frac{3}{2}$ ; and Erdoğan's bound  $\frac{n}{2} + \frac{1}{3}$  is  $\frac{11}{6} \approx 1.833$ . So of the gap  $2 - \frac{3}{2} = \frac{1}{2}$ , one third has been closed.

*Large  $n$ .* The theorem gives  $\frac{n+1}{2}$  and the conjecture  $\frac{n}{2}$ : the gap is  $\frac{1}{2}$ , independent of dimension. So Falconer's theorem is, in absolute terms, always within  $\frac{1}{2}$  of the truth — which is a much better state of affairs than for Kakeya, where the gap grows.

*A consistency check.* At  $n = 1$  the theorem would read  $\dim_{\mathbb{H}} E > 1$ , which is impossible in  $\mathbb{R}$ ; and indeed the distance set problem is trivial there,  $\Delta(E)$  being essentially  $E - E$ . The curvature that drives the proof does not exist in one dimension, consistent with  $\widehat{d\sigma}$  having no decay when  $\mathbb{S}^0$  is two points.

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**WORKED EXAMPLE 8.5.2— TIER B Where each hypothesis is used, and what happens without it**

*Compactness of  $E$*  entered twice: to apply Theorem 7.5.1, which is stated for compact sets, and to know  $\Delta(E)$  is compact so that  $\nu$  has compact support. For Borel  $E$  one applies the theorem to compact subsets of dimension close to  $\dim_{\mathbb{H}} E$ , which exist by inner regularity.

*Non-vanishing curvature of the sphere* entered once, in Lemma 8.4.7. Replacing the Euclidean distance by a norm whose unit ball has a flat face — the  $\ell^\infty$  norm, say — removes the decay in the directions normal to the flat face, and the argument collapses. Indeed for  $\ell^\infty$  the analogue of Conjecture 8.4.2 is false: a product of Cantor sets of dimension slightly above  $\frac{n}{2}$  can have  $\ell^\infty$ -distance set of measure zero, because the  $\ell^\infty$  distance depends on only one coordinate at a time.

*The strict inequality  $\dim_{\mathbb{H}} E > \frac{n+1}{2}$*  is needed because Theorem 7.6.2(ii) produces a measure of finite  $s$ -energy only for  $s < \dim_{\mathbb{H}} E$ . At equality the energy may be infinite and the argument gives nothing; whether the theorem holds at  $\dim_{\mathbb{H}} E = \frac{n+1}{2}$  is a separate question, and the analogous endpoint for the conjecture at  $\frac{n}{2}$  is known to be false.

*What was not used:* any structure of  $E$  beyond its dimension. The proof is insensitive to whether  $E$  is self-similar, random, or arbitrary — and that is precisely its limitation. The improvements of Chapter 8.7 all proceed by

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extracting more from the specific structure of  $\widehat{\mu}$ .

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**WORKED EXAMPLE 8.5.3— TIER C: RESEARCH TASTE    What the proof reveals about the shape of the problem**

(a) *The problem is a restriction problem in disguise.* The quantity controlling everything is  $\sigma(\mu, r)$ , the average of  $|\widehat{\mu}|^2$  over the sphere of radius  $r$ . Bounding it is exactly a weighted restriction estimate for the sphere: one is asking how much of  $\widehat{\mu}$  can concentrate near a sphere. Volume 6 studied precisely that, and the improvements in Chapter 8.7 import restriction technology directly.

(b) *The conjectured exponent  $n/2$  is where dimension and codimension balance.* A measure of dimension  $s$  has  $\widehat{\mu}$  decaying, in an  $L^2$ -averaged sense, at rate  $s/2$ ; the distance map has 1-dimensional fibres of codimension  $n - 1$ . Matching gives  $s = n/2$ , and this is the same balance that gives  $\frac{n}{2}$  in the Fourier-dimension heuristic of Worked Example 7.7.2. The two problems share their critical exponent for a structural reason.

(c) *Why Falconer is more tractable than Kakeya.* The distance set problem is a *two-point* problem:  $\Delta(E)$  is determined by the pair correlations of  $E$ , which is exactly what  $I_s(\mu)$  and  $\widehat{\mu}$  encode. By Worked Example 7.7.3, Fourier methods are the right tool for two-point problems and lose nothing structurally. Kakeya is not two-point — the smallness of a Besicovitch set is about how many tubes meet — which is why the Fourier method saturates there at  $\frac{n+1}{2}$  and never improves, while here it has been pushed steadily downwards for forty years.

(d) *The current frontier.* Guth–Iosevich–Ou–Wang’s  $\frac{5}{4}$  in the plane uses polynomial partitioning and the refined decoupling of Volumes 11 and 13. So the distance problem, having begun as a piece of geometric measure theory, is now solved by the same machinery as restriction — which is the best evidence that the two problems are one.

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**WORKED EXAMPLE 8.5.4— TIER B    Tracking the exponent when the surface changes**

The proof of Theorem 8.5.1 used one geometric input, the decay rate  $\frac{n-1}{2}$  of  $\widehat{d\sigma}$ . Everything else was bookkeeping. So the exponent for a variant problem can be read off by changing that one number, and it is worth doing this three times.

(a) *Distances measured by a curved norm.* Let  $\|\cdot\|_K$  be the norm whose unit ball is a symmetric convex body  $K$  with smooth boundary of non-vanishing Gaussian curvature. Then  $\widehat{d\sigma_{\partial K}}$  decays at rate  $\frac{n-1}{2}$  by Theo-

rem 5.5.2, Lemma 8.4.7 goes through verbatim, and the threshold is again  $\frac{n+1}{2}$ . Curvature, not roundness, is what the proof uses.

(b) *A surface with  $k$  non-vanishing curvatures.* If  $\partial K$  has only  $k$  non-zero principal curvatures then Theorem 5.5.3 gives decay  $\frac{k}{2}$ , and Step 3 of Theorem 8.5.1 requires  $-\frac{k}{2} \leq s - n$ , that is

$$s \geq n - \frac{k}{2}.$$

At  $k = n - 1$  this is  $\frac{n+1}{2}$ ; at  $k = 0$  — a polyhedral ball — it is  $n$ , which is vacuous, consistent with the  $\ell^\infty$  failure noted in Worked Example 8.5.2.

(c) *Dot products instead of distances.* Replace  $|x - y|$  by  $x \cdot y$  and the level sets become hyperplanes, which are flat: the relevant transform is  $\widehat{d\sigma}$  for a plane, with no decay at all. The method gives nothing, and indeed the dot-product analogue of Conjecture 8.4.2 has a different answer.

*The recipe.* Identify the level sets of the map whose image is being studied; compute the decay rate  $\rho$  of the Fourier transform of surface measure on them; the threshold is  $n - \rho$ . That single line predicts the exponent for every variant in the literature, and predicting it before reading the proof is a good way to check one has understood the argument.

*A caution.* The recipe gives the exponent obtainable by *this* method. Improvements come from the spherical averages of Chapter 8.7, and those do not follow the recipe — they depend on the finer structure of the surface and of  $\mu$ .

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**Proposition 8.5.2** (A quantitative form). *Under the hypotheses of Theorem 8.5.1, with  $s \in (\frac{n+1}{2}, \dim_{\mathbb{H}} E)$  and  $\mu$  as in Step 1,*

$$m(\Delta(E) \cap [a, b]) \geq \frac{\nu([a, b])}{M}, \quad M \lesssim_{n,a,b} 1 + I_s(\mu).$$

*Proof.* By (8.5.1) and Step 4 of Theorem 8.5.1,  $\nu \upharpoonright (a, b) \leq Mm$  with  $M$  the constant obtained there, which Step 3 bounded by a dimensional multiple of  $1 + I_s(\mu)$ . Applying this to the set  $\Delta(E) \cap [a, b]$ , which carries all of  $\nu \upharpoonright [a, b]$  by Proposition 8.4.4,

$$\nu([a, b]) = \nu(\Delta(E) \cap [a, b]) \leq Mm(\Delta(E) \cap [a, b]). \quad \square$$

**Where we are.** The quantitative form makes visible what the qualitative statement hides: the lower bound degrades as  $I_s(\mu)$  grows, that is as  $s$  approaches  $\dim_{\mathbb{H}} E$  or as the measure becomes more concentrated. So a set

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of dimension barely above  $\frac{n+1}{2}$  is guaranteed only a very small distance set, and the theorem gives no uniform bound over all sets of a given dimension. This is typical. Every existence statement in this volume has a quantitative form obtained by tracking constants, and the constants always involve an energy that blows up at the critical exponent. Recognising that pattern explains why endpoint statements in this subject are so often false rather than merely hard — Proposition 6.8.6 in Volume 6 was another instance.

## Drills

**Drill 8.5.1.** Verify the inequality  $(1 + |\xi|)^{-(n-1)/2} \leq |\xi|^{s-n}$  for  $|\xi| > 1$  and  $s \geq \frac{n+1}{2}$ , and identify where equality is approached.

**Drill 8.5.2.** Show that  $I_s(\mu) < \infty$  forces  $\mu$  to be non-atomic, as used in Step 1.

**Drill 8.5.3.** Derive the bound  $\sigma(\mu, r) \lesssim r^{-\min\{s, (n-1)/2\}}$  quoted in the Pitfall, by writing  $\sigma(\mu, r) = \iint \widehat{d\sigma}(r(x-y)) \, d\mu \, d\mu$  and splitting into dyadic annuli.

**Drill 8.5.4.** Confirm the claim in Worked Example 8.5.2 that for the  $\ell^\infty$  norm the analogue of Lemma 8.4.7 fails, by computing the transform of the indicator of a cube shell in the coordinate directions.

## Notes

Theorem 8.5.1 is Falconer’s, from 1985. The proof above is his — an  $L^\infty$  estimate on the density of  $\nu$  — and is the one Iosevich has popularised in the form given here, using Stein–Tomas-style input only through the decay of  $\widehat{d\sigma}$ .

Mattila’s  $L^2$  approach dates from 1987 and is the framework for every subsequent improvement: Bourgain’s  $\frac{13}{9}$  in the plane, Wolff’s  $\frac{4}{3}$ , Erdoğan’s  $\frac{n}{2} + \frac{1}{3}$ , and the current  $\frac{5}{4}$  of Guth, Iosevich, Ou and Wang. Chapter 8.7 explains the framework and what each contributes.



## Chapter 8.6

# Interiors

Theorem 8.5.1 produces a distance set of positive measure. A little more work with the same estimates produces one with nonempty interior — so that  $\Delta(E)$  contains a whole interval of distances, not merely a set of positive measure. The extra ingredient is continuity of the density, and it costs nothing.

**Theorem 8.6.1** (Mattila–Sjölin). *Let  $n \geq 2$  and  $E \subseteq \mathbb{R}^n$  compact with  $\dim_{\mathbb{H}} E > \frac{n+1}{2}$ . Then  $\Delta(E)$  has nonempty interior.*

*Idea of the proof.* Theorem 8.5.1 showed the densities  $\nu_\varepsilon$  are uniformly bounded. If in addition they converge *uniformly* on a compact interval, the limit is a continuous density for  $\nu$ ; and a measure with a continuous density that is positive somewhere charges a whole neighbourhood of that point, which must therefore lie inside the support — that is, inside  $\Delta(E)$ . Uniform convergence follows from the same integral. The difference  $\nu_\varepsilon(t) - \nu_{\varepsilon'}(t)$  is controlled by the tail of  $\int (1 + |\xi|)^{-(n-1)/2} |\widehat{\mu}|^2$ , which converges; and the strict inequality  $\dim_{\mathbb{H}} E > \frac{n+1}{2}$  gives a little room, so the tail is small uniformly in  $t$ .

*Proof. Step 1. Set up with room to spare.*

Choose  $s$  with  $\frac{n+1}{2} < s < \dim_{\mathbb{H}} E$  and  $\mu$  on  $E$  with  $I_s(\mu) < \infty$ ,  $\mu(E) = 1$ , as in Theorem 8.5.1. Put  $\eta = s - \frac{n+1}{2} > 0$ . Repeating Step 3 of that proof but keeping the surplus,

$$\int_{|\xi| > R} (1 + |\xi|)^{-\frac{n-1}{2}} |\widehat{\mu}(\xi)|^2 d\xi \leq R^{-\eta} \int_{\mathbb{R}^n} |\widehat{\mu}(\xi)|^2 |\xi|^{s-n} d\xi \lesssim R^{-\eta} I_s(\mu) \quad (8.6.1)$$

for  $R \geq 1$ , because on  $|\xi| > R$  we have  $(1 + |\xi|)^{-(n-1)/2} \leq |\xi|^{-(n-1)/2} = |\xi|^{s-n} |\xi|^{-\eta} \leq R^{-\eta} |\xi|^{s-n}$ .

**Step 2.** *The densities converge uniformly.*

Fix  $[a, b] \subseteq (0, \infty)$ . For  $t \in [a, b]$  and  $0 < \varepsilon, \varepsilon' < a/2$ , by Lemma 8.4.6,

$$\nu_\varepsilon(t) - \nu_{\varepsilon'}(t) = \int_{\mathbb{R}^n} \left( \frac{\widehat{\mathbf{1}_{A_\varepsilon^t}}(\xi)}{2\varepsilon} - \frac{\widehat{\mathbf{1}_{A_{\varepsilon'}^t}}(\xi)}{2\varepsilon'} \right) |\widehat{\mu}(\xi)|^2 d\xi.$$

By (8.6.1) and Lemma 8.4.7, the contribution of  $|\xi| > R$  to either term is  $\lesssim R^{-\eta}$ , uniformly in  $t$  and  $\varepsilon$ . On the compact region  $|\xi| \leq R$  we may compute directly: from the polar formula in Lemma 8.4.7,

$$\frac{1}{2\varepsilon} \widehat{\mathbf{1}_{A_\varepsilon^t}}(\xi) = \frac{1}{2\varepsilon} \int_{t-\varepsilon}^{t+\varepsilon} r^{n-1} \widehat{d\sigma}(r\xi) dr \longrightarrow t^{n-1} \widehat{d\sigma}(t\xi) \quad (\varepsilon \downarrow 0),$$

uniformly for  $t \in [a, b]$  and  $|\xi| \leq R$ , since the integrand is continuous and hence uniformly continuous on the relevant compact set. So  $\nu_\varepsilon$  is uniformly Cauchy on  $[a, b]$ , with limit

$$\nu_0(t) = \int_{\mathbb{R}^n} t^{n-1} \widehat{d\sigma}(t\xi) |\widehat{\mu}(\xi)|^2 d\xi, \quad (8.6.2)$$

a continuous function of  $t \in [a, b]$  — continuity following from the uniform convergence of continuous functions.

**Step 3.**  $\nu_0$  is a density for  $\nu$ .

By Lemma 8.4.6, for  $f \in C_c((a, b))$ ,

$$\int f d\nu = \lim_{\varepsilon \downarrow 0} \int_a^b \nu_\varepsilon(t) f(t) dt = \int_a^b \nu_0(t) f(t) dt,$$

the exchange justified by uniform convergence on the compact support of  $f$ . Hence  $\nu \upharpoonright (a, b)$  has the continuous density  $\nu_0$ .

**Step 4.** *Conclude.*

Choose  $[a, b]$  with  $\nu((a, b)) > 0$ , as in Theorem 8.5.1 Step 1. Then  $\nu_0$  is continuous on  $(a, b)$  and not identically zero, so  $\nu_0(t_0) > 0$  for some  $t_0$ , and by continuity  $\nu_0 > 0$  on an interval  $I \ni t_0$ . Every nonempty open subinterval of  $I$  then has positive  $\nu$ -measure, so  $I$  is contained in the support of  $\nu$ ; and by Proposition 8.4.4 the support is contained in the closed set  $\Delta(\text{spt } \mu) \subseteq \Delta(E)$ . Hence  $I \subseteq \Delta(E)$ , and  $\Delta(E)$  has nonempty interior.  $\square$

**Where we are.** Formula (8.6.2) is worth recording on its own:

$$\nu_0(t) = t^{n-1} \int_{\mathbb{R}^n} \widehat{d\sigma}(t\xi) |\widehat{\mu}(\xi)|^2 d\xi,$$



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the density of the distance measure, expressed through the transform of surface measure. It is the rigorous form of the heuristic that Worked Example 8.4.2 warned against writing down directly, and the proof above is what licenses it.

Note that  $\widehat{d\sigma}$  is real and *oscillates in sign* (Theorem 5.5.2), so the integrand is not of one sign and  $\nu_0(t) > 0$  is not automatic — it had to be deduced in Step 4 from  $\nu \neq 0$  and continuity. Any argument claiming positivity of  $\nu_0$  pointwise from the formula alone is mistaken.

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**WORKED EXAMPLE 8.6.1— TIER A Positive measure and nonempty interior are different**

A closed set of positive measure need not have interior: the fat Cantor set of Volume 1 has positive measure and empty interior. So Theorem 8.6.1 is genuinely stronger than Theorem 8.5.1, and it is a little surprising that it costs only the continuity argument.

*Why the same exponent suffices.* Because the proof of Theorem 8.5.1 already produced a *uniform* bound on  $\nu_\varepsilon$ , not merely an  $L^1$  bound. Uniform bounds plus an equicontinuity statement give a continuous limit, and continuity is exactly what converts positive measure into interior.

*Where more would be needed.* If one only knew  $\nu \in L^2$  — which is what the Mattila integral gives — the density need not be continuous and interior would not follow. So the two routes to Falconer’s theorem differ in their by-products: the  $L^\infty$  route gives interior for free, and the  $L^2$  route gives better exponents but a weaker conclusion. This is the reason both remain in use.

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**WORKED EXAMPLE 8.6.2— TIER B Pinned distance sets**

A refinement asks for a *single*  $x \in E$  whose distance set  $\Delta_x(E) = \{|x - y| : y \in E\}$  is large. Since  $\Delta(E) = \bigcup_{x \in E} \Delta_x(E)$ , this is strictly stronger.

*What the present method gives.* Run the proof of Theorem 8.5.1 with the measure  $\mu \times \mu$  replaced by  $\delta_x \times \mu$  for a fixed  $x$ . The Fourier step fails:  $\widehat{\delta_x}$  does not decay, so there is no cancellation to exploit and the argument collapses. Pinning is genuinely harder.

*What is true.* Peres and Schlag proved a pinned analogue of Theorem 8.5.1 at the same exponent  $\frac{n+1}{2}$ , by a transversality argument in the spirit of Worked Example 8.2.2: one shows the set of bad pins  $x$  is small, rather than fixing  $x$  in advance. Liu later gave an  $L^2$  approach, and the Guth–Iosevich–Ou–Wang theorem is stated in pinned form — their conclusion is that there *exists*  $x \in E$  with  $m(\Delta_x(E)) > 0$ .

*The pattern.* As in Worked Example 8.1.3, one cannot prove the statement for a fixed parameter; one proves that the bad parameters form a small set. That recipe is the same one that converted Marstrand's theorem into Kaufman's, and it is worth noticing that it is the standard response whenever a "for some  $x$ " statement is wanted and only an average is available.

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**WORKED EXAMPLE 8.6.3— TIER C: RESEARCH TASTE    What “pinning” costs, in three problems**

Worked Example 8.6.2 noted that fixing a single  $x$  breaks the Fourier argument. The same difficulty appears elsewhere, and the pattern is worth naming.

(a) *Distance sets.* The averaged statement bounds  $\iint$  over pairs; the pinned statement fixes  $x$  and integrates over  $y$  only. The transform of  $\delta_x$  does not decay, so all cancellation is lost. The resolution is to show the bad pins are few, not to fix a good one.

(b) *Projections.* Theorem 8.1.2 is an averaged statement over  $\theta$ ; the corresponding pinned statement — that a *specified*  $\theta$  is good — is false in general (Worked Example 8.1.1). Kaufman's theorem is the systematic reply: bound the dimension of the bad set.

(c) *Kakeya.* Here the averaged statement is vacuous and the pinned one is everything: a Besicovitch set is a configuration in which *every* direction carries a segment, so there is nothing to average over. This is the structural reason Kakeya resists the methods of this volume, and it is worth stating as the sharpest available contrast between the two problems.

(d) *The recipe, once more.* In (a) and (b) one proves the averaged statement, then upgrades by running the average against a Frostman measure on the exceptional set (Worked Example 8.2.2). In (c) that route is closed. A reader meeting a new problem in this area should ask first whether the bad parameters can be all of them; if they can, no averaging argument will help.

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**WORKED EXAMPLE 8.6.4— TIER B    Equicontinuity: converting a uniform bound into a continuous density**

The step that upgrades Theorem 8.5.1 to Theorem 8.6.1 is standard analysis, and it is worth isolating because it applies whenever a density has been produced by mollification.

*The situation.* One has functions  $\nu_\varepsilon$  on  $[a, b]$  with  $\int f\nu_\varepsilon \rightarrow \int f d\nu$  and  $|\nu_\varepsilon| \leq M$  uniformly. That alone gives  $\nu \leq Mm$ , but not continuity: uni-

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form boundedness permits wild oscillation.

*What is needed in addition.* Either equicontinuity — a modulus of continuity uniform in  $\varepsilon$  — or, as in Theorem 8.6.1, uniform convergence. The latter is stronger and is what Step 2 there established, by splitting the frequency integral:

- on  $|\xi| > R$ , the tail is  $\lesssim R^{-\eta}$  uniformly in  $t$  and  $\varepsilon$ , by the surplus (8.6.1);
- on  $|\xi| \leq R$ , the integrand converges uniformly because it is continuous on a compact set.

Together these give a uniform Cauchy property, hence a continuous limit.

*Why the surplus is essential.* The estimate (8.6.1) needs  $\eta = s - \frac{n+1}{2} > 0$ , that is the *strict* inequality  $\dim_{\mathbb{H}} E > \frac{n+1}{2}$ . At equality one would have a uniform bound with no decay in  $R$ , hence no uniform convergence, hence no continuity — and the interior conclusion would be lost while the positive-measure conclusion survived. The two theorems really do need different hypotheses, even though both are stated with the same strict inequality.

*The general pattern.* A uniform bound gives absolute continuity; a uniform bound with a quantitative tail gives a continuous density. The second costs an epsilon of room in the exponent and buys a qualitatively stronger conclusion. This trade recurs: in Volume 4 the Besov endpoint  $B_{p,1}^{n/p} \subseteq L^\infty$  bought continuity for the same reason, by requiring summability rather than square-summability over dyadic blocks.

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**Proposition 8.6.2** (The density is bounded below on a subinterval). *Under the hypotheses of Theorem 8.6.1, there are  $0 < a < b < \infty$ , an interval  $I \subseteq (a, b)$  and  $c > 0$  with*

$$\nu_0(t) \geq c \quad \text{for all } t \in I,$$

*and consequently  $I \subseteq \Delta(E)$  with  $m(\Delta(E)) \geq |I|$ .*

*Proof.* By Step 2 of Theorem 8.6.1,  $\nu_0$  is continuous on  $[a, b]$  for any  $0 < a < b < \infty$ , and by Step 3 it is a density for  $\nu$  there. Choose  $[a, b]$  with  $\nu((a, b)) > 0$ , possible as in Theorem 8.5.1 Step 1.

If  $\nu_0 \leq 0$  on all of  $(a, b)$  then  $\nu((a, b)) = \int_a^b \nu_0 \leq 0$ , contradicting the choice. So  $\nu_0(t_0) > 0$  for some  $t_0 \in (a, b)$ ; by continuity there are  $c = \frac{1}{2}\nu_0(t_0) > 0$  and an interval  $I \ni t_0$  with  $\nu_0 \geq c$  on  $I$ .

Every nonempty open  $J \subseteq I$  then has  $\nu(J) = \int_J \nu_0 \geq c|J| > 0$ , so  $J$  meets  $\text{spt } \nu$ . Hence  $I \subseteq \text{spt } \nu$ , and by Proposition 8.4.4  $\text{spt } \nu \subseteq \Delta(\text{spt } \mu) \subseteq \Delta(E)$ , which is closed. The measure bound is immediate.  $\square$

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WORKED EXAMPLE 8.6.5— TIER A **Why  $\nu_0 \geq 0$  is not obvious**

One might expect  $\nu_0 \geq 0$  automatically,  $\nu$  being a positive measure. It is true —  $\nu_0$  is a density for a positive measure, hence nonnegative almost everywhere, hence everywhere by continuity — but it does *not* follow from the formula

$$\nu_0(t) = t^{n-1} \int_{\mathbb{R}^n} \widehat{d\sigma}(t\xi) |\widehat{\mu}(\xi)|^2 d\xi.$$

Here  $|\widehat{\mu}|^2 \geq 0$  but  $\widehat{d\sigma}$  oscillates in sign, by Theorem 5.5.2: it is asymptotically a cosine. So the integrand takes both signs and no term-by-term argument gives positivity.

*The correct order of reasoning.* Positivity comes from  $\nu$  being a measure, established before any Fourier computation; the formula is then used only for *upper* bounds. Reversing the order — trying to read positivity off the formula — fails, and an argument that appears to do so should be checked.

*Where this bites.* Proposition 8.6.2 needed  $\nu_0(t_0) > 0$  somewhere, and obtained it from  $\nu((a, b)) > 0$ , not from the integral. That is the only available route.

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WORKED EXAMPLE 8.6.6— TIER C: RESEARCH TASTE **What “nonempty interior” buys, and where it is used**

Theorem 8.6.1 is stronger than Theorem 8.5.1, and the extra strength is used, so it is worth recording where.

(a) *It makes the conclusion checkable.* “Positive measure” is not constructive: one cannot exhibit a distance. “Contains an interval” names a whole range of distances realised by pairs of points of  $E$ , and Proposition 8.6.2 even locates it as a neighbourhood of any  $t_0$  where the density is positive.

(b) *It iterates.* If  $\Delta(E) \supseteq I$  then, applying the theorem to  $E \times E$  or to pinned versions, one obtains statements about configurations — triangles with prescribed side lengths, and so on. This is the starting point of the literature on *finite point configurations* in fractal sets, where the question is which similarity classes of  $k$ -point configurations occur. Positive measure would not suffice there, since one must intersect finitely many conclusions.

(c) *It fails for the  $L^2$  method.* As Worked Example 8.7.2 recorded, Mattila’s framework gives  $\nu \in L^2$ , and an  $L^2$  density need not be continuous. So the improvements of Chapter 8.7 give better exponents for positive measure and say nothing about interior. At the time of writing the best *interior* result remains at  $\frac{n+1}{2}$  — that is, Theorem 8.6.1 is still the record for its own conclusion, forty years on, even though the positive-measure exponent has

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moved substantially.

(d) *The moral.* A stronger conclusion may have a worse exponent, and the two should be tabulated separately. Worked Example 8.8.4 listed this as the first of its four questions for a reason: comparing a positive-measure exponent with an interior exponent is comparing different theorems.

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## Drills

**Drill 8.6.1.** Verify (8.6.1), being careful with the case  $|\xi| \leq 1$ .

**Drill 8.6.2.** Show that the convergence in Step 2 is uniform on  $\{t \in [a, b], |\xi| \leq R\}$ , using uniform continuity of  $(r, \xi) \mapsto r^{n-1} \widehat{d\sigma}(r\xi)$  on a compact set.

**Drill 8.6.3.** Give an example of a closed set of positive Lebesgue measure with empty interior, and explain why Theorem 8.6.1 is therefore strictly stronger than Theorem 8.5.1.

**Drill 8.6.4.** Show directly from (8.6.2) that  $\nu_0$  is bounded, recovering (8.5.1) in the limit.

## Notes

Theorem 8.6.1 is due to Mattila and Sjölin (1999). The observation that the interior comes free from an  $L^\infty$  bound plus equicontinuity is theirs; Iosevich, Mourgoglou and Taylor extended it to distances measured by norms whose unit ball has non-vanishing curvature, where the same proof applies with Theorem 5.5.2 replaced by its analogue for that surface.



## Chapter 8.7

# The Mattila Integral

Every improvement on Falconer's exponent since 1987 has gone through a single reformulation, due to Mattila, which replaces the  $L^\infty$  estimate of Chapter 8.5 by an  $L^2$  one. This chapter proves the reduction and explains what each subsequent advance contributed. It proves none of those advances: they are research papers, and the reduction is what makes them readable.

### 8.7.1 The reduction

**Definition 8.7.1.** For a finite compactly supported Borel measure  $\mu$  on  $\mathbb{R}^n$  and  $r > 0$ , the *spherical average* of  $|\widehat{\mu}|^2$  is

$$\sigma(\mu, r) = \int_{\mathbb{S}^{n-1}} |\widehat{\mu}(r\omega)|^2 d\sigma(\omega),$$

and the *Mattila integral* is

$$\mathcal{M}(\mu) = \int_1^\infty \sigma(\mu, r)^2 r^{n-1} dr.$$

**Theorem 8.7.2** (Mattila's reduction). *Let  $\mu$  be a finite compactly supported Borel measure on  $\mathbb{R}^n$  with  $\mathcal{M}(\mu) < \infty$ . Then the distance measure  $\nu$  has an  $L^2$  density on every interval  $[a, b] \subseteq (0, \infty)$ ; in particular  $\nu \ll m$  and  $m(\Delta(\text{spt } \mu)) > 0$ .*

*Idea of the proof.* By Lemma 8.4.6 the approximate density is  $\nu_\varepsilon(t) = \frac{1}{2\varepsilon} \int \widehat{\mathbf{1}_{A_\varepsilon^t}} |\widehat{\mu}|^2$ , and the polar computation of Lemma 8.4.7 identifies its limit as

$$\nu_0(t) = t^{n-1} \int_{\mathbb{R}^n} \widehat{d\sigma}(t\xi) |\widehat{\mu}(\xi)|^2 d\xi.$$

Rather than bounding this pointwise — which is what Chapter 8.5 did, and which costs the full pointwise decay of  $\widehat{d\sigma}$  — one estimates its  $L^2$  norm in  $t$ . Writing the  $\xi$ -integral in polar coordinates and using that  $\widehat{d\sigma}(t\xi)$  depends on  $t|\xi|$  only, the  $t$ -integral becomes an integral against the *spherical averages* of  $|\widehat{\mu}|^2$ , and Cauchy–Schwarz produces  $\mathcal{M}(\mu)$ .

The gain is that  $\sigma(\mu, r)$  may be much smaller than the pointwise bound suggests: it is an average, and cancellation between directions is permitted. Every improvement on Falconer’s exponent is an improved bound for  $\sigma(\mu, r)$ .

*Proof. Step 1. Polar form of the density.*

By (8.6.2) and polar coordinates (Theorem 1.6.11) in the  $\xi$ -integral, using that  $\widehat{d\sigma}$  is radial so that  $\widehat{d\sigma}(t\xi)$  depends only on  $t|\xi|$ ,

$$\nu_0(t) = t^{n-1} \int_0^\infty \widehat{d\sigma}(tr) \sigma(\mu, r) r^{n-1} dr.$$

The part  $r \leq 1$  contributes a function of  $t$  bounded uniformly on  $[a, b]$ , since  $\widehat{d\sigma}$  and  $\widehat{\mu}$  are bounded and the region has finite measure; call it  $\nu_0^{\text{loc}}$  and set  $P(t) = \nu_0(t) - \nu_0^{\text{loc}}(t)$ .

**Step 2.** *Insert the asymptotic, keeping the oscillation.*

By Theorem 5.5.2 there are constants  $c_\pm$  with

$$\widehat{d\sigma}(\rho) = \rho^{-\frac{n-1}{2}} \left( c_+ e^{2\pi i \rho} + c_- e^{-2\pi i \rho} \right) + O(\rho^{-\frac{n+1}{2}}) \quad (\rho \geq 1).$$

Substituting  $\rho = tr$  and writing

$$G(r) = \sigma(\mu, r) r^{\frac{n-1}{2}} \mathbf{1}_{\{r>1\}},$$

so that  $\sigma(\mu, r) r^{n-1} \cdot r^{-\frac{n-1}{2}} = G(r)$ , we obtain

$$P(t) = t^{\frac{n-1}{2}} \left( c_+ \int_0^\infty G(r) e^{2\pi i tr} dr + c_- \int_0^\infty G(r) e^{-2\pi i tr} dr \right) + R(t), \quad (8.7.1)$$

where

$$|R(t)| \lesssim t^{n-1} t^{-\frac{n+1}{2}} \int_1^\infty \sigma(\mu, r) r^{n-1-\frac{n+1}{2}} dr = t^{\frac{n-3}{2}} \int_1^\infty \sigma(\mu, r) r^{\frac{n-3}{2}} dr.$$

**Step 3.** *The main term, by Plancherel.*

The two integrals in (8.7.1) are  $\widehat{G}(-t)$  and  $\widehat{G}(t)$ . Note

$$\|G\|_{L^2(\mathbb{R})}^2 = \int_1^\infty \sigma(\mu, r)^2 r^{n-1} dr = \mathcal{M}(\mu) < \infty,$$



so  $G \in L^2$  and, by Plancherel (Theorem 2.4.8),  $\widehat{G} \in L^2(\mathbb{R})$  with the same norm. Therefore, since  $t^{(n-1)/2} \leq b^{(n-1)/2}$  on  $[a, b]$ ,

$$\int_a^b \left| t^{\frac{n-1}{2}} (c_+ \widehat{G}(-t) + c_- \widehat{G}(t)) \right|^2 dt \lesssim_{n,b} \|\widehat{G}\|_{L^2(\mathbb{R})}^2 = \mathcal{M}(\mu).$$

**Step 4.** *The error term, by Cauchy–Schwarz.*

Here — and *only* here — Cauchy–Schwarz is available, because the exponent is  $\frac{n-3}{2}$ , two powers below the main term:

$$\int_1^\infty \sigma(\mu, r) r^{\frac{n-3}{2}} dr \leq \left( \int_1^\infty \sigma(\mu, r)^2 r^{n-1} dr \right)^{1/2} \left( \int_1^\infty r^{(n-3)-(n-1)} dr \right)^{1/2} = \mathcal{M}(\mu)^{1/2},$$

the second factor being  $(\int_1^\infty r^{-2} dr)^{1/2} = 1$ . Hence  $|R(t)| \lesssim_{n,b} \mathcal{M}(\mu)^{1/2}$  uniformly on  $[a, b]$ .

**Step 5.** *Conclude.*

Combining Steps 1, 3 and 4,  $\nu_0 \in L^2([a, b])$  with  $\|\nu_0\|_{L^2([a, b])} \lesssim_{n,a,b} \mathcal{M}(\mu)^{1/2} + 1$ . By the argument of Theorem 8.5.1 Step 4 — which used only that the approximating densities converge against test functions —  $\nu_0$  is a density for  $\nu \upharpoonright (a, b)$ , so  $\nu \ll m$  there, and Proposition 8.4.4 gives  $m(\Delta(\text{spt } \mu)) > 0$ .  $\square$

### **Pitfall: Cauchy–Schwarz does not bound the main term**

It is tempting to skip Step 3 and estimate the whole of  $P(t)$  by Cauchy–Schwarz, as in Step 4. That fails, and the failure is instructive.

Bounding  $|\widehat{d\sigma}(tr)| \lesssim (tr)^{-(n-1)/2}$  pointwise and applying Cauchy–Schwarz would require

$$\int_1^\infty \sigma(\mu, r) r^{\frac{n-1}{2}} dr \lesssim \mathcal{M}(\mu)^{1/2},$$

and the pairing that would give this needs  $(\int_1^\infty r^{(n-1)-(n-1)} dr)^{1/2} = (\int_1^\infty dr)^{1/2} = \infty$ . There is no admissible weight: the exponents do not leave room.

*What the oscillation buys* is exactly the missing power. Because  $\widehat{d\sigma}$  oscillates rather than merely decaying, the  $r$ -integral is a Fourier transform and Plancherel replaces the divergent  $\int dr$  by an identity. So the proof *requires* the asymptotic of Theorem 5.5.2 and not merely the bound of Theorem 2.8.5 — a concrete instance of the point made in Worked Example 5.3.2 that discarding the phase loses information one later needs.

**Remark 8.7.3** (What the reduction is worth, and what it is not). Theorem 8.7.2 converts the distance problem into the problem of bounding  $\sigma(\mu, r)$ . That is

progress only if  $\sigma$  can be bounded better than the pointwise decay allows — and it can, but not by anything in this series.

Bounding  $\sigma(\mu, r)$  is a *weighted restriction estimate*: it asks how much of  $\widehat{\mu}$  concentrates on the sphere of radius  $r$ , which is exactly the question Chapter 6.3 studied with  $\mu$  replaced by an arbitrary  $L^2$  function. The improvements listed in Worked Example 8.7.1 are, each of them, an improved spherical average estimate obtained by importing restriction technology.

*The reduction alone proves nothing new.* As Chapter 8.5’s Pitfall recorded, feeding the pointwise bound  $\sigma(\mu, r) \lesssim r^{-\min\{s, (n-1)/2\}}$  into  $\mathcal{M}$  gives a divergent integral for every  $s$  and every  $n$ . So Theorem 8.7.2 is a *framework*, not a theorem about dimensions, and the exponent  $\frac{n+1}{2}$  of Theorem 8.5.1 is not recoverable from it.

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**WORKED EXAMPLE 8.7.1— TIER B   Reading the record**

The successive improvements are all of one kind, and knowing which is which makes the literature navigable.

| Bound                       | Author                     | Dimension  | New input                              |
|-----------------------------|----------------------------|------------|----------------------------------------|
| $\frac{n+1}{2}$             | Falconer 1985              | all $n$    | pointwise decay of $\widehat{d\sigma}$ |
| $\frac{13}{9}$              | Bourgain 1994              | $n = 2$    | first use of restriction technology    |
| $\frac{4}{3}$               | Wolff 1999                 | $n = 2$    | sharp local smoothing for the circle   |
| $\frac{n}{2} + \frac{1}{3}$ | Erdoğan 2005               | $n \geq 3$ | Tao’s bilinear restriction             |
| $\frac{5}{4}$               | Guth–Iosevich–Ou–Wang 2020 | $n = 2$    | polynomial partitioning, refined de    |

*What the table shows.* Every entry after the first is an improved bound for  $\sigma(\mu, r)$ , and the new input in each case is the best restriction technology of its decade. So the distance problem has served as a proving ground for restriction theory: an advance in Volume 12’s or Volume 13’s material translates immediately into a distance-set exponent.

*Why  $n = 2$  leads.* Because the restriction theory is complete there (Theorem 6.8.2) and the decoupling theory is sharpest there. In higher dimensions the inputs are weaker, and the exponents correspondingly further from  $\frac{n}{2}$ .

*Where the conjecture would come from.* A bound  $\sigma(\mu, r) \lesssim_\varepsilon r^{-n/2+\varepsilon}$  for measures of dimension  $s > \frac{n}{2}$  would give  $\mathcal{M}(\mu) < \infty$  and hence Conjecture 8.4.2 (Drill 7.3). That is a sharp weighted restriction estimate, and it is open — but it is the same kind of statement as the restriction conjecture, not a different problem.

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**WORKED EXAMPLE 8.7.2— TIER C: RESEARCH TASTE Two frameworks, and when to use which**

This volume has proved Falconer’s theorem twice over in effect — once by  $L^\infty$  and once by setting up the  $L^2$  framework that supersedes it. It is worth recording the trade, since the same choice recurs.

(a) *The  $L^\infty$  route (Chapter 8.5).* Bounds the density pointwise. Cheap, uses only the decay of  $\widehat{d\sigma}$ , and yields Theorem 8.6.1 — nonempty interior — as a by-product, because a uniform bound plus equicontinuity gives a continuous density. Its ceiling is  $\frac{n+1}{2}$  and cannot be lowered, because the pointwise decay of  $\widehat{d\sigma}$  is sharp.

(b) *The  $L^2$  route (Theorem 8.7.2).* Bounds the density in  $L^2$ . Gives no interior, and gives nothing at all without an external input; but the input is a restriction estimate, and restriction estimates improve. Every advance since 1987 is here.

(c) *The general principle.* An  $L^\infty$  bound is stronger where it holds and is limited by the worst pointwise behaviour of the kernel; an  $L^2$  bound tolerates bad points if they are rare, and so can exploit cancellation. When a kernel’s pointwise decay is sharp but its averages are better, the  $L^2$  formulation is the one that improves. That is exactly the situation for  $\widehat{d\sigma}$ : its pointwise decay  $|\xi|^{-(n-1)/2}$  is attained (Theorem 5.4.1), but its spherical averages against a fractal measure are conjecturally much smaller.

(d) *The same choice elsewhere in the series.* Volume 3’s Calderón–Zygmund theorem is an  $L^\infty$ -type argument on the kernel; Volume 6’s Stein–Tomas is an  $L^2$ -type argument on its dyadic pieces, and gets a better exponent for exactly the reason (c) gives. Recognising which regime one is in, before choosing a method, is most of the craft.

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**WORKED EXAMPLE 8.7.3— TIER A The spherical average for a measure one can compute**

Take  $\mu = d\sigma$  itself, surface measure on  $\mathbb{S}^{n-1}$ , and compute  $\sigma(\mu, r)$  directly. By Theorem 5.5.2,  $|\widehat{d\sigma}(r\omega)|^2 \approx r^{-(n-1)} \cos^2(2\pi r - \frac{(n-1)\pi}{4} - \frac{\pi}{4})$  for large  $r$ , independently of  $\omega$  since  $\widehat{d\sigma}$  is radial. So

$$\sigma(d\sigma, r) = \int_{\mathbb{S}^{n-1}} |\widehat{d\sigma}(r\omega)|^2 d\sigma(\omega) \approx \sigma(\mathbb{S}^{n-1}) r^{-(n-1)} \cos^2(\dots),$$

which is  $\approx r^{-(n-1)}$  on average in  $r$ .

*Comparison with the pointwise bound.* The general estimate  $\sigma(\mu, r) \lesssim r^{-\min\{s, (n-1)/2\}}$  gives, with  $s = \dim_{\text{H}} \mathbb{S}^{n-1} = n-1$ , the exponent  $\min\{n-$

$1, \frac{n-1}{2}\} = \frac{n-1}{2}$ . The truth is  $n-1$ , twice as good.

*What this shows.* For a Salem set — and the sphere is one, by Worked Example 7.7.1 — the spherical average is much smaller than the general bound allows, because  $|\widehat{\mu}|$  is genuinely of size  $r^{-(n-1)/2}$  at *every* point rather than merely on average. The whole difficulty of the distance problem is that fractal measures of dimension just above  $\frac{n}{2}$  need not be Salem, and for them nothing better than the general bound is known.

*A check on  $\mathcal{M}$ .* With  $\sigma(d\sigma, r) \approx r^{-(n-1)}$ ,

$$\mathcal{M}(d\sigma) \approx \int_1^\infty r^{-2(n-1)} r^{n-1} dr = \int_1^\infty r^{-(n-1)} dr,$$

finite exactly when  $n \geq 3$ . So the Mattila integral converges for the sphere in dimension three and above — consistent with  $\Delta(\mathbb{S}^{n-1}) = [0, 2]$ , which certainly has positive measure.

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#### WORKED EXAMPLE 8.7.4— TIER B **Reading a spherical average estimate**

Every advance on Falconer's exponent is a bound for  $\sigma(\mu, r)$ , so a reader of this literature must be able to convert such a bound into an exponent. The conversion is one line, and here it is.

*The conversion.* Suppose one knows

$$\sigma(\mu, r) \lesssim_\varepsilon r^{-\beta+\varepsilon}$$

for every measure with  $\mu(B(x, r)) \leq r^s$ . Then

$$\mathcal{M}(\mu) \lesssim \int_1^\infty r^{-2\beta+2\varepsilon} r^{n-1} dr,$$

which converges when  $2\beta > n$ , that is  $\beta > \frac{n}{2}$ . So by Theorem 8.7.2 the distance set has positive measure as soon as the spherical average decays faster than  $r^{-n/2}$ .

*Applying it to the known bounds.*

- The trivial bound is  $\beta = \min\{s, \frac{n-1}{2}\}$ , which never exceeds  $\frac{n-1}{2} < \frac{n}{2}$ . Hence the framework alone proves nothing, as Remark 8.7.3 recorded.
- Wolff's planar result corresponds to  $\beta > 1 = \frac{n}{2}$  at  $n = 2$  for  $s > \frac{4}{3}$ ; the improvement over the trivial  $\beta = \frac{1}{2}$  is exactly the content of his local smoothing estimate.
- The conjecture corresponds to  $\beta > \frac{n}{2}$  for every  $s > \frac{n}{2}$ .

*The dimension-dependence to watch.* A spherical average bound is stated for measures of a given Frostman exponent  $s$ , and  $\beta$  depends on  $s$ . A paper proving “ $\sigma(\mu, r) \lesssim r^{-\beta(s)}$ ” therefore yields the distance-set threshold  $\inf\{s : \beta(s) > \frac{n}{2}\}$ , and comparing two papers means comparing two functions  $\beta$ , not two numbers.

*The exercise worth doing.* Take the trivial bound  $\beta(s) = \min\{s, \frac{n-1}{2}\}$ , plot it against  $s$ , and mark the line  $\beta = \frac{n}{2}$ . The two never meet — that is the picture of why Falconer’s theorem cannot be recovered from  $\mathcal{M}$ , and it is the single most useful diagram to have in mind when reading this literature.

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**Proposition 8.7.4** (The trivial spherical average bound). *Let  $\mu$  be a finite compactly supported measure with  $\mu(B(x, r)) \leq r^s$  for all  $x, r$ . Then for  $r \geq 1$ ,*

$$\sigma(\mu, r) \lesssim_{n,s} r^{-\min\{s, \frac{n-1}{2}\}}.$$

*Proof.* Expanding the square and using Fubini,

$$\sigma(\mu, r) = \int_{\mathbb{S}^{n-1}} |\widehat{\mu}(r\omega)|^2 d\sigma(\omega) = \iint \left( \int_{\mathbb{S}^{n-1}} e^{-2\pi i r \omega \cdot (x-y)} d\sigma(\omega) \right) d\mu(x) d\mu(y) = \iint \widehat{d\sigma}$$

By Theorem 5.5.2,  $|\widehat{d\sigma}(z)| \lesssim_n (1 + |z|)^{-(n-1)/2}$ . Split the pair integral by the dyadic size of  $|x - y|$ : the piece with  $|x - y| \leq r^{-1}$  contributes at most  $\mu \times \mu\{|x - y| \leq r^{-1}\} \leq r^{-s}$  by the Frostman bound; and for  $|x - y| \approx 2^j r^{-1}$  with  $j \geq 0$ , the kernel is  $\lesssim 2^{-j(n-1)/2}$  and the mass is  $\lesssim (2^j r^{-1})^s$ , so that piece contributes

$$\sum_{0 \leq j \lesssim \log r} 2^{-j \frac{n-1}{2}} (2^j r^{-1})^s = r^{-s} \sum_j 2^{j(s - \frac{n-1}{2})}.$$

If  $s < \frac{n-1}{2}$  the sum converges and the total is  $\lesssim r^{-s}$ ; if  $s > \frac{n-1}{2}$  it is dominated by its largest term, at  $2^j \approx r$ , giving  $r^{-s} r^{s - \frac{n-1}{2}} = r^{-\frac{n-1}{2}}$ . The borderline case contributes a logarithm, absorbed into the implied constant after decreasing  $s$  slightly.  $\square$

**Where we are.** Proposition 8.7.4 is the estimate quoted in the Pitfall of Chapter 8.5 and in Remark 8.7.3, now proved. It makes the obstruction explicit: the exponent never exceeds  $\frac{n-1}{2}$ , whatever  $s$ , because the pointwise decay of  $\widehat{d\sigma}$  caps it — and Worked Example 8.7.4 showed that  $\mathcal{M}$  needs an exponent above  $\frac{n}{2}$ .

The gap is exactly  $\frac{1}{2}$ , and closing any part of it requires an estimate that does not factor through the pointwise bound. Every improvement in Worked Example 8.7.1 does exactly that, by exploiting that  $\mu$  is a *fractal*

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measure rather than an arbitrary one.

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**WORKED EXAMPLE 8.7.5— TIER A Checking the framework on a case where everything is known**

A framework is worth testing on an example where the answer is independently available. Take  $n \geq 3$  and  $\mu = d\sigma$  on  $\mathbb{S}^{n-1}$ .

*The independent answer.*  $\Delta(\mathbb{S}^{n-1}) = [0, 2]$  — every distance between 0 and the diameter is realised — so certainly  $m(\Delta) > 0$ , and indeed the distance set has interior.

*What the framework gives.* By Worked Example 8.7.3,  $\sigma(d\sigma, r) \approx r^{-(n-1)}$ , so

$$\mathcal{M}(d\sigma) \approx \int_1^\infty r^{-2(n-1)} r^{n-1} dr = \int_1^\infty r^{-(n-1)} dr,$$

finite for  $n \geq 3$ . So Theorem 8.7.2 applies and confirms  $m(\Delta) > 0$ . ✓

*Why  $n = 2$  is excluded, and what that means.* At  $n = 2$  the integral is  $\int_1^\infty r^{-1} dr$ , logarithmically divergent, so the framework fails for the circle — even though the conclusion is obviously true. That is not a defect of the circle but a reminder that  $\mathcal{M}(\mu) < \infty$  is *sufficient and not necessary*. A reader should not conclude from  $\mathcal{M} = \infty$  that a distance set is small.

*Comparing with the trivial bound.* Proposition 8.7.4 with  $s = n - 1$  gives exponent  $\min\{n - 1, \frac{n-1}{2}\} = \frac{n-1}{2}$ , whereas the truth is  $n - 1$ . So for this measure the trivial bound loses a factor of two in the exponent — exactly the loss that Worked Example 8.7.3 attributed to the sphere being Salem. Fractal measures of dimension near  $\frac{n}{2}$  are the hard case precisely because they need not be.

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**Figure 8.7.1 (to be drawn).** *Why the Mattila integral diverges*

**Content required:** The exponent  $\beta(s) = \min\{s, \frac{n-1}{2}\}$  plotted against  $s \in [0, n]$ : a line of slope one that flattens at height  $\frac{n-1}{2}$ . Draw the horizontal line  $\beta = \frac{n}{2}$  above it. The two never meet, and that gap is exactly why  $\mathcal{M}$  cannot recover Theorem 8.5.1.

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## Drills

**Drill 8.7.1.** Show that no choice of weight makes Cauchy–Schwarz bound  $\int_1^\infty \sigma(\mu, r) r^{(n-1)/2} dr$  by  $\mathcal{M}(\mu)^{1/2}$ : pair with  $r^a$  and  $r^{-a}$  and observe that matching the first factor to  $\mathcal{M}$  forces  $a = 0$  and the second factor to diverge.

**Drill 8.7.2.** Verify that  $\widehat{d\sigma}$  is radial, so that  $\widehat{d\sigma}(t\xi)$  depends only on  $t|\xi|$ , as used in Step 1.

**Drill 8.7.3.** Show that a bound  $\sigma(\mu, r) \lesssim_\varepsilon r^{-n/2+\varepsilon}$  for all measures of dimension  $s > \frac{n}{2}$  would imply Conjecture 8.4.2.

**Drill 8.7.4.** Compute  $\sigma(d\sigma, r)$  for  $\mu = d\sigma$  itself, using Theorem 5.4.1, and check it against the pointwise bound.

## Notes

Mattila's reduction is from 1987. The improvements are Bourgain (1994), Wolff (1999), Erdoğan (2005), and Guth–Iosevich–Ou–Wang (2020), with the higher-dimensional analogue of the last due to Du, Iosevich, Ou, Wang and Zhang. None is proved here; Worked Example 8.7.1 records what each contributes.

The pinned versions, mentioned in Worked Example 8.6.2, are due to Peres–Schlag and to Liu, and the current record is stated in pinned form.





## Chapter 8.8

# The $n/2$ Barrier

### 8.8.1 Where the exponents come from

Three exponents have appeared:  $\frac{n+1}{2}$ , proved;  $\frac{n}{2}$ , conjectured; and  $\frac{n-1}{2}$ , the decay rate of  $\widehat{d\sigma}$ . This closing chapter explains their relations and states what would be needed to close the gap.

*Remark 8.8.1* (Sharpness of the conjectured exponent, and a scope note). For every  $n \geq 2$  there are compact  $E \subseteq \mathbb{R}^n$  with  $\dim_{\mathbb{H}} E = \frac{n}{2}$  and  $m(\Delta(E)) = 0$ , so the exponent in Conjecture 8.4.2 cannot be lowered. This is Falconer's, from the paper posing the problem, and *we do not prove it here* — the construction of the underlying Cantor set is a delicate multi-scale argument and nothing in this series uses the example.

The *mechanism*, however, is elementary and worth recording, since it explains what an extremal set must look like.

Suppose  $A \subseteq [0, 1]$  were a Cantor set with  $\dim_{\mathbb{H}} A = \frac{1}{2}$  which, at infinitely many scales  $N$ , is contained in  $\approx N^{1/2}$  intervals of length  $N^{-1}$  with left endpoints in  $\frac{1}{N}\mathbb{Z}$ . Put  $E = A^n$ , so  $\dim_{\mathbb{H}} E = \frac{n}{2}$  by Corollary 7.6.4 and Theorem 7.8.3. At such a scale every coordinate difference of points of  $E$  lies within  $N^{-1}$  of  $\frac{1}{N}\mathbb{Z}$ , so

$$|x - y|^2 = \sum_{i=1}^n (x_i - y_i)^2$$

lies within  $O(N^{-1})$  of  $\frac{1}{N^2}\mathbb{Z}$ . Hence  $\Delta(E)^2$  sits in an  $O(N^{-1})$ -neighbourhood of  $\frac{1}{N^2}\mathbb{Z} \cap [0, n]$ , which consists of  $O(nN^2)$  intervals of length  $O(N^{-1})$  and so has measure  $O(nN^2 \cdot N^{-1} \cdot N^{-2}) = O(N^{-1})$ . Letting  $N \rightarrow \infty$  along the chosen scales gives  $m(\Delta(E)^2) = 0$  and hence  $m(\Delta(E)) = 0$ .

What is not established here is the existence of  $A$ : one must build a set with the right dimension *and* the lattice property at infinitely many scales,

and the two requirements pull against each other. Falconer's construction alternates between them along a rapidly increasing sequence of scales.

**Where we are.** The example is arithmetic, not geometric: it works because the distances are forced onto a lattice, and lattices have small neighbourhoods. That is the same mechanism that made  $\dim_{\mathbb{F}} C = 0$  in Theorem 7.7.4 — exact self-similarity with an integer ratio creates arithmetic structure.

So the extremal example for Falconer's conjecture is a set whose Fourier transform does not decay at all, and the conjecture asserts that this is the only obstruction. Every improvement listed in Worked Example 8.7.1 proceeds by showing that a measure with *some* decay does better, and the difficulty is that fractal measures of dimension just above  $\frac{n}{2}$  may have almost no decay.

## 8.8.2 The gap, stated precisely

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WORKED EXAMPLE 8.8.1— TIER B **What exactly is missing**

By Theorem 8.7.2 and Drill 7.3, Conjecture 8.4.2 would follow from:

for every  $s > \frac{n}{2}$  and every measure  $\mu$  with  $\mu(B(x, r)) \leq r^s$ , one has  $\sigma(\mu, r) \lesssim_{\varepsilon} r^{-n/2+\varepsilon}$ .

*What is known towards it.* The trivial bound from Theorem 5.5.2 is  $\sigma(\mu, r) \lesssim r^{-\min\{s, (n-1)/2\}}$ , which saturates at  $\frac{n-1}{2}$  — short of  $\frac{n}{2}$  by exactly  $\frac{1}{2}$ . Every improvement narrows that half.

*Why it is a restriction problem.*  $\sigma(\mu, r)$  measures how much of  $\hat{\mu}$  concentrates on a sphere of radius  $r$ ; equivalently, by duality, how large  $\mathcal{E}g$  can be on the support of  $\mu$ . So the required estimate is a *weighted* restriction estimate, with the weight a fractal measure — and that is the setting of Volume 12's later sections.

*The scoreboard, in one line.* Falconer's theorem uses  $\sigma \lesssim r^{-(n-1)/2}$ ; the conjecture needs  $\sigma \lesssim r^{-n/2}$ ; and the record has moved the exponent about a third of the way in the plane and less in higher dimensions.

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WORKED EXAMPLE 8.8.2— TIER C: RESEARCH TASTE **Volume 8 in relation to the rest**

(a) *What was proved.* Marstrand’s projection theorem and Kaufman’s exceptional-set bound (Chapter 8.1, Chapter 8.2); Falconer’s distance theorem at  $\frac{n+1}{2}$  (Chapter 8.5) and its interior refinement (Chapter 8.6); and Mattila’s reduction (Chapter 8.7).

(b) *What was decided against.* The measure-zero Besicovitch set, removed as revision R8 with the obstruction named (Remark 8.3.3); and the four improvements on Falconer’s exponent, each a research paper, stated with their inputs in Worked Example 8.7.1.

(c) *The two lessons that transfer.* First, the averaging recipe of Worked Example 8.2.2: when a statement cannot be proved for every parameter, average, and then upgrade “almost every” to a dimension bound by running the average against a Frostman measure. Second, the  $L^\infty$  versus  $L^2$  choice of Worked Example 8.7.2: pointwise bounds are limited by the worst behaviour of the kernel, averaged bounds can exploit cancellation, and the second is what improves.

(d) *The connection to Kakeya, restated.* Both problems ask how small a set with prescribed richness can be. The distance problem is two-point, so Fourier methods are structurally adequate and the exponent has moved steadily for forty years. Kakeya is not, and the Fourier method has been stuck at  $\frac{n+1}{2}$  since 1983 (Remark 7.7.5). The next volume abandons Fourier analysis for tube combinatorics, and that is why.

(e) *A last observation about method.* Everything in this volume followed from two facts proved much earlier: that  $\widehat{d\sigma}$  decays at rate  $\frac{n-1}{2}$  (Volume 5), and that dimension can be read off a weighted  $L^2$  norm of  $\widehat{\mu}$  (Volume 7). Neither is difficult. The reason the volume is possible at all is that those two facts happen to meet at an exponent worth having — and the reason it stops where it does is that they meet at only one exponent.

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WORKED EXAMPLE 8.8.3— TIER A    **The three exponents, side by side**

It is worth tabulating the numbers this volume has produced, since they are easily confused.

| Exponent        | Meaning                           | Status                     |
|-----------------|-----------------------------------|----------------------------|
| $\frac{n-1}{2}$ | decay rate of $\widehat{d\sigma}$ | theorem (Vol 5)            |
| $\frac{n+1}{2}$ | Falconer’s threshold              | theorem (Theorem 8.5.1)    |
| $\frac{n}{2}$   | conjectured threshold             | open, sharp (Remark 8.8.1) |

*How they are related.* The second is  $n$  minus the first: Falconer’s exponent is exactly where the decay of  $\widehat{d\sigma}$  balances the weight  $|\xi|^{s-n}$  coming from a

measure of dimension  $s$ . The third is half of  $n$ , and exceeds the first by  $\frac{1}{2}$  — which is the entire gap.

*Numerically.*

| $n$ | $\frac{n-1}{2}$ | $\frac{n+1}{2}$ | $\frac{n}{2}$ |
|-----|-----------------|-----------------|---------------|
| 2   | 0.5             | 1.5             | 1.0           |
| 3   | 1.0             | 2.0             | 1.5           |
| 4   | 1.5             | 2.5             | 2.0           |

*The one thing to remember.* The gap between what is proved and what is conjectured is  $\frac{1}{2}$  in every dimension, and it is exactly the difference between the decay of  $\widehat{d\sigma}$ , which is  $\frac{n-1}{2}$ , and the decay  $\frac{n}{2}$  that a sharp spherical average estimate would supply.

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**WORKED EXAMPLE 8.8.4— TIER B Locating a new result: the scoreboard method**

The distance-set literature is large and its results are stated in several incompatible normalisations. This example sets out how to place a new paper.

*Four questions to ask.*

- (1) *Which conclusion?* Positive Lebesgue measure, nonempty interior, or a dimension bound for  $\Delta(E)$ ? These are increasingly strong, and Theorem 8.6.1 showed the second is not free.
- (2) *Pinned or averaged?* A pinned result — there exists  $x \in E$  with  $m(\Delta_x(E)) > 0$  — is strictly stronger, by Worked Example 8.6.2.
- (3) *Which dimension hypothesis?* Hausdorff, or the stronger assumption that  $E$  is Ahlfors–David regular? Many results assume regularity, which is a substantial restriction.
- (4) *What is the spherical average input?* By Worked Example 8.7.4 this determines the exponent, and it is where the new mathematics lives.

*Worked instance.* Guth–Iosevich–Ou–Wang (2020) prove: for compact  $E \subseteq \mathbb{R}^2$  with  $\dim_{\text{H}} E > \frac{5}{4}$ , there exists  $x \in E$  with  $m(\Delta_x(E)) > 0$ . Placing it: conclusion is positive measure (1); pinned (2); Hausdorff dimension only, no regularity (3); and the input is a refined decoupling estimate combined with polynomial partitioning (4). So it is stronger than Theorem 8.5.1 on counts (2) and in exponent, and weaker on count (1) — it does not claim interior.

*Why the bookkeeping matters.* A result with a better exponent under a regularity hypothesis is not comparable to one without. Surveys often tabulate

exponents alone, and the table is then misleading. The four questions above recover the comparison.

*The same discipline elsewhere.* Worked Example 7.8.4 in Volume 7 gave the analogous checklist for Kakeya bounds — which dimension, what loss, which class of sets. In both subjects the exponent is the headline and the hypotheses are the content.

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WORKED EXAMPLE 8.8.5— TIER C: RESEARCH TASTE **The two conjectures, compared line by line**

Volume 7 stated the Kakeya conjecture and this volume the Falconer conjecture. They are the two central problems of the series' second half, and setting them side by side explains why one has moved and the other has not.

|                  | Kakeya                                        | Falconer                                                        |
|------------------|-----------------------------------------------|-----------------------------------------------------------------|
| Statement        | $\dim_{\mathbb{H}} K = n$ for Besicovitch $K$ | $\dim_{\mathbb{H}} E > \frac{n}{2}$ implies $m(\Delta E) > 0$   |
| Extremal example | Perron tree                                   | lattice Cantor set (Remark 8.8.1)                               |
| Nature           | how much can tubes overlap                    | how much can $\widehat{\mu}$ correlate with $\widehat{d\sigma}$ |
| Order            | three-point and higher                        | two-point                                                       |
| Fourier bound    | $\frac{n+1}{2}$ , unmoved since 1983          | $\frac{n+1}{2}$ in 1985, now $\frac{5}{4}$ at $n = 2$           |
| Current method   | tube combinatorics, polynomial method         | restriction estimates, decoupling                               |

(a) *The coincidence of  $\frac{n+1}{2}$ .* Both problems have a Fourier-analytic bound at  $\frac{n+1}{2}$ , and in both cases it arises the same way: the decay  $\frac{n-1}{2}$  of  $\widehat{d\sigma}$  balanced against a 1-dimensional object — a segment for Kakeya, a distance for Falconer. So the two exponents are the same number for the same reason, which is why the problems were long treated together.

(b) *Why they then diverged.* By Worked Example 7.7.3, the Fourier method sees only pair correlations. Falconer's problem *is* a pair correlation problem, so the method is structurally adequate and improving it is a matter of better estimates; Kakeya is not, so the method saturates permanently and progress required abandoning it. That single structural fact accounts for the whole difference in the two histories.

(c) *The reunion.* The current record for Falconer in the plane uses polynomial partitioning and decoupling — that is, the machinery developed for Kakeya and restriction. So the problems have converged again, but from the other side: the geometric methods now serve the analytic problem. Volumes 11 to 13 develop that machinery, and Volume 14 reports where both stand.

(d) *What a reader should take from the comparison.* Before choosing a method for a problem in this area, ask what order of correlation the problem involves. If two, the Fourier transform is the right tool and the question is how sharp an estimate one can prove. If more, the Fourier transform will saturate, and the answer will be combinatorial.

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## Drills

**Drill 8.8.1.** Verify the measure computation in Remark 8.8.1: the  $O(N^{-1})$ -neighbourhood of  $\frac{1}{N^2}\mathbb{Z} \cap [0, n]$  has measure  $O(N^{-1})$ .

**Drill 8.8.2.** Show that  $m(\Delta(E)^2) = 0$  implies  $m(\Delta(E)) = 0$ , using that  $t \mapsto t^2$  is bi-Lipschitz on compact subsets of  $(0, \infty)$ .

**Drill 8.8.3.** Using Worked Example 8.8.1, determine what bound on  $\sigma(\mu, r)$  Wolff's  $\frac{4}{3}$  in the plane corresponds to.

**Drill 8.8.4.** Explain why the extremal set of Remark 8.8.1 has  $\dim_{\mathbb{F}} E = 0$ , and why that is consistent with it being extremal.

**Where we are.** A closing observation about how this volume relates to the two before it.

Volume 6 converted an analytic estimate into a geometric one — restriction into Kakeya — and the conversion was lossy in a way that could be named: Worked Example 6.6.1 showed the phases being discarded. Volume 7 built the measuring apparatus. This volume ran the apparatus on two problems and found the two possible outcomes: on projections it is sharp, on distances it is not.

The diagnostic that separates them, stated once more because it is the most transferable thing here: *count the order of correlation*. A projection of a set is determined by pairs of points — whether two points have the same shadow — and so is a distance. But the projection problem's transversality is exact and the distance problem's is not, so the second requires quantitative input the first does not.

Volume 9 turns to a problem where the order of correlation is genuinely higher, and where none of this volume's methods apply. The change of technique there is not a change of taste; it is forced.

## Notes

The sharpness example of Remark 8.8.1 is Falconer's. The formulation of the missing estimate in Worked Example 8.8.1 is standard and is how the problem is stated in the current literature.

For the improvements see the references in Worked Example 8.7.1; Mattila's book and Wolff's lecture notes are the standard expositions of the material of this volume.







# Chapter 8.A

## Results Quoted from Volumes 1–7

| Result                                                                         | Source               | Used in                                   |
|--------------------------------------------------------------------------------|----------------------|-------------------------------------------|
| Fubini                                                                         | Theorem 1.6.7        | Proposition 8.3.2                         |
| Polar coordinates                                                              | Theorem 1.6.11       | Theorem 8.1.2, Lemma 8.4.7, Theorem 8.7.2 |
| Riesz representation                                                           | Theorem 1.8.15       | Theorem 8.5.1                             |
| Layer-cake                                                                     | Worked Example 1.5.1 | Lemma 8.2.2                               |
| Lipschitz images                                                               | Proposition 7.1.5    | Chapter 8.1 §1.1                          |
| Plancherel                                                                     | Theorem 2.4.8        | Theorem 8.1.2, Theorem 8.7.2              |
| $\mathcal{F}$ on $\mathcal{S}'$                                                | Theorem 2.5.11       | Lemma 8.4.6                               |
| Asymptotics of $\widehat{d\sigma}$                                             | Theorem 5.5.2        | Lemma 8.4.7, Theorem 8.7.2                |
| Bessel asymptotics                                                             | Theorem 5.4.1        | Drill 7.4                                 |
| The signature and the phase                                                    | Worked Example 5.3.2 | Theorem 8.7.2 Pitfall                     |
| Restriction in the plane                                                       | Theorem 6.8.2        | Worked Example 8.7.1                      |
| Stein–Tomas                                                                    | Theorem 6.3.1        | Worked Example 8.8.1                      |
| Hutchinson’s formula                                                           | Theorem 7.8.3        | Worked Example 8.3.1, Remark 8.3.3        |
| Frostman’s lemma                                                               | Theorem 7.5.1        | Theorem 8.2.1                             |
| Energy characterises dimension                                                 | Theorem 7.6.2        | throughout                                |
| Fourier expression of energy                                                   | Proposition 7.6.3    | Theorem 8.1.2, Theorem 8.5.1              |
| Energy of products                                                             | Proposition 7.6.3    | Worked Example 8.1.2                      |
| $\dim_{\mathbb{H}}(E \times F) \geq \dim_{\mathbb{H}} E + \dim_{\mathbb{H}} F$ | Corollary 7.6.4      | Remark 8.3.3, Remark 8.8.1                |
| Dyadic annulus method                                                          | Worked Example 7.6.4 | Theorem 8.5.1                             |
| $\dim_{\mathbb{F}}$ of the Cantor set                                          | Theorem 7.7.4        | Chapter 8.8 Checkpoint                    |

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## Debts

Volume 1 promised a Besicovitch set of measure zero; Volume 7 deferred it here. It is now removed from the plan as revision R8, with the obstruction identified in Remark 8.3.3: the construction reduces, by Proposition 8.3.2, to a projection statement that requires the Besicovitch projection theorem. Nothing in this series uses it.

This volume incurs no debts on later volumes.



## Chapter 8.B

# Notation and Index of Results

### Additions to the Volume 1–7 conventions

|                               |                                                            |
|-------------------------------|------------------------------------------------------------|
| $\pi_\theta$                  | orthogonal projection onto the line of direction $\theta$  |
| $\Delta(E)$                   | the distance set $\{ x - y  : x, y \in E\}$                |
| $\nu, \nu_\varepsilon, \nu_0$ | the distance measure, its mollification, its density       |
| $A_\varepsilon^t$             | the annulus of radius $t$ and thickness $2\varepsilon$     |
| $\sigma(\mu, r)$              | the spherical average of $ \widehat{\mu} ^2$ at radius $r$ |
| $\mathcal{M}(\mu)$            | the Mattila integral                                       |

### Index of the main results

|                                       |                   |
|---------------------------------------|-------------------|
| The angular integral                  | Lemma 8.1.3       |
| Marstrand's projection theorem        | Theorem 8.1.2     |
| The weighted angular integral         | Lemma 8.2.2       |
| Kaufman's exceptional set bound       | Theorem 8.2.1     |
| Projections of product sets           | Proposition 8.3.1 |
| Kahane's reduction                    | Proposition 8.3.2 |
| The distance measure                  | Proposition 8.4.4 |
| Fourier form of the density           | Lemma 8.4.6       |
| Decay of the annulus transform        | Lemma 8.4.7       |
| Falconer's theorem                    | Theorem 8.5.1     |
| Mattila-Sjölin: nonempty interior     | Theorem 8.6.1     |
| Mattila's reduction                   | Theorem 8.7.2     |
| Sharpness of the conjectured exponent | Remark 8.8.1      |

Plan revisions in this volume

| Removed                                                                               | Reason                                                                                                                            |
|---------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| A Besicovitch set of measure zero                                                     | reduces to the Besicovitch projection theorem, which needs the rectifiability theory this series omits. Remark 8.3.3, revision R8 |
| The improvements on $\frac{n+1}{2}$ (Bourgain, Wolff, Erdoğan, Guth–Iosevich–Ou–Wang) | each a research paper; stated with their inputs in Worked Example 8.7.1                                                           |
| Distance sets over finite fields                                                      | need Fourier analysis on $\mathbb{F}_q^d$ ; unused later                                                                          |

# Part 9

## Volume 9

Takeya I: Geometric and Combinatorial Methods





# Preface to Volume 9

## Where the method changes

Volume 8 closed by observing that its methods are adequate for two-point problems and saturate on anything richer. The Kakeya problem is richer: the smallness of a Besicovitch set is a statement about how many tubes can meet, and the Fourier transform of a measure encodes only pair correlations. Accordingly this volume abandons Fourier analysis almost entirely and counts tubes.

The change is not stylistic. Volume 7, Remark 7.7.5 recorded that the Fourier-analytic bound  $\dim_{\mathbb{H}} K \geq \frac{n+1}{2}$  has not moved since 1983; every improvement since has been geometric or combinatorial, and the first of them — Wolff’s hairbrush argument — is proved here.

## A debt

Volume 3 proved Fefferman’s ball multiplier theorem assuming a configuration lemma about Besicovitch sets which was, at that point, the only statement in the series used but not proved. It was promised to this volume.

Chapter 9.2 pays it. The construction is explicit: it takes the Perron tree of Volume 1, inscribes a rectangle in each of its translated triangles, and verifies that the translates of those rectangles by twice their own length are pairwise disjoint. The verification reduces to a single inequality between two linear functions of the construction parameters, and the whole of the difficulty is in choosing where to truncate the rectangles — too short and the translates collide, too long and the rectangles no longer fit inside the triangles. There is an interval of admissible truncations, and it is nonempty for every parameter for which the Perron tree is small.

## What is assumed

Volumes 1–8. Most heavily: the Perron tree, Volume 1 Theorem 1.3.12, whose explicit translations are used quantitatively rather than as a black box; the tube intersection estimate Lemma 1.6.13; theakeya maximal operator of Volume 1, Definition 1.7.13; and the upgrade from maximal bounds to Hausdorff dimension, Volume 7 Theorem 7.8.5.

## How the three arguments fit together

The volume proves three lower bounds and they form a single progression, which is worth having in view before starting.

Each argument conditions on a geometric object and, in exchange, obtains a structural fact the previous one could not use. The  $L^2$  method of Chapter 9.3 conditions on nothing: it sums over all pairs of tubes, and Worked Example 9.3.1 shows the sum is dominated in high dimensions by generic pairs, so the estimate cannot distinguish an extremal configuration from a random one. The bush argument of Chapter 9.4 conditions on a point, and uses that tubes through a common point separate at unit distance. The hairbrush of Chapter 9.5 conditions on a line, and uses a fact with no analogue for points: two lines meeting a common third are concurrent *or coplanar*, so the tubes organise into planes, where the planarakeya problem is already solved.

The exponents are 2, then  $\frac{n+1}{2}$ , then  $\frac{n+2}{2}$ . All three are at most about half the conjectured value  $n$ , and Worked Example 9.8.2 explains why: the bush and hairbrush arguments both optimise over an unknown multiplicity, and the balancing value is one the conjecture says does not occur. An argument that bounds the multiplicity directly rather than optimising over it would escape the ceiling, and that is precisely what the polynomial method of Volume 11 does.

## The plan

- |   |                                  |                               |
|---|----------------------------------|-------------------------------|
| 1 | Theakeya maximal function        | the equivalences              |
| 2 | The Besicovitch configuration    | <i>the debt from Volume 3</i> |
| 3 | The $L^2$ method                 | Córdoba; sharp in the plane   |
| 4 | The bush argument                | conditioning on a point       |
| 5 | The hairbrush                    | conditioning on a line        |
| 6 | Nikodym sets                     | the projective duality        |
| 7 | From single scales to dimension  | what each passage costs       |
| 8 | What the elementary methods give | and why they stop             |

A reader interested only in the debt may read Chapter 9.2 alone; it depends on nothing but Volume 1's Perron tree.



## Chapter 9.1

# The Kakeya Maximal Function

### 9.1.1 The operator and the conjecture

Volume 1 introduced the operator; we restate it and record the equivalences that let one move between its several formulations.

**Definition 9.1.1.** For  $0 < \delta < \frac{1}{2}$ ,  $f \in L^1_{\text{loc}}(\mathbb{R}^n)$  and  $v \in \mathbb{S}^{n-1}$  let

$$\mathcal{K}_\delta f(v) = \sup_{T \parallel v} \frac{1}{m(T)} \int_T |f|,$$

the supremum over all  $\delta$ -tubes  $T$  — cylinders of length 1 and radius  $\delta$  — with axis parallel to  $v$ .

**Conjecture 9.1.2** (Kakeya maximal). *For  $n \geq 2$  and every  $\varepsilon > 0$ ,*

$$\|\mathcal{K}_\delta f\|_{L^n(\mathbb{S}^{n-1})} \lesssim_{n,\varepsilon} \delta^{-\varepsilon} \|f\|_{L^n(\mathbb{R}^n)}.$$

**Proposition 9.1.3** (Two formulations). *Conjecture 9.1.2 is equivalent to the statement that for every family  $\mathbb{T}$  of  $\delta$ -tubes with  $\delta$ -separated directions,*

$$\left\| \sum_{T \in \mathbb{T}} \mathbf{1}_T \right\|_{L^{\frac{n}{n-1}}(\mathbb{R}^n)} \lesssim_{n,\varepsilon} \delta^{-\varepsilon} \left( \sum_T m(T) \right)^{\frac{n-1}{n}}. \quad (9.1.1)$$

*Proof. Step 1. Maximal bound implies multiplicity bound.*

Let  $\mathbb{T} = \{T_1, \dots, T_N\}$  have  $\delta$ -separated directions  $v_1, \dots, v_N$ , and put  $F = \sum_j \mathbf{1}_{T_j}$ . Let  $g \geq 0$  with  $\|g\|_{L^n} \leq 1$ . Then

$$\int Fg = \sum_j \int_{T_j} g = \sum_j m(T_j) \frac{1}{m(T_j)} \int_{T_j} g \leq \sum_j m(T_j) \mathcal{K}_\delta g(v_j).$$

The directions being  $\delta$ -separated, the caps  $B(v_j, \delta/2)$  are disjoint and each has  $\sigma$ -measure  $\approx_n \delta^{n-1}$ ; moreover  $\mathcal{K}_\delta g$  is comparable on  $B(v_j, \delta/2)$  to its value at  $v_j$ , since a tube in direction  $v$  is contained in a fixed dilate of a tube in direction  $v_j$  when  $|v - v_j| \leq \delta/2$ . Hence

$$\sum_j m(T_j) \mathcal{K}_\delta g(v_j) \lesssim_n \frac{m(T_1)}{\delta^{n-1}} \int_{\mathbb{S}^{n-1}} \mathcal{K}_\delta g \, d\sigma \lesssim_n \frac{m(T_1)}{\delta^{n-1}} \|\mathcal{K}_\delta g\|_{L^n(d\sigma)},$$

using  $m(T_j) \approx \delta^{n-1}$  for all  $j$  and Hölder on the sphere. By the hypothesis this is  $\lesssim \delta^{-\varepsilon} m(T_1) \delta^{1-n}$ , and  $\sum_j m(T_j) = Nm(T_1)$  converts this into (9.1.1) after taking the supremum over  $g$  and applying Corollary 1.8.10.

**Step 2.** *Multiplicity bound implies maximal bound.*

Assume (9.1.1) and let  $f \geq 0$  with  $\|f\|_{L^n} = 1$ . Fix  $\lambda > 0$  and let  $\{v_j\}_{j \in J}$  be a maximal  $\delta$ -separated subset of  $\{v : \mathcal{K}_\delta f(v) > \lambda\}$ . For each  $j \in J$  choose a  $\delta$ -tube  $T_j \parallel v_j$  with

$$\frac{1}{m(T_j)} \int_{T_j} f > \lambda.$$

The  $T_j$  have  $\delta$ -separated directions, so (9.1.1) applies. Summing the defining inequality and using Hölder,

$$\lambda \sum_{j \in J} m(T_j) < \int_{\mathbb{R}^n} f \sum_{j \in J} \mathbf{1}_{T_j} \leq \|f\|_{L^n} \left\| \sum_{j \in J} \mathbf{1}_{T_j} \right\|_{L^{\frac{n}{n-1}}} \lesssim_{n,\varepsilon} \delta^{-\varepsilon} \left( \sum_{j \in J} m(T_j) \right)^{\frac{n-1}{n}}.$$

Writing  $A = \sum_{j \in J} m(T_j)$  and dividing by  $A^{(n-1)/n}$ ,

$$\lambda A^{1/n} \lesssim_{n,\varepsilon} \delta^{-\varepsilon}, \quad \text{so} \quad A \lesssim_{n,\varepsilon} \lambda^{-n} \delta^{-n\varepsilon}.$$

Since  $m(T_j) \approx_n \delta^{n-1}$  and the caps  $B(v_j, \delta/2)$  are disjoint with  $\sigma$ -measure  $\approx_n \delta^{n-1}$ , and since  $\mathcal{K}_\delta f$  is comparable across such a cap,

$$\sigma\{v : \mathcal{K}_\delta f(v) > \lambda\} \lesssim_n \#J \cdot \delta^{n-1} \approx_n A \lesssim_{n,\varepsilon} \lambda^{-n} \delta^{-n\varepsilon}.$$

This is the weak-type bound  $\mathcal{K}_\delta : L^n(\mathbb{R}^n) \rightarrow L^{n,\infty}(d\sigma)$  with constant  $\delta^{-\varepsilon}$ . Interpolating it against the trivial bound  $\mathcal{K}_\delta : L^\infty(\mathbb{R}^n) \rightarrow L^\infty(d\sigma)$  of Worked Example 9.1.1(a) by the Marcinkiewicz theorem (Theorem 2.2.4) gives the strong bound at any exponent above  $n$ ; applying the argument with  $n$  replaced by  $n - \eta$  and interpolating gives it at  $n$  itself, with  $\delta^{-\varepsilon}$  replaced by  $\delta^{-C\varepsilon}$ . Since  $\varepsilon > 0$  was arbitrary this is Conjecture 9.1.2.  $\square$

**Where we are.** Both formulations will be used. The maximal form is the one that interpolates and the one stated in the literature; the multiplicity

form (9.1.1) is the one that geometry produces, since every tube-counting argument bounds  $\sum \mathbf{1}_T$  directly.

Note that  $\sum_T m(T) \approx N\delta^{n-1}$ , and that a maximal  $\delta$ -separated set of directions has  $N \approx \delta^{-(n-1)}$ , so the right side of (9.1.1) is  $\approx \delta^{-\varepsilon}$  — the normalisation used in Volume 6, Theorem 6.6.2, and in Volume 7, Theorem 7.8.5.

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WORKED EXAMPLE 9.1.1— TIER A **What the trivial bounds give**

Two estimates are free, and knowing them fixes the scale of the problem.

(a)  $L^\infty$ .  $\mathcal{K}_\delta f(v) \leq \|f\|_\infty$  for every  $v$ , so  $\|\mathcal{K}_\delta f\|_{L^\infty(d\sigma)} \leq \|f\|_{L^\infty}$  with no loss.

(b)  $L^1$ . A tube has measure  $\approx \delta^{n-1}$ , so  $\mathcal{K}_\delta f(v) \leq \delta^{1-n}\|f\|_{L^1}$ , giving  $\|\mathcal{K}_\delta f\|_{L^\infty(d\sigma)} \lesssim \delta^{1-n}\|f\|_{L^1}$ .

(c) *Interpolating*. Theorem 2.2.1 applied to (a) and (b) gives

$$\|\mathcal{K}_\delta f\|_{L^\infty(d\sigma)} \lesssim \delta^{-\frac{n-1}{p}} \|f\|_{L^p(\mathbb{R}^n)},$$

and combining with the trivial  $L^\infty(d\sigma) \subseteq L^n(d\sigma)$  on a finite measure space, one obtains Conjecture 9.1.2 with  $\delta^{-\varepsilon}$  replaced by  $\delta^{-(n-1)/n}$ . That is the trivial bound, and the conjecture asserts the exponent can be reduced to 0.

*The scale of the problem.* So the whole subject lives between exponent  $\frac{n-1}{n}$ , free, and exponent 0, conjectured. Every theorem in this volume is a partial reduction of that exponent, and it is worth stating each result in those terms when comparing them.

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WORKED EXAMPLE 9.1.2— TIER B **Why  $L^n$  and not another exponent**

Conjecture 9.1.2 is stated at  $L^n$ , and the choice is forced by two examples.

*The example forcing  $p \leq n$ .* Take  $f = \mathbf{1}_{B(0,\delta)}$ , a ball of radius  $\delta$ . Every direction admits a tube through the ball, and  $\mathcal{K}_\delta f(v) \approx \delta^n/\delta^{n-1} = \delta$  for every  $v$ . So  $\|\mathcal{K}_\delta f\|_{L^q(d\sigma)} \approx \delta$  while  $\|f\|_{L^p} = \delta^{n/p}$ , and a bound with loss  $\delta^{-\varepsilon}$  requires  $\delta \lesssim \delta^{-\varepsilon}\delta^{n/p}$ , that is  $n/p \leq 1 + \varepsilon$ . Letting  $\varepsilon \rightarrow 0$  gives  $p \geq n$ .

*The example forcing  $p \geq n$ .* Take  $f = \mathbf{1}_{K_\delta}$  with  $K$  a Besicovitch set, so  $m(K_\delta)$  is small. Every direction has a tube inside  $K_\delta$ , so  $\mathcal{K}_\delta f \equiv 1$  and  $\|\mathcal{K}_\delta f\|_{L^q(d\sigma)} \approx 1$ , while  $\|f\|_{L^p} = m(K_\delta)^{1/p}$ . If  $m(K_\delta) \approx \delta^\eta$  then the bound requires  $1 \lesssim \delta^{-\varepsilon}\delta^{\eta/p}$ , so  $\eta/p \geq -\varepsilon$  — no constraint — but the *conclusion*  $\eta \leq p\varepsilon$  is exactly the Minkowski dimension statement, and it is strongest when  $p$  is smallest. Since the first example forces  $p \geq n$ , the strongest usable statement is at  $p = n$ .

*The moral.* The exponent  $n$  is where the two extremal examples — a single ball, and a Besicovitch set — balance. That is the same phenomenon as

in Volume 6, where the Knapp cap and the whole sphere balanced at the conjectured restriction exponent, and it is the standard way conjectures in this subject are pinned down.

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**WORKED EXAMPLE 9.1.3— TIER C: RESEARCH TASTE   Why a maximal function at all**

The Kakeya conjecture is a statement about sets. Conjecture 9.1.2 is a statement about an operator. It is worth being clear why the subject works with the second.

(a) *Operators interpolate; sets do not.* A bound for  $\mathcal{K}_\delta$  at one exponent can be combined with a bound at another to give bounds in between, by Theorem 2.2.1 or Theorem 2.2.4. Worked Example 9.1.1 used exactly this to get the trivial bound. There is no corresponding operation on statements about the dimension of a set.

(b) *Operators localise.*  $\mathcal{K}_\delta$  is defined at each scale separately, so a proof may work at one  $\delta$  and the conclusions be assembled afterwards — which is what Chapter 9.7 does. A statement about Hausdorff dimension refers to all scales at once and cannot be attacked scale by scale without the machinery of Volume 7.

(c) *Operators connect to the rest of analysis.* Volume 6, Theorem 6.6.2 produced a Kakeya statement from a restriction estimate, and the object it produced was a bound on  $\sum \mathbf{1}_T$  — the multiplicity form of Proposition 9.1.3. Had the target been a statement about Hausdorff dimension, the implication could not have been stated, let alone proved.

(d) *The cost.* The operator formulation is *a priori* stronger than the set statement, since it quantifies over all  $f$  rather than over indicator functions of neighbourhoods of Besicovitch sets. That the two are nonetheless equivalent — via Proposition 9.1.3 and Volume 7, Theorem 7.8.5 — is a theorem, not a definition, and it is what licenses the whole approach.

(e) *The historical remark.* Bourgain introduced the maximal operator in 1991 and the exponent moved for the first time in fifteen years shortly afterwards. That is not a coincidence: the reformulation made the tools of harmonic analysis applicable to a problem that had been the province of geometric measure theory.

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**WORKED EXAMPLE 9.1.4— TIER B   Linearising a maximal operator**



Step 2 of Proposition 9.1.3 began by choosing, for each direction, a tube nearly attaining the supremum. That move — *linearisation* — is used whenever a maximal operator must be estimated, and it repays being done slowly once.

*The problem.*  $\mathcal{K}_\delta$  is a supremum, so it is not linear in  $f$  and none of the machinery of Volumes 2 and 3 applies directly. Duality, in particular, is unavailable: there is no adjoint.

*The device.* Fix a measurable selection  $v \mapsto T(v)$  of tubes with

$$\frac{1}{m(T(v))} \int_{T(v)} |f| \geq \frac{1}{2} \mathcal{K}_\delta f(v),$$

and define the *linearised* operator  $Lf(v) = m(T(v))^{-1} \int_{T(v)} |f|$ . Then  $L$  is linear in  $|f|$ ,  $Lf \leq \mathcal{K}_\delta f \leq 2Lf$ , and any bound for  $L$  *uniform in the selection* is a bound for  $\mathcal{K}_\delta$ .

*Why uniformity is the whole point.* The selection depends on  $f$ . A bound whose constant depended on the selection would say nothing. So one proves the estimate for an arbitrary fixed assignment  $v \mapsto T(v)$  and only afterwards observes that the constant did not use which assignment was chosen. Every step of Proposition 9.1.3 Step 2 has this character — the tubes  $T_j$  are arbitrary subject to being  $\delta$ -separated in direction.

*The dual formulation that results.* With  $L$  linear, duality applies:  $\|Lf\|_{L^p(d\sigma)} \lesssim \|f\|_{L^p}$  is equivalent to a bound on  $L^*$ , and  $L^*$  applied to a sum of point masses is exactly  $\sum_j \mathbf{1}_{T_j}$  up to normalisation. That is where the multiplicity form (9.1.1) comes from, and it explains why the two formulations of Proposition 9.1.3 are adjoint to one another rather than merely equivalent.

*Where the technique recurs.* The Hardy–Littlewood maximal function of Volume 1 was handled by covering lemmas rather than linearisation, because there the geometry was simple enough. For  $\mathcal{K}_\delta$  no covering lemma is available — that is essentially the content of the *Takeya* problem — so linearisation is the only route, and it is used in every paper in this area.

## Drills

**Drill 9.1.1.** Verify that  $\mathcal{K}_\delta g$  is comparable on  $B(v_j, \delta/2)$  to its value at  $v_j$ , as used in Step 1.

**Drill 9.1.2.** Complete Step 2 of Proposition 9.1.3, carrying out the duality between the  $L^n \rightarrow L^n$  bound and the  $L^{n/(n-1)}$  bound on  $\sum \mathbf{1}_{T_j}$ .

**Drill 9.1.3.** Carry out Worked Example 9.1.1(c) explicitly and confirm the exponent  $\frac{n-1}{n}$  at  $p = n$ .

**Drill 9.1.4.** Show that Conjecture 9.1.2 fails for  $p < n$  using the ball example, and identify the largest  $q$  for which the  $L^p \rightarrow L^q$  statement survives.

**Where we are.** Two conventions, fixed here and used for the rest of the series.

*Tube families are always normalised.* A family  $\mathbb{T}$  has  $\delta$ -separated directions and  $N \approx_n \delta^{-(n-1)}$  members, so that  $\sum_T m(T) \approx_n 1$  by (9.3.1). Every estimate is stated against that normalisation, and a family with fewer tubes must be renormalised before the estimates apply — Worked Example 9.3.4 shows what goes wrong if it is not.

*Losses are always  $\delta^{-\varepsilon}$ .* No estimate in this volume tracks constants, and every one is stated with an arbitrarily small power loss. By Proposition 9.7.3 that is exactly the accuracy needed for the dimension conclusions and no more, so nothing is being given away.

With those in place the whole subject can be phrased as a single question: *how small can  $m(\bigcup_{T \in \mathbb{T}} T)$  be?* The conjecture says  $\delta^\varepsilon$ ; the trivial bound is  $\delta^{(n-1)/n}$  by Worked Example 9.1.1; and the rest of this volume closes part of the gap.

## Notes

The maximal operator was introduced by Bourgain in 1991, though the underlying counting had appeared earlier in Córdoba's work. Proposition 9.1.3 is standard and is the reason the literature moves freely between the operator and the multiplicity function.

## Chapter 9.2

# The Besicovitch Configuration

Volume 3 proved Fefferman's ball multiplier theorem from a geometric input which it stated and did not prove: a family of rectangles, overlapping so much that their union is tiny, whose translates by twice their own length are pairwise disjoint. This chapter constructs the family, discharging the only statement in Volumes 1 to 8 used without proof.

### 9.2.1 The statement

**Theorem 9.2.1** (Besicovitch configuration). *For every  $\varepsilon > 0$  there exist  $N \in \mathbb{N}$  and rectangles  $R_1, \dots, R_N \subseteq \mathbb{R}^2$ , with  $R_j$  of dimensions  $\ell_j \times w_j$  and long axis in direction  $v_j \in \mathbb{S}^1$ , such that, writing*

$$\tilde{R}_j = R_j + 2\ell_j v_j$$

*for the translate of  $R_j$  by twice its length along its own axis,*

*(i) the  $\tilde{R}_j$  are pairwise disjoint;*

$$(ii) \quad m\left(\bigcup_{j=1}^N R_j\right) \leq \varepsilon \sum_{j=1}^N m(R_j).$$

This is Volume 3, Lemma 3.8.4, and with it that volume's Theorem 3.8.2 is unconditional.

*Idea of the proof.* Everything comes from the Perron tree of Volume 1, used quantitatively rather than as a black box.

Recall the construction: a triangle is bisected  $k$  times into  $2^k$  elementary triangles sharing an apex, and each is translated horizontally by an amount determined by its position, so that the union has area at most  $\alpha^{2k} + 4(1 -$

$\alpha)/(1 + \alpha)$ . Property (ii) will follow immediately once each  $R_j$  is placed *inside* the corresponding translated triangle.

For (i) one must compute. Each triangle has a median running from the midpoint of its base to the apex; take  $R_j$  to be a thin rectangle about the lower part of that median, truncated at height  $h$ . Translating  $R_j$  by twice its length moves it to the band between heights  $2h$  and  $3h$ , and the question is whether the translates separate there.

They do, and the computation is a single line: the horizontal positions of the translates at a given height are an affine function of the binary expansion indexing the triangle, with slope  $y + 2h + 1 - 2\alpha$ . Since the rectangles have horizontal width  $(1 - h)2^{-k}$  and the positions are spaced  $2^{-k}$  apart, the translates are disjoint precisely when

$$2h + 1 - 2\alpha > 1 - h, \quad \text{that is} \quad h > \frac{2\alpha}{3}.$$

And the rectangles fit inside the triangles precisely when their width does not exceed the triangle's width at height  $h$ , which is  $(1 - h)2^{-k}$  — so  $h < 1$ . The interval  $(\frac{2\alpha}{3}, 1)$  is nonempty for every  $\alpha < 1$ , and that is the whole proof.

The tension is worth naming: *a longer rectangle separates better but fits worse*. The construction survives because the two constraints leave a gap, and the gap is what makes Fefferman's theorem true.

## 9.2.2 The construction

Fix  $\alpha \in (\frac{1}{2}, 1)$  and  $k \in \mathbb{N}$ , to be chosen in §9.3.

**Definition 9.2.2** (The Perron data). Let  $\Delta$  be the triangle with vertices  $(0, 0)$ ,  $(1, 0)$ ,  $(\frac{1}{2}, 1)$ , and for  $0 \leq i < 2^k$  let  $\Delta_i$  be the elementary triangle with base  $[i2^{-k}, (i+1)2^{-k}]$  and apex  $(\frac{1}{2}, 1)$ . Write the binary digits of  $i$  as  $b_1, \dots, b_k$ , so  $i = \sum_j b_j 2^{k-j}$ , and set

$$S_i = \sum_{j=1}^k b_j 2^{-j} \in [0, 1), \quad t_i = -2(1 - \alpha)S_i, \quad m_i = 2^{-k-1} + S_i,$$

so that  $m_i$  is the midpoint of the base of  $\Delta_i$  and  $\tau_i(x) = x + (t_i, 0)$  is the translation produced by iterating the two-triangle lemma. By Volume 1, Theorem 1.3.12,

$$m\left(\bigcup_i \tau_i \Delta_i\right) \leq \alpha^{2k} + \frac{4(1 - \alpha)}{1 + \alpha}. \quad (9.2.1)$$

The numbers  $S_i$  are exactly the  $2^k$  multiples of  $2^{-k}$  in  $[0, 1)$ , each occurring once; in particular

$$|S_i - S_{i'}| \geq 2^{-k} \quad (i \neq i'). \quad (9.2.2)$$

**Definition 9.2.3** (The rectangles). Fix  $h \in (\frac{2\alpha}{3}, 1)$ . For each  $i$  let

$$M_i = (m_i + t_i, 0), \quad C_i = (\tfrac{1}{2} + t_i, 1)$$

be the base midpoint and apex of  $\tau_i \Delta_i$ , and let

$$Q_i(y) = (m_i + t_i + y(\tfrac{1}{2} - m_i), y), \quad y \in [0, 1],$$

parametrise the median. Define  $R_i$  to be the closed rectangle with axis along the segment from  $Q_i(0) = M_i$  to  $Q_i(h)$ , of that length, and of width

$$w = \frac{2}{\sqrt{5}} (1 - h) 2^{-k}$$

measured perpendicular to the axis.

**Lemma 9.2.4** (The rectangles fit).  $R_i \subseteq \tau_i \Delta_i$  for every  $i$ , and

$$\sum_i m(R_i) \geq \frac{2}{\sqrt{5}} h(1 - h).$$

*Proof.* At height  $y$  the triangle  $\tau_i \Delta_i$  meets the horizontal line in an interval of length  $(1 - y)2^{-k}$ , centred at  $Q_i(y)$  — the median joins the midpoints of the horizontal cross-sections, by similar triangles. So the triangle contains every point within horizontal distance  $\frac{1}{2}(1 - y)2^{-k}$  of the median, for  $0 \leq y \leq 1$ .

The axis of  $R_i$  is the segment  $Q_i(0)Q_i(h)$ , whose direction is

$$v_i = \frac{(\frac{1}{2} - m_i, 1)}{|(\frac{1}{2} - m_i, 1)|}, \quad |\tfrac{1}{2} - m_i| \leq \tfrac{1}{2},$$

so the angle  $\theta_i$  between  $v_i$  and the vertical satisfies  $\cos \theta_i \geq 2/\sqrt{5}$ . A point at perpendicular distance  $s$  from the axis lies at horizontal distance  $s/\cos \theta_i$  from the median, so  $R_i$  lies within horizontal distance

$$\frac{w/2}{\cos \theta_i} \leq \frac{w}{2} \cdot \frac{\sqrt{5}}{2} = \tfrac{1}{2}(1 - h)2^{-k}$$

of the median, and its points have heights in  $[0, h]$ . Since  $(1 - y)2^{-k} \geq (1 - h)2^{-k}$  for  $y \leq h$ , the containment follows.

For the areas,  $R_i$  has length  $|Q_i(h) - Q_i(0)| \geq h$  — the vertical displacement alone is  $h$  — and width  $w$ , so  $m(R_i) \geq hw = \frac{2}{\sqrt{5}} h(1 - h)2^{-k}$ . Summing over the  $2^k$  values of  $i$  gives the stated bound.  $\square$

**Lemma 9.2.5** (The translates separate). *With  $\tilde{R}_i = R_i + 2\ell_i v_i$ , where  $\ell_i$  is the length of  $R_i$ , the rectangles  $\tilde{R}_1, \dots, \tilde{R}_{2^k}$  are pairwise disjoint.*

*Proof. Step 1. Locate the translates.*

Since  $\ell_i v_i = Q_i(h) - Q_i(0)$ , the translate  $\tilde{R}_i$  is the rectangle about the segment from  $Q_i(0) + 2(Q_i(h) - Q_i(0))$  to  $Q_i(h) + 2(Q_i(h) - Q_i(0))$ . Its points therefore lie at heights in  $[2h, 3h]$ , and the centre of its horizontal cross-section at height  $y + 2h$ , for  $y \in [0, h]$ , is the point

$$Q_i(y) + 2(Q_i(h) - Q_i(0)) = \left( m_i + t_i + (y + 2h)\left(\frac{1}{2} - m_i\right), y + 2h \right).$$

All the  $\tilde{R}_i$  occupy exactly the same band of heights, so it suffices to separate them at each height.

**Step 2. Compute the separation.**

Write  $x_i(y)$  for the first coordinate above. Using  $m_i = 2^{-k-1} + S_i$  and  $t_i = -2(1 - \alpha)S_i$  from Definition 9.2.2,

$$x_i(y) = 2^{-k-1} + S_i - 2(1 - \alpha)S_i + (y + 2h)\left(\frac{1}{2} - 2^{-k-1} - S_i\right),$$

so that, subtracting the same expression for  $i'$ , every term free of  $S$  cancels and

$$x_i(y) - x_{i'}(y) = \left(1 - 2(1 - \alpha) - (y + 2h)\right)(S_i - S_{i'}) = -(y + 2h + 1 - 2\alpha)(S_i - S_{i'}). \quad (9.2.3)$$

For  $y \in [0, h]$  the coefficient has absolute value at least  $2h + 1 - 2\alpha$ , which is positive since  $h > \frac{2\alpha}{3} > \alpha - \frac{1}{2}$ . Combining with (9.2.2),

$$|x_i(y) - x_{i'}(y)| \geq (2h + 1 - 2\alpha)2^{-k} \quad (i \neq i', y \in [0, h]).$$

**Step 3. Compare with the width.**

The horizontal extent of  $\tilde{R}_i$  at a given height is  $w/\cos\theta_i \leq \frac{\sqrt{5}}{2}w = (1 - h)2^{-k}$ , by the computation in Lemma 9.2.4. So the two cross-sections at height  $y + 2h$  are disjoint as soon as

$$(2h + 1 - 2\alpha)2^{-k} > (1 - h)2^{-k}, \quad \text{that is} \quad 3h > 2\alpha,$$

which is the hypothesis  $h > \frac{2\alpha}{3}$ . As this holds at every height in the common band, and the  $\tilde{R}_i$  meet no other heights, they are pairwise disjoint.  $\square$

### 9.2.3 Proof of the theorem

*Proof of Theorem 9.2.1.* Let  $\varepsilon > 0$ . Choose in order:

$$\alpha = 1 - \frac{\varepsilon}{100}, \quad h = \frac{2\alpha + 1}{3} \in \left(\frac{2\alpha}{3}, 1\right), \quad k \geq \frac{\log(\varepsilon/100)}{2 \log \alpha},$$

the last so that  $\alpha^{2k} \leq \varepsilon/100$ . Note  $h \geq \frac{2}{3}$  and  $1 - h = \frac{2(1-\alpha)}{3} \leq \frac{\varepsilon}{150}$ ; we shall also use  $h \leq 1$ .

Take  $N = 2^k$  and the rectangles  $R_1, \dots, R_N$  of Definition 9.2.3.

**Step 1.** *Disjointness of the translates.*

This is Lemma 9.2.5, whose hypothesis  $h > \frac{2\alpha}{3}$  holds by construction.

**Step 2.** *The union is small.*

By Lemma 9.2.4,  $R_i \subseteq \tau_i \Delta_i$ , so by (9.2.1)

$$m\left(\bigcup_i R_i\right) \leq m\left(\bigcup_i \tau_i \Delta_i\right) \leq \alpha^{2k} + \frac{4(1-\alpha)}{1+\alpha} \leq \frac{\varepsilon}{100} + 4(1-\alpha) \leq \frac{\varepsilon}{100} + \frac{4\varepsilon}{100} = \frac{\varepsilon}{20},$$

using  $1 + \alpha \geq 1$  and  $1 - \alpha = \varepsilon/100$ .

**Step 3.** *The total area is bounded below.*

By Lemma 9.2.4,

$$\sum_i m(R_i) \geq \frac{2}{\sqrt{5}} h(1-h).$$

This is where the choice of  $h$  must be examined, since  $1 - h$  is small: with  $1 - h = \frac{2(1-\alpha)}{3} = \frac{\varepsilon}{150}$  and  $h \geq \frac{2}{3}$ ,

$$\sum_i m(R_i) \geq \frac{2}{\sqrt{5}} \cdot \frac{2}{3} \cdot \frac{\varepsilon}{150} \geq \frac{\varepsilon}{300}.$$

**Step 4.** *Combine.*

By Steps 2 and 3,

$$\frac{m\left(\bigcup_i R_i\right)}{\sum_i m(R_i)} \leq \frac{\varepsilon/20}{\varepsilon/300} = 15,$$

which is a constant and not small — so the choice above does *not* give (ii) directly. The remedy is to decouple the two parameters:  $h$  was tied to  $\alpha$  only for definiteness, and any  $h \in (\frac{2\alpha}{3}, 1)$  is admissible. Take instead

$$h = \frac{9}{10},$$

which satisfies  $h > \frac{2\alpha}{3}$  for every  $\alpha < 1$  since  $\frac{2}{3} < \frac{9}{10}$ . Then  $1 - h = \frac{1}{10}$  is a fixed constant, and

$$\sum_i m(R_i) \geq \frac{2}{\sqrt{5}} \cdot \frac{9}{10} \cdot \frac{1}{10} \geq \frac{1}{13},$$

independent of  $\varepsilon$ . Now choose  $\alpha = 1 - \varepsilon/200$  and  $k$  with  $\alpha^{2k} \leq \varepsilon/200$ , so that Step 2 gives  $m(\bigcup R_i) \leq \varepsilon/200 + 4\varepsilon/200 = \varepsilon/40$ , and therefore

$$m\left(\bigcup_i R_i\right) \leq \frac{\varepsilon}{40} \leq \frac{\varepsilon}{40} \cdot 13 \sum_i m(R_i) \leq \varepsilon \sum_i m(R_i),$$

which is (ii). □

**Where we are.** Step 4 records a false start deliberately, because the trap is a natural one. It is tempting to tie  $h$  to  $\alpha$  — both control the same tension — and doing so makes  $1 - h$  shrink with  $\varepsilon$ , which destroys the lower bound on  $\sum m(R_i)$  exactly as fast as the union shrinks.

The two parameters must be decoupled. The constraint linking them,  $h > \frac{2\alpha}{3}$ , is satisfied by the *fixed* choice  $h = \frac{9}{10}$  for every  $\alpha < 1$ , because  $\frac{2\alpha}{3} < \frac{2}{3}$  always. So  $h$  may be frozen and  $\alpha$  sent to 1, which is what the proof does. That  $\frac{2\alpha}{3}$  stays below  $\frac{2}{3}$  is the one piece of luck in the construction. Had the constraint been  $h > \alpha$ , no fixed  $h$  would work and the theorem would be false as stated.

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WORKED EXAMPLE 9.2.1— TIER A **The numbers, for a concrete case**

Take  $\alpha = 0.95$  and  $h = 0.9$ , so  $k$  is whatever makes  $\alpha^{2k}$  small.

*The separation coefficient.* From (9.2.3) at  $y = 0$ ,

$$2h + 1 - 2\alpha = 1.8 + 1 - 1.9 = 0.9.$$

*The width, in the same units.*  $1 - h = 0.1$ .

*The margin.*  $0.9 > 0.1$  by a factor of 9 — the translates are not merely disjoint but separated by nine times their own width. The construction is not delicate at these parameters.

*Where it does become delicate.* At  $h$  just above  $\frac{2\alpha}{3}$  the two sides converge: with  $\alpha = 0.95$  and  $h = 0.64$ , the coefficient is  $2(0.64) + 1 - 1.9 = 0.38$  and the width is 0.36, a margin of only 6%. Taking  $h$  below  $\frac{2\alpha}{3} \approx 0.633$  reverses the inequality and the translates overlap.

*The lesson.* The admissible interval  $(\frac{2\alpha}{3}, 1)$  is genuinely open at the left end, and the geometric content of the construction is that it is nonempty. Drill 2.2 asks for the corresponding statement when the rectangles are truncated at the top rather than the bottom.

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**WORKED EXAMPLE 9.2.2— TIER B Why the translates separate: the mechanism, without coordinates**

The computation in Lemma 9.2.5 is three lines of algebra and can look like an accident. It is not, and the reason is worth having.

*The two motions.* As  $i$  increases by one, two things happen to the triangle  $\tau_i\Delta_i$ : its base slides *right* by  $2^{-k}$ , because  $m_i$  increases by that amount; and the Perron translation slides it *left* by  $2(1-\alpha)2^{-k}$ . The net motion of the base is right by  $(1-2(1-\alpha))2^{-k} = (2\alpha-1)2^{-k}$ .

*The apex barely moves.* The apex is at  $(\frac{1}{2} + t_i, 1)$ , so it moves *left* by  $2(1-\alpha)2^{-k}$  only — the  $m_i$  term is absent. This is the whole point of the Perron construction: the triangles are made to overlap by pinning their apexes close together while their bases spread out.

*Extending the median therefore amplifies.* A line through a base point moving right at rate  $2\alpha-1$  and an apex moving left at rate  $2(1-\alpha)$  must, when extended beyond the apex, move left at a rate that grows with the extension. At height  $y+2h$  the rate is  $y+2h+1-2\alpha$  — increasing in  $y$ , as (9.2.3) shows.

*So overlap below forces separation above.* The very mechanism that makes the union small — apexes pinned, bases spread — makes the extended tubes diverge. The construction does not balance two competing effects; it uses one effect twice, with opposite signs.

*Why this matters beyond the lemma.* Fefferman’s theorem is often described as “a Besicovitch set disproves the ball multiplier”. The description is accurate but hides the point: what is used is not merely that a Besicovitch set is small, but that its smallness is achieved by a mechanism which simultaneously makes translated copies disjoint. A different small set need not have that property, and the theorem would not follow from it.

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**WORKED EXAMPLE 9.2.3— TIER C: RESEARCH TASTE The debt, and what its payment settles**

(a) *What was outstanding.* From the writing of Volume 3 until now, the series contained exactly one statement used but not proved: Lemma 3.8.4, on which Theorem 3.8.2 — that the ball multiplier is bounded on  $L^p(\mathbb{R}^n)$  only for  $p=2$  — depended. Theorem 9.2.1 discharges it, and Volume 3’s theorem is now unconditional.

(b) *Why it took until Volume 9.* Not because the proof is long — it is three pages — but because it requires the Perron tree in *quantitative* form. Volume 1 proved the area bound (9.2.1) and, having no further use for the transla-

tions, did not record them. The present chapter needs the explicit formula  $t_i = -2(1 - \alpha)S_i$ , and it is that formula, not the area bound, which makes (9.2.3) computable.

*The transferable observation:* when a construction is used as a black box, record its internal parameters anyway. The second application will need them.

(c) *What remains unproved in Volumes 1–9.* With this chapter, nothing that is used. Nine results across Volumes 4 to 8 were removed from the plan by announced revisions R1 to R8, each stated where it would have appeared, attributed, and used nowhere; and Chapter 9.1 and this chapter add none.

(d) *What the configuration is good for besides Fefferman.* The same family, with the translates disjoint, is the model for every “overlapping tubes, separated extensions” argument in the subject. In Volume 12 the wave packets associated to a Besicovitch configuration are used to test multilinear estimates, and the disjointness of the extensions is what makes the test informative. The lemma is a tool, not only a step in one theorem.

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WORKED EXAMPLE 9.2.4— TIER B **Reading a construction for its parameters**

Worked Example 9.2.3 observed that the chapter needed the Perron tree’s internal translations, which Volume 1 had not recorded. The episode is worth generalising, since the same situation will arise again.

(a) *What Volume 1 recorded.* A single number: the area bound  $\alpha^{2k} + 4(1 - \alpha)/(1 + \alpha)$ . That is what Theorem 1.3.12 was for, and for the purposes of Volume 1 it sufficed — the Besicovitch set exists, and its measure is small.

(b) *What this chapter needed.* The formula  $t_i = -2(1 - \alpha)S_i$  for the translation applied to the  $i$ -th elementary triangle. Without it, (9.2.3) cannot be computed and the disjointness of the translates is inaccessible.

(c) *Why the difference matters.* An existence statement — “there is a set of small measure containing segments in all directions” — supports only arguments that use the set as a black box. A construction with recorded parameters supports arguments about the *mechanism*, and Worked Example 9.2.2 showed that Fefferman’s theorem needs the mechanism, not merely the conclusion.

(d) *The practical rule.* When a construction is given, record: the recursion, the parameter that controls it, the explicit form of every choice made at each stage, and the range of parameters for which the estimate holds. The first three are usually written down; the fourth almost never is, and it is what Theorem 9.2.1 needed most — the admissible interval  $h \in (\frac{2\alpha}{3}, 1)$  is exactly

a statement about parameter ranges.

(e) *A second instance, ahead.* Volume 10 will construct Kakeya sets from arithmetic progressions, and Volume 14 will need to know which progressions. The same discipline applies, and the manifest now carries a note to that effect.

**Proposition 9.2.6** (The configuration in the form Volume 3 uses). *For every  $\varepsilon > 0$  there are rectangles  $R_1, \dots, R_N$  as in Theorem 9.2.1 which in addition satisfy*

$$\sum_j m(R_j) \geq \frac{1}{13}, \quad m\left(\bigcup_j R_j\right) \leq \varepsilon,$$

*with all  $R_j$  contained in a fixed bounded region and all  $\tilde{R}_j$  contained in another.*

*Proof.* Take  $h = \frac{9}{10}$  and  $\alpha, k$  as in Step 4 of Theorem 9.2.1. The lower bound on  $\sum m(R_j)$  was computed there. The upper bound on the union is Step 2 with  $\varepsilon$  in place of  $\varepsilon/40$ , which only requires taking  $\alpha$  closer to 1.

For the containments: every  $R_j$  lies in  $\tau_j \Delta_j \subseteq \Delta'$ , where  $\Delta'$  is the triangle  $\Delta$  enlarged to absorb the translations, which are bounded by  $2(1 - \alpha) \leq 2$  in absolute value; so  $\Delta' \subseteq [-2, 3] \times [0, 1]$ . Each  $\tilde{R}_j$  lies at heights in  $[2h, 3h] \subseteq [\frac{9}{5}, \frac{27}{10}]$  and, by the computation in Lemma 9.2.5, at horizontal position  $m_j + t_j + (y + 2h)(\frac{1}{2} - m_j)$  with all factors bounded, hence in  $[-4, 5] \times [\frac{9}{5}, \frac{27}{10}]$ .  $\square$

**Where we are.** Volume 3's Lemma 3.8.4 was stated with the normalisation  $m(\bigcup R_j) \leq \varepsilon \sum m(R_j)$ , and Proposition 9.2.6 gives the two factors separately — which is what the proof of Theorem 3.8.2 actually consumes, since it applies the Rademacher–Khinchine argument to the family  $\{R_j\}$  and needs the total mass bounded below independently of  $\varepsilon$ .

The boundedness of the regions matters too: Fefferman's argument places the multiplier's symbol on a ball containing all the  $\tilde{R}_j$ , and a family whose translates escaped to infinity would not serve. Both requirements are implicit in Volume 3 and are recorded here so that the interface between the two volumes is explicit.

**Figure 9.2.1 (to be drawn).** *The Besicovitch configuration and its translates*

**Content required:** The Perron tree at a small  $k$ : the translated triangles  $\tau_i \Delta_i$  overlapping heavily near their common apex height, with the inscribed rectangles  $R_i$  shaded inside them, truncated at height  $h$ . Above, in the band between heights  $2h$  and  $3h$ , the translates  $\tilde{R}_i$  drawn *disjoint* and fanning apart. Mark the separation  $(2h + 1 - 2\alpha)2^{-k}$  against the width  $(1 - h)2^{-k}$ . This is the single most important figure in Volume 9.

## Drills

**Drill 9.2.1.** Verify (9.2.2): the numbers  $S_i$  are exactly the multiples of  $2^{-k}$  in  $[0, 1)$ , each occurring once.

**Drill 9.2.2.** Repeat Lemma 9.2.5 for rectangles truncated at the *top* — about the median from height  $h$  to height 1 — and determine the admissible range of  $h$ .

**Drill 9.2.3.** Check the claim in Lemma 9.2.4 that the median joins the midpoints of the horizontal cross-sections of a triangle, and that  $\cos \theta_i \geq 2/\sqrt{5}$ .

**Drill 9.2.4.** Track the constants in the proof of Theorem 9.2.1 to obtain an explicit  $N = N(\varepsilon)$ , and estimate its growth as  $\varepsilon \rightarrow 0$ .

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**WORKED EXAMPLE 9.2.5— TIER C: RESEARCH TASTE    The shape of a construction proof**

Theorem 9.2.1 is the only purely constructive theorem in this volume, and construction proofs have a different rhythm from estimate proofs. It is worth naming it, since Volumes 10 and 14 contain several more.

(a) *The object comes first, the verification second.* One writes down  $R_i$  explicitly — Definition 9.2.3 — and only then checks the two properties. There is no search and no optimisation in the statement; all the choosing happens in fixing  $\alpha$ ,  $k$  and  $h$ .

(b) *The parameters carry the difficulty.* Lemma 9.2.4 needs  $h < 1$ ; Lemma 9.2.5 needs  $h > \frac{2\alpha}{3}$ ; and the theorem is true because the interval between them is nonempty. A construction proof is, almost always, the assertion that a system of constraints on parameters is consistent, and the mathematics is in exhibiting a solution.

(c) *The false start is instructive and was kept.* Step 4 first ties  $h$  to  $\alpha$  and fails. That failure is not a defect of exposition: it is the natural first attempt, and seeing why it fails is what teaches the reader that the two parameters play different roles. A construction proof that presents only the successful choice conceals where the difficulty lay.

(d) *Verification before drafting.* The construction here was checked numerically — the separation coefficient computed for a range of  $\alpha$  and  $h$ , and compared against the rectangle width — before any of it was written. Two earlier candidates failed that check: constant-width tubes do not fit inside the triangles, and half-height tubes fit but their translates collide as  $\alpha \rightarrow 1$ .

Neither failure is visible from the statement, and both would have survived a plausible-looking write-up.

(e) *The general recommendation.* For a construction, verify a small case before writing. For an estimate, check the exponents in the extreme cases. In both, the aim is to fail fast: the cost of discovering an error after the proof is written is far higher than the cost of the check.

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## Notes

The configuration is due to Besicovitch, and its use for the ball multiplier is Fefferman's (1971). The explicit form given here — with the truncation parameter  $h$  isolated and the admissibility condition  $h > \frac{2\alpha}{3}$  made visible — follows the structure of Fefferman's argument while keeping the Perron parameters of Volume 1.

Readers comparing with other treatments will find the rectangles described as “a Besicovitch set of rectangles” with the disjointness of translates asserted as part of the construction. Making it a computation, as here, has the advantage that the constraint linking the truncation to the Perron parameter becomes explicit, and with it the reason the construction is possible at all.



## Chapter 9.3

# The $L^2$ Method

The simplest way to bound the union of a tube family is to expand a square. This chapter carries that out, obtains the sharp planar result, and identifies precisely why the method stalls at dimension 2 in every ambient dimension.

### 9.3.1 The estimate

Throughout,  $\mathbb{T} = \{T_1, \dots, T_N\}$  is a family of  $\delta$ -tubes with  $\delta$ -separated directions  $v_1, \dots, v_N$ , and  $N \approx_n \delta^{-(n-1)}$  so that

$$\sum_j m(T_j) \approx_n N \delta^{n-1} \approx_n 1. \quad (9.3.1)$$

We write  $E = \bigcup_j T_j$  and  $F = \sum_j \mathbf{1}_{T_j}$  for the multiplicity function.

**Lemma 9.3.1** (Angle counting). *For each  $j$  and each dyadic  $\theta = 2^m \delta \leq 1$ ,*

$$\#\{j' : \angle(v_j, v_{j'}) \approx \theta\} \lesssim_n \left(\frac{\theta}{\delta}\right)^{n-1}.$$

*Proof.* The directions  $v_{j'}$  in question lie in a spherical annulus about  $v_j$  of angular radius  $\approx \theta$ , of  $\sigma$ -measure  $\lesssim_n \theta^{n-1}$ . They are  $\delta$ -separated, so the caps  $B(v_{j'}, \delta/2)$  are disjoint and each has measure  $\approx_n \delta^{n-1}$ . Comparing measures gives the bound.  $\square$

**Theorem 9.3.2** (The  $L^2$  bound). *With the notation above,*

$$\|F\|_{L^2(\mathbb{R}^n)}^2 \lesssim_n \begin{cases} \log \frac{1}{\delta}, & n = 2, \\ \delta^{2-n}, & n \geq 3. \end{cases}$$

*Proof. Step 1. Expand.*

$$\|F\|_{L^2}^2 = \int \left( \sum_j \mathbf{1}_{T_j} \right)^2 = \sum_{j,j'} m(T_j \cap T_{j'}).$$

**Step 2.** *Insert the intersection estimate.*

By Volume 1, Lemma 1.6.13, two  $\delta$ -tubes whose directions differ by angle  $\theta$  satisfy

$$m(T \cap T') \lesssim_n \frac{\delta^n}{\theta + \delta}.$$

Fix  $j$  and group the  $j'$  by dyadic angle  $\theta = 2^m \delta$ ,  $0 \leq m \leq \log_2(1/\delta)$ . By Lemma 9.3.1 the group has  $\lesssim (2^m)^{n-1}$  members, and each contributes  $\lesssim \delta^n / (2^m \delta) = \delta^{n-1} 2^{-m}$ . Hence

$$\sum_{j'} m(T_j \cap T_{j'}) \lesssim_n \sum_{m=0}^{\log_2(1/\delta)} 2^{m(n-1)} \delta^{n-1} 2^{-m} = \delta^{n-1} \sum_{m=0}^{\log_2(1/\delta)} 2^{m(n-2)}. \quad (9.3.2)$$

**Step 3.** *Evaluate the sum.*

If  $n = 2$  the summand is 1 and the sum is  $\approx \log \frac{1}{\delta}$ . If  $n \geq 3$  the sum is geometric with ratio  $2^{n-2} > 1$ , so it is dominated by its last term, at  $2^m \approx \delta^{-1}$ , giving  $\approx \delta^{-(n-2)}$ . Substituting into (9.3.2),

$$\sum_{j'} m(T_j \cap T_{j'}) \lesssim_n \begin{cases} \delta \log \frac{1}{\delta}, & n = 2, \\ \delta^{n-1} \delta^{-(n-2)} = \delta, & n \geq 3. \end{cases}$$

**Step 4.** *Sum over  $j$ .*

Multiplying by  $N \approx \delta^{-(n-1)}$  gives  $\|F\|_{L^2}^2 \lesssim \delta^{-(n-1)} \delta \log \frac{1}{\delta} = \log \frac{1}{\delta}$  when  $n = 2$ , and  $\delta^{-(n-1)} \delta = \delta^{2-n}$  when  $n \geq 3$ .  $\square$

**Corollary 9.3.3** (Córdoba's bound). *For every  $\delta$ -tube family as above,*

$$m(E) \gtrsim_n \begin{cases} (\log \frac{1}{\delta})^{-1}, & n = 2, \\ \delta^{n-2}, & n \geq 3. \end{cases}$$

*Consequently every Besicovitch set in  $\mathbb{R}^n$  has upper Minkowski dimension at least 2, and in the plane it has Minkowski dimension exactly 2.*

*Proof.* By (9.3.1) and Cauchy–Schwarz,  $F$  being supported in  $E$ ,

$$1 \approx_n \sum_j m(T_j) = \int_E F \leq m(E)^{1/2} \|F\|_{L^2},$$



so  $m(E) \geq \|F\|_{L^2}^{-2}$ , and Theorem 9.3.2 gives the two cases.

For the dimension: a Besicovitch set  $K$  has  $K_\delta$  containing a  $\delta$ -tube in each of  $\approx \delta^{-(n-1)}$  separated directions, so  $m(K_\delta) \geq m(E)$  for that family. By Proposition 7.2.3,  $\overline{\dim}_M K \geq n - \liminf \frac{\log m(K_\delta)}{\log \delta}$ , and the two bounds give  $\geq n - 0 = n$  for  $n = 2$  — a logarithm contributes nothing to the exponent — and  $\geq n - (n - 2) = 2$  for  $n \geq 3$ .  $\square$

**Where we are.** The planar case is sharp: Corollary 9.3.3 gives Minkowski dimension 2, which is the Kakeya conjecture for  $n = 2$  at the level of Minkowski dimension, and the logarithm is genuinely there — Volume 1's Perron tree produces sets with  $m(K_\delta) \approx 1/\log \frac{1}{\delta}$ , so the estimate cannot be improved.

In every higher dimension the method gives dimension 2 and no more, which is worse than the bush bound  $\frac{n+1}{2}$  as soon as  $n \geq 4$ . The next section explains where the loss is.

## 9.3.2 Why the method stalls

### WORKED EXAMPLE 9.3.1— TIER B The exact point of failure

The whole of Theorem 9.3.2 turns on the sum  $\sum_m 2^{m(n-2)}$  in (9.3.2), and its behaviour changes character at  $n = 2$ .

*$n = 2$ : every scale contributes equally.* The summand is 1, so all  $\log \frac{1}{\delta}$  dyadic angles contribute the same amount. No single angular scale dominates, and the answer is the number of scales.

*$n \geq 3$ : the largest angles dominate.* The summand grows geometrically, so the sum is carried by  $\theta \approx 1$  — tubes that are nearly perpendicular to  $T_j$ . But those are exactly the pairs whose intersection is *smallest*,  $m(T \cap T') \approx \delta^n$ . The sum is dominated by the most numerous pairs rather than the most overlapping ones.

*Why that is a loss.* The quantity  $\|F\|_{L^2}^2$  counts all intersections with equal weight, and in high dimensions the count is dominated by a generic pair, about which nothing is being assumed. The estimate is therefore insensitive to the *arrangement* of the tubes: it gives the same answer for a Besicovitch configuration as for a random one. Since a random arrangement has  $m(E) \approx 1$ , an estimate that cannot distinguish the two cannot prove anything strong.

*What is needed instead.* An argument that uses the tubes' incidence structure — which tubes meet which — rather than only the total count of intersections. The bush argument of Chapter 9.4 is the first such: it asks how many tubes pass through a single point, which  $\|F\|_{L^2}^2$  averages away.

*A general moral.* An  $L^2$  argument is a second-moment argument, and second moments see only pair correlations. This is the same limitation identified in Volume 8, Worked Example 7.7.3, in Fourier language, and it is the same reason the Fourier bound stalls at  $\frac{n+1}{2}$ . Both methods are two-point; Kakeya is not.

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**WORKED EXAMPLE 9.3.2— TIER A   Checking the planar bound against the Perron tree**

Corollary 9.3.3 says  $m(E) \gtrsim 1/\log \frac{1}{\delta}$  in the plane. It is worth confirming this is attained.

Volume 1, Theorem 1.3.12 gives, for each  $\alpha$  and  $k$ , a union of translated triangles of area at most  $\alpha^{2k} + 4(1 - \alpha)/(1 + \alpha)$ . Taking  $\delta = 2^{-k}$  — so that the elementary triangles have base  $\delta$  — and optimising  $\alpha$ , one finds the area is  $\approx 1/k = 1/\log_2 \frac{1}{\delta}$ : choose  $1 - \alpha \approx 1/k$ , so that the second term is  $\approx 1/k$  and  $\alpha^{2k} \approx e^{-2}$  is bounded, then iterate the construction  $\log k$  times to remove the constant.

So the Perron tree achieves  $m(E) \approx 1/\log \frac{1}{\delta}$  and Corollary 9.3.3 is sharp up to constants. In particular the planar Kakeya problem is completely settled at the level of Minkowski dimension, with matching upper and lower bounds — one of the few places in this subject where that is true.

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**WORKED EXAMPLE 9.3.3— TIER C: RESEARCH TASTE   Second moments, and the limits of averaging**

Worked Example 9.3.1 diagnosed the  $L^2$  method's ceiling. The diagnosis generalises, and it is worth stating in the form that applies across the series.

(a) *What a second moment is.*  $\|F\|_{L^2}^2 = \sum_{T, T'} m(T \cap T')$  is a sum over *pairs*. It is determined by the pair correlation structure of the family and by nothing else: two configurations with the same multiset of pairwise intersections have the same  $L^2$  norm, however different they look.

(b) *Why Kakeya is not determined by pairs.* A Besicovitch set is small because *many* tubes concentrate in a small region — a statement about high-order coincidences. Drill 3.4 asks for the computation showing that a random family and an extremal family have comparable second moments while their unions differ by a power of  $\delta$ .

(c) *The same limitation, three times over.* This is the third appearance of the identical obstruction. In Volume 7, Worked Example 7.7.3 showed the Fourier method sees only  $I_t(\mu)$ , a two-point quantity, and stalls at  $\frac{n+1}{2}$ . In Volume 8 the distance set problem *was* two-point and the Fourier method

was accordingly the right tool. Here the  $L^2$  method is the same thing in physical space. Fourier transform, energy integral, and second moment are three names for one restriction on what an argument can see.

(d) *What escapes it.* An argument conditioning on a configuration — Chapter 9.4 on a point, Chapter 9.5 on a line — extracts information no average carries. So does an argument using algebraic structure, as Volume 11 does. The common feature is that neither can be written as a sum over pairs.

(e) *A test to apply.* Given a proposed method, ask: could it distinguish a Besicovitch configuration from a random one with the same pairwise statistics? If not, its ceiling is already visible, and it will be whatever the second moment gives.

**WORKED EXAMPLE 9.3.4— TIER B Running the  $L^2$  computation for a sub-family** Theorem 9.3.2 was stated for a full family of  $\approx \delta^{-(n-1)}$  tubes. Theorem 9.5.3 needed the planar case for a sub-family of  $m$  tubes, and the modification is worth doing carefully, since the naive version gives the wrong answer.

*The naive attempt.* Apply Theorem 9.3.2 with  $n = 2$  to conclude  $\|\sum_{j \leq m} \mathbf{1}_{T_j}\|_{L^2}^2 \lesssim \log \frac{1}{\delta}$ . This is *false* reasoning: the bound was obtained by multiplying a per-tube estimate by  $N \approx \delta^{-1}$ , and with only  $m$  tubes the multiplication is by  $m$ .

*The correct computation.* For a sub-family of  $m$  tubes in the plane with  $\delta$ -separated directions, the angle count of Lemma 9.3.1 becomes

$$\#\{j' : \theta \approx 2^p \delta\} \leq \min\{2^p, m\},$$

since there are at most  $2^p$  directions available at that angular scale and at most  $m$  tubes in total. So, per tube,

$$\sum_{j'} m(T_j \cap T_{j'}) \lesssim \sum_p \min\{2^p, m\} \frac{\delta^2}{2^p \delta} = \delta \sum_p \min\{1, m 2^{-p}\} \lesssim \delta \log \frac{1}{\delta},$$

the sum being  $\lesssim \log m + O(1) \leq \log \frac{1}{\delta}$ . Multiplying by  $m$ ,

$$\left\| \sum_{j \leq m} \mathbf{1}_{T_j} \right\|_{L^2}^2 \lesssim m \delta \log \frac{1}{\delta}.$$

*The conclusion.* With  $\sum_j m_2(T_j) = m\delta$  and Cauchy–Schwarz,

$$m\delta \leq m_2\left(\bigcup T_j\right)^{1/2} (m\delta \log \frac{1}{\delta})^{1/2}, \quad \text{so} \quad m_2\left(\bigcup T_j\right) \gtrsim \frac{m\delta}{\log \frac{1}{\delta}},$$

which is the estimate used in Theorem 9.5.3 Step 3.

*The check.* At  $m \approx \delta^{-1}$ , a full family, this gives  $m \gtrsim 1/\log \frac{1}{\delta}$  — Corollary 9.3.3. ✓ At  $m = 1$  it gives  $m \gtrsim \delta/\log$ , slightly worse than the truth  $\delta$ ; the logarithm is an artefact of the method for small families and costs nothing where it is used.

*The habit worth forming.* Whenever a stated estimate is applied to a subfamily, re-derive it rather than quoting it: the normalisations almost always change, and the error is invisible in the final statement.

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## Drills

**Drill 9.3.1.** Verify Lemma 9.3.1 in the plane, where the directions form a circle and the count is  $\approx \theta/\delta$ .

**Drill 9.3.2.** Carry out Step 3 for  $n = 3$  explicitly and confirm the exponent  $\delta^{2-n} = \delta^{-1}$ .

**Drill 9.3.3.** Show that the bound of Corollary 9.3.3 for  $n \geq 3$  is weaker than the bush bound  $\delta^{(n-1)/2}$  exactly when  $n \geq 4$ , and that the two agree at  $n = 3$ .

**Drill 9.3.4.** Compute  $\|F\|_{L^2}^2$  for a *random* family of  $\delta$ -separated tubes and confirm that Theorem 9.3.2 is attained, so the estimate cannot distinguish random from extremal configurations.

**Where we are.** One point of technique before leaving the chapter. The estimate  $m(T \cap T') \lesssim \delta^n/(\theta + \delta)$  of Lemma 1.6.13 is used in every argument in this volume, and it is worth remembering in the form: the intersection of two  $\delta$ -tubes is a box of dimensions  $\delta \times \cdots \times \delta \times \min\{1, \delta/\theta\}$ , the last factor being the length along which they stay together.

At  $\theta \approx \delta$  they are parallel to within their own width and overlap along their whole length, giving  $\delta^{n-1}$ . At  $\theta \approx 1$  they cross transversally and meet in a cube of side  $\delta$ , giving  $\delta^n$ . Every intermediate case interpolates, and the interpolation is exactly the factor  $\delta/\theta$ .

Reconstructing the estimate from that picture is quicker than recalling it, and it makes the angle-counting of Lemma 9.3.1 feel inevitable rather than technical.

## Notes

The  $L^2$  argument is Córdoba's (1977), and in the plane it remains the standard proof. The observation that it stalls at dimension 2 in higher dimensions — and that the obstruction is the dominance of nearly perpendicular pairs — is the starting point of Bourgain's and Wolff's work.



## Chapter 9.4

# The Bush Argument

Volume 1 proved the bound  $\dim_{\mathbb{H}} K \geq \frac{n+1}{2}$  by an argument about tubes through a common point. This chapter restates it in the language of Chapter 9.1, where the mechanism is clearer, and identifies what it uses that the  $L^2$  method of Chapter 9.3 does not.

### 9.4.1 Two bounds in competition

As before  $\mathbb{T} = \{T_1, \dots, T_N\}$  is a family of  $\delta$ -tubes with  $\delta$ -separated directions,  $N \approx_n \delta^{-(n-1)}$ ,  $E = \bigcup_j T_j$  and  $F = \sum_j \mathbf{1}_{T_j}$ .

**Definition 9.4.1.** The *multiplicity* of the family is  $\mu = \|F\|_{L^\infty} = \sup_x \#\{j : x \in T_j\}$ .

**Lemma 9.4.2** (The bush estimate). *Suppose  $x_0$  lies in  $\mu$  of the tubes. Then*

$$m\left(\bigcup_{j: x_0 \in T_j} T_j\right) \gtrsim_n \mu \delta^{n-1}.$$

*Proof.* Let  $J = \{j : x_0 \in T_j\}$ , so  $\#J = \mu$ . Each  $T_j$  with  $j \in J$  has length 1 and contains  $x_0$ , so it contains a sub-tube  $T'_j$  of length  $\frac{1}{4}$  lying at distance between  $\frac{1}{4}$  and  $\frac{1}{2}$  from  $x_0$  — one of the two halves of  $T_j$  has this property. Then  $m(T'_j) \approx_n \delta^{n-1}$ .

We claim the  $T'_j$ ,  $j \in J$ , have bounded overlap. Let  $j \neq j'$  in  $J$  and let  $\theta = \angle(v_j, v_{j'}) \geq \delta$ , the directions being  $\delta$ -separated. If  $y \in T'_j \cap T'_{j'}$ , then  $y$  is within  $\delta$  of the line through  $x_0$  in direction  $v_j$  and within  $\delta$  of the line through  $x_0$  in direction  $v_{j'}$ , and  $|y - x_0| \geq \frac{1}{4}$ . The two lines meet only at  $x_0$  and separate at rate  $\theta$ , so

$$2\delta \geq \text{dist}(y, \text{line } j) + \text{dist}(y, \text{line } j') \gtrsim |y - x_0| \theta \gtrsim \theta,$$

forcing  $\theta \lesssim \delta$ . Since the directions are  $\delta$ -separated, only boundedly many  $j'$  can satisfy this for a given  $j$ . Hence

$$m\left(\bigcup_{j \in J} T_j\right) \geq m\left(\bigcup_{j \in J} T'_j\right) \gtrsim_n \sum_{j \in J} m(T'_j) \approx_n \mu \delta^{n-1}. \quad \square$$

**Lemma 9.4.3** (The multiplicity estimate).  $m(E) \geq \mu^{-1} \sum_j m(T_j) \approx_n \mu^{-1}$ .

*Proof.*  $F \leq \mu$  pointwise and  $F$  is supported in  $E$ , so  $\sum_j m(T_j) = \int_E F \leq \mu m(E)$ .  $\square$

**Theorem 9.4.4** (Bush bound). *For every  $\delta$ -tube family as above,*

$$m(E) \gtrsim_n \delta^{\frac{n-1}{2}}.$$

*Consequently every Besicovitch set in  $\mathbb{R}^n$  has upper Minkowski dimension at least  $\frac{n+1}{2}$ .*

*Proof. Step 1. The two bounds.*

Let  $\mu$  be the multiplicity and let  $x_0$  be a point lying in  $\mu$  tubes, which exists by definition of  $\mu = \|F\|_{L^\infty}$  and the fact that  $F$  is integer-valued. Lemma 9.4.2 applied at  $x_0$  gives

$$m(E) \geq m\left(\bigcup_{j: x_0 \in T_j} T_j\right) \gtrsim_n \mu \delta^{n-1}, \quad (9.4.1)$$

the first inequality because the bush is a subfamily of  $\mathbb{T}$ . And Lemma 9.4.3 gives

$$m(E) \gtrsim_n \mu^{-1}. \quad (9.4.2)$$

**Step 2. Eliminate  $\mu$ .**

The parameter  $\mu$  is not under our control: it depends on the family, and both extremes are conceivable *a priori*. But (9.4.1) is strong when  $\mu$  is large and (9.4.2) is strong when  $\mu$  is small, so their product is free of  $\mu$ :

$$m(E)^2 \gtrsim_n (\mu \delta^{n-1}) \cdot \mu^{-1} = \delta^{n-1}.$$

Equivalently,  $m(E) \geq \max\{c\mu\delta^{n-1}, c\mu^{-1}\}$  and the right side is minimised over  $\mu > 0$  where the two agree, at  $\mu = \delta^{-(n-1)/2}$ , with common value  $c\delta^{(n-1)/2}$ . Either way,

$$m(E) \gtrsim_n \delta^{\frac{n-1}{2}}.$$

**Step 3. Pass to dimension.**

A Besicovitch set  $K$  has  $K_\delta$  containing an admissible family, so  $m(K_\delta) \gtrsim_n \delta^{(n-1)/2}$ ; by Proposition 7.2.3,  $\overline{\dim}_M K \geq n - \frac{n-1}{2} = \frac{n+1}{2}$ .  $\square$



**Where we are.** The proof is two inequalities and a multiplication, and it is worth seeing why multiplying is the right move. The two bounds pull in opposite directions: a large multiplicity makes the bush large, a small multiplicity makes the union large. Neither alone is useful, because  $\mu$  is not under our control. Taking the geometric mean eliminates  $\mu$  entirely. This is the standard shape of an extremal argument in the subject — produce two bounds whose product is  $\mu$ -free — and it recurs in Chapter 9.5, where the two bounds are more refined and the product is correspondingly better.

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WORKED EXAMPLE 9.4.1— TIER B **What the bush uses that the  $L^2$  method does not**

Both arguments bound  $m(E)$  from below, and the bush is better as soon as  $n \geq 4$  (Drill 3.3). The difference is worth isolating, since it is the first instance of a pattern that governs the rest of the subject.

*The  $L^2$  method uses only  $\|F\|_{L^2}$ , a second moment.* By Worked Example 9.3.1 that quantity is dominated, in high dimensions, by pairs of nearly perpendicular tubes — generic pairs, carrying no information about the configuration.

*The bush uses  $\|F\|_{L^\infty}$ , the maximum multiplicity, together with a geometric fact about the tubes realising it.* That is a statement about a *single point* and all the tubes through it: an incidence, not an average.

*Why the incidence is worth more.* Lemma 9.4.2 says that  $\mu$  tubes through a common point cannot be packed — at distance  $\approx 1$  from the point they have separated, because  $\delta$ -separated directions diverge at rate  $\delta$  and a unit distance amplifies that to  $\delta$ , which is exactly the tube width. The estimate is sharp and it is genuinely three-point in character: it concerns the point, and two tubes, and the distance between them.

*The general principle. Averages see pairs; incidences see configurations.* Every improvement in this volume and the next replaces an average by a more refined incidence count:

| Method    | What is counted       | Exponent        |
|-----------|-----------------------|-----------------|
| $L^2$     | pairs of tubes        | 2               |
| bush      | tubes through a point | $\frac{n+1}{2}$ |
| hairbrush | tubes through a line  | $\frac{n+2}{2}$ |

The progression is not accidental: each step conditions on a richer geometric object.

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**WORKED EXAMPLE 9.4.2— TIER A The bush bound in low dimensions, and where it is beaten**

$n = 2$ . The bush gives  $\frac{n+1}{2} = \frac{3}{2}$ , while Corollary 9.3.3 gives 2 — the truth. So in the plane the  $L^2$  method is strictly better, and the bush is never used.  
 $n = 3$ . Both give 2. The bush and the  $L^2$  method agree, and Wolff’s hairbrush will give  $\frac{n+2}{2} = \frac{5}{2}$ .

$n = 4$ . The bush gives  $\frac{5}{2}$ , the  $L^2$  method 2. The bush wins from here on.

*Large  $n$ .* The bush gives  $\frac{n+1}{2}$ , growing linearly, while the  $L^2$  method is stuck at 2. But the conjecture asks for  $n$ , and  $\frac{n+1}{2}$  is asymptotically half of it — so in high dimensions even the better elementary bound recovers only half the truth.

*The lesson for reading the literature.* A bound of the form  $\dim_{\mathbb{H}} K \geq cn$  with  $c < 1$  fixed is, in high dimensions, qualitatively far from the conjecture, however impressive the constant. The bounds that matter asymptotically are those of the form  $n - o(n)$ , and the first of those came from the arithmetic methods of Volume 10.

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**WORKED EXAMPLE 9.4.3— TIER C: RESEARCH TASTE The bush as the first genuinely geometric argument**

Theorem 9.4.4 is short, and its brevity can obscure that it is the first argument in this series to use the geometry of tubes rather than their statistics.

(a) *The one geometric input.* Lemma 9.4.2 rests on a single fact: two lines through a common point at angle  $\theta$  are, at distance  $r$  from that point, about  $r\theta$  apart. With  $r \approx 1$  and  $\theta \geq \delta$  this exceeds the tube width  $\delta$ , so the tubes have separated. Nothing else about the configuration is used.

(b) *Why the unit distance matters.* The tubes have length 1 and the separation is  $r\theta$ , so the argument needs  $r \approx 1$  to convert  $\theta \geq \delta$  into separation  $\geq \delta$ . Shorter tubes would not do: a family of  $\delta$ -tubes of length  $\rho$  gives separation  $\rho\delta < \delta$  and the bush estimate fails. This is why the Kakeya problem is stated with unit segments, and why rescaling arguments must be careful.

(c) *What it does not use.* The bush argument never uses that the family has  $\approx \delta^{-(n-1)}$  tubes, nor that the directions fill the sphere. It applies verbatim to any family with  $\delta$ -separated directions, and gives  $m \gtrsim \mu\delta^{n-1}$  for the subfamily through a point. That generality is why it survives into the later volumes as a lemma.

(d) *Its role later.* Volume 11’s polynomial partitioning argument repeatedly reduces to bounding a bush; Volume 14’s analysis of sticky configurations uses the bush estimate as the base case of an induction. The argument of this chapter is therefore not superseded by the later ones — it is absorbed

into them.

(e) *The honest assessment.* As a bound it is weak: half the conjectured exponent, and beaten by the  $L^2$  method in the plane. As a lemma it is permanent. Distinguishing these two kinds of value is worth doing when reading any result in this area.

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**WORKED EXAMPLE 9.4.4— TIER B   Producing the two competing bounds: a template**

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The bush argument, the hairbrush argument, and several later ones all have the shape “two bounds, multiply”. The template is worth stating so that the next instance can be recognised.

*The setup.* One wants a lower bound on a quantity  $V$  (here  $m(E)$ ) and there is an unknown structural parameter  $\mu$  (here the multiplicity). One produces:

$$(i) \quad V \gtrsim \mu A, \qquad (ii) \quad V \gtrsim \mu^{-1} B,$$

with  $A, B$  known. Then  $V^2 \gtrsim AB$  and  $V \gtrsim (AB)^{1/2}$ .

*Where each bound comes from.* Bound (i) is *geometric*: it says a large  $\mu$  forces a large configuration — here, that  $\mu$  tubes through a point cannot be packed (Lemma 9.4.2). Bound (ii) is *measure-theoretic*: it is Chebyshev,  $\int F \leq \|F\|_\infty m(\text{spt } F)$  (Lemma 9.4.3). The two always have this character.

*How to improve the result.* Only by improving  $A$  or  $B$ . In Chapter 9.5 the improvement is to  $A$ : collecting tubes along a line rather than at a point raises  $A$  from  $\delta^{n-1}$  to  $\delta^{n-2}$ . Improving  $B$  would mean a better Chebyshev step, and there is none — (ii) is an identity up to constants.

*How to know when the template is exhausted.* Compute the balancing value  $\mu_* = (B/A)^{1/2}$  and ask whether a configuration with that multiplicity exists. Worked Example 9.8.2 carries this out and finds that for large  $n$  it does not — so the bound is being determined by a hypothetical worst case. At that point the template must be abandoned rather than refined.

*Two later instances.* Volume 11’s polynomial partitioning produces bounds of exactly this shape, with  $\mu$  the number of tubes in a cell; and Volume 12’s broad–narrow decomposition splits according to a parameter and optimises, which is the same device with the roles of the two bounds played by the broad and narrow parts. Recognising the template makes those arguments readable in outline before working through them line by line.

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**Figure 9.4.1 (to be drawn).** *A bush of tubes*

**Content required:** Six to eight  $\delta$ -tubes through a common point  $x_0$ , with  $\delta$ -separated directions, shown converging at  $x_0$  and visibly separated at distance  $\approx 1$ . Mark one pair with its angle  $\theta$  and the separation  $\approx r\theta$  at distance  $r$ , illustrating why the estimate needs  $r \approx 1$  to convert  $\theta \geq \delta$  into separation exceeding the tube width.

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## Drills

**Drill 9.4.1.** Verify the separation computation in Lemma 9.4.2: two lines through  $x_0$  at angle  $\theta$  are at distance  $\gtrsim |y - x_0|\theta$  apart at the point  $y$ .

**Drill 9.4.2.** Show that Lemma 9.4.2 is sharp, by exhibiting  $\mu$  tubes through a point whose union has measure  $\approx \mu\delta^{n-1}$ .

**Drill 9.4.3.** Explain why the two bounds of Theorem 9.4.4 cannot both be improved by the same argument, and identify the configuration that makes each sharp.

**Drill 9.4.4.** Combine Corollary 9.3.3 and Theorem 9.4.4 to state the best elementary bound in each dimension  $n = 2, \dots, 6$ .

**Where we are.** Worth recording explicitly, since it is easy to miss: Theorem 9.4.4 makes no use of the tubes' *positions*. The bush estimate needs only that the directions are  $\delta$ -separated and that some point lies in  $\mu$  tubes; the multiplicity estimate is Chebyshev. So the theorem holds for any family whatever, Besicovitch or not.

That generality is a strength and a diagnosis at once. A strength, because the result is reusable as a lemma — Worked Example 9.4.3(d) lists where. A diagnosis, because an argument insensitive to position cannot detect a Besicovitch configuration, which is defined by how its tubes are positioned. The bush improves on the  $L^2$  method by conditioning on a point, but it still does not see the arrangement, and Chapter 9.5 is the first argument here that does.

## Notes

The bush argument is Bourgain's (1991), though the geometric content — that tubes through a point separate at unit distance — is older and appears in Córdoba's work. Volume 1, Theorem 1.7.15 gave the same bound; the present formulation, in terms of the two competing estimates on the multiplicity, is the one that generalises.

The table in Worked Example 9.4.1 is the outline of this volume: each row is a chapter, and each conditions on a richer object than the last.



## Chapter 9.5

# The Hairbrush

The bush argument of Chapter 9.4 conditions on a point and the tubes through it. Wolff's improvement conditions on a *tube* and the tubes meeting it. The gain comes from a fact of elementary geometry which has no analogue for points: two lines that meet a common third line, and meet each other, are forced into a plane.

### 9.5.1 The geometry

**Lemma 9.5.1** (Concurrent or coplanar). *Let  $\ell_0, \ell_1, \ell_2 \subseteq \mathbb{R}^n$  be distinct lines such that  $\ell_1$  and  $\ell_2$  each meet  $\ell_0$ , and  $\ell_1$  meets  $\ell_2$ . Then either  $\ell_0, \ell_1, \ell_2$  have a common point, or they lie in a common 2-plane.*

*Proof.* Let  $p = \ell_1 \cap \ell_2$ , and let  $q_i = \ell_i \cap \ell_0$  for  $i = 1, 2$ .

If  $q_1 = q_2$  then that point lies on all three lines and we are in the first case.

Suppose  $q_1 \neq q_2$ . Both lie on  $\ell_0$ , so  $\ell_0$  is the line through  $q_1$  and  $q_2$ . If  $p \in \ell_0$  then  $p, q_1, q_2$  are collinear; since  $p, q_1 \in \ell_1$  and  $\ell_1 \neq \ell_0$ , this forces  $p = q_1$ , and likewise  $p = q_2$ , contradicting  $q_1 \neq q_2$ . So  $p \notin \ell_0$ .

The three points  $p, q_1, q_2$  are therefore not collinear and span a unique 2-plane  $P$ . Now  $\ell_1$  contains  $p$  and  $q_1$ , hence  $\ell_1 \subseteq P$ ; similarly  $\ell_2 \subseteq P$ ; and  $\ell_0$  contains  $q_1, q_2$ , so  $\ell_0 \subseteq P$ .  $\square$

**Where we are.** This is the entire source of Wolff's improvement, and it is worth pausing on why there is no analogue for the bush.

In a bush, all tubes pass through one point, and two of them meet only there — Lemma 9.4.2 used exactly that. The lemma gives no additional structure, because the first alternative always holds.

In a hairbrush the tubes meet a *line*, and two of them can meet each other away from it. When they do, the second alternative applies and they

are coplanar with  $\ell_0$ . So the tubes of a hairbrush that interact with one another are organised into *planes through  $\ell_0$* , and within each such plane the configuration is two-dimensional — where, by Corollary 9.3.3, the Kakeya problem is solved with only a logarithmic loss.

The hairbrush therefore reduces an  $n$ -dimensional problem to a family of planar ones, and planar Kakeya is sharp. That is the whole idea.

## 9.5.2 Transverse hairbrushes

**Definition 9.5.2.** Let  $\mathbb{T}$  be a family of  $\delta$ -tubes with  $\delta$ -separated directions and let  $T_0 \in \mathbb{T}$  have axis  $\ell_0$ . For  $0 < \kappa \leq 1$  the  $\kappa$ -*transverse hairbrush* of  $T_0$  is

$$H_\kappa(T_0) = \{T \in \mathbb{T} : T \cap T_0 \neq \emptyset, \angle(T, T_0) \geq \kappa\}.$$

**Theorem 9.5.3** (Hairbrush volume estimate). *Let  $H = H_\kappa(T_0)$  with  $\kappa \geq \frac{1}{2}$ , and write  $m = \#H$ . Then*

$$m\left(\bigcup_{T \in H} T\right) \gtrsim_n \frac{m \delta^{n-1}}{\log \frac{1}{\delta}}.$$

*Idea of the proof.* Sort the tubes of  $H$  by which 2-plane through  $\ell_0$  they lie near — Lemma 9.5.1 says this is the right invariant, because two tubes assigned to different planes cannot interact away from  $\ell_0$ . Within one plane the tubes form a planar Kakeya configuration inside a thin slab, and Corollary 9.3.3 bounds its area from below with only a logarithmic loss. Summing over planes gives the theorem, provided the slabs are essentially disjoint — which is where transversality is needed, since a tube nearly parallel to  $T_0$  hugs  $\ell_0$  and never leaves the region where all the slabs meet.

*Proof. Step 1. Sort into leaves.*

Fix coordinates with  $\ell_0$  the  $x_n$ -axis. Each  $T \in H$  meets  $T_0$ , so its axis lies within  $2\delta$  of a line  $\ell$  meeting  $\ell_0$ ; let  $P(T)$  be the 2-plane spanned by  $\ell_0$  and  $\ell$ . The set of such planes is parametrised by the direction of  $\ell$  in the orthogonal complement of  $\ell_0$ , that is by a point of  $\mathbb{S}^{n-2}$  modulo sign. Choose a maximal  $\delta$ -separated set  $\{P_1, \dots, P_M\}$  of such planes, so  $M \approx_n \delta^{-(n-2)}$ , and assign each  $T \in H$  to a nearest  $P_i$ , obtaining a partition  $H = \bigsqcup_i H_i$  with  $m = \sum_i m_i$ ,  $m_i = \#H_i$ .

**Step 2.** *Each leaf lies in a slab.*

If  $T \in H_i$  then every point of  $T$  lies within  $O(\delta)$  of  $P_i$ : the axis of  $T$  meets  $\ell_0$ , has length 1, and its plane differs from  $P_i$  by angle  $\leq \delta$ , so it deviates



from  $P_i$  by at most  $1 \cdot \delta$ ; adding the tube radius gives  $O(\delta)$ . Write  $S_i$  for the  $C\delta$ -neighbourhood of  $P_i$ , a slab of thickness  $\approx \delta$  in  $n - 2$  directions.

**Step 3.** *Planar Kakeya inside a leaf.*

Let  $\pi_i : \mathbb{R}^n \rightarrow P_i \cong \mathbb{R}^2$  be the orthogonal projection. For  $T \in H_i$ ,  $\pi_i(T)$  contains a planar  $\delta$ -tube, and distinct  $T, T' \in H_i$  have  $\delta$ -separated directions, hence so do their projections — the projection is an isometry on  $P_i$  and the tubes lie within  $O(\delta)$  of it. By the planar case of Theorem 9.3.2, applied to the  $m_i$  projected tubes,

$$m_2\left(\bigcup_{T \in H_i} \pi_i(T)\right) \gtrsim \frac{m_i \delta}{\log \frac{1}{\delta}}.$$

Here the planar  $L^2$  computation is run for a subfamily of  $m_i$  tubes rather than a full one: in (9.3.2) with  $n = 2$  the count  $\#\{j' : \theta \approx 2^m \delta\}$  is at most  $\min\{2^m, m_i\}$ , so the sum is  $\lesssim \delta \log \frac{1}{\delta}$  per tube and  $\|\sum_{H_i} \mathbf{1}\|_{L^2}^2 \lesssim m_i \delta \log \frac{1}{\delta}$ ; combining with  $\sum_{H_i} m_2 = m_i \delta$  and Cauchy–Schwarz gives the display.

Since  $\bigcup_{H_i} T \subseteq S_i$  and  $S_i$  has thickness  $\approx \delta$  transverse to  $P_i$ ,

$$m\left(\bigcup_{T \in H_i} T\right) \gtrsim_n \delta^{n-2} \cdot \frac{m_i \delta}{\log \frac{1}{\delta}} = \frac{m_i \delta^{n-1}}{\log \frac{1}{\delta}}. \quad (9.5.1)$$

**Step 4.** *Different leaves separate — here transversality is used.*

Let  $A = \{x : \text{dist}(x, \ell_0) \geq \frac{1}{8}\}$ . If  $T \in H$  has  $\angle(T, T_0) \geq \kappa \geq \frac{1}{2}$  and  $T$  meets  $T_0$ , then  $T$  leaves the  $\frac{1}{8}$ -neighbourhood of  $\ell_0$  within distance  $\frac{1}{8}/\sin \kappa \leq \frac{1}{4}$  of the crossing point, so at least half of  $T$  lies in  $A$ ; hence (9.5.1) holds with  $\bigcup_{H_i} T$  replaced by  $(\bigcup_{H_i} T) \cap A$ , at the cost of an absolute constant.

Now two distinct planes  $P_i, P_{i'}$  make an angle  $\geq \delta$  along  $\ell_0$ , so at distance  $t$  from  $\ell_0$  their  $C\delta$ -neighbourhoods are disjoint once  $t\delta \gg \delta$ , that is once  $t \gtrsim 1$ . Shrinking the constant in  $A$  if necessary, the sets  $S_i \cap A$  have bounded overlap.

**Step 5.** *Sum.*

By Steps 3 and 4,

$$m\left(\bigcup_{T \in H} T\right) \geq m\left(\bigcup_i \left(\bigcup_{T \in H_i} T\right) \cap A\right) \gtrsim_n \sum_i \frac{m_i \delta^{n-1}}{\log \frac{1}{\delta}} = \frac{m \delta^{n-1}}{\log \frac{1}{\delta}}. \quad \square$$

**Theorem 9.5.4** (Wolff’s bound, transverse case). *Let  $\mathbb{T}$  be a family of  $\delta$ -tubes with  $\delta$ -separated directions,  $N \approx_n \delta^{-(n-1)}$ , and suppose that every point of  $E = \bigcup T$  lies in at most  $\mu$  tubes and that some  $T_0 \in \mathbb{T}$  satisfies  $\#H_{1/2}(T_0) \gtrsim \mu \delta^{-1}$ . Then*

$$m(E) \gtrsim_n \frac{\delta^{\frac{n-2}{2}}}{\log \frac{1}{\delta}}.$$

*Proof. Step 1. The hairbrush bound.*

By hypothesis some  $T_0$  has  $m = \#H_{1/2}(T_0) \gtrsim \mu\delta^{-1}$ . Since  $\kappa = \frac{1}{2}$ , Theorem 9.5.3 applies and gives

$$m(E) \geq m\left(\bigcup_{T \in H_{1/2}(T_0)} T\right) \gtrsim_n \frac{m\delta^{n-1}}{\log \frac{1}{\delta}} \gtrsim_n \frac{\mu\delta^{-1}\delta^{n-1}}{\log \frac{1}{\delta}} = \frac{\mu\delta^{n-2}}{\log \frac{1}{\delta}}.$$

This is the analogue of (9.4.1), improved by one power of  $\delta$ : collecting tubes along the whole of  $T_0$  rather than at a single point has raised the exponent from  $n-1$  to  $n-2$ .

**Step 2.** *The multiplicity bound.*

Unchanged from the bush argument: Lemma 9.4.3 gives  $m(E) \gtrsim_n \mu^{-1}$ . Nothing in that estimate refers to the geometry, so no improvement is available there — as Worked Example 9.4.4 observes, it is Chebyshev and is sharp.

**Step 3.** *Multiply.*

As in Theorem 9.4.4 the product eliminates  $\mu$ :

$$m(E)^2 \gtrsim_n \frac{\mu\delta^{n-2}}{\log \frac{1}{\delta}} \cdot \mu^{-1} = \frac{\delta^{n-2}}{\log \frac{1}{\delta}}.$$

Taking square roots,  $m(E) \gtrsim_n \delta^{(n-2)/2} (\log \frac{1}{\delta})^{-1/2}$ , and since  $(\log \frac{1}{\delta})^{-1/2} \geq (\log \frac{1}{\delta})^{-1}$  for  $\delta < \frac{1}{2}$  the stated form follows.  $\square$

*Remark 9.5.5* (Scope decision: the small-angle case). Wolff's theorem is that *every* Besicovitch set in  $\mathbb{R}^n$  satisfies  $\dim_{\text{H}} K \geq \frac{n+2}{2}$ , with no transversality hypothesis and no assumption that a tube meets  $\mu\delta^{-1}$  others. *We do not prove that here*, and the gap is exactly the two hypotheses of Theorem 9.5.4. This is revision R9.

The obstruction is a single mechanism, and it is worth naming because the arithmetic of it is transparent.

Suppose one tries to produce the hypothesis  $\#H \gtrsim \mu\delta^{-1}$  by counting. The axis  $T_0$  has unit length, so it is covered by  $\approx \delta^{-1}$  balls of radius  $\delta$ ; if each has multiplicity  $\mu$  the incidences number  $\approx \mu\delta^{-1}$ . But a tube making angle  $\theta$  with  $T_0$  meets  $\approx \theta^{-1}$  of those balls, so

$$\#H \approx \frac{\mu\delta^{-1}}{\theta^{-1}} = \mu\theta\delta^{-1},$$

and running Theorem 9.5.4 with this gives  $m(E)^2 \gtrsim \theta\delta^{n-2}$ . At  $\theta \approx 1$  this is Wolff's bound; at  $\theta \approx \delta$  it degrades to  $\delta^{n-1}$ , which is exactly Theorem 9.4.4 and no improvement at all.

So the near-parallel tubes must be treated separately, and Wolff does so by an induction on scales: a family of tubes confined to a narrow angular sector is, after rescaling, a Kakeya family in lower dimension, and the bound is applied recursively. Theorem 9.5.3 also requires transversality — Step 4 fails for small  $\kappa$ , since a tube nearly parallel to  $T_0$  never leaves the region where all the leaves overlap.

Both difficulties are the same one seen twice, and resolving it is the content of Wolff's paper rather than a lemma.

*What depends on the general case.* When this remark was first written it asserted that nothing did. That is no longer accurate, and the correction is worth making precisely.

Volume 14, Proposition 14.4.6 — the base of the bootstrap Theorem 14.4.7, and hence of the proof of the Kakeya conjecture in  $\mathbb{R}^3$  — is Wolff's bound in the form  $\mathcal{D}(\frac{1}{2}, 0)$ . It is *not* Theorem 9.5.4: the transversality hypothesis is replaced there by the *Frostman Slab Wolff axiom* of Volume 14, Definition 14.4.1, which bounds how many tubes can lie in a slab.

*Why that substitution works.* The obstruction above is that near-parallel tubes contribute  $\#H \approx \mu\theta\delta^{-1}$  rather than  $\mu\delta^{-1}$ . Tubes making angle  $\leq \theta$  with  $T_0$  and meeting it are confined to a slab of thickness  $O(\theta)$  about the plane they span, and the Frostman Slab axiom bounds their number by  $O(\theta \cdot \#\mathbb{T})$  — exactly compensating the loss. *So the axiom is designed to make the hairbrush argument survive the small-angle case*, and Volume 14 adopts it for that reason.

*What this means for R9.* The general theorem remains unproved here. But the version Volume 14 needs is the Frostman Slab version, and the mechanism by which that hypothesis repairs the small-angle difficulty is the paragraph above. Volume 10's arithmetic bound and Volume 11's polynomial method supersede Wolff's theorem in the dimensions where they apply, and Volume 14 reports the current record.

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WORKED EXAMPLE 9.5.1— TIER A **The three bounds compared**

| $n$ | $L^2$ | bush $\frac{n+1}{2}$ | hairbrush $\frac{n+2}{2}$ |
|-----|-------|----------------------|---------------------------|
| 2   | 2     | 1.5                  | 2                         |
| 3   | 2     | 2                    | 2.5                       |
| 4   | 2     | 2.5                  | 3                         |
| 5   | 2     | 3                    | 3.5                       |
| 10  | 2     | 5.5                  | 6                         |

*Reading the table.* In the plane the  $L^2$  method is already sharp and nothing else is needed. From  $n = 3$  the hairbrush is the best of the three, and it

beats the bush by exactly  $\frac{1}{2}$  in every dimension — the gain is additive, not multiplicative.

*What that means asymptotically.* All three bounds are of the form  $cn + O(1)$  with  $c \leq \frac{1}{2}$ , while the conjecture asks for  $n$ . So none of the methods of this chapter reaches even a fixed fraction better than one half, and the first bounds of the form  $n - o(n)$  came from elsewhere — the arithmetic method of Volume 10, which gives  $\frac{n}{2} + 1$  in a form that improves for large  $n$ , and the polynomial method of Volume 11.

*A caution about the table.* These are *Minkowski* bounds as proved here; the Hausdorff versions follow from Volume 7, Theorem 7.8.5, but only because the estimates carry  $\delta^{-\varepsilon}$  losses for every  $\varepsilon$ . A bound with a fixed power loss would give a strictly weaker Hausdorff conclusion, by Drill 8.4 of Volume 7.

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**WORKED EXAMPLE 9.5.2— TIER B Why conditioning on a line beats conditioning on a point**

The progression  $L^2 \rightarrow \text{bush} \rightarrow \text{hairbrush}$  is a progression in what the argument conditions on, and the mechanism is worth isolating since Volumes 11 and 14 continue it.

*Conditioning on nothing ( $L^2$ ).* One counts all pairs of tubes. By Worked Example 9.3.1 the count is dominated by generic transverse pairs and is blind to the configuration.

*Conditioning on a point (bush).* One fixes  $x_0$  and considers the tubes through it. These separate at unit distance, giving  $m \gtrsim \mu\delta^{n-1}$ . But the tubes through a point interact with each other only at that point — Lemma 9.5.1’s first alternative — so no further structure is available.

*Conditioning on a line (hairbrush).* One fixes  $T_0$  and considers the tubes meeting it. Now Lemma 9.5.1’s second alternative becomes available: tubes that interact are coplanar with  $\ell_0$ , and the configuration decomposes into planar problems. Since planar *Takeya* is *solved*, each piece is handled sharply, and the only loss is the logarithm.

*The pattern.* Each step conditions on a higher-dimensional object and extracts more structure, and the reason it works is that the low-dimensional *Takeya* problem is better understood than the high-dimensional one. That is precisely the design of the polynomial method in Volume 11: partition space by a polynomial surface, and induct on the dimension of the pieces.

*Where the pattern stops.* Conditioning on a 2-plane would require understanding three-dimensional *Takeya* sharply, which was open until Wang and Zahl (2025). Volume 14 discusses whether their result enables such an argument in  $\mathbb{R}^4$ ; at the time of writing it does not, and the reason is that their

proof is not of the form the induction would need.

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**WORKED EXAMPLE 9.5.3— TIER C: RESEARCH TASTE    What a full proof of Wolff’s theorem would require**

Remark 9.5.5 descoped the small-angle case. It is worth setting out what the missing argument looks like, both to make the boundary of this volume precise and because the technique — induction on scales — is used throughout Volumes 11 to 13.

(a) *The situation.* One has a hairbrush whose tubes make angle  $\theta \ll 1$  with  $T_0$ . They are confined to a slab of angular width  $\theta$  about the direction of  $T_0$ , and Theorem 9.5.3 gives nothing, since Step 4 fails.

(b) *The rescaling.* Apply the linear map that stretches the  $\theta$ -narrow directions to unit width. The tubes become  $\delta/\theta$ -tubes with  $\delta/\theta$ -separated directions, in a configuration that is again ofakeya type but at a coarser scale. So the problem at angular scale  $\theta$  is the original problem at scale  $\delta' = \delta/\theta$ .

(c) *The induction.* Assume the desired bound at all scales coarser than  $\delta$  and deduce it at  $\delta$ . Each application of the rescaling reduces the number of scales remaining by a factor, so the induction terminates after  $\approx \log \log \frac{1}{\delta}$  steps — and, as Worked Example 6.5.3 in Volume 6 showed, an induction of that depth costs a power of a logarithm, which  $\delta^{-\varepsilon}$  absorbs.

(d) *Why it is delicate.* The rescaled family need not have the same structure as the original: the tubes may no longer be  $\delta'$ -separated in all directions, and the multiplicity may change. Controlling those changes is the technical content of Wolff’s paper, and it is why the argument runs to several pages rather than the half-page the sketch above suggests.

(e) *What the reader should take from the omission.* Not that the theorem is doubtful — it is forty years old and universally used — but that the transverse case proved here contains the idea and the general case contains the bookkeeping. Lemma 9.5.1 is the mathematics; the induction is the price of generality.

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**WORKED EXAMPLE 9.5.4— TIER B    Verifying the leaf decomposition on a concrete configuration**

Step 1 of Theorem 9.5.3 sorts the hairbrush into planes through  $\ell_0$ . It is worth checking the count and the slab geometry in  $\mathbb{R}^3$ , where everything can be pictured.

*The setting.*  $n = 3$ ,  $\ell_0$  the  $z$ -axis,  $T_0$  the  $\delta$ -tube about it. Planes containing  $\ell_0$  are parametrised by an angle  $\varphi \in [0, \pi)$  — the direction of the plane in the  $xy$ -plane. So the leaves are indexed by  $\mathbb{S}^{n-2} = \mathbb{S}^1$  modulo sign, and a  $\delta$ -separated set has  $M \approx \delta^{-1} = \delta^{-(n-2)}$  members. ✓

*A tube's assignment.* A tube  $T$  meeting  $T_0$  at height  $z_0$  with direction making angle  $\theta$  with the  $z$ -axis lies near the plane containing  $\ell_0$  and the direction of  $T$ ; that plane's parameter  $\varphi$  is the azimuth of  $T$ 's direction. Two tubes are in the same leaf when their azimuths agree to within  $\delta$ .

*The slab.* A tube of length 1 whose plane differs from  $P_\varphi$  by angle  $\leq \delta$  deviates from  $P_\varphi$  by at most  $\delta$  over its length; adding the radius  $\delta$  gives a slab of half-thickness  $\approx 2\delta$ . ✓ In  $\mathbb{R}^3$  this slab is a genuine plate: two-dimensional in extent,  $\delta$ -thick.

*The separation of leaves.* Two planes at azimuthal separation  $\delta$  meet only along  $\ell_0$  and diverge linearly: at horizontal distance  $r$  from the  $z$ -axis they are  $\approx r\delta$  apart. So their  $2\delta$ -slabs are disjoint once  $r \gtrsim 1$ , which is why Step 4 restricts to  $\text{dist}(x, \ell_0) \geq \frac{1}{8}$ . ✓

*Where transversality enters, visibly.* A tube at angle  $\theta$  to the  $z$ -axis, meeting it at height  $z_0$ , reaches horizontal distance  $\approx \theta$  over its unit length. For  $\theta \approx 1$  it exits the region  $r \leq \frac{1}{8}$  almost immediately; for  $\theta \approx \delta$  it never does, and lives entirely in the region where all  $\delta^{-1}$  leaves overlap. That is Step 4's hypothesis, seen concretely, and it is the whole of Remark 9.5.5.

*The count, assembled.* With  $m_i$  tubes in leaf  $i$ : planar area  $\gtrsim m_i\delta/\log$ , times slab thickness  $\delta^{n-2} = \delta$ , gives volume  $\gtrsim m_i\delta^2/\log = m_i\delta^{n-1}/\log$ . ✓ Summing over the  $\approx \delta^{-1}$  leaves recovers Theorem 9.5.3.

#### WORKED EXAMPLE 9.5.5— TIER A **The coplanarity lemma, tested**

Lemma 9.5.1 is the pivot of the chapter, so it is worth testing against the cases where one expects it to be tight.

(a) *Three lines through a point.*  $\ell_0, \ell_1, \ell_2$  all through the origin, spanning  $\mathbb{R}^3$ . They pairwise meet, and the first alternative holds. The second fails — they are not coplanar — so the “or” is genuinely exclusive here, and the lemma is not asserting both.

(b) *Three lines in a plane, in general position.* A triangle's three sides: each pair meets, at three distinct points. The second alternative holds and the first fails. So both alternatives occur, and neither implies the other.

(c) *Both hypotheses are needed.* Take  $\ell_0$  the  $z$ -axis and  $\ell_1$  the  $x$ -axis, which meet at the origin. Let  $\ell_2 = \{(1, t, 1) : t \in \mathbb{R}\}$ . Then  $\ell_2$  meets neither  $\ell_0$  — it has constant height 1 and constant first coordinate 1 — nor  $\ell_1$ , which sits

at height 0. Adjusting to  $\ell'_2 = \{(1, t, 0)\}$  makes  $\ell'_2$  meet  $\ell_1$  at  $(1, 0, 0)$  but still miss  $\ell_0$ , and now  $\ell_0$  is not coplanar with  $\ell_1, \ell'_2$ : the latter two span the plane  $z = 0$ , which does not contain the  $z$ -axis. So dropping the hypothesis that *both*  $\ell_1$  and  $\ell_2$  meet  $\ell_0$  destroys the conclusion, and the lemma is not a statement about three pairwise-meeting lines in general.

(d) *The quantitative version needed in practice.* Tubes, not lines. Two  $\delta$ -tubes meeting a common  $\delta$ -tube and meeting each other are *approximately* coplanar with it — within an angle depending on how far from  $T_0$  they meet. Step 1 of Theorem 9.5.3 handles this by assigning each tube to a nearest plane and absorbing the error into the slab thickness. That the error is  $O(\delta)$  over unit length is what makes the leaf decomposition work, and it is the only quantitative input.

(e) *Why there is no analogue for the bush.* In a bush all tubes meet at one point, so alternative (a) always holds and the lemma yields nothing. That is the precise sense in which conditioning on a line is strictly more informative than conditioning on a point.

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**Figure 9.5.1 (to be drawn).** *The hairbrush and its leaves*

**Content required:** In  $\mathbb{R}^3$ : the tube  $T_0$  along the  $z$ -axis, with tubes meeting it at various heights and angles. Sort them into two or three planes through  $\ell_0$ , drawn as slabs, and show the slabs separating once the horizontal distance from  $\ell_0$  exceeds  $\approx 1$ . Include one near-parallel tube that never leaves the overlap region — the configuration Step 4 excludes and Remark 9.5.5 is about.

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## Drills

**Drill 9.5.1.** Verify Lemma 9.5.1 in  $\mathbb{R}^3$  by writing the three lines parametrically and solving.

**Drill 9.5.2.** Check the count  $M \approx \delta^{-(n-2)}$  of  $\delta$ -separated 2-planes through a fixed line in  $\mathbb{R}^n$ .

**Drill 9.5.3.** Carry out the modified planar  $L^2$  computation of Step 3 for a subfamily of  $m$  tubes, confirming the bound  $m\delta \log \frac{1}{\delta}$ .

**Drill 9.5.4.** Show that Step 4 fails for  $\kappa = \delta$ : exhibit tubes meeting  $T_0$  at angle  $\delta$  whose leaves overlap along their whole length.

## Notes

The hairbrush argument is Wolff's (1995), and Lemma 9.5.1 is the observation it rests on. The bound  $\frac{n+2}{2}$  was the best known for general  $n$  until Katz–Labat–Tao's improvement in  $\mathbb{R}^3$  and the arithmetic method of Volume 10.

Remark 9.5.5 records what is not proved here. Readers wanting the full argument should consult Wolff's paper or his lecture notes, where the small-angle induction is carried out.



## Chapter 9.6

# Nikodym Sets

A Besicovitch set contains a segment in every *direction*. A Nikodym set contains a segment through every *point*. The two look like different problems and are the same one, related by a projective transformation which interchanges position and direction. This chapter proves the equivalence and draws the consequences.

### 9.6.1 The two definitions

**Definition 9.6.1.** A compact set  $N \subseteq \mathbb{R}^n$  is a *Nikodym set* if for every  $x \in [0, 1]^n$  there is a line  $\ell_x$  through  $x$  such that  $\ell_x \cap N$  contains a segment of unit length having  $x$  as an endpoint.

**Definition 9.6.2.** The *Nikodym maximal function* at scale  $\delta$  is

$$\mathcal{N}_\delta f(x) = \sup_{T \ni x} \frac{1}{m(T)} \int_T |f|,$$

the supremum over all  $\delta$ -tubes containing  $x$ .

The contrast with Definition 9.1.1 is exactly the contrast between the two set problems:  $\mathcal{K}_\delta$  is a function of a *direction* and takes the supremum over positions, while  $\mathcal{N}_\delta$  is a function of a *position* and takes the supremum over directions.

### 9.6.2 The projective duality

**Definition 9.6.3.** For  $x = (x', x_n) \in \mathbb{R}^{n-1} \times (0, \infty)$  set

$$\Phi(x', x_n) = \left( \frac{x'}{x_n}, \frac{1}{x_n} \right).$$

**Lemma 9.6.4** (Position and direction are interchanged). *Let  $a, b \in \mathbb{R}^{n-1}$  and let  $\ell_{a,b}$  be the line  $\{(a + tb, t) : t \in \mathbb{R}\}$ , which passes through the point  $(a, 0)$  with direction  $(b, 1)$ . Then for  $t > 0$ ,*

$$\Phi(\ell_{a,b}) = \{(b + sa, s) : s > 0\} = \ell_{b,a} \cap \{x_n > 0\}.$$

*That is,  $\Phi$  carries the line with position  $a$  and direction  $b$  to the line with position  $b$  and direction  $a$ .*

*Proof.* A point of  $\ell_{a,b}$  with  $t > 0$  is  $(a + tb, t)$ , and

$$\Phi(a + tb, t) = \left( \frac{a + tb}{t}, \frac{1}{t} \right) = \left( \frac{a}{t} + b, \frac{1}{t} \right).$$

Substituting  $s = 1/t$ , which ranges over  $(0, \infty)$  as  $t$  does, this is  $(sa + b, s)$ , a point of  $\ell_{b,a}$ . The correspondence  $t \leftrightarrow s$  is a bijection of  $(0, \infty)$ , so the images coincide.  $\square$

**Lemma 9.6.5** ( $\Phi$  preserves dimension on slabs). *For  $0 < c < C < \infty$ , the restriction of  $\Phi$  to  $\{x : c \leq x_n \leq C, |x'| \leq C\}$  is bi-Lipschitz. Consequently  $\dim_{\mathbb{H}} \Phi(A) = \dim_{\mathbb{H}} A$  for every  $A$  in that region.*

*Proof. Step 1. Compute the derivative.*

$\Phi$  is smooth on  $\{x_n > 0\}$ , and differentiating  $\Phi(x', x_n) = (x'/x_n, 1/x_n)$  componentwise,

$$d\Phi = \begin{pmatrix} x_n^{-1} I_{n-1} & -x_n^{-2} x' \\ 0 & -x_n^{-2} \end{pmatrix}, \quad |\det d\Phi| = x_n^{-(n-1)} \cdot x_n^{-2} = x_n^{-(n+1)},$$

the determinant being that of a block triangular matrix.

**Step 2.** *Bound the derivative above and below on the region.*

Let  $R = \{x : c \leq x_n \leq C, |x'| \leq C\}$ . On  $R$  every entry of  $d\Phi$  is bounded in modulus by  $\max\{c^{-1}, Cc^{-2}\}$ , so  $\|d\Phi\| \leq L_1(c, C)$ . And  $|\det d\Phi| = x_n^{-(n+1)} \geq C^{-(n+1)}$ , so by Cramer's rule the inverse matrix also has bounded entries and  $\|(d\Phi)^{-1}\| \leq L_2(c, C)$ .

**Step 3.** *From the derivative to a Lipschitz bound.*

$R$  is convex — it is a product of a ball and an interval — so for  $x, y \in R$  the segment  $[x, y]$  lies in  $R$  and the mean value inequality gives

$$|\Phi(x) - \Phi(y)| \leq L_1 |x - y|.$$

Applying the same argument to  $\Phi^{-1} = \Phi$  on the image  $\Phi(R)$ , which is again of the form treated in Step 2 after adjusting  $c$  and  $C$ , gives  $|\Phi(x) - \Phi(y)| \geq L_2^{-1} |x - y|$ .

**Step 4. Conclude.**

So  $\Phi \upharpoonright R$  is bi-Lipschitz. By Proposition 7.1.5(iii) applied to  $\Phi$ ,  $\dim_{\mathbb{H}} \Phi(A) \leq \dim_{\mathbb{H}} A$ ; applied to  $\Phi^{-1}$  on  $\Phi(A)$ , the reverse. Hence  $\dim_{\mathbb{H}} \Phi(A) = \dim_{\mathbb{H}} A$  for  $A \subseteq R$ .  $\square$

**Theorem 9.6.6** (Kakeya and Nikodym are equivalent). *Let  $s \in [0, n]$ . The following are equivalent.*

- (i) *Every Besicovitch set in  $\mathbb{R}^n$  has  $\dim_{\mathbb{H}} \geq s$ .*
- (ii) *Every Nikodym set in  $\mathbb{R}^n$  has  $\dim_{\mathbb{H}} \geq s$ .*

*Proof. Step 1. From a Nikodym set to a Besicovitch set.*

Assume (i) and let  $N$  be a Nikodym set. By Definition 9.6.1 every  $x \in [0, 1]^n$  lies on a line meeting  $N$  in a unit segment. Restrict attention to the slab  $Q = \{x : \frac{1}{2} \leq x_n \leq 1, |x'| \leq 1\}$ ; the points of  $Q$  still have this property, and we may take the segments to lie in the region  $c \leq x_n \leq C$  after discarding a bounded portion and rescaling, since a unit segment through a point of  $Q$  meets that region in a segment of length  $\gtrsim 1$ .

Consider  $\Phi(N)$ . Let  $(b, 1)$  be any direction with  $|b| \leq 1$ . The point  $(b, 0)$  — or rather a nearby point of  $Q$  after the normalisation above — lies on a line  $\ell_{b,a}$  meeting  $N$  in a segment, for some  $a$ . By Lemma 9.6.4,  $\Phi$  maps  $\ell_{b,a}$  to  $\ell_{a,b}$ , which has direction  $(b, 1)$ . So  $\Phi(N)$  contains a segment in the direction  $(b, 1)$ , and as  $b$  ranges over  $|b| \leq 1$  these directions fill an open subset of  $\mathbb{S}^{n-1}$ .

A compact set containing a segment in every direction of an open subset of the sphere has the same dimension bound as a Besicovitch set: cover the sphere by boundedly many rotated copies of that subset and take the union of the corresponding sets, using countable stability (Proposition 7.1.5(ii)). So  $\dim_{\mathbb{H}} \Phi(N) \geq s$  by (i), and  $\dim_{\mathbb{H}} N = \dim_{\mathbb{H}} \Phi(N) \geq s$  by Lemma 9.6.5.

**Step 2. From a Besicovitch set to a Nikodym set.**

Assume (ii) and let  $K$  be a Besicovitch set. Running the same argument with  $\Phi^{-1}$  — which is  $\Phi$  itself, since  $\Phi(\Phi(x)) = x$  by direct computation — and with the roles of  $a$  and  $b$  exchanged,  $\Phi(K)$  contains, through every point of a suitable cube, a segment; so  $\Phi(K)$  contains a Nikodym set up to the same covering argument, and  $\dim_{\mathbb{H}} K = \dim_{\mathbb{H}} \Phi(K) \geq s$ .  $\square$

**Where we are.** The map  $\Phi$  is an involution —  $\Phi \circ \Phi = \text{id}$  — which is why the two directions of Theorem 9.6.6 are the same argument. It is the affine chart expression of the projective duality between points and hyperplanes, and the identity

$$\text{position } a, \text{ direction } b \quad \longleftrightarrow \quad \text{position } b, \text{ direction } a$$

of Lemma 9.6.4 is the whole content.

The reason the equivalence is worth having is that the two problems suggest different attacks. The *Takeya* formulation makes the directions the parameter and is what the maximal function of Chapter 9.1 is built on. The *Nikodym* formulation makes the points the parameter and is closer to the maximal operators of Volume 1, where the supremum is over sets containing a given point. Some arguments are more natural on one side — the polynomial method of Volume 11 is usually run on the *Nikodym* side — and the duality lets one choose.

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**WORKED EXAMPLE 9.6.1— TIER A Nikodym’s original example**

*Nikodym* constructed, in 1927, a set  $N \subseteq [0, 1]^2$  with  $m(N) = 1$  such that for every  $x \in N$  there is a line  $\ell_x$  with  $\ell_x \cap N = \{x\}$ . The complement of  $N$  in the square therefore has measure zero and contains, through every point of  $N$ , a punctured segment.

*Why this is startling.* It says a set of *full* measure can be arranged so that every one of its points is isolated along some line. The intuition that a full-measure set must be “thick in every direction at almost every point” is simply false.

*Its relation to Definition 9.6.1.* The complement is essentially a *Nikodym* set in the sense used here: it has measure zero and meets a line through every point of  $N$  in a segment. The definitions in the literature differ in whether the segment must contain the point or merely approach it, and the variants are equivalent for dimension purposes.

*Why it does not settle anything.* As with *Besicovitch* sets, measure zero is compatible with full dimension — *Worked Example 7.1.4* in Volume 7 made this point — and *Nikodym*’s construction says nothing about the dimension. By Theorem 9.6.6 the dimension question is exactly as hard as *Takeya*.

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**WORKED EXAMPLE 9.6.2— TIER B Transferring an estimate across the duality**

Theorem 9.6.6 is stated for dimensions. The corresponding statement for maximal functions requires more care, because  $\Phi$  distorts measure, and it is worth seeing where the exponent  $n$  comes from.

*The distortion.* By Lemma 9.6.5,  $|\det d\Phi| = x_n^{-(n+1)}$ . So a change of variables  $y = \Phi(x)$  in an integral introduces a weight  $x_n^{n+1}$ , which is bounded above and below only on a slab.

*Why  $L^n$  is the natural exponent.* Suppose one wants to transfer a bound  $\|\mathcal{K}_\delta f\|_{L^p(d\sigma)} \lesssim \|f\|_{L^p}$  to  $\|\mathcal{N}_\delta g\|_{L^p(\mathbb{R}^n)} \lesssim \|g\|_{L^p}$ . On the *Takeya* side the left

norm is over an  $(n - 1)$ -dimensional sphere and on the Nikodym side over an  $n$ -dimensional cube; the tube geometry contributes  $\delta^{n-1}$  on both sides. Matching the homogeneities under the scaling  $f \mapsto f(\lambda \cdot)$  forces  $p = n$ , which is the same computation that Worked Example 9.1.2 performed directly on theakeya side.

*The transfer, in words.* Decompose the region into slabs  $2^{-k-1} \leq x_n \leq 2^{-k}$ , on each of which  $\Phi$  is bi-Lipschitz with constants  $\approx 2^k$ ; apply theakeya bound after rescaling the slab to unit size; and sum over  $k$ , which costs a factor  $\log \frac{1}{\delta}$ . Since every statement in this subject carries  $\delta^{-\varepsilon}$ , the logarithm is invisible.

*The upshot.*  $\mathcal{K}_\delta$  and  $\mathcal{N}_\delta$  satisfy the same  $L^n$  bounds up to  $\delta^{-\varepsilon}$ , and Conjecture 9.1.2 has an exactly equivalent Nikodym form. Tao’s paper on the equivalence carries out the bookkeeping in full; the point for a reader here is that the equivalence is not formal — it uses that  $\Phi$  maps lines to lines, and no other feature.

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#### WORKED EXAMPLE 9.6.3— TIER C: RESEARCH TASTE **Which side to work on**

The duality is free, so the only question is which formulation makes a given argument easier. Four considerations.

(a) *Separation.* On theakeya side the tubes have  $\delta$ -separated *directions*, and their positions are arbitrary. On the Nikodym side the positions are spread and the directions are arbitrary. An argument that needs one kind of separation should be run on the side that supplies it. The bush argument of Chapter 9.4 used direction-separation and is naturally aakeya argument.

(b) *Incidence geometry.* Statements about incidences between points and lines — which is what the polynomial method of Volume 11 counts — are symmetric in points and lines, so the duality is invisible there. That is why those arguments are usually presented in whichever form is notationally lighter.

(c) *The ambient measure.* The Nikodym formulation lives entirely in  $\mathbb{R}^n$ , with no sphere; for arguments that decompose space — polynomial partitioning, induction on scales — this is a genuine simplification.

(d) *The extremal examples.* Besicovitch’s construction is explicit (Volume 1) and Nikodym’s is not much harder, but they are not images of each other under  $\Phi$  in any illuminating way: the duality preserves dimension, not the construction. So intuition about extremal configurations should be developed on theakeya side, where the Perron tree is available, and results transferred afterwards.

*A caution.* The equivalence is for the *dimension* problems and, with the

bookkeeping of Worked Example 9.6.2, for the maximal estimates at  $L^n$ . It is not an equivalence of every associated question: the Nikodym maximal operator has a different behaviour at exponents other than  $n$ , and statements about specific sets do not transfer, since  $\Phi$  does not map Besicovitch sets to Nikodym sets in a canonical way.

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**WORKED EXAMPLE 9.6.4— TIER B Dualising an argument: the bush becomes what?**

The duality of Theorem 9.6.6 lets any Kakeya argument be transported to the Nikodym side. Doing this once, on the bush, shows what the transport does to the geometry.

*The Kakeya statement.* Lemma 9.4.2:  $\mu$  tubes with  $\delta$ -separated *directions* through a common *point* have union of measure  $\gtrsim \mu\delta^{n-1}$ , because they separate at unit distance from the point.

*Applying  $\Phi$ .* By Lemma 9.6.4, position and direction are exchanged. A family of lines through a common point, with separated directions, becomes a family of lines with a common direction, through separated points — that is, a family of *parallel* tubes at separated positions.

*The dual statement.*  $\mu$  parallel  $\delta$ -tubes whose positions are  $\delta$ -separated have union of measure  $\approx \mu\delta^{n-1}$  — which is trivially true, since they are disjoint.

*What this shows.* The bush estimate is, on the Nikodym side, a triviality. That is not a defect of the duality: it means the difficulty of the bush argument lies entirely in the *quantifier structure* — which point, which tubes — and not in the geometry of the separation. The content of Theorem 9.4.4 is the combination with Lemma 9.4.3, and that combination does not simplify under  $\Phi$ .

*The general moral.* A duality that makes one lemma trivial has not made the theorem easier; it has relocated the difficulty. When transporting an argument, identify which step carries the content before deciding the transport was worthwhile. Here the content is the optimisation over  $\mu$  (Worked Example 9.8.2), and it is invariant under  $\Phi$ .

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**WORKED EXAMPLE 9.6.5— TIER A The duality in the plane, drawn out**

In  $\mathbb{R}^2$  the transformation  $\Phi(x_1, x_2) = (x_1/x_2, 1/x_2)$  can be followed explicitly, and doing so makes Lemma 9.6.4 concrete.

*A single line.* Take  $a = 0$ ,  $b = 1$ : the line through the origin with direction  $(1, 1)$ , that is  $\{(t, t)\}$ . Then  $\Phi(t, t) = (1, 1/t)$ , the *vertical* line  $x_1 = 1$ . So

a line of slope 1 through the origin becomes a vertical line through  $(1, 0)$  — position and direction have swapped, and “slope 1” has become “position 1”.

*A family through a point.* The lines through the origin, one per direction, are  $\{(t, \lambda t)\}$  for  $\lambda \in \mathbb{R}$ . Their images are  $\Phi(t, \lambda t) = (1/\lambda, 1/(\lambda t))$  — the vertical line  $x_1 = 1/\lambda$ . So a *pencil through a point* becomes a *family of parallel lines*, which is exactly the dual statement computed abstractly in Worked Example 9.6.4.

*A family in all directions.* Conversely, lines with all slopes but through different points — the raw material of a Besicovitch set — map to lines through all points but with restricted slopes, the raw material of a Nikodym set.

*The involution.*  $\Phi(\Phi(x_1, x_2)) = \Phi(x_1/x_2, 1/x_2) = ((x_1/x_2) \cdot x_2, x_2) = (x_1, x_2)$ . ✓

*What to retain.* The single sentence:  $\Phi$  turns pencils into parallel families. Everything else in this chapter is that observation with bookkeeping.

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**WORKED EXAMPLE 9.6.6— TIER A Nikodym’s example, and the intuition it destroys**

Worked Example 9.6.1 recorded Nikodym’s set. The example deserves a second look, because the intuition it refutes is one most readers hold.

*The false intuition.* If  $E \subseteq [0, 1]^2$  has full measure, then almost every point of  $E$  should be a point of density 1 (Theorem 1.7.4) — so  $E$  should look, near almost every one of its points, like the whole plane. In particular one expects that through almost every point of  $E$ , *every* line meets  $E$  in a set of positive length near that point.

*What Nikodym showed.* There is a full-measure  $E$  such that through *every* point of  $E$  there is a line meeting  $E$  only at that point. So the expectation is false for every point, not merely almost every one.

*Reconciling the two.* The Lebesgue density theorem is a statement about balls:  $m(E \cap B(x, r))/m(B(x, r)) \rightarrow 1$ . A line is a null set, and the density theorem says nothing about null sets. The intuition slides from “ $E$  fills almost all of every small ball” to “ $E$  fills almost all of every small segment”, and the second does not follow.

*The general warning.* Almost-everywhere statements with respect to Lebesgue measure carry no information along any fixed lower-dimensional family. Volume 8’s projection theorems are the systematic response: they establish what *is* true along almost every member of such a family, and Worked Example 8.1.1 there showed that “almost every” cannot be improved to “every”.

*Why it belongs in this chapter.* By Theorem 9.6.6 the dimension question for Nikodym sets is exactly Kakeya. So the counterintuitive example and the hard conjecture are two faces of the same phenomenon: lines see something that balls do not, and measure theory alone cannot control it.

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**Figure 9.6.1 (to be drawn).** *The projective duality*

**Content required:** Two panels. Left: a pencil of lines through the origin in the plane, one per slope. Right: their images under  $\Phi$ , a family of parallel vertical lines at positions  $1/\lambda$ . Label the correspondence “slope  $\lambda$ ”  $\leftrightarrow$  “position  $1/\lambda$ ”, which is Lemma 9.6.4 in one picture.

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## Drills

**Drill 9.6.1.** Verify that  $\Phi \circ \Phi = \text{id}$  on  $\{x_n > 0\}$ .

**Drill 9.6.2.** Compute  $d\Phi$  and confirm  $|\det d\Phi| = x_n^{-(n+1)}$ .

**Drill 9.6.3.** Show that a compact set containing a unit segment in every direction of an open subset of  $\mathbb{S}^{n-1}$  has the same dimension lower bound as a Besicovitch set, by the covering argument of Step 1.

**Drill 9.6.4.** Carry out the scaling computation of Worked Example 9.6.2 and confirm that  $p = n$  is forced.

**Where we are.** The duality of this chapter is the third structural equivalence the series has met, and the three are worth comparing.

Volume 6, Theorem 6.1.3 related restriction to extension by adjointness: a formal identity, with the content lying in the observation that one side is computable and the other is not.

Volume 7, Theorem 7.5.1 related a covering condition to the existence of a measure: not formal at all, and the content is the construction.

Theorem 9.6.6 relates two geometric problems by an explicit change of variable. It is formal in the sense that  $\Phi$  is written down and verified in two lines, but it is not trivial: Lemma 9.6.5 must check that  $\Phi$  preserves dimension, and that requires restricting to a slab, which in turn requires the covering argument of Step 1.

The common lesson is that a useful equivalence is one that moves a problem into a category where different tools apply. Restriction became extension so that  $\mathcal{E}g$  could be written down; a dimension hypothesis became a measure so that integrals could be estimated; and Kakeya becomes Nikodym so that



arguments which decompose  $\mathbb{R}^n$  — rather than the sphere — may be run. Volume 11 uses the last of these throughout.

## Notes

Nikodym's example is from 1927. The equivalence of the Kakeya and Nikodym maximal conjectures is due to Tao (1999); the dimension statement, Theorem 9.6.6, is the easier half and the projective duality underlying it is classical.

The transformation  $\Phi$  is the standard affine chart for projective duality and appears throughout incidence geometry; Volume 11 uses it again when translating between point-line incidence bounds and tube estimates.



## Chapter 9.7

# From Single Scales to Dimension

Every estimate in this volume bounds  $m(E)$  for a family of  $\delta$ -tubes at one fixed  $\delta$ . The Kakeya conjecture is about Hausdorff dimension, which is a statement about all scales at once. Volume 7 supplied the bridge; this chapter walks across it, records exactly what is lost, and assembles the resulting dimension bounds.

### 9.7.1 The dictionary

**Proposition 9.7.1** (Single-scale bounds become Minkowski bounds). *Suppose that for every family of  $\delta$ -tubes with  $\delta$ -separated directions one has  $m(\bigcup T) \gtrsim_n \delta^\beta$ , uniformly in  $\delta$ . Then every Besicovitch set  $K \subseteq \mathbb{R}^n$  satisfies  $\overline{\dim}_M K \geq n - \beta$ .*

*Proof. Step 1. Produce an admissible family inside  $K_\delta$ .*

Let  $\{v_1, \dots, v_N\}$  be a maximal  $\delta$ -separated subset of  $\mathbb{S}^{n-1}$ , so  $N \approx_n \delta^{-(n-1)}$ . Since  $K$  is a Besicovitch set it contains a unit segment  $\ell_j$  in each direction  $v_j$ . Let  $T_j$  be the  $\delta$ -tube about  $\ell_j$ ; then  $T_j \subseteq K_\delta$  by definition of the neighbourhood, and  $\{T_j\}$  has  $\delta$ -separated directions, so the hypothesis applies to it.

**Step 2.** *Transfer the bound.*

$$m(K_\delta) \geq m\left(\bigcup_j T_j\right) \gtrsim_n \delta^\beta.$$

**Step 3.** *Convert to a dimension statement.*

By Proposition 7.2.3,

$$\overline{\dim}_M K = n - \liminf_{\delta \rightarrow 0} \frac{\log m(K_\delta)}{\log \delta}.$$

Step 2 gives  $m(K_\delta) \geq c\delta^\beta$  for small  $\delta$ . Taking logarithms and dividing by  $\log \delta$ , *which is negative*, reverses the inequality:

$$\frac{\log m(K_\delta)}{\log \delta} \leq \frac{\log c + \beta \log \delta}{\log \delta} = \beta + \frac{\log c}{\log \delta} \rightarrow \beta.$$

Hence  $\liminf_{\delta \rightarrow 0} \frac{\log m(K_\delta)}{\log \delta} \leq \beta$  and  $\overline{\dim}_M K \geq n - \beta$ . □

**Theorem 9.7.2** (The bounds of this volume, in dimension form). *Every Besicovitch set  $K \subseteq \mathbb{R}^n$  satisfies*

$$\overline{\dim}_M K \geq \begin{cases} 2, & \text{by Corollary 9.3.3, all } n, \\ \frac{n+1}{2}, & \text{by Theorem 9.4.4,} \end{cases}$$

and, under the transversality hypotheses of Theorem 9.5.4,  $\overline{\dim}_M K \geq \frac{n+2}{2}$ . Moreover the same bounds hold for  $\dim_H K$ , and for every Nikodym set by Theorem 9.6.6.

*Proof.* The Minkowski statements are Proposition 9.7.1 applied with  $\beta = n-2$ ,  $\beta = \frac{n-1}{2}$  and  $\beta = \frac{n-2}{2}$  respectively; logarithmic factors do not affect the exponent.

For the Hausdorff statements, each of the three estimates holds with a loss  $\delta^{-\varepsilon}$  for every  $\varepsilon > 0$  — indeed with at most a logarithmic loss, which is stronger. By Proposition 9.1.3 each is equivalent to the corresponding Kakeya maximal bound  $K(s)$  of Volume 7, Theorem 7.8.5, with  $s = n - \beta$ , and that theorem converts it into  $\dim_H K \geq s$ . The Nikodym statements follow from Theorem 9.6.6. □

**Where we are.** Theorem 9.7.2 is the reason this volume was allowed to work entirely at a single scale. It is worth being precise about what the licence requires: *the single-scale estimate must hold with a loss  $\delta^{-\varepsilon}$  for every  $\varepsilon > 0$ .*

All three estimates here satisfy that — Corollary 9.3.3 and Theorem 9.5.3 lose a logarithm and Theorem 9.4.4 loses nothing. Had any of them lost a fixed power  $\delta^{-c}$ , Volume 7's Theorem 7.8.5 would give only  $\dim_H K \geq s - sc$  for the corresponding  $s$ , by Drill 8.4 there, and the Hausdorff bound would be strictly weaker than the Minkowski one.

This is worth checking whenever a new estimate appears. A paper proving a tube-counting bound with an unspecified constant is proving a Minkowski statement; the Hausdorff statement needs the  $\varepsilon$ .

## 9.7.2 What is lost at each step

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### WORKED EXAMPLE 9.7.1— TIER B **Tracing a bound from tubes to dimension**

It is instructive to follow one estimate the whole way and see where the losses occur. Take the bush bound in  $\mathbb{R}^4$ .

*Step 1: the tube estimate.* Theorem 9.4.4 gives  $m(\bigcup T) \gtrsim \delta^{(n-1)/2} = \delta^{3/2}$ , with an absolute implied constant and *no* loss.

*Step 2: to the neighbourhood of a Besicovitch set.* Proposition 9.7.1 gives  $m(K_\delta) \gtrsim \delta^{3/2}$ . No loss — the tubes are inside  $K_\delta$  by definition.

*Step 3: to Minkowski dimension.*  $\overline{\dim}_M K \geq 4 - \frac{3}{2} = \frac{5}{2}$ . No loss, but note this is an lim sup statement: it says nothing about  $\underline{\dim}_M$ , and by Proposition 7.2.5 the two can differ.

*Step 4: to the maximal function.* Proposition 9.1.3 converts the multiplicity bound into a bound for  $\mathcal{K}_\delta$ , at the cost of a constant. The equivalence is not free in general — Step 2 of that proof used Marcinkiewicz interpolation — but the losses are constants, not powers.

*Step 5: to Hausdorff dimension.* Volume 7, Theorem 7.8.5 gives  $\dim_H K \geq \frac{5}{2}$ . *This step costs the  $\varepsilon$ :* the proof there pigeonholes over  $\approx \log \frac{1}{\delta}$  dyadic scales and over directions, each costing a power of a logarithm, and the  $\delta^{-\varepsilon}$  absorbs them.

*The tally.* Four of the five steps are lossless, and the one that is not loses only logarithms. So the chain is tight, and the exponent  $\frac{5}{2}$  in the Hausdorff statement is exactly the exponent in the tube estimate. That is why the literature quotes tube-counting exponents and dimension exponents interchangeably, and it is legitimate precisely because of this accounting.

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### WORKED EXAMPLE 9.7.2— TIER A **The best elementary bound in each dimension**

Assembling Corollary 9.3.3, Theorem 9.4.4 and, where its hypotheses apply, Theorem 9.5.4:

| $n$ | best proved here | method        | conjecture  |
|-----|------------------|---------------|-------------|
| 2   | 2                | $L^2$         | 2 (settled) |
| 3   | 2                | $L^2$ or bush | 3           |
| 4   | 2.5              | bush          | 4           |
| 5   | 3                | bush          | 5           |
| 10  | 5.5              | bush          | 10          |

With Wolff's theorem in full — not proved here, Remark 9.5.5 — the entries for  $n \geq 3$  become  $\frac{n+2}{2}$ : 2.5, 3, 3.5, 6.

*Reading the table.* The planar case is settled. Every other entry is at most about half the conjectured value, and the ratio does not improve with  $n$ . So the elementary methods, taken together, establish that a Besicovitch set is at least half-dimensional and no more.

*Where the next improvements come from.* Volume 10's arithmetic method gives bounds exceeding  $\frac{n+2}{2}$  in high dimensions by combining sumset inequalities with the tube geometry; Volume 11's polynomial method gives the sharp result in the finite-field model and strong bounds in  $\mathbb{R}^n$ ; Volume 14 reports Wang and Zahl's theorem, which settles  $n = 3$ . None of them is elementary in the sense of this volume.

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**WORKED EXAMPLE 9.7.3— TIER C: RESEARCH TASTE Minkowski, Hausdorff, and what a paper is actually claiming**

Theorem 9.7.2 produced both Minkowski and Hausdorff bounds from the same tube estimates. The two are not interchangeable, and the literature is not always explicit about which is meant.

(a) *The logical order.* A tube estimate gives a Minkowski bound immediately (Proposition 9.7.1) and a Hausdorff bound only through Volume 7, Theorem 7.8.5, which requires the loss to be  $\delta^{-\varepsilon}$  for every  $\varepsilon$ . So Minkowski is the primitive statement and Hausdorff the derived one.

(b) *The gap is real.* By Worked Example 7.2.1 in Volume 7 the two dimensions can differ by the whole range. For Besicovitch sets they are believed equal — both conjecturally  $n$  — but that is a conjecture, not a theorem, and in dimensions where only Minkowski bounds are known the Hausdorff question is formally open.

(c) *Historically the Minkowski bounds came first.* Davies proved  $\dim_{\text{H}} = 2$  in the plane in 1971; the corresponding Minkowski statement is Córdoba's  $L^2$  argument and is easier. In higher dimensions the pattern repeats: a Minkowski bound appears, and the Hausdorff version follows once the estimate is up-

graded to carry  $\delta^{-\varepsilon}$ .

(d) *Three questions to ask of any stated bound.* Which dimension? What is the loss —  $\delta^{-\varepsilon}$  for every  $\varepsilon$ , or a fixed power? And is it for all Besicovitch sets or a restricted class? Worked Example 7.8.4 in Volume 7 gave the same checklist; this chapter has now supplied the machinery that makes the answers computable.

(e) *The convention adopted here.* Every bound in this volume is proved as a tube estimate with at most a logarithmic loss, so both dimension statements follow and Theorem 9.7.2 asserts both. Where a later volume proves something weaker, it will say so.

**Proposition 9.7.3** (The equivalence is not lossy). *Let  $\beta \geq 0$ . The following are equivalent, up to losses of  $\delta^{-\varepsilon}$  for every  $\varepsilon > 0$ :*

- (i)  $m(\bigcup T) \gtrsim \delta^\beta$  for every  $\delta$ -tube family with  $\delta$ -separated directions;
- (ii)  $\|\sum_T \mathbf{1}_T\|_{L^q} \lesssim \delta^{-\varepsilon}$  with  $q = \frac{n-\beta}{n-\beta-1}$ , for every such family;
- (iii)  $\|\mathcal{K}_\delta f\|_{L^{n-\beta}(\mathrm{d}\sigma)} \lesssim \delta^{-\varepsilon} \|f\|_{L^{n-\beta}}$ ;
- (iv) every Besicovitch set has  $\dim_{\mathrm{H}} \geq n - \beta$ .

*Proof.* (ii) $\Leftrightarrow$ (iii) is Proposition 9.1.3 with  $n$  replaced by  $s = n - \beta$ . (ii) $\Rightarrow$ (i): by Hölder on the support,  $\sum_T m(T) = \int_E F \leq m(E)^{1/s} \|F\|_{L^q}$ , and  $\sum_T m(T) \approx 1$ , giving  $m(E) \gtrsim \delta^{\varepsilon s}$ ; the exponent matches after relabelling  $\varepsilon$ .

(i) $\Rightarrow$ (ii) is the reverse Hölder step, valid because  $F$  takes values in  $\{0, 1, \dots\}$  and its distribution is controlled by  $m(E)$  together with  $\int F \approx 1$ ; the argument is the weak-type computation of Proposition 9.1.3 Step 2 run backwards.

(iii) $\Rightarrow$ (iv) is Volume 7, Theorem 7.8.5. (iv) $\Rightarrow$ (i) follows since a Besicovitch set's  $\delta$ -neighbourhood contains such a family and  $\dim_{\mathrm{H}} \leq \overline{\dim_{\mathrm{M}}}$  (Theorem 7.2.4).  $\square$

**Where we are.** Proposition 9.7.3 is the formal statement of what Worked Example 9.7.1 traced by hand: all four formulations of a Kakeya bound carry the same exponent, and the passages between them cost only  $\delta^{-\varepsilon}$ . That is why the literature quotes whichever form is convenient and why a reader may translate freely. The single caveat, repeated once more because it is the one that catches people: the equivalence with (iv) requires the loss to be  $\delta^{-\varepsilon}$  for *every*  $\varepsilon$ . Formulations (i)–(iii) remain equivalent to one another under a fixed power loss; only the dimension statement degrades.

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**WORKED EXAMPLE 9.7.4— TIER B Converting between the four formulations, in practice**

Proposition 9.7.3 says the four statements are equivalent. Using the equivalence requires knowing which direction is cheap and which needs care.

*Tube estimate  $\rightarrow$  multiplicity bound.* This is Hölder, and it is the step most often written without comment. From  $m(E) \gtrsim \delta^\beta$  one gets  $\|\sum \mathbf{1}_T\|_{L^q} \lesssim \delta^{-\varepsilon}$  with  $q = \frac{s}{s-1}$ ,  $s = n - \beta$ , provided the multiplicity function's distribution is controlled — which it is, since  $\int F \approx 1$  pins the first moment and  $m(E)$  the support.

*Multiplicity bound  $\rightarrow$  maximal bound.* Proposition 9.1.3 Step 2. This requires linearisation (Worked Example 9.1.4) and produces a *weak*-type bound; the strong-type statement needs Marcinkiewicz interpolation against the trivial  $L^\infty$  bound, and costs another  $\varepsilon$ . Papers frequently state the weak-type version and leave the upgrade implicit.

*Maximal bound  $\rightarrow$  Hausdorff dimension.* Volume 7, Theorem 7.8.5. Two pigeonholings, over scales and over directions, each costing a power of a logarithm. *This is the step that requires the loss to be  $\delta^{-\varepsilon}$  for every  $\varepsilon$ .*

*Hausdorff dimension  $\rightarrow$  tube estimate.* The cheapest direction, and the one used to derive necessary conditions: if some Besicovitch set had small dimension, its neighbourhood would furnish a tube family violating the estimate.

*A worked conversion.* Suppose a paper proves  $\|\sum \mathbf{1}_T\|_{L^{5/3}} \lesssim \delta^{-\varepsilon}$  in  $\mathbb{R}^4$ . Then  $q = \frac{5}{3}$  gives  $s = \frac{q}{q-1} = \frac{5}{2}$ , so  $\beta = n - s = \frac{3}{2}$ , and the tube estimate is  $m(E) \gtrsim \delta^{3/2-\varepsilon}$ ; the dimension conclusion is  $\dim_{\mathbb{H}} K \geq \frac{5}{2}$ . That is the bush bound in  $\mathbb{R}^4$  (Worked Example 9.7.2). ✓

*The habit.* On meeting any Kakeya-type estimate, compute  $s$  immediately and compare with  $n$ : the ratio  $s/n$  is the fraction of the conjecture achieved, and it is the only scale-free way to compare results across dimensions.

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## Drills

**Drill 9.7.1.** Verify the inequality  $\log m(K_\delta)/\log \delta \leq \beta + o(1)$  used in Proposition 9.7.1, being careful with the sign of  $\log \delta$ .

**Drill 9.7.2.** Show that a single-scale estimate with a fixed power loss  $\delta^{-c}$  gives  $\dim_{\mathbb{H}} K \geq s - sc$  and not  $s$ , and compute the loss for  $c = \frac{1}{10}$  at  $n = 4$ .

**Drill 9.7.3.** Explain why Proposition 9.7.1 bounds  $\overline{\dim_{\mathbb{M}}}$  and not  $\underline{\dim_{\mathbb{M}}}$ , and what additional information would be needed for the latter.



**Drill 9.7.4.** Assemble the Nikodym versions of the table in Worked Example 9.7.2, citing Theorem 9.6.6.

**Where we are.** A last reminder about the direction of the implications. Everything in this chapter converts a *lower* bound on the volume of a tube union into a *lower* bound on a dimension. There is no useful converse: knowing that some particular Besicovitch set has large dimension says nothing about tube families in general, because the families arising from a single set are a vanishingly small subfamily of all admissible ones. So the chain of Proposition 9.7.3 should be read as four equivalent ways of stating one quantitative fact about *all* families, and not as a bridge between statements about individual sets.

## Notes

The passage from maximal estimates to dimension is standard and is carried out in Volume 7, Theorem 7.8.5. The observation that the whole chain is lossless except for logarithms — so that tube exponents and dimension exponents may be quoted interchangeably — is folklore, and is what allows the literature to be read as it is written.



# Chapter 9.8

## What the Elementary Methods Give

### 9.8.1 The three arguments, side by side

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|               | $L^2$                                                    | bush                                            | hairbrush                                            |
|---------------|----------------------------------------------------------|-------------------------------------------------|------------------------------------------------------|
| Conditions on | nothing                                                  | a point                                         | a line                                               |
| Counts        | pairs of tubes                                           | tubes through $x_0$                             | tubes meeting $T_0$                                  |
| Key fact      | $m(T \cap T') \lesssim \frac{\delta^n}{\theta + \delta}$ | tubes through a point separate at unit distance | two lines meeting a third are concurrent or coplanar |
| Exponent      | 2                                                        | $\frac{n+1}{2}$                                 | $\frac{n+2}{2}$                                      |
| Sharp when    | $n = 2$                                                  | never                                           | never                                                |
| Loss          | log                                                      | none                                            | log, plus transversality                             |

*The progression.* Each argument conditions on an object of one higher dimension and, in exchange, obtains a structural fact unavailable to its predecessor. The  $L^2$  method has no structural input at all — which is why Worked Example 9.3.1 could show it cannot distinguish a Besicovitch configuration from a random one.

*The common shape.* The bush and hairbrush arguments both produce two bounds on  $m(E)$  whose product eliminates the unknown multiplicity  $\mu$ :

$$m(E) \gtrsim \mu \cdot A \quad \text{and} \quad m(E) \gtrsim \mu^{-1} \quad \implies \quad m(E) \gtrsim A^{1/2}.$$

The bush has  $A = \delta^{n-1}$ ; the hairbrush improves it to  $A = \delta^{n-2}$  by collecting tubes along a whole line rather than at a point. Any further improvement of this type must increase  $A$  again, and Chapter 9.5 showed where that stalls.

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**WORKED EXAMPLE 9.8.2— TIER B Why the geometric mean is the right move, and when it is not**

The device of multiplying two bounds to eliminate a parameter recurs throughout this subject, and it is worth understanding what it costs.

*What it does.* Given  $m(E) \geq \max\{\mu A, \mu^{-1}\}$  with  $\mu$  unknown, the worst case is where the two agree, at  $\mu = A^{-1/2}$ , giving  $m(E) \geq A^{1/2}$ . The bound is therefore attained only if the extremal configuration really has multiplicity  $A^{-1/2}$ .

*When it is sharp.* For the bush at  $n = 3$ :  $A = \delta^2$ , so  $\mu = \delta^{-1}$ , and indeed there are tube families with multiplicity  $\approx \delta^{-1}$  — a bush of  $\delta^{-1}$  tubes through a point in  $\mathbb{R}^3$  is realisable. So the argument is not obviously wasteful there.

*When it is not.* For large  $n$  the balancing multiplicity is  $\delta^{-(n-1)/2}$ , and it is not known whether a Besicovitch configuration can have multiplicity that large — indeed the Kakeya conjecture asserts that  $m(E) \gtrsim \delta^\varepsilon$ , which by the second bound forces  $\mu \lesssim \delta^{-\varepsilon}$ : the multiplicity should be almost bounded. So in high dimensions the geometric mean is balancing against a case that conjecturally does not occur, and the loss is severe.

*What would replace it.* An argument that bounds  $\mu$  directly, rather than optimising over it. That is exactly what the finite-field polynomial method achieves — Dvir’s theorem in Volume 10 gives the sharp bound with no multiplicity parameter at all — and it is the structural reason the polynomial method broke a barrier that had stood for a decade.

*The transferable lesson.* When an argument optimises over an unknown parameter, ask whether the balancing value is attainable. If it is not, the argument is proving a theorem about configurations that do not exist, and the loss is the distance between the balancing value and the true one.

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**WORKED EXAMPLE 9.8.3— TIER C: RESEARCH TASTE Where this volume leaves the subject**

(a) *What was proved.* The Besicovitch configuration with disjoint translates (Theorem 9.2.1), discharging Volume 3’s debt; Córdoba’s  $L^2$  bound, sharp in the plane (Corollary 9.3.3); the bush bound  $\frac{n+1}{2}$  (Theorem 9.4.4); the geometry of the hairbrush and Wolff’s bound in the transverse case (Theorem 9.5.3,

Theorem 9.5.4); and the Kakeya–Nikodym equivalence (Theorem 9.6.6).

(b) *What was decided against.* Wolff’s theorem in full generality (Remark 9.5.5, revision R9), with the obstruction — near-parallel tubes — identified arithmetically rather than gestured at.

(c) *The state of the problem after this volume.* Settled for  $n = 2$ ; about half the conjectured dimension for every  $n \geq 3$ ; and no method here has any prospect of reaching  $n$ , because all three are limited by the same optimisation over an unattainable multiplicity (Worked Example 9.8.2).

(d) *What must change.* Three things, each supplied by a later volume. Volume 10 replaces the geometry by arithmetic, studying the additive structure of the parameter set rather than the incidences of the tubes. Volume 11 introduces the polynomial method, which bounds the multiplicity directly rather than optimising over it. Volumes 12 and 13 return to the analytic side with multilinear and decoupling estimates, importing the geometric input those volumes supply.

(e) *A remark on the shape of the subject.* It is worth noticing that the three arguments of this volume are all from before 1995 and that the exponent  $\frac{n+2}{2}$  stood as the general record for a decade. The barrier was not technical: it was the limit of what incidence counting at a single scale can deliver. Every subsequent advance changed the category of the argument — arithmetic, algebraic, or analytic — rather than refining the geometry. That is the pattern this series has met repeatedly, most explicitly in Volume 8, Worked Example 8.8.5, and it is worth carrying into the volumes ahead.

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**WORKED EXAMPLE 9.8.4— TIER B Estimating what a new method can reach, before reading it**

A practical skill, and one this volume has now supplied the ingredients for: given a proposed approach to Kakeya, predict its ceiling.

*Step 1: identify the order.* Does the method’s central quantity involve pairs of tubes, or configurations of three or more? A second moment, an energy integral, a Fourier transform of a measure: all pairs, and by Worked Example 9.3.3 the ceiling is what the  $L^2$  method gives — dimension 2, or  $\frac{n+1}{2}$  if a point is also conditioned on.

*Step 2: identify the free parameter.* Most arguments optimise over something — a multiplicity, a scale, an angular width. Compute the balancing value, as in Worked Example 9.8.2, and ask whether a configuration attaining it exists. If not, the method is bounded by a case that does not occur and the loss is quantifiable.

*Step 3: check the losses.* A method producing  $\delta^{-\varepsilon}$  for every  $\varepsilon$  gives Hausdorff bounds; a fixed power gives only Minkowski, weakened by Volume 7, Drill 8.4. This decides whether the method addresses the conjecture at all.

*Worked prediction: the polynomial method.* Its central object is a polynomial vanishing on a set of points, and the count is of *incidences* between tubes and cells — not pairs. So Step 1 gives no ceiling. There is no free multiplicity parameter: Dvir’s argument bounds the set directly. And in the finite-field model there is no  $\delta$  and hence no loss. Prediction: the method should be sharp in the finite-field setting and strong in  $\mathbb{R}^n$ . That is what Volume 11 finds.

*Worked prediction: decoupling.* Its central object is an  $L^p$  norm of a sum over caps, and the estimate is  $\ell^2$ -based — a second moment in the frequency variable. Step 1 therefore predicts a ceiling, and Volume 13 confirms it: decoupling gives sharp results for problems whose extremisers are “flat” but does not by itself resolveakeya.

*The value of the exercise.* Not to avoid reading the papers, but to know what to expect from them, and to recognise quickly when a claimed approach cannot reach the conjecture — which is the commonest failure mode of proposed attacks on this problem.

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WORKED EXAMPLE 9.8.5— TIER A **The scoreboard forakeya, as it stands**

Collecting what the series has established and what it has reported.

| Bound                 | Source         | Dimensions    | Status here                    |
|-----------------------|----------------|---------------|--------------------------------|
| 2                     | Córdoba        | all $n$       | proved, Corollary 9.3.3        |
| $\frac{n+1}{2}$       | bush           | all $n$       | proved, Theorem 9.4.4          |
| $\frac{n+1}{2}$       | Drury, Fourier | all $n$       | reported, Vol 7 Remark 7.7.5   |
| $\frac{n+2}{2}$       | Wolff          | all $n$       | transverse case, Theorem 9.5.4 |
| $\frac{n+2}{2} + c_n$ | Katz–Laba–Tao  | $n = 3$       | Volume 10                      |
| $q^n$ points          | Dvir           | finite fields | Volume 10, sharp               |
| $n$                   | Wang–Zahl      | $n = 3$       | Volume 14, settled             |

*Reading the middle rows.* That the bush and Drury’s Fourier bound give the same exponent  $\frac{n+1}{2}$  is not a coincidence: Worked Example 7.7.2 in Volume 7 showed both balance a 1-dimensional object against a decay of order  $\frac{n-1}{2}$ . Two methods, one obstruction.

*Reading the last rows.* The finite-field problem is *solved*, sharply, by an argument of a page. The real problem in  $\mathbb{R}^3$  is solved by a long paper. And  $n \geq 4$

remains open. The gap between the finite-field and real cases is the subject of Volume 10, Chapter 8, and is where a reader should look before believing any short argument for the real case.

*The single number to remember.* Before 2005 the general- $n$  record was  $\frac{n+2}{2}$  — half the conjecture, plus one. Everything in this volume is that record and how it was reached.

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## Drills

**Drill 9.8.1.** For each of the three methods, identify the configuration that makes its key estimate sharp, and check that no single configuration is extremal for all three.

**Drill 9.8.2.** Compute the balancing multiplicity  $\mu = A^{-1/2}$  for the bush and the hairbrush in  $\mathbb{R}^4$ , and compare with the multiplicity the Kakeya conjecture would permit.

**Drill 9.8.3.** Show that if a Besicovitch set had a tube family of bounded multiplicity, then  $m(E) \gtrsim 1$  and the conjecture would follow immediately for that family.

**Drill 9.8.4.** Using Worked Example 9.7.2, determine the smallest  $n$  for which the bush bound exceeds the  $L^2$  bound, and the smallest for which the hairbrush exceeds the bush.

**Where we are.** A closing note on where Volumes 1–9 leave the series.

The first five volumes built the analysis: measure and integration, the Fourier transform, singular integrals, function spaces, oscillatory integrals. Volume 6 posed the restriction problem and showed it implies Kakeya. Volumes 7 and 8 built the geometric measure theory and applied it to projections and distance sets, reaching the limit of what two-point methods give. This volume attacked Kakeya directly with incidence geometry and reached the limit of what single-scale counting gives.

Three limits, all reached, and all the same limit seen from different angles: an argument that sums over pairs cannot detect the high-order coincidences that make a Besicovitch set small. Worked Example 9.3.3 states this as a test to apply to any proposed method, and Worked Example 9.8.4 turns it into a procedure.

What follows changes category rather than refining technique. Volume 10 studies the additive structure of the parameter set and proves the finite-field case outright; Volume 11 introduces the polynomial method, which bounds

the multiplicity instead of optimising over it; Volumes 12 and 13 return to the analytic side with multilinear and decoupling estimates. Volume 14 reports where all of it stands, including Wang and Zahl’s resolution of the three-dimensional case.

The debt from Volume 3 is discharged (Theorem 9.2.1), and nothing in Volumes 1 to 9 is now used without proof.

## Notes

The three arguments are Córdoba’s (1977), Bourgain’s (1991) and Wolff’s (1995). The observation that they share the structure of Worked Example 9.8.2 — two bounds multiplied to eliminate a multiplicity — is standard, and the diagnosis in that example of why the device is lossy in high dimensions is the usual explanation for the stagnation of the exponent before the arithmetic and algebraic methods arrived.



# Chapter 9.A

## Results Quoted from Volumes 1–8

| Result                                          | Source               | Used in                |
|-------------------------------------------------|----------------------|------------------------|
| The Perron tree, with its explicit translations | Theorem 1.3.12       | Chapter 9.2 throughout |
| Tube intersection estimate                      | Lemma 1.6.13         | Theorem 9.3.2          |
| The bush bound                                  | Theorem 1.7.15       | Chapter 9.4            |
| Takeya maximal operator                         | Definition 1.7.13    | Chapter 9.1            |
| Norms by testing                                | Corollary 1.8.10     | Proposition 9.1.3      |
| Marcinkiewicz interpolation                     | Theorem 2.2.4        | Proposition 9.1.3      |
| Riesz–Thorin                                    | Theorem 2.2.1        | Worked Example 9.1.1   |
| The ball multiplier theorem                     | Theorem 3.8.2        | Worked Example 9.2.3   |
| The configuration lemma, as assumed             | Lemma 3.8.4          | Theorem 9.2.1          |
| Lipschitz images and dimension                  | Proposition 7.1.1    | Lemma 9.6.5            |
| Minkowski content                               | Proposition 7.2.3    | Proposition 9.7.1      |
| Maximal bounds give Hausdorff bounds            | Theorem 7.8.5        | Theorem 9.7.2          |
| Stability of Minkowski dimension                | Proposition 7.2.5    | Worked Example 9.7.1   |
| The Takeya conjecture                           | Conjecture 7.1.1     | throughout             |
| Two-point versus richer problems                | Worked Example 7.7.3 | Worked Example 9.3.1   |
| Takeya against Falconer                         | Worked Example 8.8.5 | Worked Example 9.8.3   |

## Debts

**Discharged.** Volume 3, Lemma 3.8.4 — the Besicovitch configuration with pairwise disjoint translates — is proved as Theorem 9.2.1. It was the only statement in Volumes 1 to 8 used without proof, and Volume 3’s Theorem 3.8.2 is now unconditional.

**Incurred.** None. Nothing in this volume is owed to a later one.

**Removed.** Wolff’s hairbrush bound in full generality (Remark 9.5.5, revision R9); the transverse case is proved.

# Chapter 9.B

## Notation and Index of Results

### Additions to the Volume 1–8 conventions

|                                        |                                                            |
|----------------------------------------|------------------------------------------------------------|
| $\mathbb{T}, N$                        | a family of $\delta$ -tubes and its cardinality            |
| $E = \bigcup T, F = \sum \mathbf{1}_T$ | the union and the multiplicity function                    |
| $\mu = \ F\ _{L^\infty}$               | the maximum multiplicity                                   |
| $\mathcal{N}_\delta$                   | the Nikodym maximal operator                               |
| $H_\kappa(T_0)$                        | the $\kappa$ -transverse hairbrush of $T_0$                |
| $S_i, t_i, m_i$                        | the Perron data of Definition 9.2.2                        |
| $\Phi$                                 | the projective duality $(x', x_n) \mapsto (x'/x_n, 1/x_n)$ |

### Index of the main results

|                                         |                   |
|-----------------------------------------|-------------------|
| Maximal and multiplicity forms agree    | Proposition 9.1.3 |
| The rectangles fit inside the triangles | Lemma 9.2.4       |
| The translates are pairwise disjoint    | Lemma 9.2.5       |
| The Besicovitch configuration           | Theorem 9.2.1     |
| Angle counting                          | Lemma 9.3.1       |
| The $L^2$ bound                         | Theorem 9.3.2     |
| Córdoba's bound                         | Corollary 9.3.3   |
| The bush estimate                       | Lemma 9.4.2       |
| The bush bound                          | Theorem 9.4.4     |
| Concurrent or coplanar                  | Lemma 9.5.1       |
| Hairbrush volume estimate               | Theorem 9.5.3     |
| Wolff's bound, transverse case          | Theorem 9.5.4     |
| Position and direction interchanged     | Lemma 9.6.4       |
| Takeya and Nikodym are equivalent       | Theorem 9.6.6     |
| Single scales give Minkowski bounds     | Proposition 9.7.1 |
| The bounds in dimension form            | Theorem 9.7.2     |

## Plan revisions in this volume

| Removed                               | Reason                                                                                                                                                    |
|---------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| Wolff's bound $\frac{n+2}{2}$ in full | the transverse case is proved; the near-parallel case degrades exactly to the bush bound and needs Wolff's induction on scales. Remark 9.5.5, revision R9 |

# Part 10

## Volume 10

Takeya II: Arithmetic and Algebraic Methods



# Preface to Volume 10

## Two changes of category

Volume 9 reached the limit of incidence counting: the  $L^2$  method, the bush and the hairbrush all optimise over an unknown multiplicity, and Worked Example 9.8.2 showed the balancing value is one the Kakeya conjecture says does not occur. Refining the geometry cannot help. What is needed is an argument of a different kind, and this volume supplies two.

The first is *algebraic*. Replace  $\mathbb{R}^n$  by  $\mathbb{F}_q^n$ , a vector space over a finite field, and the Kakeya problem still makes sense: a Kakeya set contains a line in every direction, and the question is how few points it can have. Dvir’s theorem answers it completely, sharply, and in half a page. The argument constructs a polynomial vanishing on the set and derives a contradiction from its degree — there is no multiplicity parameter to optimise over, which is precisely why it escapes Volume 9’s ceiling.

The second is *arithmetic*. A Besicovitch set built from a parameter set has its size controlled, through Fubini, by the additive structure of that parameter set. Sumset inequalities — Ruzsa’s triangle inequality, the Plünnecke–Ruzsa estimates — then bound the geometry. This route gives the first bounds of the form  $cn$  with  $c > \frac{1}{2}$ , exceeding what Volume 9 achieved for every large  $n$ .

## A warning about the finite field model

Dvir’s proof is short and complete, and it is tempting to conclude that the real Kakeya problem must have a similarly short solution. It does not, and Chapter 10.7 explains why in detail. The essential point can be stated now: over  $\mathbb{F}_q$  a line has  $q$  points and two distinct lines meet in at most one, so “a line” is a combinatorial object with no scale. Over  $\mathbb{R}$  a tube has a *width*, two tubes can be nearly parallel and overlap along their whole length, and every argument must track how much. The polynomial method has real analogues — Volume 11 develops them — but they are quantitative and hard, and Dvir’s argument is not among them.

## What is assumed

Volumes 1–9, though less heavily than usual: Chapter 10.2 depends on nothing but linear algebra. From Volume 9, the framing of the Kakeya problem and the comparison of methods in Worked Example 9.8.1; from Volume 7, the passage from single-scale bounds to dimension.

## Why both methods escape Volume 9’s ceiling

Volume 9 ended with a diagnosis: the  $L^2$  method, the bush and the hairbrush all produce two bounds on  $m(E)$  and multiply them to eliminate an unknown multiplicity  $\mu$ , and Worked Example 9.8.2 showed the resulting bound is determined by a balancing value of  $\mu$  that the Kakeya conjecture says is never attained. The methods are proving theorems about configurations that do not exist.

Both methods here avoid that, in different ways.

*Dvir’s argument has no free parameter at all.* It builds a polynomial, restricts it to lines, and contradicts a degree count. There is nothing to optimise over, and consequently nothing lost — which is why it is the only result in this series that attains the truth exactly (Proposition 10.1.4 and Theorem 10.2.1 give the same number in the plane).

*The arithmetic method replaces the optimisation.* Instead of balancing against a multiplicity it balances against additive structure, and the balance point is the fixed point of a recursion —  $2 - \sqrt{2}$ , as Worked Example 10.6.5 explains. That is still a loss, and the constant is still below 1, but it exceeds  $\frac{1}{2}$ , which no method in Volume 9 does.

The general principle, stated once: *a method’s ceiling is set by what it optimises over.* Volume 9, Worked Example 9.8.4 turns this into a test, and applying it to the two methods of this volume predicts exactly what they deliver.

## The plan

- |   |                        |                                |
|---|------------------------|--------------------------------|
| 1 | The finite field model | and the Schwartz–Zippel lemma  |
| 2 | Dvir’s theorem         | the polynomial method, sharply |

Chapters 3–7, planned: sumset inequalities, with Ruzsa’s triangle inequality and Petridis’s proof of Plünnecke; the arithmetic reduction of the Kakeya problem to a sumset problem; the bounds of Bourgain and of Katz–Labat–Tao; why the finite field argument does not transfer (Chapter 10.7); and an assessment.



## Chapter 10.1

# The Finite Field Model

### 10.1.1 The problem over $\mathbb{F}_q$

Let  $q$  be a prime power and  $\mathbb{F}_q$  the field with  $q$  elements. The space  $\mathbb{F}_q^n$  has  $q^n$  points, and a *line* through  $a$  in direction  $b \neq 0$  is

$$\ell_{a,b} = \{a + tb : t \in \mathbb{F}_q\},$$

a set of exactly  $q$  points. Two distinct lines meet in at most one point, since their difference determines  $t$  uniquely.

**Definition 10.1.1.** A set  $K \subseteq \mathbb{F}_q^n$  is a *Kakeya set* if for every direction  $b \in \mathbb{F}_q^n \setminus \{0\}$  there is  $a \in \mathbb{F}_q^n$  with  $\ell_{a,b} \subseteq K$ .

Directions are naturally indexed by  $b$  up to scalar multiplication, so there are  $\frac{q^n-1}{q-1} \approx q^{n-1}$  of them — the finite projective space  $\mathbb{P}^{n-1}(\mathbb{F}_q)$ .

**Proposition 10.1.2** (The trivial bounds). *Every Kakeya set  $K \subseteq \mathbb{F}_q^n$  satisfies*

$$q \leq |K| \leq q^n,$$

*and there exist Kakeya sets with  $|K| \lesssim_n q^n$ ; in the plane there are Kakeya sets with  $|K| \leq \frac{q(q+1)}{2} + \frac{q-1}{2}$ .*

*Proof.* The lower bound holds because  $K$  contains at least one line, of  $q$  points; the upper because  $K \subseteq \mathbb{F}_q^n$ . That  $\mathbb{F}_q^n$  itself is a Kakeya set gives the second claim with constant 1.

For the planar construction, take

$$K = \left\{ (x, y) : y = \frac{x^2}{4} + bx \text{ for some } b \right\} \cup \{\text{points at infinity}\},$$

or more simply the union over  $b \in \mathbb{F}_q$  of the lines  $\ell_{a(b),b}$  with  $a(b) = (-b^2/4, 0)$  chosen so that the lines are tangent to a common parabola. Two such lines

meet in one point, and there are  $q$  of them, so the union has at most  $q \cdot q - \binom{q}{2}$  points; evaluating gives the stated bound. The vertical direction contributes one further line.  $\square$

**Where we are.** So in the plane a Kakeya set has between  $q$  and about  $\frac{q^2}{2}$  points, and the upper bound is achieved. The question is the true order of the lower bound: is it  $q$ , or  $q^2$ , or something between? Theorem 10.2.1 answers it in every dimension:  $|K| \gtrsim_n q^n$ , the maximum possible order. Before proving that, we need one lemma about polynomials — and it is the only input.

## 10.1.2 The Schwartz–Zippel lemma

**Lemma 10.1.3** (Schwartz–Zippel). *Let  $P \in \mathbb{F}_q[x_1, \dots, x_n]$  be a non-zero polynomial of total degree  $d$ . Then*

$$\#\{x \in \mathbb{F}_q^n : P(x) = 0\} \leq dq^{n-1}.$$

*In particular, if  $d < q$  then  $P$  does not vanish identically on  $\mathbb{F}_q^n$ .*

*Proof.* Induct on  $n$ . For  $n = 1$  a non-zero polynomial of degree  $d$  has at most  $d$  roots in a field, which is the claim.

Let  $n \geq 2$  and write

$$P(x_1, \dots, x_n) = \sum_{j=0}^k P_j(x_1, \dots, x_{n-1}) x_n^j,$$

where  $k$  is the largest power of  $x_n$  occurring and  $P_k \neq 0$ . Then  $\deg P_k \leq d - k$ .

Fix  $x' = (x_1, \dots, x_{n-1})$  and consider two cases.

If  $P_k(x') \neq 0$ : the polynomial  $t \mapsto P(x', t)$  is non-zero of degree  $k$ , so it has at most  $k$  roots. The number of  $x'$  in this case is at most  $q^{n-1}$ , contributing at most  $kq^{n-1}$  zeros.

If  $P_k(x') = 0$ : the whole line  $\{(x', t)\}$  may consist of zeros, giving  $q$  each. By the inductive hypothesis applied to  $P_k$ , which is non-zero of degree at most  $d - k$  in  $n - 1$  variables, there are at most  $(d - k)q^{n-2}$  such  $x'$ , contributing at most  $(d - k)q^{n-2} \cdot q = (d - k)q^{n-1}$  zeros.

Adding, the total is at most  $kq^{n-1} + (d - k)q^{n-1} = dq^{n-1}$ .

For the last claim, if  $d < q$  then  $dq^{n-1} < q^n$ , so  $P$  cannot vanish everywhere.  $\square$

---

WORKED EXAMPLE 10.1.1—TIER A **The lemma is sharp, and where**

*Sharpness.* Take  $P(x) = \prod_{j=1}^d (x_1 - c_j)$  with  $c_1, \dots, c_d$  distinct elements of

$\mathbb{F}_q$ ,  $d \leq q$ . Its zero set is the union of  $d$  parallel hyperplanes, of size exactly  $dq^{n-1}$ . So no better bound holds in general.

*What sharpness costs.* The extremal example is a union of hyperplanes — maximally degenerate. For an irreducible  $P$  the count is typically much smaller,  $\approx q^{n-1}$  by the Lang–Weil estimate, but that is a theorem of a different order and is not needed here.

*The form actually used.* Theorem 10.2.1 uses only the last sentence of Lemma 10.1.3: a non-zero polynomial of degree less than  $q$  cannot vanish on all of  $\mathbb{F}_q^n$ . That is the finite-field replacement for the statement, obvious over  $\mathbb{R}$ , that a non-zero polynomial is non-zero somewhere — and over a finite field it genuinely requires the degree hypothesis, since  $x^q - x$  vanishes identically on  $\mathbb{F}_q$ .

*The example to keep in mind.*  $x^q - x$  has degree  $q$  and vanishes on all of  $\mathbb{F}_q$ . So the bound  $d < q$  is not an artefact: at  $d = q$  the conclusion is false.

#### WORKED EXAMPLE 10.1.2— TIER B Counting dimensions of polynomial spaces

The other ingredient of Chapter 10.2 is a dimension count, and it is worth doing carefully since off-by-one errors here are easy.

*The space.* Let  $\mathcal{P}_d^n$  be the  $\mathbb{F}_q$ -vector space of polynomials in  $n$  variables of total degree at most  $d$ . A basis is the set of monomials  $x^\alpha$  with  $|\alpha| \leq d$ , so

$$\dim \mathcal{P}_d^n = \#\{\alpha \in \mathbb{N}_0^n : |\alpha| \leq d\} = \binom{d+n}{n},$$

by the standard stars-and-bars count: choosing  $\alpha$  with  $\alpha_1 + \cdots + \alpha_n \leq d$  is choosing  $\alpha_1 + \cdots + \alpha_n + \alpha_{n+1} = d$  in  $n+1$  non-negative integers.

*The value used.* With  $d = q-1$ ,

$$\dim \mathcal{P}_{q-1}^n = \binom{q-1+n}{n} = \binom{q+n-1}{n}.$$

*The comparison with  $q^n$ .* Since  $\binom{q+n-1}{n} = \frac{(q+n-1)(q+n-2)\cdots q}{n!} \geq \frac{q^n}{n!}$ , the dimension is at least  $q^n/n!$  — and at most  $q^n$  for  $q$  large, as it must be, since  $\mathcal{P}_{q-1}^n$  injects into the space of all functions  $\mathbb{F}_q^n \rightarrow \mathbb{F}_q$  of dimension  $q^n$ .

*Numerically.*

| $q$ | $n$ | $\binom{q+n-1}{n}$ | $q^n/n!$ | $q^n$ |
|-----|-----|--------------------|----------|-------|
| 3   | 2   | 6                  | 4.5      | 9     |
| 5   | 2   | 15                 | 12.5     | 25    |
| 7   | 3   | 84                 | 57.2     | 343   |
| 11  | 3   | 286                | 221.8    | 1331  |

The middle column is the bound Theorem 10.2.1 will deliver.

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**WORKED EXAMPLE 10.1.3— TIER C: RESEARCH TASTE** **Why a finite field model at all**

Replacing  $\mathbb{R}^n$  by  $\mathbb{F}_q^n$  is a standard device in this subject, and it is worth being clear about what it is for.

(a) *It removes scale.* Over  $\mathbb{R}$ , a tube has a width  $\delta$  and two tubes may be at any angle; the whole difficulty of Volume 9 was bookkeeping over angles and scales. Over  $\mathbb{F}_q$  a line is a set of  $q$  points, two distinct lines meet in at most one, and there is nothing to track. The combinatorics is exposed.

(b) *It preserves the extremal examples — sometimes.* The planar construction of Proposition 10.1.2 is a finite-field Perron tree: lines tangent to a parabola, overlapping as much as incidence geometry allows. So the model is not vacuous; it has its own extremal configurations.

(c) *It is a genuine test of a method.* Before Dvir, every approach to the finite field Kakeya problem gave bounds matching the real case —  $q^{(n+1)/2}$  from the bush,  $q^{(n+2)/2}$  from the hairbrush. That the answer turned out to be  $q^n$ , by an argument with no real analogue, is the strongest evidence that the real problem's difficulty is *metric* rather than combinatorial.

(d) *What it is not.* It is not a step towards the real problem. Chapter 10.7 sets out why: Dvir's argument uses that a line is  $q$  points with no thickness, and every attempt to give it thickness destroys the degree count. Wolff proposed the finite field problem in 1999 precisely as a model, and its resolution in 2008 did not resolve the real case, nor has it since.

(e) *What it did change.* It introduced the polynomial method to the subject. Volume 11 develops the quantitative version — polynomial partitioning — which *does* apply over  $\mathbb{R}$  and which underlies both the Guth–Katz solution of the Erdős distinct distances problem and the current records for restriction. So the model's influence was methodological rather than logical.

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**Proposition 10.1.4** (The planar construction, precisely). *Let  $q$  be an odd prime power and put*

$$L_b = \{(t, bt - b^2) : t \in \mathbb{F}_q\} \quad (b \in \mathbb{F}_q), \quad K = \left( \bigcup_{b \in \mathbb{F}_q} L_b \right) \cup (\{0\} \times \mathbb{F}_q).$$

*Then  $K$  contains a line in every direction and*

$$|K| \leq \frac{q(q+1)}{2} + q.$$

*Proof. Step 1. Every direction occurs.*

$L_b$  is the line of slope  $b$  through  $(0, -b^2)$ , so as  $b$  ranges over  $\mathbb{F}_q$  the lines  $L_b$  realise all  $q$  non-vertical slopes; the vertical direction is realised by  $\{0\} \times \mathbb{F}_q$ .

**Step 2.** *Count the union of the  $L_b$  exactly.*

A point  $(t, y)$  lies in  $\bigcup_b L_b$  precisely when  $y = bt - b^2$  for some  $b$ . Completing the square,

$$bt - b^2 = \frac{t^2}{4} - \left(b - \frac{t}{2}\right)^2,$$

which is legitimate since  $q$  is odd, so 2 is invertible. As  $b$  ranges over  $\mathbb{F}_q$ , the quantity  $(b - \frac{t}{2})^2$  ranges over the squares of  $\mathbb{F}_q$ , of which there are exactly  $\frac{q+1}{2}$  — namely 0 and the  $\frac{q-1}{2}$  non-zero squares. Hence for each fixed  $t$  there are exactly  $\frac{q+1}{2}$  admissible values of  $y$ , and

$$\left| \bigcup_b L_b \right| = q \cdot \frac{q+1}{2}.$$

**Step 3.** *Add the vertical line.*

It contributes at most  $q$  further points, giving the stated bound.  $\square$

**Where we are.** Step 2 is exact, not an estimate: the union has exactly  $\frac{q(q+1)}{2}$  points. Direct computation confirms it — at  $q = 3, 5, 7, 11, 13$  the union has 6, 15, 28, 66, 91 points, matching  $\frac{q(q+1)}{2}$  in every case.

Compare Theorem 10.2.1, which gives the lower bound  $\binom{q+1}{2} = \frac{q(q+1)}{2}$  in the plane. So the construction and the theorem agree to the leading term, and the true minimum sits between them, exceeding Dvir’s bound by exactly  $\frac{q-1}{2}$  (Worked Example 10.2.1).

The mechanism of the construction is worth naming: the lines are tangent to a common parabola, so each pair meets in one point and the union is as small as incidence geometry allows. It is the finite-field analogue of the Perron tree, with the tangency playing the role of the overlapping translates.

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WORKED EXAMPLE 10.1.4— TIER A **Kekeya sets in  $\mathbb{F}_3^2$ , by hand**

The smallest case is small enough to work out completely, and doing so makes every definition concrete.

*The setting.*  $\mathbb{F}_3^2$  has 9 points. Directions are non-zero vectors up to scalar, and there are  $\frac{3^2-1}{3-1} = 4$  of them:  $(1, 0)$ ,  $(0, 1)$ ,  $(1, 1)$ ,  $(1, 2)$ . Each line has 3 points, and there are 3 parallel lines in each direction, 12 lines in all.

*A Kekeya set must contain one line from each of the four parallel classes.* Four lines of three points each would give 12 points if disjoint, but lines

in different directions meet, so the union is smaller. Two lines in distinct directions share exactly one point, and with four lines there are  $\binom{4}{2} = 6$  pairs, so the union has at least  $12 - 6 = 6$  points.

*Dvir's bound.*  $\binom{q+n-1}{n} = \binom{4}{2} = 6$ . So the theorem gives exactly the count above — which is reassuring, since the theorem's proof uses nothing about how lines meet.

*The truth is 7.* Exhaustive search over all  $3^4 = 81$  choices of one line per direction shows the minimum union has 7 points, not 6. The discrepancy is because the six intersection points cannot all be distinct: four lines in general position in the plane have six intersection points, but here some must coincide. The extremal configuration has three lines through a common point and the fourth crossing them elsewhere, giving  $3 + 2 + 2 = 7$ .

*The general gap.* Worked Example 10.2.1 records that the minimum in the plane is  $\frac{q(q+1)}{2} + \frac{q-1}{2}$ , exceeding Dvir's  $\frac{q(q+1)}{2}$  by exactly  $\frac{q-1}{2}$ . At  $q = 3$  that is 1, which is the discrepancy just found.

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#### WORKED EXAMPLE 10.1.5— TIER B Using Schwartz–Zippel

The lemma is applied in three distinct ways in this volume, and separating them prevents confusion about what it does.

*Use 1: a non-zero polynomial is non-zero somewhere.* If  $\deg P < q$  then  $P$  does not vanish on all of  $\mathbb{F}_q^n$ . This is the form used in Theorem 10.2.1 Step 4, and it is the weakest consequence — but over a finite field it is not automatic, since  $x^q - x$  vanishes identically.

*Use 2: a quantitative bound on a zero set.*  $\#\{P = 0\} \leq dq^{n-1}$ . This is used in Corollary 10.2.2, where one knows  $P_d$  vanishes on a large cone and deduces a lower bound on  $d$ .

*Use 3: a line meets a variety in few points.* Restricting to a line  $\ell_{a,b}$  and applying the case  $n = 1$ : a non-zero  $P$  of degree  $d$  has at most  $d$  zeros on any line it does not contain. This is the form that *transfers to*  $\mathbb{R}$ , and Worked Example 10.7.1 identifies it as the whole of what Volume 11 uses.

*Which uses transfer.* Use 3 is a statement about incidences and holds over any field, including  $\mathbb{R}$ . Uses 1 and 2 are about the finiteness of the field and have no real analogue — over  $\mathbb{R}$  a non-zero polynomial is non-zero on a set of full measure, which is far stronger and correspondingly useless as a contradiction.

*The diagnostic.* When reading a polynomial-method argument, ask which use is in play. If Use 1 or 2, the argument is finite-field specific. If Use 3, it may transfer, and Volume 11 will show how.

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**WORKED EXAMPLE 10.1.6— TIER B Translating a real statement into the finite field model**

Before trusting a finite-field analogue one must know how the dictionary works, and where it has no entry.

| $\mathbb{R}^n$                                 | $\mathbb{F}_q^n$             |
|------------------------------------------------|------------------------------|
| $\delta$ -tube, $\delta = 1/q$                 | line ( $q$ points)           |
| number of directions $\approx \delta^{-(n-1)}$ | $\approx q^{n-1}$            |
| $m(K_\delta)$                                  | $ K /q^n$                    |
| $\dim_{\mathbb{H}} K \geq s$                   | $ K  \gtrsim q^s$            |
| $\delta$ -separated directions                 | distinct projective points   |
| two tubes at angle $\theta$                    | two lines meeting in a point |
| <i>angle between tubes</i>                     | <i>no entry</i>              |
| <i>nearly parallel tubes</i>                   | <i>no entry</i>              |
| <i>tube width</i>                              | <i>no entry</i>              |

*Reading the top half.* The dictionary is faithful for anything expressible in terms of incidences and counts. The Kakeya conjecture itself translates exactly:  $\dim_{\mathbb{H}} K = n$  becomes  $|K| \gtrsim q^n$ , which is Theorem 10.2.1.

*Reading the bottom half.* There is no notion of angle, because two distinct lines in  $\mathbb{F}_q^n$  either meet or do not, with no intermediate case. There is no near-parallelism, and no width. Every entry in the bottom half is precisely what Volume 9’s arguments spent their effort on — Lemma 9.3.1, Lemma 1.6.13, and the small-angle case that Remark 9.5.5 could not handle.

*The consequence for reading results.* A finite-field theorem whose proof uses only the top half may suggest a real analogue. A real theorem whose proof uses the bottom half has no finite-field shadow at all — Wolff’s hairbrush, for instance, is about angles and simply does not translate.

*The consequence for Dvir.* His proof uses only the top half, which is why it is short. Chapter 10.7 shows that the missing bottom half is exactly what blocks the return journey.

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**Where we are.** A remark on the role of  $q$  being a prime power rather than merely a prime.

Nothing in this volume uses more than that  $\mathbb{F}_q$  is a field: a non-zero polynomial in one variable of degree  $d$  has at most  $d$  roots (Lemma 10.1.3, base case), and every non-zero element is invertible (used in Proposition 10.1.4 to divide by 2 and 4). Both hold for any finite field, so  $q = p^k$  is as good as  $q = p$ .

What would fail over a ring:  $\mathbb{Z}/6\mathbb{Z}$  has zero divisors,  $x^2 - x$  has four roots there rather than two, and the root count that drives Step 3 of Theorem 10.2.1 collapses. The Kakeya problem over  $\mathbb{Z}/N\mathbb{Z}$  for composite  $N$  is genuinely different and is not treated here.

## Drills

**Drill 10.1.1.** Verify that there are  $\frac{q^n-1}{q-1}$  directions in  $\mathbb{F}_q^n$ , and that this is  $\approx q^{n-1}$ .

**Drill 10.1.2.** Check the base case and the inductive step of Lemma 10.1.3 for  $n = 2$ ,  $q = 3$ ,  $d = 1$ .

**Drill 10.1.3.** Confirm the stars-and-bars count in Worked Example 10.1.2 and evaluate  $\dim \mathcal{P}_{q-1}^n$  for  $q = 5$ ,  $n = 3$ .

**Drill 10.1.4.** Show that  $x^q - x$  vanishes on all of  $\mathbb{F}_q$ , and explain why this does not contradict Lemma 10.1.3.

**Where we are.** Two conventions for this volume.

*q plays the role of  $1/\delta$ .* A line has  $q$  points where a  $\delta$ -tube has  $\approx \delta^{-1}$  separated points; the number of directions is  $\approx q^{n-1}$  against  $\approx \delta^{-(n-1)}$ . So a finite-field bound  $|K| \gtrsim q^s$  corresponds to a real bound  $m(K_\delta) \gtrsim \delta^{n-s}$ , and the Kakeya conjecture is  $s = n$  in both.

*Constants are dimensional.* All implied constants depend on  $n$  and never on  $q$ ; a bound with a  $q$ -dependent constant says nothing, since  $q \rightarrow \infty$  is the regime of interest. This is the exact analogue of the rule in Volume 9 that constants must not depend on  $\delta$ .

## Notes

The finite field Kakeya problem was posed by Wolff in his 1999 survey, as a model in which the combinatorial difficulties might be isolated from the metric ones. The Schwartz–Zippel lemma is due independently to Schwartz, Zippel and DeMillo–Lipton in the late 1970s, and in the form used here goes back to Ore.



## Chapter 10.2

# Dvir's Theorem

**Theorem 10.2.1** (Dvir). *Let  $K \subseteq \mathbb{F}_q^n$  be a Kakeya set. Then*

$$|K| \geq \binom{q+n-1}{n} \geq \frac{q^n}{n!}.$$

*Idea of the proof.* Suppose  $K$  were smaller than the dimension of the space of polynomials of degree at most  $q-1$ . Then linear algebra — more unknowns than equations — produces a non-zero polynomial  $P$  of degree at most  $q-1$  vanishing on  $K$ .

Now use the lines. Along a line  $\ell_{a,b} \subseteq K$ , the function  $t \mapsto P(a+tb)$  is a polynomial in one variable of degree at most  $q-1$  with  $q$  roots, hence identically zero. Its leading coefficient is  $P_d(b)$ , where  $P_d$  is the top-degree homogeneous part of  $P$ ; so  $P_d(b) = 0$ .

Since  $K$  has a line in *every* direction,  $P_d$  vanishes at every  $b \neq 0$ , and at 0 too by homogeneity. So  $P_d$  is a non-zero polynomial of degree  $d \leq q-1$  vanishing on all of  $\mathbb{F}_q^n$ , contradicting Lemma 10.1.3.

The whole argument is three moves: build a polynomial, restrict it to the lines, and read off its top-degree part. There is no parameter to optimise and no geometry beyond “a line has  $q$  points”.

*Proof.* Suppose for contradiction that

$$|K| < \binom{q+n-1}{n} = \dim \mathcal{P}_{q-1}^n, \quad (10.2.1)$$

the dimension being computed in Worked Example 10.1.2.

**Step 1.** *A polynomial vanishing on  $K$ .*

Consider the linear map

$$\text{ev} : \mathcal{P}_{q-1}^n \longrightarrow \mathbb{F}_q^K, \quad \text{ev}(P) = (P(x))_{x \in K},$$

evaluating a polynomial at the points of  $K$ . Its target has dimension  $|K|$ , which by (10.2.1) is strictly less than the dimension of the source. So  $\text{ev}$  has non-trivial kernel: there is  $P \in \mathcal{P}_{q-1}^n$ ,  $P \neq 0$ , with  $P \equiv 0$  on  $K$ .

Let  $d = \deg P$ , so  $0 \leq d \leq q-1$ , and let  $P_d$  be the homogeneous part of degree  $d$ , which is non-zero by definition of the degree. Note  $d \geq 1$ : a non-zero constant does not vanish on the non-empty set  $K$ .

**Step 2.** *Restrict to a line.*

Fix a direction  $b \in \mathbb{F}_q^n \setminus \{0\}$  and let  $\ell_{a,b} \subseteq K$  be a line in that direction, which exists by Definition 10.1.1. Define

$$g(t) = P(a + tb) \in \mathbb{F}_q[t].$$

Expanding  $P$  into homogeneous parts  $P = \sum_{j \leq d} P_j$  and using homogeneity,

$$g(t) = \sum_{j=0}^d P_j(a + tb), \quad \text{and} \quad P_j(a + tb) \text{ has degree } \leq j \text{ in } t,$$

with the coefficient of  $t^j$  in  $P_j(a + tb)$  equal to  $P_j(b)$  — take the term of  $P_j$  in which every factor contributes  $tb$  rather than  $a$ . Hence

$$\deg_t g \leq d, \quad [t^d]g = P_d(b). \quad (10.2.2)$$

**Step 3.** *The restriction vanishes identically.*

Since  $\ell_{a,b} \subseteq K$  and  $P \equiv 0$  on  $K$ , we have  $g(t) = 0$  for every  $t \in \mathbb{F}_q$  — that is,  $g$  has  $q$  roots. But  $\deg_t g \leq d \leq q-1 < q$ , so a non-zero  $g$  could have at most  $q-1$  roots. Therefore  $g$  is the zero polynomial, and in particular every coefficient vanishes. By (10.2.2),

$$P_d(b) = 0.$$

**Step 4.** *Conclude.*

Step 3 holds for every  $b \neq 0$ . And  $P_d(0) = 0$  since  $P_d$  is homogeneous of degree  $d \geq 1$ . So  $P_d$  vanishes on all of  $\mathbb{F}_q^n$ .

But  $P_d$  is a non-zero polynomial of total degree  $d \leq q-1 < q$ , so by Lemma 10.1.3 it cannot vanish identically on  $\mathbb{F}_q^n$ . This contradiction refutes (10.2.1), and the first inequality of the theorem follows. The second is the computation in Worked Example 10.1.2.  $\square$

**Where we are.** It is worth pausing on how little was used.

*About  $K$ :* only that it contains a line in each direction. No metric, no scale, no separation of directions.

*About lines:* only that  $\ell_{a,b}$  has  $q$  points and is parametrised affinely by  $t$ , so that restricting a polynomial gives a polynomial in one variable of no higher degree.

*About the field:* only that a non-zero polynomial in one variable of degree  $< q$  has fewer than  $q$  roots, and Lemma 10.1.3 in  $n$  variables.

There is no multiplicity, no optimisation, no case division. Compare Volume 9, where every argument produced two bounds and multiplied them (Worked Example 9.4.4), and where Worked Example 9.8.2 diagnosed the resulting loss as the distance between the balancing multiplicity and the true one. Dvir's argument has no such parameter, and that is exactly why it is sharp.

**Corollary 10.2.2.** *The same conclusion holds if  $K$  contains a line in each of only  $\geq \frac{1}{2}|\mathbb{P}^{n-1}(\mathbb{F}_q)|$  directions, with  $\binom{q+n-1}{n}$  replaced by a constant multiple.*

*Proof.* Step 4 used that  $P_d$  vanishes at every  $b \neq 0$ . If instead  $P_d$  vanishes on a set  $D$  of directions with  $|D| \geq \frac{1}{2} \cdot \frac{q^n-1}{q-1}$ , then the cone over  $D$  — the union of the lines  $\mathbb{F}_q b$  for  $b \in D$  — has at least  $\frac{1}{2}(q^n - 1)$  points, on all of which  $P_d$  vanishes by homogeneity. By Lemma 10.1.3 this forces  $dq^{n-1} \geq \frac{1}{2}(q^n - 1)$ , that is  $d \gtrsim q$ . Running the argument with  $\mathcal{P}_{cq}^n$  for a suitable  $c < 1$  then gives the conclusion with a smaller constant.  $\square$

**Proposition 10.2.3** (Dvir for cylinders over a subset of directions). *Let  $D \subseteq \mathbb{F}_q^n \setminus \{0\}$  be a set of directions closed under scalar multiplication, and suppose  $K \subseteq \mathbb{F}_q^n$  contains a line in every direction of  $D$ . Then for every  $d < q$  with  $dq^{n-1} < |D|$ ,*

$$|K| \geq \binom{d+n}{n}.$$

*Proof.* Suppose  $|K| < \binom{d+n}{n} = \dim \mathcal{P}_d^n$ . As in Step 1 of Theorem 10.2.1, evaluation at the points of  $K$  has non-trivial kernel, so there is a non-zero  $P \in \mathcal{P}_d^n$  vanishing on  $K$ ; let  $e = \deg P \leq d$  and  $P_e$  its top-degree part.

Steps 2 and 3 of Theorem 10.2.1 apply verbatim to each direction  $b \in D$ : the restriction of  $P$  to a line of  $K$  in direction  $b$  is a polynomial in  $t$  of degree  $\leq e \leq d < q$  with  $q$  roots, hence zero, so  $P_e(b) = 0$ .

Thus  $P_e$  vanishes on  $D$ , a set of size  $|D| > dq^{n-1} \geq eq^{n-1}$ . But  $P_e$  is non-zero of degree  $e$ , so by Lemma 10.1.3 it has at most  $eq^{n-1}$  zeros — a contradiction.  $\square$

**Where we are.** Proposition 10.2.3 recovers Theorem 10.2.1 by taking  $D$  to be all non-zero directions, so  $|D| = q^n - 1$ , and  $d = q - 1$ : the condition  $dq^{n-1} < |D|$  reads  $(q-1)q^{n-1} < q^n - 1$ , which holds. It also yields Corollary 10.2.2 by taking  $|D| \geq \frac{1}{2}(q^n - 1)$  and  $d \approx \frac{q}{2}$ .

The proposition isolates what the argument really needs: not that *every* direction occurs, but that the directions occurring are too numerous for a low-degree form to vanish on all of them. That is a statement about the *dimension of the set of directions*, and it is the form in which the polynomial method generalises — to Furstenberg sets, where only a lower-dimensional family of directions is prescribed.

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**WORKED EXAMPLE 10.2.1— TIER A The bound against the truth, in small cases**

Theorem 10.2.1 gives  $|K| \geq \binom{q+n-1}{n}$ . How close is that?

*In the plane.* The bound is  $\binom{q+1}{2} = \frac{q(q+1)}{2}$ , and Proposition 10.1.2 exhibits Kakeya sets of size about  $\frac{q(q+1)}{2}$ . So the bound is sharp up to lower-order terms, and in fact the exact minimum is known:

$$\min|K| = \frac{q(q+1)}{2} + \frac{q-1}{2} \quad (q \text{ odd}).$$

Exhaustive search confirms this for small  $q$ : at  $q = 3$  the minimum is 7 against the bound 6; at  $q = 5$  it is 17 against the bound 15. The gap is  $\frac{q-1}{2}$ , exactly.

*In higher dimensions.* The bound is  $\approx \frac{q^n}{n!}$  and the best known constructions have  $\approx \frac{q^n}{2^{n-1}}$  points. Both are  $c_n q^n$ , so the *order* is settled and only the constant is open. That is a completely different state of affairs from the real problem, where the exponent itself is unknown for  $n \geq 4$ .

*The moral.* Over  $\mathbb{F}_q$  the Kakeya problem is solved: the answer is  $q^n$  up to a dimensional constant, and the remaining question — the sharp constant — is a finite combinatorial one. Nothing of the kind is available over  $\mathbb{R}$ .

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**WORKED EXAMPLE 10.2.2— TIER B The polynomial method: the pattern to recognise**

Dvir's argument is the first instance in this series of a technique used repeatedly in Volume 11. The pattern has three steps and it is worth naming them.

*Step A: parameter count.* A set  $S$  of  $N$  points imposes  $N$  linear conditions on polynomials. The space of polynomials of degree  $\leq d$  in  $n$  variables has dimension  $\approx d^n/n!$ . So if  $N < d^n/n!$  there is a non-zero polynomial of degree  $\leq d$  vanishing on  $S$ . *Vanishing is cheap when the set is small.*

*Step B: propagate along structure.* The set is not arbitrary — here it contains lines. A polynomial vanishing at  $q > d$  points of a line vanishes on the whole line as a polynomial, and one reads off information about its top-degree part.

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*Structure turns pointwise vanishing into algebraic vanishing.*

*Step C: contradiction from degree.* The conclusion of Step B is that some non-zero polynomial of controlled degree vanishes too often, contradicting Schwartz–Zippel. *Degree is the resource being spent.*

*Where each step reappears.* Volume 11’s polynomial partitioning replaces Step A by a partitioning statement — rather than vanishing on the set, the polynomial *separates* it into cells — and Steps B and C become an induction on the degree. The Guth–Katz solution of the distinct distances problem is that variant, and Guth’s proof of the multilinear restriction estimate is another.

*The diagnostic for whether the method can apply.* Ask whether the configuration has algebraic structure that a polynomial can detect. Lines, yes. Tubes of positive width, *no* — a polynomial does not vanish on a tube, and Chapter 10.7 makes this precise. That single observation explains both the success over  $\mathbb{F}_q$  and the difficulty over  $\mathbb{R}$ .

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WORKED EXAMPLE 10.2.3— TIER C: RESEARCH TASTE **What Dvir’s theorem settled, and what it did not**

(a) *It settled the finite field Kakeya conjecture.* Wolff posed it in 1999 as a model problem; before 2008 the best bounds were  $q^{(n+2)/2}$ , matching the hairbrush. Dvir’s proof gives  $q^n$  and is half a page.

(b) *It did not settle the real problem, and gave no route to it.* The argument uses that a line has no thickness. Chapter 10.7 sets out the obstruction; the short version is that the polynomial in Step 1 is built from a *finite* set of vanishing conditions, and a tube of positive width imposes infinitely many.

(c) *It changed which methods the field takes seriously.* Before 2008 the polynomial method was known in combinatorics — Alon’s Combinatorial Nullstellensatz, the Cap Set problem — but not in harmonic analysis. Afterwards Guth and Katz adapted it to  $\mathbb{R}^n$  via partitioning, and within five years it had solved the Erdős distinct distances problem and produced the best known restriction estimates. Volume 11 is that development.

(d) *It sharpened what the real difficulty is.* Since the combinatorics of lines is now completely understood and the real problem remains open, the difficulty must be metric — the width of tubes, the near-parallel configurations that Volume 9’s Remark 9.5.5 could not handle. That is a genuinely useful conclusion, and it is why Volume 14’s account of Wang and Zahl’s work emphasises their treatment of sticky configurations rather than any algebraic input.

(e) *A caution about model problems generally.* The finite field model was proposed on the hypothesis that it would isolate the essential difficulty. It did the opposite: it turned out to be easy, and the difficulty it isolated was the one it had removed. That is worth remembering when a model problem is offered as a stepping stone.

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**WORKED EXAMPLE 10.2.4— TIER B Running the proof on a specific polynomial**

The three steps of Theorem 10.2.1 are short enough to trace on an example, which is the quickest way to see what each does.

*Setting.*  $q = 3$ ,  $n = 2$ , and suppose someone proposes a Kakeya set  $K \subseteq \mathbb{F}_3^2$  with  $|K| = 5 < 6 = \binom{4}{2}$ .

*Step 1 produces a polynomial.*  $\mathcal{P}_2^2$  has basis  $\{1, x, y, x^2, xy, y^2\}$ , dimension 6. Evaluation at the 5 points of  $K$  is a linear map  $\mathbb{F}_3^6 \rightarrow \mathbb{F}_3^5$ , so its kernel has dimension at least 1: there is a non-zero  $P$  of degree  $\leq 2$  vanishing on  $K$ .

*Step 2 restricts to a line.* Say  $P = \alpha x^2 + \beta xy + \gamma y^2 + (\text{lower order})$ , so  $P_2 = \alpha x^2 + \beta xy + \gamma y^2$  if not all of  $\alpha, \beta, \gamma$  vanish. Along the line  $\{(a_1 + tb_1, a_2 + tb_2)\}$  the coefficient of  $t^2$  in  $P(a + tb)$  is

$$\alpha b_1^2 + \beta b_1 b_2 + \gamma b_2^2 = P_2(b). \checkmark$$

*Step 3 forces  $P_2(b) = 0$ .* The restriction is a polynomial in  $t$  of degree  $\leq 2$  with 3 roots, hence zero; so every coefficient vanishes, including  $P_2(b)$ .

*Step 4 contradicts.*  $P_2$  is a non-zero binary quadratic form vanishing at all four directions  $(1, 0), (0, 1), (1, 1), (1, 2)$  and hence, by homogeneity, on all 9 points. But a non-zero form of degree  $2 < 3$  has at most  $2 \cdot 3 = 6$  zeros by Lemma 10.1.3. Contradiction.

*What if  $\deg P < 2$ ?* Then  $P_d$  is linear or constant, and the same argument applies with  $d = 1$ : a non-zero linear form has 3 zeros, not 9. The proof does not care which  $d$  occurs, only that  $d \leq q - 1$ .

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**WORKED EXAMPLE 10.2.5— TIER B The method of multiplicities, and why it sharpens only here**

Theorem 10.2.1 gives  $|K| \geq \frac{q^n}{n!}$ ; the sharp constant is  $\frac{q^n}{2^n}$  up to lower order, and the improvement is due to Dvir, Kopparty, Saraf and Sudan. The mechanism is worth knowing, and so is the reason it does not help over  $\mathbb{R}$ .

*The idea.* Instead of requiring  $P$  merely to vanish on  $K$ , require it to vanish to order  $m$  — all partial derivatives up to order  $m - 1$  vanish. That is  $\binom{m+n-1}{n}$

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conditions per point instead of one, so the degree must rise; but the propagation step strengthens correspondingly, because a one-variable polynomial vanishing to order  $m$  at  $q$  points has degree  $\geq mq$ .

*The count.* With multiplicity  $m$ , Step 1 needs  $\frac{d^n}{n!} > |K| \binom{m+n-1}{n}$  and Step 3 needs  $d < mq$ . Balancing as  $m \rightarrow \infty$  gives  $|K| \gtrsim \frac{q^n}{2^n}$  — the factor  $n!$  is replaced by  $2^n$ .

*Why it does not transfer.* Proposition 10.7.2 shows that over  $\mathbb{R}$  the obstruction is that the points of a  $\delta$ -net inside a tube are *not collinear*. Raising the order of vanishing does not make them collinear; the polynomial exhibited there vanishes to any prescribed order on a curve inside the tube while missing the axis entirely. The repair addresses the wrong difficulty.

*The general point.* A refinement that improves a constant in one setting need not touch the obstruction in another. Before adapting a technique, identify which step it strengthens and whether that step is the one that fails.

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**WORKED EXAMPLE 10.2.6 — TIER C: RESEARCH TASTE**    **Why nobody found this for sixty years**

Dvir's proof uses linear algebra and the fact that a polynomial has finitely many roots. Both were available to Besicovitch. It is worth asking why the argument waited until 2008, because the answer is about how techniques migrate rather than about difficulty.

(a) *The problem was posed in 1999, not 1928.* The finite field version is Wolff's, and only nine years elapsed. The real problem is old; the model is not.

(b) *The method was in a different field.* The polynomial method was established in combinatorics — Alon's Combinatorial Nullstellensatz (1999), the Chevalley–Warning theorem, and the cap-set literature. Harmonic analysts working on Kakeya were using geometric and Fourier-analytic tools, and the two communities overlapped little.

(c) *The prevailing intuition pointed away.* Every bound before 2008 — bush, hairbrush, arithmetic — was of the form  $cn$  with  $c \leq \frac{1}{2}$  or slightly above, in both settings. That the finite field answer was  $q^n$  exactly, with no loss, was not expected; the numerical evidence from small  $q$  was consistent with it but had not been read that way.

(d) *The proof is short but not obvious.* The move that unlocks it — pass to the top-degree homogeneous part, whose vanishing at  $b$  encodes the whole line — is natural in hindsight and was not attempted. Step 3's observation that a degree- $(q-1)$  polynomial with  $q$  roots is *identically* zero, rather than

merely small, is the hinge.

(e) *What to take from this.* Not that hard problems have short solutions — the real Kakeya problem does not — but that a problem's difficulty is relative to the techniques being applied to it. Worked Example 9.8.4 in Volume 9 gives a test for whether a method can reach a target; Dvir's argument passes it trivially because it has no free parameter, and no method with a free parameter was ever going to find it.

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#### WORKED EXAMPLE 10.2.7— TIER A **The three steps, on one page**

A compressed statement of the argument, for reference.

*Hypothesis.*  $K \subseteq \mathbb{F}_q^n$  contains a line in every direction, and  $|K| < \binom{q+n-1}{n}$ .

*Step 1 — build.*  $\dim \mathcal{P}_{q-1}^n = \binom{q+n-1}{n} > |K|$ , so evaluation at  $K$  has a kernel: there is  $P \neq 0$  of degree  $d \leq q-1$  vanishing on  $K$ . *Resource spent: the smallness of  $K$ .*

*Step 2 — restrict.* On a line  $\ell_{a,b} \subseteq K$ , the map  $t \mapsto P(a+tb)$  is a polynomial of degree  $\leq d$  whose  $t^d$ -coefficient is  $P_d(b)$ . *Resource spent: the affine parametrisation of a line.*

*Step 3 — conclude.* That polynomial has  $q > d$  roots, so it is zero and  $P_d(b) = 0$ ; this for every  $b$ , so  $P_d$  vanishes on  $\mathbb{F}_q^n$ , contradicting Lemma 10.1.3. *Resource spent:  $d < q$ , twice.*

*The degree budget.* Everything turns on  $d \leq q-1$ . Step 1 wants  $d$  large, to make the polynomial space big enough; Steps 2 and 3 want  $d$  small, to make the root counts bite. The theorem is the statement that  $d = q-1$  satisfies both, and the resulting bound  $\binom{q+n-1}{n}$  is exactly the dimension at that degree.

*Compare Volume 9.* There the competing demands were on the multiplicity  $\mu$ , and Worked Example 9.8.2 showed the balance point was unattainable. Here they are on the degree  $d$ , and the balance point  $d = q-1$  is not only attainable but forced — there is no optimisation, only a single admissible choice. That is the structural difference between the two volumes.

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## Drills

**Drill 10.2.1.** Verify (10.2.2): the coefficient of  $t^d$  in  $P(a+tb)$  is  $P_d(b)$ .

**Drill 10.2.2.** Check that  $d \geq 1$  in Step 1, and identify where the argument would fail if  $K$  were empty.



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**Drill 10.2.3.** Confirm the numbers in Worked Example 10.2.1 for  $q = 3$ ,  $n = 2$ : compute  $\binom{q+1}{2}$  and exhibit a Kakeya set in  $\mathbb{F}_3^2$  of 7 points.

**Drill 10.2.4.** Prove Corollary 10.2.2 with explicit constants, tracking the choice of  $c$ .

**Where we are.** Worth recording explicitly: Theorem 10.2.1 is not merely sharp in order, it is sharp in a stronger sense than anything else in this series. The constructions of Proposition 10.1.4 give  $\frac{q(q+1)}{2}$  in the plane and the theorem gives  $\binom{q+1}{2} = \frac{q(q+1)}{2}$  — the same number.

No other bound in Volumes 6 to 10 has this property. Stein–Tomas is sharp only in its open range; the bush and hairbrush are off by a factor of two in the exponent; the arithmetic bounds are off by more. That a single half-page argument attains the truth exactly is the strongest available evidence that the finite field problem is of a different character from the real one.

## Notes

Dvir’s proof appeared in 2008, resolving Wolff’s problem. The form given here, using the top-degree homogeneous part, is Dvir’s; a variant of Alon and Tao replaces it by an application of the Combinatorial Nullstellensatz.

The sharp constant in the plane is due to Blokhuis and Mazzocca; in higher dimensions the best constructions and the best lower bounds still differ, and closing that gap is a finite combinatorial problem rather than an analytic one.



## Chapter 10.3

# Sumset Inequalities

The second method of this volume is arithmetic: a Besicovitch set built from a parameter set has its measure controlled, through Fubini, by the additive structure of that set. This chapter assembles the additive-combinatorial tools. They are used in Chapter 10.5; here they are proved for their own sake, and each is elementary.

### 10.3.1 Notation and the basic counts

**Definition 10.3.1.** For finite subsets  $A, B$  of an abelian group,

$$A + B = \{a + b\}, \quad A - B = \{a - b\}, \quad kA = \underbrace{A + \cdots + A}_k.$$

**Proposition 10.3.2** (Trivial bounds). *For non-empty finite  $A, B$  in an abelian group,*

$$\max\{|A|, |B|\} \leq |A + B| \leq |A||B|,$$

*with the left equality when one of them is a coset of a subgroup containing the other's differences, and the right when all sums  $a + b$  are distinct.*

*Proof.* Fixing  $b \in B$ , the map  $a \mapsto a + b$  is injective, so  $|A + B| \geq |A|$ ; symmetrically  $\geq |B|$ . The upper bound counts pairs.  $\square$

**Definition 10.3.3.** The *doubling constant* of  $A$  is  $K = |A + A|/|A|$ .

Small doubling means  $A$  is additively structured; the extreme case  $K = 1$  forces  $A$  to be a coset of a subgroup. The inequalities below say that small doubling propagates: if  $A + A$  is small then so is every iterated sumset.

### 10.3.2 Ruzsa's triangle inequality

**Theorem 10.3.4** (Ruzsa). *For non-empty finite  $A, B, C$  in an abelian group,*

$$|A| |B - C| \leq |B - A| |A - C|.$$

*Idea of the proof.* Read the statement as  $d(B, C) \leq d(B, A) + d(A, C)$  after taking logarithms, with  $d(X, Y) = \log \frac{|X-Y|}{\sqrt{|X||Y|}}$  — the Ruzsa distance. The proof is an injection, and the point is that an element of  $B - C$  can be routed through any element of  $A$ .

*Proof.* For each  $d \in B - C$  fix once and for all a representation

$$d = b(d) - c(d), \quad b(d) \in B, \quad c(d) \in C.$$

Define

$$\Phi : A \times (B - C) \longrightarrow (B - A) \times (A - C), \quad \Phi(a, d) = (b(d) - a, a - c(d)).$$

The two coordinates do lie in the stated sets, since  $b(d) \in B$ ,  $a \in A$ ,  $c(d) \in C$ .

$\Phi$  is injective. Suppose  $\Phi(a, d) = (u, v)$ . Then

$$u + v = (b(d) - a) + (a - c(d)) = b(d) - c(d) = d,$$

so  $d$  is determined by  $(u, v)$ . Knowing  $d$  determines  $b(d)$ , since the representation was fixed in advance; and then  $a = b(d) - u$  is determined. So  $(a, d)$  is recovered from  $(u, v)$ .

Comparing cardinalities of an injection gives the claim.  $\square$

**Where we are.** Everything in this chapter is of this type: a map between products of sumsets, shown injective by exhibiting the inverse. No inequality is used, and no structure of the group beyond associativity and cancellation.

The device to notice is *fixing representations in advance*. The map  $\Phi$  is well defined only because  $b(d)$  and  $c(d)$  were chosen once; had they been allowed to vary with  $a$ , the inverse would not exist. That trick recurs in every proof below.

**Pitfall: the triangle inequality does not convert differences into sums**

It is tempting to deduce from Theorem 10.3.4 that  $|A - A| \leq K|A|$  implies  $|A + A| \lesssim K^2|A|$ . Every direct substitution fails, and the failure is structural rather than a lack of ingenuity: the triangle inequality relates *difference* sets

to difference sets, and each choice of  $B, C$  that puts  $A + A$  on the left also puts it on the right, giving  $x \leq x$ .

The conversion is true — it is Corollary 10.3.7 below — but it requires Theorem 10.3.5, which is genuinely deeper. A reader who believes they have derived it from Theorem 10.3.4 alone should write out both sides and check that they differ.

### 10.3.3 The Plünnecke–Ruzsa inequality

**Theorem 10.3.5** (Plünnecke–Ruzsa, after Petridis). *Let  $A, B$  be finite non-empty subsets of an abelian group with  $|A + B| \leq K|A|$ . Then for all integers  $k, l \geq 0$ ,*

$$|kB - lB| \leq K^{k+l} |A|.$$

**Lemma 10.3.6** (The minimising subset). *With  $A, B$  as above, choose a non-empty  $X \subseteq A$  minimising*

$$K' = \frac{|X + B|}{|X|}.$$

*Then  $K' \leq K$  and, for every finite set  $C$ ,*

$$|X + B + C| \leq K' |X + C|. \quad (10.3.1)$$

*Proof.* That  $K' \leq K$  follows by taking  $X = A$ , which is admissible.

We prove (10.3.1) by induction on  $|C|$ .

**Step 1.** *Base case.*

For  $C = \{c\}$  both sides equal  $|X + B|$  and  $K'|X|$  respectively, which agree by the definition of  $K'$ .

**Step 2.** *The inductive step: a set of “already covered” elements.*

Let  $C = C' \cup \{c\}$  with  $c \notin C'$ , and assume the claim for  $C'$ . Put

$$Z = \{x \in X : x + B + c \subseteq X + B + C'\}.$$

Two containments will be used, and both are immediate once written down.

(i) *If  $x \in X \setminus Z$  then  $x + c \notin X + C'$ .* Contrapositively, suppose  $x + c = x' + c'$  with  $x' \in X, c' \in C'$ . Then  $x + B + c = x' + B + c' \subseteq X + B + C'$ , so  $x \in Z$ .

(ii)  *$((X \setminus Z) + B + c) \setminus (X + B + C')$ , translated by  $-c$ , is contained in  $(X + B) \setminus (Z + B)$ .* Indeed  $Z + B + c \subseteq X + B + C'$  by definition of  $Z$ , so  $Z + B \subseteq (X + B + C') - c$ ; hence any element of  $((X \setminus Z) + B)$  lying outside  $(X + B + C') - c$  lies outside  $Z + B$ , and it lies in  $X + B$ .

**Step 3.** *Count.*

Since  $X + B + C = (X + B + C') \cup ((X \setminus Z) + B + c)$ , containment (ii) gives

$$|X + B + C| \leq |X + B + C'| + (|X + B| - |Z + B|).$$

Now apply the definition of  $K'$  twice:  $|X + B| = K'|X|$ , and  $|Z + B| \geq K'|Z|$  because  $Z \subseteq X \subseteq A$  and  $K'$  was chosen minimal — this is the *only* place minimality is used, and it is used as a lower bound. Hence

$$|X + B| - |Z + B| \leq K'|X| - K'|Z| = K'|X \setminus Z|.$$

Combining with the inductive hypothesis  $|X + B + C'| \leq K'|X + C'|$ ,

$$|X + B + C| \leq K'(|X + C'| + |X \setminus Z|).$$

**Step 4.** *Compare with  $|X + C|$ .*

By containment (i), the points  $x + c$  for  $x \in X \setminus Z$  lie in  $X + C$  and outside  $X + C'$ , and they are distinct. Therefore

$$|X + C| \geq |X + C'| + |X \setminus Z|,$$

and substituting into Step 3 gives  $|X + B + C| \leq K'|X + C|$ .

(If  $Z = \emptyset$  the bound  $|Z + B| \geq K'|Z|$  reads  $0 \geq 0$  and the argument is unchanged.)  $\square$

*Proof of Theorem 10.3.5.* Let  $X$  be as in Lemma 10.3.6. Applying (10.3.1) repeatedly with  $C = jB$  gives, by induction on  $j$ ,

$$|X + jB| \leq K'^j |X| \quad (j \geq 0).$$

Now apply Theorem 10.3.4 with  $A$  there equal to  $X$ ,  $B$  equal to  $kB$  and  $C$  equal to  $lB$ :

$$|X| |kB - lB| \leq |kB - X| |X - lB| = |X + kB| |X + lB| \text{ up to reflection,}$$

where the last step replaces  $B$  by  $-B$  where needed, which changes no cardinality. Hence

$$|kB - lB| \leq \frac{K'^k |X| \cdot K'^l |X|}{|X|} = K'^{k+l} |X| \leq K^{k+l} |A|,$$

using  $K' \leq K$  and  $|X| \leq |A|$ .  $\square$

**Corollary 10.3.7** (Difference sets control sums). *If  $|A - A| \leq K|A|$  then  $|A + A| \leq K^2|A|$ .*

*Proof.* Apply Theorem 10.3.5 with  $B = -A$ . Its hypothesis reads

$$|A + B| = |A - A| \leq K|A|,$$

which holds. The conclusion with  $k = 0, l = 2$  gives

$$|-2B| = |2A| = |A + A| \leq K^2|A|,$$

since  $-2B = -2(-A) = 2A$  and cardinality is unchanged by negation.  $\square$

**Theorem 10.3.8** (Ruzsa covering lemma). *Let  $A, B$  be finite non-empty subsets of an abelian group with  $|A + B| \leq K|B|$ . Then there is  $X \subseteq A$  with  $|X| \leq K$  and*

$$A \subseteq X + B - B.$$

*Proof.* Choose  $X \subseteq A$  maximal subject to the sets  $\{x + B : x \in X\}$  being pairwise disjoint. Such an  $X$  exists because  $A$  is finite.

**Step 1.**  $X$  is small.

The sets  $x + B$ ,  $x \in X$ , are disjoint subsets of  $A + B$ , each of size  $|B|$ . Hence  $|X||B| \leq |A + B| \leq K|B|$ , so  $|X| \leq K$ .

**Step 2.**  $X + B - B$  covers  $A$ .

Let  $a \in A$ . If  $a \in X$  then  $a \in X + B - B$ , taking the same element of  $B$  twice. If  $a \notin X$  then by maximality  $X \cup \{a\}$  fails the disjointness condition, so  $(a + B) \cap (x + B) \neq \emptyset$  for some  $x \in X$ . Pick  $b, b' \in B$  with  $a + b = x + b'$ ; then

$$a = x + b' - b \in X + B - B. \quad \square$$

**Where we are.** The covering lemma is the third of the three elementary tools, and it plays a different role from the other two. Theorem 10.3.4 and Theorem 10.3.5 bound the *size* of sumsets; Theorem 10.3.8 bounds the *structure*, saying that  $A$  is covered by boundedly many translates of  $B - B$ . That is the form needed whenever one wants to conclude that a set is contained in something structured rather than merely that it is small — which is exactly the step from small doubling to a generalised arithmetic progression in Freiman’s theorem. Worked Example 10.5.1 identified that step as one of the two this series does not carry out; the covering lemma is the part of it that is elementary.

Note the proof: a maximal disjoint family, then a counting bound and a maximality argument. It is the same stopping-time pattern as Volume 3, Worked Example 3.1.5, and as the  $5r$ -covering lemma of Volume 7, Theorem 7.4.1. Three appearances of one idea.

**WORKED EXAMPLE 10.3.1— TIER A Sumsets of arithmetic progressions and of random sets**

Two extremes, which bracket everything.

*An arithmetic progression.*  $A = \{0, 1, \dots, N-1\}$  has  $A+A = \{0, \dots, 2N-2\}$ , so  $|A+A| = 2|A| - 1$  and  $K \approx 2$ . By Theorem 10.3.5,  $|kA - lA| \leq 2^{k+l}|A|$  — and directly,  $kA - lA = \{-l(N-1), \dots, k(N-1)\}$  has  $(k+l)(N-1) + 1$  elements, so the truth is  $\approx (k+l)|A|$ . Plünnecke is exponential in  $k+l$  where the truth is linear: sharp in its dependence on  $K$ , lossy in  $k, l$ .

*A random set.* Let  $A$  be a random subset of  $\{1, \dots, N\}$  of size  $N^{1/2}$ . Then almost all sums  $a + a'$  are distinct, so  $|A+A| \approx \frac{1}{2}|A|^2$  and  $K \approx \frac{1}{2}|A|$  — as large as Proposition 10.3.2 permits. Such a set has no additive structure, and the inequalities say nothing useful about it.

*The dichotomy.* Small  $K$  means the set is close to a progression — made precise by Freiman's theorem, which this series does not prove. Large  $K$  means it is essentially random. The arithmetic method for Kakeya works by showing that a small Besicovitch set forces small  $K$  for an associated set, and then contradicting the structure that small  $K$  entails.

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**WORKED EXAMPLE 10.3.2— TIER B Reading a sumset hypothesis**

Additive combinatorics states hypotheses in several equivalent-looking ways, and the differences matter.

$|A+A| \leq K|A|$  — *small doubling*. The standard hypothesis. Controls all  $|kA - lA|$  by Theorem 10.3.5.

$|A-A| \leq K|A|$  — *small difference*. Formally different, but by Theorem 10.3.5 applied with  $B = A$  each implies the other with  $K$  replaced by  $K^2$ . So the two are equivalent up to squaring the constant, and a paper using one may be compared with a paper using the other only after that adjustment.

$|\{(a, a') \in A \times A : a + a' = x\}|$  *large for many  $x$  — large additive energy*. Not equivalent: a set can have large energy and large doubling, by consisting of a structured piece plus a random piece. The Balog–Szemerédi–Gowers theorem repairs this, producing a large subset with small doubling; it is not proved in this series.

*Which one an application needs.* The arithmetic Kakeya argument produces an *energy* hypothesis from the geometry — many triples of tubes meeting — and must pass to doubling before Plünnecke applies. That passage is Balog–Szemerédi–Gowers, and it is where the constants degrade badly. It is the reason the arithmetic bounds, though better than  $\frac{n+2}{2}$  for large  $n$ , are far from  $n$ .

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**WORKED EXAMPLE 10.3.3— TIER A The Ruzsa distance**

Theorem 10.3.4 is named for a reason, and the metric interpretation is worth making explicit.

*The definition.* For non-empty finite  $A, B$  set

$$d(A, B) = \log \frac{|A - B|}{\sqrt{|A||B|}}.$$

*It is symmetric* —  $|A - B| = |B - A|$  after negation — *and satisfies the triangle inequality*: taking logarithms in  $|A||B - C| \leq |B - A||A - C|$  and rearranging,

$$\log \frac{|B - C|}{\sqrt{|B||C|}} \leq \log \frac{|B - A|}{\sqrt{|B||A|}} + \log \frac{|A - C|}{\sqrt{|A||C|}},$$

which is  $d(B, C) \leq d(B, A) + d(A, C)$ . ✓

*It is not a metric.*  $d(A, A) = \log \frac{|A - A|}{|A|} \geq 0$  with equality only when  $A$  is a coset of a subgroup, so distinct sets can be at distance zero from themselves and  $d(A, B) = 0$  does not force  $A = B$ . It is a pseudo-metric on the space of sets modulo translation and subgroup structure.

*Why the normalisation is  $\sqrt{|A||B|}$ .* Because it makes  $d$  symmetric and translation-invariant, and because Proposition 10.3.2 then gives  $d(A, B) \geq 0$  always. Any other normalisation breaks one of these.

*What it is for.* It converts additive-combinatorial statements into geometric ones: “ $A$  and  $B$  have similar additive structure” becomes “ $A$  and  $B$  are close”. Chains of inequalities become paths, and Theorem 10.3.5 becomes a statement that a short step in one direction bounds all longer steps.

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**WORKED EXAMPLE 10.3.4— TIER B Applying Plünnecke: the checklist**

Theorem 10.3.5 is used constantly and misapplied often. Four things to check.

(1) *Which set carries the hypothesis?* The hypothesis is  $|A + B| \leq K|A|$  — the bound is relative to  $|A|$ , and the conclusion  $|kB - lB| \leq K^{k+l}|A|$  is *also* relative to  $|A|$ , not to  $|B|$ . Confusing the two is the commonest error, and it matters when  $A$  and  $B$  have very different sizes.

(2) *Which set is iterated?* The conclusion iterates  $B$ , not  $A$ . To control  $|kA - lA|$  one applies the theorem with  $B = A$ , and then the hypothesis reads  $|A + A| \leq K|A|$ , the doubling condition.

(3) *Signs.*  $|kB - lB|$  covers both sums and differences: taking  $l = 0$  gives

$|kB|$ , taking  $k = 0$  gives  $|lB|$  after negation. So a single application controls every iterated sumset and difference set simultaneously.

(4) *The exponent is  $k + l$ , not max.* The bound  $K^{k+l}$  is exponential in the total number of copies. Worked Example 10.3.1 showed the truth for an arithmetic progression is linear in  $k + l$ , so the theorem is lossy there — but it is sharp in its dependence on  $K$ , which is what applications need.

*A worked application.* Suppose  $|A - A| \leq K|A|$  and one wants  $|3A - 2A|$ . Apply the theorem with  $B = -A$ : the hypothesis is  $|A + B| = |A - A| \leq K|A|$  ✓, and the conclusion with  $k = 2, l = 3$  gives  $|2B - 3B| = |3A - 2A| \leq K^5|A|$ .

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**WORKED EXAMPLE 10.3.5— TIER C: RESEARCH TASTE   Where additive combinatorics enters this subject**

Sumset inequalities look remote from Besicovitch sets, and it is worth setting out the three places they enter, since only the first belongs to this volume.

(a) *The Kakeya problem, through slices.* Lemma 10.5.2 turns lines into three-term progressions, so a Besicovitch set becomes a configuration with many additive triples. That is Chapter 10.5, and it is the route Bourgain took.

(b) *The sum-product phenomenon.* A finite set of reals cannot have both  $A + A$  and  $A \cdot A$  small; Erdős and Szemerédi conjectured  $\max\{|A + A|, |A \cdot A|\} \gtrsim |A|^{2-\epsilon}$ . Sum-product estimates feed into incidence bounds, and incidence bounds feed into Kakeya — a longer chain, exploited in the finite-field setting by Bourgain, Katz and Tao.

(c) *The Furstenberg set problem.* A set meeting each line of a family in dimension  $s$ ; the extremal examples are arithmetically structured, and the solution of Ren and Wang reported in Volume 14 uses additive-combinatorial input.

*What the three have in common.* In each, a geometric configuration forces an additive relation, and the additive relation constrains the geometry. The inequalities of this chapter are the constraint half; the geometric half differs in each case.

*Why the losses are severe in every case.* The passage from geometry to additive structure is through *energy* — many triples — and the passage from energy to *doubling*, which is what Theorem 10.3.5 needs, costs polynomial factors by Balog–Szemerédi–Gowers. Worked Example 10.6.2 tracks this for the Kakeya application, and the same accounting applies to (b) and (c).

*A caution for the reader heading into this literature.* The inequalities here are stated for abelian groups; applications to  $\mathbb{R}^n$  use them for  $\mathbb{Z}^n$  or for  $\delta$ -discretised sets, and the discretisation introduces error terms that must

be tracked. A statement about finite subsets of  $\mathbb{Z}$  does not transfer to  $\delta$ -separated subsets of  $\mathbb{R}$  without care.

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**WORKED EXAMPLE 10.3.6— TIER B Proving a sumset inequality: the injection recipe**

Every result in this chapter is proved by exhibiting an injection. The recipe is uniform and worth practising, since the literature contains dozens of such inequalities and they are all proved this way.

*The shape.* To prove  $|X_1||X_2| \leq |Y_1||Y_2|$ , construct  $\Phi : X_1 \times X_2 \rightarrow Y_1 \times Y_2$  and show it injective by exhibiting the inverse explicitly.

*Step 1: fix representations in advance.* Elements of a sumset generally have many representations. Choose one for each, once and for all, before defining  $\Phi$ . In Theorem 10.3.4 this is  $d \mapsto (b(d), c(d))$ , and without it the inverse does not exist.

*Step 2: make the map recover a distinguished element.* Design  $\Phi$  so that some combination of the output coordinates — usually their sum or difference — returns an element of the input. In Theorem 10.3.4,  $u + v = d$ .

*Step 3: unwind.* With  $d$  recovered, the fixed representation gives  $b(d)$ , and then the remaining input coordinate follows.

*Worked practice: the Ruzsa sum triangle inequality.* Prove  $|A||B + C| \leq |A + B||A + C|$ . Fix for each  $s \in B + C$  a representation  $s = b(s) + c(s)$ , and set

$$\Phi(a, s) = (a + b(s), a + c(s)) \text{ —}$$

but this is not injective, since  $a$  cannot be recovered from the two coordinates: their difference is  $b(s) - c(s)$ , which gives  $s$  only if that determines it. The correct map uses the *difference* form, which is why Theorem 10.3.4 is stated with  $B - C$  rather than  $B + C$ . The sum version requires Theorem 10.3.5 and is genuinely deeper — a useful thing to discover by attempting it.

*The habit.* Before believing a claimed sumset inequality, try to write the injection. If Step 2 has no natural candidate, the inequality is probably false or probably deep.

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**WORKED EXAMPLE 10.3.7— TIER A Doubling constants, computed**

Six sets, with their doubling constants, to calibrate what  $K$  measures.

| $A \subseteq \mathbb{Z}$                                              | $ A $   | $ A + A $               | $K$                   |
|-----------------------------------------------------------------------|---------|-------------------------|-----------------------|
| $\{0, 1, \dots, N - 1\}$                                              | $N$     | $2N - 1$                | $\approx 2$           |
| $\{0, d, 2d, \dots, (N - 1)d\}$                                       | $N$     | $2N - 1$                | $\approx 2$           |
| $\{0, \dots, N - 1\} \times \{0, \dots, N - 1\}$ (in $\mathbb{Z}^2$ ) | $N^2$   | $(2N - 1)^2$            | $\approx 4$           |
| $\{1, 2, 4, 8, \dots, 2^{N-1}\}$                                      | $N$     | $\binom{N}{2} + N$      | $\approx \frac{N}{2}$ |
| $\{0, \dots, N - 1\} \cup \{M\}$ , $M \gg N$                          | $N + 1$ | $\approx 3N$            | $\approx 3$           |
| random subset of $[N^2]$ of size $N$                                  | $N$     | $\approx \frac{N^2}{2}$ | $\approx \frac{N}{2}$ |

*What the table shows.* Doubling is small exactly for sets that are progressions, or products of progressions, or progressions with a bounded number of extra points. It is as large as possible —  $\approx \frac{|A|}{2}$  — for geometric progressions and for random sets, which have no additive structure.

*The dilation invariance.* Rows one and two have the same  $K$ : doubling is unchanged by an affine map  $x \mapsto ax + b$ , since such a map is a bijection commuting with addition up to translation. So  $K$  measures structure, not size or position.

*The rank shows up in the constant.* A  $d$ -dimensional progression has  $K \approx 2^d$ , as row three illustrates with  $d = 2$ . So  $\log_2 K$  estimates the rank, which is the quantitative content of Freiman's theorem.

*Why row five matters.* Adding one far-away point to a progression roughly triples the sumset but barely changes the size. So  $K$  is sensitive to outliers, and applications must either remove them — which is what Theorem 10.3.8 and Balog–Szemerédi–Gowers do — or tolerate the degradation.

## Drills

**Drill 10.3.1.** Verify the injectivity of  $\Phi$  in Theorem 10.3.4 by writing out the inverse explicitly.

**Drill 10.3.2.** Check Theorem 10.3.4 numerically on  $A = B = C = \{0, 1, 3\}$  in  $\mathbb{Z}$ .

**Drill 10.3.3.** Verify the base case  $|C| = 1$  of Lemma 10.3.6, and identify where the minimality of  $K'$  is used in the inductive step.

**Drill 10.3.4.** For  $A$  an arithmetic progression of length  $N$ , compute  $|kA - lA|$  exactly and compare with the bound of Theorem 10.3.5.

**Where we are.** Three inequalities, three roles, worth separating before leaving the chapter.

Theorem 10.3.4 is a *comparison*: it relates the sizes of three difference sets and is the metric backbone, as Worked Example 10.3.3 makes explicit.

Theorem 10.3.5 is a *propagation*: one hypothesis about  $A+B$  controls every iterated sumset of  $B$ . It is the workhorse, and Worked Example 10.3.4 lists the four ways it is misapplied.

Theorem 10.3.8 is a *structure* statement:  $A$  is covered by few translates of  $B-B$ . It is the only one of the three whose conclusion is about containment rather than cardinality, and it is the elementary part of the passage to Freiman’s theorem.

All three are proved by injections and maximal families, and none uses anything about the ambient group beyond cancellation.

## Notes

Ruzsa’s triangle inequality is from 1996 and the distance interpretation is his. Plünnecke’s original proof (1970) used a graph-theoretic argument; Petridis’s proof, given here, appeared in 2012 and is substantially shorter.

Freiman’s theorem and the Balog–Szemerédi–Gowers theorem are the two deeper results of the area; neither is proved in this series, and Worked Example 10.3.2 records where the second would be needed.



## Chapter 10.4

# Balog–Szemerédi–Gowers

Chapter 10.3 developed the sumset toolkit — Ruzsa’s triangle inequality and the Plünnecke–Ruzsa inequality — under the hypothesis that a sumset  $A + B$  is small. In applications one frequently knows less: that  $a + b$  takes few values for *many* pairs  $(a, b)$ , not all. This chapter proves that the weaker hypothesis suffices after passing to large subsets.

The result is used twice in this series: in Chapter 10.5 for the arithmetic approach to Kakeya, and in Volume 14, Proposition 14.14.12 for Bourgain’s projection theorem. It was the last additive-combinatorial input recorded as quoted.

### 10.4.1 Partial sumsets

**Definition 10.4.1** (Partial sumset). Let  $A, B$  be finite subsets of an abelian group and  $G \subseteq A \times B$ . The *partial sumset* is

$$A +_G B = \{a + b : (a, b) \in G\}.$$

We regard  $G$  as the edge set of a bipartite graph with parts  $A$  and  $B$ .

**Where we are.** The hypothesis  $|A +_G B|$  small does *not* imply  $|A + B|$  small, and the gap can be extreme.

*The standard example.* Let  $A = B = \{1, \dots, n\} \cup S$  where  $S$  is a set of  $n$  integers in general position — say  $\{2^{n+1}, \dots, 2^{2n}\}$ . Take  $G$  to be the pairs with both entries in  $\{1, \dots, n\}$ . Then  $|G| = n^2 = \frac{1}{4}|A||B|$  and  $|A +_G B| = 2n - 1$ , while  $|A + B| \geq |S + S| \approx \frac{1}{2}n^2$ .

*So no conclusion about  $A+B$  itself is possible*, and the theorem must instead produce subsets. What is remarkable is how little is lost: the subsets retain a  $K^{-O(1)}$  fraction, and the constant in the sumset bound is  $K^{O(1)}$ .

**Theorem 10.4.2** (Balog–Szemerédi–Gowers). *There is an absolute constant  $C$  such that the following holds. Let  $A, B$  be finite subsets of an abelian group and  $G \subseteq A \times B$  with*

$$|G| \geq \frac{|A||B|}{K}, \quad |A +_G B| \leq K|A|^{1/2}|B|^{1/2}.$$

*Then there are  $A' \subseteq A$  and  $B' \subseteq B$  with*

$$|A'| \gtrsim K^{-C}|A|, \quad |B'| \gtrsim K^{-C}|B|, \quad |A' + B'| \lesssim K^C|A|^{1/2}|B|^{1/2},$$

*and moreover  $|G \cap (A' \times B')| \gtrsim K^{-C}|A||B|$ .*

## 10.4.2 The graph lemma

Everything rests on a statement about bipartite graphs with no arithmetic in it.

**Lemma 10.4.3** (Paths of length two). *Let  $H$  be a bipartite graph with parts  $A, B$  of size  $n$  and at least  $cn^2$  edges. Then there is  $A_1 \subseteq A$  with  $|A_1| \geq cn/\sqrt{2}$  such that at most half the pairs  $(a, a') \in A_1^2$  satisfy  $|N(a) \cap N(a')| < \frac{c^2}{4}n$ .*

*Proof.* Choose  $b \in B$  uniformly at random and set  $A_1 = N(b)$ . Then

$$\mathbb{E}|A_1| = \frac{1}{n} \sum_{b \in B} \deg(b) = \frac{|E|}{n} \geq cn,$$

so by Cauchy–Schwarz  $\mathbb{E}|A_1|^2 \geq (\mathbb{E}|A_1|)^2 \geq c^2n^2$ .

Call  $(a, a')$  *bad* if  $|N(a) \cap N(a')| < \varepsilon n$ . A bad pair lies in  $A_1^2$  precisely when  $b$  is a common neighbour, which happens with probability  $|N(a) \cap N(a')|/n < \varepsilon$ . Hence

$$\mathbb{E}[\#\{\text{bad pairs in } A_1^2\}] = \sum_{\text{bad } (a, a')} \frac{|N(a) \cap N(a')|}{n} < \varepsilon n^2.$$

Taking  $\varepsilon = c^2/4$ ,

$$\mathbb{E}\left[|A_1|^2 - 2\#\{\text{bad in } A_1^2\}\right] \geq c^2n^2 - 2 \cdot \frac{c^2}{4}n^2 = \frac{1}{2}c^2n^2 > 0,$$

so some  $b$  realises at least the average. For that  $b$ ,  $|A_1|^2 \geq \frac{1}{2}c^2n^2$  and  $\#\{\text{bad in } A_1^2\} \leq \frac{1}{2}|A_1|^2$ .  $\square$



**Where we are.** This is *dependent random choice*: one selects a random vertex and keeps its neighbourhood, and the point is that a set arising this way is *biased towards having large common neighbourhoods*. A uniformly random subset of  $A$  of the same size would have no such property.

The mechanism is the second display: a pair with few common neighbours is unlikely to survive, because surviving requires  $b$  to be one of them. So selecting by a random neighbourhood automatically suppresses the bad pairs, and the two effects are balanced by choosing  $\varepsilon = c^2/4$ .

**Lemma 10.4.4** (Paths of length three). *Let  $H$  be a bipartite graph with parts  $A, B$  of size  $n$  and at least  $cn^2$  edges. Then there are  $A' \subseteq A$  and  $B' \subseteq B$  with  $|A'|, |B'| \gtrsim cn$  such that for every  $a \in A'$  and  $b \in B'$  there are  $\gtrsim c^5 n^2$  paths of length three from  $a$  to  $b$ .*

*Proof.* **Step 1.** *A set with mostly-good pairs.*

By Lemma 10.4.3 take  $A_1$  with  $|A_1| \geq cn/\sqrt{2}$  in which at most half the pairs are bad, “bad” meaning fewer than  $\frac{c^2}{4}n$  common neighbours. Discarding the vertices lying in the most bad pairs, we obtain  $A_2 \subseteq A_1$  with  $|A_2| \geq \frac{1}{2}|A_1|$  in which at most a  $\frac{1}{5}$  fraction of pairs are bad — a Markov argument, since a vertex in many bad pairs is rare.

**Step 2.** *Vertices of  $B$  seeing  $A_2$  often.*

Let

$$B' = \left\{ b \in B : \deg_{A_2}(b) \geq \frac{c}{4}|A_2| \right\}.$$

Counting edges between  $A_2$  and  $B$  from the  $A_2$  side and using that  $H$  has  $\geq cn^2$  edges,  $|B'| \gtrsim cn$ .

**Step 3.** *Vertices of  $A$  with many good companions.*

Call  $a \in A$  *friendly* to  $a' \in A_2$  if  $a$  and  $a'$  have  $\geq \frac{c^2}{4}n$  common neighbours. Let

$$A' = \left\{ a \in A : a \text{ is friendly to at least } \left(1 - \frac{c}{5}\right)|A_2| \text{ vertices of } A_2 \right\}.$$

Since at most  $\frac{1}{5}$  of pairs within  $A_2$  are bad, a Markov argument gives  $|A'| \gtrsim cn$ ; note  $A'$  need not lie in  $A_2$ .

**Step 4.** *Count the paths.*

Fix  $a \in A'$  and  $b \in B'$ . The vertices of  $A_2$  friendly to  $a$  number at least  $(1 - \frac{c}{5})|A_2|$ ; the neighbours of  $b$  in  $A_2$  number at least  $\frac{c}{4}|A_2|$ . These two sets must overlap in at least

$$\left(\frac{c}{4} - \frac{c}{5}\right)|A_2| = \frac{c}{20}|A_2| \gtrsim c^2 n$$

vertices. *This is the whole point of Steps 2 and 3:* the thresholds  $\frac{c}{4}$  and  $1 - \frac{c}{5}$  are chosen so that the overlap is forced.

For each  $a'$  in the overlap,  $a$  and  $a'$  have  $\geq \frac{c^2}{4}n$  common neighbours  $b'$ , and  $a'b$  is an edge. Each choice of  $(b', a')$  gives a distinct path  $a \rightarrow b' \rightarrow a' \rightarrow b$  of length three, so the number of such paths is

$$\gtrsim c^2 n \cdot \frac{c^2}{4} n \gtrsim c^4 n^2,$$

which is at least the claimed  $c^5 n^2$ .  $\square$

**Where we are.** *Why paths of length three and not one.* The conclusion one would like — that  $A' \times B'$  is entirely contained in the edge set — is false. A random bipartite graph of density  $\frac{1}{2}$  contains no complete bipartite subgraph with parts of size  $2 \log n$ , so no bound of the form  $|A'|, |B'| \gtrsim cn$  can force even one edge between every pair.

Paths of length *two* fail as well: Step 1 gives only that *most* pairs have many common neighbours, not all. Length three is the first case where one can pass from “most” to “all”, and Step 4 is where that passage happens — by intersecting two large subsets of  $A_2$  rather than by any property of a single pair.

### 10.4.3 Proof of the theorem

*Proof of Theorem 10.4.2.* We may assume  $|A| = |B| = n$ ; the general case follows by padding with elements far from everything, as in the Checkpoint after Definition 10.4.1.

Apply Lemma 10.4.4 to the graph  $G$  with  $c = K^{-1}$ , obtaining  $A' \subseteq A$  and  $B' \subseteq B$  with  $|A'|, |B'| \gtrsim K^{-1}n$  and, for every  $a \in A'$ ,  $b \in B'$ , at least  $\gamma n^2$  paths of length three from  $a$  to  $b$ , where  $\gamma \gtrsim K^{-5}$ .

Fix such  $a, b$ . A path of length three is a triple  $(b_1, a_1, b_2)$  with  $ab_1$ ,  $a_1b_1$ ,  $a_1b_2$  edges of  $G$  and  $b_2 = b$ ; write it as

$$a + b = \underbrace{(a + b_1)}_{\in A +_G B} - \underbrace{(a_1 + b_1)}_{\in A +_G B} + \underbrace{(a_1 + b)}_{\in A +_G B}.$$

*Each path exhibits  $a + b$  as an element of  $(A +_G B) - (A +_G B) + (A +_G B)$ , and distinct paths give distinct triples of summands, since  $b_1$  and  $a_1$  are recoverable from them.*

Therefore the map sending a path to its triple of summands is injective, and counting over all  $(a, b) \in A' \times B'$ ,

$$|A' + B'| \cdot \gamma n^2 \leq \#\{\text{paths of length three in } G\} \leq |A +_G B|^3.$$

Hence, using  $|A +_G B| \leq Kn$ ,

$$|A' + B'| \leq \frac{|A +_G B|^3}{\gamma n^2} \lesssim \frac{K^3 n^3}{K^{-5} n^2} = K^8 n,$$

which is the stated bound with  $C = 8$ .

For the final assertion, the number of edges of  $G$  inside  $A' \times B'$  is at least  $|A'| |B'| - \#\{\text{non-edges}\}$ ; but every pair of  $A' \times B'$  is joined by  $\gamma n^2 > 0$  paths, and the same counting applied to  $G \cap (A' \times B')$  gives  $|G \cap (A' \times B')| \gtrsim K^{-C} n^2$  after shrinking  $A', B'$  by a further constant factor.  $\square$

*Remark 10.4.5* (R10, discharged). *Theorem 10.4.2 is now proved in full.* Revision R10 recorded the Balog–Szemerédi–Gowers theorem and Freiman’s theorem as the two quoted inputs of Chapter 10.5; the first is discharged here.

*Freiman’s theorem remains quoted.* It asserts that a set of small doubling is contained in a generalised arithmetic progression of bounded rank and comparable size — a structure theorem, not an inequality, and a substantially deeper result. Nothing in this series depends on it except Theorem 10.6.1, whose omission R10 also records.

*Where this is used.* Chapter 10.5, and Volume 14, Proposition 14.14.12, in the proof of Bourgain’s projection theorem. With this chapter, *that theorem rests entirely on results proved in this series*: Marstrand and Kaufman from Volume 8, Ruzsa and Plünnecke–Ruzsa from Chapter 10.3, Blichfeldt’s principle, and Theorem 10.4.2.

#### WORKED EXAMPLE 10.4.1— TIER A **The three counting steps, side by side**

The proof is three counting arguments and nothing else. Isolating them:

(1) *Dependent random choice* (Lemma 10.4.3). Selecting a random neighbourhood biases towards large common neighbourhoods. Balanced by  $\varepsilon = c^2/4$ , and verified:  $\mathbb{E}[|A_1|^2 - 2\#\text{bad}] \geq \frac{1}{2}c^2n^2$ .

(2) *Forced overlap* (Step 4 of Lemma 10.4.4). Two subsets of  $A_2$  of relative sizes  $\geq \frac{c}{4}$  and  $\geq 1 - \frac{c}{5}$  must meet in  $\geq \frac{c}{20}$  of it. Elementary, and it is what upgrades “most pairs” to “all pairs”.

(3) *Injective encoding* (Theorem 10.4.2). Each path of length three writes  $a + b$  as  $x - y + z$  with  $x, y, z \in A +_G B$ , injectively. So many paths force  $|A' + B'|$  small.

*The shape.* A hypothesis about *many* pairs becomes one about *all* pairs by (1) and (2), and then (3) converts the graph statement into an arithmetic one. *Only step (3) knows any arithmetic*, and it occupies four lines.

*The exponent.*  $C = 8$  above, from  $\gamma \gtrsim K^{-5}$  and  $|A +_G B|^3$ . The optimal exponent is not known; what matters for every application in this series is only that it is polynomial in  $K$ .

**WORKED EXAMPLE 10.4.2— TIER B When to reach for BSG**

The theorem answers one specific difficulty, and recognising it is most of the skill.

(a) *The symptom.* One has good control of  $a + b$  for many pairs, and no control for the rest — typically because the pairs came from a pigeonhole, a popularity argument, or an incidence count.

(b) *What BSG gives.* Subsets on which the control is total, at polynomial cost. One may then apply Ruzsa and Plünnecke–Ruzsa (Chapter 10.3), which require full control.

(c) *The instance in Volume 14.* Proposition 14.14.12 has a set  $E$  that is only dense in a product  $A_1 \times A_2$ , not equal to it. BSG converts this into genuine subsets  $B_i \subseteq A_i$  with  $|B_1 + B_2|$  controlled, and the argument proceeds.

(d) *The instance in this volume.* Chapter 10.5 obtains small-doubling information only on a positive proportion of a set, and BSG supplies the subset on which Freiman’s theorem can be applied.

(e) *The cost to watch.* The subsets are smaller by  $K^{-O(1)}$ . In an iterative argument this compounds, and it is the reason Theorem 10.6.1 yields a fixed fraction of  $n$  rather than  $n - O(1)$  — Volume 10, Worked Example 10.6.2.

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## Drills

**Drill 10.4.1.** Verify the example in the Checkpoint after Definition 10.4.1: compute  $|A +_G B|$  and  $|A + B|$ .

**Drill 10.4.2.** In Lemma 10.4.3, check that the probability a bad pair lies in  $A_1^2$  equals  $|N(a) \cap N(a')|/n$ , and confirm the choice  $\varepsilon = c^2/4$ .

**Drill 10.4.3.** Verify the forced overlap in Step 4 of Lemma 10.4.4, and explain why the thresholds  $\frac{c}{4}$  and  $1 - \frac{c}{5}$  must be chosen with  $\frac{c}{4} > \frac{c}{5}$ .

**Drill 10.4.4.** Confirm the injectivity of the encoding in Theorem 10.4.2: recover  $b_1$  and  $a_1$  from the triple of summands.

## Notes

Balog and Szemerédi proved the theorem with weaker bounds in 1994; Gowers obtained polynomial dependence in 1998. The proof here follows Sudakov, Szemerédi and Vu, as presented by Tao and Vu. Remark 10.4.5 records that Freiman’s theorem, the other input of Chapter 10.5, remains quoted.

## Chapter 10.5

# The Arithmetic Reduction

The sumset inequalities of Chapter 10.3 concern subsets of an abelian group. This chapter connects them to Besicovitch sets. The bridge is a single observation: *a line meets three parallel hyperplanes in an arithmetic progression*, so a set containing lines in many directions contains many additive triples.

### 10.5.1 Slices and progressions

**Definition 10.5.1.** For  $E \subseteq \mathbb{R}^n$  and  $t \in \mathbb{R}$  write  $E_t = \{y \in \mathbb{R}^{n-1} : (y, t) \in E\}$  for the horizontal slice at height  $t$ .

**Lemma 10.5.2** (Three-slice lemma). *Let  $\ell$  be a line in  $\mathbb{R}^n$  meeting the hyperplanes  $x_n = 0$ ,  $x_n = \frac{1}{2}$ ,  $x_n = 1$ , at the points  $(a, 0)$ ,  $(b, \frac{1}{2})$ ,  $(c, 1)$ . Then*

$$a + c = 2b.$$

*Conversely, given  $a, b, c \in \mathbb{R}^{n-1}$  with  $a + c = 2b$ , the line through  $(a, 0)$  and  $(c, 1)$  passes through  $(b, \frac{1}{2})$ .*

*Proof.* A line meeting all three hyperplanes is not horizontal, so it may be written  $\{(a + tv, t) : t \in \mathbb{R}\}$  for some  $a, v \in \mathbb{R}^{n-1}$ . At  $t = 0, \frac{1}{2}, 1$  the first coordinates are  $a, a + \frac{v}{2}, a + v$ . Hence  $b = a + \frac{v}{2}$  and  $c = a + v$ , so  $a + c = 2a + v = 2b$ . The converse is the same computation with  $v = c - a$ .  $\square$

**Where we are.** This is the whole geometric input of the chapter. It says the three slices of a line, at equally spaced heights, form a three-term arithmetic progression in  $\mathbb{R}^{n-1}$  — and conversely every such progression arises from a line.

So the family of lines through a set is in bijection with the family of additive triples among its slices. A Besicovitch set has a line in every direction, hence many triples; and Chapter 10.3 says that a small set cannot have

many triples unless it is additively structured. Playing those against each other is the arithmetic method.

**Definition 10.5.3** (Additive triples). For finite  $A, B, C$  in an abelian group, let

$$T(A, B, C) = \#\{(a, b, c) \in A \times B \times C : a + c = 2b\}.$$

**Proposition 10.5.4** (A Besicovitch set has many additive triples). *Let  $K \subseteq \mathbb{R}^n$  be a Besicovitch set and  $0 < \delta < \frac{1}{2}$ . Let  $A, B, C \subseteq \delta\mathbb{Z}^{n-1}$  be the sets of  $\delta$ -cubes meeting  $K_\delta$  in the slabs  $|x_n| \leq \delta$ ,  $|x_n - \frac{1}{2}| \leq \delta$  and  $|x_n - 1| \leq \delta$  respectively. Then*

$$T'(A, B, C) := \#\{(a, b, c) \in A \times B \times C : |a + c - 2b| \leq 4\delta\} \gtrsim_n \delta^{-(n-1)},$$

while

$$|A|, |B|, |C| \lesssim_n \delta^{-n} m(K_\delta).$$

*Proof. Step 1. Count the triples.*

Let  $v$  range over a maximal  $\delta$ -separated set of directions in  $\{(w, 1) : |w| \leq 1\}$ , of which there are  $\approx_n \delta^{-(n-1)}$ . For each  $v$  the set  $K$  contains a unit segment in that direction; extending and rescaling so that it meets the three hyperplanes — which costs only a dimensional constant, since a unit segment in a direction bounded away from horizontal crosses a fixed proportion of the slab  $0 \leq x_n \leq 1$  — we obtain points  $(a_v, 0), (b_v, \frac{1}{2}), (c_v, 1)$  on the segment. By Lemma 10.5.2,  $a_v + c_v = 2b_v$  exactly.

Rounding each to the nearest  $\delta$ -cube gives  $a \in A, b \in B, c \in C$  with  $|a + c - 2b| \leq 4\delta$ . Distinct  $v$  give distinct triples: the direction is recovered as  $c - a$  up to  $O(\delta)$ , and the  $v$  are  $\delta$ -separated. So there are  $\gtrsim_n \delta^{-(n-1)}$  triples.

**Step 2. Bound the sizes of the slices.**

Each  $\delta$ -cube counted in  $A$  contributes a set of measure  $\approx \delta^n$  to  $K_\delta$  — the cube thickened in the  $x_n$  direction by  $\delta$  — and distinct cubes contribute disjointly up to bounded overlap. Hence  $|A| \delta^n \lesssim_n m(K_\delta)$ , that is  $|A| \lesssim_n \delta^{-n} m(K_\delta)$ . The same holds for  $B$  and  $C$ .  $\square$

## 10.5.2 What the triples buy

**Proposition 10.5.5** (The trivial consequence). *With  $A, B, C$  as above and  $N = \max\{|A|, |B|, |C|\}$ ,*

$$T'(A, B, C) \lesssim_n N^2.$$

*Consequently  $\delta^{-(n-1)} \lesssim_n N^2$ , and*

$$m(K_\delta) \gtrsim_n \delta^n \delta^{-(n-1)/2} = \delta^{(n+1)/2}.$$

*Proof.* A triple is determined by  $a$  and  $b$ , since  $c = 2b - a$  up to the  $O(\delta)$  rounding, which permits boundedly many choices of  $c$ . So  $T' \lesssim_n |A||B| \leq N^2$ .

Combining with Proposition 10.5.4 Step 1,  $\delta^{-(n-1)} \lesssim N^2$ , so  $N \gtrsim \delta^{-(n-1)/2}$ ; and by Proposition 10.5.4,  $m(K_\delta) \gtrsim \delta^n N \gtrsim \delta^{n-(n-1)/2} = \delta^{(n+1)/2}$ .  $\square$

**Where we are.** Proposition 10.5.5 gives  $m(K_\delta) \gtrsim \delta^{(n+1)/2}$ , which is *weaker* than the bush bound  $\delta^{(n-1)/2}$  of Volume 9, Theorem 9.4.4, by a factor  $\delta$ . So the arithmetic reduction, used with only the trivial count  $T' \leq N^2$ , is worse than the geometry it replaces.

That is exactly as it should be. The trivial count throws away all the structure: it uses only that a triple is determined by two of its entries. The content of the method is that  $T'$  being *large* — comparable to  $N^2$  — forces  $A, B, C$  to be additively structured, and structured sets cannot arise from Besicovitch configurations. Extracting that is the subject of Chapter 10.6, and it is where the sumset inequalities enter.

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**WORKED EXAMPLE 10.5.1— TIER B Where the additive structure would come from**

The reduction produces an *energy* statement: many additive triples. Turning it into a *doubling* statement, to which Theorem 10.3.5 applies, is the technical heart, and it is worth seeing the shape.

*Step 1: energy to doubling, via Balog–Szemerédi–Gowers.* If  $T(A, B, C) \geq \eta|A||B|$  then there are subsets  $A' \subseteq A$ ,  $C' \subseteq C$  with  $|A'| \gtrsim \eta^{O(1)}|A|$  and  $|A' + C'| \lesssim \eta^{-O(1)}|A'|$ . So a large-energy pair contains a small-doubling pair, at the cost of polynomial losses in  $\eta$ . This theorem is *not* proved in this series; Worked Example 10.3.2 recorded that.

*Step 2: doubling to structure, via Plünnecke.* Theorem 10.3.5 then bounds all iterated sumsets of  $A'$ , and Freiman’s theorem — also not proved here — says  $A'$  is contained in a generalised arithmetic progression of bounded rank and comparable size.

*Step 3: structure contradicts the geometry.* A Besicovitch configuration whose slices lie in low-rank progressions can be analysed directly, and one shows the directions cannot be  $\delta$ -separated in the required number. This is where the dimension bound comes out.

*Why the losses are severe.* Step 1 costs a power of  $\eta$ , and  $\eta$  is itself a power of  $\delta$ ; so the exponent obtained is worse than the naive count suggests. That is the reason the arithmetic bounds, though better than  $\frac{n+2}{2}$  for large  $n$ , are far from  $n$  — and why they are stated with unattractive constants like  $\frac{13}{25}$ .

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**Proposition 10.5.6** (Slices and the maximal function). *Let  $K \subseteq \mathbb{R}^n$  be a Besicovitch set and  $0 < \delta < \frac{1}{2}$ . Then for every  $t \in [0, 1]$ ,*

$$m_{n-1}((K_\delta)_t) \lesssim_n \delta^{-1} m_n(K_\delta),$$

*and there is a set of  $t$  of measure  $\geq \frac{1}{2}$  on which*

$$m_{n-1}((K_\delta)_t) \gtrsim_n m_n(K_\delta).$$

*Proof.* For the first bound, note  $(K_\delta)_t \times [t - \delta, t + \delta] \subseteq K_{2\delta}$  up to a dimensional constant, since a point within  $\delta$  of  $K$  at height  $t$  remains within  $2\delta$  of  $K$  at nearby heights. Hence  $2\delta m_{n-1}((K_\delta)_t) \leq m_n(K_{2\delta}) \lesssim_n m_n(K_\delta)$ , the last step because doubling the neighbourhood increases its measure by at most a dimensional factor.

For the second, Fubini gives  $\int_0^1 m_{n-1}((K_\delta)_t) dt \leq m_n(K_\delta)$ , so the average slice has measure at most  $m_n(K_\delta)$ ; but  $K$  contains segments crossing every height, so every slice is non-empty and, by the first bound applied in reverse on a large set of  $t$ , comparable to the average. Chebyshev on the complement gives the stated set of  $t$ .  $\square$

**Where we are.** The proposition says the slices are all comparable, up to a factor  $\delta^{-1}$ , to the measure of the whole neighbourhood. That is what licenses Proposition 10.5.4's bound on  $|A|, |B|, |C|$ : the three particular heights  $0, \frac{1}{2}, 1$  are not special, and any three would do.

It also shows where the reduction is wasteful. The Besicovitch set is a three-dimensional object of information — position, direction, and scale — and the reduction retains only three slices. Everything about how the segments are arranged *between* the slices is discarded, which is why Worked Example 10.5.2 found the planar bound so poor.

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**WORKED EXAMPLE 10.5.2— TIER A The reduction in the plane, drawn out**

For  $n = 2$  the three-slice picture is concrete and worth following.

*The slices.*  $K \subseteq \mathbb{R}^2$  and the three horizontal lines  $y = 0, \frac{1}{2}, 1$ . Each slice  $K_t$  is a subset of  $\mathbb{R}$ .

*The lemma.* A line of slope  $1/v$  meets them at  $a, a + \frac{v}{2}, a + v$  — a three-term arithmetic progression in  $\mathbb{R}$  with common difference  $\frac{v}{2}$ .

*The Besicovitch hypothesis.*  $K$  contains a segment in every direction, so for every  $v$  in a range there is  $a$  with  $a \in K_0, a + \frac{v}{2} \in K_{1/2}, a + v \in K_1$ . Discretising at scale  $\delta$ , that is  $\approx \delta^{-1}$  progressions.



*The count.* If  $m(K_\delta) = \delta^\beta$  then each slice has  $\lesssim \delta^{\beta-2}$  cubes, by Proposition 10.5.4. A progression is determined by its first two terms, so

$$\delta^{-1} \lesssim (\delta^{\beta-2})^2 = \delta^{2\beta-4},$$

giving  $-1 \geq 2\beta - 4$ , that is  $\beta \leq \frac{3}{2}$ . So  $m(K_\delta) \gtrsim \delta^{3/2}$  and  $\overline{\dim}_M K \geq 2 - \frac{3}{2} = \frac{1}{2}$ .

*Which is terrible.* The truth in the plane is 2, and Córdoba's  $L^2$  argument gives it. The arithmetic reduction with the trivial count gives  $\frac{1}{2}$  — worse even than the bush.

*The point.* The reduction discards the geometry entirely: it retains only that certain triples exist, not where they sit. That is a genuine loss in low dimensions, and it is recovered only in high dimensions, where the additive structure forced by *many* triples becomes restrictive. Worked Example 10.6.1 shows the crossover is around  $n = 25$ .

**WORKED EXAMPLE 10.5.3— TIER C: RESEARCH TASTE    What the reduction discards, and whether that is recoverable**

Worked Example 10.5.2 found that the arithmetic reduction gives  $\frac{1}{2}$  in the plane where the truth is 2. That is a large loss and it is worth locating.

(a) *It keeps only three slices.* A Besicovitch set is determined by its segments; the reduction records where they meet three hyperplanes and forgets everything else. Proposition 10.5.6 shows the slices are all comparable, so no single slice is special — but the *joint* information across all heights is discarded.

(b) *It keeps only the existence of triples, not their location.*  $T'(A, B, C) \gtrsim \delta^{-(n-1)}$  says many progressions occur; it says nothing about which. Two Besicovitch sets with identical triple-counts but different geometry are indistinguishable to the method.

(c) *It discards the separation of directions.* The count uses that  $\delta$ -separated directions give distinct triples, but afterwards the directions play no role. In Volume 9 the separation was the whole content of Lemma 9.4.2.

(d) *Is the loss recoverable?* Partly. Bourgain and Katz–Tao recover enough to beat  $\frac{n+2}{2}$  asymptotically by exploiting that *many* triples force structure — so (b) is partially undone. Nothing recovers (a) or (c), which is why the method is weak in low dimensions where the geometry is rich.

(e) *The general shape.* A reduction that maps a rich structure to a poorer one trades information for tractability, and the trade is favourable only where the poorer structure is still constraining. Additive structure constrains strongly when the ambient dimension is large — there is more room for triples to

force relations — and weakly when it is small. That is exactly the crossover Worked Example 10.6.1 located around  $n = 25$ .

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**WORKED EXAMPLE 10.5.4— TIER B Discretisation: turning a geometric statement into an arithmetic one**

Proposition 10.5.4 passes from a continuous Besicovitch set to finite subsets of  $\delta\mathbb{Z}^{n-1}$ . That passage is routine and error-prone, and the four checks worth making are these.

(1) *What is the ambient group?* Here  $\delta\mathbb{Z}^{n-1}$ , isomorphic to  $\mathbb{Z}^{n-1}$  after rescaling. The sumset inequalities of Chapter 10.3 are stated for abelian groups and apply directly. Had the discretisation produced a set in  $\mathbb{R}^{n-1}$  with no group structure, they would not.

(2) *Are the relations exact or approximate?* Lemma 10.5.2 gives  $a + c = 2b$  exactly for the continuous points; rounding to  $\delta$ -cubes makes it  $|a + c - 2b| \leq 4\delta$ . So the arithmetic statement is about *approximate* triples, and every inequality applied to them must tolerate that — which is why  $T'$  is defined with the tolerance built in.

(3) *How much does the tolerance cost?* An approximate relation  $|a + c - 2b| \leq C\delta$  permits  $O_n(1)$  choices of  $c$  given  $a, b$ , so counts change by a dimensional constant only. That is the reason the reduction survives discretisation at all; had the tolerance permitted  $\delta^{-1}$  choices, the count of Proposition 10.5.5 would be vacuous.

(4) *Does the conclusion transfer back?* The arithmetic conclusion bounds  $|A|$ , and Proposition 10.5.4 relates  $|A|$  to  $m(K_\delta)$ . Both directions of that relation must be available, and the second — from a bound on the number of cubes to a bound on the measure — uses that the contributing cubes are essentially disjoint.

*The general caution.* A discretisation that loses a power of  $\delta$  at any of the four steps destroys the argument, since the quantities being compared differ by powers of  $\delta$ . Checking all four before trusting a discretised argument is quick and usually decisive.

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**WORKED EXAMPLE 10.5.5— TIER A Three-term progressions, and why three**

Lemma 10.5.2 uses three hyperplanes. It is worth asking why not two, or four.

*Two is not enough.* Two slices at heights 0 and 1 give points  $a$  and  $c$  on a line, with  $c - a$  the direction. Every pair  $(a, c)$  arises from some line, so the

condition is vacuous: there is no relation to exploit, and the count gives only  $|A||C| \geq \#\text{directions}$ , which is Proposition 10.3.2 and says nothing.

*Three is exactly enough.* The third slice is determined by the first two:  $b = \frac{a+c}{2}$ . So the triple  $(a, b, c)$  satisfies one linear relation, and requiring  $b$  to lie in a *small* set is a genuine constraint. That is the first non-vacuous case, and it is why the reduction produces additive triples rather than pairs.

*Four is no better.* Slices at  $0, \frac{1}{3}, \frac{2}{3}, 1$  give  $a, b, c, d$  with  $b = \frac{2a+d}{3}$  and  $c = \frac{a+2d}{3}$  — two relations, but both determined by the same pair  $(a, d)$ . No new information: the additional constraint is implied. One could use unequally spaced slices to get genuinely new relations, and the arithmetic literature does, but the gain is in constants rather than in kind.

*The general principle.* A line in  $\mathbb{R}^n$  has  $2(n-1)$  degrees of freedom — position and direction — and each slice records  $n-1$  numbers. Two slices determine the line; every further slice is a function of them. So three is the smallest number giving a relation, and the relation is necessarily *one* equation, which is why the problem becomes additive rather than multiplicative.

*The connection to Volume 9.* Worked Example 9.3.3 there classified methods by the order of correlation they see. Two slices is a two-point method, and Proposition 10.5.5 confirms it gives nothing. Three slices is a three-point method, which is why the arithmetic route can in principle beat the  $L^2$  ceiling — and Theorem 10.6.1 confirms it does.

## Drills

**Drill 10.5.1.** Verify Lemma 10.5.2 for a line in  $\mathbb{R}^3$  given parametrically, and check the converse.

**Drill 10.5.2.** Confirm the counting in Step 1 of Proposition 10.5.4: distinct  $\delta$ -separated directions give distinct triples.

**Drill 10.5.3.** Derive the bound  $|A| \lesssim_n \delta^{-n} m(K_\delta)$  of Proposition 10.5.4 from the disjointness of the contributing cubes.

**Drill 10.5.4.** Check that Proposition 10.5.5 gives  $\delta^{(n+1)/2}$  and compare with the bush bound  $\delta^{(n-1)/2}$  of Volume 9, Theorem 9.4.4.

## Notes

The reduction of Kakeya to an arithmetic problem is Bourgain's (1999), refined by Katz and Tao. The three-slice observation is elementary and old; what is

not elementary is the passage from many triples to additive structure, which requires the Balog–Szemerédi–Gowers theorem.

## Chapter 10.6

# The Arithmetic Bounds

Chapter 10.5 reduced the Kakeya problem to a question about additive triples, and Proposition 10.5.5 showed that the reduction used naively gives less than the geometry of Volume 9. This chapter records what the reduction gives when the additive structure is exploited properly, states the resulting bounds, and is explicit about which parts are proved here.

### 10.6.1 The statements

**Theorem 10.6.1** (Bourgain; Katz–Tao). *There is  $c > \frac{1}{2}$  such that every Besicovitch set  $K \subseteq \mathbb{R}^n$  satisfies*

$$\dim_{\mathrm{H}} K \geq cn - O(1).$$

*Bourgain’s argument gives  $c = \frac{13}{25}$ ; the refinement of Katz and Tao gives  $c = 2 - \sqrt{2} \approx 0.5858$ .*

*Remark 10.6.2* (Scope decision). *Theorem 10.6.1 is not proved in this series.* This is revision R10, and the reason is that the argument needs two theorems of additive combinatorics which the series does not develop.

*What is proved here.* The geometric reduction — Lemma 10.5.2 and Proposition 10.5.4, which convert a Besicovitch set into a configuration with  $\gtrsim \delta^{-(n-1)}$  additive triples among sets of size  $\lesssim \delta^{-n}m(K_\delta)$  — together with the sumset machinery of Chapter 10.3: Ruzsa’s triangle inequality and the Plünnecke–Ruzsa inequality, both with complete proofs.

*What is missing.* The passage from *many triples* to *additive structure*, which is the Balog–Szemerédi–Gowers theorem, and the identification of small-doubling sets with generalised arithmetic progressions, which is Freiman’s theorem. Worked Example 10.5.1 set out where each is used. Both are substantial results with proofs longer than this chapter, and neither is needed anywhere else in the series.

*Why the omission is safe.* Nothing later depends on Theorem 10.6.1. Volume 11’s polynomial method supersedes it in every dimension where both apply, and the current records reported in Volume 14 come from elsewhere entirely. The theorem is included because it was, for several years, the only bound exceeding  $\frac{n+2}{2}$  for large  $n$ , and because the reduction that produces it is elementary and worth having.

*Update.* Of the two inputs, Balog–Szemerédi–Gowers is now proved: Theorem 10.4.2 of Chapter 10.4. Freiman’s theorem remains quoted, and it is the only additive-combinatorial result this series does not establish.

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WORKED EXAMPLE 10.6.1— TIER A **The bounds compared**

| $n$ | Vol 9 best | Bourgain $\frac{13}{25}n$ | Katz–Tao $(2 - \sqrt{2})n$ |
|-----|------------|---------------------------|----------------------------|
| 3   | 2.5        | —                         | —                          |
| 5   | 3.5        | 2.6                       | 2.9                        |
| 10  | 6          | 5.2                       | 5.9                        |
| 25  | 13.5       | 13                        | 14.6                       |
| 50  | 26         | 26                        | 29.3                       |
| 100 | 51         | 52                        | 58.6                       |

(The Volume 9 column is  $\frac{n+2}{2}$ , Wolff’s bound, which Remark 9.5.5 proves only in the transverse case; the constants  $O(1)$  in Theorem 10.6.1 are suppressed.)

*Reading the table.* The arithmetic bounds beat  $\frac{n+2}{2}$  only for large  $n$  — around  $n = 25$  for Bourgain’s constant, earlier for Katz–Tao’s. In low dimensions the geometry wins, which is why Chapter 10.5’s reduction is not the method of choice for  $n = 3$  or 4.

*The significance.* The constant  $c$  exceeds  $\frac{1}{2}$ , so asymptotically these are the first bounds establishing that a Besicovitch set has dimension *more* than half the ambient. Every method in Volume 9 gives exactly  $\frac{n}{2} + O(1)$ , and the barrier at one half was the state of the art for a decade.

*What would be needed for  $n - o(n)$ .* A constant  $c$  arbitrarily close to 1. No known method achieves this, and Worked Example 9.8.4 in Volume 9 gives the reason: the arithmetic method, like the geometric ones, optimises over a parameter whose extremal value is not attained.

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WORKED EXAMPLE 10.6.2— TIER B **Tracing the losses**

It is instructive to see why  $c$  comes out near  $\frac{1}{2}$  rather than near 1, since the arithmetic is transparent even though the theorem is not proved.

*Start.* Proposition 10.5.4 gives  $T' \gtrsim \delta^{-(n-1)}$  triples among sets of size  $N \lesssim \delta^{-n} m(K_\delta)$ . Write  $m(K_\delta) = \delta^\beta$ , so  $N \approx \delta^{\beta-n}$  and the target is to force  $\beta$  small.

*The energy density.*  $T'/N^2 \approx \delta^{-(n-1)} \delta^{-2(\beta-n)} = \delta^{n+1-2\beta}$ . For the Balog–Szemerédi–Gowers theorem to be useful this must not be too small, that is  $\beta$  must not be much below  $\frac{n+1}{2}$  — which is exactly the regime the trivial count of Proposition 10.5.5 already reaches.

*The cost of BSG.* The theorem returns subsets of relative size  $\eta^{O(1)}$  with doubling  $\eta^{-O(1)}$ , where  $\eta = T'/N^2$ . Both are powers of  $\delta$ , and the exponents in the  $O(1)$  are not small — in the standard proof they are around 4 or 5.

*The cost of Plünnecke.* Theorem 10.3.5 gives  $|kB - lB| \leq K^{k+l}|A|$ , exponential in  $k+l$ . The argument uses  $k+l$  growing with  $n$ , so this too costs a power.

*The upshot.* Each step multiplies the exponent by a factor slightly less than 1, and  $c = 2 - \sqrt{2}$  is the value at which the multiplicative losses balance — Katz and Tao obtain it by optimising a recursive inequality. The constant is not natural; it is what the arithmetic yields.

*The general lesson.* A chain of powerful but lossy tools gives a bound determined by the losses rather than by the problem. That is a signature of a method being pushed past its natural range, and it is the same diagnosis Worked Example 9.8.2 gave for the geometric arguments.

**WORKED EXAMPLE 10.6.3 — TIER C: RESEARCH TASTE** **What the arithmetic method contributed**

(a) *It broke the one-half barrier.* Before 1999 every bound was  $\frac{n}{2} + O(1)$ . Bourgain’s was the first with a constant exceeding  $\frac{1}{2}$ , and that was its point — not the numerical value but the demonstration that the barrier was not intrinsic.

(b) *It identified the right auxiliary problem.* The reduction of Chapter 10.5 showed that Kakeya is, at bottom, a question about sets with many three-term arithmetic progressions. That reformulation is due to Bourgain and has been reused since — in the work on the Furstenberg problem reported in Volume 14, and in the sum-product literature.

(c) *It did not scale.* The constants degrade with each additional tool, and no refinement has pushed  $c$  past  $2 - \sqrt{2}$ . The method is, in the terminology of Worked Example 9.8.4, at its ceiling.

(d) *It was superseded, but not made irrelevant.* Volume 11's polynomial method gives better bounds in low dimensions and, in the finite-field setting, the sharp answer. But the arithmetic reduction remains the only route that sees the *additive* structure of a Besicovitch set, and problems where that structure is the point — the Furstenberg set problem, and the sum-product phenomenon — still use it.

(e) *A note on what this volume has done.* It has proved two things completely: Dvir's theorem, which settles the finite field problem, and the sumset inequalities, which are the arithmetic engine. It has proved the geometric reduction connecting Kakeya to additive triples. And it has been explicit that the remaining step — from triples to structure — is not carried out here. Given that nothing later depends on it, that seems the right division.

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WORKED EXAMPLE 10.6.4— TIER C: RESEARCH TASTE **Announced omissions: the discipline this series follows**

Remark 10.6.2 is the tenth announced omission in this series. The practice is uniform and worth stating once in full, since a reader encountering R10 without context may wonder what it licenses.

(a) *The rule.* Every stated result is either proved, or its omission is announced at the point where it is stated, with three things recorded: what exactly is missing, why it is missing, and what depends on it.

(b) *Removal versus deferral.* A *deferral* names the volume that supplies the argument and is a promise — Volume 3's configuration lemma was deferred to Volume 9 and Theorem 9.2.1 discharged it. A *removal* states that no volume will supply it. Volume 4, Remark 4.4.5 sets out why inventing a home for an omitted result would be a fiction, and R10 is a removal on those grounds.

(c) *The ten.* R1–R3 removed function-space results from Volume 4; R4 removed Stein interpolation and the sharp  $L^p$  theory of oscillatory operators from Volume 5; R5 removed the implication Bochner–Riesz  $\Rightarrow$  restriction; R6 removed the Besicovitch covering theorem; R7 removed Drury's Fourier bound, already available by the bush; R8 removed the measure-zero Besicovitch set; R9 removed Wolff's theorem in full generality, the transverse case being proved; and R10 removes the arithmetic bounds.

(d) *What they have in common.* Each is a result that would require machinery used nowhere else — Stein interpolation, rectifiability theory, Balog–Szemerédi–Gowers — and on which nothing later depends. None is a result the series uses.



(e) *The one that was not an omission.* Lemma 3.8.4 was used without proof for six volumes, and that was a genuine debt, not a scope decision. It is recorded differently in the manifest for exactly that reason, and its discharge in Volume 9 is the only entry marked as paid.

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WORKED EXAMPLE 10.6.5— TIER A **The recursive inequality behind  $2 - \sqrt{2}$**

The constant  $2 - \sqrt{2}$  looks arbitrary. It is the fixed point of a recursion, and seeing the shape — without proving the theorem — makes it intelligible.

*The setup.* Katz and Tao track a quantity measuring how much additive structure a hypothetical small Besicovitch set must have, and derive an inequality of the form

$$\alpha \geq f(\alpha)$$

where  $\alpha$  is the exponent to be bounded and  $f$  arises from one round of the reduction: apply Lemma 10.5.2, extract structure, and re-apply.

*The fixed point.* For the particular  $f$  they obtain, the inequality is equivalent to  $\alpha^2 - 4\alpha + 2 \leq 0$ , whose roots are  $2 \pm \sqrt{2}$ . The relevant root is  $2 - \sqrt{2} \approx 0.5858$ , and the bound is  $\dim_{\text{H}} K \geq (2 - \sqrt{2})n - O(1)$ .

*Why a quadratic.* Because the recursion applies the reduction twice before closing: each application converts a bound on the set into a bound on its slices, and two applications return to the original type of statement with the exponent transformed. A single application would give a linear inequality and a worse constant; three would not improve it further, the recursion having already reached its fixed point.

*Sanity check.*  $2 - \sqrt{2} > \frac{1}{2}$  ✓ — the whole point — and  $2 - \sqrt{2} < 1$ , so the method does not reach the conjecture. Both are as expected.

*The transferable observation.* When a bound is an algebraic irrationality, look for a fixed-point equation. The number is then not arbitrary but the solution of the recursion the method generates, and its distance from the conjectured value measures the loss per round.

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WORKED EXAMPLE 10.6.6— TIER B **What a proof of Theorem 10.6.1 would look like**

The theorem is stated and not proved, so a reader may reasonably want the shape of the argument, if only to judge the size of the omission.

*Step 1 — geometry to triples.* Proposition 10.5.4: a Besicovitch set with  $m(K_\delta) = \delta^\beta$  yields sets  $A, B, C$  of size  $\lesssim \delta^{\beta-n}$  with  $\gtrsim \delta^{-(n-1)}$  approximate

additive triples. *Proved here.*

*Step 2 — triples to energy.* Rewrite the triple count as an additive energy:  $E(A, C) \gtrsim T'^2/|B|$ , by Cauchy–Schwarz over the fibres. This is routine and could be included; it is omitted only because Step 3 is not.

*Step 3 — energy to doubling.* Balog–Szemerédi–Gowers: large energy yields large subsets  $A' \subseteq A$ ,  $C' \subseteq C$  with  $|A' + C'| \lesssim \eta^{-O(1)}|A'|$ . *Not proved here*; this is the substantial omission.

*Step 4 — doubling to iterated sumsets.* Theorem 10.3.5 bounds  $|kA' - lA'|$  by  $K^{k+l}|A'|$ . *Proved here.*

*Step 5 — structure to a contradiction.* The iterated sumsets are constrained by the geometry, since they too must be realised by slices of  $K$ ; comparing gives an inequality in  $\beta$ , and optimising the number of iterations gives the recursion of Worked Example 10.6.5.

*The accounting.* Of five steps, three are proved here, one is routine, and one is a theorem of another subject. So the omission is a single step, and Remark 10.6.2 names it precisely. A reader wanting the full argument needs Balog–Szemerédi–Gowers and nothing else from outside this series.

*Why that matters for the reader.* An omission of one named theorem is a different thing from an omission of an argument, and the distinction is worth preserving in any treatment that leaves gaps. The manifest records which of the ten revisions are of each kind.

**WORKED EXAMPLE 10.6.7— TIER C: RESEARCH TASTE   Why the arithmetic route was abandoned**

Between 1999 and 2008 the arithmetic method gave the best asymptotic bounds. It is now little used, and the reasons are worth recording.

(a) *Its ceiling was reached quickly.* Bourgain’s  $\frac{13}{25}$  became Katz–Tao’s  $2 - \sqrt{2}$  within three years, and nothing has improved it since. By Worked Example 10.6.5 the constant is a fixed point, so incremental refinement of the individual steps cannot move it — only a change in the recursion would.

(b) *The losses are structural, not technical.* Worked Example 10.6.2 attributes them to Balog–Szemerédi–Gowers and to the exponential dependence in Theorem 10.3.5. Both are sharp as stated, so there is no slack to recover.

(c) *The polynomial method arrived.* Volume 11’s partitioning gives better bounds in low dimensions and a genuinely different mechanism. Where two methods compete, effort follows the one with room to grow.

(d) *But the reduction survives.* Lemma 10.5.2 and Proposition 10.5.4 are still the standard way of extracting additive information from a geometric configuration, and they reappear — in the Furstenberg set problem, in sum-product estimates, and in the discretised incidence bounds of Volume 14. The method was superseded; the reduction was not.

(e) *The lesson about methods.* A technique may be abandoned because its ceiling is low without its component ideas being abandoned. Distinguishing the two is worth doing before concluding that a line of work is exhausted — and it is why this chapter proves the reduction in full while omitting the theorem it was built for.

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## Drills

**Drill 10.6.1.** Using Worked Example 10.6.1, find the smallest  $n$  for which  $(2 - \sqrt{2})n$  exceeds  $\frac{n+2}{2}$ .

**Drill 10.6.2.** Verify the energy density computation  $T'/N^2 \approx \delta^{n+1-2\beta}$  in Worked Example 10.6.2.

**Drill 10.6.3.** Explain why Remark 10.6.2 classifies the omission as a removal rather than a deferral, using the criterion of Volume 4, Remark 4.4.5.

**Drill 10.6.4.** State the Balog–Szemerédi–Gowers theorem precisely and identify the two places Worked Example 10.5.1 would use it.

**Where we are.** A closing remark on what this chapter records. It states a theorem it does not prove, which the series does sparingly and always with the three disclosures of Worked Example 10.6.4: what is missing, why, and what depends on it.

Here the missing input is Balog–Szemerédi–Gowers, the reason is that it is a substantial theorem used nowhere else, and the answer to the third is nothing. What *is* proved — the three-slice reduction and the sumset inequalities — is the part a reader is likely to reuse, since the reduction reappears in the Furstenberg problem and the inequalities are standard equipment.

**Where we are.** The five-step anatomy of Worked Example 10.6.6 is worth keeping: three steps proved here, one routine, one imported. An omission of that shape is a citation, not a gap.

## Notes

Bourgain's arithmetic bound is from 1999; the constant  $2 - \sqrt{2}$  is due to Katz and Tao (2002), who also introduced the recursive optimisation that produces it. The Balog–Szemerédi–Gowers theorem dates from 1994 in Balog and Szemerédi's form and 1998 in Gowers's quantitative version; Freiman's theorem is from 1966, with Ruzsa's proof from 1994.

None of these is proved here; Remark 10.6.2 records the decision.

## Chapter 10.7

# Why the Finite Field Argument Does Not Transfer

Dvir's proof is half a page and settles the Kakeya problem over  $\mathbb{F}_q$  completely. The real problem has been open since 1928. This chapter explains the gap precisely, because the explanation is genuinely informative — it identifies where the difficulty of the real problem lives.

### 10.7.1 The naive attempt, and where it fails

Suppose one tries to run Theorem 10.2.1 over  $\mathbb{R}$ . There are three steps to transfer.

**Proposition 10.7.1** (The parameter count survives). *Let  $K \subseteq [0, 1]^n$  be a Besicovitch set and  $0 < \delta < \frac{1}{2}$ . Let  $S$  be a maximal  $\delta$ -separated subset of  $K$ , so that*

$$|S| \approx_n \frac{m(K_\delta)}{\delta^n}.$$

*Write  $m(K_\delta) = \delta^\beta$  with  $0 \leq \beta \leq n$ . Then there is a non-zero polynomial of degree*

$$d \lesssim_n |S|^{1/n} \approx \delta^{(\beta-n)/n}$$

*vanishing on  $S$ ; and the number of points of  $S$  on any single  $\delta$ -tube is  $\approx \delta^{-1}$ , which exceeds  $d$  whenever  $\beta > 0$ .*

*Proof.* The dimension count of Worked Example 10.1.2 gives  $\dim \mathcal{P}_d^n = \binom{d+n}{n} \geq \frac{d^n}{n!}$ , so a non-zero polynomial vanishing on  $S$  exists as soon as  $\frac{d^n}{n!} > |S|$ , that is  $d \gtrsim_n |S|^{1/n}$ . The estimate  $|S| \approx m(K_\delta)\delta^{-n}$  is the standard comparison between a  $\delta$ -net and the measure of the  $\delta$ -neighbourhood.

A  $\delta$ -tube has length 1 and the points of  $S$  inside it are  $\delta$ -separated, so there are  $\lesssim \delta^{-1}$  of them, and  $\approx \delta^{-1}$  if  $K$  contains the corresponding segment. Finally

$$d \approx \delta^{(\beta-n)/n} < \delta^{-1} \iff \frac{\beta-n}{n} > -1 \iff \beta > 0. \quad \square$$

**Where we are.** So Steps A and the arithmetic of Step B of Worked Example 10.2.2 both survive: a polynomial of low degree exists, and each tube carries more points of  $S$  than the degree. Over  $\mathbb{F}_q$  that was exactly the configuration that produced the contradiction.

The transfer nevertheless fails, and it fails at the one step whose finite-field version is trivial.

**Proposition 10.7.2** (The propagation step fails). *Let  $\ell$  be a line and  $T$  the  $\delta$ -tube around it. There exist non-zero polynomials  $P$  of degree 2 vanishing at  $\approx \delta^{-1}$   $\delta$ -separated points of  $T$  but not vanishing at any point of  $\ell$ .*

*Proof.* Take  $n = 2$ ,  $\ell$  the  $x$ -axis,  $T = \{(x, y) : 0 \leq x \leq 1, |y| \leq \delta\}$ . Let  $P(x, y) = y - \frac{\delta}{2}$ , of degree 1, which vanishes on the horizontal line  $y = \frac{\delta}{2}$ . That line lies inside  $T$  and contains  $\approx \delta^{-1}$   $\delta$ -separated points of  $T$ . But  $P$  vanishes nowhere on  $\ell$ , where  $y = 0$  and  $P = -\frac{\delta}{2} \neq 0$ .

For a degree-2 example not vanishing on any line, take  $P(x, y) = (y - \frac{\delta}{2})^2 + \varepsilon x^2$  with  $\varepsilon$  small; it vanishes only at finitely many points, so instead perturb:  $P(x, y) = y - \frac{\delta}{2} - \varepsilon x^2$  vanishes on a parabola lying in  $T$  for  $\varepsilon < \frac{\delta}{2}$ , which again contains  $\approx \delta^{-1}$  separated points of  $T$  and misses  $\ell$  entirely.  $\square$

**Where we are.** This is the whole obstruction, and it is worth stating without hedging.

Over  $\mathbb{F}_q$ , a polynomial of degree  $d < q$  vanishing at  $q$  points of a line vanishes on the line *as a polynomial in the line's parameter* — because a polynomial in one variable with more roots than its degree is zero. The conclusion is algebraic and exact.

Over  $\mathbb{R}$ , the points of  $S$  inside a tube are *not collinear*. They lie within  $\delta$  of a line, and a polynomial can vanish on a curve threading through them — as Proposition 10.7.2 exhibits — without vanishing anywhere on the line itself. There is no one-variable polynomial to which the root count applies. So the failure is not a matter of constants or of technique. *A tube is not a line*, and the algebraic step that makes Dvir's proof work has no approximate version.

## 10.7.2 What survives

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WORKED EXAMPLE 10.7.1— TIER B **The three repairs, and what each costs**

Three ways of adapting the polynomial method to  $\mathbb{R}^n$  have been tried. Two work and one does not.

*Repair 1: shrink the tubes to lines.* Replace  $K$  by the union of the actual segments, a set of measure zero, and ask for a polynomial vanishing on it. Now Step B works — the points on a segment *are* collinear. But Step A fails: the set is infinite, so no finite-degree polynomial vanishes on it unless it vanishes identically. *This repair is empty.*

*Repair 2: polynomial partitioning.* Do not ask the polynomial to vanish on the set. Ask it to *divide space* into cells each containing a controlled fraction of the set, then induct on the cells. A line meets the zero set of a degree- $d$  polynomial in at most  $d$  points unless it lies inside, so each line visits at most  $d + 1$  cells — a statement about lines that survives discretisation, because it counts *crossings* rather than requiring exact vanishing. *This is the repair that works*, and Volume 11 develops it.

*Repair 3: vanish to high order.* Require the polynomial to vanish to order  $m$  at each point, which strengthens the propagation at the cost of raising the degree. Over  $\mathbb{F}_q$  this is Dvir’s method of multiplicities, and it improves the constant to the sharp  $q^n$ . Over  $\mathbb{R}$  it does not address Proposition 10.7.2 at all, since the obstruction is the non-collinearity of the points and not the order of vanishing. *This repair helps only in the finite field setting.*

*The pattern.* What transfers is the statement “a line meets a degree- $d$  variety in  $\leq d$  points”, which is about *incidences* and is stable. What does not transfer is “a polynomial vanishing at enough points of a line vanishes on it”, which is about *exact algebraic identities* and is not. Volume 11 is built entirely on the first.

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WORKED EXAMPLE 10.7.2— TIER C: RESEARCH TASTE **What this says about the real Kakeya problem**

The finite field model was proposed to isolate the difficulty. Its resolution tells us where the difficulty is *not*, and by elimination, where it is.

(a) *Not in the combinatorics of lines.* Over  $\mathbb{F}_q$  the incidence structure of lines is exactly as rich as over  $\mathbb{R}$  — two points determine a line, two lines meet in at most a point, there are  $\approx q^{n-1}$  directions. And the problem is easy. So the combinatorics alone does not make Kakeya hard.

(b) *Not in the dimension count.* Proposition 10.7.1 shows the parameter count transfers exactly: a Besicovitch set of any positive measure exponent admits a polynomial of degree below the number of points per tube.

(c) *In the width of the tubes.* What is left is Proposition 10.7.2: the points lie *near* a line rather than on one. Everything the real problem has that the model lacks is contained in that  $\delta$ .

(d) *Which is consistent with everything else in the series.* Volume 9's Remark 9.5.5 descoped Wolff's theorem precisely at the near-parallel tubes, where widths interact. Volume 6's Knapp example is a statement about the *thickness* of a cap's dual box. Volume 14's account of Wang and Zahl centres on sticky configurations, which are exactly the ones where tubes of positive width nest inside one another. Three independent lines of evidence pointing at the same place.

(e) *The methodological moral.* A model problem is informative when it is solved, whichever way it comes out. Had the finite field problem been hard, one would have learned that the combinatorics is the obstruction. It was easy, so one learns that the metric is. Either outcome advances understanding; what would not have advanced it is leaving the model unresolved.

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WORKED EXAMPLE 10.7.3— TIER B **Testing whether an argument transfers**

Chapter 10.7 diagnosed one specific failure. The diagnosis generalises into a test, worth applying to any finite-field argument offered as a model for the real case.

*The test.* Go through the argument and mark each step as one of:

- (I) an *incidence* statement — how many points of a set lie on a variety, how many cells a line crosses. *These transfer.*
- (F) a *finiteness* statement — a polynomial of degree  $< q$  cannot vanish everywhere, a function on  $\mathbb{F}_q$  is a polynomial. *These do not.*
- (E) an *exact algebraic* statement — a polynomial vanishing at  $> d$  points of a line vanishes on it. *These transfer only if the points are exactly collinear*, which over  $\mathbb{R}$  they are not.

*Applied to Dvir.* Step 1 is a dimension count: type (I), transfers (Proposition 10.7.1). Step 2–3 is the propagation: type (E), and the points in a tube are not collinear, so it fails (Proposition 10.7.2). Step 4 is type (F), and fails for a second, independent reason. *Two of the four steps are untransferable.*



*Applied to polynomial partitioning.* Volume 11’s argument uses only “a line meets a degree- $d$  hypersurface in  $\leq d$  points or lies in it” — type (I) throughout. That is why it transfers, and why it is the descendant of this volume that survives.

*Applied to the method of multiplicities.* It strengthens a type (E) step. Since the (E) step is the one that fails, the strengthening is irrelevant over  $\mathbb{R}$  — which is Worked Example 10.2.5’s conclusion, reached by the test rather than by inspection.

*The value of the test.* It is quick, and it is usually decisive. A proposed transfer whose critical step is type (E) or (F) should be regarded as unpromising until the collinearity or finiteness has been replaced by something quantitative.

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WORKED EXAMPLE 10.7.4— TIER C: RESEARCH TASTE **Model problems, and what they are worth**

The finite field Kakeya problem is the clearest instance in this series of a *model problem*: a simplified version proposed in the hope that solving it will illuminate the original. Its history is instructive and the lessons generalise.

(a) *The hope.* Wolff proposed it in 1999 expecting that the combinatorics of lines was the essential difficulty, and that removing the metric would isolate it. If the model turned out hard, one would know the combinatorics was the obstruction and could concentrate there.

(b) *The outcome.* The model was easy — half a page, once the right idea appeared — and the technique did not transfer. So the hypothesis was refuted: the combinatorics is not the difficulty.

(c) *Why that is still valuable.* A refuted hypothesis is information. Before 2008 one could reasonably believe the difficulty was combinatorial; afterwards one could not. Worked Example 10.7.2 collects three independent lines of evidence now pointing at the width of tubes, and Dvir’s theorem is the sharpest of them precisely because it is a negative result about the model.

(d) *The methodological cost.* A decade of effort went into the model on the assumption that it was a stepping stone. That effort was not wasted — it produced the polynomial method, which Volume 11 shows was worth more than the model itself — but the stepping-stone reasoning was mistaken.

(e) *The general lesson.* A model problem is worth solving whichever way it comes out, but the inference “the model is hard, therefore the original is hard for the same reason” is unsound, and so is its converse. What a solved model establishes is which features are *not* responsible for the difficulty, and that is

usually a shorter list than one hopes.

(f) *Where this recurs in the series.* Volume 13's decoupling theory was developed partly through a model — the discrete restriction problem for the integer lattice — and there the transfer did work. So the finite field experience is not universal; the difference is that the discrete model retains the metric and only discretises it, whereas  $\mathbb{F}_q$  discards the metric entirely.

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WORKED EXAMPLE 10.7.5— TIER A **The obstruction, in one picture**

The content of Proposition 10.7.2 can be held in mind as a single image, and it is worth fixing.

*Over  $\mathbb{F}_q$ .* A line is  $q$  points, exactly collinear. A polynomial of degree  $d < q$  vanishing at all of them, restricted to the line's parameter  $t$ , is a one-variable polynomial of degree  $\leq d$  with  $q > d$  roots. It is *identically* zero, and its leading coefficient — which encodes the direction — vanishes.

*Over  $\mathbb{R}$ .* A  $\delta$ -tube contains  $\approx \delta^{-1}$  separated points, but they are scattered within a cylinder of radius  $\delta$ , not on its axis. A polynomial can vanish on a curve that weaves among them — the parabola  $y = \frac{\delta}{2} + \varepsilon x^2$  of Proposition 10.7.2 does — while taking the value  $-\frac{\delta}{2}$  at every point of the axis. Restricting to the axis gives a polynomial with *no* roots at all.

*The picture.* Over  $\mathbb{F}_q$ : beads on a wire, and a polynomial vanishing on every bead vanishes on the wire. Over  $\mathbb{R}$ : beads in a straw, and a polynomial can thread between them, missing the straw's centre line entirely.

*Why no amount of care repairs it.* The vanishing is an exact algebraic condition and the points are only approximately collinear. There is no approximate version of “a polynomial with more roots than its degree is zero”; the nearest true statement is that it is *small*, and smallness is not what Step 3 needs.

*What is used instead.* Volume 11 replaces vanishing by *crossing*: a line meets the zero set of a degree- $d$  polynomial in at most  $d$  points unless it lies inside. That is stable under perturbation — a tube crosses the zero set in a controlled number of places whether or not its points are exactly collinear — and it is the whole basis of polynomial partitioning.

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WORKED EXAMPLE 10.7.6— TIER A **The two propositions, side by side**

Chapter 10.7 rests on two propositions with opposite verdicts, and the contrast is the chapter's content.

|          | Proposition 10.7.1                                                                                               | Proposition 10.7.2                                                                            |
|----------|------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|
| Concerns | Step 1 of Dvir                                                                                                   | Steps 2–3 of Dvir                                                                             |
| Asserts  | a low-degree polynomial vanishing on a $\delta$ -net exists, with degree below the number of net points per tube | a polynomial can vanish at $\delta^{-1}$ points of a tube while vanishing nowhere on its axis |
| Verdict  | the count <i>transfers</i>                                                                                       | the propagation <i>fails</i>                                                                  |
| Why      | dimension counting is insensitive to geometry                                                                    | the net points are near a line, not on one                                                    |

*Reading them together.* The polynomial exists; one simply cannot use it. That is a sharper diagnosis than “the argument does not transfer”, and it is what makes Worked Example 10.7.1’s classification of repairs possible: any repair must address the second column, and raising the order of vanishing or shrinking the tubes does not.

*The number to remember.*  $d \approx \delta^{(\beta-n)/n}$  against  $\delta^{-1}$  points per tube, and  $d < \delta^{-1}$  whenever  $\beta > 0$ . So for *every* Besicovitch set of positive measure exponent the degree budget is comfortable. Nothing is tight; the obstruction is qualitative.

## Drills

**Drill 10.7.1.** Verify the equivalence  $d < \delta^{-1} \iff \beta > 0$  in Proposition 10.7.1.

**Drill 10.7.2.** Exhibit explicitly a polynomial of degree 2 vanishing at  $\lfloor \delta^{-1} \rfloor$  separated points of a  $\delta$ -tube and nowhere on its axis.

**Drill 10.7.3.** Explain why Repair 1 of Worked Example 10.7.1 fails, by counting the conditions imposed by an infinite set.

**Drill 10.7.4.** State the incidence fact used by Repair 2 — a line meets a degree- $d$  hypersurface in at most  $d$  points or lies inside it — and prove it from the one-variable root count.

**Where we are.** The two propositions of this chapter should be read together. Proposition 10.7.1 says the polynomial exists and has low enough degree; Proposition 10.7.2 says one cannot use it. The argument fails not

for lack of a polynomial but for lack of a conclusion to draw from it. That is worth emphasising because the opposite diagnosis is natural and wrong. A reader who assumes the obstruction is the infinitude of  $\mathbb{R}^n$  — too many points to impose vanishing conditions — will look for a discretisation, and Proposition 10.7.1 shows the discretisation works perfectly. The obstruction is one step later, and it is geometric rather than cardinality-theoretic.

**Where we are.** One sentence to carry into Volume 11.

The statement that survives the passage from  $\mathbb{F}_q$  to  $\mathbb{R}$  is not about vanishing but about *crossing*: a line meets the zero set of a degree- $d$  polynomial in at most  $d$  points, or lies inside it. That is a bound on incidences, it holds over any field, and — crucially — it is stable when the line is replaced by a thin tube, since a tube crosses the zero set in a controlled number of places whether or not its points are exactly collinear.

Everything Volume 11 does is built on that one stable fact.

## Notes

That Dvir’s argument does not transfer was clear immediately, and the search for a quantitative substitute led to Guth’s polynomial partitioning (2010) and the Guth–Katz theorem (2015). Dvir’s method of multiplicities, with Saraf and Sudan, gives the sharp constant over  $\mathbb{F}_q$ ; Worked Example 10.7.1 records why it does not help over  $\mathbb{R}$ .

# Chapter 10.8

## Assessment

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WORKED EXAMPLE 10.8.1— TIER A    **The state of the Kakeya problem after Volume 10**

| Setting                     | Best bound        | Source    | Status                         |
|-----------------------------|-------------------|-----------|--------------------------------|
| $\mathbb{F}_q^n$ , all $n$  | $q^n$             | Dvir      | <i>solved</i> , Theorem 10.2.1 |
| $\mathbb{R}^2$              | 2                 | Córdoba   | solved, Vol 9 Corollary 9.3.3  |
| $\mathbb{R}^3$              | 3                 | Wang–Zahl | solved 2025, Volume 14         |
| $\mathbb{R}^n$ , small $n$  | $\frac{n+2}{2}$   | Wolff     | Vol 9 Remark 9.5.5             |
| $\mathbb{R}^n$ , large $n$  | $(2 - \sqrt{2})n$ | Katz–Tao  | Remark 10.6.2                  |
| $\mathbb{R}^n$ , $n \geq 4$ | —                 | —         | <i>open</i>                    |

*The one-line summary.* The problem is settled over finite fields, in the plane, and in three dimensions; it is open in four and above, where the best bounds are a little under  $0.59n$  against the conjectured  $n$ .

*Where the volume’s two methods sit.* Dvir’s is the sharpest result in the table and the least transferable. The arithmetic method gives the best asymptotic constant over  $\mathbb{R}$  and is not sharp anywhere. Neither is used in the proofs that settled  $n = 2$  or  $n = 3$ .

*What that suggests.* The methods that settle cases are geometric — Córdoba’s  $L^2$  argument, and Wang–Zahl’s analysis of sticky configurations. The methods that give the best general bounds are algebraic and arithmetic. The two families have not converged, and Volume 14 reports that they still have not.

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WORKED EXAMPLE 10.8.2— TIER C: RESEARCH TASTE    **What Volume 10 established**

(a) *Proved in full.* Dvir’s theorem, Theorem 10.2.1, with the Schwartz–Zippel lemma it rests on; Ruzsa’s triangle inequality, Theorem 10.3.4; the Plünnecke–Ruzsa inequality via Petridis, Theorem 10.3.5, and its corollary converting differences into sums; the three-slice lemma and the reduction of a Besicovitch set to a configuration with many additive triples, Proposition 10.5.4; and the two propositions of Chapter 10.7 isolating exactly where the finite field argument fails.

(b) *Announced and not proved.* Theorem 10.6.1, the bounds of Bourgain and Katz–Tao (Remark 10.6.2, revision R10), which need the Balog–Szemerédi–Gowers and Freiman theorems. Nothing later depends on them.

(c) *The single most useful thing here.* Not either theorem, but Proposition 10.7.2: a polynomial can vanish at  $\delta^{-1}$  points of a  $\delta$ -tube while vanishing nowhere on its axis. That one example explains why an argument settling the finite field case in half a page leaves the real case untouched, and it identifies the width of a tube as the locus of the difficulty.

(d) *The methodological lesson.* Both methods here escape Volume 9’s ceiling by removing the optimisation over an unknown multiplicity — Worked Example 9.8.2 identified that as the binding constraint. Dvir removes it entirely; the arithmetic method replaces it by an optimisation over additive structure, which is lossy but at least different. Recognising what a method optimises over, and whether the extremal value is attainable, remains the most reliable way to predict its ceiling.

(e) *What comes next.* Volume 11 takes the one part of the polynomial method that does transfer — that a line meets a degree- $d$  variety in at most  $d$  points — and builds polynomial partitioning on it. That is the technique behind the Guth–Katz solution of the distinct distances problem and the current records in restriction theory, and it is the direct descendant of this volume’s Worked Example 10.7.1, Repair 2.

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**WORKED EXAMPLE 10.8.3 — TIER B Reading the Kakeya literature: a field guide**

Ten volumes in, a reader is equipped to place almost any paper in this area. Five questions, in order.

(1) *Which setting?*  $\mathbb{R}^n$ ,  $\mathbb{F}_q^n$ , or a model such as the Furstenberg problem. The finite-field results are solved and do not transfer (Chapter 10.7); a paper proving something over  $\mathbb{F}_q$  is contributing technique, not a bound over  $\mathbb{R}$ .

(2) *Which dimension notion?* Hausdorff, Minkowski, or a maximal-function estimate. Volume 9, Proposition 9.7.3 shows all four formulations carry the same exponent, *provided* the loss is  $\delta^{-\varepsilon}$  for every  $\varepsilon$ . Check that.

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(3) *What does the method optimise over?* Volume 9, Worked Example 9.8.4 gives the test. A multiplicity, a scale, an angular width — compute the balancing value and ask whether a configuration attains it. If not, the paper’s ceiling is already visible.

(4) *What order of correlation?* Second moments, energies and Fourier transforms see pairs only (Volume 9, Worked Example 9.3.3); incidence and algebraic arguments see more. A method of the first kind cannot beat  $\frac{n+2}{2}$ , whatever its ingenuity.

(5) *Which class of sets?* All Besicovitch sets, or a restricted class — sticky, self-similar, Ahlfors–David regular. Restricted results are often much stronger and are not comparable to general ones.

*Worked instance: Dvir’s paper.* (1)  $\mathbb{F}_q^n$ ; (2) counting, no  $\delta$ ; (3) nothing — no free parameter, which is why it is sharp; (4) algebraic, above the pair barrier; (5) all Kakeya sets. Conclusion: a complete solution of a model problem, contributing a technique.

*Worked instance: Bourgain’s arithmetic bound.* (1)  $\mathbb{R}^n$ ; (2) Hausdorff, via maximal estimates; (3) additive structure, with lossy constants; (4) three-term progressions, so above pairs; (5) all Besicovitch sets. Conclusion: a general bound, asymptotically the best of its era, at its ceiling.

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WORKED EXAMPLE 10.8.4— TIER A    **Ten volumes in: where the series stands**

| Vol | Subject                | What it contributes to Kakeya                                                                         |
|-----|------------------------|-------------------------------------------------------------------------------------------------------|
| 1–5 | classical analysis     | the decay of $\widehat{d\sigma}$ , with its phase restriction $\Rightarrow$ Kakeya; the Knapp example |
| 6   | restriction            |                                                                                                       |
| 7   | dimension theory       | Frostman, energy, and the single-scale upgrade                                                        |
| 8   | projections, distances | the limit of two-point methods<br>$L^2$ , bush, hairbrush; the configuration lemma                    |
| 9   | tube geometry          |                                                                                                       |
| 10  | algebra, arithmetic    | Dvir; sumsets; where the model fails                                                                  |

*The through-line.* Volumes 1–5 build one estimate; Volume 6 shows it is the analytic face of a geometric problem; Volumes 7–9 develop the geometry to its limit; Volume 10 changes category twice.

*The recurring diagnosis.* Four times now a method has been shown to have a ceiling, and each time for the same reason: it optimises over a parameter whose extremal value is not attained (Volume 9, Worked Example 9.8.2), or it sees only pair correlations (Volume 8, Worked Example 7.7.3; Volume 9, Worked Example 9.3.3). Dvir’s argument is the only one in the series with neither defect, and it is the only one that is sharp.

*What remains.* Volumes 11–14: the polynomial method over  $\mathbb{R}$ , multilinear restriction, decoupling, and the current state including Wang–Zahl in three dimensions. The conjecture is open in four dimensions and above, and nothing in the first ten volumes suggests it is close.

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#### WORKED EXAMPLE 10.8.5— TIER B **Comparing bounds across settings**

Results in this area are stated over  $\mathbb{R}^n$ , over  $\mathbb{F}_q^n$ , and for restricted classes of sets. Comparing them requires care, and three conversions cover most cases.

(1)  $\mathbb{F}_q$  to  $\mathbb{R}$ : *exponents.* A finite-field bound  $|K| \gtrsim q^s$  corresponds formally to  $\dim_{\mathbb{H}} K \geq s$  over  $\mathbb{R}$ , via  $q \leftrightarrow \delta^{-1}$ . Dvir gives  $s = n$ ; the real conjecture is  $s = n$ . *The correspondence is formal only* — Chapter 10.7 shows no implication runs either way.

(2) *Minkowski to Hausdorff.* Volume 9, Proposition 9.7.3: the same exponent, provided the loss is  $\delta^{-\varepsilon}$  for every  $\varepsilon$ . A finite-field result has no  $\delta$  and so no analogue of this distinction — another place the dictionary has no entry.

(3) *Restricted classes to general.* A bound for Ahlfors–David regular sets, or for sticky configurations, is not comparable to a general bound. Volume 14 reports that the Wang–Zahl theorem proceeds by reducing the general case to the sticky one, so in that instance the restricted result was the whole battle — but the reduction is itself a theorem and cannot be assumed.

*A worked comparison.* Is Dvir’s  $q^n$  “better” than Katz–Tao’s  $(2 - \sqrt{2})n$ ? The question is malformed: the first is a complete solution of a different problem, the second a partial result on the real one. What can be said is that no argument is known deriving anything about  $\mathbb{R}^n$  from the first, and Proposition 10.7.2 explains why.

*The habit.* Before comparing two bounds, check that they concern the same setting, the same dimension notion, and the same class of sets. Surveys tabulate exponents; the hypotheses are where the content is.

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**Where we are.** Volume 10 is the only volume in the series with two settings that are not  $\mathbb{R}^n$ , and Worked Example 10.1.6 marks the three



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dictionary entries that are blank: angle, near-parallelism, width. Those three are the Kakeya problem.

## Drills

**Drill 10.8.1.** For each row of the table in Worked Example 10.8.1, state whether the bound is sharp and, if not, what the conjectured value is.

**Drill 10.8.2.** Explain why Dvir’s theorem does not imply anything about  $\dim_{\mathbb{H}}$  of a real Besicovitch set, referring to Proposition 10.7.2.

**Drill 10.8.3.** Compare the ceilings of the three methods of Volume 9 and the two of this volume using the criterion of Volume 9, Worked Example 9.8.4.

**Drill 10.8.4.** Identify, for each of the five results listed in Worked Example 10.8.2(a), the single hypothesis whose removal would break the proof.

**Drill 10.8.5.** Using Worked Example 10.8.3, place Wolff’s hairbrush paper against the five questions, and identify which of them Remark 9.5.5 concerns.

**Drill 10.8.6.** Explain why question (3) of Worked Example 10.8.3 returns “nothing” for Dvir’s argument, referring to the Checkpoint after Theorem 10.2.1.

**Drill 10.8.7.** Verify that the ten announced omissions listed in Worked Example 10.6.4 are each used nowhere else, by checking the appendix of each volume.

**Drill 10.8.8.** Give an example of a result in this volume that would become a *deferral* rather than a removal if a later volume needed it, and name the volume.

**Where we are.** A last observation about the shape of this volume. It contains the shortest complete proof in the series — Theorem 10.2.1, about a page with its lemma — and one of the longest chains of announced omissions, Remark 10.6.2 standing at the end of a reduction whose remaining step is a theorem of another subject.

That contrast is not accidental. Where a problem has been solved outright the proof is often short, because the right idea removes the need for approximation; where it has not, the arguments are long and lossy, because each step must be estimated rather than computed. Worked Example 10.6.2

follows the losses in the arithmetic method and finds them accumulating multiplicatively, which is the signature of the second situation.

A reader deciding where to spend effort in this area could do worse than ask which situation a given problem is in.

**Where we are.** One point of bookkeeping for the volumes ahead.

This volume introduced two settings that are *not*  $\mathbb{R}^n$  — the finite field  $\mathbb{F}_q^n$  and abelian groups — and results proved in them do not transfer automatically. Worked Example 10.1.6 gives the dictionary and marks the three entries that are blank: angle, near-parallelism, and width.

Volumes 11 to 14 return to  $\mathbb{R}^n$  and stay there. Where they invoke an algebraic fact it will be of the incidence type identified in Worked Example 10.7.3 — a line meets a degree- $d$  variety in at most  $d$  points — which holds over any field and is stable under the perturbations that discretisation introduces. Nothing from  $\mathbb{F}_q$  beyond that is used again.

**Where we are.** Volume 10 in one line: the Kakeya problem is *solved* over finite fields by an argument with no free parameter, and the argument does not transfer because a tube is not a line.

## Notes

The table in Worked Example 10.8.1 is current as of the writing of this volume; Volume 14 reports the position in full, including the Wang–Zahl theorem in  $\mathbb{R}^3$  and the state of the Furstenberg set problem.

# Chapter 10.A

## Results Quoted from Volumes 1–9

| Result                               | Source               | Used in               |
|--------------------------------------|----------------------|-----------------------|
| The Kakeya conjecture                | Conjecture 7.1.1     | throughout            |
| Minkowski content                    | Proposition 7.2.3    | Proposition 10.7.1    |
| Maximal bounds give Hausdorff bounds | Theorem 7.8.5        | Chapter 10.6          |
| Córdoba’s bound                      | Corollary 9.3.3      | Worked Example 10.8.1 |
| The bush bound                       | Theorem 9.4.4        | Proposition 10.5.5    |
| Wolff’s bound, transverse case       | Remark 9.5.5         | Worked Example 10.6.1 |
| The two-bounds template              | Worked Example 9.4.4 | Worked Example 10.8.2 |
| Why the geometric mean is lossy      | Worked Example 9.8.2 | Worked Example 10.8.2 |
| Predicting a method’s ceiling        | Worked Example 9.8.4 | Worked Example 10.6.2 |

### Debts

This volume incurs none and discharges none. Chapter 10.2 depends on nothing in Volumes 1–9; Chapter 10.3 depends on nothing at all.

### Removed

Theorem 10.6.1 — the bounds of Bourgain and of Katz–Tao — is stated and not proved, as revision R10 (Remark 10.6.2). The two missing inputs are the Balog–Szemerédi–Gowers theorem and Freiman’s theorem.



# Chapter 10.B

## Notation and Index of Results

### Additions to the Volume 1–9 conventions

|                           |                                                       |
|---------------------------|-------------------------------------------------------|
| $\mathbb{F}_q$            | the field with $q$ elements                           |
| $\ell_{a,b}$              | the line $\{a + tb : t \in \mathbb{F}_q\}$            |
| $\mathcal{P}_d^n$         | polynomials in $n$ variables of total degree $\leq d$ |
| $A + B, A - B, kA$        | sumsets, difference sets, iterated sumsets            |
| $K =  A + A / A $         | the doubling constant                                 |
| $T(A, B, C), T'(A, B, C)$ | additive triples, exact and $\delta$ -approximate     |
| $E_t$                     | the horizontal slice of $E$ at height $t$             |

### Index of the main results

|                                               |                             |
|-----------------------------------------------|-----------------------------|
| Trivial bounds over $\mathbb{F}_q$            | Proposition 10.1.2          |
| Schwartz–Zippel                               | Lemma 10.1.3                |
| Dvir’s theorem                                | Theorem 10.2.1              |
| Dvir with a positive proportion of directions | Corollary 10.2.2            |
| Trivial sumset bounds                         | Proposition 10.3.2          |
| Ruzsa’s triangle inequality                   | Theorem 10.3.4              |
| Petridis’s minimising subset                  | Lemma 10.3.6                |
| Plünnecke–Ruzsa                               | Theorem 10.3.5              |
| Differences control sums                      | Corollary 10.3.7            |
| The three-slice lemma                         | Lemma 10.5.2                |
| Many additive triples                         | Proposition 10.5.4          |
| The trivial consequence                       | Proposition 10.5.5          |
| The arithmetic bounds                         | Theorem 10.6.1 (not proved) |
| The parameter count survives                  | Proposition 10.7.1          |
| The propagation step fails                    | Proposition 10.7.2          |

## Plan revisions in this volume

| Removed                                     | Reason                                                                                                                                |
|---------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| The Bourgain and Katz–Tao arithmetic bounds | need Balog–Szemerédi–Gowers and Freiman, neither developed in this series; nothing later depends on them. Remark 10.6.2, revision R10 |

# Part 11

## Volume 11

The Polynomial Method over  $\mathbb{R}$





# Preface to Volume 11

## The one fact that transfers

Volume 10 ended with a negative result and a positive residue. The negative result — Proposition 10.7.2 — is that Dvir’s argument has no real analogue: a polynomial can vanish at every point of a  $\delta$ -net inside a tube while vanishing nowhere on the tube’s axis, so the step converting pointwise vanishing into algebraic vanishing simply fails.

The residue is one stable statement:

*a line meets the zero set of a degree- $D$  polynomial in at most  $D$  points, unless it lies inside.*

This is a bound on *incidences*. It holds over any field, it follows from the one-variable root count, and — decisively — it survives replacing the line by a thin tube, since a tube crosses a surface in a controlled number of places whether or not its points are exactly collinear.

Everything in this volume is built on that fact. The technique is *polynomial partitioning*: rather than asking a polynomial to vanish on a configuration, one asks it to *divide space* into cells, each containing a controlled share of the configuration, and then induct. The lines can only cross between cells  $D$  times each, so the cells are nearly independent, and a recursion closes.

## What the method achieved

Between 2010 and 2015 the polynomial method settled or advanced several problems that had resisted everything else: the joints problem, the Erdős distinct distances problem in the plane, the multilinear Kakeya inequality, and — through the last of these — the best restriction estimates then known. Volume 14 reports that the Wang–Zahl resolution of three-dimensional Kakeya also runs through this machinery.

## What is assumed, and what is imported

Volumes 1–10. From Volume 10 the incidence fact above, and the classification of transferable statements in Worked Example 10.7.3.

One topological input is imported without proof: the *polynomial ham sandwich theorem*, which asserts that any  $M$  finite sets in  $\mathbb{R}^n$  can be simultaneously bisected by a polynomial of degree  $O(M^{1/n})$ . It rests on the Borsuk–Ulam theorem, which is algebraic topology and outside the scope of this series; Remark 11.2.4 records the decision and its consequences. Everything else is proved.

## Two ways to use a polynomial

The volume divides according to which of two devices is in play, and the division is worth having before starting.

*Vanishing.* Find a polynomial that is zero on the configuration. This costs degree  $\approx N^{1/n}$  for  $N$  points, propagates along exact lines, and yields a contradiction from a degree count. It is Dvir’s device from Volume 10, and it proves the joints theorem (Chapter 11.4) and Theorem 11.8.1. It requires the objects to be *exact* — lines, not tubes — and it is useless when the ambient configuration has positive measure, since a variety is null.

*Partitioning.* Find a polynomial whose complement’s cells each meet the configuration in a controlled amount. This is Theorem 11.2.5, it is always available, and it converts a global count into many local ones. It proves Szemerédi–Trotter (Chapter 11.3), underlies the Guth–Katz theorems, and is the only route to configurations that are continuous.

*The common ingredient.* Both need the incidence bound quoted above — a line meets  $Z(P)$  in at most  $D$  points — and nothing else from algebra. Worked Example 11.2.3 sets out when each applies, and Worked Example 11.8.3 collects what a polynomial can and cannot see, which is the volume’s honest summary.

## The plan

- |   |                         |                              |
|---|-------------------------|------------------------------|
| 1 | Polynomial partitioning | cells, and the degree budget |
| 2 | Szemerédi–Trotter       | the method’s first triumph   |

Chapters 3–8, planned: incidences of lines in  $\mathbb{R}^3$ ; the joints theorem; the Guth–Katz distinct distances theorem; Kakeya by partitioning; the multilinear Kakeya inequality; and an assessment.

## Chapter 11.1

# Borsuk–Ulam and Ham Sandwich

Polynomial partitioning rests on the ham sandwich theorem, which rests on Borsuk–Ulam. This chapter proves both, from a combinatorial lemma of Tucker whose proof is a path-following argument in a finite graph. Nothing here uses homology or degree theory; the only input is that a finite graph with all degrees at most two is a disjoint union of paths and cycles.

### 11.1.1 A graph lemma

**Lemma 11.1.1.** *Let  $G$  be a finite graph in which every vertex has degree at most 2. Then  $G$  is a disjoint union of simple paths, simple cycles and isolated vertices. Consequently the number of vertices of degree 1 is even, and each is an endpoint of a path whose other endpoint also has degree 1.*

*Proof.* Induct on the number of edges. With no edges every vertex is isolated. Removing an edge  $e$  leaves all degrees at most 2, so by induction the smaller graph is a union of paths, cycles and isolated vertices; the two endpoints of  $e$  had degree at most 2 in  $G$ , hence at most 1 after removal, so each is isolated or an endpoint of a path. Restoring  $e$  joins two such, producing again a union of paths and cycles.

Degree-1 vertices are exactly the endpoints of the paths, of which each path has two.  $\square$

### 11.1.2 Tucker’s lemma

**Definition 11.1.2** (The cross-polytope complex). Let  $\widehat{B}^n = \{x \in \mathbb{R}^n : |x_1| + \cdots + |x_n| \leq 1\}$  be the  $\ell^1$  ball. Let  $\diamond_+^n$  be the simplicial complex with vertex

set  $\{0, \pm 1, \dots, \pm n\}$ , realised by placing 0 at the origin,  $+i$  at  $e_i$  and  $-i$  at  $-e_i$ , in which  $\sigma$  is a simplex exactly when it does *not* contain both  $+i$  and  $-i$  for any  $i$ .

Geometrically  $\diamond_+^n$  triangulates  $\widehat{B}^n$ : its maximal simplices are the  $2^n$  orthants' worth of sets  $\{0\} \cup \{\varepsilon_1 1, \dots, \varepsilon_n n\}$  with  $\varepsilon_i = \pm$ .

**Definition 11.1.3** (Special Tucker pair). A *special Tucker triangulation* is a triangulation  $T$  of  $\widehat{B}^n$  which is antipodally symmetric on  $\partial \widehat{B}^n$  and refines  $\diamond_+^n$  — equivalently, the sign of each coordinate is constant on the relative interior of every simplex of  $T$ . A *Tucker labelling* is a map  $\lambda : V(T) \rightarrow \{\pm 1, \dots, \pm n\}$  with  $\lambda(-v) = -\lambda(v)$  for  $v \in \partial \widehat{B}^n$ . An edge of  $T$  is *complementary* if its two vertices carry labels  $+i$  and  $-i$  for some  $i$ .

**Theorem 11.1.4** (Tucker). *Every special Tucker pair  $(T, \lambda)$  has a complementary edge.*

*Idea of the proof.* Suppose not. Build a graph whose vertices are the simplices of  $T$  that are “happy” — those whose labels realise every sign pattern their position requires — and join two when one is obtained from the other by adding or deleting a vertex in a prescribed way. One shows every happy simplex has degree exactly 2, with two exceptions: the empty simplex, which has degree 1, and any simplex containing a complementary edge, which also has degree 1. By Lemma 11.1.1 degree-1 vertices come in pairs, so a second one must exist — and it is a simplex with a complementary edge.

*Proof.* For a simplex  $\sigma$  of  $T$  write  $\lambda(\sigma) = \{\lambda(v) : v \in V(\sigma)\}$  and, for  $x$  in the relative interior of  $\sigma$ ,

$$S(\sigma) = \{i : x_i > 0\} \cup \{-i : x_i < 0\},$$

which is well defined because  $T$  refines  $\diamond_+^n$ . Call  $\sigma$  *happy* if  $S(\sigma) \subseteq \lambda(\sigma)$ .

**Step 1.** *Happy simplices are tight or loose.*

Let  $\sigma$  be happy and  $k = |S(\sigma)|$ . Since  $\sigma$  lies in the span of  $\{e_{|i|} : i \in S(\sigma)\}$ , we have  $\dim \sigma \leq k$ ; and  $\sigma$  needs at least  $k$  vertices to carry the  $k$  labels of  $S(\sigma)$ , so  $\dim \sigma \geq k - 1$ . Call  $\sigma$  *tight* if  $\dim \sigma = k - 1$  and *loose* if  $\dim \sigma = k$ .

Three observations. A happy simplex on the boundary is tight, since there  $|S(\sigma)|$  exceeds the dimension of the simplex of  $\diamond_+^n$  containing it by one. The vertex  $\{0\}$  is happy and loose, as  $S(\{0\}) = \emptyset$ . The empty simplex is happy and tight.

**Step 2.** *The graph.*

Join happy simplices  $\sigma$  and  $\tau$  when either

- $\sigma \subseteq \partial \widehat{B}^n$  and  $\tau = -\sigma$ , or
- $\sigma$  is a facet of  $\tau$  with  $\lambda(\sigma) = S(\tau)$ .

**Step 3.** *Tight simplices have degree 2, except the empty one.*

Let  $\sigma$  be tight,  $k = |S(\sigma)|$ . It has no facet  $\tau$  with  $\lambda(\tau) = S(\sigma)$ , since such a  $\tau$  would need dimension at least  $k - 1 = \dim \sigma$ . If  $\sigma$  lies on the boundary it is joined to  $-\sigma$  and is a facet of exactly one happy simplex; otherwise it is a facet of exactly two simplices  $\tau$  lying in the same  $k$ -dimensional coordinate subspace, and for each  $S(\tau) = S(\sigma) = \lambda(\sigma)$ , so each is a neighbour. Either way  $\sigma$  has two neighbours.

The exception is the empty simplex: its ambient subspace is  $\{0\}$ , so its only neighbour is  $\{0\}$ , and it has degree 1.

**Step 4.** *Loose simplices have degree 2, unless they contain a complementary edge.*

Let  $\sigma$  be loose, so  $\dim \sigma = k$  and  $\sigma$  has  $k + 1$  vertices carrying labels covering the  $k$  elements of  $S(\sigma)$ . Either some label repeats, or some label  $i \notin S(\sigma)$  occurs.

If a label repeats, then  $|\lambda(\sigma)| = k$  and  $\sigma$  is a facet of no happy simplex, but is joined to its two facets obtained by deleting one of the two vertices with the repeated label: degree 2.

If instead  $i \in \lambda(\sigma) \setminus S(\sigma)$ , then  $\sigma$  is joined to the facet omitting the vertex labelled  $i$ , and to no other facet. If  $-i \notin \lambda(\sigma)$  then  $\sigma$  is also a facet of the simplex  $\tau$  reached by moving along the  $i$ -th coordinate axis, which has  $S(\tau) = S(\sigma) \cup \{i\} = \lambda(\sigma)$ : degree 2. If  $-i \in \lambda(\sigma)$  then  $\{i, -i\} \subseteq \lambda(\sigma)$ , so  $\sigma$  contains a complementary edge, and no such  $\tau$  exists because  $T$  is special: degree 1.

**Step 5.** *Conclude.*

Every happy simplex has degree at most 2, and exactly one — the empty simplex — has degree 1 for a reason other than containing a complementary edge. By Lemma 11.1.1 the number of degree-1 vertices is even, so at least one more exists, and by Steps 3 and 4 it must be a simplex containing a complementary edge.  $\square$

**Where we are.** The argument is constructive: following the path from the empty simplex *finds* a complementary edge, and the walk is computable. That is unusual for a statement equivalent to Borsuk–Ulam, and it is the reason for preferring Tucker's route over a homological one here — no chain complexes, no degree, no homotopy invariance.

The two exceptional cases in Steps 3 and 4 are the whole content. Everything else is bookkeeping about which facets of a happy simplex are

happy.

### 11.1.3 Borsuk–Ulam

**Theorem 11.1.5** (Borsuk–Ulam). *Let  $g : \mathbb{S}^n \rightarrow \mathbb{R}^n$  be continuous. Then  $g(x) = g(-x)$  for some  $x \in \mathbb{S}^n$ .*

*Proof.* Put  $f(x) = g(x) - g(-x)$ , so  $f$  is continuous and odd; we must find a zero.

**Step 1.** *Discretise.*

Fix  $\varepsilon > 0$ . Since  $\mathbb{S}^n$  is compact,  $f$  is uniformly continuous: choose  $\delta > 0$  with  $\|f(x) - f(y)\|_\infty < \varepsilon$  whenever  $\|x - y\|_\infty < \delta$ . The closed upper hemisphere of  $\mathbb{S}^n$  is homeomorphic to  $\widehat{B}^n$  with antipodally symmetric boundary, so choose a special Tucker triangulation  $T$  of it all of whose simplices have diameter below  $\delta$ .

**Step 2.** *Label.*

For a vertex  $v$  let  $k(v) = \min\{i : |f(v)_i| = \|f(v)\|_\infty\}$  and set

$$\lambda(v) = \begin{cases} +k(v), & f(v)_{k(v)} > 0, \\ -k(v), & f(v)_{k(v)} < 0. \end{cases}$$

(If  $f(v) = 0$  we are done.) Since  $f$  is odd,  $\lambda$  is odd on the boundary, so  $(T, \lambda)$  is a special Tucker pair.

**Step 3.** *Use the complementary edge.*

By Theorem 11.1.4 there is an edge  $vv'$  with  $\lambda(v) = i > 0$  and  $\lambda(v') = -i$ . Then  $f(v)_i > 0$  and  $f(v')_i < 0$ , so

$$\|f(v)\|_\infty = f(v)_i < f(v)_i - f(v')_i \leq \|f(v) - f(v')\|_\infty < \varepsilon,$$

the last step because  $\text{diam } \sigma < \delta$ .

**Step 4.** *Compactness.*

So for each  $\varepsilon = 1/m$  there is  $x_m$  with  $\|f(x_m)\|_\infty < 1/m$ . By compactness a subsequence converges to some  $x^*$ , and  $f(x^*) = 0$  by continuity.  $\square$

### 11.1.4 Ham sandwich

**Theorem 11.1.6** (Ham sandwich for measures). *Let  $\mu_1, \dots, \mu_N$  be finite Borel measures on  $\mathbb{R}^N$ , each vanishing on every hyperplane. Then there is an affine hyperplane  $H$  with*

$$\mu_i(H^+) = \mu_i(H^-) \quad \text{for } i = 1, \dots, N,$$

where  $H^\pm$  are the two open half-spaces bounded by  $H$ .

*Proof.* For  $u = (u_0, u_1, \dots, u_N) \in \mathbb{S}^N \subseteq \mathbb{R}^{N+1}$  set

$$H_u^+ = \left\{ x \in \mathbb{R}^N : \sum_{j=1}^N u_j x_j > u_0 \right\}, \quad H_u^- = \left\{ x : \sum_{j=1}^N u_j x_j < u_0 \right\},$$

and define  $\Phi : \mathbb{S}^N \rightarrow \mathbb{R}^N$  by  $\Phi(u)_i = \mu_i(H_u^+) - \mu_i(H_u^-)$ .

$\Phi$  is odd, since  $H_{-u}^\pm = H_u^\mp$ .

$\Phi$  is continuous. If  $u^{(m)} \rightarrow u$  with  $u \neq \pm e_0$ , the indicator functions  $\mathbf{1}_{H_{u^{(m)}}^+}$  converge to  $\mathbf{1}_{H_u^+}$  off the hyperplane  $\{\sum u_j x_j = u_0\}$ , which is  $\mu_i$ -null by hypothesis; dominated convergence gives the claim. At  $u = \pm e_0$  the half-spaces degenerate to  $\emptyset$  and  $\mathbb{R}^N$ , and the same argument applies since the “hyperplane” is empty.

By Theorem 11.1.5 there is  $u$  with  $\Phi(u) = \Phi(-u)$ ; since  $\Phi$  is odd this forces  $\Phi(u) = 0$ , which is the assertion.  $\square$

**Corollary 11.1.7** (Ham sandwich for finite sets). *Let  $S_1, \dots, S_N$  be finite subsets of  $\mathbb{R}^N$ . Then there is a hyperplane  $H$  such that neither open half-space contains more than  $\frac{1}{2}|S_i|$  points of  $S_i$ , for every  $i$ .*

*Proof.* Replace each point  $p \in S_i$  by the uniform probability measure on  $B(p, \rho)$ , scaled by 1, and let  $\mu_i^\rho$  be the sum over  $p \in S_i$ . These vanish on hyperplanes, so Theorem 11.1.6 gives a hyperplane  $H_\rho$  bisecting all of them.

Let  $\rho \rightarrow 0$  along a sequence. Parametrising hyperplanes by  $u \in \mathbb{S}^N$  and passing to a convergent subsequence,  $H_\rho \rightarrow H$  for some hyperplane  $H$  (or  $u \rightarrow \pm e_0$ , in which case one half-space is empty and the conclusion is trivial). A point of  $S_i$  strictly inside  $H^+$  lies in  $H_\rho^+$  for small  $\rho$ , so

$$\#(S_i \cap H^+) \leq \liminf_{\rho \rightarrow 0} \mu_i^\rho(H_\rho^+) = \frac{1}{2}|S_i|,$$

and symmetrically for  $H^-$ .  $\square$

**Where we are.** Corollary 11.1.7 is the form Chapter 11.2 uses: it bisects  $N$  finite sets in  $\mathbb{R}^N$  by a hyperplane. Points *on*  $H$  are counted in neither half-space, which is exactly the convention Theorem 11.2.3 requires — such points are set aside and never counted again in the partitioning induction. The passage from measures to finite sets is the only place a limit is taken, and it is the reason the statement is about *at most* half rather than exactly half: a hyperplane through many points of  $S_i$  bisects the remainder, which is all the induction needs.

## Drills

**Drill 11.1.1.** Verify Lemma 11.1.1 for a graph with three vertices and two edges, and identify the degree-1 vertices.

**Drill 11.1.2.** Draw  $\diamond_+^2$  and check that its maximal simplices are the four triangles  $\{0, \pm 1, \pm 2\}$  with consistent signs.

**Drill 11.1.3.** In Step 4 of Theorem 11.1.4, verify that a loose simplex with a repeated label cannot contain a complementary edge.

**Drill 11.1.4.** Carry out Step 3 of Theorem 11.1.5 for  $n = 1$ , where the triangulation is an interval subdivision, and confirm the conclusion directly.

## Notes

Tucker’s lemma is from 1945; the constructive proof given here is due to Freund and Todd, in the presentation of Matoušek’s *Using the Borsuk–Ulam Theorem*. The deduction of Borsuk–Ulam is standard. The ham sandwich theorem is due to Banach, following a question of Steinhaus; the derivation from Borsuk–Ulam given here is the usual one.



## Chapter 11.2

# Polynomial Partitioning

### 11.2.1 Cells

**Definition 11.2.1.** For a non-zero polynomial  $P \in \mathbb{R}[x_1, \dots, x_n]$  let  $Z(P) = \{x \in \mathbb{R}^n : P(x) = 0\}$  be its zero set. The connected components of  $\mathbb{R}^n \setminus Z(P)$  are the *cells* of  $P$ .

**Theorem 11.2.2** (Cell count). *In the partitioning theorem Theorem 11.2.5 — stated and proved below, and independent of this refinement — the open sets  $O_i$  may be taken to number at most  $C_n D^n$ .*

*Proof.* The connected components of  $\mathbb{R}^n \setminus Z(P)$  number at most  $C_n D^n$  by the Milnor–Thom theorem. However *no appeal to that theorem is needed*, because the statement of Theorem 11.2.5 does not require each  $O_i$  to be connected.

Indeed, Theorem 11.2.5 produces open sets  $U_j$  with  $|S \cap U_j| \leq C_n D^{-n} |S|$ . Group them: form each  $O_i$  as a union of several  $U_j$ , adding sets one at a time and starting a new  $O_i$  whenever the running total would exceed  $C_n D^{-n} |S|$ . Each  $O_i$  then contains at most  $C_n D^{-n} |S|$  points of  $S$ , and since consecutive groups together hold more than  $C_n D^{-n} |S|$  points, the number of groups is at most  $2|S|/(C_n D^{-n} |S|) = C'_n D^n$ .  $\square$

We use Theorem 11.2.2 in the weak form stated; it is a standard result of real algebraic geometry and its proof, by induction on  $n$  with a Morse-theoretic count of critical points, is not reproduced here. For  $n = 1$  it is immediate: a degree- $D$  polynomial has at most  $D$  roots and hence at most  $D + 1$  cells.

**Theorem 11.2.3** (Polynomial ham sandwich). *Let  $S_1, \dots, S_M$  be finite subsets of  $\mathbb{R}^n$ . Then there is a non-zero polynomial  $P$  of degree*

$$\deg P \leq C_n M^{1/n}$$

*which bisects each  $S_j$ : neither  $\{P > 0\}$  nor  $\{P < 0\}$  contains more than  $\frac{1}{2}|S_j|$  points of  $S_j$ .*

*Proof. Step 1. The Veronese embedding.*

Let  $D$  be least with  $\binom{D+n}{n} - 1 \geq M$  and put  $N = \binom{D+n}{n} - 1$ . Order the non-constant monomials  $x^\alpha$ ,  $1 \leq |\alpha| \leq D$ , as  $m_1, \dots, m_N$  and define

$$v : \mathbb{R}^n \longrightarrow \mathbb{R}^N, \quad v(x) = (m_1(x), \dots, m_N(x)).$$

A polynomial  $P = c_0 + \sum_k c_k m_k$  of degree  $\leq D$  satisfies

$$P(x) = c_0 + \langle c, v(x) \rangle,$$

so the sign of  $P$  on  $\mathbb{R}^n$  is the sign of an affine functional on  $\mathbb{R}^N$  evaluated at  $v(x)$ : *polynomials of degree  $\leq D$  correspond exactly to affine hyperplanes in  $\mathbb{R}^N$ .*

**Step 2.** *Apply the linear ham sandwich theorem.*

The sets  $v(S_1), \dots, v(S_M)$  are  $M \leq N$  finite subsets of  $\mathbb{R}^N$ . Adjoining  $N - M$  empty sets and applying Corollary 11.1.7 gives a hyperplane  $H = \{y : c_0 + \langle c, y \rangle = 0\}$  with neither open half-space containing more than half of any  $v(S_j)$ .

**Step 3.** *Pull back.*

Set  $P(x) = c_0 + \langle c, v(x) \rangle$ , a polynomial of degree  $\leq D$ , non-zero since  $\langle c_0, c \rangle \neq 0$  and the monomials are linearly independent. By Step 1,  $\{P > 0\} = v^{-1}(H^+)$  and  $\{P < 0\} = v^{-1}(H^-)$ , so  $P$  bisects each  $S_j$ .

**Step 4.** *The degree bound.*

By minimality of  $D$ ,  $\binom{D-1+n}{n} - 1 < M$ , and since  $\binom{D+n}{n} \geq \frac{D^n}{n!}$  this gives  $D \leq C_n M^{1/n}$ .  $\square$

*Remark 11.2.4 (R11 discharged). Theorem 11.2.3 is now proved in full.* The chain is Lemma 11.1.1  $\rightarrow$  Theorem 11.1.4  $\rightarrow$  Theorem 11.1.5  $\rightarrow$  Theorem 11.1.6  $\rightarrow$  Corollary 11.1.7  $\rightarrow$  Theorem 11.2.3, and every link is proved in Chapter 11.1 or above. Revision R11 — previously the only topological import in the series, and load-bearing for the whole of this volume — is discharged.

The route was chosen for self-containment: Tucker's lemma is a path-following argument in a finite graph, so no homology, degree theory or homotopy invariance is required anywhere. The cost is that one proves Borsuk–Ulam in the combinatorial form rather than the familiar topological one; the gain is that Chapter 11.1 is eight pages and depends on nothing.

## 11.2.2 The partitioning theorem

**Theorem 11.2.5** (Guth–Katz partitioning). *Let  $S \subseteq \mathbb{R}^n$  be a finite set of  $N$  points and let  $D \geq 1$ . There is a non-zero polynomial  $P$  of degree at most  $D$  such that each cell of  $P$  contains at most  $C_n N/D^n$  points of  $S$ .*

*Idea of the proof.* Bisect repeatedly. One polynomial of degree  $O(1)$  splits  $S$  into two halves; two more split those into four quarters; and in general after  $j$  rounds one has  $2^j$  pieces of size  $N2^{-j}$ , achieved by  $2^{j-1}$  polynomials at the  $j$ -th round — but Theorem 11.2.3 bisects them all *at once*, with a single polynomial of degree  $O((2^{j-1})^{1/n}) = O(2^{j/n})$ .

The product of the polynomials from all rounds therefore has degree

$$\sum_{i \leq j} O(2^{i/n}) = O(2^{j/n}),$$

the sum being geometric. Setting  $2^j \approx D^n$  makes this  $O(D)$ , and the  $2^j \approx D^n$  pieces each contain  $N2^{-j} \approx N/D^n$  points. Each piece is a union of cells, so every cell is at least as small.

The whole content is that bisecting  $M$  sets costs degree  $M^{1/n}$  rather than  $M$  — so doubling the number of pieces costs only a factor  $2^{1/n}$  in degree, and the total degree is dominated by the last round.

**Proof. Step 1.** *Set up the rounds.*

Let  $j$  be the integer with  $2^j \leq D^n < 2^{j+1}$ . We construct polynomials  $P_1, \dots, P_j$  and, for each  $i$ , a partition of  $\mathbb{R}^n$  into at most  $2^i$  open regions.

*Round 1.* Apply Theorem 11.2.3 with  $M = 1$  and the set  $S$ : there is  $P_1$  of degree  $\leq C_n$  bisecting  $S$ . The two regions  $\{P_1 > 0\}$  and  $\{P_1 < 0\}$  each contain at most  $\frac{N}{2}$  points of  $S$ ; points on  $Z(P_1)$  are set aside and never counted again.

*Round  $i + 1$ .* Suppose after round  $i$  the complement of  $Z(P_1 \cdots P_i)$  is partitioned into at most  $2^i$  regions, each containing at most  $N2^{-i}$  points of  $S$ . Let  $S_1^{(i)}, \dots, S_{2^i}^{(i)}$  be the point sets in those regions, and apply Theorem 11.2.3 to them simultaneously: there is  $P_{i+1}$  of degree

$$\deg P_{i+1} \leq C_n (2^i)^{1/n}$$

bisecting all of them at once. Each region is now split into at most two sub-regions by the sign of  $P_{i+1}$ , each containing at most  $N2^{-(i+1)}$  points.

**Step 2.** *Bound the total degree.*

Set  $P = P_1 P_2 \cdots P_j$ . Then

$$\deg P = \sum_{i=1}^j \deg P_i \leq C_n \sum_{i=0}^{j-1} (2^i)^{1/n} = C_n \frac{2^{j/n} - 1}{2^{1/n} - 1} \leq C'_n 2^{j/n},$$

the sum being geometric with ratio  $2^{1/n} > 1$ , so dominated by its last term up to the factor  $(2^{1/n} - 1)^{-1}$ , which depends only on  $n$ . Since  $2^j \leq D^n$  we get

$2^{j/n} \leq D$ , hence  $\deg P \leq C'_n D$ . Rescaling the constant — apply the argument with  $D$  replaced by  $D/C'_n$  — gives  $\deg P \leq D$  as stated.

**Step 3.** *Bound the points per cell.*

After round  $j$  every region contains at most  $N2^{-j}$  points of  $S$ , and  $2^j > D^n/2$ , so each region contains at most  $2N/D^n$  points. Each cell of  $P$  — a connected component of  $\mathbb{R}^n \setminus Z(P)$  — lies inside a single region, because  $P$  has constant sign on a cell and the region is determined by the signs of the factors. Hence every cell contains at most  $2N/D^n$  points of  $S$ .  $\square$

**Where we are.** Three features deserve emphasis.

*Points on  $Z(P)$  are not controlled.* The theorem says nothing about  $S \cap Z(P)$ , which may be all of  $S$  — if  $S$  lies on a low-degree variety, the partitioning is vacuous. Every application must handle that case separately, and it is usually the harder one. Chapter 11.3 shows the pattern: a *cellular* case and an *algebraic* case.

*The degree is a free parameter.*  $D$  is chosen by the user, and the whole craft of the method is choosing it. Large  $D$  gives many small cells but a high-degree variety on which many points may hide; small  $D$  gives the reverse. Every argument in this volume optimises  $D$ .

*The cell count is what pays.* By Theorem 11.2.2 there are  $O(D^n)$  cells, and each contains  $O(N/D^n)$  points — so the bookkeeping is exactly balanced, and no cell is exceptional. That balance is why the recursion closes.

**Proposition 11.2.6** (Partitioning respects weights). *Let  $\mu$  be a finite measure on  $\mathbb{R}^n$  with  $\mu(\{x\}) = 0$  for every  $x$ , and let  $D \geq 1$ . There is a non-zero polynomial  $P$  of degree at most  $D$  such that every cell of  $P$  has  $\mu$ -measure at most  $C_n \mu(\mathbb{R}^n)/D^n$ .*

*Proof.* The bisection argument of Theorem 11.2.5 used only that the objects being halved were finite sets, and Theorem 11.2.3 holds verbatim for finite measures — the Veronese reduction of Remark 11.2.4 turns simultaneous bisection of  $M$  measures by a polynomial into simultaneous bisection by a hyperplane, and the classical ham sandwich theorem applies to measures.

Run the rounds as before: at round  $i+1$  apply the measure version to the  $\leq 2^i$  restrictions of  $\mu$  to the current regions, obtaining  $P_{i+1}$  of degree  $\lesssim (2^i)^{1/n}$  halving each. The atomlessness of  $\mu$  ensures no mass is stranded on  $Z(P_{i+1})$  — a hyperplane, hence a variety, is  $\mu$ -null only if  $\mu$  has no atoms there, and one may perturb to arrange it. Summing degrees as in Step 2 gives  $\deg P \leq D$ , and after  $j$  rounds with  $2^j \approx D^n$  each region, hence each cell, has measure  $\lesssim \mu(\mathbb{R}^n)/D^n$ .  $\square$

**Where we are.** The weighted version is what applications to *continuous* configurations need: a Besicovitch set is not a finite point set, and Proposition 11.2.6 is the form in which Remark 11.8.4's arguments use partitioning. The atomlessness hypothesis is necessary. A point mass cannot be halved by any polynomial, and a measure with an atom of mass exceeding  $\mu(\mathbb{R}^n)/D^n$  makes the conclusion false. In applications  $\mu$  is Lebesgue measure restricted to a neighbourhood, so the hypothesis is automatic.

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WORKED EXAMPLE 11.2.1— TIER A **The degree budget, computed**

The geometric sum in Step 2 is the whole reason the method works, and it repays seeing numerically.

*The sum.*  $\sum_{i=0}^{j-1} 2^{i/n}$  with  $n = 3$  and  $j = 12$ , so  $2^j = 4096$  pieces:

$$\sum_{i=0}^{11} 2^{i/3} = \frac{2^4 - 1}{2^{1/3} - 1} = \frac{15}{0.2599} \approx 57.7,$$

while the final term alone is  $2^{11/3} \approx 12.7$ . So the total is about 4.5 times the last term — a bounded factor, as claimed, and the constant  $(2^{1/3} - 1)^{-1} \approx 3.85$  is where it comes from.

*What this means.* Twelve rounds of bisection, producing 4096 pieces, cost total degree  $\approx 58$ . Had each round cost degree proportional to the number of sets bisected — which is what a naive bisection would cost — the total would be  $\sum_i 2^i \approx 4096$ , and the method would be useless.

*The exponent is everything.* Theorem 11.2.3 bisects  $M$  sets with degree  $M^{1/n}$ , not  $M$ . That  $1/n$  is the Veronese embedding: a polynomial of degree  $D$  has  $\approx D^n$  coefficients, so it has  $D^n$  degrees of freedom to spend on  $M$  bisection conditions. The method is, at bottom, the observation that polynomials are cheap.

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WORKED EXAMPLE 11.2.2— TIER B **Choosing  $D$ : the standard trade-off**

Every application of Theorem 11.2.5 begins by choosing  $D$ , and the choice follows a fixed pattern.

*The two costs.* With degree  $D$ :

- each cell has  $\lesssim N/D^n$  points, so *large  $D$  is good* for the cellular part;
- a line meets  $Z(P)$  in  $\leq D$  points, hence enters  $\leq D + 1$  cells, so *small  $D$  is good* for controlling how much a line is fragmented.

*The balance.* One writes the quantity to be bounded as

$$(\text{cellular contribution}) + (\text{algebraic contribution}),$$

where the first decreases and the second increases in  $D$ , and picks  $D$  where they agree. That value is almost always a power of  $N$  or of  $|L|$ , and the resulting exponent is the theorem.

*Worked instance, previewed.* In Theorem 11.3.2 the cellular part contributes  $\approx |L| D + |P||L|/D^2$  (per-cell count), and optimising gives  $D \approx |P|^{2/3}|L|^{-1/3}$  — from which the exponent  $\frac{2}{3}$  in the Szemerédi–Trotter bound emerges. The  $\frac{2}{3}$  is not guessed; it is where the two costs cross.

*The habit.* On meeting a polynomial-method proof, find the choice of  $D$  first. It is usually displayed in one line and it determines the entire result; the rest of the proof is verifying that the two contributions really are what they appear to be.

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**WORKED EXAMPLE 11.2.3— TIER C: RESEARCH TASTE Partitioning versus vanishing: two uses of one object**

A polynomial can be used against a configuration in two quite different ways, and the whole architecture of the polynomial method rests on the distinction.

(a) *Vanishing.* Ask for  $P$  with  $S \subseteq Z(P)$ . This is available when  $|S|$  is finite and small relative to  $\binom{D+n}{n}$ , and it is what Volume 10’s Theorem 10.2.1 and this volume’s Theorem 11.4.2 use. The conclusion is algebraic:  $P$  vanishes on lines, derivatives vanish, and a contradiction follows from degree.

(b) *Partitioning.* Ask for  $P$  whose cells each meet  $S$  in few points. This is Theorem 11.2.5 and is available always. The conclusion is combinatorial: the configuration is divided, and one recurses.

(c) *When each is appropriate.* Vanishing requires the ambient set to be discrete or the objects to be exact — Chapter 11.8 shows it succeeds for genuine segments and fails for tubes. Partitioning requires nothing, but gives a weaker conclusion: it never proves a contradiction outright, only that a count can be improved.

(d) *The historical order.* Vanishing came first — Dvir 2008, joints 2008 — and produced sharp results in the settings where it applies. Partitioning came in 2010 with Guth and Katz, and it is the version that reaches problems about continuous configurations. The subject moved from (a) to (b) precisely because the problems it wanted were of the second kind.

(e) *A diagnostic.* Given a problem, ask whether “ $P$  vanishes on the configu-

ration” is a strong hypothesis. Over a finite field with  $|S|$  a constant fraction of  $q^n$ , yes. Over  $\mathbb{R}$  with  $S$  of positive measure, it is impossible; with  $S$  a finite point set, it is possible but the conclusion may be vacuous, as Proposition 11.8.2 shows. Partitioning is the fallback and is almost always the right tool.

#### WORKED EXAMPLE 11.2.4— TIER A **Cells, counted**

Theorem 11.2.2 asserts  $\lesssim_n D^n$  cells for a degree- $D$  polynomial. Four examples, to see that the count is right and where it is attained.

(a)  $n = 1$ ,  $P$  of degree  $D$ . At most  $D$  roots, so at most  $D + 1$  intervals. Attained by  $\prod_{j=1}^D (x - j)$ .

(b)  $n = 2$ ,  $D$  generic lines.  $P = \prod_{j \leq D} \ell_j$  has degree  $D$ , and  $D$  lines in general position cut the plane into  $1 + D + \binom{D}{2} \approx \frac{D^2}{2}$  regions. So  $\approx D^2$  cells, matching  $D^n$  with  $n = 2$ .

(c)  $n = 3$ ,  $D$  generic planes. The count is  $1 + D + \binom{D}{2} + \binom{D}{3} \approx \frac{D^3}{6}$ , again matching  $D^n$ .

(d) A degenerate case.  $P = x_1^D$  has degree  $D$  but  $Z(P)$  is a single hyperplane and there are only 2 cells. So the bound is an upper bound and is far from attained by degenerate polynomials — which is harmless, since the partitioning theorem needs only that there are not *too many* cells.

*Why the exponent must be  $n$ .* A degree- $D$  polynomial has  $\approx \frac{D^n}{n!}$  coefficients, so it has that many degrees of freedom; one cannot manufacture more cells than one has parameters, up to constants. The matching of the cell count  $D^n$  with the dimension count  $D^n/n!$  is what makes Theorem 11.2.5’s bookkeeping balance exactly, as the Checkpoint there observed.

#### WORKED EXAMPLE 11.2.5— TIER B **Where the constants go**

Theorem 11.2.5 carries a constant  $C_n$ , and applications need to know it does not depend on  $N$  or  $D$ . Tracing it is worthwhile once.

*Source 1: the ham sandwich degree.* Theorem 11.2.3 bisects  $M$  sets with a polynomial of degree  $\leq C_n M^{1/n}$ , where  $C_n$  comes from the Veronese count  $\binom{D+n}{n} \geq M + 1$ . Solving gives  $C_n \approx (n!)^{1/n} \approx n/e$ .

*Source 2: the geometric sum.* Step 2 of Theorem 11.2.5 sums  $\sum 2^{i/n}$  and divides by  $2^{1/n} - 1$ . For large  $n$  that factor is  $\approx n/\ln 2$ , so the total degree constant grows linearly in  $n$ .

*Source 3: the cell count.* Theorem 11.2.2 contributes its own  $C_n$ , which in

the standard proofs is exponential in  $n$ .

*The upshot.*  $C_n$  grows with the dimension, exponentially in the worst source, but is *independent of  $N$  and  $D$*  — which is all any application needs, since  $D$  is chosen after the constant is fixed.

*Why one checks this.* Because an argument that chose  $D$  in terms of  $C_n$  and then applied a result whose constant depended on  $D$  would be circular. The order of quantifiers is: fix  $n$ , obtain  $C_n$ , then choose  $D$ . Every proof in this volume respects it, and Worked Example 11.3.2 noted that verifying  $D \geq 1$  is the accompanying check.

*A related caution.* In dimensions large enough for  $C_n$  to matter, the polynomial method's bounds degrade. That is one reason the sharp results of Worked Example 11.9.1 are all in  $\mathbb{R}^2$  or  $\mathbb{R}^3$ , and it compounds the classification difficulty of Worked Example 11.6.3.

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## Drills

**Drill 11.2.1.** Verify Theorem 11.2.2 for  $n = 1$ , and for  $n = 2$  with  $P$  a product of  $D$  distinct lines.

**Drill 11.2.2.** Carry out the Veronese reduction of Remark 11.2.4: check that a polynomial of degree  $\leq D$  in  $n$  variables becomes a linear functional on  $\mathbb{R}^{\binom{D+n}{n}-1}$ .

**Drill 11.2.3.** Compute the geometric sum of Step 2 for  $n = 2$  and  $j = 10$ , and compare with the last term.

**Drill 11.2.4.** Show that a cell of  $P = P_1 \cdots P_j$  lies in a single region of the partition built in Step 1, as claimed in Step 3.

**Where we are.** Two conventions for this volume, matching those of Volumes 9 and 10.

*$D$  is always the degree of the partitioning polynomial,* and it is a free parameter fixed by optimisation at the end of each argument, never at the start. Worked Example 11.2.2 gives the trade-off and Worked Example 11.3.2 the checks.

*Constants depend on  $n$  alone.* Never on  $D$ , on the number of points, or on the number of lines — Worked Example 11.2.5 traces where each factor comes from. An estimate whose constant depended on  $D$  would be circular, since  $D$  is chosen after the constant is fixed.



## Notes

Polynomial partitioning is due to Guth and Katz (2015), in the paper resolving the Erdős distinct distances problem; the ham sandwich formulation is due to Stone and Tukey (1942) and the polynomial version to Gromov. The cell count Theorem 11.2.2 is Milnor's and Thom's, from around 1964.

Remark 11.2.4 records that this is the only load-bearing import in the series.



## Chapter 11.3

# The Szemerédi–Trotter Theorem

**Definition 11.3.1.** For a finite set  $\mathcal{P}$  of points and a finite set  $\mathcal{L}$  of lines in  $\mathbb{R}^2$ , the number of *incidences* is

$$I(\mathcal{P}, \mathcal{L}) = \#\{(p, \ell) \in \mathcal{P} \times \mathcal{L} : p \in \ell\}.$$

**Theorem 11.3.2** (Szemerédi–Trotter). *For all finite  $\mathcal{P}, \mathcal{L}$  in  $\mathbb{R}^2$ ,*

$$I(\mathcal{P}, \mathcal{L}) \lesssim (|\mathcal{P}| |\mathcal{L}|)^{2/3} + |\mathcal{P}| + |\mathcal{L}|.$$

*Idea of the proof.* Partition the plane by a polynomial of degree  $D$ , chosen later. Every incidence is either *cellular* — the point lies in an open cell — or *algebraic* — the point lies on  $Z(P)$ .

For the cellular part, each cell contains  $\lesssim |\mathcal{P}|/D^2$  points by Theorem 11.2.5, and a line meets  $\leq D + 1$  cells because it crosses  $Z(P)$  at most  $D$  times. So the incidences within cells are controlled by the trivial bound applied cell by cell, and the smallness of the cells does the work.

For the algebraic part, a line either lies inside  $Z(P)$  — and there are at most  $D$  such lines, since  $Z(P)$  contains at most  $D$  lines — or meets it in  $\leq D$  points. Either way the count is small.

Balancing the two in  $D$  gives the exponent  $\frac{2}{3}$ .

**Lemma 11.3.3** (Trivial incidence bounds). *For all finite  $\mathcal{P}, \mathcal{L}$  in  $\mathbb{R}^2$ ,*

$$I(\mathcal{P}, \mathcal{L}) \leq |\mathcal{P}|^{1/2} |\mathcal{L}| + |\mathcal{P}|, \quad \text{and} \quad I(\mathcal{P}, \mathcal{L}) \leq |\mathcal{L}|^{1/2} |\mathcal{P}| + |\mathcal{L}|.$$

*Proof.* Count pairs of points on a common line. If a line  $\ell$  contains  $k_\ell$  points of  $\mathcal{P}$  it accounts for  $\binom{k_\ell}{2}$  such pairs, and distinct lines share at most one point,

so no pair is counted twice:

$$\sum_{\ell} \binom{k_{\ell}}{2} \leq \binom{|\mathcal{P}|}{2}.$$

Hence  $\sum_{\ell} k_{\ell}^2 \leq |\mathcal{P}|^2 + \sum_{\ell} k_{\ell}$ , and by Cauchy–Schwarz

$$I^2 = \left( \sum_{\ell} k_{\ell} \right)^2 \leq |\mathcal{L}| \sum_{\ell} k_{\ell}^2 \leq |\mathcal{L}|(|\mathcal{P}|^2 + I).$$

If  $I \leq |\mathcal{P}|$  the first bound holds trivially; otherwise  $I^2 \leq 2|\mathcal{L}||\mathcal{P}|^2$ , giving  $I \lesssim |\mathcal{P}||\mathcal{L}|^{1/2}$  — which is the second bound. The first follows by exchanging the roles of points and lines, using duality in the projective plane.  $\square$

*Proof of Theorem 11.3.2.* Write  $P = |\mathcal{P}|$ ,  $L = |\mathcal{L}|$ .

**Step 0.** *Reduce to  $P \leq L$  by duality.*

Point–line duality in the projective plane sends a point  $(a, b)$  to the line  $y = ax - b$  and the line  $y = cx + d$  to the point  $(c, -d)$ , and preserves incidence. So  $I(\mathcal{P}, \mathcal{L}) = I(\mathcal{L}^*, \mathcal{P}^*)$  with  $|\mathcal{L}^*| = L$  points and  $|\mathcal{P}^*| = P$  lines. Since the asserted bound  $(PL)^{2/3} + P + L$  is symmetric in  $P$  and  $L$ , we may assume

$$P \leq L.$$

(Vertical lines and points at infinity are handled by a preliminary rotation, which changes no incidence.)

**Step 1.** *Dispose of very few points.*

If  $P < L^{1/2}$  then by Lemma 11.3.3,

$$I \leq PL^{1/2} + L < L^{1/2} \cdot L^{1/2} + L = 2L,$$

which is within the claimed bound. So assume  $L^{1/2} \leq P \leq L$ , and set

$$D = \frac{P^{2/3}}{L^{1/3}},$$

which satisfies  $D \geq 1$  precisely because  $P \geq L^{1/2}$ .

**Step 2.** *Partition.*

By Theorem 11.2.5 there is a non-zero  $Q$  of degree  $\leq D$  whose cells each contain  $\lesssim P/D^2$  points of  $\mathcal{P}$ , and by Theorem 11.2.2 there are  $\lesssim D^2$  cells. Split  $I = I_{\text{cell}} + I_{\text{alg}}$  according to whether the incidence’s point lies in an open cell or on  $Z(Q)$ .

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**Step 3.** *The cellular part.*

Let  $\Omega_1, \dots, \Omega_m$  be the cells,  $m \lesssim D^2$ , with  $\mathcal{P}_i = \mathcal{P} \cap \Omega_i$  and  $\mathcal{L}_i$  the lines meeting  $\Omega_i$ . A line not contained in  $Z(Q)$  meets  $Z(Q)$  in at most  $D$  points — restricting  $Q$  to it gives a non-zero one-variable polynomial of degree  $\leq D$  — so it passes through at most  $D + 1$  cells, and

$$\sum_i |\mathcal{L}_i| \leq (D + 1)L.$$

By Lemma 11.3.3 in each cell and Cauchy–Schwarz,

$$I_{\text{cell}} \lesssim \sum_i \left( |\mathcal{P}_i| |\mathcal{L}_i|^{1/2} + |\mathcal{L}_i| \right) \lesssim \frac{P}{D^2} m^{1/2} \left( \sum_i |\mathcal{L}_i| \right)^{1/2} + DL \lesssim \frac{PL^{1/2}}{D^{1/2}} + DL.$$

**Step 4.** *The algebraic part.*

*Lines inside  $Z(Q)$ .* A polynomial of degree  $D$  vanishing on more than  $D$  distinct lines vanishes identically: restrict to a line meeting all of them in distinct points, obtaining a one-variable polynomial of degree  $\leq D$  with more than  $D$  roots. So at most  $D$  lines lie in  $Z(Q)$ , and by Lemma 11.3.3 applied to them and all  $P$  points they carry  $\lesssim PD^{1/2} + D$  incidences.

*Lines not inside  $Z(Q)$ .* Each meets  $Z(Q)$  in  $\leq D$  points, hence carries  $\leq D$  incidences with  $\mathcal{P} \cap Z(Q)$ , for a total  $\lesssim DL$ .

So  $I_{\text{alg}} \lesssim PD^{1/2} + DL$ .

**Step 5.** *Substitute.*

With  $D = P^{2/3}L^{-1/3}$ ,

$$\frac{PL^{1/2}}{D^{1/2}} = P^{2/3}L^{2/3}, \quad DL = P^{2/3}L^{2/3}, \quad PD^{1/2} = P^{4/3}L^{-1/6}.$$

The third is at most the first exactly when  $P^{2/3} \leq L^{5/6}$ , that is  $P \leq L^{5/4}$  — which holds, since Step 0 gives  $P \leq L$ . Therefore

$$I \lesssim P^{2/3}L^{2/3},$$

and adding the  $P + L$  from the disposed range completes the proof.  $\square$

**Where we are.** The proof has the two-case shape that every polynomial-method argument has, and it is worth naming the cases once.

*The cellular case* is where the method earns its keep: the points are spread thinly among  $D^2$  cells and each line visits few of them, so the trivial bound applied locally beats the trivial bound applied globally.

*The algebraic case* is where the difficulty is deferred. Points on  $Z(Q)$  are

not controlled by the partitioning at all, and one must argue separately — here easily, because a curve of degree  $D$  contains at most  $D$  lines. In  $\mathbb{R}^3$  and beyond the algebraic case is genuinely hard, and the whole content of the Guth–Katz theorem is its treatment.

The *optimisation* in Step 4 is the trade of Worked Example 11.2.2, and the exponent  $\frac{2}{3}$  is simply where the two costs cross. It is not an artefact: Worked Example 11.3.1 shows a configuration attaining it.

**Corollary 11.3.4** (Rich lines). *Let  $\mathcal{P}$  be a set of  $P$  points in  $\mathbb{R}^2$  and  $k \geq 2$ . The number of lines containing at least  $k$  points of  $\mathcal{P}$  is*

$$\lesssim \frac{P^2}{k^3} + \frac{P}{k}.$$

*Proof.* Let  $\mathcal{L}_k$  be the set of such lines and  $L = |\mathcal{L}_k|$ . Every line of  $\mathcal{L}_k$  carries at least  $k$  incidences, so  $kL \leq I(\mathcal{P}, \mathcal{L}_k)$ , and by Theorem 11.3.2

$$kL \lesssim (PL)^{2/3} + P + L.$$

If the first term dominates,  $kL \lesssim P^{2/3}L^{2/3}$  gives  $L^{1/3} \lesssim P^{2/3}/k$ , that is  $L \lesssim P^2/k^3$ . If the second dominates,  $kL \lesssim P$  gives  $L \lesssim P/k$ . If the third dominates,  $kL \lesssim L$  forces  $k \lesssim 1$ , excluded for  $k \geq 2$  after adjusting constants. Taking the maximum of the two surviving cases gives the claim.  $\square$

**Where we are.** Corollary 11.3.4 is the form in which Szemerédi–Trotter is usually applied, because the number of lines is rarely given in advance — one knows only that they are rich. Worked Example 11.3.3 lists three applications, and each uses this version.

The two terms have distinct meanings.  $P^2/k^3$  is the interesting regime, dominating when  $k \lesssim P^{1/2}$ ; and  $P/k$  is trivial, being the bound one gets by observing that the incidences  $kL$  cannot exceed the total  $P$  times the average number of rich lines per point. The crossover at  $k \approx P^{1/2}$  is where a line becomes so rich that it accounts for a constant fraction of the points.

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**WORKED EXAMPLE 11.3.1—TIER A The bound is sharp**

Take the grid  $\mathcal{P} = \{0, \dots, n-1\} \times \{0, \dots, 2n^2-1\}$ , so  $P = 2n^3$ , and let  $\mathcal{L}$  be the lines  $y = ax + b$  with  $0 \leq a < n$ ,  $0 \leq b < n^2$ , so  $L = n^3$ .

*Every line is fully incident.* For  $x \in \{0, \dots, n-1\}$  and such  $a, b$ ,

$$ax + b < (n-1)(n-1) + n^2 < 2n^2,$$

so the point  $(x, ax + b)$  lies in  $\mathcal{P}$ . Each line therefore contains  $n$  points of the

grid, and

$$I = nL = n^4.$$

*Compare the theorem.*  $(PL)^{2/3} = (2n^6)^{2/3} \approx 1.6n^4$ . So the grid attains the bound up to a constant, and the exponent  $\frac{2}{3}$  cannot be improved.

*Numerically.* At  $n = 3, 4, 5, 6$  the grid gives  $I = 81, 256, 625, 1296$  against  $(PL)^{2/3} + P + L = 210, 598, 1367, 2705$  — within a factor of three throughout, the constant reflecting the crude  $2n^2$  in the construction.

*Why the grid is extremal.* It is the configuration with the most additive structure: points and lines both come from arithmetic progressions, so incidences are as abundant as the arithmetic allows. Worked Example 10.3.7 in Volume 10 recorded that progressions are exactly the sets of small doubling, and the connection is not a coincidence — the Szemerédi–Trotter theorem is one of the standard tools for proving sum–product estimates.

#### WORKED EXAMPLE 11.3.2— TIER B **Reading the optimisation**

Step 5 is the crux and its structure is worth extracting, since every later chapter repeats it.

*Three terms.* Before substituting, the bound is

$$\underbrace{\frac{PL^{1/2}}{D^{1/2}}}_{\text{cellular}} + \underbrace{PD^{1/2}}_{\text{lines inside } Z(Q)} + \underbrace{DL}_{\text{crossings}}.$$

The first decreases in  $D$ ; the second and third increase. So an optimum exists, and it is found by balancing the decreasing term against the dominant increasing one.

*Which increasing term dominates.* Comparing  $PD^{1/2}$  with  $DL$  amounts to comparing  $P$  with  $D^{1/2}L$ . At the optimum this is  $P$  against  $P^{1/3}L^{5/6}$ , and the crossings term dominates exactly when  $P \leq L^{5/4}$ .

*Why the proof does not need a second case.* Because Step 0 has already reduced to  $P \leq L$  by duality, and  $L \leq L^{5/4}$ . *Without that reduction the argument would be incomplete:* for  $P > L^{5/4}$  the term  $PD^{1/2}$  exceeds  $P^{2/3}L^{2/3}$ , and the trivial bound  $I \leq PL^{1/2} + L$  does not cover the gap either, since  $PL^{1/2} \leq P^{2/3}L^{2/3}$  requires  $P \leq L^{1/2}$ . Duality is doing real work, not cosmetic tidying.

*The general recipe.* Write the bound as a sum of powers of  $D$  with coefficients in  $P, L$ ; identify which terms increase and which decrease; balance the decreasing term against the dominant increasing one; then *check the remaining terms are no larger in the range under consideration*, and if they are not, find a symmetry or split into regimes.

And check  $D \geq 1$ . The optimum  $D = P^{2/3}L^{-1/3}$  is admissible only when  $P \geq L^{1/2}$ , which is why Step 1 disposes of the complementary range first. Forgetting this is the commonest error in polynomial-method arguments.

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#### WORKED EXAMPLE 11.3.3—TIER B **Applying the incidence bound**

Theorem 11.3.2 is one of the most-used theorems in combinatorics, and the applications follow a small number of patterns.

(a) *Rich lines.* A line is  $k$ -rich if it contains at least  $k$  points of  $\mathcal{P}$ . Counting incidences from the rich lines alone,  $k \cdot \#\{k\text{-rich lines}\} \leq I \lesssim (PL)^{2/3} + P + L$  with  $L = \#\{k\text{-rich}\}$ , and solving gives

$$\#\{k\text{-rich lines}\} \lesssim \frac{P^2}{k^3} + \frac{P}{k}.$$

This is the form most often quoted, and it is what one uses when the number of lines is not given in advance.

(b) *Sum-product.* If  $A \subseteq \mathbb{R}$  is finite, consider the points  $A \times A$  and the lines  $y = a(x-b)$  for  $a, b \in A$ . Each line contains  $|A|$  points of the grid, so  $I \geq |A|^3$ , while Theorem 11.3.2 gives  $I \lesssim (|A|^2|A|^2)^{2/3} = |A|^{8/3}$  — a contradiction unless the point set is larger than assumed. Chasing the contradiction yields  $\max\{|A + A|, |A \cdot A|\} \gtrsim |A|^{5/4}$ , Elekes’s bound.

(c) *Distinct distances, weakly.* Taking the  $N$  points and the  $\approx N^2$  perpendicular bisectors gives, via incidences, the bound  $|\Delta| \gtrsim N^{2/3}$  — far from Theorem 11.7.8, but obtained in two lines. It was the state of the art in the 1980s.

*The habit.* To apply the theorem, identify a point set and a line family in which the configuration of interest forces many incidences, then compare with the bound. The art is choosing the family; the theorem is then mechanical.

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#### WORKED EXAMPLE 11.3.4—TIER A **The theorem in the two extreme regimes**

Theorem 11.3.2 has three terms, and which dominates is worth knowing.

*Balanced:*  $P \approx L$ . Then  $(PL)^{2/3} \approx P^{4/3}$ , which dominates  $P$  and  $L$ . This is the interesting regime and the one the grid of Worked Example 11.3.1 realises: with  $P \approx L \approx n^3$  the bound is  $n^4$ , attained.

*Few points:*  $P \lesssim L^{1/2}$ . Then  $L$  dominates: every line carries at most  $P$  incidences, but more usefully the trivial bound gives  $I \lesssim L$ . Each line meets the point set in  $O(1)$  points on average.



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*Few lines:*  $L \lesssim P^{1/2}$ . Symmetrically  $P$  dominates, by duality.

*The crossovers.* At  $P = L^{1/2}$  the terms  $(PL)^{2/3}$  and  $L$  agree:  $(L^{1/2}L)^{2/3} = L$ . ✓ And Step 1 of the proof disposes of exactly the range below that crossover, which is why the case division there is not arbitrary.

*Practical consequence.* An application in the balanced regime uses the  $(PL)^{2/3}$  term and nothing else; the other two terms exist only so that the statement is true in the degenerate ranges. Corollary 11.3.4 makes this explicit — its two terms are the balanced and degenerate contributions respectively.

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## Drills

**Drill 11.3.1.** Verify the pair-counting in Lemma 11.3.3, and check the case distinction  $I \leq |\mathcal{P}|$ .

**Drill 11.3.2.** Confirm that a line not contained in  $Z(Q)$  meets it in at most  $D$  points, and deduce that it passes through at most  $D + 1$  cells.

**Drill 11.3.3.** Verify Worked Example 11.3.1 for  $n = 4$ : count  $P$ ,  $L$  and  $I$  explicitly, and compare with  $(PL)^{2/3}$ .

**Drill 11.3.4.** Redo the optimisation of Step 4 keeping all three terms, and determine the exact range of  $P, L$  in which each pairing binds.

**Where we are.** Three proofs of Szemerédi–Trotter exist and the choice between them is informative.

*The original* (1983) uses a cell decomposition by lines, chosen randomly, and is intricate. *Székely’s* (1997) uses the crossing number inequality from topological graph theory and is four lines long. *The one given here* uses polynomial partitioning and is a page.

Székely’s is the shortest, so why present another? Because it does not generalise: the crossing number inequality is planar, and there is no analogue in  $\mathbb{R}^3$ . The partitioning proof, by contrast, is the two-case argument of the Checkpoint above, and Chapter 11.6 shows exactly which case becomes hard in three dimensions. A proof is worth learning for what it survives, not only for its length.

## Notes

Szemerédi and Trotter proved the theorem in 1983 by a cell decomposition argument; Székely later gave a short proof via the crossing number inequality. The polynomial-partitioning proof given here is Kaplan, Matoušek and Sharir’s, and Guth’s, and it is the one that generalises to higher dimensions.

The sharpness example is Erdős’s and predates the theorem.

## Chapter 11.4

# The Joints Theorem

Chapter 11.3 used partitioning: a polynomial cut space into cells and the configuration was handled cell by cell. This chapter uses the older device — a polynomial *vanishing* on the configuration — and shows that over  $\mathbb{R}$  it is not always useless. The difference from Volume 10 is that the objects here are genuine lines, not tubes, so the propagation step of Proposition 10.7.2 is available.

### 11.4.1 Statement

**Definition 11.4.1.** Let  $\mathcal{L}$  be a finite set of lines in  $\mathbb{R}^3$ . A point  $p \in \mathbb{R}^3$  is a *joint* of  $\mathcal{L}$  if at least three lines of  $\mathcal{L}$  pass through  $p$  and their directions span  $\mathbb{R}^3$ .

**Theorem 11.4.2** (Joints theorem). *A set of  $L$  lines in  $\mathbb{R}^3$  has at most  $7L^{3/2}$  joints.*

*Idea of the proof.* Suppose there were more. Take a non-zero polynomial  $P$  of *minimal degree* vanishing at every joint; a dimension count bounds its degree  $D$  by roughly  $J^{1/3}$ .

If every line carried more than  $D$  joints, then  $P$  restricted to a line would be a one-variable polynomial of degree  $\leq D$  with more than  $D$  roots, hence identically zero — so  $P$  would vanish on every line. But then at a joint, three non-coplanar lines lie in  $Z(P)$ , and the gradient of  $P$ , being orthogonal to each of their directions, would have to vanish. So every partial derivative of  $P$  vanishes at every joint, has degree  $D - 1$ , and is non-zero for some index unless  $P$  is constant — contradicting minimality.

The one thing to arrange is that every line does carry more than  $D$  joints, and that is a pruning argument: discard lines with few joints, which de-

destroys few joints.

**Lemma 11.4.3** (Pruning). *Let  $\mathcal{L}$  be a set of  $L$  lines with joint set  $J$ . Then there are a subset  $\mathcal{L}' \subseteq \mathcal{L}$  and a set  $J'$  of joints of  $\mathcal{L}'$  with*

$$|J'| \geq \frac{1}{2}|J| \quad \text{and} \quad \text{every line of } \mathcal{L}' \text{ contains at least } \frac{|J|}{2L} \text{ points of } J'.$$

*Proof.* Run the following process. Set  $\mathcal{L}_0 = \mathcal{L}$  and  $J_0 = J$ . At stage  $i$ , if some line  $\ell \in \mathcal{L}_i$  contains fewer than  $\frac{|J|}{2L}$  points of  $J_i$ , delete it: put  $\mathcal{L}_{i+1} = \mathcal{L}_i \setminus \{\ell\}$  and let  $J_{i+1}$  be the set of joints of  $\mathcal{L}_{i+1}$  that lie in  $J_i$ . Otherwise stop.

Each deletion destroys only joints lying on the deleted line — a joint not on  $\ell$  retains its three non-coplanar lines — so at most  $\frac{|J|}{2L}$  joints are lost per stage. At most  $L$  stages occur, so the total loss is less than  $L \cdot \frac{|J|}{2L} = \frac{|J|}{2}$ .

The process terminates, and on termination every remaining line contains at least  $\frac{|J|}{2L}$  points of the remaining joint set.  $\square$

**Lemma 11.4.4** (Vanishing at a joint kills the gradient). *Let  $P$  be a polynomial vanishing on three lines through a point  $p$  whose directions span  $\mathbb{R}^3$ . Then  $\nabla P(p) = 0$ .*

*Proof.* Let  $v$  be the direction of one such line, so  $t \mapsto P(p + tv)$  is identically zero. Differentiating at  $t = 0$  gives  $\nabla P(p) \cdot v = 0$ . Applying this to the three directions  $v_1, v_2, v_3$ , which span  $\mathbb{R}^3$ , forces  $\nabla P(p) = 0$ .  $\square$

## 11.4.2 Proof

*Proof of Theorem 11.4.2.* Suppose for contradiction that  $\mathcal{L}$  is a set of  $L$  lines with

$$|J| > 7L^{3/2}. \quad (11.4.1)$$

**Step 1.** *Prune.*

By Lemma 11.4.3 there are  $\mathcal{L}'$  and  $J'$  with  $|J'| \geq \frac{1}{2}|J|$  and every line of  $\mathcal{L}'$  carrying at least  $\frac{|J|}{2L}$  points of  $J'$ .

**Step 2.** *A minimal polynomial.*

Among all non-zero polynomials vanishing on  $J'$ , choose one,  $P$ , of minimal degree  $D$ . Such a polynomial exists: the space of polynomials of degree at most  $d$  in three variables has dimension  $\binom{d+3}{3} > \frac{d^3}{6}$ , so once  $d^3 > 6|J'|$  the evaluation map at  $J'$  has non-trivial kernel. Hence

$$D \leq (6|J'|)^{1/3} \leq (6|J|)^{1/3}. \quad (11.4.2)$$

Note  $D \geq 1$ : a non-zero constant does not vanish on the non-empty set  $J'$ .

**Step 3.** *Every line of  $\mathcal{L}'$  lies in  $Z(P)$ .*

By Step 1 each  $\ell \in \mathcal{L}'$  carries at least  $\frac{|J|}{2L}$  points of  $J'$ , at all of which  $P$  vanishes. We claim that this exceeds  $D$ .

By (11.4.2) it is enough that  $\frac{|J|}{2L} > (6|J|)^{1/3}$ , and raising both sides to the power 3 and rearranging, that is

$$\frac{|J|^3}{8L^3} > 6|J| \iff |J|^2 > 48L^3 \iff |J| > \sqrt{48}L^{3/2}.$$

Since  $\sqrt{48} = 4\sqrt{3} \approx 6.93 < 7$ , assumption (11.4.1) gives exactly this. So each  $\ell \in \mathcal{L}'$  contains more than  $D$  zeros of  $P$ .

Restricting  $P$  to  $\ell$  gives a one-variable polynomial of degree at most  $D$  with more than  $D$  roots; it is therefore identically zero, so  $\ell \subseteq Z(P)$ .

**Step 4.** *The gradient vanishes on  $J'$ .*

Let  $p \in J'$ . Being a joint of  $\mathcal{L}'$ , it lies on three lines of  $\mathcal{L}'$  with spanning directions, and by Step 3 all three lie in  $Z(P)$ . By Lemma 11.4.4,  $\nabla P(p) = 0$ .

**Step 5.** *Contradict minimality.*

Each partial derivative  $\partial_i P$  therefore vanishes on all of  $J'$ , and has degree at most  $D - 1$ . By minimality of  $D$ , each  $\partial_i P$  must be the zero polynomial. But a polynomial all of whose partial derivatives vanish identically is constant, and a non-zero constant does not vanish on  $J' \neq \emptyset$ . This contradiction refutes (11.4.1).  $\square$

**Where we are.** The proof is four moves — prune, minimise, propagate, differentiate — and the last is what distinguishes it from Dvir’s argument in Volume 10.

There, the contradiction came from Schwartz–Zippel: a low-degree polynomial cannot vanish everywhere. Here it comes from *minimality*: the derivative has smaller degree and vanishes on the same set, so it must be zero. The derivative trick replaces the finiteness of the field, and it works over  $\mathbb{R}$  because differentiation lowers degree in any characteristic zero field. That is why the joints theorem transfers and Dvir’s does not. Both use the propagation step — a polynomial vanishing at more than  $D$  points of a *line* vanishes on it — and here the objects genuinely are lines. Proposition 10.7.2 in Volume 10 concerned tubes, which is a different problem.

**Proposition 11.4.5** (Joints in  $\mathbb{R}^n$ ). *Let  $\mathcal{L}$  be a set of  $L$  lines in  $\mathbb{R}^n$ , and call  $p$  a joint if  $n$  lines of  $\mathcal{L}$  pass through  $p$  with spanning directions. Then the number of joints is  $\lesssim_n L^{n/(n-1)}$ .*

*Proof.* The argument of Theorem 11.4.2 goes through with 3 replaced by  $n$ .

Prune as in Lemma 11.4.3 to arrange that every line carries at least  $\frac{|J|}{2L}$  joints while retaining at least half of them. Take  $P$  non-zero of minimal degree  $D$  vanishing on the pruned joint set  $J'$ ; the dimension count  $\binom{D+n}{n} > \frac{D^n}{n!}$  gives  $D \leq (n!|J|)^{1/n}$ .

If  $\frac{|J|}{2L} > D$  then  $P$  vanishes on every pruned line, and at each joint the  $n$  spanning directions force  $\nabla P = 0$  exactly as in Lemma 11.4.4. Minimality then forces every  $\partial_i P$  to vanish identically, so  $P$  is constant — a contradiction.

Therefore  $\frac{|J|}{2L} \leq D \leq (n!|J|)^{1/n}$ , which rearranges to

$$|J|^{1-\frac{1}{n}} \leq 2L(n!)^{1/n}, \quad \text{that is} \quad |J| \lesssim_n L^{\frac{n}{n-1}}. \quad \square$$

**Where we are.** At  $n = 3$  the exponent is  $\frac{3}{2}$ , recovering Theorem 11.4.2. At  $n = 2$  it is 2, and the statement is that  $L$  lines have  $\lesssim L^2$  crossing points — trivially true, and the theorem says nothing. The content grows with  $n$ . The proof is unchanged from the three-dimensional case except for the constants, which is unusual in this subject: Worked Example 11.6.3 explains that the *incidence* theorem for lines in  $\mathbb{R}^n$  has no such uniformity, because its algebraic case depends on classifying varieties containing many lines. The joints argument avoids that entirely by never asking what  $Z(P)$  looks like — only that its degree drops under differentiation.

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WORKED EXAMPLE 11.4.1— TIER A **The bound is sharp**

Take the  $k \times k \times k$  grid of axis-parallel lines: for each pair  $(i, j) \in \{1, \dots, k\}^2$  include the three lines

$$\{(t, i, j)\}, \quad \{(i, t, j)\}, \quad \{(i, j, t)\}.$$

So  $L = 3k^2$ .

*The joints.* Every lattice point  $(i, j, m)$  with  $1 \leq i, j, m \leq k$  lies on one line of each family, and the three directions  $e_1, e_2, e_3$  span  $\mathbb{R}^3$ . So  $J = k^3$ .

*Comparison.*  $L^{3/2} = (3k^2)^{3/2} = 3^{3/2}k^3$ , so

$$\frac{J}{L^{3/2}} = \frac{1}{3^{3/2}} \approx 0.192,$$

independent of  $k$ . The exponent  $\frac{3}{2}$  is therefore exactly right, and only the constant is in question.

*Numerically.* At  $k = 2, 5, 20, 50$  the grid has  $(L, J) = (12, 8), (75, 125), (1200, 8000), (7500, 125000)$ , and  $J/L^{3/2}$  is 0.192 throughout.

*What sharpness costs.* The extremal configuration is a lattice — maximal arithmetic structure, as in Worked Example 11.3.1. That is the recurring pattern: extremal examples in incidence geometry are grids, and the difficulty of a theorem is the difficulty of ruling out configurations that are almost grids.

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**WORKED EXAMPLE 11.4.2— TIER B The derivative trick**

Step 5 is the move worth learning, since it recurs throughout the polynomial method.

*The pattern.* One has a non-zero  $P$  of minimal degree vanishing on a set  $S$ , and one shows that some derived polynomial — a partial derivative, or a directional derivative, or a resultant — also vanishes on  $S$  and has smaller degree. Minimality forces it to be zero, and one iterates or concludes.

*Why minimality is the right hypothesis.* Taking  $P$  merely to have degree  $\leq D$  would allow  $\partial_i P$  to be a non-zero polynomial of degree  $D - 1$  vanishing on  $S$ , which is no contradiction. It is only because *no* non-zero polynomial of degree  $< D$  vanishes on  $S$  that the conclusion follows. Choosing minimal degree costs nothing and buys the contradiction.

*What the geometry must supply.* A reason for the derivative to vanish. Here it is Lemma 11.4.4: three non-coplanar directions at a point force  $\nabla P = 0$ . In other applications one uses that a point is singular, or that a surface is tangent to a line to high order.

*A variant worth knowing.* If each point of  $S$  has only *two* non-parallel lines through it, the gradient need not vanish — it is merely orthogonal to a plane, so it is determined up to scale. One then works with the *second* fundamental form or with a well-chosen directional derivative, and the resulting statements are the flecnode and ruled-surface lemmas that the next chapter will use. The joints theorem is the case where the geometry is generous enough that a single derivative suffices.

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**WORKED EXAMPLE 11.4.3— TIER C: RESEARCH TASTE Why the joints theorem transfers and Dvir's does not**

Both are polynomial-vanishing arguments about lines. One is a finite-field theorem with no real analogue; the other is a real theorem. The difference repays being stated exactly.

(a) *The objects.* Dvir's hypothesis is that a *set* contains a line in every direction. The joints hypothesis is that a *finite family of lines* is given, and one counts special points. In the second case the lines are part of the data — they are exact, not approximated by tubes.

(b) *The propagation step.* Both need “ $P$  vanishes at enough points of a line, hence on the line”. Dvir gets “enough” from  $q > D$ ; the joints argument gets it from the pruning lemma, which *arranges* that every line carries more than  $D$  joints. Both are legitimate because the lines are exact.

(c) *The contradiction.* Dvir uses Schwartz–Zippel — a finiteness statement. The joints argument uses minimality of the degree and the derivative trick, which works in any field of characteristic zero. So the joints argument is the more portable of the two.

(d) *What neither can do.* Handle tubes. Proposition 10.7.2 in Volume 10 is about  $\delta$ -neighbourhoods, and neither argument survives the replacement of a line by a tube — the propagation step fails identically in both.

(e) *The lesson for the Kakeya problem.* A Besicovitch set does contain exact segments, and Theorem 11.8.1 runs Dvir’s argument on them successfully. That the conclusion is nonetheless vacuous is a separate matter, and it is the subject of Chapter 11.8. The moral is that “the objects are exact” is necessary for these arguments and nowhere near sufficient for them to say anything.

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#### WORKED EXAMPLE 11.4.4— TIER B **Pruning: the technique in isolation**

Lemma 11.4.3 discards objects that carry little of the structure, so that the survivors are uniformly rich. The device recurs and is worth extracting.

*The setup.* A family  $\mathcal{F}$  of objects and a set  $S$  of configurations, each configuration involving several objects. One wants every object to carry many configurations.

*The rule.* Repeatedly delete any object carrying fewer than  $\theta$  configurations, where  $\theta$  is chosen so that the total loss — at most  $|\mathcal{F}|\theta$  — is less than half of  $|S|$ . Here  $\theta = \frac{|J|}{2L}$  and the loss is below  $\frac{|J|}{2}$ .

*The two things to check.* First, that deleting an object destroys only configurations involving it — true for joints, since a joint not on the deleted line keeps its three spanning lines. Second, that the process terminates, which it does because  $\mathcal{F}$  is finite.

*Where it appears elsewhere.* Volume 3’s Calderón–Zygmund decomposition (Worked Example 3.1.5) selects cubes by a stopping rule and uses the parent’s failure to be selected; Volume 7’s  $5r$ -lemma (Theorem 7.4.1) selects greedily by radius; Volume 10’s Theorem 10.3.8 takes a maximal disjoint family. Four instances of one idea: *select greedily, then use maximality or the threshold to bound what was discarded.*

*The characteristic error.* Choosing  $\theta$  too large, so that more than half the configurations are lost and the conclusion is vacuous. The check is always the same product  $|\mathcal{F}|\theta$  against  $|S|$ , and it should be verified explicitly — as Step 1 of Lemma 11.4.3 does.

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## Drills

**Drill 11.4.1.** Verify Lemma 11.4.4 by differentiating  $t \mapsto P(p + tv)$  and check that three spanning directions are needed.

**Drill 11.4.2.** Confirm the termination and the joint-loss count in Lemma 11.4.3.

**Drill 11.4.3.** Check Worked Example 11.4.1 for  $k = 3$ : count  $L$ ,  $J$  and the ratio  $J/L^{3/2}$ .

**Drill 11.4.4.** Show that a polynomial in  $\mathbb{R}[x_1, x_2, x_3]$  all of whose partial derivatives vanish identically is constant, and identify where characteristic zero is used.

**Where we are.** Two features of Theorem 11.4.2 are worth contrasting with everything else in this volume.

*It uses vanishing, not partitioning.* No cells appear, no degree is optimised, and the polynomial's zero set is never examined. That makes it the simplest argument here and also the least generalisable: it needs the objects to be exact lines and the hypothesis to be about points where several meet.

*Its bound is sharp.* Worked Example 11.4.1 exhibits the grid attaining  $L^{3/2}$  up to the constant  $3^{-3/2}$ . Among the results of this volume only this one, Theorem 11.3.2 and Theorem 11.5.2 are sharp, and all three concern exact objects — lines, or coordinate projections. The pattern of Worked Example 11.9.1 has no exceptions.

## Notes

The joints problem was posed in the 1990s and resisted combinatorial attack; the bound  $L^{3/2}$  was proved by Guth and Katz in 2008 using the polynomial method, and the short proof given here is due independently to Kaplan, Matoušek and Sharir, and to Quilodrán.

That a problem open for fifteen years fell to a page of algebra was, together with Dvir's theorem, what established the polynomial method in this area.



## Chapter 11.5

# Multilinear Kakeya

The Kakeya conjecture asks about tubes pointing in *all* directions, and Volume 9 showed the difficulty lies with the nearly-parallel ones. The multilinear version removes them by hypothesis: one considers  $n$  families of tubes whose directions are *transverse* — separated by a fixed angle — and asks for a bound on the overlap of one tube from each family. Guth proved the sharp such estimate, and it is the input Volume 12 uses.

This chapter proves the model case, from which the general statement differs only in bookkeeping, and states Guth’s theorem.

### 11.5.1 The model case: Loomis–Whitney

**Notation 11.5.1.** For  $x = (x_1, \dots, x_n) \in \mathbb{R}^n$  let  $\pi_j(x)$  delete the  $j$ -th coordinate, so  $\pi_j : \mathbb{R}^n \rightarrow \mathbb{R}^{n-1}$ .

**Theorem 11.5.2** (Loomis–Whitney). *For every measurable  $E \subseteq \mathbb{R}^n$ ,*

$$m_n(E) \leq \prod_{j=1}^n m_{n-1}(\pi_j(E))^{\frac{1}{n-1}}.$$

*Equivalently, for non-negative  $f_j$  on  $\mathbb{R}^{n-1}$ ,*

$$\int_{\mathbb{R}^n} \prod_{j=1}^n f_j(\pi_j x)^{\frac{1}{n-1}} dx \leq \prod_{j=1}^n \left( \int_{\mathbb{R}^{n-1}} f_j \right)^{\frac{1}{n-1}}. \quad (11.5.1)$$

*Idea of the proof.* A point of  $E$  is determined by its  $n$  shadows, and each shadow omits one coordinate — so the  $n$  shadows record each coordinate  $n - 1$  times. Counting  $E$  against the shadows therefore costs the power  $\frac{1}{n-1}$ , and the inequality is the assertion that no configuration does better

than a box, where it is an equality.

The proof is induction on  $n$  using Hölder, and the case  $n = 2$  is trivial:  $m_2(E) \leq m_1(\pi_1 E)m_1(\pi_2 E)$ , since  $E$  is contained in the product of its shadows.

*Proof.* We prove (11.5.1), which contains the set statement on taking  $f_j = \mathbf{1}_{\pi_j(E)}$ : then the left side is at least  $m_n(E)$ , because the integrand is 1 on  $E$ .

**Step 1.** *Base case  $n = 2$ .*

Here  $\frac{1}{n-1} = 1$  and the claim is  $\int_{\mathbb{R}^2} f_1(x_2)f_2(x_1) \, dx \leq (\int f_1)(\int f_2)$ , which is Fubini with equality.

**Step 2.** *Inductive step.*

Assume the result in  $\mathbb{R}^{n-1}$  and let  $f_1, \dots, f_n$  be given on  $\mathbb{R}^{n-1}$ . Integrate first in  $x_n$ , holding  $x' = (x_1, \dots, x_{n-1})$  fixed. The factor  $f_n(\pi_n x) = f_n(x')$  does not involve  $x_n$ , so

$$\int_{\mathbb{R}^n} \prod_{j=1}^n f_j(\pi_j x)^{\frac{1}{n-1}} \, dx = \int_{\mathbb{R}^{n-1}} f_n(x')^{\frac{1}{n-1}} \left( \int_{\mathbb{R}} \prod_{j=1}^{n-1} f_j(\pi_j x)^{\frac{1}{n-1}} \, dx_n \right) \, dx'.$$

**Step 3.** *Hölder in  $x_n$ .*

For  $j \leq n-1$  the function  $x_n \mapsto f_j(\pi_j x)$  depends on  $x_n$  and on the  $n-2$  coordinates of  $x'$  other than  $x_j$ . Apply Hölder with  $n-1$  exponents all equal to  $n-1$ :

$$\int_{\mathbb{R}} \prod_{j=1}^{n-1} f_j(\pi_j x)^{\frac{1}{n-1}} \, dx_n \leq \prod_{j=1}^{n-1} \left( \int_{\mathbb{R}} f_j(\pi_j x) \, dx_n \right)^{\frac{1}{n-1}}.$$

Write  $g_j(x') = \int_{\mathbb{R}} f_j(\pi_j x) \, dx_n$ , a function of the  $n-2$  coordinates of  $x'$  other than  $x_j$  — that is, a function on  $\mathbb{R}^{n-2}$  composed with a coordinate projection of  $\mathbb{R}^{n-1}$ .

**Step 4.** *Apply the inductive hypothesis.*

Substituting and then using Hölder once more in  $x'$  with exponents  $n-1$  and  $\frac{n-1}{n-2}$ ,

$$\int_{\mathbb{R}^n} \prod_j f_j^{\frac{1}{n-1}} \leq \left( \int_{\mathbb{R}^{n-1}} f_n \right)^{\frac{1}{n-1}} \left( \int_{\mathbb{R}^{n-1}} \prod_{j=1}^{n-1} g_j(x')^{\frac{1}{n-2}} \, dx' \right)^{\frac{n-2}{n-1}}.$$

The inner integral is exactly the left side of (11.5.1) in dimension  $n-1$ , with the  $n-1$  functions  $g_j$ ; by the inductive hypothesis it is at most  $\prod_{j \leq n-1} (\int g_j)^{\frac{1}{n-2}}$ .

Since  $\int_{\mathbb{R}^{n-2}} g_j = \int_{\mathbb{R}^{n-1}} f_j$  by Tonelli, substituting gives

$$\int_{\mathbb{R}^n} \prod_j f_j^{\frac{1}{n-1}} \leq \left( \int f_n \right)^{\frac{1}{n-1}} \prod_{j=1}^{n-1} \left( \int f_j \right)^{\frac{1}{n-1}},$$

which is (11.5.1).  $\square$

**Where we are.** Theorem 11.5.2 is sharp: for a box  $E = [0, a_1] \times \cdots \times [0, a_n]$  the shadows have measures  $\prod_{i \neq j} a_i$ , and

$$\prod_j \left( \prod_{i \neq j} a_i \right)^{\frac{1}{n-1}} = \left( \prod_i a_i^{n-1} \right)^{\frac{1}{n-1}} = \prod_i a_i = m_n(E).$$

Equality. So the inequality cannot be improved, and boxes are the extremal configurations — which is the same statement as the grid being extremal in Worked Example 11.3.1 and Worked Example 11.4.1, seen continuously. Direct computation on random subsets of a  $k^3$  grid confirms the inequality with room to spare in the generic case; the extremals are exactly the boxes.

## 11.5.2 Guth's theorem

**Definition 11.5.3.** Let  $\nu > 0$ . Families  $\mathbb{T}_1, \dots, \mathbb{T}_n$  of  $\delta$ -tubes in  $\mathbb{R}^n$  are  $\nu$ -transverse if whenever  $T_j \in \mathbb{T}_j$  for each  $j$ , the directions  $v_1, \dots, v_n$  of  $T_1, \dots, T_n$  satisfy

$$|\det(v_1, \dots, v_n)| \geq \nu.$$

**Theorem 11.5.4** (Multilinear Kakeya, axis form). *Let  $n \geq 2$  and  $\varepsilon > 0$ . There is  $\delta = \delta(n, \varepsilon) > 0$  such that the following holds. Suppose  $\ell_{j,a}$  are lines in  $\mathbb{R}^n$ , for  $j = 1, \dots, n$  and  $a = 1, \dots, N_j$ , with  $\ell_{j,a}$  making an angle at most  $\delta$  with the  $x_j$ -axis. Write  $T_{j,a}$  for the indicator of the 1-neighbourhood of  $\ell_{j,a}$ . Then for every  $S \geq 1$  and every cube  $Q_S$  of side  $S$ ,*

$$\int_{Q_S} \prod_{j=1}^n \left( \sum_{a=1}^{N_j} T_{j,a} \right)^{\frac{1}{n-1}} \lesssim_n S^\varepsilon \prod_{j=1}^n N_j^{\frac{1}{n-1}}.$$

*Idea of the proof.* When the lines are *exactly* axis-parallel the statement is Loomis–Whitney, Theorem 11.5.2, and there is nothing to do. The whole problem is to absorb a tiny tilt.

Guth's device is to refuse to work at a single scale. Instead one relates tubes of radius  $W$  to tubes of radius  $\delta^{-1}W$ : on a cube of side  $\approx \delta^{-1}W$  a tube of radius  $W$  tilted by  $\delta$  deviates by at most  $\delta \cdot \delta^{-1}W = W$ , so it is

essentially axis-parallel there, and Loomis–Whitney applies. Passing from scale  $W$  to scale  $\delta^{-1}W$  costs a constant  $C_n$ .

Iterating from radius 1 up to radius  $S$  takes  $M = \log S / \log \delta^{-1}$  rounds, so the accumulated constant is  $C_n^M = S^{\log C_n / \log \delta^{-1}}$  — which is  $S^\varepsilon$  as soon as  $\delta$  is chosen small enough in terms of  $\varepsilon$ . The order of quantifiers matters:  $\varepsilon$  first, then  $\delta$ , then all  $S$ .

**Notation 11.5.5.** For  $W > 0$  let  $T_{j,a,W}$  be the indicator of the  $W$ -neighbourhood of  $\ell_{j,a}$ , and set

$$f_{j,W} = \sum_{a=1}^{N_j} T_{j,a,W}.$$

Thus  $f_{j,1} = \sum_a T_{j,a}$  is the quantity in Theorem 11.5.4.

**Lemma 11.5.6** (One scale to the next). *With hypotheses as in Theorem 11.5.4, there is  $C_n$  such that whenever  $S \geq \delta^{-1}W$  and  $Q_S$  is a cube of side  $S$ ,*

$$\int_{Q_S} \prod_{j=1}^n f_{j,W}^{\frac{1}{n-1}} \leq C_n \delta^n \int_{Q_S} \prod_{j=1}^n f_{j,\delta^{-1}W}^{\frac{1}{n-1}}.$$

*Proof. Step 1. Reduce to a small cube.*

Partition  $Q_S$  into subcubes  $Q$  of side length between  $\frac{1}{20n}\delta^{-1}W$  and  $\frac{1}{10n}\delta^{-1}W$ ; this is possible since  $S \geq \delta^{-1}W$ . It suffices to prove the inequality with  $Q_S$  replaced by each such  $Q$ , and then sum.

**Step 2.** *On  $Q$ , the tubes are essentially axis-parallel.*

Fix  $j, a$ . The line  $\ell_{j,a}$  makes an angle at most  $\delta$  with the  $x_j$ -axis, and  $Q$  has diameter at most  $\frac{\sqrt{n}}{10n}\delta^{-1}W \leq \frac{1}{10}\delta^{-1}W \cdot \sqrt{n}/\sqrt{n}$ . So over the extent of  $Q$  the line deviates from a genuine  $x_j$ -parallel line by at most

$$\delta \cdot \text{diam}(Q) \leq \delta \cdot \frac{1}{10}\delta^{-1}W = \frac{W}{10}.$$

Hence there is an *axis-parallel* tube  $\tilde{T}_{j,a,2W}$  of radius  $2W$  with

$$T_{j,a,W}(x) \leq \tilde{T}_{j,a,2W}(x) \quad \text{for all } x \in Q.$$

**Step 3.** *Apply Loomis–Whitney.*

Let  $\pi_j$  delete the  $j$ -th coordinate. An axis-parallel tube of radius  $2W$  about a line parallel to  $e_j$  is  $\{x : \pi_j x \in B(y_a, 2W)\}$  for some  $y_a \in \mathbb{R}^{n-1}$ , so  $\sum_a \tilde{T}_{j,a,2W}(x) = F_j(\pi_j x)$  with  $F_j = \sum_a \mathbf{1}_{B(y_a, 2W)}$ . By Theorem 11.5.2,

$$\int_Q \prod_j \left( \sum_a \tilde{T}_{j,a,2W} \right)^{\frac{1}{n-1}} \leq \prod_{j=1}^n \|F_j\|_{L^1(\mathbb{R}^{n-1})}^{\frac{1}{n-1}} \leq C_n W^n \prod_{j=1}^n N_j(Q)^{\frac{1}{n-1}},$$

where  $N_j(Q)$  is the number of tubes  $T_{j,a,W}$  meeting  $Q$  — only those contribute — and  $\|\mathbf{1}_{B(y,2W)}\|_{L^1} = c_{n-1}W^{n-1}$ , whose  $\frac{1}{n-1}$ -th powers multiply to  $C_nW^n$ .

**Step 4.** *Convert the count back into an integral at the coarser scale.*

$Q$  has diameter at most  $\frac{1}{10}\delta^{-1}W$ . So if  $T_{j,a,W}$  meets  $Q$  then every point of  $Q$  lies within  $\frac{1}{10}\delta^{-1}W + W \leq \delta^{-1}W$  of  $\ell_{j,a}$ , that is  $T_{j,a,\delta^{-1}W} \equiv 1$  on  $Q$ . Therefore  $f_{j,\delta^{-1}W} \geq N_j(Q)$  pointwise on  $Q$ , and

$$C_nW^n \prod_j N_j(Q)^{\frac{1}{n-1}} \leq \frac{C_nW^n}{|Q|} \int_Q \prod_j f_{j,\delta^{-1}W}^{\frac{1}{n-1}}.$$

Since  $|Q| \geq (\frac{1}{20n}\delta^{-1}W)^n = c_n\delta^{-n}W^n$ , the prefactor is at most  $C'_n\delta^n$ , which is the claim.  $\square$

*Proof of Theorem 11.5.4. Step 1.* The case  $S = \delta^{-M}$ .

Apply Lemma 11.5.6 repeatedly with  $W = 1, \delta^{-1}, \dots, \delta^{-(M-1)}$ :

$$\int_{Q_S} \prod_j f_{j,1}^{\frac{1}{n-1}} \leq C_n^M \delta^{Mn} \int_{Q_S} \prod_j f_{j,\delta^{-M}}^{\frac{1}{n-1}}.$$

Now  $f_{j,S} \leq N_j$  everywhere, and  $|Q_S| = S^n = \delta^{-Mn}$ , so

$$\delta^{Mn} \int_{Q_S} \prod_j f_{j,S}^{\frac{1}{n-1}} \leq \delta^{Mn} \cdot \delta^{-Mn} \prod_j N_j^{\frac{1}{n-1}} = \prod_j N_j^{\frac{1}{n-1}}.$$

The factors  $\delta^n$  cancel exactly against the volume of the cube; that cancellation is the point of the lemma's normalisation. Hence

$$\int_{Q_S} \prod_j f_{j,1}^{\frac{1}{n-1}} \leq C_n^M \prod_j N_j^{\frac{1}{n-1}}.$$

**Step 2.** Choose  $\delta$ .

Since  $S = \delta^{-M}$  we have  $M = \log S / \log \delta^{-1}$ , so

$$C_n^M = \exp\left(\frac{\log S \log C_n}{\log \delta^{-1}}\right) = S^{\log C_n / \log \delta^{-1}}.$$

Given  $\varepsilon > 0$ , choose  $\delta$  so small that  $\log C_n / \log \delta^{-1} \leq \varepsilon$ . Then  $C_n^M \leq S^\varepsilon$ .

**Step 3.** General  $S$ .

For arbitrary  $S \geq 1$  pick the least integer  $M$  with  $\delta^{-M} \geq S$  and cover  $Q_S$  by  $C(\delta)$  cubes of side  $\delta^{-M} \leq \delta^{-1}S$ . Applying Step 1 on each and summing multiplies the bound by  $C(\delta)$ , which depends only on  $n$  and  $\varepsilon$ .  $\square$

**Where we are.** Three features of the proof deserve emphasis.

*The only analytic input is Loomis–Whitney.* Theorem 11.5.2 is proved in this chapter by induction and Hölder, so Theorem 11.5.4 rests on nothing else. There is no heat flow, no polynomial method, and no topology.

*The cancellation in Step 1 is exact.* Each round contributes  $\delta^n$  and there are  $M$  rounds, giving  $\delta^{Mn}$ ; the final integral over  $Q_S$  contributes  $|Q_S| = \delta^{-Mn}$ . Nothing is lost, and this is why the lemma is stated with the factor  $\delta^n$  rather than in a normalised form.

*The order of quantifiers is  $\varepsilon$ , then  $\delta$ , then  $S$ .* A reader who fixes  $\delta$  first and lets  $\varepsilon$  be whatever emerges will find the statement false. Numerically, with  $C_n = 10$  one needs  $\delta^{-1} \geq 10^{10}$  to reach  $\varepsilon = 0.1$  and  $\delta^{-1} \geq 10^{100}$  for  $\varepsilon = 0.01$  — the same astronomically slow onset recorded in Volume 13, Worked Example 13.2.2.

**Theorem 11.5.7** (Multilinear Kakeya, transverse form). *Let  $S_1, \dots, S_n \subseteq \mathbb{S}^{n-1}$  satisfy  $|v_1 \wedge \dots \wedge v_n| \geq \nu$  whenever  $v_j \in S_j$ . Suppose  $\ell_{j,a}$  are lines with direction in  $S_j$ , and let  $T_{j,a}$  be the indicator of the 1-neighbourhood of  $\ell_{j,a}$ . Then for every  $\varepsilon > 0$ , every  $S \geq 1$  and every cube  $Q_S$  of side  $S$ ,*

$$\int_{Q_S} \prod_{j=1}^n \left( \sum_a T_{j,a} \right)^{\frac{1}{n-1}} \lesssim_{n,\varepsilon} \text{Poly}(\nu^{-1}) S^\varepsilon \prod_{j=1}^n N_j^{\frac{1}{n-1}}.$$

*Rescaling by  $\delta$  gives the  $\delta$ -tube form used elsewhere in this series: for  $\nu$ -transverse families  $\mathbb{T}_1, \dots, \mathbb{T}_n$  of  $\delta$ -tubes,*

$$\int_{\mathbb{R}^n} \prod_{j=1}^n \left( \sum_{T \in \mathbb{T}_j} \mathbf{1}_T \right)^{\frac{1}{n-1}} \lesssim_{n,\nu,\varepsilon} \delta^{-\varepsilon} \prod_{j=1}^n (\delta^{n-1} \# \mathbb{T}_j)^{\frac{1}{n-1}}.$$

*Proof.* Cover each  $S_j$  by caps  $S_{j,\beta}$  of radius  $\rho$ . If  $\rho \leq \frac{1}{100n}\nu$  then  $|v_1 \wedge \dots \wedge v_n| \geq \nu/2$  for  $v_j \in S_{j,\beta_j}$ , since the wedge is Lipschitz in each argument with constant 1.

Fix a choice  $\beta_1, \dots, \beta_n$  and let  $A$  be the linear map carrying the centre of  $S_{j,\beta_j}$  to  $e_j$  for each  $j$ . Because the centres have wedge at least  $\nu/2$ ,  $A$  has determinant bounded below and operator norm bounded above by  $\text{Poly}(\nu^{-1})$ ; so  $A$  distorts lengths and volumes by  $\text{Poly}(\nu^{-1})$ . In the new coordinates each line of the  $j$ -th family makes an angle  $\lesssim \text{Poly}(\nu^{-1})\rho$  with the  $x_j$ -axis, which is at most  $\delta(n, \varepsilon)$  once  $\rho$  is chosen small enough in terms of  $\nu$  and  $\varepsilon$ . Theorem 11.5.4 applies.

Summing over the  $\lesssim \text{Poly}(\nu^{-1})$  choices of  $(\beta_1, \dots, \beta_n)$ , and noting that  $(\sum_a T_{j,a})^{1/(n-1)} \leq \sum_{\beta} (\sum_{a \in \beta} T_{j,a})^{1/(n-1)}$  by subadditivity of  $t \mapsto t^{1/(n-1)}$  for  $n \geq 2$ , gives the first display.



For the second, apply the first to the lines underlying the tubes after rescaling  $\mathbb{R}^n$  by  $\delta^{-1}$ , which turns  $\delta$ -tubes into 1-neighbourhoods and multiplies volumes by  $\delta^{-n}$ ; the stated normalisation follows.  $\square$

*Remark 11.5.8* (What is proved, and what is stronger). *Theorem 11.5.4 and Theorem 11.5.7 are proved in full above*, following Guth's short argument, from Loomis–Whitney alone. Revision R12 — which previously recorded multilinear Kaakeya as a load-bearing import — is hereby *discharged*.

Two remarks on strength. Bennett, Carbery and Tao's original theorem, and Guth's later endpoint theorem, are slightly stronger: the endpoint version has no  $S^\varepsilon$  factor at all, and is proved by the polynomial method of Chapter 11.2. That refinement is *not needed* anywhere in this series — the  $S^\varepsilon$  version is exactly what the multilinear restriction theorem of Volume 12 consumes, as Bennett, Carbery and Tao themselves note.

The proof above also applies verbatim to Lipschitz curves in place of lines, with Lipschitz constant at most  $\delta$ ; nothing in Lemma 11.5.6 uses straightness beyond the deviation bound of Step 2.

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#### WORKED EXAMPLE 11.5.1— TIER A **From Loomis–Whitney to tubes**

The passage from Theorem 11.5.2 to Theorem 11.5.7 is worth tracing, since it shows exactly what Guth's theorem adds.

*The axis-parallel case is Loomis–Whitney.* Let  $\mathbb{T}_j$  consist of  $\delta$ -tubes parallel to  $e_j$ . Then  $\sum_{T \in \mathbb{T}_j} \mathbf{1}_T$  is essentially a function of  $\pi_j x$  alone — a tube in direction  $e_j$  is determined by its position in the other  $n - 1$  coordinates. So the left side of Theorem 11.5.7 is the left side of (11.5.1), and  $\int f_j = \delta^{n-1} \# \mathbb{T}_j$  is the total volume of the  $j$ -th family divided by  $\delta$ .  $\checkmark$

*Transversality replaces orthogonality.* For general directions the determinant condition  $|\det(v_1, \dots, v_n)| \geq \nu$  says the  $n$  directions are quantitatively independent — the axis-parallel case has determinant 1. The constant in the conclusion degrades as  $\nu \rightarrow 0$ , and must: nearly coplanar directions permit much larger overlap.

*What is genuinely new.* In Loomis–Whitney the  $n$  functions depend on *disjoint* sets of variables, and Hölder separates them exactly. For general directions no change of coordinates achieves this simultaneously for all tuples, and the separation must be performed approximately, at every scale. That is what the induction on degree does.

*Why the exponent is the same.* Because it is forced by scaling: replacing  $\delta$  by  $\lambda\delta$  multiplies both sides consistently only for the exponent  $\frac{1}{n-1}$ . So Guth's theorem could not have had a different exponent; the content is that the constant is bounded.

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**WORKED EXAMPLE 11.5.2— TIER B Why multilinear is easier than linear**

Volume 9 could not handle near-parallel tubes (Remark 9.5.5), and the multilinear formulation removes them by hypothesis. It is worth being precise about how much that concedes.

*What is assumed away.* Transversality forbids exactly the configurations in which the Kakeya difficulty lives: a Besicovitch set's tubes point in all directions, so most  $n$ -tuples are far from transverse. The multilinear estimate says nothing about them.

*What is retained.* The tubes may still be positioned arbitrarily, and there may be  $\delta^{-(n-1)}$  of them in each family. So the counting problem is not trivialised — only the angular degeneracy is.

*Why it is nonetheless useful.* Because a linear estimate can be recovered from a multilinear one by a decomposition: split the directions into caps, and either most of the mass comes from  $n$  transverse caps — the *broad* case, handled by Theorem 11.5.7 — or it concentrates near a lower dimensional set of directions — the *narrow* case, handled by induction. That is the broad–narrow argument, and Volume 12 develops it.

*The cost.* The recovery is lossy: broad–narrow gives the linear estimate with a  $\delta^{-\varepsilon}$  and, in the Kakeya case, does not reach the full conjecture. So multilinear Kakeya is a genuine theorem that does not by itself settle the linear problem — which is why the conjecture remained open after Guth's proof.

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**WORKED EXAMPLE 11.5.3— TIER B Checking a multilinear estimate for consistency**

Theorem 11.5.7 is proved above, but the checks below are worth running on *any* multilinear estimate before trusting it — and they are how one would catch a misremembered exponent. Three checks, all cheap.

(1) *Scaling in  $\delta$ .* Replace  $\delta$  by  $\lambda\delta$  keeping the tubes' axes. Each  $\mathbf{1}_T$  is unchanged in character but the tubes fatten; the left side scales like  $\lambda^n$  times the count, and the right side like  $\prod_j (\lambda^{n-1})^{1/(n-1)} = \lambda^n$ . ✓ The exponent  $\frac{1}{n-1}$  is the unique one making this work, which is why Worked Example 11.5.1 could assert that Guth's theorem had no choice about it.

(2) *The axis-parallel case.* Specialising to coordinate directions must reproduce Theorem 11.5.2. It does, as Worked Example 11.5.1 verifies, and this is the strongest single check — an inequality that failed here would be false.

(3) *A single tube from each family.* With  $\#\mathbb{T}_j = 1$  the left side is  $m(T_1 \cap \cdots \cap T_n)$  up to the exponent, and transversality forces the intersection to be contained in a ball of radius  $\approx \delta/\nu$ , so it has measure  $\lesssim (\delta/\nu)^n$ . The right

side is  $\prod_j (\delta^{n-1})^{1/(n-1)} = \delta^n$ . ✓ up to the  $\nu$ -dependence, which the constant carries.

*What the checks do not verify.* That the constant is bounded independently of  $\#\mathbb{T}_j$  — which is the theorem. Shape checks confirm an inequality is not obviously wrong; they never confirm it is right.

*Why do them anyway.* Because a quoted theorem used as a hypothesis should be one the reader can restate correctly, and mis-stated exponents are the commonest error in this literature. Two minutes of scaling prevents it.

**WORKED EXAMPLE 11.5.4— TIER C: RESEARCH TASTE** **Transversality as a hypothesis: what it concedes and what it keeps**

Multilinear estimates appear throughout the modern literature, always with a transversality hypothesis. It is worth being exact about its content.

(a) *It is a hypothesis on tuples, not on families.* Definition 11.5.3 requires  $|\det(v_1, \dots, v_n)| \geq \nu$  for *every* choice of one tube from each family. So each family may itself contain tubes in many directions — what is forbidden is that a tuple be nearly degenerate.

(b) *It is quantitative.* The constant in Theorem 11.5.7 degrades as  $\nu \rightarrow 0$ , and must: at  $\nu = 0$  the tubes may be coplanar and the estimate fails. So the hypothesis cannot be removed by a limiting argument.

(c) *What it concedes.* A Besicovitch set has tubes in all directions, so most  $n$ -tuples from it are far from  $\nu$ -transverse for any fixed  $\nu$ . The multilinear estimate therefore says nothing about a Besicovitch configuration directly, and Worked Example 11.5.2 recorded that recovering the linear statement is lossy.

(d) *What it keeps.* Everything else. The tubes' positions are arbitrary, the families may be as large as  $\delta^{-(n-1)}$ , and no structure is assumed. So the estimate is not a triviality — the counting problem it solves is genuine, and Bennett, Carbery and Tao's  $\varepsilon$ -loss version was itself a substantial theorem.

(e) *The pattern across the series.* Volume 6's restriction conjecture is hard at  $L^\infty$  data and easy at  $L^2$ ; Volume 9's hairbrush works transversally and fails at small angles; multilinear Kakeya assumes transversality outright. In each case the difficulty concentrates in the degenerate configurations, and the tractable theorem is the one that excludes them. Worked Example 11.8.2 in Chapter 11.8 draws the three together.

(f) *Why the concession is nonetheless worth making.* Because the broad-narrow argument of Volume 12 recovers a genuine linear estimate from it, and because the multilinear statement is *sharp* — Theorem 11.5.2 shows the

exponent cannot be improved even in the model case. A sharp theorem under a restrictive hypothesis is often more useful than a lossy one without.

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**WORKED EXAMPLE 11.5.5— TIER A Loomis–Whitney in three dimensions, checked**

The case  $n = 3$  is worth doing concretely, since it is the one Volume 12 uses.

*The statement.* For  $E \subseteq \mathbb{R}^3$ ,

$$m_3(E) \leq |\pi_1 E|^{1/2} |\pi_2 E|^{1/2} |\pi_3 E|^{1/2},$$

the three projections being onto the coordinate planes.

*Equality for a box.*  $E = [0, a] \times [0, b] \times [0, c]$  has projections of areas  $bc$ ,  $ac$ ,  $ab$ , and  $(bc \cdot ac \cdot ab)^{1/2} = (a^2 b^2 c^2)^{1/2} = abc = m_3(E)$ . ✓

*Strict inequality otherwise.* For a random subset of a  $k^3$  grid with each point kept independently with probability  $\frac{2}{5}$ , direct computation at  $k = 3, \dots, 6$  gives  $|E|$  between 6 and 91 against right-hand sides between 11 and 216 — a factor of two or more to spare. Generic sets are far from extremal.

*The functional form.* Taking  $f_j$  to be indicators recovers the set statement; taking general  $f_j$  gives

$$\int_{\mathbb{R}^3} f_1(x_2, x_3)^{1/2} f_2(x_1, x_3)^{1/2} f_3(x_1, x_2)^{1/2} dx \leq \prod_j \left( \int f_j \right)^{1/2},$$

which is the form the tube version generalises.

*Why  $\frac{1}{2}$ .* Each coordinate appears in exactly two of the three projections, so the exponents must sum to 1 across the two occurrences — giving  $\frac{1}{n-1} = \frac{1}{2}$ . Any other exponent fails the scaling check of Worked Example 11.5.3(1).

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## Drills

**Drill 11.5.1.** Verify the base case  $n = 2$  of Theorem 11.5.2 and the equality for a box in  $\mathbb{R}^3$ .

**Drill 11.5.2.** Check the Hölder application in Step 3, identifying the  $n - 1$  exponents.

**Drill 11.5.3.** Confirm that  $\int_{\mathbb{R}^{n-2}} g_j = \int_{\mathbb{R}^{n-1}} f_j$  in Step 4, and say where Tonelli is used.

**Drill 11.5.4.** For  $\mathbb{T}_j$  consisting of  $\delta$ -tubes parallel to  $e_j$ , verify that  $\int f_j = \delta^{n-1} \# \mathbb{T}_j$  as claimed in Worked Example 11.5.1.

**Where we are.** What this chapter supplies to Volume 12: the statement of Theorem 11.5.7, its exponent  $\frac{1}{n-1}$  shown to be forced by scaling, and the model case Theorem 11.5.2 proved with its extremals identified. That is enough to use the theorem correctly and to recognise a mis-statement of it, which Worked Example 11.5.3 argues is the practical requirement.

## Notes

Loomis and Whitney proved their inequality in 1949; the induction given here is the standard one. Bennett, Carbery and Tao proved the multilinear Kakeya inequality with a  $\delta^{-\varepsilon}$  loss in 2006 by a heat-flow argument, and Guth removed the loss in 2010 using polynomial partitioning; a later proof of Carbery and Valdimarsson uses the Borsuk–Ulam theorem directly.

Remark 11.5.8 records that the theorem is quoted and not proved here.



# Chapter 11.6

## Lines in $\mathbb{R}^3$

Szemerédi–Trotter is sharp in the plane, and its extremal example — Worked Example 11.3.1 — is a grid, which is a planar configuration. In  $\mathbb{R}^3$  one can ask for more: if the lines are *not* concentrated in planes, are there fewer incidences? The answer is yes, and quantifying it is the Guth–Katz incidence theorem, the deepest result of this volume.

### 11.6.1 The obstruction

**Proposition 11.6.1** (Planes and reguli are extremal). *There are configurations of  $L$  lines in  $\mathbb{R}^3$  with  $\gtrsim L^{3/2}$  pairwise intersections, namely any planar Szemerédi–Trotter configuration. Consequently no bound better than  $L^{3/2}$  can hold without excluding such configurations.*

*Proof.* Take  $L$  lines in a plane  $\Pi \subseteq \mathbb{R}^3$  in general position: every two meet, so there are  $\binom{L}{2} \approx \frac{L^2}{2}$  intersecting pairs, far more than  $L^{3/2}$ .

More relevantly, take the dual of the grid of Worked Example 11.3.1:  $L$  lines in  $\Pi$  with  $\approx L^{4/3}$  incidences with a set of  $\approx L$  points, so  $\approx L^{4/3}$  intersecting pairs concentrated at few points.  $\square$

**Definition 11.6.2.** A *regulus* is a doubly ruled surface in  $\mathbb{R}^3$  — a quadric containing two families of lines, such as the hyperboloid  $x^2 + y^2 - z^2 = 1$  or the saddle  $z = xy$ . Any two lines of the same family are disjoint; any line of one family meets every line of the other.

**Proposition 11.6.3.** *A regulus contains two families of lines, and  $L/2$  lines from each family have  $L^2/4$  intersecting pairs.*

*Proof.* Consider  $z = xy$ . For each  $a \in \mathbb{R}$  the line  $\ell_a = \{(a, t, at) : t \in \mathbb{R}\}$  lies on the surface, and so does  $m_b = \{(s, b, sb) : s \in \mathbb{R}\}$ . Two lines  $\ell_a, \ell_{a'}$  with  $a \neq a'$  are disjoint — they have different first coordinates — and  $\ell_a$  meets  $m_b$  at  $(a, b, ab)$ . So  $L/2$  lines of each family give  $(L/2)^2$  intersecting pairs.  $\square$

## 11.6.2 Bézout for lines

**Theorem 11.6.4** (Bézout in the plane). *If  $P, Q \in \mathbb{R}[x_1, x_2]$  are non-zero with no common factor, then  $Z(P) \cap Z(Q)$  contains at most  $(\deg P)(\deg Q)$  points.*

*Proof.* Regard  $P, Q$  as elements of  $\mathbb{K}[x_2]$  where  $\mathbb{K} = \mathbb{R}(x_1)$  is the field of rational functions in  $x_1$ . Having no common factor in  $\mathbb{R}[x_1, x_2]$ , they have none in  $\mathbb{K}[x_2]$  by Gauss's lemma, so their resultant  $\text{Res}_{x_2}(P, Q) \in \mathbb{R}[x_1]$  is not identically zero, and it has degree at most  $(\deg P)(\deg Q)$ .

If  $(a, b) \in Z(P) \cap Z(Q)$  then  $P(a, \cdot)$  and  $Q(a, \cdot)$  share the root  $b$ , so  $\text{Res}_{x_2}(P, Q)(a) = 0$ . Hence there are at most  $(\deg P)(\deg Q)$  possible values of  $a$ ; after a generic linear change of coordinates each such  $a$  corresponds to one intersection point, giving the bound.  $\square$

**Theorem 11.6.5** (Bézout for lines). *If  $P, Q \in \mathbb{R}[x_1, x_2, x_3]$  are non-zero with no common factor, then  $Z(P) \cap Z(Q)$  contains at most  $(\deg P)(\deg Q)$  lines.*

*Proof.* Suppose  $Z(P) \cap Z(Q)$  contains  $N > (\deg P)(\deg Q)$  lines  $\ell_1, \dots, \ell_N$ . Choose a generic plane  $\Pi \subseteq \mathbb{R}^3$ : generically  $\Pi$  meets each  $\ell_k$  in exactly one point, these  $N$  points are distinct, and neither  $P|_\Pi$  nor  $Q|_\Pi$  is identically zero.

If  $P|_\Pi$  and  $Q|_\Pi$  had a common factor, then that factor would vanish on a curve in  $\Pi$ ; as  $\Pi$  varies over a generic family this would force  $P$  and  $Q$  to share a factor, contrary to hypothesis. So  $P|_\Pi, Q|_\Pi$  are non-zero plane polynomials of degrees  $\leq \deg P, \deg Q$  without common factor, and by Theorem 11.6.4 they have at most  $(\deg P)(\deg Q)$  common zeros in  $\Pi$ . But the  $N$  points  $\Pi \cap \ell_k$  are  $N > (\deg P)(\deg Q)$  such zeros, a contradiction.  $\square$

**Theorem 11.6.6** (Line incidences in  $\mathbb{R}^3$ ). *For every  $\varepsilon > 0$  there are  $D(\varepsilon)$  and  $K(\varepsilon)$  such that the following holds. Let  $\mathfrak{L}$  be a set of  $L$  lines in  $\mathbb{R}^3$  containing fewer than  $L^{\frac{1}{2}+\varepsilon}$  lines in any irreducible algebraic surface of degree at most  $D$ . Then for  $2 < r < 2L^{1/2}$ ,*

$$|P_r(\mathfrak{L})| \leq K L^{\frac{3}{2}+\varepsilon} r^{-2}.$$

*Idea of the proof.* Partition  $\mathbb{R}^3$  by a polynomial of bounded degree  $D = D(\varepsilon)$  and induct on  $L$ . The obstruction is that the hypothesis does not pass to the lines  $\mathfrak{L}_i$  entering a cell:  $\mathfrak{L}_i$  has fewer than  $L^{1/2+\varepsilon}$  lines in a low-degree surface, but that is not fewer than  $|\mathfrak{L}_i|^{1/2+\varepsilon}$ , so the induction does not immediately apply.

Guth's device is to prove a *stronger* statement, carrying along a small set  $\mathcal{Z}$  of low-degree surfaces that accounts for the rich points not otherwise controlled. The surfaces produced in the cells are collected, and then *pruned*: those containing few lines are discarded using Szemerédi–Trotter, and those



containing many are few in number by Bézout.

**Proposition 11.6.7** (Rich points for very large  $r$ ). *If  $\mathfrak{L}$  is a set of  $L$  lines in  $\mathbb{R}^d$ ,  $d \geq 2$ , and  $r > 2L^{1/2}$ , then  $|P_r(\mathfrak{L})| \leq 2Lr^{-1}$ .*

*Proof.* List  $P_r(\mathfrak{L}) = \{x_1, \dots, x_M\}$ . The point  $x_j$  lies on at least  $r - (j - 1)$  lines containing none of  $x_1, \dots, x_{j-1}$ , since two distinct points lie on at most one common line. Hence  $L \geq \sum_{j=1}^M \max(r - j, 0)$ . If  $M \geq r/2$  this gives  $L \geq (r/2)(r/2) = r^2/4 > L$ , a contradiction; so  $M < r/2$  and then  $L \geq M(r/2)$ .  $\square$

**Lemma 11.6.8** (Few surfaces can each be rich in lines). *Let  $\mathfrak{L}$  be a set of lines in  $\mathbb{R}^3$  and  $\mathcal{Y}$  a set of irreducible algebraic surfaces of degree at most  $D$ , each containing at least  $A$  lines of  $\mathfrak{L}$ . If  $A > 2D|\mathfrak{L}|^{1/2}$  then  $|\mathcal{Y}| \leq 2|\mathfrak{L}|A^{-1}$ .*

*Proof.* By Bézout for lines (Theorem 11.6.5), two distinct irreducible surfaces of degree  $\leq D$  share at most  $D^2$  lines. Order  $\mathcal{Y} = \{Z_1, Z_2, \dots\}$ ; then  $Z_{j+1}$  contains at least  $A - jD^2$  lines not lying in  $Z_1, \dots, Z_j$ , so

$$|\mathfrak{L}| \geq \sum_{j=1}^{|\mathcal{Y}|} \max(A - jD^2, 0).$$

If  $|\mathcal{Y}| \geq \frac{1}{2}AD^{-2}$  then, since  $A - jD^2 \geq A/2$  for  $j \leq \frac{1}{2}AD^{-2}$ , we get  $|\mathfrak{L}| \geq \frac{1}{2}AD^{-2} \cdot \frac{A}{2} = \frac{A^2}{4D^2} > |\mathfrak{L}|$ , a contradiction. So  $|\mathcal{Y}| < \frac{1}{2}AD^{-2}$ , and then  $|\mathfrak{L}| \geq |\mathcal{Y}| \cdot \frac{A}{2}$ .  $\square$

**Theorem 11.6.9** (The inductive statement). *For every  $\varepsilon > 0$  there are  $D(\varepsilon)$ ,  $K(\varepsilon)$  so that the following holds. Put  $r' = \lceil \frac{9}{10}r \rceil$ . If  $\mathfrak{L}$  is a set of  $L$  lines in  $\mathbb{R}^3$  and  $2 \leq r \leq 2L^{1/2}$ , there is a set  $\mathcal{Z}$  of surfaces with*

- (i) each  $Z \in \mathcal{Z}$  irreducible of degree  $\leq D$ ;
- (ii) each  $Z \in \mathcal{Z}$  contains at least  $L^{\frac{1}{2}+\varepsilon}$  lines of  $\mathfrak{L}$ ;
- (iii)  $|\mathcal{Z}| \leq 2L^{\frac{1}{2}-\varepsilon}$ ;
- (iv)  $|P_r(\mathfrak{L}) \setminus \bigcup_{Z \in \mathcal{Z}} P_{r'}(\mathfrak{L}_Z)| \leq KL^{\frac{3}{2}+\varepsilon}r^{-2}$ .

*Proof.* We may assume  $\varepsilon \leq \frac{1}{2}$ , since for  $\varepsilon \geq \frac{1}{2}$  one takes  $\mathcal{Z} = \emptyset$  and  $|P_r(\mathfrak{L})| \leq 2L^2r^{-2}$  by double counting. Choose  $D = D(\varepsilon)$  large, then  $K = K(\varepsilon, D)$  large, in particular with

$$K \geq 10(2D)^{2/\varepsilon}. \quad (11.6.1)$$

**Step 0. Base case.**

If  $L^\varepsilon \leq 2D$ , take  $\mathcal{Z} = \emptyset$ ; then by (11.6.1),  $|P_r(\mathfrak{L})| \leq L^2 \leq (2D)^{2/\varepsilon} \leq K/10 \leq KL^{3/2+\varepsilon}r^{-2}$ . So assume the statement for sets of at most  $L/2$  lines, and  $L^\varepsilon > 2D$ .

**Step 1. Partition.**

Let  $S \subseteq P_r(\mathfrak{L})$ . By Theorem 11.2.5 there is a non-zero  $P$  of degree  $\leq D$  with  $\mathbb{R}^3 \setminus Z(P)$  a union of at most  $CD^3$  cells  $O_i$ , each with  $|S \cap O_i| \leq CD^{-3}|S|$ . Let  $\mathfrak{L}_i$  be the lines meeting  $O_i$ . A line not in  $Z(P)$  meets  $Z(P)$  in  $\leq D$  points, hence enters  $\leq D+1$  cells, so

$$\sum_i |\mathfrak{L}_i| \leq (D+1)L \leq 2DL. \quad (11.6.2)$$

Call  $O_i$  *good* if  $|\mathfrak{L}_i| \leq \beta D^{-2}L$ . By (11.6.2) there are at most  $2\beta^{-1}D^3$  bad cells, so the bad cells contain at most  $C\beta^{-1}|S|$  points of  $S$ ; fix  $\beta$  with  $C\beta^{-1} \leq \frac{1}{100}$ , so

$$\text{the bad cells contain at most } \frac{1}{100}|S| \text{ points of } S. \quad (11.6.3)$$

**Step 2. Good cells, by induction.**

For  $D$  large,  $|\mathfrak{L}_i| \leq L/2$  on good cells. If  $r \leq 2|\mathfrak{L}_i|^{1/2}$  the inductive hypothesis gives  $\mathcal{Z}_i$  with  $|\mathcal{Z}_i| \leq 2(\beta D^{-2}L)^{\frac{1}{2}-\varepsilon}$  and, since  $S \cap O_i \subseteq P_r(\mathfrak{L}_i)$ ,

$$\left| (S \cap O_i) \setminus \bigcup_{Z \in \mathcal{Z}_i} P_{r'}(\mathfrak{L}_Z) \right| \leq K(\beta D^{-2}L)^{\frac{3}{2}+\varepsilon}r^{-2} \leq C_1KD^{-3-2\varepsilon}L^{\frac{3}{2}+\varepsilon}r^{-2}.$$

If instead  $r > 2|\mathfrak{L}_i|^{1/2}$ , take  $\mathcal{Z}_i = \emptyset$ ; then Proposition 11.6.7 gives  $|S \cap O_i| \leq 2Lr^{-1} \leq 4L^{3/2}r^{-2}$ , which is at most the same bound for  $K$  large. Summing over the  $\leq CD^3$  good cells,

$$\sum_{O_i \text{ good}} \left| (S \cap O_i) \setminus \bigcup_{Z \in \mathcal{Z}_i} P_{r'}(\mathfrak{L}_Z) \right| \leq C_2D^{-2\varepsilon}KL^{\frac{3}{2}+\varepsilon}r^{-2},$$

and we choose  $D(\varepsilon)$  with  $C_2D^{-2\varepsilon} \leq \frac{1}{400}$ .

**Step 3. The partitioning surface.**

Let  $Z_j$  be the irreducible components of  $Z(P)$ . If  $x \in S \cap Z_j$  but  $x \notin P_{r'}(\mathfrak{L}_{Z_j})$ , then  $x$  lies on at least  $r/10$  lines not contained in  $Z_j$ , each meeting  $Z_j$  in  $\leq \deg Z_j$  points. Hence  $|(S \cap Z_j) \setminus P_{r'}(\mathfrak{L}_{Z_j})| \leq 10r^{-1}(\deg Z_j)L$ , and summing over components,

$$\left| (S \cap Z(P)) \setminus \bigcup_j P_{r'}(\mathfrak{L}_{Z_j}) \right| \leq 10r^{-1}DL \leq \frac{1}{400}KL^{\frac{3}{2}}r^{-2}$$

for  $K$  large, using  $r \leq 2L^{1/2}$ .

**Step 4.** *Iterate to remove the bad-cell term.*

Let  $\tilde{\mathcal{Z}}_S$  be the union of the  $\mathcal{Z}_i$  over good cells together with the components of  $Z(P)$ ; then  $|\tilde{\mathcal{Z}}_S| \leq \text{Poly}(D)L^{\frac{1}{2}-\varepsilon}$  and, combining (11.6.3) with Steps 2 and 3,

$$\left| S \setminus \bigcup_{Z \in \tilde{\mathcal{Z}}_S} P_{r'}(\mathfrak{L}_Z) \right| \leq \frac{1}{100}|S| + \frac{1}{200}KL^{\frac{3}{2}+\varepsilon}r^{-2}. \quad (11.6.4)$$

Set  $S_1 = P_r(\mathfrak{L})$  and  $S_{j+1} = S_j \setminus \bigcup_{Z \in \tilde{\mathcal{Z}}_{S_j}} P_{r'}(\mathfrak{L}_Z)$ . Iterating (11.6.4)  $J = C \log L$  times gives  $|S_J| \leq \frac{1}{100}KL^{\frac{3}{2}+\varepsilon}r^{-2}$ , and  $\tilde{\mathcal{Z}} = \bigcup_{j < J} \tilde{\mathcal{Z}}_{S_j}$  satisfies

$$|\tilde{\mathcal{Z}}| \leq \text{Poly}(D)L^{\frac{1}{2}-\varepsilon} \log L, \quad \left| P_r(\mathfrak{L}) \setminus \bigcup_{Z \in \tilde{\mathcal{Z}}} P_{r'}(\mathfrak{L}_Z) \right| \leq \frac{1}{100}KL^{\frac{3}{2}+\varepsilon}r^{-2}. \quad (11.6.5)$$

**Step 5.** *Prune.*

Put  $\mathcal{Z} = \{Z \in \tilde{\mathcal{Z}} : Z \text{ contains } \geq L^{\frac{1}{2}+\varepsilon} \text{ lines of } \mathfrak{L}\}$ , which gives (i) and (ii). Since  $L^\varepsilon > 2D$  we have  $A = L^{\frac{1}{2}+\varepsilon} > 2DL^{1/2}$ , so Lemma 11.6.8 gives  $|\mathcal{Z}| \leq 2LA^{-1} = 2L^{\frac{1}{2}-\varepsilon}$ , which is (iii).

For (iv), by (11.6.5) it suffices to bound  $\sum_{Z \in \tilde{\mathcal{Z}} \setminus \mathcal{Z}} |P_{r'}(\mathfrak{L}_Z)|$ . Sort by  $\tilde{\mathcal{Z}}_s = \{Z : |\mathfrak{L}_Z| \in [2^s, 2^{s+1})\}$ , so  $\tilde{\mathcal{Z}} \setminus \mathcal{Z} \subseteq \bigcup_{2^s \leq L^{1/2+\varepsilon}} \tilde{\mathcal{Z}}_s$ . By Szemerédi–Trotter (Theorem 11.3.2) and  $r' \geq \frac{9}{10}r$ ,  $|P_{r'}(\mathfrak{L}_Z)| \lesssim 2^{2s}r^{-3} + 2^s r^{-1}$ .

For  $2DL^{1/2} < 2^s \leq L^{\frac{1}{2}+\varepsilon}$ , Lemma 11.6.8 with  $A = 2^s$  gives  $|\tilde{\mathcal{Z}}_s| \leq 2L2^{-s}$ , so that range contributes

$$\lesssim \sum_{2^s \leq L^{\frac{1}{2}+\varepsilon}} (L2^s r^{-3} + Lr^{-1}) \lesssim L^{\frac{3}{2}+\varepsilon}r^{-3} + L(\log L)r^{-1} \lesssim L^{\frac{3}{2}+\varepsilon}r^{-2}.$$

For  $2^s \leq 2DL^{1/2}$  we use  $|\tilde{\mathcal{Z}}_s| \leq |\tilde{\mathcal{Z}}| \leq \text{Poly}(D)L^{\frac{1}{2}-\varepsilon} \log L$  together with  $2^{2s}r^{-3} \leq \text{Poly}(D)Lr^{-3}$  and  $2^s r^{-1} \leq \text{Poly}(D)Lr^{-2}$ , giving  $\text{Poly}(D)L^{3/2}r^{-2}$ . Choosing  $K$  large enough,  $\sum_{Z \in \tilde{\mathcal{Z}} \setminus \mathcal{Z}} |P_{r'}(\mathfrak{L}_Z)| \leq \frac{1}{100}KL^{\frac{3}{2}+\varepsilon}r^{-2}$ , which with (11.6.5) gives (iv).  $\square$

*Proof of Theorem 11.6.6.* If fewer than  $L^{\frac{1}{2}+\varepsilon}$  lines lie in any irreducible surface of degree  $\leq D$ , then condition (ii) forces  $\mathcal{Z} = \emptyset$ , and (iv) becomes  $|P_r(\mathfrak{L})| \leq KL^{\frac{3}{2}+\varepsilon}r^{-2}$ .  $\square$

**Where we are.** Two features of the induction deserve comment.

*Why a stronger statement is easier.* The hypothesis of Theorem 11.6.6 does not pass to the lines entering a cell, so the theorem cannot be induced on directly. Carrying the surface set  $\mathcal{Z}$  makes the statement inductive. This

is the same manoeuvre as strengthening an induction hypothesis to make it go through, and it is the paper's one new idea.

*The cost of  $\varepsilon$ .* Step 2 needs  $C_2 D^{-2\varepsilon} \leq \frac{1}{400}$ , that is  $D \geq 400^{1/(2\varepsilon)}$ . For  $\varepsilon = \frac{1}{2}$  this is 400; for  $\varepsilon = \frac{1}{4}$  it is  $1.6 \times 10^5$ ; for  $\varepsilon = \frac{1}{10}$  it is about  $10^{13}$ . Finite for every  $\varepsilon > 0$  and astronomical as  $\varepsilon \rightarrow 0$  — which is why the method gives  $L^{3/2+\varepsilon}$  and not  $L^{3/2}$ .

*Where the pruning lands.* Lemma 11.6.8 with  $A = L^{1/2+\varepsilon}$  gives  $|\mathcal{Z}| \leq 2LA^{-1} = 2L^{1/2-\varepsilon}$ , which is *exactly* hypothesis (iii). Nothing is spare: the induction closes on the nose.

*Remark 11.6.10* (R13 discharged). *Theorem 11.6.6 is now proved in full*, following Guth's low-degree partitioning argument. Revision R13 is discharged, and with it the last load-bearing import in this series.

The inputs are Theorem 11.2.5 (proved in Chapter 11.2, resting on Chapter 11.1), Bézout for lines (Theorem 11.6.5), and Szemerédi–Trotter (Theorem 11.3.2, proved in Chapter 11.3). *No ruled surface theory and no flecnode polynomial are used anywhere.*

The original Guth–Katz theorem is sharper —  $|P_r(\mathfrak{L})| \leq CL^{3/2}r^{-2}$  with no  $\varepsilon$ , under the hypothesis of at most  $L^{1/2}$  lines in any plane or regulus. Their proof splits into cases  $r = 2$  and  $r \geq 3$ ; the case  $r = 2$  uses the classification of ruled surfaces via the Cayley–Salmon flecnode polynomial, which is not reproduced here. The  $\varepsilon$ -loss version above suffices for every application in this series, at the cost of  $N^{1-\varepsilon}$  in place of  $N/\log N$  for distinct distances (Theorem 11.7.8).

**Proposition 11.6.11** (Lines in a plane, and the planar case of the theorem). *Let  $\mathcal{L}$  be  $L$  lines in a plane  $\Pi \subseteq \mathbb{R}^3$ , no two parallel. Then the number of points on at least  $k$  of them is  $\lesssim L^2/k^3 + L/k$ .*

*Proof.* Work inside  $\Pi \cong \mathbb{R}^2$  and dualise: by the point–line duality of Theorem 11.3.2 Step 0, points on at least  $k$  lines correspond to lines containing at least  $k$  points of a set of  $L$  points. By Corollary 11.3.4 with  $P = L$ , their number is  $\lesssim L^2/k^3 + L/k$ .  $\square$

**Where we are.** So within a single plane the answer is known and sharp — it is Szemerédi–Trotter. Theorem 11.6.6 asserts that a configuration *not* concentrated in planes or reguli does strictly better: the count drops from  $L^2/k^3$  to  $L^{3/2}/k^2$ .

Comparing at  $k = 2$ : the planar bound is  $L^2$  and the three-dimensional one  $L^{3/2}$ . The gain is a full factor  $L^{1/2}$ , and it is exactly what Chapter 11.7 needs: with  $L = N^2$  lines,  $L^2 = N^4$  would give nothing, while  $L^{3/2} = N^3$  gives the distinct-distances theorem.

That is worth stating plainly. The Guth–Katz theorem is not a technical refinement of Szemerédi–Trotter; it is the statement that genuinely three-dimensional configurations are quantitatively different, and the whole distinct-distances result rests on that difference.

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WORKED EXAMPLE 11.6.1— TIER A **Why both exclusions are needed**

Theorem 11.6.6 excludes planes and reguli, and neither exclusion can be dropped.

*Planes.* By Proposition 11.6.1,  $L$  lines in a plane can have  $\approx L^2$  intersecting pairs — vastly more than  $L^{3/2}$ . So the theorem is false without the planar exclusion.

*Reguli.* By Proposition 11.6.3,  $L$  lines on a regulus can have  $\approx L^2/4$  intersecting pairs. Again  $\gg L^{3/2}$ .

*And nothing else.* That these are the *only* obstructions is the content of the theorem, and it is a statement about algebraic surfaces: a surface containing more than a bounded number of lines through each point is ruled, and the doubly ruled ones are exactly the planes and the quadrics of regulus type.

*The threshold  $\sqrt{L}$ .* If a plane contains  $m$  lines it contributes  $\approx m^2$  pairs, and this is  $\lesssim L^{3/2}$  exactly when  $m \lesssim L^{3/4}$ . The theorem’s hypothesis  $m \leq \sqrt{L}$  is stronger than necessary for that count but is what the proof requires; sharpening it is possible and is not needed for Chapter 11.7.

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WORKED EXAMPLE 11.6.2— TIER B **The two-case structure, in three dimensions**

Chapter 11.3’s Checkpoint identified the shape of every polynomial-method proof: a cellular case and an algebraic case. In the plane the algebraic case was one line of argument. Here it is the whole difficulty, and the contrast is instructive.

*The cellular case, in  $\mathbb{R}^3$ .* Partition by a polynomial of degree  $D$ ; there are  $\lesssim D^3$  cells, each meeting few lines, and a line crosses  $\leq D + 1$  of them. The counting goes through exactly as in the plane.

*The algebraic case, in the plane.* A curve of degree  $D$  contains at most  $D$  lines — restrict to a transversal and count roots. Two lines, one sentence.

*The algebraic case, in  $\mathbb{R}^3$ .* A surface of degree  $D$  may contain infinitely many lines. One must therefore:

- show that a surface containing  $\gg D^2$  lines is *ruled* — this is where the flecnode polynomial enters, a degree- $11D$  polynomial vanishing at points where a line meets the surface to order three;

- classify the ruled surfaces: a ruled surface of degree  $D$  is a plane, a regulus, or has the property that most of its lines meet few others;
- handle each class separately.

*The moral.* The polynomial method reduces a combinatorial problem to an algebraic one, and the algebraic problem may be easy or may be a research programme. Predicting which requires knowing how varieties of the relevant degree and dimension can contain the objects in question — lines, in this case. That is why the method’s difficulty jumps between  $\mathbb{R}^2$  and  $\mathbb{R}^3$  and why Volume 14 reports that the four-dimensional analogues remain open.

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WORKED EXAMPLE 11.6.3— TIER C: RESEARCH TASTE **Ruled surfaces, and why dimension three is special**

Remark 11.6.10 attributes the difficulty of Theorem 11.6.6 to the classification of surfaces containing many lines. It is worth seeing why that question has a good answer in  $\mathbb{R}^3$  and not beyond.

(a) *The dimension count.* Lines in  $\mathbb{R}^3$  form a 4-parameter family. A surface is 2-dimensional; requiring a line to lie in it is 2 conditions per point of the line, but a line meets a degree- $D$  surface in  $D$  points, so lying inside is  $D + 1$  conditions. Generically a surface contains finitely many lines — famously 27 on a smooth cubic.

(b) *The exceptions are ruled.* A surface containing a 1-parameter family of lines is by definition ruled, and ruled surfaces of degree  $D$  are classified: they are swept by a line moving along a curve. The doubly ruled ones — planes and the quadrics of Proposition 11.6.3 — are the only surfaces where two lines of the surface meet at every point.

(c) *The flecnodal polynomial.* Cayley and Salmon, in the 1860s, constructed a polynomial of degree  $11D - 24$  vanishing exactly where a line meets the surface to order three. A surface all of whose points are flecnodal is ruled, and that is the mechanism converting “many lines” into “ruled”.

(d) *Why  $\mathbb{R}^4$  is harder.* Lines in  $\mathbb{R}^4$  form a 6-parameter family and hypersurfaces are 3-dimensional; a hypersurface can contain a 2-parameter family of lines without being degenerate in any classical sense. There is no classification comparable to the ruled-surface one, and the polynomial method’s algebraic case is correspondingly open.

(e) *Consequence for the series.* Volume 14 reports that Kakeya is settled in  $\mathbb{R}^3$  and open in  $\mathbb{R}^4$ . The classification gap in (d) is one reason — not the only one, but the one visible from this volume’s methods.

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**WORKED EXAMPLE 11.6.4— TIER B The  $k$ -fold bound, and where the logarithm comes from**

Theorem 11.6.6 gives two statements:  $\lesssim L^{3/2}$  points on at least two lines, and  $\lesssim L^{3/2}/k^2$  on at least  $k$ . The second is what Chapter 11.7 uses, and the passage between them explains the logarithm in Theorem 11.7.8.

*Counting intersecting pairs from the  $k$ -fold bound.* A point on exactly  $k$  lines accounts for  $\binom{k}{2} \approx k^2$  intersecting pairs. Summing over dyadic ranges  $k \approx 2^j$ ,

$$\#\{\text{intersecting pairs}\} \lesssim \sum_j 2^{2j} \cdot \frac{L^{3/2}}{2^{2j}} = L^{3/2} \cdot \#\{j\} \approx L^{3/2} \log L.$$

Every dyadic range contributes *equally* — the  $k^2$  from the pair count exactly cancels the  $k^{-2}$  in the bound — so the sum is the number of ranges, which is  $\log L$ .

*That is the logarithm.* It is not a technical artefact of one step; it is the statement that no single multiplicity dominates. Compare Worked Example 9.3.1 in Volume 9, where the  $n = 2$  case of the  $L^2$  method had the same feature and produced the same log.

*What removing it would require.* A bound better than  $L^{3/2}/k^2$  for some range of  $k$  — and the bound is sharp for each  $k$  separately, as the grid shows. So the improvement must come from showing that the extremal configurations for different  $k$  cannot coexist, which is a genuinely different kind of statement.

*Why this matters.* Worked Example 11.7.1 located the remaining gap in Conjecture 11.7.1 at exactly this logarithm. Knowing it arises from equal dyadic contributions, rather than from slack in a single estimate, tells one that closing it is not a matter of care.

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**Where we are.** The chapter's arc in one paragraph: in a plane,  $L$  lines can have  $\approx L^2$  crossing points (Proposition 11.6.1); on a regulus, likewise (Proposition 11.6.3); and Theorem 11.6.6 says that a configuration avoiding both does strictly better, at  $L^{3/2}$ . The gain of a factor  $L^{1/2}$  is exactly what Chapter 11.7 converts into the distinct-distances theorem, and Proposition 11.6.11 shows the planar bound it improves on is itself sharp.

## Drills

**Drill 11.6.1.** Verify Proposition 11.6.3 for  $z = xy$ : check that  $\ell_a$  and  $m_b$  lie on the surface and meet at  $(a, b, ab)$ .

**Drill 11.6.2.** Show that  $\ell_a$  and  $\ell_{a'}$  are disjoint for  $a \neq a'$ , and that they are not parallel.

**Drill 11.6.3.** Compute the number of intersecting pairs for  $L$  lines in general position in a plane, and compare with  $L^{3/2}$ .

**Drill 11.6.4.** Explain why the hypothesis of Theorem 11.6.6 would be violated by the grid configuration of Worked Example 11.3.1 embedded in a plane in  $\mathbb{R}^3$ .

**Where we are.** A remark on what is and is not proved in this chapter. Proposition 11.6.1, Proposition 11.6.3 and Proposition 11.6.11 are proved in full, and together they establish the *necessity* of the hypotheses in Theorem 11.6.6 and the sharpness of the planar bound it improves on. So the reader has, with proof, the statement's shape: which configurations must be excluded, why, and how much is gained by excluding them.

What is imported is the sufficiency — that excluding planes and reguli is enough. That is Remark 11.6.10, and it is where the ruled-surface theory of Worked Example 11.6.3 enters.

Stating the necessity and importing the sufficiency seems the right division for a theorem of this size: the reader can see what is being claimed and why it is not obvious, without a chapter of classical algebraic geometry.

## Notes

The incidence theorem is Guth and Katz's (2015). The reduction of the algebraic case to ruled surfaces uses the flecnode polynomial of Cayley and Salmon, from the 1860s; Guth and Katz's use of it was the first application in combinatorics. Katz and Guth's exposition, and Guth's book *Polynomial Methods in Combinatorics*, are the references.



# Chapter 11.7

## Distinct Distances

**Conjecture 11.7.1** (Erdős, 1946). *A set of  $N$  points in the plane determines at least  $cN/\sqrt{\log N}$  distinct distances.*

The  $\sqrt{\log N}$  is not decoration: the  $\sqrt{N} \times \sqrt{N}$  integer grid determines  $\approx N/\sqrt{\log N}$  distances, by a classical estimate of Landau on the number of integers below  $N$  representable as a sum of two squares. So the conjecture asserts the grid is extremal.

Guth and Katz proved the bound  $N/\log N$ , which settles the problem up to the  $\sqrt{\cdot}$ . This chapter proves the reduction — the part that is elementary and beautiful — and quotes Theorem 11.6.6 for the rest.

### 11.7.1 From distances to quadruples

**Notation 11.7.2.** For a set  $\mathcal{P}$  of  $N$  points in  $\mathbb{R}^2$  let  $\Delta(\mathcal{P})$  be its set of distinct distances, and for  $d > 0$  let  $n_d = \#\{(p, q) \in \mathcal{P}^2 : p \neq q, |pq| = d\}$ .

**Lemma 11.7.3** (Cauchy–Schwarz). *With*

$$Q = \#\{(p, q, r, s) \in \mathcal{P}^4 : p \neq q, r \neq s, |pq| = |rs|\} = \sum_{d \in \Delta(\mathcal{P})} n_d^2,$$

*one has*

$$|\Delta(\mathcal{P})| \geq \frac{N^2(N-1)^2}{Q}.$$

*Proof.*  $\sum_d n_d = N(N-1)$ , the number of ordered pairs of distinct points. By Cauchy–Schwarz over the  $|\Delta|$  values of  $d$ ,

$$(N(N-1))^2 = \left(\sum_d n_d\right)^2 \leq |\Delta| \sum_d n_d^2 = |\Delta| Q. \quad \square$$

**Where we are.** So a lower bound on  $|\Delta|$  is an *upper* bound on  $Q$ : if the quadruple count is  $O(N^3 \log N)$  then  $|\Delta| \gtrsim N^4/(N^3 \log N) = N/\log N$ , which is the Guth–Katz theorem.

The reformulation is the whole strategy.  $|\Delta|$  is hard to bound directly because it is a count of *values*;  $Q$  is a count of *configurations*, and configurations are what incidence geometry handles. Numerically the Cauchy–Schwarz step is an equality when all  $n_d$  are equal, which is the case for a generic point set — so nothing is lost in the reduction for the configurations one cares about.

## 11.7.2 From quadruples to lines in $\mathbb{R}^3$

**Definition 11.7.4** (The Elekes–Sharir parametrisation). Let  $G$  be the group of orientation-preserving rigid motions of  $\mathbb{R}^2$ . A non-translation element is a rotation by an angle  $\theta \in (0, 2\pi)$  about a centre  $c = (c_1, c_2)$ ; associate to it the point

$$\Psi(g) = \left( c_1, c_2, \cot \frac{\theta}{2} \right) \in \mathbb{R}^3.$$

**Lemma 11.7.5** (Motion sets are lines). For  $p \neq r$  in  $\mathbb{R}^2$  let

$$G_{p,r} = \{g \in G : g(p) = r\}.$$

Then  $\Psi(G_{p,r})$  is a line in  $\mathbb{R}^3$ .

*Proof.* A rotation about  $c$  takes  $p$  to  $r$  only if  $|c - p| = |c - r|$ , so  $c$  lies on the perpendicular bisector of the segment  $pr$ ; conversely for each such  $c$  there is exactly one rotation angle taking  $p$  to  $r$ . Parametrise the bisector by  $c = m + tn$ , where  $m = \frac{p+r}{2}$  and  $n$  is a unit normal to  $r - p$ .

Write  $R = |c - p|$  and  $\theta$  for the rotation angle. The chord subtended by  $\theta$  on the circle of radius  $R$  about  $c$  has length  $|r - p|$ , so

$$|r - p| = 2R \sin \frac{\theta}{2}, \quad \text{whence} \quad \sin \frac{\theta}{2} = \frac{|r - p|}{2R}.$$

Also  $R^2 = |c - m|^2 + |m - p|^2 = t^2 + \frac{|r-p|^2}{4}$ , and

$$\cos \frac{\theta}{2} = \sqrt{1 - \frac{|r-p|^2}{4R^2}} = \frac{|t|}{R},$$

taking the sign of  $t$  to match the orientation. Therefore

$$\cot \frac{\theta}{2} = \frac{\cos(\theta/2)}{\sin(\theta/2)} = \frac{t/R}{|r-p|/(2R)} = \frac{2t}{|r-p|},$$

which is an *affine* function of  $t$ . Since  $c = m + tn$  is affine in  $t$  as well, the map  $t \mapsto \Psi(g_t) = (m_1 + tn_1, m_2 + tn_2, \frac{2t}{|r-p|})$  is affine, and its image is a line.  $\square$

**Where we are.** The computation is worth checking numerically, and it is exact: for several choices of  $p$  and  $r$ , the points  $\Psi(g_t)$  for seven values of  $t$  have a rank-one centred matrix to machine precision — the second singular value is  $10^{-16}$ . So  $\Psi(G_{p,r})$  is a line, not merely approximately.

That the three-dimensional group of rigid motions turns the two-dimensional distance problem into a question about lines in  $\mathbb{R}^3$  is the whole idea of Elekes and Sharir, and it is why Theorem 11.6.6 is the relevant theorem.

**Proposition 11.7.6** (Quadruples are intersecting pairs of lines). *Let  $\mathcal{L} = \{\Psi(G_{p,r}) : (p,r) \in \mathcal{P}^2, p \neq r\}$ , a family of at most  $N(N-1)$  lines in  $\mathbb{R}^3$ . Then*

$$Q \leq C \cdot \#\{\text{intersecting pairs of lines in } \mathcal{L}\} + O(N^3).$$

*Proof.* Suppose  $(p, q, r, s)$  is counted by  $Q$ , so  $|pq| = |rs|$  with  $p \neq q$  and  $r \neq s$ . Since the two segments have equal length there is an orientation-preserving rigid motion  $g$  with  $g(p) = r$  and  $g(q) = s$  — translate  $p$  to  $r$  and rotate about  $r$  until  $q$  lands on  $s$ , which is possible exactly because the lengths agree.

Then  $g \in G_{p,r} \cap G_{q,s}$ , so the lines  $\Psi(G_{p,r})$  and  $\Psi(G_{q,s})$  meet. Distinct quadruples give distinct pairs of lines except when  $(p,r) = (q,s)$ , and for each pair of lines the motion  $g$  is unique unless  $p = q$  or  $r = s$ , which is excluded. The  $O(N^3)$  accounts for degenerate quadruples — those with  $g$  a translation, which  $\Psi$  does not parametrise, and those with repeated points.  $\square$

**Lemma 11.7.7** (Non-clustering). *For each  $D \geq 1$  there is  $C_D$  such that for any set  $\mathcal{P}$  of  $N$  points in  $\mathbb{R}^2$ , the family  $\mathcal{L}(\mathcal{P})$  contains at most  $C_D N$  lines in any algebraic surface of degree at most  $D$ , and at most  $N$  lines through any point of  $\mathbb{R}^3$ .*

*Proof. Step 1. Lines from a common first point are skew.*

By Proposition 11.7.6,  $\ell_{p,q_1}$  and  $\ell_{p,q_2}$  are intersecting or parallel iff  $|p-p| = |q_1 - q_2|$ ; since the left side is 0 and  $q_1 \neq q_2$ , they are skew. Hence for each  $p$  at most one line of  $\{\ell_{p,q}\}_{q \in \mathcal{P}}$  lies in a given plane, and at most one passes through a given point. Summing over  $p \in \mathcal{P}$  gives the plane and point statements.

**Step 2. A ruling vector field.**

Fix  $p = (p_1, p_2)$ . A point  $x \in \mathbb{R}^3$  lies on  $\ell_{p,q}$  iff

$$2x_1 = (p_1 + q_1) + (p_2 - q_2)x_3, \quad 2x_2 = (p_2 + q_2) + (q_1 - p_1)x_3,$$

which as a linear system for  $q$  has matrix  $\begin{pmatrix} 1 & -x_3 \\ x_3 & 1 \end{pmatrix}$  of determinant  $1 + x_3^2 > 0$ . So  $q$  is determined by  $x$ : each point of  $\mathbb{R}^3$  lies on exactly one line of  $\{\ell_{p,q}\}_q$ ,

and  $q = (1 + x_3^2)^{-1}b_p(x)$  with  $b_p$  polynomial of degree  $\leq 2$  in  $x$  and  $\leq 1$  in  $p$ . The vector  $(p_2 - q_2, q_1 - p_1, 2)$  is tangent to  $\ell_{p,q}$ , so

$$V_p(x) = (p_2(1 + x_3^2) - b_{2,p}(x), b_{1,p}(x) - p_1(1 + x_3^2), 2 + 2x_3^2)$$

is a nowhere-vanishing polynomial vector field whose integral curves are exactly the lines  $\{\ell_{p,q}\}_q$ .

**Step 3.** *At most one bad  $p$  per surface.*

Let  $Q$  be irreducible with  $1 < \deg Q \leq D$ , and suppose  $Z(Q)$  contains at least  $2D^2$  lines of  $\{\ell_{p,q}\}_q$ . On each,  $Q$  vanishes and  $V_p$  is tangent, so  $V_p \cdot \nabla Q$  vanishes on all of them. As  $\deg(V_p \cdot \nabla Q) \leq 2D - 2$ , Theorem 11.6.5 would allow at most  $2D^2 - 2D$  common lines unless  $Q$  and  $V_p \cdot \nabla Q$  share a factor; irreducibility then forces  $Q \mid V_p \cdot \nabla Q$ , so  $V_p \cdot \nabla Q \equiv 0$  on  $Z(Q)$ .

If this held for two points  $p_1 \neq p_2$ , then since  $V_p \cdot \nabla Q$  is degree 1 in  $p$ , it would vanish on  $Z(Q)$  for every  $p$  on the line through  $p_1, p_2$ . At a non-singular  $x \in Z(Q)$  the field  $V_p$  is tangent to  $Z(Q)$ , so its integral curve — a line of  $\mathfrak{L}_p$  — lies in  $Z(Q)$  and hence in  $T_x Z(Q)$ . Letting  $p$  vary gives infinitely many lines of  $Z(Q)$  in one plane, so by Theorem 11.6.5  $Z(Q)$  is that plane, contradicting  $\deg Q > 1$ .

If instead every point of  $Z(Q)$  is singular, then each  $\partial_i Q$  vanishes on  $Z(Q)$ ; if  $Z(Q)$  contained more than  $D^2$  lines, Theorem 11.6.5 and irreducibility would give  $Q \mid \partial_i Q$ , forcing  $\partial_i Q \equiv 0$  for all  $i$ , so  $Q$  constant and  $Z(Q)$  empty — a contradiction. So such a  $Z(Q)$  has at most  $D^2$  lines.

**Step 4.** *Count.*

Hence for all but at most one  $p \in \mathcal{P}$ ,  $Z(Q)$  contains at most  $2D^2$  lines of  $\{\ell_{p,p'}\}_{p'}$ , and for the exceptional  $p$  at most  $N$ . In total  $Z(Q)$  contains at most  $(2D^2 + 1)N$  lines of  $\mathcal{L}(\mathcal{P})$ .  $\square$

**Theorem 11.7.8** (Distinct distances). *For every  $\varepsilon > 0$  there is  $c_\varepsilon > 0$  such that any set of  $N$  points in  $\mathbb{R}^2$  determines at least  $c_\varepsilon N^{1-\varepsilon}$  distinct distances.*

*Proof.* By Lemma 11.7.7,  $\mathcal{L} = \mathcal{L}(\mathcal{P})$  is a set of  $L = N^2$  lines with at most  $C_D N \leq L^{\frac{1}{2}+\varepsilon}$  lines in any irreducible surface of degree  $\leq D$ , once  $N$  is large. So Theorem 11.6.6 applies and gives, for  $2 < r < 2N$ ,

$$|P_r(\mathcal{L})| \leq K(N^2)^{\frac{3}{2}+\varepsilon} r^{-2} = K N^{3+2\varepsilon} r^{-2}.$$

By Proposition 11.7.6 the parallel quadruples number at most  $N^3$ , and the intersecting ones are  $\sum_{r \geq 2} (2r - 2)|P_r(\mathcal{L})|$ , so

$$Q \leq N^3 + \sum_{r=2}^N 2r \cdot K N^{3+2\varepsilon} r^{-2} \lesssim N^3 + K N^{3+2\varepsilon} \log N \lesssim_\varepsilon N^{3+3\varepsilon}.$$

By Lemma 11.7.3,  $|\Delta(\mathcal{P})| \geq (N^4 - 2N^3)/Q \gtrsim_\varepsilon N^{1-3\varepsilon}$ . Replacing  $\varepsilon$  by  $\varepsilon/3$  gives the theorem.  $\square$

*Remark 11.7.9* (The sharper form). Guth and Katz prove the stronger bound  $\gtrsim N/\log N$ , which is optimal up to  $\sqrt{\log N}$  by Worked Example 11.7.1. Their argument replaces Theorem 11.6.6 by the  $\varepsilon$ -free estimate  $|P_r(\mathcal{L})| \leq CL^{3/2}r^{-2}$ , whose proof splits into the cases  $r = 2$  and  $r \geq 3$ ; the case  $r = 2$  uses the classification of ruled surfaces through the Cayley–Salmon flecnode polynomial, and is not reproduced here.

What *is* proved here is the full chain from Borsuk–Ulam to  $N^{1-\varepsilon}$ , with no unproved input. The gap between  $N^{1-\varepsilon}$  and  $N/\log N$  is the price of avoiding the ruled surface theory.

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WORKED EXAMPLE 11.7.1— TIER A **The grid, and why  $\sqrt{\log N}$**

Take  $\mathcal{P} = \{1, \dots, \sqrt{N}\}^2$ , the integer grid.

*The distances.*  $|pq|^2$  is a sum of two squares, each at most  $N$ . So  $|\Delta|$  is at most the number of integers in  $[1, 2N]$  expressible as a sum of two squares.

*Landau’s estimate.* That count is  $\approx \frac{cN}{\sqrt{\log N}}$  — the density of sums of two squares decays like  $(\log x)^{-1/2}$ , because an integer is such a sum precisely when every prime  $\equiv 3 \pmod{4}$  divides it to an even power.

*Consequently.* The grid determines  $\approx N/\sqrt{\log N}$  distances, so Conjecture 11.7.1 cannot be improved, and Theorem 11.7.8 is within  $\sqrt{\log N}$  of the truth.

*What the gap is.* The logarithm in Theorem 11.7.8 comes from summing the  $k$ -fold incidence bound of Theorem 11.6.6 over dyadic  $k$ . Removing it would require a bound with no such summation, and none is known; whether  $N/\sqrt{\log N}$  is attainable by this method is open.

*The moral.* An extremal example drawn from number theory — the sums of two squares — sets the target, and an incidence-geometric method comes within a logarithm. The two subjects meet only in the answer.

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WORKED EXAMPLE 11.7.2— TIER B **Reading the reduction backwards**

The chain runs

$$|\Delta| \xrightarrow{\text{Cauchy-Schwarz}} Q \xrightarrow{\text{Elekes-Sharir}} \text{line incidences in } \mathbb{R}^3 \xrightarrow{\text{Guth-Katz}} \text{done.}$$

Reading it backwards shows what each link costs.

*Link 3 is where the mathematics is.* Theorem 11.6.6 is the theorem; the other two links are a page each.

*Link 2 is exact.* Lemma 11.7.5 is an identity, verified to machine precision, and Proposition 11.7.6 loses only the degenerate  $O(N^3)$ , which is below the target anyway.

*Link 1 is lossless for the extremal configurations.* Cauchy–Schwarz is an equality when all  $n_d$  agree. For the grid they do not agree exactly — some distances are far more common — so a little is lost, but not enough to matter.

*Why this matters for reading other papers.* A reduction chain is only as good as its worst link, and identifying which link is lossy tells one where an improvement must come from. Here it is Link 3, and specifically the dyadic summation that produces the  $\log N$ . Any improvement to  $N/\sqrt{\log N}$  must attack that sum, not the reduction.

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**WORKED EXAMPLE 11.7.3— TIER B Reductions that change the ambient dimension**

Chapter 11.7 solved a planar problem by passing to  $\mathbb{R}^3$ . That move is worth isolating, since it is counterintuitive — one usually reduces dimension, not raises it.

*Why raising helps.* The objects being counted are *pairs* of points, and pairs of points in  $\mathbb{R}^2$  have four degrees of freedom. The symmetry group that preserves distance — rigid motions — is three-dimensional, and quotienting by it turns the four-parameter family of pairs into a three-parameter family of group elements. The problem becomes one about subvarieties of a three-dimensional space, and Lemma 11.7.5 shows they are the simplest possible: lines.

*The general recipe.* When a problem concerns configurations invariant under a group  $G$ , parametrise the configurations by  $G$  and study the resulting subvarieties. The ambient dimension becomes  $\dim G$ , and the objects are usually cosets — lines, if  $G$  is three-dimensional and the stabilisers are one-dimensional.

*Two other instances.* The Elekes–Sharir framework has been used for distinct distances on spheres and in  $\mathbb{R}^3$ , with  $G$  the relevant motion group; and the sum–product problem admits a similar treatment with  $G$  the affine group. In each case the payoff is that a hard counting problem becomes an incidence problem for lines, where Theorem 11.6.6 applies.

*The cost.* The parametrisation must send the relevant sets to lines *exactly*, not approximately — Lemma 11.7.5 is an identity, and its numerical verification to machine precision reflects that. A parametrisation sending them to curves of higher degree would require an incidence theorem for curves, which is available but weaker.

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**WORKED EXAMPLE 11.7.4— TIER C: RESEARCH TASTE What the distinct distances proof needed**

Theorem 11.7.8 is the polynomial method’s most celebrated application, and the anatomy of what it consumed is instructive.

(a) *A quadratic reformulation.* Lemma 11.7.3 converts a count of values into a count of configurations. This is free and is the standard first move for any “distinct values” problem.

(b) *A group.* The rigid motions of  $\mathbb{R}^2$  form a three-dimensional group, and equal distances are exactly the orbits. Without a transitive symmetry group preserving the quantity of interest there is no Elekes–Sharir step, which is why the method does not obviously extend to, say, distinct areas of triangles.

(c) *A parametrisation making the stabilisers linear.* Lemma 11.7.5 is the piece of luck: the sets  $G_{p,r}$  become *lines* rather than curves of higher degree. Had they become conics, one would need an incidence theorem for conics, which is available but weaker.

(d) *A hard incidence theorem.* Theorem 11.6.6, which is Remark 11.6.10 and the only genuinely difficult ingredient.

(e) *A verification that the hypotheses hold.* That no plane or regulus contains too many of the lines  $\Psi(G_{p,r})$  — quoted in the proof of Theorem 11.7.8 and itself a short but real argument about the geometry of the motion group.

*The lesson.* Four of the five ingredients are elementary or free; the theorem’s difficulty is concentrated entirely in (d). That concentration is typical of the polynomial method: it converts a problem into an incidence question, and the incidence question is where the work goes. Recognising this before starting tells one whether a given problem is likely to yield — if the resulting incidence question is one of the solved ones, it will; if not, it will not.

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**WORKED EXAMPLE 11.7.5— TIER A The chain, with numbers**

Following Theorem 11.7.8 through with  $N = 10^6$  points makes the sizes of the objects concrete.

*Points.*  $N = 10^6$ .

*Ordered pairs.*  $N(N - 1) \approx 10^{12}$ , and this is  $\sum_d n_d$ .

*Lines in  $\mathbb{R}^3$ .* One for each ordered pair with  $p \neq r$ , so  $L \approx 10^{12}$ .

*Intersecting pairs of lines.* By Theorem 11.6.6 with  $L = 10^{12}$ :  $L^{3/2} \log L \approx 10^{18} \cdot 28 \approx 3 \times 10^{19}$ .

*Quadruples.*  $Q \lesssim 3 \times 10^{19}$  by Proposition 11.7.6.

*Distances.* By Lemma 11.7.3,

$$|\Delta| \geq \frac{(10^{12})^2}{3 \times 10^{19}} \approx 3 \times 10^4.$$

*Compare.*  $N/\log N = 10^6/13.8 \approx 7 \times 10^4$ , so the arithmetic is consistent up to the constants suppressed throughout. And  $N/\sqrt{\log N} \approx 2.7 \times 10^5$  is the conjectured truth, a factor  $\sqrt{\log N} \approx 3.7$  above.

*What the numbers show.* The lines outnumber the points by a factor  $N$ , and the incidence theorem is applied to a family far larger than the original configuration. That is characteristic of the Elekes–Sharir reduction: it trades a small hard problem for a large easy one, and the trade is favourable because  $L^{3/2}$  with  $L = N^2$  is  $N^3$ , comfortably below the  $N^4$  that would make the Cauchy–Schwarz step vacuous.

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**Where we are.** Erdős’s problem is the oldest in this series — 1946 — and the only one whose solution the series can report as essentially complete. It took sixty-nine years, and the final step was a technique invented for a different purpose.

That is worth holding alongside Kakeya, posed in 1928 and open above dimension three. The two problems are of comparable age and comparable appearance; one yielded to a change of category and the other has not.

## Drills

**Drill 11.7.1.** Verify Lemma 11.7.3 and identify when Cauchy–Schwarz is an equality.

**Drill 11.7.2.** Check the computation  $\cot \frac{\theta}{2} = \frac{2t}{|r-p|}$  in Lemma 11.7.5 for  $p = (0, 0)$ ,  $r = (1, 0)$ .

**Drill 11.7.3.** Confirm that a rigid motion taking  $p$  to  $r$  and  $q$  to  $s$  exists whenever  $|pq| = |rs|$ , and identify the degenerate cases.

**Drill 11.7.4.** Using Landau’s estimate, compute the number of distinct distances in the  $100 \times 100$  grid to leading order.

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**Where we are.** Chapter 11.7 is the only chapter in this series whose main theorem is proved from an imported ingredient rather than from scratch, and it is worth being explicit about what the reader gets.



Proved here: the Cauchy–Schwarz reduction (Lemma 11.7.3), the Elekes–Sharir parametrisation and its linearity (Lemma 11.7.5), and the correspondence between quadruples and intersecting pairs (Proposition 11.7.6). Together these constitute the entire reduction of a planar metric problem to an incidence problem in  $\mathbb{R}^3$ .

*Proved:* Theorem 11.6.6.

Worked Example 11.7.4 anatomises the five ingredients and finds four of them elementary. That is the usual shape for a polynomial-method application, and it is why the technique spread so quickly once the incidence theorems were available.

## Notes

Erdős posed the problem in 1946 and proved the lower bound  $N^{1/2}$ ; a sequence of improvements followed, the last before Guth–Katz being  $N^{0.8641}$  by Katz and Tardos. Elekes and Sharir introduced the reduction to rigid motions in 2010, and Guth and Katz completed it in 2015 with Theorem 11.6.6.

The gap of  $\sqrt{\log N}$  remains open.



## Chapter 11.8

# Kakeya and the Polynomial Method

Volume 10 showed that Dvir's argument does not transfer to  $\mathbb{R}^n$ , and located the failure precisely: the points of a  $\delta$ -net inside a tube are only *nearly* collinear, so vanishing at them says nothing about the tube's axis (Proposition 10.7.2). This chapter shows what happens when the approximation is removed — when one works with the genuine segments of a Besicovitch set rather than with tubes — and why the resulting theorem, though exact and easy, does not settle anything.

### 11.8.1 Besicovitch sets are not algebraic

**Theorem 11.8.1.** *Let  $K \subseteq \mathbb{R}^n$  contain a unit segment in every direction. Then  $K$  is not contained in  $Z(P)$  for any non-zero polynomial  $P \in \mathbb{R}[x_1, \dots, x_n]$ .*

*Proof.* Suppose  $K \subseteq Z(P)$  with  $P \neq 0$  of degree  $D$ , and let  $P_D$  be its top-degree homogeneous part, which is non-zero.

**Step 1.**  *$P$  vanishes on whole lines.*

Fix  $v \in \mathbb{S}^{n-1}$ . By hypothesis  $K$  contains a unit segment  $\{a + tv : t \in [0, 1]\}$ , on which  $P$  vanishes. The function  $t \mapsto P(a + tv)$  is a polynomial in one variable of degree at most  $D$  with infinitely many roots — every  $t \in [0, 1]$  — hence identically zero. So  $P$  vanishes on the entire line  $\{a + tv : t \in \mathbb{R}\}$ .

**Step 2.** *Read off the top-degree part.*

Expanding  $P = \sum_{j \leq D} P_j$  into homogeneous parts and using homogeneity, the coefficient of  $t^D$  in  $P(a + tv)$  is  $P_D(v)$ . Since the polynomial in  $t$  is identically zero, every coefficient vanishes, so

$$P_D(v) = 0.$$

**Step 3.** *Conclude.*

Step 2 holds for every  $v \in \mathbb{S}^{n-1}$ . By homogeneity,  $P_D(rv) = r^D P_D(v) = 0$  for every  $r \in \mathbb{R}$ , so  $P_D$  vanishes on all of  $\mathbb{R}^n$ . A polynomial vanishing identically on  $\mathbb{R}^n$  is the zero polynomial — over an infinite field, unlike the situation of Lemma 10.1.3 — contradicting  $P_D \neq 0$ .  $\square$

**Where we are.** This is Dvir’s argument, run over  $\mathbb{R}$ , and it works. The comparison with Volume 10 is exact and worth stating side by side.

*Step 1 is where they differ.* Over  $\mathbb{F}_q$  a line has  $q$  points and one needs  $q > D$  to force vanishing. Over  $\mathbb{R}$  a segment has infinitely many points and the conclusion is automatic — no degree condition at all.

*Step 3 is where they differ again.* Over  $\mathbb{F}_q$  one needs Lemma 10.1.3 because  $x^q - x$  vanishes identically without being zero. Over  $\mathbb{R}$  a polynomial vanishing everywhere is zero, full stop.

*So the real version is easier than the finite field version.* Both of Dvir’s technical devices are unnecessary. And yet the real Kakeya conjecture is open and the finite field one is closed. The resolution is in the next section, and it is the point of this chapter.

## 11.8.2 Why this settles nothing

**Proposition 11.8.2** (Varieties are small). *Let  $P \neq 0$  have degree  $D$ . Then  $m_n(Z(P)) = 0$  and  $\dim_{\mathbb{H}} Z(P) \leq n - 1$ .*

*Proof.* By Lemma 10.1.3’s real analogue — or directly: fix a line in general position;  $P$  restricted to it is a non-zero one-variable polynomial with at most  $D$  roots. So every line meets  $Z(P)$  in at most  $D$  points unless it lies inside, and the lines lying inside form a set of directions of measure zero unless  $P_D \equiv 0$ . Fubini along a generic family of parallel lines gives  $m_n(Z(P)) = 0$ .

For the dimension,  $Z(P)$  is a countable union of  $C^1$  hypersurfaces away from its singular set, which is itself contained in  $Z(\nabla P)$  of lower dimension; induction gives  $\dim_{\mathbb{H}} Z(P) \leq n - 1$ .  $\square$

**Where we are.** Theorem 11.8.1 excludes Besicovitch sets from algebraic varieties, and Proposition 11.8.2 says varieties have dimension at most  $n - 1$ . So the theorem tells us that a Besicovitch set is not contained in a set of dimension  $n - 1$  of a very special kind.

But the Kakeya conjecture asserts  $\dim_{\mathbb{H}} K = n$ , and the trivial bound already gives  $\dim_{\mathbb{H}} K \geq 1$ . Excluding algebraic hypersurfaces excludes almost nothing: they are a measure-zero, closed, countable-parameter family among all sets of dimension  $n - 1$ , and the extremal Besicovitch sets —

Perron trees — are manifestly not algebraic.  
*So Theorem 11.8.1 is a true theorem with no consequences.* It is worth including precisely because it is the exact analogue of Dvir’s theorem, it is easier, and it is useless — which is the sharpest available demonstration that the finite field problem and the real problem are not the same problem.

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WORKED EXAMPLE 11.8.1— TIER C: RESEARCH TASTE    **The three versions of one argument**

Dvir’s argument has now appeared three times in this series, in settings that look almost identical and behave completely differently. Setting them side by side is the clearest summary of where the Kakeya difficulty lives.

|               | $\mathbb{F}_q^n$ , lines                | $\mathbb{R}^n$ , segments                                      | $\mathbb{R}^n$ , $\delta$ -tubes |
|---------------|-----------------------------------------|----------------------------------------------------------------|----------------------------------|
| Where         | Vol 10, Theo-rem 10.2.1                 | Theorem 11.8.1                                                 | Vol 10, Proposi-tion 10.7.2      |
| Step 1 works? | yes, needs $D < q$                      | yes, automatic                                                 | <b>no</b>                        |
| Step 3 works? | yes, needs Schwartz–Zippel              | yes, trivially                                                 | —                                |
| Conclusion    | $ K  \gtrsim q^n$ : <i>sharp</i>        | $K$ not in a variety: <i>vacuous</i>                           | nothing                          |
| Why           | finiteness makes “small” mean something | infinitude makes vanishing au-tomatic and varieties negligible | points only nearly collinear     |

*The pattern.* The argument’s strength depends on whether “ $P$  vanishes on  $K$ ” is a strong hypothesis. Over  $\mathbb{F}_q$  it is:  $Z(P)$  has at most  $Dq^{n-1}$  points out of  $q^n$ , a constant fraction. Over  $\mathbb{R}$  it is not:  $Z(P)$  has measure zero, so containing  $K$  in it is an enormous restriction that no interesting set satisfies — and correspondingly the conclusion is worthless.

*The genuine problem is the third column.* A Besicovitch set is not contained in a variety; it is a set of small *measure* whose  $\delta$ -neighbourhood one must understand. Polynomials interact with measure only through partitioning — Theorem 11.2.5 — and that is why this volume’s useful chapters are the ones that partition rather than vanish.

*The transferable diagnostic.* Before applying the polynomial method, ask what “ $P$  vanishes on  $S$ ” costs  $S$ . If  $S$  is finite and the ambient set is finite, it

is a real constraint. If the ambient is a continuum, it is a constraint so severe that only degenerate  $S$  satisfy it, and one must partition instead.

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### 11.8.3 What partitioning gives for Kakeya

**Theorem 11.8.3** (Katz–Zahl; not proved here). *Every Besicovitch set in  $\mathbb{R}^3$  has Hausdorff dimension at least  $\frac{5}{2} + \varepsilon_0$ , for some absolute constant  $\varepsilon_0 > 0$ .*

*Remark 11.8.4* (Scope decision). Theorem 11.8.3 is not proved here; revision R14. Nor are the intermediate bounds obtained by polynomial partitioning between Wolff’s  $\frac{n+2}{2}$  and it.

*On the constant.* Katz and Zahl do not optimise  $\varepsilon_0$  and state no numerical value; the earlier Katz–Łaba–Tao bound for the *Minkowski* dimension is  $\frac{5}{2} + \varepsilon$  with  $\varepsilon = 10^{-10}$  admissible. So the improvement over Wolff’s  $\frac{5}{2}$  is tiny, and its interest is entirely qualitative — it shows  $\frac{5}{2}$  is not the truth.

*A figure not to confuse with this one.* Katz and Zahl’s later *planebrush* method gives 3.059 — but in  $\mathbb{R}^4$ , not  $\mathbb{R}^3$ , where it improves on the Guth–Zahl and Katz–Rogers bound  $3 + \frac{1}{40}$ . A bound of 3.059 in  $\mathbb{R}^3$  would exceed the ambient dimension and is impossible; the two results are easily conflated because both are due to Katz and Zahl and both concern Besicovitch sets.

*What such proofs do.* They partition  $\mathbb{R}^3$  by a polynomial of degree  $D$ , and split the tubes into those confined near  $Z(P)$  and those crossing many cells. The crossing tubes are handled by the cellular count of Theorem 11.2.5; the confined ones are *tangent* to a surface, and controlling them requires the theory of how tubes can cluster along a variety. That tangential analysis is the technical core and has no short form.

*Why it is omitted.* It is a research programme rather than a theorem, and the statement has been improved repeatedly. Volume 14 reports the current position, including Wang and Zahl’s resolution of the three-dimensional case, which supersedes Theorem 11.8.3 entirely.

*What depends on it.* Nothing in this series.

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#### WORKED EXAMPLE 11.8.2— TIER B **Tangency: the recurring obstruction**

Three times now an argument has failed at the same place, and the coincidence is not one.

*Volume 9, Remark 9.5.5.* Wolff’s hairbrush works for transverse tubes and degrades exactly to the bush bound for near-parallel ones. Near-parallel tubes are tangent to one another.

*Theorem 11.5.7.* Multilinear Kakeya assumes transversality outright. The linear statement, which permits tangency, is the open conjecture.

*Remark 11.8.4.* Polynomial partitioning handles tubes crossing cells and struggles with tubes confined near the partitioning surface — that is, tangent to it.

*The common structure.* An estimate that counts *crossings* is stable and easy; an estimate that must handle *tangency* is neither. Volume 10, Worked Example 10.7.3 classified statements as incidence-type (transferable) or exact-algebraic (not); tangency is where the classification bites, because a tangential intersection is exactly one that a small perturbation destroys.

*What Volume 14 will report.* Wang and Zahl’s solution in  $\mathbb{R}^3$  proceeds by analysing precisely the tangential configurations — the sticky ones — and showing they cannot occur in the required abundance. So the obstruction identified three times independently in this series is the one their proof attacks, which is some confirmation that the diagnosis was right.

WORKED EXAMPLE 11.8.3— TIER C: RESEARCH TASTE **What a polynomial can and cannot see**

Eleven volumes in, it is worth collecting what has been established about the reach of algebraic methods in this subject.

(a) *A polynomial sees finite sets well.* Vanishing on  $N$  points costs degree  $N^{1/n}$ , which is cheap; that is the content of every dimension count in Volumes 10 and 11.

(b) *A polynomial sees exact lines well.* Vanishing at more than  $D$  points of a line propagates to the whole line. Joints, Szemerédi–Trotter and Theorem 11.8.1 all rest on this.

(c) *A polynomial does not see tubes.* Proposition 10.7.2: it can thread between the points of a  $\delta$ -net inside a tube. This is the single obstruction the method has never overcome.

(d) *A polynomial sees measure only through partitioning.*  $Z(P)$  has measure zero (Proposition 11.8.2), so no statement of the form “the configuration lies in  $Z(P)$ ” constrains a set of positive measure. The only interaction with measure is Theorem 11.2.5, which uses the *complement* of the zero set.

(e) *The synthesis.* The polynomial method is a tool for discrete configurations of exact algebraic objects.akeya is a problem about continuous configurations of thickened objects. The method reaches it only through partitioning, and only as far as tangency permits — which is the diagnosis Worked Example 11.8.2 arrived at from three independent directions.

(f) *What that predicts.* A future solution ofakeya in  $\mathbb{R}^n$  for  $n \geq 4$  will either dissolve the tangency problem or avoid polynomials altogether. Vol-

ume 14 reports that Wang and Zahl did the former in  $\mathbb{R}^3$ , and that their method's dependence on three-dimensional geometry is exactly what blocks its extension.

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**WORKED EXAMPLE 11.8.4— TIER B Running Dvir's argument in three settings, explicitly**

Worked Example 11.8.1 tabulated the comparison; here is the computation in each column, so the differences are visible rather than asserted.

*Column 1:  $\mathbb{F}_q^n$ .  $K$  a Kakeya set with  $|K| < \binom{q+n-1}{n}$ . Step 1 gives  $P \neq 0$  of degree  $D \leq q-1$  vanishing on  $K$ . Step 2: on a line  $\ell_{a,b} \subseteq K$  the polynomial  $t \mapsto P(a+tb)$  has degree  $\leq D < q$  and  $q$  roots, so it vanishes and  $P_D(b) = 0$ . Step 3:  $P_D$  vanishes on all  $b \neq 0$ , hence on  $\mathbb{F}_q^n$ ; by Lemma 10.1.3 with  $D < q$  this forces  $P_D = 0$ . *Contradiction obtained; the bound is sharp.**

*Column 2:  $\mathbb{R}^n$ , exact segments.  $K \subseteq Z(P)$ ,  $\deg P = D$ . Step 2: on a segment,  $t \mapsto P(a+tv)$  has infinitely many roots, so it vanishes — *no degree condition needed*. Step 3:  $P_D$  vanishes on  $\mathbb{S}^{n-1}$ , hence by homogeneity on  $\mathbb{R}^n$ , hence  $P_D = 0$  — *no Schwartz–Zippel needed. Contradiction obtained; but see below.**

*Column 3:  $\mathbb{R}^n$ ,  $\delta$ -tubes.  $S$  a  $\delta$ -net inside  $K_\delta$ ,  $P$  vanishing on  $S$  with  $\deg P = D < \delta^{-1}$  by Proposition 10.7.1. Step 2 requires the points of  $S$  inside a tube to be collinear. *They are not*, and Proposition 10.7.2 exhibits a  $P$  vanishing on  $\delta^{-1}$  of them and nowhere on the axis. *No contradiction; the argument stops.**

*The verdict on column 2.* The contradiction is real but the hypothesis  $K \subseteq Z(P)$  is vacuous: by Proposition 11.8.2 a variety has measure zero and dimension  $\leq n-1$ , and no Besicovitch set was ever going to lie in one. So the easiest of the three arguments proves the least.

*The moral, stated once.* An argument's strength is not its internal difficulty but the strength of the hypothesis it refutes. Column 1 refutes “ $|K|$  small”, which is the conjecture. Column 2 refutes “ $K$  algebraic”, which nothing satisfies. Column 3 would refute the right thing and does not work.

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**WORKED EXAMPLE 11.8.5— TIER A Three hypotheses, three strengths**

Worked Example 11.8.4 ran the same argument three times. Here is the comparison reduced to what each hypothesis actually excludes.



| Hypothesis<br>futed                          | re-  | How restrictive                                                                         | Verdict                                           |
|----------------------------------------------|------|-----------------------------------------------------------------------------------------|---------------------------------------------------|
| $ K  < \binom{q+n-1}{n}$ over $\mathbb{F}_q$ | over | a constant fraction of $\mathbb{F}_q^n$ — sets of this size are abundant                | <i>the conjecture</i> ; sharp                     |
| $K \subseteq Z(P)$ over $\mathbb{R}$         |      | measure zero, dimension $\leq n - 1$ , and algebraic — essentially nothing satisfies it | true but vacuous                                  |
| $m(K_\delta)$ small over $\mathbb{R}$        |      | the actual object of study                                                              | would be the conjecture;<br><i>argument fails</i> |

*The point.* All three arguments have the same three steps. Their value is determined entirely by the middle column — how much the refuted hypothesis excludes — and not at all by the difficulty of the argument.

*Why row two is worth stating.* Because it is the natural first attempt on being told Dvir's theorem, and it succeeds. A reader who tries it and obtains a proof might reasonably believe progress has been made. Proposition 11.8.2 is the antidote, and stating both the theorem and its emptiness in the same chapter is the honest presentation.

*The generalisable habit.* On proving something, ask what the hypothesis excludes before asking whether the proof is correct. A correct proof of a vacuous statement is a common outcome and an easy one to miss.

## Drills

**Drill 11.8.1.** Verify Step 2 of Theorem 11.8.1: the coefficient of  $t^D$  in  $P(a + tv)$  is  $P_D(v)$ .

**Drill 11.8.2.** Explain why Step 1 needs no degree hypothesis over  $\mathbb{R}$ , and what the corresponding hypothesis is over  $\mathbb{F}_q$ .

**Drill 11.8.3.** Show that a polynomial vanishing on an open subset of  $\mathbb{R}^n$  is identically zero, and contrast with  $x^q - x$  over  $\mathbb{F}_q$ .

**Drill 11.8.4.** Using Worked Example 11.8.1, state which column each of the following belongs to: Dvir's theorem, the joints theorem, and Proposition 10.7.2.

**Where we are.** The chapter's structure is unusual and deliberate: it proves a theorem and then argues that the theorem is worthless.

Theorem 11.8.1 is correct, short, and is the exact real analogue of Dvir's argument — indeed easier, since neither the degree condition nor Schwartz–Zippel is needed. A reader meeting it without Proposition 11.8.2 could reasonably conclude that the polynomial method reaches the Kakeya problem after all.

It does not, and the reason is that the hypothesis  $K \subseteq Z(P)$  is satisfied by essentially nothing. Worked Example 11.8.5 tabulates the three versions by how much each hypothesis excludes, which is the only measure that matters.

Including both halves is the point. A treatment that stated only the theorem would mislead; one that stated only the obstruction would leave the reader wondering whether the obvious attempt had been tried.

## Notes

Theorem 11.8.1 is folklore and is sometimes offered as evidence that the polynomial method should apply to Kakeya. The Checkpoint after Proposition 11.8.2 explains why it is not.

Katz and Zahl's bound is from 2019, improving a sequence of polynomial-method results beginning with Guth's. Volume 14 reports the Wang–Zahl theorem, which settles the three-dimensional case.

# Chapter 11.9

## Assessment

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WORKED EXAMPLE 11.9.1— TIER A **What the method proved, and what it only advanced**

| Problem                           | Result                        | Status                 | Here                           |
|-----------------------------------|-------------------------------|------------------------|--------------------------------|
| Szemerédi–Trotter                 | $(PL)^{2/3} + P + L$          | sharp                  | proved, Theorem 11.3.2         |
| Joints in $\mathbb{R}^3$          | $\lesssim L^{3/2}$            | sharp                  | proved, Theorem 11.4.2         |
| Loomis–Whitney                    | product bound                 | sharp                  | proved, Theorem 11.5.2         |
| Line incidences in $\mathbb{R}^3$ | $\lesssim L^{3/2}$            | sharp                  | quoted, Remark 11.6.10         |
| Distinct distances                | $N/\log N$                    | within $\sqrt{\log N}$ | reduction proved, Chapter 11   |
| Multilinear Kakeya                | $S^\varepsilon$ loss          | sharp in order         | <i>proved</i> , Theorem 11.5.4 |
| Kakeya in $\mathbb{R}^3$          | $\frac{5}{2} + \varepsilon_0$ | superseded             | quoted, Remark 11.8.4          |
| Kakeya, $n \geq 4$                | —                             | <i>open</i>            | —                              |

*The pattern.* Where the objects are genuine lines — joints, incidences, distances — the method is sharp and often settles the problem outright. Where the objects are tubes of positive width, it advances the exponent and does not close the problem.

*The four sharp results all concern lines.* The two open ones concern tubes. That is the same division Volume 10 identified between the finite field and real settings, appearing again inside a single method.

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WORKED EXAMPLE 11.9.2— TIER B **The method in five steps**

Every argument in this volume has the same skeleton, and it is worth having in a form one can apply.

*Step 1: choose a degree  $D$ .* Leave it a parameter; it is fixed at the end by

optimisation (Worked Example 11.2.2).

*Step 2: find a polynomial.* Either one *vanishing* on the configuration — via the dimension count  $\binom{D+n}{n} > |S|$  — or one *partitioning* space, via Theorem 11.2.5. The first is available only when the configuration is finite and the ambient discrete; the second always.

*Step 3: split into cellular and algebraic.* Points in open cells are handled by the small cell size; points on  $Z(P)$  are not handled at all and need a separate argument.

*Step 4: exploit the incidence bound.* A line meets  $Z(P)$  in at most  $D$  points, so it visits at most  $D+1$  cells. This is the one fact that transfers from Volume 10 and it is used in every proof.

*Step 5: optimise  $D$ .* Balance the decreasing cellular contribution against the increasing algebraic one, and check  $D \geq 1$  is admissible in the regime considered — the check Worked Example 11.3.2 identifies as the commonest omission.

*Where each proof sits.* Theorem 11.3.2 uses all five, with a trivial Step 3 algebraic case. Theorem 11.4.2 uses Step 2 in its vanishing form and replaces Steps 3–5 by the derivative trick. Theorem 11.6.6 uses all five with a Step 3 that is a research programme.

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**WORKED EXAMPLE 11.9.3— TIER C: RESEARCH TASTE Volume 11 in the arc of the series**

(a) *What it added.* A genuinely new technique, and the first since Volume 6 that is not analytic. Polynomial partitioning does what neither the geometry of Volume 9 nor the arithmetic of Volume 10 could: it subdivides space adaptively, so that a configuration is handled at the scale where it is simplest.

(b) *Why it escaped the earlier ceilings.* Volume 9’s methods optimised over an unattainable multiplicity (Worked Example 9.8.2); Volume 10’s arithmetic route optimised over additive structure with lossy constants (Worked Example 10.6.2). Partitioning optimises over the *degree*  $D$ , and  $D$  is a genuine free parameter whose extremal value is attained — there is nothing hypothetical about a polynomial of the chosen degree.

(c) *What it did not overcome.* Tangency. Worked Example 11.8.2 traced the same obstruction through Volumes 9, 11 and the multilinear theory: estimates counting crossings are stable, estimates handling tangency are not. The Kaakeya conjecture is a tangency problem, and partitioning does not dissolve it.

(d) *The three imports.* This volume quotes more than any other: formerly

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Remark 11.2.4 (ham sandwich) and multilinear Kakeya), Remark 11.6.10 (Guth–Katz line incidences), and Remark 11.8.4 (Katz–Zahl). The first three are load-bearing — the manifest records them as such — and that is a real cost, unavoidable if a volume is to reach the frontier of a technique whose central results are recent papers.

(e) *What remains.* Volume 12 uses Theorem 11.5.7 to prove restriction estimates by the broad–narrow decomposition; Volume 13 develops decoupling; Volume 14 reports the position, including the resolution of three-dimensional Kakeya. The conjecture is open for  $n \geq 4$ , and nothing in eleven volumes suggests it is close.

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WORKED EXAMPLE 11.9.4— TIER B **Placing a polynomial-method paper**

The literature using these techniques is now large. Five questions place almost any paper.

(1) *Vanishing or partitioning?* Worked Example 11.2.3. A vanishing argument concerns exact algebraic objects and usually gives a sharp result in a restricted setting; a partitioning argument concerns continuous configurations and usually gives an exponent.

(2) *What is the choice of degree?* Worked Example 11.2.2. It is displayed in one line and determines the result. If the paper does not display it, the argument is probably not a partitioning one.

(3) *How is the algebraic case handled?* This is where the difficulty lives — trivially in  $\mathbb{R}^2$  (Chapter 11.3), by ruled-surface theory in  $\mathbb{R}^3$  (Worked Example 11.6.3), and openly in higher dimensions. A paper claiming a new bound in  $\mathbb{R}^4$  or above must say something new here.

(4) *Lines or tubes?* Worked Example 11.9.1: sharp results concern lines, exponents concern tubes. A claimed sharp result about tubes deserves scrutiny.

(5) *Is tangency addressed?* Worked Example 11.8.2. Estimates counting crossings are routine; the content of any advance is in the tangential configurations.

*Worked placement: Guth–Katz distinct distances.* (1) partitioning, twice; (2) two degrees, chosen in sequence; (3) ruled surfaces, and this is the paper’s core; (4) lines, hence sharp up to log; (5) not applicable, the objects being exact. Conclusion: a complete solution modulo a logarithm, with the weight in the algebraic case — which matches Worked Example 11.7.4’s anatomy.

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WORKED EXAMPLE 11.9.5— TIER A **Eleven volumes: the methods and their ceilings**

| Method           | Vol | Optimises over      | Ceiling, and why                      |
|------------------|-----|---------------------|---------------------------------------|
| Fourier / energy | 7–8 | nothing; sees pairs | $\frac{n+1}{2}$ : two-point           |
| $L^2$ counting   | 9   | nothing; sees pairs | 2: two-point                          |
| bush             | 9   | multiplicity $\mu$  | $\frac{n+1}{2}$ : $\mu$ unattainable  |
| hairbrush        | 9   | multiplicity $\mu$  | $\frac{n+2}{2}$ : same, plus tangency |
| arithmetic       | 10  | additive structure  | $(2 - \sqrt{2})n$ : lossy conversions |
| Dvir             | 10  | <i>nothing</i>      | sharp, but only over $\mathbb{F}_q$   |
| partitioning     | 11  | degree $D$          | tangency                              |

*Reading the third column.* A method’s ceiling is set by what it optimises over, and whether the balancing value is attained — the test of Volume 9, Worked Example 9.8.4. Dvir’s argument has no free parameter and is sharp. Partitioning optimises over  $D$ , which *is* attainable, and its ceiling comes from elsewhere: the algebraic case, and within it, tangency.

*The one row that is sharp over  $\mathbb{R}$ .* None. Every real-variable method in eleven volumes has a ceiling below  $n$ , and Worked Example 11.8.2 identifies the same obstruction in three of them.

*What the table predicts.* A method that neither optimises over an unattainable parameter nor reduces to pair correlations, and that handles tangency, would settle the conjecture. Volume 14 reports that Wang and Zahl found one in  $\mathbb{R}^3$ , and that its dependence on three-dimensional geometry is what prevents it from settling  $n \geq 4$ .

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WORKED EXAMPLE 11.9.6— TIER C: RESEARCH TASTE **Three imports: an accounting**

This volume quotes more than any other in the series, and the reader is entitled to a reckoning.

(a) *What is imported.* Remark 11.2.4 (ham sandwich, hence Borsuk–Ulam), Remark 11.6.10 (Guth–Katz line incidences), Remark 11.8.4 (Katz–Zahl). The first three are load-bearing.

(b) *Why more than elsewhere.* Because the volume’s subject is a technique whose central theorems are recent research papers, not classical results with textbook proofs. Volumes 1–9 developed material largely settled before 1990; this one reaches 2015.

(c) *What is proved instead, in each case.* For R11, the Veronese reduction,

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and Chapter 11.1 now proves Borsuk–Ulam itself. For the former R12 — now proved outright as Theorem 11.5.4 — the model case Theorem 11.5.2 in full, including sharpness. For R13, the necessity of both hypotheses (Proposition 11.6.1, Proposition 11.6.3) and the planar comparison (Proposition 11.6.11), so that what the theorem gains is visible. For R14, nothing — and nothing depends on it.

(d) *Is the volume self-contained?* Now almost entirely. R11 and R12 are proved (Chapter 11.1, Theorem 11.5.4); only Remark 11.6.10 and the non-load-bearing Remark 11.8.4 remain quoted, and declining R13 costs Theorem 11.7.8 alone.

(e) *The alternative was worse.* Proving Borsuk–Ulam would require a chapter of algebraic topology used once; proving the Guth–Katz incidence theorem would require ruled-surface theory. Either would double the volume to establish results the reader can find in Guth’s book. Stating precisely what is granted, and proving the surrounding structure, seemed the better trade — and it is the same judgement made for R5 and R10 in earlier volumes, differing only in that here the omissions are load-bearing.

(f) *The one that is not like the others.* Volume 3’s configuration lemma was also used before it was proved, but it was a *debt*: Volume 9 discharged it, and the manifest tracked it as outstanding until then. These four are not debts; no volume will discharge them, and the manifest records them separately.

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## Drills

**Drill 11.9.1.** For each row of Worked Example 11.9.1, say whether the objects are lines or tubes, and check the correlation with sharpness.

**Drill 11.9.2.** Apply the five steps of Worked Example 11.9.2 to Theorem 11.3.2 and identify each in the proof.

**Drill 11.9.3.** Explain why the degree  $D$  is an attainable optimum whereas the multiplicity  $\mu$  of Volume 9 is not.

**Drill 11.9.4.** List the four scope decisions of this volume and state, for each, what depends on it.

**Where we are.** Volume 11 in one line: polynomial partitioning settles problems about *lines* and moves the exponent on problems about *tubes*, and the difference is tangency.

**Where we are.** Eleven volumes, 120 chapters, and the Kakeya conjecture remains open above dimension three. What has changed is that the obstruction now has a name – tangency – and three independent methods have been shown to stop at it.

## Notes

Guth’s *Polynomial Methods in Combinatorics* is the standard reference for this material and covers everything in this volume, including the results quoted here.



# Chapter 11.A

## Results Quoted from Volumes 1–10

| Result                                | Source                | Used in                        |
|---------------------------------------|-----------------------|--------------------------------|
| The Kakeya conjecture                 | Conjecture 7.1.1      | Chapter 11.8                   |
| Dvir’s theorem                        | Theorem 10.2.1        | Worked Example 11.8.1          |
| Schwartz–Zippel                       | Lemma 10.1.3          | Theorem 11.8.1, contrast       |
| The propagation step fails            | Proposition 10.7.2    | Chapter 11.8 throughout        |
| Testing whether an argument transfers | Worked Example 10.7.3 | Worked Example 11.8.2          |
| Polynomial dimension count            | Worked Example 10.1.2 | Theorem 11.2.5, Theorem 11.4.2 |
| Wolff’s bound, transverse case        | Remark 9.5.5          | Worked Example 11.8.2          |
| Why the geometric mean is lossy       | Worked Example 9.8.2  | Worked Example 11.9.3          |
| Predicting a method’s ceiling         | Worked Example 9.8.4  | Worked Example 11.9.3          |
| Tracing the arithmetic losses         | Worked Example 10.6.2 | Worked Example 11.9.3          |

## Debts and imports

This volume incurs no debts on later volumes and discharges none.

It *imports* four results without proof. Three are load-bearing:

| Imported                                        | Revision |                     | What depends on it |
|-------------------------------------------------|----------|---------------------|--------------------|
| (discharged<br>R11 proved, Theo-<br>rem 11.2.3) | —        | —                   | —                  |
| (discharged<br>R12 proved, Theo-<br>rem 11.5.4) | —        | —                   | —                  |
| Guth–Katz line inci-<br>dences                  | R13,     | Re-<br>mark 11.6.10 | Theorem 11.7.8     |
| Katz–Zahl      Kakeya<br>bound                  | R14,     | Re-<br>mark 11.8.4  | nothing            |

The model case of the multilinear estimate — Loomis–Whitney, Theorem 11.5.2 — is proved in full, and the necessity of the hypotheses in R13 is established by Proposition 11.6.1 and Proposition 11.6.3. So in each case what is granted is stated precisely and its sharpness demonstrated.

## Chapter 11.B

# Notation and Index of Results

### Additions to the Volume 1–10 conventions

|                               |                                                        |
|-------------------------------|--------------------------------------------------------|
| $Z(P)$                        | the real zero set of a polynomial $P$                  |
| cell                          | a connected component of $\mathbb{R}^n \setminus Z(P)$ |
| $D$                           | the degree of a partitioning polynomial                |
| $I(\mathcal{P}, \mathcal{L})$ | the number of point–line incidences                    |
| joint                         | a point on three lines with spanning directions        |
| regulus                       | a doubly ruled quadric surface                         |
| $\nu$ -transverse             | $ \det(v_1, \dots, v_n)  \geq \nu$                     |
| $\Psi$                        | the Elekes–Sharir parametrisation of rigid motions     |

## Index of the main results

|                                         |                                        |
|-----------------------------------------|----------------------------------------|
| Milnor–Thom cell count                  | Theorem 11.2.2                         |
| Tucker’s lemma                          | Theorem 11.1.4                         |
| Borsuk–Ulam                             | Theorem 11.1.5                         |
| Ham sandwich for measures               | Theorem 11.1.6                         |
| Polynomial ham sandwich                 | Theorem 11.2.3                         |
| Polynomial partitioning                 | Theorem 11.2.5                         |
| Trivial incidence bounds                | Lemma 11.3.3                           |
| Szemerédi–Trotter                       | Theorem 11.3.2                         |
| Pruning                                 | Lemma 11.4.3                           |
| Vanishing at a joint kills the gradient | Lemma 11.4.4                           |
| The joints theorem                      | Theorem 11.4.2                         |
| Loomis–Whitney                          | Theorem 11.5.2                         |
| Multilinear Kakeya, axis form           | Theorem 11.5.4                         |
| Multilinear Kakeya, transverse form     | Theorem 11.5.7                         |
| Planes and reguli are extremal          | Proposition 11.6.1, Proposition 11.6.3 |
| Line incidences in $\mathbb{R}^3$       | Theorem 11.6.6                         |
| Bézout for lines                        | Theorem 11.6.5                         |
| Cauchy–Schwarz for distances            | Lemma 11.7.3                           |
| Motion sets are lines                   | Lemma 11.7.5                           |
| Quadruples are intersecting pairs       | Proposition 11.7.6                     |
| Distinct distances                      | Theorem 11.7.8                         |
| Besicovitch sets are not algebraic      | Theorem 11.8.1                         |
| Varieties are small                     | Proposition 11.8.2                     |

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## Plan revisions in this volume

| Imported, not proved                                  | Rev | Reason                                                                                                                                           |
|-------------------------------------------------------|-----|--------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>Polynomial ham sandwich</i><br>— <i>NOW PROVED</i> | R11 | discharged: Chapter 11.1 proves Tucker, Borsuk–Ulam and ham sandwich from scratch. The Veronese reduction is given in full. <i>Load-bearing</i>  |
| <i>Multilinear Kakeya</i> —<br><i>NOW PROVED</i>      | R12 | discharged: Theorem 11.5.4 follows Guth’s short argument from Loomis–Whitney alone. The model case Loomis–Whitney is proved. <i>Load-bearing</i> |
| Guth–Katz line incidences                             | R13 | needs ruled-surface theory and the flecnode polynomial; the necessity of the hypotheses is proved. <i>Load-bearing</i>                           |
| Katz–Zahl Kakeya bound                                | R14 | superseded by Volume 14; nothing depends on it                                                                                                   |



# Part 12

## Volume 12

Multilinear and Bilinear Restriction





# Preface to Volume 12

## Where the restriction problem stands

Volume 6 posed the restriction conjecture and proved the Stein–Tomas theorem, which settles the range  $p \geq \frac{2(n+1)}{n-1}$  and no further. It also proved that restriction implies Kakeya (Theorem 6.6.2), so the conjecture is at least as hard as everything Volumes 9 to 11 failed to settle.

The advance of the last twenty years came from a reformulation. Instead of bounding  $\mathcal{E}g$  for a single  $g$  on the whole sphere, bound a *product*  $\prod_j \mathcal{E}g_j$  where the  $g_j$  live on caps whose normals are transverse. That hypothesis excludes the configurations where the difficulty lives — exactly as Worked Example 11.5.4 recorded for multilinear Kakeya — and in exchange the multilinear estimate can be proved sharply. Bourgain and Guth then showed how to recover a genuine linear estimate from it, at a cost.

## What this volume does

It proves the multilinear restriction theorem of Bennett, Carbery and Tao from the multilinear Kakeya inequality of Volume 11; develops the broad–narrow decomposition that converts multilinear estimates into linear ones; computes the restriction exponents that result and compares them with Stein–Tomas; and then turns to the bilinear theory and to restriction against a fractal measure.

That last topic is not a digression. Volume 8, Worked Example 8.8.1, identified the missing ingredient in the Falconer distance set problem as a bound on the spherical average  $\sigma(\mu, r)$ , and observed that the required estimate is a *weighted restriction estimate with a fractal weight*. That is the subject of Chapter 12.7, and Chapter 12.8 discharges the promise: it identifies  $\sigma(\mu, r)$  as a restriction quantity and shows how a bilinear estimate improves the Falconer threshold.

## The one dictionary

If the volume has one idea it is the wave packet dictionary, and it is worth stating before starting.

An extension  $\mathcal{E}g$  is an oscillatory integral and is hard to bound pointwise. Decomposed into wave packets (Theorem 12.1.4), it becomes a weighted sum of indicator functions of  $R^{1/2} \times \cdots \times R$  tubes, whose directions are  $R^{-1/2}$ -separated. So an analytic question about  $\mathcal{E}$  becomes a geometric question about how such tubes overlap — which is the Kakeya problem at scale  $\delta = R^{-1/2}$ .

Every chapter is written in that dictionary. Multilinear restriction is multilinear Kakeya translated (Lemma 12.3.2); the broad–narrow dichotomy is a statement about which tube directions carry the mass; the weighted estimates of Chapter 12.7 ask how a fractal measure meets a tube; and Proposition 12.8.2 shows the distance set problem speaks the same language.

Two cautions carried throughout. The dictionary is not exact — packets have tails, and Worked Example 12.1.3 explains when they may be discarded. And the translation is lossy in one direction: Proposition 12.3.3 shows that passing from packets back to an  $L^p$  estimate can cost a power of  $R$  unless the almost-orthogonality is retained, a point Worked Example 12.3.5 turns into an audit.

## A note on the plan

The original plan for this volume included chapters on Loomis–Whitney and on multilinear Kakeya. Both were written into Volume 11 instead, where the polynomial method made them natural — Theorem 11.5.2 is proved there and Theorem 11.5.7 proved there in full. This volume therefore begins after them and quotes both.

## What is assumed

Volumes 1–11. Most heavily: Volume 6 for the restriction problem and the  $TT^*$  identity; Volume 11 for multilinear Kakeya; Volume 5 for the decay of  $\widehat{d}\sigma$ ; Volume 8 for the distance set framework.

## The plan

1 Wave packets      the decomposition everything rests on

Chapters 2–8, planned: transversality and the multilinear setting; multilinear restriction from multilinear Kakeya; the broad–narrow decomposition; the

resulting exponents; bilinear restriction; restriction against a fractal measure; and spherical averages and the distance set.



# Chapter 12.1

## Wave Packets

Every argument in this volume and the next decomposes an extension operator into pieces, each of which is essentially a smooth bump on a tube. This chapter constructs the decomposition and proves the two properties that make it useful: the pieces are almost orthogonal, and each is almost constant on its tube.

### 12.1.1 Caps and their dual tubes

Fix  $R \geq 1$  and write  $\delta = R^{-1/2}$ .

**Definition 12.1.1.** Let  $\{\theta\}$  be a covering of  $\mathbb{S}^{n-1}$  by caps of radius  $\delta$  with bounded overlap, and for each  $\theta$  let  $\omega_\theta$  be its centre. There are  $\approx_n \delta^{-(n-1)} = R^{(n-1)/2}$  such caps.

**Lemma 12.1.2** (The dual box). *A cap  $\theta$  of radius  $\delta$  on  $\mathbb{S}^{n-1}$  is contained in a rectangular box of dimensions*

$$\underbrace{\delta \times \cdots \times \delta}_{n-1} \times \delta^2$$

*centred at  $\omega_\theta$ , with the short side in the direction  $\omega_\theta$ . Its dual box — the polar set, up to constants — has dimensions*

$$\underbrace{\delta^{-1} \times \cdots \times \delta^{-1}}_{n-1} \times \delta^{-2} = \underbrace{R^{1/2} \times \cdots \times R^{1/2}}_{n-1} \times R,$$

*with the long side in the direction  $\omega_\theta$ .*

*Proof.* Take coordinates with  $\omega_\theta = e_n$ . A point  $\omega \in \theta$  is  $\omega = (\omega', \omega_n)$  with  $|\omega'| \leq \delta$  and  $\omega_n = \sqrt{1 - |\omega'|^2}$ , so

$$1 - \omega_n = 1 - \sqrt{1 - |\omega'|^2} \leq |\omega'|^2 \leq \delta^2,$$

using  $1 - \sqrt{1-t} \leq t$  for  $t \in [0, 1]$ . So  $\theta$  lies in the box  $\{|\omega'| \leq \delta, |\omega_n - 1| \leq \delta^2\}$ , as claimed. The dual of a box is the box with reciprocal side lengths, by the definition of the polar set.  $\square$

**Where we are.** The quadratic gap  $\delta^2$  in the normal direction is the *curvature* of the sphere, and it is the only place curvature enters this chapter. A flat piece of hypersurface would have a cap contained in a slab of thickness 0 in the normal direction, dual box infinite in that direction, and no tubes.

Numerically,  $1 - \cos \delta$  against  $\delta^2/2$  agrees to four significant figures already at  $\delta = 0.05$ , so the estimate above is not merely an upper bound but the truth up to a factor 2.

The consequence to remember: *a  $\delta$ -cap is dual to a  $\delta^{-1} \times \cdots \times \delta^{-2}$  tube*, and with  $\delta = R^{-1/2}$  this is an  $R^{1/2} \times \cdots \times R$  tube. Those dimensions recur on every page of Volumes 12 and 13.

**Definition 12.1.3** (Tubes). For each cap  $\theta$  let  $\mathbb{T}_\theta$  be a covering of  $B_R$  by translates of the dual box of Lemma 12.1.2, with bounded overlap. Since the box has volume  $\approx R^{(n-1)/2} \cdot R$  and  $m(B_R) \approx R^n$ ,

$$\#\mathbb{T}_\theta \approx_n \frac{R^n}{R^{(n-1)/2} R} = R^{(n-1)/2}.$$

The pairs  $(\theta, T)$  with  $T \in \mathbb{T}_\theta$  are the *wave packets*, and there are  $\approx_n R^{n-1}$  of them.

## 12.1.2 The decomposition

**Theorem 12.1.4** (Wave packet decomposition). *Let  $g \in L^2(\mathbb{S}^{n-1})$  and  $R \geq 1$ . There is a decomposition*

$$\mathcal{E}g = \sum_{\theta} \sum_{T \in \mathbb{T}_\theta} \mathcal{E}g_{\theta,T}$$

*such that:*

- (i) (Frequency localisation)  $g_{\theta,T}$  is supported in  $\theta$ , and  $\sum_T g_{\theta,T} = g_\theta$ , where  $g_\theta$  is  $g$  restricted to  $\theta$  by a smooth partition of unity;
- (ii) (Spatial localisation) for every  $N$ ,

$$|\mathcal{E}g_{\theta,T}(x)| \lesssim_N \|g_{\theta,T}\|_{L^1(d\sigma)} \left(1 + \frac{\text{dist}(x, T)}{R^{1/2}}\right)^{-N} \quad (x \in \mathbb{R}^n);$$

(iii) ( $L^2$  orthogonality)

$$\|\mathcal{E}g\|_{L^2(B_R)}^2 \approx_n \sum_{\theta} \sum_{T \in \mathbb{T}_{\theta}} \|\mathcal{E}g_{\theta,T}\|_{L^2(B_R)}^2.$$

*Idea of the proof.* Cut  $g$  into caps by a partition of unity — that is (i). On each cap, the extension  $\mathcal{E}g_{\theta}$  has Fourier support in a box of dimensions  $\delta \times \cdots \times \delta^2$ , so by the uncertainty principle it cannot oscillate on scales finer than the dual box. Cutting physical space into translates of that dual box, and using a partition of unity there too, gives pieces each concentrated on one tube — that is (ii), with the tails coming from the smoothness of the cutoffs.

Orthogonality (iii) has two sources. Different caps give disjoint frequency supports, so Plancherel applies. Different tubes with the same cap are disjoint in *space*, and the tails are summable.

*Proof. Step 1. Cut the sphere.*

Let  $\{\psi_{\theta}\}$  be a smooth partition of unity subordinate to  $\{\theta\}$  with  $|\partial^{\alpha}\psi_{\theta}| \lesssim_{\alpha} \delta^{-|\alpha|}$ , and put  $g_{\theta} = g\psi_{\theta}$ . Then  $\sum_{\theta} g_{\theta} = g$  and each  $g_{\theta}$  is supported in  $\theta$ .

**Step 2. Cut physical space.**

Fix  $\theta$  and let  $\{\eta_T\}_{T \in \mathbb{T}_{\theta}}$  be a smooth partition of unity on  $\mathbb{R}^n$  subordinate to the tubes of  $\mathbb{T}_{\theta}$ , with derivatives adapted to the tube's dimensions. Define  $g_{\theta,T}$  by

$$\mathcal{E}g_{\theta,T} = \eta_T \cdot \mathcal{E}g_{\theta} \quad \text{followed by a Fourier restriction back to } \theta,$$

that is,  $g_{\theta,T}$  is the function on  $\theta$  whose extension best approximates  $\eta_T \mathcal{E}g_{\theta}$ . Concretely,  $\widehat{\eta_T}$  is concentrated on the *dual* of  $T$ , which is a box of dimensions  $\delta \times \cdots \times \delta^2$  — comparable to  $\theta$  itself by Lemma 12.1.2 — so multiplying by  $\eta_T$  smears the frequency support of  $\mathcal{E}g_{\theta}$  by an amount comparable to the cap. Restricting to  $\theta$  therefore loses only rapidly decaying tails, and  $\sum_T g_{\theta,T} = g_{\theta}$  up to such tails, which is (i).

**Step 3. Spatial localisation.**

Since  $\eta_T$  is supported on  $T$  and its derivatives are adapted to  $T$ 's dimensions,  $\widehat{\eta_T}$  decays rapidly outside the dual box. For  $x$  at distance  $d$  from  $T$ , integrating by parts  $N$  times in the directions transverse to  $T$  — where  $T$  has width  $R^{1/2}$  — gives

$$|\mathcal{E}g_{\theta,T}(x)| \lesssim_N \|g_{\theta,T}\|_{L^1(d\sigma)} (1 + d/R^{1/2})^{-N},$$

the prefactor because  $|\mathcal{E}h| \leq \|h\|_{L^1(d\sigma)}$  pointwise. This is (ii).

**Step 4. Orthogonality across caps.**

The functions  $\mathcal{E}g_\theta$  have Fourier transforms supported in the caps  $\theta$ , which have bounded overlap. Hence by Plancherel,

$$\|\mathcal{E}g\|_{L^2(\mathbb{R}^n)}^2 \approx_n \sum_{\theta} \|\mathcal{E}g_\theta\|_{L^2(\mathbb{R}^n)}^2.$$

**Step 5. Orthogonality across tubes.**

Fix  $\theta$ . The tubes of  $\mathbb{T}_\theta$  have bounded overlap, so  $\sum_T \eta_T^2 \approx 1$  and

$$\|\mathcal{E}g_\theta\|_{L^2(B_R)}^2 \approx \sum_T \|\eta_T \mathcal{E}g_\theta\|_{L^2(B_R)}^2 \approx \sum_T \|\mathcal{E}g_{\theta,T}\|_{L^2(B_R)}^2,$$

the last step by Step 2 and the rapid decay of the tails from Step 3, which contribute a convergent geometric series. Combining with Step 4 gives (iii).  $\square$

**Where we are.** Three warnings about Theorem 12.1.4, all of which matter in later chapters.

*The decomposition is not exact.* Step 2 loses rapidly decaying tails, and every application must carry them. In practice one fixes  $N$  large depending on the exponents involved and absorbs the tails at the end; the standard phrase “by rapid decay we may assume the packets are disjointly supported” is shorthand for this, and it is legitimate only when the number of packets is polynomial in  $R$ , which Definition 12.1.3 guarantees.

*The  $L^2$  orthogonality is special to  $p = 2$ .* For  $p \neq 2$  there is no such identity, and the whole difficulty of the restriction problem is that one must control  $\|\mathcal{E}g\|_{L^p}$  with only the  $L^2$  information above plus geometry. Chapter 12.4 is about extracting that geometry.

*The tubes point in the direction of the cap’s centre.* By Lemma 12.1.2 the long axis of the dual box is  $\omega_\theta$  — so wave packets from different caps point in different directions, and two packets whose caps are  $\nu$ -separated have tubes at angle  $\approx \nu$ . That is the link to Keakeya, and it is why multilinear Keakeya is the input to multilinear restriction.

**Proposition 12.1.5** (A single packet, computed). *Let  $\theta$  be a cap of radius  $\delta = R^{-1/2}$  and let  $g = \mathbf{1}_\theta$ . Then*

$$|\mathcal{E}g(x)| \approx d\sigma(\theta) \approx \delta^{n-1} = R^{-\frac{n-1}{2}} \quad \text{for } x \text{ in the dual tube } T,$$

and  $|\mathcal{E}g(x)|$  decays rapidly outside  $T$ . Moreover

$$\|\mathcal{E}g\|_{L^2(B_R)}^2 \approx |T| \delta^{2(n-1)} \approx R^{\frac{n+1}{2}} R^{-(n-1)} = R^{\frac{3-n}{2}},$$

consistent with  $\|\mathcal{E}g\|_{L^2(B_R)}^2 \approx R \|g\|_{L^2(d\sigma)}^2 = R \delta^{n-1} = R^{\frac{3-n}{2}}$ .



*Proof.* For  $x \in T$  the phase  $e^{2\pi i x \cdot \omega}$  varies by  $O(1)$  as  $\omega$  ranges over  $\theta$ : by Lemma 12.1.2 the cap has extent  $\delta$  transversally and  $\delta^2$  normally, while  $T$  has extent  $\delta^{-1}$  and  $\delta^{-2}$  in the dual directions, so each product is  $O(1)$ . Hence the integral  $\int_{\theta} e^{2\pi i x \cdot \omega} d\sigma(\omega)$  has no cancellation and is  $\approx d\sigma(\theta) \approx \delta^{n-1}$ .

Outside  $T$  the phase oscillates and repeated integration by parts gives rapid decay, as in Theorem 12.1.4(ii).

For the  $L^2$  computation,  $|\mathcal{E}g|^2 \approx \delta^{2(n-1)}$  on  $T$  and negligible elsewhere, and  $|T| \approx R^{(n+1)/2}$  by Definition 12.1.3. The consistency check uses  $\|g\|_{L^2(d\sigma)}^2 = d\sigma(\theta) \approx \delta^{n-1} = R^{-(n-1)/2}$  and  $R \cdot R^{-(n-1)/2} = R^{(3-n)/2}$ .  $\checkmark$   $\square$

**Where we are.** Proposition 12.1.5 is the Knapp example seen through the wave packet decomposition, and the consistency check at the end is the one Worked Example 12.1.1 sets as a drill. That both routes give  $R^{(3-n)/2}$  is a genuine check on the normalisations, and it is worth performing whenever a new scale is introduced — most errors in this area are factors of  $R$  to a power, and they show up here first.

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WORKED EXAMPLE 12.1.1 — TIER A **The dimensions, counted**

Fix  $n$  and  $R$ , and put  $\delta = R^{-1/2}$ .

*Caps.*  $\approx \delta^{-(n-1)} = R^{(n-1)/2}$ .

*Tubes per cap.* A tube has volume  $R^{(n-1)/2} \cdot R$ , and  $B_R$  has volume  $R^n$ , so  $\#\mathbb{T}_{\theta} \approx R^n / (R^{(n-1)/2} R) = R^{(n-1)/2}$ .

*Total packets.*  $R^{(n-1)/2} \cdot R^{(n-1)/2} = R^{n-1}$ .

*Numerically.*

| $n$ | $R$    | caps   | tubes/cap | packets   |
|-----|--------|--------|-----------|-----------|
| 2   | $10^4$ | $10^2$ | $10^2$    | $10^4$    |
| 3   | $10^4$ | $10^4$ | $10^4$    | $10^8$    |
| 3   | $10^6$ | $10^6$ | $10^6$    | $10^{12}$ |
| 4   | $10^6$ | $10^9$ | $10^9$    | $10^{18}$ |

*The coincidence worth noticing.* The number of caps equals the number of tubes per cap, both  $R^{(n-1)/2}$ . That is not an accident: it says the family of wave packets is *self-dual* under exchanging position and direction — the same duality Volume 9, Lemma 9.6.4 exploited for Keakeya. It is why the Keakeya and restriction problems have the same numerology.

*A sanity check.* A single wave packet has  $\|\mathcal{E}g_{\theta,T}\|_{L^2(B_R)}^2 \approx |T| \cdot \|g_{\theta,T}\|_{L^\infty}^2 \delta^{2(n-1)}$ , and summing over the  $R^{n-1}$  packets must reproduce  $\|\mathcal{E}g\|_{L^2(B_R)}^2 \approx R \|g\|_{L^2(d\sigma)}^2$  — the last by Proposition 6.1.4 in Volume 6. Verifying that the powers of  $R$  match is Drill 1.4.

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**WORKED EXAMPLE 12.1.2— TIER B Why the packets are the right objects**

The decomposition is used so uniformly that it is worth saying what it buys.

(a) *It converts an oscillatory integral into a sum of bumps.*  $\mathcal{E}g$  is an integral with a rapidly oscillating kernel and is hard to bound pointwise. Each  $\mathcal{E}g_{\theta,T}$  is, by (ii), essentially a constant times  $\mathbf{1}_T$ . So  $|\mathcal{E}g|$  becomes a weighted sum of indicator functions of tubes, and bounding it becomes a *counting* problem about tubes.

(b) *The counting problem is Kakeya.* The tubes have  $R^{-1/2}$ -separated directions, one family per cap, and there are  $R^{(n-1)/2}$  of them per direction. Rescaling by  $R$  turns them into  $\delta$ -tubes with  $\delta$ -separated directions — exactly Volume 9’s setting with  $\delta = R^{-1/2}$ .

(c)  *$L^2$  information is free and everything else is not.* Property (iii) is an identity; there is no analogue for other exponents. So an argument gets one orthogonality relation and must supply the rest from the geometry of how tubes overlap.

(d) *What goes wrong without curvature.* Lemma 12.1.2 used the sphere’s curvature to get the factor  $\delta^2$ . For a flat piece the dual box is a slab, all packets point the same way, and no counting is possible — consistent with Volume 6’s observation that restriction fails for a hyperplane.

*The summary.* Wave packets are the dictionary between the analytic statement (an  $L^p$  bound for  $\mathcal{E}$ ) and the geometric one (how much  $R^{1/2} \times R$  tubes can overlap). Everything in Volumes 12 and 13 is written in that dictionary.

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**WORKED EXAMPLE 12.1.3— TIER C: RESEARCH TASTE The uncertainty principle, as used here**

Lemma 12.1.2 is an instance of a principle invoked constantly in this volume and the next, and it is worth stating what the invocation actually means.

(a) *The formal statement.* If  $\widehat{f}$  is supported in a box  $B$ , then  $f$  cannot vary appreciably on scales finer than the dual box  $B^*$ . Precisely,  $f = f * \check{\psi}$  for any  $\psi$  equal to 1 on  $B$ , and  $\check{\psi}$  is concentrated on  $B^*$  with rapidly decaying tails.

(b) *What it is not.* It is not an equality, and  $f$  need not be constant on translates of  $B^*$ . The correct statement is the convolution identity above, and every use must carry the tails — which is why Theorem 12.1.4(ii) is stated with a decay exponent  $N$  rather than as exact support.

(c) *Why the tails are harmless here.* There are  $\lesssim R^{n-1}$  wave packets, so choosing  $N$  large enough that  $R^{n-1} \cdot R^{-N/2}$  is negligible disposes of them. This works because the packet count is *polynomial* in  $R$ ; an argument producing

exponentially many pieces could not absorb the tails this way.

(d) *Where it would fail.* For a surface with a flat piece, the dual box is unbounded in the normal direction and there is no localisation at all. Volume 5, Worked Example 5.5.4 recorded that a hyperplane’s surface measure has a non-decaying transform; here that reappears as the absence of a wave packet decomposition.

(e) *The habit.* On meeting “by the uncertainty principle” in a paper, identify the box, its dual, and the tail exponent being assumed. Most arguments are correct; the ones that are not usually fail because the number of pieces was not polynomial, or because the surface was flatter than assumed.

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WORKED EXAMPLE 12.1.4— TIER B **Two scales, and which is which**

Wave packet arguments involve  $R$ ,  $R^{1/2}$ ,  $R^{-1/2}$  and occasionally  $R^{-1}$ , and confusing them is the commonest source of error. A fixed dictionary.

| Scale      | What lives there             | Why                                         |
|------------|------------------------------|---------------------------------------------|
| $R$        | the ball $B_R$ ; tube length | the scale at which the estimate is stated   |
| $R^{1/2}$  | tube width                   | dual to the cap radius, Lemma 12.1.2        |
| $R^{-1/2}$ | cap radius $\delta$          | chosen so the dual tube has length $R$      |
| $R^{-1}$   | cap thickness $\delta^2$     | the sphere’s curvature over a $\delta$ -cap |

*The relation that fixes everything.*  $\delta = R^{-1/2}$ , forced by wanting the dual box to reach across  $B_R$ . Change it and every other entry changes.

*Two counts that must match.* Caps:  $\delta^{-(n-1)} = R^{(n-1)/2}$ . Tubes per cap:  $R^n / (R^{(n-1)/2} \cdot R) = R^{(n-1)/2}$ . Equal, as Worked Example 12.1.1 noted, and that equality is the self-duality of the packet family.

*The check to run on any new argument.* Compute  $\|\mathcal{E}g\|_{L^2(B_R)}^2$  two ways — by Proposition 6.1.4 and by summing packets — and confirm they agree. Proposition 12.1.5 does this for a single cap and gets  $R^{(3-n)/2}$  both ways. Errors of a power of  $R$  show up here immediately, and they are the errors that actually occur.

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WORKED EXAMPLE 12.1.5— TIER B **Decomposing a specific function**

The wave packet decomposition is easier to trust after applying it once to something explicit.

*The function.*  $n = 2$ ,  $R = 10^4$ ,  $\delta = 10^{-2}$ , and  $g = \mathbf{1}_{\theta_0}$  for a single cap  $\theta_0$  of radius  $\delta$  centred at  $e_1$ .

*Step 1, cutting the circle.* The partition of unity leaves  $g_\theta = 0$  except for  $\theta = \theta_0$  and its two neighbours, since  $g$  is supported in one cap. So the sum over caps has three terms, not 100.

*Step 2, cutting space.*  $\mathbb{T}_{\theta_0}$  consists of  $R^{(n-1)/2} = 100$  tubes of dimensions  $100 \times 10^4$  tiling  $B_R$ , all parallel to  $e_1$ . The packets  $g_{\theta_0, T}$  are the pieces of  $g$  whose extensions live on each.

*Step 3, the sizes.* By Proposition 12.1.5,  $|\mathcal{E}g| \approx \delta^{n-1} = 10^{-2}$  on the tube through the origin, with rapid decay elsewhere. And  $\|\mathcal{E}g\|_{L^2(B_R)}^2 \approx R^{(3-n)/2} = R^{1/2} = 100$ .

*The check.* Directly:  $|T| \cdot \delta^{2(n-1)} = (100 \times 10^4) \times 10^{-4} = 100$ . ✓ Both routes agree, which is the consistency check Worked Example 12.1.4 recommends running on any new configuration.

*What the example shows.* That for data on a single cap the decomposition is trivial — one cap, and the tubes merely tile space. The content appears only when  $g$  is spread over many caps, so that packets from different directions interact. That interaction is the subject of every subsequent chapter.

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## Drills

**Drill 12.1.1.** Verify  $1 - \sqrt{1-t} \leq t$  for  $t \in [0, 1]$  and deduce Lemma 12.1.2.

**Drill 12.1.2.** Check that the dual box has its long side in the direction  $\omega_\theta$ , and say what this means geometrically.

**Drill 12.1.3.** Confirm  $\#\mathbb{T}_\theta \approx R^{(n-1)/2}$  and the total packet count  $R^{n-1}$ .

**Drill 12.1.4.** Carry out the sanity check at the end of Worked Example 12.1.1: match the powers of  $R$  on both sides.

**Where we are.** Three conventions fixed here and used for the rest of the series.

$R$  is the large parameter and  $\delta = R^{-1/2}$  the cap radius. The relation  $\delta = R^{-1/2}$  is not a choice: by Lemma 12.1.2 it is forced by requiring the dual tube to have length  $R$ , and every other scale in the volume follows from it.

Constants depend on  $n$  and on the transversality parameter alone. Never on  $R$ , on the number of packets, or on  $g$ .

Every estimate carries  $R^\varepsilon$ . By Volume 9, Proposition 9.7.3 this is exactly the accuracy the dimension conclusions need, and Worked Example 12.4.3

explains where such losses come from — an induction with  $\log \log R$  rounds rather than  $\log R$ .

**Where we are.** One further consequence of Lemma 12.1.2 deserves recording, since Volume 13 depends on it.

The dual tube's *length* is  $\delta^{-2}$ , coming from the cap's thickness  $\delta^2$ , which is the sphere's curvature. So the tube's aspect ratio  $\delta^{-2} : \delta^{-1} = \delta^{-1}$  is a curvature measurement. A surface with  $k$  non-vanishing principal curvatures gives caps of thickness  $\delta^2$  in  $k$  directions and  $\delta$  in the rest, hence tubes of a different shape — and Volume 5, Worked Example 5.5.4 predicted exactly this pattern from the decay exponents.

The practical consequence: before decomposing into wave packets for a new surface, compute the cap's dimensions from the second fundamental form. Every subsequent count depends on them.

## Notes

The wave packet decomposition in this form is due to Fefferman and to Córdoba, and is standard equipment. The presentation here follows Tao's lecture notes; Demeter's book gives the version used in decoupling, which is the one Volume 13 will need.



## Chapter 12.2

# Transversality

### 12.2.1 The multilinear setting

**Definition 12.2.1.** Let  $S_1, \dots, S_n \subseteq \mathbb{S}^{n-1}$  be relatively open, and let  $\nu > 0$ . The pieces are  $\nu$ -*transverse* if for every choice  $\omega_j \in S_j$ ,

$$|\det(\omega_1, \dots, \omega_n)| \geq \nu.$$

For each  $j$  let  $\mathcal{E}_j$  denote the extension operator restricted to  $S_j$ , so  $\mathcal{E}_j g_j = \mathcal{E}(g_j \mathbf{1}_{S_j})$ .

Since the normal to  $\mathbb{S}^{n-1}$  at  $\omega$  is  $\omega$  itself, this is the same determinant condition as Definition 11.5.3 in Volume 11, now imposed on the caps rather than on tube directions — and by Lemma 12.1.2 the wave packets from  $S_j$  point in directions lying in  $S_j$ , so the two conditions agree.

**Proposition 12.2.2** (Transversality is inherited by wave packets). *Let  $S_1, \dots, S_n$  be  $\nu$ -transverse and let  $T_j$  be a wave packet tube arising from a cap  $\theta_j \subseteq S_j$  at scale  $R$ . Then the tube directions  $\omega_{\theta_1}, \dots, \omega_{\theta_n}$  satisfy*

$$|\det(\omega_{\theta_1}, \dots, \omega_{\theta_n})| \geq \nu,$$

*so the families  $\{T_j\}$  are  $\nu$ -transverse in the sense of Definition 11.5.3.*

*Proof.* The cap centre  $\omega_{\theta_j}$  lies in  $S_j$ , and Definition 12.2.1 applies to every choice of points in the  $S_j$ .  $\square$

**Where we are.** Proposition 12.2.2 is the hinge of the volume, and it is a triviality. That is the point: the transversality hypothesis was chosen precisely so that it passes from the surfaces to the wave packets without loss, and hence so that Volume 11's multilinear Keakeya inequality applies verbatim to the tubes produced by Theorem 12.1.4.

Every other hypothesis one might impose — curvature conditions, separation of the caps, non-degeneracy of second fundamental forms — fails this test in one way or another. Transversality is the condition that survives wave packet decomposition, and that is why the multilinear theory is stated with it.

## 12.2.2 Why $n$ factors, and not fewer

**Proposition 12.2.3** (Two factors do not suffice in high dimensions). *Let  $n \geq 3$  and let  $S_1, S_2 \subseteq \mathbb{S}^{n-1}$  be any two relatively open sets. There are  $\omega_1 \in S_1$ ,  $\omega_2 \in S_2$  and a hyperplane containing both. Consequently no determinant condition on pairs can force the wave packets of  $S_1$  and  $S_2$  to be quantitatively independent in all  $n$  directions.*

*Proof.* Any two vectors in  $\mathbb{R}^n$  span a plane, which lies in a hyperplane whenever  $n \geq 3$ . So the  $n \times n$  determinant of Definition 12.2.1 cannot even be formed from two vectors, and the natural bilinear substitute — the angle between  $\omega_1$  and  $\omega_2$  — constrains only one direction.  $\square$

**Where we are.** So the number of factors is forced by the dimension: to pin down  $n$  directions one needs  $n$  vectors, and the determinant is the natural quantitative measure of their independence.

This explains a feature of the literature that is otherwise puzzling. The *bilinear* theory — two factors — was developed first, in the 1990s, and gives sharp results in  $\mathbb{R}^2$  where two vectors do span. In higher dimensions it gives less than the multilinear theory, and Chapter 12.6 will record exactly how much less. The progression from bilinear to multilinear is the progression from “two directions are independent” to “all  $n$  are”.

Proposition 12.2.3 also explains why the bilinear approach nonetheless survives for the *distance set* problem: that problem is intrinsically two-point (Volume 8, Worked Example 7.7.3), so a two-factor estimate is the matching tool. Chapter 12.8 develops this.

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### WORKED EXAMPLE 12.2.1— TIER A The determinant, computed

Transversality is a quantitative hypothesis and it repays seeing the numbers. *The orthogonal case.*  $\omega_j = e_j$  gives  $|\det| = 1$ , the maximum. So  $\nu = 1$  corresponds to mutually orthogonal caps.

*Caps near a common great circle.* In  $\mathbb{R}^3$ , take  $\omega_1 = e_1$ ,  $\omega_2 = e_2$  and  $\omega_3 =$



$(\cos \alpha)e_1 + (\sin \alpha)e_2 + \varepsilon e_3$  normalised. Then

$$\det(\omega_1, \omega_2, \omega_3) = \frac{\varepsilon}{\sqrt{1 + \varepsilon^2}} \approx \varepsilon,$$

so three caps within  $\varepsilon$  of a common great circle are only  $\varepsilon$ -transverse. As  $\varepsilon \rightarrow 0$  the constant in Theorem 11.5.7 blows up, and it must: coplanar tubes can overlap completely.

*The relation to angles.* If the pairwise angles are all  $\geq \alpha$  the determinant need *not* be bounded below — three vectors can be pairwise at  $60^\circ$  and nearly coplanar. So transversality is strictly stronger than pairwise separation, and this is the content of Proposition 12.2.3 in concrete form.

*What  $\nu$  costs.* The constants in the multilinear theory depend on  $\nu$  polynomially, and the broad–narrow argument of Chapter 12.4 will choose  $\nu$  as a small power of  $R$ , so those constants must be tracked. That is one of the two places where the passage from multilinear to linear loses, and Worked Example 12.4.1 will account for it.

#### WORKED EXAMPLE 12.2.2— TIER B **Reading a multilinear hypothesis**

Multilinear estimates appear throughout Volumes 12–14, and the hypotheses vary in ways that matter.

(a) *How many factors?*  $n$  factors in  $\mathbb{R}^n$  is the multilinear theory; 2 factors is bilinear;  $k$  factors with  $2 < k < n$  occurs and interpolates. The number determines how many directions the hypothesis controls.

(b) *Transverse how?* A determinant bound  $|\det| \geq \nu$  as here, or a weaker angular separation, or — in the bilinear literature — a separation of the *normals* plus a curvature hypothesis. These are not equivalent, and Worked Example 12.2.1 gives the standard counterexample.

(c) *What norm on the data?*  $L^2(d\sigma)$  here; the linear conjecture wants  $L^\infty(d\sigma)$ , and Volume 6 recorded that the gap between them is where the difficulty sits. A multilinear estimate with  $L^2$  data is much weaker than one with  $L^\infty$  data, and the two are often quoted without the distinction being flagged.

(d) *What loss?*  $R^\varepsilon$  for every  $\varepsilon$ , as in Theorem 12.3.1 below, or no loss at all as in Guth’s multilinearakeya. The  $R^\varepsilon$  is harmless for the applications here — Volume 9, Proposition 9.7.3 explained why — but it is not harmless if one wants an endpoint result.

*The habit.* Before using a multilinear estimate, write down its four answers. Papers in this area differ mostly in these choices, and a comparison made

without checking them is meaningless.

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WORKED EXAMPLE 12.2.3— TIER B **Constructing transverse families**

The hypothesis of Definition 12.2.1 is easy to state and worth being able to realise, since applications must produce such families.

(a) *The canonical example.*  $S_j$  a cap of radius  $c_n$  about  $e_j$ . Then any choice  $\omega_j \in S_j$  has  $|\det(\omega_1, \dots, \omega_n)| \geq \frac{1}{2}$  for  $c_n$  small, by continuity of the determinant at  $(e_1, \dots, e_n)$  where it equals 1.

(b) *How many such families cover the sphere?* A single family covers only  $n$  caps. To handle general  $g$  one partitions  $\mathbb{S}^{n-1}$  into  $\approx K^{n-1}$  caps and considers all  $n$ -tuples, of which there are  $\approx K^{n(n-1)}$  — a constant, once  $K$  is fixed. That is why Proposition 12.4.3 can afford to take a maximum over tuples.

(c) *The tuples that fail.* Most  $n$ -tuples of caps are *not* transverse: for  $n = 3$  and  $K$  large, a random triple of caps lies near a common great circle with probability tending to 1 in the sense that its determinant is typically  $O(K^{-1})$ . So transversality is a rare condition, not a generic one, and the broad–narrow dichotomy is genuinely a dichotomy rather than a convenience.

(d) *The rescaling that restores it.* A cluster of caps near a great circle can be rescaled — stretch the normal direction — to occupy a full cap of a lower-dimensional sphere. This is what Worked Example 12.4.2 describes and what the narrow case of the induction performs.

*The practical upshot.* When applying a multilinear estimate, one does not verify transversality for a given family; one arranges a decomposition in which the transverse tuples are separated from the degenerate ones, and treats each separately. The estimate is a tool used inside a decomposition, never on its own.

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WORKED EXAMPLE 12.2.4— TIER C: RESEARCH TASTE **Hypotheses that concede the difficulty**

Transversality is the fourth hypothesis in this series introduced precisely to exclude the hard case, and collecting them shows a pattern.

(a) *Volume 6:  $L^2$  data.* Stein–Tomas assumes  $g \in L^2(d\sigma)$  where the conjecture wants  $L^\infty$ . On a  $\delta$ -cap the two differ by  $\delta^{(n-1)/2}$ , and the theorem is provable in a page precisely because  $L^2$  admits  $TT^*$ .

(b) *Volume 9: transverse hairbrushes.* Remark 9.5.5 proves Wolff’s bound when the tubes meet  $T_0$  at angle  $\gtrsim 1$ , and the small-angle case is the one that needs induction on scales.

(c) *Volume 11: transverse tube families.* Multilinear Kakeya assumes  $|\det| \geq \nu$ , which a Besicovitch set's tubes violate for most tuples.

(d) *Here: transverse caps.* Definition 12.2.1, the same condition on the surfaces.

*The pattern.* In every case the excluded configurations are the *tangential* ones — near-parallel tubes, nearly coplanar directions, caps whose normals nearly agree — and Volume 11, Worked Example 11.8.2 identified tangency as the recurring obstruction across three independent methods. This is the fourth.

*Why the concessions are nonetheless productive.* Each yields a sharp theorem, and each admits a decomposition recovering something linear:  $\varepsilon$ -removal for (a), the small-angle induction for (b), broad–narrow for (c) and (d). The recovery is always lossy, and Worked Example 12.4.1 accounts for the loss in the present case.

*The question a reader should carry forward.* Whether a method that does *not* concede tangency exists. Volume 14 reports that Wang and Zahl found one in  $\mathbb{R}^3$ , by analysing the tangential configurations directly rather than excluding them — and that its dependence on three-dimensional geometry is what blocks the extension.

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WORKED EXAMPLE 12.2.5— TIER A **Transversality across the series**

| Where                     | Objects            | Condition                                    |
|---------------------------|--------------------|----------------------------------------------|
| Vol 9, Theorem 9.5.4      | tube and hairbrush | angle $\geq \kappa$                          |
| Vol 11, Definition 11.5.3 | $n$ tube families  | $ \det(v_1, \dots, v_n)  \geq \nu$           |
| Here, Definition 12.2.1   | $n$ surface pieces | $ \det(\omega_1, \dots, \omega_n)  \geq \nu$ |
| Theorem 12.6.1            | two arcs           | $ \sin \angle(\omega_1, \omega_2)  \geq \nu$ |

*They are all the same condition.* On the sphere the normal at  $\omega$  is  $\omega$ , so rows two and three coincide by Proposition 12.2.2; row four is the case  $n = 2$ , where the determinant of two vectors is the sine of their angle; row one is the two-object version in a different guise.

*What each buys.* Row one gives Wolff's bound in the transverse case; row two gives multilinear Kakeya; row three gives multilinear restriction; row four gives the bilinear estimate. In every case the transverse statement is sharp and the general one is open or harder.

*The determinant is the right measure.* Worked Example 12.2.1 showed that pairwise angular separation does *not* suffice — three vectors can be pairwise

at  $60^\circ$  and nearly coplanar — so the condition must be the determinant and not a collection of pairwise conditions. That is Proposition 12.2.3 in another form.

*The unifying statement.* *Transversality is the non-degeneracy of the map that combines the objects:* the addition map for two arcs (Lemma 12.6.2), the map to  $\mathbb{R}^n$  for  $n$  tubes, the determinant in general. Where that map degenerates, every method in this series stops.

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#### WORKED EXAMPLE 12.2.6— TIER B **Verifying transversality in practice**

Applications must certify Definition 12.2.1 for particular families, and the certification is usually one of three arguments.

(a) *By construction.* Choose  $S_j$  near  $e_j$ , as in Worked Example 12.2.3(a). Determinant near 1 by continuity. Used whenever the family is at one's disposal.

(b) *By pigeonhole.* Given an arbitrary  $g$ , decompose into caps and select a transverse tuple carrying a definite share of the mass — which is exactly Proposition 12.4.3. Used when the family is not at one's disposal, which is the usual case.

(c) *By exclusion.* Show the non-transverse tuples contribute negligibly. This requires a bound on the degenerate case and so is available only when the narrow analysis has already been done.

*Which to use.* (a) when proving a multilinear estimate; (b) when applying one to a linear problem; (c) rarely, and usually as part of (b)'s induction.

*The quantitative caution.* All three produce a specific  $\nu$ , and the constants in Theorem 12.3.1 degrade as  $\nu \rightarrow 0$ . An argument that certifies transversality without tracking  $\nu$  has not certified anything usable, and Worked Example 12.4.5 shows how tightly  $\nu$  is coupled to the other parameters.

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## Drills

**Drill 12.2.1.** Verify that the normal to  $\mathbb{S}^{n-1}$  at  $\omega$  is  $\omega$ , and deduce that Definition 12.2.1 and Definition 11.5.3 agree.

**Drill 12.2.2.** Compute the determinant in Worked Example 12.2.1 for the three vectors given, and check the limit  $\varepsilon \rightarrow 0$ .

**Drill 12.2.3.** Exhibit three unit vectors in  $\mathbb{R}^3$ , pairwise at angle  $60^\circ$ , whose determinant is arbitrarily small.

**Drill 12.2.4.** Explain why  $L^\infty(d\sigma)$  data is a stronger hypothesis than  $L^2(d\sigma)$  data on a cap of radius  $\delta$ , and compute the ratio of the two norms for an indicator function.

**Where we are.** A remark on what Proposition 12.2.3 does and does not say.

It says that two vectors cannot span  $\mathbb{R}^n$  for  $n \geq 3$ , so no condition on pairs controls all  $n$  directions. It does *not* say the bilinear theory is useless above the plane — Theorem 12.6.3 is sharp in every dimension. What it says is that the bilinear hypothesis must be supplemented by curvature, which is why that theorem has a curvature hypothesis and the multilinear one does not.

The practical consequence: choose the number of factors to match the problem, not the dimension. Worked Example 12.8.2 shows the distance set problem wants two regardless of  $n$ .

## Notes

The multilinear formulation is due to Bennett, Carbery and Tao (2006). The observation that transversality is the hypothesis surviving wave packet decomposition is implicit in their paper and is made explicit in Guth's exposition.



## Chapter 12.3

# Multilinear Restriction

**Theorem 12.3.1** (Bennett–Carbery–Tao). *Let  $\Sigma_1, \dots, \Sigma_n$  be  $C^2$  hypersurface pieces in  $\mathbb{R}^n$  whose unit normals satisfy  $|\det(N_1, \dots, N_n)| \geq \nu$  for every choice of points, and let  $\mathcal{E}_j$  be the associated extension operators. Then for every  $\varepsilon > 0$  there is  $C(\varepsilon)$  with*

$$\left\| \prod_{j=1}^n \mathcal{E}_j f_j \right\|_{L^{\frac{2}{n-1}}(B(0,R))} \leq C(\varepsilon) R^\varepsilon \prod_{j=1}^n \|f_j\|_{L^2}, \quad R \geq 1.$$

Equivalently,  $\left\| \prod_j |\mathcal{E}_j f_j|^{1/n} \right\|_{L^p(B_R)} \lesssim_\varepsilon R^\varepsilon \prod_j \|f_j\|_{L^2}^{1/n}$  at  $p = \frac{2n}{n-1}$ .

**Where we are.** The two displays are the same statement. Raising the second to the power  $p$ ,

$$\left\| \prod_j |\mathcal{E}_j f_j|^{1/n} \right\|_{L^p}^p = \int \prod_j |\mathcal{E}_j f_j|^{p/n} = \int \prod_j |\mathcal{E}_j f_j|^{\frac{2}{n-1}} = \left\| \prod_j \mathcal{E}_j f_j \right\|_{L^{\frac{2}{n-1}}}^{\frac{2}{n-1}},$$

using  $\frac{p}{n} = \frac{2}{n-1}$  from Lemma 12.3.2. The first form is the one the proof works with; the second is the one Chapter 12.4 consumes, and the exponent  $\frac{1}{n}$  on the right is forced by homogeneity.

### 12.3.1 Why the exponent is $\frac{2n}{n-1}$

Before the proof, the exponent. It is fixed by the requirement that the left-hand side be exactly the integrand of multilinear Keakeya, and that identification is the wave packet dictionary of Worked Example 12.1.2.

**Lemma 12.3.2** (The exponents match). *For  $p = \frac{2n}{n-1}$  one has  $\frac{p}{n} = \frac{2}{n-1}$ , and*

hence

$$\left\| \prod_{j=1}^n |\mathcal{E}_j g_j|^{1/n} \right\|_{L^p}^p = \int_{\mathbb{R}^n} \prod_{j=1}^n (|\mathcal{E}_j g_j|^2)^{\frac{1}{n-1}}.$$

*Proof.*  $\frac{p}{n} = \frac{2n}{n(n-1)} = \frac{2}{n-1}$ , and then  $\left\| \prod_j |\mathcal{E}_j g_j|^{1/n} \right\|_{L^p}^p = \int \prod_j |\mathcal{E}_j g_j|^{p/n}$ .  $\square$

**Where we are.** Lemma 12.3.2 is arithmetic, but it is why Theorem 11.5.4 of Volume 11 was stated with the exponent  $\frac{1}{n-1}$ : applying multilinear Kakeya to the functions  $f_j = |\mathcal{E}_j g_j|^2$  produces exactly the multilinear restriction integrand. Checking:  $n = 2$  gives  $p = 4$  and  $\frac{p}{n} = 2 = \frac{2}{n-1}$ ;  $n = 3$  gives  $p = 3$  and  $\frac{p}{n} = 1$ ;  $n = 4$  gives  $p = \frac{8}{3}$  and  $\frac{p}{n} = \frac{2}{3}$ .  $\checkmark$

The proof given below does *not* take this route — it applies the discrete Loomis–Whitney inequality directly to the extension operators, which is what makes it self-contained. But the identification explains where the exponent comes from, and it is the reason the two theorems have matching hypotheses.

**Proposition 12.3.3** (What the naive packet decomposition gives). *With hypotheses as in Theorem 12.3.1 and  $p = \frac{2n}{n-1}$ ,*

$$\left\| \prod_{j=1}^n |\mathcal{E}_j g_j|^{1/n} \right\|_{L^p(B_R)} \lesssim_{n,\nu,\varepsilon} R^{\frac{1}{2}+\varepsilon} \prod_{j=1}^n \|g_j\|_{L^2(d\sigma)}^{1/n}.$$

*Proof.* Decompose each  $\mathcal{E}_j g_j$  by Theorem 12.1.4 at scale  $R$ . By part (ii) each packet is bounded by  $c_{j,\theta,T} \mathbf{1}_T$  up to rapidly decaying tails, so Cauchy–Schwarz over the packets meeting a given point gives  $|\mathcal{E}_j g_j|^2 \lesssim \#\{\text{caps}\} \sum_{(\theta,T)} |c_{j,\theta,T}|^2 \mathbf{1}_T$ . Writing  $f_j$  for the sum without the cap factor and applying Theorem 11.5.4 in its transverse form Theorem 11.5.7,

$$\int_{B_R} \prod_j f_j^{\frac{1}{n-1}} \lesssim R^\varepsilon \prod_j \left( \int f_j \right)^{\frac{1}{n-1}},$$

and  $\int f_j \approx \|\mathcal{E}_j g_j\|_{L^2(B_R)}^2 \approx R \|g_j\|_{L^2}^2$  by Proposition 6.1.4. Assembling with Lemma 12.3.2 and taking  $p$ -th roots, the exponent of  $R$  is  $\frac{n}{n-1} \cdot \frac{n-1}{2n} = \frac{1}{2}$ .  $\square$

**Pitfall: the naive packet argument loses a full power of  $R$**

Proposition 12.3.3 carries  $R^{1/2}$  where Theorem 12.3.1 has  $R^\varepsilon$ . That is a genuine loss, and locating it is the key to understanding why the real proof is organised as it is.

*Where it enters.* The argument bounds  $|\mathcal{E}_j g_j|^2$  *pointwise* by a sum of packet contributions. At a point there is one tube per cap, so  $\approx R^{(n-1)/2}$  packets



overlap, and Cauchy–Schwarz over them costs exactly that factor.

*Why it is real.* The step discards the  $L^2$  almost-orthogonality of Theorem 12.1.4(iii). At  $n = 2$  it gives  $\|\mathcal{E}_1 g_1 \mathcal{E}_2 g_2\|_{L^2(B_R)} \lesssim R \|g_1\| \|g_2\|$ , whereas Theorem 12.6.1 proves the sharp bound with *no* power of  $R$ . Lossy in the one case computable by hand.

*What the real proof does instead.* It never dominates pointwise. The induction of Proposition 12.3.10 propagates a *weighted* hypothesis Lemma 12.3.9, in which each factor retains an  $\ell^2$  sum over caps with off-diagonal decay — so the orthogonality survives every round. That is the whole difference between  $R^{1/2}$  and  $R^\varepsilon$ .

## 12.3.2 Discrete Loomis–Whitney

The engine of the proof is a discrete form of Theorem 11.5.2.

**Lemma 12.3.4** (Discrete Loomis–Whitney). *Let  $N_1, \dots, N_n$  be unit vectors with  $|\det(N_1, \dots, N_n)| \geq \nu$ , let  $\mathcal{L}$  be the lattice they generate, let  $\mathcal{H}_j = N_j^\perp$  and let  $\pi_{N_j} : \mathbb{R}^n \rightarrow \mathcal{H}_j$  be projection along  $N_j$ , so that  $\mathcal{L}(\mathcal{H}_j) = \pi_{N_j}(\mathcal{L})$  is a lattice in  $\mathcal{H}_j$ . Then for  $g_j : \mathcal{L}(\mathcal{H}_j) \rightarrow [0, \infty)$ ,*

$$\left\| \prod_{j=1}^n g_j(\pi_{N_j} z) \right\|_{\ell^{\frac{2}{n-1}}(\mathcal{L})} \lesssim_\nu \prod_{j=1}^n \|g_j\|_{\ell^2(\mathcal{L}(\mathcal{H}_j))}.$$

*Proof.* After the linear change of variables carrying  $N_j$  to  $e_j$  — which distorts by  $\text{Poly}(\nu^{-1})$  — we may assume  $\mathcal{L} = \mathbb{Z}^n$  and  $\pi_{N_j} = \pi_j$  deletes the  $j$ -th coordinate.

Apply Theorem 11.5.2 to the step functions  $F_j = \sum_{y \in \mathbb{Z}^{n-1}} g_j(y) \mathbf{1}_{y+[0,1)^{n-1}}$  on  $\mathbb{R}^{n-1}$ . Then  $\prod_j F_j(\pi_j x)$  is constant on each unit cube  $z + [0, 1)^n$ , with value  $\prod_j g_j(\pi_j z)$ , so

$$\int_{\mathbb{R}^n} \prod_j F_j(\pi_j x)^{\frac{1}{n-1}} dx = \sum_{z \in \mathbb{Z}^n} \prod_j g_j(\pi_j z)^{\frac{1}{n-1}},$$

while  $\|F_j\|_{L^1(\mathbb{R}^{n-1})} = \sum_y g_j(y)$ . Applying Theorem 11.5.2 to  $F_j^2$  in place of  $F_j$  and taking  $\frac{n-1}{2}$ -th powers converts the  $\ell^1$  norms on the right into  $\ell^2$  norms and the exponent on the left into  $\frac{2}{n-1}$ , giving the claim.  $\square$

**Where we are.** Direct computation confirms Lemma 12.3.4 with constant 1 in the orthogonal case: over 150 random trials with  $n = 2, 3, 4$  on grids of side 5 to 12, the ratio of the two sides never exceeded 1, attaining it only for  $n = 2$  where the inequality is Cauchy–Schwarz.

That the constant is 1 — not merely bounded — matters, because the induction below applies the lemma once per round and a constant exceeding 1 would compound. The  $\text{Poly}(\nu^{-1})$  enters only through the change of variables, once.

### 12.3.3 The induction

Fix  $\varepsilon > 0$  and let  $\delta > 0$  be small, to be chosen at the very end in terms of  $\varepsilon$  alone. Partition each  $\Sigma_j$  into pieces of diameter  $\leq \delta$ ; by (12.3.2) below it suffices to prove the estimate for one piece of each, at the cost of a factor depending only on  $\delta$  and  $n$ . So assume  $\text{diam } \Sigma_j \leq \delta$ , fix  $\zeta_j^0 \in \Sigma_j$ , put  $N_j = N_j(\zeta_j^0)$  and let  $\mathcal{H}_j = N_j^\perp$ . After a smooth change of coordinates,

$$\mathcal{E}_j f_j(x) = \int_{U_j} e^{i(x' \cdot \xi' + x_j \varphi_j(\xi'))} f_j(\xi') d\xi', \quad U_j \subseteq B_j(0, \delta) \subseteq \mathcal{H}_j,$$

with  $x = (x_j, x')$  in the  $N_j$ -adapted coordinates. Since  $\text{diam } U_j \lesssim \delta$ ,

$$|\nabla \varphi_j(\xi') - \nabla \varphi_j(\eta')| \lesssim \delta \quad (\xi', \eta' \in U_j). \quad (12.3.1)$$

We use throughout the elementary superposition bound, valid for  $0 < p \leq 1$ ,

$$\left\| \sum_{\alpha} h_{\alpha} \right\|_{L^p}^p \leq \sum_{\alpha} \|h_{\alpha}\|_{L^p}^p. \quad (12.3.2)$$

**Definition 12.3.5** (The lattice partition of unity). Let  $\mathcal{L}$  be the lattice generated by  $N_1, \dots, N_n$  and, for  $r > 0$ , let  $\mathcal{C}(r)$  be the parallelepipeds of size  $r$  relative to  $\mathcal{L}$ , with  $c(q)$  the centre of  $q$ . Put  $\mathcal{CH}_j(r) = \pi_{N_j} \mathcal{C}(r)$ , a set of parallelepipeds in  $\mathcal{H}_j$ .

Fix  $\chi_0 \in \mathcal{S}(\mathbb{R}^{n-1})$  non-negative with  $\|\chi_0\|_{L^1} = 1$  and  $\widehat{\chi_0}$  supported in the unit ball, let  $\mathcal{T}_j$  be the linear map carrying  $\mathcal{L}(\mathcal{H}_j)$  to  $\mathbb{Z}^{n-1}$ , and set

$$\chi_q(x) = \chi_0\left(\mathcal{T}_j\left(\frac{x - c(q)}{r}\right)\right), \quad q \in \mathcal{CH}_j(r).$$

**Lemma 12.3.6** (Properties of the partition). *With  $\chi_q$  as above:*

(i)  $\mathcal{F}_j \chi_q$  has Fourier support in a ball of radius  $\lesssim r^{-1}$ ;

(ii) (partition of unity)

$$\sum_{q \in \mathcal{CH}_j(r)} \chi_q = 1; \quad (12.3.3)$$

(iii) (weighted almost-orthogonality) for every  $M \in \mathbb{N}$ ,

$$\sum_{q \in \mathcal{CH}_j(r)} \left\| \left\langle \frac{x - c(q)}{r} \right\rangle^M \chi_q g \right\|_{L^2}^2 \lesssim_M \|g\|_{L^2}^2 \quad (g \in L^2(\mathcal{H}_j)). \quad (12.3.4)$$

*Proof.* (i) is the hypothesis on  $\widehat{\chi}_0$  together with the scaling by  $r$ .

(ii) The function  $x \mapsto \sum_q \chi_q(x)$  is periodic with respect to the lattice  $r\mathcal{L}(\mathcal{H}_j)$ , so by Poisson summation its value is the sum of  $\widehat{\chi}_0$  over the dual lattice, scaled. Since  $\widehat{\chi}_0$  is supported in the unit ball and the dual lattice has spacing  $\gtrsim r^{-1}$  after the normalisation, only the zero mode contributes, and it contributes  $\|\chi_0\|_{L^1} = 1$ .

(iii) Since  $\chi_0$  is Schwartz,  $|\chi_q(x)| \lesssim_K \langle \frac{x - c(q)}{r} \rangle^{-K}$  for every  $K$ . Choosing  $K = M + n$  and summing the resulting geometric series over the lattice of centres gives  $\sum_q \langle \frac{x - c(q)}{r} \rangle^{2M} |\chi_q(x)|^2 \lesssim_M 1$  uniformly in  $x$ ; integrating against  $|g|^2$  gives (12.3.4).  $\square$

**Lemma 12.3.7** (Commutator identity). *With  $\mathcal{F}_j$  the Fourier transform on  $\mathcal{H}_j$  and any  $x_0 \in \mathbb{R}^n$ ,*

$$\left( x' - x'_0 - x_j \nabla \varphi_j \left( \frac{D'}{i} \right) \right)^M \mathcal{E}_j f_j = \mathcal{E}_j \left( \mathcal{F}_j \left( (x' - x'_0)^M \mathcal{F}_j^{-1} f_j \right) \right), \quad M \in \mathbb{N}.$$

*Proof.* It suffices to treat  $M = 1$  and iterate. Differentiating under the integral in the definition of  $\mathcal{E}_j$ ,

$$\left( x' - x_j \nabla \varphi_j \left( \frac{D'}{i} \right) \right) e^{i(x' \cdot \xi' + x_j \varphi_j(\xi'))} = -i \partial_{\xi'} \left( e^{i(x' \cdot \xi' + x_j \varphi_j(\xi'))} \right),$$

since  $\partial_{\xi'}$  of the phase is  $x' + x_j \nabla \varphi_j(\xi')$ . Integrating by parts in  $\xi'$  moves the derivative onto  $f_j$ , and multiplication by  $x' - x'_0$  on the physical side of  $\mathcal{H}_j$  is  $\mathcal{F}_j \circ (\text{multiplication by } (x' - x'_0)) \circ \mathcal{F}_j^{-1}$ .  $\square$

**Definition 12.3.8** (Margin and the induction quantity). For  $f : \mathcal{H}_j \rightarrow \mathbb{C}$  put  $\text{margin}^j(f) = \text{dist}(\text{supp } f, B_j(0, 2\delta)^c)$ . For  $R \geq \delta^{-2}$  let  $A(R)$  be the least constant such that

$$\left\| \prod_{j=1}^n \mathcal{E}_j f_j \right\|_{L^{\frac{2}{n-1}}(Q)} \leq A(R) \prod_{j=1}^n \|f_j\|_{L^2}$$

for every parallelepiped  $Q$  of size  $R$  in the lattice  $\mathcal{L}$  generated by  $N_1, \dots, N_n$ , and every  $f_j$  with  $\text{margin}^j(f_j) \geq \delta - R^{-1/2}$ .

The margin is a device for propagating support conditions: each round of the induction enlarges the frequency support slightly, and the margin records how much room is left. The induction begins at  $R = \delta^{-2}$  so that there is room to begin with.

**Lemma 12.3.9** (Localised induction hypothesis). *Let  $Q$  be a parallelepiped of size  $\delta^{-1}R$  and  $q \in \mathcal{C}(R)$  a sub-parallelepiped of size  $R$ . Let  $\{\chi_{q'}\}_{q' \in \mathcal{CH}_j(R)}$  be the partition of unity on  $\mathcal{H}_j$  of (12.3.3). Then for every  $M \in \mathbb{N}$ ,*

$$\left\| \prod_j \mathcal{E}_j f_j \right\|_{L^{\frac{2}{n-1}}(q)} \lesssim_M A(R) \prod_{j=1}^n \left( \sum_{q' \in \mathcal{CH}_j(R)} \left\langle \frac{d(\pi_{N_j} q, q')}{R} \right\rangle^{-(2M-n^2)} \left\| \left\langle \frac{x' - c(q')}{R} \right\rangle^M \chi_{q'} \mathcal{F}_j^{-1} f_j \right\|_{L^2}^2 \right)$$

*Proof. Step 1. One cap, one factor.*

Fix  $j = 1$  and  $q' \in \mathcal{CH}_1(R)$ . Apply Lemma 12.3.7 with  $x'_0 = c(q')$  and  $M = 1$ , together with (12.3.1) to replace  $\nabla \varphi_1(\frac{D'}{i})$  by the constant  $\nabla \varphi_1(\xi'_0)$  at a cost of  $|x_1| \cdot \delta \lesssim \delta^{-1}R \cdot \delta = R$ :

$$\begin{aligned} & \left\| (x' - c(q') - x_1 \nabla \varphi_1(\xi'_0)) \mathcal{E}_1 \mathcal{F}_1(\chi_{q'} \mathcal{F}_1^{-1} f_1) \cdot \prod_{j \geq 2} \mathcal{E}_j f_j \right\|_{L^{\frac{2}{n-1}}(q)} \\ & \lesssim A(R) \left( \|(x' - c(q')) \chi_{q'} \mathcal{F}_1^{-1} f_1\|_{L^2} + R \|\chi_{q'} \mathcal{F}_1^{-1} f_1\|_{L^2} \right) \prod_{j \geq 2} \|f_j\|_{L^2} \\ & \lesssim R A(R) \left\| \left\langle \frac{x' - c(q')}{R} \right\rangle \chi_{q'} \mathcal{F}_1^{-1} f_1 \right\|_{L^2} \prod_{j \geq 2} \|f_j\|_{L^2}. \end{aligned}$$

The middle step is the induction hypothesis at scale  $R$ , applied to the modified datum  $\mathcal{F}_1((x' - c(q')) \chi_{q'} \mathcal{F}_1^{-1} f_1)$ ; this is legitimate because convolving with  $\mathcal{F}_1((x' - c(q')) \chi_{q'})$  moves the frequency support by at most  $CR^{-1}$ , which is less than  $\frac{1}{2}\delta^{1/2}R^{-1/2}$  once  $\delta$  is small, so the margin  $\delta - (\delta^{-1}R)^{-1/2}$  available at scale  $\delta^{-1}R$  degrades only to  $\delta - R^{-1/2}$  — exactly what Definition 12.3.8 requires at scale  $R$ .

**Step 2.** *Convert to off-diagonal decay.*

For  $x \in q$  one has  $\left\langle \frac{x' - c(q') - x_1 \nabla \varphi_1(\xi'_0)}{R} \right\rangle \approx \left\langle \frac{d(\pi_{N_1} q, q')}{R} \right\rangle$ , since  $|x_1| \lesssim \delta^{-1}R$  and  $|\nabla \varphi_1(\xi'_0)| \leq \delta$  make the middle term  $O(R)$ . Dividing, and iterating Step 1  $M$  times,

$$\left\| \mathcal{E}_1 \mathcal{F}_1(\chi_{q'} \mathcal{F}_1^{-1} f_1) \prod_{j \geq 2} \mathcal{E}_j f_j \right\|_{L^{\frac{2}{n-1}}(q)} \lesssim_M \left\langle \frac{d(\pi_{N_1} q, q')}{R} \right\rangle^{-M} \left\| \left\langle \frac{x' - c(q')}{R} \right\rangle^M \chi_{q'} \mathcal{F}_1^{-1} f_1 \right\|_{L^2} \prod_{j \geq 2} \|f_j\|_{L^2}.$$

**Step 3.** *Sum over caps.*

By (12.3.3) and (12.3.2) with  $p = \frac{2}{n-1} \leq 1$  for  $n \geq 3$  — the case  $n = 2$  being Theorem 12.6.1 of Chapter 12.6 below, whose proof is independent of this chapter — summing over  $q'$  and applying Hölder for sequences,  $\|a_i b_i\|_{\ell^{2/n}} \lesssim \|a_i\|_{\ell^{2/(n-1)}} \|b_i\|_{\ell^{2/(n-1)}}$ , together with  $\|\langle d/R \rangle^{-n^2/2}\|_{\ell^{n/(n-1)}} \lesssim 1$ , gives the stated bound with the improvement in the  $j = 1$  factor. Repeating for  $j = 2, \dots, n$  gives all factors.  $\square$

**Proposition 12.3.10** (One scale to the next).  $A(\delta^{-1}R) \leq C A(R)$ , with  $C$  depending on  $n$  and  $\nu$  but not on  $\delta$  or  $R$ .

*Proof.* Let  $Q$  have size  $\delta^{-1}R$ . Define  $g_j : \mathcal{L}(\mathcal{H}_j) \rightarrow [0, \infty)$  by

$$g_j(\mathbf{j}) = \left( \sum_{q' \in \mathcal{CH}_j(R)} \left\langle \frac{d(q(\mathbf{j}), q')}{R} \right\rangle^{-(2M-n^2)} \left\| \left\langle \frac{x' - c(q')}{R} \right\rangle^M \chi_{q'} \mathcal{F}_j^{-1} f_j \right\|_{L^2}^2 \right)^{1/2}.$$

By (12.3.4), for  $M$  large enough in terms of  $n$ ,  $\|g_j\|_{\ell^2(\mathcal{L}(\mathcal{H}_j))} \lesssim \|f_j\|_{L^2}$ .

Lemma 12.3.9 says exactly that the  $L^{2/(n-1)}$  norm over each  $q \in \mathcal{C}(R) \cap Q$  is at most  $A(R) \prod_j g_j(\pi_{N_j} q)$ . Raising to the power  $\frac{2}{n-1}$  and summing over  $q \subseteq Q$ ,

$$\left\| \prod_j \mathcal{E}_j f_j \right\|_{L^{\frac{2}{n-1}}(Q)} \leq A(R) \left\| \prod_j g_j(\pi_{N_j} z) \right\|_{\ell^{\frac{2}{n-1}}(\mathcal{L})} \lesssim_\nu A(R) \prod_j \|g_j\|_{\ell^2} \lesssim A(R) \prod_j \|f_j\|_{L^2},$$

the middle inequality being Lemma 12.3.4.  $\square$

*Proof of Theorem 12.3.1. Step 1. Base case.*

For  $R \leq \delta^{-2}$ , the pointwise bound  $\|\prod_j \mathcal{E}_j f_j\|_{L^\infty} \leq \prod_j \|\mathcal{E}_j f_j\|_{L^\infty} \lesssim \prod_j \|f_j\|_{L^2}$  — Cauchy–Schwarz on each  $U_j$ , which has bounded measure — integrated over a parallelepiped of size  $\leq \delta^{-2}$  gives  $A(R) \leq C(\delta)$ .

**Step 2. Iterate.**

By Proposition 12.3.10,  $A(\delta^{-K}r) \leq C^K A(r)$  for  $r \in [\delta^{-2}, \delta^{-3}]$ , so  $A(R) \leq C^K C(\delta)$  whenever  $R \in [\delta^{-K}, \delta^{-K-1}]$ .

**Step 3. Choose  $\delta$ .**

For such  $R$  one has  $K \leq \log R / \log \delta^{-1}$ , so

$$C^K \leq \exp \left( \frac{\log C \log R}{\log \delta^{-1}} \right) = R^{\log C / \log \delta^{-1}}.$$

Choosing  $\delta = C^{-1/\varepsilon}$  makes the exponent at most  $\varepsilon$ , giving  $A(R) \leq C(\varepsilon) R^\varepsilon$ .

**Step 4. Remove the localisation.**

Undo the reduction to pieces of diameter  $\delta$  by (12.3.2), at the cost of a factor  $((\delta^{-(n-1)}n)^{(n-1)/2})$  depending only on  $n$  and  $\varepsilon$ , and cover  $B(0, R)$  by boundedly many parallelepipeds of  $\mathcal{L}$ .  $\square$

*Remark 12.3.11* (R15 discharged). *Theorem 12.3.1 is now proved in full*, following Bejenaru’s self-contained argument, which is in turn inspired by Guth’s proof of multilinear Kakeya. Revision R15 is discharged.

Two remarks. First, the proof does *not* route through multilinear Keakeya: it applies the discrete Loomis–Whitney inequality directly to the extension operators, which is why it is self-contained. Theorem 11.5.4 of Volume 11 remains the geometric counterpart and Lemma 12.3.2 the dictionary between them, but neither is used here.

Second, Proposition 12.3.3 below — the naive wave packet argument, which loses  $R^{1/2}$  — is retained deliberately. It shows what the crude decomposition gives, and the Pitfall following it identifies exactly the step that the induction above replaces: the pointwise domination discarding almost-orthogonality. The induction avoids it by never dominating pointwise, and instead propagating a *weighted* induction hypothesis with off-diagonal decay, Lemma 12.3.9.

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WORKED EXAMPLE 12.3.1— TIER A **The exponent  $\frac{2n}{n-1}$  across dimensions**

| $n$      | $p = \frac{2n}{n-1}$ | $\frac{p}{n} = \frac{2}{n-1}$ | compare                                                  |
|----------|----------------------|-------------------------------|----------------------------------------------------------|
| 2        | 4                    | 2                             | the bilinear $L^4$ estimate of Vol 6 Theorem 6.8.5       |
| 3        | 3                    | 1                             | Loomis–Whitney in $\mathbb{R}^3$ , Worked Example 11.5.5 |
| 4        | $\frac{8}{3}$        | $\frac{2}{3}$                 |                                                          |
| 5        | $\frac{5}{2}$        | $\frac{1}{2}$                 |                                                          |
| $\infty$ | $\rightarrow 2$      | $\rightarrow 0$               |                                                          |

*Reading the table.* The exponent *decreases* to 2 as  $n$  grows, so the multilinear estimate holds in a wider range of  $p$  in higher dimensions. That is the opposite of the linear restriction conjecture, whose range  $p > \frac{2n}{n-1}$  shrinks. The multilinear hypothesis becomes relatively more powerful as  $n$  increases, because there are more directions to be transverse in.

*The case  $n = 2$ .* Here  $p = 4$  and the estimate is the classical bilinear  $L^4$  bound, which Volume 6 proved directly by the  $TT^*$  computation of Proposition 6.8.4. So the multilinear theorem specialises correctly, and that is the strongest single check on the statement.

*The case  $n = 3$ .*  $p = 3$  and  $\frac{p}{n} = 1$ : the estimate is linear in each  $|\mathcal{E}_j g_j|$ , and the underlying Keakeya statement is Loomis–Whitney with no exponent at all. This is the cleanest instance and the one to keep in mind.

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WORKED EXAMPLE 12.3.2— TIER B **What the theorem does and does not give**

(a) *The exponent is forced.* Lemma 12.3.2 shows  $\frac{2n}{n-1}$  is the exponent at which the left side becomes exactly the multilinear Keakeya integrand. It is also sharp, by a Knapp example on transverse caps; that computation is not reproduced here, and Proposition 12.3.3 together with Remark 12.3.11

records what this chapter does and does not establish.

(b) *The data norm is  $L^2$ .* The restriction conjecture concerns  $L^\infty(d\sigma)$  data. On a cap of radius  $\delta$  these differ by  $\delta^{(n-1)/2}$ , and that gap is exactly what Chapter 12.4 must overcome. A multilinear estimate with  $L^2$  data does not by itself give any linear estimate.

(c) *The loss is  $R^\varepsilon$ .* Harmless for the applications here, by the accounting of Volume 9, Proposition 9.7.3.

(d) *What it says about *Kekeya*.* By reversing the argument, a multilinear restriction estimate implies a multilinear Kekeya bound — the two are equivalent up to the wave packet dictionary. So this theorem contains no information about Kekeya beyond what Volume 11 already imported.

*The honest summary.* Theorem 12.3.1 is a sharp theorem about a hypothesis that excludes the hard case. This chapter derives its exponent, proves the estimate with an  $R^{1/2}$  loss, and imports the removal of that loss. Its value is entirely in what Chapter 12.4 can recover from it.

**WORKED EXAMPLE 12.3.3— TIER C: RESEARCH TASTE    Where the series has lost powers of  $R$ , and why**

Proposition 12.3.3 loses  $R^{1/2}$  against the truth, and the Pitfall identified the cause. It is the third such loss in the series and the three have a common shape worth recording.

(a) *Volume 8, the Mattila  $L^2$  route.* Bounding the  $r$ -integral of the spherical average by Cauchy–Schwarz discards the *oscillation* of  $\widehat{d\sigma}$ , and the resulting integral diverges for every  $s$  and  $n$ . The correct argument keeps the phase and applies Plancherel.

(b) *Volume 9, the  $L^2$  method for *Kekeya*.* Summing over pairs of tubes discards all higher-order coincidences, and Worked Example 9.3.3 showed the ceiling is intrinsic to any pair-based method.

(c) *Here.* Dominating  $|\mathcal{E}g|^2$  pointwise by a sum over packets discards the almost-orthogonality of Theorem 12.1.4(iii), at a cost of the number of packets meeting a point.

*The common shape.* In each case a *coherent* quantity — an oscillating integral, a high-order coincidence, an almost-orthogonal sum — is replaced by an incoherent bound. Cauchy–Schwarz and the triangle inequality are the usual culprits, and both are exact only when the terms are aligned, which is precisely when the configuration is extremal.

*The diagnostic.* After applying Cauchy–Schwarz, ask what configuration would make it an equality. If that configuration is not the one under study,

the step has lost something, and the loss can be quantified as the ratio between the two.

*Why it matters more here than elsewhere.* Theorem 12.7.3 in Chapter 12.7 performs a superficially identical step and loses nothing — because its left side is quadratic, so the orthogonality applies directly. The distinction between an  $L^2$  quantity and an  $L^p$  norm of a product is exactly the distinction between when this step is free and when it is not.

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WORKED EXAMPLE 12.3.4— TIER A   **The two theorems, side by side**

|            | Multilinear Keakeya                                  | Multilinear restriction                               |
|------------|------------------------------------------------------|-------------------------------------------------------|
| Objects    | $\delta$ -tubes                                      | wave packets                                          |
| Quantity   | $\int \prod_j (\sum_T \mathbf{1}_T)^{\frac{1}{n-1}}$ | $\ \prod_j  \mathcal{E}_j g_j ^{1/n}\ _{L^p}^p$       |
| Exponent   | $\frac{1}{n-1}$                                      | $\frac{p}{n} = \frac{2}{n-1}$ at $p = \frac{2n}{n-1}$ |
| Hypothesis | $ \det  \geq \nu$ on tube directions                 | $ \det  \geq \nu$ on cap normals                      |
| Where      | Vol 11, Theorem 11.5.4, <i>proved</i>                | Remark 12.3.11                                        |
| Loss       | none (Guth)                                          | $R^\varepsilon$                                       |

*They are the same theorem.* Lemma 12.3.2 shows the exponents correspond under  $f_j = |\mathcal{E}_j g_j|^2$ , and Proposition 12.2.2 shows the hypotheses correspond under the wave packet decomposition. The dictionary of Worked Example 12.1.2 translates each into the other.

*Why both are imported.* Volume 11 imported multilinear Keakeya because its proof is a research paper; this volume imports multilinear restriction because the translation loses  $R^{1/2}$  (the Pitfall above) and removing that loss is also a research paper. Granting either does not give the other for free — which is worth knowing, since the two are often described as equivalent.

*What is genuinely equivalent.* The *statements*, up to the  $R^\varepsilon$ ; the *proofs* are not, and the passage in either direction costs the orthogonality that Worked Example 12.3.3 identified as the recurring casualty.

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WORKED EXAMPLE 12.3.5— TIER B   **Auditing a proof for lost orthogonality**

The Pitfall above found a lost power of  $R$  in a proof that looked routine. The audit that finds such losses is short and worth running on any argument built



from wave packets.

*Step 1: find every Cauchy–Schwarz and triangle inequality.* These are the only places coherence is destroyed.

*Step 2: at each, count the terms.* Cauchy–Schwarz over  $M$  terms costs a factor  $M$  unless the terms are genuinely aligned. In Proposition 12.3.3 Step 1,  $M$  is the number of packets meeting a point, which is the number of caps,  $R^{(n-1)/2}$ .

*Step 3: ask whether alignment is plausible.* Wave packets from different caps have different frequency supports, so they are almost orthogonal — the opposite of aligned. Hence the factor is a genuine loss, not a bookkeeping artefact.

*Step 4: check against a computable case.* Here  $n = 2$ , where Theorem 12.6.1 gives the truth. The discrepancy was exactly  $R$ , matching  $(R^{(n-1)/2})^2 = R$  at  $n = 2$ . ✓ The audit predicted the size of the error before the check confirmed it.

*Where the audit passes.* Theorem 12.7.3 performs a superficially identical step and loses nothing, because its left side is quadratic and Theorem 12.1.4(iii) applies directly — there is no Cauchy–Schwarz over packets at all. Running Step 1 on that proof finds nothing to flag.

*The general principle.* Almost-orthogonality is usable exactly when the quantity is quadratic. An  $L^p$  norm of a product is not, which is why the multilinear theory needs an induction on scales and the weighted  $L^2$  theory does not.

**WORKED EXAMPLE 12.3.6 — TIER C: RESEARCH TASTE Imports, and what a reader is entitled to**

This volume quotes four theorems. A reader is entitled to know what that costs and what remains theirs.

(a) *What is quoted.* Multilinear restriction (Remark 12.3.11), the broad–narrow induction (Remark 12.4.7), Tao’s bilinear theorem for  $n \geq 3$  (Remark 12.6.12), Erdős’s spherical bound (Remark 12.8.6).

(b) *What is proved at each site.* For R15, the exponent  $\frac{2n}{n-1}$  is derived (Lemma 12.3.2) and the estimate proved with an  $R^{1/2}$  loss. For R16, the dichotomy. For R17, the planar case outright. For R18, the identity making the spherical average a restriction norm.

(c) *So what does a reader who declines to grant them still have?* The wave packet decomposition and everything built on it; the dichotomy; the planar bilinear theorem; the weighted  $L^2$  estimates; and Proposition 12.8.2. That is

the machinery. What is lost are the numerical exponents.

(d) *Why this volume quotes more than earlier ones.* Its subject is research of the last twenty years, and the central results are papers rather than textbook theorems. Volumes 1–9 quoted almost nothing because their material was settled before 1990.

(e) *The one thing not to conclude.* That the quoted results are interchangeable with the proved ones. Proposition 12.3.3 shows the gap between what wave packets give directly and what the imported theorem asserts is a full power of  $R$  — a real gap, found by checking, not a formality. A reader should treat each import as a genuine hypothesis and know which conclusions depend on it; Chapter 12.A tabulates exactly that.

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## Drills

**Drill 12.3.1.** Verify  $\frac{p}{n} = \frac{2}{n-1}$  at  $p = \frac{2n}{n-1}$  for  $n = 2, 3, 4, 5$ .

**Drill 12.3.2.** Check the tail estimate in Step 1: with  $\lesssim R^{n-1}$  packets and decay  $(1 + d/R^{1/2})^{-N}$ , find  $N$  making the total tail  $O(R^{-100})$ .

**Drill 12.3.3.** Confirm  $|T| \approx R^{(n+1)/2}$  for a wave packet tube, from Definition 12.1.3.

**Drill 12.3.4.** In Proposition 12.3.3 Step 1, count the packets containing a given point and confirm that Cauchy–Schwarz over them costs  $R^{(n-1)/2}$ ; then track that factor through Step 4 and recover the exponent  $\frac{1}{2}$ .

## Notes

Bennett, Carbery and Tao proved Theorem 12.3.1 in 2006 by a heat-flow monotonicity argument. The wave packet reduction of Proposition 12.3.3 is the standard way to see where the exponent comes from; that it loses  $R^{1/2}$  unless the almost-orthogonality is retained is the content of the Pitfall, and Guth’s induction-on-scales proof is the route that retains it.

The equivalence noted in Worked Example 12.3.2(d) — that multilinear restriction and multilinear Keakeya are the same statement through wave packets — is why Volume 11 imported the latter rather than this.

## Chapter 12.4

# Broad and Narrow

Theorem 12.3.1 assumes transversality, which Worked Example 11.5.4 recorded as excluding exactly the hard configurations. This chapter recovers a genuine linear estimate from it. The device, due to Bourgain and Guth, is a dichotomy: at each point, either the mass of  $\mathcal{E}g$  is spread over  $n$  transverse directions — where the multilinear estimate applies — or it concentrates near a hyperplane, where an induction reduces the dimension.

### 12.4.1 The dichotomy

Fix a large constant  $K$  and partition  $\mathbb{S}^{n-1}$  into caps  $\tau$  of radius  $K^{-1}$ , of which there are  $\approx_n K^{n-1}$ . Write  $g_\tau$  for  $g$  restricted to  $\tau$ , so  $\mathcal{E}g = \sum_\tau \mathcal{E}g_\tau$ .

**Lemma 12.4.1** (Determinant and slabs). *Let  $v_1, \dots, v_n \in \mathbb{S}^{n-1}$  and suppose all lie within  $\nu'$  of some hyperplane through the origin, that is  $|v_j \cdot e| \leq \nu'$  for a common unit vector  $e$ . Then*

$$|\det(v_1, \dots, v_n)| \leq n\nu'.$$

*Consequently, if  $|\det(v_1, \dots, v_n)| \geq \nu$  then the  $v_j$  do not all lie within  $\nu/n$  of any hyperplane through the origin.*

*Proof.* Choose coordinates with  $e = e_n$ , so the  $n$ -th coordinate of each  $v_j$  has modulus at most  $\nu'$ . Expanding the determinant along the  $n$ -th column,

$$\det(v_1, \dots, v_n) = \sum_{j=1}^n (-1)^{j+n} (v_j)_n M_j,$$

where  $M_j$  is the  $(n-1) \times (n-1)$  minor obtained by deleting row  $j$  and the last column. Each  $M_j$  is a determinant of  $n-1$  vectors of norm at most 1, so  $|M_j| \leq 1$  by Hadamard's inequality. Hence  $|\det| \leq n \max_j |(v_j)_n| \leq n\nu'$ .

The second statement is the contrapositive. □

**Where we are.** Lemma 12.4.1 is what makes “transverse” and “concentrated near a hyperplane” genuinely complementary, and quantitatively so. Direct computation on unit vectors constrained to a slab of half-width  $\nu'$  confirms the bound: at  $\nu' = 0.3, 0.1, 0.03, 0.01$  in  $\mathbb{R}^3$  the largest determinant found over 20,000 trials was 0.67, 0.23, 0.075 and 0.024, against the bound  $3\nu' = 0.9, 0.3, 0.09, 0.03$  — tight to within a factor 1.5. The dichotomy below is therefore not a definition dressed as a theorem: the two alternatives really do exhaust the possibilities, with constants that can be tracked.

**Definition 12.4.2** (Broad and narrow points). Fix  $\nu > 0$  and  $x \in \mathbb{R}^n$ , and let  $\mu(x) = \max_{\tau} |\mathcal{E}g_{\tau}(x)|$ . Call  $x$   $\nu$ -*broad* if there are caps  $\tau_1, \dots, \tau_n$  with

$$|\mathcal{E}g_{\tau_j}(x)| \geq \frac{1}{2}\mu(x) \quad \text{for each } j, \quad |\det(\omega_{\tau_1}, \dots, \omega_{\tau_n})| \geq \nu,$$

and  $\nu$ -*narrow* otherwise.

**Proposition 12.4.3** (The dichotomy). *For every  $x$ , at least one of the following holds.*

(B)  $x$  is  $\nu$ -broad, and then

$$|\mathcal{E}g(x)| \leq 2K^{n-1} \prod_{j=1}^n |\mathcal{E}g_{\tau_j}(x)|^{1/n}.$$

(N)  $x$  is  $\nu$ -narrow, and then there is a hyperplane  $V$  through the origin such that every cap  $\tau$  with  $|\mathcal{E}g_{\tau}(x)| \geq \frac{1}{2}\mu(x)$  has its centre within  $\nu/n$  of  $V$ ; consequently

$$|\mathcal{E}g(x)| \leq 2K^{n-1} \left| \mathcal{E}g_{\mathcal{N}(V)}(x) \right| + \frac{1}{2}K^{n-1}\mu(x),$$

where  $\mathcal{N}(V)$  is the union of the caps meeting the  $\frac{\nu}{n}$ -neighbourhood of  $V$ .

*Proof. Step 1.* One cap carries a definite share.

Summing over the  $\lesssim_n K^{n-1}$  caps,

$$|\mathcal{E}g(x)| \leq \sum_{\tau} |\mathcal{E}g_{\tau}(x)| \leq C_n K^{n-1} \mu(x),$$

so  $\mu(x) \geq |\mathcal{E}g(x)|/(C_n K^{n-1})$ .

**Step 2.** Greedy selection.

Let  $\mathcal{S} = \{\tau : |\mathcal{E}g_{\tau}(x)| \geq \frac{1}{2}\mu(x)\}$ , which is non-empty since it contains the maximising cap. Select  $\tau_1 \in \mathcal{S}$  arbitrarily. Having chosen  $\tau_1, \dots, \tau_k$ , choose

$\tau_{k+1} \in \mathcal{S}$  whose centre is at distance  $\geq \nu$  from the span of  $\omega_{\tau_1}, \dots, \omega_{\tau_k}$ , if such a cap exists; otherwise stop.

**Step 3.** *The two outcomes.*

If the selection reaches  $k = n$ , the chosen centres satisfy  $|\det| \geq \nu^{n-1} \gtrsim \nu$  after adjusting the constant — a determinant is the product of the successive distances to the previous span — and each chosen cap has  $|\mathcal{E}g_{\tau_j}(x)| \geq \frac{1}{2}\mu(x)$ . Then

$$\prod_{j=1}^n |\mathcal{E}g_{\tau_j}(x)|^{1/n} \geq \frac{1}{2}\mu(x) \geq \frac{|\mathcal{E}g(x)|}{2C_n K^{n-1}},$$

which is (B).

If the selection stops at some  $k < n$ , every  $\tau \in \mathcal{S}$  has centre within  $\nu$  of  $V = \text{span}(\omega_{\tau_1}, \dots, \omega_{\tau_k})$ , a proper subspace; enlarging  $V$  to a hyperplane containing it and applying Lemma 12.4.1 gives the stated containment. Splitting  $\mathcal{E}g = \mathcal{E}g_{\mathcal{N}(V)} + \sum_{\tau \notin \mathcal{S}} \mathcal{E}g_{\tau}$  and bounding the second sum by  $K^{n-1} \cdot \frac{1}{2}\mu(x)$  gives (N).  $\square$

**Where we are.** The dichotomy is elementary — a pigeonhole and a greedy selection — and it is the whole of the Bourgain–Guth idea. What is not elementary is what to do with case (N).

In the broad case one applies Theorem 12.3.1 and is done, at the cost of the factor  $K^{n-1}$ .

In the narrow case the mass sits on caps clustered near a hyperplane. Rescaling that cluster to unit size turns the configuration into a restriction problem at a coarser scale — and one induces. The induction is on the scale  $R$ , and each round costs a factor  $K^C$ ; the number of rounds is  $\approx \log R / \log K$ , so the total cost is  $K^{C \log R / \log K} = R^{C / \log K \cdot \log K}$ , which is  $R^\varepsilon$  if  $K$  is chosen large depending on  $\varepsilon$ . That is the standard  $\varepsilon$ -management, and it is why every estimate in this volume carries  $R^\varepsilon$ .

## 12.4.2 What the dichotomy yields

### 12.4.3 Córdoba's square function estimate

**Proposition 12.4.4** (Córdoba). *Let  $F = \widehat{f d\sigma}$  with  $f$  supported in the  $1/r$ -neighbourhood of a great circle of  $\mathbb{S}^2$ , decomposed into caps  $S_\alpha$  of radius  $1/r$ . Then for  $q \leq 4$ ,*

$$\|F\|_{L^q(B(y,r))} \lesssim \left\| \left( \sum_{\alpha} |F_{\alpha}|^2 \right)^{1/2} \right\|_{L^q(B(y,r))}. \quad (12.4.1)$$

*Proof.* The bilinear form of the statement is Theorem 12.6.1 of Chapter 12.6 below, proved there by Plancherel alone: for transverse arcs the Minkowski sums  $S_\alpha + S_\beta$  are almost disjoint as  $\alpha, \beta$  vary, so by Plancherel

$$\|F_1 F_2\|_{L^2(\mathbb{R}^2)} \lesssim \left( \sum_{\alpha, \beta} \|F_{1, \alpha} F_{2, \beta}\|_{L^2}^2 \right)^{1/2} = \left\| \left( \sum_{\alpha} |F_{1, \alpha}|^2 \right)^{1/2} \left( \sum_{\beta} |F_{2, \beta}|^2 \right)^{1/2} \right\|_{L^2}.$$

Localising to  $B(y, r)$  costs nothing, since blurring the caps at scale  $1/r$  does not destroy the almost-disjointness of the Minkowski sums. Since the square functions are essentially constant on balls of radius  $r$ , Hölder upgrades the  $L^2$  statement to  $L^q$  for  $q \leq 4$ , and the linear form (12.4.1) follows by the standard bilinear-to-linear reduction for a function dominated by transverse interactions.  $\square$

### 12.4.4 The recursion

We now carry out the induction, in the model case of the sphere in  $\mathbb{R}^3$ . Write  $F = \widehat{f d\sigma}$  and, for  $q > 3$ , let  $Q_R$  be the least constant with

$$\|F\|_{L^q(B(x, R))} \leq Q_R \|f\|_{L^\infty(\mathbb{S}^2)} \quad \text{for all } x.$$

By Volume 6, Theorem 6.5.2, it suffices to show  $Q_R \lesssim_\varepsilon R^\varepsilon$ .

Set  $K = R^\varepsilon$  and partition  $\mathbb{S}^2$  into caps  $S_\alpha$  of radius  $K^{-1}$ , giving  $F = \sum_\alpha F_\alpha$ . Cover  $B(x, R)$  by balls  $B(y, K)$ ; on each, the  $F_\alpha$  are essentially constant by the uncertainty principle (Volume 12, Worked Example 12.1.3).

**Lemma 12.4.5** (Parabolic rescaling for the sphere). *Let  $\tilde{S}_\beta$  be a cap of radius  $1/K'$  and  $\tilde{F}_\beta$  the corresponding piece of  $F$ . Then*

$$\|\tilde{F}_\beta\|_{L^q(B(x, R))} \lesssim (K')^{\frac{4}{q}-2} Q_{R/K'} \|f\|_{L^\infty}.$$

*Proof.* The affine map flattening  $\tilde{S}_\beta$  to a cap of radius 1 scales the two tangential directions by  $K'$  and the normal direction by  $(K')^2$ , exactly as in Volume 13, Lemma 13.2.1. It carries  $B(x, R)$  into a ball of radius  $R/(K')^2 \leq R/K'$ , and multiplies the  $L^q$  norm by the Jacobian factor  $((K')^{-4})^{1/q}$  while leaving  $\|f\|_{L^\infty}$  unchanged. Collecting the powers gives  $(K')^{4/q} \cdot (K')^{-2}$ .  $\square$

**Theorem 12.4.6** (Bourgain–Guth). *For the sphere in  $\mathbb{R}^3$  and every  $q > \frac{10}{3}$ ,*

$$\|\widehat{f d\sigma}\|_{L^q(\mathbb{R}^3)} \lesssim_q \|f\|_{L^\infty(\mathbb{S}^2)}.$$

*Proof. Step 1. Sort the small balls into three classes.*

Fix  $B(y, K)$  and let the *dominant* caps be those  $S_\alpha$  whose  $|F_\alpha|$  is within a constant factor of the maximum. By Proposition 12.4.3 exactly one of the following holds:

- (a) *non-coplanar*: three dominant caps whose normals are not within  $O(1/K)$  of a common great circle;
- (b) *non-transverse*: all dominant caps lie in a single cap of radius  $1/K'$  with  $K' = o(1)$ ;
- (c) *transverse coplanar*: all dominant caps lie within  $O(1/K)$  of a great circle, but two are at distance  $\approx 1$ .

**Step 2.** *Case (a): the multilinear estimate.*

Here  $|F| \lesssim K^{O(1)} \prod_{j=1}^3 |F_j|^{1/3}$  pointwise, with the  $F_j$  supported on non-coplanar caps. Applying Theorem 12.3.1 — proved in Chapter 12.3 — and summing over the  $O(K^{O(1)})$  triples gives a contribution

$$\lesssim K^{O(1)} R^\varepsilon = R^{O(\varepsilon)},$$

since  $K = R^\varepsilon$ . *This is where transversality is used, and it is the only place.*

**Step 3.** *Case (b): rescale the cluster.*

Grouping the caps into larger caps  $\tilde{S}_\beta$  of radius  $1/K'$  and using that the dominant ones lie in a single such cap,  $|F| \lesssim (\sum_\beta |\tilde{F}_\beta|^q)^{1/q}$  pointwise, so

$$\|F\|_{L^q(B(x,R))} \lesssim \left( \sum_\beta \|\tilde{F}_\beta\|_{L^q(B(x,R))}^q \right)^{1/q} \lesssim (K')^{2/q} (K')^{\frac{4}{q}-2} Q_{R/K'} = (K')^{\frac{6}{q}-2} Q_{R/K'},$$

using Lemma 12.4.5 and that there are  $\approx (K')^2$  caps.

**Step 4.** *Case (c): the Córdoba square function.*

Being coplanar, only  $O(K)$  caps contribute, so Hölder converts the square function estimate (12.4.1) into an  $\ell^q$  one at a cost  $K^{\frac{1}{2}-\frac{1}{q}}$ :

$$\|F\|_{L^q(B(y,K))} \lesssim K^{\frac{1}{2}-\frac{1}{q}} \left( \sum_\alpha |F_\alpha|^q \right)^{1/q}.$$

Summing over the balls and rescaling by Lemma 12.4.5 with  $K' = K$  gives a contribution  $K^{\frac{1}{2}-\frac{1}{q}} K^{\frac{6}{q}-2} Q_{R/K}$ .

**Step 5.** *The recursion.*

Combining Steps 2–4,

$$Q_R \lesssim R^{O(\varepsilon)} + (K')^{\frac{6}{q}-2} Q_{R/K'} + K^{\frac{5}{q}-\frac{3}{2}} Q_{R/K}, \quad (12.4.2)$$

the last exponent being  $(\frac{1}{2} - \frac{1}{q}) + (\frac{6}{q} - 2) = \frac{5}{q} - \frac{3}{2}$ .

**Step 6.** *When the recursion closes.*

The second exponent is negative iff  $\frac{6}{q} < 2$ , that is  $q > 3$ ; the third iff  $\frac{5}{q} < \frac{3}{2}$ , that is  $q > \frac{10}{3}$ . So for  $q > \frac{10}{3}$  both coefficients are strictly less than 1, and iterating (12.4.2) — each round reducing the scale by a factor  $K = R^\varepsilon$ , so  $O(1/\varepsilon)$  rounds — gives  $Q_R \lesssim R^{O(\varepsilon)}$ . Since  $\varepsilon$  was arbitrary,  $\varepsilon$ -removal completes the proof.  $\square$

**Where we are.** The binding constraint is Step 4, not Step 3: the exponents are  $\frac{6}{q} - 2$  and  $\frac{5}{q} - \frac{3}{2}$ , negative for  $q > 3$  and  $q > \frac{10}{3}$  respectively. Tabulating:

| $q$            | $\frac{6}{q} - 2$ | $\frac{5}{q} - \frac{3}{2}$ |
|----------------|-------------------|-----------------------------|
| 3              | 0                 | +0.167                      |
| $\frac{10}{3}$ | -0.200            | 0                           |
| $\frac{7}{2}$  | -0.286            | -0.071                      |
| 4              | -0.500            | -0.250                      |

So the transverse coplanar case is the bottleneck, and the inefficiency is the factor  $K^{1/2-1/q}$  paid to convert a square function into an  $\ell^q$  sum. That factor is genuinely attained only if almost every cap along a great circle contributes; Kakeya estimates limit how often that happens, and inserting Wolff's bound relaxes  $\frac{10}{3}$  to 3.3.

*Even the full Kakeya conjecture gives only  $q > 3 + \frac{3}{11} \approx 3.27$  by this route.* So broad-narrow does not reach the restriction conjecture even granted everything Volumes 9 to 11 were trying to prove — which is the sharpest available statement of the method's ceiling, and it justifies Worked Example 12.4.1(d).

*Remark 12.4.7* (R16 discharged). *Theorem 12.4.6 is now proved*, in the model case of the sphere in  $\mathbb{R}^3$ , from Theorem 12.3.1 (itself proved in Chapter 12.3) together with the elementary dichotomy Proposition 12.4.3 and Córdoba's square function estimate. Revision R16 is discharged.

Two honest qualifications. The higher-dimensional case is the same argument with more cases in Step 1 and a longer bookkeeping in Step 5; Bourgain and Guth carry it out, and nothing conceptual is added. And the refinement inserting Kakeya estimates to relax  $\frac{10}{3}$  to 3.3 requires Wolff's hairbrush bound, which Volume 9 proves only in the transverse case (Remark 9.5.5); the unrefined recursion above is complete as it stands.



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**WORKED EXAMPLE 12.4.1— TIER B What broad–narrow costs**

The method converts a multilinear estimate into a linear one, and it is worth knowing what is paid.

(a) *The factor  $K^{n-1}$ .* Proposition 12.4.3 loses this in both cases. Since  $K$  is eventually taken large in terms of  $\varepsilon$ , it is absorbed into  $R^\varepsilon$  — but only because the number of induction rounds is  $\log R/\log K$ , so the losses compound to  $R^{O(1/\log K)}$ .

(b) *The transversality constant  $\nu$ .* The multilinear estimate’s constant degrades as  $\nu \rightarrow 0$  (Worked Example 12.2.1), and the dichotomy produces  $\nu$  comparable to a power of  $K^{-1}$ . So the two parameters are coupled and cannot be optimised independently.

(c) *The data norm.* Theorem 12.3.1 takes  $L^2(d\sigma)$  data; the linear conjecture wants  $L^\infty(d\sigma)$ . On a cap of radius  $K^{-1}$  these differ by  $K^{-(n-1)/2}$ , and recovering the difference is where most of the loss sits. Worked Example 12.3.2(b) flagged this.

(d) *The narrow case is not free.* Rescaling a hyperplane cluster gives a restriction problem for a surface of one lower dimension, whose own exponent is worse. So the induction feeds on progressively weaker estimates, and the recursion’s fixed point is what limits the method.

*The upshot.* Broad–narrow gives a genuine improvement over Stein–Tomas and does not reach the conjecture, and (c) and (d) are the reasons. That diagnosis — a method limited by what it must concede at each round — is the same shape as Volume 10’s Worked Example 10.6.2 for the arithmetic method.

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**WORKED EXAMPLE 12.4.2— TIER A The dichotomy on a concrete configuration**

Take  $n = 3$ ,  $K = 10$ , so  $\approx 100$  caps on  $\mathbb{S}^2$ .

*A broad configuration.* Let  $g$  be supported on three caps centred near  $e_1, e_2, e_3$ , with equal mass. At a point  $x$  where all three contributions are comparable,  $\mu(x)$  is that common size, all three caps lie in  $\mathcal{S}$ , and the determinant of the centres is  $\approx 1$ . So  $x$  is  $\nu$ -broad for any  $\nu \lesssim 1$ , and Proposition 12.4.3(B) applies with the loss  $K^2 = 100$ .

*A narrow configuration.* Let  $g$  be supported on caps whose centres lie within 0.01 of the equator  $\{\omega_3 = 0\}$ . By Lemma 12.4.1 every determinant is at most  $3 \times 0.01 = 0.03$ , so no point is  $\nu$ -broad for  $\nu > 0.03$ , and every point falls into case (N).

*What happens next in the narrow case.* The caps occupy a 0.01-neighbourhood of a great circle. Rescaling that neighbourhood by 100 in

the normal direction turns it into a full neighbourhood of  $\mathbb{S}^1$  inside a plane — a two-dimensional restriction problem, whose exponent is the one Volume 6 established for the circle. The induction then runs at the coarser scale.

*The moral.* The narrow case is not a degenerate nuisance; it is a genuine lower-dimensional instance of the same problem, and the method's strength is exactly the strength of the estimates available in that lower dimension.

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**WORKED EXAMPLE 12.4.3— TIER B  $\varepsilon$ -management: how the losses are absorbed**

Every estimate in Volumes 12 and 13 carries  $R^\varepsilon$ , and the bookkeeping that produces it is uniform. It is worth doing once explicitly.

*The setup.* An induction on scales with  $\approx \log R / \log K$  rounds, each costing a factor  $K^C$  for a fixed  $C$  depending only on  $n$ .

*The total cost.*

$$(K^C)^{\log R / \log K} = \exp\left(C \log K \cdot \frac{\log R}{\log K}\right) = \exp(C \log R) = R^C.$$

That is *not*  $R^\varepsilon$  — the  $\log K$  cancels and the losses compound to a fixed power. So the naive accounting fails.

*What actually happens.* The per-round loss is not  $K^C$  but  $K^C$  multiplying a quantity that *improves* by a factor at each round. Writing the induction as  $F(R) \leq K^C F(R/K^2)$  with  $F$  the best constant at scale  $R$ , and iterating  $m = \log R / (2 \log K)$  times,

$$F(R) \leq K^{Cm} F(1) = R^{C/2} F(1),$$

which is still a power. The  $R^\varepsilon$  arises only when the recursion has the form  $F(R) \leq K^C F(R^{1-\delta})$  — a *sublinear* reduction of the exponent — for then the number of rounds is  $\log \log R$  and the total is  $K^{C \log \log R}$ , which is  $R^\varepsilon$  for  $K$  large.

*The lesson.*  $R^\varepsilon$  losses come from inductions with  $\log \log R$  rounds, not  $\log R$  rounds. Volume 9, Worked Example 9.5.3 observed the same for Wolff's induction, and the distinction is worth checking whenever an argument claims an  $\varepsilon$ -loss: count the rounds.

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**WORKED EXAMPLE 12.4.4— TIER C: RESEARCH TASTE **Dichotomies as a proof technique****

Proposition 12.4.3 splits every point into one of two cases and handles them

separately. That is the fourth such split in the series, and the pattern is worth naming.

(a) *Volume 3, Calderón–Zygmund.* Split  $f$  into a bounded part and a sum of mean-zero pieces; handle the first by  $L^2$  and the second by the kernel. The stopping time chooses the split.

(b) *Volume 11, cellular and algebraic.* Split a configuration into points in cells and points on  $Z(P)$ ; the first is handled by the partitioning, the second needs separate work — and in  $\mathbb{R}^3$  that work is the whole theorem.

(c) *Volume 9, bush and hairbrush.* Split according to whether a configuration concentrates at a point or along a line.

(d) *Here, broad and narrow.* Split according to whether the mass spreads over transverse directions or clusters near a hyperplane.

*The common structure.* One case is the “generic” one and is handled by the method’s main estimate; the other is degenerate and is handled by induction, by a lower-dimensional instance, or by a separate argument. *The degenerate case is almost always the hard one* — (b) and (d) both have the property that the theorem’s difficulty lives entirely there.

*What makes a dichotomy useful.* That the two cases are quantitatively complementary, so that failing one gives usable information about the other. Lemma 12.4.1 supplies exactly this here, and its absence is why some natural-looking splits do not work: if the negation of “broad” were merely “not broad”, nothing would follow.

*The question to ask of any dichotomy.* What does the degenerate case reduce to, and is that reduction to something already known? Here it is a lower-dimensional restriction problem, whose exponent is worse — which Worked Example 12.4.1(d) identified as the method’s ceiling.

#### WORKED EXAMPLE 12.4.5— TIER A **The dichotomy’s two costs, quantified**

Proposition 12.4.3 loses a factor  $K^{n-1}$  and produces a transversality parameter  $\nu$  tied to  $K$ . Both are worth seeing numerically.

| $n$ | $K$ | $K^{n-1}$ | $\nu$ available        |
|-----|-----|-----------|------------------------|
| 3   | 10  | 100       | $\approx K^{-1} = 0.1$ |
| 3   | 100 | $10^4$    | $\approx 0.01$         |
| 4   | 10  | $10^3$    | $\approx 0.1$          |
| 4   | 100 | $10^6$    | $\approx 0.01$         |

*The tension.* Larger  $K$  gives finer caps and hence a better dichotomy, but a

worse constant  $K^{n-1}$  and a smaller  $\nu$ , which degrades the multilinear estimate. The two costs pull in the same direction, so  $K$  cannot be taken large freely.

*Why  $K$  is nonetheless taken large.* Because the losses enter only through  $R^{O(1/\log K)}$  once the induction is set up correctly (Worked Example 12.4.3), while the gain from a finer dichotomy is a genuine improvement in the exponent. The trade is favourable, but only because of the  $\log \log R$  round count.

*The check.* At  $K$  fixed and  $R \rightarrow \infty$ , the losses  $K^{n-1}$  per round are constants and the number of rounds grows — so a  $\log R$  induction would give a fixed power of  $R$  and be useless. That Worked Example 12.4.3 identifies this as the distinction between  $\log R$  and  $\log \log R$  inductions is the single most important piece of bookkeeping in Volumes 12 and 13.

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## Drills

**Drill 12.4.1.** Verify Hadamard’s inequality for a  $2 \times 2$  determinant and complete the proof of Lemma 12.4.1 for  $n = 3$ .

**Drill 12.4.2.** Check Step 1 of Proposition 12.4.3: count the caps and confirm  $\mu(x) \geq |\mathcal{E}g(x)|/(C_n K^{n-1})$ .

**Drill 12.4.3.** In Step 3, justify that successive distances to the previous span bound the determinant from below.

**Drill 12.4.4.** For the narrow configuration of Worked Example 12.4.2, compute the largest possible determinant and confirm no point is 0.1-broad.

**Where we are.** The chapter proves the dichotomy and imports the induction, and it is worth being clear which half carries the mathematics. The dichotomy is elementary: a pigeonhole, a greedy selection, and Lemma 12.4.1. It is also the half that explains *why* multilinear estimates bear on the linear problem at all, and a reader who takes only that away has the idea.

The induction is where the exponents come from, and it is long. More to the point, its output has been improved repeatedly since 2011 (Remark 12.5.3), so the number a textbook would quote is the least durable part of the subject. Proving the dichotomy and reporting the exponents seemed the right division, and it is the same one made for R13 in Volume 11 — prove the structure, import the estimate.

## Notes

The broad–narrow decomposition is due to Bourgain and Guth (2011). The version given here isolates the elementary dichotomy from the induction on scales; the induction, and the exponents it produces, are in their paper and have been refined since, most notably by Guth using polynomial partitioning.



# Chapter 12.5

## What This Gives

### 12.5.1 The three exponents

For the sphere in  $\mathbb{R}^n$  the restriction problem has three exponents worth keeping apart.

**Proposition 12.5.1** (The benchmarks). *For the estimate  $\|\mathcal{E}g\|_{L^p(\mathbb{R}^n)} \lesssim \|g\|_{L^\infty(d\sigma)}$ :*

- (i) *it fails for  $p \leq \frac{2n}{n-1}$ , by the Knapp example;*
- (ii) *it holds for  $p \geq \frac{2(n+1)}{n-1}$ , by Stein–Tomas;*
- (iii) *the restriction conjecture asserts it holds for all  $p > \frac{2n}{n-1}$ .*

*Proof.* (i) is Theorem 6.2.3 and (ii) is Theorem 6.3.1, both of Volume 6; (iii) is Conjecture 6.2.6 there.  $\square$

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WORKED EXAMPLE 12.5.1— TIER A **The gap, in numbers**

| $n$ | necessary $p >$             | Stein–Tomas $p \geq$        | gap          |
|-----|-----------------------------|-----------------------------|--------------|
| 2   | 4                           | 6                           | 4 to 6       |
| 3   | 3                           | 4                           | 3 to 4       |
| 4   | $\frac{8}{3} \approx 2.67$  | $\frac{10}{3} \approx 3.33$ | 2.67 to 3.33 |
| 5   | $\frac{5}{2} = 2.5$         | 3                           | 2.5 to 3     |
| 10  | $\frac{20}{9} \approx 2.22$ | $\frac{22}{9} \approx 2.44$ | 2.22 to 2.44 |

*The gap narrows with dimension.*  $\frac{2(n+1)}{n-1} - \frac{2n}{n-1} = \frac{2}{n-1}$ , which tends to 0. So in high dimensions Stein–Tomas is close to sharp in absolute terms — though never sharp.

*The planar case is settled.* For  $n = 2$  the conjecture is a theorem, proved by

Fefferman and Zygmund; Volume 6, Theorem 6.8.2 gives it. So the first row is closed, and every open case has  $n \geq 3$ .

*Where the interest is.*  $n = 3$ : the conjecture asks for  $p > 3$  and Stein–Tomas gives  $p \geq 4$ . Every improvement since 1991 has been a reduction of that 4, and the current record is a little above 3.

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**Theorem 12.5.2** (The broad–narrow exponents; not proved here). *Assuming Theorem 12.3.1 and the induction of Remark 12.4.7, the estimate of Proposition 12.5.1 holds for  $p > p_n$  with  $p_n < \frac{2(n+1)}{n-1}$  for every  $n \geq 3$ . In  $\mathbb{R}^3$  Bourgain and Guth obtained  $p > \frac{33}{10} = 3.3$ , improving Stein–Tomas’s 4; subsequent work using polynomial partitioning has reduced this further, and the record is a moving target.*

The refined exponents are not proved here. *Theorem 12.3.1 and the base recursion of Theorem 12.4.6 are both proved (Chapter 12.3, Chapter 12.4); what is quoted is the sharpened bookkeeping producing numerical values below  $\frac{10}{3}$ , and nothing in this series depends on it beyond the table of Worked Example 12.5.1.*

**Remark 12.5.3** (On quoting exponents). Theorem 12.5.2 is not proved here — it rests on Remark 12.3.11 and Remark 12.4.7 — and it names one number while declining to name the current record. Both are deliberate.

The restriction exponent in  $\mathbb{R}^3$  has been improved at least five times since 2011, by Bourgain–Guth, Guth, Hickman–Rogers, Wang, and others, each by a fraction. A textbook that quotes the record dates itself within a year and misleads a reader who does not check.

What does not change is the *structure*: the conjecture is  $p > 3$ , the classical benchmark is  $p \geq 4$ , and every modern improvement lies between, obtained by the broad–narrow scheme of Chapter 12.4 with progressively better treatments of the narrow case. Volume 14 reports the position as of its writing, with the same caveat.

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**WORKED EXAMPLE 12.5.2— TIER B Comparing restriction exponents across sources**

Restriction estimates are stated in several inequivalent forms and comparison requires care. Four things to check.

(1) *Which operator?*  $\mathcal{R}$  (restriction) or  $\mathcal{E}$  (extension). They are adjoint (Volume 6, Theorem 6.1.3), so an estimate for one is an estimate for the other at the conjugate exponent — but the conjugate must be taken.

(2) *Which data norm?*  $L^\infty(d\sigma)$ ,  $L^2(d\sigma)$ , or  $L^q(d\sigma)$ . The conjecture concerns  $L^\infty$ ; Stein–Tomas is an  $L^2$  statement, which is why it is provable by  $TT^*$  and



why it stops where it does.

(3) *Global or local?*  $L^p(\mathbb{R}^n)$  or  $L^p(B_R)$  with an  $R^\varepsilon$  loss. Volume 6, Theorem 6.5.2 shows these are equivalent for the sphere, so the distinction is usually harmless — but the equivalence is a theorem and uses compactness of the sphere.

(4) *Which surface?* Sphere, paraboloid, or cone. Sphere and paraboloid have the same exponents, both having non-vanishing Gaussian curvature; the cone has one vanishing principal curvature and its exponents differ, as Volume 5’s Worked Example 5.5.4 predicts.

*A worked comparison.* Stein–Tomas is often quoted as “ $\mathcal{R} : L^p(\mathbb{R}^n) \rightarrow L^2(d\sigma)$  for  $p \leq \frac{2(n+1)}{n+3}$ ”. Taking conjugates,  $\frac{2(n+1)}{n+3}$  has dual exponent  $\frac{2(n+1)}{n-1}$ , which is the form used here. ✓ The two statements agree, and a reader who does not take the conjugate will think they disagree.

**WORKED EXAMPLE 12.5.3— TIER C: RESEARCH TASTE** **Why the restriction conjecture is hard**

Six volumes have circled this problem. It is worth collecting what is now known about *why* it resists.

(a) *It implies Kakeya.* Volume 6, Theorem 6.6.2. So it is at least as hard as a problem open since 1928, and no proof can avoid the tube geometry of Volumes 9 and 11.

(b) *The data norm is  $L^\infty$ .* Stein–Tomas is an  $L^2$  theorem and is proved by  $TT^*$  in a page. The conjecture asks for  $L^\infty$  data, where no orthogonality is available, and Worked Example 12.4.1(c) identified this as where most of the broad–narrow loss occurs.

(c) *The extremal examples are not unique.* Knapp caps show the exponent cannot be lowered, but they are not the only near-extremal configurations, and a proof must handle all of them simultaneously. Volume 14 reports that the recent progress in  $\mathbb{R}^3$  consists largely of classifying them.

(d) *Every known method concedes something.* Bilinear and multilinear estimates assume transversality; broad–narrow recovers the linear statement but recurses into lower dimensions with worse exponents; polynomial partitioning handles crossings and struggles with tangency. The concessions differ, and no combination has closed the gap.

(e) *What would suffice.* By Proposition 12.5.1 the target is  $p > \frac{2n}{n-1}$ , and by the reductions of Volume 6 a sharp Kakeya maximal bound would not by itself deliver it — the implication runs the other way. So restriction is strictly harder than Kakeya on current evidence, and Kakeya is open above

dimension three.

*The honest position after twelve volumes.* The problem is settled in the plane, open in every higher dimension, and the best methods improve the classical exponent by a fraction while leaving the structure of the difficulty —  $L^\infty$  data over a curved surface, with tangential configurations uncontrolled — exactly where Stein found it.

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WORKED EXAMPLE 12.5.4— TIER B **Interpolating restriction estimates**

Restriction estimates are routinely combined by interpolation, and the mechanics are worth having.

(a) *What interpolates.* The operator  $\mathcal{E}$  is linear, so Riesz–Thorin (Volume 2, Theorem 2.2.1) applies to the pair of exponents  $(p, q)$  in  $\mathcal{E} : L^q(d\sigma) \rightarrow L^p(\mathbb{R}^n)$ . A bound at  $(p_0, q_0)$  and one at  $(p_1, q_1)$  give bounds on the segment between.

(b) *The two anchors.* The trivial bound  $\|\mathcal{E}g\|_{L^\infty} \leq \|g\|_{L^1(d\sigma)}$ , valid always; and Stein–Tomas at  $(\frac{2(n+1)}{n-1}, 2)$ . Interpolating gives a triangle of admissible pairs, and Proposition 12.5.1 states its edge.

(c) *What does not interpolate.* The multilinear estimates, which are not bounds on a single linear operator. One interpolates them by Hölder in the product, which is weaker, and the broad–narrow argument must be run at each exponent separately.

(d) *A worked instance.* From Stein–Tomas at  $q = 2$  and the trivial bound at  $q = 1$ , interpolation gives  $\mathcal{E} : L^q \rightarrow L^p$  for  $\frac{1}{q} = \frac{1-\vartheta}{1} + \frac{\vartheta}{2}$  and  $\frac{1}{p} = \frac{\vartheta}{2(n+1)/(n-1)}$ . Eliminating  $\vartheta$  traces the edge of the triangle, and the conjecture asserts the region extends past it to  $q = \infty$ .

*The caution.* Interpolation never proves a new endpoint; it fills in between known ones. Every advance in this subject has been a new endpoint, and the interpolation is bookkeeping afterwards.

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WORKED EXAMPLE 12.5.5— TIER A **The four conjectures, and their implications**

Volume 6, Theorem 6.7.4 established a chain of implications among the central conjectures. With twelve volumes written it can be stated in full.

| Conjecture    | Asserts                                                                                 | Status               |
|---------------|-----------------------------------------------------------------------------------------|----------------------|
| Bochner–Riesz | $S^\delta$ bounded for $\delta > \max\{0, n \frac{1}{p} - \frac{1}{2}  - \frac{1}{2}\}$ | $n = 2$ only         |
| Restriction   | $\mathcal{E} : L^\infty(d\sigma) \rightarrow L^p$ , $p > \frac{2n}{n-1}$                | $n = 2$ only         |
| Takeya        | $\dim_{\mathbb{H}} K = n$                                                               | $n \leq 3$           |
| Falconer      | $\dim_{\mathbb{H}} E > \frac{n}{2} \Rightarrow m(\Delta E) > 0$                         | open, all $n \geq 2$ |

*The implications.* Bochner–Riesz  $\Rightarrow$  restriction  $\Rightarrow$  Takeya, by Volume 5, Worked Example 5.8.2 and Volume 6, Theorem 6.6.2. So the list is in decreasing order of strength, and Takeya is the weakest — which is why its resolution in  $\mathbb{R}^3$  did not settle the others.

*Where Falconer sits.* Not on the chain. It is not known to imply or be implied by any of the three, though Proposition 12.8.2 shows it is a weighted restriction question and so shares their machinery. Volume 8, Worked Example 8.8.5 compared it with Takeya and found the methods transfer but the statements do not.

*What is settled where.* Every conjecture on the list is a theorem in the plane except Falconer, which is open even there — the best planar threshold being  $\frac{5}{4}$  against the conjectured 1. That is worth noting: the *easiest* ambient dimension does not make Falconer easy.

*Reading the table forward.* Volume 13 addresses decoupling, which is not on this list but implies results about all four; Volume 14 reports the current position.

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#### WORKED EXAMPLE 12.5.6— TIER B **Reading an exponent claim**

Papers in this area announce numerical exponents, and Remark 12.5.3 declined to quote the current ones. A reader who does consult the literature needs five checks.

(1) *Which conjecture?* Restriction, Bochner–Riesz, Takeya and decoupling all have exponents, and Worked Example 12.5.5 shows they are related but distinct. An improvement to one need not touch the others.

(2) *Which formulation?* Worked Example 12.5.2 lists four things to fix: operator, data norm, global or local, and surface. Two papers can quote different numbers for the same theorem.

(3) *Conditional or not?* Many exponents are conditional on multilinear or bilinear estimates. Theorem 12.5.2 is conditional on Remark 12.3.11 and Remark 12.4.7, and says so.

(4) *What is the loss?*  $R^\varepsilon$  for every  $\varepsilon$ , or a fixed power. The first is harmless by Volume 9, Proposition 9.7.3; the second is not.

(5) *Superseded?* The  $\mathbb{R}^3$  restriction exponent has moved at least five times since 2011. A number from a paper is the record as of that paper.

*A worked reading.* “Restriction holds for  $p > 3.25$  in  $\mathbb{R}^3$ ”: (1) restriction; (2) presumably  $L^\infty$  data, local, sphere — check; (3) unconditional if it is Guth’s, conditional in some later refinements; (4)  $R^\varepsilon$ ; (5) improved since. Placing it: between the conjectured 3 and Stein–Tomas’s 4, closing three-quarters of the gap.

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**WORKED EXAMPLE 12.5.7— TIER B Locating a result on the exponent line**

For the sphere in  $\mathbb{R}^3$  the exponent line runs from 3 (conjectured) to 4 (Stein–Tomas), and placing any result on it takes three steps.

*Step 1: normalise the statement.* Convert to the form  $\mathcal{E} : L^\infty(d\sigma) \rightarrow L^p(B_R)$  with  $R^\varepsilon$  loss, using Worked Example 12.5.2’s four checks.

*Step 2: read off  $p$ .* Then  $\frac{4-p}{4-3} = 4 - p$  is the fraction of the gap closed.

*Step 3: check the hypotheses.* Unconditional, or conditional on a multilinear or bilinear input? Theorem 12.5.2 is conditional on Remark 12.3.11 and Remark 12.4.7.

*Worked placements.*

| $p$  | fraction of gap closed |                               |
|------|------------------------|-------------------------------|
| 4    | 0                      | Stein–Tomas                   |
| 3.3  | 0.7                    | Bourgain–Guth                 |
| 3.25 | 0.75                   | Guth, polynomial partitioning |
| 3    | 1                      | the conjecture                |

*What the table shows.* Two decades of work have closed roughly three quarters of the gap in  $\mathbb{R}^3$ , and the remaining quarter has resisted several distinct methods. Remark 12.5.3 explains why no current figure is quoted; the shape of the line does not change.

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## Drills

**Drill 12.5.1.** Verify  $\frac{2(n+1)}{n-1} - \frac{2n}{n-1} = \frac{2}{n-1}$  and tabulate for  $n = 2, \dots, 6$ .

**Drill 12.5.2.** Confirm that  $\frac{2(n+1)}{n+3}$  and  $\frac{2(n+1)}{n-1}$  are conjugate exponents.

**Drill 12.5.3.** For  $n = 3$ , locate  $\frac{33}{10}$  between the conjectured and classical exponents, and compute what fraction of the gap it closes.

**Drill 12.5.4.** Explain, using Volume 5 Worked Example 5.5.4, why the cone's restriction exponents differ from the sphere's.

## Notes

Bourgain and Guth's paper is from 2011. The subsequent improvements in  $\mathbb{R}^3$  use polynomial partitioning and are due to Guth, and to Hickman, Rogers and others; Remark 12.5.3 explains why no current number is quoted here.



## Chapter 12.6

# Bilinear Restriction

The multilinear theory of Chapter 12.3 uses  $n$  factors because  $n$  vectors are needed to span  $\mathbb{R}^n$  (Proposition 12.2.3). The bilinear theory uses two, and is therefore weaker in high dimensions — but it is older, sharper in the plane, and it is the tool the distance set problem needs, because that problem is intrinsically two-point.

### 12.6.1 The planar estimate, proved

**Theorem 12.6.1** (Bilinear  $L^2$  estimate for transverse arcs). *Let  $S_1, S_2 \subseteq \mathbb{S}^1$  be arcs whose unit normals satisfy  $|\sin \angle(\omega_1, \omega_2)| \geq \nu$  for all  $\omega_j \in S_j$ . Then for  $g_j \in L^2(S_j)$ ,*

$$\|\mathcal{E}_1 g_1 \mathcal{E}_2 g_2\|_{L^2(\mathbb{R}^2)} \lesssim \nu^{-1/2} \|g_1\|_{L^2(d\sigma)} \|g_2\|_{L^2(d\sigma)}.$$

There is no power of  $R$ : the estimate is global.

*Idea of the proof.* The Fourier transform of a product is a convolution, so by Plancherel the left side is the  $L^2$  norm of  $(g_1 d\sigma_1) * (g_2 d\sigma_2)$ . That convolution is the pushforward of  $g_1 \otimes g_2$  under the addition map  $(\omega_1, \omega_2) \mapsto \omega_1 + \omega_2$ , and the Jacobian of that map is exactly  $\sin \angle(\omega_1, \omega_2)$  — which transversality bounds below. A bounded Jacobian means the pushforward has bounded density, and Cauchy–Schwarz finishes.

The entire content is that *transversality is the non-degeneracy of the addition map*.

**Lemma 12.6.2** (The addition map). *Let  $\Phi : \mathbb{S}^1 \times \mathbb{S}^1 \rightarrow \mathbb{R}^2$  be  $\Phi(\omega_1, \omega_2) = \omega_1 + \omega_2$ . In the parametrisation  $\omega_j = (\cos \theta_j, \sin \theta_j)$ ,*

$$|\det d\Phi| = |\sin(\theta_1 - \theta_2)| = |\sin \angle(\omega_1, \omega_2)|.$$

*Proof.* Differentiating  $\Phi(\theta_1, \theta_2) = (\cos \theta_1 + \cos \theta_2, \sin \theta_1 + \sin \theta_2)$ ,

$$d\Phi = \begin{pmatrix} -\sin \theta_1 & -\sin \theta_2 \\ \cos \theta_1 & \cos \theta_2 \end{pmatrix},$$

whose determinant is  $-\sin \theta_1 \cos \theta_2 + \sin \theta_2 \cos \theta_1 = -\sin(\theta_1 - \theta_2)$ .  $\square$

*Proof of Theorem 12.6.1. Step 1. Plancherel.*

$\mathcal{E}_j g_j = \widehat{g_j d\sigma_j}$  up to a reflection, so  $\mathcal{E}_1 g_1 \cdot \mathcal{E}_2 g_2 = (g_1 d\sigma_1) * (g_2 d\sigma_2)$  and by Plancherel

$$\|\mathcal{E}_1 g_1 \mathcal{E}_2 g_2\|_{L^2(\mathbb{R}^2)} = \|(g_1 d\sigma_1) * (g_2 d\sigma_2)\|_{L^2(\mathbb{R}^2)}.$$

**Step 2.** *Identify the convolution as a pushforward.*

For a test function  $\varphi$ ,

$$\int \varphi d((g_1 d\sigma_1) * (g_2 d\sigma_2)) = \iint \varphi(\omega_1 + \omega_2) g_1(\omega_1) g_2(\omega_2) d\theta_1 d\theta_2,$$

so the convolution is  $\Phi_*(g_1 \otimes g_2)$ . By Lemma 12.6.2 and the co-area formula, it is absolutely continuous with density

$$F(\xi) = \sum_{(\omega_1, \omega_2) \in \Phi^{-1}(\xi)} \frac{g_1(\omega_1) g_2(\omega_2)}{|\sin \angle(\omega_1, \omega_2)|}.$$

For  $\xi$  with  $0 < |\xi| < 2$  the fibre  $\Phi^{-1}(\xi)$  consists of at most two points — given the sum of two unit vectors, the pair is determined up to swapping — so the sum has at most two terms.

**Step 3.** *Estimate the density.*

By transversality  $|\sin \angle| \geq \nu$  on  $S_1 \times S_2$ , so

$$|F(\xi)| \leq \frac{2}{\nu} \max_{\Phi^{-1}(\xi)} |g_1(\omega_1) g_2(\omega_2)|.$$

Squaring and integrating, and changing variables back through  $\Phi$  — whose Jacobian contributes  $|\sin \angle| \geq \nu$  once more —

$$\int |F|^2 \lesssim \frac{1}{\nu^2} \iint |g_1(\omega_1)|^2 |g_2(\omega_2)|^2 |\sin \angle| d\theta_1 d\theta_2 \lesssim \frac{1}{\nu} \|g_1\|_{L^2}^2 \|g_2\|_{L^2}^2,$$

using  $|\sin \angle| \leq 1$  and then  $\nu^{-2} \cdot \nu \cdot \nu^0$ ; tracking the powers gives  $\nu^{-1}$ , and taking square roots gives the stated  $\nu^{-1/2}$ .  $\square$



**Where we are.** Two features distinguish Theorem 12.6.1 from everything in Chapter 12.3.

*It is global.* There is no  $R$ , no  $R^\varepsilon$ , and no localisation to a ball. The estimate holds on all of  $\mathbb{R}^2$ .

*It is elementary.* Plancherel, a Jacobian, and Cauchy–Schwarz.

This is the estimate that exposed the defect in Proposition 12.3.3: the wave packet argument, specialised to  $n = 2$ , gives  $\|\mathcal{E}_1 g_1 \mathcal{E}_2 g_2\|_{L^2(B_R)} \lesssim R \|g_1\| \|g_2\|$ , whereas the truth has no power of  $R$  at all. A method that is lossy in the one case one can compute by hand is lossy in general, and the Pitfall there identified where.

The moral is worth stating: *keep an independently computable case.* Every method in this series has one — the plane for Keakeya,  $n = 2$  for restriction, the box for Loomis–Whitney — and checking against it is the cheapest available error detection.

## 12.6.2 Higher dimensions

**Theorem 12.6.3** (Tao). *Work in  $\mathbb{R} \times \mathbb{R}^m$ , so that the ambient dimension is  $n = m + 1$ , and let*

$$S = \left\{ (\tau, \xi) \in \mathbb{R} \times \mathbb{R}^m : \tau = -\frac{1}{2}|\xi|^2 \right\}$$

*be the paraboloid. Let  $S_1, S_2 \subseteq S$  be disjoint compact subsets. Then for every  $q > \frac{m+3}{m+1} = \frac{n+2}{n}$ ,*

$$\|\widehat{f_1 d\sigma_1} \widehat{f_2 d\sigma_2}\|_{L^q(\mathbb{R} \times \mathbb{R}^m)} \lesssim_{q, S_1, S_2} \|f_1\|_{L^2(d\sigma_1)} \|f_2\|_{L^2(d\sigma_2)}.$$

**Where we are.** The exponent agrees with the one stated earlier in this chapter:  $\frac{m+3}{m+1} = \frac{n+2}{n}$ , giving  $2, \frac{5}{3}, \frac{3}{2}, \frac{7}{5}$  for  $n = 2, 3, 4, 5$ . At  $n = 2$  it is  $q = 2$  and the theorem is Theorem 12.6.1, proved in Section 12.6.1 by Plancherel alone.

The proof below is Tao’s adaptation of Wolff’s cone argument. It occupies the rest of this chapter and its shape is: reduce to a localised estimate; induct on scales; decompose into wave packets; split off a part handled by the induction; decompose the remainder at the fine scale  $\sqrt{R}$ ; prove a local  $L^2$  estimate exploiting a second frequency constraint; and finish with a Keakeya-type bush count.

## 12.6.3 Reduction to a localised estimate

**Definition 12.6.4.** Write  $\mathcal{R}^*(2 \times 2 \rightarrow q, \alpha)$  for the statement

$$\|\widehat{f_1 d\sigma_1} \widehat{f_2 d\sigma_2}\|_{L^q(B(z_0, R))} \leq C R^\alpha \|f_1\|_{L^2} \|f_2\|_{L^2}$$

for all  $R \geq 1$  and all spacetime balls  $B(z_0, R)$ .

By a finite partition and a rotation, scaling and Galilean transformation — the last translating  $\xi$  while preserving  $S$  — we may assume

$$S_1 \subseteq \{|\xi - e_1| \leq \frac{1}{100m}\}, \quad S_2 \subseteq \{|\xi + e_1| \leq \frac{1}{100m}\}.$$

By the  $\varepsilon$ -removal lemma of Volume 6, Theorem 6.5.2, it suffices to prove  $\mathcal{R}^*(2 \times 2 \rightarrow \frac{m+3}{m+1}, \alpha)$  for every  $\alpha > 0$ .

**Proposition 12.6.5** (The inductive step). *If  $\mathcal{R}^*(2 \times 2 \rightarrow \frac{m+3}{m+1}, \alpha)$  holds for some  $\alpha > 0$ , then it holds with  $\alpha$  replaced by  $\max\{(1 - \delta)\alpha, C\delta\} + C\varepsilon$ , for all  $0 < \delta, \varepsilon \ll 1$ , with  $C$  independent of  $\delta, \varepsilon$ .*

*Proof of Theorem 12.6.3 from Proposition 12.6.5.* The crude bound  $\|\widehat{f_j d\sigma_j}\|_{L^\infty} \lesssim \|f_j\|_{L^2}$  gives the localised estimate for some large  $\alpha$ . Choosing  $\delta$  and  $\varepsilon$  so that  $\max\{(1 - \delta)\alpha, C\delta\} + C\varepsilon \leq \alpha - c\alpha^2$  and iterating, the infimum of admissible  $\alpha$  is 0.  $\square$

## 12.6.4 Wave packets and the two-ends reduction

Fix  $R \gg 1$ , let  $X = R^{1/2}\mathbb{Z}^m$  and  $V = R^{-1/2}\mathbb{Z}^m$ , and let  $V_j$  be those  $v \in V$  with  $(-\frac{1}{2}|v|^2, v) \in \tilde{S}_j$ . An  $\tilde{S}_j$ -tube is

$$T = \{(t, x) : \frac{R}{2} \leq t \leq R, |x - (x(T) + tv(T))| \leq R^{1/2}\}$$

with  $x(T) \in X$ ,  $v(T) \in V_j$ . Because of the normalisation  $\tau = -\frac{1}{2}|\xi|^2$ , the group velocity equals the spatial frequency.

**Lemma 12.6.6** (Wave packet decomposition). *For smooth  $f_j$  on  $S_j$  there is a decomposition  $\widehat{f_j d\sigma_j} = \sum_{T_j} c_{T_j} \phi_{T_j}$  over  $\tilde{S}_j$ -tubes with*

$$\left(\sum_{T_j} |c_{T_j}|^2\right)^{1/2} \lesssim \|f_j\|_{L^2},$$

where each  $\phi_{T_j}$  solves the free Schrödinger equation  $4\pi i u_t - \Delta u = 0$ , has  $\widehat{\phi_{T_j}(t)}$  supported in  $\{\xi = v(T_j) + O(R^{-1/2})\}$ , and satisfies

$$|\phi_{T_j}(t, x)| \leq C_N R^{-m/4} \left(1 + \frac{|x - (x(T_j) + tv(T_j))|}{R^{1/2}}\right)^{-N}.$$

Moreover  $P(\sum_{T_j \in \mathbb{T}_j} \phi_{T_j}) \lesssim \#\mathbb{T}_j$ , where  $P(u) = \|u(t)\|_{L_x^2}^2$  is the conserved probability.

*Proof.* Assume first  $f_j$  supported in a cap  $\{\xi = v + O(R^{-1/2})\}$ . Take  $\eta$  Schwartz with  $\widehat{\eta}$  supported in a ball and  $\sum_{k \in \mathbb{Z}^m} \eta(x - k) \equiv 1$ , and let  $F(x) = \widehat{f_j d\sigma_j}(0, x)$ . Then  $F = \sum_{x_0 \in X} \eta(\frac{x-x_0}{R^{1/2}})F$ , and each summand has spatial Fourier support in the same cap. Let  $u_{x_0}$  be the Schrödinger evolution of that summand, put  $c_T = R^{m/4} \mathcal{M}F(x(T))$  with  $\mathcal{M}$  the Hardy–Littlewood maximal function, and  $\phi_T = u_{x(T)}/c_T$ .

Since  $\widehat{F}$  is supported in a ball of radius  $R^{-1/2}$ ,  $F = F * \psi$  with  $|\psi|(x) \lesssim_N R^{-m/2}(1 + |x|/R^{1/2})^{-N}$ , so  $\mathcal{M}F(x) \approx \mathcal{M}F(x')$  for  $|x - x'| \lesssim R^{1/2}$ . Hence  $\sum_T |c_T|^2 \lesssim \|\mathcal{M}F\|_{L^2}^2 \lesssim \|F\|_{L^2}^2 \lesssim \|f_j\|_{L^2}^2$  by the maximal inequality (Volume 1, Theorem 1.7.3) and Plancherel.

The pointwise bound follows from the fundamental solution together with a stationary phase computation (Volume 5, Theorem 5.3.3) applied to the kernel  $K_v$ ; the probability bound follows since tubes with a fixed velocity have  $R^{1/2}$ -separated initial positions, and then by summing over  $v$  using frequency orthogonality.

For general  $f_j$ , decompose into caps and sum, using  $\sum_v \|f_{j,v}\|_{L^2}^2 \approx \|f_j\|_{L^2}^2$ .  $\square$

After discarding tubes far from  $B(0, CR)$  and dyadic pigeonholing in the coefficient sizes — both costing only  $R^{O(\varepsilon)}$  — it suffices to prove:

**Proposition 12.6.7.** *For collections  $\mathbb{T}_1, \mathbb{T}_2$  of tubes meeting  $B(0, CR)$ ,*

$$\left\| \sum_{T_1 \in \mathbb{T}_1} \sum_{T_2 \in \mathbb{T}_2} \phi_{T_1} \phi_{T_2} \right\|_{L^{\frac{m+3}{m+1}}(Q_R)} \lesssim (R^{(1-\delta)\alpha} + R^{C\delta}) (\#\mathbb{T}_1)^{1/2} (\#\mathbb{T}_2)^{1/2},$$

where  $Q_R = \{\frac{R}{2} \leq t \leq R, |x| \leq R\}$  and  $\lesssim$  allows a factor  $C_\varepsilon R^\varepsilon$ .

*Reduction to a global estimate.* Cover  $Q_R$  by  $O(R^{C\delta})$  balls  $B$  of radius  $R^{1-\delta}$ . Given any relation  $\sim$  between tubes and balls satisfying

$$\#\{B : T \sim B\} \lesssim 1 \quad \text{for every } T, \quad (12.6.1)$$

split the sum into the *local* part, over pairs with  $T_1 \sim B$  and  $T_2 \sim B$ , and the *global* part.

For the local part, Lemma 12.6.6 gives  $P(\sum_{T_j \sim B} \phi_{T_j}) \lesssim \#\{T_j \sim B\}$ , so the induction hypothesis applies on each  $B$  and, after Cauchy–Schwarz and (12.6.1), contributes  $\lesssim R^{(1-\delta)\alpha} (\#\mathbb{T}_1)^{1/2} (\#\mathbb{T}_2)^{1/2}$ .

So it remains to bound the global part on a single  $B$ , and by the triangle inequality it suffices to treat the terms with  $T_1 \not\sim B$ . Interpolating with the  $L^1$  bound

$$\left\| \sum_{T_1 \not\sim B} \sum_{T_2} \phi_{T_1} \phi_{T_2} \right\|_{L^1(B)} \lesssim R (\#\mathbb{T}_1)^{1/2} (\#\mathbb{T}_2)^{1/2},$$

which follows from Hölder and the probability bound, the task reduces to the  $L^2$  estimate

$$\left\| \sum_{T_1 \not\sim B} \sum_{T_2} \phi_{T_1} \phi_{T_2} \right\|_{L^2(B)} \lesssim R^{C\delta} R^{-\frac{m-1}{4}} (\#\mathbb{T}_1)^{1/2} (\#\mathbb{T}_2)^{1/2}. \quad (12.6.2)$$

*This is where the exponent  $\frac{m+3}{m+1}$  is used, and the power  $R^{-(m-1)/4}$  is sharp — see Worked Example 12.6.1 below.*  $\square$

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**WORKED EXAMPLE 12.6.1— TIER A The  $L^2$  exponent is sharp**

The power  $R^{-(m-1)/4}$  in (12.6.2) cannot be improved, and the example showing it is the same one that forces the exponent  $\frac{m+3}{m+1}$ .

*The configuration.* Let  $\pi = \{(t, x_1 e_1) : t, x_1 = O(R)\}$  be a spacetime plane. Let  $\mathbb{T}_1$  consist of the  $O(\sqrt{R})$  tubes with velocity  $e_1$  meeting  $\pi$ , and  $\mathbb{T}_2$  the  $O(\sqrt{R})$  tubes with velocity  $-e_1$  meeting  $\pi$ .

*The two sides.* By Lemma 12.6.6 each  $\phi_{T_j}$  has size  $\approx R^{-m/4}$  on its tube, so the double sum has magnitude  $\approx R^{-m/2}$  on the  $\sqrt{R}$ -neighbourhood of  $\pi$ , whose volume is  $R^{(m+3)/2}$ . Hence

$$\text{LHS} \approx R^{-\frac{m}{2}} \cdot R^{\frac{m+3}{4}} = R^{\frac{3-m}{4}}, \quad \text{RHS} \approx R^{-\frac{m-1}{4}} \cdot R^{\frac{1}{4}} \cdot R^{\frac{1}{4}} = R^{\frac{3-m}{4}}.$$

Equal.  $\checkmark$

*Numerically.* At  $m = 2$  (so  $n = 3$ ) both sides are  $R^{1/4}$ ; at  $m = 3$  both are  $R^0$ ; at  $m = 4$  both are  $R^{-1/4}$ .

*What the example is.* Two families of tubes, oppositely directed, all piled onto a single plane — the configuration in which a bilinear expression concentrates as much as transversality permits. It is the bilinear analogue of the Knapp example, and it is why no argument can do better than  $\frac{m+3}{m+1}$ .

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## 12.6.5 The improved local estimate

Cover  $Q_R$  by balls  $q$  of radius  $\sqrt{R}$  and pigeonhole dyadically: let  $\mathbf{q}(\mu_1, \mu_2)$  be the balls with  $\#\mathbb{T}_j(q) \approx \mu_j$ , where  $\mathbb{T}_j(q) = \{T_j : T_j \cap R^\delta q \neq \emptyset\}$ , and let  $\mathbb{T}_1[\lambda_1, \mu_1, \mu_2]$  be the tubes meeting  $\approx \lambda_1$  such balls.

**Definition 12.6.8.** For  $\xi_1 \in \Omega_1$ ,  $\xi'_2 \in \Omega_2$  put

$$\pi(\xi_1, \xi'_2) = \left\{ \xi'_1 \in \Omega_1 : \xi_1 + \xi_2 = \xi'_1 + \xi'_2, |\xi_1|^2 + |\xi_2|^2 = |\xi'_1|^2 + |\xi'_2|^2 \text{ for some } \xi_2 \in \Omega_2 \right\}.$$

**Lemma 12.6.9** (The second constraint is a hyperplane).  *$\pi(\xi_1, \xi'_2)$  is contained in the hyperplane through  $\xi_1$  orthogonal to  $\xi'_2 - \xi_1$ ; that is,*

$$\langle \xi'_1 - \xi_1, \xi'_2 - \xi_1 \rangle = 0 \quad \text{for } \xi'_1 \in \pi(\xi_1, \xi'_2).$$

*Proof.* From  $\xi_1 + \xi_2 = \xi'_1 + \xi'_2$  we get  $\xi_2 - \xi'_2 = \xi'_1 - \xi_1$ . Then

$$0 = |\xi_1|^2 + |\xi_2|^2 - |\xi'_1|^2 - |\xi'_2|^2 = -\langle \xi'_1 - \xi_1, \xi'_1 + \xi_1 \rangle + \langle \xi_2 - \xi'_2, \xi_2 + \xi'_2 \rangle,$$

and substituting  $\xi_2 - \xi'_2 = \xi'_1 - \xi_1$  and  $\xi_2 + \xi'_2 = \xi_1 + \xi'_2 + \xi'_2 - \xi'_1 + \xi'_1 - \xi_1$  — equivalently eliminating  $\xi_2$  — the two inner products combine to  $\langle \xi'_1 - \xi_1, \xi_2 + \xi'_2 - \xi_1 - \xi'_1 \rangle = 0$ . Using  $\xi_2 = \xi'_1 + \xi'_2 - \xi_1$  this reads  $\langle \xi'_1 - \xi_1, 2(\xi'_2 - \xi_1) \rangle = 0$ .

Equivalently:  $\xi_1, \xi'_1, \xi_2, \xi'_2$  form a rectangle in  $\mathbb{R}^m$ .  $\square$

**Where we are.** Lemma 12.6.9 is Tao's innovation over Wolff. For the *cone* the corresponding geometric fact is that a light ray meets the light cone through a point in at most one place, which follows from the single vanishing principal curvature. The paraboloid has no vanishing curvature and that argument is unavailable.

What replaces it is the observation that the *two* frequency constraints — conservation of  $\xi_1 + \xi_2$  and of  $|\xi_1|^2 + |\xi_2|^2$  — together confine  $\xi'_1$  to a hyperplane, not merely to a hypersurface. That extra degree of freedom removed is exactly what makes the collection  $\mathbb{T}_2$  behave like a family of null directions, so that Wolff's counting argument applies after all.

**Lemma 12.6.10** (Improved local estimate). *With  $\nu(q, \lambda_1, \mu_1, \mu_2) = \sup_{\xi_1, \xi'_2} \#\mathbb{T}_1^{\mathcal{A}B}(q, \lambda_1, \mu_1)$  where the latter is the set of  $T_1 \in \mathbb{T}_1^{\mathcal{A}B}(q) \cap \mathbb{T}_1[\lambda_1, \mu_1, \mu_2]$  with  $v(T_1)$  within  $O(R^{C\delta}R^{-1/2})$  of  $\pi(\xi_1, \xi'_2)$ ,*

$$\left\| \sum_{T_1 \in \mathbb{T}'_1} \sum_{T_2 \in \mathbb{T}'_2} \phi_{T_1} \phi_{T_2} \right\|_{L^2(q)}^2 \lesssim R^{C\delta} R^{-\frac{m-1}{2}} \nu(\#\mathbb{T}'_1)(\#\mathbb{T}'_2).$$

*Proof.* The spacetime Fourier transform of  $\phi_{T_1} \phi_{T_2}$  is supported in a ball of radius  $CR^{-1/2}$  about  $(-\frac{1}{2}|v(T_1)|^2, v(T_1)) + (-\frac{1}{2}|v(T_2)|^2, v(T_2))$  and has magnitude  $O(R^{1/2})$ , so  $\|\phi_{T_1} \phi_{T_2}\|_{L^2} \lesssim R^{-(m-1)/4}$ .

Expanding the square, the inner product  $\langle \phi_{T_1} \phi_{T_2}, \phi_{T'_1} \phi_{T'_2} \rangle$  vanishes unless

$$v(T_1) + v(T_2) = v(T'_1) + v(T'_2) + O(R^{-1/2}), \quad |v(T_1)|^2 + |v(T_2)|^2 = |v(T'_1)|^2 + |v(T'_2)|^2 + O(R^{-1/2})$$

By Lemma 12.6.9, for fixed  $T_1, T'_2$  the velocity  $v(T'_1)$  lies within  $O(R^{-1/2})$  of the hyperplane  $\pi(v(T_1), v(T'_2))$ , so there are  $O(R^{C\delta}\nu)$  choices of  $T'_1$ ; and then  $T_2$  is determined up to  $O(R^{C\delta})$  choices by the first constraint. Multiplying the number of quadruples by the bound  $O(R^{-(m-1)/2})$  on each inner product gives the claim.  $\square$

## 12.6.6 The combinatorial estimate

Define  $\sim$  as follows: for  $T_1 \in \mathbb{T}_1[\lambda_1, \mu_1, \mu_2]$  let  $B(T_1)$  maximise  $\#\{q \in \mathbf{q}(\mu_1, \mu_2) : T_1 \cap R^\delta q \neq \emptyset, q \cap B(T_1) \neq \emptyset\}$ , and declare  $T_1 \sim B'$  when  $B' \subseteq 10B(T_1)$ . Then

(12.6.1) holds, and by pigeonhole

$$\#\{q \in \mathbf{q}(\mu_1, \mu_2) : T_1 \cap R^\delta q \neq \emptyset, q \cap B(T_1) \neq \emptyset\} \gtrsim R^{-C\delta} \lambda_1. \quad (12.6.3)$$

**Lemma 12.6.11** (The geometric observation). *For each  $T_2 \in \mathbb{T}_2$ ,*

$$\#\{(q, T_1) \in \mathbf{q} \times \mathbb{T}'_1 : T_1 \cap R^\delta q \neq \emptyset, T_2 \cap R^\delta q \neq \emptyset, \text{dist}(q_0, q) \gtrsim R^{-C\delta} R\} \lesssim R^{C\delta}.$$

*Proof.* Let  $(t_0, x_0)$  and  $(t, x)$  be the centres of  $q_0$  and  $q$ . Since  $T_1$  meets both  $R^\delta q_0$  and  $R^\delta q$ ,

$$x - x_0 = v(T_1)(t - t_0) + O(R^{C\delta} R^{1/2}), \quad R^{-C\delta} R \lesssim |t - t_0| \lesssim R.$$

Since  $T_1 \in \mathbb{T}'_1$ ,  $v(T_1)$  lies within  $O(R^{C\delta} R^{-1/2})$  of  $\pi(\xi_1, \xi'_2)$ , so with  $e = \xi'_2 - \xi_1$ , Lemma 12.6.9 gives

$$\left\langle \frac{x - x_0}{t - t_0} - \xi_1, e \right\rangle \lesssim R^{C\delta} R^{-1/2},$$

that is  $\langle (t - t_0, x - x_0), (-\langle \xi_1, e \rangle, e) \rangle \lesssim R^{C\delta} R^{1/2}$ . So  $(t, x)$  lies within  $O(R^{C\delta} R^{1/2})$  of a fixed hyperplane  $\Pi$  through  $(t_0, x_0)$ .

Now  $e$  is within  $\frac{1}{5m}$  of  $-2e_1$  and  $-\langle \xi_1, e \rangle$  within  $\frac{1}{5m}$  of  $+2$ , while  $v(T_2)$  is within  $\frac{1}{5m}$  of  $-e_1$ ; hence  $T_2$  meets  $\Pi$  at angle  $\approx 1$ . Since also  $\text{dist}((t, x), T_2) \lesssim R^{C\delta} R^{1/2}$ , the point  $(t, x)$  is confined to a ball of radius  $R^{C\delta} R^{1/2}$ , leaving  $O(R^{C\delta})$  choices of  $q$  and then  $O(R^{C\delta})$  of  $T_1$ .  $\square$

*Conclusion of the proof of Theorem 12.6.3.* By Lemma 12.6.10 it suffices to prove

$$\sum_{q \subseteq 2B} \nu(q, \lambda_1, \mu_1, \mu_2) \#(\mathbb{T}'_1 \not\sim^B(q) \cap \mathbb{T}_1[\lambda_1, \mu_1, \mu_2]) \# \mathbb{T}_2(q) \lesssim R^{C\delta} (\#\mathbb{T}_1)(\#\mathbb{T}_2).$$

Now  $\#\mathbb{T}_2(q) \lesssim \mu_2$ , and by Fubini and the definition of  $\mathbb{T}_1[\lambda_1, \mu_1, \mu_2]$ ,  $\sum_q \#(\mathbb{T}'_1 \not\sim^B(q) \cap \mathbb{T}_1[\lambda_1]) \lesssim (\#\mathbb{T}_1) \lambda_1$ . So it suffices to show  $\nu \lesssim R^{C\delta} \frac{\#\mathbb{T}_2}{\lambda_1 \mu_2}$ .

Fix  $q_0$  and  $\xi_1, \xi'_2$ , and let  $\mathbb{T}'_1$  be as in Lemma 12.6.10. Each  $T_1 \in \mathbb{T}'_1$  has  $T_1 \not\sim B$ , so  $\text{dist}(q_0, 2B(T_1)) \gtrsim R^{-C\delta} R$ , and by (12.6.3) it meets  $\gtrsim R^{-C\delta} \lambda_1$  balls  $q$  at distance  $\gtrsim R^{-C\delta} R$  from  $q_0$ . Each such  $q$  meets  $\geq \mu_2$  tubes of  $\mathbb{T}_2$ . Summing over  $T_1 \in \mathbb{T}'_1$ ,

$$\#\{(q, T_1, T_2)\} \gtrsim R^{-C\delta} \lambda_1 \mu_2 \#\mathbb{T}'_1.$$

By Lemma 12.6.11 the same count is  $\lesssim R^{C\delta} \#\mathbb{T}_2$ . Comparing gives  $\#\mathbb{T}'_1 \lesssim R^{C\delta} \frac{\#\mathbb{T}_2}{\lambda_1 \mu_2}$ , as required.  $\square$

*Remark 12.6.12* (R17 discharged). *Theorem 12.6.3 is now proved in full*, following Tao’s paper. Revision R17 is discharged.

The one genuinely new ingredient over Wolff’s cone argument is Lemma 12.6.9: for the cone one uses that a light ray meets a light cone through a point once, a consequence of the vanishing principal curvature; for the paraboloid no such fact holds, and its place is taken by the *second* frequency constraint, which confines  $\xi'_1$  to a hyperplane. Everything else — induction on scales, wave packets, the two-ends localisation, the fine-scale decomposition, the bush count — is Wolff’s.

The proof gives the non-endpoint range  $q > \frac{m+3}{m+1}$ ; the endpoint remains open for the paraboloid, and Tao notes that the pigeonholing which produces the parameter  $\nu$  appears to obstruct its removal.

**Corollary 12.6.13** (The bilinear estimate on a ball). *Under the hypotheses of Theorem 12.6.1, for every  $R \geq 1$ ,*

$$\|\mathcal{E}_1 g_1 \mathcal{E}_2 g_2\|_{L^1(B_R)} \lesssim \nu^{-1/2} R \|g_1\|_{L^2(d\sigma)} \|g_2\|_{L^2(d\sigma)},$$

*and this is sharp.*

*Proof.* By Cauchy–Schwarz on  $B_R$ ,  $\|F\|_{L^1(B_R)} \leq |B_R|^{1/2} \|F\|_{L^2(B_R)}$  with  $|B_R| \approx R^2$  in the plane, and Theorem 12.6.1 bounds the  $L^2$  norm.

For sharpness, take  $g_j = \mathbf{1}_{S_j}$ , so  $\|g_j\|_{L^2(d\sigma)} \approx 1$ . In  $\mathbb{R}^2$  the decay of  $\widehat{d\sigma}$  from Volume 5 gives  $|\mathcal{E}_j g_j(x)| \approx |x|^{-1/2}$  for large  $|x|$ , so  $|\mathcal{E}_1 g_1 \mathcal{E}_2 g_2| \approx |x|^{-1}$  and

$$\int_{B_R} |x|^{-1} dx = 2\pi \int_0^R r^{-1} r dr = 2\pi R,$$

matching the right-hand side up to the constant  $2\pi$ , uniformly in  $R$ .

*The single-cap example is not extremal here.* Taking  $g_j$  to be indicators of  $R^{-1/2}$ -caps gives, by Proposition 12.1.5, a left side of  $\approx 1$  against a right side of  $\approx R^{1/2}$  — short by  $R^{1/2}$ . So the extremiser for the ball estimate is the spread configuration, not the Knapp one, which is the reverse of the situation for the linear restriction problem.  $\square$

|             | <b>bilinear</b>                | <b>multilinear</b>            |
|-------------|--------------------------------|-------------------------------|
| factors     | 2                              | $n$                           |
| hypothesis  | transverse <i>and</i> curved   | transverse only               |
| exponent    | $q > \frac{n+2}{n}$ (Tao)      | $p \geq \frac{2n}{n-1}$ (BCT) |
| loss        | none                           | $R^\varepsilon$               |
| proof input | curvature, induction on scales | multilinear Keakeya           |
| sharp in    | all $n$                        | all $n$                       |
| best for    | two-point problems             | restriction exponents         |

*Why bilinear needs curvature and multilinear does not.* With only two directions, transversality controls one direction; the remaining  $n - 2$  must be controlled by the surfaces bending, which is why Theorem 12.6.3 has a curvature hypothesis and Theorem 12.3.1 does not. Proposition 12.2.3 is the formal statement.

*Why bilinear is nonetheless the right tool sometimes.* A problem whose natural object is a *pair* — two points, one distance — matches a two-factor estimate. Volume 8, Worked Example 7.7.3 classified the distance set problem as two-point, and Chapter 12.8 will exploit exactly that.

*The historical order.* Bilinear came first, in the 1990s, and gave the best restriction exponents until 2011; multilinear then overtook it for restriction, while bilinear remained the tool of choice for distance sets. Both are current.

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**WORKED EXAMPLE 12.6.3 — TIER B Reading the transversality hypothesis in the bilinear setting**

The bilinear literature states transversality in several ways and the differences are substantive.

(a) *Separated supports.*  $\text{dist}(S_1, S_2) \geq \nu$  on the sphere. Weakest; does not by itself give Lemma 12.6.2’s Jacobian bound, since two nearby-in-angle points can still have parallel normals on a general surface.

(b) *Transverse normals.*  $|\sin \angle(\omega_1, \omega_2)| \geq \nu$ , as in Theorem 12.6.1. This is what the Jacobian computation needs. On the sphere it coincides with (a) since the normal at  $\omega$  is  $\omega$ .

(c) *Transverse and curved.* Required for  $n \geq 3$ : the surfaces must additionally have non-vanishing second fundamental form. Without it the theorem is false — two transverse hyperplanes give no decay at all, consistent with Volume 5, Worked Example 5.5.4.

*The check to run.* Compute the Jacobian of the addition map for the surfaces in question. If it is bounded below, one is in case (b); if it degenerates but



the surfaces curve, one is in case (c) and needs the full theorem; if neither, no bilinear estimate holds.

*An instance where it fails.*  $S_1$  and  $S_2$  two transverse pieces of the cone in  $\mathbb{R}^3$ . The cone has a vanishing principal curvature, and the bilinear exponent differs from the sphere's — the cone's bilinear theory is Wolff's, with its own exponent, and is what Volume 13's local smoothing application will need.

WORKED EXAMPLE 12.6.4— TIER C: RESEARCH TASTE **Keeping a computable case**

Theorem 12.6.1 caught a defect in Proposition 12.3.3, and the episode generalises into the most useful working practice in this subject.

(a) *The principle.* For every method, keep one instance where the answer is known independently — by an elementary argument, a closed-form computation, or a numerical check — and test every new estimate against it.

(b) *The instances used in this series.* The plane for Kakeya, where Córdoba's  $L^2$  argument gives the sharp answer (Corollary 9.3.3). The case  $n = 2$  for restriction, settled by Fefferman and Zygmund. The box for Loomis–Whitney, where Theorem 11.5.2 is an equality. The grid for Szemerédi–Trotter and for joints (Worked Example 11.3.1, Worked Example 11.4.1). Small  $q$  for Dvir (Worked Example 10.1.4). In each case the test is cheap.

(c) *What a failed test tells you.* Not merely that something is wrong, but usually *where*. Proposition 12.3.3 failed at  $n = 2$  by exactly a factor  $R$ , and the count of packets meeting a point is exactly  $R^{(n-1)/2}$ , which at  $n = 2$  is  $R^{1/2}$  — squaring to  $R$  in the final estimate. The size of the discrepancy identified the step.

(d) *What a passed test does not tell you.* That the estimate is correct. A formula can match at one endpoint and fail elsewhere; the exponent formula withdrawn in Chapter 12.8 passed the check at  $s = \frac{n}{2}$  and failed the one that mattered. Two independent checks are worth more than one.

(e) *Why this is worth stating in a textbook.* Because the checks are not part of the finished mathematics and so are invisible in the literature. A reader who sees only polished proofs may conclude that the authors did not need them. They did.

WORKED EXAMPLE 12.6.5— TIER A **The bilinear route to Falconer, in outline**

Chapter 12.8 will use the bilinear theory rather than the multilinear, and it is worth seeing the shape of that use before meeting it.

*Step 1.* By Proposition 12.8.2,  $\sigma(\mu, r)$  is an  $L^2(d\sigma)$  norm of  $\mathcal{R}_r\mu$  — a restriction quantity with  $\mu$  as data.

*Step 2.* Decompose  $\mathbb{S}^{n-1}$  into caps and split the resulting double sum into transverse pairs and near-parallel pairs. The transverse part is what a bilinear estimate controls.

*Step 3.* Apply Theorem 12.6.3 to each transverse pair, integrated against  $\mu$  — the fractal weight of Chapter 12.7.

*Step 4.* Handle the near-parallel pairs by rescaling and induction, as in Chapter 12.4's narrow case.

*Why two factors and not  $n$ .* Step 1 produces a *double* integral over  $\mu \times \mu$ , so the natural decomposition is into pairs. An  $n$ -fold decomposition would have nothing to attach to. Worked Example 12.8.2 makes this precise.

*What each step costs.* Step 2 is free; Step 3 costs the bilinear constant, which depends on the transversality parameter; Step 4 costs the induction, and this is where the exponent is determined. Remark 12.8.6 records that Erdős's optimisation of these costs is not reproduced here.

*The pattern to recognise.* It is the broad–narrow shape again — transverse part by the sharp estimate, degenerate part by induction — applied to a two-point problem instead of a linear one. Volume 13 will apply the same shape once more, with local smoothing in place of the bilinear estimate.

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## Drills

**Drill 12.6.1.** Verify Lemma 12.6.2 and check the Jacobian at angles  $\frac{\pi}{2}$ ,  $\frac{\pi}{3}$  and 0.05.

**Drill 12.6.2.** Confirm that  $\Phi^{-1}(\xi)$  has at most two points for  $0 < |\xi| < 2$ , and describe what happens at  $|\xi| = 0$  and  $|\xi| = 2$ .

**Drill 12.6.3.** Track the powers of  $\nu$  in Step 3 and confirm the final  $\nu^{-1/2}$ .

**Drill 12.6.4.** Check that  $\frac{n+2}{n} = 2$  at  $n = 2$ , and compare Theorem 12.6.3 with Theorem 12.6.1 there.

**Where we are.** The two estimates of this chapter differ in what they need from the surface.

Theorem 12.6.1 needs only that the addition map is non-degenerate, which in the plane is transversality alone — the arcs may be any curves with transverse tangents.

Theorem 12.6.3 needs transversality *and* curvature, because for  $n \geq 3$  the fibres of the addition map are positive-dimensional and their geometry enters. Worked Example 12.6.3 lists the three hypotheses in use and warns that they are not equivalent.

The lesson generalises: in the plane, transversality of two curves controls everything, because two directions span. Above the plane it does not, and curvature must supply the rest — which is Proposition 12.2.3 again.

## Notes

The planar bilinear estimate is classical, going back to Fefferman and Córdoba in the early 1970s. Tao's sharp bilinear restriction theorem is from 2003, building on work of Wolff for the cone and of Moyua, Vargas and Vega. Remark 12.6.12 records that only the planar case is proved here.



## Chapter 12.7

# Restriction Against a Fractal Measure

Every restriction estimate so far has measured  $\mathcal{E}g$  in  $L^p(\mathbb{R}^n)$  or on a ball. This chapter replaces Lebesgue measure by a fractal measure  $\mu$ . The resulting estimates are what the Falconer distance set problem requires, and Chapter 12.8 will show precisely why.

### 12.7.1 The setting

**Definition 12.7.1.** Let  $0 < s \leq n$ . A finite Borel measure  $\mu$  on  $\mathbb{R}^n$  is *s-dimensional with constant  $C_\mu$*  if

$$\mu(B(x, r)) \leq C_\mu r^s \quad \text{for all } x \in \mathbb{R}^n, r > 0.$$

By Frostman's lemma — Volume 7, Theorem 7.5.1 — a compact set  $E$  carries such a measure for every  $s < \dim_{\text{H}} E$ , and conversely the existence of one forces  $\dim_{\text{H}} E \geq s$ . So  $s$ -dimensional measures are the analytic substitute for the hypothesis  $\dim_{\text{H}} E > s$ .

**Proposition 12.7.2** (The two extremes). *Let  $\mu$  be  $s$ -dimensional and supported in the unit ball.*

- (i) *If  $s = n$  then  $\mu$  is absolutely continuous with bounded density, and a restriction estimate against  $\mu$  follows from the corresponding  $L^p(\mathbb{R}^n)$  estimate.*
- (ii) *If  $s = 0$  the condition is vacuous —  $\mu$  may be a point mass — and no restriction estimate against  $\mu$  can hold beyond the trivial one.*

*Proof.* (i) The condition  $\mu(B(x, r)) \leq C_\mu r^n$  says exactly that  $\mu$  is dominated by  $C_\mu$  times Lebesgue measure on every ball, hence  $d\mu \leq C_\mu dx$  by differentiation (Volume 1, Theorem 1.7.4).

(ii) A point mass  $\delta_{x_0}$  satisfies  $\mu(B(x, r)) \leq 1 = r^0$ . Then  $\int |\mathcal{E}g|^2 d\mu = |\mathcal{E}g(x_0)|^2$ , which is controlled only by  $\|g\|_{L^1(d\sigma)}$  and by nothing sharper.  $\square$

**Where we are.** So the interesting range is  $0 < s < n$ , and the question is how much of the  $L^p(\mathbb{R}^n)$  theory survives when Lebesgue measure is replaced by something of dimension  $s$ .

The heuristic is that a fractal measure of dimension  $s$  “sees” a fraction  $R^{s-n}$  of a ball of radius  $R$ , so an estimate valid against Lebesgue measure should degrade by about that factor. Making this precise is the content of the weighted restriction theory, and the sharp form is not known — which is exactly why the Falconer problem is open.

## 12.7.2 The basic estimate

**Theorem 12.7.3** (Weighted restriction,  $L^2$  form). *Let  $\mu$  be a finite Borel measure on  $\mathbb{R}^n$ , let  $R \geq 1$ , and put*

$$M_R(\mu) = \sup_{x \in \mathbb{R}^n} \mu(B(x, R^{1/2})).$$

*Then for  $g \in L^2(\mathbb{S}^{n-1})$ ,*

$$\int_{\mathbb{R}^n} |\mathcal{E}g|^2 d\mu \lesssim_{n,N} R^{\frac{2-n}{2}} M_R(\mu) \|g\|_{L^2(d\sigma)}^2 + R^{-N} \|g\|_{L^1(d\sigma)}^2.$$

*Proof. Step 1. Wave packets.*

Decompose  $\mathcal{E}g = \sum_{(\theta, T)} \mathcal{E}g_{\theta, T}$  by Theorem 12.1.4 at scale  $R$ . By part (ii) each packet is bounded by  $c_{\theta, T} \mathbf{1}_T$  up to tails decaying faster than any power, with  $c_{\theta, T} = \|g_{\theta, T}\|_{L^1(d\sigma)}$ ; the tails contribute the final term.

**Step 2.** *The measure of one tube.*

A tube  $T$  has dimensions  $R^{1/2} \times \cdots \times R^{1/2} \times R$ , so it is covered by  $\approx R^{1/2}$  balls of radius  $R^{1/2}$ . Hence

$$\mu(T) \lesssim R^{1/2} M_R(\mu).$$

**Step 3.** *Sum, keeping the orthogonality.*

Since the left side is quadratic, Theorem 12.1.4(iii) applies directly and

$$\int |\mathcal{E}g|^2 d\mu \lesssim \sum_{(\theta, T)} |c_{\theta, T}|^2 \mu(T) \lesssim R^{1/2} M_R(\mu) \sum_{(\theta, T)} |c_{\theta, T}|^2.$$

Now  $\sum |c_{\theta,T}|^2 |T| \approx \|\mathcal{E}g\|_{L^2(B_R)}^2 \approx R \|g\|_{L^2(d\sigma)}^2$  by Proposition 6.1.4 of Volume 6, and  $|T| \approx R^{(n+1)/2}$ , so

$$\sum_{(\theta,T)} |c_{\theta,T}|^2 \approx \frac{R \|g\|_{L^2}^2}{R^{(n+1)/2}} = R^{\frac{1-n}{2}} \|g\|_{L^2}^2.$$

**Step 4. Collect.**

Multiplying, the power of  $R$  is  $\frac{1}{2} + \frac{1-n}{2} = \frac{2-n}{2}$ , which is the claim.  $\square$

**Corollary 12.7.4** (The  $s$ -dimensional form). *If in addition  $\mu$  is  $s$ -dimensional with constant  $C_\mu$ , then  $M_R(\mu) \leq C_\mu R^{s/2}$  and hence*

$$\int |\mathcal{E}g|^2 d\mu \lesssim C_\mu R^{\frac{2-n+s}{2}} \|g\|_{L^2(d\sigma)}^2 + R^{-N} \|g\|_{L^1(d\sigma)}^2.$$

*Proof.* Immediate from Definition 12.7.1 with  $r = R^{1/2}$  and Theorem 12.7.3.  $\square$

**Where we are.** Theorem 12.7.3 is crude — it uses only the size of  $\mu$  on a single tube — and it is exactly the estimate that produces Falconer’s classical threshold. Chapter 12.8 will show the arithmetic.

Note what Step 3 does differently from Proposition 12.3.3: it keeps the  $L^2$  orthogonality instead of dominating pointwise. Here that is legitimate because the left side is itself an  $L^2$  quantity, so the orthogonality applies directly. In the multilinear setting the left side was an  $L^p$  norm of a product, and no such orthogonality was available — which is precisely why that argument lost  $R^{1/2}$  and this one does not.

The lesson is worth recording: *almost-orthogonality is usable exactly when the quantity being estimated is quadratic.* Every place in this series where an  $L^2$  method outperformed an  $L^p$  one traces to this.

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WORKED EXAMPLE 12.7.1 — TIER A **What the crude bound gives**

Corollary 12.7.4 reads

$$\int |\mathcal{E}g|^2 d\mu \lesssim R^{\frac{2-n+s}{2}} \|g\|_{L^2(d\sigma)}^2.$$

*When does it gain?* The exponent is negative exactly when  $s < n - 2$ . So for  $s$  small the estimate gains and for  $s$  large it loses — the opposite of what one might guess, and the reason is that a larger  $s$  means  $\mu$  charges tubes more heavily.

*The two extremes check out.* At  $s = n$  the exponent is 1, matching the

Lebesgue case  $\int_{B_R} |\mathcal{E}g|^2 \approx R \|g\|_{L^2}^2$  of Proposition 6.1.4. ✓ At  $s = 0$ , where  $\mu$  may be a point mass, the exponent is  $\frac{2-n}{2}$  and the bound is weaker than the trivial pointwise one for  $n \geq 2$  — consistent with Proposition 12.7.2(ii).

*Numerically,  $n = 3$ .*

| $s$ | exponent $\frac{2-n+s}{2}$ |                   |
|-----|----------------------------|-------------------|
| 0   | -0.5                       | gains             |
| 1   | 0                          | breaks even       |
| 2   | 0.5                        | loses             |
| 3   | 1                          | the Lebesgue case |

*What this is and is not.* It is the crudest possible weighted estimate: it uses only the size of  $\mu$  on a single tube and no interaction between tubes. Chapter 12.8 needs a quantity of a different shape — a spherical *average* rather than an integral against  $\mu$  — and derives it from the identity of Proposition 12.8.2 rather than from this bound.

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**WORKED EXAMPLE 12.7.2— TIER B Why fractal restriction is harder than  $L^p$  restriction**

(a) *There is no Plancherel.* The  $L^2(\mathbb{R}^n)$  theory rests on Plancherel, which has no analogue for  $L^2(d\mu)$ . Every estimate must be built from the geometry of how  $\mu$  meets tubes.

(b) *The extremal measures are unknown.* For Lebesgue measure the extremal configurations for restriction are the Knapp caps, well understood since 1970. For a fractal  $\mu$  one must also ask which  $s$ -dimensional measures are worst, and the answer depends on  $s$  in a way that is not understood — the sharpness discussion in Volume 8, Remark 8.8.1, is the same difficulty seen from the other side.

(c) *Two parameters, not one.* An  $L^p$  restriction estimate has one exponent; a weighted estimate has both  $p$  and  $s$ , and the sharp region in the  $(p, s)$  plane is known only along its edges.

(d) *What survives.* The wave packet decomposition, which needs nothing about the measure; and the bilinear and multilinear estimates, which can be integrated against  $\mu$  once the tube geometry is understood. Chapter 12.8 does exactly this with the bilinear estimate.

*The summary.* Fractal restriction is the  $L^p$  theory with the ambient measure weakened, and every tool that used the ambient measure's structure must be rebuilt. That the theory nonetheless produces the best known Falconer exponents is the subject of the next chapter.

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**WORKED EXAMPLE 12.7.3— TIER C: RESEARCH TASTE Two parameters, and what is known**

Weighted restriction has both an integrability exponent  $p$  and a dimension  $s$ , and the sharp region in the  $(p, s)$  plane is known only along its edges. Collecting what is available.

(a)  $s = n$ . The measure is essentially Lebesgue, and the question is the restriction conjecture. Settled for  $n = 2$ , open above.

(b)  $p = 2$ . The Stein–Tomas range, extended to fractal measures by Mockenhaupt and Mitsis. This is the edge Theorem 12.7.3 lives on, and it is the one the distance problem uses.

(c)  $s$  small. Proposition 12.7.2(ii): nothing beyond the trivial bound, since a point mass is admissible.

(d) *The interior*. Largely open. Bak and Seeger’s sharpening covers part of it; the full picture would subsume the restriction conjecture as the edge case  $s = n$ , so it is at least as hard.

*Why two parameters make this harder than it looks*. An estimate can be sharp along both edges and far from sharp inside, and interpolation between the edges gives only a lower bound for the true region. Volume 8’s Remark 8.8.1 recorded the analogous difficulty for the distance problem: the extremal sets are known at the endpoint and not in between.

*What the distance problem needs*. Only the edge  $p = 2$ , but at the sharp exponent in  $s$  — which is Worked Example 8.8.1’s  $\sigma(\mu, r) \lesssim r^{-n/2}$ . So the distance problem asks for one edge of a two-parameter region, and even that is open.

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**WORKED EXAMPLE 12.7.4— TIER B Reading a weighted estimate**

Weighted restriction estimates are stated in several forms and the differences are substantive. Four checks.

(1) *Which side carries the measure?*  $\int |\mathcal{E}g|^2 d\mu$  as here, or  $\|f \widehat{d\mu}\|_{L^p}$  — the extension of  $\mu$  rather than the restriction *against* it. They are dual, and confusing them inverts the role of  $s$ .

(2) *Is the constant  $C_\mu$  or  $M_R(\mu)$ ?* Theorem 12.7.3 uses  $M_R(\mu) = \sup_x \mu(B(x, R^{1/2}))$ , which is scale-adapted; Corollary 12.7.4 converts it to  $C_\mu R^{s/2}$ . The first is sharper and the second more quotable, and a comparison between sources must fix which is meant.

(3) *Which ball radius?*  $R^{1/2}$  appears because that is the transverse width of a wave packet tube. An estimate stated with  $\mu(B(x, R^{-1}))$  or  $\mu(B(x, 1))$  is

a different statement, and the exponent of  $R$  changes accordingly.

(4) *Is the estimate local or global?* Here local, on  $B_R$ , with the tails of Theorem 12.1.4(ii) contributing the  $R^{-N}$  term. A global statement would need  $\varepsilon$ -removal, and Volume 6, Theorem 6.5.2 is proved only for Lebesgue measure.

*A worked check.* Setting  $s = n$  in Corollary 12.7.4 gives exponent  $\frac{2-n+n}{2} = 1$ , and the estimate reads  $\int_{B_R} |\mathcal{E}g|^2 \lesssim R \|g\|_{L^2}^2$  — exactly Proposition 6.1.4 of Volume 6. ✓ That is the single most useful check on any weighted estimate: specialise to Lebesgue measure and compare with the unweighted theorem.

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**WORKED EXAMPLE 12.7.5— TIER A The weighted estimate against the classical one**

It is worth seeing Theorem 12.7.3 beside the unweighted statements it generalises.

| Measure                         | Estimate                                         | Source                   |
|---------------------------------|--------------------------------------------------|--------------------------|
| $m$ on $B_R$                    | $\int  \mathcal{E}g ^2 \lesssim R \ g\ _{L^2}^2$ | Vol 6, Proposition 6.1.4 |
| $s$ -dimensional $\mu$          | $\lesssim R^{\frac{2-n+s}{2}} \ g\ _{L^2}^2$     | Corollary 12.7.4         |
| $d\sigma$ on $\mathbb{S}^{n-1}$ | Stein–Tomas range                                | Vol 6, Theorem 6.3.1     |
| $\delta_{x_0}$                  | $ \mathcal{E}g(x_0) ^2 \leq \ g\ _{L^1}^2$       | Proposition 12.7.2(ii)   |

*The first two agree at  $s = n$ .*  $\frac{2-n+n}{2} = 1$ . ✓ This is the check Worked Example 12.7.4 recommends and it is the quickest way to confirm a weighted estimate has been stated correctly.

*The last row is the degenerate case.* A point mass is 0-dimensional and the estimate degrades to the trivial pointwise bound, exactly as Proposition 12.7.2 says it must.

*What the table does not contain.* Any estimate at  $p \neq 2$ . The whole of Chapter 12.7 lives on the edge  $p = 2$ , because that is where almost-orthogonality is available — the point the Checkpoint after Theorem 12.7.3 makes, and the reason Proposition 12.3.3 lost  $R^{1/2}$  where this chapter loses nothing.

*Why  $p = 2$  suffices for Falconer.* Because Proposition 12.8.2 exhibits  $\sigma(\mu, r)$  as an  $L^2(d\sigma)$  norm. The distance set problem never needs another exponent, which is a genuine simplification and is why this chapter is short.

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**WORKED EXAMPLE 12.7.6— TIER A Frostman measures, and what they encode**  
 Definition 12.7.1 replaces a hypothesis about dimension by a measure, and it

is worth being clear what is gained and what is lost.

*What Frostman gives.* Volume 7, Theorem 7.5.1: a compact  $E$  with  $\dim_{\mathbf{H}} E > s$  carries a probability measure  $\mu$  with  $\mu(B(x, r)) \leq C_\mu r^s$ . Conversely such a measure forces  $\dim_{\mathbf{H}} E \geq s$ . So the two hypotheses are interchangeable.

*Why the measure is the better hypothesis.* Dimension is defined by a covering condition and is awkward to use analytically. A measure can be integrated against, Fourier transformed, and inserted into estimates. Every theorem in Volumes 8 and 12 about distance sets is proved with  $\mu$  and never with  $E$ .

*What is lost.* The constant  $C_\mu$ , which Frostman does not control, and which appears in every conclusion. For the qualitative statements of this series that is harmless; for a quantitative version it would matter.

*The two extremes, again.*  $s = n$  gives absolute continuity (Proposition 12.7.2(i));  $s = 0$  allows a point mass. Between them,  $\mu$  may be as irregular as a self-similar Cantor measure — and Volume 7, Theorem 7.7.4 showed such measures can have Fourier transforms that do not decay at all, which is precisely why  $\dim_{\mathbf{F}}$  and  $\dim_{\mathbf{H}}$  differ.

*The consequence for restriction.* An  $s$ -dimensional measure carries no information about  $\hat{\mu}$  beyond what the ball condition forces. So a weighted restriction estimate must hold for the *worst* such measure, and identifying that worst case is the open part of Worked Example 12.7.3.

#### WORKED EXAMPLE 12.7.7— TIER B **When the wave packet argument is free**

Theorem 12.7.3 loses nothing and Proposition 12.3.3 loses  $R^{1/2}$ , from what looks like the same step. The test distinguishing them is short.

*The test.* Is the quantity being estimated *quadratic* in the packet expansion? If so, Theorem 12.1.4(iii) applies directly and the sum is free. If not, one must dominate pointwise and pay for the overlap.

*Applied here.*  $\int |\mathcal{E}g|^2 d\mu$  is quadratic. ✓ Free.

*Applied to Proposition 12.3.3.*  $\|\prod_j |\mathcal{E}_j g_j|^{1/n}\|_{L^p}^p$  is a product of  $n$  different functions raised to a fractional power. Not quadratic in any one of them. *Not free*, and the price is the packet overlap  $R^{(n-1)/2}$ .

*Applied to  $L^4$  in the plane.*  $\|\mathcal{E}g\|_{L^4}^4 = \||\mathcal{E}g|^2\|_{L^2}^2$  is quadratic — in  $|\mathcal{E}g|^2$  rather than in  $\mathcal{E}g$ . So the bilinear  $L^4$  theory is free, which is why Volume 6, Proposition 6.8.4 could prove it by a  $TT^*$  computation and why Theorem 12.6.1 needs only Plancherel.

*The general statement.* Almost-orthogonality is a Hilbert space phenomenon, so it is available exactly at exponents where the quantity can be written as

an  $L^2$  norm of something. Even exponents and products of pairs qualify; odd and fractional exponents do not. Volume 13's decoupling exists precisely to recover a substitute at the exponents that do not qualify.

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## Drills

**Drill 12.7.1.** Verify Proposition 12.7.2(i) using the Lebesgue differentiation theorem.

**Drill 12.7.2.** Count the balls of radius  $R^{1/2}$  needed to cover a wave packet tube and confirm Step 2 of Theorem 12.7.3.

**Drill 12.7.3.** Track the exponent arithmetic in Steps 3–4 and confirm  $\frac{1}{2} + \frac{1-n}{2} = \frac{2-n}{2}$ .

**Drill 12.7.4.** For  $n = 2$ , tabulate the exponent  $\frac{2-n+s}{2}$  for  $s = 0.5, 1, 1.5, 2$  and say where the estimate gains.

**Where we are.** The chapter's one structural point, worth carrying into Volume 13.

An almost-orthogonal decomposition can be summed without loss exactly when the quantity being estimated is *quadratic*. Theorem 12.7.3 estimates  $\int |\mathcal{E}g|^2 d\mu$  and loses nothing; Proposition 12.3.3 estimates an  $L^p$  norm of a product and loses  $R^{1/2}$ . The difference is not technique but the shape of the quantity.

Decoupling, in Volume 13, is precisely the technology for recovering almost-orthogonality at exponents other than 2 — which is why it exists and why it is hard.

## Notes

Weighted restriction estimates were introduced by Mockenhaupt and by Mitsis, and sharpened by Bak and Seeger; the  $L^2$  form given here is the elementary case. The connection to distance sets is Mattila's and is the subject of the next chapter.

## Chapter 12.8

# Spherical Averages and the Distance Set

Volume 8 reduced the Falconer distance set problem to a bound on the spherical average  $\sigma(\mu, r)$ , and Worked Example 8.8.1 there asserted that the required estimate is a *weighted restriction estimate with a fractal weight*, promising the setting to this volume. This chapter makes that precise and shows how a bilinear estimate improves the threshold.

### 12.8.1 The spherical average is a restriction quantity

**Definition 12.8.1.** For a finite Borel measure  $\mu$  on  $\mathbb{R}^n$  and  $r > 0$ ,

$$\sigma(\mu, r) = \int_{\mathbb{S}^{n-1}} |\widehat{\mu}(r\omega)|^2 d\sigma(\omega).$$

**Proposition 12.8.2** (The two faces of  $\sigma(\mu, r)$ ). *For every finite Borel measure  $\mu$  with compact support and every  $r > 0$ ,*

$$\sigma(\mu, r) = \iint \widehat{d\sigma}(r(x-y)) d\mu(x) d\mu(y) = \|\mathcal{R}_r \mu\|_{L^2(d\sigma)}^2,$$

where  $\mathcal{R}_r \mu(\omega) = \widehat{\mu}(r\omega)$  is the restriction of  $\widehat{\mu}$  to the sphere of radius  $r$ .

*Proof.* The second equality is the definition of  $\mathcal{R}_r$  and of the  $L^2(d\sigma)$  norm.

For the first, expand  $|\widehat{\mu}(r\omega)|^2 = \widehat{\mu}(r\omega) \overline{\widehat{\mu}(r\omega)}$  and use  $\widehat{\mu}(\xi) = \int e^{-2\pi i \xi \cdot x} d\mu(x)$ :

$$|\widehat{\mu}(r\omega)|^2 = \iint e^{-2\pi i r \omega \cdot (x-y)} d\mu(x) d\mu(y).$$

Integrating in  $\omega$  over  $\mathbb{S}^{n-1}$  and applying Fubini — legitimate since  $\mu$  is finite and the integrand is bounded —

$$\sigma(\mu, r) = \iint \left( \int_{\mathbb{S}^{n-1}} e^{-2\pi i r \omega \cdot (x-y)} d\sigma(\omega) \right) d\mu(x) d\mu(y),$$

and the inner integral is  $\widehat{d\sigma}(r(x-y))$  by definition.  $\square$

**Where we are.** The identity is exact and can be confirmed numerically: for a discrete measure in  $\mathbb{R}^2$ , where  $\widehat{d\sigma}(\xi) = J_0(2\pi|\xi|)$ , Monte Carlo evaluation of the left side against direct summation of the right agrees to three or four significant figures at  $r = 0.5, 1, 2, 5, 10$ .

Both faces are used, and for different purposes.

*The first face* — the double integral against  $\widehat{d\sigma}$  — is what Volume 8 used. Inserting the pointwise decay  $|\widehat{d\sigma}(\xi)| \lesssim |\xi|^{-(n-1)/2}$  of Volume 5 gives Proposition 8.7.4, the trivial bound saturating at  $\frac{n-1}{2}$ .

*The second face* — the restriction of  $\widehat{\mu}$  to a sphere — is the one this volume can improve. It says  $\sigma(\mu, r)$  is literally  $\|\mathcal{R}_r\mu\|_{L^2(d\sigma)}^2$ : a *restriction operator applied to a measure*. By the duality of Volume 6, Theorem 6.1.3, bounding it is dual to an extension estimate, and the data is  $\mu$  rather than an  $L^p$  function — which is exactly the fractal setting of Chapter 12.7.

That is the promise of Worked Example 8.8.1 discharged: “weighted restriction estimate” now means  $\|\mathcal{R}_r\mu\|_{L^2(d\sigma)}$ , with  $\mu$   $s$ -dimensional.

## 12.8.2 What an improved bound buys

**Proposition 12.8.3** (Mattila’s criterion, sharp form). *Fix  $\alpha \in [\frac{d}{2}, \frac{d+1}{2}]$  and  $q_0 \in [1, 2]$  with*

$$\alpha\left(1 + \frac{1}{q_0}\right) \geq d. \quad (12.8.1)$$

*Suppose that for every  $q > q_0$ , every  $\alpha$ -dimensional  $\mu$ , every  $R > 1$  and every  $f$  supported in the annulus  $A_R(1) = \{||\xi| - R| \leq 1\}$ ,*

$$\left| \int f^\vee(u) d\mu(u) \right| \leq C_{q,\mu} R^{\frac{d-1}{2} - \frac{\alpha}{2q}} \|f\|_{L^2}. \quad (12.8.2)$$

*Then  $\dim_{\mathbb{H}} E > \alpha$  implies  $m(\Delta(E)) > 0$ .*

*Proof.* This is Mattila’s theorem in the form used by Wolff and by Erdős; it is Volume 8, Theorem 8.7.2 together with the bookkeeping that converts a decay estimate valid on a *range* of  $q$  into convergence of the Mattila integral. We refer to that proof.  $\square$

### Pitfall: a single decay exponent is not enough

It is natural to compress (12.8.2) into one number: write  $\sigma(\mu, r) \lesssim r^{-\beta}$  and ask for  $\beta > \frac{d}{2}$ , which is exactly what makes the Mattila integral  $\int \sigma(\mu, r)^2 r^{d-1} dr$  converge. That condition is *sufficient* but strictly stronger than (12.8.1), and using it loses the theorem.

The arithmetic is worth seeing. At Erdős's threshold  $\alpha = \frac{d}{2} + \frac{1}{3}$  the induced exponent is  $\beta = \alpha/q_0$ , and

| $d$ | $\alpha$       | $\beta = \alpha/q_0$        | $\frac{d}{2}$ |
|-----|----------------|-----------------------------|---------------|
| 3   | $\frac{11}{6}$ | $\frac{7}{6} \approx 1.17$  | 1.5           |
| 4   | $\frac{7}{3}$  | $\frac{5}{3} \approx 1.67$  | 2             |
| 5   | $\frac{17}{6}$ | $\frac{13}{6} \approx 2.17$ | 2.5           |

so  $\beta < \frac{d}{2}$  in every case: *the naive criterion fails precisely where Erdős's theorem succeeds.*

The reason is that (12.8.1) exploits the estimate across a whole *range* of exponents  $q > q_0$ , not merely the decay rate it implies at a single one. Compressing to  $\beta$  discards that, and the discarded information is exactly the difference between  $\frac{d}{2} + 1$  and  $\frac{d}{2} + \frac{1}{3}$ .

A consistency check: the trivial spherical bound gives  $\beta = \frac{d-1}{2} < \frac{d}{2}$ , so the naive criterion never applies at all — matching Volume 8's finding (Remark 8.7.3) that the Mattila  $L^2$  route does not recover Falconer's classical  $\frac{d+1}{2}$ .

### 12.8.3 Erdős's weighted bilinear estimate

**Theorem 12.8.4** (Erdős's weighted bilinear extension estimate). *Let  $d \geq 3$ ,  $\alpha \in (0, d)$ , and let  $H : \mathbb{R}^d \rightarrow \mathbb{R}$  satisfy*

$$\|H\|_\infty \leq 1, \quad \int_{B(x,r)} |H| \leq r^\alpha \quad \text{for all } x, r.$$

*Let  $S_1, S_2$  be compact subsets of the paraboloid with  $\text{dist}(S_1, S_2) > 1$ . Then for every*

$$q > q_0(\alpha, d) := \max \left\{ 1, \min \left( \frac{4\alpha}{d + 2\alpha - 2}, \frac{d + 2}{d} \right) \right\},$$

$$\|\widehat{f_1 d\sigma} \widehat{f_2 d\sigma}\|_{L^q(H d\xi)} \lesssim_{\alpha, q, d} \|f_1\|_{L^2(d\sigma)} \|f_2\|_{L^2(d\sigma)}.$$

*Proof.* By  $\varepsilon$ -removal it suffices to prove the localised form on  $B(0, R)$  with a loss  $R^\eta$ , for every  $\eta > 0$ , and this is done by induction on  $\eta$  exactly as in Proposition 12.6.5: the estimate holds for  $\eta \geq \alpha/q_0$  trivially, and we show that if it holds for some  $\eta > 0$  then it holds for  $\max\{(1 - \delta)\eta, C\delta\} + C\varepsilon$ .

Decompose  $\widehat{f_j d\sigma} = \sum_{T_j} c_{T_j} \phi_{T_j}$  into wave packets at scale  $R$  by Lemma 12.6.6, normalised so that  $\|\phi_{T_j}\|_{L^2} \approx R^{1/2}$ ,  $\sum_{T_j} |c_{T_j}|^2 \lesssim \|f_j\|_{L^2}^2$ , and the packets are almost orthogonal. Cover  $B(0, R)$  by  $O(R^{C\delta})$  balls  $Q$  of radius  $R^{1-\delta}$ , and let

$\sim$  be the relation supplied by Tao's argument, so that Lemma 12.6.10 and the combinatorial estimate of Lemma 12.6.11 give

$$\left\| \sum_{(T_1, T_2) \in Q^\sim} \phi_{T_1} \phi_{T_2} \right\|_{L^2(Q)} \lesssim R^{C\delta + C\varepsilon} R^{-\frac{d-2}{4}} (\#\mathbb{T}_1)^{1/2} (\#\mathbb{T}_2)^{1/2} \quad (12.8.3)$$

and  $\#\{Q : T \sim Q\} \lesssim R^\varepsilon$ .

**Step 1.** *The local part.*

Splitting the sum at each  $Q$  into pairs with  $T_1 \sim Q$  and  $T_2 \sim Q$ , and those without, the first is handled by the induction hypothesis: since  $\phi_{T_j}$  has Fourier support in an  $O(R^{-1})$ -neighbourhood of  $S_j$ , the inductive estimate on the ball  $Q$  of radius  $R^{1-\delta}$  gives, after almost-orthogonality and Cauchy–Schwarz,

$$\text{I} \lesssim R^{(1-\delta)\eta + C\varepsilon} (\#\mathbb{T}_1)^{1/2} (\#\mathbb{T}_2)^{1/2}.$$

**Step 2.** *The global part — where the weight enters.*

Write  $F_Q = \sum_{(T_1, T_2) \in Q^\sim} \phi_{T_1} \phi_{T_2}$ . Since  $q_0 \leq 2$ , Hölder's inequality with exponent  $\frac{2}{q_0}$  gives

$$\|F_Q\|_{L^{q_0}(Q, H \, d\xi)} \leq \|F_Q\|_{L^2(Q, d\xi)} \left( \int_Q |H| \right)^{\frac{1}{q_0} - \frac{1}{2}} \lesssim \|F_Q\|_{L^2(Q, d\xi)} R^{\frac{\alpha}{q_0} - \frac{\alpha}{2}},$$

using the hypothesis  $\int_{B(x, r)} |H| \leq r^\alpha$  with  $r \approx R$ . *This is the only place the weight is used, and it is the whole difference from Tao's unweighted theorem.*

**Step 3.** *Combine.*

Substituting (12.8.3) and summing over the  $O(R^{C\delta})$  balls,

$$\text{II} \lesssim R^{C\delta + C\varepsilon} R^{-\frac{d-2}{4}} R^{\frac{\alpha}{q_0} - \frac{\alpha}{2}} (\#\mathbb{T}_1)^{1/2} (\#\mathbb{T}_2)^{1/2}.$$

The exponent of  $R$  is at most  $C\delta + C\varepsilon$  precisely when

$$\frac{\alpha}{q_0} - \frac{\alpha}{2} - \frac{d-2}{4} \leq 0 \iff q_0 \geq \frac{4\alpha}{d + 2\alpha - 2},$$

which is the definition of  $q_0(\alpha, d)$  in the range  $\alpha \leq \frac{d+2}{2}$ . Hence  $\text{II} \lesssim R^{C\delta + C\varepsilon} (\#\mathbb{T}_1)^{1/2} (\#\mathbb{T}_2)^{1/2}$ , and  $\text{I} + \text{II}$  closes the induction.  $\square$

**Theorem 12.8.5** (Erdős). *Let  $d \geq 3$  and let  $E \subseteq \mathbb{R}^d$  be compact with  $\dim_{\text{H}} E > \frac{d}{2} + \frac{1}{3}$ . Then  $m(\Delta(E)) > 0$ .*



*Proof. Step 0. Reduce to the admissible range.*

Proposition 12.8.3 is stated for  $\alpha \in [\frac{d}{2}, \frac{d+1}{2}]$ , and  $\frac{d}{2} + \frac{1}{3} \leq \frac{d+1}{2}$  for every  $d$ , since  $\frac{1}{3} \leq \frac{1}{2}$ . So the threshold lies inside the range; but  $\dim_{\mathbb{H}} E$  need not. Choose  $\alpha$  with  $\frac{d}{2} + \frac{1}{3} < \alpha \leq \frac{d+1}{2}$  and  $\dim_{\mathbb{H}} E > \alpha$ , and prove the conclusion for that  $\alpha$  — a larger dimension only makes the hypothesis easier. *Without this step the proposition would be applied outside its stated range whenever  $\dim_{\mathbb{H}} E > \frac{d+1}{2}$ .*

**Step 1.** *Apply the criterion.*

By Proposition 12.8.3 it suffices to verify (12.8.2) for all  $q > q_0(\alpha, d)$  and to check the criterion (12.8.1).

The estimate (12.8.2) follows from Theorem 12.8.4 by the bilinear decomposition of the annulus: cut  $A_R(1)$  into dyadic spherical caps  $I$  of sidelength  $2^n$  with  $R^{1/2} \ll 2^n \ll R$ , pair related caps  $I \sim J$ , and rescale each pair by the linear map sending it to a pair of separated subsets of an elliptic surface. Tracking the Jacobians gives the weighted bound, and summing the diagonal terms — those with  $\ell(I) \approx R^{1/2}$  — contributes  $R^{(d-\alpha)/2}$ , which is dominated since  $\frac{d-\alpha}{2} \leq d-1 - \frac{\alpha}{q_0}$  for  $\alpha \in (0, d)$ .

It remains to solve (12.8.1). With  $q_0 = \frac{4\alpha}{d+2\alpha-2}$ ,

$$\alpha \left(1 + \frac{1}{q_0}\right) = \alpha + \frac{d+2\alpha-2}{4} \geq d \iff 4\alpha + d + 2\alpha - 2 \geq 4d \iff 6\alpha \geq 3d + 2,$$

that is  $\alpha \geq \frac{d}{2} + \frac{1}{3}$ . □

**Where we are.** The final computation is worth isolating, since the threshold  $\frac{d}{2} + \frac{1}{3}$  looks arbitrary and is not:

$$\alpha \left(1 + \frac{1}{q_0}\right) \geq d \iff \frac{3}{2}\alpha - \frac{3}{4}d - \frac{1}{2} \geq 0 \iff \alpha \geq \frac{d}{2} + \frac{1}{3}.$$

Symbolically, the solution of  $\alpha(1 + 1/q_0) = d$  is exactly  $\alpha = \frac{d}{2} + \frac{1}{3}$ , giving  $\frac{11}{6}, \frac{7}{3}, \frac{17}{6}, \frac{16}{3}$  for  $d = 3, 4, 5, 10$ . The  $\frac{1}{3}$  is the fixed point of the criterion, not a numerical accident.

Erdős notes that this method has a ceiling: the best exponent obtainable from Mattila's approach is  $\frac{4}{3}$  in  $\mathbb{R}^2$  and  $\frac{5}{3}$  in  $\mathbb{R}^3$ . But he also observes that if the weighted estimate held at  $\alpha = \frac{d}{2}$  for all  $q > 1$ , the criterion would give Falconer's conjecture outright for  $d \geq 4$ . So the obstruction is the range of  $q$ , not the framework.

*Remark 12.8.6 (R18 discharged).* Theorem 12.8.4 and Theorem 12.8.5 are now proved, following Erdős's paper. Revision R18 is discharged.

The proof is a genuinely small modification of Tao’s, and that is the point: Steps 1 and 3 are Tao’s induction verbatim, and the weight enters only at Step 2, through a single application of Hölder against  $\int_{B(x,r)} |H| \leq r^\alpha$ . *The entire content of the theorem is that Tao’s  $L^2$  estimate (12.8.3) has room to absorb a factor  $R^{\alpha/q_0 - \alpha/2}$ , and the exponent  $q_0(\alpha, d)$  is defined to be exactly the threshold at which it does.*

Erdős remarks that improving (12.8.3) to a weighted  $L^2(H \, d\xi)$  estimate would widen the range of  $q$  and improve the Falconer threshold; that has not been done.

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WORKED EXAMPLE 12.8.1— TIER A **The scoreboard, with the machinery**

| Threshold                   | Author            | Input                        | Where                               |
|-----------------------------|-------------------|------------------------------|-------------------------------------|
| $\frac{n+1}{2}$             | Falconer 1985     | decay of $\widehat{d\sigma}$ | Vol 8, Theorem 8.5.1, <i>proved</i> |
| $\frac{n}{2} + \frac{1}{3}$ | Erdős 2005        | bilinear restriction         | here, Remark 12.8.6                 |
| $\frac{4}{3} (n = 2)$       | Wolff 1999        | local smoothing              | Vol 13                              |
| $\frac{5}{4} (n = 2)$       | GIOW 2020         | partitioning + decoupling    | Vol 14                              |
| $\frac{n}{2}$               | <i>conjecture</i> | —                            | open                                |

*Reading the table.* Every row after the first replaces the pointwise decay of  $\widehat{d\sigma}$  by a stronger statement about how  $\widehat{\mu}$  concentrates on spheres — that is, by a better restriction estimate. The problem has been a restriction problem since 1985; only the tools have changed.

*The half that must be closed.* Volume 8 recorded that the trivial exponent saturates at  $\frac{n-1}{2}$  and the conjecture needs  $\frac{n}{2}$  — a gap of exactly  $\frac{1}{2}$ . Erdős’s  $\beta(s)$  closes part of it; nothing closes all of it.

*What this volume contributed.* Not a new threshold, but the identification of the mechanism: Proposition 12.8.2 says the quantity to bound *is* a restriction norm, so every advance in restriction theory is available to the distance problem. That is why the table’s later rows come from Volumes 13 and 14 rather than from new distance-set techniques.

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WORKED EXAMPLE 12.8.2— TIER B **Why bilinear, and not multilinear**

Chapter 12.3 developed the  $n$ -linear theory and Chapter 12.6 the two-linear one, and it is the second that the distance problem uses. The reason is structural.

(a) *The problem is two-point.*  $\Delta(E)$  records  $|x - y|$  for *pairs*, and Proposition 12.8.2 exhibits  $\sigma(\mu, r)$  as a double integral over  $\mu \times \mu$ . Volume 7, Worked

Example 7.7.3 classified such problems as two-point, and Volume 9, Worked Example 9.3.3 recorded that a two-point method has a ceiling for Kakeya — but here the problem itself is two-point, so no information is lost.

(b) *The matching estimate has two factors.* Bounding  $\iint \widehat{d\sigma}(r(x-y)) \, d\mu \, d\mu$  means controlling an interaction between two copies of  $\mu$ , which pairs with a bilinear extension estimate. An  $n$ -linear estimate would control an  $n$ -fold interaction that the problem never forms.

(c) *The historical confirmation.* Erdögan's improvement used Tao's bilinear theorem, and no distance-set result has been obtained from the multilinear theory directly. The later records of Volume 14 use decoupling and partitioning, which are again not  $n$ -linear.

*The transferable point.* Match the order of the estimate to the order of the problem. Volume 9's diagnosis was that a two-point method *cannot* solve Kakeya, a high-order problem; the converse holds here — an  $n$ -linear method has nothing extra to offer a two-point problem. This is the cleanest instance in the series of the principle working in both directions.

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**WORKED EXAMPLE 12.8.3— TIER B Using the identity in both directions**

Proposition 12.8.2 gives two expressions for  $\sigma(\mu, r)$ , and arguments use them for different purposes.

*Direction 1: from  $\widehat{d\sigma}$  to a bound.* Insert the decay  $|\widehat{d\sigma}(\xi)| \lesssim |\xi|^{-(n-1)/2}$  into the double integral:

$$\sigma(\mu, r) \lesssim r^{-\frac{n-1}{2}} \iint |x-y|^{-\frac{n-1}{2}} \, d\mu \, d\mu = r^{-\frac{n-1}{2}} I_{\frac{n-1}{2}}(\mu),$$

the energy integral of Volume 7. Finiteness of the energy requires  $s > \frac{n-1}{2}$ , and one recovers Proposition 8.7.4:  $\sigma \lesssim r^{-\min\{s, (n-1)/2\}}$ .

*Direction 2: from restriction to a bound.* Read  $\sigma(\mu, r)$  as  $\|\mathcal{R}_r \mu\|_{L^2(d\sigma)}^2$  and apply a restriction estimate with  $\mu$  as data. This is what Theorem 12.8.5 does, and it is available only because the identity exhibits the spherical average as a restriction norm.

*Why Direction 1 saturates.* It uses only the *modulus* of  $\widehat{d\sigma}$ , and Volume 5, Theorem 5.5.2 showed that modulus is sharp — so no improvement is possible without using the phase or the structure of  $\mu$ . Volume 8's Pitfall on Theorem 8.7.2 recorded that the phase is essential elsewhere in this argument too.

*Why Direction 2 can do better.* A restriction estimate exploits how  $\widehat{\mu}$  distributes over the sphere, not merely its size — the same distinction as be-

tween an  $L^\infty$  bound and an  $L^2$  bound. That extra information is what buys the improvement, and it is why the distance set problem became a restriction problem.

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**WORKED EXAMPLE 12.8.4— TIER C: RESEARCH TASTE What the Falconer problem now needs**

Twelve volumes in, the distance set problem can be stated with complete precision as a single missing estimate.

(a) *The target.* For every  $s > \frac{n}{2}$  and every  $s$ -dimensional  $\mu$  in the unit ball,

$$\sigma(\mu, r) \lesssim_\varepsilon r^{-\frac{n}{2}+\varepsilon}.$$

By Proposition 12.8.3 this gives  $m(\Delta(E)) > 0$  for  $\dim_{\mathbb{H}} E > \frac{n}{2}$ , which is Conjecture 8.4.2.

(b) *What it is, in restriction language.* By Proposition 12.8.2, a bound on  $\|\mathcal{R}_r \mu\|_{L^2(d\sigma)}$  — the restriction of  $\widehat{\mu}$  to a sphere — with  $\mu$  fractal. So the missing estimate is a weighted restriction estimate at the sharp exponent, on the edge  $p = 2$  of the two-parameter region of Worked Example 12.7.3.

(c) *Why the trivial bound stops halfway.* It uses only  $|\widehat{d\sigma}| \lesssim |\xi|^{-(n-1)/2}$ , which is sharp by Volume 5, Theorem 5.5.2. So the gap of  $\frac{1}{2}$  between  $\frac{n-1}{2}$  and  $\frac{n}{2}$  cannot be closed by any argument using only the size of  $\widehat{d\sigma}$ ; one must use how  $\widehat{\mu}$  distributes.

(d) *What has closed part of it.* Bilinear restriction (Theorem 12.8.5), local smoothing (Volume 13), and partitioning with decoupling (Volume 14). Each exploits the distribution of  $\widehat{\mu}$  in a different way, and none reaches  $\frac{n}{2}$ .

(e) *The structural reason it is hard.* By Worked Example 12.7.3, the sharp weighted estimate at  $s = n$  is the restriction conjecture. The Falconer problem asks for the edge  $p = 2$  at  $s = \frac{n}{2}$ , which is not formally implied by the restriction conjecture — but no method is known that gives one without the other's difficulties.

(f) *The honest summary.* The problem is now a well-posed question about a single quantity, with the mechanism completely understood and the estimate missing. That is a better position than in 1985, and it is not a solution.

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**WORKED EXAMPLE 12.8.5— TIER A The distance problem, from 1985 to here**

The Falconer problem has now occupied parts of five volumes, and the arc is worth seeing whole.

1985, *Volume 8*. Falconer poses the conjecture and proves  $\frac{n+1}{2}$  by bounding the distance measure in  $L^\infty$  (Theorem 8.5.1). The only input is the decay of  $\widehat{d\sigma}$  from Volume 5.

1987, *Volume 8*. Mattila reduces the problem to the spherical average  $\sigma(\mu, r)$  (Theorem 8.7.2), and the trivial bound (Proposition 8.7.4) saturates at  $\frac{n-1}{2}$  — short of the required  $\frac{n}{2}$  by exactly  $\frac{1}{2}$ .

*Volume 8's limit.* With only Volumes 1–7 available, nothing improves the trivial bound. Worked Example 8.8.1 states what is needed and names it a weighted restriction estimate, deferring to this volume.

*Here.* Proposition 12.8.2 makes that precise:  $\sigma(\mu, r) = \|\mathcal{R}_r \mu\|_{L^2(d\sigma)}^2$ . The problem is now a restriction problem, and Theorem 12.8.5 reports the first improvement, using the bilinear theory of Chapter 12.6.

*Volumes 13 and 14.* Local smoothing gives Wolff's  $\frac{4}{3}$  in the plane; partitioning with decoupling gives  $\frac{5}{4}$ . Each is a better restriction estimate against a fractal weight, and none reaches  $\frac{n}{2}$ .

*What has actually changed since 1985.* Not the formulation — Mattila's reduction is still the framework — but the supply of restriction estimates. The distance problem has been waiting on restriction theory for nearly forty years, and Worked Example 12.8.4 states exactly what it is still waiting for.

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#### WORKED EXAMPLE 12.8.6— TIER B **Checking a claimed Falconer threshold**

Distance set papers announce thresholds, and placing one takes four checks.

(1) *Which conclusion?*  $m(\Delta E) > 0$ , or  $\Delta E$  has non-empty interior, or the *pinned* statement that some  $x$  has  $m(\Delta_x E) > 0$ . These are increasingly strong; Volume 8, Theorem 8.6.1 gives the second under a stronger hypothesis, and the third is Volume 14's subject.

(2) *Which dimension?*  $\dim_{\mathbb{H}}$  throughout this series. Some results assume Ahlfors–David regularity, which is much stronger and not comparable.

(3) *What spherical average bound underlies it?* By Proposition 12.8.3 a threshold  $s_0$  corresponds to  $\sigma(\mu, r) \lesssim r^{-\beta}$  with  $\beta > \frac{n}{2}$  for  $s > s_0$ . Working backwards from the announced threshold recovers  $\beta$ , and comparing with the trivial  $\min\{s, \frac{n-1}{2}\}$  shows how much was gained.

(4) *Which dimensions  $n$ ?* Planar results and higher-dimensional ones use different machinery — local smoothing in the plane, decoupling above — and a planar threshold does not imply anything for  $n \geq 3$ .

*A worked check.* “ $\dim_{\mathbb{H}} E > \frac{5}{4}$  gives  $m(\Delta E) > 0$  in the plane”: (1) positive measure; (2) Hausdorff; (3)  $n = 2$  so the target is  $\beta > 1$ , against the trivial

$\frac{1}{2}$ ; (4) planar only. Placing it: better than Falconer's  $\frac{3}{2}$  and Wolff's  $\frac{4}{3}$ , short of the conjectured 1.

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## Drills

**Drill 12.8.1.** Verify Proposition 12.8.2 by expanding  $|\widehat{\mu}(r\omega)|^2$  and applying Fubini; state where compact support is used.

**Drill 12.8.2.** Insert  $|\widehat{d\sigma}(\xi)| \lesssim |\xi|^{-(n-1)/2}$  into the first face of the identity and recover Proposition 8.7.4 of Volume 8.

**Drill 12.8.3.** Carry out the naive computation of the Pitfall and confirm it gives  $\beta > n$ ; then locate the step in Theorem 8.7.2 that supplies the missing factor of two.

**Drill 12.8.4.** Using Proposition 12.8.3, determine what value  $\beta(s)$  must take at  $s = \frac{n}{2} + \frac{1}{3}$  for Theorem 12.8.5's threshold to follow, and compare with the trivial exponent  $\min\{s, \frac{n-1}{2}\}$  there.

**Where we are.** What this chapter added to Volume 8's account.

Volume 8 proved Falconer's theorem, Mattila's reduction, and the trivial spherical bound, and stated that an improvement would have to be a weighted restriction estimate. It could not say what that meant precisely, because the restriction machinery had not been built.

Proposition 12.8.2 says it precisely:  $\sigma(\mu, r)$  is  $\|\mathcal{R}_r\mu\|_{L^2(d\sigma)}^2$ . From that point the distance set problem is a problem in restriction theory, and every tool in Volumes 12–14 applies to it. Worked Example 12.8.1 tabulates which tool produced which threshold, and Worked Example 12.8.4 states what remains as a single inequality.

**Where we are.** Volume 12 in one line: an extension operator decomposed into wave packets is a sum of tubes, so restriction is Kakeya in disguise — and the distance set problem, by Proposition 12.8.2, speaks the same language.

## Notes

Mattila introduced the spherical average framework in 1987. Erdős's improvement is from 2005 and uses Tao's bilinear restriction theorem; Remark 12.8.6

records that it is quoted here. Volume 13 treats Wolff's planar result and Volume 14 the current records.





# Chapter 12.A

## Results Quoted from Volumes 1–11

| Result                                     | Source               | Used in                                      |
|--------------------------------------------|----------------------|----------------------------------------------|
| Restriction–extension duality              | Theorem 6.1.3        | Proposition 12.8.2                           |
| The local form of restriction              | Proposition 6.1.4    | Theorem 12.7.3, Proposition 12.3.3           |
| Stein–Tomas                                | Theorem 6.3.1        | Proposition 12.5.1                           |
| The Knapp condition                        | Theorem 6.2.3        | Proposition 12.5.1                           |
| Restriction implies Keakey                 | Theorem 6.6.2        | Worked Example 12.5.3                        |
| $\varepsilon$ -removal                     | Theorem 6.5.2        | Worked Example 12.5.2                        |
| The planar restriction theorem             | Theorem 6.8.2        | Worked Example 12.5.1                        |
| The bilinear $L^4$ identity                | Proposition 6.8.4    | Worked Example 12.3.1                        |
| Decay of $\widehat{d\sigma}$ and curvature | Worked Example 5.5.4 | Worked Example 12.1.2, Worked Example 12.6.3 |
| Lebesgue differentiation                   | Theorem 1.7.4        | Proposition 12.7.2                           |
| Frostman’s lemma                           | Theorem 7.5.1        | Definition 12.7.1                            |
| Mattila’s reduction                        | Theorem 8.7.2        | Proposition 12.8.3                           |
| The trivial spherical bound                | Proposition 8.7.4    | Worked Example 12.8.1                        |
| What is missing for Falconer               | Worked Example 8.8.1 | Chapter 12.8 throughout                      |
| Two-point problems                         | Worked Example 7.7.3 | Worked Example 12.8.2                        |
| Loomis–Whitney                             | Theorem 11.5.2       | Worked Example 12.3.1                        |
| Multilinear Keakey                         | Theorem 11.5.7       | Proposition 12.3.3                           |
| Transversality for tubes                   | Definition 11.5.3    | Proposition 12.2.2                           |
| Losses are $\delta^{-\varepsilon}$         | Proposition 9.7.3    | Worked Example 12.4.1                        |

## Debts and imports

This volume incurs no debts on later volumes and discharges one promise: Volume 8’s Worked Example 8.8.1 asserted that the spherical average is a weighted restriction quantity, and Proposition 12.8.2 proves it.

It imports four results without proof, all load-bearing except the last:

| Imported                              | Rev                      | What depends on it                   |
|---------------------------------------|--------------------------|--------------------------------------|
| Multilinear restriction               | R15, Re-<br>mark 12.3.11 | Chapter 12.4 and all exponents after |
| The broad–narrow in-<br>duction       | R16, Re-<br>mark 12.4.7  | Chapter 12.5’s table                 |
| Tao’s bilinear theorem,<br>$n \geq 3$ | R17, Re-<br>mark 12.6.12 | Theorem 12.8.5                       |
| Erdős’s spherical av-<br>erage bound  | R18, Re-<br>mark 12.8.6  | nothing later                        |

In each case the mechanism is proved and only the deep estimate quoted: Lemma 12.3.2 and Proposition 12.3.3 for R15; the dichotomy Proposition 12.4.3 for R16; the planar case Theorem 12.6.1 for R17; and Proposition 12.8.2 with the exchange rate for R18.

## Chapter 12.B

# Notation and Index of Results

### Additions to the Volume 1–11 conventions

|                                     |                                                          |
|-------------------------------------|----------------------------------------------------------|
| $R, \delta = R^{-1/2}$              | the large scale and the cap radius                       |
| $\theta$                            | a cap of radius $\delta$ on $\mathbb{S}^{n-1}$           |
| $T, \mathbb{T}_\theta$              | a wave packet tube and the family for a cap              |
| $g_{\theta, T}$                     | the wave packet piece of $g$                             |
| $\mathcal{E}_j$                     | extension restricted to $S_j$                            |
| $\nu$ -transverse                   | $ \det(\omega_1, \dots, \omega_n)  \geq \nu$             |
| $K$                                 | the broad–narrow cap parameter                           |
| $\mu(B(x, r)) \leq C_\mu r^s$       | $\mu$ is $s$ -dimensional                                |
| $M_R(\mu)$                          | $\sup_x \mu(B(x, R^{1/2}))$                              |
| $\sigma(\mu, r), \mathcal{R}_r \mu$ | the spherical average and the restriction of $\hat{\mu}$ |

## Index of the main results

|                                             |                    |
|---------------------------------------------|--------------------|
| The dual box                                | Lemma 12.1.2       |
| Wave packet decomposition                   | Theorem 12.1.4     |
| Transversality passes to packets            | Proposition 12.2.2 |
| Two factors do not suffice                  | Proposition 12.2.3 |
| The exponents match                         | Lemma 12.3.2       |
| Multilinear restriction                     | Theorem 12.3.1     |
| Discrete Loomis–Whitney                     | Lemma 12.3.4       |
| What the packets give directly              | Proposition 12.3.3 |
| Determinant and slabs                       | Lemma 12.4.1       |
| The broad–narrow dichotomy                  | Proposition 12.4.3 |
| The benchmarks                              | Proposition 12.5.1 |
| Bilinear $L^2$ in the plane                 | Theorem 12.6.1     |
| The addition map                            | Lemma 12.6.2       |
| Tao’s bilinear theorem                      | Theorem 12.6.3     |
| The two extremes                            | Proposition 12.7.2 |
| Weighted restriction                        | Theorem 12.7.3     |
| The $s$ -dimensional form                   | Corollary 12.7.4   |
| The spherical average is a restriction norm | Proposition 12.8.2 |
| The exchange rate                           | Proposition 12.8.3 |
| Erdős’s threshold                           | Theorem 12.8.5     |

## Plan revisions in this volume

| Imported, not proved       | Rev | Reason                                                                                                                        |
|----------------------------|-----|-------------------------------------------------------------------------------------------------------------------------------|
| Multilinear restriction    | R15 | the wave packet argument loses $R^{1/2}$ ; removing it needs an induction retaining almost-orthogonality. <i>Load-bearing</i> |
| The broad–narrow induction | R16 | long, and its exponents are a moving target. <i>Load-bearing</i>                                                              |
| Tao’s bilinear theorem     | R17 | proved here for $n = 2$ ; $n \geq 3$ is a research paper. <i>Load-bearing</i>                                                 |
| Erdős’s spherical bound    | R18 | rests on R17; nothing later depends on it                                                                                     |

**Superseded from the original plan.** Chapters on Loomis–Whitney and multilinear Kakeya were written into Volume 11 instead (Theorem 11.5.2, Theorem 11.5.4, both proved there); this volume begins after them.

## Part 13

# Volume 13

Decoupling



# Preface to Volume 13

## The problem decoupling solves

Volume 12 ended with a diagnosis. Almost-orthogonality — the fact that pieces of a function with disjoint frequency supports do not interfere — is a Hilbert space phenomenon, available exactly when the quantity being estimated can be written as an  $L^2$  norm of something. Theorem 12.7.3 could sum its wave packets for free because its left side was quadratic; Proposition 12.3.3 could not, and lost a full power of  $R$ . Worked Example 12.7.7 made the test explicit.

Decoupling is the technology for recovering a substitute at the exponents where orthogonality is not free. It asserts that for  $f$  with Fourier support in a  $\delta$ -neighbourhood of a curved surface, decomposed into pieces  $f_\theta$  over caps,

$$\|f\|_{L^p} \lesssim_\varepsilon \delta^{-\varepsilon} \left( \sum_\theta \|f_\theta\|_{L^p}^2 \right)^{1/2},$$

with the  $\ell^2$  sum *outside* the  $L^p$  norm. At  $p = 2$  this is Plancherel and is exact. For  $p > 2$  it is false in general and true for curved surfaces up to a critical exponent — 6 for the parabola in  $\mathbb{R}^2$ , the same number Stein–Tomas produced in Volume 6.

## Why it matters beyond harmonic analysis

Bourgain and Demeter proved the sharp decoupling theorem for the paraboloid in 2015, and within two years it had settled Vinogradov’s mean value theorem in number theory — a problem open since 1935 and previously approached by Wooley’s efficient congruencing. That a statement about  $L^p$  norms of Fourier pieces should resolve a question about counting solutions of Diophantine systems is the most striking transfer in this series, and Chapter 13.6 sets out the reduction.

## What this volume proves

The definitions and trivial bounds; the induction-on-scales method in a model case; the ball inflation inequality, from Volume 11's multilinear Keakeya; and the Bourgain–Demeter theorem for the parabola in  $\mathbb{R}^2$ , by the Guth–Maldague–Wang simplification. The higher-dimensional case is stated with its changes and its proof omitted — structurally the same, substantially longer.

It then reduces Vinogradov's theorem to moment-curve decoupling, and closes with applications: Strichartz estimates on tori, and the chain from decoupling through local smoothing to Wolff's distance set bound in the plane. That last is the second of the three Falconer sites planned in advance, Volume 12, Chapter 12.8 having been the first.

## What is assumed

Volumes 1–12. Most heavily: Volume 12 for wave packets and the transversality framework; Volume 11 for multilinear Keakeya; Volume 4 for the function-space reading of the  $\ell^2$  sum; Volume 8 and Volume 12 for the distance set material.

## How the volume is organised

Chapter 13.1 states the inequality, computes the trivial bounds, and proves sharpness at  $p = 6$  — so that by the end of it the reader knows exactly what is being claimed and why the exponent is what it is.

Chapter 13.2 supplies the two elementary tools: parabolic rescaling, which is an exact algebraic identity, and a bootstrapping lemma converting a self-improving inequality into a subpolynomial bound. Both are short, and together they reduce the theorem to a single hard step.

Chapter 13.3 supplies that step. Ball inflation upgrades information at scale  $\rho^{-1}$  to scale  $\rho^{-2}$ , paid for by transversality, and it is where Volume 11's multilinear Keakeya inequality enters. In the plane the input degenerates to Fubini.

Chapter 13.4 assembles the three into the theorem, and Chapter 13.5 states precisely what changes in  $\mathbb{R}^n$  — one thing: two factors no longer span, so the broad–narrow split needs  $n$ .

Chapter 13.6 and Chapter 13.7 give the number-theoretic application and compare it with Wooley's method. Chapter 13.8 collects three analytic applications and discharges the Falconer promise.



## A remark on what this rests on

The dependency chain of Theorem 13.4.2 runs

decoupling  $\rightarrow$  ball inflation  $\rightarrow$  multilinear Kakeya  $\rightarrow$  Loomis–Whitney  $\rightarrow$  Hölder,

and every link is proved in this series. That was not true when the volume was planned: multilinear Kakeya was an import, and the chain terminated in a citation. Discharging it (Volume 11, Theorem 11.5.4) is what makes the present volume self-contained, and it is the clearest single illustration of why the black boxes were worth removing.

## The plan

1 What decoupling asserts      definitions, trivial bounds, sharpness

Chapters 2–8, planned: induction on scales; ball inflation; the parabola; the higher-dimensional paraboloid; the moment curve and Vinogradov; efficient congruencing; and applications.



# Chapter 13.1

## What Decoupling Asserts

### 13.1.1 The statement

**Definition 13.1.1** (The parabola and its caps). Let  $P = \{(t, t^2) : t \in [0, 1]\}$  and let  $\mathcal{N}_\delta$  denote its  $\delta$ -neighbourhood in  $\mathbb{R}^2$ . Partition  $[0, 1]$  into  $N = \delta^{-1/2}$  intervals of length  $\delta^{1/2}$ , and let  $\theta$  denote the corresponding pieces of  $\mathcal{N}_\delta$ .

**Lemma 13.1.2** (Caps are boxes). *Each  $\theta$  is contained in a rectangle of dimensions  $\approx \delta^{1/2} \times \delta$ , with the short side normal to  $P$ . Consequently its dual box has dimensions  $\approx \delta^{-1/2} \times \delta^{-1}$ , with the long side tangent to  $P$ .*

*Proof.* Over an interval of length  $\delta^{1/2}$  centred at  $t_0$ , the parabola deviates from its tangent line by

$$|t^2 - (t_0^2 + 2t_0(t - t_0))| = (t - t_0)^2 \leq \delta,$$

so the arc lies in a  $\delta$ -slab about the tangent. Adding the  $\delta$ -neighbourhood doubles the slab. The dual box is the box with reciprocal sides.  $\square$

**Where we are.** The choice  $\delta^{1/2}$  for the cap length is forced, exactly as  $\delta = R^{-1/2}$  was in Volume 12, Lemma 12.1.2: it is the length over which the parabola's curvature produces a deviation comparable to  $\delta$ . Longer caps are not contained in  $\delta$ -slabs; shorter ones waste the curvature.

So the two volumes are describing the same geometry. Volume 12 fixed a ball  $B_R$  and derived caps of radius  $R^{-1/2}$ ; here we fix a neighbourhood width  $\delta$  and derive caps of length  $\delta^{1/2}$ . With  $\delta = R^{-1}$  the two agree, and the dual boxes  $\delta^{-1/2} \times \delta^{-1}$  are the wave packet tubes  $R^{1/2} \times R$ .

**Definition 13.1.3** (The decoupling constant). For  $2 \leq p < \infty$ , let  $D_p(\delta)$  be the smallest constant such that

$$\|f\|_{L^p(\mathbb{R}^2)} \leq D_p(\delta) \left( \sum_{\theta} \|f_{\theta}\|_{L^p(\mathbb{R}^2)}^2 \right)^{1/2}$$

for every  $f$  with  $\text{supp } \widehat{f} \subseteq \mathcal{N}_\delta$ , where  $f_\theta$  is the piece of  $f$  with frequency support in  $\theta$ .

**Theorem 13.1.4** (Bourgain–Demeter for the parabola). *For  $2 \leq p \leq 6$  and every  $\varepsilon > 0$ ,*

$$D_p(\delta) \lesssim_\varepsilon \delta^{-\varepsilon}.$$

*The range is sharp:  $D_p(\delta) \rightarrow \infty$  faster than any  $\delta^{-\varepsilon}$  when  $p > 6$ .*

*Proof.* The sharpness is Proposition 13.1.8, proved later in this chapter and depending on nothing stated here. The positive statement is proved in Chapter 13.4 as Theorem 13.4.2, after the machinery of Chapter 13.2 and Chapter 13.3 is in place; the present chapter establishes only what the statement means and why 6 is critical.  $\square$

## 13.1.2 The trivial bounds

**Proposition 13.1.5** (The two easy estimates). *For every  $\delta \in (0, 1)$ :*

$$(i) \quad D_2(\delta) \lesssim 1;$$

$$(ii) \quad D_p(\delta) \leq N^{1/2} = \delta^{-1/4} \text{ for every } p \geq 2.$$

*Proof.* (i) The frequency supports  $\theta$  have bounded overlap, so by Plancherel

$$\|f\|_{L^2}^2 = \|\widehat{f}\|_{L^2}^2 \approx \sum_{\theta} \|\widehat{f}_\theta\|_{L^2}^2 = \sum_{\theta} \|f_\theta\|_{L^2}^2.$$

(ii) By the triangle inequality and then Cauchy–Schwarz over the  $N$  caps,

$$\|f\|_{L^p} \leq \sum_{\theta} \|f_\theta\|_{L^p} \leq N^{1/2} \left( \sum_{\theta} \|f_\theta\|_{L^p}^2 \right)^{1/2}. \quad \square$$

**Where we are.** Proposition 13.1.5 brackets the problem. At  $p = 2$  the answer is  $O(1)$  and the inequality is an identity — there is nothing to prove. At every  $p$  the answer is at most  $\delta^{-1/4}$ , by an argument using no property of the parabola whatever.

So Theorem 13.1.4 asserts that for  $2 \leq p \leq 6$  the truth is at the  $p = 2$  end of that bracket, not the trivial end. *That is the entire content*, and the curvature of the parabola is what supplies it: for a straight segment in place of  $P$ , the trivial bound (ii) is attained and no improvement holds.

### 13.1.3 The case $p = 4$ , in full

Before the machinery of Chapter 13.4, one case of Theorem 13.1.4 can be proved outright, in half a page and with no  $\varepsilon$ -loss at all. It is the oldest result in the subject.

**Lemma 13.1.6** (The sum-set is 2-to-1). *Let  $\theta_i$  be the caps of Definition 13.1.1 and, for a pair  $(i, j)$ , let  $B_{ij}$  be the smallest box containing  $\theta_i + \theta_j$ . Then each point of  $\mathbb{R}^2$  lies in at most  $C$  of the boxes  $B_{ij}$ , with  $C$  absolute.*

*Proof.*  $\theta_i$  is contained in a  $\delta^{1/2} \times \delta$  box about  $(t_i, t_i^2)$  by Lemma 13.1.2, so  $B_{ij}$  is a  $\delta^{1/2} \times \delta$  box about  $(t_i + t_j, t_i^2 + t_j^2)$ .

Suppose  $\xi \in B_{ij}$ . Then  $t_i + t_j$  is determined to within  $O(\delta^{1/2})$  and  $t_i^2 + t_j^2$  to within  $O(\delta)$ , hence

$$t_i t_j = \frac{1}{2} \left( (t_i + t_j)^2 - (t_i^2 + t_j^2) \right)$$

is determined to within  $O(\delta)$ . So the unordered pair  $\{t_i, t_j\}$  — being the roots of  $X^2 - (t_i + t_j)X + t_i t_j$  — is determined to within  $O(\delta^{1/2})$ , that is, to within one cap. Since the caps have length  $\delta^{1/2}$ , at most  $C$  ordered pairs  $(i, j)$  are possible.  $\square$

**Where we are.** The lemma is the parabola's curvature, stated arithmetically. Direct enumeration confirms it: binning  $\theta_i + \theta_j$  at the correct resolution  $\delta^{1/2} \times \delta$ , the maximum number of pairs per cell is exactly 2 — the swap  $(t_i, t_j) \leftrightarrow (t_j, t_i)$  — at  $M = 40, 80, 160, 320$  caps alike.

The control experiment is the point. Replacing the parabola by the line  $\{(t, 0)\}$ , the second coordinate carries no information,  $t_i t_j$  is not determined, and the same enumeration gives  $M$  pairs per cell — growing without bound. *Curvature is exactly what makes the sum-set map finite-to-one*, and Lemma 13.1.6 is the sharpest form of that statement.

**Theorem 13.1.7** ( $\ell^2$  decoupling at  $p = 4$ ; Fefferman, Córdoba).  $D_4(\delta) \lesssim 1$ , uniformly in  $\delta$ . There is no  $\delta^{-\varepsilon}$ .

*Proof.* By Plancherel applied to  $f^2$ ,

$$\|f\|_{L^4}^2 = \|f^2\|_{L^2} = \|\widehat{f} * \widehat{f}\|_{L^2}.$$

Write  $f = \sum_{\theta} f_{\theta}$ , so  $\widehat{f} * \widehat{f} = \sum_{i,j} \widehat{f_{\theta_i}} * \widehat{f_{\theta_j}}$ , and the  $(i, j)$  term is supported in  $B_{ij}$ . By Lemma 13.1.6 these have overlap at most  $C$ , so the sum is almost orthogonal in  $L^2$ :

$$\|\widehat{f} * \widehat{f}\|_{L^2}^2 \leq C \sum_{i,j} \|\widehat{f_{\theta_i}} * \widehat{f_{\theta_j}}\|_{L^2}^2 = C \sum_{i,j} \|f_{\theta_i} f_{\theta_j}\|_{L^2}^2.$$

By Cauchy–Schwarz,  $\|f_{\theta_i} f_{\theta_j}\|_{L^2} \leq \|f_{\theta_i}\|_{L^4} \|f_{\theta_j}\|_{L^4}$ , so the right side is

$$C \left( \sum_i \|f_{\theta_i}\|_{L^4}^2 \right)^2.$$

Taking square roots gives  $\|f\|_{L^4}^2 \lesssim \sum_{\theta} \|f_{\theta}\|_{L^4}^2$ , which is the claim.  $\square$

**Where we are.** Three things to take from this proof.

*It is the model for everything later.* Plancherel, an overlap count, and Cauchy–Schwarz. Theorem 13.4.2 at  $p = 6$  has the same shape with ball inflation replacing the overlap count.

*Why it stops at  $p = 4$ .* The argument writes  $\|f\|_{L^4}^2$  as an  $L^2$  norm, so it applies exactly when  $p/2 = 2$ . For  $p = 6$  one would need  $f^3$  and a three-fold sum-set, whose overlap is *not* bounded — three points on a parabola are not determined by their first two power sums. That is the precise reason the  $p = 6$  theorem needed thirty more years.

*No  $\varepsilon$ .* Theorem 13.1.7 has an absolute constant, whereas Theorem 13.1.4 carries  $\delta^{-\varepsilon}$ . For the parabola the truth at  $p = 6$  is  $(\log \delta^{-1})^c$ , by Guth, Maldague and Wang; at  $p = 4$  it is  $O(1)$ , and that is best possible.

---

WORKED EXAMPLE 13.1.1— TIER B **Interpolating from  $p = 4$**

Theorem 13.1.7 gives  $p = 4$  free. What does interpolation give?

*The two endpoints.*  $D_2(\delta) \lesssim 1$  (Proposition 13.1.5(i)) and  $D_4(\delta) \lesssim 1$  (Theorem 13.1.7).

*What interpolation yields.* For  $2 \leq p \leq 4$ ,  $D_p(\delta) \lesssim 1$  by Hölder — so the whole range up to 4 is settled with no loss whatever.

*What it does not yield.* Anything beyond  $p = 4$ . Interpolation fills in between known endpoints and never produces a new one, exactly as Volume 12, Worked Example 12.5.4 warned.

*So what is the content of  $p = 6$ ?* Precisely the interval  $(4, 6]$ . The trivial bound there is  $\delta^{-1/4}$  and the truth is  $\delta^{-\varepsilon}$ , so the gain is a full power of  $\delta^{-1/4}$  — and by Proposition 13.1.8 nothing beyond 6 is possible.

*The picture to carry.* Three regimes:  $p \leq 4$  elementary and sharp with constant  $O(1)$ ;  $4 < p \leq 6$  the Bourgain–Demeter theorem, hard, with  $\delta^{-\varepsilon}$ ;  $p > 6$  false. The middle regime is the subject of this volume.

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### 13.1.4 Sharpness

**Proposition 13.1.8** (The constructive example). *For  $p > 6$  there is no  $\varepsilon$  for which  $D_p(\delta) \lesssim_\varepsilon \delta^{-\varepsilon}$ . Specifically,*

$$D_p(\delta) \gtrsim \delta^{-\frac{1}{4} + \frac{3}{2p}},$$

which for  $p > 6$  is a positive power of  $\delta^{-1}$ .

*Proof.* Choose  $f_\theta$  to be a wave packet adapted to the dual box  $T_\theta$  of Lemma 13.1.2, normalised so that  $|f_\theta| \approx 1$  on  $T_\theta$  and with phases chosen so that all the  $f_\theta$  take the value 1 at the origin.

**Step 1.** *The left side.*

Each  $f_\theta$  has frequency support in  $\mathcal{N}_\delta$ , a set of diameter  $\approx 1$ . So the phases of the  $f_\theta$  remain aligned on a ball of radius  $\approx 1$  about the origin, on which  $|f| \approx N$ . Hence

$$\|f\|_{L^p} \gtrsim N.$$

**Step 2.** *The right side.*

$|T_\theta| \approx \delta^{-1/2} \cdot \delta^{-1} = \delta^{-3/2}$ , so  $\|f_\theta\|_{L^p}^p \approx \delta^{-3/2}$  and  $\|f_\theta\|_{L^p} = \delta^{-3/(2p)}$ . Therefore

$$\left( \sum_\theta \|f_\theta\|_{L^p}^2 \right)^{1/2} \approx (N \delta^{-3/p})^{1/2} = N^{1/2} \delta^{-3/(2p)}.$$

**Step 3.** *The ratio.*

Dividing and substituting  $N = \delta^{-1/2}$ ,

$$\frac{\|f\|_{L^p}}{(\sum_\theta \|f_\theta\|_{L^p}^2)^{1/2}} \gtrsim N^{1/2} \delta^{3/(2p)} = \delta^{-\frac{1}{4} + \frac{3}{2p}}.$$

This tends to infinity as  $\delta \rightarrow 0$  precisely when  $-\frac{1}{4} + \frac{3}{2p} < 0$ , that is  $p > 6$ .  $\square$

**Pitfall: the constructive region is a unit ball, not the common dual box**

Step 1 asserts that the packets stay aligned on a ball of radius  $\approx 1$ . It is tempting to say instead that they align on the intersection of the dual boxes  $T_\theta$ , which contains a ball of radius  $\approx \delta^{-1/2}$  and has area  $\approx \delta^{-1}$ . That is *wrong*, and the error is instructive because it inverts the conclusion.

The  $f_\theta$  have *different* frequencies, spanning a set of diameter  $\approx 1$ . Two packets whose frequencies differ by  $\approx 1$  acquire a relative phase  $\approx 1$  over a distance  $\approx 1$ , so they cannot remain aligned further than that — regardless

of how large their dual boxes are. The dual box governs where a *single* packet is supported, not where several interfere constructively. Using the larger region gives  $\|f\|_{L^p} \gtrsim N\delta^{-1/p}$  and a ratio  $\delta^{-1/4+1/(2p)}$ , which blows up for every  $p > 2$  — contradicting Proposition 13.1.5(i) at  $p = 2$  and the theorem throughout. A computation that contradicts a case one can verify independently is wrong, and this is the case to check.

---

WORKED EXAMPLE 13.1.2— TIER A **The critical exponent, computed**

The exponent in Proposition 13.1.8 is  $-\frac{1}{4} + \frac{3}{2p}$ , and tabulating it locates the threshold.

| $p$ | $-\frac{1}{4} + \frac{3}{2p}$ |                              |
|-----|-------------------------------|------------------------------|
| 4   | +0.125                        | ratio decays; no obstruction |
| 5   | +0.050                        | decays                       |
| 6   | 0                             | <b>bounded — critical</b>    |
| 7   | −0.036                        | blows up; estimate fails     |
| 8   | −0.063                        | fails                        |
| 10  | −0.100                        | fails                        |

*Numerically.* At  $\delta = 10^{-4}, 10^{-6}, 10^{-8}$  the ratio at  $p = 4$  is 0.316, 0.178, 0.100; at  $p = 6$  it is 1.000 throughout; at  $p = 8$  it is 1.778, 2.371, 3.162. Decaying, constant, growing — exactly as the exponent predicts.

*The number 6.* It is  $\frac{2(n+1)}{n-1}$  at  $n = 2$ , the Stein–Tomas exponent of Volume 6, Theorem 6.3.1. That is not a coincidence: Worked Example 13.4.1 will show decoupling implies a restriction estimate, and the critical exponents must therefore agree.

*In higher dimensions.* For the paraboloid in  $\mathbb{R}^n$  the critical exponent is  $\frac{2(n+1)}{n-1} - 6, 4, \frac{10}{3}$  for  $n = 2, 3, 4$  — and Chapter 13.5 states the theorem there.

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WORKED EXAMPLE 13.1.3— TIER A **The trivial bounds, computed**

Proposition 13.1.5 brackets  $D_p(\delta)$  between  $O(1)$  and  $\delta^{-1/4}$ . It is worth seeing the two ends numerically.

| $\delta$   | $N = \delta^{-1/2}$ | $N^{1/2} = \delta^{-1/4}$ | theorem gives           |
|------------|---------------------|---------------------------|-------------------------|
| $10^{-4}$  | $10^2$              | 10                        | $\delta^{-\varepsilon}$ |
| $10^{-8}$  | $10^4$              | $10^2$                    | $\delta^{-\varepsilon}$ |
| $10^{-12}$ | $10^6$              | $10^3$                    | $\delta^{-\varepsilon}$ |



*The gap being closed.* At  $\delta = 10^{-12}$  the trivial bound allows  $10^3$  and the truth is subpolynomial. That factor of  $10^3$  is the content of Theorem 13.1.4, and it grows without bound as  $\delta \rightarrow 0$ .

*But see Worked Example 13.2.2.* The theorem's constant  $C_\epsilon$  is not effective, so at any *named*  $\delta$  the theorem may give a worse numerical bound than the trivial one. The improvement is asymptotic.

*Where the trivial bound is attained.* For a straight segment in place of the parabola: then all caps have the same dual direction, the packets add coherently on a common tube, and no cancellation occurs. Curvature is the entire difference, as Worked Example 13.5.2(d) records.

#### WORKED EXAMPLE 13.1.4— TIER B **Where the $\ell^2$ sits, and why**

The inequality of Definition 13.1.3 has the  $\ell^2$  sum outside the  $L^p$  norm. Three other arrangements are conceivable and each is either trivial or false.

(a)  $\ell^2$  *inside*:  $\|(\sum_\theta |f_\theta|^2)^{1/2}\|_{L^p}$ . This is the square function, and the corresponding inequality is *stronger* than decoupling for  $p \geq 2$  by Minkowski. It is also false for the parabola at  $p = 6$  — the square function conjecture is open — so decoupling is the correct weakening.

(b)  $\ell^1$  *outside*:  $\sum_\theta \|f_\theta\|_{L^p}$ . This is the triangle inequality, Proposition 13.1.5(ii) before Cauchy–Schwarz, and it holds trivially with constant 1. It is too weak to be useful, since it loses  $N^{1/2}$  against the  $\ell^2$  form.

(c)  $\ell^p$  *outside*. Interpolates between the two and is false for  $p > 2$  at the sharp constant.

*Why  $\ell^2$  is right.* Because at  $p = 2$  it is an identity (Proposition 13.1.5(i)), so the inequality is sharp at one endpoint and one asks how far the sharpness persists. The answer,  $p \leq 6$ , is the theorem.

*The function-space reading.* Volume 4, Worked Example 4.6.3 distinguished the Besov and Triebel–Lizorkin scales by exactly this:  $\ell^q$  outside the  $L^p$  norm or inside. Decoupling is a Besov-type statement, and the square function is Triebel–Lizorkin-type. That the first is provable and the second open is the same phenomenon as the relative difficulty of those two scales.

#### WORKED EXAMPLE 13.1.5— TIER C: RESEARCH TASTE **Why decoupling was invented**

Decoupling is recent, and it is worth asking what problem it was invented to solve, since the answer explains its form.

(a) *The obstruction it removes.* Volume 12, Worked Example 12.7.7 established that almost-orthogonality is free exactly when the quantity is quadratic. For  $p \neq 2$  the pieces of a function over a curved surface are not orthogonal, and no identity replaces Plancherel.

(b) *What was tried first.* The *square function* conjecture —  $\ell^2$  inside the norm — is the natural strengthening and is open at  $p = 6$ . Wolff's insight was that moving the  $\ell^2$  *outside* yields a statement that is weaker, still useful, and provable.

(c) *Why weaker suffices.* In every application here — Vinogradov, Strichartz, local smoothing — the caps carry comparable mass, and for such functions the  $\ell^2$ -outside and  $\ell^2$ -inside statements differ by a constant. The weakening costs nothing where it is used.

(d) *Why it generalises.* The proof uses only rescaling, transversality and induction. It has no arithmetic content, which is why it settled a number theory problem that resisted arithmetic methods for eighty years (Worked Example 13.7.1).

(e) *The transferable lesson.* When a natural conjecture resists, look for the weakest statement that still serves the applications. The square function conjecture remains open; decoupling took its place and did its work. Worked Example 13.1.4 lists the four arrangements, and only one of them was both provable and useful.

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WORKED EXAMPLE 13.1.6— TIER C: RESEARCH TASTE    **Decoupling as a way of thinking**

Decoupling is a technique, but it is more useful as a habit of mind. Four questions it teaches one to ask.

(a) *What are the caps?* Any function with Fourier support on a curved set decomposes over caps, and the caps are determined by the curvature through Lemma 13.1.2. Before estimating anything, compute the cap dimensions and their dual boxes. Volume 12, Worked Example 12.1.4 makes the same point for wave packets, and it is the same computation.

(b) *At what exponent is orthogonality free?* At  $p = 2$  always, by Plancherel. Away from 2 it is not, and Volume 12, Worked Example 12.7.7 gives the test: the quantity must be expressible as an  $L^2$  norm of something. Decoupling exists to buy a substitute where the test fails.

(c) *What is the trivial bound, and what would sharpness cost?* Here the trivial bound is  $N^{1/2} = \delta^{-1/4}$  (Proposition 13.1.5(ii)) and the truth is  $\delta^{-\varepsilon}$ .

Knowing both endpoints before starting tells one how much is at stake, and Proposition 13.1.8 shows where the boundary lies.

(d) *Is the loss  $\varepsilon$ , polylog, or a power?* These differ. Volume 12, Worked Example 12.4.3 and Proposition 13.2.3 both trace the answer to the round count of the induction:  $\log R$  rounds give a fixed power,  $\log \log R$  rounds give  $\delta^{-\varepsilon}$ . Counting rounds is the fastest way to predict the shape of the answer.

*The wider pattern.* Every major estimate in Volumes 9 to 14 is an induction on scales, and they differ mainly in what is being propagated: tube multiplicities forakeya, a weighted  $\ell^2$  hypothesis for multilinear restriction (Volume 12, Lemma 12.3.9), a surface set for the Guth–Katz incidence bound (Volume 11, Theorem 11.6.9), and here a decoupling constant. Recognising which object is being carried is the quickest route into any of these papers.

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**WORKED EXAMPLE 13.1.7— TIER B The four objects, and their scales**

Decoupling arguments manipulate four objects at once, and keeping their scales straight is most of the difficulty for a first reader.

| Object               | Lives in  | Scale                              | Introduced        |
|----------------------|-----------|------------------------------------|-------------------|
| $\mathcal{N}_\delta$ | frequency | thickness $\delta$                 | Definition 13.1.1 |
| cap $\theta$         | frequency | $\delta^{1/2} \times \delta$       | Lemma 13.1.2      |
| dual box $T_\theta$  | physical  | $\delta^{-1/2} \times \delta^{-1}$ | Lemma 13.1.2      |
| ball $B$             | physical  | $\delta^{-1}$                      | Definition 13.1.3 |

*The two pairings.* Frequency and physical scales are reciprocal:  $\delta \leftrightarrow \delta^{-1}$  and  $\delta^{1/2} \leftrightarrow \delta^{-1/2}$ . Every quantity in the volume is one of these four or a product of them.

*The counts that follow.*  $\#\theta = \delta^{-1/2}$ ;  $|T_\theta| = \delta^{-3/2}$ ;  $|B| = \delta^{-2}$ ; and  $\#\theta \cdot |T_\theta| = \delta^{-2} = |B|$ , so the dual boxes tile the ball exactly once. That identity is the check Worked Example 13.4.4 recommends, and it fails for any other cap length.

*Where each appears.* The neighbourhood defines the problem; the caps index the sum; the dual boxes carry the wave packets of Volume 12, Theorem 12.1.4; the ball is where the norm is taken. Confusing the last two is the commonest error, and Worked Example 13.3.1 explains why they differ by a square.

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## Drills

**Drill 13.1.1.** Verify Lemma 13.1.2: check that  $(t - t_0)^2 \leq \delta$  for  $|t - t_0| \leq \delta^{1/2}$ , and identify where curvature is used.

**Drill 13.1.2.** Prove Proposition 13.1.5(ii) and confirm that no property of the parabola is used.

**Drill 13.1.3.** Carry out the erroneous computation of the Pitfall — constructive region of area  $\delta^{-1}$  — and confirm it contradicts Proposition 13.1.5(i) at  $p = 2$ .

**Drill 13.1.4.** Compute  $\frac{2(n+1)}{n-1}$  for  $n = 2, \dots, 6$  and compare with the Stein–Tomas exponents of Volume 6.

## Notes

Decoupling inequalities were introduced by Wolff for the cone in 2000 and developed by Bourgain and Demeter, whose sharp paraboloid theorem is from 2015. The formulation with  $\ell^2$  outside the norm is theirs; Demeter’s book is the standard reference.

## Chapter 13.2

# Induction on Scales

Every proof of a decoupling inequality is an induction on scales, and the method has two ingredients: a *rescaling* identity saying that a piece of the problem at one scale is the whole problem at another, and a *bootstrapping* lemma converting a self-improving inequality into a subpolynomial bound. Both are elementary and both are proved here.

### 13.2.1 Parabolic rescaling

**Lemma 13.2.1** (Parabolic rescaling). *Let  $\tau \subseteq [0, 1]$  be an interval of length  $\sigma$  with left endpoint  $t_0$ , and let  $L_\tau : \mathbb{R}^2 \rightarrow \mathbb{R}^2$  be the affine map*

$$L_\tau(x_1, x_2) = \left( \frac{x_1 - t_0}{\sigma}, \frac{x_2 - t_0^2 - 2t_0(x_1 - t_0)}{\sigma^2} \right).$$

*Then  $L_\tau$  maps the arc of  $P$  over  $\tau$  onto all of  $P$ , and*

$$L_\tau(\mathcal{N}_\delta \cap \{x_1 \in \tau\}) = \mathcal{N}_{\delta/\sigma^2}(P).$$

*Proof.* Write  $t = t_0 + \sigma s$  with  $s \in [0, 1]$ , so the arc is  $\{(t_0 + \sigma s, (t_0 + \sigma s)^2)\}$ . Then

$$\frac{t - t_0}{\sigma} = s, \quad \frac{t^2 - t_0^2 - 2t_0\sigma s}{\sigma^2} = \frac{(t_0 + \sigma s)^2 - t_0^2 - 2t_0\sigma s}{\sigma^2} = \frac{\sigma^2 s^2}{\sigma^2} = s^2,$$

so  $L_\tau$  carries the arc to  $\{(s, s^2) : s \in [0, 1]\} = P$  exactly.

$L_\tau$  is affine with linear part  $\text{diag}(\sigma^{-1}, \sigma^{-2})$  composed with a shear, and the shear preserves the second coordinate's scaling. The  $\delta$ -neighbourhood is measured in the direction normal to  $P$ , which for a short arc is within a bounded factor of the  $x_2$ -direction; that direction is scaled by  $\sigma^{-2}$ , so the neighbourhood width becomes  $\delta/\sigma^2$ .  $\square$

**Where we are.** Lemma 13.2.1 is exact, not asymptotic: the image is *precisely* the parabola, because  $s \mapsto s^2$  is the same function after the substitution. That exactness is special to the parabola and to the moment curve; for a general curved surface the analogous map is only approximate and the induction acquires error terms.

The consequence to remember is the identity

$$D_p(\delta) \text{ on an arc of length } \sigma = D_p(\delta/\sigma^2) \text{ on the whole parabola.}$$

So a decoupling problem at scale  $\delta$ , restricted to an arc of length  $\sigma$ , is the *same* problem at scale  $\delta/\sigma^2$ . Taking  $\sigma = \delta^{1/2}$  gives  $\delta/\sigma^2 = 1$ : a single cap is the trivial problem, which is why  $\delta^{1/2}$  is the right cap size.

## 13.2.2 Bernstein's inequality

One further elementary tool is used repeatedly and is worth proving once.

**Proposition 13.2.2** (Bernstein's inequality on a box). *Let  $R$  be a box in  $\mathbb{R}^n$  of dimensions  $a_1 \times \cdots \times a_n$  and let  $\widehat{g}$  be supported in  $R$ . Then for  $1 \leq p \leq q \leq \infty$ ,*

$$\|g\|_{L^q(\mathbb{R}^n)} \lesssim_n |R|^{\frac{1}{p} - \frac{1}{q}} \|g\|_{L^p(\mathbb{R}^n)}.$$

*Proof.* Let  $\psi$  be Schwartz with  $\widehat{\psi} = 1$  on  $R$  and  $\widehat{\psi}$  supported in  $2R$ , obtained by translating and dilating a fixed bump. Then  $g = g * \psi$ , and  $\|\psi\|_{L^r} \approx |R|^{1-1/r}$  for every  $r$ , since  $\psi$  is essentially  $|R|$  on the dual box  $R^*$  of volume  $|R|^{-1}$  and decays rapidly beyond.

By Young's inequality with  $\frac{1}{q} = \frac{1}{p} + \frac{1}{r} - 1$ ,

$$\|g\|_{L^q} = \|g * \psi\|_{L^q} \leq \|g\|_{L^p} \|\psi\|_{L^r} \approx |R|^{1-\frac{1}{r}} \|g\|_{L^p} = |R|^{\frac{1}{p} - \frac{1}{q}} \|g\|_{L^p}. \quad \square$$

**Where we are.** Bernstein converts one exponent into another at a cost measured by the *volume of the frequency support*. It is the reason cap decompositions are useful at all: a single cap  $\theta$  has  $|\theta| = \delta^{(n+1)/2}$ , so Bernstein on one cap is cheap, while on the whole neighbourhood  $|\mathcal{N}_\delta| \approx \delta$  it is expensive.

Two cautions. It is sharp only when  $g$  is a single bump on the dual box; for oscillating  $g$  it is very lossy. And it is not an orthogonality statement — it holds for any  $g$  with the given support, which is precisely why it cannot substitute for decoupling.

---

**WORKED EXAMPLE 13.2.1— TIER A Bernstein against decoupling**

Both estimate  $\|f\|_{L^p}$  from cap data, and comparing them shows what decoupling adds.

*Bernstein, applied crudely.* Each  $f_\theta$  has frequency support of volume  $\delta^{3/2}$  in  $\mathbb{R}^2$ , so  $\|f_\theta\|_{L^6} \lesssim \delta^{3/2(1/2-1/6)} \|f_\theta\|_{L^2} = \delta^{1/2} \|f_\theta\|_{L^2}$ . Summing over the  $\delta^{-1/2}$  caps by the triangle inequality gives

$$\|f\|_{L^6} \lesssim \delta^{-1/2} \cdot \delta^{1/2} \max_\theta \|f_\theta\|_{L^2} = \max_\theta \|f_\theta\|_{L^2},$$

which is far weaker than decoupling: it controls by a maximum, losing all information about how the caps combine.

*Decoupling.*  $\|f\|_{L^6} \lesssim_\varepsilon \delta^{-\varepsilon} (\sum_\theta \|f_\theta\|_{L^6}^2)^{1/2}$ : an  $\ell^2$  sum, which for  $\delta^{-1/2}$  comparable caps is  $\delta^{-1/4}$  times the individual size.

*The comparison.* Decoupling is better exactly when the caps are *spread*, and Bernstein is competitive only when one cap dominates. Since the extremal configurations for restriction are single Knapp caps (Volume 6, Theorem 6.2.3), both are needed — and every argument in this volume uses Bernstein for the narrow case and decoupling for the broad one.

*The rule.* Bernstein trades integrability for frequency volume; decoupling trades a sum for an  $\ell^2$  sum. They are not comparable, and knowing which to reach for is a matter of asking whether the caps are concentrated or spread.

---

### 13.2.3 Bootstrapping

**Proposition 13.2.3** (A self-improving inequality gives  $\delta^{-\varepsilon}$ ). *Let  $D : (0, 1) \rightarrow [1, \infty)$  be non-increasing and suppose there are constants  $C \geq 1$  and  $\delta_0 \in (0, 1)$  with*

$$D(\delta) \leq C D(\delta^{1/2}) \quad \text{for all } \delta \in (0, \delta_0). \quad (13.2.1)$$

*Then for every  $\varepsilon > 0$  there is  $C_\varepsilon$  with  $D(\delta) \leq C_\varepsilon \delta^{-\varepsilon}$  for all  $\delta \in (0, 1)$ .*

*Proof. Step 1. Iterate.*

Applying (13.2.1)  $m$  times,

$$D(\delta) \leq C^m D(\delta^{2^{-m}}).$$

**Step 2.** *Choose the number of rounds.*

Let  $m$  be least with  $\delta^{2^{-m}} \geq \delta_0$ , that is

$$2^{-m} \log \frac{1}{\delta} \leq \log \frac{1}{\delta_0}, \quad \text{so} \quad m = \left\lceil \log_2 \frac{\log(1/\delta)}{\log(1/\delta_0)} \right\rceil.$$

Then  $D(\delta^{2^{-m}}) \leq D(\delta_0)$  since  $D$  is non-increasing.

**Step 3.** *Estimate the accumulated constant.*

Writing  $L = \log(1/\delta)$ ,

$$C^m \leq C \cdot C^{\log_2(L/\log(1/\delta_0))} = C \left( \frac{L}{\log(1/\delta_0)} \right)^{\log_2 C} \lesssim_{C, \delta_0} \left( \log \frac{1}{\delta} \right)^{\log_2 C}.$$

**Step 4.** *Conclude.*

A power of  $\log \frac{1}{\delta}$  is  $O_\varepsilon(\delta^{-\varepsilon})$  for every  $\varepsilon > 0$ , since  $(\log u)^a/u^\varepsilon \rightarrow 0$  as  $u \rightarrow \infty$ . Hence  $D(\delta) \leq C_\varepsilon \delta^{-\varepsilon}$ .  $\square$

**Where we are.** Step 2 is where the  $\varepsilon$  comes from, and it is worth comparing with Volume 12, Worked Example 12.4.3. There the point was that an induction with  $\log R$  rounds produces a fixed power and only one with  $\log \log R$  rounds produces  $R^\varepsilon$ . Here the round count is  $m \approx \log_2 \log \frac{1}{\delta}$  — a log log — because each round *squares* the scale rather than multiplying it by a constant.

That is exactly why Lemma 13.2.1 is stated with  $\delta \mapsto \delta/\sigma^2$ : the square is what makes the induction terminate in log log steps.

---

**WORKED EXAMPLE 13.2.2— TIER A How slowly the bound actually kicks in**

Proposition 13.2.3 concludes  $D(\delta) \lesssim_\varepsilon \delta^{-\varepsilon}$ , and it is worth seeing how weak that is for any  $\delta$  one might name.

Take  $C = 4$  and  $\delta_0 = \frac{1}{2}$ , so  $C^m = 4^{\lceil \log_2 \log_2(1/\delta) \rceil}$ .

| $\delta$    | rounds $m$ | $C^m$             | $\delta^{-0.1}$      |
|-------------|------------|-------------------|----------------------|
| $10^{-6}$   | 5          | $1.0 \times 10^3$ | 4.0                  |
| $10^{-12}$  | 6          | $4.1 \times 10^3$ | 16                   |
| $10^{-30}$  | 7          | $1.6 \times 10^4$ | $1.0 \times 10^3$    |
| $10^{-100}$ | 9          | $2.6 \times 10^5$ | $1.0 \times 10^{10}$ |

*Reading the table.* The polylogarithmic bound  $C^m$  exceeds  $\delta^{-0.1}$  until  $\delta$  is somewhere below  $10^{-30}$ . The theorem is true — polylog is eventually smaller than any power — but the crossover is astronomically late.

*Why this is worth saying.* An  $\varepsilon$ -loss is often described as “harmless”. It is harmless *asymptotically*, and Volume 9, Proposition 9.7.3 showed it is exactly the accuracy dimension statements need. It is not harmless for any numerical purpose, and a reader who imagines the constants are moderate will be surprised.

*The honest formulation.*  $D_p(\delta)$  grows slower than any power of  $\delta^{-1}$ ; the



implicit constant  $C_\varepsilon$  blows up as  $\varepsilon \rightarrow 0$ , and no proof in this subject controls it usefully.

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**WORKED EXAMPLE 13.2.3— TIER B The shape of a decoupling proof**

Lemma 13.2.1 and Proposition 13.2.3 are the frame; every decoupling argument fills it in the same way.

*Step 1: split the frequency support.* Partition  $\mathcal{N}_\delta$  into caps at an intermediate scale  $\rho$  with  $\delta^{1/2} < \rho < 1$ .

*Step 2: separate transverse from parallel.* Tuples of caps that are transverse are handled by a multilinear estimate — the broad case, exactly as in Volume 12, Proposition 12.4.3. Tuples that cluster are handled by rescaling.

*Step 3: rescale the clustered part.* By Lemma 13.2.1 a cluster inside an arc of length  $\sigma$  is the same problem at scale  $\delta/\sigma^2$ , so it contributes  $D_p(\delta/\sigma^2)$ .

*Step 4: assemble into a self-improving inequality.* Steps 2 and 3 give  $D_p(\delta) \leq C \cdot D_p(\delta/\sigma^2) \cdot (\text{transverse factor})$ , and choosing  $\sigma$  so that  $\delta/\sigma^2 = \delta^{1/2}$  — that is  $\sigma = \delta^{1/4}$  — puts it in the form of (13.2.1).

*Step 5: bootstrap.* Proposition 13.2.3.

*What is hard.* Only Step 2. The transverse estimate is *ball inflation*, Chapter 13.3, and it is where multilinear Keakeya enters. Everything else in this chapter is bookkeeping, and that is why the chapter is short.

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**WORKED EXAMPLE 13.2.4— TIER B Rescaling on other surfaces**

Lemma 13.2.1 is exact for the parabola. That exactness is special, and knowing where it fails is what tells one which surfaces are tractable.

(a) *The parabola and paraboloid.* Exact, by the identity  $|\xi|^2 - |\xi_0|^2 - 2\xi_0 \cdot (\xi - \xi_0) = |\xi - \xi_0|^2$ . The affine map is available because the defining function is a quadratic form.

(b) *The moment curve.* Exact, by the same mechanism: the map  $t \mapsto t_0 + \sigma t$  carries  $(t, t^2, \dots, t^k)$  to an affine image of itself, since each  $t^j$  expands in lower powers.

(c) *The cone.* Exact under *Lorentz* rescaling rather than parabolic, because the cone is invariant under the Lorentz group rather than under parabolic dilations. Remark 13.8.4 records that the induction is otherwise the same.

(d) *The sphere.* Only approximate: a cap of the sphere is affinely equivalent to a cap of the paraboloid up to errors of higher order. In practice one passes to the paraboloid first, which is legitimate because Worked Example 13.5.2(a)

makes every elliptic surface locally a paraboloid graph.

(e) *A general curved surface.* Approximate, with errors that must be tracked through the induction. This is the technical reason decoupling theorems are usually stated for model surfaces and transferred by affine equivalence.

*The check.* Given a surface, ask whether the group acting on it — affine, Lorentz, or none — carries caps to the whole surface. If yes, the induction is clean; if no, expect error terms and a longer argument.

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## 13.2.4 A worked induction

The abstract statements of this chapter are easier to trust after running one induction concretely. We prove a decoupling-type estimate that is far weaker than Theorem 13.1.4 but is genuinely obtained by induction on scales, with every constant explicit.

**Proposition 13.2.4** (A crude self-improvement). *For  $2 \leq p \leq 6$ ,*

$$D_p(\delta) \leq C D_p(\delta^{1/2}) D_p^{\text{loc}}(\delta^{1/2}),$$

where  $D_p^{\text{loc}}(\sigma)$  is the decoupling constant for arcs of length  $\sigma$  subdivided into arcs of length  $\sigma \cdot \sigma^{1/2}$ .

*Proof.* Let  $\{\tau\}$  be the arcs of length  $\delta^{1/4}$  and  $\{\theta\}$  those of length  $\delta^{1/2}$ , so each  $\tau$  contains  $\delta^{-1/4}$  of the  $\theta$ .

**Step 1.** *Decouple into the intermediate arcs.*

The function  $f$  has Fourier support in  $\mathcal{N}_\delta \subseteq \mathcal{N}_{\delta^{1/2}}(P)$ , and the arcs  $\tau$  have length  $\delta^{1/4} = (\delta^{1/2})^{1/2}$ , so they are the natural caps at scale  $\delta^{1/2}$ . By definition of the decoupling constant at that scale,

$$\|f\|_{L^p} \leq D_p(\delta^{1/2}) \left( \sum_{\tau} \|f_\tau\|_{L^p}^2 \right)^{1/2}.$$

**Step 2.** *Decouple each intermediate arc.*

Fix  $\tau$ . By Lemma 13.2.1 the arc of length  $\delta^{1/4}$  at scale  $\delta$  is an affine image of the whole parabola at scale  $\delta/(\delta^{1/4})^2 = \delta^{1/2}$ , and under that image the arcs  $\theta \subseteq \tau$  become the caps of length  $(\delta^{1/2})^{1/2}$ . Hence

$$\|f_\tau\|_{L^p} \leq D_p^{\text{loc}}(\delta^{1/2}) \left( \sum_{\theta \subseteq \tau} \|f_\theta\|_{L^p}^2 \right)^{1/2}.$$

**Step 3.** *Combine.*

Substituting Step 2 into Step 1 and interchanging the sums,

$$\|f\|_{L^p} \leq D_p(\delta^{1/2}) D_p^{\text{loc}}(\delta^{1/2}) \left( \sum_{\theta} \|f_{\theta}\|_{L^p}^2 \right)^{1/2}. \quad \square$$

**Where we are.** Proposition 13.2.4 is the *multiplicativity* of decoupling constants across scales, and it is the skeleton of every argument in the volume. Note what it does and does not give.

*What it gives.* If one can show  $D_p^{\text{loc}}(\sigma) \leq C$  uniformly — a bounded constant for one step — then Proposition 13.2.4 is exactly the hypothesis (13.2.1) of Proposition 13.2.3, and  $D_p(\delta) \lesssim_{\varepsilon} \delta^{-\varepsilon}$  follows.

*What it does not give.* The bound on  $D_p^{\text{loc}}$ . That is the entire content of the theorem, and it is what Chapter 13.3 supplies.

*Why state it separately.* Because it isolates the bookkeeping from the mathematics. A reader who has understood Proposition 13.2.4 and Proposition 13.2.3 knows the shape of the proof of Theorem 13.1.4 completely, and needs only the one transverse estimate.

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WORKED EXAMPLE 13.2.5— TIER A **Running the induction numerically**

Take  $C = 4$  in (13.2.1) and  $\delta_0 = \frac{1}{2}$ , and follow the scales.

| round | scale                 | $D$ bound                 |           |
|-------|-----------------------|---------------------------|-----------|
| 0     | $\delta = 10^{-16}$   | —                         | start     |
| 1     | $10^{-8}$             | $\leq 4 D(10^{-8})$       |           |
| 2     | $10^{-4}$             | $\leq 4^2 D(10^{-4})$     |           |
| 3     | $10^{-2}$             | $\leq 4^3 D(10^{-2})$     |           |
| 4     | $10^{-1}$             | $\leq 4^4 D(10^{-1})$     |           |
| 5     | $\approx \frac{1}{2}$ | $\leq 4^5 D(\frac{1}{2})$ | base case |

*Five rounds* for  $\delta = 10^{-16}$ , and  $4^5 = 1024$ . Against  $\delta^{-1/4} = 10^4$ , the trivial bound, the induction has gained a factor 10. Against  $\delta^{-\varepsilon}$  with  $\varepsilon = 0.1$ , which is  $10^{1.6} \approx 40$ , it has *not* yet won — consistent with Worked Example 13.2.2, which shows the crossover occurs far below  $10^{-30}$ .

*The lesson.* The theorem is asymptotic and the constants are bad. What the induction guarantees is that the exponent is 0; it says nothing useful at any scale one might compute with.

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## Drills

**Drill 13.2.1.** Verify Lemma 13.2.1 by direct substitution, and confirm the image is exactly  $P$  rather than approximately.

**Drill 13.2.2.** Check that  $\sigma = \delta^{1/2}$  gives  $\delta/\sigma^2 = 1$ , and interpret.

**Drill 13.2.3.** Carry out Steps 1–3 of Proposition 13.2.3 with  $C = 2$ ,  $\delta_0 = e^{-1}$  and  $\delta = e^{-16}$ , computing  $m$  and  $C^m$  explicitly.

**Drill 13.2.4.** Determine, for  $C = 4$  and  $\varepsilon = 0.1$ , the value of  $\delta$  below which  $C^m \leq \delta^{-\varepsilon}$ , and compare with Worked Example 13.2.2.

## Notes

Induction on scales in this form is due to Bourgain, and the parabolic rescaling identity is classical. The observation that the round count is  $\log \log$  rather than  $\log$ , and that this is what produces the  $\varepsilon$ -loss, is standard bookkeeping made explicit here and in Volume 12, Worked Example 12.4.3.

## Chapter 13.3

# Ball Inflation

Worked Example 13.2.3 identified the one hard step: the transverse case. This chapter supplies it. The name *ball inflation* refers to the mechanism: information about  $f$  on balls of radius  $\rho^{-1}$  is upgraded to information on the much larger balls of radius  $\rho^{-2}$ , and the upgrade is paid for by transversality.

### 13.3.1 The bilinear input

**Lemma 13.3.1** (The addition map on the parabola). *Let  $\Phi(t, s) = (t, t^2) + (s, s^2)$ . Then*

$$|\det d\Phi(t, s)| = 2|t - s|.$$

*Proof.*  $\Phi(t, s) = (t + s, t^2 + s^2)$ , so  $d\Phi = \begin{pmatrix} 1 & 1 \\ 2t & 2s \end{pmatrix}$ , of determinant  $2s - 2t$ .  $\square$

**Where we are.** This is the same computation as Volume 12, Lemma 12.6.2, where the Jacobian of the addition map on the circle was  $|\sin \angle(\omega_1, \omega_2)|$ . In both cases *transversality is exactly the non-degeneracy of the addition map*, and in both cases that is the whole of the bilinear estimate.

For the parabola the transversality parameter is  $|t - s|$ , the separation of the arcs in the parameter. Two arcs at distance  $\gtrsim \nu$  have Jacobian  $\gtrsim \nu$ .

**Proposition 13.3.2** (Bilinear  $L^2$  for transverse arcs). *Let  $\tau_1, \tau_2 \subseteq [0, 1]$  be intervals with  $\text{dist}(\tau_1, \tau_2) \geq \nu$ , and let  $f_j$  have Fourier support in the part of  $\mathcal{N}_\delta$  over  $\tau_j$ . Then*

$$\|f_1 f_2\|_{L^2(\mathbb{R}^2)} \lesssim \nu^{-1/2} \|f_1\|_{L^2} \|f_2\|_{L^2}.$$

*Proof.* By Plancherel,  $\|f_1 f_2\|_{L^2} = \|\widehat{f_1} * \widehat{f_2}\|_{L^2}$ . The convolution is supported in  $\Phi(\tau_1 \times \tau_2)$  thickened by  $O(\delta)$ , and by Lemma 13.3.1 the map  $\Phi$  has Jacobian

$\geq 2\nu$  there, so it is a local diffeomorphism with boundedly many preimages. Changing variables,

$$\|\widehat{f}_1 * \widehat{f}_2\|_{L^2}^2 \lesssim \nu^{-1} \|\widehat{f}_1\|_{L^2}^2 \|\widehat{f}_2\|_{L^2}^2,$$

exactly as in Volume 12, Theorem 12.6.1.  $\square$

### 13.3.2 Ball inflation

**Definition 13.3.3** (Averaged norms). For a ball  $\Delta$  and  $g \geq 0$  write  $\text{avg}_\Delta g = |\Delta|^{-1} \int_\Delta g$ . For  $f$  with Fourier support in  $\mathcal{N}_\delta$  and  $\rho \geq \delta^{1/2}$ , let  $\{\tau\}$  be the arcs of length  $\rho$  and  $\{\theta\}$  those of length  $\delta^{1/2}$ , and put

$$\mathcal{S}_\tau f = \left( \sum_{\theta \subseteq \tau} |f_\theta|^2 \right)^{1/2}.$$

**Theorem 13.3.4** (Ball inflation, bilinear form). *Let  $\tau_1, \tau_2$  be arcs of length  $\rho$  with  $\text{dist}(\tau_1, \tau_2) \geq \nu\rho$ , and let  $B$  be a ball of radius  $\rho^{-2}$ . Then*

$$\frac{1}{|\mathcal{D}|} \sum_{\Delta \in \mathcal{D}} \prod_{j=1}^2 \left( \text{avg}_\Delta (\mathcal{S}_{\tau_j} f)^2 \right)^{1/2} \lesssim_{\nu, \varepsilon} \rho^{-\varepsilon} \prod_{j=1}^2 \left( \text{avg}_B (\mathcal{S}_{\tau_j} f)^2 \right)^{1/2},$$

where  $\mathcal{D}$  is the family of balls of radius  $\rho^{-1}$  tiling  $B$ .

*Proof.* Each  $f_\theta$  has Fourier support in a  $\delta^{1/2} \times \delta$  box, so by Volume 12, Lemma 12.1.2,  $|f_\theta|^2$  is essentially constant on translates of the dual box, a tube  $T_\theta$  of dimensions  $\delta^{-1/2} \times \delta^{-1}$  oriented normal to  $\theta$ . Since  $\rho \geq \delta^{1/2}$ , on the scale  $\rho^{-1}$  the relevant tubes are the  $\rho^{-1} \times \rho^{-2}$  tubes normal to  $\tau_j$ .

Write  $g_j = (\mathcal{S}_{\tau_j} f)^2$ , a sum of such tube indicators with coefficients. The left-hand side is then, up to constants,

$$\frac{1}{|\mathcal{D}|} \sum_{\Delta} \prod_{j=1}^2 \left( \text{avg}_\Delta g_j \right)^{1/2},$$

and the tubes contributing to  $g_1$  and to  $g_2$  point in directions separated by  $\gtrsim \nu\rho$ , because the normals to  $\tau_1$  and  $\tau_2$  differ by the arc separation. Rescaling  $B$  to the unit ball turns them into two  $\nu$ -transverse families of tubes of radius  $\rho^{-1} \cdot \rho^2 = \rho$ .

Now apply the multilinear Kakeya inequality in the plane — Volume 11, Theorem 11.5.7 with  $n = 2$ , where the exponent  $\frac{1}{n-1}$  is 1 and the statement reduces to Loomis–Whitney, that is to Fubini:

$$\int \prod_{j=1}^2 \left( \sum_{T \in \mathbb{T}_j} \mathbf{1}_T \right) \lesssim_{\nu, \varepsilon} \rho^{-\varepsilon} \prod_{j=1}^2 \left( \rho \# \mathbb{T}_j \right).$$

Translating back through the rescaling gives the claim.  $\square$

**Where we are.** Two remarks on what this proof does and does not need. *It uses multilinear Kakeya, which is proved.* Volume 11, Theorem 11.5.4 was established from Loomis–Whitney by Guth’s short argument, so Theorem 13.3.4 rests on nothing unproved. In the plane the input degenerates to Fubini, which is why the two-dimensional case is elementary; in higher dimensions the full strength of Theorem 11.5.7 is required, and that is the sense in which decoupling for the paraboloid in  $\mathbb{R}^n$  is a genuinely  $n$ -dimensional theorem.

*The name.* The average on the left is over balls of radius  $\rho^{-1}$ ; the average on the right is over the single ball  $B$  of radius  $\rho^{-2}$ . The inequality says the finer averages are controlled by the coarser one — the ball has been *inflated* from  $\rho^{-1}$  to  $\rho^{-2}$ . Since  $\rho^{-2}$  is the square of  $\rho^{-1}$ , iterating inflates scales by squaring, which is exactly the log log round count of Proposition 13.2.3.

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WORKED EXAMPLE 13.3.1— TIER A **Why the exponent  $\rho^{-2}$  and not  $\rho^{-1}$**

The inflation goes from  $\rho^{-1}$  to  $\rho^{-2}$ , and the reason is the geometry of the caps, not a choice.

*The dual box.* By Lemma 13.1.2, an arc of length  $\rho$  on the parabola sits in a  $\rho \times \rho^2$  box. Its dual is  $\rho^{-1} \times \rho^{-2}$ .

*What that means.* A function with Fourier support in one arc of length  $\rho$  is essentially constant on balls of radius  $\rho^{-1}$  — the short direction of the dual box — but retains structure out to  $\rho^{-2}$ , the long direction. So  $\rho^{-1}$  is the scale at which the caps are indistinguishable and  $\rho^{-2}$  the scale at which they are fully resolved.

*The numerology.* With  $\rho = \delta^{1/2}$  one has  $\rho^{-2} = \delta^{-1}$ , the largest scale in the problem. So one round of inflation at  $\rho = \delta^{1/2}$  already reaches the top, and the induction is over intermediate  $\rho$ .

*Numerically,*  $\delta = 10^{-6}$ .

| $\rho$    | $\rho^{-1}$ | $\rho^{-2}$ |                       |
|-----------|-------------|-------------|-----------------------|
| $10^{-1}$ | 10          | $10^2$      | first round           |
| $10^{-2}$ | $10^2$      | $10^4$      | second                |
| $10^{-3}$ | $10^3$      | $10^6$      | reaches $\delta^{-1}$ |

Three rounds, each squaring — and  $\log_2 \log_{10} 10^6 \approx 2.6$ , matching Proposition 13.2.3.

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**WORKED EXAMPLE 13.3.2— TIER B Ball inflation against multilinear Makeya**

The two statements look different and are the same.

*Multilinear Makeya* bounds  $\int \prod_j (\sum_T \mathbf{1}_T)^{1/(n-1)}$  for transverse tube families: a purely geometric statement about overlaps.

*Ball inflation* bounds an average of products of square functions over small balls by the same over a large ball: an analytic statement about functions.

*The dictionary.* Volume 12, Worked Example 12.1.2 converts a function with Fourier support in a cap into a sum over tubes. Under it,  $\mathcal{S}_\tau f$  becomes  $\sum_T \mathbf{1}_T$  and the two statements coincide. The transversality hypotheses correspond because the normal to an arc determines the tube direction.

*Which is more general.* Neither: they are equivalent given the dictionary. But multilinear Makeya is the one with the short proof (Theorem 11.5.4), so it is the one to prove and the other to derive — exactly as Volume 12 derived multilinear restriction from discrete Loomis–Whitney rather than the reverse.

*The transferable point.* When an analytic estimate resists, look for the geometric statement it encodes. Three times in this series the geometric form has been the tractable one: Makeya for restriction, Loomis–Whitney for multilinear restriction, and multilinear Makeya for ball inflation.

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**WORKED EXAMPLE 13.3.3— TIER B Reading a ball inflation statement**

Ball inflation is stated in several forms and the differences matter.

(1) *Which two scales?* Here  $\rho^{-1}$  and  $\rho^{-2}$ , and Worked Example 13.3.1 explains why the square. A statement relating  $\rho^{-1}$  to  $\rho^{-3/2}$  is a different and weaker assertion.

(2) *How many factors?* The bilinear form given here suffices in the plane; in  $\mathbb{R}^n$  one needs  $n$ , by Volume 12, Proposition 12.2.3.

(3) *Which exponent on the square function?* The statement involves  $(\text{avg}_\Delta(\mathcal{S}_\tau f)^2)^{1/2}$ ; other sources use  $(\text{avg}_\Delta(\mathcal{S}_\tau f)^p)^{1/p}$  and interpolate. The  $L^2$  form is the one that pairs with multilinear Makeya directly.

(4) *Is the loss  $\rho^{-\varepsilon}$  or a constant?* Here  $\rho^{-\varepsilon}$ , inherited from Theorem 11.5.7. Since inflation is applied  $O(\log \log \delta^{-1})$  times, an  $\varepsilon$ -loss per round is affordable and a fixed constant per round would also be — but a power of  $\rho$  would not.

*A worked check.* Set  $\rho = \delta^{1/2}$ . Then  $\rho^{-1} = \delta^{-1/2}$  and  $\rho^{-2} = \delta^{-1}$ : one round carries the finest caps to the full scale, so the statement reduces to a single application. Any formulation that does not have this property at  $\rho = \delta^{1/2}$  has been misquoted.

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**WORKED EXAMPLE 13.3.4— TIER C: RESEARCH TASTE    The three inductions on scales, compared**

By this point the series contains three fully proved inductions on scales. They have the same skeleton and differ in what they propagate, and seeing them side by side is the fastest way to read a fourth.

|            | <b>multilinear<br/>Kakeya</b> | <b>multilinear re-<br/>striction</b> | <b>decoupling</b>                 |
|------------|-------------------------------|--------------------------------------|-----------------------------------|
| Where      | Vol 11, Theo-<br>rem 11.5.4   | Vol 12, Proposi-<br>tion 12.3.10     | Proposition 13.2.3                |
| Propagated | tube multiplicities           | a weighted $\ell^2$ hy-<br>pothesis  | the constant<br>$D_p(\delta)$     |
| Scale step | $W \rightarrow \delta^{-1}W$  | $R \rightarrow \delta^{-1}R$         | $\delta \rightarrow \delta^{1/2}$ |
| Rounds     | $\log S / \log \delta^{-1}$   | $\log R / \log \delta^{-1}$          | $\log \log \delta^{-1}$           |
| Engine     | Loomis–Whitney                | discrete Loomis–<br>Whitney          | ball inflation                    |
| Loss       | $S^\varepsilon$               | $R^\varepsilon$                      | $\delta^{-\varepsilon}$           |

*The common skeleton.* Fix a small parameter; show one scale controls the next at bounded cost; iterate; choose the parameter at the end so the accumulated cost is subpolynomial.

*The one real difference.* What is carried. Guth carries a geometric quantity, Bejenaru an analytic hypothesis with off-diagonal decay, and decoupling a single number. The third is easiest, which is why Chapter 13.4 is the shortest of the three proofs despite proving the newest theorem.

*Where the round count differs.* The first two square the *scale*  $\delta^{-1}$  per round and so take  $\log / \log$  rounds; decoupling squares  $\delta$  itself and takes  $\log \log$ . Both give  $\varepsilon$ -losses, but by different bookkeeping — and Volume 12, Worked Example 12.4.3 warns that getting this wrong produces a fixed power instead.

*Reading a fourth.* Given a new induction-on-scales paper, identify these six rows before reading any proof. In every case in this series they determine the shape of the argument, and usually the difficulty.

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### 13.3.3 The transverse estimate, assembled

It is worth stating explicitly what Theorem 13.3.4 delivers when iterated, since Chapter 13.4 uses it as a single step.

**Proposition 13.3.5** (Iterated inflation). *Let  $\tau_1, \tau_2$  be  $\nu$ -transverse arcs of*

length  $K^{-1}$  and let  $B$  be a ball of radius  $\delta^{-1}$ . Then for every  $\varepsilon > 0$ ,

$$\frac{1}{|\mathcal{D}_{\delta^{-1/2}}|} \sum_{\Delta} \prod_{j=1}^2 \left( \text{avg}_{\Delta}(\mathcal{S}_{\tau_j} f)^2 \right)^{1/2} \lesssim_{K, \nu, \varepsilon} \delta^{-\varepsilon} \prod_{j=1}^2 \left( \text{avg}_B(\mathcal{S}_{\tau_j} f)^2 \right)^{1/2},$$

where  $\mathcal{D}_{\delta^{-1/2}}$  tiles  $B$  by balls of radius  $\delta^{-1/2}$ .

*Proof.* Apply Theorem 13.3.4 at scales  $\rho = K^{-1}, K^{-2}, K^{-4}, \dots$ , each round replacing  $\rho$  by  $\rho^2$  and costing  $\rho^{-\varepsilon'}$ . Starting from  $\rho = K^{-1}$  and squaring, after  $m$  rounds one has  $\rho = K^{-2^m}$ , so reaching  $\rho = \delta^{1/2}$  requires

$$2^m \log K = \frac{1}{2} \log \delta^{-1}, \quad \text{that is} \quad m = \log_2 \left( \frac{\log \delta^{-1}}{2 \log K} \right).$$

The accumulated cost is  $\prod_{i < m} K^{-2^i \varepsilon'} \leq \delta^{-\varepsilon'}$ , and choosing  $\varepsilon' = \varepsilon$  gives the claim. *The round count is  $\log \log \delta^{-1}$ , matching Proposition 13.2.3.*  $\square$

**Where we are.** The round count is worth pausing on, because it is the third appearance of the same phenomenon.

Volume 12, Worked Example 12.4.3 observed that an induction with  $\log R$  rounds gives a fixed power and only  $\log \log R$  rounds give  $R^\varepsilon$ . Proposition 13.2.3 produced  $\log \log \delta^{-1}$  by squaring the scale. Proposition 13.3.5 produces it again, by squaring  $\rho$ .

*Why squaring is natural here.* The dual box of a  $\rho$ -cap is  $\rho^{-1} \times \rho^{-2}$  (Worked Example 13.3.1), so the two scales present in any cap differ by a square. An induction that moves from one to the other necessarily squares, and hence necessarily terminates in  $\log \log$  steps. *The  $\varepsilon$ -loss in decoupling is a consequence of the parabola's second order contact*, not of any inefficiency in the argument.

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WORKED EXAMPLE 13.3.5— TIER C: RESEARCH TASTE **Reading an induction-on-scales paper**

By the end of this volume the reader has seen four induction-on-scales proofs in full. A checklist for the fifth.

(1) *What is propagated?* A constant, a geometric count, or a weighted hypothesis. Worked Example 13.3.4 tabulates the three cases in this series; the answer determines how the induction step is stated.

(2) *What is the scale step?* Multiplying by a constant, or squaring. This determines the round count and hence the loss: Proposition 13.2.3 and Volume 12, Worked Example 12.4.3. A paper claiming an  $\varepsilon$ -loss from a log-round induction has an error.

(3) *What is the base case?* Usually a scale below which the statement is trivial. In Theorem 11.5.4 it is  $L^\varepsilon \leq 2D$ ; in Theorem 12.3.1 it is  $R \leq \delta^{-2}$ . Checking that the base case is genuinely trivial is quick and occasionally catches a gap.

(4) *Where is the non-degeneracy?* Every such proof has a determinant or separation hypothesis, and identifying it tells you what the degenerate case costs. Worked Example 12.2.5 of Volume 12 lists four instances.

(5) *In what order are the parameters chosen?* Almost always:  $\varepsilon$  first, then the auxiliary parameter, then all scales. Reading them in the wrong order makes the argument look circular, and this is the single commonest source of confusion. Theorem 11.5.4's Checkpoint states it explicitly for that theorem.

(6) *What would break in higher dimensions?* Usually the transverse estimate; see Worked Example 13.5.5.

*The payoff.* These six questions can be answered from the statement of the main lemma and the choice of parameters, typically in the first three pages. If they can be answered, the rest of the paper is verification, and it can be read quickly or skipped.

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## Drills

**Drill 13.3.1.** Verify Lemma 13.3.1 and compare with Volume 12, Lemma 12.6.2 for the circle.

**Drill 13.3.2.** Check that at  $n = 2$  the exponent  $\frac{1}{n-1}$  in Theorem 11.5.7 equals 1, and that the statement is then Fubini.

**Drill 13.3.3.** For  $\delta = 10^{-8}$ , tabulate the inflation rounds as in Worked Example 13.3.1 and count them.

**Drill 13.3.4.** Explain why  $\rho \geq \delta^{1/2}$  is required in Definition 13.3.3.

## Notes

Ball inflation is Theorem 6.6 of Bourgain and Demeter's paper, where it is derived from the multilinear Kakeya inequality of Bennett, Carbery and Tao. The planar case given here degenerates to Fubini; the general case needs Theorem 11.5.7, proved in Volume 11.



## Chapter 13.4

# Decoupling for the Parabola

Everything is now in place. Lemma 13.2.1 handles clustered caps, Proposition 13.2.3 converts a self-improving inequality into a subpolynomial bound, and Theorem 13.3.4 supplies the transverse case. This chapter assembles them.

### 13.4.1 Broad and narrow, again

**Lemma 13.4.1** (The bilinear reduction). *Fix  $K \geq 2$  and partition  $[0, 1]$  into  $K$  arcs  $\tau$ . For every  $x$ ,*

$$|f(x)| \leq K \max_{\tau} |f_{\tau}(x)| \quad \text{and} \quad |f(x)| \leq K^2 \max_{\text{dist}(\tau_1, \tau_2) \geq K^{-1}} |f_{\tau_1}(x) f_{\tau_2}(x)|^{1/2} + 3 \max_{\tau} |f_{\tau}(x)|.$$

*Proof.* The first is the triangle inequality over the  $K$  arcs.

For the second, let  $\tau^*$  maximise  $|f_{\tau}(x)|$ . If some  $\tau$  with  $\text{dist}(\tau, \tau^*) \geq K^{-1}$  has  $|f_{\tau}(x)| \geq K^{-2}|f(x)|$ , then

$$|f_{\tau^*}(x) f_{\tau}(x)|^{1/2} \geq (K^{-1}|f(x)| \cdot K^{-2}|f(x)|)^{1/2} \geq K^{-2}|f(x)|,$$

using  $|f_{\tau^*}(x)| \geq K^{-1}|f(x)|$  from the first bound; this gives the first term. Otherwise every arc not adjacent to  $\tau^*$  contributes at most  $K^{-2}|f(x)|$ , so summing over the at most  $K$  of them leaves  $|f(x)| \leq 3 \max_{\tau} |f_{\tau}(x)| + K^{-1}|f(x)|$ , and absorbing gives the second term.  $\square$

**Where we are.** This is the broad–narrow dichotomy of Volume 12, Proposition 12.4.3, in the form decoupling needs. Two differences.

*Two factors, not  $n$ .* In the plane two directions span, so Proposition 12.2.3 of Volume 12 imposes no obstruction and the bilinear form suffices. In  $\mathbb{R}^n$  one needs  $n$  factors and Chapter 13.5 says so.

*The narrow term is a single arc, not a cluster.* Because the second bound compares against  $\max_{\tau} |f_{\tau}|$  directly, the narrow case is handled by one ap-

plication of Lemma 13.2.1, not by an induction on dimension. That is why decoupling for the parabola is markedly simpler than restriction for the sphere.

## 13.4.2 The theorem

**Theorem 13.4.2** (Bourgain–Demeter for the parabola). *For  $2 \leq p \leq 6$  and every  $\varepsilon > 0$ ,*

$$D_p(\delta) \lesssim_{\varepsilon} \delta^{-\varepsilon}.$$

*Proof.* By interpolation with Proposition 13.1.5(i) it suffices to treat  $p = 6$ .

**Step 1.** *Split by Lemma 13.4.1.*

Raising the second bound to the sixth power and integrating over  $B$ ,

$$\|f\|_{L^6(B)}^6 \lesssim K^{12} \max_{\tau_1, \tau_2 \text{ transverse}} \| |f_{\tau_1} f_{\tau_2}|^{1/2} \|_{L^6(B)}^6 + 3^6 \sum_{\tau} \|f_{\tau}\|_{L^6(B)}^6.$$

**Step 2.** *The narrow term, by rescaling.*

Each  $\tau$  has length  $K^{-1}$ , so by Lemma 13.2.1 the arc is an affine image of the whole parabola at scale  $\delta K^2$ . Hence

$$\|f_{\tau}\|_{L^6} \leq D_6(\delta K^2) \left( \sum_{\theta \subseteq \tau} \|f_{\theta}\|_{L^6}^2 \right)^{1/2},$$

and summing over the  $K$  arcs,

$$\left( \sum_{\tau} \|f_{\tau}\|_{L^6}^6 \right)^{1/6} \leq D_6(\delta K^2) \left( \sum_{\theta} \|f_{\theta}\|_{L^6}^2 \right)^{1/2}.$$

**Step 3.** *The broad term, by ball inflation.*

Fix transverse  $\tau_1, \tau_2$ . The quantity  $\| |f_{\tau_1} f_{\tau_2}|^{1/2} \|_{L^6}^6 = \|f_{\tau_1} f_{\tau_2}\|_{L^3}^3$  is estimated by interpolating Proposition 13.3.2 — which controls the  $L^2$  norm with no loss — against the trivial  $L^\infty$  bound, and then applying Theorem 13.3.4 at each intermediate scale  $\rho = K^{-1}, K^{-2}, \dots$  down to  $\delta^{1/2}$ .

Each round of inflation replaces averages over  $\rho^{-1}$ -balls by averages over  $\rho^{-2}$ -balls at a cost  $\rho^{-\varepsilon}$ , and passes from the square function at scale  $\rho$  to the square function at scale  $\rho^2$ . After  $O(\log \log \delta^{-1})$  rounds one reaches scale  $\delta^{1/2}$ , and the total cost is  $\delta^{-O(\varepsilon)}$ . The outcome is

$$\| |f_{\tau_1} f_{\tau_2}|^{1/2} \|_{L^6(B)} \lesssim_{K, \varepsilon} \delta^{-\varepsilon} \left( \sum_{\theta} \|f_{\theta}\|_{L^6}^2 \right)^{1/2}.$$

**Step 4.** *The recursion.*

Combining Steps 1–3,

$$D_6(\delta) \leq C(K, \varepsilon) \delta^{-\varepsilon} + C D_6(\delta K^2). \quad (13.4.1)$$

Choosing  $K = \delta^{-1/4}$  makes  $\delta K^2 = \delta^{1/2}$ , so (13.4.1) is exactly the self-improving inequality (13.2.1) of Proposition 13.2.3, and that proposition gives  $D_6(\delta) \lesssim_\varepsilon \delta^{-\varepsilon}$ .  $\square$

**Where we are.** The proof is short because every ingredient was prepared. It is worth recording which volume each came from.

| Ingredient          | Where proved           | Rests on               |
|---------------------|------------------------|------------------------|
| Parabolic rescaling | Lemma 13.2.1           | algebra                |
| Bootstrapping       | Proposition 13.2.3     | log log round count    |
| Bilinear $L^2$      | Proposition 13.3.2     | Plancherel, a Jacobian |
| Ball inflation      | Theorem 13.3.4         | multilinear Keakeya    |
| Multilinear Keakeya | Vol 11, Theorem 11.5.4 | Loomis–Whitney         |
| Loomis–Whitney      | Vol 11, Theorem 11.5.2 | Hölder                 |

*The whole chain terminates in Hölder’s inequality.* That is the payoff of the work in Volume 11: decoupling for the parabola, one of the central theorems of the last decade, rests on nothing deeper.

*Where  $p = 6$  enters.* Step 3, in the interpolation:  $L^6$  of a half-product is  $L^3$  of the product, and 3 is where the bilinear  $L^2$  estimate and the trivial  $L^\infty$  bound meet after ball inflation. Beyond  $p = 6$  the interpolation fails, consistently with Proposition 13.1.8, whose exponent  $-\frac{1}{4} + \frac{3}{2p}$  is  $+\frac{1}{8}$ , 0,  $-\frac{1}{16}$  at  $p = 4, 6, 8$ .

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WORKED EXAMPLE 13.4.1— TIER A **Decoupling implies a restriction estimate**

Worked Example 13.1.2 promised that the critical exponent 6 agrees with Stein–Tomas because decoupling implies restriction. Here is the deduction.

*Setup.* Let  $g$  be supported on the parabola and  $f = \mathcal{E}g$  its extension, localised to  $B_R$  with  $\delta = R^{-1}$ . Then  $\hat{f}$  is supported in  $\mathcal{N}_\delta$ .

*Apply decoupling.* By Theorem 13.4.2 at  $p = 6$ ,

$$\|\mathcal{E}g\|_{L^6(B_R)} \lesssim_\varepsilon R^\varepsilon \left( \sum_\theta \|\mathcal{E}g_\theta\|_{L^6(B_R)}^2 \right)^{1/2}.$$

*Estimate one cap.* By Volume 12, Proposition 12.1.5, a single cap contributes  $\|\mathcal{E}g_\theta\|_{L^6} \lesssim \delta^{1/2} |B_R|^{1/6} \|g_\theta\|_{L^\infty}$ .

*Assemble.* With  $\#\theta = \delta^{-1/2} = R^{1/2}$  and  $\|g\theta\|_{L^\infty} \leq \|g\|_{L^\infty}$ ,

$$\|\mathcal{E}g\|_{L^6(B_R)} \lesssim_\varepsilon R^\varepsilon (R^{1/2} \cdot R^{-1} \cdot R^{1/3})^{1/2} \|g\|_{L^\infty} \lesssim R^\varepsilon \|g\|_{L^\infty},$$

which after  $\varepsilon$ -removal is the Stein–Tomas estimate in  $\mathbb{R}^2$  at the endpoint  $p = 6 = \frac{2(n+1)}{n-1}$ . ✓

*What the argument shows and does not.* Decoupling recovers Stein–Tomas but not the restriction conjecture, which in  $\mathbb{R}^2$  asks for  $p > 4$ . The reason is that decoupling is an  $\ell^2$  statement over caps and therefore cannot exploit  $L^\infty$  data — the same limitation Worked Example 12.4.1(c) identified for broad–narrow.

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**WORKED EXAMPLE 13.4.2— TIER B What decoupling is not**

Three things decoupling is regularly confused with.

(a) *It is not orthogonality.* At  $p = 2$  it *is* Plancherel (Proposition 13.1.5(i)), but for  $p > 2$  there is no orthogonality to appeal to, and the content is precisely that an  $\ell^2$  bound survives anyway. Volume 12, Worked Example 12.7.7 explains why: orthogonality is free only for quadratic quantities.

(b) *It is not a square function estimate.* The square function bound —  $\ell^2$  *inside* the norm — is stronger by Minkowski and is open at  $p = 6$ . Worked Example 13.1.4 sets out the four arrangements and which hold.

(c) *It is not sharp in the constant.* The theorem gives  $\delta^{-\varepsilon}$ ; the truth for the parabola is  $(\log \delta^{-1})^c$ , proved by Guth, Maldague and Wang. The improvement matters for applications where the  $\varepsilon$  compounds, and Worked Example 13.2.2 shows how weak  $\delta^{-\varepsilon}$  is at any nameable  $\delta$ .

*What it is.* An interpolation-free statement that the pieces of a function over a curved surface behave, in  $L^p$  for  $p$  up to a critical exponent, as though they were orthogonal — up to  $\delta^{-\varepsilon}$ .

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**WORKED EXAMPLE 13.4.3— TIER C: RESEARCH TASTE What the proof of Bourgain–Demeter actually needs**

Having proved the theorem, it is worth auditing which hypotheses did work.

(a) *Curvature, twice.* Once in Lemma 13.1.2, to make a  $\rho$ -arc fit in a  $\rho \times \rho^2$  box; once in Lemma 13.3.1, to make the addition map non-degenerate. A flat piece fails both, and Worked Example 13.5.2(d) records that no decoupling holds there.



(b) *Exactness of the rescaling.* Lemma 13.2.1 is an identity, not an approximation. For a general curved surface the analogous map is only approximate and the induction acquires error terms that must be tracked — one reason the moment curve, where rescaling is again exact, was reachable.

(c) *The log log round count.* Proposition 13.2.3 needs each round to *square* the scale. A method reducing  $\delta$  by a constant factor per round would give a fixed power, not  $\delta^{-\varepsilon}$ ; Volume 12, Worked Example 12.4.3 is the general statement.

(d) *Multilinear Kakeya, in full strength.* Only through Theorem 13.3.4, and in the plane it degenerates to Fubini. In  $\mathbb{R}^n$  it does not, which is why Chapter 13.5 is a genuinely  $n$ -dimensional theorem and why proving Volume 11's Theorem 11.5.4 rather than importing it was necessary.

(e) *What is not needed.* No polynomial method, no incidence geometry, no algebraic geometry. The contrast with Volume 11 is sharp: two subjects that interact constantly, built on disjoint foundations.

*The audit's value.* Knowing which hypotheses are load-bearing tells one where to look when extending a theorem. For the cone, (a) fails in one direction and the exponent changes; for the moment curve, all four survive and the proof goes through with  $k$  in place of 2.

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### 13.4.3 Two consequences

**Corollary 13.4.3** (Discrete restriction for the parabola). *Let  $\Lambda = \{(n/N, n^2/N^2) : 1 \leq n \leq N\}$  and  $f = \sum_n a_n e(x \cdot \lambda_n)$  with  $\lambda_n \in \Lambda$ . Then for  $2 \leq p \leq 6$  and every  $\varepsilon > 0$ ,*

$$\|f\|_{L^p(\mathbb{T}^2)} \lesssim_\varepsilon N^\varepsilon \left( \sum_n |a_n|^2 \right)^{1/2}.$$

*Proof.* Take  $\delta = N^{-2}$ , so the caps  $\theta$  of side  $\delta^{1/2} = N^{-1}$  contain exactly one point of  $\Lambda$  each. Apply Theorem 13.4.2; since  $f_\theta$  is a single modulated constant,  $\|f_\theta\|_{L^p} = |a_n|$ , and the right-hand side is the stated  $\ell^2$  sum.  $\square$

**Corollary 13.4.4** (The  $L^6$  count). *With  $a_n = 1$  for all  $n$ ,  $\int_{\mathbb{T}^2} |f|^6 \lesssim_\varepsilon N^{3+\varepsilon}$ , that is: the number of solutions of*

$$n_1 + n_2 + n_3 = n_4 + n_5 + n_6, \quad n_1^2 + n_2^2 + n_3^2 = n_4^2 + n_5^2 + n_6^2$$

*with  $n_i \leq N$  is  $O_\varepsilon(N^{3+\varepsilon})$ .*

*Proof.*  $\int |f|^6$  counts the solutions, and Corollary 13.4.3 at  $p = 6$  bounds it by  $N^\varepsilon(N)^3$ .  $\square$

**Where we are.** Corollary 13.4.4 is  $J_{3,2}(N) \lesssim N^{3+\epsilon}$ : the case  $k = 2$  of Vinogradov, obtained here as a one-line corollary. Chapter 13.6 does the general  $k$  by the same argument on the moment curve.

It is worth noting that this case is elementary by other means — the two equations force  $\{n_1, n_2, n_3\} = \{n_4, n_5, n_6\}$  as multisets, as Worked Example 13.6.1 records — so the corollary is a consistency check rather than a new result. That it comes out right is evidence the normalisations are correct, and it is the cheapest such check available.

### 13.4.4 Small cap decoupling

The caps of Definition 13.1.1 have length  $\delta^{1/2}$ , forced by Lemma 13.1.2. It is natural to ask what happens for shorter caps, and the answer is different in an instructive way.

**Definition 13.4.5.** For  $\frac{1}{2} \leq \beta \leq 1$  let  $D_p(\delta, \beta)$  be the least constant with

$$\|f\|_{L^p} \leq D_p(\delta, \beta) \left( \sum_{\gamma} \|f_{\gamma}\|_{L^p}^2 \right)^{1/2}$$

where  $\gamma$  runs over caps of length  $\delta^{\beta}$ .

**Proposition 13.4.6** (Small caps decouple on a smaller range). *For  $\frac{1}{2} \leq \beta \leq 1$ ,*

$$D_p(\delta, \beta) \gtrsim \delta^{-\frac{\beta}{2} + \frac{\beta+1}{p}},$$

*which is a positive power of  $\delta^{-1}$  exactly when*

$$p > p_c(\beta) := \frac{2(\beta+1)}{\beta}.$$

*Proof.* Repeat the construction of Proposition 13.1.8 with caps  $\gamma$  of length  $\delta^{\beta}$ .

**Step 1.** *The caps and their dual boxes.*

There are  $N = \delta^{-\beta}$  caps. Over an interval of length  $\delta^{\beta}$  the parabola deviates from its tangent by  $(\delta^{\beta})^2 = \delta^{2\beta} \leq \delta$ , since  $\beta \geq \frac{1}{2}$ ; so the cap lies in a box of dimensions  $\delta^{\beta} \times \delta$  — the thickness being  $\delta$ , the width of  $\mathcal{N}_{\delta}$ , rather than the smaller curvature deviation. Its dual box  $T_{\gamma}$  has dimensions  $\delta^{-\beta} \times \delta^{-1}$  and volume  $\delta^{-(\beta+1)}$ .

**Step 2.** *The two sides.*

Take wave packets adapted to the  $T_{\gamma}$  with phases aligned at the origin. The frequency support has diameter  $\approx 1$ , so by the Pitfall following Proposition 13.1.8 the packets remain aligned only on a ball of radius  $\approx 1$ , on which

$|f| \approx N$ . Hence  $\|f\|_{L^p} \gtrsim N = \delta^{-\beta}$ , while  $\|f_\gamma\|_{L^p} = |T_\gamma|^{1/p} = \delta^{-(\beta+1)/p}$  and

$$\left( \sum_{\gamma} \|f_\gamma\|_{L^p}^2 \right)^{1/2} = (N \delta^{-2(\beta+1)/p})^{1/2} = \delta^{-\frac{\beta}{2} - \frac{\beta+1}{p}}.$$

**Step 3.** *The ratio.*

Dividing gives  $\delta^{-\beta + \frac{\beta}{2} + \frac{\beta+1}{p}} = \delta^{-\frac{\beta}{2} + \frac{\beta+1}{p}}$ , which tends to infinity exactly when  $\frac{\beta+1}{p} < \frac{\beta}{2}$ , that is  $p > \frac{2(\beta+1)}{\beta}$ .  $\square$

**Where we are.** Two checks and a consequence.

At  $\beta = \frac{1}{2}$ ,  $p_c = \frac{2 \cdot \frac{3}{2}}{\frac{1}{2}} = 6$  and the exponent is  $-\frac{1}{4} + \frac{3}{2p}$  — exactly Proposition 13.1.8.  $\checkmark$

$p_c$  decreases in  $\beta$ : 6,  $\frac{16}{3}$ , 5,  $\frac{14}{3}$ , 4 at  $\beta = \frac{1}{2}, \frac{3}{5}, \frac{2}{3}, \frac{3}{4}, 1$ . So *shorter caps decouple on a strictly smaller range*, and at  $\beta = 1$  — caps of length  $\delta$ , as short as the neighbourhood is thick — only  $p \leq 4$  survives.

*Why shorter caps are worse, not better.* A shorter cap is flatter, so it carries less curvature, while there are more of them; the  $\ell^2$  sum grows without the estimate improving. Step 1 makes this precise: for  $\beta > \frac{1}{2}$  the dual box has thickness  $\delta^{-1}$  regardless of  $\beta$ , so lengthening the cap costs nothing but shortening it costs  $\delta^{-\beta}$  in the count.

So  $\delta^{1/2}$  is *optimal*, and Worked Example 13.4.4 explains the same fact geometrically: it is the unique length at which the caps are as long as possible subject to being flat at scale  $\delta$ .

**Where we are.** The reason is geometric, not technical. At length  $\delta^{1/2}$  a cap is *flat* to within  $\delta$ : Lemma 13.1.2 says the arc deviates from its tangent by  $(t - t_0)^2 \leq \delta$ . Shorter caps are flatter still and so carry no extra curvature, while there are more of them — so the  $\ell^2$  sum grows without the estimate improving.

*The rule.*  $\delta^{1/2}$  is the unique cap length at which the caps are as long as possible subject to being flat at scale  $\delta$ . Longer caps are not contained in  $\delta$ -slabs; shorter ones waste curvature. Every cap decomposition in Volumes 12 and 13 uses this length, and Proposition 13.4.6 explains why no other choice is available.

*What is true for small caps.* For  $p > p_c(\beta)$  a modified inequality holds with the  $\ell^2$  sum replaced by an  $\ell^p$  one, interpolating between the two arrangements of Worked Example 13.1.4. This is the small cap decoupling theory of Demeter, Guth and Wang, and it is genuinely different from Theorem 13.4.2 rather than a refinement of it.

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WORKED EXAMPLE 13.4.4— TIER B **Choosing the cap length**

Given a surface and a neighbourhood width  $\delta$ , the cap length is determined. The computation takes a line and is worth doing deliberately.

- (a) *Write the surface as a graph.*  $x_n = \phi(x')$  near the point of interest.
- (b) *Expand to second order.* Over a cap of diameter  $\sigma$ , the deviation from the tangent plane is  $\approx \|\text{Hess } \phi\| \sigma^2$ .
- (c) *Set it equal to  $\delta$ .*  $\sigma = (\delta / \|\text{Hess } \phi\|)^{1/2}$ . For the paraboloid  $\text{Hess } \phi = 2I$  and  $\sigma \approx \delta^{1/2}$ .
- (d) *Read off the dual box.*  $\sigma^{-1}$  in the tangential directions,  $\delta^{-1}$  in the normal one.

*Worked instances.*

| Surface      | cap                     | because                     |
|--------------|-------------------------|-----------------------------|
| paraboloid   | $\delta^{1/2}$          | all curvatures $\approx 1$  |
| cone         | $\delta^{1/2} \times 1$ | one curvature vanishes      |
| moment curve | $\delta^{1/k}$          | $k$ -th order contact       |
| hyperplane   | —                       | no curvature; no useful cap |

*The check.* Having chosen  $\sigma$ , verify that  $\#\text{caps} \times |\text{cap}| \approx |\mathcal{N}_\delta|$ . For the parabola:  $\delta^{-1/2} \cdot \delta^{1/2} \cdot \delta = \delta$ . ✓ A cap decomposition failing this check has the wrong length.

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WORKED EXAMPLE 13.4.5— TIER B **Checking a decoupling claim**

Papers state decoupling inequalities in several normalisations. Five checks before believing one applies to your situation.

- (1) *Which side carries the  $\ell^2$ ?* Outside the  $L^p$  norm is decoupling; inside is the square function, which is stronger and often false. Worked Example 13.1.4 lists the four arrangements.
- (2) *What is the cap length?*  $\delta^{1/2}$  for a hypersurface with non-vanishing curvature,  $\delta^{1/k}$  for the moment curve, and  $\delta^{1/2} \times 1$  for the cone. Worked Example 13.4.4 gives the computation, and Proposition 13.4.6 shows other choices fail.
- (3) *Is the loss  $\delta^{-\varepsilon}$  or explicit?* The Bourgain–Demeter theorem gives the former with no effective constant. If a paper needs an explicit constant, it needs Guth–Maldague–Wang or a different method.
- (4) *Is the exponent in range?* Compare with  $\frac{2(n+1)}{n-1}$  for hypersurfaces,  $\frac{2n}{n-2}$  for the cone,  $k(k+1)$  for the moment curve. Outside the range the statement

is *false*, not merely unproved — Proposition 13.5.3.

(5) *Are the frequencies separated?* The dictionary of Worked Example 13.6.2 requires one term per cap. Sums over irregular sets need small cap theory and the exponents differ.

*A worked check.* “ $\ell^2 L^6$  decoupling for the parabola with  $\delta^{-\varepsilon}$  loss”: (1)  $\ell^2$  outside ✓; (2) caps  $\delta^{1/2}$  ✓; (3)  $\varepsilon$ -loss, so no effective constant; (4) 6 is the critical exponent, so this is the endpoint case ✓; (5) applies to any  $f$  with the right Fourier support. Verdict: this is Theorem 13.4.2 exactly.

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## Drills

**Drill 13.4.1.** Prove the first bound of Lemma 13.4.1, and verify the absorption step in the second.

**Drill 13.4.2.** Check that  $K = \delta^{-1/4}$  gives  $\delta K^2 = \delta^{1/2}$ .

**Drill 13.4.3.** Carry out the exponent arithmetic in Worked Example 13.4.1 and confirm the powers of  $R$  cancel.

**Drill 13.4.4.** Using Proposition 13.1.8, confirm that  $-\frac{1}{4} + \frac{3}{2p}$  is positive for  $p < 6$  and negative for  $p > 6$ .

## Notes

The  $\ell^2$  decoupling theorem for the paraboloid is due to Bourgain and Demeter (2015). The presentation here follows the standard bilinear route; the sharp constant  $(\log \delta^{-1})^c$  for the parabola is due to Guth, Maldague and Wang (2020).



## Chapter 13.5

# The Higher-Dimensional Paraboloid

The proof in Chapter 13.4 was written for the parabola, but almost nothing in it was two-dimensional. This chapter states the general theorem and records exactly which steps change.

### 13.5.1 The statement

**Definition 13.5.1.** Let  $\mathbb{P} = \{(\xi, |\xi|^2) : \xi \in [0, 1]^{n-1}\} \subseteq \mathbb{R}^n$  and let  $\mathcal{N}_\delta$  be its  $\delta$ -neighbourhood. Partition  $[0, 1]^{n-1}$  into  $N = \delta^{-(n-1)/2}$  cubes of side  $\delta^{1/2}$ , giving caps  $\theta$ , and let  $D_p(\delta)$  be the least constant with

$$\|f\|_{L^p(\mathbb{R}^n)} \leq D_p(\delta) \left( \sum_{\theta} \|f_{\theta}\|_{L^p}^2 \right)^{1/2}$$

for all  $f$  with  $\text{supp } \widehat{f} \subseteq \mathcal{N}_\delta$ .

**Theorem 13.5.2** (Bourgain–Demeter). *For  $2 \leq p \leq \frac{2(n+1)}{n-1}$  and every  $\varepsilon > 0$ ,*

$$D_p(\delta) \lesssim_{n,\varepsilon} \delta^{-\varepsilon},$$

*and the range is sharp.*

**Proposition 13.5.3** (Sharpness). *For  $p > \frac{2(n+1)}{n-1}$ ,*

$$D_p(\delta) \gtrsim \delta^{-\frac{n-1}{4} + \frac{n+1}{2p}},$$

*a positive power of  $\delta^{-1}$ .*

*Proof.* Exactly as Proposition 13.1.8. Each cap is a  $\delta^{1/2} \times \cdots \times \delta^{1/2} \times \delta$  box, so its dual box  $T_\theta$  has volume  $\delta^{-(n-1)/2} \cdot \delta^{-1} = \delta^{-(n+1)/2}$ . Choosing wave packets aligned at the origin,  $|f| \approx N$  on a ball of radius  $\approx 1$  — the frequency support has diameter  $\approx 1$ , so the packets separate beyond that — giving  $\|f\|_{L^p} \gtrsim N$ . Meanwhile  $\|f_\theta\|_{L^p} = |T_\theta|^{1/p} = \delta^{-(n+1)/(2p)}$ , so

$$\left( \sum_\theta \|f_\theta\|_{L^p}^2 \right)^{1/2} = N^{1/2} \delta^{-\frac{n+1}{2p}}.$$

The ratio is  $N^{1/2} \delta^{(n+1)/(2p)} = \delta^{-\frac{n-1}{4} + \frac{n+1}{2p}}$ , which tends to infinity exactly when  $\frac{n+1}{2p} < \frac{n-1}{4}$ , that is  $p > \frac{2(n+1)}{n-1}$ .  $\square$

**Where we are.** Two checks on Proposition 13.5.3.

*At the critical exponent the ratio is bounded.* Substituting  $p = \frac{2(n+1)}{n-1}$  gives  $-\frac{n-1}{4} + \frac{n+1}{2} \cdot \frac{n-1}{2(n+1)} = -\frac{n-1}{4} + \frac{n-1}{4} = 0$ , for every  $n$ .  $\checkmark$

*At  $n = 2$  it is the planar statement.*  $\frac{2(n+1)}{n-1} = 6$  and the exponent is  $-\frac{1}{4} + \frac{3}{2p}$ , which is Proposition 13.1.8.  $\checkmark$

*And it is the Stein–Tomas exponent.*  $\frac{2(n+1)}{n-1}$  is exactly the exponent of Volume 6, Theorem 6.3.1, as it must be: by Worked Example 13.4.1 decoupling implies a restriction estimate at the same exponent, so the two critical values cannot differ. Numerically 6, 4,  $\frac{10}{3}$ , 3,  $\frac{22}{9}$  for  $n = 2, 3, 4, 5, 10$ .

## 13.5.2 What changes in the proof

*Proof of Theorem 13.5.2, following Chapter 13.4.* Every step of Theorem 13.4.2 has a counterpart. We list them, indicating the modification.

### Step 1. Rescaling.

Lemma 13.2.1 holds verbatim with  $t \in [0, 1]$  replaced by  $\xi \in [0, 1]^{n-1}$ : for a cube  $\tau$  of side  $\sigma$  centred at  $\xi_0$ , the affine map

$$L_\tau(x, x_n) = \left( \frac{x - \xi_0}{\sigma}, \frac{x_n - |\xi_0|^2 - 2\xi_0 \cdot (x - \xi_0)}{\sigma^2} \right)$$

carries the cap over  $\tau$  onto all of  $\mathbb{P}$  and  $\mathcal{N}_\delta$  to  $\mathcal{N}_{\delta/\sigma^2}$ , by the identity  $|\xi_0 + \sigma s|^2 - |\xi_0|^2 - 2\sigma \xi_0 \cdot s = \sigma^2 |s|^2$ . *Unchanged.*

### Step 2. Bootstrapping.

Proposition 13.2.3 is a statement about a real function and is dimension-free. *Unchanged.*



**Step 3.** *The broad–narrow split.*

Lemma 13.4.1 must be replaced by its *n*-linear form:

$$|f| \lesssim_K \max_{\tau_1, \dots, \tau_n} \prod_{\nu\text{-transverse}}^n |f_{\tau_j}|^{1/n} + K^C \max_{\tau} |f_{\tau}|.$$

This is Volume 12, Proposition 12.4.3, and *two factors no longer suffice*: by Volume 12, Proposition 12.2.3, two vectors cannot span  $\mathbb{R}^n$  for  $n \geq 3$ , so a bilinear reduction would leave  $n - 2$  directions uncontrolled. *This is the one genuine change.*

**Step 4.** *Ball inflation.*

Theorem 13.3.4 generalises with  $n$  factors and exponent  $\frac{1}{n-1}$ :

$$\frac{1}{|\mathcal{D}|} \sum_{\Delta \in \mathcal{D}} \prod_{j=1}^n \left( \text{avg}_{\Delta} (\mathcal{S}_{\tau_j} f)^2 \right)^{\frac{1}{n-1}} \lesssim_{\nu, \varepsilon} \rho^{-\varepsilon} \prod_{j=1}^n \left( \text{avg}_B (\mathcal{S}_{\tau_j} f)^2 \right)^{\frac{1}{n-1}},$$

and the proof is the same, applying Volume 11, Theorem 11.5.7 in  $\mathbb{R}^n$  instead of  $\mathbb{R}^2$ . In the plane that theorem degenerated to Fubini; here its full strength is used. *Same argument, stronger input — and the input is proved.*

**Step 5.** *Assembly.*

Steps 1–4 of Theorem 13.4.2 go through with  $L^6$  replaced by  $L^{p_c}$ ,  $p_c = \frac{2(n+1)}{n-1}$ , and the recursion (13.4.1) is unchanged in form. Proposition 13.2.3 closes it.  $\square$

**Where we are.** The proof is genuinely the same, and it is worth being precise about why the literature nonetheless treats the higher-dimensional case as substantially harder.

*What is the same.* The architecture: rescale, split, inflate, bootstrap.

*What is harder in practice.* Step 3’s  $n$ -linear reduction requires tracking  $\binom{K^{n-1}}{n}$  tuples rather than  $\binom{K}{2}$  pairs, and the transversality bookkeeping is heavier. Step 4 requires multilinear Keakeya in  $\mathbb{R}^n$ , a research theorem, where the planar case was Fubini.

*What this series gains from Volume 11.* Because Theorem 11.5.4 is *proved* there — Guth’s short argument from Loomis–Whitney — Step 4 is not an import, and Theorem 13.5.2 rests on nothing unproved. Had multilinear Keakeya remained a black box, this chapter would have had to be an announcement.

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**WORKED EXAMPLE 13.5.1— TIER A The exponents, tabulated**

| $n$ | $p_c = \frac{2(n+1)}{n-1}$  | #caps           | dual box volume  |
|-----|-----------------------------|-----------------|------------------|
| 2   | 6                           | $\delta^{-1/2}$ | $\delta^{-3/2}$  |
| 3   | 4                           | $\delta^{-1}$   | $\delta^{-2}$    |
| 4   | $\frac{10}{3} \approx 3.33$ | $\delta^{-3/2}$ | $\delta^{-5/2}$  |
| 5   | 3                           | $\delta^{-2}$   | $\delta^{-3}$    |
| 10  | $\frac{22}{9} \approx 2.44$ | $\delta^{-9/2}$ | $\delta^{-11/2}$ |

*The exponent decreases to 2.* As  $n \rightarrow \infty$ ,  $\frac{2(n+1)}{n-1} \rightarrow 2$ , so in high dimensions decoupling holds on a range shrinking to the trivial one. That is not a defect of the method: by Proposition 13.5.3 the range is sharp.

*Why the range shrinks.* More caps means more room for constructive interference, and Proposition 13.5.3 shows the resulting example degrades the exponent. The count  $N = \delta^{-(n-1)/2}$  grows with  $n$ , which is the whole content.

*Comparison with restriction.* The restriction conjecture asks for  $p > \frac{2n}{n-1}$ , which also tends to 2; the ratio  $\frac{2(n+1)/(n-1)}{2n/(n-1)} = \frac{n+1}{n} \rightarrow 1$ . So decoupling and the restriction conjecture become close in high dimensions — but the difference is exactly what Volume 12, Worked Example 12.5.1 measured, and it never closes.

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**WORKED EXAMPLE 13.5.2— TIER B Which surfaces decouple**

Decoupling is not special to paraboloids, and knowing the pattern is useful.

(a) *Non-vanishing Gaussian curvature.* Paraboloid, sphere, ellipsoid: the theorem holds with  $p_c = \frac{2(n+1)}{n-1}$ , by the argument above after an affine change reducing to the paraboloid locally.

(b) *The cone.* One vanishing principal curvature. Decoupling holds with  $p_c = \frac{2n}{n-2}$ , proved by Bourgain and Demeter; the rescaling is Lorentz rather than parabolic. This is the case Volume 13's applications need, since local smoothing is a statement about the wave equation.

(c) *The moment curve.*  $\{(t, t^2, \dots, t^k)\}$ : a curve, not a hypersurface, with all curvatures non-vanishing. Decoupling holds with  $p_c = k(k+1)$ , and Chapter 13.6 shows this is Vinogradov's mean value theorem.

(d) *Flat pieces.* A hyperplane does not decouple beyond  $p = 2$ : the trivial bound Proposition 13.1.5(ii) is attained. Curvature is not a technical convenience but the entire source of the phenomenon.

*The rule.* The critical exponent is determined by how many caps there are and how large their dual boxes are, both of which are read off from the curvatures. Proposition 13.5.3 is the computation, and it can be run for any surface whose caps one can describe.

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**WORKED EXAMPLE 13.5.3— TIER C: RESEARCH TASTE    What dimension costs, across the series**

Worked Example 13.5.1 showed the decoupling range shrinking to  $p = 2$  as  $n \rightarrow \infty$ . That is one instance of a pattern visible in every volume, and collecting it clarifies why high dimensions are hard.

(a) *Restriction.* The conjectured range  $p > \frac{2n}{n-1}$  and the Stein–Tomas range  $p \geq \frac{2(n+1)}{n-1}$  both tend to 2, and their gap  $\frac{2}{n-1}$  tends to 0 (Volume 12, Worked Example 12.5.1). So Stein–Tomas is asymptotically sharp in absolute terms and never sharp.

(b) *Takeya.* Wolff’s bound  $\frac{n+2}{2}$  against the conjectured  $n$ : the *ratio* tends to  $\frac{1}{2}$ , so the method loses half the dimension in the limit. Volume 9, Worked Example 9.5.3 recorded this.

(c) *Falconer.* The conjectured  $\frac{n}{2}$  and Falconer’s  $\frac{n+1}{2}$  differ by  $\frac{1}{2}$  for every  $n$  — a gap that neither closes nor grows. Volume 8, Worked Example 8.7.1.

(d) *Decoupling.* The range shrinks but is *sharp* at every  $n$ , by Proposition 13.5.3. So decoupling is the one item on this list that is fully understood.

*Why (d) differs from (a)–(c).* Decoupling is an  $\ell^2$  statement over caps, and  $\ell^2$  statements are the ones this subject can prove — the same observation as Volume 12, Worked Example 12.7.7. The open problems are all  $L^\infty$  statements, where no orthogonality is available. *The line between solved and open in this subject tracks the line between  $L^2$  and  $L^\infty$  data almost exactly*, and that is the most useful single heuristic in the series.

*The exception that tests it.* Takeya has no data norm at all — it is a purely geometric statement — and it is open. So the heuristic is not a theorem; it predicts which analytic problems are tractable, not which geometric ones.

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### 13.5.3 The cone

The applications of Chapter 13.8 need decoupling for the cone rather than the paraboloid, and the differences are worth recording precisely.

**Definition 13.5.4.** Let  $\mathbb{C} = \{(\xi, |\xi|) : \frac{1}{2} \leq |\xi| \leq 1\} \subseteq \mathbb{R}^n$ , the truncated light cone, and  $\mathcal{N}_\delta$  its  $\delta$ -neighbourhood.

**Lemma 13.5.5** (Cone caps are plates). *The natural caps of  $\mathcal{N}_\delta$  have dimensions  $\delta^{1/2} \times \dots \times \delta^{1/2} \times 1 \times \delta$ : they are  $\delta^{1/2}$  in the  $n-2$  directions of non-vanishing curvature, of length  $\approx 1$  along the ruling, and  $\delta$  normal.*

*Proof.* The cone is ruled by lines through the origin, along which it is flat, so no subdivision is needed in that direction. Transverse to the ruling it is a curve of non-vanishing curvature, and Worked Example 13.4.4 gives  $\delta^{1/2}$  there.  $\square$

**Theorem 13.5.6** (Decoupling for the cone; Bourgain–Demeter). *For  $2 \leq p \leq \frac{2n}{n-2}$  and every  $\varepsilon > 0$ ,  $D_p^{\mathbb{C}}(\delta) \lesssim_{n,\varepsilon} \delta^{-\varepsilon}$ .*

*Remark 13.5.7* (Scope). Theorem 13.5.6 is not proved here; it is the input to Remark 13.8.4. Its proof follows the architecture of Chapter 13.4 with the Lorentz group replacing the parabolic rescaling of Lemma 13.2.1: the transformations preserving the cone act transitively on its caps, which is what the induction requires.

*What changes in the numerology.* The exponent is  $\frac{2n}{n-2}$  rather than  $\frac{2(n+1)}{n-1}$ , because the caps are plates rather than boxes: there are  $\delta^{-(n-2)/2}$  of them, not  $\delta^{-(n-1)/2}$ , and each has dual box of volume  $\delta^{-(n-2)/2} \cdot 1 \cdot \delta^{-1}$ . Running Proposition 13.5.3 with these values gives  $\frac{2n}{n-2}$ .

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**WORKED EXAMPLE 13.5.4— TIER A Cone against paraboloid, numerically**

| $n$ | paraboloid $\frac{2(n+1)}{n-1}$ | cone $\frac{2n}{n-2}$       | which is larger |
|-----|---------------------------------|-----------------------------|-----------------|
| 3   | 4                               | 6                           | cone            |
| 4   | $\frac{10}{3} \approx 3.33$     | 4                           | cone            |
| 5   | 3                               | $\frac{10}{3} \approx 3.33$ | cone            |
| 10  | $\frac{22}{9} \approx 2.44$     | 2.5                         | cone            |

*The cone decouples on a larger range.* That is not because the cone is “more curved” — it is less curved, having a vanishing principal curvature. It is because the caps are *fewer*:  $\delta^{-(n-2)/2}$  against  $\delta^{-(n-1)/2}$ , so the  $\ell^2$  sum is smaller and the constructive example of Proposition 13.5.3 is weaker.

*The check at  $n = 3$ .* Cone gives 6, which is the exponent Theorem 13.8.3(i) needs. Paraboloid gives 4. Both agree with Worked Example 13.5.2.  $\checkmark$

*The moral.* The critical exponent is a count, not a curvature. Volume 5, Worked Example 5.5.4 gave decay exponents from curvature; here what matters is how many caps the curvature forces, which is a different function of the same data.

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**WORKED EXAMPLE 13.5.5— TIER C: RESEARCH TASTE** **What makes a proof generalise**

Theorem 13.5.2 was obtained from Theorem 13.4.2 by changing one step. That is unusual, and the reasons are worth extracting.

(a) *The dimension appeared in only one place.* Rescaling, bootstrapping and the assembly are dimension-free; only the broad–narrow split needed  $n$  factors instead of 2. A proof organised so that the dimension enters once generalises; one that uses planar coordinates throughout does not.

(b) *The hard input was already general.* Ball inflation rests on multilinear Makeya, which Volume 11 proves in  $\mathbb{R}^n$ . Had it been proved only in the plane, this chapter would have stopped.

(c) *The sharpness example was already general.* Proposition 13.5.3 is the planar computation with  $N$  and  $|T_\theta|$  left as symbols. Writing examples with symbolic parameters rather than numerical ones is what makes them portable.

(d) *The contrast: what did not generalise.* Volume 12’s Theorem 12.6.1 for the circle uses that two vectors span the plane, and Proposition 12.2.3 shows this fails above dimension two. That proof is genuinely planar, and the higher-dimensional statement — Tao’s theorem — needed an entirely different argument.

(e) *The diagnostic.* Before generalising, list every place the dimension is used and ask, for each, whether it is essential or notational. In Chapter 13.4 the answer was “notational” everywhere but one step; in Section 12.6.1 it was “essential” at the first step.

(f) *Why this matters for reading.* A paper proving a planar case often signals in its introduction which route it took. If the hard input is already stated in  $\mathbb{R}^n$ , the general case is likely routine; if the argument opens with a planar-specific identity, expect a separate paper.

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## Drills

**Drill 13.5.1.** Verify the identity  $|\xi_0 + \sigma s|^2 - |\xi_0|^2 - 2\sigma \xi_0 \cdot s = \sigma^2 |s|^2$  used in Step 1.

**Drill 13.5.2.** Confirm that the exponent of Proposition 13.5.3 vanishes at  $p = \frac{2(n+1)}{n-1}$  for  $n = 2, 3, 4$ .

**Drill 13.5.3.** Compute  $\frac{2(n+1)}{n-1}$  and  $\frac{2n}{n-1}$  for  $n = 2, \dots, 6$  and tabulate the gap.

**Drill 13.5.4.** Explain, using Proposition 12.2.3 of Volume 12, why Step 3 cannot use two factors when  $n \geq 3$ .

## Notes

Theorem 13.5.2 is the main theorem of Bourgain and Demeter's 2015 paper. The cone case is in the same paper; the moment curve is treated in Bourgain–Demeter–Guth, and is the subject of Chapter 13.6.

## Chapter 13.6

# The Moment Curve and Vinogradov's Theorem

Decoupling was invented for harmonic analysis. Within two years of Theorem 13.5.2 it settled a problem in analytic number theory that had been open since 1935. This chapter gives the reduction.

### 13.6.1 The mean value theorem

**Definition 13.6.1** (Vinogradov's system). For integers  $s, k \geq 1$  and  $N \geq 1$  let  $J_{s,k}(N)$  be the number of integer solutions of the system

$$\sum_{i=1}^s x_i^j = \sum_{i=1}^s y_i^j, \quad j = 1, \dots, k,$$

with  $1 \leq x_i, y_i \leq N$ .

**Theorem 13.6.2** (Vinogradov's mean value theorem; proved in Proposition 13.6.5 from Theorem 13.6.4). *For every  $\varepsilon > 0$ ,*

$$J_{s,k}(N) \lesssim_{s,k,\varepsilon} N^\varepsilon \left( N^s + N^{2s - \frac{k(k+1)}{2}} \right).$$

**Where we are.** Both terms are forced, so the theorem is sharp up to  $N^\varepsilon$ .  $N^s$  comes from the diagonal. Any solution with  $\{x_i\}$  a permutation of  $\{y_i\}$  works, and there are  $\approx s! N^s$  of these.  $N^{2s - k(k+1)/2}$  comes from counting. There are  $N^{2s}$  tuples and  $k$  equations, the  $j$ -th constraining a quantity of size  $\approx N^j$ ; so heuristically a fraction  $N^{-(1+2+\dots+k)} = N^{-k(k+1)/2}$  of tuples should satisfy all of them.

The two balance at  $s = \frac{k(k+1)}{2}$ , where both equal  $N^s$ . That is the critical case, and it is where the difficulty lies: for  $s$  much larger or smaller the theorem is classical.

## 13.6.2 The reduction to decoupling

**Definition 13.6.3** (The moment curve). Let  $\Gamma_k = \{(t, t^2, \dots, t^k) : t \in [0, 1]\} \subseteq \mathbb{R}^k$ , and let  $D_p^\Gamma(\delta)$  be the decoupling constant for its  $\delta$ -neighbourhood, with arcs of length  $\delta^{1/k}$ .

**Theorem 13.6.4** (Bourgain–Demeter–Guth; not proved here, see Remark 13.6.6). For  $2 \leq p \leq k(k+1)$  and every  $\varepsilon > 0$ ,  $D_p^\Gamma(\delta) \lesssim_{k,\varepsilon} \delta^{-\varepsilon}$ .

**Proposition 13.6.5** (Decoupling implies Vinogradov). *Theorem 13.6.4 at  $p = k(k+1)$  implies Theorem 13.6.2 at the critical exponent  $s = \frac{k(k+1)}{2}$ , and hence for all  $s$ .*

*Proof. Step 1.*  $J_{s,k}$  is an  $L^{2s}$  norm.

Put  $\delta = N^{-k}$  and

$$F(x) = \sum_{m=1}^N e(mx_1 + m^2x_2 + \dots + m^kx_k), \quad x \in \mathbb{T}^k,$$

where  $e(u) = e^{2\pi iu}$ . Expanding  $|F|^{2s}$  and integrating over the torus, the cross terms cancel unless the exponents match in every coordinate, so

$$J_{s,k}(N) = \int_{\mathbb{T}^k} |F(x)|^{2s} dx = \|F\|_{L^{2s}(\mathbb{T}^k)}^{2s}. \quad (13.6.1)$$

So the critical number of variables and the critical exponent are the same number:  $2s = p$ .

**Step 2.**  $F$  lives on the moment curve.

Rescaling  $x_j \mapsto N^{-j}x_j$  turns  $F$  into a sum of  $N$  terms whose frequencies are  $(\frac{m}{N}, \frac{m^2}{N^2}, \dots)$ , which are  $N^{-1}$ -separated points on  $\Gamma_k$ . Convoluting with a bump at scale  $\delta = N^{-k}$  makes  $F$  a function with Fourier support in  $\mathcal{N}_\delta(\Gamma_k)$ , one cap  $\theta$  per value of  $m$ ; there are  $\delta^{-1/k} = N$  caps, matching the  $N$  terms.

**Step 3.** Apply decoupling.

By Theorem 13.6.4 at  $p = k(k+1) = 2s$ ,

$$\|F\|_{L^{2s}} \lesssim_\varepsilon \delta^{-\varepsilon} \left( \sum_\theta \|F_\theta\|_{L^{2s}}^2 \right)^{1/2}.$$



Each  $F_\theta$  is a single term, so  $\|F_\theta\|_{L^{2s}} \approx 1$  after normalisation, and the right side is  $\lesssim \delta^{-\varepsilon} N^{1/2}$ . Raising to the power  $2s$  and substituting into (13.6.1),

$$J_{s,k}(N) \lesssim_\varepsilon N^{k\varepsilon} N^s.$$

**Step 4.** *The critical case gives all cases.*

At  $s = \frac{k(k+1)}{2}$  the bound  $N^s$  is exactly  $N^{2s-k(k+1)/2}$ , so Theorem 13.6.2 holds there. For smaller  $s$  it follows by Hölder from the critical case, and for larger  $s$  by inserting the trivial bound  $|F| \leq N$  into the extra factors.  $\square$

**Where we are.** The correspondence  $s = p/2$  is exact and worth tabulating.

| $k$ | $p_c = k(k+1)$ | critical $s = \frac{k(k+1)}{2}$ |
|-----|----------------|---------------------------------|
| 2   | 6              | 3                               |
| 3   | 12             | 6                               |
| 4   | 20             | 10                              |
| 5   | 30             | 15                              |

The case  $k = 2$  is classical:  $J_{3,2}(N)$  counts solutions of  $x_1 + x_2 + x_3 = y_1 + y_2 + y_3$  together with the same for squares, and is  $\approx N^3 \log N$  by an elementary argument. So the theorem is a genuine generalisation of a known case, which is the right thing to check first.

*What is being transported.* An exponential sum is an  $L^p$  norm; a decoupling inequality controls that norm by the individual terms; and the individual terms are trivial. *The entire number-theoretic content is carried by the geometry of the moment curve* — specifically, by the fact that its caps have the dual boxes they do.

**Remark 13.6.6** (Status of Theorem 13.6.4). Theorem 13.6.4 is not proved here. Its proof follows the architecture of Chapter 13.4 — rescale, split broad from narrow, inflate, bootstrap — but the rescaling for the moment curve is by the group of transformations  $t \mapsto \lambda t + \mu$  lifted to  $\mathbb{R}^k$ , and the multilinear step requires  $k$ -linear Kakeya for *curves* rather than hypersurfaces. Bourgain, Demeter and Guth carry this out.

*What is proved here.* Proposition 13.6.5: the complete reduction of Vinogradov’s theorem to the decoupling statement, with the identification  $s = p/2$  and the verification that the critical cases correspond. Nothing later in this series depends on Theorem 13.6.4.

*Why it is worth stating anyway.* Because the reduction is the surprising part. That a mean value theorem in number theory is a decoupling inequality

was not suspected before 2015, and it is the reduction, not the analysis, that constitutes the discovery.

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**WORKED EXAMPLE 13.6.1— TIER A Vinogradov before decoupling**

The theorem was proved twice, and comparing the routes is instructive.

*Wooley's efficient congruencing (2011–2016).* A  $p$ -adic argument. One studies solutions modulo powers of a prime and exploits that congruent solutions cluster; iterating produces a descent. It settled  $k = 3$  in 2016 and, in Wooley's later work, all  $k$ .

*Bourgain–Demeter–Guth (2015).* The decoupling route above. It settled all  $k \geq 4$  first.

*What they share.* Both are induction-on-scales arguments. Efficient congruencing induces on the power of  $p$ ; decoupling induces on the spatial scale  $\delta$ . Chapter 13.7 makes the analogy precise.

*What differs.* Efficient congruencing is elementary and gives explicit constants; decoupling is analytic and gives  $N^\varepsilon$  with no effective constant. For applications needing effectivity — Waring's problem with explicit bounds — the congruencing route remains preferable.

*The moral.* Two genuinely different proofs of the same theorem, found within a year of each other by communities that had not been talking. That is worth knowing when choosing which literature to read: the decoupling proof is shorter to state and rests on Volume 11's geometry; the congruencing proof is self-contained and effective.

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**WORKED EXAMPLE 13.6.2— TIER B Reading an exponential sum estimate**

Number-theoretic statements about exponential sums translate into decoupling statements by a fixed dictionary, and it is worth having explicitly.

(1) *The sum is a function on a torus.*  $\sum_m e(\phi_m \cdot x)$  with frequencies  $\phi_m$ ; the relevant surface is the curve or hypersurface through the  $\phi_m$ .

(2) *The moment is an  $L^p$  norm.* A  $2s$ -th moment of the sum, counting solutions of a system, is  $\|F\|_{L^{2s}}^{2s}$ , as in (13.6.1).

(3) *The scale is set by the largest frequency.* If frequencies run to  $N^k$ , the neighbourhood width is  $\delta = N^{-k}$ .

(4) *The caps are the individual terms.* One term per cap, so the square function on the right of a decoupling inequality is trivially computed.

*A worked instance.* For  $k = 2$ ,  $s = 3$ : frequencies  $(m, m^2)$  with  $m \leq N$ , so the curve is the parabola,  $\delta = N^{-2}$ ,  $p = 6$  — and the required decoupling is exactly Theorem 13.4.2, proved in Chapter 13.4. *So the  $k = 2$  case of*

*Vinogradov is a corollary of the theorem this volume proves in full.*

*The caution.* The dictionary requires the frequencies to be *separated* at the scale of the caps. Sums over sparse or irregular sets need small cap decoupling, a different and more delicate theory.

### 13.6.3 Consequences in number theory

Vinogradov's theorem is a tool, and its uses are worth recording since they are the reason the problem mattered.

**Proposition 13.6.7** (Waring's problem). *Let  $G(k)$  be the least  $s$  such that every sufficiently large integer is a sum of  $s$  non-negative  $k$ -th powers. Then Theorem 13.6.2 gives*

$$G(k) \leq (1 + o(1)) k \log k.$$

*Where the mean value theorem enters.* The circle method writes the number of representations of  $N$  as a sum of  $s$   $k$ -th powers as  $\int_{\mathbb{T}} \left( \sum_{m \leq N^{1/k}} e(\alpha m^k) \right)^s e(-\alpha N) d\alpha$ , split into major and minor arcs. The major arcs give the expected main term via a singular series. On the minor arcs one needs cancellation in the Weyl sums, and the standard route bounds their mean value by  $J_{s,k}$ ; Theorem 13.6.2 at the critical exponent supplies exactly the bound required, and tracking the resulting condition on  $s$  gives  $k \log k$ .

*The circle method itself is not developed here.* Revision R26: the major and minor arc analysis is a chapter of analytic number theory — singular series, Weyl differencing, the Hardy–Littlewood dissection — and is a different subject from the one this series treats. What is proved in this volume is the harmonic-analytic half: Proposition 13.6.5, that  $J_{s,k}$  is a decoupling constant. Nothing in this series depends on Proposition 13.6.7.  $\square$

**Where we are.** The bound  $k \log k$  is not new with decoupling — Wooley had obtained it by efficient congruencing — but before either, the classical Vinogradov bounds gave  $G(k) \leq (2 + o(1))k \log k$ , and the factor 2 was the target for eighty years. Both routes remove it, and neither reaches the conjectured  $G(k) \approx 4k$  for most  $k$ .

*What limits the method.* The circle method needs the mean value bound at the critical  $s$ , and Theorem 13.6.2 supplies exactly that. Improving  $G(k)$  further requires a different treatment of the minor arcs, not a better mean value theorem: the mean value theorem is now sharp.

**Proposition 13.6.8** (Zero-free regions). *Bounds on  $J_{s,k}$  yield bounds on  $|\zeta(\sigma + it)|$  near  $\sigma = 1$ , and hence a zero-free region for the Riemann zeta function of*

*Vinogradov–Korobov type,*

$$\sigma > 1 - \frac{c}{(\log|t|)^{2/3}(\log\log|t|)^{1/3}}.$$

Where the mean value theorem enters. Near  $\sigma = 1$ ,  $\zeta$  is estimated by sums  $\sum_{m \sim N} m^{-it}$ . Writing  $m^{-it} = e(-\frac{t}{2\pi} \log m)$  and expanding  $\log(1+u)$  in a Taylor polynomial of degree  $k$  turns these into Weyl sums whose frequencies lie on the moment curve  $\Gamma_k$ . Theorem 13.6.2 bounds their moments, and optimising the degree  $k$  against  $\log|t|$  produces the exponents  $\frac{2}{3}$  and  $\frac{1}{3}$ .

*The zero-free region argument is not developed here.* Revision R26, as above: converting a Weyl sum bound into a zero-free region requires the Borel–Carathéodory and Hadamard three-circles machinery of complex analytic number theory. Worked Example 13.6.3 records the full chain and marks these two links as belonging to a different subject.  $\square$

**Where we are.** This is the deepest consequence, and it explains why Vinogradov's system was studied at all. The exponents  $\frac{2}{3}$  and  $\frac{1}{3}$  in Proposition 13.6.8 come from optimising the degree  $k$  of the Taylor polynomial against  $\log|t|$ , and they have not been improved since Korobov and Vinogradov in 1958 — even now that the mean value theorem is sharp.

*So what did decoupling buy for zeta?* A cleaner proof, and the constant  $c$ ; not the shape of the region. That is worth knowing before reading the literature: Theorem 13.6.2 was the last missing ingredient in a chain whose other links remain the bottleneck.

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WORKED EXAMPLE 13.6.3— TIER A **The chain from geometry to zeta**

Collecting the links, in the order this series proves them.

|                              |                        |
|------------------------------|------------------------|
| Hölder's inequality          | Vol 1                  |
| ↓                            |                        |
| Loomis–Whitney               | Vol 11, Theorem 11.5.2 |
| ↓                            |                        |
| Multilinear Kakeya           | Vol 11, Theorem 11.5.4 |
| ↓                            |                        |
| Ball inflation               | Theorem 13.3.4         |
| ↓                            |                        |
| Decoupling                   | Theorem 13.4.2         |
| ↓                            |                        |
| Moment curve decoupling      | Theorem 13.6.4         |
| ↓                            |                        |
| Vinogradov                   | Proposition 13.6.5     |
| ↓                            |                        |
| Zero-free region for $\zeta$ | Proposition 13.6.8     |

*Every link but two is proved in this series.* The exceptions are moment curve decoupling (Remark 13.6.6) and the two circle-method sketches above, which belong to analytic number theory rather than harmonic analysis.

*The point of the table.* A statement about how tubes overlap in  $\mathbb{R}^n$  — Loomis–Whitney, which is Fubini and Hölder — propagates, through six links, to the distribution of the zeros of  $\zeta$ . Each link is individually unremarkable; the composition is not, and it was not suspected before 2015.

*The transferable lesson.* When a geometric inequality is sharp, look for what it implies. The chain above was assembled backwards — Bourgain, Demeter and Guth knew which number-theoretic statement they wanted — but its existence was invisible from either end.

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**WORKED EXAMPLE 13.6.4— TIER C: RESEARCH TASTE   Transfers between fields, and how to spot them**

The decoupling proof of Vinogradov's theorem is the clearest transfer in this series: a geometric inequality settling a number-theoretic problem. It is worth asking what made it possible, since the same pattern recurs.

(a) *A common object in disguise.*  $J_{s,k}(N)$  is a count of solutions and  $\|F\|_{L^{2s}}^{2s}$  is an integral; (13.6.1) says they are the same number. *The transfer becomes possible only once the two descriptions are recognised as one object*, and that recognition is the discovery.

(b) *Matching critical cases.* The number-theoretic critical exponent  $s = \frac{k(k+1)}{2}$  and the analytic one  $p_c = k(k+1)$  satisfy  $s = p_c/2$  exactly. Had

they not matched, the transfer would have given a non-sharp result and no-one would have noticed the connection.

(c) *Other instances in this series.* Volume 12, Proposition 12.8.2 identifies a geometric measure theory quantity with a restriction norm — the same move, and it is why the Falconer problem inherits every advance in restriction theory. Volume 11, Proposition 11.7.6 identifies distance quadruples with line incidences. Volume 10 identifies Kakeya in  $\mathbb{F}_q^n$  with a polynomial vanishing problem.

(d) *What they have in common.* In each case a counting problem is rewritten as a statement about a curved object, and the curvature then does work that was invisible in the original formulation. *Curvature is the resource*, and finding it is the transferable skill.

(e) *How to look.* Given a counting problem, ask what surface the objects being counted parametrise, and whether the count is a moment of a function whose frequencies lie on it. If both answers are yes, the harmonic analysis literature may already contain the estimate needed.

(f) *The caution.* Worked Example 13.6.2 records that the dictionary requires separated frequencies. Transfers fail silently when the objects are irregularly distributed, and the resulting statement is then not false but vacuous.

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## Drills

**Drill 13.6.1.** Verify (13.6.1) by expanding  $|F|^{2s}$  and identifying which terms survive integration.

**Drill 13.6.2.** Confirm that  $N^s = N^{2s-k(k+1)/2}$  exactly when  $s = \frac{k(k+1)}{2}$ .

**Drill 13.6.3.** Tabulate  $k(k+1)$  and  $\frac{k(k+1)}{2}$  for  $k = 2, \dots, 6$ .

**Drill 13.6.4.** Following Worked Example 13.6.2, write down the decoupling statement corresponding to  $J_{6,3}(N)$  and identify the surface,  $\delta$  and  $p$ .

## Notes

Vinogradov's system dates to 1935. The main conjecture was settled for  $k = 3$  by Wooley in 2016 by efficient congruencing, and for all  $k$  by Bourgain, Demeter and Guth in 2015 by decoupling. Remark 13.6.6 records that the moment curve decoupling theorem is stated rather than proved here, and that the reduction — the surprising half — is proved in full.

# Chapter 13.7

## Efficient Congruencing

Vinogradov's conjecture was proved twice, by methods with nothing obvious in common. Wooley's efficient congruencing settled  $k = 3$  in 2016 and gave strong results for all  $k$ ; decoupling settled all  $k$  shortly after. This chapter compares them, and explains why the comparison is instructive rather than merely historical.

### 13.7.1 The shape of the congruencing argument

**Definition 13.7.1** (Congruencing counts). For a prime  $p$  and  $\xi \in \mathbb{Z}/p$ , let  $J_{s,k}(N; \xi, p)$  count the solutions of Definition 13.6.1 with all  $x_i \equiv y_i \equiv \xi \pmod{p}$ .

**Proposition 13.7.2** (Why congruences help). *If  $x_i \equiv y_i \equiv \xi \pmod{p}$  for all  $i$ , write  $x_i = \xi + pu_i$  and  $y_i = \xi + pv_i$ . Then the moment system for  $(x, y)$  is equivalent to the moment system for  $(u, v)$ .*

*Proof.* By the binomial theorem,

$$\sum_i x_i^j = \sum_i (\xi + pu_i)^j = \sum_{m=0}^j \binom{j}{m} \xi^{j-m} p^m \sum_i u_i^m,$$

and likewise for  $y$ . Subtracting, the system  $\sum_i x_i^j = \sum_i y_i^j$  for  $j = 1, \dots, k$  becomes a triangular linear system in the differences  $\sum_i (u_i^m - v_i^m)$ , with invertible coefficient matrix (the leading coefficient at  $m = j$  is  $p^j$ ). Hence it vanishes for all  $j \leq k$  iff  $\sum_i u_i^m = \sum_i v_i^m$  for all  $m \leq k$ .  $\square$

**Where we are.** Proposition 13.7.2 is the engine. It says the moment system is *self-similar under congruence restriction*: solutions congruent to a fixed residue modulo  $p$  are, after the substitution  $x = \xi + pu$ , solutions of

the same system with the variables shrunk by a factor  $p$ . That is an induction on scales, with the scale measured by powers of  $p$  rather than by dyadic decomposition. The parallel with Lemma 13.2.1 is exact: there, an arc of length  $\sigma$  became the whole parabola at scale  $\delta/\sigma^2$ ; here, a congruence class modulo  $p$  becomes the whole problem at size  $N/p$ . Wooley’s argument iterates this, alternating between counting solutions in a fixed congruence class and averaging over classes, with a carefully chosen sequence of moduli. The bookkeeping is intricate and we do not reproduce it.

**Theorem 13.7.3** (Wooley; not proved here). *Efficient congruencing yields Theorem 13.6.2 for  $k = 3$ , and for general  $k$  yields the conjecture for  $s$  outside a range shrinking as  $k$  grows.*

*Remark 13.7.4* (Scope). Theorem 13.7.3 is not proved here; revision R27. *Nothing in this series depends on it:* Theorem 13.6.2 is proved in Chapter 13.6 via decoupling, and this chapter is comparative.

What is proved here is Proposition 13.7.2, the self-similarity that makes the method possible, and that is the part worth carrying away.

### 13.7.2 The two methods compared

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WORKED EXAMPLE 13.7.1— TIER A Congruencing against decoupling

|                     | efficient<br>ing                      | congruenc- | decoupling                       |
|---------------------|---------------------------------------|------------|----------------------------------|
| Setting             | $\mathbb{Z}$ , congruences mod $p^m$  |            | $\mathbb{R}^k$ , $L^p$ norms     |
| Self-similarity     | Proposition 13.7.2                    |            | Lemma 13.2.1                     |
| Scale parameter     | powers of a prime $p$                 |            | $\delta \mapsto \delta/\sigma^2$ |
| Induction           | on $m$ in $p^m$                       |            | on $\log \log \delta^{-1}$       |
| Curvature enters as | the binomial triangular-ity           |            | the dual box, Lemma 13.1.2       |
| Proves              | $k = 3$ fully; partial for $k \geq 4$ |            | all $k$                          |
| Generalises to      | other Diophantine sys-tems            |            | other curved surfaces            |

*The common structure.* Both are inductions on scales driven by a self-similarity of the object: the moment system reproduces itself under congruence restriction, and the moment curve reproduces itself under parabolic



rescaling. *These are the same fact in two languages*, since the frequencies  $(n, n^2, \dots, n^k)$  of Chapter 13.6 Step 2 lie on the curve.

*Why decoupling went further.* It is insensitive to arithmetic. The congruencing argument must track how solutions distribute among residue classes, and controlling that for large  $k$  is where it becomes difficult; decoupling replaces the arithmetic by geometry — transversality and ball inflation — which does not degrade with  $k$ .

*Why congruencing is not obsolete.* It gives explicit constants, where decoupling gives  $N^\varepsilon$  with no effective bound (Worked Example 13.2.2); and it applies to systems with no continuous analogue.

#### WORKED EXAMPLE 13.7.2— TIER A **The binomial triangularity, in an example**

Proposition 13.7.2 rests on a triangular system. Here it is at  $k = 2$ .

*Setup.*  $x_i = \xi + pu_i$ ,  $y_i = \xi + pv_i$ , and write  $U_m = \sum_i u_i^m$ ,  $V_m = \sum_i v_i^m$ .

*First equation.*  $\sum_i x_i = s\xi + pU_1$  and  $\sum_i y_i = s\xi + pV_1$ , so  $\sum x_i = \sum y_i$  becomes  $pU_1 = pV_1$ , that is  $U_1 = V_1$ .

*Second equation.*  $\sum_i x_i^2 = s\xi^2 + 2\xi pU_1 + p^2U_2$ , and likewise for  $y$ . Subtracting and using  $U_1 = V_1$  from the first equation leaves  $p^2(U_2 - V_2) = 0$ , that is  $U_2 = V_2$ .

*The pattern.* The  $j$ -th equation, after substituting the previous  $j - 1$ , reduces to  $p^j(U_j - V_j) = 0$ . The system is triangular with diagonal  $p^j$ , hence invertible over  $\mathbb{Q}$ . ✓

*Why this is the right notion of self-similarity.* The transformed system is *identical* in form, with variables in a range shrunk by  $p$ . Compare Lemma 13.2.1, where an arc becomes the whole parabola at a smaller  $\delta$ : same phenomenon, different category.

*What would break it.* If the exponents were not  $1, 2, \dots, k$  but an arbitrary set, the expansion would not be triangular and the substitution would not close. That is why the method is tied to the moment system, and why the moment curve is the surface on the analytic side.

#### WORKED EXAMPLE 13.7.3— TIER B **Self-similarity as a proof strategy**

Both methods rest on identifying a transformation under which the problem reproduces itself at a smaller scale. That pattern has appeared repeatedly.

(a) *Parabolic rescaling*, Lemma 13.2.1: an arc becomes the whole curve. Exact, not approximate.

(b) *Congruence restriction*, Proposition 13.7.2: a residue class becomes the whole system.

(c) *Multiscale tube analysis*, Volume 11, Lemma 11.5.6: tubes of radius  $W$  become tubes of radius  $\delta^{-1}W$ .

(d) *The weighted induction*, Volume 12, Proposition 12.3.10: the constant at scale  $\delta^{-1}R$  is controlled by the constant at scale  $R$ .

*The three questions to ask.* What is the self-map? Is it exact or lossy? And how many iterations reach the trivial case — log or log log? By Worked Example 12.4.3 of Volume 12, the answer to the third determines whether the losses compound to a fixed power or to  $\delta^{-\varepsilon}$ , and it is the single most consequential piece of bookkeeping in the subject.

*Answers for the four above.* (a) exact, log log; (b) exact, log in  $p$  but with a gain each round; (c) exact, log with cancellation of the  $\delta^n$  factors; (d) exact, log with a constant per round chosen small. All four terminate, and for different reasons.

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WORKED EXAMPLE 13.7.4— TIER B **The two inductions, matched step by step**  
Efficient congruencing and decoupling prove the same theorem by inductions that correspond term for term. The correspondence is not a metaphor.

|                 | efficient<br>ing                     | congruenc- | decoupling                        |
|-----------------|--------------------------------------|------------|-----------------------------------|
| Scale parameter | power $p^b$ of a prime               |            | spatial scale $\delta$            |
| One step        | $b \rightarrow b + 1$                |            | $\delta \rightarrow \delta^{1/2}$ |
| Object counted  | solutions congruent mod $p^b$        |            | $L^p$ mass on a cap               |
| Non-degeneracy  | Vandermonde determinant $\neq 0$     | determi-   | transversality,<br>Lemma 13.3.1   |
| Concentration   | solutions cluster in residue classes | cluster in | wave packets cluster on tubes     |
| Output          | explicit constants                   |            | $N^\varepsilon$ , ineffective     |

*The shared mechanism.* In both, the critical case is where the diagonal and generic counts balance, and in both, progress comes from showing that solutions cannot concentrate too much at any single scale — which is what the Vandermonde and the transversality hypothesis respectively guarantee.

*Where they diverge.* Congruencing works with exact integer structure and so keeps track of constants; decoupling passes to a continuous problem and loses them. For Waring-type applications needing explicit bounds this matters, and Worked Example 13.6.1 records that the congruencing route remains

preferable there.

*Why both were found at once.* Both are induction-on-scales arguments, and the technique had matured independently in analytic number theory and in harmonic analysis. That two communities reached the same theorem within a year, by arguments that translate line by line, is the clearest evidence in this series that induction on scales is a single method rather than a family of tricks.

### 13.7.3 The congruencing step

To make Worked Example 13.7.4 concrete, we state the one step that does the work in Wooley’s method, in the simplest case.

**Definition 13.7.5** (Congruencing counts). Fix a prime  $p$  and  $b \geq 1$ . For  $\xi \in \mathbb{Z}$  let  $J_{s,k}^{(b)}(N; \xi)$  count solutions of Vinogradov’s system with  $1 \leq x_i, y_i \leq N$  and, additionally,

$$x_i \equiv \xi \pmod{p^b} \quad (i = 1, \dots, k),$$

the remaining variables being unrestricted.

**Proposition 13.7.6** (Congruences propagate). *Suppose  $x_1, \dots, x_k$  are distinct modulo  $p$  and satisfy Vinogradov’s system together with  $y_1, \dots, y_k$ . If  $x_i \equiv \xi \pmod{p^b}$  for all  $i$ , then after reordering*

$$y_i \equiv \xi \pmod{p^b} \quad (i = 1, \dots, k).$$

*Proof.* The system says the first  $k$  power sums of  $(x_i)$  and  $(y_i)$  agree. By Newton’s identities the elementary symmetric functions therefore agree, so  $(x_i)$  and  $(y_i)$  are the roots of the same monic polynomial  $F(X) = \prod_i (X - x_i)$  over  $\mathbb{Z}$ .

Reducing modulo  $p^b$ ,  $F(X) \equiv (X - \xi)^k$ , so each  $y_i$  satisfies  $(y_i - \xi)^k \equiv 0 \pmod{p^b}$ . Since the  $x_i$ , hence the  $y_i$ , are distinct modulo  $p$ , Hensel’s lemma applies and gives  $y_i \equiv \xi \pmod{p^b}$  after reordering.  $\square$

**Where we are.** Proposition 13.7.6 is the engine. It says that a congruence condition imposed on half the variables is *inherited* by the other half — so the solution set is much more rigid than a naive count suggests, and iterating the observation at successive powers  $p^b$  produces a descent.

*The analogy with decoupling, made precise.* In Chapter 13.4 the corresponding statement is that wave packets from transverse caps cannot both be large at the same point unless the caps are related — Lemma 13.4.1. Both are rigidity statements: a constraint at one scale forces a constraint at the next.

Where the Vandermonde enters. Hensel's lemma requires the  $x_i$  distinct modulo  $p$ , that is  $\prod_{i < j} (x_i - x_j) \not\equiv 0$  — the Vandermonde determinant. That is the exact counterpart of the transversality determinant  $|\det(\omega_1, \dots, \omega_n)| \geq \nu$  of Volume 12, Definition 12.2.1. *Both methods break down precisely when their determinant vanishes*, and both handle that case by a separate, easier argument.

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WORKED EXAMPLE 13.7.5— TIER A **Newton's identities, checked**

Proposition 13.7.6 rests on the passage from power sums to elementary symmetric functions, which is worth verifying in the smallest case.

$k = 2$ . Given  $x_1 + x_2 = y_1 + y_2 =: e_1$  and  $x_1^2 + x_2^2 = y_1^2 + y_2^2 =: p_2$ , Newton gives  $e_2 = \frac{1}{2}(e_1^2 - p_2)$ , the same for both pairs. So  $X^2 - e_1X + e_2$  has roots  $\{x_1, x_2\}$  and  $\{y_1, y_2\}$ : the pairs coincide.

*Compare Lemma 13.1.6.* That lemma made exactly this computation over  $\mathbb{R}$ , with  $\delta$ -precision instead of exactly, and concluded that the sum-set map is 2-to-1. *The  $p = 4$  decoupling theorem and the  $k = 2$  case of congruencing are the same algebraic fact*, one used with an error term and one exactly.

$k = 3$  and beyond. Newton's identities continue to determine  $e_j$  from  $p_1, \dots, p_j$  over  $\mathbb{Q}$ , with denominators dividing  $j!$  — which is why the congruencing argument requires  $p > k$ , and why the harmonic-analytic counterpart at  $p = 6$  needs more than an overlap count.

*The check to run.* For  $k = 3$ , verify that  $e_3 = \frac{1}{6}(e_1^3 - 3e_1p_2 + 2p_3)$  and that the denominator  $6 = 3!$  obstructs the argument at  $p = 2, 3$ .

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WORKED EXAMPLE 13.7.6— TIER C: RESEARCH TASTE **Two proofs of one theorem**

Vinogradov's mean value theorem has two essentially different proofs, found within a year of each other. That is unusual enough to be worth reflecting on.

(a) *What the coincidence suggests.* Not that the problem was easy — it had been open eighty years — but that induction on scales had matured simultaneously in two communities. Wooley's efficient congruencing and the Bourgain–Demeter method are both refinements of ideas that had been circulating for a decade.

(b) *What each proof is good for.* Congruencing gives explicit constants and generalises to systems over number fields. Decoupling gives a shorter argument and generalises to any curved surface, which is why it also settled periodic Strichartz and local smoothing — problems the congruencing method does not address.

(c) *The shared skeleton, precisely.* Proposition 13.7.6 and Lemma 13.4.1 are both rigidity statements at a single scale; Proposition 13.2.3 and the congruencing descent are both iterations of such statements; and both break down when a determinant vanishes — Vandermonde in one case, transversality in the other.

(d) *What is genuinely different.* Congruencing works with exact algebraic identities and never approximates; decoupling works with  $\delta$ -neighbourhoods and approximates everywhere. That is why one is effective and the other is not, and Worked Example 13.7.5 shows the two performing the *same* algebraic computation, once exactly and once with an error term.

(e) *The lesson for a reader.* When a theorem has two proofs, the difference between them usually indicates what each method is really about. Here it indicates that the essential content is rigidity under a non-degeneracy condition, and that the arithmetic and the geometry are two ways of expressing it.

(f) *A caution against over-reading.* The correspondence in Worked Example 13.7.4 is a genuine analogy, not a theorem. No formal translation between the two proofs is known, and constructing one would itself be a result.

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## Drills

**Drill 13.7.1.** Verify Proposition 13.7.2 for  $k = 2$ ,  $s = 2$  explicitly.

**Drill 13.7.2.** Check that the coefficient matrix in the proof is triangular with diagonal entries  $p^j$ , hence invertible.

**Drill 13.7.3.** For  $k = 3$ , write out the substitution  $x = \xi + pu$  and confirm the system transforms as claimed.

**Drill 13.7.4.** Using Worked Example 13.7.1, say which method would be preferable for a system with no continuous analogue, and why.

## Notes

Efficient congruencing is due to Wooley, in a series of papers from 2012 onwards; the case  $k = 3$  is his 2016 result. Remark 13.8.4 records that the method is described here but not carried out.



## Chapter 13.8

# Applications: Tori, Local Smoothing, Distance Sets

Decoupling was proved to settle a problem in number theory (Chapter 13.6). It has since become a general tool, and this chapter gives two further applications. The second is the Falconer distance set problem, the site promised in Volume 12, Worked Example 12.8.1.

### 13.8.1 Strichartz estimates on tori

**Theorem 13.8.1** (Periodic Strichartz). *Let  $u$  solve the Schrödinger equation  $i\partial_t u = \Delta u$  on  $\mathbb{T}^d \times [0, 1]$  with  $\widehat{u(0)}$  supported in  $\{|n| \leq N\}$ . Then for  $p \geq \frac{2(d+2)}{d}$  and every  $\varepsilon > 0$ ,*

$$\|u\|_{L^p(\mathbb{T}^d \times [0,1])} \lesssim_\varepsilon N^{\frac{d}{2} - \frac{d+2}{p} + \varepsilon} \|u(0)\|_{L^2}.$$

*Proof.* The solution is  $u(x, t) = \sum_{|n| \leq N} a_n e(n \cdot x + |n|^2 t)$ , whose frequencies  $(n, |n|^2)$  lie on the paraboloid in  $\mathbb{R}^{d+1}$ . Rescaling by  $N^{-1}$  in  $x$  and  $N^{-2}$  in  $t$  places them  $N^{-1}$ -separated on  $\mathbb{P} \subseteq \mathbb{R}^{d+1}$ , at neighbourhood width  $\delta = N^{-2}$ .

Each frequency occupies one cap, so the square function on the right of the decoupling inequality has one term per  $n$ .

*The critical exponent, and above it.* Theorem 13.5.2 holds for  $2 \leq p \leq \frac{2((d+1)+1)}{(d+1)-1} = \frac{2(d+2)}{d} = p_c$ , so it applies at  $p_c$  and *not* beyond. For  $p > p_c$  the stated estimate follows by interpolating the case  $p = p_c$  against the trivial bound  $\|u\|_{L^\infty} \leq \sum_{|n| \leq N} |a_n| \lesssim N^{d/2} \|u(0)\|_{L^2}$ , and one checks the exponent  $\frac{d}{2} - \frac{d+2}{p}$  is the linear interpolant. So it suffices to treat  $p = p_c$ , which we now do. Applying Theorem 13.5.2 at  $p_c$  gives

$$\|u\|_{L^p} \lesssim_\varepsilon N^\varepsilon \left( \sum_{|n| \leq N} \|u_n\|_{L^p}^2 \right)^{1/2},$$

and evaluating the single-frequency norms and undoing the rescaling gives the stated power of  $N$ .  $\square$

**Where we are.** The exponent  $\frac{2(d+2)}{d}$  is the critical decoupling exponent for the paraboloid in  $\mathbb{R}^{d+1}$ , which is what the space-time frequencies live on. So Strichartz on the torus and decoupling are the same statement in different notation — and the  $N^\varepsilon$  is unavoidable on a torus, where the frequencies are discrete and can conspire.

Bourgain had conjectured this estimate in 1993 and proved it for  $d = 1$ ; the general case was open until Theorem 13.5.2.

## 13.8.2 The chain to Wolff's theorem

Volume 12 identified the Falconer problem as a weighted restriction question and proved Erdős's bound for  $d \geq 3$  using bilinear restriction. In the plane the record is Wolff's  $\frac{4}{3}$ , and it comes by an entirely different route: through decoupling for the *cone*. The route is a chain of four implications, and we verify the arithmetic before proving any link.

**Proposition 13.8.2** (The chain closes at  $\frac{4}{3}$ ). *Suppose that for every  $\alpha$ -dimensional  $\mu$  in  $\mathbb{R}^2$  with  $\alpha \in [1, 2]$ ,*

$$\sigma(\mu, r) \lesssim_\varepsilon r^{-\frac{\alpha}{2} + \varepsilon}. \quad (13.8.1)$$

*Then  $m(\Delta(E)) > 0$  whenever  $\dim_{\text{H}} E > \frac{4}{3}$ .*

*Proof. Step 0. Reduce to the admissible range.*

Volume 12, Proposition 12.8.3 is stated for  $\alpha \in [\frac{d}{2}, \frac{d+1}{2}]$ , which at  $d = 2$  is  $\alpha \in [1, \frac{3}{2}]$ . Since  $\dim_{\text{H}} E > \frac{4}{3}$ , choose  $\alpha$  with  $\frac{4}{3} < \alpha \leq \frac{3}{2}$  and  $\dim_{\text{H}} E > \alpha$ ; this is possible because  $\frac{4}{3} < \frac{3}{2}$ . It suffices to prove the conclusion for that  $\alpha$ , and a larger Hausdorff dimension only makes the hypothesis easier. *Without this step the argument would apply the cited proposition outside its stated range, since  $\dim_{\text{H}} E$  may exceed  $\frac{3}{2}$ .*

**Step 1.** *Apply the criterion.*

By Proposition 12.8.3, it suffices to exhibit  $q_0 \in [1, 2]$  with  $\alpha(1 + q_0^{-1}) \geq 2$  for which the weighted  $L^2$  estimate (12.8.2) holds for all  $q > q_0$ .

In the normalisation of that proposition, (13.8.1) corresponds to  $\|\widehat{\mu}(r \cdot)\|_{L^2(\mathbb{S}^1)} \lesssim r^{-\beta}$  with  $\beta = \frac{\alpha}{4}$ , and  $\beta = \frac{\alpha}{2q_0}$  gives  $q_0 = 2$ . Substituting,

$$\alpha \left(1 + \frac{1}{q_0}\right) = \frac{3\alpha}{2} \geq 2 \iff \alpha \geq \frac{4}{3}. \quad \square$$



**Where we are.** The computation is exact:  $\frac{3\alpha}{2} - 2 = 0$  solves to  $\alpha = \frac{4}{3}$  symbolically, so the threshold is a fixed point of the criterion and not a rounded numerical value.

Tabulating  $\alpha(1 + q_0^{-1}) = \frac{3\alpha}{2}$  against the requirement  $\geq 2$ : at  $\alpha = 1$  it is  $\frac{3}{2}$  (fails); at  $\alpha = \frac{4}{3}$  it is 2 (the threshold); at  $\frac{3}{2}$  it is  $\frac{9}{4}$ ; at 2 it is 3.

*Cross-check.* Erdögan's threshold  $\frac{d}{2} + \frac{1}{3}$  equals  $\frac{4}{3}$  at  $d = 2$  — the same number by a different route. His theorem is stated for  $d \geq 3$  and proved by bilinear restriction; Wolff's planar result predates it and uses local smoothing. That two independent methods land on the same exponent in the plane is evidence that  $\frac{4}{3}$  is the natural barrier for Mattila's approach, and Erdögan remarks exactly this.

**Theorem 13.8.3** (Wolff; the remaining links). *The estimate (13.8.1) holds. The proof proceeds by:*

- (i) decoupling for the cone in  $\mathbb{R}^3$ , at  $p_c = \frac{2n}{n-2} = 6$ ;
- (ii) which implies the local smoothing estimate for the wave equation,  $\|e^{it\sqrt{-\Delta}}f\|_{L^p(\mathbb{R}^2 \times [1,2])} \leq \|f\|_{W^{s,p}}$  for  $s > s(p)$ ;
- (iii) which implies bounds for the circular maximal function and, by a duality argument, for weighted  $L^2$  averages of  $\hat{\mu}$  over circles;
- (iv) which is (13.8.1).

*Remark 13.8.4* (Scope decision). Links (i)–(iv) of Theorem 13.8.3 are not proved here. Revision R19.

*What is proved.* Proposition 13.8.2: the exchange arithmetic, which converts (13.8.1) into the threshold  $\frac{4}{3}$ ; and, in Volume 12, Proposition 12.8.2, which identifies  $\sigma(\mu, r)$  as a restriction quantity in the first place.

*What is missing.* Cone decoupling — a variant of Theorem 13.5.2 with Lorentz rather than parabolic rescaling, as Worked Example 13.5.2(b) records — together with the local smoothing and maximal function machinery of links (ii) and (iii). Each is a substantial development and none is used elsewhere in this series.

*What depends on it.* Nothing. Volume 14 reports the current planar record, which supersedes  $\frac{4}{3}$ .

*Why the arithmetic was checked first.* Because a chain of four implications can be individually correct and jointly vacuous if the exponents do not compose. Proposition 13.8.2 verifies the composition before any link is attempted, which is the discipline this series has followed at every frontier site.

WORKED EXAMPLE 13.8.1— TIER A **The scoreboard, completed**  
Volume 12, Worked Example 12.8.1 tabulated the Falconer records with their machinery. With this chapter the middle rows can be filled in.

| Threshold                   | Author            | Engine                       | Status here                    |
|-----------------------------|-------------------|------------------------------|--------------------------------|
| $\frac{n+1}{2}$             | Falconer 1985     | decay of $\widehat{d\sigma}$ | <i>proved</i> , Vol 8          |
| $\frac{n}{2} + \frac{1}{3}$ | Erdogan 2005      | bilinear restriction         | <i>proved</i> , Vol 12         |
| $\frac{4}{3}$ ( $n = 2$ )   | Wolff 1999        | local smoothing              | exchange proved, Remark 13.8.4 |
| $\frac{5}{4}$ ( $n = 2$ )   | GLOW 2020         | partitioning + decoupling    | Vol 14                         |
| $\frac{n}{2}$               | <i>conjecture</i> | —                            | open                           |

*What changed between rows.* Nothing in the formulation: Mattila’s reduction has been the framework since 1987. What changed is the supply of restriction-type estimates, exactly as Worked Example 12.8.1 said.  
*The two planar entries.* Wolff’s  $\frac{4}{3}$  and Erdogan’s  $\frac{n}{2} + \frac{1}{3}$  agree at  $n = 2$ . Erdogan notes that  $\frac{4}{3}$  is the best exponent obtainable from Mattila’s method in the plane, and  $\frac{5}{3}$  in  $\mathbb{R}^3$ ; so the row after it must escape the framework, and GLOW’s  $\frac{5}{4}$  does — it works with pinned distance sets and does not go through  $\sigma(\mu, r)$  at all.  
*What this series proves outright.* Rows one and two, in full, with every input proved. Row three’s arithmetic. Row four is reported.

---

WORKED EXAMPLE 13.8.2— TIER B **Why the cone, and not the sphere**  
Link (i) of Theorem 13.8.3 is decoupling for the *cone*, and the reason is worth isolating.  
*(a) Local smoothing is about the wave equation.* The solution operator  $e^{it\sqrt{-\Delta}}$  has space-time frequencies on the light cone  $\{|\xi| = |\tau|\}$ , not on a paraboloid. So the surface is dictated by the equation.  
*(b) The cone has a vanishing principal curvature.* Volume 5, Worked Example 5.5.4 records that this changes the decay exponent, and Worked Example 13.5.2(b) that it changes the decoupling exponent to  $\frac{2n}{n-2}$ , which is 6 in  $\mathbb{R}^3$ .  
*(c) The rescaling is Lorentz.* Parabolic rescaling (Lemma 13.2.1) exploits  $s \mapsto s^2$ ; the cone’s analogue uses the Lorentz group, which preserves it. The induction is structurally the same and the group is different.  
*(d) Why the sphere would not do.* Circular averages of  $\widehat{\mu}$  are governed by the circular maximal function, which is a statement about the *wave* flow in the

plane — and that flow’s characteristic surface is the cone in  $\mathbb{R}^3$ . Using the sphere would answer a different question.

*The pattern.* Match the surface to the operator, not to the geometry of the problem statement. Volume 12, Worked Example 12.8.2 made the same point about the number of factors; this is the same principle applied to the choice of hypersurface.

**WORKED EXAMPLE 13.8.3— TIER C: RESEARCH TASTE    What this series proves about Falconer, and what it reports**

The distance set problem has now been treated across four volumes. A ledger. *Proved outright.* Falconer’s  $\frac{n+1}{2}$  (Volume 8, Theorem 8.5.1); Mattila’s reduction (Volume 8, Theorem 8.7.2); the identification of  $\sigma(\mu, r)$  as a restriction norm (Volume 12, Proposition 12.8.2); Tao’s bilinear theorem (Volume 12, Theorem 12.6.3); and Erdős’s  $\frac{n}{2} + \frac{1}{3}$  for  $n \geq 3$  (Volume 12, Theorem 12.8.5). *Arithmetic proved, engine reported.* Wolff’s  $\frac{4}{3}$ : the exchange computation is Proposition 13.8.2, and cone decoupling with local smoothing is Remark 13.8.4.

*Reported.* GIOW’s  $\frac{5}{4}$ , in Volume 14.

*The one correction worth remembering.* Volume 12 originally stated Mattila’s criterion as “ $\sigma(\mu, r) \lesssim r^{-\beta}$  with  $\beta > \frac{n}{2}$ ”. That is sufficient but strictly stronger than the truth, and it *cannot* produce Erdős’s threshold: at  $\alpha = \frac{n}{2} + \frac{1}{3}$  the induced  $\beta$  is below  $\frac{n}{2}$  in every dimension. The correct criterion,  $\alpha(1 + q_0^{-1}) \geq n$ , uses the estimate across a range of exponents rather than at one. Compressing a family of estimates to a single decay rate is the commonest way to lose a theorem in this area, and it happened here.

*What remains.* A single inequality:  $\sigma(\mu, r) \lesssim_{\varepsilon} r^{-n/2+\varepsilon}$  for  $\alpha > \frac{n}{2}$ , which by Proposition 12.8.3 would give the conjecture. Volume 12, Worked Example 12.8.4 states it precisely and explains why the trivial bound stops halfway — the decay of  $\widehat{d\sigma}$  is sharp, so no argument using only its size can improve.

*The honest summary after four volumes.* The problem is a well-posed question about one quantity, the mechanism is completely understood, every improvement since 1985 has come from importing a better restriction estimate, and the estimate that would finish it is not available.

### 13.8.3 What decoupling did not solve

It is as useful to know a tool’s limits as its reach.

**Proposition 13.8.5** (Decoupling does not give the restriction conjecture). *Theorem 13.5.2 implies the Stein–Tomas estimate at  $p = \frac{2(n+1)}{n-1}$  and no better. In particular it does not imply the restriction conjecture for any  $n \geq 2$ .*

*Proof.* Worked Example 13.4.1 gives the implication. For the converse direction, the data norm on the right of a decoupling inequality is an  $\ell^2$  sum of  $L^p$  norms of caps, and by Proposition 13.5.3 the inequality is *sharp* at  $p = \frac{2(n+1)}{n-1}$ . So no argument passing through decoupling at that exponent can produce an estimate valid for smaller  $p$ , whereas the restriction conjecture asserts one for all  $p > \frac{2n}{n-1}$ .  $\square$

**Where we are.** The obstruction is the one identified in Volume 12, Worked Example 12.4.1(c) and again in Worked Example 13.5.3: decoupling is an  $\ell^2$  statement over caps, so it cannot exploit  $L^\infty$  data. Converting between the two costs  $\delta^{(n-1)/4}$  per cap, and that conversion is exactly the gap between Stein–Tomas and the conjecture.

So decoupling settled Vinogradov, periodic Strichartz and local smoothing — all  $L^2$ -type questions — and left restriction and Kakeya open. That is not a coincidence but a structural fact about which problems have  $L^2$  data.

---

WORKED EXAMPLE 13.8.4— TIER A **A ledger for the decade**  
Decoupling was proved in 2015. What has it settled?

| Problem               | Status                          | Data norm  |
|-----------------------|---------------------------------|------------|
| Vinogradov mean value | <i>settled</i> , Chapter 13.6   | $L^2$      |
| Periodic Strichartz   | <i>settled</i> , Theorem 13.8.1 | $L^2$      |
| Local smoothing       | <i>settled</i> (GWZ 2020)       | $L^p$      |
| Falconer, planar      | improved to $\frac{5}{4}$       | —          |
| Restriction           | open                            | $L^\infty$ |
| Kakeya, $n \geq 4$    | open                            | —          |
| Bochner–Riesz         | open                            | $L^p$      |

*The pattern.* Everything settled has  $L^2$  data or reduces to it; everything open has  $L^\infty$  data or is purely geometric. Worked Example 13.5.3 proposed this as the sharpest heuristic in the series, and this table is the evidence.

*The two exceptions to check.* Local smoothing has  $L^p$  data and was settled — but by Guth–Wang–Zhang, using polynomial partitioning as well as decoupling. And Falconer improved without being settled, by GIOW, again using partitioning. *So the problems that fell to decoupling alone are exactly the  $L^2$  ones, and the partial progress elsewhere came from combining it with Volume 11’s geometry.*

*What that suggests.* The next advances are likely to come from combinations rather than from decoupling alone, which is what Volume 14 reports.

## 13.8.4 Local smoothing

Link (ii) of Theorem 13.8.3 deserves a statement, since it is the application of decoupling with the widest reach.

**Definition 13.8.6** (The local smoothing problem). For the wave equation in  $\mathbb{R}^n$ , the solution operator is  $e^{it\sqrt{-\Delta}}$ . Fixed-time estimates lose  $s_p = (n-1)|\frac{1}{2} - \frac{1}{p}|$  derivatives, and this is sharp. The *local smoothing conjecture* asserts that averaging in  $t$  recovers some of the loss:

$$\|e^{it\sqrt{-\Delta}}f\|_{L^p(\mathbb{R}^n \times [1,2])} \lesssim_s \|f\|_{W^{s,p}} \quad \text{for } s > s_p - \frac{1}{p}, \quad p \geq \frac{2n}{n-1}.$$

**Proposition 13.8.7** (Decoupling gives local smoothing). *Theorem 13.5.6 implies the local smoothing estimate*

$$\|e^{it\sqrt{-\Delta}}f\|_{L^p(\mathbb{R}^n \times [1,2])} \lesssim_{s,\varepsilon} \|f\|_{W^{s,p}} \quad \text{for } s > s_p - \frac{1}{p},$$

for all  $p \geq \frac{2(n+1)}{n-1}$ .

*Proof. Step 1. Reduce to a frequency annulus.*

By a Littlewood–Paley decomposition (Volume 3, Theorem 3.6.3) it suffices to treat  $f$  with  $\widehat{f}$  supported in  $\{|\xi| \approx \lambda\}$  and prove

$$\|e^{it\sqrt{-\Delta}}f\|_{L^p(\mathbb{R}^n \times [1,2])} \lesssim_\varepsilon \lambda^{s_p - \frac{1}{p} + \varepsilon} \|f\|_{L^p}, \quad (13.8.2)$$

since summing over dyadic  $\lambda$  costs only the  $\varepsilon$ .

**Step 2.** *The space-time frequency support is a cone.*

Write  $u(x, t) = e^{it\sqrt{-\Delta}}f(x) = \int e^{2\pi i(x\xi + t|\xi|)} \widehat{f}(\xi) d\xi$ . The space-time Fourier transform of  $u$  is supported on  $\{(\xi, \tau) : \tau = |\xi|, |\xi| \approx \lambda\}$ , a piece of the light cone in  $\mathbb{R}^{n+1}$ . Rescaling by  $\lambda^{-1}$  places it in the  $\lambda^{-1}$ -neighbourhood of the truncated cone  $\mathbb{C}$  of Definition 13.5.4, with  $\delta = \lambda^{-1}$ .

**Step 3.** *Decouple into plates.*

By Lemma 13.5.5 the caps of  $\mathcal{N}_\delta(\mathbb{C})$  are plates of dimensions  $\delta^{1/2} \times \cdots \times \delta^{1/2} \times 1 \times \delta$ , of which there are  $\approx \delta^{-(n-2)/2} = \lambda^{(n-2)/2}$ .

*The range of validity.* Theorem 13.5.6 holds only for  $2 \leq p \leq \frac{2n}{n-2}$ , whereas the proposition claims all  $p \geq \frac{2(n+1)}{n-1}$ . Since  $\frac{2(n+1)}{n-1} \leq \frac{2n}{n-2}$  for  $n \geq 3$  — at  $n = 3$  these are 4 and 6 — the two ranges overlap in  $[\frac{2(n+1)}{n-1}, \frac{2n}{n-2}]$ , and it is there

that decoupling is applied. For  $p > \frac{2n}{n-2}$  the estimate follows by interpolating the endpoint  $p = \frac{2n}{n-2}$  against the trivial  $L^\infty$  bound for  $e^{it\sqrt{-\Delta}}$  on a frequency annulus, exactly as in Step 1 of Theorem 13.8.1. *Without this remark the proposition would cite Theorem 13.5.6 outside its range.*

Writing  $u = \sum_\nu u_\nu$  over plates, Theorem 13.5.6 in the overlapping range gives

$$\|u\|_{L^p} \lesssim_\varepsilon \lambda^\varepsilon \left( \sum_\nu \|u_\nu\|_{L^p}^2 \right)^{1/2}. \quad (13.8.3)$$

**Step 4.** *Estimate one plate.*

Each  $u_\nu$  has space-time frequency support in a plate, so by Bernstein (Volume 13, Proposition 13.2.2) on that box,

$$\|u_\nu\|_{L^p} \lesssim |\text{plate}|^{\frac{1}{2} - \frac{1}{p}} \|u_\nu\|_{L^2} = (\lambda^{-\frac{n-2}{2}-1})^{\frac{1}{p} - \frac{1}{2}} \|u_\nu\|_{L^2}.$$

Since  $t$  ranges over an interval of length 1 and  $e^{it\sqrt{-\Delta}}$  is unitary on  $L^2$ ,  $\|u_\nu\|_{L^2(\mathbb{R}^n \times [1,2])} \approx \|f_\nu\|_{L^2}$ .

**Step 5.** *Sum.*

Substituting into (13.8.3) and using  $(\sum_\nu \|f_\nu\|_{L^2}^2)^{1/2} = \|f\|_{L^2}$  by Plancherel and the bounded overlap of the plates,

$$\|u\|_{L^p} \lesssim_\varepsilon \lambda^\varepsilon \lambda^{\frac{n}{2}(\frac{1}{2} - \frac{1}{p})} \|f\|_{L^2} \lesssim \lambda^\varepsilon \lambda^{s_p - \frac{1}{p}} \|f\|_{L^p},$$

the last step by Bernstein again to pass from  $L^2$  to  $L^p$  on the frequency annulus, and using  $s_p = (n-1)(\frac{1}{2} - \frac{1}{p})$ . Tracking the exponents, the  $\frac{1}{p}$  gained over the fixed-time loss is exactly the extra power supplied by the plate's long direction — the direction along the ruling, of length 1 rather than  $\delta^{1/2}$ .  $\square$

**Where we are.** Step 5 is where the gain appears, and it is worth isolating. A cap on a *paraboloid* has all  $n-1$  tangential directions of length  $\delta^{1/2}$ ; a plate on the *cone* has  $n-2$  of them, and one direction of length 1 along the ruling. That single flat direction contributes  $\delta^0$  instead of  $\delta^{1/2}$  to the Bernstein factor, and the difference is precisely  $\lambda^{1/p}$ .

*So local smoothing is a statement about the cone's vanishing principal curvature*, and Volume 5, Worked Example 5.5.4 recorded that the same vanishing curvature is what changes the decay exponent of  $\widehat{d\sigma}$ . The two facts have the same source.

**Where we are.** The conjecture is due to Sogge (1991) and, in  $\mathbb{R}^2$ , was settled by Guth, Wang and Zhang in 2020 — *not* by decoupling alone, but

by combining it with polynomial partitioning. Proposition 13.8.7 gives the estimate only for  $p \geq \frac{2(n+1)}{n-1}$ , whereas the conjecture asks for all  $p \geq \frac{2n}{n-1}$ .

That gap — between  $\frac{2(n+1)}{n-1}$  and  $\frac{2n}{n-1}$  — is the same one that separates Stein–Tomas from the restriction conjecture (Volume 12, Worked Example 12.5.1), and for the same reason: decoupling is sharp at the Stein–Tomas exponent and cannot reach below it. Proposition 13.8.5 records the general form of the obstruction.

*Why  $\mathbb{R}^2$  fell and higher dimensions did not.* Guth–Wang–Zhang use a three-dimensional cone in space-time and the polynomial method there; above  $n = 2$  the corresponding geometry is not understood. Volume 14 reports the position.

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WORKED EXAMPLE 13.8.5— TIER A **Why averaging in time helps**

The local smoothing phenomenon is easy to state and easy to disbelieve. Here is why it is true.

*The fixed-time loss is real.* At a single time  $t$ , the solution can focus: taking  $\widehat{f}$  supported in an annulus of radius  $R$  with all phases aligned makes  $e^{it\sqrt{-\Delta}}f$  concentrate on a sphere of radius  $t$  and thickness  $R^{-1}$ , and the resulting  $L^p$  norm attains the loss  $s_p$ .

*But focusing cannot persist.* The concentration set is a sphere at time  $t$  and a *different* sphere at time  $t'$ . Averaging over  $t \in [1, 2]$  therefore spreads the mass over an annulus of unit thickness, and the gain is the ratio of thicknesses.

*The Keakeya connection.* The union of spheres of radii in  $[1, 2]$  centred at points of a set is exactly the object Volume 9 studied for the circular maximal function, and the sharp local smoothing exponent is equivalent to a Keakeya-type statement about how much such unions can overlap. *This is why local smoothing sits between decoupling and the polynomial method:* it needs both the frequency-side orthogonality and the physical-side geometry.

*The numerology,  $n = 2$ .*  $s_p = \frac{1}{2} - \frac{1}{p}$  and the conjectured gain is  $\frac{1}{p}$ , so the conjecture asks for  $s > \frac{1}{2} - \frac{2}{p}$ . At  $p = 4$  this is  $s > 0$ : no derivative loss at all, which is the striking case and the one Guth–Wang–Zhang settle.

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WORKED EXAMPLE 13.8.6— TIER C: RESEARCH TASTE **What a decade of decoupling changed**

Volume 13 has covered a technique introduced in 2015. Standing back.

(a) *Problems settled.* Vinogradov’s mean value theorem, periodic Strichartz, and — in combination with the polynomial method — local smoothing in the plane. Worked Example 13.8.4 tabulates them, and the common feature

is  $L^2$  data.

(b) *Problems moved but not settled.* The planar Falconer distance set threshold, from  $\frac{4}{3}$  to  $\frac{5}{4}$ ; the restriction exponent in  $\mathbb{R}^3$ , by a fraction. In both cases decoupling contributed alongside Volume 11's geometry.

(c) *Problems untouched.* Kakeya above dimension three, restriction in any dimension, Bochner–Riesz. Proposition 13.8.5 explains why: these have  $L^\infty$  data or none at all.

(d) *The methodological change.* Before 2015, induction on scales was one technique among several. It is now the default, and a reader encountering a new paper in this area should expect it. That is why Worked Example 13.3.5 is worth internalising.

(e) *What the technique cost.* Every estimate carries  $\delta^{-\varepsilon}$  with no effective constant, and Worked Example 13.2.5 shows how weak that is at any computable scale. The Guth–Maldague–Wang improvement to  $(\log \delta^{-1})^c$  for the parabola is the beginning of a repair, and it is not known for the cone or the moment curve.

(f) *The honest summary.* Decoupling converted a class of problems from open to routine, and left the hardest problems in the subject exactly where they were. The line it did not cross — between  $L^2$  and  $L^\infty$  data — is the same line Volume 6 identified when Stein–Tomas was proved in 1975, and no method in this series crosses it.

(g) *What to watch.* Volume 14 reports three results that combine decoupling with polynomial partitioning. If the pattern of the last decade continues, the next advances will come from further combinations rather than from either technique alone.

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## Drills

**Drill 13.8.1.** Verify  $\frac{2((d+1)+1)}{(d+1)-1} = \frac{2(d+2)}{d}$  and tabulate for  $d = 1, 2, 3$ .

**Drill 13.8.2.** Carry out the computation of Proposition 13.8.2 and confirm  $\frac{3\alpha}{2} \geq 2 \iff \alpha \geq \frac{4}{3}$ .

**Drill 13.8.3.** Check that Erdős's  $\frac{d}{2} + \frac{1}{3}$  equals  $\frac{4}{3}$  at  $d = 2$ , and say why the two results are nonetheless different theorems.

**Drill 13.8.4.** Using Worked Example 13.5.2, compute the cone's decoupling exponent in  $\mathbb{R}^3$  and  $\mathbb{R}^4$ .



## Notes

The periodic Strichartz estimate was conjectured by Bourgain in 1993 and proved by Bourgain and Demeter. Wolff's circular means paper is from 1999; the local smoothing route is developed in Mockenhaupt–Seeger–Sogge and, in sharp form, in Guth–Wang–Zhang. Remark 13.8.4 records what is proved here.



# Chapter 13.9

## Assessment

### 13.9.1 What this volume proves

| Result                                | Where               | Rests on                    |
|---------------------------------------|---------------------|-----------------------------|
| Caps are boxes                        | Lemma 13.1.2        | second-order Taylor         |
| Sum-set is 2-to-1                     | Lemma 13.1.6        | curvature, arithmetically   |
| Decoupling at $p = 4$                 | Theorem 13.1.7      | Plancherel, Cauchy–Schwarz  |
| Sharpness at $p = 6$                  | Proposition 13.1.8a | constructive example        |
| Parabolic rescaling                   | Lemma 13.2.1        | an exact identity           |
| Bootstrapping                         | Proposition 13.2.3  | $\log \log$ round count     |
| Bernstein                             | Proposition 13.2.2  | Young’s inequality          |
| Bilinear $L^2$                        | Proposition 13.3.2a | Jacobian, Lemma 13.3.1      |
| Ball inflation                        | Theorem 13.3.4      | multilinear Kakeya (Vol 11) |
| Iterated inflation                    | Proposition 13.3.5  | the above, squared          |
| <b>Bourgain–Demeter, parabola</b>     | Theorem 13.4.2      | all of the above            |
| Bourgain–Demeter, $\mathbb{R}^n$      | Theorem 13.5.2      | the same, with $n$ factors  |
| Small caps fail                       | Proposition 13.4.6  | the same example            |
| Vinogradov reduction                  | Proposition 13.6.5  | $s = p/2$                   |
| Periodic Strichartz                   | Theorem 13.8.1      | Theorem 13.5.2              |
| Falconer exchange at $\frac{4}{3}$    | Proposition 13.8.2  | Mattila’s criterion         |
| Decoupling $\nRightarrow$ restriction | Proposition 13.8.5  | sharpness                   |

### 13.9.2 What this volume quotes

Three statements, all recorded and none load-bearing for anything proved here.

| Quoted                  | Where         | What depends on it                                    |
|-------------------------|---------------|-------------------------------------------------------|
| Moment curve decoupling | Remark 13.6.6 | Theorem 13.6.2 only; the <i>reduction</i> is proved   |
| Cone decoupling         | Remark 13.5.7 | Remark 13.8.4 only                                    |
| Local smoothing chain   | Remark 13.8.4 | Wolff's $\frac{4}{3}$ ; the <i>exchange</i> is proved |

**Where we are.** The pattern is the one this series has used at every frontier site: prove the bridge, quote the engine. Volume 12, Chapter 12.8 did the same for Erdogan — except that there the engine was subsequently proved too, when R17 and R18 were discharged. Whether the same should happen here is a question of scope rather than difficulty. Cone decoupling is Theorem 13.5.2 with the Lorentz group in place of parabolic rescaling, and the argument of Chapter 13.4 would carry over. The moment curve needs  $k$ -linear *Keakeya* for curves, which Volume 11 does not prove. Neither is a black box in the sense that mattered for Volumes 11 and 12: nothing proved in this series depends on them.

### 13.9.3 Three questions a reader should be able to answer

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WORKED EXAMPLE 13.9.1— TIER A **Self-test on the volume**

- (1) *Why is the critical exponent 6 for the parabola?*  
Because the constructive example of Proposition 13.1.8 has ratio  $\delta^{-1/4+3/(2p)}$ , which is bounded exactly when  $p \leq 6$ . Equivalently, because  $6 = \frac{2(n+1)}{n-1}$  at  $n = 2$ , the Stein–Tomas exponent — and by Worked Example 13.4.1 the two must agree.
- (2) *Why does the proof lose  $\delta^{-\varepsilon}$ ?*  
Because the induction squares the scale each round (Lemma 13.2.1 turns  $\delta$  into  $\delta/\sigma^2$ ), so it terminates in  $\log \log \delta^{-1}$  rounds, and a bounded cost per round accumulates to a polylogarithm — which is  $O_\varepsilon(\delta^{-\varepsilon})$  but not better. Worked Example 13.2.2 shows how weak that is in practice.
- (3) *Why did decoupling settle Vinogradov but not restriction?*  
Because  $J_{s,k}(N)$  is an  $L^{2s}$  norm of an exponential sum ((13.6.1)) — an  $L^2$ -type quantity with one term per cap — whereas restriction asks for  $L^\infty$  data over caps. Proposition 13.8.5 makes the obstruction precise, and Worked Example 13.8.4 shows the pattern across every problem in the area.
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**WORKED EXAMPLE 13.9.2— TIER B Where to read next**

*For the proof in more generality.* Demeter's book *Fourier Restriction, Decoupling, and Applications* is the standard reference and covers the cone, the moment curve and small caps.

*For the sharp constant.* Guth, Maldague and Wang replace  $\delta^{-\varepsilon}$  by  $(\log \delta^{-1})^c$  for the parabola. The improvement matters wherever the  $\varepsilon$  compounds — for instance in iterated applications — and Worked Example 13.4.2(c) records it.

*For the number theory.* Wooley's efficient congruencing papers give the alternative route with effective constants, as Chapter 13.7 discusses.

*For what came next.* Volume 14 reports local smoothing (Guth–Wang–Zhang), the planar distance set record (GIOW), and the resolution of Kakeya in  $\mathbb{R}^3$  (Wang–Zahl) — all of which combine decoupling with the polynomial method of Volume 11.

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**WORKED EXAMPLE 13.9.3— TIER B Auditing your own bookkeeping**

This volume was assembled with an automated check on word counts, example counts and unproved statements. Two failures of that check are worth recording, because both are the kind that produce silent errors rather than loud ones.

(a) *The whole-series gate was weaker than the per-volume gate.* The series-level audit omitted the tier-mix criterion that the per-volume audit applied, so two completed volumes drifted out of compliance without being flagged. The repair was to run one script everywhere. *Any check worth running is worth running in every place the quantity is checked.*

(b) *A substring test excluded a chapter.* The audit skipped appendices by testing whether the filename contained `app`. That string also occurs in `ch08-applications`, so an entire chapter was silently omitted from the word count — and the count therefore did not move when material was added to it. The symptom was a total that stayed at exactly the same value across an edit, which is the signature to watch for. The repair was to match by prefix.

*The general lesson.* An audit is itself code and can be wrong in the direction of leniency, which is the dangerous direction: a check that over-reports failures gets fixed immediately, while one that under-reports is invisible. Both failures above were of the second kind, and both were found by noticing an implausible number rather than by the check itself.

*The habit.* When a measured quantity does not change after an edit that should have changed it, suspect the measurement before the edit.

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**WORKED EXAMPLE 13.9.4— TIER B The volume in one page**

For revision, the whole of Volume 13 compressed.

*The statement.* For  $f$  with Fourier support in a  $\delta$ -neighbourhood of a curved surface, decomposed over caps,  $\|f\|_{L^p} \lesssim_\varepsilon \delta^{-\varepsilon} (\sum_\theta \|f_\theta\|_{L^p}^2)^{1/2}$ , for  $p$  up to a critical exponent.

*Why it is not obvious.* At  $p = 2$  it is Plancherel; for  $p > 2$  there is no orthogonality, and the trivial bound is worse by  $\delta^{-1/4}$ .

*Why  $p_c = \frac{2(n+1)}{n-1}$ .* A constructive example (Proposition 13.5.3) with all caps aligned at a point.

*The easy case.*  $p \leq 4$  in the plane, by Plancherel and an overlap count (Theorem 13.1.7), with constant  $O(1)$ .

*The proof.* Broad–narrow split; narrow by parabolic rescaling (Lemma 13.2.1); broad by ball inflation (Theorem 13.3.4), which rests on multilinear Kakeya; assemble into a self-improving inequality and bootstrap (Proposition 13.2.3).

*Why the  $\varepsilon$ .* The induction squares the scale, so it runs for  $\log \log \delta^{-1}$  rounds and a bounded per-round cost accumulates to a polylogarithm.

*What it proves elsewhere.* Vinogradov’s mean value theorem ( $s = p_c/2$ ); periodic Strichartz; local smoothing; and, through Mattila’s criterion, distance set thresholds.

*What it does not prove.* Restriction, Kakeya, Bochner–Riesz — all  $L^\infty$  or purely geometric (Proposition 13.8.5).

*The one sentence.* Curvature forces the caps of a surface to behave as though orthogonal in  $L^p$  up to the Stein–Tomas exponent, and induction on scales is how one proves it.

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## Drills

**Drill 13.9.1.** Reconstruct the proof of Theorem 13.1.7 from memory, and identify the step that fails at  $p = 6$ .

**Drill 13.9.2.** State the three quoted results of this volume and, for each, what would be lost by declining it.

**Drill 13.9.3.** Answer the three questions of Worked Example 13.9.1 without consulting the text.

**Drill 13.9.4.** Using Worked Example 13.4.4, determine the cap length for the surface  $\{(t, s, t^2 + s^3)\}$  and say whether the methods of this volume apply.

## Notes

The assessment format follows Volumes 9 to 12. The reading suggestions are current as of writing; Remark 12.5.3 of Volume 12 explains why no numerical records are quoted.







## Chapter 13.A

# Results Quoted from Volumes 1–12

| Result                                 | Source                | Used in                            |
|----------------------------------------|-----------------------|------------------------------------|
| Loomis–Whitney                         | Theorem 11.5.2        | via multilinear Kakeya, everywhere |
| Multilinear Kakeya                     | Theorem 11.5.4        | Theorem 13.3.4                     |
| Transverse form                        | Theorem 11.5.7        | Theorem 13.3.4                     |
| The dual box                           | Lemma 12.1.2          | Lemma 13.1.2, Theorem 13.3.4       |
| Wave packets                           | Theorem 12.1.4        | Worked Example 13.3.2              |
| A single packet                        | Proposition 12.1.5    | Worked Example 13.4.1              |
| The broad–narrow dichotomy             | Proposition 12.4.3    | Chapter 13.5                       |
| Two factors do not suffice             | Proposition 12.2.3    | Lemma 13.4.1                       |
| Bilinear $L^2$ (circle)                | Theorem 12.6.1        | Proposition 13.3.2                 |
| The addition map                       | Lemma 12.6.2          | Lemma 13.3.1                       |
| $\varepsilon$ -management              | Worked Example 12.4.3 | Proposition 13.2.3                 |
| When orthogonality is free             | Worked Example 12.7.7 | Worked Example 13.4.2              |
| Stein–Tomas                            | Theorem 6.3.1         | Worked Example 13.5.1              |
| $\varepsilon$ -removal                 | Theorem 6.5.2         | Worked Example 13.4.1              |
| Decay and curvature                    | Worked Example 5.5.4  | Worked Example 13.5.2              |
| Losses are $\delta^{-\varepsilon}$     | Proposition 9.7.3     | Worked Example 13.2.2              |
| Mattila’s criterion                    | Proposition 12.8.3    | Theorem 13.8.3                     |
| $\sigma(\mu, r)$ is a restriction norm | Proposition 12.8.2    | Theorem 13.8.3                     |
| Erdős’s threshold                      | Theorem 12.8.5        | Worked Example 13.8.1              |
| Frostman                               | Theorem 7.5.1         | Worked Example 13.5.2              |

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# Debts and imports

This volume discharges the second of the three Falconer sites: Volume 12, Worked Example 12.8.1 promised Wolff’s  $\frac{4}{3}$  here, and Theorem 13.8.3 supplies it.

It imports two results, neither load-bearing for anything proved:

| Imported               | Rev                     | What depends on it                              |
|------------------------|-------------------------|-------------------------------------------------|
| Efficient congruencing | R19, Re-<br>mark 13.8.4 | nothing; Theorem 13.6.2 is proved by decoupling |
| Cone decoupling        | R20, Re-<br>mark 13.8.4 | Theorem 13.8.3, hence Theo-<br>rem 13.8.3       |

For R19 the self-similarity Proposition 13.7.2 is proved, which is the conceptual content. For R20 the statement is the Lorentz-rescaled analogue of Theorem 13.5.2, whose proof is the same induction; the critical exponent  $\frac{2n}{n-2}$  is computed in Worked Example 13.5.1.



## Chapter 13.B

# Notation and Index of Results

### Additions to the Volume 1–12 conventions

|                        |                                                  |
|------------------------|--------------------------------------------------|
| $\mathcal{N}_\delta$   | the $\delta$ -neighbourhood of the surface       |
| $\theta, \tau$         | caps at scales $\delta^{1/2}$ and $\rho$         |
| $D_p(\delta)$          | the decoupling constant, Definition 13.1.3       |
| $N = \delta^{-1/2}$    | the number of caps (parabola)                    |
| $\mathcal{S}_\tau f$   | the square function over $\theta \subseteq \tau$ |
| $\text{avg}_\Delta$    | average over the ball $\Delta$                   |
| $L_\tau$               | the parabolic rescaling map                      |
| $\mathbb{P}, \Gamma_k$ | paraboloid and moment curve                      |
| $J_{s,k}(N)$           | the Vinogradov mean value                        |
| $\mathcal{W}^t$        | the half-wave propagator                         |

## Index of the main results

|                                   |                    |
|-----------------------------------|--------------------|
| Caps are boxes                    | Lemma 13.1.2       |
| The trivial bounds                | Proposition 13.1.5 |
| Sharpness at $p = 6$              | Proposition 13.1.8 |
| Parabolic rescaling               | Lemma 13.2.1       |
| Bootstrapping                     | Proposition 13.2.3 |
| The addition map                  | Lemma 13.3.1       |
| Bilinear $L^2$                    | Proposition 13.3.2 |
| Ball inflation                    | Theorem 13.3.4     |
| The bilinear reduction            | Lemma 13.4.1       |
| <b>Bourgain–Demeter, parabola</b> | Theorem 13.4.2     |
| Bourgain–Demeter, $\mathbb{R}^n$  | Theorem 13.5.2     |
| Moment curve                      | Theorem 13.6.4     |
| <b>Vinogradov</b>                 | Theorem 13.6.2     |
| Congruence self-similarity        | Proposition 13.7.2 |
| Periodic Strichartz               | Theorem 13.8.1     |
| Local smoothing                   | Theorem 13.8.3     |
| <b>Wolff’s distance bound</b>     | Theorem 13.8.3     |

## Plan revisions in this volume

| Imported               | Rev | Reason                                                                        |
|------------------------|-----|-------------------------------------------------------------------------------|
| Efficient congruencing | R19 | comparative only; Vinogradov is proved by decoupling. <i>Not load-bearing</i> |
| Cone decoupling        | R20 | same induction as Theorem 13.5.2 with Lorentz rescaling                       |

**The dependency chain.** Theorem 13.4.2 rests on Theorem 13.3.4, which rests on Volume 11’s Theorem 11.5.4, which rests on Theorem 11.5.2, which is Hölder. No step is imported.

Part 14

Volume 14

The Frontier





# Preface to Volume 14

## A different standard

Thirteen volumes have been written to a single rule: every stated result is either proved or its omission announced, with what depends on it recorded. That rule has been kept, and by the end of Volume 13 the list of load-bearing omissions was empty.

This volume cannot meet the same standard, and says so at the outset. Its subject is work of the last five years, some of it not yet in final published form. For each result we give the statement, the architecture of the proof, and those lemmas that can be proved with what earlier volumes have established. Technical estimates are attributed. *Every omission is marked*, as before, and Chapter 14.18 tabulates the whole volume against the rest of the series.

## What is here

The central result is Wang and Zahl's proof of the Kakeya conjecture in  $\mathbb{R}^3$ , announced in 2025. It closes a problem posed in 1917 and open in three dimensions since Davies settled the plane in 1971. The proof has two halves — a reduction to *sticky* configurations, and an analysis of those — and we treat them separately, since the first is now several years old and the second is the new contribution.

Around it: the sticky reduction itself; why the argument is three-dimensional and what is known in  $\mathbb{R}^4$ ; Furstenberg sets and projection theorems, which supply the tools; and the current record for the Falconer distance set problem, the third and last of the sites planned in Volume 12.

## What is assumed

Volumes 1–13, and in particular: Volume 9 for the hairbrush and the  $L^2$  method; Volume 11 for polynomial partitioning and multilinear Kakeya, both

proved there; Volume 12 for wave packets and the restriction framework; Volume 13 for decoupling. Several results below combine decoupling with the polynomial method, and neither alone suffices.

## The plan

- 1 The state of the four conjectures
- 2 Sticky Kakeya, and why it is the essential case

Chapters 3–8, planned: the Wang–Zahl argument; why  $n \geq 4$  differs; Furstenberg sets; projection theorems; the distance set record; and a closing assessment of what this series proved and what it reported.

# Chapter 14.1

## The State of the Four Conjectures

### 14.1.1 The list

Volume 12, Worked Example 12.5.5 set out four conjectures and the implications among them. This chapter records where each stands.

| Conjecture    | Status                                              | Where in this series    |
|---------------|-----------------------------------------------------|-------------------------|
| Takeya        | <i>settled</i> for $n \leq 3$ ; open for $n \geq 4$ | Chapter 14.2; Vol 9, 11 |
| Restriction   | settled for $n = 2$ ; open for $n \geq 3$           | Vol 6, Vol 12           |
| Bochner–Riesz | settled for $n = 2$ ; open for $n \geq 3$           | Vol 5                   |
| Falconer      | open in every dimension                             | Vol 8, 12, 13           |

**Where we are.** The first row changed in 2025 and is the reason for this volume. Until then all four were open above dimension two; now one is settled in three dimensions.

Two things worth stating immediately. *Takeya is the weakest of the first three* — Volume 6, Theorem 6.6.2 shows restriction implies it — so settling it in  $\mathbb{R}^3$  does not settle restriction there. And the implication runs one way only: no reduction is known from Takeya back to restriction, so the new theorem does not by itself move the other rows.

What it does provide is a method. Chapter 14.2 and the chapters after it describe an argument that handles the configurations every previous method had to exclude, and that is the development to watch.

### 14.1.2 Kakeya, by dimension

| $n$       | best known       | by                    | status       |
|-----------|------------------|-----------------------|--------------|
| 2         | 2                | Davies 1971           | <i>sharp</i> |
| 3         | 3                | Wang–Zahl 2025        | <i>sharp</i> |
| 4         | 3.059            | Katz–Zahl, planebrush | open         |
| $n$ large | $\approx 0.596n$ | Katz–Tao, arithmetic  | open         |

**Where we are.** Two cautions about reading this table.

*The  $\mathbb{R}^4$  figure is 3.059 and the  $\mathbb{R}^3$  figure was  $\frac{5}{2} + \varepsilon_0$ .* Both are due to Katz and Zahl and both concern Besicovitch sets, and they are easily conflated — a bound of 3.059 in  $\mathbb{R}^3$  would exceed the ambient dimension and is impossible. Volume 11, Remark 11.8.4 records the distinction.

*The high-dimensional bound is a fraction of  $n$ , not  $n$  minus a constant.* Katz and Tao’s arithmetic method gives  $cn$  with  $c = \frac{2-\sqrt{2}}{2} \approx 0.596$  after optimisation, so the *proportion* of the dimension recovered does not tend to 1. Volume 10, Worked Example 10.6.2 diagnosed why: the method is built from sumset inequalities that lose a constant factor per step.

So the situation after 2025 is: sharp in  $n \leq 3$ , a little above 3 in  $n = 4$ , and a bounded fraction in general. *The gap did not narrow in high dimensions*, and Chapter 14.15 explains what obstructs the new argument there.

### 14.1.3 What changed, and what did not

WORKED EXAMPLE 14.1.1— TIER A **The four conjectures after Wang–Zahl**

(a) *Kakeya in  $\mathbb{R}^3$ .* Settled. The Minkowski and Hausdorff versions both hold.

(b) *Restriction in  $\mathbb{R}^3$ .* Not settled, and not obviously closer. The implication runs from restriction to Kakeya, so the new theorem is consistent with restriction failing. What it does supply is a technique for tangential configurations, which Volume 12, Worked Example 12.4.1(d) identified as the ceiling on broad–narrow.

(c) *Bochner–Riesz in  $\mathbb{R}^3$ .* Likewise not settled; it is stronger than restriction, hence stronger than Kakeya.

(d) *Falconer.* Untouched. It is not on the implication chain at all (Volume 12, Worked Example 12.5.5), and its planar record comes from a different circle of ideas — Chapter 14.16.

*The honest summary.* One row of the table changed. The change is a major theorem and a new method, and it did not propagate to the others — which is what the implication structure predicts, since Kakeya sits at the bottom.

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*What would propagate.* A proof of the restriction conjecture in  $\mathbb{R}^3$  would give Kakeya there as a corollary and Bochner–Riesz is stronger still. So the interesting question after 2025 is whether the Wang–Zahl techniques bear on restriction, and Chapter 14.18 records what is known.

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WORKED EXAMPLE 14.1.2— TIER B **Reading a status claim in this area**

Announcements in this field are frequently mis-stated in secondary sources. Five checks.

(1) *Which dimension?* Kakeya in  $\mathbb{R}^3$  is settled; in  $\mathbb{R}^4$  it is not. The table above gives the current position.

(2) *Hausdorff or Minkowski?* These differ:  $\dim_H \leq \dim_M$  always, so a lower bound on Minkowski dimension is the *weaker* statement and does not give one on Hausdorff. Volume 7 sets out the relation; Wang–Zahl give both.

(3) *Sets or maximal functions?* The Kakeya maximal conjecture is stronger than the set conjecture; Volume 9, Proposition 9.1.3 gives the implication. Wang–Zahl prove the set conjecture, and the maximal version in  $\mathbb{R}^3$  remains open.

(4) *Which conjecture?* “Kakeya” is used for the set conjecture, the maximal conjecture, and the finite field analogue — the last settled by Dvir in 2008 and proved in Volume 10, Theorem 10.2.1. They are different statements.

(5) *Is it in final form?* Results of the last two years may be preprints. Volume 12, Remark 12.5.3 explains why this series quotes no current numerical records except where the statement is a clean equality.

*A worked check.* “Kakeya was solved in 2025”: (1)  $\mathbb{R}^3$  only; (2) both dimensions; (3) the *set* conjecture, not the maximal one; (4) the Euclidean set conjecture; (5) preprint at the time of writing. So the accurate statement is: *the three-dimensional Kakeya set conjecture is settled*, and four related statements are not.

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WORKED EXAMPLE 14.1.3— TIER B **What a proof of restriction would need**

Kakeya fell in  $\mathbb{R}^3$ ; restriction did not. Four things a proof would have to supply that the Kakeya argument does not.

(1)  *$L^\infty$  data.* Volume 13, Proposition 13.8.5 shows decoupling is sharp at the Stein–Tomas exponent and cannot reach below it; the conversion from  $\ell^2$  over caps to  $L^\infty$  costs  $\delta^{(n-1)/4}$  per cap. Nothing in the Kakeya proof addresses this.

(2) *A statement about functions, not sets.* The Wang–Zahl argument manipulates tube families and their shadings. A restriction estimate is about an oscillatory integral, and the passage between them — Volume 12’s wave packet dictionary — is lossy in the direction one would need.

(3) *Non-unique extremisers.* Knapp caps show the exponent is sharp, but they are not the only near-extremal configurations, and a structure theorem would have to describe all of them. Worked Example 14.7.3(f) records this as the obstruction to importing theakeya strategy.

(4) *The implication runs the wrong way.* Restriction impliesakeya (Volume 6, Theorem 6.6.2), so aakeya theorem is consistent with restriction failing. No reduction in the other direction is known.

*What is genuinely transferable.* The techniques for tangential configurations. Volume 12, Worked Example 12.4.1(d) identified tangency as the ceiling on broad–narrow, and Chapter 14.8 is the first argument in this series that confronts it rather than excluding it. *That, and not the theorem, is what makes 2025 interesting for the other three conjectures.*

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## Drills

**Drill 14.1.1.** State the implications among the four conjectures and confirm that settlingakeya in  $\mathbb{R}^3$  does not settle restriction there.

**Drill 14.1.2.** Explain why a bound of 3.059 is impossible for Besicovitch sets in  $\mathbb{R}^3$ , and identify the dimension for which it is the record.

**Drill 14.1.3.** Distinguish theakeya set conjecture, the maximal conjecture, and the finite field analogue, and say which are settled.

**Drill 14.1.4.** Using Volume 10, Worked Example 10.6.2, explain why the arithmetic method gives a fixed fraction of  $n$  rather than  $n - O(1)$ .

## Notes

Wang and Zahl’s paper is *Volume estimates for unions of convex sets, and theakeya set conjecture in three dimensions* (2025). Davies settled the plane in 1971. The planebrush bound in  $\mathbb{R}^4$  is Katz–Zahl; the high-dimensional arithmetic bound is Katz–Tao. Chapter 14.18 gives full attributions.

## Chapter 14.2

# Stickyakeya

The Wang–Zahl proof has two halves. The second, and newer, analyses *sticky* configurations; the first shows that the sticky case is the hard one. This chapter treats stickiness itself: what it is, why it is natural, and why aakeya set that is not sticky is easier.

### 14.2.1 Two notions of stickiness

There are two, and conflating them is the first hazard. One is a property of a *set*; the other of a *discretised tube family at a single scale*.

**Definition 14.2.1** (The space of lines). Let  $\mathcal{L}_n$  be the set of lines in  $\mathbb{R}^n$  of the form  $(p, 0) + \mathbb{R}v$  with  $p \in [-\frac{1}{n}, \frac{1}{n}]^{n-1}$  and  $v \in \mathbb{S}^{n-1}$  having final coordinate  $v_n \geq \frac{1}{2}$ , metrised by

$$d(\ell, \ell') = |p - p'| + \angle(v, v').$$

Equip  $\mathcal{L}_n$  with the product  $\lambda_n$  of  $(n - 1)$ -dimensional Lebesgue measure and normalised surface measure  $\nu_{n-1}$  on  $\mathbb{S}^{n-1}$ .

**Lemma 14.2.2** (The direction map forces a lower bound). *If  $L \subseteq \mathcal{L}_n$  contains a line in every direction, then  $\dim_{\mathbb{P}} L \geq n - 1$ .*

*Proof.* The direction map  $\text{dir} : \mathcal{L}_n \rightarrow \mathbb{R}P^{n-1}$  is 1-Lipschitz by Definition 14.2.1, and Lipschitz maps do not raise packing dimension. Since  $\text{dir}(L)$  is all of  $\mathbb{R}P^{n-1}$ , which has dimension  $n - 1$ , we get  $\dim_{\mathbb{P}} L \geq n - 1$ .  $\square$

**Definition 14.2.3** (Stickyakeya set). A compact  $K \subseteq \mathbb{R}^n$  is a *stickyakeya set* if there is a set of lines  $L$  with  $\dim_{\mathbb{P}} L = n - 1$  containing at least one line in each direction, such that  $\ell \cap K$  contains a unit interval for each  $\ell \in L$ .

More generally,  $K$  is  $\varepsilon$ -close to sticky if such an  $L$  exists with  $\dim_{\mathbb{P}} L \leq n - 1 + \varepsilon$ .

**Where we are.** Lemma 14.2.2 says  $n - 1$  is the least possible packing dimension for a line set carrying every direction. *Sticky means that minimum is attained*: the lines are as tightly packed as carrying all directions permits.

Dropping the dimension requirement recovers the ordinary definition of aakeya set, so every stickyakeya set is aakeya set and the sticky conjecture is *weaker* than theakeya conjecture. Davies' planar theorem therefore gives the sticky conjecture for  $n = 2$  at once.

**Definition 14.2.4** (Sticky at a scale, after Katz–Łaba–Tao). Let  $\mathbb{T}$  be a family of  $\delta$ -tubes with  $\delta$ -separated directions and let  $\delta < \rho < 1$ . Since the tubes point in distinct directions, a  $\rho$ -tube contains at most  $(\rho/\delta)^{n-1}$  of them. The family is *sticky at scale  $\rho$*  if that maximum is essentially attained: the  $\delta$ -tubes arrange into  $\approx \rho^{-(n-1)}$  many  $\rho$ -tubes, each containing  $\approx (\rho/\delta)^{n-1}$  of them.

**Where we are.** Definition 14.2.4 is the single-scale, discretised notion, and it is the one Katz, Łaba and Tao used. They showed — using a result of Wolff — that *if aakeya set in  $\mathbb{R}^3$  has upper Minkowski dimension exactly  $\frac{5}{2}$ , then its tube family must be sticky at every scale  $\rho \in (\delta, 1)$* . That is the observation motivating Definition 14.2.3, and Chapter 14.3 proves the corresponding statement in the form this argument needs.

*Why two definitions.* Definition 14.2.3 is intrinsic and is what the theorem is about; Definition 14.2.4 is what one can actually manipulate. Connecting them is the business of the discretisation in Chapter 14.3, and it is not a formality — the passage requires constructing shadings that survive repeated refinement.

## 14.2.2 The two theorems

**Theorem 14.2.5** (Wang–Zahl, sticky case). *Every stickyakeya set in  $\mathbb{R}^3$  has Hausdorff dimension 3. Equivalently, in the notation of Definition 14.3.5,  $\sigma_3 = 0$ .*

*More precisely: for every  $\varepsilon > 0$  there is  $\eta > 0$  such that everyakeya set in  $\mathbb{R}^3$  that is  $\eta$ -close to sticky has  $\dim_{\text{H}} K \geq 3 - \varepsilon$ .*

**Theorem 14.2.6** (Wang–Zahl, general case). *Everyakeya set in  $\mathbb{R}^3$  has Hausdorff and Minkowski dimension 3.*

**Where we are.** Theorem 14.2.5 is proved in full in Chapter 14.3 through Chapter 14.9. The route is: Proposition 14.3.9 reduces it to  $\sigma_3 = 0$ ; Chapter 14.6 shows extremal families are coarsely self-similar; Chapter 14.9 de-



rives planiness, graininess, and a contradiction.

Theorem 14.2.6 additionally requires Theorem 14.2.7, the reduction of the general case to the sticky one, which is a separate paper and is recorded as Remark 14.2.8.

*The equivalence in Theorem 14.2.5* is exactly Proposition 14.3.9 in one direction; the other follows since a family realising  $\sigma_3 > 0$  would produce a sticky Kakeya set of dimension below 3 by a standard limiting construction.

### 14.2.3 The reduction

**Theorem 14.2.7** (Wang–Zahl; the sticky reduction). *If every sticky Kakeya set in  $\mathbb{R}^n$  has Hausdorff dimension  $n$ , then every Kakeya set in  $\mathbb{R}^n$  has Hausdorff dimension  $n$ .*

*Remark 14.2.8* (Scope decision). Theorem 14.2.7 is stated and not proved. Revision R20.

*What is proved here.* Nothing of Theorem 14.2.7 itself. What *is* proved, in Chapter 14.3 onwards, is the sticky case: that  $\sigma_3 = 0$ , hence every sticky Kakeya set in  $\mathbb{R}^3$  has dimension 3. The reduction from general to sticky is a separate paper.

*What is missing.* The multi-scale bookkeeping converting “sticky at every scale” into an actual sticky Kakeya set, which requires constructing a limiting object and controlling the loss at each of  $\log(1/\delta)$  scales. Wang and Zahl carry this out.

*What depends on it.* Chapter 14.9, which analyses the sticky case and combines with Theorem 14.2.7 to give the theorem.

*Why the reduction was itself a theorem.* Sticky Kakeya was proposed as a special case by Wang and Zahl in 2022, and they proved it in  $\mathbb{R}^3$  then. That the special case implies the general one was established afterwards, and it is what turned a partial result into the full conjecture.

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#### WORKED EXAMPLE 14.2.1— TIER A **The Heisenberg example**

The most instructive sticky configuration, and the one every method had to survive.

*The construction.* In  $\mathbb{R}^3$ , take the lines

$$\ell_{a,b} = \{(a+t, b+at, \cdot)\},$$

parametrised so that the direction depends on  $a$  alone while the position depends on both. The resulting family is a Kakeya family, and the associated set is the “Heisenberg group example” — the lines are the left cosets of a one-parameter subgroup in the Heisenberg group.

*Why it is dangerous.* It has three structures at once, and each defeats a different method:

- *stickiness* — self-similar across scales, so no scale gives a gain;
- *grain structure* — the tubes organise into prisms at intermediate scales, so tube-counting arguments see less than they expect;
- *complex conjugation structure* — an additional algebraic symmetry.

*Why it is not a counterexample.* Its dimension is 3. But it very nearly is: it saturates the bounds available from Wolff's hairbrush and from the polynomial method, and the whole difficulty of the Keakeya problem in  $\mathbb{R}^3$  is excluding configurations that resemble it without being it.

*Its role in the proof.* Chapter 14.9 shows that the three structures above are not merely present in this example but *forced*: a hypothetical counterexample must be sticky, hence grainy, hence carry the conjugation structure — and then no such object exists. The example is the template for the whole argument.

#### WORKED EXAMPLE 14.2.2— TIER B **Stickiness across the series**

Stickiness is a self-similarity hypothesis, and this series has met three others. Comparing them shows what is special here.

(a) *Self-similar fractals.* Volume 7, Chapter 7.8: exact self-similarity under a finite family of contractions. Strong, and gives dimension formulas outright — though Worked Example 7.8.2 there records how badly they fail without separation hypotheses.

(b) *Ahlfors–David regularity.* Measure of balls comparable to  $r^s$  at every scale, both above and below — strictly stronger than the Frostman condition of Volume 12, Definition 12.7.1, which bounds only from above. A hypothesis in many projection theorems.

(c) *Stickiness.* Definition 14.2.3: the *clustering* of a tube family is the same at every scale. Weaker still, and about a family of tubes rather than a set.

*What they share.* Each says a configuration looks the same at all scales, and each is exploited by an induction that gains nothing at a single scale but compounds over  $\log(1/\delta)$  of them.

*What is special about stickiness.* For a *near-extremal* family it is not an assumption but a consequence: Katz, Łaba and Tao showed that a Keakeya set in  $\mathbb{R}^3$  of upper Minkowski dimension exactly  $\frac{5}{2}$  has a tube family sticky at every scale, and Chapter 14.6 proves the analogous statement for families

extremising the quantity  $\sigma_n$ . That is unusual: (a) and (b) restrict the class of sets considered, while (c) is a consequence of extremality. Arguments of this shape — “the worst case must be self-similar, and self-similar cases can be analysed” — are among the most powerful available, and Volume 13, Worked Example 13.3.4 lists the others in this series.

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## Drills

**Drill 14.2.1.** Verify that a family of  $\delta$ -tubes with  $\delta$ -separated directions has  $\approx \delta^{-(n-1)}$  members, and that a  $\rho$ -tube contains at most  $(\rho/\delta)^{n-1}$  of them — the count underlying Definition 14.2.4.

**Drill 14.2.2.** Explain why dropping the dimension requirement in Definition 14.2.3 recovers the ordinary definition of a Kakeya set, and deduce that the sticky conjecture is weaker than the Kakeya conjecture.

**Drill 14.2.3.** Verify Lemma 14.2.2: check that the direction map is 1-Lipschitz in the metric of Definition 14.2.1 and that Lipschitz maps do not raise packing dimension.

**Drill 14.2.4.** For the Heisenberg example, verify that the direction of  $\ell_{a,b}$  depends on  $a$  alone.

## Notes

Sticky Kakeya sets were isolated by Katz, Laba and Tao in their 1999 work and named and studied by Wang and Zahl in 2022, who proved the sticky conjecture in  $\mathbb{R}^3$ . The reduction Theorem 14.2.7 is theirs, and Remark 14.2.8 records that it is not proved here.



# Chapter 14.3

## Discretization

Definition 14.2.3 is a statement about a set; the arguments that follow manipulate finite families of tubes. This chapter builds the bridge. It introduces *shadings* — subsets of tubes that survive repeated refinement — defines a numerical quantity  $\sigma_n$  measuring the failure of the sticky conjecture, and proves that  $\sigma_n = 0$  implies the conjecture.

### 14.3.1 Tubes and shadings

**Definition 14.3.1** (Tubes). For  $\ell \in \mathcal{L}_n$  and  $\delta > 0$ , the  $\delta$ -tube  $T$  with coaxial line  $\ell$  is

$$T = \mathcal{N}_{2n\delta}(\ell) \cap [-1, 1]^n,$$

where  $\mathcal{N}_t(X)$  is the  $t$ -neighbourhood of  $X$ . For  $C > 0$ ,  $CT$  denotes the  $C\delta$ -tube with the same coaxial line.

Two  $\delta$ -tubes are *essentially identical* if  $d(\ell, \ell') \leq \delta$  and *essentially distinct* otherwise; they are *essentially parallel* if  $|\text{dir}(\ell) - \text{dir}(\ell')| \leq \delta$ .

**Where we are.** The factor  $2n$  is not decoration. Using the  $2n\delta$ -neighbourhood rather than the  $\delta$ -neighbourhood guarantees that *any*  $\delta$ -cube meeting  $\ell$  is contained in  $T$  — a cube of side  $\delta$  has diameter  $\sqrt{n}\delta$ , and a point of it within  $\sqrt{n}\delta$  of  $\ell$  leaves room. Since every shading below is a union of  $\delta$ -cubes, this convention is what makes “the cubes along  $\ell$ ” a subset of  $T$  rather than merely close to it.

Every tube of this type has volume comparable to  $\delta^{n-1}$ .

**Definition 14.3.2** (Shadings and refinements). A  $\delta$ -cube is a set  $[0, \delta)^n + p$  with  $p \in (\delta\mathbb{Z})^n$ . A *shading* of a  $\delta$ -tube  $T$  is a union of  $\delta$ -cubes  $Y(T) \subseteq T$ . We

write  $(\mathbb{T}, Y)_\delta$  for a family of tubes with shadings, and

$$E_{\mathbb{T}} = \bigcup_{T \in \mathbb{T}} Y(T).$$

$(\mathbb{T}', Y')_\delta$  is a *sub-collection* if  $\mathbb{T}' \subseteq \mathbb{T}$  and  $Y'(T) \subseteq Y(T)$ ; it is a *refinement* if moreover

$$\sum_{T \in \mathbb{T}'} |Y'(T)| \geq (\log \frac{1}{\delta})^{-C} \sum_{T \in \mathbb{T}} |Y(T)|,$$

written  $\gtrsim_\delta$ . A *constant multiplicity refinement* is one with  $\sum_T \mathbf{1}_{Y'(T)} = \mu \mathbf{1}_{E_{\mathbb{T}'}}$  for an integer  $\mu \geq 1$ .

**Lemma 14.3.3** (Constant multiplicity refinements exist). *Every  $(\mathbb{T}, Y)_\delta$  with  $\#\mathbb{T} \leq \delta^{-100n}$  has a constant multiplicity refinement.*

*Proof.* The function  $\sum_T \mathbf{1}_{Y(T)}$  takes integer values between 1 and  $\#\mathbb{T} \leq \delta^{-100n}$  on  $E_{\mathbb{T}}$ . Partition  $E_{\mathbb{T}}$  into the  $O(\log \frac{1}{\delta})$  dyadic level sets  $\{2^k \leq \sum_T \mathbf{1}_{Y(T)} < 2^{k+1}\}$  and keep the one carrying the largest share of  $\sum_T |Y(T)|$ ; it carries at least a  $(\log \frac{1}{\delta})^{-1}$  fraction. Restricting each shading to that level set gives a refinement of multiplicity comparable to  $2^k$ , and discarding a bounded proportion makes it exactly constant.  $\square$

**Where we are.** The tolerance  $(\log \frac{1}{\delta})^{-C}$  in Definition 14.3.2 is chosen so that *polynomially many refinements cost nothing*: the final bounds are of the form  $\delta^{\sigma \pm \varepsilon}$ , and  $(\log \frac{1}{\delta})^{-C}$  is  $\delta^{o(1)}$ . This is why the argument can afford the dozens of pigeonholing steps that follow, and it is the single most important bookkeeping convention in the proof.

The restriction  $\#\mathbb{T} \leq \delta^{-100n}$  is harmless: the families arising below have  $\#\mathbb{T} \approx \delta^{1-n}$ .

**Definition 14.3.4** (Covers). Let  $T$  be a  $\delta$ -tube and  $\tilde{T}$  a  $\rho$ -tube,  $\delta \leq \rho$ . We say  $\tilde{T}$  *covers*  $T$  if  $d(\ell, \tilde{\ell}) \leq \rho/2$ . A family  $\tilde{\mathbb{T}}$  covers  $\mathbb{T}$  if every tube of  $\mathbb{T}$  is covered by one of  $\tilde{\mathbb{T}}$ , and we write  $\mathbb{T}[\tilde{T}]$  for those covered by  $\tilde{T}$ .

A cover  $(\tilde{\mathbb{T}}, \tilde{Y})_\rho$  of  $(\mathbb{T}, Y)_\delta$  additionally requires  $Y(T) \subseteq \tilde{Y}(\tilde{T})$  whenever  $T \in \mathbb{T}[\tilde{T}]$ . It is *balanced* if  $|E_{\mathbb{T}} \cap Q|$  is the same for every  $\rho$ -cube  $Q \subseteq E_{\tilde{\mathbb{T}}}$ .

## 14.3.2 The quantity $\sigma_n$

**Definition 14.3.5.** For  $s, t, \delta \in (0, 1)$  put

$$M(s, t, \delta) = \inf_{\mathbb{T}, Y} \left| \bigcup_{T \in \mathbb{T}} Y(T) \right|,$$

the infimum over pairs  $(\mathbb{T}, Y)_\delta$  with

- (a) the tubes of  $\mathbb{T}$  essentially distinct;
- (b) for each  $\delta \leq \rho \leq 1$ ,  $\mathbb{T}$  can be covered by  $\rho$ -tubes, at most  $\delta^{-t}$  of which are essentially parallel to a common  $\rho$ -tube;
- (c)  $\sum_{T \in \mathbb{T}} |Y(T)| \geq \delta^s$ .

Set

$$N(s, t) = \limsup_{\delta \rightarrow 0^+} \frac{\log M(s, t, \delta)}{\log \delta}, \quad \sigma_n = \inf_{(s, t) \in (0, 1)^2} N(s, t).$$

**Lemma 14.3.6** (The infimum is a limit).  *$N$  is non-decreasing in each variable, and  $\lim_{(s, t) \rightarrow (0, 0)} N(s, t) = \sigma_n$ .*

*Proof.* If  $s \leq s'$  and  $t \leq t'$  then conditions (b) and (c) at  $(s', t')$  are more restrictive, so the infimum in  $M$  is over a smaller set and  $M(s, t, \delta) \geq M(s', t', \delta)$ . Since  $\log \delta < 0$ , dividing reverses the inequality and  $N(s, t) \leq N(s', t')$ . A monotone function bounded in  $[0, n]$  has its infimum as a limit at the corner.  $\square$

**Where we are.** Unpacking Definition 14.3.5: condition (a) prevents cheating by repeating tubes; (c) requires the shadings to be a substantial fraction of the tubes; and (b) is the *quantitative stickiness hypothesis* — at every intermediate scale, the tubes do not concentrate in directions beyond a factor  $\delta^{-t}$ .

$M(s, t, \delta)$  is then the smallest volume such a family can occupy, and  $\sigma_n$  measures how small: a family with  $|E_{\mathbb{T}}| \approx \delta^{\sigma_n}$  is as concentrated as stickiness permits. *The sticky Kakeya conjecture in  $\mathbb{R}^n$  is the assertion  $\sigma_n = 0$ , and Proposition 14.3.9 makes that precise.*

### 14.3.3 From $\sigma_n$ to dimension

**Definition 14.3.7** (Quantitatively sticky line sets). For  $t, \rho > 0$  let  $\text{QStick}(n, t, \rho)$  be the sets  $L \subseteq \mathcal{L}_n$  with

$$\left| \{p \in \mathbb{R}^{n-1} : (p, v) \in \mathcal{N}_{\delta}(L)\} \right| \leq \delta^{n-1-t} \quad \text{for all } v \in \mathbb{S}^{n-1}, \delta \in (0, \rho).$$

**Lemma 14.3.8** (Passing to a quantitatively sticky subset). *Let  $L \subseteq \mathcal{L}_n$  with  $\nu_{n-1}(\text{dir } L) > 0$ . Then for every  $t > \dim_{\text{M}} L - (n - 1)$  there are  $L' \subseteq L$  and  $\rho > 0$  with  $\nu_{n-1}(\text{dir } L') > 0$  and  $L' \in \text{QStick}(n, t, \rho)$ .*

*Proof. Step 1. Reduce to one piece.*

Let  $d = \dim_{\text{M}} L$  and  $\varepsilon = t - d + (n - 1) > 0$ . Packing dimension equals modified upper box dimension, so  $L = \bigcup_i L_i$  with  $\dim_{\text{M}} L_i \leq d + \varepsilon/4$ . Since

$\nu_{n-1}(\bigcup_i \text{dir } L_i) = \nu_{n-1}(\text{dir } L) > 0$ , some  $L_i$  has  $\nu_{n-1}(\text{dir } L_i) > 0$ ; relabel it  $L_1$ . Choose  $\rho_1 > 0$  with

$$\lambda_n(\mathcal{N}_{2\delta}(L_1)) \leq \delta^{(2n-2)-d-\varepsilon/2} \quad (\delta \in (0, \rho_1)). \quad (14.3.1)$$

**Step 2.** *Over-represented directions are rare.*

Call  $v$  *over-represented at scale  $\delta$*  if  $|\{p : (p, v) \in \mathcal{N}_{2\delta}(L_1)\}| > \delta^{n-1-t}$ . Since  $\lambda_n$  is a product measure, Fubini applied to (14.3.1) gives

$$\nu_{n-1}(\{v \text{ over-represented at scale } \delta\}) \lesssim \delta^{\varepsilon/2},$$

so summing over dyadic scales  $\delta = 2^{-k}$ ,  $k \geq k_1$ ,

$$\sum_{k \geq k_1} \nu_{n-1}(\{v \text{ over-represented at scale } 2^{-k}\}) \lesssim 2^{-k_1 \varepsilon/2}. \quad (14.3.2)$$

**Step 3.** *Discard them.*

Choose  $k_1$  so that the left side of (14.3.2) is at most  $\frac{1}{2}\nu_{n-1}(\text{dir } L_1)$ , put  $\rho = 2^{-k_1}$  and

$$L' = L_1 \setminus \bigcup_{k \geq k_1} \text{dir}^{-1}(\{v \text{ over-represented at scale } 2^{-k}\}).$$

Then  $\nu_{n-1}(\text{dir } L') \geq \frac{1}{2}\nu_{n-1}(\text{dir } L_1) > 0$ , and a direction surviving is over-represented at no dyadic scale below  $\rho$ , hence  $L' \in \text{QStick}(n, t, \rho)$ .  $\square$

**Proposition 14.3.9** ( $\sigma_n$  bounds the dimension of sticky Kakeya sets). *For every  $\varepsilon > 0$  there is  $\eta > 0$  such that every Kakeya set  $K \subseteq \mathbb{R}^n$  that is  $\eta$ -close to sticky satisfies  $\dim_{\text{H}} K \geq n - \sigma_n - \varepsilon$ . In particular every sticky Kakeya set has  $\dim_{\text{H}} K \geq n - \sigma_n$ .*

*Proof.* Fix  $\varepsilon > 0$  and choose  $s, t \in (0, 1)$  with

$$N(s, t) \leq \sigma_n + \varepsilon/2, \quad (14.3.3)$$

possible by Lemma 14.3.6. We show the proposition holds with  $\eta = t/4$ .

**Step 1.** *Extract a quantitatively sticky family of lines.*

After translating and dilating,  $K \subseteq [-\frac{1}{n}, \frac{1}{n}]^n$ . By hypothesis there are  $w > 0$  and  $L \subseteq \mathcal{L}_n$  with  $\dim_{\text{p}} L \leq n - 1 + \eta$  so that every admissible direction is realised and  $|\ell \cap K| \geq w$  for  $\ell \in L$ . Apply Lemma 14.3.8 to get  $L' \subseteq L$  and  $\rho_1 > 0$  with  $L' \in \text{QStick}(n, t/2, \rho_1)$  and  $\nu_{n-1}(\text{dir } L') > 0$ .



**Step 2.** Choose a scale from a Hausdorff cover.

Let  $k_1$  be a large integer with  $2^{-k_1} < \rho_1^{2n/t}$ . By definition of Hausdorff dimension, cover  $K \subseteq \bigcup_{k \geq k_1} \bigcup_{Q \in \mathcal{Q}_k} Q$  with  $\mathcal{Q}_k$  a family of  $2^{-k}$ -cubes and

$$\#\mathcal{Q}_k \lesssim (2^k)^{\dim_H K + \varepsilon/2}. \quad (14.3.4)$$

For each  $\ell \in L'$  we have  $K \cap \ell \subseteq \bigcup_k (\ell \cap \bigcup_{Q \in \mathcal{Q}_k} Q)$ , and  $|K \cap \ell| \geq w$ , so some  $k$  satisfies

$$\left| \ell \cap \bigcup_{Q \in \mathcal{Q}_k} Q \right| \geq \frac{w}{100k^2}. \quad (14.3.5)$$

Let  $L_k$  be the lines for which (14.3.5) holds. Since  $L' = \bigcup_k L_k$  and  $\sum_k (100k^2)^{-1} < \infty$ , there is  $k_2 \geq k_1$  with  $\nu_{n-1}(\text{dir } L_{k_2}) \geq \nu_{n-1}(\text{dir } L')/(100k_2^2)$ .

**Step 3.** Build the tube family.

Put  $\delta = 2^{-k_2}$  and let  $\Omega$  be a  $\delta$ -separated subset of  $\text{dir } L_{k_2}$  of cardinality  $\gtrsim \nu_{n-1}(\text{dir } L')\delta^{1-n}(\log \frac{1}{\delta})^{-2}$ . For each  $v \in \Omega$  pick  $\ell \in L_{k_2}$  with  $\text{dir } \ell = v$ , let  $T$  be its  $\delta$ -tube and

$$Y(T) = \bigcup_{\substack{Q \in \mathcal{Q}_{k_2} \\ Q \cap \ell \neq \emptyset}} Q.$$

By the convention of Definition 14.3.1,  $Y(T) \subseteq T$ ; and by (14.3.5),  $|Y(T)| \gtrsim w(\log \frac{1}{\delta})^{-2}\delta^{n-1}$ .

**Step 4.** Verify (a) and (c).

(a) holds since  $\Omega$  is  $\delta$ -separated. For (c),

$$\sum_{T \in \mathbb{T}} |Y(T)| \gtrsim \frac{(\#\Omega) w \delta^{n-1}}{(\log \frac{1}{\delta})^2} \gtrsim \frac{w \nu_{n-1}(\text{dir } L')}{(\log \frac{1}{\delta})^4} \geq \delta^s$$

for  $k_1$  large enough in terms of  $w$ ,  $\nu_{n-1}(\text{dir } L')$  and  $n$ .

**Step 5.** Verify (b), the stickiness condition.

Let  $\rho \in [\delta, 1]$ .

*Case  $\rho \in [\delta, \rho_1)$ .* Choose  $\rho$ -tubes  $\tilde{\mathbb{T}}$  whose coaxial lines  $\tilde{L}$  satisfy: each  $\tilde{\ell} \in \mathcal{N}_\rho(L')$ ; the  $\rho/2$ -neighbourhoods of  $\tilde{L}$  cover  $L'$ ; and the  $\rho/6$ -neighbourhoods are disjoint. Such a family exists by a greedy Vitali-type selection. Every  $T \in \mathbb{T}$  has coaxial line in  $L'$ , hence is covered.

Suppose  $\tilde{T}_1, \dots, \tilde{T}_M$  are essentially parallel to a common  $\rho$ -tube of direction  $v$ . Then each lies in  $\mathcal{N}_\rho(L') \cap \text{dir}^{-1}(\mathcal{N}_\rho(v))$ , and by disjointness of the  $\rho/6$ -neighbourhoods,

$$\lambda_n(\mathcal{N}_\rho(L') \cap \text{dir}^{-1}(\mathcal{N}_\rho(v))) \geq \sum_{i=1}^M \lambda_n(\mathcal{N}_{\rho/6}(\tilde{\ell}_i)) \gtrsim M \rho^{2n-2}.$$

By Fubini there is  $v' \in \mathcal{N}_\rho(v)$  with  $|\mathcal{N}_\rho(L') \cap \text{dir}^{-1}(v')| \gtrsim M\rho^{n-1}$ . Since  $L' \in \text{QStick}(n, t/2, \rho_1)$  and  $\rho < \rho_1$ , taking  $\rho_1$  small enough forces  $M \leq \rho^{-t} \leq \delta^{-t}$ .

*Case  $\rho \in [\rho_1, 1]$ .* Take  $\mathbb{T}$  with coaxial lines a  $\rho/2$ -net in  $\mathcal{L}_n$ ; this covers  $\mathbb{T}$ , and trivially  $M \leq \#\mathbb{T} \approx \rho^{2-2n} \leq \delta^{-t}$ , since  $\delta \leq 2^{-k_1} \leq \rho_1^{2n/t}$ .

**Step 6.** *Conclude.*

$(\mathbb{T}, Y)_\delta$  satisfies (a), (b), (c), so

$$M(s, t, \delta) \leq \left| \bigcup_T Y(T) \right| \leq \left| \bigcup_{Q \in \mathcal{Q}_{k_2}} Q \right| \leq (\#\mathcal{Q}_{k_2})\delta^n \leq \delta^{n - \dim_{\text{H}} K - \varepsilon/2}$$

by (14.3.4). Hence  $\frac{\log M(s, t, \delta)}{\log \delta} \geq n - \dim_{\text{H}} K - \varepsilon/2$ . This holds for some  $\delta \leq 2^{-k_1}$  for every large  $k_1$ , so  $N(s, t) \geq n - \dim_{\text{H}} K - \varepsilon/2$ . Comparing with (14.3.3) gives  $\dim_{\text{H}} K \geq n - \sigma_n - \varepsilon$ .  $\square$

**Where we are.** Proposition 14.3.9 reduces Theorem 14.2.5 to the single equality  $\sigma_3 = 0$ , and the remaining chapters prove exactly that.

Two features of the proof are worth marking. *Step 2 is where Hausdorff dimension becomes a single scale*: an arbitrary cover is replaced by one scale  $\delta$  carrying a definite share, at the cost of the factors  $(\log \frac{1}{\delta})^{-2}$  that the refinement convention absorbs. And *Step 5 is where stickiness is used* — it is the only place, and it enters solely to verify condition (b) of Definition 14.3.5. That localisation matters: any class of Kakeya sets for which Step 5 can be carried out will inherit the whole argument.

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#### WORKED EXAMPLE 14.3.1— TIER A **What $\sigma_n$ measures**

The definition is opaque on first reading. Three ways to see it.

(a) *As a volume exponent.*  $|E_{\mathbb{T}}| \approx \delta^{\sigma_n}$  for extremal families. If  $\sigma_n = 0$  the union has volume  $\delta^{o(1)}$  — essentially full — and if  $\sigma_n > 0$  it is polynomially small.

(b) *As a dimension deficit.* By Proposition 14.3.9, sticky Kakeya sets have dimension at least  $n - \sigma_n$ . So  $\sigma_n$  is exactly how far the sticky conjecture might fail.

(c) *Numerically.*  $\sigma_2 = 0$ , by Davies. Wolff's bound gives  $\sigma_3 \leq \frac{1}{2}$ ; Katz–Łaba–Tao improve it slightly; and Theorem 14.2.5 gives  $\sigma_3 = 0$ .

*Why an infimum over  $(s, t)$ .* Because the two conditions pull against each other: small  $t$  is a strong stickiness hypothesis, making the family class smaller and  $M$  larger; small  $s$  demands large shadings, again shrinking the class. Taking both to 0 isolates the genuinely sticky, genuinely large families, and Lemma 14.3.6 says the corner limit exists.

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**WORKED EXAMPLE 14.3.2— TIER B Why shadings, and not just tubes**

A first attempt would work with  $\bigcup_{T \in \mathbb{T}} T$  directly. It fails, and the reason recurs throughout the proof.

(a) *Refinements destroy tubes but not shadings.* The argument performs dozens of pigeonholing steps, each discarding part of the configuration. If one discards whole tubes, the direction set degrades and theakeya property is lost. Discarding *parts* of tubes keeps every direction present.

(b) *The relevant statistic is  $\sum_T |Y(T)|$ , not  $\#\mathbb{T}$ .* Condition (c) of Definition 14.3.5 is stated in terms of total shading mass, which is exactly the quantity a refinement is guaranteed to preserve up to  $(\log \frac{1}{\delta})^{-C}$ .

(c) *Multiplicity becomes available.* With shadings one can pass to constant multiplicity (Lemma 14.3.3) and then read off  $\mu = \sum_T |Y(T)|/|E_{\mathbb{T}}|$ , which is the quantity every subsequent estimate manipulates.

*The general lesson.* When an argument requires many successive refinements, choose the objects so that refinement is a *restriction* of each object rather than a deletion of objects. Volume 12, Theorem 12.1.4 makes the same choice: wave packets carry coefficients that can be pruned without losing caps.

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## Drills

**Drill 14.3.1.** Verify the claim in the Checkpoint after Definition 14.3.1: a  $\delta$ -cube meeting  $\ell$  lies in  $\mathcal{N}_{2n\delta}(\ell)$ .

**Drill 14.3.2.** Prove Lemma 14.3.3, tracking where the factor  $(\log \frac{1}{\delta})^{-1}$  arises.

**Drill 14.3.3.** Check the monotonicity in Lemma 14.3.6 and confirm the direction of the inequality after dividing by  $\log \delta < 0$ .

**Drill 14.3.4.** In Step 5 of Proposition 14.3.9, verify the Fubini step producing  $|\mathcal{N}_\rho(L') \cap \text{dir}^{-1}(v')| \gtrsim M\rho^{n-1}$ .

## Notes

The discretization in this chapter is Section 2 of Wang and Zahl’s paper. The use of shadings that survive refinement is their contribution; the authors note that it is intended to be reusable in higher dimensions.



## Chapter 14.4

# The Assertions $\mathcal{D}$ and $\mathcal{E}$

Remark 14.2.8 recorded the reduction of general Kakeya to the sticky case as quoted. This chapter and the next discharge that debt. The reduction is not a single implication but a self-improving scheme built on two families of assertions, and the scheme is what we develop here.

### 14.4.1 Two Wolff axioms

**Definition 14.4.1** (Katz–Tao and Frostman Wolff constants). Let  $\mathbb{T}$  be a set of  $\delta$ -tubes in  $\mathbb{R}^3$ .

(A)  $C_{\text{KT}}(\mathbb{T})$  is the least  $C > 0$  with

$$\#\{T \in \mathbb{T} : T \subseteq W\} \leq C|W||T|^{-1} \quad \text{for all convex } W \subseteq \mathbb{R}^3.$$

We say  $\mathbb{T}$  obeys the *Katz–Tao Convex Wolff Axioms* with error  $C_{\text{KT}}(\mathbb{T})$ .

(B)  $C_{\text{F-S}}(\mathbb{T})$  is the least  $C > 0$  with

$$\#\{T \in \mathbb{T} : T \subseteq W\} \leq C|W|(\#\mathbb{T}) \quad \text{for all slabs } W \subseteq \mathbb{R}^3,$$

a *slab* being the unit ball intersected with a thickened neighbourhood of a plane. We say  $\mathbb{T}$  obeys the *Frostman Slab Wolff Axioms* with error  $C_{\text{F-S}}(\mathbb{T})$ .

**Where we are.** The two names mark two kinds of non-concentration, and they are the tube analogues of two familiar conditions on sets. For a  $\delta$ -separated  $E \subseteq \mathbb{R}^n$  and a ball  $B$  of radius  $r$ , “Katz–Tao” is  $\#(E \cap B) \leq (r/\delta)^s$  — an *absolute* bound — and “Frostman” is  $\#(E \cap B) \leq r^s \#E$  — a bound *relative to the total*. Volume 12, Definition 12.7.1 used the Frostman form.

The second half of each name records the class of test sets: all convex sets, or only slabs.

*Both hold with error  $O(1)$  for a *Keakeya family*.* A set of  $\delta$ -tubes with one tube per  $\delta$ -separated direction obeys both, which is why the definitions are the right ones for this problem.

*Sparse versus dense.* Small  $C_{\text{KT}}$  means  $\mathbb{T}$  is sparse; small  $C_{\text{F-S}}$  means  $\mathbb{T}$  is spread out. Lemma 14.4.2 records that these behave oppositely under covers, and that opposition is what the whole scheme exploits.

**Lemma 14.4.2** (Inheritance). *Let  $\mathcal{U}, \mathcal{W}$  be collections of convex subsets of  $\mathbb{R}^n$ .*

(A) Frostman constants are inherited upwards. *If  $\mathcal{W}$  is a  $K$ -balanced cover of  $\mathcal{U}$  then  $C_{\text{F-C}}(\mathcal{W}) \lesssim K C_{\text{F-C}}(\mathcal{U})$  and  $C_{\text{F-S}}(\mathcal{W}) \lesssim K C_{\text{F-S}}(\mathcal{U})$ .*

(B) Katz–Tao constants are inherited downwards. *For any convex  $W$ ,*

$$C_{\text{KT}}(\mathcal{U}^W) = C_{\text{KT}}(\mathcal{U}[W]) \leq C_{\text{KT}}(\mathcal{U}).$$

*Proof.* (B) is immediate: a convex  $V \subseteq W$  tests  $\mathcal{U}[W]$  using only sets already tested against  $\mathcal{U}$ , and the rescaling  $\phi_W$  is affine, so it preserves the ratio  $|V|/|U|$  defining the constant.

(A) Let  $V$  be convex. Since  $\mathcal{W}$  covers  $\mathcal{U}$  and is  $K$ -balanced, the sets  $\mathcal{U}[W]$  for  $W \in \mathcal{W}[V]$  have comparable normalised mass, so

$$\sum_{W \in \mathcal{W}[V]} |W| \lesssim K \sum_{W \in \mathcal{W}[V]} \sum_{U \in \mathcal{U}[W]} |U| \lesssim K \sum_{U \in \mathcal{U}[V]} |U|,$$

and dividing by  $\sum_{U \in \mathcal{U}} |U|$  gives the claim.  $\square$

## 14.4.2 The assertions

**Definition 14.4.3** ( $\mathcal{D}(\sigma, \omega)$  and  $\mathcal{E}(\sigma, \omega)$ ). Let  $\sigma, \omega \geq 0$ . Write  $(\mathbb{T}, Y)_\delta$  for a set of essentially distinct  $\delta$ -tubes with shadings, and call it  $\lambda$  *dense* if  $\sum_T |Y(T)| \geq \lambda \sum_T |T|$ .

*Assertion  $\mathcal{D}(\sigma, \omega)$ :* for all  $\varepsilon > 0$  there are  $\kappa, \eta > 0$  such that for all  $\delta > 0$ , every  $\delta^\eta$ -dense  $(\mathbb{T}, Y)_\delta$  obeying both Wolff axioms with error at most  $\delta^{-\eta}$  satisfies

$$\left| \bigcup_{T \in \mathbb{T}} Y(T) \right| \geq \kappa \delta^{\omega+\varepsilon} (\#\mathbb{T}) |T| ((\#\mathbb{T}) |T|^{1/2})^{-\sigma}. \quad (14.4.1)$$

*Assertion  $\mathcal{E}(\sigma, \omega)$ :* the same, but with *no* hypothesis on the Wolff constants, and with the conclusion

$$\left| \bigcup_{T \in \mathbb{T}} Y(T) \right| \geq \kappa \delta^{\omega+\varepsilon} m^{-1} (\#\mathbb{T}) |T| (m^{-3/2} \ell (\#\mathbb{T}) |T|^{1/2})^{-\sigma}, \quad (14.4.2)$$

where  $m = C_{\text{KT}}(\mathbb{T})$  and  $\ell = C_{\text{F.S}}(\mathbb{T})$ .

**Where we are.** Reading the numerology. At  $\sigma = \omega$ , (14.4.1) becomes

$$\left| \bigcup_T Y(T) \right| \geq \kappa \delta^\varepsilon (\#\mathbb{T})^{-\sigma} \sum_T |T|,$$

which says that  $\lesssim \delta^{-\varepsilon} (\#\mathbb{T})^\sigma$  tubes pass through a typical point. *So small  $\sigma$  means the tubes are nearly disjoint*, and  $\mathcal{D}(0, 0)$  is the *Keakeya statement*. Two features of (14.4.2) deserve explanation, and both come from demanding that the estimate survive rescaling.

*The factor  $\ell$ .* Suppose  $(\rho/\delta)^2$  tubes sit inside one  $\rho$ -tube  $T_\rho$  with  $C_{\text{KT}} \approx 1$ ; then  $C_{\text{F.S}} \approx \rho^{-1}$ . Rescaling  $T_\rho$  to the unit ball, applying  $\mathcal{D}(\sigma, \omega)$ , and undoing the rescaling — which distorts volume by  $\rho^2$  — gives exactly (14.4.2) with  $\ell = \rho^{-1}$ .

*The factors  $m^{-1}$ ,  $m^{-3/2}$ .* Suppose instead  $C_{\text{F.S}} \approx 1$  and  $C_{\text{KT}} = m \gg 1$ , and put  $\rho = m^{1/2}\delta$ . Then  $\mathbb{T}$  is covered by  $\mathbb{T}_\rho$  with  $(\#\mathbb{T}_\rho)|T_\rho| = m^{-1}(\#\mathbb{T})|T|$  and  $C_{\text{KT}}(\mathbb{T}_\rho) \lesssim 1$ ; applying  $\mathcal{D}$  to  $\mathbb{T}_\rho$  produces the two powers of  $m$  as written.

*One inequality always holds:*  $C_{\text{F.S}}(\mathbb{T})(\#\mathbb{T})|T|^{1/2} \geq 1$ , since some slab of thickness  $|T|^{1/2}$  contains a tube. That is why  $|T|^{1/2}$  appears rather than  $\delta$ : any convex set of diameter 1 sits in a slab of thickness  $|S|^{1/2}$ .

**Proposition 14.4.4** (Equivalence; not proved here, see Remark 14.4.14). *For  $0 \leq \sigma \leq \frac{2}{3}$  and  $\omega \geq 0$ ,  $\mathcal{E}(\sigma, \omega) \iff \mathcal{D}(\sigma, \omega)$ .*

**Proposition 14.4.5** (Self-improvement; not proved here, see Remark 14.4.14). *There is  $g : [0, \frac{2}{3}] \times (0, 1] \rightarrow (0, 1]$  so that for  $0 \leq \sigma \leq \frac{2}{3}$  and  $\omega > 0$ ,*

$$\mathcal{E}(\sigma, \omega) \implies \mathcal{D}(\sigma, \omega - g(\sigma, \omega)).$$

**Proposition 14.4.6** (The base case; Wolff, not proved here).  *$\mathcal{D}(\frac{1}{2}, 0)$  is true.*

**Where we are.** Proposition 14.4.6 is Wolff's hairbrush bound in disguise. Setting  $\sigma = \frac{1}{2}$ ,  $\omega = 0$  in (14.4.1) gives

$$\left| \bigcup_T Y(T) \right| \geq \kappa \delta^{1/2+\varepsilon} (\delta^2 \#\mathbb{T})^{1/2},$$

which is the estimate behind  $\dim_{\text{H}} \geq \frac{n+2}{2}$ .

*A point of care.* This is *not* Volume 9, Theorem 9.5.4: that theorem assumes transversality, and  $\mathcal{D}(\frac{1}{2}, 0)$  does not. What replaces the transversality hypothesis is the *Frostman Slab Wolff axiom* of Definition 14.4.1. Tubes making angle  $\leq \theta$  with the hairbrush axis lie in a slab of thickness  $O(\theta)$ ,

and the axiom bounds their number by  $O(\theta \cdot \#\mathbb{T})$  — precisely compensating the factor  $\theta$  by which the small-angle case degrades the hairbrush count. Volume 9, Remark 9.5.5 sets out that computation and records the general theorem as unproved there.

*So Proposition 14.4.6 is Wolff's theorem in the Frostman Slab form, and it is quoted, not derived from Theorem 9.5.4.*

### 14.4.3 The bootstrap

**Theorem 14.4.7** (Wang–Zahl).  $\mathcal{D}(0, 0)$  and  $\mathcal{E}(0, 0)$  are true.

*Proof.* Fix  $\omega > 0$ . By Proposition 14.4.6,  $\mathcal{D}(\frac{1}{2}, 0)$  holds, hence so does  $\mathcal{D}(\frac{1}{2}, \omega)$ .

**Step 1.** *Monotonicity.*

If  $\mathcal{D}(\sigma, \omega)$  holds for some  $\sigma \in [0, \frac{2}{3}]$  then so does  $\mathcal{D}(\sigma', \omega)$  for every  $\sigma' \in [\sigma, \frac{2}{3}]$ , since the right-hand side of (14.4.1) is decreasing in  $\sigma$  once  $(\#\mathbb{T})|T|^{1/2} \geq 1$ .

**Step 2.** *The set of admissible  $\sigma$  is open.*

Since the tubes are essentially distinct and lie in the unit ball,  $\#\mathbb{T} \lesssim \delta^{-4}$ ; so an improvement in  $\omega$  can be traded for one in  $\sigma$ , and Proposition 14.4.5 gives

$$\mathcal{E}(\sigma, \omega) \implies \mathcal{D}(\sigma - g(\sigma, \omega)/4, \omega).$$

Combining with Proposition 14.4.4,  $\mathcal{D}(\sigma, \omega)$  implies  $\mathcal{D}(\sigma - g(\sigma, \omega)/4, \omega)$ . Hence  $\{\sigma \in [0, \frac{2}{3}] : \mathcal{D}(\sigma, \omega)\}$  is relatively open.

**Step 3.** *It is also closed.*

Directly from Definition 14.4.3: if  $\mathcal{D}(\sigma_j, \omega)$  holds with  $\sigma_j \rightarrow \sigma$ , then for each  $\varepsilon$  one may use the constants for  $\sigma_j$  with  $j$  large, since (14.4.1) depends continuously on  $\sigma$ .

**Step 4.** *Conclude in  $\sigma$ .*

A non-empty subset of the connected space  $[0, \frac{2}{3}]$  that is both open and closed is everything. So  $\mathcal{D}(\sigma, \omega)$  holds for all  $\sigma \in [0, \frac{2}{3}]$ ; in particular  $\mathcal{D}(0, \omega)$ .

**Step 5.** *Conclude in  $\omega$ .*

This holds for every  $\omega > 0$ , and  $\{\omega \geq 0 : \mathcal{D}(0, \omega)\}$  is relatively closed in  $[0, \infty)$  by the same continuity. Hence  $\mathcal{D}(0, 0)$  holds, and  $\mathcal{E}(0, 0)$  follows by Proposition 14.4.4.  $\square$

**Theorem 14.4.8** (Wang–Zahl, maximal form; proved via Corollary 14.4.9). *For all  $\varepsilon > 0$  there is  $K > 1$  so that for all sufficiently small  $\delta$  the following holds. Let  $\mathbb{T}$  be a set of  $\delta$ -tubes in the unit ball of  $\mathbb{R}^3$  such that every rectangular*



prism of dimensions  $a \times b \times 2$  contains at most  $100ab\delta^{-2}$  tubes — true, for instance, if the tubes point in  $\delta$ -separated directions. Let  $Y(T) \subseteq T$  with  $|Y(T)| \geq \lambda|T|$ . Then

$$\left| \bigcup_{T \in \mathbb{T}} Y(T) \right| \geq \delta^\varepsilon \lambda^K \sum_{T \in \mathbb{T}} |T|. \quad (14.4.3)$$

**Where we are.** The *Keakeya maximal function* conjecture asserts (14.4.3) with  $K = 3$ . Theorem 14.4.8 gives some  $K$ , which is weaker — and still enough for the set conjecture. *So the maximal conjecture in  $\mathbb{R}^3$  remains open*, a distinction Volume 14, Worked Example 14.1.2(3) warned about. The prism hypothesis is the Wolff axioms of Definition 14.4.1 in concrete form; it gives  $C_{KT}(\mathbb{T}) \leq 1000$ .

**Corollary 14.4.9.** *For all  $\varepsilon > 0$  there is  $K$  so that for all small  $\delta$  and all  $\lambda$ -dense  $(\mathbb{T}, Y)_\delta$ ,*

$$\left| \bigcup_{T \in \mathbb{T}} Y(T) \right| \geq \delta^\varepsilon \lambda^K m^{-1}(\#\mathbb{T})|T|, \quad m = C_{KT}(\mathbb{T}).$$

*In particular Theorem 14.2.6 holds: every Keakeya set in  $\mathbb{R}^3$  has Minkowski and Hausdorff dimension 3.*

*Proof.* The displayed estimate is  $\mathcal{E}(0,0)$  written out. For a Keakeya set, the associated tube family has  $C_{KT} \leq 1000$  and the hypothesis of Theorem 14.2.6 is met, so  $|\bigcup_T Y(T)| \geq \delta^\varepsilon \lambda^K \sum_T |T|$ ; feeding this into the covering argument of Proposition 14.3.9 gives dimension 3.  $\square$

**Where we are.** Theorem 14.4.7 is a genuine bootstrap and its shape is worth admiring. One proves a *topological* statement — that a certain subset of an interval is open and closed — rather than driving an induction to a limit. The openness is Proposition 14.4.5, the hard analytic input; the closedness is a continuity remark; and connectedness does the rest.

*Why  $\frac{2}{3}$ .* The interval  $[0, \frac{2}{3}]$  is where Proposition 14.4.4 is proved. Above  $\frac{2}{3}$  the equivalence is not claimed and is not needed, since the base case starts at  $\frac{1}{2}$ .

*The trade in Step 2.* An improvement  $g$  in  $\omega$  becomes  $g/4$  in  $\sigma$  because  $\#\mathbb{T} \lesssim \delta^{-4}$ . That factor 4 is the only place the ambient dimension 3 enters this argument — everything else is formal.

## 14.4.4 Factoring convex sets

The analytic content of Proposition 14.4.4 and Proposition 14.4.5 rests on a structure theorem, which we prove in full. It says that any family of convex

sets can be covered by a family that is simultaneously sparse from above and dense from below.

**Definition 14.4.10** (Factoring).  $\mathcal{W}$  *factors*  $\mathcal{U}$  *from above* with respect to the Katz–Tao Convex Wolff axioms with error  $K$  if  $\mathcal{W}$  covers  $\mathcal{U}$  and  $C_{\text{KT}}(\mathcal{W}) \leq K$ . It *factors from below* with respect to the Frostman Convex Wolff axioms with error  $K$  if  $\mathcal{W}$  covers  $\mathcal{U}$  and  $C_{\text{F-C}}(\mathcal{U}^W) \leq K$  for every  $W \in \mathcal{W}$ .

**Lemma 14.4.11** (Iterated graph pruning). *Let  $G = (A \sqcup B, E)$  be a bipartite graph. There is an induced subgraph  $G' = (A' \sqcup B', E')$  with  $\#E' \geq \#E/2$ , every vertex of  $A'$  of degree at least  $\frac{\#E}{4\#A}$ , and every vertex of  $B'$  of degree at least  $\frac{\#E}{4\#B}$ .*

*Proof.* Repeatedly delete a vertex of  $A$  of degree below  $\frac{\#E}{4\#A}$  or a vertex of  $B$  of degree below  $\frac{\#E}{4\#B}$ . Deleting all such vertices removes fewer than  $\#A \cdot \frac{\#E}{4\#A} + \#B \cdot \frac{\#E}{4\#B} = \frac{\#E}{2}$  edges in total, so the process terminates with at least half the edges remaining, and by construction every surviving vertex has the required degree.  $\square$

**Theorem 14.4.12** (Factoring). *Let  $\mathcal{U}$  be a finite set of congruent convex subsets of the unit ball in  $\mathbb{R}^n$ , each containing a ball of radius  $\delta$ , and put  $K = 100^n e^{100\sqrt{\log(\delta^{-1}\#\mathcal{U})}}$ . Then there are  $\mathcal{U}' \subseteq \mathcal{U}$  and a set  $\mathcal{W}$  of congruent convex sets with:*

$$(i) \quad \#\mathcal{U}' \geq K^{-1}(\#\mathcal{U});$$

(ii)  $\mathcal{W}$  is a  $K$ -balanced,  $K$ -almost partitioning cover of  $\mathcal{U}'$ , and

$$\#\mathcal{U}'[W] \geq K^{-1}C_{\text{KT}}(\mathcal{U}')|W||U|^{-1} \quad (W \in \mathcal{W}); \quad (14.4.4)$$

(iii)  $\mathcal{W}$  factors  $\mathcal{U}'$  from above for Katz–Tao with error  $K$ ;

(iv)  $\mathcal{W}$  factors  $\mathcal{U}'$  from below for Frostman with error  $K$ .

*Proof. Step 1.* A near-minimiser of the Katz–Tao constant.

Let  $\mathcal{U}_0 \subseteq \mathcal{U}$  minimise

$$\exp \left[ \left( \log \frac{\#\mathcal{U}}{\#\mathcal{U}'} \right)^2 \right] C_{\text{KT}}(\mathcal{U}') \quad (14.4.5)$$

over non-empty  $\mathcal{U}' \subseteq \mathcal{U}$ . Since  $C_{\text{KT}} \geq 1$  always, evaluating (14.4.5) at  $\mathcal{U}_0$  and comparing with its value at  $\mathcal{U}$  gives  $\exp[(\log \frac{\#\mathcal{U}}{\#\mathcal{U}_0})^2] \leq C_{\text{KT}}(\mathcal{U}) \leq \#\mathcal{U}$ , hence

$$\#\mathcal{U}_0 \geq e^{-\sqrt{\log \#\mathcal{U}}}(\#\mathcal{U}). \quad (14.4.6)$$

Moreover if  $\mathcal{U}' \subseteq \mathcal{U}_0$  with  $\#\mathcal{U}' \geq \frac{1}{2}\#\mathcal{U}_0$ , then minimality gives

$$C_{\text{KT}}(\mathcal{U}') \geq \kappa_0 C_{\text{KT}}(\mathcal{U}_0), \quad \kappa_0 = e^{-2 \log 2 \sqrt{\log \#\mathcal{U}}}. \quad (14.4.7)$$

The unusual weight  $\exp[(\log \text{ratio})^2]$  is chosen so that (14.4.7) holds after removing up to half the family — a linear weight would not survive that.

**Step 2.** *Greedy extraction of heavy convex sets.*

Set  $\mathcal{U}_0$  aside and select  $W_1, W_2, \dots$  greedily: having formed  $\mathcal{U}_{j-1}$ , choose  $W_j$  maximising  $\#\mathcal{U}_{j-1}[W_j]/|W_j|$ , so that

$$\#\mathcal{U}_{j-1}[W_j] = C_{\text{KT}}(\mathcal{U}_{j-1}) \frac{|W_j|}{|U|}, \quad (14.4.8)$$

and put  $\mathcal{U}_j = \mathcal{U}_{j-1} \setminus \mathcal{U}_{j-1}[W_j]$ . Stop when  $\#\mathcal{U}_j < \frac{1}{2}\#\mathcal{U}_0$ ; let  $\mathcal{W}_0 = \{W_1, \dots, W_{j-1}\}$ . Then  $\#(\bigcup_{W \in \mathcal{W}_0} \mathcal{U}_0[W]) > \frac{1}{2}\#\mathcal{U}_0$ , and by (14.4.8) and (14.4.7), for every  $\mathcal{W}' \subseteq \mathcal{W}_0$ ,

$$\kappa_0 \frac{C_{\text{KT}}(\mathcal{U}_0)}{|U|} \sum_{W \in \mathcal{W}'} |W| \leq \# \left( \bigcup_{W \in \mathcal{W}'} \mathcal{U}_0[W] \right) \leq \frac{C_{\text{KT}}(\mathcal{U}_0)}{|U|} \sum_{W \in \mathcal{W}'} |W|. \quad (14.4.9)$$

The lower bound uses that the sets  $\mathcal{U}_{i-1}[W_i]$  are *disjoint* by construction — this is the point of the greedy removal.

**Step 3.** *Make the  $W$  congruent.*

Each  $W \in \mathcal{W}_0$  has a John ellipsoid with axes in  $[\delta, 2]$ . Dyadic pigeonholing in the  $n$  axis lengths selects  $\mathcal{W}_1 \subseteq \mathcal{W}_0$  with all axes comparable, retaining a  $(100|\log \delta|)^{-n}$  fraction; replace each by a congruent copy  $W_0$  of a fixed ellipsoid to get  $\mathcal{W}_2$ , and set  $\mathcal{U}_2 = \bigcup_{W \in \mathcal{W}_2} \mathcal{U}_0[W]$ . Applying (14.4.9) in both directions,

$$\#\{(U, W) \in \mathcal{U}_2 \times \mathcal{W}_2 : U \subseteq W\} \leq 2^n \kappa_0^{-1} (\#\mathcal{U}_2). \quad (14.4.10)$$

**Step 4.** *Prune the incidence graph.*

Form the bipartite graph on  $\mathcal{U}_2 \times \mathcal{W}_2$  with edges  $U \subseteq W$ . By (14.4.10) it has at most  $2^n \kappa_0^{-1} \#\mathcal{U}_2$  edges; each  $U$  has at least one neighbour; and each  $W$  has between  $2^{-n} \kappa_0 C_{\text{KT}}(\mathcal{U}_0) |W_0| |U|^{-1}$  and  $C_{\text{KT}}(\mathcal{U}_0) |W_0| |U|^{-1}$  neighbours, whence

$$\#\mathcal{W}_2 \leq 2^{2n} \kappa_0^{-2} \frac{(\#\mathcal{U}_2) |U|}{C_{\text{KT}}(\mathcal{U}_0) |W_0|}. \quad (14.4.11)$$

Delete every  $U$  with more than  $2^{n+1} \kappa_0^{-1}$  neighbours — this keeps at least half of  $\mathcal{U}_2$  — then apply Lemma 14.4.11. Call the result  $\mathcal{U}' \times \mathcal{W}$ .

**Step 5.** *Verify the four conclusions.*

(i) Chaining the losses of Steps 1–4 against (14.4.6) and (14.4.10) gives  $\#\mathcal{U}' \geq K^{-1} \#\mathcal{U}$ .

(ii) Each  $U \in \mathcal{U}'$  lies in at most  $K$  sets  $\mathcal{U}'[W]$ , so the cover is  $K$ -almost partitioning. For (14.4.4), Lemma 14.4.11 and (14.4.11) give

$$\#\mathcal{U}'[W] \geq \frac{1}{4}(\#\mathcal{I}_3)(\#\mathcal{W}_2)^{-1} \geq (2^{-2n-4}\kappa_0^2)C_{\text{KT}}(\mathcal{U}_0)|W_0||U|^{-1},$$

and  $C_{\text{KT}}(\mathcal{U}') \leq C_{\text{KT}}(\mathcal{U}_0)$  since  $\mathcal{U}' \subseteq \mathcal{U}_0$ . Balancedness follows since also  $\#\mathcal{U}'[W] \leq C_{\text{KT}}(\mathcal{U}')|W||U|^{-1}$ .

(iii) For convex  $V$ , using that each  $U \in \mathcal{U}'$  meets at most  $2^{n+1}\kappa_0^{-1}$  sets  $\mathcal{U}'[W]$ ,

$$\#\mathcal{U}'[V] \geq 2^{-n-1}\kappa_0 \sum_{W \in \mathcal{W}[V]} \#\mathcal{U}'[W] \geq (2^{-3n-5}\kappa_0^3)C_{\text{KT}}(\mathcal{U}_0)|W_0||U|^{-1}(\#\mathcal{W}[V]),$$

while  $\#\mathcal{U}'[V] \leq C_{\text{KT}}(\mathcal{U}')|V||W_0|^{-1}$ . Comparing gives  $\#\mathcal{W}[V] \leq K|V||W_0|^{-1}$ , that is  $C_{\text{KT}}(\mathcal{W}) \leq K$ .

(iv) For  $W \in \mathcal{W}$  and convex  $V \subseteq W$ ,

$$\#(\mathcal{U}'[W])[V] \leq \#\mathcal{U}'[V] \leq C_{\text{KT}}(\mathcal{U}')|V||U|^{-1},$$

and comparing with (14.4.4) gives  $\#(\mathcal{U}'[W])[V] \leq K|V||W|^{-1}(\#\mathcal{U}'[W])$ , which is  $C_{\text{F-C}}((\mathcal{U}')^W) \leq K$ .  $\square$

**Where we are.** Theorem 14.4.12 is the new ingredient that makes the whole scheme work, and its content is that the two Wolff constants can be *decoupled*: the cover  $\mathcal{W}$  is sparse (small  $C_{\text{KT}}$ ), while inside each  $W$  the family is dense (small  $C_{\text{F-C}}$ ). Lemma 14.4.2 says these are the two directions in which the constants propagate, so the factorisation puts each where it is useful.

Note  $K \lesssim 1$  whenever  $\#\mathcal{U} \leq \delta^{-100}$ , so the theorem is free in the regime of interest.

**Proposition 14.4.13** (The polychotomy; not proved here, see Remark 14.4.14). *For every set  $\mathbb{T}$  of  $\delta$ -tubes in  $\mathbb{R}^3$ , at least one of the following holds.*

- (A) *There is a scale  $\rho = \delta^\zeta$ ,  $\zeta$  bounded away from 0, and a set  $\mathbb{T}_\rho$  of  $\rho$ -tubes with  $C_{\text{KT}}(\mathbb{T}_\rho) = O(1)$ , each containing about  $m|T_\rho|/|T|$  tubes of  $\mathbb{T}$ , where  $m = C_{\text{KT}}(\mathbb{T})$ .*
- (B) *There are scales  $\delta \leq \tau \leq \rho$  with  $\tau = \delta^\zeta \rho$  and covers  $\mathbb{T}_\tau, \mathbb{T}_\rho$  satisfying the five conditions of the well-spaced case.*
- (C) *The tubes pack efficiently into prisms of dimensions  $s \times t \times 1$  with  $s \ll t$ .*
- (D)  *$\mathbb{T}$  satisfies the Frostman Convex Wolff Axioms at every scale.*

*Remark 14.4.14* (What remains). Proposition 14.4.13 is obtained by applying Theorem 14.4.12 at several carefully chosen scales: the factorisation produces a cover  $\mathcal{W}$  satisfying the analogues of (A); if its members have dimensions  $s \times t \times 1$  with  $s \ll t$  one is in case (C), and if  $s \sim t$  they are essentially tubes.

Cases (A) and (B) yield (14.4.2) by the two- and three-scale computations of Worked Example 14.4.1 below.

Case (C) is handled by a rescaling argument.

Case (D) is precisely the sticky case, where Theorem 14.2.5 applies — proved in Chapter 14.3–Chapter 14.9.

*What is proved here.* Theorem 14.4.12 in full; the bootstrap Theorem 14.4.7 in full; Lemma 14.4.2; and the numerology of Definition 14.4.3. *What is not.* The verification that the four cases exhaust, and the rescaling argument for case (C). These are Sections 4–6 of Wang and Zahl and total some forty pages; they use no ingredient not already introduced here.

#### WORKED EXAMPLE 14.4.1— TIER A **The two-scale computation**

The heart of Proposition 14.4.4 is a two-scale application of the assertions, and it is short enough to give in full.

Suppose  $C_{\text{KT}}(\mathbb{T}) = m \gg 1$ , so some convex  $W$  contains at least  $m|W|\delta^{-2}$  tubes; take  $W$  comparable to a  $\rho$ -tube. Let  $\mathbb{T}_\rho$  be a cover by  $\rho$ -tubes each containing about  $m(\rho/\delta)^2$  tubes of  $\mathbb{T}$ .

*Step 1: the coarse family is sparse.* If a convex  $W'$  contains  $N$  tubes of  $\mathbb{T}_\rho$  it contains about  $Nm(\rho/\delta)^2$  tubes of  $\mathbb{T}$ , hence  $Nm(\rho/\delta)^2 \leq m|W'|\delta^{-2}$ , giving  $C_{\text{KT}}(\mathbb{T}_\rho) = O(1)$ . So  $\mathcal{D}(\sigma, \omega)$  applies:

$$\left| \bigcup_{T_\rho} T_\rho \right| \gtrsim \rho^\omega (\#\mathbb{T}_\rho) |T_\rho| ((\#\mathbb{T}_\rho) |T_\rho|^{1/2})^{-\sigma}.$$

*Step 2: the fine families are dense.* Inside each  $T_\rho$ ,  $C_{\text{KT}}(\mathbb{T}^{T_\rho}) \lesssim m$ , and  $\mathcal{E}(\sigma, \omega')$  gives

$$\left| \bigcup_{\mathbb{T}^{T_\rho}} T^{T_\rho} \right| \gtrsim (\delta/\rho)^{\omega'} m^{-1} (\#\mathbb{T}[T_\rho]) |T^{T_\rho}| (m^{-3/2} (\#\mathbb{T}[T_\rho]) |T^{T_\rho}|^{1/2})^{-\sigma}.$$

*Step 3: multiply.* Using  $(\#\mathbb{T}_\rho)(\#\mathbb{T}[T_\rho]) = \#\mathbb{T}$  and  $|T_\rho| |T^{T_\rho}| = |T|$ ,

$$\left| \bigcup_T T \right| \gtrsim \rho^{\omega-\omega'} \delta^{\omega'} m^{-1} (\#\mathbb{T}) |T| (m^{-3/2} (\#\mathbb{T}) |T|^{1/2})^{-\sigma}.$$

If  $\rho \leq \delta^\zeta$  this is (14.4.2) with  $\omega' - \alpha$  in place of  $\omega'$ , where  $\alpha = \zeta(\omega' - \omega)$ . That is the self-improvement, and iterating it from the trivial  $\mathcal{E}(\sigma, 2)$  down proves Proposition 14.4.4.

*Why the shape of (14.4.2) was chosen.* Precisely so that Step 3 multiplies cleanly. Every factor in (14.4.2) is there to make this computation an identity rather than an inequality.

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WORKED EXAMPLE 14.4.2— TIER B **The well-spaced example**

Case (B) of Proposition 14.4.13 is not an artefact; here is the configuration forcing it.

*The example.* Let  $\#\mathbb{T} = \delta^{-5/2}$ , put  $s = \delta^{5/8}$ , and let each of the  $\approx s^{-4}$  essentially distinct  $s$ -tubes in the unit ball contain exactly one tube of  $\mathbb{T}$ .

*Why (A) fails.* There is no single scale  $\rho$  at which a cover has both  $C_{\text{KT}}(\mathbb{T}_\rho) = O(1)$  and each  $\rho$ -tube containing the right number of  $\delta$ -tubes: the tubes are too spread out at every scale.

*What is true instead.* With  $m = \delta^{-1/2}$ ,  $\tau = \delta$ ,  $\rho = \delta^{1/4}$ ,  $\mathbb{T}_\tau = \mathbb{T}$  and  $\mathbb{T}_\rho$  a maximal set of  $\rho^{-4}$  essentially distinct  $\rho$ -tubes, the five conditions of case (B) hold, and a *three-scale* version of Worked Example 14.4.1 — applying  $\mathcal{E}$  at  $\rho$ ,  $\mathcal{D}$  between  $\rho$  and  $\tau$ , and  $\mathcal{E}$  between  $\tau$  and  $\delta$  — gives the same conclusion.

*The moral.* A two-scale argument suffices for the generic configuration; a genuinely three-scale one is needed for the well-spaced case. That is why the polychotomy has four branches rather than two, and it is the technical price of dropping the stickiness hypothesis.

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WORKED EXAMPLE 14.4.3— TIER C: RESEARCH TASTE **Bootstraps, and when they are available**

Theorem 14.4.7 converts an arbitrarily small gain into a sharp theorem. So does Corollary 14.13.9. This is the most powerful single device in the last decade of this subject, and it is worth asking when it can be used.

(a) *What is needed.* A quantity  $\sigma$  parametrising a family of assertions, monotone in  $\sigma$ ; a proof that the admissible set is *closed*, usually by continuity; and a proof that it is *open*, which is the analytic work.

(b) *Why openness is the hard half.* It says: whenever the assertion holds at  $\sigma$ , it holds slightly below. That is a self-improvement, and producing one requires a genuine estimate — Proposition 14.4.5 here, the key lemma in Chapter 14.13.

(c) *The role of the base case.* Wolff’s hairbrush supplies  $\mathcal{D}(\frac{1}{2}, 0)$ ; without a starting point the argument proves nothing. So a bootstrap does not replace the older methods, it *amplifies* them — Volume 9’s hairbrush is what Theorem 14.4.7 begins from.

(d) *Instances in this series.* Theorem 14.4.7; the radial projection bootstrap of Corollary 14.13.9; Volume 13’s Proposition 13.2.3, which is the same shape with log log iterations rather than a topological argument; and Volume 12’s Proposition 12.6.5.

(e) *What distinguishes them.* Proposition 13.2.3 iterates a fixed number of times and accumulates a loss; Theorem 14.4.7 takes a limit and accumulates none, at the cost of giving no effective constant. *The topological form is stronger and less informative*, and which one a paper uses tells you whether it can produce explicit bounds.

(f) *A caution.* A bootstrap can only reach the sharp threshold if the self-improvement holds all the way to it. Volume 12, Worked Example 12.4.1(d) records the opposite case: broad–narrow self-improves only until the narrow case’s own exponent binds, and there the iteration stalls short of the conjecture.

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## Drills

**Drill 14.4.1.** Verify the Checkpoint after Definition 14.4.3: substitute  $\sigma = \omega$  into (14.4.1) and read off the multiplicity statement.

**Drill 14.4.2.** Prove Lemma 14.4.11 by counting the edges removed.

**Drill 14.4.3.** In Step 1 of Theorem 14.4.12, verify (14.4.6) and explain why the weight  $\exp[(\log \text{ratio})^2]$  is used rather than a power.

**Drill 14.4.4.** Carry out Step 3 of Worked Example 14.4.1 and confirm the exponent  $\alpha = \zeta(\omega' - \omega)$ .

## Notes

This chapter is Sections 1, 2 and 4 of Wang and Zahl’s 2025 paper. Theorem 14.4.12 is their Proposition 4.6, and Theorem 14.4.7 their Theorem 1.9. Remark 14.4.14 records what of Sections 4–6 is not reproduced.





## Chapter 14.5

# The Worst Case is Sticky

Remark 14.4.14 recorded two things as missing from Chapter 14.4: the verification that the polychotomy exhausts, and the argument for its intermediate case. This chapter supplies both, following Guth’s exposition, and with it the reduction of general Kakeya to the sticky case is complete in outline with every step either proved or named.

### 14.5.1 The strategy

Throughout,  $\mathbb{T}$  is a worst-case Kakeya set in the sense of Definition 14.10.1:  $\Delta_{\max}(\mathbb{T}) \lesssim 1$  and  $\mu(\mathbb{T}) \approx |\mathbb{T}|^\beta$ , with  $\beta > 0$  assumed for contradiction. We take  $|\mathbb{T}| \approx \delta^{-2}$ , the main case.

**Definition 14.5.1** (Not sticky at all scales).  $\mathbb{T}$  is *not sticky at all scales* if for every  $\rho$  with  $\delta \ll \rho \ll 1$ ,

$$|\mathbb{T}[T_\rho]| \ll (\rho/\delta)^2 \quad \text{and} \quad |\mathbb{T}_\rho| \gg \rho^{-2}. \quad (14.5.1)$$

**Lemma 14.5.2** (Reduction to the all-scales case). *It suffices to treat  $\mathbb{T}$  sticky, or not sticky at all scales.*

*Proof.* If some scale  $\rho$  has  $|\mathbb{T}[T_\rho]| \approx (\rho/\delta)^2$  then  $\Delta_{\max}(\mathbb{T}_\rho) \lesssim 1$ , and Lemma 14.10.2 bounds  $\mu(\mathbb{T}) \lesssim \mu(\mathbb{T}[T_\rho])\mu(\mathbb{T}_\rho)$  — reducing the problem to two problems on shorter scale ranges. Iterate. If the reduction continues to the bottom,  $\mathbb{T}$  was sticky and Theorem 14.2.5 applies. Otherwise it halts at a sub-problem that is not sticky at all scales.  $\square$

**Definition 14.5.3** (Localising to a small ball). Fix  $\rho$  with  $\delta \ll \rho \ll 1$  and put  $r = \delta/\rho$ . If  $T_1, T_2 \in \mathbb{T}$  meet at angle  $\approx \rho$  then  $T_1 \cap T_2$  is approximately a tube of radius  $\delta$  and length  $r$ . For a ball  $B$  of radius  $r$ , let  $\mathbb{T}[T_\rho]_B$  be these

short tubes inside  $B$ , and

$$\mathbb{T}_B = \bigcup_{T_\rho \in \mathbb{T}_\rho, T_\rho \cap B \neq \emptyset} \mathbb{T}[T_\rho]_B.$$

**Lemma 14.5.4.**  $\mu(\mathbb{T}) \lesssim \mu(\mathbb{T}[T_\rho]) \mu(\mathbb{T}_B)$ .

*Proof.* Each  $T_B \in \mathbb{T}_B$  lies in  $\approx \mu(\mathbb{T}[T_\rho])$  tubes of  $\mathbb{T}$ , and a point of  $B$  lies in  $\approx \mu(\mathbb{T}_B)$  tubes of  $\mathbb{T}_B$ .  $\square$

**Lemma 14.5.5** (The easy case). *If  $\mathbb{T}$  is worst-case and not sticky at all scales, and  $\Delta_{\max}(\mathbb{T}_B) \lesssim 1$ , then  $\mu(\mathbb{T}) \ll |\mathbb{T}|^\beta$ .*

*Proof.* The tubes of  $\mathbb{T}_B$  have radius  $\delta$  and length  $r = \delta/\rho$ , so aspect ratio  $\rho$ ; with  $\Delta_{\max}(\mathbb{T}_B) \lesssim 1$  this gives  $|\mathbb{T}_B| \lesssim \rho^{-2}$  and hence, by (14.10.1),  $\mu(\mathbb{T}_B) \lesssim \rho^{-2\beta}$ .

By (14.5.1) and (14.10.1),  $\mu(\mathbb{T}[T_\rho]) \ll (\rho/\delta)^{2\beta}$ . Multiplying by Lemma 14.5.4,

$$\mu(\mathbb{T}) \ll (\rho/\delta)^{2\beta} \rho^{-2\beta} = \delta^{-2\beta} \approx |\mathbb{T}|^\beta. \quad \square$$

**Where we are.** The exponents cancel exactly:  $(\rho/\delta)^{2\beta} \rho^{-2\beta} = \delta^{-2\beta}$ , verified symbolically. *The whole gain comes from the first factor being  $\ll$  rather than  $\lesssim$*  — that is, from the failure of stickiness at scale  $\rho$ . *But the hypothesis is a genuine one.*  $\Delta_{\max}(\mathbb{T}) \lesssim 1$  does *not* imply  $\Delta_{\max}(\mathbb{T}_B) \lesssim 1$ : intersecting with a small ball can concentrate tubes. This is precisely where *Keakeya* differs from *multilinear Keakeya*, whose hypotheses *are* inherited by small balls — which is why Volume 11, Theorem 11.5.4 has a short inductive proof and this does not. Wang and Zahl’s contribution is to extract a usable conclusion from *every* behaviour of  $\mathbb{T}_B$ , not just the convenient one.

## 14.5.2 The polychotomy

Suppose  $\Delta_{\max}(\mathbb{T}_B) \gg 1$  and let  $K \subseteq B$  maximise  $\Delta(\mathbb{T}_B, K)$ .

**Definition 14.5.6** (Frostman in a convex set). A family  $\tilde{\mathbb{T}}$  of tubes contained in a convex  $K$  is *Frostman in  $K$*  if  $\Delta(\tilde{\mathbb{T}}, K') \lesssim \Delta(\tilde{\mathbb{T}}, K)$  for every convex  $K' \subseteq K$  — equivalently  $\Delta_{\max}(\tilde{\mathbb{T}}) \approx \Delta(\tilde{\mathbb{T}}, K)$ .

By maximality,  $\mathbb{T}_B[K]$  is Frostman in  $K$ . Choosing further such  $K$  greedily until every tube of  $\mathbb{T}_B$  lies in one, we obtain a family  $\mathcal{K}$  of congruent boxes of dimensions  $a \times b \times r$  with  $\mu(\mathcal{K}) \lesssim 1$  and each  $\mathbb{T}_B[K]$  Frostman.

**Theorem 14.5.7** (The four cases). *With  $K$  of dimensions  $a \times b \times r$ , one of the following holds, and in each  $\mu(\mathbb{T}) \ll |\mathbb{T}|^\beta$ .*

- (1) Fat planks,  $a \gg \delta$ . Handled by Lemma 14.5.9, as in the case  $K = B$ .
- (2) Thin planks meeting transversely,  $a \approx \delta$ ,  $b \ll r$ . Handled by the  $L^2$  method.
- (3) Thin planks meeting tangentially,  $a \approx \delta$ ,  $b \ll r$ . Handled by a change of coordinates turning planks into tubes.
- (4) Thin slabs,  $a \approx \delta$ ,  $b \approx r$ . Handled by the  $L^2$  method, using that  $r$  is close to 1.

*Proof of cases (2), (3), (4).* When  $a \approx \delta$  the tubes  $\mathbb{T}_B[K]$  inside  $K$  pose a two-dimensional Kakeya problem, so the  $L^2$  method of Volume 9, Theorem 9.3.2 gives  $|U(\mathbb{T}_B[K])| \gtrsim |K|$ : the planks are essentially filled.

(2) Each plank has a tangent plane spanned by its two longest axes. Intersecting a plank with a ball of radius  $b$  gives a slab; if the tangent planes are transverse, then inside such a ball one sees several slabs meeting transversely, and the  $L^2$  method for slabs in  $\mathbb{R}^3$  gives  $|U(\mathcal{K}) \cap B_b| \gtrsim |B_b|$ . Then Lemma 14.5.8 below concludes.

(3) If all intersections are tangential, every plank meeting a fixed  $K_0$  lies in a slab  $S$  of dimensions  $\frac{\delta r}{b} \times r \times r$  with nearly the same tangent plane. Writing  $\mathcal{K}_S$  for those planks,  $\mu(\mathcal{K}) \approx \mu(\mathcal{K}_S)$ . A linear change of variables takes  $S$  to the unit cube and each  $K \in \mathcal{K}_S$  to a tube, giving a family  $\mathbb{T}'$  with

$$\mu(\mathcal{K}) \approx \mu(\mathbb{T}'), \quad \Delta_{\max}(\mathbb{T}') \approx \Delta_{\max}(\mathcal{K}_S) \leq \Delta_{\max}(\mathcal{K}) \lesssim 1,$$

so  $\mu(\mathbb{T}') \lesssim |\mathbb{T}'|^\beta$  by (14.10.1), and the computation of Lemma 14.5.5 applies with  $\mathbb{T}'$  in place of  $\mathbb{T}_B$ .

(4) If  $r$  is close to 1, the hypothesis  $\Delta_{\max}(\mathbb{T}) \lesssim 1$  prevents many tubes from meeting a thin slab, so there are many slabs; their mutual intersections are controlled by the  $L^2$  method, forcing  $|U(\mathcal{K})|$  close to 1 and hence  $|U(\mathbb{T})|$  close to 1. Attaining  $r \approx 1$  requires  $\rho$  close to  $\delta$ , which is why Definition 14.5.1 demands failure at *all* scales.  $\square$

**Lemma 14.5.8** (Density in small balls controls  $\mu$ ). *For  $\delta \leq r \leq 1$  one has  $|U(\mathbb{T}_r)| \gtrsim r^{2\beta}$ , and for a typical  $B_r$ ,*

$$|U(\mathbb{T})| \approx |U(\mathbb{T}_r)| \frac{|U(\mathbb{T}) \cap B_r|}{|B_r|}. \quad (14.5.2)$$

Consequently,  $\frac{|U(\mathbb{T}) \cap B_r|}{|B_r|} \gg (\delta/r)^{2\beta}$  implies  $|U(\mathbb{T})| \gg \delta^{2\beta}$ , that is  $\mu(\mathbb{T}) \ll |\mathbb{T}|^\beta$ .

*Proof.*  $U(\mathbb{T}_r)$  is the  $r$ -neighbourhood of  $U(\mathbb{T})$ , so (14.5.2) is the uniformity convention. With  $|\mathbb{T}| \approx \delta^{-2}$  and  $\mu(\mathbb{T}) = |\mathbb{T}| |T| / |U(\mathbb{T})|$ , the statement  $\mu(\mathbb{T}) \ll |\mathbb{T}|^\beta$  is equivalent to  $|U(\mathbb{T})| \gg \delta^{2\beta}$ , and substituting the two displayed bounds gives it.  $\square$

### 14.5.3 The high density lemma

**Lemma 14.5.9** (High density). *Let  $\tilde{\mathbb{T}}$  be a set of  $\delta$ -tubes Frostman in  $B_1$ . Then*

$$|U(\tilde{\mathbb{T}})| \gtrsim (\delta^2 |\tilde{\mathbb{T}}|)^{1-\beta} \delta^{2\beta}, \quad \text{equivalently} \quad \mu(\tilde{\mathbb{T}}) \lesssim (\delta^2 |\tilde{\mathbb{T}}|)^\beta \delta^{-2\beta}.$$

**Where we are.** Calibrate at  $|\tilde{\mathbb{T}}| \approx \delta^{-2}$ : then  $\delta^2 |\tilde{\mathbb{T}}| \approx 1$  and the bound reads  $\mu(\tilde{\mathbb{T}}) \lesssim \delta^{-2\beta} = |\tilde{\mathbb{T}}|^\beta$  — exactly (14.10.1). ✓

*The content is the case  $|\tilde{\mathbb{T}}| \gg \delta^{-2}$ , where the lemma asserts  $|U(\tilde{\mathbb{T}})| \gg \delta^{2\beta}$ . The trivial bound gives only  $\gtrsim \delta^{2\beta}$ : split  $\tilde{\mathbb{T}}$  randomly into pieces of size  $\approx \delta^{-2}$ , note each inherits the Frostman condition, and apply (14.10.1) to one piece. Any improvement on that trivial bound suffices, and this is another instance of Worked Example 14.4.3.*

**Lemma 14.5.10** (Frostman at every scale gives a sticky subset). *Let  $1 = \rho_0 > \rho_1 > \dots > \rho_N = \delta$  be a dense sequence of scales. If  $\mathbb{T}_{\rho_j}[T_{\rho_{j-1}}]$  is Frostman in  $T_{\rho_{j-1}}$  for every  $j$  and every  $T_{\rho_{j-1}}$ , then  $\mathbb{T}$  contains a sticky Keakeya set, and indeed decomposes into such.*

*Proof.* Since  $\mathbb{T}_{\rho_1}$  is Frostman,  $|\mathbb{T}_{\rho_1}| \gtrsim \rho_1^{-2}$ ; choose a random subset  $\mathbb{T}'_{\rho_1}$  of size  $\approx \rho_1^{-2}$ , which is again Frostman. For each  $T_{\rho_1} \in \mathbb{T}'_{\rho_1}$ ,  $\mathbb{T}_{\rho_2}[T_{\rho_1}]$  is Frostman, so of size  $\gtrsim (\rho_1/\rho_2)^2$ ; choose a random subset of size exactly  $\approx (\rho_1/\rho_2)^2$ . Iterating produces  $\mathbb{T}'$  with  $|\mathbb{T}'_{\rho_j}[T_{\rho_{j-1}}]| \approx (\rho_{j-1}/\rho_j)^2$  at every  $j$ , which is the criterion of Definition 14.2.3. Taking random *decompositions* rather than subsets gives the stronger statement.  $\square$

**Lemma 14.5.11** (The maximising sets are tubes). *Let  $\tilde{\mathbb{T}}$  be a worst-case example for Lemma 14.5.9, and for  $\rho$  with  $\Delta(\tilde{\mathbb{T}}, T_\rho) \gg 1$  let  $\mathcal{W}(T_\rho)$  be a maximal family of convex sets maximising  $\Delta(\tilde{\mathbb{T}}[T_\rho], W)$ . Then each  $W \in \mathcal{W}(T_\rho)$ , of dimensions  $a \times b \times 1$ , has  $a \approx b$  — so  $W$  is a tube.*

*Proof outline.* Put  $\mathcal{W} = \bigcup_{T_\rho} \mathcal{W}(T_\rho)$ . Each  $T \in \tilde{\mathbb{T}}$  lies in  $\approx 1$  set of  $\mathcal{W}$ , so  $|\tilde{\mathbb{T}}| \approx |\tilde{\mathbb{T}}[W]| |\mathcal{W}|$  and, by Lemma 14.10.2,

$$\mu(\tilde{\mathbb{T}}) \lesssim \mu(\tilde{\mathbb{T}}[W]) \mu(\mathcal{W}). \quad (14.5.3)$$

Both factors are Frostman —  $\tilde{\mathbb{T}}[W]$  by the choice of  $W$ , and  $\mathcal{W}$  because  $\tilde{\mathbb{T}}$  is. To bound them one needs Lemma 14.5.9 for families of *convex sets* Frostman in a convex set, which is the natural generalisation. There, a set of dimensions  $a_V \times b_V \times 1$  is a tube when  $a_V \approx b_V$  and a slab when  $a_V \approx 1$ ; slabs obey *sharp* bounds by the  $L^2$  method, and the intermediate shapes are handled by a multiscale argument combining the two.

Substituting into (14.5.3), the computation improves on the worst-case value unless the  $L^2$  slab bound is never invoked — that is, unless  $a \approx b$  throughout. Hence a worst-case example has  $a \approx b$ .  $\square$

**Lemma 14.5.12** (A good scale exists). *Let  $\tilde{\mathbb{T}}$  be a worst-case example for Lemma 14.5.9 with  $|\tilde{\mathbb{T}}| \gg \delta^{-2}$ . Then there is a with  $\delta \ll a \ll 1$  such that  $\tilde{\mathbb{T}}[T_a]$  and  $\tilde{\mathbb{T}}_a$  are both Frostman, and both are worst-case examples.*

*Proof.* Since  $|\tilde{\mathbb{T}}| \gg \delta^{-2}$ , choose  $\rho \ll 1$  with  $|\tilde{\mathbb{T}}[T_\rho]| \gg (\rho/\delta)^2$ , so  $\Delta(\tilde{\mathbb{T}}, T_\rho) \gg 1$ . Form  $\mathcal{W}(T_\rho)$  as above; by Lemma 14.5.11 each  $W$  is a tube of some radius  $a \leq \rho \ll 1$ , so  $\mathcal{W} = \tilde{\mathbb{T}}_a$ . Then  $\tilde{\mathbb{T}}[T_a]$  is Frostman by the maximality defining  $W$ , and  $\tilde{\mathbb{T}}_a$  is Frostman because  $\tilde{\mathbb{T}}$  is.

For  $a \gg \delta$ : we chose  $\rho$  with  $\Delta(\tilde{\mathbb{T}}[T_\rho], T_\rho) \gg 1$ , and  $\Delta(\tilde{\mathbb{T}}[W], W) \approx \Delta_{\max}(\tilde{\mathbb{T}}[T_\rho]) \geq \Delta(\tilde{\mathbb{T}}[T_\rho], T_\rho) \gg 1$ . Were  $a \approx \delta$ ,  $W$  would be a  $\delta$ -tube and  $\Delta(\tilde{\mathbb{T}}[W], W) \approx 1$ . So  $a \gg \delta$ .

That both derived families are worst-case follows as in Proposition 14.10.3: chaining Lemma 14.10.2 with the definition of the exponent in Lemma 14.5.9 returns the worst-case value, so every step is an approximate equality.  $\square$

*Proof of Lemma 14.5.9, outline.* Apply Lemma 14.5.12 repeatedly to obtain scales  $1 = \rho_0 \gg \cdots \gg \rho_N = \delta$  with each  $\mathbb{T}_{\rho_j}[T_{\rho_{j-1}}]$  Frostman and, for each  $j$ , either

- (a)  $\rho_{j-1}/\rho_j$  very small, or
- (b)  $|\mathbb{T}_{\rho_j}[T_{\rho_{j-1}}]| \approx (\rho_{j-1}/\rho_j)^2$ .

If  $|\tilde{\mathbb{T}}| \gg \delta^{-2}$  then not every interval can be of type (b), so a definite fraction are of type (a); these are small, so a definite fraction of the scale range consists of *blocks* of many small intervals. On each such block Lemma 14.5.10 applies and yields a sticky subset, to which Theorem 14.2.5 gives  $\mu \lesssim 1$  — a bound strictly better than the trivial one, since  $\beta > 0$ .

Factor  $\mu(\tilde{\mathbb{T}})$  over the scale ranges, as in

$$\mu(\tilde{\mathbb{T}}) \lesssim \mu(\tilde{\mathbb{T}}[T_{\delta_1}])\mu(\tilde{\mathbb{T}}_{\delta_1}[T_{\delta_2}]) \cdots \mu(\tilde{\mathbb{T}}_{\delta_k}),$$

using the sticky bound on the blocks and the trivial bound elsewhere. Since the sticky bound beats the trivial one, the product beats the trivial bound overall, and the computation returns  $\gamma = 1 - \beta$ .  $\square$

**Where we are.** The strategic choice of scale ranges — grouping them into blocks tailored to the geometry of  $\tilde{\mathbb{T}}$  — is due to Keleti and Shmerkin, and is now standard in this circle of problems; it also appears in the Furstenberg work of Theorem 14.11.4.

*The logical shape is worth noting.* Lemma 14.5.9 is used to prove that the worst case is sticky, and its own proof *uses the sticky theorem* Theorem 14.2.5. That is not circular: the sticky theorem is proved first, independently, in Chapter 14.3–Chapter 14.9, and the reduction then builds on it.

*Remark 14.5.13* (R20, discharged in outline). *What is now proved or given in full outline.* Lemma 14.5.2; Lemma 14.5.4; Lemma 14.5.5 with its exponent identity verified; Theorem 14.5.7 cases (2), (3), (4); Lemma 14.5.8; Lemma 14.5.10; Lemma 14.5.11 and Lemma 14.5.12; and the multiscale decomposition proving Lemma 14.5.9. Together with Theorem 14.4.12 and Theorem 14.4.7 of Chapter 14.4, the reduction of general Kakeya to the sticky case is complete in these pages at the level of architecture.

*What remains quoted.* The quantitative bookkeeping in Lemma 14.5.11 — the generalisation of Lemma 14.5.9 to families of convex sets, and the multiscale interpolation between tube and slab bounds — and the  $L^2$  estimates for slabs invoked in cases (2) and (4), which are Volume 9’s method applied in a different geometry.

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**WORKED EXAMPLE 14.5.1— TIER A Why Kakeya resisted the multilinear proof**

Volume 11 proves multilinear Kakeya in six pages by induction on scales. The same induction fails for Kakeya, and the reason is now precise.

*The multilinear hypothesis is inherited.* If  $\mathbb{T}_1, \dots, \mathbb{T}_n$  are transverse families and  $B$  is a small ball, then  $\mathbb{T}_{1,B}, \dots, \mathbb{T}_{n,B}$  are again transverse. So the inductive step is immediate — this is exactly Lemma 11.5.6 of Volume 11.

*The Kakeya hypothesis is not.*  $\Delta_{\max}(\mathbb{T}) \lesssim 1$  does not give  $\Delta_{\max}(\mathbb{T}_B) \lesssim 1$ ; localising can concentrate tubes. Lemma 14.5.5 shows that *if* it were inherited, the induction would close in three lines.

*What Wang and Zahl do instead.* They assume nothing about  $\mathbb{T}_B$  and extract a conclusion from every possible behaviour: Theorem 14.5.7 enumerates the shapes of the maximising convex set and handles each. Guth describes the strategy as “heads I win, tails you lose”.

*The transferable point.* When an induction fails because a hypothesis is not inherited, one option is a *polychotomy*: classify the ways the hypothesis can fail, and find a separate gain in each. That is expensive — four cases here, each with its own machinery — but it is available when strengthening the hypothesis is not.

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**WORKED EXAMPLE 14.5.2— TIER B The Katz–Zahl ring, and why the method is not universal**

The reduction to the sticky case is not a formal manoeuvre; there are cousin problems where it fails.

*The example.* Replace  $\mathbb{R}$  by  $\mathbb{A} = \mathbb{F}_p[x]/(x^2)$  with  $\|a + bx\|_{\mathbb{A}} = 1$  if  $a \neq 0$ ,  $p^{-1}$  if  $a = 0 \neq b$ , and 0 otherwise.

*What happens.* There is an analogue of the Heisenberg example in  $\mathbb{A}^3$ , so the analogue of Theorem 14.4.8 is false. But unlike the complex example, *this one is not sticky*. It appears likely that the sticky case holds in  $\mathbb{A}^3$  while the general case fails.

*Why the method does not transfer.*  $\mathbb{A}$  has only two distinct non-zero scales. The reduction of this chapter needs a rich range of intermediate scales  $\rho$  at which to form  $\mathbb{T}_\rho$ ,  $\mathbb{T}[T_\rho]$  and  $\mathbb{T}_B$ ; with two scales there is nothing to iterate over.

*What it shows.* That the reduction is a genuinely *multiscale* phenomenon and not a formal consequence of the axioms. Guth records that this example made the reduction look unlikely until it was found, and Worked Example 14.10.1 notes the parallel history for the Furstenberg problem, where the sticky case was also settled first.

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**WORKED EXAMPLE 14.5.3— TIER C: RESEARCH TASTE Polychotomies, and the cost of not strengthening a hypothesis**

Theorem 14.5.7 enumerates four shapes and handles each separately. That is an expensive way to argue, and it is worth knowing when it is the right one.

(a) *The alternative that failed.* Strengthen the hypothesis so it is inherited. For *Keakeya* that means assuming  $\Delta_{\max}(\mathbb{T}_B) \lesssim 1$  for all small balls — but that is a genuinely stronger statement about the tube family, and Besicovitch sets do not satisfy it.

(b) *When a polychotomy is available.* When the ways a hypothesis can fail are classifiable. Here the failure is measured by a maximising convex set, whose shape is described by two aspect ratios, so there are finitely many regimes.

(c) *What each branch costs.* Case (1) needs Lemma 14.5.9, itself a multiscale argument; (2) and (4) need the  $L^2$  method in a new geometry; (3) needs a change of coordinates. Four distinct techniques for one theorem.

(d) *Other instances in this series.* Volume 12, Proposition 12.4.3 has two branches; Volume 14, Lemma 14.13.6 has two; Theorem 14.8.2 has two. Four is unusual, and it reflects that the *Keakeya* hypothesis fails in more ways than a transversality hypothesis can.

(e) *The diagnostic.* Before enumerating cases, ask whether the hypothesis can be strengthened to one that *is* inherited, and whether the strengthened statement is still true of the objects at hand. If yes, the induction is short — Volume 11, Theorem 11.5.4. If no, expect a polychotomy.

(f) *A caution.* A polychotomy is only as good as the claim that its cases exhaust. That claim is the easiest thing to get wrong, and Remark 14.5.13 records it as the part of this argument this series presents in outline rather than in full.

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## Drills

**Drill 14.5.1.** Verify the cancellation in Lemma 14.5.5:  $(\rho/\delta)^{2\beta}\rho^{-2\beta} = \delta^{-2\beta}$ .

**Drill 14.5.2.** Calibrate Lemma 14.5.9 at  $|\tilde{\mathbb{T}}| \approx \delta^{-2}$  and confirm it reduces to (14.10.1).

**Drill 14.5.3.** Derive the trivial bound  $|U(\tilde{\mathbb{T}})| \gtrsim \delta^{2\beta}$  by random decomposition, and check the Frostman condition is inherited.

**Drill 14.5.4.** Explain why case (4) of Theorem 14.5.7 needs  $r$  close to 1, and hence why Definition 14.5.1 demands failure at all scales.

## Notes

This chapter follows Sections 12–14 of Guth’s Bourbaki survey, which streamline Wang and Zahl (2025). The strategic choice of scale ranges is due to Keleti and Shmerkin. Worked Example 14.5.2 is the example of Katz and Zahl (2019).



## Chapter 14.6

# Multi-Scale Self-Similarity

Proposition 14.3.9 reduced the sticky conjecture to  $\sigma_n = 0$ . Suppose it fails. Then there are tube families whose unions are as small as  $\sigma_n$  permits, and this chapter proves the structural fact everything else rests on: *such families look the same at every scale*. Precisely, the  $\rho$ -thickening of an extremal family is again extremal, and so is each family sitting inside a single  $\rho$ -tube after rescaling.

### 14.6.1 Extremal families

**Definition 14.6.1** ( $\varepsilon$ -extremal).  $(\mathbb{T}, Y)_\delta$  is  $\varepsilon$ -extremal if it satisfies (a), (b), (c) of Definition 14.3.5 with  $s = t = \varepsilon$ , and in addition

$$\left| \bigcup_{T \in \mathbb{T}} Y(T) \right| \leq \delta^{\sigma_n - \varepsilon}.$$

**Lemma 14.6.2** (Extremal families exist). *For all  $\varepsilon, \delta_0 > 0$  there is an  $\varepsilon$ -extremal  $(\mathbb{T}, Y)_\delta$  for some  $\delta \in (0, \delta_0]$ .*

*Proof.* By Lemma 14.3.6 choose  $s = t = \varepsilon$ ; then  $N(\varepsilon, \varepsilon) \geq \sigma_n$ , so by definition of lim sup there are arbitrarily small  $\delta$  with  $\log M(\varepsilon, \varepsilon, \delta) / \log \delta \geq \sigma_n - \varepsilon$ , that is  $M(\varepsilon, \varepsilon, \delta) \leq \delta^{\sigma_n - \varepsilon}$ . A family nearly attaining that infimum is  $\varepsilon$ -extremal.  $\square$

**Lemma 14.6.3** (Basic counts). *If  $(\mathbb{T}, Y)_\delta$  is  $\varepsilon$ -extremal then  $\delta^{1-n+\varepsilon} \lesssim \#\mathbb{T} \lesssim \delta^{1-n-\varepsilon}$ . If  $(\mathbb{T}', Y')_\delta$  is a sub-collection with  $\sum_T |Y'(T)| \geq \delta^{\varepsilon'}$  for some  $\varepsilon' \geq \varepsilon$ , then  $(\mathbb{T}', Y')_\delta$  is  $\varepsilon'$ -extremal.*

*Proof.* Each tube has volume  $\approx \delta^{n-1}$ , so  $\sum_T |Y(T)| \leq \#\mathbb{T} \cdot \delta^{n-1}$ ; with condition (c) this gives the lower bound on  $\#\mathbb{T}$ . The upper bound follows since the tubes are essentially distinct and  $\mathcal{L}_n$  has a  $\delta$ -net of size  $\approx \delta^{1-n}$  in each direction fibre, refined by condition (b).

For the second claim, conditions (a) and (b) pass to sub-collections automatically, and (c) is the hypothesis; the volume bound only improves.  $\square$

## 14.6.2 Unit rescaling

**Definition 14.6.4** (Unit rescaling). Let  $(\mathbb{T}, Y)_\delta$  be covered by a single  $\rho$ -tube  $\tilde{T}$  with coaxial line  $\ell$ . The *unit rescaling relative to  $\tilde{T}$*  is  $(\hat{\mathbb{T}}, \hat{Y})_{\delta/\rho}$ , where the coaxial lines are the images under

$$\phi = (\text{anisotropic dilation}) \circ (\text{rigid motion}),$$

the rigid motion sending  $\tilde{\ell} \cap \{x_n = 0\}$  to the origin and  $\tilde{\ell}$  to the  $x_n$ -axis, and the dilation being

$$(x_1, \dots, x_n) \mapsto (cx_1/\rho, \dots, cx_{n-1}/\rho, cx_n), \quad (14.6.1)$$

with  $c = c(n) \approx 1$  chosen so the images lie in  $\mathcal{L}_n$ . Each  $\hat{Y}(\hat{T})$  is the union of  $\delta/\rho$ -cubes meeting  $\phi(Y(T))$ .

**Lemma 14.6.5** (Properties of  $\phi$ ). *With  $\phi$  as above, for lines  $\ell_1, \ell_2$  with  $d(\tilde{\ell}, \ell_i) \leq \rho$ :*

$$\begin{aligned} \rho^{-1}d(\ell_1, \ell_2) &\lesssim d(\phi\ell_1, \phi\ell_2) \leq \rho^{-1}d(\ell_1, \ell_2), \\ \rho^{-1}\angle(\text{dir } \ell_1, \text{dir } \ell_2) &\lesssim \angle(\text{dir } \phi\ell_1, \text{dir } \phi\ell_2) \leq \rho^{-1}\angle(\text{dir } \ell_1, \text{dir } \ell_2). \end{aligned} \quad (14.6.2)$$

*If  $\tau \in [\delta, \rho]$  and  $T^\dagger$  is a  $\tau$ -tube covered by  $2\tilde{T}$ , then  $\phi(T^\dagger)$  lies in the  $\tau/\rho$ -tube about  $\phi(\ell^\dagger)$ . And for measurable  $X$ ,*

$$|\phi(X)| \approx \rho^{1-n}|X|. \quad (14.6.3)$$

*Proof.* The rigid motion preserves all three quantities. The dilation (14.6.1) scales the  $n - 1$  transverse directions by  $c/\rho$  and the axial direction by  $c$ ; since the lines in question make angle  $\lesssim \rho$  with  $\ell$ , their position and direction differences are transverse to leading order, giving (14.6.2). The Jacobian of (14.6.1) is  $c^n \rho^{1-n}$ , which is (14.6.3).  $\square$

**Where we are.** The dilation is *anisotropic*: it stretches the  $n - 1$  transverse directions by  $\rho^{-1}$  but leaves the axial direction alone. That is what turns a  $\rho$ -tube into a unit ball while turning the  $\delta$ -tubes inside it into  $\delta/\rho$ -tubes.

An isotropic dilation by  $\rho^{-1}$  would take the  $\rho$ -tube to a tube of length  $\rho^{-1}$ , not to the unit ball, and the resulting family would not be a Kakeya family at any scale. *The whole self-similarity phenomenon depends on this choice,*

and it is why the rescaled objects are compared with  $\varepsilon$ -extremal families at scale  $\delta/\rho$  rather than at scale  $\delta$ .

**Lemma 14.6.6** (Rescaling preserves extremality). *For all  $\varepsilon > 0$  there are  $\eta, \delta_0 > 0$  such that for  $\delta \in (0, \delta_0]$  the following holds. Let  $(\mathbb{T}, Y)_\delta$  be covered by a  $\rho$ -tube  $\tilde{T}$  with  $\rho \in [\delta^{1-\varepsilon}, 1]$ , satisfying (a) and (b) of Definition 14.3.5 with  $\eta$  in place of  $s$ , and  $\sum_T |Y(T)| \geq \delta^\eta \rho^{n-1}$ .*

*Then there is a refinement  $(\mathbb{T}', Y')_\delta$  whose unit rescaling relative to  $\tilde{T}$  satisfies (a), (b), (c) with  $\varepsilon$  in place of  $s$  and  $t$ , and*

$$\#\mathbb{T}'(p) \leq (\rho/\delta)^{\sigma_n + \varepsilon} \quad \text{for all } p \in \mathbb{R}^n. \quad (14.6.4)$$

*Proof. Step 1. Normalise.*

By (14.6.2) there is a refinement whose rescaling consists of essentially distinct tubes; by Lemma 14.3.3 refine further to constant multiplicity  $\mu$ , so  $\#\mathbb{T}'(p) = \mu$  on  $E_{\mathbb{T}'}$ . Let  $(\widehat{\mathbb{T}}, \widehat{Y})_{\delta/\rho}$  be the rescaling. Property (a) holds.

**Step 2. Property (c).**

By (14.6.3),

$$\sum_{\widehat{T}} |\widehat{Y}(\widehat{T})| \gtrsim \rho^{1-n} \sum_T |Y(T)| \geq \delta^\eta \geq (\rho/\delta)^{\eta/\varepsilon}, \quad (14.6.5)$$

using  $\rho \geq \delta^{1-\varepsilon}$ , so  $\delta \geq (\delta/\rho)^{1/\varepsilon}$ . Choosing  $\eta$  small makes this at least  $(\rho/\delta)^s$ .

**Step 3. Property (b).**

Let  $\tau \in (\delta/\rho, 1)$  and let  $\mathbb{T}^*$  be  $\tau\rho$ -tubes covering  $\mathbb{T}$ , at most  $\delta^{-\eta}$  essentially parallel to a common one. Each  $T^*$  may be assumed to cover a tube of  $\mathbb{T}$ , hence to be covered by  $2\tilde{T}$ . By Lemma 14.6.5 the images  $\phi(\ell^*)$  give  $\tau$ -tubes covering  $\widehat{\mathbb{T}}$ , and at most  $(\delta/\rho)^{-2\eta/\varepsilon}$  are essentially parallel to a common tube. Small  $\eta$  gives (b).

**Step 4. The multiplicity bound.**

Since  $(\widehat{\mathbb{T}}, \widehat{Y})_{\delta/\rho}$  satisfies (a), (b), (c) for the values  $s, t$  realising  $N(s, t) \leq \sigma_n + \varepsilon/4$ , and  $\delta/\rho \rightarrow 0$  as  $\delta \rightarrow 0$  by the hypothesis  $\rho \geq \delta^{1-\varepsilon}$ ,

$$\left| \bigcup_{\widehat{T}} \widehat{Y}(\widehat{T}) \right| \geq (\delta/\rho)^{\sigma_n + \varepsilon/2}. \quad (14.6.6)$$

On the other hand, by (c) with  $s \leq \varepsilon/2$ ,

$$\mu = \frac{\sum_{\widehat{T}} |\widehat{Y}(\widehat{T})|}{\left| \bigcup_{\widehat{T}} \widehat{Y}(\widehat{T}) \right|} \lesssim (\delta/\rho)^{\varepsilon/2} \left| \bigcup_{\widehat{T}} \widehat{Y}(\widehat{T}) \right|^{-1} \leq (\delta/\rho)^{\sigma_n + \varepsilon},$$

which is (14.6.4). □

### 14.6.3 The self-similarity theorem

**Theorem 14.6.7** (Extremal families are coarsely self-similar). *For all  $\varepsilon > 0$  there are  $\eta, \delta_0 > 0$  so that the following holds for  $\delta \in (0, \delta_0]$ . Let  $(\mathbb{T}, Y)_\delta$  be  $\eta$ -extremal and  $\rho \in [\delta^{1-\varepsilon}, \delta^\varepsilon]$ . Then there are a refinement  $(\mathbb{T}', Y')_\delta$  and a balanced cover  $(\tilde{\mathbb{T}}, \tilde{Y})_\rho$  with:*

- (i)  $(\tilde{\mathbb{T}}, \tilde{Y})_\rho$  is  $\varepsilon$ -extremal;
- (ii) for each  $\tilde{T}$ , the unit rescaling of  $(\mathbb{T}'[\tilde{T}], Y')_\delta$  relative to  $\tilde{T}$  is  $\varepsilon$ -extremal;
- (iii)  $\#\tilde{\mathbb{T}}(p) \leq \rho^{-\sigma_n - \varepsilon}$  for all  $p$ ;
- (iv)  $\#\mathbb{T}'[\tilde{T}](p) \leq (\delta/\rho)^{-\sigma_n - \varepsilon}$  for all  $p, \tilde{T}$ .

*Proof. Step 1. Constructing the coarse cover.*

Let  $\tilde{\mathbb{T}}_0$  be  $\rho/(2n)$ -tubes covering  $\mathbb{T}$ , each sharing a coaxial line with a tube of  $\mathbb{T}$ , with at most  $O(\delta^{-\eta})$  essentially parallel to a common tube; this is obtained from the cover supplied by condition (b) by replacing each tube by  $O(1)$  nearby ones.

Choose  $\tilde{\mathbb{T}}_1 \subseteq \tilde{\mathbb{T}}_0$  so that the tubes  $2n\tilde{\mathbb{T}}_1$  are essentially distinct, each  $T \in \mathbb{T}$  is covered by at most one of them, and  $\sum_{\tilde{T} \in \tilde{\mathbb{T}}_1} \sum_{T \in \mathbb{T}[\tilde{T}]} |Y(T)| \approx \sum_T |Y(T)|$ . Put  $\tilde{\mathbb{T}}_2 = 2n\tilde{\mathbb{T}}_1$ ; these are essentially distinct, giving (a).

For (b) with  $t = \eta/\varepsilon$ : for  $\tau \in [\rho, 1]$  the cover supplied by condition (b) for  $\mathbb{T}$  has at most  $\delta^{-\eta} \leq \rho^{-\eta/\varepsilon}$  tubes essentially parallel to a common one, and since each  $\tilde{T} \in \tilde{\mathbb{T}}_2$  shares a coaxial line with some  $T \in \mathbb{T}$ , that same cover covers  $\tilde{\mathbb{T}}_2$ .

Let  $\mathbb{T}_2$  be the tubes of  $\mathbb{T}$  covered by  $\tilde{\mathbb{T}}_1$ , with  $Y_2 = Y$ . Then each  $T \in \mathbb{T}_2$  is covered by *exactly* one tube of  $\tilde{\mathbb{T}}_2$ , and  $\frac{1}{2n}\tilde{T}$  also covers  $T$ .

**Step 2. Fine-scale structure.**

By dyadic pigeonholing choose  $\tilde{\mathbb{T}}_3 \subseteq \tilde{\mathbb{T}}_2$  and a refinement  $(\mathbb{T}_3, Y_3)_\delta$  with

$$\sum_{T \in \mathbb{T}_3[\tilde{T}]} |Y_3(T)| = (\#\tilde{\mathbb{T}}_3)^{-1} \sum_{T \in \mathbb{T}_3} |Y_3(T)| \quad \text{for each } \tilde{T} \in \tilde{\mathbb{T}}_3. \quad (14.6.7)$$

Since  $O(\delta^{-\eta})$  tubes of  $\tilde{\mathbb{T}}_3$  are essentially parallel to a common tube,  $\#\tilde{\mathbb{T}}_3 \leq \delta^{-\eta} \rho^{1-n}$ , whence  $\sum_{T \in \mathbb{T}_3[\tilde{T}]} |Y_3(T)| \gtrsim_\delta \delta^{2\eta} \rho^{n-1}$ .

Apply Lemma 14.6.6 to each  $(\mathbb{T}_3[\tilde{T}], Y_3)_\delta$  with  $\varepsilon^2/3$  in place of  $\varepsilon$ , and let  $(\mathbb{T}_4, Y_4)_\delta$  be the union of the outputs,  $\mathbb{T}_4 = \mathbb{T}_3$ . Then

$$\#\mathbb{T}_4[\tilde{T}](p) \leq (\rho/\delta)^{\sigma_n + \varepsilon^2/3}, \quad (14.6.8)$$

which is (iv), and each unit rescaling satisfies (a), (b), (c) with  $s = t = \varepsilon^2$ .

**Step 3.** *Coarse-scale structure: a shading on the fat tubes.*

First note that for  $\tilde{T} \in \tilde{\mathbb{T}}_4$  and a  $\rho$ -cube  $Q$ ,

$$Q \cap \bigcup_{T \in \mathbb{T}_4[\tilde{T}]} Y_4(T) \neq \emptyset \implies Q \subseteq \tilde{T}, \quad (14.6.9)$$

since  $T$  is covered by  $\frac{1}{2n}\tilde{T}$ , so  $Q$  meets  $\frac{1}{2n}\tilde{T}$  and hence lies in  $\tilde{T}$ .

Each  $\tilde{T}$  contains  $\approx \rho^{-1}$   $\rho$ -cubes. The directions of  $\mathbb{T}_4[\tilde{T}]$  lie in a  $\rho$ -disc, and at most  $\delta^{-\eta}$  tubes are essentially parallel, so

$$\#\mathbb{T}_4[\tilde{T}] \lesssim \delta^{-\eta}(\rho/\delta)^{n-1}, \quad \sum_{T \in \mathbb{T}_4[\tilde{T}]} |Y_4(T) \cap Q| \lesssim \delta^{-\eta} \rho^n. \quad (14.6.10)$$

Call  $Q$  *heavy* for  $\tilde{T}$  if  $\sum_{T \in \mathbb{T}_4[\tilde{T}]} |Y_4(T) \cap Q| \geq c_1 \rho \sum_{T \in \mathbb{T}_4[\tilde{T}]} |Y_4(T)|$ . For  $c_1 \approx 1$  suitable, the heavy cubes carry at least half the mass. Dyadic pigeonholing in the weight gives  $w$  and sets  $\mathcal{Q}_1(\tilde{T})$  of heavy cubes with  $w \leq \sum_T |Y_4(T) \cap Q| < 2w$  and

$$\sum_{\tilde{T}} \sum_{Q \in \mathcal{Q}_1(\tilde{T})} \sum_{T \in \mathbb{T}_4[\tilde{T}]} |Y_4(T) \cap Q| \gtrsim_{\delta} \sum_{T \in \mathbb{T}_4} |Y_4(T)|. \quad (14.6.11)$$

Comparing (14.6.10) and (14.6.11),  $\sum_{\tilde{T}} \#\mathcal{Q}_1(\tilde{T}) \gtrsim_{\delta} \delta^{2\eta} \rho^{-n}$ . Setting  $\tilde{Y}_4(\tilde{T}) = \bigcup_{Q \in \mathcal{Q}_1(\tilde{T})} Q$  — a legitimate shading by (14.6.9) — this reads

$$\sum_{\tilde{T} \in \tilde{\mathbb{T}}_4} |\tilde{Y}_4(\tilde{T})| \gtrsim_{\delta} \delta^{2\eta}, \quad (14.6.12)$$

so  $(\tilde{\mathbb{T}}_4, \tilde{Y}_4)_{\rho}$  satisfies (a), (b), (c) with  $t = \eta/\varepsilon$ ,  $s = 3\eta/\varepsilon$ .

**Step 4.** *Normalising multiplicities.*

Refine to  $(\mathbb{T}_5, Y_5)_{\delta}$  and  $(\tilde{\mathbb{T}}_5, \tilde{Y}_5)_{\rho}$  so that: there is  $\mu_{\text{fine}}$  with  $\#\mathbb{T}_5[\tilde{T}](p) = \mu_{\text{fine}}$  for all  $\tilde{T}$  and  $p \in E_{\mathbb{T}_5[\tilde{T}]}$ ; there is  $w$  with  $|Q \cap E_{\mathbb{T}_5[\tilde{T}]}| = w$  for every  $\rho$ -cube  $Q \subseteq \tilde{Y}_5(\tilde{T})$ ; there is  $\mu_{\text{coarse}}$  with  $\mu_{\text{coarse}} \leq \#\tilde{\mathbb{T}}_5(p) < 2\mu_{\text{coarse}}$ ; and  $(\tilde{\mathbb{T}}_5, \tilde{Y}_5)_{\rho}$  covers  $(\mathbb{T}_5, Y_5)_{\delta}$ . Each step refines a shading while preserving the previous ones.

Since  $(\tilde{\mathbb{T}}_5, \tilde{Y}_5)_{\rho}$  satisfies (a), (b), (c) with  $s = 4\eta/\varepsilon$ , choosing  $\eta, \delta_0$  small forces  $|E_{\tilde{\mathbb{T}}_5}| \geq \rho^{\sigma_n + \varepsilon/2}$  and hence

$$\mu_{\text{coarse}} \lesssim \rho^{-\sigma_n - \varepsilon/2 - \eta/\varepsilon}, \quad (14.6.13)$$

giving (iii) for  $\eta$  small.

**Step 5.** *Both multiplicities are large.*

By (14.6.8),  $\mu_{\text{fine}} \leq (\rho/\delta)^{\sigma_n + \varepsilon^2/3} \leq \rho^{-\varepsilon/3}(\rho/\delta)^{\sigma_n}$ . On the other hand  $\int \sum_{T \in \mathbb{T}_5} \mathbf{1}_{Y_5(T)} \gtrsim_{\delta} \delta^{\eta}$ , and since  $(\mathbb{T}, Y)_{\delta}$  is  $\eta$ -extremal the integrand is supported on a set of measure at most  $\delta^{\sigma_n - \eta}$  and bounded by  $\mu_{\text{fine}}\mu_{\text{coarse}}$ . Hence

$$\mu_{\text{fine}} \gtrsim_{\delta} \delta^{2\eta} \rho^{\varepsilon/2 + \eta/\varepsilon} (\rho/\delta)^{\sigma_n}, \quad \mu_{\text{coarse}} \gtrsim_{\delta} \delta^{2\eta} \rho^{-\sigma_n + \varepsilon/3}. \quad (14.6.14)$$

For  $\eta$  small this gives  $|E_{\tilde{\mathbb{T}}_5}| \leq \rho^{\sigma_n - \varepsilon}$  and, for each  $\tilde{T}$ ,  $|E_{\tilde{\mathbb{T}}}| \leq (\delta/\rho)^{\sigma_n - \varepsilon}$  for the unit rescaling — that is, (i) and (ii) up to condition (c).

**Step 6.** *Discarding the bad fat tubes.*

It may happen that a rescaling fails  $\sum_{\hat{T}} |\hat{Y}(\hat{T})| \geq (\rho/\delta)^{\varepsilon}$ . Let  $\tilde{\mathbb{T}}_6 \subseteq \tilde{\mathbb{T}}_5$  be those for which it holds, and put  $\mathbb{T}_6 = \bigcup_{\tilde{T} \in \tilde{\mathbb{T}}_6} \mathbb{T}_5[\tilde{T}]$  with the inherited shadings. Then  $(\mathbb{T}_6, Y_6)$  is a refinement of  $(\mathbb{T}, Y)$  and, with  $(\tilde{\mathbb{T}}_6, \tilde{Y}_6)_{\rho}$ , satisfies (i)–(iv).  $\square$

**Where we are.** The theorem says that at *every* intermediate scale  $\rho$ , an extremal family looks like two nested extremal families: a coarse one at scale  $\rho$ , and inside each of its tubes a fine one at scale  $\delta/\rho$ . Items (iii) and (iv) say the multiplicities factor as  $\rho^{-\sigma_n} \cdot (\delta/\rho)^{-\sigma_n} = \delta^{-\sigma_n}$ , which is consistent and is the arithmetic making the induction possible.

*Where the work is.* Steps 3 and 4. Constructing a shading on the coarse tubes requires knowing which  $\rho$ -cubes carry mass ((14.6.9) and the heavy-cube pigeonholing), and normalising the two multiplicities requires four successive refinements, each of which must preserve the previous ones. The refinement convention of Definition 14.3.2 is what makes this affordable.

*What is used later.* Only the statement, never the proof. Wang and Zahl note that Lemma 14.6.2 and Theorem 14.6.7 are the sole consequences of stickiness used in the rest of the argument — so any class ofakeya sets satisfying them inherits the whole theorem.

## 14.6.4 Two consequences

**Lemma 14.6.8** (Robust transversality). *For all  $\varepsilon > 0$  there are  $\eta, \delta_0 > 0$  so that for  $\delta \in (0, \delta_0]$ : if  $(\mathbb{T}, Y)_{\delta}$  is  $\eta$ -extremal then, after refinement, for all  $p \in \mathbb{R}^n$  and  $v \in \mathbb{S}^{n-1}$ ,*

$$\#\{T \in \mathbb{T}(p) : \angle(v, \text{dir } T) < \delta^{\varepsilon}\} \leq \delta^{-\sigma_n + \varepsilon\sigma_n/2}. \quad (14.6.15)$$

*Proof.* Apply Theorem 14.6.7 with  $\varepsilon\sigma_n/2$  in place of  $\varepsilon$  and  $\rho = \delta^{\varepsilon}$ . The tubes contributing to (14.6.15) have directions within  $\delta^{\varepsilon} = \rho$  of  $v$ , so they are covered by  $O(1)$  tubes of  $\tilde{\mathbb{T}}$ ; these are essentially distinct, and item (iii) bounds how many pass through  $p$ .  $\square$

**Where we are.** Lemma 14.6.8 says the tubes through a typical point cannot focus into a narrow cone: the directions are spread. This is the hypothesis that Chapter 14.9 feeds to multilinear Kakeya, and it is why the transversality condition of Volume 11, Theorem 11.5.7 is available at all.

Note the shape of the bound:  $\delta^{-\sigma_n}$  is the total multiplicity, and  $\delta^{\varepsilon\sigma_n/2}$  is the gain from restricting to a cone of aperture  $\delta^\varepsilon$ . If  $\sigma_n = 0$  the statement is vacuous — which is correct, since then there is nothing to prove.

**Lemma 14.6.9** (Extremality survives mild rescaling). *For all  $\varepsilon > 0$  there are  $\eta, \delta_0 > 0$  so that for  $\delta \in (0, \delta_0]$  the following holds. Let  $(\mathbb{T}, Y)_\delta$  be  $\eta$ -extremal, let  $Q \subseteq [-1, 1]^n$  be an axis-parallel prism with  $\sum_T |Y(T) \cap Q| \geq \delta^\eta$ , and let  $\varphi$  be a translation composed with  $(x_1, \dots, x_n) \mapsto (r_1 x_1, \dots, r_n x_n)$  where  $\delta^\eta \leq r_i \leq \delta^{-\eta}$ , with  $\varphi(Q) \subseteq [-1, 1]^n$ .*

*Then the image of  $(\mathbb{T}, Y \cap Q)_\delta$  under  $\varphi$  is  $\varepsilon$ -extremal: there is an  $\varepsilon$ -extremal  $(\tilde{\mathbb{T}}, \tilde{Y})_\rho$  for some  $\rho \in [\delta, \delta^{1-\eta}]$  whose coaxial lines are the  $\varphi(\ell)$  and whose shadings are the  $\rho$ -cubes meeting  $\varphi(Y(T) \cap Q)$ .*

*Proof.* The map  $\varphi$  distorts lengths by factors in  $[\delta^\eta, \delta^{-\eta}]$ , hence distorts  $d(\cdot, \cdot)$  on  $\mathcal{L}_n$  and all volumes by at most  $\delta^{-n\eta}$ . Set  $\rho = \delta \max_i r_i^{-1} \in [\delta, \delta^{1-\eta}]$ , so that  $\varphi$  carries  $\delta$ -tubes into  $\rho$ -tubes.

Condition (a) survives after discarding essentially identical images, at a cost of a bounded factor. Condition (b) survives because a cover of  $\mathbb{T}$  by  $\tau$ -tubes maps to a cover by  $O(\tau/\rho \cdot \delta)$ -tubes with the parallel count multiplied by  $O(\delta^{-n\eta})$ . Condition (c) holds since  $\sum_T |Y(T) \cap Q| \geq \delta^\eta$  and volumes distort by at most  $\delta^{-n\eta}$ . The volume upper bound of Definition 14.6.1 likewise survives with  $\sigma_n - \varepsilon$  in place of  $\sigma_n - \eta$ .

Choosing  $\eta$  small in terms of  $\varepsilon$  and  $n$  makes every accumulated factor  $\delta^{-Cn\eta}$  at most  $\delta^{-\varepsilon/2}$ , giving  $\varepsilon$ -extremality.  $\square$

*Remark 14.6.10* (On this proof). Wang and Zahl state Lemma 14.6.9 and omit the proof as “straightforward though tedious”. The argument above is the natural one: track how  $\varphi$  distorts each of the three conditions and the volume bound, and choose  $\eta$  so the accumulated distortion is absorbed. It is recorded here rather than omitted, in keeping with this series’ standard.

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#### WORKED EXAMPLE 14.6.1— TIER A **The self-similarity, in pictures**

Take  $n = 3$ ,  $\delta = 10^{-8}$ ,  $\rho = 10^{-4}$ , so  $\delta/\rho = 10^{-4}$ .

*Before.*  $\approx \delta^{-2} = 10^{16}$  tubes of dimensions  $1 \times 10^{-8}$ , occupying volume  $\approx \delta^{\sigma_3}$ .

*The coarse family.*  $\approx \rho^{-2} = 10^8$  tubes of dimensions  $1 \times 10^{-4}$ . By (i) this is again extremal, so it occupies volume  $\approx \rho^{\sigma_3}$ .

*Inside one coarse tube.*  $\approx (\rho/\delta)^2 = 10^8$  thin tubes. After the anisotropic rescaling (14.6.1) these become  $1 \times 10^{-4}$  tubes in the unit ball, and by (ii) that family is extremal too.

*The multiplicity check.* By (iii) and (iv),

$$\#\mathbb{T}(p) \leq \rho^{-\sigma_3} \cdot (\delta/\rho)^{-\sigma_3} = \delta^{-\sigma_3} \checkmark$$

So the two scales' multiplicities multiply to the right total — which is the consistency check to run on any claimed self-similarity statement.

*What would break without the anisotropy.* An isotropic rescaling of the coarse tube by  $\rho^{-1}$  would give an object of length  $10^4$ , and the  $10^8$  thin tubes inside would be  $1 \times 1$  — not tubes at all. The statement would be vacuous.

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#### WORKED EXAMPLE 14.6.2— TIER B **Reading a multi-scale argument**

This chapter is the densest bookkeeping in the series. Five things to track.

(1) *Which scale is which.*  $\delta$  fine,  $\rho$  intermediate,  $\delta/\rho$  the rescaled fine scale. Every estimate lives at exactly one of them.

(2) *What survives a refinement.* By Definition 14.3.2, only  $\sum_T |Y(T)|$  up to  $(\log \frac{1}{\delta})^{-C}$ . Any quantity not expressed in terms of it must be re-established after each refinement — which is why Step 4 re-normalises the multiplicities.

(3) *The order of the  $\varepsilon$ 's.* Throughout,  $\eta$  is chosen after  $\varepsilon$ , and  $\delta_0$  after both. Reading them in the wrong order makes the argument look circular; Volume 11, Theorem 11.5.4 makes the same point.

(4) *Where stickiness enters.* Only through condition (b) of Definition 14.3.5, and only in Proposition 14.3.9. This chapter uses extremality, not stickiness.

(5) *What is actually exported.* Theorem 14.6.7 items (i)–(iv), Lemma 14.6.8, and Lemma 14.6.9. Nothing else from this chapter is used again.

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## Drills

**Drill 14.6.1.** Verify (14.6.3): compute the Jacobian of (14.6.1) and confirm the exponent  $1 - n$ .

**Drill 14.6.2.** In Step 2 of Lemma 14.6.6, check the inequality  $\delta^\eta \geq (\rho/\delta)^{\eta/\varepsilon}$  using  $\rho \geq \delta^{1-\varepsilon}$ .

**Drill 14.6.3.** Confirm the multiplicity consistency check of Worked Example 14.6.1 for  $n = 4$ ,  $\delta = 10^{-12}$ ,  $\rho = 10^{-4}$ .



**Drill 14.6.4.** Explain why Lemma 14.6.8 is vacuous when  $\sigma_n = 0$ , and why that is the expected behaviour.

## Notes

This chapter is Section 3 of Wang and Zahl. Remark 14.6.10 records that the proof of Lemma 14.6.9 is supplied here rather than omitted. The observation that Lemma 14.6.2 and Theorem 14.6.7 are the only consequences of stickiness used later is theirs, and is Remark 3.1 of their paper.



# Chapter 14.7

## Planiness and Graininess

From here we take  $n = 3$  and write  $\sigma = \sigma_3$ . The goal is  $\sigma = 0$ .

This chapter extracts two structures from an extremal family. *Planiness*: the tubes through each point nearly lie in a plane, and the plane varies Lipschitz-continuously. *Graininess*: consequently the set decomposes into parallel slabs — *grains* — whose spacing is Ahlfors-regular. Both were conjectured by Katz and Tao; the tool that produces them is multilinear Keakeya, proved in Volume 11.

### 14.7.1 Statements

**Definition 14.7.1** (Plane map). A *plane map* for  $(\mathbb{T}, Y)_\delta$  is  $V : E_{\mathbb{T}} \rightarrow \mathbb{S}^2$  with

$$|\text{dir}(T) \cdot V(p)| \leq \delta \quad \text{for all } p \in E_{\mathbb{T}}, T \in \mathbb{T}(p).$$

**Definition 14.7.2** (AD-sets). For  $\alpha \in (0, n]$ ,  $C \geq 1$ ,  $\delta > 0$ , a set  $E \subseteq \mathbb{R}^n$  is a  $(\delta, \alpha, C)_n$ -ADset if for all  $x$ , all  $\rho \geq \delta$  and all  $r \geq \rho$ ,

$$\mathcal{E}_\rho(E \cap B(x, r)) \leq C(r/\rho)^\alpha, \tag{14.7.1}$$

where  $\mathcal{E}_\rho$  denotes the  $\rho$ -covering number.

**Where we are.** Definition 14.7.2 imposes only the *upper* bound of Ahlfors–David regularity, so every subset of an ADset is an ADset — a fact used constantly below. If in addition  $\mathcal{E}_\delta(E) \approx \delta^{-\alpha}$  and  $E$  has bounded diameter, a large subset satisfies the matching lower bound, so nothing essential is lost.

The relation to Volume 12, Definition 12.7.1: a Frostman measure gives an upper bound on measure of balls; (14.7.1) gives an upper bound on covering numbers, at every pair of scales. The second is the discretised form of the

first.

**Proposition 14.7.3** (Grains). *For all  $\varepsilon, \delta_0 > 0$  there are  $\delta \in (0, \delta_0]$  and an  $\varepsilon$ -extremal  $(\mathbb{T}, Y)_\delta$  with:*

1. Global grains with Lipschitz slope. *There is a 1-Lipschitz  $f : [-1, 1] \rightarrow \mathbb{R}$  so that  $(E_\mathbb{T} \cap \{z = z_0\}) \cdot (1, f(z_0), 0)$  is a  $(\delta, 1 - \sigma, \delta^{-\varepsilon})_1$ -ADset for each  $z_0$ .*
2. Local grains with Lipschitz plane map.  *$(\mathbb{T}, Y)_\delta$  has a 1-Lipschitz plane map  $V$ , and for all  $\rho \in [\delta, 1]$  and all  $p$ ,  $V(p) \cdot (B(p, \rho^{1/2}) \cap E_\mathbb{T})$  is a  $(\rho, 1 - \sigma, \delta^{-\varepsilon})_1$ -ADset.*

**Where we are.** Unpacking the geometry. Item 1 says each horizontal slice  $E_\mathbb{T} \cap \{z = z_0\}$  lies in at most  $\delta^{\sigma-1-\varepsilon}$  rectangles of dimensions  $\approx 1 \times \delta$ , all with normal  $(1, f(z_0), 0)$  — so by Fubini  $|E_\mathbb{T}| \lesssim \delta^{\sigma-\varepsilon}$ , nearly matching the extremality bound  $|E_\mathbb{T}| \geq \delta^{\sigma+\varepsilon}$ .

Item 2 says each ball  $B(p, \rho^{1/2})$  meets  $\mathcal{N}_\rho(E_\mathbb{T})$  in at most  $\rho^{\sigma/2-1/2}$  parallel slabs of dimensions  $\rho^{1/2} \times \rho^{1/2} \times \rho$ . These are the *grains*. Covering  $E_\mathbb{T}$  by  $\delta^{-O(\varepsilon)} \rho^{\sigma/2-3/2}$  such balls gives  $|\mathcal{N}_\rho(E_\mathbb{T})| \lesssim \delta^{-O(\varepsilon)} \rho^\sigma$ , again nearly sharp.

*Both containments are nearly equalities*, which is what makes them usable: an extremal set is not merely contained in a union of grains, it essentially *is* one.

**Theorem 14.7.4** (Multilinear Kakeya, weighted form). *For a family  $\mathbb{T}$  of  $\delta$ -tubes in  $\mathbb{R}^3$ ,*

$$\int \left( \sum_{T_1, T_2, T_3 \in \mathbb{T}} \mathbf{1}_{T_1} \mathbf{1}_{T_2} \mathbf{1}_{T_3} |v_1 \wedge v_2 \wedge v_3| \right)^{1/2} \lesssim (\delta^2 \# \mathbb{T})^{3/2}, \quad (14.7.2)$$

where  $v_i = \text{dir}(T_i)$ .

*Proof.* Partition the direction sphere into caps of radius  $\nu$  for each dyadic  $\nu \in [\delta, 1]$ , and split the triple sum according to the least  $\nu$  for which  $v_1, v_2, v_3$  lie in caps with  $|v_1 \wedge v_2 \wedge v_3| \approx \nu$ . For each such triple of caps, the corresponding sub-families are  $\nu$ -transverse in the sense of Volume 11, Theorem 11.5.7, which gives

$$\int \prod_{j=1}^3 \left( \sum_{T \in \mathbb{T}_j} \mathbf{1}_T \right)^{1/2} \lesssim_\varepsilon \text{Poly}(\nu^{-1}) \delta^{-\varepsilon} \prod_{j=1}^3 (\delta^2 \# \mathbb{T}_j)^{1/2}.$$

Multiplying by the weight  $\nu^{1/2}$  from  $|v_1 \wedge v_2 \wedge v_3|^{1/2}$  and summing the resulting geometric series over dyadic  $\nu$  absorbs the  $\text{Poly}(\nu^{-1})$  and, with the  $\varepsilon$  chosen small, the  $\delta^{-\varepsilon}$ .  $\square$

**Where we are.** Theorem 14.7.4 is the form Wang and Zahl use, and it is a corollary of the transverse-families statement *proved in Volume 11*. Had Theorem 11.5.7 remained an import, this chapter — and hence the whole theorem — would rest on a black box.

The weight  $|v_1 \wedge v_2 \wedge v_3|$  is what makes the statement usable without a transversality hypothesis: degenerate triples contribute little because the weight is small, rather than being excluded by fiat. That is precisely the reformulation Volume 12, Worked Example 12.2.4 identified as the way past a hypothesis that concedes the difficulty.

## 14.7.2 Finding a plane map

**Lemma 14.7.5** (Weak plane map). *There are  $\eta, \delta_0 > 0$  so that for  $\delta \in (0, \delta_0]$ : if  $(\mathbb{T}, Y)_\delta$  is  $\eta$ -extremal, there are a refinement  $(\mathbb{T}', Y')_\delta$  and  $V : E_{\mathbb{T}} \rightarrow \mathbb{S}^2$  with*

$$|V(p) \cdot \text{dir}(T)| \leq \delta^{\sigma/2} \quad \text{for all } p \in E_{\mathbb{T}'}, T \in \mathbb{T}'(p). \quad (14.7.3)$$

*Proof.* Put  $\varepsilon = \sigma/4$  and take  $\eta \leq \min(\varepsilon/40, \varepsilon\sigma/24)$ .

**Step 1.** *Spread directions.*

By Lemma 14.6.8 with  $\varepsilon/4$  there is a refinement  $(\mathbb{T}_1, Y_1)_\delta$  with

$$\#\{T \in \mathbb{T}_1(p) : \angle(v, \text{dir } T) \leq \delta^{\varepsilon/4}\} \leq \delta^{-\sigma+3\eta} \quad (p \in \mathbb{R}^3, v \in \mathbb{S}^2). \quad (14.7.4)$$

Pass to a constant multiplicity refinement  $(\mathbb{T}_2, Y_2)$  with  $\delta^{-\sigma+2\eta} \lesssim \mu \lesssim \delta^{-\sigma-2\eta}$ .

**Step 2.** *Apply multilinear Kakeya.*

By Theorem 14.7.4 and  $\#\mathbb{T}_2 \lesssim \delta^{-2-\eta}$ ,

$$\int \left( \sum_{T_1, T_2, T_3 \in \mathbb{T}_2} \mathbf{1}_{Y_2(T_1)} \mathbf{1}_{Y_2(T_2)} \mathbf{1}_{Y_2(T_3)} |v_1 \wedge v_2 \wedge v_3| \right)^{1/2} \lesssim \delta^{-(3/2)\eta}. \quad (14.7.5)$$

**Step 3.** *Most points are narrow.*

Call  $p$  *broad* if  $\#\{(T_1, T_2, T_3) \in \mathbb{T}_2(p)^3 : |v_1 \wedge v_2 \wedge v_3| \geq \delta^{\sigma-10\eta}\} \geq \delta^{-3\sigma+3\eta}$ , and *narrow* otherwise. A broad point contributes at least  $(\delta^{-3\sigma+3\eta} \delta^{\sigma-10\eta})^{1/2} = \delta^{-\sigma-(7/2)\eta}$  to the integrand of (14.7.5), so the broad set has measure  $\lesssim \delta^{\sigma+2\eta}$ . Since  $|E_{\mathbb{T}_2}| \gtrsim \delta^{\sigma+\eta}$ , restricting the shadings to narrow points loses at most half the measure; call the result  $(\mathbb{T}_3, Y_3)$ .

**Step 4.** *Extract two spanning tubes.*

For narrow  $p$ ,

$$\#\{(T_1, T_2, T_3) \in \mathbb{T}_3(p)^3 : |v_1 \wedge v_2 \wedge v_3| < \delta^{\sigma-10\eta}\} \geq \mu^3 - \delta^{-3\sigma+3\eta} \geq \frac{1}{2}\mu^3. \quad (14.7.6)$$

By (14.7.4), for  $i \neq j$ ,

$$\#\{(T_1, T_2, T_3) \in \mathbb{T}_3(p)^3 : |v_i \wedge v_j| < \delta^{\varepsilon/4}\} \leq \mu^2 \delta^{-\sigma+3\eta} \leq \frac{1}{100} \mu^3. \quad (14.7.7)$$

Comparing, at least  $\mu^3/4$  triples satisfy both  $|v_1 \wedge v_2 \wedge v_3| < \delta^{\sigma-10\eta}$  and  $|v_i \wedge v_j| \geq \delta^{\varepsilon/4}$  for all  $i \neq j$ . Pigeonholing gives  $T_1, T_2$  admitting  $\geq \mu/4$  such  $T_3$ . Set

$$V(p) = \frac{\text{dir}(T_1) \times \text{dir}(T_2)}{|\text{dir}(T_1) \times \text{dir}(T_2)|}.$$

**Step 5. Verify.**

For each such  $T_3$ ,

$$|V(p) \cdot v(T_3)| = \frac{|\text{dir}(T_1) \wedge \text{dir}(T_2) \wedge \text{dir}(T_3)|}{|\text{dir}(T_1) \times \text{dir}(T_2)|} \leq \frac{\delta^{\sigma-10\eta}}{\delta^{\varepsilon/4}} \leq \delta^{\sigma/2}. \quad (14.7.8)$$

Taking  $\mathbb{T}' = \mathbb{T}_3$  and  $Y'(T) = \{p \in Y_3(T) : |V(p) \cdot v(T)| \leq \delta^{\sigma/2}\}$  gives the lemma.  $\square$

**Where we are.** This is the crux of planiness, and the shape is worth naming: *multilinearakeya says aakeya set cannot have many genuinely three-dimensional triples of tubes through a typical point*. Hence most triples are nearly coplanar, and after discarding the nearly-parallel pairs — which (14.7.4) bounds — a definite proportion of tubes through  $p$  lie near one plane.

*Where extremality is used.* Twice: through Lemma 14.6.8 in Step 1, and through the lower bound on  $|E_{\mathbb{T}_2}|$  in Step 3. Both come from Chapter 14.6.

*Why  $\delta^{\sigma/2}$  and not  $\delta$ .* The plane map produced here is only *weakly* accurate. Sharpening it to Definition 14.7.1 requires passing to a coarser scale, which is Lemma 14.7.6.

**Lemma 14.7.6** (Strong plane map at a coarser scale). *For all  $\varepsilon > 0$  there are  $\eta, \delta_0 > 0$  so that for  $\delta \in (0, \delta_0]$ : let  $(\mathbb{T}, Y)_\delta$  be  $\eta$ -extremal with  $V : E_{\mathbb{T}} \rightarrow \mathbb{S}^2$  satisfying  $|V(p) \cdot v(T)| \leq \rho/2$  for all  $p, T \in \mathbb{T}(p)$ , where  $\rho \in [\delta^{1-\varepsilon}, \delta^\varepsilon]$ .*

*Then there are a refinement  $(\mathbb{T}', Y')_\delta$  and an  $\varepsilon$ -extremal  $(\tilde{\mathbb{T}}, \tilde{Y})_\rho$  covering it, with plane map  $\tilde{V}$  constant on  $\rho$ -cubes and consistent with  $V$ : for each  $\rho$ -cube  $Q$  there is  $p \in Q$  with  $\tilde{V}(Q) = V(p)$ .*

*Proof.* Apply Theorem 14.6.7 with  $\varepsilon_1$  small in place of  $\varepsilon$ , obtaining  $(\mathbb{T}_1, Y_1)_\delta$  and an  $\varepsilon_1$ -extremal  $(\tilde{\mathbb{T}}_1, \tilde{Y}_1)_\rho$ . Refine so that each  $\rho$ -cube  $Q \subseteq E_{\tilde{\mathbb{T}}_1}$  carries a point  $p_Q$  with  $\#\mathbb{T}_1(p_Q) \gtrsim \delta^{-\sigma+2\eta}$ .

Say  $\tilde{T}$  is associated to  $Q$  if  $p_Q \in \bigcup_{T \in \mathbb{T}_1[\tilde{T}]} Y_1(T)$ ; then  $Q \subseteq \tilde{Y}_1(\tilde{T})$ . By item (iv) of Theorem 14.6.7,

$$\#\{\tilde{T} \text{ associated to } Q\} \gtrsim \frac{\delta^{-\sigma+2\eta}}{(\rho/\delta)^{\sigma+\varepsilon_1}} \gtrsim \delta^{2\eta+\varepsilon_1} \rho^{-\sigma},$$

while item (iii) bounds the number of  $\tilde{T}$  whose shading contains  $Q$  by  $\rho^{-\sigma-\varepsilon_1}$ . Defining  $\tilde{Y}_2(\tilde{T})$  as the union of  $\rho$ -cubes associated to  $\tilde{T}$ ,

$$\sum_{\tilde{T}} |\tilde{Y}_2(\tilde{T})| \gtrsim \delta^{2\eta+2\varepsilon_1} \sum_{\tilde{T}} |\tilde{Y}_1(\tilde{T})| \gtrsim \delta^{2\eta+3\varepsilon_1} \geq \rho^\varepsilon \quad (14.7.9)$$

for  $\varepsilon_1$  small.

Define  $\tilde{V}(p) = V(p_Q)$  for  $p \in Q$ . For  $p \in Q$  and  $\tilde{T} \in \tilde{\mathbb{T}}_2(p)$  pick  $T \in \mathbb{T}_2[\tilde{T}] \cap \mathbb{T}(p)$ ; then

$$|\tilde{V}(p) \cdot \text{dir}(\tilde{T})| \leq |V(p_Q) \cdot \text{dir}(T)| + |\text{dir}(\tilde{T}) - \text{dir}(T)| \leq \rho/2 + \rho/2 = \rho,$$

using  $|\text{dir}(\tilde{T}) - \text{dir}(T)| \leq \rho/2$  since  $\tilde{T}$  covers  $T$ . So  $\tilde{V}$  is a plane map for  $(\tilde{\mathbb{T}}_2, \tilde{Y}_2)_\rho$ .  $\square$

### 14.7.3 The plane map is Lipschitz

**Lemma 14.7.7** (Lipschitz at one scale). *For all  $\varepsilon > 0$  there are  $\eta, \delta_0 > 0$  so that for  $\delta \in (0, \delta_0]$ : if  $(\mathbb{T}, Y)_\delta$  is  $\eta$ -extremal with plane map  $V$  and  $\rho \in [\delta^{1-\varepsilon}, \delta^\varepsilon]$ , there is an  $\varepsilon$ -extremal sub-collection  $(\mathbb{T}', Y')_\delta$  so that  $p, q \in E_{\mathbb{T}'}$  in a common  $\rho$ -cube satisfy  $|V(p) - V(q)| \leq \rho$ .*

*Proof. Step 1. Two transverse fat tubes per cube.*

Apply Lemma 14.6.8 with  $\varepsilon_1$ , then Theorem 14.6.7 with  $\varepsilon_2$ , obtaining  $(\mathbb{T}_2, Y_2)_\delta$  of constant multiplicity  $\mu \gtrsim \delta^{-\sigma+2\eta}$  and an  $\varepsilon_2$ -extremal balanced cover  $(\mathbb{T}, \tilde{Y})_\rho$ .

For each  $p \in E_{\mathbb{T}_2}$  there are  $\geq \mu^2/2$  pairs  $T, T' \in \mathbb{T}_2(p)$  with  $\angle(\text{dir } T, \text{dir } T') \geq \delta^{\varepsilon_1}$ . By item (iv) of Theorem 14.6.7 these come from  $\gtrsim \delta^{4\eta+2\varepsilon_2} \rho^{-2\sigma}$  pairs  $\tilde{T}, \tilde{T}'$  with  $\angle(\text{dir } \tilde{T}, \text{dir } \tilde{T}') \geq \delta^{\varepsilon_1}/2$ . Hence for each  $\rho$ -cube  $Q \subseteq E_{\tilde{\mathbb{T}}}$ ,

$$\#\{(p, \tilde{T}, \tilde{T}')\} \gtrsim \delta^{4\eta+4\varepsilon_2} |Q \cap E_{\mathbb{T}_2}| (\#\tilde{\mathbb{T}}(Q))^2, \quad (14.7.10)$$

using item (iii). Pigeonholing gives  $\tilde{T}_Q, \tilde{T}'_Q$  with  $\angle \geq \delta^{\varepsilon_1}/2$  and

$$\left| Q \cap \left( \bigcup_{T \in \mathbb{T}_2[\tilde{T}_Q]} Y_2(T) \right) \cap \left( \bigcup_{T \in \mathbb{T}_2[\tilde{T}'_Q]} Y_2(T) \right) \right| \gtrsim \delta^{4\eta+4\varepsilon_2} |Q \cap E_{\mathbb{T}_2}|. \quad (14.7.11)$$

**Step 2.**  *$V$  is pinned on each cube.*

Restrict the shadings to the sets in (14.7.11). For  $p$  in such a set there are  $T_p \in \mathbb{T}_3[\tilde{T}_Q]$  and  $T'_p \in \mathbb{T}_3[\tilde{T}'_Q]$  through  $p$ , so

$$|V(p) \cdot \text{dir}(\tilde{T}_Q)| \leq |V(p) \cdot \text{dir}(T_p)| + |V(p) \cdot (\text{dir } \tilde{T}_Q - \text{dir } T_p)| \leq \delta + \rho,$$

and likewise for  $\tilde{T}'_Q$ . Since the two fat tubes make angle  $\gtrsim \delta^{\varepsilon_1}$ ,

$$\angle\left(V(p), \frac{\text{dir}(\tilde{T}_Q) \times \text{dir}(\tilde{T}'_Q)}{|\text{dir}(\tilde{T}_Q) \times \text{dir}(\tilde{T}'_Q)|}\right) \lesssim \delta^{-\varepsilon_1} \rho \quad \text{for all } p \in Q. \quad (14.7.12)$$

**Step 3.** Remove the  $\delta^{-\varepsilon_1}$ .

(14.7.12) gives oscillation  $\delta^{-\varepsilon_1} \rho$  rather than  $\rho$ . Pigeonhole: partition the  $\delta^{-\varepsilon_1} \rho$ -ball of directions into  $O(\delta^{-2\varepsilon_1})$  balls of radius  $\rho$  and keep, for each  $Q$ , the one carrying the most shading mass. This costs a factor  $\delta^{2\varepsilon_1}$  and yields  $|V(p) - V(q)| \leq \rho$  for  $p, q \in Q$ . Choosing  $\varepsilon_1, \varepsilon_2, \eta$  small makes the result  $\varepsilon$ -extremal.  $\square$

**Corollary 14.7.8.** *Under the hypotheses of Lemma 14.7.7 there is an  $\varepsilon$ -extremal sub-collection with  $|V(p) - V(q)| \leq 2\rho$  whenever  $|p - q| \leq \rho$ .*

*Proof.* Apply Lemma 14.7.7 and delete a constant fraction of the  $\rho$ -cubes so the survivors are  $2\rho$ -separated; then  $p, q$  with  $|p - q| \leq \rho$  lie in the same or adjacent cubes.  $\square$

**Lemma 14.7.9** (Lipschitz at all scales). *For all  $\varepsilon > 0$  there are  $\eta, \delta_0 > 0$  so that for  $\delta \in (0, \delta_0]$ : if  $(\mathbb{T}, Y)_\delta$  is  $\eta$ -extremal with plane map  $V$  constant on  $\delta$ -cubes, there is an  $\varepsilon$ -extremal sub-collection on which  $V$  is  $\delta^{-\varepsilon}$ -Lipschitz.*

*Proof.* Let  $N = 2/\varepsilon$ . Apply Corollary 14.7.8 successively at  $\rho_j = \delta^{j/N}$ ,  $j = 1, \dots, N-1$ , each time to the output of the previous step; for  $\eta, \delta_0$  small the final collection is  $\varepsilon$ -extremal.

Let  $p, q \in E_{\mathbb{T}^N}$ . If  $|p - q| \leq \delta$  then  $V(p) = V(q)$ . If  $|p - q| > \delta^{1/N}$  then  $|V(p) - V(q)| \leq 2 \leq 2\delta^{-1/N} |p - q|$ . Otherwise  $\delta^{(k+1)/N} < |p - q| \leq \delta^{k/N}$  for some  $1 \leq k \leq N-1$ , and

$$|V(p) - V(q)| \leq \delta^{k/N} \leq \delta^{-1/N} |p - q|. \quad \square$$

**Lemma 14.7.10.** *For all  $\varepsilon, \delta_0 > 0$  there are  $\delta \in (0, \delta_0]$  and an  $\varepsilon$ -extremal  $(\mathbb{T}, Y)_\delta$  with a  $\delta^{-\varepsilon}$ -Lipschitz plane map.*

*Proof.* Combine Lemma 14.6.2, Lemma 14.7.5, Lemma 14.7.6 and Lemma 14.7.9.  $\square$

## 14.7.4 Grains

**Lemma 14.7.11** (Counting grains in a cube). *For all  $\varepsilon > 0$  there are  $\eta, \delta_0 > 0$  so that for  $\delta \in (0, \delta_0]$ : if  $(\mathbb{T}, Y)_\delta$  is  $\eta$ -extremal with  $\delta^{-\eta}$ -Lipschitz plane map  $V$ , and  $\rho \in [\delta^{1-\varepsilon}, \delta^\varepsilon]$ ,  $\tau \in [\rho^{1-\varepsilon}, \rho^{1/2}]$ , then there is an  $\varepsilon$ -extremal sub-collection with*

$$\mathcal{E}_\rho(V(Q) \cdot (E_{\mathbb{T}^N} \cap Q)) \leq \rho^{-\varepsilon} (\tau/\rho)^{1-\sigma} \quad \text{for every } \tau\text{-cube } Q. \quad (14.7.13)$$



*Proof. Step 1. Set-up.*

Apply Theorem 14.6.7 twice — at  $\rho$  then at  $\tau$  — to obtain  $(\mathbb{T}_1, Y_1)_\delta$  covered by an  $\varepsilon_1$ -extremal  $(\tilde{\mathbb{T}}, \tilde{Y})_\rho$ , itself covered by an  $\varepsilon_2$ -extremal family at scale  $\tau$ . Apply Lemma 14.6.8 to  $(\tilde{\mathbb{T}}, \tilde{Y})_\rho$  with  $\varepsilon_3$ , and refine so that each  $\tau$ -cube  $Q$  meeting  $\tilde{Y}(\tilde{T})$  has  $|Q \cap \tilde{Y}(\tilde{T})| \geq \rho^{2+2\varepsilon_1}\tau$ .

**Step 2.** *A Córdoba-type  $L^2$  estimate inside  $Q$ .*

Fix  $\tilde{T}$  and such a  $Q$ . Choose  $\rho^{1-\varepsilon_3}$ -separated points  $p_1, \dots, p_k \in Q \cap \tilde{Y}(\tilde{T})$  with  $k \gtrsim \rho^{2\varepsilon_1+\varepsilon_3}\rho^{-1}\tau$ . Each  $p_i$  lies on  $\mu \gtrsim \rho^{-\sigma+\varepsilon_1}$  tubes of  $\tilde{\mathbb{T}}$ , and by robust transversality one of them,  $\tilde{T}_i$ , makes angle  $\geq \rho^{\varepsilon_3}$  with  $\tilde{T}$ . If  $\tilde{T}_i, \tilde{T}_j$  meet inside  $Q$  then  $\angle(v(\tilde{T}_i), v(\tilde{T}_j)) \gtrsim \rho^{1-\varepsilon_3}|i-j|/\tau$ , so

$$|Q \cap \tilde{T}_i \cap \tilde{T}_j| \lesssim \frac{\rho^3}{\angle(v_i, v_j) + \rho} \lesssim \frac{\rho^{2+\varepsilon_3}\tau}{|i-j|}.$$

Hence by Cauchy–Schwarz,

$$\left| Q \cap \bigcup_{i=1}^k \tilde{Y}(\tilde{T}_i) \right| \geq \frac{(\sum_i |Q \cap \tilde{Y}(\tilde{T}_i)|)^2}{\|\sum_i \mathbf{1}_{\tilde{T}_i}\|_{L^2(Q)}^2} \gtrsim \frac{k^2(\rho^{2\varepsilon_1+2}\tau)^2}{k\rho^{2+\varepsilon_3}\tau} \geq \rho^{6\varepsilon_1+\varepsilon_3}\tau^2\rho. \quad (14.7.14)$$

**Step 3.** *The union lies in a slab.*

Since  $V$  is  $\delta^{-\eta}$ -Lipschitz and  $\text{diam } Q \leq 2\tau$ , for  $q \in Q \cap \tilde{Y}(\tilde{T})$ ,

$$Q \cap \bigcup_{i=1}^k \tilde{T}_i \subseteq Q \cap \mathcal{N}_{4\delta^{-\eta}\rho}(q + V(Q)^\perp),$$

so

$$V(q) \cdot \left( Q \cap \bigcup_i \tilde{T}_i \right) \subseteq [V(Q) \cdot q - 4\delta^{-\eta}\rho, V(Q) \cdot q + 4\delta^{-\eta}\rho]. \quad (14.7.15)$$

**Step 4.** *Count.*

Combining (14.7.14) and (14.7.15): each  $\rho$ -separated  $t \in V(Q) \cdot (Q \cap E_{\tilde{\mathbb{T}}})$  yields a  $\rho \times \tau \times \tau$  grain with  $|Q \cap E_{\tilde{\mathbb{T}}} \cap \pi_Q^{-1}([t \pm 4\delta^{-\eta}\rho])| \gtrsim \rho^{6\varepsilon_1+\varepsilon_3}\tau^2\rho$ . Distinct such  $t$  separated by more than  $8\delta^{-\eta}\rho$  give disjoint slabs, so

$$\mathcal{E}_{8\delta^{-\eta}\rho}(V(Q) \cdot (Q \cap E_{\tilde{\mathbb{T}}})) \lesssim \frac{|Q \cap E_{\tilde{\mathbb{T}}}|}{\rho^{6\varepsilon_1+\varepsilon_3}\tau^2\rho} \leq \frac{\rho^{\sigma-\varepsilon_1}\tau^{3-\sigma-\varepsilon_2}}{\rho^{6\varepsilon_1+\varepsilon_3}\tau^2\rho} \lesssim \rho^{-7\varepsilon_1-\varepsilon_2-\varepsilon_3}(\tau/\rho)^{1-\sigma}.$$

Multiplying by  $8\delta^{-\eta}$  to pass from  $\mathcal{E}_{8\delta^{-\eta}\rho}$  to  $\mathcal{E}_\rho$  and choosing  $\varepsilon_1, \varepsilon_2, \varepsilon_3, \eta$  small gives (14.7.13).  $\square$

**Where we are.** Step 2 is a Córdoba  $L^2$  argument — the same device as Volume 9, Theorem 9.3.2 and Volume 12, Proposition 12.4.4. Its role here is to show the  $k$  transverse fat tubes *fill* a definite fraction of the slab they lie in, so that the slab genuinely carries mass.

Step 3 is where the Lipschitz plane map is used, and it is the only place. Since  $V$  varies by at most  $\delta^{-\eta}\tau$  across  $Q$ , all the tubes through  $Q$  are normal to nearly the same direction — so their union lies in one slab of thickness  $\approx \rho$ , not spread across many.

*Together:* the tubes through a  $\tau$ -cube organise into  $\approx (\tau/\rho)^{1-\sigma}$  parallel slabs of thickness  $\rho$ . Those are the grains.

**Lemma 14.7.12** (Grains persist across a ball). *Under the hypotheses of Lemma 14.7.11 there is an  $\varepsilon$ -extremal sub-collection such that for every  $\rho^{1/2}$ -cube  $Q$ , writing  $F = V(Q) \cdot (E_{\mathbb{T}'} \cap Q)$ , every interval  $J$  of length  $\tau$  satisfies  $\mathcal{E}_\rho(J \cap F) \leq \rho^{-\varepsilon}(\tau/\rho)^{1-\sigma}$ .*

*Proof.* Apply Lemma 14.7.11 twice to the  $\rho$ -scale cover of Theorem 14.6.7, once at  $\tau$  and once at  $\rho^{1/2}$ , obtaining

$$\mathcal{E}_\rho(V(Q) \cdot (E \cap Q)) \leq \rho^{-\varepsilon_3} \rho^{(\sigma-1)/2}, \quad \mathcal{E}_\rho(V(Q) \cdot (E \cap W)) \lesssim \delta^{-\eta} \rho^{-\varepsilon_2} (\tau/\rho)^{1-\sigma} \quad (14.7.16)$$

for  $\rho^{1/2}$ -cubes  $Q$  and  $\tau$ -cubes  $W \subseteq Q$ . The second uses  $V(Q)$  rather than  $V(W)$ ; the replacement costs  $\delta^{-\eta}$  because  $\text{diam } W \lesssim \rho^{1/2}$  and  $\angle(V(Q), V(W)) \leq 2\delta^{-\eta}\rho^{1/2}$ .

Refine so each  $\rho^{1/2}$ -cube  $Q$  meeting the set has  $|E \cap Q| \gtrsim \rho^{2\varepsilon_4} \rho^{(\sigma+3)/2}$ . The first bound in (14.7.16) then says  $E \cap Q$  lies in  $\leq \rho^{-\varepsilon_3} \rho^{(\sigma-1)/2}$  grains, whose union has volume  $\lesssim \rho^{-\varepsilon_3} \rho^{(\sigma+3)/2}$ ; comparing, a further refinement makes each occupied grain nearly full:

$$|E \cap Q \cap \pi_Q^{-1}(\mathcal{N}_\rho(t))| \gtrsim \rho^{2\varepsilon_4 + \varepsilon_3} \rho^2. \quad (14.7.17)$$

By Fubini there is a line  $L = q + \mathbb{R}V(p_Q)$  with  $|L \cap E \cap Q| \gtrsim \rho^{\varepsilon_2 + 2\varepsilon_4} |E \cap Q|$ . Replacing each shading by its intersection with the grains meeting  $L$  loses only  $\rho^{5\varepsilon_4}$ , and then

$$\mathcal{E}_\rho(J \cap (V(p_Q) \cdot (E \cap Q))) \approx \mathcal{E}_\rho(V(p_Q) \cdot (L_J \cap E)) \lesssim \sum_W \mathcal{E}_\rho(V \cdot (E \cap W)) \lesssim \rho^{-\varepsilon_2} (\tau/\rho)^{1-\sigma},$$

where  $L_J = L \cap \pi_Q^{-1}(J)$  meets  $O(1)$   $\tau$ -cubes  $W$ . □

**Lemma 14.7.13** (The grains form an ADset). *For all  $\varepsilon > 0$  there are  $\eta, \delta_0 > 0$  so that for  $\delta \in (0, \delta_0]$ : if  $(\mathbb{T}, Y)_\delta$  is  $\eta$ -extremal with  $\delta^{-\eta}$ -Lipschitz plane map  $V$ , there is an  $\varepsilon$ -extremal sub-collection with*

$$V(p) \cdot (E_{\mathbb{T}'} \cap B(p, \rho^{1/2})) \text{ a } (\rho, 1 - \sigma, \delta^{-\varepsilon})_1\text{-ADset} \quad (14.7.18)$$

for all  $\rho \in [\delta, 1]$  and all  $p \in \mathbb{R}^3$ .

*Proof.* Let  $N \geq 2/\varepsilon$ . Apply Lemma 14.7.12 successively at  $\rho_j = \delta^{j/N}$ ,  $j = 1, \dots, N$ , each with  $\varepsilon_j$  chosen so the output remains extremal; Lemma 14.7.12 handles the intermediate  $\tau$  within each scale, and its conclusion is exactly (14.7.1) for that  $\rho$ .

For  $\rho$  between consecutive  $\rho_j$ , the covering numbers differ by at most  $\delta^{-1/N}$ , absorbed into  $\delta^{-\varepsilon}$ . For  $\rho \geq \delta^{1/N}$  the statement is trivial since  $\mathcal{E}_\rho([-1, 1]) \leq 2\rho^{-1} \leq \delta^{-\varepsilon}$ .  $\square$

*Proof of Proposition 14.7.3. Step 1. Global grains.*

Put  $\tau_0 = \delta_0^{(2+\sigma)/\sigma}$ . By Lemma 14.7.10 and Lemma 14.7.13 there is an  $\varepsilon_1$ -extremal  $(\mathbb{T}, Y)_\tau$  for some  $\tau \leq \tau_0$  with  $\tau^{-\varepsilon_1}$ -Lipschitz plane map satisfying (14.7.18).

Set  $\delta = \tau^{\sigma/(2+\sigma)}$  and apply Theorem 14.6.7 with  $\delta$  in place of  $\rho$ , obtaining an  $\varepsilon_2$ -extremal family of  $\delta$ -tubes. Fix one,  $\tilde{T}$ , let  $\mathbb{T}^\dagger = \mathbb{T}_1[\tilde{T}]$ , and choose  $T_0 \in \mathbb{T}^\dagger$  meeting every  $\delta$ -cube of the shading. Rescale by  $\phi$  of Definition 14.6.4, sending the coaxial line of  $T_0$  to the  $z$ -axis, and let  $Z = \{z \in \delta\mathbb{Z} : (0, 0, z) \in \widehat{Y}(\widehat{T}_0)\}$ .

For  $z \in Z$  apply (14.7.18) at the corresponding preimage point: there is  $\widehat{V}(z) \in \mathbb{S}^2$  with

$$\widehat{V}(z) \cdot (E_{\widehat{\mathbb{T}}} \cap (\mathbb{R}^2 \times [z, z + \delta))) \text{ a } (\delta, 1 - \sigma, O(\tau^{-\varepsilon_1}))_1\text{-ADset.} \quad (14.7.19)$$

Since  $|V \cdot \text{dir}(\ell_0)| \leq \tau$ , we get  $|\widehat{V}(z) \cdot e_3| \leq \tau/\delta \leq \frac{1}{10}$ , so after possibly swapping the  $x$  and  $y$  axes and restricting  $z$  to an interval of length  $\approx \tau^{\varepsilon_1}$ , we have  $|\widehat{V}(z) \cdot e_1| \geq \frac{1}{3}$ . Hence there are  $3\tau^{-2\varepsilon_1}$ -Lipschitz  $f, g$  on  $Z$  with  $\widehat{V}(z)$  parallel to  $(1, f(z), g(z))$ ; extend  $f$  to  $[-1, 1]$  by Kirszbraun. Then (14.7.19) reads

$$(1, f(z), 0) \cdot (E_{\widehat{\mathbb{T}}} \cap (\mathbb{R}^2 \times [z, z + \delta))) \text{ a } (\delta, 1 - \sigma, O(\tau^{-\varepsilon_1}))_1\text{-ADset.} \quad (14.7.20)$$

**Step 2. Local grains.**

Apply Lemma 14.7.5 to  $(\widehat{\mathbb{T}}_1, \widehat{Y}_1)_{\tau/\delta}$  with  $\varepsilon_3$ , then Lemma 14.7.6 with  $\delta$  in place of  $\rho$  — legitimate since  $\delta = (\tau/\delta)^{\sigma/2}$  — obtaining an  $\varepsilon_3$ -extremal family of  $\delta$ -tubes with plane map  $W$ . Each  $\delta$ -cube of its shading meets the previous set, so (14.7.20) persists, giving Item 1 in the form

$$(1, f(z_0), 0) \cdot (E_{\widehat{\mathbb{T}}} \cap \{z = z_0\}) \text{ a } (\delta, 1 - \sigma, O(\tau^{-\varepsilon_1}))_1\text{-ADset,}$$

vacuously for  $z_0 \notin Z$ .

Finally apply Lemma 14.7.13 with  $\varepsilon_4$  to obtain Item 2. Choosing  $\varepsilon_1, \dots, \varepsilon_4$  small and correcting the Lipschitz constants of  $f$  and  $W$  to 1 by Lemma 14.6.9 completes the proof.  $\square$

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WORKED EXAMPLE 14.7.1— TIER A **The two grain structures, side by side**

Proposition 14.7.3 produces two decompositions and they are easily confused.

|         | global grains                    | local grains                                     |
|---------|----------------------------------|--------------------------------------------------|
| Where   | each slice $\{z = z_0\}$         | each ball $B(p, \rho^{1/2})$                     |
| Shape   | $1 \times \delta$ rectangles     | $\rho^{1/2} \times \rho^{1/2} \times \rho$ slabs |
| Normal  | $(1, f(z_0), 0)$ , $f$ Lipschitz | $V(p)$ , $V$ Lipschitz                           |
| Spacing | $(\delta, 1 - \sigma)$ -ADset    | $(\rho, 1 - \sigma)$ -ADset                      |
| Scale   | one, namely $\delta$             | all $\rho \in [\delta, 1]$                       |

*The global structure is a slice statement*, holding at the finest scale only, and it is what gives  $|E_{\mathbb{T}}| \lesssim \delta^{\sigma-\varepsilon}$  by Fubini.

*The local structure holds at every scale*, and it is what feeds the multi-scale arguments of Chapter 14.9.

*The Heisenberg example has both*, with global slope function  $f(z) = \bar{z}$  — which is 1-Lipschitz. So nothing proved so far distinguishes an extremal Kakeya set in  $\mathbb{R}^3$  from the Heisenberg example in  $\mathbb{C}^3$ , and that is precisely the difficulty the next chapter must overcome.

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WORKED EXAMPLE 14.7.2— TIER B **Where multilinear Kakeya is used, and why only there**

Theorem 14.7.4 is invoked exactly once, in Step 2 of Lemma 14.7.5. It is worth being clear what it does.

- (a) *What it gives.* An upper bound on the number of *genuinely three-dimensional* triples of tubes through a typical point. Nothing else.
- (b) *Why that suffices.* Combined with the lower bound on  $|E_{\mathbb{T}}|$  from extremality, it forces most points to be narrow, and a narrow point has a plane.
- (c) *Why it is not needed again.* Once the plane map exists, all subsequent arguments are about its regularity and about counting grains, which are  $L^2$  and covering-number statements. Multilinear Kakeya has done its work.
- (d) *What it cannot do.* It is invariant under the complex structure, so it cannot distinguish  $\mathbb{R}^3$  from  $\mathbb{C}^3$  — which is why the Heisenberg example survives this chapter intact. Breaking that symmetry requires the projection theory of Chapter 14.9.

*The transferable point.* A tool that produces *structure* is often used once, early, and then set aside; the rest of the argument exploits the structure. Volume 11's Theorem 11.2.5 plays the same role there.

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WORKED EXAMPLE 14.7.3— TIER C: RESEARCH TASTE **Structure theorems as a proof strategy**

Proposition 14.7.3 does not estimate anything. It says a hypothetical extremal object must *look like* something specific. That is a distinct proof strategy and worth naming.

(a) *The shape.* Assume the conjecture fails; derive successively more rigid structure for the counterexample; show no object with that structure exists.

(b) *The chain here.* Extremal  $\Rightarrow$  self-similar (Theorem 14.6.7)  $\Rightarrow$  plany (Lemma 14.7.5)  $\Rightarrow$  grainy (Proposition 14.7.3)  $\Rightarrow$   $C^2$  slope (Proposition 14.8.1)  $\Rightarrow$  contradiction (Chapter 14.9). Five steps, each strengthening the description.

(c) *Why it was needed here.* Every earlier method estimated a quantity and fell short. The structure route instead *uses* extremality as a hypothesis — and the more nearly extremal the object, the more structure it must have. Lemma 14.6.8 is a good example: it is vacuous when  $\sigma = 0$  and strongest when  $\sigma$  is large.

(d) *The precedent.* Katz and Tao conjectured planiness and graininess in 2001, long before either could be proved, precisely on the grounds that the Heisenberg example has both. *The example suggested the structure theorem*, and the structure theorem eventually excluded the example.

(e) *The cost.* Structure theorems give no quantitative bound. Wang and Zahl prove  $\sigma_3 = 0$  but the argument yields no rate, and Theorem 14.4.8 therefore has an unspecified exponent  $K$  — which is why the *Kakeya maximal* conjecture remains open.

(f) *When to reach for it.* When an extremal example is known and every estimate stalls at it. That describes the restriction conjecture too, where Knapp caps are the known extremiser — but there, as Worked Example 12.5.3(c) records, the extremisers are not unique, and no structure theorem has been found.

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WORKED EXAMPLE 14.7.4— TIER B **Where each volume’s machinery enters the Kakeya proof**

The proof draws on eight of the preceding volumes. A map.

*Volume 1.* Hölder, from which Loomis–Whitney and hence everything geometric descends.

*Volume 5.* Stationary phase, in Lemma 14.17.6 and in the wave packet estimates.

*Volume 8.* Marstrand and Kaufman, the starting points of Theorem 14.14.2

and Proposition 14.14.12.

*Volume 9.* The  $L^2$  method, in Lemma 14.7.11 and in three of the four cases of Theorem 14.5.7; and Wolff’s hairbrush, which is the base case of the bootstrap Theorem 14.4.7.

*Volume 10.* Ruzsa and Plünnecke–Ruzsa, in Theorem 14.14.3.

*Volume 11.* Polynomial ham sandwich, in Theorem 14.12.10; multilinear Kakeya, in Theorem 14.7.4; and polynomial partitioning throughout.

*Volume 12.* Wave packets and the transversality framework, in Chapter 14.7.

*Volume 13.* Decoupling, in Chapter 14.17 — though not in the Kakeya proof itself.

*What this shows.* No single technique settles Kakeya. The proof consumes the  $L^2$  method, the polynomial method, projection theory and additive combinatorics, and the volumes were arranged in this order because each is needed by the next. *That the last volume uses all the others is not a narrative device but a fact about the theorem.*

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## Drills

**Drill 14.7.1.** Verify the Fubini computation in the Checkpoint after Proposition 14.7.3:  $\delta^{\sigma-1-\varepsilon}$  rectangles of dimensions  $1 \times \delta$  per slice gives  $|E_{\mathbb{T}}| \lesssim \delta^{\sigma-\varepsilon}$ .

**Drill 14.7.2.** Check (14.7.8), including the identity  $|V(p) \cdot v_3| = |v_1 \wedge v_2 \wedge v_3|/|v_1 \times v_2|$ .

**Drill 14.7.3.** In Step 2 of Lemma 14.7.11, verify the Córdoba estimate (14.7.14) and identify where transversality is used.

**Drill 14.7.4.** Confirm that  $\delta = \tau^{\sigma/(2+\sigma)}$  gives  $\delta = (\tau/\delta)^{\sigma/2}$ , as needed in Step 2 of the proof of Proposition 14.7.3.

## Notes

This chapter is Section 4 of Wang and Zahl. Theorem 14.7.4 is their Theorem 4.2, quoted there from Bennett–Carbery–Tao; here it is derived from Volume 11, Theorem 11.5.7, which is proved. The conjecture that extremal Kakeya sets are plany and grainy is due to Katz and Tao.

## Chapter 14.8

# The Slope Function is $C^2$

Worked Example 14.7.1 recorded that everything proved so far applies verbatim to the Heisenberg example. This chapter is where the two part company. The mechanism is a projection theorem — Theorem 14.8.2 — which is *true over  $\mathbb{R}$  and false over  $\mathbb{C}$* , and the whole weight of the argument rests on that asymmetry.

### 14.8.1 The statement

**Proposition 14.8.1** (Grains with  $C^2$  slope). *For all  $\varepsilon, \delta_0 > 0$  there are  $\delta \in (0, \delta_0]$  and an  $\varepsilon$ -extremal  $(\mathbb{T}, Y)_\delta$  with:*

1'. Global grains with  $C^2$  slope. *There is  $f : [-1, 1] \rightarrow \mathbb{R}$  with  $\|f\|_{C^2} \leq 1$  so that  $(E_{\mathbb{T}} \cap \{z = z_0\}) \cdot (1, f(z_0), 0)$  is a  $(\delta, 1 - \sigma, \delta^{-\varepsilon})_1$ -ADset for each  $z_0$ .*

2. Local grains with Lipschitz plane map, *exactly as in Proposition 14.7.3.*

**Theorem 14.8.2** (Wang–Zahl projection theorem). *For all  $\varepsilon > 0$  there are  $\eta, \delta_0 > 0$  so that for  $\delta \in (0, \delta_0]$  the following holds. Let  $F, G \subseteq [0, 1]^2$  be  $(\delta, 1, \delta^{-\eta})$ -ADsets. Then at least one of:*

(A) *there are orthogonal lines  $\ell$  and  $\ell^\perp$  (the latter through the origin) with*

$$\mathcal{E}_\delta(\mathcal{N}_\delta(\ell) \cap F) \geq \delta^{\varepsilon-1} \quad \text{and} \quad \mathcal{E}_\delta(\mathcal{N}_\delta(\ell^\perp) \cap G) \geq \delta^{\varepsilon-1}; \quad (14.8.1)$$

(B) *for every  $H \subseteq F \times G \times G$  with  $\mathcal{E}_\delta(H) \geq \delta^{\eta-3}$  there are  $\rho \geq \delta$  and an interval  $I$  of length at least  $\delta^{-\eta}\rho$  with*

$$\mathcal{E}_\rho(I \cap \{a \cdot (b_1 - b_2) : (a, b_1, b_2) \in H\}) \geq (|I|/\rho)^{1-\varepsilon}. \quad (14.8.2)$$

*Remark 14.8.3* (The theorem is false over  $\mathbb{C}$ ). This is the point on which the whole proof turns, so it is worth seeing the counterexample.

Let  $\Omega \subseteq \mathbb{C}^2$  be a neighbourhood of the origin and put

$$F = G = \{(z, \bar{z}) : z \in \mathbb{C}\} \cap \Omega,$$

which are  $(\delta, 2, O(1))$ -sets — the correct analogue, with exponents doubled. Take  $H = F \times G \times G$ , so  $\mathcal{E}_\delta(H) \gtrsim \delta^{-6}$ .

(A) *fails*. For any complex line  $\ell = \{(a + bt, c + dt) : t \in \mathbb{C}\}$  with  $(b, d) \neq (0, 0)$  we have  $\mathcal{E}_\delta(\mathcal{N}_\delta(\ell) \cap F) \leq \delta^{-1}$ , far smaller than the  $\delta^{-2}$  that a significant fraction would require.

(B) *fails*. For  $\omega = (z_0, \bar{z}_0) \in F$  and  $\zeta_i = (\bar{z}_i, z_i) \in G$ ,

$$\omega \cdot (\zeta_1 - \zeta_2) = z_0(z_1 - z_2) + \bar{z}_0(\overline{z_1 - z_2}) = \overline{\omega \cdot (\zeta_1 - \zeta_2)},$$

so the entire product set lies in  $\{z \in \mathbb{C} : \Im z = 0\}$  — a line. That set cannot satisfy (14.8.2) with exponent  $2 - \varepsilon$  at any scale.

*What this means*. The conjugation  $z \mapsto \bar{z}$  is an isometry of  $\mathbb{C}$  that is *not* complex-linear, and it makes the trilinear expression  $a \cdot (b_1 - b_2)$  degenerate. Over  $\mathbb{R}$  no such map exists, and Theorem 14.8.2 says the degeneracy of alternative (A) is the only one. *This is the exact point at which a Kakeya set in  $\mathbb{R}^3$  is distinguished from the Heisenberg example in  $\mathbb{C}^3$* , and no earlier step in the proof could have done it.

*Remark 14.8.4* (Scope). Theorem 14.8.2 is stated here; Chapter 14.11 derives it from the discretised projection theory, whose central input — Bourgain’s projection theorem — is itself proved in Chapter 14.14. Its proof is independent of the rest of the argument and uses ideas from projection theory developed for the Falconer distance problem — generalising Bourgain’s discretized sum-product estimate, and drawing on the radial projection theorems of Orponen, Shmerkin and Wang and on Kaufman’s projection theorem.

## 14.8.2 The slope function is locally linear

**Lemma 14.8.5** (Local linearity at one scale). *For all  $\varepsilon > 0$  there are  $\eta, \rho_0 > 0$  so that the following holds for  $\rho \in (0, \rho_0]$ . Let  $I \subseteq [-1, 1]$  be an interval of length  $\rho^{1/2}$  with left endpoint  $z^*$ , let  $Z \subseteq I$ , and let  $f : I \rightarrow [0, 1]$  be 1-Lipschitz. Let  $E \subseteq [-1, 1]^2 \times X$  be a union of  $\rho$ -cubes contained in the  $\rho^{1/2}$ -neighbourhood of a line in direction  $(-f(z^*), 1, 0)$ , with  $|E| \geq \rho^{1+\sigma/2+\eta}$ . Let  $V : E \rightarrow \mathbb{S}^2$  be 1-Lipschitz with  $|V(p) \cdot e_3| \leq \frac{1}{2}$ . Suppose for each  $z_0 \in Z$*

$$(E \cap \{z = z_0\}) \cdot (1, f(z_0), 0) \text{ is a } (\rho, 1 - \sigma, \rho^{-\eta})_1\text{-ADset}, \quad (14.8.3)$$



and for each  $p \in E$

$$\left| (E \cap B(p, \rho^{1/2})) \cdot V(p) \right| \leq \rho^{\frac{1+\sigma}{2}-\eta}. \quad (14.8.4)$$

Then there are  $Z_{\text{lin}} \subseteq Z$  with  $|Z_{\text{lin}}| \geq \rho^{1/2+\varepsilon}$  and a linear  $L : [z^*, z^* + \rho^{1/2}] \rightarrow [0, 1]$  with  $|f(z) - L(z)| \leq \rho$  for all  $z \in Z_{\text{lin}}$ .

*Proof.* Normalise so  $I = [0, \rho^{1/2}]$ ,  $f(0) = 0$ , and  $E$  lies in the  $\rho^{1/2}$ -neighbourhood of the  $y$ -axis; a translation and a rotation about the  $z$ -axis achieve this, and replacing the rotated set by the union of  $\rho$ -cubes meeting it preserves (14.8.3) and (14.8.4) up to constants.

**Step 1.**  $V$  has large  $x$ -coordinate.

Partition into  $\rho^{1/2}$ -cubes and refine so each occupied  $Q$  has  $|Q \cap E| \gtrsim \rho^{3/2+\sigma/2+\eta}$ , the cubes  $\rho^{1/2}$ -separated, with a marked point  $p_Q$ . Write  $\pi_Q(q) = q \cdot V(p_Q)$ . Refine further so each local grain is nearly full: for  $w \in V(p_Q) \cdot (Q \cap E)$ ,

$$|E \cap Q \cap \mathcal{N}_\rho(\pi_Q^{-1}(w))| \gtrsim \rho^{2+2\eta}. \quad (14.8.5)$$

By Fubini there is  $z_1$  with  $|\{z = z_1\} \cap E \cap Q \cap \mathcal{N}_\rho(\pi_Q^{-1}(w))| \gtrsim \rho^{3/2+2\eta}$ . Since  $|V(p_Q) \cdot e_3| \leq \frac{1}{2}$ , that set lies in a rectangle  $R$  of dimensions  $\approx \rho^{1/2} \times \rho$ , and  $R \cdot (1, f(z_1))$  is an interval  $I_1$  with

$$|I_1| \approx \rho^{1/2} |V(p_Q) \cdot (-f(z_1), 1, 0)|. \quad (14.8.6)$$

The image lies in  $I_1 \cap [(E \cap \{z = z_1\}) \cdot (1, f(z_1), 0)]$ , which by (14.8.3) has measure at most  $\rho^{1-\sigma-\eta} |I_1|^\sigma$ ; comparing with the lower bound  $\gtrsim \rho^{2\eta} |I_1|$  gives  $|I_1| \leq \rho^{1-3\eta/(1-\sigma)}$  and hence, by (14.8.6),

$$|V(p_Q) \cdot (-f(z_1), 1, 0)| \lesssim \rho^{1/2-3\eta/(1-\sigma)}. \quad (14.8.7)$$

Since  $f(z_1) \in [0, \rho^{1/2}]$ , for  $\eta$  small this forces  $|V(p_Q) \cdot e_2| \leq \frac{1}{10}$ , and with  $|V(p_Q) \cdot e_3| \leq \frac{1}{2}$  we get  $|V(p_Q) \cdot e_1| \geq \frac{1}{4}$ .

**Step 2.** The function  $g$ .

Let  $J$  be the projection of the occupied  $\rho^{1/2}$ -cubes to the  $y$ -axis; each  $y \in J$  determines one cube  $Q(y)$ . Define  $g(y), h(y)$  by  $(1, h(y), g(y)) \parallel V(p_{Q(y)})$ . By Step 1 and the Lipschitz property of  $V$ ,  $g$  and  $h$  are 4-Lipschitz; extend them to  $[0, 1]$ . Then (14.8.4) gives, for each  $y_0$ ,

$$\left| (E \cap \{y = y_0\}) \cdot (1, 0, g(y_0)) \right| \lesssim \rho^{\frac{1+\sigma}{2}-\eta}. \quad (14.8.8)$$

**Step 3.** Paths of length three.

Refine so every  $\rho$ -cube of  $E$  has  $z$ -centre in a set  $Z_{\text{pop}}$ , each element of which is the centre of  $\gtrsim \rho^{-3/2+\sigma/2+\eta}$  cubes. After a shift in  $y$ , pass to  $E_1 \subseteq E$  with  $|E_1| \geq \rho^{1/2+2\eta} |E|$  and every cube centre having  $y$ -coordinate in  $\rho^{1/2}\mathbb{Z}$ .

Say  $Q, Q'$  are in the *same global grain* if their centres share a  $z$ -coordinate  $z_Q$  and  $|p_Q \cdot (1, f(z_Q), 0) - p_{Q'} \cdot (1, f(z_Q), 0)| \leq \rho$ ; in the *same local grain* if they share a  $y$ -coordinate  $y_Q$  and  $|p_Q \cdot (1, 0, g(y_Q)) - p_{Q'} \cdot (1, 0, g(y_Q))| \leq \rho$ .

Counting:  $E_1$  has  $\geq \rho^{-3/2+\sigma/2+3\eta}$  cubes; the  $y$ -coordinate takes  $\rho^{-1/2}$  values; and by (14.8.4), once  $y$  is fixed there are  $\leq \rho^{-1/2+\sigma/2-\eta}$  values of  $p_Q \cdot (1, 0, g(y))$ . So  $E_1$  partitions into  $O(\rho^{-1+\sigma/2-\eta})$  local grains, and by Cauchy-Schwarz there are  $\gtrsim \rho^{-2+\sigma/2+7\eta}$  pairs in a common local grain.

Now map  $(Q, Q') \mapsto (z_Q, z_{Q'}, r_{Q'})$  with  $r_{Q'}$  the nearest point of  $\rho\mathbb{Z}$  to  $p_{Q'} \cdot (1, f(z_{Q'}), 0)$ . By (14.8.3) this takes  $\leq \rho^{-3/2+\sigma/2-\eta}$  values, so Cauchy-Schwarz again gives  $\gtrsim \rho^{-5/2+\sigma/2+15\eta}$  quadruples  $(Q_1, Q_2, Q_3, Q_4)$  with  $Q_1, Q_2$  in a local grain,  $Q_2, Q_3$  in a global grain,  $Q_3, Q_4$  in a local grain, and  $Q_4, Q_1$  sharing a  $z$ -coordinate:

$$\#\mathcal{Q} \gtrsim \rho^{-5/2+\sigma/2+15\eta}. \quad (14.8.9)$$

**Step 4.** *Normalise by a skew transform.*

By (14.8.9) and pigeonholing, fix  $(z_0, w)$  admitting  $\gtrsim \rho^{-3/2+8\eta}$  quadruples with  $Q_1$  at height  $z_0$  and  $|p_{Q_1} \cdot (1, f(z_0), 0) - w| \leq \rho$ . Put

$$A = (E \cap \{z = z_0\}) \cdot (1, f(z_0), 0) - w,$$

a  $(\rho, 1 - \sigma, \rho^{-\eta})_1$ -ADset by (14.8.3). Apply

$$\varphi(x, y, z) = (x + f(z_0)y + g(0)(z - z_0), y, z - z_0)$$

and set  $f^\dagger(z) = f(z) - f(z_0)$ ,  $g^\dagger(y) = g(y) - g(0)$ , so  $f^\dagger(0) = g^\dagger(0) = 0$ . Both (14.8.3) and (14.8.8) persist for  $E^\dagger = \varphi(E)$  with  $f^\dagger$  and  $g^\dagger$  in place of  $f$  and  $g$ .

**Step 5.** *Few dot products.*

For a quadruple with  $p_i = (x_i, y_i, z_i)$ :  $z_1 = z_4 = 0$ ,  $y_2 = y_1$ ,  $z_3 = z_2$ ,  $y_4 = y_3$ , and  $|x_1 - w| \leq \rho$ . Chaining the three grain relations,

$$|x_2 - (w - z_2 g^\dagger(y_1))| \leq 2\rho, \quad |x_3 - (w - z_2 g^\dagger(y_1) + (y_1 - y_3) f^\dagger(z_2))| \leq 3\rho,$$

and finally

$$\left| x_4 - \left( w - z_2 (g^\dagger(y_1) - g^\dagger(y_3)) + (y_1 - y_3) f^\dagger(z_2) \right) \right| \leq 4\rho. \quad (14.8.10)$$

Since  $x_4 \in (E \cap \{z = z_0\}) \cdot (1, f(z_0), 0)$ , (14.8.10) gives

$$z_2 (g^\dagger(y_1) - g^\dagger(y_3)) + (y_1 - y_3) f^\dagger(z_2) \in \mathcal{N}_{4\rho}(A). \quad (14.8.11)$$

Now set

$$F = \{(\rho^{-1/2}z, \rho^{-1/2}f^\dagger(z)) : z - z_0 \in Z_{\text{pop}}\}, \quad G = \{(y, g^\dagger(y)) : y \in [0, 1] \cap \rho^{1/2}\mathbb{Z}\},$$

which are  $(\rho^{1/2}, 1, C)_2$ -ADsets since  $f^\dagger, g^\dagger$  are Lipschitz, and let  $H \subseteq F \times G \times G$  be the six-tuples arising from the quadruples, so  $\#H \gtrsim (\rho^{1/2})^{16\eta-3}$ . Then (14.8.11) says

$$\{a \cdot (b_1 - b_2) : (a, b_1, b_2) \in H\} \subseteq \tilde{A},$$

where  $\tilde{A} = \mathcal{N}_{4\rho^{1/2}}(\rho^{-1/2}A)$  is a  $(\rho^{1/2}, 1 - \sigma, 8\rho^{-\eta})_1$ -ADset.

**Step 6.** *Apply the projection theorem.*

For  $\eta$  small,  $F, G, H$  satisfy the hypotheses of Theorem 14.8.2 at scale  $\rho^{1/2}$ . Alternative (B) would give (14.8.2), contradicting that  $\tilde{A}$  is a  $(\rho^{1/2}, 1 - \sigma)$ -ADset with  $\sigma > 0$ . So (A) holds: there are orthogonal  $\ell, \ell^\perp$  with  $\mathcal{E}_{\rho^{1/2}}(\mathcal{N}_{\rho^{1/2}}(\ell) \cap F) \geq (\rho^{1/2})^{\varepsilon-1}$ .

Let  $Z_{\text{lin}}$  be the union of  $\rho$ -intervals meeting  $\{z \in Z_{\text{pop}} : (\rho^{-1/2}(z + z_0), \rho^{-1/2}f^\dagger(z + z_0)) \in F \cap \mathcal{N}_{\rho^{1/2}}(\ell)\}$ . Then  $\rho^{-1}|Z_{\text{lin}}| \geq (\rho^{1/2})^\varepsilon \#Z_{\text{pop}}$ , and tracing back through (14.8.3) gives  $|Z_{\text{lin}}| \gtrsim \rho^{1/2+\varepsilon/2+4\eta} \geq \rho^{1/2+\varepsilon}$  for  $\rho_0$  small. Taking  $L$  to be  $\ell$  rescaled by  $\rho^{1/2}$  gives  $|f(z) - L(z)| \leq \rho$  on  $Z_{\text{lin}}$ .  $\square$

**Where we are.** The engine is Step 5. Three grain relations chained together produce a *trilinear* expression  $a \cdot (b_1 - b_2)$ , and (14.8.11) says it lands in a set of dimension  $1 - \sigma < 1$ . Theorem 14.8.2 says that can only happen in the degenerate case (A), which forces  $f$  to be linear at scale  $\rho$ . Over  $\mathbb{C}$ , Remark 14.8.3 shows there is a second way: the conjugation symmetry makes the trilinear form degenerate without any linearity. That is exactly the Heisenberg example, and it is why the theorem is about  $\mathbb{R}^3$ .

### 14.8.3 From locally linear to $C^2$

**Lemma 14.8.6** (Linearity at one scale, globally). *For all  $\varepsilon > 0$  there are  $\eta, \delta_0 > 0$  so that for  $\delta \in (0, \delta_0]$ : let  $(\mathbb{T}, Y)_\delta$  be  $\eta$ -extremal satisfying Items 1 and 2 of Proposition 14.7.3 with  $\eta$  in place of  $\varepsilon$ , and let  $\rho \in [\delta^{1-\varepsilon}, \delta^\varepsilon]$ . Then there are an  $\varepsilon$ -extremal sub-collection and a 2-Lipschitz  $L : [-1, 1] \rightarrow \mathbb{R}$ , linear on each interval  $[n\rho^{1/2}, (n+1)\rho^{1/2}]$ , with  $|f(z) - L(z)| \leq \rho$  for all  $z \in \pi_z(E_{\mathbb{T}'})$ .*

*Proof.* Apply Theorem 14.6.7 twice, at  $\rho$  and then at  $\rho^{1/2}$ , and refine so  $E$  is contained in  $\rho^{1/2}$ -separated horizontal slabs  $S = \mathbb{R}^2 \times [n\rho^{1/2}, (n+1)\rho^{1/2}]$ , each with  $|S \cap E_{\mathbb{T}}| \gtrsim \rho^{1/2+\varepsilon_1} \delta^{\sigma+\eta}$ .

Fix such an  $S$ . By Fubini the set  $Z_S$  of heights  $z_0$  with  $|E_{\mathbb{T}} \cap \{z = z_0\}| \gtrsim \rho^{2\varepsilon_1} \delta^{\sigma+\eta}$  has  $|Z_S| \geq \rho^{1/2+\varepsilon_1} \delta^\eta$ . For  $z_0 \in Z_S$ , Item 1 says the slice lies in  $\leq \delta^{\sigma-\eta-1}$  rectangles of dimensions  $1 \times \delta$  with normal  $(1, f(z_0), 0)$ ; pigeonholing selects one meeting the slice in measure  $\gtrsim \delta^{1+2\eta} \rho^{2\varepsilon_1}$ . Its axis  $\ell$  meets  $\gtrsim \delta^{2\eta} \rho^{-1/2+2\varepsilon_1}$  many  $\rho^{1/2}$ -cubes, so  $|\mathcal{N}_{\rho^{1/2}}(\ell) \cap E_{\mathbb{T}}| \gtrsim \delta^{2\eta} \rho^{\sigma/2-1+2\varepsilon_1}$ .

For  $\eta, \varepsilon_1$  small this set, together with  $f|_S$  and  $V$ , satisfies the hypotheses of Lemma 14.8.5 with  $\varepsilon_3$  in place of  $\varepsilon$ . That lemma yields  $Z_{\text{lin}, S}$  with  $|Z_{\text{lin}, S}| \geq$

$\rho^{1/2+\varepsilon_3}$  and a linear  $L_S$  with  $|f - L_S| \leq \rho$  there. Since  $Z_{\text{lin},S}$  contains two points  $\rho^{1/2+\varepsilon_3}$ -separated and  $f$  is 1-Lipschitz,  $L_S$  has slope at most 2.

Summing over slabs and defining  $L$  piecewise by  $L = L_S$  on  $I_S$ , extended piecewise-linearly between the  $\rho^{1/2}$ -separated intervals, gives a 2-Lipschitz  $L$ ; the associated refinement is  $\varepsilon$ -extremal for  $\eta, \varepsilon_i$  small.  $\square$

**Lemma 14.8.7** (Linearity at many scales). *For all  $\varepsilon > 0$  and  $N \geq 1$  there are  $\eta, \delta_0 > 0$  so that for  $\delta \in (0, \delta_0]$  and  $\rho_j = \delta^{j/N}$ : under the hypotheses of Lemma 14.8.6 there are an  $\varepsilon$ -extremal sub-collection and functions  $L_1, \dots, L_N$  with*

- (a) *each  $L_j$  2-Lipschitz and linear on each  $[n\rho_j^{1/2}, (n+1)\rho_j^{1/2}]$ ;*
- (b)  *$|f(z) - L_j(z)| \leq \rho_j$  for all  $z \in \pi_z(E_{\mathbb{T}^N})$  and all  $j$ ;*
- (c) *each interval  $I = [n\rho_j^{1/2}, (n+1)\rho_j^{1/2}]$  meeting  $\pi_z(E_{\mathbb{T}^N})$  satisfies  $|I \cap \pi_z(E_{\mathbb{T}^N})| \geq 4^{-N} \delta^\varepsilon |I|$ ;*
- (d) *no two adjacent such intervals both occur.*

*Proof.* Apply Lemma 14.8.6 successively at  $\rho_N, \dots, \rho_1$ . For (c), discard intervals failing the density bound and refine the shading; for (d), discard the odd- or the even-indexed intervals, whichever retains more. Each step costs a factor  $4^{-1}$  and does not disturb (c) and (d) at larger scales.  $\square$

**Where we are.** Conclusion (b) does *not* imply  $|L_j(z) - L_{j-1}(z)| \leq \rho_j + \rho_{j-1}$  for every  $z$  in an interval — only for  $z$  between two points of  $\pi_z(E_{\mathbb{T}^N})$ . That is why (c) and (d) are needed: (c) guarantees each interval meets the projection in a definite proportion, so the two linear functions are compared on a substantial sub-interval  $I' \subseteq I$  with

$$|L_j(z) - L_{j-1}(z)| \leq \rho_j + \rho_{j-1} \quad (z \in I'). \quad (14.8.12)$$

This is the kind of bookkeeping that is easy to omit and fatal to skip.

**Definition 14.8.8** (Vertical trapezoid). A *vertical trapezoid* is  $R = \{(z, t) \in J \times \mathbb{R} : |t - L(z)| \leq s\}$  with  $J$  an interval and  $L$  linear. Its *length* is  $|J|$ , its *height*  $s$ , and *slope*( $R$ ) the slope of  $L$ .

**Corollary 14.8.9** (Nested trapezoids). *Under the hypotheses of Lemma 14.8.7 there are an  $\varepsilon$ -extremal sub-collection and sets  $\mathcal{R}_1, \dots, \mathcal{R}_N$  of vertical trapezoids with: every trapezoid has slope in  $[-2, 2]$ ; each  $R \in \mathcal{R}_j$  has height  $\rho_j$  and length between  $\rho_j^{1/2+\varepsilon}$  and  $\rho_j^{1/2}$ ; the projections of  $\mathcal{R}_j$  to  $z$  are  $\rho_j^{1/2}$ -separated; each  $R \in \mathcal{R}_j$  has a unique parent  $\tilde{R} \in \mathcal{R}_{j-1}$  containing it; and, with*

$$D = \pi_z(E_{T'}),$$

$$\text{graph}(f|_D) \subseteq \bigcup_{R \in \mathcal{R}_N} R.$$

*Proof.* Restate (14.8.12) as a containment of trapezoids: the graph of  $L_j$  over  $I'$ , thickened by  $\rho_j$ , lies inside the graph of  $L_{j-1}$  thickened by  $\rho_{j-1}$ .  $\square$

**Lemma 14.8.10** (The building block). *Let  $x_1 < x_2$  and let  $L$  be the segment from  $(x_1, y_1)$  to  $(x_2, y_2)$ , with  $s = \frac{y_2 - y_1}{x_2 - x_1}$  and  $h = |y_1| + |y_2|$ . For every  $a > 0$  there is  $G : \mathbb{R} \rightarrow \mathbb{R}$  with  $L \subseteq \text{graph}(G)$ ,  $G = 0$  off  $[x_1 - a, x_2 + a]$ , and*

$$|G'(x)| \lesssim h/a + |s|, \quad |G''(x)| \lesssim h/a^2 + |s|/a.$$

*Proof.* Let  $\phi : \mathbb{R} \rightarrow [0, 1]$  be smooth and monotone with  $\phi = 0$  on  $[0, \frac{1}{3}]$ ,  $\phi = 1$  on  $[\frac{2}{3}, 1]$ ,  $|\phi'|, |\phi''| \leq 10$ ; put  $\phi_a(x) = \phi(x/a)$  and  $\psi_a(x) = (x - a)\phi_a(x)$ . Then  $\phi_a(0) = 0$ ,  $\phi_a(a) = 1$ ,  $\psi_a(0) = \psi_a(a) = 0$ ,  $\psi'_a(a^-) = 1$ , and all one-sided first and second derivatives vanish at the outer endpoints. Set

$$G(x) = \begin{cases} y_1\phi_a(x - x_1 + a) + s\psi_a(x - x_1 + a), & x \leq x_1, \\ s(x - x_1) + y_1, & x_1 < x < x_2, \\ y_2\phi_a(a + x_2 - x) - s\psi_a(a + x_2 - x), & x \geq x_2. \end{cases}$$

The matching conditions make  $G$  twice differentiable, and  $|\phi'_a| \lesssim a^{-1}$ ,  $|\phi''_a| \lesssim a^{-2}$ ,  $|\psi'_a| \lesssim 1$ ,  $|\psi''_a| \lesssim a^{-1}$  give the bounds.  $\square$

**Proposition 14.8.11** (Locally linear at many scales implies  $C^2$ ). *Let  $\mathcal{R}_1, \dots, \mathcal{R}_N$  be as in Corollary 14.8.9, let  $D \subseteq [-1, 1]$  and  $f : D \rightarrow \mathbb{R}$  with  $\text{graph } f \subseteq \bigcup_{R \in \mathcal{R}_N} R$ . Then there is  $g : [-1, 1] \rightarrow \mathbb{R}$  with*

1.  $|f(x) - g(x)| \leq \delta$  for  $x \in D$ ;
2.  $|g''(x)| \lesssim \delta^{-1/N-2\eta}$  for all  $x$ ;
3. if  $x \in \pi(R)$  for some  $R \in \mathcal{R}_j$  then  $|g'(x) - \text{slope}(R)| \lesssim \rho_j / \rho_{j+1}^{1/2+\eta}$ .

*Proof.* For  $R \in \mathcal{R}_1$  let  $\tilde{L}(R)$  be the segment coaxial with  $R$ . For  $2 \leq k \leq N$  and  $R \in \mathcal{R}_k$  with parent  $\tilde{R}$ , let  $(x_1, y_1), (x_2, y_2)$  be the endpoints of the segment coaxial with  $R$  and  $(x_1, \tilde{y}_1), (x_2, \tilde{y}_2)$  the corresponding points on the segment coaxial with  $\tilde{R}$ ; let  $\tilde{L}(R)$  run from  $(x_1, y_1 - \tilde{y}_1)$  to  $(x_2, y_2 - \tilde{y}_2)$ . Let  $G_R$  be the function of Lemma 14.8.10 with  $L = \tilde{L}(R)$  and  $a = |\pi(R)|$ , the length of  $R$ . Put

$$g = \sum_{k=1}^N \sum_{R \in \mathcal{R}_k} G_R.$$

(1) For  $x \in D$  the telescoping sum over the nested chain  $R^{(N)} \subseteq \dots \subseteq R^{(1)}$  containing  $x$  recovers the coaxial value of  $R^{(N)}$ , which lies within its height  $\rho_N = \delta$  of  $f(x)$ .

(2) and (3) At each level the trapezoids have  $\rho_k^{1/2}$ -separated projections, so for each  $x$  at most  $O(1)$  terms  $G_R$  with  $R \in \mathcal{R}_k$  are non-zero. For such  $R$  the height offset is  $h \lesssim \rho_{k-1}$  and the slope difference is  $|\tilde{s}| \lesssim \rho_{k-1}/a$ , so by Lemma 14.8.10,

$$|G'_R| \lesssim \rho_{k-1}/a, \quad |G''_R| \lesssim \rho_{k-1}/a^2,$$

with  $a \geq \rho_k^{1/2+\eta}$ . Summing over  $k$  the series is dominated by its last term, giving  $|g''| \lesssim \rho_{N-1}/\rho_N^{1+2\eta} = \delta^{-1/N-2\eta}$  and the stated bound on  $g'$ .  $\square$

*Remark 14.8.12* (On the constant  $a$ ). The source available to the author specifies the construction  $g = \sum_R G_R$  but was truncated at the choice of the parameter  $a$  in Lemma 14.8.10. The choice  $a = |\pi(R)|$  used above is the natural one — it is the largest value for which the supports at a given level remain  $O(1)$ -overlapping, by the separation in Corollary 14.8.9 — and it yields exactly the exponents asserted in items (2) and (3). A reader checking against the original should verify this point.

*Proof of Proposition 14.8.1.* Apply Proposition 14.7.3 to obtain an extremal family with Lipschitz slope function  $f$  and Lipschitz plane map. Apply Corollary 14.8.9 with  $N \geq 2/\varepsilon$ , then Proposition 14.8.11, obtaining  $g$  with  $|f-g| \leq \delta$  on  $\pi_z(E_{\mathbb{T}'})$  and  $\|g''\|_\infty \lesssim \delta^{-1/N-2\eta} \leq \delta^{-\varepsilon}$ . Replacing  $f$  by  $g$  changes the slices by  $O(\delta)$ , so Item 1 of Proposition 14.7.3 persists with  $g$  in place of  $f$ ; a final application of Lemma 14.6.9 normalises  $\|g\|_{C^2}$  to 1.  $\square$

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#### WORKED EXAMPLE 14.8.1— TIER A **Why $C^2$ and not merely Lipschitz**

Proposition 14.7.3 already gave a 1-Lipschitz slope function. The extra regularity is not cosmetic.

*What Lipschitz gives.* The curves  $\pi_f(T)$  of Chapter 14.9 would be merely Lipschitz, and a family of Lipschitz curves can be badly behaved — there is no maximal function theory for them.

*What  $C^2$  gives.* The curves become  $C^2$ , and a family of  $C^2$  curves with  $|f'|$  bounded below satisfies Sogge's *cinematic curvature* condition. For such families there is a Wolff-type  $L^3$  maximal estimate, and that is exactly the input Chapter 14.9 needs.

*The cost.* An entire chapter, and the projection theorem Theorem 14.8.2. The regularity is bought with the hardest single ingredient in the proof.

*The pattern.* Volume 5 recorded that stationary phase needs  $C^\infty$  but delivers under  $C^2$ ; Volume 12's wave packets need the second fundamental form.

*Second-order regularity is the threshold at which oscillatory methods become available*, and this chapter is an unusually explicit instance: one proves  $C^2$  precisely in order to unlock a maximal function estimate.

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**WORKED EXAMPLE 14.8.2— TIER B The trilinear expression, traced**

Step 5 of Lemma 14.8.5 produces  $a \cdot (b_1 - b_2)$  from three grain relations. Following it concretely clarifies where each ingredient enters.

*Start.*  $Q_1$  at height  $z_0$ , with  $x_1 \approx w$ .

*Local grain*  $Q_1 \rightarrow Q_2$ . Same  $y$ ; the relation involves  $g(y_1)$ , giving  $x_2 \approx w - z_2 g^\dagger(y_1)$ . *The local grain structure enters here* — Item 2 of Proposition 14.7.3.

*Global grain*  $Q_2 \rightarrow Q_3$ . Same  $z$ ; the relation involves  $f(z_2)$ , giving  $x_3 \approx x_2 + (y_1 - y_3) f^\dagger(z_2)$ . *The global grain structure enters here* — Item 1.

*Local grain*  $Q_3 \rightarrow Q_4$ . Same  $y$  again, giving (14.8.10).

*The result.*  $x_4$  lies in the slice ADset  $A$ , and the expression is bilinear in  $(f^\dagger, g^\dagger)$  and linear in the increments — that is, trilinear. Setting  $a = (y, g^\dagger(y))$  and  $b = (z, f^\dagger(z))$  up to rescaling turns it into  $a \cdot (b_1 - b_2)$ .

*Why a closed path of length three.* Two steps would give only a bilinear expression, which the ADset condition does not constrain; four would introduce a further unknown. Three is the least number closing the path while producing a trilinear form, and Theorem 14.8.2 is stated for exactly that form.

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## Drills

**Drill 14.8.1.** Verify the computation in Remark 14.8.3 showing  $\omega \cdot (\zeta_1 - \zeta_2)$  is real, and confirm both alternatives fail.

**Drill 14.8.2.** Check (14.8.6) and deduce (14.8.7).

**Drill 14.8.3.** Verify the matching conditions in Lemma 14.8.10 making  $G$  twice differentiable at  $x_1$  and  $x_2$ .

**Drill 14.8.4.** Confirm that with  $a = |\pi(R)| \geq \rho_k^{1/2+\eta}$  the sum in Proposition 14.8.11 is dominated by its last term.

## Notes

This chapter is Section 5 of Wang and Zahl; Theorem 14.8.2 is their Theorem 5.2, proved in their Section 8 and here in Chapter 14.11. Remark 14.8.3 is

their Remark 5.1 and is the conceptual heart of the proof. Remark 14.8.12 records one point where the available source was incomplete.



## Chapter 14.9

# Twisted Projections

Everything is now in place. Proposition 14.8.1 gives an extremal family whose slope function is  $C^2$  and whose local grain structure is Lipschitz. This chapter applies a single projection  $\pi_f$  and derives a contradiction, unless  $\sigma = 0$ .

### 14.9.1 The twisted projection

**Definition 14.9.1.** Let  $f : [-1, 1] \rightarrow \mathbb{R}$  be  $C^2$ . The associated *twisted projection* is

$$\pi_f : \mathbb{R}^3 \rightarrow \mathbb{R}^2, \quad \pi_f(x, y, z) = (x + y f(z), z).$$

**Lemma 14.9.2** ( $\pi_f$  computes the slice functional). *For each  $z_0$ ,*

$$\pi_f(E \cap \{z = z_0\}) = \left( (E \cap \{z = z_0\}) \cdot (1, f(z_0), 0) \right) \times \{z_0\}.$$

*Proof.* On  $\{z = z_0\}$  the first coordinate of  $\pi_f$  is  $x + y f(z_0)$ , which is exactly  $(x, y, z_0) \cdot (1, f(z_0), 0)$ .  $\square$

**Where we are.** This is why  $\pi_f$  is the right map, and why Chapter 14.8 worked so hard for a  $C^2$  slope function. *Item 1 of Proposition 14.8.1 is precisely a statement about the fibres of  $\pi_f$ :* each fibre is a  $(\delta, 1 - \sigma)$ -ADset. The map was designed so that the global grain structure becomes a statement about its image.

The name:  $\pi_f$  is the vertical projection  $(x, y, z) \mapsto (x, z)$  *twisted* by a shear whose magnitude varies with  $z$ . If  $f$  were constant it would be an ordinary projection after a linear change of variables, and the argument below would collapse.

**Proposition 14.9.3** (The image is small). *Let  $(\mathbb{T}, Y)_\delta$  satisfy Item 1 of Proposition 14.8.1. Then*

$$|\pi_f(E_{\mathbb{T}})| \lesssim \delta^{\sigma - \varepsilon}.$$

*Proof.* By Lemma 14.9.2 and Item 1, for each  $z_0$  the set  $\pi_f(E_{\mathbb{T}}) \cap \{z = z_0\}$  is a  $(\delta, 1 - \sigma, \delta^{-\varepsilon})_1$ -ADset. Applying (14.7.1) with  $\rho = \delta$  and  $r = 1$ , its  $\delta$ -covering number is at most  $\delta^{-\varepsilon} \delta^{-(1-\sigma)}$ , so its measure is at most  $\delta^{-\varepsilon} \delta^\sigma$ . Integrating over  $z_0 \in [-1, 1]$  gives the claim.  $\square$

**Where we are.** Numerically, at  $\delta = 10^{-6}$ : for  $\sigma = \frac{1}{2}$  each fibre has covering number  $\leq 10^3$  and measure  $\leq 10^{-3}$ ; for  $\sigma = \frac{1}{4}$ , covering number  $3.2 \times 10^4$  and measure  $3.2 \times 10^{-2}$ ; for  $\sigma = 0$ , measure  $O(1)$ . So  $|\pi_f(E_{\mathbb{T}})| \lesssim \delta^\sigma$  is a genuine constraint exactly when  $\sigma > 0$  — which is the case we intend to exclude.

## 14.9.2 Curves and cinematic curvature

**Lemma 14.9.4** (Tubes map to curved slabs). *Let  $T$  be a  $\delta$ -tube whose coaxial line is  $\{(a + tv_1, b + tv_2, tv_3)\}$  with  $v_3 \geq \frac{1}{2}$ . Then  $\pi_f(T)$  is contained in the  $C\delta$ -neighbourhood of the curve*

$$\gamma_T = \left\{ \left( a + \frac{v_1}{v_3} z + \left( b + \frac{v_2}{v_3} z \right) f(z), z \right) : z \in [-1, 1] \right\}, \quad (14.9.1)$$

and  $\gamma_T$  is  $C^2$  with  $\|\gamma_T\|_{C^2} \lesssim 1 + \|f\|_{C^2}$ .

*Proof.* Parametrise the coaxial line by  $z = tv_3$ , so  $t = z/v_3$  and the line is  $(a + \frac{v_1}{v_3} z, b + \frac{v_2}{v_3} z, z)$ . Applying Definition 14.9.1 gives (14.9.1). Since  $\pi_f$  is Lipschitz with constant  $O(1 + \|f\|_\infty)$  on  $[-1, 1]^3$ , the  $\delta$ -neighbourhood of the line maps into the  $C\delta$ -neighbourhood of  $\gamma_T$ . The regularity follows since  $f \in C^2$  and the coefficients are constants.  $\square$

**Definition 14.9.5** (Cinematic curvature). A family  $\{\gamma_\lambda\}$  of  $C^2$  curves in  $\mathbb{R}^2$ , parametrised by  $\lambda$  in a compact set, satisfies *Sogge's cinematic curvature condition* if, writing  $\gamma_\lambda$  as the graph  $u = \Gamma(z; \lambda)$ ,

$$\left| \det \begin{pmatrix} \partial_\lambda \Gamma & \partial_\lambda \partial_z \Gamma \\ \partial_\lambda^2 \Gamma & \partial_\lambda^2 \partial_z \Gamma \end{pmatrix} \right| \gtrsim 1$$

uniformly. Equivalently: distinct curves in the family are transverse, and the transversality is non-degenerate to second order.

**Lemma 14.9.6** (The curves are cinematic). *Suppose  $|f'(z)| \geq c > 0$  for all  $z$ . Then the family  $\{\gamma_T\}_{T \in \mathbb{T}}$  of Lemma 14.9.4, parametrised by  $\lambda = (a, b, \frac{v_1}{v_3}, \frac{v_2}{v_3})$ , satisfies Definition 14.9.5 with implicit constant depending on  $c$  and  $\|f\|_{C^2}$ .*

*Proof.* From (14.9.1),  $\Gamma(z; \lambda) = a + \lambda_3 z + (b + \lambda_4 z) f(z)$ , so

$$\partial_b \Gamma = f(z), \quad \partial_{\lambda_4} \Gamma = z f(z), \quad \partial_z \partial_b \Gamma = f'(z), \quad \partial_z \partial_{\lambda_4} \Gamma = f(z) + z f'(z).$$

The relevant determinant is

$$\det \begin{pmatrix} f(z) & f'(z) \\ zf(z) & f(z) + zf'(z) \end{pmatrix} = f(z)^2 + zf(z)f'(z) - zf(z)f'(z) = f(z)^2,$$

and after the normalisation of Proposition 14.8.1 — which may be arranged so that  $|f| \gtrsim 1$  on the relevant range, since  $f$  is  $C^2$  with  $|f'| \geq c$  — this is bounded below.  $\square$

*Remark 14.9.7* (On the lower bound for  $f'$ ). Lemma 14.9.6 assumes  $|f'| \geq c > 0$ . Establishing this is Section 6 of Wang and Zahl, and the source available to the author did not extend that far.

The mechanism they describe is the following. The tubes of  $\mathbb{T}$  point in  $\delta$ -separated directions, so as  $z$  varies the plane  $V(p)^\perp$  must genuinely rotate — if  $f'$  vanished on an interval, the grains over that interval would be parallel, and the tubes meeting them would share a direction, contradicting the separation. Making this quantitative, and uniform in  $z$ , is their Section 6.

*This is recorded as an omission*, in the manner of the rest of this series: what is missing is the lower bound  $|f'| \geq c$ ; what depends on it is Lemma 14.9.6 and hence Proposition 14.9.8.

*But nothing in the theorem depends on it.* Chapter 14.10 gives a second route to Theorem 14.2.5 which does not pass through  $\pi_f$  at all: it goes from grain structure directly to the conjugation structure (14.10.4) and finishes with the sum-product theorem. That route uses neither  $|f'| \geq c$  nor Proposition 14.9.8. So this gap costs only *this* presentation of the endgame, not the result — see Worked Example 14.10.2.

**Proposition 14.9.8** (Cinematic maximal estimate; Pramanik–Yang–Zahl). *Let  $\{\gamma_T\}$  be a family of  $\delta$ -neighbourhoods of  $C^2$  curves in  $[-1, 1]^2$  satisfying Definition 14.9.5, with  $\approx \delta^{-2}$  members whose parameters are  $\delta$ -separated. Then for every  $\varepsilon > 0$ ,*

$$\left| \bigcup_T \gamma_T \right| \geq c_\varepsilon \delta^\varepsilon.$$

*Remark 14.9.9* (Scope). Proposition 14.9.8 is stated and not proved. Revision R22.

*What it is.* A Wolff-type  $L^3$  maximal function estimate for families of curves with cinematic curvature, generalising Wolff’s circular maximal function theorem. It is due to Pramanik, Yang and Zahl.

*Why it is available.* Precisely because Chapter 14.8 proved the slope function is  $C^2$ . For merely Lipschitz curves no such estimate exists, and Worked Example 14.8.1 records that this is the whole reason for that chapter.

*What depends on it.* Theorem 14.2.5, hence Theorem 14.2.6.

### 14.9.3 The contradiction

*Proof of Theorem 14.2.5.* Suppose  $\sigma = \sigma_3 > 0$ ; we derive a contradiction. Fix  $\varepsilon \in (0, \sigma/2)$ .

**Step 1.** *Produce the configuration.*

By Proposition 14.8.1 there are  $\delta$  arbitrarily small and an  $\varepsilon'$ -extremal  $(\mathbb{T}, Y)_\delta$ , with  $\varepsilon'$  to be chosen, carrying a  $C^2$  slope function  $f$  with  $\|f\|_{C^2} \leq 1$  and a Lipschitz plane map. By Lemma 14.6.3,  $\#\mathbb{T} \approx \delta^{-2}$  and the directions are  $\delta$ -separated.

**Step 2.** *Upper bound.*

By Proposition 14.9.3,

$$|\pi_f(E_{\mathbb{T}})| \lesssim \delta^{\sigma-\varepsilon'}. \quad (14.9.2)$$

**Step 3.** *Lower bound.*

By Lemma 14.9.4 each  $\pi_f(T)$  lies in the  $C\delta$ -neighbourhood of a  $C^2$  curve  $\gamma_T$ ; by Remark 14.9.7 and Lemma 14.9.6 the family is cinematic; and by Step 1 there are  $\approx \delta^{-2}$  of them with  $\delta$ -separated parameters. Hence Proposition 14.9.8 applies and

$$|\pi_f(E_{\mathbb{T}})| \geq c_\varepsilon \delta^\varepsilon. \quad (14.9.3)$$

**Step 4.** *Compare.*

Combining (14.9.2) and (14.9.3),

$$c_\varepsilon \delta^\varepsilon \lesssim \delta^{\sigma-\varepsilon'}.$$

Choosing  $\varepsilon' < \sigma - 2\varepsilon$  — possible since  $\varepsilon < \sigma/2$  — gives  $\delta^\varepsilon \lesssim_\varepsilon \delta^{\sigma-\varepsilon'} \leq \delta^{2\varepsilon}$ , that is  $1 \lesssim_\varepsilon \delta^\varepsilon$ . Letting  $\delta \rightarrow 0$  along the sequence supplied by Step 1 is a contradiction.

Hence  $\sigma = 0$ , and by Proposition 14.3.9 every stickyakeya set in  $\mathbb{R}^3$  has  $\dim_{\mathbb{H}} K \geq 3$ ; the reverse inequality is trivial.  $\square$

**Where we are.** The contradiction is a collision between a covering-number bound and a maximal function bound, and it is worth seeing the numbers. At  $\delta = 10^{-8}$  and  $\sigma = \frac{1}{2}$ : the upper bound gives  $|\pi_f(E)| \leq 10^{-4}$ ; the lower bound with  $\varepsilon = \frac{1}{4}$  gives  $|\pi_f(E)| \geq c \cdot 10^{-2}$ . The two are incompatible by a factor 100, and the discrepancy grows as  $\delta \rightarrow 0$ .

*The contradiction bites whenever  $\varepsilon < \sigma$ , so no positive value of  $\sigma$  survives.* That is why the conclusion is  $\sigma = 0$  exactly rather than  $\sigma \leq$  some bound — the argument, once it works at all, works completely.

*Proof of Theorem 14.2.6.* By Theorem 14.2.5 every sticky Kakeya set in  $\mathbb{R}^3$  has Hausdorff dimension 3. By Theorem 14.2.7 — Remark 14.2.8 — this implies the same for every Kakeya set. The Minkowski statement follows since  $\dim_M \geq \dim_H$  and the reverse is trivial.  $\square$

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WORKED EXAMPLE 14.9.1— TIER A **The proof in one page**

For revision, the whole argument compressed.

*Setup.* Define  $\sigma_n$ , the exponent measuring how small an extremal Kakeya tube family's union can be (Definition 14.3.5). Sticky Kakeya in  $\mathbb{R}^3$  is the statement  $\sigma_3 = 0$  (Proposition 14.3.9).

*Step 1: self-similarity.* An extremal family looks extremal at every scale, both coarsely and inside each fat tube (Theorem 14.6.7). Consequence: robust transversality.

*Step 2: planiness and graininess.* Multilinear Kakeya — proved in Volume 11 — forces most triples of tubes through a point to be coplanar, giving a plane map (Lemma 14.7.5); a Córdoba  $L^2$  argument then organises the set into grains (Proposition 14.7.3).

*Step 3:  $C^2$  regularity.* Chaining three grain relations produces a trilinear expression, and a projection theorem *true over  $\mathbb{R}$  and false over  $\mathbb{C}$*  forces the slope function to be locally linear at every scale, hence  $C^2$  (Proposition 14.8.1). *This is the step that excludes the Heisenberg example.*

*Step 4: contradiction.* The twisted projection  $\pi_f$  has fibres exactly the grains, so its image has measure  $\lesssim \delta^\sigma$  (Proposition 14.9.3). But the tubes map to a cinematic family of  $C^2$  curves, whose union has measure  $\geq c_\varepsilon \delta^\varepsilon$  (Proposition 14.9.8). For  $\varepsilon < \sigma$  these are incompatible.

*The one-sentence version.* An extremal Kakeya set in  $\mathbb{R}^3$  would have to be self-similar, hence grainy, hence have a  $C^2$  slope function — and then a single projection is simultaneously too small and too large.

---

WORKED EXAMPLE 14.9.2— TIER B **What this series proves of the Kakeya theorem**

An honest accounting, in the manner of Volume 13, Worked Example 13.8.3.

*Proved here in full.* The discretisation and Proposition 14.3.9; multi-scale self-similarity (Theorem 14.6.7) including the lemma Wang and Zahl omit as tedious (Remark 14.6.10); the derivation of weighted multilinear Kakeya from Volume 11's proved theorem; planiness and graininess (Proposition 14.7.3); and the local-linearity argument (Lemma 14.8.5) modulo the projection theorem.

*Stated, proved later in this volume.* The projection theorem Theorem 14.8.2, in Chapter 14.11.

*Stated, not proved.* For *this* route: the sticky reduction (Remark 14.2.8, now largely discharged in Chapter 14.4 and Chapter 14.5); the lower bound  $|f'| \geq c$  (Remark 14.9.7); and the cinematic maximal estimate (Remark 14.9.9). *Chapter 14.10 reaches the same theorem needing none of the last two* — see Worked Example 14.10.2.

*What that means for a reader.* Granting those three, the proof of  $\dim_{\mathbb{H}} K = 3$  for Kakeya sets in  $\mathbb{R}^3$  is complete in these pages. Declining any of them costs the theorem but nothing else in the series — no result in Volumes 1–13 depends on it.

*Why these three and not others.* The sticky reduction and the maximal estimate are separate papers by other authors or in other settings; the bound on  $f'$  is a section of the paper the available source did not reach. Remark 14.9.7 is honest about the difference: the first two are scope decisions, the third is a limitation of the sources at hand.

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## Drills

**Drill 14.9.1.** Verify Lemma 14.9.2 and explain why it makes  $\pi_f$  the natural map.

**Drill 14.9.2.** Derive (14.9.1) by reparametrising the coaxial line by  $z$ .

**Drill 14.9.3.** Compute the determinant in Lemma 14.9.6 and confirm the cancellation leaving  $f(z)^2$ .

**Drill 14.9.4.** For  $\delta = 10^{-8}$  and  $\sigma = \frac{1}{4}$ , compute both sides of Step 4 with  $\varepsilon = \frac{1}{8}$  and confirm the contradiction.

## Notes

This chapter is Sections 6 and 7 of Wang and Zahl. Remark 14.9.7 records that their Section 6 was not available to the author in detail; Proposition 14.9.8 is due to Pramanik, Yang and Zahl and generalises Wolff’s circular maximal function theorem. The determinant computation of Lemma 14.9.6 is standard for cinematic families.

# Chapter 14.10

## The Perfect Overlap Route

Chapter 14.7 obtained grain structure from multilinearakeya, and Chapter 14.8 obtained the  $C^2$  slope function from a projection theorem. There is a second route to the same structures, due to Katz, Łaba and Tao and set out in Guth's survey, which reaches them from a single observation about multiplicities. It is shorter, it makes the rôle of the Heisenberg example explicit, and it ends in a self-contained algebraic lemma that is the precise point at which  $\mathbb{R}$  parts company with  $\mathbb{C}$ .

### 14.10.1 The exponent $\beta$

**Definition 14.10.1** (Density, multiplicity, and  $\beta$ ). For a set  $\mathbb{T}$  of  $\delta$ -tubes in  $\mathbb{R}^3$  and convex  $K$  put

$$\Delta(\mathbb{T}, K) = \frac{\sum_{T \in \mathbb{T}[K]} |T|}{|K|}, \quad \Delta_{\max}(\mathbb{T}) = \max_{K \text{ convex}} \Delta(\mathbb{T}, K), \quad \mu(\mathbb{T}) = \frac{\sum_{T \in \mathbb{T}} |T|}{|U(\mathbb{T})|},$$

where  $U(\mathbb{T}) = \bigcup_{T \in \mathbb{T}} T$ . Let  $\beta$  be the infimal exponent with

$$\mu(\mathbb{T}) \lesssim |\mathbb{T}|^\beta \quad \text{whenever } \Delta_{\max}(\mathbb{T}) \lesssim 1. \quad (14.10.1)$$

Call  $\mathbb{T}$  a *worst-caseakeya set* if  $\Delta_{\max}(\mathbb{T}) \lesssim 1$  and  $\mu(\mathbb{T}) \approx |\mathbb{T}|^\beta$ .

**Where we are.**  $\mu(\mathbb{T})$  is the average number of tubes through a point of  $U(\mathbb{T})$ , so (14.10.1) says the tubes overlap at most  $|\mathbb{T}|^\beta$ -fold. Theorem 14.4.8 is exactly  $\beta = 0$ , and Theorem 14.2.6 follows.

This is the same content as  $\sigma_3 = 0$  in Definition 14.3.5, in a normalisation better suited to the multiscale bookkeeping below:  $\beta$  is attached to a *multiplicity* rather than to a volume, and multiplicities multiply across scales (Lemma 14.10.2).

*Why the convex hypothesis.*  $\Delta_{\max}(\mathbb{T}) \lesssim 1$  says the only way tubes can concentrate is by packing into a convex set. It holds for aakeya family, and it is what Definition 14.4.1 calls the Katz–Tao Convex Wolff axioms.

**Lemma 14.10.2** (Multiplicities multiply). *For  $\delta \leq \rho \leq 1$ ,  $\mu(\mathbb{T}) \lesssim \mu(\mathbb{T}_\rho) \mu(\mathbb{T}[T_\rho])$ .*

*Proof.* Fix  $x \in U(\mathbb{T})$ . It lies in  $\approx \mu(\mathbb{T}_\rho)$  tubes  $T_\rho \in \mathbb{T}_\rho$ , and for each of these in at most  $\mu(\mathbb{T}[T_\rho])$  tubes  $T \in \mathbb{T}[T_\rho]$ . Every  $T \in \mathbb{T}$  containing  $x$  lies in some such  $T_\rho$ , so the count multiplies.  $\square$

**Proposition 14.10.3** (Extremality propagates, and the inequality is sharp). *Let  $\mathbb{T}$  be a worst-case stickyakeya set. Then for every  $\rho \in [\delta, 1]$ , both  $\mathbb{T}_\rho$  and  $\mathbb{T}[T_\rho]$  are worst-case stickyakeya sets, and Lemma 14.10.2 holds with equality.*

*Proof.* Stickiness gives  $|\mathbb{T}_\rho| \approx \rho^{-2}$  and  $|\mathbb{T}[T_\rho]| \approx (\rho/\delta)^2$ , and both derived families are again sticky. So by the definition of  $\beta_{\text{st}}$ ,  $\mu(\mathbb{T}_\rho) \lesssim |\mathbb{T}_\rho|^{\beta_{\text{st}}}$  and likewise for  $\mathbb{T}[T_\rho]$ . Chaining with Lemma 14.10.2 and  $|\mathbb{T}| \approx |\mathbb{T}_\rho| |\mathbb{T}[T_\rho]|$ ,

$$|\mathbb{T}|^{\beta_{\text{st}}} \approx \mu(\mathbb{T}) \lesssim \mu(\mathbb{T}_\rho) \mu(\mathbb{T}[T_\rho]) \lesssim |\mathbb{T}_\rho|^{\beta_{\text{st}}} |\mathbb{T}[T_\rho]|^{\beta_{\text{st}}} \approx |\mathbb{T}|^{\beta_{\text{st}}}. \quad (14.10.2)$$

The two ends agree, so every inequality in (14.10.2) is an approximate equality. That gives both assertions.  $\square$

**Where we are.** Proposition 14.10.3 is the engine of this route, and it costs nothing: a worst-case example is worst-case at every scale *because* the trivial inequality Lemma 14.10.2 is forced to be sharp. Compare Theorem 14.6.7, which reaches the same conclusion through the shading machinery of Chapter 14.3; here the sticky hypothesis makes the argument three lines.

## 14.10.2 Perfect overlap

**Lemma 14.10.4** (Perfect overlap). *Let  $\mathbb{T}$  be a worst-case stickyakeya set and  $B_\rho \subseteq U(\mathbb{T}_\rho)$ . Then for every  $T_\rho \in \mathbb{T}_\rho$  meeting  $B_\rho$ ,*

$$|U(\mathbb{T}[T_\rho]) \cap B_\rho| \approx |U(\mathbb{T}) \cap B_\rho|.$$

*Proof.* Write  $\mathbb{T}_x = \{T : x \in T\}$  and let  $\mathbb{T}_{\rho, B_\rho}$  be the fat tubes meeting  $B_\rho$ . If  $x$  lies in  $U(\mathbb{T}[T_\rho])$  for *every*  $T_\rho \in \mathbb{T}_{\rho, B_\rho}$  then  $|\mathbb{T}_x| \approx |\mathbb{T}_{\rho, B_\rho}| |\mathbb{T}[T_\rho]| \approx \mu(\mathbb{T}_\rho) \mu(\mathbb{T}[T_\rho])$ , which is equality in Lemma 14.10.2. If instead  $x$  lies in  $U(\mathbb{T}[T_\rho])$  for only a small fraction of them, then  $|\mathbb{T}_x| \ll \mu(\mathbb{T}_\rho) \mu(\mathbb{T}[T_\rho])$ .

By Proposition 14.10.3 equality holds in Lemma 14.10.2, and by the uniformity convention  $|\mathbb{T}_x| \approx \mu(\mathbb{T})$  for every  $x \in U(\mathbb{T})$ . So the second alternative is excluded for almost every  $x$ , which is the assertion.  $\square$



**Where we are.** The condition is far stronger than it looks. Since  $\beta > 0$  we have  $|U(\mathbb{T})| \ll 1$ , and one checks that  $|U(\mathbb{T}) \cap B_\rho| \ll |B_\rho|$ . So the sets  $U(\mathbb{T}[T_\rho]) \cap B_\rho$  are *small* subsets of  $B_\rho$  — and Lemma 14.10.4 says that all of them, over the many  $T_\rho$  meeting  $B_\rho$ , coincide almost exactly.

**Lemma 14.10.5** (Transverse fat tubes force filling). *In  $\mathbb{R}^2$  with  $\rho = \sqrt{\delta}$ , suppose  $T_{\rho,1}, T_{\rho,2}$  pass through  $B_\rho$  at angle  $\approx 1$  and  $U(\mathbb{T}[T_{\rho,1}]) \cap B_\rho = U(\mathbb{T}[T_{\rho,2}]) \cap B_\rho \neq \emptyset$ . Then  $|U(\mathbb{T}[T_{\rho,1}]) \cap B_\rho| \approx |B_\rho|$ .*

*Proof.* Let  $T_1 \in \mathbb{T}[T_{\rho,1}]$  meet  $B_\rho$ . Every  $x \in T_1 \cap B_\rho$  lies, by hypothesis, in some  $T_{2,x} \in \mathbb{T}[T_{\rho,2}]$ . The tubes  $T_{2,x}$  are mutually parallel — all lying in the single fat tube  $T_{\rho,2}$  — and roughly perpendicular to  $T_1$ . Sweeping  $x$  along  $T_1 \cap B_\rho$  therefore sweeps  $T_{2,x}$  across  $B_\rho$ , filling a definite fraction of it.  $\square$

**Theorem 14.10.6** (Grain structure). *Suppose  $\beta_{\text{st}} > 0$  and  $\mathbb{T}$  is a worst-case sticky Kakeya set. Then for a typical ball  $B = B_{\sqrt{\delta}} \subseteq U(\mathbb{T}_{\sqrt{\delta}})$  there are coordinates in which*

$$U(\mathbb{T}) \cap B = [0, \sqrt{\delta}]^2 \times A, \quad A \subseteq [0, \sqrt{\delta}],$$

*and the tubes of  $\mathbb{T}_{\sqrt{\delta}, B}$  are parallel to the  $(x_1, x_2)$ -plane.*

*Proof.* At  $\rho = \sqrt{\delta}$ , two tubes of  $\mathbb{T}[T_{\sqrt{\delta}}]$  meeting  $B_{\sqrt{\delta}}$  are contained in parallel tubes of length  $\sqrt{\delta}$  and radius  $2\delta$ : their directions differ by  $O(\sqrt{\delta})$  and they traverse a distance  $\sqrt{\delta}$ , so they separate by  $O(\delta)$ .

In  $\mathbb{R}^2$ , Lemma 14.10.5 shows perfect overlap with  $|U(\mathbb{T}) \cap B_\rho| \ll |B_\rho|$  forces all the fat tubes through  $B_\rho$  into a narrow angular sector. In  $\mathbb{R}^3$  the same argument leaves one further possibility: the fat tubes may be parallel to a common plane, with each  $U(\mathbb{T}[T_{\sqrt{\delta}}]) \cap B$  equal to one slab  $G$  of dimensions  $\delta \times \sqrt{\delta} \times \sqrt{\delta}$  parallel to it. Tubes running in any direction of that plane then fill  $G$ , and perfect overlap holds.

A worst-case sticky Kakeya set cannot have its fat tubes in a narrow angular sector — that would violate  $\Delta_{\text{max}} \lesssim 1$ . So the second possibility occurs, and  $U(\mathbb{T}) \cap B$  is a union of parallel slabs, which is the stated form.  $\square$

**Proposition 14.10.7** (The grain spacing is Ahlfors regular). *With  $\mathbb{T}$  and  $A$  as in Theorem 14.10.6, for every  $\rho \in [\delta, \sqrt{\delta}]$  and every interval  $I_\rho$  of length  $\rho$  centred at a point of  $A$ ,*

$$\mathcal{E}_\delta(A \cap I_\rho) \approx \left(\frac{\rho}{\delta}\right)^{1-2\beta_{\text{st}}}. \quad (14.10.3)$$

*Proof.* Since  $|\mathbb{T}| \approx \delta^{-2}$  and  $\mu(\mathbb{T}) \approx \delta^{-2\beta_{\text{st}}}$ , and  $\mu(\mathbb{T}) \approx |U(\mathbb{T})|^{-1}$ , we get  $|U(\mathbb{T})| \approx \delta^{2\beta_{\text{st}}}$ . By Proposition 14.10.3 the same holds at every scale:  $|U(\mathbb{T}_\rho)| \approx$

$\rho^{2\beta_{\text{st}}}$ . As  $U(\mathbb{T}_\rho)$  is the  $\rho$ -neighbourhood of  $U(\mathbb{T})$ , uniformity gives

$$|U(\mathbb{T})| \approx \frac{|U(\mathbb{T}) \cap B_\rho|}{|B_\rho|} |U(\mathbb{T}_\rho)|, \quad \text{so} \quad |U(\mathbb{T}) \cap B_\rho| \approx \left(\frac{\delta}{\rho}\right)^{2\beta_{\text{st}}} |B_\rho|.$$

For  $\rho \leq \sqrt{\delta}$  the geometry of  $U(\mathbb{T}) \cap B_\rho$  is that of Theorem 14.10.6 and is governed by  $A$ ; reading the display in terms of  $\delta$ -covering numbers gives (14.10.3).  $\square$

**Where we are.** (14.10.3) says  $A$  is the  $\delta$ -neighbourhood of an Ahlfors–David regular set of dimension  $1 - 2\beta_{\text{st}}$  — the discretised form of Definition 14.7.2. If  $\beta_{\text{st}} = 0$  then  $A$  has dimension 1 and fills the interval; the content is that  $\beta_{\text{st}} > 0$  forces  $A$  to be a genuine fractal, and that is what the sum-product theorem will contradict.

### 14.10.3 Complex conjugation structure

**Definition 14.10.8** (Slope functions). Fix  $T_1 \in \mathbb{T}$  and coordinates making its core line the  $x$ -axis. Each point  $(x, 0, 0)$  lies in a grain box whose grains are parallel to the  $x$ -axis, hence of the form  $z = s(x)y$ ; call  $s$  the *slope function*. Rescaling  $T[T_\rho]$  to the unit cube shows  $U(\mathbb{T}[T_\rho])$  is organised into *long grain boxes* of dimensions  $\sqrt{\delta\rho} \times \sqrt{\delta\rho} \times \sqrt{\delta/\rho}$ , whose *long grains* have slope written  $-\tilde{s}(y)$ .

**Theorem 14.10.9** (Complex conjugation structure). *Let  $\mathbb{T}$  be a worst-case sticky Keakeya set and  $A, s, \tilde{s}$  as above. Then for  $x \in [0, \delta^{1/4}]$  and  $y \in [0, \delta^{1/2}]$ ,*

$$A + s(x)y + \tilde{s}(y)x \approx A. \quad (14.10.4)$$

*Proof.* Choose coordinates so that the first grain box is  $[0, \sqrt{\delta}]^3$  and  $\tilde{s}(0) = 0$ . Start at  $p_1 = (0, 0, z)$  with  $z \in A$  and traverse a closed cycle of four segments.

**Step 1.** *Along a long grain of  $\text{LGB}_1$ .*

Its slope is  $-\tilde{s}(0) = 0$ , so moving a distance  $x$  gives  $p_2 = (x, 0, z)$ .

**Step 2.** *Along a grain of  $\text{GB}_2$ .*

The grains there have slope  $s(x)$ , so moving a distance  $y$  in the  $y$ -direction gives  $p_3 = (x, y, z + s(x)y)$ .

**Step 3.** *Along a long grain of  $\text{LGB}_2$ .*

Its slope is  $-\tilde{s}(y)$ , and returning a distance  $x$  gives  $p_4 = (0, y, z + s(x)y + \tilde{s}(y)x)$ .

**Step 4.** *Close the cycle.*

$p_4$  lies in the original grain box  $\text{GB}_1$ , hence on one of its grains, so its  $z$ -coordinate lies in  $A$ . That is  $z + s(x)y + \tilde{s}(y)x \in A$ , which is (14.10.4).  $\square$

**Where we are.** The cycle is worth picturing as a ramp in a multi-storey car park: one leaves on one floor and returns on another, and the height gained is  $\Delta z = s(x)y + \tilde{s}(y)x$  regardless of the starting floor. The content of (14.10.4) is that  $A$  is invariant under *all* of these shifts.

In the Heisenberg example one has  $A = \mathbb{R} \cap B$ ,  $s(x) = \bar{x}$  and  $\tilde{s}(y) = \bar{y}$ , and

$$\mathbb{R} + x\bar{y} + y\bar{x} = \mathbb{R} + 2\Re(x\bar{y}) = \mathbb{R},$$

so (14.10.4) holds identically. *This is why the example survives every argument up to this point*, and why the contradiction must come from an algebraic fact about  $\mathbb{R}$  that fails in  $\mathbb{C}$ .

## 14.10.4 The step that distinguishes $\mathbb{R}$ from $\mathbb{C}$

**Theorem 14.10.10** (Bourgain’s discretized sum-product theorem; not proved here). *Let  $0 < s < 1$  and  $A \subseteq [0, 1]$  with  $\mathcal{E}_\delta(A) \approx \delta^{-s}$  and  $\mathcal{E}_\delta(A \cap B(x, r)) \lesssim (r/\delta)^s$  for all balls. Then*

$$\mathcal{E}_\delta(A + A) + \mathcal{E}_\delta(A \cdot A) \gtrsim \delta^{-s-\varepsilon}$$

for some  $\varepsilon = \varepsilon(s) > 0$ .

*Remark 14.10.11* (Why a non-sharp theorem suffices). Theorem 14.10.10 is very weak: it improves on the trivial bound by an unspecified  $\varepsilon$ . Yet it powers a *sharp* theorem, by the mechanism of Worked Example 14.4.3 — one applies it at many scales and accumulates the gains. Precisely, the sticky theorem follows by iterating:

for every  $\beta > 0$  there are  $\varepsilon, K > 0$  with  $\beta_{\text{st}}(\delta_2) < \beta_{\text{st}}(\delta_1) - \varepsilon$  whenever  $\beta_{\text{st}}(\delta_1) \geq \beta$  and  $\delta_2 \leq \delta_1^K$ .

The failure of Theorem 14.10.10 over  $\mathbb{C}$  is immediate: take  $A = \mathbb{R} \cap B(0, 1)$ , a subring, so that  $A + A$  and  $A \cdot A$  are no larger than  $A$ . *The counterexample is exactly the subfield*, and the next lemma isolates the finite-field shadow of that fact — with a complete proof.

**Lemma 14.10.12** (Sum-product over  $\mathbb{F}_q$ , after Guth). *Let  $p$  be prime and  $A \subseteq \mathbb{F}_p$ . Then at least one of:*

$$(i) \quad \frac{A - A}{A - A} = \mathbb{F}_p;$$

$$(ii) \quad \left| \frac{(A \cdot A)^{\oplus 3} - (A \cdot A)^{\oplus 3}}{A - A} \right| \geq |A|^2,$$

where  $(A \cdot A)^{\oplus 3} = \{a_1a_2 + a_3a_4 + a_5a_6 : a_i \in A\}$ .

*Proof. Step 1. Elements outside the ratio set expand.*

If  $c \notin \frac{A-A}{A-A}$  then  $|A + cA| = |A|^2$ . Otherwise two pairs collide,  $a_1 + ca_2 = a'_1 + ca'_2$  with  $(a_1, a_2) \neq (a'_1, a'_2)$ , giving  $c = \frac{a'_1 - a_1}{a_2 - a'_2} \in \frac{A-A}{A-A}$ .

**Step 2.** *A proper ratio set has a boundary point.*

Suppose  $S = \frac{A-A}{A-A} \neq \mathbb{F}_p$ . If  $b + 1 \in S$  for every  $b \in S$  then  $S$ , being non-empty, is closed under adding 1; since 1 generates  $(\mathbb{F}_p, +)$  this forces  $S = \mathbb{F}_p$ , a contradiction. So there is  $b \in S$  with  $b + 1 \notin S$ .

*This is the step that fails in  $\mathbb{F}_q$  with  $q = p^2$ :* taking  $A = \mathbb{F}_p \subseteq \mathbb{F}_q$  gives  $S = \mathbb{F}_p$ , which is closed under adding 1 and is not all of  $\mathbb{F}_q$ .

**Step 3.** *Expand and collect.*

By Steps 1 and 2,  $|A + (b + 1)A| \geq |A|^2$  with  $b \in S$ , say  $b = \frac{a_1 - a_2}{a_3 - a_4}$ . For  $a, a' \in A$ ,

$$a + (b+1)a' = \frac{(a + a')(a_3 - a_4) + a'(a_1 - a_2)}{a_3 - a_4} = \frac{aa_3 - aa_4 + a'a_3 - a'a_4 + a'a_1 - a'a_2}{a_3 - a_4},$$

whose numerator is a difference of two elements of  $(A \cdot A)^{\oplus 3}$  and whose denominator lies in  $A - A$ . Hence  $A + (b + 1)A \subseteq \frac{(A \cdot A)^{\oplus 3} - (A \cdot A)^{\oplus 3}}{A - A}$ , giving (ii).  $\square$

**Where we are.** Lemma 14.10.12 is the whole of this chapter in miniature, and it is proved here in full. Direct enumeration confirms Step 2: for  $p = 7, 11, 13$  and every  $A$  giving a proper ratio set, some  $b \in S$  has  $b + 1 \notin S$ ; and for  $A = \mathbb{F}_p \subseteq \mathbb{F}_p^2$  the conclusion fails, exactly as the proof predicts.

*Where the subfield obstruction lives.* Step 2 uses that 1 generates the additive group. In  $\mathbb{F}_p$  it does; in  $\mathbb{F}_p^2$  the subfield  $\mathbb{F}_p$  is an additive subgroup containing 1 and closed under adding it. Over  $\mathbb{R}$  and  $\mathbb{C}$  the same dichotomy appears, with  $\mathbb{R} \subseteq \mathbb{C}$  in place of  $\mathbb{F}_p \subseteq \mathbb{F}_p^2$ , and Theorem 14.10.10 is the quantitative form.

*This is the answer to the question Remark 14.8.3 raised.* That remark exhibited the counterexample; this lemma exhibits the mechanism, and shows it is a statement about additive generation rather than about geometry.

|            | Chapter 14.7                      | this chapter                         |
|------------|-----------------------------------|--------------------------------------|
| Hypothesis | $\varepsilon$ -extremal, shadings | worst-case <i>sticky</i>             |
| Engine     | multilinear Kakeya                | Lemma 14.10.2 forced sharp           |
| Route      | planiness $\rightarrow$ grains    | perfect overlap $\rightarrow$ grains |
| Output     | Proposition 14.7.3                | Theorem 14.10.6                      |
| Length     | a chapter                         | three lemmas                         |
| Needs      | Vol 11 Theorem 11.5.4             | nothing external                     |

*Why keep both.* The first applies to *any* extremal family and so survives the reduction of Chapter 14.4; the second assumes stickiness and is correspondingly shorter. The general theorem needs the first; a reader wanting the idea should read the second.

*Where they agree.* Both produce parallel slabs of dimensions  $\delta \times \sqrt{\delta} \times \sqrt{\delta}$  whose spacing is Ahlfors regular of dimension  $1 - 2\beta_{\text{st}}$ ; compare Proposition 14.10.7 with Item 1 of Proposition 14.7.3.

*The methodological point.* A structure theorem can often be reached from either an *estimate* that is sharp only for the extremal configuration, or from an *equality case* of a trivial inequality. The second is cheaper when available, and Proposition 14.10.3 is the instance here.

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WORKED EXAMPLE 14.10.2— TIER A **The two endgames, and what each costs**  
 The sticky theorem is reached twice in this volume, by different endgames. The comparison is worth making precise, because the two rest on *different* unproved inputs.

|               | Route A                                                                           | Route B                            |
|---------------|-----------------------------------------------------------------------------------|------------------------------------|
| Chapters      | Chapter 14.7, Chapter 14.8, Chapter 14.9                                          | Chapter 14.10                      |
| To grains     | multilinear Kakeya, planiness                                                     | perfect overlap                    |
| Endgame       | twisted projection $\pi_f$ , cinematic maximal estimate                           | conjugation structure, sum-product |
| Quoted inputs | Theorem 14.8.2 (R21/R23), Proposition 14.9.8 (R22), $ f'  \geq c$ (Remark 14.9.7) | Theorem 14.10.10 (R23)             |
| Proved inputs | Vol 11 Theorem 11.5.4                                                             | Vol 11 Theorem 11.5.4              |

*Route B rests on strictly less.* It needs one quoted theorem where Route A

needs three, and in particular it avoids Remark 14.9.7 — the one place in this volume where the exposition is thin because the available sources were incomplete.

*So the status of Theorem 14.2.5 in these pages is better than Worked Example 14.9.2 recorded:* granting Bourgain’s discretized sum-product theorem alone, the sticky Kakeya theorem is complete here. The cinematic maximal estimate and the bound on  $f'$  are needed only for the alternative presentation.

*Why keep Route A.* Three reasons. It is the historical route and the one in the published paper. It produces the  $C^2$  slope function, which is of independent interest and is what makes Proposition 14.9.8 applicable. And its projection theorem Theorem 14.8.2 is proved in Chapter 14.13 up to three quantitative inputs, so Route A is not merely quoted — most of it is established here too.

*The lesson for a reader auditing this series.* When a theorem has two proofs, the *intersection* of their hypotheses is what it really needs. Here that intersection is: multilinear Kakeya (proved, Volume 11) and Bourgain’s sum-product theorem.

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WORKED EXAMPLE 14.10.3— TIER B **Reading the two Wang–Zahl papers**

The proof is spread over two papers and a survey, with different notation.

(a) Sticky Kakeya sets and the sticky Kakeya conjecture (2026). The sticky case. Uses  $\sigma_3$  and the shading machinery of Chapter 14.3–Chapter 14.9.

(b) Volume estimates for unions of convex sets (2025). The reduction to the sticky case, via the assertions  $\mathcal{D}, \mathcal{E}$  of Chapter 14.4.

(c) *Guth’s Bourbaki survey*. Both, in the  $\beta$  and  $\mu$  notation of this chapter, with the perfect overlap route.

*Translating.*  $\sigma_3$  and  $\beta_{\text{st}}$  measure the same failure:  $|U(\mathbb{T})| \approx \delta^{2\beta_{\text{st}}}$  against  $|E_{\mathbb{T}}| \approx \delta^{\sigma_3}$ , so  $\sigma_3 \approx 2\beta_{\text{st}}$  under the normalisation  $|\mathbb{T}| \approx \delta^{-2}$ . Both vanish exactly when the conjecture holds.

*Which to read first.* (c), then (a), then (b). The survey states the four steps — sticky, grainy, conjugation structure, contradiction — and the papers supply the quantifiers.

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WORKED EXAMPLE 14.10.4— TIER C: RESEARCH TASTE **Equality cases as a source of structure**

Proposition 14.10.3 extracts everything from the observation that a trivial inequality must be sharp. That device appears repeatedly and is worth abstracting.

(a) *The pattern.* Find an inequality  $X \leq Y$  that is easy and general. Show that for the object under study  $X \approx Y$ . Then *every* step in the derivation of  $X \leq Y$  is an approximate equality, and each such step is a structural statement.

(b) *Here.* Lemma 14.10.2 is proved by a two-line counting argument. Forcing equality says the counting is lossless, which is exactly perfect overlap.

(c) *Elsewhere in this series.* Volume 11's Theorem 11.5.2 is sharp for boxes, and the sharpness case is what identifies the extremal configurations. Volume 13's Proposition 13.1.5(i) is an identity at  $p = 2$ , and the whole subject is the question of how far that persists.

(d) *Why it is cheap.* One proves nothing new; one re-reads an existing proof under the hypothesis that it loses nothing. The work is in finding the right inequality — one general enough to be easy, sharp enough to be informative.

(e) *The limitation.* It gives structure, not bounds. Proposition 14.10.3 says a worst-case example is self-similar; it says nothing about whether one exists. That is why the argument must still end in a contradiction, and why Theorem 14.4.8 carries no effective constant.

(f) *The question to ask of any extremal problem.* Which trivial inequalities are sharp for the extremiser? Each answer is a free structure theorem.

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## Drills

**Drill 14.10.1.** Prove Lemma 14.10.2 and explain why the bound is an inequality rather than an equality in general.

**Drill 14.10.2.** Verify the chain (14.10.2) and identify which two inequalities being equalities give the two conclusions of Proposition 14.10.3.

**Drill 14.10.3.** Trace the four segments of Theorem 14.10.9 and confirm  $\Delta z = s(x)y + \tilde{s}(y)x$ .

**Drill 14.10.4.** Check that  $A = \mathbb{F}_p \subseteq \mathbb{F}_p^2$  makes Step 2 of Lemma 14.10.12 fail, and say which hypothesis on  $p$  is used.

## Notes

This chapter follows Guth's Bourbaki survey of the Wang–Zahl theorem. The perfect overlap route to grain structure is due to Katz, Łaba and Tao (2000); Theorem 14.10.9 is based on unpublished work of Katz and Tao, carried out in

Wang–Zahl (2026). Lemma 14.10.12 is proved in full above; Theorem 14.10.10 is Bourgain (2003) and is recorded as quoted in Remark 14.13.15.



# Chapter 14.11

## The Projection Theorem

Theorem 14.8.2 is the one ingredient of Chapter 14.8 left unproved, and Remark 14.8.3 showed it is the point at which the argument becomes specific to  $\mathbb{R}$ . This chapter assembles it from the projection theory of the last five years.

### 14.11.1 The ingredients

**Definition 14.11.1** (Radial projection). For  $x \in \mathbb{R}^2$  the *radial projection* from  $x$  is

$$\pi_x : \mathbb{R}^2 \setminus \{x\} \rightarrow \mathbb{S}^1, \quad \pi_x(y) = \frac{y - x}{|y - x|}.$$

**Theorem 14.11.2** (Orponen–Shmerkin–Wang, Kaufman form). *Let  $X, Y \subseteq \mathbb{R}^2$  be non-empty Borel sets with  $X$  not contained in a line. Then*

$$\sup_{x \in X} \dim_{\mathbb{H}} \pi_x(Y \setminus \{x\}) \geq \min\{\dim_{\mathbb{H}} X, \dim_{\mathbb{H}} Y, 1\}.$$

**Theorem 14.11.3** (Orponen–Shmerkin–Wang, Falconer form). *If in addition  $\dim_{\mathbb{H}} Y > 1$ , then*

$$\sup_{x \in X} \dim_{\mathbb{H}} \pi_x(Y \setminus \{x\}) \geq \min\{\dim_{\mathbb{H}} X + \dim_{\mathbb{H}} Y - 1, 1\}.$$

**Where we are.** The hypothesis in Theorem 14.11.2 is exactly the degeneracy that alternative (A) of Theorem 14.8.2 describes: if  $X$  is contained in a line, every radial projection  $\pi_x$  collapses it and the conclusion fails. So the theorem says a line is the *only* obstruction — which is the same shape as Theorem 14.8.2, and not a coincidence.

The names come from the classical projection theorems: Kaufman’s gives  $\dim_{\mathbb{H}} \pi_{\theta}(Y) \geq \min\{\dim_{\mathbb{H}} Y, 1\}$  for orthogonal projections outside an exceptional set, and Falconer’s gives the improved bound when  $\dim_{\mathbb{H}} Y > 1$ .

Theorem 14.11.2 and Theorem 14.11.3 are the radial analogues, and they are strictly harder: the family  $\{\pi_x\}_{x \in X}$  is parametrised by a set that may itself be fractal.

**Theorem 14.11.4** (Ren–Wang; Furstenberg sets in the plane). *Let  $s \in (0, 1]$ ,  $t \in (0, 2]$ , and let  $F \subseteq \mathbb{R}^2$  be an  $(s, t)$ -Furstenberg set: there is a family  $\mathfrak{L}$  of lines with  $\dim_{\mathrm{H}} \mathfrak{L} \geq t$  such that  $\dim_{\mathrm{H}}(F \cap \ell) \geq s$  for every  $\ell \in \mathfrak{L}$ . Then*

$$\dim_{\mathrm{H}} F \geq \min \left\{ s + t, \frac{3s + t}{2}, s + 1 \right\},$$

and this is sharp.

**Theorem 14.11.5** (Bourgain; discretized projection theorem). *For all  $\alpha \in (0, 2)$  and  $\kappa > 0$  there are  $\varepsilon, \delta_0 > 0$  so that for  $\delta \in (0, \delta_0]$ : if  $A \subseteq [0, 1]^2$  is a  $(\delta, \alpha, \delta^{-\varepsilon})$ -set and  $\Theta \subseteq \mathbb{S}^1$  is a  $(\delta, \kappa, \delta^{-\varepsilon})$ -set, then there is  $\theta \in \Theta$  with*

$$\mathcal{E}_{\delta}(\pi_{\theta}(A)) \geq \delta^{-\alpha/2-\varepsilon}.$$

*Remark 14.11.6* (Scope). When first written this remark recorded four quoted theorems. Three are now proved: Theorem 14.11.2 and Theorem 14.11.3 in Chapter 14.13, and Theorem 14.11.5 in Chapter 14.14. *What remains under R23 is Theorem 14.11.4* — Ren and Wang’s sharp Furstenberg estimate — together with the two quantitative inputs recorded in Remark 14.13.15 and Remark 14.14.13.

*Why grouped together.* They form a single body of work — the discretized projection theory developed between 2019 and 2023 — and each is a research paper in its own right. Ren and Wang’s Furstenberg theorem alone is over eighty pages, and it rests on the Orponen–Shmerkin–Wang estimates, which in turn rest on Bourgain’s.

*What depends on them.* Theorem 14.8.2, hence Chapter 14.8, hence Theorem 14.2.5. Nothing in Volumes 1–13.

*What is proved here.* Proposition 14.11.7: the derivation of Theorem 14.8.2 from these inputs, which is where the trilinear structure and the dichotomy come from.

## 14.11.2 Assembling the theorem

**Proposition 14.11.7** (Architecture of Theorem 14.8.2). *Theorem 14.8.2 follows from Theorem 14.11.3 and Theorem 14.11.4 by the following scheme.*

*Scheme. Step 1.* Reformulate the trilinear expression.

For  $a = (a_1, a_2) \in F$  and  $b_i = (b_{i1}, b_{i2}) \in G$ ,

$$a \cdot (b_1 - b_2) = a_1(b_{11} - b_{21}) + a_2(b_{12} - b_{22}).$$

Writing  $u = b_1 - b_2$ , the map  $a \mapsto a \cdot u$  is the orthogonal projection of  $a$  onto the direction  $u/|u|$ , scaled by  $|u|$ . So (14.8.2) asserts that *some* projection of  $F$ , in a direction arising as a difference of two points of  $G$ , is large at some scale.

**Step 2.** *Identify the direction set.*

As  $b_1, b_2$  range over  $G$ , the directions  $u/|u|$  range over the union of the radial projections  $\pi_{b_2}(G \setminus \{b_2\})$ . By Theorem 14.11.2 applied with  $X = Y = G$ , if  $G$  is not contained in a line then this direction set has dimension at least  $\min\{\dim_H G, 1\} = 1$ , since  $G$  is a  $(\delta, 1)$ -ADset.

**Step 3.** *Apply the projection theorem.*

With a direction set of full dimension and  $F$  a  $(\delta, 1)$ -set, Theorem 14.11.5 with  $\alpha = 1$  gives a direction  $\theta$  in that set with  $\mathcal{E}_\delta(\pi_\theta F) \geq \delta^{-1/2-\varepsilon}$  — and the  $\varepsilon$ -improvement, in the sharper form supplied by Theorem 14.11.4, upgrades this to  $\delta^{-1+\varepsilon}$  under the additional hypothesis that  $F$  is not concentrated on a line.

**Step 4.** *The dichotomy.*

Two things can go wrong. Either  $G$  is contained in a line — so Step 2 fails — or  $F$  is concentrated near a line, so Step 3 fails. In the first case the directions  $u$  are confined to a single direction  $\theta_0$  and its negative; in the second  $F$  concentrates on a line  $\ell$ . Combining, both failures occur simultaneously only in the configuration described by alternative (A):  $F$  concentrated on a line  $\ell$  and  $G$  concentrated on a line  $\ell^\perp$  through the origin, the orthogonality arising because  $a \cdot u$  degenerates precisely when  $u \perp \ell$ .

If neither failure occurs, Step 3 supplies a scale  $\rho$  and an interval  $I$  for which (14.8.2) holds, which is alternative (B).  $\square$

**Where we are.** The scheme explains the shape of Theorem 14.8.2 but is not a substitute for its proof: the quantitative bookkeeping — producing the *same* scale  $\rho$  and interval  $I$  for every large  $H \subseteq F \times G \times G$ , rather than for one — is the technical content of Wang and Zahl’s Section 8, and it is what Remark 14.11.6 records as omitted.

*Where the complex counterexample sits in the scheme.* Over  $\mathbb{C}$ , Step 2 fails in a way not visible to alternative (A): the direction set  $\{u/|u|\}$  is a full-dimensional subset of  $\mathbb{S}^3$ , so no line contains  $G$ ; but the *pairing*  $a \cdot u$  is degenerate because conjugation makes it real. So the failure is in the bilinear form, not in the dimensions — and no purely dimensional argument can detect it. Remark 14.8.3 is the concrete witness.

WORKED EXAMPLE 14.11.1— TIER A   **The projection theorems, tabulated**

| Theorem   | Projections                             | Conclusion                                                                                  | Stat  |
|-----------|-----------------------------------------|---------------------------------------------------------------------------------------------|-------|
| Marstrand | orthogonal, a.e. $\theta$               | $\dim_{\mathbb{H}} \pi_{\theta} Y = \min\{\dim_{\mathbb{H}} Y, 1\}$                         | Vol 8 |
| Kaufman   | orthogonal, exceptional set             | $\dim_{\mathbb{H}}\{\theta : \text{small}\} \leq \dim_{\mathbb{H}} Y$                       | Vol 8 |
| Bourgain  | discretized, $\varepsilon$ -improvement | $\mathcal{E} \geq \delta^{-\alpha/2-\varepsilon}$                                           | Rema  |
| OSW       | radial, from a set $X$                  | $\sup_x \dim_{\mathbb{H}} \pi_x Y \geq \min\{\dim_{\mathbb{H}} X, \dim_{\mathbb{H}} Y, 1\}$ | Rema  |
| Ren–Wang  | Furstenberg sets                        | $\dim_{\mathbb{H}} F \geq \min\{s+t, \frac{3s+t}{2}, s+1\}$                                 | Rema  |
| Wang–Zahl | trilinear dichotomy                     | (A) or (B)                                                                                  | Rema  |

*The progression.* Marstrand fixes a single family and asks about almost every member; Kaufman bounds the exceptional set; Bourgain makes it quantitative at a single scale with an  $\varepsilon$ -gain; OSW replaces orthogonal by radial projections, so the family is parametrised by a fractal; Ren–Wang settles the Furstenberg problem that OSW’s method needs; and Wang–Zahl extract the trilinear dichotomy.

*The pattern.* Each step weakens the hypothesis on the family of projections and pays for it with a harder proof. *The Keakeya theorem needed the whole progression*, and that is why it was not proved earlier: the tools did not exist until 2023.

*What Volume 8 proved.* Marstrand and Kaufman, in full. The remaining four are this volume’s Remark 14.11.6.

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WORKED EXAMPLE 14.11.2— TIER B   **Why radial projections**

Orthogonal projections would be simpler. Three reasons the argument needs radial ones.

(a) *The direction set is determined by the data.* In Step 2 the directions available are  $u = b_1 - b_2$  with  $b_i \in G$  — differences of points of the set itself. That is precisely a radial projection of  $G$  from a point of  $G$ , not an orthogonal projection in a prescribed direction.

(b) *The parametrising set is fractal.*  $G$  is a  $(\delta, 1)$ -ADset in  $[0, 1]^2$ , so the family  $\{\pi_b\}_{b \in G}$  is indexed by a set of dimension 1, not by an interval. Kaufman’s theorem, which bounds an exceptional set of directions, gives nothing when the admissible directions form a set of the same dimension as the exceptional one.

(c) *The Beck-type conclusion is what is wanted.* OSW’s theorems are stated as: unless  $X$  lies in a line, some radial projection is large. That is exactly the dichotomy shape Theorem 14.8.2 needs — a single degenerate configuration,

otherwise a quantitative gain.

*The connection to Volume 8.* Worked Example 8.1.1 there recorded that projection theorems give “almost every” and not “every”, and that improving this requires controlling the exceptional set. The OSW theorems are the modern form of that control, and they are what made the Kakeya proof possible.

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## Drills

**Drill 14.11.1.** Verify that if  $X$  lies in a line then Theorem 14.11.2 fails, and identify the corresponding degenerate case of Theorem 14.8.2.

**Drill 14.11.2.** Evaluate the Ren–Wang bound at  $s = t = 1$  and at  $s = \frac{1}{2}$ ,  $t = 1$ , and confirm which of the three terms is binding.

**Drill 14.11.3.** Carry out Step 1 of Proposition 14.11.7 and confirm that  $a \cdot u$  is  $|u|$  times an orthogonal projection.

**Drill 14.11.4.** Explain, using the Checkpoint after Proposition 14.11.7, why no purely dimensional argument can detect the complex counterexample.

## Notes

Theorem 14.11.2 and Theorem 14.11.3 are Orponen, Shmerkin and Wang (2024); Theorem 14.11.4 is Ren and Wang (2023); Theorem 14.11.5 is Bourgain (2003, 2010), with the non-linear version due to Shmerkin. Remark 14.11.6 records that all four are quoted, and Proposition 14.11.7 gives the derivation of Theorem 14.8.2 from them.



## Chapter 14.12

# The Cinematic Maximal Estimate

Remark 14.9.9 recorded the maximal estimate for cinematic families as quoted. This chapter discharges it. The result descends from Wolff’s circular maximal function theorem, and the route runs through a tangency-counting bound whose modern proof uses *polynomial ham sandwich* — proved in Volume 11, Theorem 11.2.3.

### 14.12.1 The maximal function

**Definition 14.12.1** (Cinematic curvature). Let  $\Phi : U_1 \times U_2 \rightarrow \mathbb{R}$  be smooth,  $U_i \subseteq \mathbb{R}^2$ . The family of curves  $\Gamma(x, r) = \{y : \Phi(x, y) = r\}$  satisfies *Sogge’s cinematic curvature condition* at  $(a, b)$  if  $\nabla_y \Phi(a, b) \neq 0$  and

$$\det \left( \nabla_x \left[ \begin{array}{c} e \cdot \nabla_y \Phi(x, y) \\ e \cdot \nabla_y \left( \frac{e \cdot \nabla_y \Phi(x, y)}{|\nabla_y \Phi(x, y)|} \right) \end{array} \right] \right|_{(a,b)} \right) \neq 0, \quad (14.12.1)$$

where  $e \perp \nabla_y \Phi(a, b)$  is a unit vector. The two choices of  $e$  differ by a sign, so (14.12.1) does not depend on the choice.

**Where we are.** Informally, (14.12.1) says *two distinct curves of the family cannot be tangent to second order*: if  $\Gamma$  and  $\tilde{\Gamma}$  meet at  $y$ , then either their normals or their curvatures at  $y$  differ, by at least the distance between them in parameter space.

*The model case.*  $\Phi(x, y) = |x - y|$ , giving circles. By rotation, translation and scaling it suffices to check  $a = (0, 0)$ ,  $b = (1, 0)$ , where  $e = (0, 1)$  and the determinant in (14.12.1) equals 1. So *circles are cinematic*, and everything below generalises Wolff.

*Stability.* If  $\Phi(x, y) = |x - y| + P(x, y)$  with  $\|P\|_{C^3}$  small, (14.12.1) holds uniformly. So the condition is open, and Lemma 14.9.6 of Chapter 14.9 — where the curves are graphs  $a + \lambda_3 z + (b + \lambda_4 z)f(z)$  with  $f \in C^2$  — falls under it.

**Definition 14.12.2.** For  $\Gamma^\delta(x, r)$  the  $\delta$ -neighbourhood of  $\Gamma(x, r)$ , put

$$\mathcal{M}_\Phi^\delta f(r) = \sup_{x \in U_1} \frac{1}{|\Gamma^\delta(x, r)|} \int_{\Gamma^\delta(x, r)} |f|.$$

**Theorem 14.12.3** (Wolff; Zahl; Pramanik–Yang–Zahl; see Remark 14.12.11). *Let  $\Phi$  satisfy Definition 14.12.1 at  $(a, b)$ . Then for every  $\varepsilon > 0$  and every  $f$  supported near  $b$ ,*

$$\|\mathcal{M}_\Phi^\delta f\|_{L^3([1/2, 1])} \leq C_\varepsilon \delta^{-\varepsilon} \|f\|_{L^3(\mathbb{R}^2)}. \quad (14.12.2)$$

**Where we are.** Three remarks on the shape of (14.12.2).

*The  $\delta^{-\varepsilon}$  cannot be removed.* If it could, every Besicovitch–Rado–Kinney set — a planar set containing a circle of every radius in  $[\frac{1}{2}, 1]$  — would have positive measure, and that is false. So  $\varepsilon$ -losses are intrinsic here, exactly as in Volume 13, Worked Example 13.4.2(c).

*$L^3$  is the right exponent.* Interpolating (14.12.2) with the trivial  $\|\mathcal{M}^\delta f\|_\infty \lesssim \delta^{-1} \|f\|_1$  recovers the earlier Kolasa–Wolff range  $p < \frac{8}{3}$ ,  $q \leq 2p'$  — so the  $L^3$  bound is strictly stronger than what preceded it.

*Consequence.* (14.12.2) implies every BRK set has Hausdorff dimension 2, and the analogous statement for cinematic families is what Proposition 14.9.8 asserts.

## 14.12.2 Reduction to a covering statement

**Lemma 14.12.4** (The discretised form). *Theorem 14.12.3 follows from: for  $\eta > 0$  and  $\delta$  small, given a family  $\mathcal{A}$  of  $\Phi$ -circles with  $\delta$ -separated radii in  $(1 - \tau, 1)$ , there is  $\tilde{\mathcal{A}} \subseteq \mathcal{A}$  with  $\#\tilde{\mathcal{A}} \geq C^{-1} \#\mathcal{A}$  such that for all  $\Gamma \in \tilde{\mathcal{A}}$  and  $\delta < \lambda < 1$ ,*

$$\left| \left\{ y \in \Gamma^\delta : \sum_{\tilde{\Gamma} \in \tilde{\mathcal{A}}} \mathbf{1}_{\tilde{\Gamma}^\delta}(y) > \delta^{-\eta} \lambda^{-2} \right\} \right| \leq \lambda |\Gamma^\delta|. \quad (14.12.3)$$

*Reduction.* Standard: dualise  $\mathcal{M}_\Phi^\delta$ , discretise the radii, and apply (14.12.3) at each dyadic multiplicity level. The point is that (14.12.3) controls *how large the set of high-multiplicity points on a typical curve can be*, which is precisely what an  $L^3$  maximal bound needs.  $\square$



**Definition 14.12.5** (Parameter distance and tangency). For  $\Phi$ -circles  $\Gamma = \Gamma(x_0, r_0)$  and  $\tilde{\Gamma} = \Gamma(\tilde{x}_0, \tilde{r}_0)$  put  $d(\Gamma, \tilde{\Gamma}) = |x_0 - \tilde{x}_0| + |r_0 - \tilde{r}_0|$ , and for  $X \subseteq \mathbb{R}^2$ ,

$$\Delta_X(\Gamma, \tilde{\Gamma}) = \inf \left( |y - \tilde{y}| + \left| \frac{\nabla_y \Phi(x_0, y)}{\|\nabla_y \Phi(x_0, y)\|} - \frac{\nabla_y \Phi(\tilde{x}_0, \tilde{y})}{\|\nabla_y \Phi(\tilde{x}_0, \tilde{y})\|} \right| \right),$$

the infimum over  $y, \tilde{y} \in \overline{X}$  on the respective curves.

**Where we are.**  $\Delta_X$  measures *how far*  $\Gamma$  *must move to become tangent to*  $\tilde{\Gamma}$  *somewhere in*  $X$ : it is small exactly when the curves nearly touch with nearly equal normals. For circles,  $\Delta_{\mathbb{R}^2}(\Gamma, \tilde{\Gamma}) = ||x_0 - \tilde{x}_0| - |r_0 - \tilde{r}_0||$ , which vanishes precisely on the light cone  $\{|x - x_0| = |r - r_0|\}$  — internal tangency.

This is the geometry the whole argument tracks:  $d$  measures separation in parameter space,  $\Delta$  measures tangency, and cinematic curvature is what relates them.

**Definition 14.12.6** (Bipartite pairs and rectangles).  $(\mathcal{W}, \mathcal{B})$  is a *t-bipartite pair* if all radii are  $\delta$ -separated,  $d(\Gamma, \tilde{\Gamma}) \in (t, 2t)$  for  $\Gamma \in \mathcal{W}, \tilde{\Gamma} \in \mathcal{B}$ , and  $d < t$  within each family.

A  $(\delta, t)$ -*rectangle* is the  $\delta$ -neighbourhood of an arc of length  $\sqrt{\delta/t}$  of a  $\Phi$ -circle;  $\Gamma$  is *incident* to  $R$  if  $R$  lies in the  $C_1\delta$ -neighbourhood of  $\Gamma$ .  $R$  has *type*  $(\geq \mu, \geq \nu)$  if it is incident to at least  $C\mu$  curves of  $\mathcal{W}$  and  $C\nu$  of  $\mathcal{B}$ .

**Theorem 14.12.7** (Schlag's criterion; not proved here, see Remark 14.12.11). *Lemma 14.12.4 holds for  $\mathcal{A}$  provided:*

(i) Intersection area.

$$|\Gamma^\delta \cap \tilde{\Gamma}^\delta \cap B| \lesssim \frac{\delta^2}{(d(\Gamma, \tilde{\Gamma}) + \delta)^{1/2} (\Delta(\Gamma, \tilde{\Gamma}) + \delta)^{1/2}}; \quad (14.12.4)$$

(ii) Tangency count. *For every t-bipartite pair with  $\#\mathcal{W} = m$ ,  $\#\mathcal{B} = n$  and every  $\varepsilon > 0$ , the number of pairwise incomparable  $(\geq \mu, \geq \nu)$  rectangles is*

$$\lesssim_\varepsilon (mn)^\varepsilon \left( \left( \frac{mn}{\mu\nu} \right)^{3/4} + \frac{m}{\mu} + \frac{n}{\nu} \right). \quad (14.12.5)$$

**Where we are.** Condition (i) is the easy half and is a direct consequence of cinematic curvature: two curves separated by  $d$  in parameter space and tangent to within  $\Delta$  meet in a set of area  $\delta^2/(d\Delta)^{1/2}$ . Kolasa and Wolff proved it.

*Condition (ii) is the whole difficulty.* It is a quantitative form of the incidence statement that  $N$  circles in the plane admit at most  $C_\varepsilon N^{3/2+\varepsilon}$

*tangencies.* The exponent  $\frac{3}{4}$  in (14.12.5) is  $\frac{3}{2} \cdot \frac{1}{2}$ : with  $\mu = \nu = 1$  it reads  $\#\mathcal{R} \lesssim (mn)^{3/4+\varepsilon}$ , which at  $m = n$  is  $N^{3/2+\varepsilon}$ .

### 14.12.3 Counting tangencies

**Lemma 14.12.8** (Two curves meet few rectangles). *For  $\Gamma \in \mathcal{W}$ ,  $\tilde{\Gamma} \in \mathcal{B}$  there are  $O(1)$  pairwise incomparable  $(\delta, t)$ -rectangles tangent to both.*

*Proof.* Since  $d(\Gamma, \tilde{\Gamma}) \approx t$ , (14.12.4) gives  $|\Gamma^\delta \cap \tilde{\Gamma}^\delta| \lesssim \delta^{3/2} t^{-1/2}$ . A  $(\delta, t)$ -rectangle has area  $\approx \delta^{3/2} t^{-1/2}$  and incomparable ones are disjoint.  $\square$

**Lemma 14.12.9** (Kővári–Sós–Turán bound). *Any set of  $(\sim \mu, \sim \nu)$  rectangles has cardinality  $\lesssim \frac{mn^{2/3}}{\mu\nu^{2/3}}$ ; any set of  $(\gtrsim \mu, \gtrsim \nu)$  rectangles has cardinality  $\lesssim \frac{mn^{2/3}}{\mu\nu^{2/3}} + \frac{n}{\nu} \log \frac{m}{\mu}$ .*

*Proof.* By Lemma 14.12.8 and the Apollonius-type bound — three curves almost tangent to a common curve at well-separated places confine that curve to a set of diameter  $\lesssim t$  — the incidence matrix has no  $2 \times 3$  submatrix of ones. The Kővári–Sós–Turán theorem then bounds the number of ones by  $mn^{2/3}$ . The second statement is dyadic summation.  $\square$

**Theorem 14.12.10** (Tangency counting). *Condition (ii) of Theorem 14.12.7 holds for every family of  $\Phi$ -circles with  $\delta$ -separated radii.*

*Structure of the proof.* It suffices to treat  $\mu = \nu = 1$  and show

$$\#\mathcal{R} \lesssim_\varepsilon (mn)^\varepsilon ((mn)^{3/4} + m \log n + n \log m); \quad (14.12.6)$$

the general case follows by random sampling. The proof is an induction on  $mn$ .

**Step 1.** *Split off the clusters.*

A *cluster* is a subfamily all of whose members are tangent to comparable  $(\delta, t)$ -rectangles. Given  $\mu_0$ , write  $\mathcal{W} = \mathcal{W}_g \sqcup \mathcal{W}_b$  where  $(\mathcal{W}_g, \mathcal{B})$  has no  $(\gtrsim \mu_0, \gtrsim 1)$  rectangles and  $\mathcal{W}_b$  is a union of  $\lesssim \frac{m}{\mu_0} (\log m)(\log n)$  clusters; likewise for  $\mathcal{B}$ . With  $\mu_0 = \nu_0 = (mn)^{1/4}$ , the cluster parts contribute  $\lesssim (\log m)(\log n)(mn)^{3/4}$ , which is acceptable.

**Step 2.** *Cell decomposition.*

Associate to each  $\Gamma$  the surface

$$S_\Gamma = \{(x, r) : \Gamma(x, r) \text{ is tangent to } \Gamma\} \subseteq \mathbb{R}^3,$$

a two-dimensional set — for circles, a light cone with vertex  $(x_0, r_0)$ . Choose a random sample  $\mathcal{D} \subseteq \mathcal{B}$  of size  $N$  and partition  $\mathbb{R}^3$  into  $O(N^3 \log N)$  cells, none meeting any  $S_\Gamma, \Gamma \in \mathcal{D}$ .

**Step 3.** *Random sampling.*

With high probability each cell meets  $\lesssim \frac{n \log n}{N}$  of the surfaces  $S_\Gamma, \Gamma \in \mathcal{B}$ . This is the standard Clarkson-type argument: for a cell  $\Omega$  avoiding all sampled surfaces,  $\mathbb{P}(Z(\Omega) \geq \lambda) \leq (1 - \lambda/n)^N$ , and  $\lambda = Cn \log n/N$  makes this  $n^{-C}$ .

**Step 4.** *Induct.*

Split  $\mathcal{W}_g = \mathcal{W}_g^* \sqcup \bigsqcup_i \mathcal{W}_g^i$  with  $\mathcal{W}_g^i$  the curves in cell  $\Omega_i$  and  $\mathcal{W}_g^*$  those close to a dividing surface. By Step 3, the curves of  $\mathcal{B}_g$  interacting with  $\mathcal{W}_g^i$  number  $\lesssim \frac{n \log n}{N}$ , so the induction hypothesis applies to each pair  $(\mathcal{W}_g^i, \mathcal{B}_g^i)$ . Summing over the  $O(N^3 \log N)$  cells and applying Hölder gives (14.12.6), provided  $N$  is chosen in the range

$$C \log(mn)^{1/\varepsilon} < N < C^{-1} \min(n^{3/4} m^{-1/4}, m^{1/4} n^{-1/12}) \log(mn).$$

The exceptional family  $\mathcal{W}_g^*$  is small by a first-moment computation and is handled by the induction hypothesis directly.  $\square$

**Where we are.** Step 2 is where the method has changed, and the change matters for this series.

*Zahl's 2010 argument* obtained the cell decomposition by *vertical algebraic decomposition* — a technique from computational geometry requiring  $\Phi$  to be *algebraic*, so that the surfaces  $S_\Gamma$  are semi-algebraic of bounded complexity and Tarski–Seidenberg applies.

*Zahl's 2011 argument* replaces it with the *discrete polynomial ham sandwich theorem*, and the algebraicity hypothesis disappears entirely: Theorem 14.12.3 holds for every  $\Phi$  satisfying Definition 14.12.1.

*Why that matters here.* Polynomial ham sandwich is Theorem 11.2.3 of Volume 11 — *proved* there, from Tucker's lemma, with no topology quoted (Chapter 11.1). So the engine of Theorem 14.12.10 is available in this series, and Proposition 14.9.8 is not an appeal to machinery developed elsewhere.

*Remark 14.12.11* (R22, discharged in part). *What is proved here.* The reduction Lemma 14.12.4; Lemma 14.12.8; Lemma 14.12.9; the structure of Theorem 14.12.10, including the cluster splitting, the cell decomposition and the random sampling; and the identification of polynomial ham sandwich as the engine, which Volume 11 supplies.

*What is not.* Schlag's criterion Theorem 14.12.7, whose proof is an induction on scales in his paper; the Kolasa–Wolff differential properties (Properties 16–22 of Zahl's paper) establishing (14.12.4) and the Apollonius bound;

and the fractal refinement of Pramanik, Yang and Zahl needed for Proposition 14.9.8 in the form Chapter 14.9 uses — that version replaces  $[\frac{1}{2}, 1]$  by a fractal set of radii.

*The honest position.* Theorem 14.12.3 is now traced to its ingredients, and its one deep ingredient is proved in Volume 11. The remaining gaps are the differential-geometric properties of cinematic families, which are routine but lengthy, and the fractal upgrade, which is a separate paper.

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WORKED EXAMPLE 14.12.1— TIER A **The tangency exponent  $3/2$**

The number  $\frac{3}{4}$  in (14.12.5) is the whole content, and it is worth locating.

*The combinatorial statement.*  $N$  circles in the plane have at most  $C_\varepsilon N^{3/2+\varepsilon}$  tangencies. Setting  $m = n = N$  and  $\mu = \nu = 1$  in (14.12.5) gives  $\#\mathcal{R} \lesssim N^{3/2+\varepsilon}$ . ✓

*Why  $3/2$  and not  $4/3$ .* Tangency is a codimension-one condition on pairs, so one expects fewer tangencies than incidences. For *incidences* between  $N$  points and  $N$  circles the Szemerédi–Trotter-type bound is  $N^{4/3}$  — Volume 11, Theorem 11.3.2. Tangencies are rarer, yet the exponent is *larger*:  $\frac{3}{2} > \frac{4}{3}$ . The reason is that a tangency is an incidence between two curves rather than a point and a curve, so the count is over pairs.

*Sharpness.*  $N^{3/2}$  is attained, by a construction of Ellenberg, Solymosi and Zahl. So (14.12.5) is sharp up to  $N^\varepsilon$  and the  $L^3$  exponent in Theorem 14.12.3 cannot be improved by this route.

*The chain of exponents.*

|                              |                         |                  |
|------------------------------|-------------------------|------------------|
| Tangencies among $N$ circles | $N^{3/2}$               | Theorem 14.12.10 |
| Maximal function exponent    | $L^3$                   | Theorem 14.12.3  |
| BRK sets                     | $\dim_{\mathrm{H}} = 2$ | corollary        |
| Keakeya in $\mathbb{R}^3$    | $\dim_{\mathrm{H}} = 3$ | Theorem 14.2.6   |

Each row is deduced from the one above, and the first is a statement in discrete geometry.

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WORKED EXAMPLE 14.12.2— TIER B **Why circles, and why this is a Keakeya problem**

The appearance of a *circular* maximal function inside a proof about *lines* deserves an explanation.

(a) *The twisted projection makes curves.* Lemma 14.9.4 showed that  $\pi_f$  maps a  $\delta$ -tube to the neighbourhood of a  $C^2$  curve. So after applying  $\pi_f$ , a family of tubes *becomes* a family of curves.

(b) *The curves are cinematic.* By Lemma 14.9.6, provided  $|f'| \geq c$ . So the relevant maximal function is  $\mathcal{M}_{\Phi}^{\delta}$  for the family  $\{\gamma_T\}$ , and Theorem 14.12.3 applies.

(c) *BRK andakeya are cousins.* A BRK set contains a circle of every radius; aakeya set contains a segment in every direction. Both are “1-parameter families of curves with prescribed parameter”, and Wolff’s theorem for the first is the model for the second. Volume 9, Chapter 9.1 developed the linear case.

(d) *Where the analogy breaks.* For lines, two distinct directions give transverse tubes; for circles, two distinct radii can give *tangent* circles, and tangency is the hard case. That is why the circular problem needs Theorem 14.12.10 while the linear one needs only the bush and hairbrush of Volume 9.

*The transferable point.* Tangency is the recurring obstruction — Volume 12, Worked Example 12.2.4 listed four instances, and this is a fifth. Wolff’s circular maximal theorem is the one place in this series where tangency is confronted directly rather than excluded by hypothesis.

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## Drills

**Drill 14.12.1.** Verify that  $\Phi(x, y) = |x - y|$  satisfies (14.12.1) at  $a = (0, 0)$ ,  $b = (1, 0)$ .

**Drill 14.12.2.** Confirm that a  $(\delta, t)$ -rectangle has area  $\approx \delta^{3/2}t^{-1/2}$  and deduce Lemma 14.12.8 from (14.12.4).

**Drill 14.12.3.** Set  $m = n = N$ ,  $\mu = \nu = 1$  in (14.12.5) and recover the  $N^{3/2+\varepsilon}$  tangency bound.

**Drill 14.12.4.** Explain why the  $\delta^{-\varepsilon}$  in (14.12.2) cannot be removed, using the existence of measure-zero BRK sets.

## Notes

Wolff’s circular maximal theorem is from 1997; the cinematic curvature condition is Sogge’s. Theorem 14.12.10 follows Zahl (2010, 2011); the replacement of vertical algebraic decomposition by polynomial ham sandwich is his 2011 paper, and it removes the algebraicity hypothesis. Remark 14.12.11 records what remains.



## Chapter 14.13

# Radial Projections and Thin Tubes

Remark 14.11.6 grouped four theorems as quoted. This chapter discharges the two that matter most — the Orponen–Shmerkin–Wang radial projection theorems — and identifies precisely what the other two contribute. The organising concept is *thin tubes*, and the mechanism is a bootstrap.

### 14.13.1 Thin tubes

**Definition 14.13.1** (Thin tubes). Let  $K, t \geq 0$ ,  $c \in (0, 1]$ , and let  $\mu, \nu \in \mathbb{P}(\mathbb{R}^d)$  with supports  $X, Y$ . The pair  $(\mu, \nu)$  has  $(t, K, c)$ -*thin tubes* if there is a Borel  $G \subseteq X \times Y$  with  $(\mu \times \nu)(G) \geq c$  such that for every  $x \in X$ ,

$$\nu(T \cap G|_x) \leq K r^t \quad \text{for all } r > 0 \text{ and all } r\text{-tubes } T \ni x. \quad (14.13.1)$$

We say  $(\mu, \nu)$  has  $t$ -*thin tubes* if this holds for some  $K, c > 0$ .

**Lemma 14.13.2** (Thin tubes give a large radial projection). *If  $(\mu, \nu)$  has  $t$ -thin tubes for some  $t > 0$ , then there is  $x \in \text{spt } \mu$  with  $\mathcal{H}^t(\pi_x(Y \setminus \{x\})) > 0$ .*

*Proof.* Pick  $x$  with  $\nu(G|_x) > 0$ . Then  $\nu(G|_x \setminus \{x\}) > 0$ , since otherwise (14.13.1) fails at exponent  $t > 0$ : a point mass at  $x$  lies in every  $r$ -tube through  $x$ . Choose compact  $K' \subseteq G|_x \setminus \{x\}$  with  $\nu(K') > 0$ .

For a set  $A \subseteq \mathbb{S}^{d-1}$  of diameter  $r$ , the preimage  $\pi_x^{-1}(A)$  is contained in an  $O(r)$ -tube through  $x$ . So (14.13.1) says the pushforward  $\pi_x(\nu|_{K'})$  satisfies a  $t$ -dimensional Frostman condition, and hence  $\mathcal{H}^t(\pi_x(K')) > 0$  by Frostman’s lemma — Volume 7, Theorem 7.5.1.  $\square$

**Where we are.** Lemma 14.13.2 is why the whole chapter is phrased in terms of thin tubes rather than dimensions directly. *A tube through  $x$  is the preimage under  $\pi_x$  of an arc*, so a bound on how much  $\nu$ -mass any thin tube through  $x$  can carry is exactly a Frostman condition on the radial projection.

The set  $G$  is a device for discarding a small exceptional part of  $X \times Y$ : one cannot expect (14.13.1) for *every* pair, since  $\mu$  and  $\nu$  may share mass on a line.

**Definition 14.13.3** ( $(\delta, s, C)$ -sets). A set  $P$  in a metric space is a  $(\delta, s, C)$ -set if

$$\mathcal{E}_\delta(P \cap B(x, r)) \leq C r^s \mathcal{E}_\delta(P) \quad \text{for all } x \text{ and } r \geq \delta.$$

A family of  $\delta$ -tubes is a  $(\delta, s, C)$ -set if the corresponding line set is one in the metric  $d(\ell_1, \ell_2) = \|\pi_{L_1} - \pi_{L_2}\| + |a_1 - a_2|$ .

**Where we are.** This is the *Frostman* form of non-concentration — relative to the total — and it differs from the Katz–Tao form  $\mathcal{E}_\delta(P \cap B(x, r)) \leq C(r/\delta)^s$  used in Definition 14.4.1. Lemma 14.13.13 below shows a  $(\delta, s)$ -set decomposes into few Katz–Tao sets, which is how one passes between them. Volume 12, Definition 12.7.1 used the Frostman form for measures; this is its discrete counterpart.

## 14.13.2 The Furstenberg input

**Theorem 14.13.4** (Orponen–Shmerkin; not proved here, see Remark 14.13.15). *Given  $s \in (0, 1)$  and  $t \in (s, 2)$  there is  $\varepsilon = \varepsilon(s, t) > 0$  such that for  $0 < \delta \leq \delta_0(s, t)$ : if  $X \subseteq B^2$  is a  $(\delta, t, \delta^{-\varepsilon})$ -set and each  $x \in X$  carries a  $(\delta, s, \delta^{-\varepsilon})$ -set  $\mathbb{T}'_x$  of  $\delta$ -tubes through  $x$ , then*

$$\left| \bigcup_{x \in X} \mathbb{T}'_x \right|_\delta \geq \delta^{-2s-\varepsilon}. \quad (14.13.2)$$

*Moreover  $\varepsilon$  is uniform on compact subsets of  $\{(s, t) : s \in (0, 1), t \in (s, 2)\}$ .*

**Where we are.** The trivial bound is  $\delta^{-2s}$ : a  $(\delta, s)$ -set of tubes through each of a  $(\delta, t)$ -set of points gives at least that many, by double counting. The content of (14.13.2) is the  $\varepsilon$ , and that is all the argument needs.

*This is the observation the whole paper starts from.* Beck’s theorem in discrete geometry rests on Szemerédi–Trotter, but it does not need its full strength — any  $\varepsilon$ -improvement over double counting suffices. The same is true here, and Theorem 14.13.4 supplies the continuous analogue.



**Lemma 14.13.5** (Furstenberg sets from tube families). *For every  $\xi > 0$  there are  $\delta_0, \varepsilon > 0$  such that for  $\delta \in (0, \delta_0]$ : if  $P \subseteq B^2$  is a  $(\delta, t, \delta^{-\varepsilon})$ -set and each  $x \in P$  carries a  $(\delta, s, \delta^{-\varepsilon})$ -set  $\mathbb{T}_x$  of tubes through  $x$ , then  $\mathbb{T} = \bigcup_{x \in P} \mathbb{T}_x$  contains a  $(\delta, \gamma(s, t), \delta^{-\xi})$ -set, where*

$$\gamma(s, t) = s + \min\{s, t\}. \quad (14.13.3)$$

*Proof by duality.* The dual statement — a  $(\delta, t)$ -set of tubes each carrying a  $(\delta, s)$ -set of points has union containing a  $(\delta, \gamma(s, t))$ -set — is the Hausdorff-content form of the known bound  $\dim_{\mathbb{H}} F \geq \gamma(s, t)$  for  $(s, t)$ -Furstenberg sets. The case  $t \leq s$  is Lutz–Stull;  $t \geq s$  goes back to Wolff.

The passage between the two is point-line duality: send  $(a, b) \mapsto D(a, b) = \{y = ax + b\}$  and  $\ell = \{y = cx + d\} \mapsto D^*(\ell) = (-c, d)$ . Then

$$p \in \ell \iff D^*(\ell) \in D(p), \quad (14.13.4)$$

and  $D, D^*$  are bilipschitz between the Euclidean metric and the metric of Definition 14.13.3, so the  $(\delta, s)$ -set property is preserved.  $\square$

### 14.13.3 The bootstrap

**Lemma 14.13.6** (The key lemma). *Let  $0 < s \leq 1$ ,  $\varepsilon \in (0, \frac{1}{10})$ , and let  $\mu, \nu \in \mathbb{P}(B^2)$  satisfy  $\mu(B_r), \nu(B_r) \leq Cr^s$  and  $\text{dist}(\text{spt } \mu, \text{spt } \nu) \geq C^{-1}$ . Let  $\beta = \beta(s)$  be the parameter of Proposition 14.13.7. If both  $(\mu, \nu)$  and  $(\nu, \mu)$  have  $(\sigma, K, 1 - \varepsilon)$ -thin tubes for some  $\beta \leq \sigma < s$ , then there is  $\eta = \eta(s, \sigma) > 0$  and  $K'$  such that both have  $(\sigma + \eta, K', 1 - 5\varepsilon)$ -thin tubes.*

*Proof.* Suppose not. Then, writing  $G$  for the good set given by the  $(\sigma, K, 1 - \varepsilon)$ -hypothesis, one pigeonholes a single scale  $r$  and a set  $\bar{X} \subseteq X$  with  $\mu(\bar{X}) \geq r^\eta$ , together with, for each  $x \in \bar{X}$ , a family  $\mathbb{T}_x$  of  $r$ -tubes through  $x$  with

$$r^{\sigma+3\eta} \leq \nu(T \cap Y_x) \leq r^{\sigma-\eta}, \quad T \in \mathbb{T}_x, \quad (14.13.5)$$

and  $\mathbb{T}_x$  an  $(r, \sigma, r^{-5\eta})$ -set with  $r^{-\sigma+3\eta} \lesssim \#\mathbb{T}_x \lesssim r^{-\sigma-\eta}$ .

**Step 1.** *Dichotomy.*

Set  $\kappa = \frac{14\eta}{s-\sigma}$ . Call  $T \in \mathbb{T}_x$  *concentrated* if some ball  $B_T$  of radius  $r^\kappa$  carries  $\nu(T \cap B_T \cap Y_x) \geq \frac{1}{3}\nu(T \cap Y_x)$ .

**Step 2.** *The non-concentrated branch.*

Suppose half the tubes are non-concentrated for  $x$  in a set of  $\mu$ -measure  $\geq \frac{1}{2}\mu(\bar{X})$ . Extract an  $(r, \sigma, r^{-2\eta})$ -set  $P$  of such  $x$  by the discrete Frostman lemma; take  $\eta \leq \varepsilon(s, \sigma)^2$  for the function of Theorem 14.13.4. That theorem gives

$$\#\mathbb{T}' \gtrsim r^{-2\sigma-\sqrt{\eta}}, \quad \mathbb{T}' = \bigcup_{x \in P} \mathbb{T}'_x. \quad (14.13.6)$$

On the other hand, each non-concentrated  $T$  contains two  $r^\kappa$ -separated sets  $Y_{T,1}, Y_{T,2}$  of  $\nu$ -measure  $\geq r^{\sigma+4\eta}$ , so counting pairs and using that at most  $|y_1 - y_2|^{-1}$  tubes contain a given pair,

$$r^{2\sigma+8\eta} \# \mathbb{T}' \leq \sum_T (\nu \times \nu)(Y_{T,1} \times Y_{T,2}) \lesssim \iint r^{-\kappa} d(\nu \times \nu) = r^{-\kappa}.$$

Thus  $\# \mathbb{T}' \lesssim r^{-2\sigma-8\eta-\kappa}$ , contradicting (14.13.6) once  $\eta$  is small in terms of  $s - \sigma$ .

**Step 3.** *The concentrated branch.*

Otherwise most tubes are concentrated. Double counting produces a  $y$  and a scale  $r \leq \xi \leq r^\kappa$  such that many tubes  $T \ni y$  satisfy  $\nu(A(y, \xi, 2\xi) \cap T) \geq r^{\sigma+4\eta}$ , where  $A$  is an annulus. Using (14.13.1) for  $(\nu, \mu)$ , at least  $\gtrsim r^{9\eta}(\xi/r)^\sigma$  of these have  $(r/\xi)$ -separated directions. Such tubes have bounded overlap outside  $B(y, \xi)$ , so

$$r^{13\eta} \xi^\sigma \lesssim r^{\sigma+4\eta} \# \mathbb{T}'_y \lesssim \nu(B(y, 2\xi)) \leq C(2\xi)^s,$$

that is  $r^{13\eta} \lesssim \xi^{s-\sigma} \leq r^{\kappa(s-\sigma)}$ . With  $\kappa = 14\eta/(s-\sigma)$  this is a contradiction.  $\square$

**Where we are.** The choice  $\kappa = \frac{14\eta}{s-\sigma}$  is what makes both branches close, and it is worth seeing why. Step 2 needs  $\kappa$  *large* — the two separated pieces must be far enough apart that few tubes contain both. Step 3 needs  $\kappa$  *small* — the annulus must sit inside the ball where the Frostman condition bites. The two requirements are compatible precisely because  $\sigma < s$ , and the gap  $s - \sigma$  is what one has to spend.

*So the lemma degenerates as  $\sigma \uparrow s$ , which is correct: at  $\sigma = s$  the conclusion would exceed the Frostman exponent and be false.*

**Proposition 14.13.7** (Starting point; Shmerkin, not proved here). *There is  $\beta(s) \in (0, s)$  such that if  $\mu, \nu \in \mathbb{P}(B^2)$  have separated supports, satisfy  $s$ -dimensional Frostman conditions, and  $\nu(T) \leq \tau$  for all  $\delta$ -tubes  $T$ , then  $(\mu, \nu)$  has  $(\beta, K, 1 - \varepsilon)$ -thin tubes.*

**Theorem 14.13.8** (Orponen–Shmerkin–Wang, Kaufman form). *Let  $X \subseteq \mathbb{R}^2$  be Borel and not contained in a line. Then for every Borel  $Y \subseteq \mathbb{R}^2$ ,*

$$\sup_{x \in X} \dim_{\mathbb{H}} \pi_x(Y \setminus \{x\}) \geq \min\{\dim_{\mathbb{H}} X, \dim_{\mathbb{H}} Y, 1\}.$$

*Proof.* Write  $t = \min\{\dim_{\mathbb{H}} X, \dim_{\mathbb{H}} Y, 1\}$  and assume  $t > 0$ .

**Step 1.** *A degenerate case.*

If  $\sup_{\ell} \dim_{\mathbb{H}}(Y \cap \ell) \geq t$ , choose  $\ell_\varepsilon$  with  $\dim_{\mathbb{H}}(Y \cap \ell_\varepsilon) \geq t - \varepsilon$ . Since  $X$  is not contained in a line, pick  $x_\varepsilon \in X \setminus \ell_\varepsilon$ ; then  $\pi_{x_\varepsilon}$  is locally bi-Lipschitz on  $\ell_\varepsilon$ , so  $\dim_{\mathbb{H}} \pi_{x_\varepsilon}(Y \cap \ell_\varepsilon) = \dim_{\mathbb{H}}(Y \cap \ell_\varepsilon)$  and we are done.

**Step 2.** *The main case.*

Otherwise  $\dim_{\mathbb{H}}(Y \cap \ell) \leq t - \varepsilon_0$  for all  $\ell$ . Fix  $t - \varepsilon_0 < s < t$  and Frostman measures  $\mu$  on  $X$ ,  $\nu$  on  $Y$  of exponent  $s$ .

If  $\mu(\ell) = 1$  for some line  $\ell$ , then  $\nu(\ell) = 0$  by the case hypothesis, so  $\nu(K') > 0$  for some compact  $K' \subseteq \mathbb{R}^2 \setminus \ell$ ; Lemma 14.13.10 below gives  $\sigma$ -thin tubes for all  $\sigma < s$ .

Otherwise  $\mu(\ell) < 1$  for every  $\ell$ , and the bootstrap Corollary 14.13.9 — stated immediately below, and proved from Proposition 14.13.7 and Lemma 14.13.6 alone — gives  $\sigma$ -thin tubes for all  $\sigma < s$ . By Lemma 14.13.2,  $\sup_x \dim_{\mathbb{H}} \pi_x(Y \setminus \{x\}) \geq s$ ; let  $s \uparrow t$ .  $\square$

**Corollary 14.13.9** (The bootstrap). *For  $0 < \sigma < s \leq 1$  and  $C, \varepsilon, \delta > 0$  there are  $\tau, K > 0$  such that: if  $\mu, \nu \in \mathbb{P}(B^2)$  satisfy  $s$ -dimensional Frostman conditions, have supports at distance  $\geq C^{-1}$ , and  $\max\{\mu(T), \nu(T)\} \leq \tau$  for all  $\delta$ -tubes, then  $(\mu, \nu)$  and  $(\nu, \mu)$  have  $(\sigma, K, 1 - \varepsilon)$ -thin tubes.*

*Proof.* Proposition 14.13.7 gives  $(\beta, K_1, 1 - \bar{\varepsilon})$ -thin tubes. If  $\beta \geq \sigma$  we are done; otherwise apply Lemma 14.13.6 repeatedly. Each application raises the exponent by  $\eta = \eta(s, \sigma) > 0$  and multiplies the failure probability by 5, so after  $N \approx \eta^{-1}$  steps one reaches  $\sigma$ . Choosing  $\bar{\varepsilon}$  initially so small that  $5^N \bar{\varepsilon} \leq \varepsilon$  gives the claim.  $\square$

**Lemma 14.13.10** (The collinear case). *If  $\text{spt } \mu \subseteq \ell$  and  $\text{spt } \nu \subseteq \mathbb{R}^2 \setminus \ell$ , with both measures  $s$ -Frostman and separated supports, then  $(\mu, \nu)$  has  $\sigma$ -thin tubes for every  $\sigma < s$ .*

*Proof.* Fix  $\sigma < s$ , so  $I_{\sigma}(\nu) < \infty$ . The standard proof of Kaufman's projection theorem gives

$$\int_{\ell} I_{\sigma}(\pi_x \nu) d\mu(x) \lesssim I_{\sigma}(\nu) < \infty, \quad (14.13.7)$$

the only input being  $\mu(\{x \in \ell : |\pi_x(y_1) - \pi_x(y_2)| \leq \delta\}) \lesssim (\delta/|y_1 - y_2|)^s$ , which is elementary here.

Alternatively, apply a projective transformation sending  $\ell$  to the line at infinity: the radial projections  $\pi_x$ ,  $x \in \ell$ , become *orthogonal* projections, and (14.13.7) is literally Kaufman's computation. Then (14.13.7) gives a  $\sigma$ -Frostman condition on  $\pi_x \nu$  for  $\mu$ -a.e.  $x$ .  $\square$

**Where we are.** Lemma 14.13.10 explains the hypothesis of Theorem 14.13.8. If *both*  $X$  and  $Y$  lie on a common line then  $\pi_x(Y)$  is a single point for every  $x \in X$  and the conclusion is false. The theorem excludes this by assuming  $X$  is not contained in a line — and the proof shows that is the *only* obstruction.

This is exactly the shape of alternative (A) in Theorem 14.8.2, and Proposition 14.11.7 used it in Step 2. So the dichotomy there is inherited from here.

### 14.13.4 The Falconer form

**Theorem 14.13.11** (OSW, Falconer form; from Theorem 14.13.14 by a second bootstrap). *Let  $X, Y \subseteq \mathbb{R}^2$  be Borel with  $X \neq \emptyset$  and  $\dim_{\mathbb{H}} Y > 1$ . Then*

$$\sup_{x \in X} \dim_{\mathbb{H}} \pi_x(Y \setminus \{x\}) \geq \min\{\dim_{\mathbb{H}} X + \dim_{\mathbb{H}} Y - 1, 1\}.$$

**Theorem 14.13.12** (Fu–Ren; not proved here, see Remark 14.13.15). *For  $0 \leq s, t \leq 2$  and every  $\varepsilon > 0$  there is  $\delta_0$  such that for  $\delta \in (0, \delta_0]$ : if  $P \subseteq B^2$  is a  $\delta$ -separated  $(\delta, s, \delta^{-\varepsilon})$ -set and  $\mathbb{T}$  a  $\delta$ -separated  $(\delta, t, \delta^{-\varepsilon})$ -set of tubes, then*

$$|\mathcal{I}(P, \mathbb{T})| \leq |P||\mathbb{T}| \delta^{\kappa(s+t-1)-5\varepsilon}, \quad \kappa = \min\left\{\frac{1}{2}, \frac{1}{s+t-1}\right\}. \quad (14.13.8)$$

**Lemma 14.13.13** (Katz–Tao decomposition). *Let  $(X, d)$  be doubling. For every  $\varepsilon, t > 0$  there is  $\delta_0$  such that any  $\delta$ -separated  $(\delta, t, C)$ -set  $P$  in a unit ball splits as  $P = P_1 \sqcup \dots \sqcup P_N$  with each  $P_j$  a Katz–Tao  $(\delta, t, 1)$ -set and  $N \leq C|P|\delta^{t-\varepsilon}$ .*

*Proof.* Put  $H = 4^{t+1}C|P|\delta^t \geq 1$ . For each dyadic  $r \in [\delta, 1]$  cover the ball by families  $\mathcal{B}_r$  of  $r$ -balls with bounded overlap, and inside each  $B \in \mathcal{B}_r$  divide  $P \cap B$  into  $\lceil |P \cap B|/H \rceil$  groups of size at most  $H$ . Form a graph on  $P$  joining members of a common group.

Each  $x$  lies in  $O(1)$  balls per scale and there are  $O(\log \frac{1}{\delta})$  scales, so the maximum degree is  $\lesssim H \log \frac{1}{\delta}$ . By Brooks' theorem the graph is  $N$ -colourable with  $N \lesssim H \log \frac{1}{\delta}$ , and each colour class is a Katz–Tao  $(\delta, t, 1)$ -set: for  $B \in \mathcal{B}_r$  with  $\rho \approx r$ ,

$$\#\{\text{groups in } B\} \leq \lceil 4^t 4^{-t-1}(\rho/\delta)^t \rceil \leq (\rho/\delta)^t,$$

and a colour class meets each group once.  $\square$

**Theorem 14.13.14** (Discretised Falconer form). *For  $t \in (1, 2]$ ,  $\sigma \in [0, 1]$ ,  $\zeta > 0$  there are  $\varepsilon, \delta_0 > 0$  such that for  $\delta \in (0, \delta_0]$ : let  $P_K$  be a  $(\delta, t, \delta^{-\varepsilon})$ -set,  $P_E$  a  $(\delta, s, \delta^{-\varepsilon})$ -set, and suppose each  $x \in P_E$  carries a  $(\delta, \sigma, \delta^{-\varepsilon})$ -set  $\mathbb{T}_x$  of tubes through  $x$  with  $|T \cap P_K| \geq \delta^{\sigma+\varepsilon}|P_K|$ . Then  $\sigma \geq s + t - 1 - \zeta$ .*

*Proof.* By Lemma 14.13.5,  $\bigcup_{x \in P_E} \mathbb{T}_x$  contains a  $(\delta, \gamma(\sigma, s), \delta^{-\varepsilon})$ -set  $\mathbb{T}$  with  $\gamma(\sigma, s) = \sigma + \min\{s, \sigma\}$ . The hypothesis gives

$$|P_K||\mathbb{T}| \delta^{\sigma+\varepsilon} \leq \sum_{T \in \mathbb{T}} |T \cap P_K| = |\mathcal{I}(P_K, \mathbb{T})|. \quad (14.13.9)$$

Compare with (14.13.8), applied with exponents  $t$  and  $\gamma(\sigma, s)$ , and set  $\bar{\kappa} = \min\{\frac{1}{2}, \frac{1}{t+\gamma(\sigma, s)-1}\}$ . Four cases:

(i)  $s \leq \sigma$ ,  $\bar{\kappa} = \frac{1}{2}$ . Then  $\delta^{2\sigma+2\varepsilon} \leq \delta^{t+s+\sigma-1-10\xi}$ , giving  $\sigma \geq s+t-1-10\xi-2\varepsilon$ , which is the claim for  $10\xi+2\varepsilon \leq \zeta$ .

(ii)  $s \leq \sigma$ ,  $\bar{\kappa} = \frac{1}{t+s+\sigma-1}$ . Then  $\delta^\sigma \leq \delta^{1-5\xi-\varepsilon}$ , impossible for  $\sigma < 1-5\xi-\varepsilon$ .

(iii)  $s > \sigma$ ,  $\bar{\kappa} = \frac{1}{2}$ . Then  $1 \leq \delta^{(t-1)/2-5\xi-\varepsilon}$ , impossible for  $(t-1)/2 > 5\xi+\varepsilon$ .

(iv)  $s > \sigma$ ,  $\bar{\kappa} = \frac{1}{t+2\sigma-1}$ . As in (ii), impossible.

Only case (i) survives, and it gives the conclusion.  $\square$

**Where we are.** The four-case analysis is the whole proof, and the structure is worth noting: *three of the four cases are impossible* and the fourth is the theorem. The impossibilities come from  $\sigma < 1$  and  $t > 1$ , the two standing hypotheses.

The Fu–Ren bound is where Fourier analysis enters: their proof uses the incidence estimate of Guth, Solomon and Wang for well-spaced tubes. So although this chapter contains no Fourier transforms, they are present one level down — which is the same phenomenon Volume 13, Worked Example 13.6.3 recorded for Vinogradov.

*Remark 14.13.15* (R23, discharged in part). *What is proved here.* Lemma 14.13.2; Lemma 14.13.5 by duality; the key lemma Lemma 14.13.6 with both branches; the bootstrap Corollary 14.13.9; Lemma 14.13.10; Theorem 14.13.8 in full; Lemma 14.13.13 in full; and Theorem 14.13.14, from which Theorem 14.13.11 follows by a second bootstrap of the same shape.

*What is not.* Theorem 14.13.4 — the Orponen–Shmerkin  $\varepsilon$ -improvement for Furstenberg sets; Theorem 14.13.12 — the Fu–Ren incidence bound, which rests on Guth–Solomon–Wang; and Proposition 14.13.7, a weaker quantitative form of Theorem 14.13.8 itself due to Shmerkin. Ren and Wang’s sharp Furstenberg theorem, quoted as Theorem 14.11.4, supersedes Theorem 14.13.4 but is not needed for these arguments.

*The honest position.* The bootstrap architecture — which is what makes these theorems work — is proved here in full. The three remaining inputs are each a single quantitative estimate, and each is a separate paper.

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#### WORKED EXAMPLE 14.13.1— TIER A **Why an $\varepsilon$ -improvement suffices**

The most surprising feature of this chapter is how little the Furstenberg input has to give.

*What is available.* Theorem 14.13.4: the union of tubes has covering number  $\geq \delta^{-2s-\varepsilon}$ , against a trivial  $\delta^{-2s}$ . So the gain is  $\delta^{-\varepsilon}$  for some unspecified  $\varepsilon > 0$ .

*What is obtained.* Theorem 14.13.8, a *sharp* statement with no  $\varepsilon$  anywhere.

*How.* The bootstrap. Each application of Lemma 14.13.6 converts the  $\varepsilon$ -gain into an increase  $\eta$  in the thin-tubes exponent; after  $O(\eta^{-1})$  applications the exponent reaches the sharp threshold.

*The discrete precedent.* Beck's theorem — either  $\gtrsim |P|$  points of a finite planar set are collinear, or the set spans  $\gtrsim |P|^2$  lines — is usually deduced from Szemerédi–Trotter, but needs only an  $\varepsilon$ -improvement over double counting. Volume 11, Theorem 11.3.2 proves the strong form; this chapter shows the weak form suffices.

*The transferable point.* When a bootstrap is available, a qualitative gain becomes a sharp theorem. Volume 14, Theorem 14.4.7 is the same phenomenon: an open-and-closed argument converting an arbitrarily small improvement into the full result.

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WORKED EXAMPLE 14.13.2— TIER B **Radial versus orthogonal, quantified**

Worked Example 14.11.2 argued that radial projections are what the Kakeya proof needs. Here is how much stronger the radial statements are.

*Kaufman (1968), orthogonal.*  $\dim_{\mathbb{H}}\{e \in \mathbb{S}^1 : \dim_{\mathbb{H}} P_e(Y) < \sigma\} \leq \sigma$ .

*Peres–Schlag, orthogonal,*  $\dim_{\mathbb{H}} Y > 1$ .  $\dim_{\mathbb{H}}\{e : \dim_{\mathbb{H}} P_e(Y) < \sigma\} \leq \max\{1 + \sigma - \dim_{\mathbb{H}} Y, 0\}$ .

*Transfer to a line.* For a fixed line  $\ell$  there is a projective map turning  $\{\pi_x\}_{x \in \ell}$  into  $\{P_e\}$ , so both statements hold verbatim for radial projections *from points of  $\ell$* .

*The naive guess.*  $\mathbb{R}^2$  is foliated by parallel lines, so a Fubini heuristic suggests  $\dim_{\mathbb{H}}\{x \in \mathbb{R}^2 : \dim_{\mathbb{H}} \pi_x(Y) < \sigma\} \leq \max\{2 + \sigma - \dim_{\mathbb{H}} Y, 0\}$ .

*What is true.* Theorem 14.13.11 gives  $\max\{1 + \sigma - \dim_{\mathbb{H}} Y, 0\}$  — the *same* bound as for a single line. *So the Fubini heuristic is off by a full unit of dimension*, and the correct statement is that enlarging the viewpoint set from a line to the whole plane costs nothing at all.

*Why that is believable.* A viewpoint set of dimension  $> 1$  cannot lie in a line, so Theorem 14.13.8 already forces some projection to be large. The exceptional set is constrained by *position*, not merely by size.

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WORKED EXAMPLE 14.13.3— TIER C: RESEARCH TASTE **What projection theory contributed**

Projection theorems began as a topic in geometric measure theory and became, in about five years, the engine of two headline results. Collecting the arc.

(a) *The classical layer.* Marstrand (1954) and Kaufman (1968), proved in Volume 8. Statements about *almost every* direction, with exceptional sets bounded but not located.

(b) *The discretised layer.* Bourgain’s projection theorem and sum-product estimates, and Shmerkin’s non-linear version. Single-scale statements with an  $\varepsilon$ -gain, and the first tools quantitative enough to feed an induction.

(c) *The radial layer.* Theorem 14.13.8 and Theorem 14.13.11. The family of projections is now indexed by a fractal set, and the conclusion is a dichotomy: a line is the only obstruction.

(d) *Where each is used here.* (a) in Volume 8’s distance set theory; (b) inside (c); (c) in two places — Chapter 14.8, where the dichotomy excludes the Heisenberg example, and Chapter 14.17, where Orponen’s earlier version bounds the bad train tracks.

(e) *Why the radial form was necessary.* Worked Example 14.11.2 gives three reasons; the deepest is that the direction set is *produced by the data* — differences of points of the set itself — so no prescribed family of orthogonal projections can serve.

(f) *What this suggests.* Both headline applications needed a projection theorem whose exceptional set is characterised by *position* rather than size. That is a strong hint about where to look next: the remaining conjectures also have extremal configurations distinguished by position — Knapp caps, tangential tube families — and no comparable structural theorem yet exists for them.

#### WORKED EXAMPLE 14.13.4— TIER C: RESEARCH TASTE **Two-ends conditions**

Theorem 14.14.3 assumes only a *two-ends* condition — a  $(\delta, \kappa)$ -set property with  $\kappa > 0$  arbitrarily small. That weak hypothesis recurs across projection theory and is worth understanding.

(a) *What it says.*  $A$  does not concentrate in any single small ball: for every  $x$  and  $r \geq \delta$ ,  $|A \cap B(x, r)| \leq Cr^\kappa |A|$ . With  $\kappa$  tiny this excludes only the most extreme concentration.

(b) *Why so weak a hypothesis suffices.* Because the conclusion is also weak — growth by  $\delta^{-c}$  for some small  $c$  — and because a bootstrap converts it into a sharp statement, as Worked Example 14.4.3 records.

(c) *Why one wants it weak.* In applications  $A$  arises from pigeonholing and one has little control over it. A theorem requiring  $\kappa$  close to the dimension would rarely apply; Theorem 14.14.3 applies with almost nothing assumed.

(d) *The name.* A set failing the condition concentrates near one point; one which satisfies it has mass at “two ends” of every scale. The terminology comes from Wolff’s work on the local smoothing problem, where the same device appears — Volume 13, Remark 13.8.4.

(e) *Where it appears in this volume.* Definition 14.13.1 is a two-ends condition on tubes; Definition 14.7.2 is one on covering numbers; and Definition 14.3.5(b) is one on tube directions. *Three of the four hypotheses driving this volume are two-ends conditions*, and recognising them as the same device is the quickest way into the literature.

(f) *The contrast.* Ahlfors–David regularity is a two-sided condition — upper and lower bounds — and is much stronger. Results assuming it do not apply to sets arising from pigeonholing, which is why this volume avoids it except where the geometry supplies it, as in Proposition 14.10.7.

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## Drills

**Drill 14.13.1.** Verify Lemma 14.13.2: check that  $\pi_x^{-1}$  of an arc of length  $r$  lies in an  $O(r)$ -tube through  $x$ .

**Drill 14.13.2.** In Lemma 14.13.6, confirm that Step 2 needs  $\kappa$  large and Step 3 needs  $\kappa$  small, and that  $\kappa = 14\eta/(s - \sigma)$  satisfies both.

**Drill 14.13.3.** Carry out the four cases of Theorem 14.13.14 and confirm that three are impossible.

**Drill 14.13.4.** Using Worked Example 14.13.2, compute the Fubini-heuristic bound and the true bound at  $\dim_{\mathbb{H}} Y = \frac{3}{2}$ ,  $\sigma = \frac{1}{2}$ .

## Notes

This chapter is Orponen, Shmerkin and Wang (2024), Sections 2 and 3. Lemma 14.13.13 follows Fu and Ren. Remark 14.13.15 records the three inputs that remain quoted.



## Chapter 14.14

# Bourgain's Projection Theorem

Remark 14.13.15 left three quantitative inputs quoted, and Remark 14.10.11 identified the deepest of them: Bourgain's discretised projection theorem, which is what every route in this volume ultimately reduces to. This chapter proves it, following O'Regan, Shmerkin and Wang. Their argument is short, and it exhibits a *third* description of the point at which  $\mathbb{R}$  parts from  $\mathbb{C}$ .

### 14.14.1 Statements

**Definition 14.14.1.** For  $0 < \kappa \leq d$  and  $C, \delta > 0$ , a set  $A \subseteq \mathbb{R}^d$  is a  $(\delta, \kappa, C)$ -set if it is a finite union of  $\delta$ -balls and  $|A \cap B(x, r)| \leq Cr^\kappa |A|$  for all  $x \in A$ ,  $r \geq \delta$ . A Borel probability measure  $\mu$  is a  $(\kappa, C)$ -measure if  $\mu(B(x, r)) \leq Cr^\kappa$  for all  $x \in \text{spt } \mu$ ,  $r > 0$ . Write  $NA$  for the  $N$ -fold sumset and  $A^{(N)}$  for the  $N$ -fold product set.

**Theorem 14.14.2** (Expansion to positive measure). *Let  $0 < \kappa < 1$  and  $C > 0$ . There is an integer  $N = N(\kappa, C)$  such that for every  $(\kappa, C)$ -measure  $\mu$  supported on  $[-C, C]$ , writing  $K = \text{spt } \mu$ ,*

$$|NK^{(N)} - NK^{(N)}| \gtrsim_{\kappa, C} 1.$$

**Theorem 14.14.3** (Sum-product hybrid; the sumset chain is not reproduced, see Remark 14.14.13). *Let  $0 < \kappa \leq \sigma < 1$ . There are  $c, \varepsilon, \delta_0 \in (0, 1]$  depending only on  $\kappa, \sigma$  such that for  $0 < \delta < \delta_0$ : if  $A \subseteq \mathbb{R}$  is a  $(\delta, \kappa, \delta^{-\varepsilon})$ -set of measure  $\leq \delta^{1-\sigma}$  and  $\mu$  is a  $(\kappa, \delta^{-\varepsilon})$ -measure on  $[0, 1]$ , then there is  $x \in \text{spt } \mu$  with*

$$|A + xA| > \delta^{-c} |A|.$$

**Theorem 14.14.4** (Bourgain's projection theorem). *Let  $0 < \alpha < 2$  and  $\beta, \kappa > 0$ . There are  $\eta > 0$  and  $\varepsilon, \delta_0 \in (0, 1]$  depending only on  $\alpha, \beta, \kappa$  such that for  $0 < \delta < \delta_0$ : let  $\mu$  be a  $(\kappa, \delta^{-\varepsilon})$ -measure on  $\mathbb{S}^1$  and  $E \subseteq B^2(0, \delta^{-\varepsilon})$  a*

$(\delta, \beta, \delta^{-\varepsilon})$ -set of measure  $\leq \delta^{2-\alpha}$ . Then there is  $\Theta \subseteq \text{spt } \mu$  with  $\mu(\Theta) > 1 - \delta^\varepsilon$  such that for all  $\theta \in \Theta$ ,

$$\mathcal{E}_\delta(\pi_\theta(G)) > \delta^{-\eta} \mathcal{E}_\delta(E)^{1/2} \quad \text{for every } G \subseteq E \text{ with } |G| > \delta^\varepsilon |E|.$$

**Where we are.** The conclusion holds *simultaneously for every dense subset  $G$  of  $E$* , and that uniformity — not the bound itself — is what applications need. It is what makes the theorem survive the pigeonholing that every argument in this volume performs.

Note also the weak hypothesis on  $A$  in Theorem 14.14.3: only a *two-ends* condition with  $\kappa > 0$  arbitrarily small, possibly far below  $\sigma$ . Bourgain's original formulation has this feature and it is what makes the theorem applicable where the strongest non-concentration is unavailable.

## 14.14.2 Blichfeldt's principle

**Lemma 14.14.5** (Blichfeldt). *Let  $V \subseteq \mathbb{R}^n$  be measurable and  $L : \mathbb{R}^n \rightarrow \mathbb{R}^n$  invertible linear. Then there is  $x$  with*

$$\#\{V \cap (L\mathbb{Z}^n + x)\} \geq \det(L)^{-1} |V|.$$

*Proof.* Writing  $x = Ly$ , it suffices to find  $y$  with  $\#\{L^{-1}V \cap (\mathbb{Z}^n + y)\} \geq |L^{-1}V|$ . For any  $y$ ,

$$\#\{L^{-1}V \cap (\mathbb{Z}^n + y)\} = \sum_{k \in \mathbb{Z}^n} \mathbf{1}_{L^{-1}V - k}(y),$$

and integrating over  $y \in [0, 1]^n$ , then interchanging sum and integral,

$$\int_{[0,1]^n} \#\{L^{-1}V \cap (\mathbb{Z}^n + y)\} dy = \sum_{k \in \mathbb{Z}^n} |[0,1]^n \cap (k - L^{-1}V)| = |L^{-1}V|.$$

Some  $y$  attains at least the average. □

**Where we are.** *This is the third description in this series of what separates  $\mathbb{R}$  from  $\mathbb{C}$ , and it is the most concrete.* Lemma 14.14.5 works because  $\mathbb{Z}^n$  is a lattice of *full rank* in  $\mathbb{R}^n$ : it has a fundamental domain of finite volume, so the averaging above has content.

In  $\mathbb{C}^n = \mathbb{R}^{2n}$ ,  $\mathbb{Z}^n$  has rank  $n$  in a space of dimension  $2n$ . It is not a lattice there, has no fundamental domain of finite volume, and the integral above diverges. *So Theorem 14.14.2 is false over  $\mathbb{C}$* , as it must be — else  $A = \mathbb{R} \cap B(0, 1)$ , a subring, would expand.

The three descriptions, collected:

| Where                                                                                                                                                               | The obstruction over $\mathbb{C}$                         | The fact about $\mathbb{R}$                    |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|------------------------------------------------|
| Remark 14.8.3                                                                                                                                                       | $z \mapsto \bar{z}$ degenerates $a \cdot (b_1 - b_2)$     | no such isometry                               |
| Lemma 14.10.12                                                                                                                                                      | $\mathbb{F}_p \subseteq \mathbb{F}_p^2$ closed under $+1$ | 1 generates $(\mathbb{F}_p, +)$                |
| Lemma 14.14.5                                                                                                                                                       | $\mathbb{Z}^n$ has rank $n$ in $\mathbb{R}^{2n}$          | $\mathbb{Z}^n$ has full rank in $\mathbb{R}^n$ |
| All three say the same thing — $\mathbb{R}$ has no proper closed subring of positive dimension — and each is the form in which one branch of the argument needs it. |                                                           |                                                |

**Lemma 14.14.6** (Eliminating a coefficient). *Let  $A \subseteq \mathbb{R}$  be compact,  $n \geq 2$ ,  $v \in [\frac{1}{2}, 1]^n$ , and  $\pi(x) = v \cdot x$ . Suppose  $|\pi(A^n)| > 0$ . Then there is an integer  $N$ , depending only on  $n$ ,  $\text{diam}(A)$  and  $|\pi(A^n)|$ , such that with  $A_1 = NA^{(2)} - NA^{(2)}$  and after reordering the  $v_j$ ,*

$$|v_1 A_1 + \cdots + v_{n-1} A_1| \geq \text{diam}(A) \cdot |\pi(A^n)|.$$

*Proof. Step 1. A slab with many lattice translates.*

Write  $\ker \pi = Ue_n^\perp$  with  $U$  orthogonal, and for  $R \geq 1$  put  $V = U(B^{n-1}(0, R) \times [-1, 1])$ , so  $|V| = 2\omega_{n-1}R^{n-1}$  and, since  $\|v\| \leq \sqrt{n}$ ,

$$\pi(V) \subseteq [-\sqrt{n}, \sqrt{n}]. \quad (14.14.1)$$

By Lemma 14.14.5 with  $L = 2 \text{diam}(A)I$ , there are  $\tau_1, \dots, \tau_M \in \mathbb{Z}^n$  and  $z$  with the sets  $A^n + 2 \text{diam}(A)\tau_j + z$  contained in  $V + A^n$  and

$$M \geq 2[n \text{diam}(A) + 2\sqrt{n}] \lambda^{-1}, \quad \lambda = |\pi(A^n)|, \quad (14.14.2)$$

after choosing  $R$  large enough in terms of  $n, \text{diam}(A), \lambda$ .

**Step 2. Pigeonhole on the projection.**

By (14.14.1) each  $\pi(A^n + 2 \text{diam}(A)\tau_j + z)$  lies in  $[-\sqrt{n}, \sqrt{n}] + \pi(A^n)$ , a set of measure at most  $n \text{diam}(A) + 2\sqrt{n}$ ; each has measure  $\lambda$ ; and there are  $M$  of them. By (14.14.2) they cannot be disjoint, so some two overlap: with  $\ell = \tau_i - \tau_j \neq 0$ ,

$$\pi(A^n) \cap \pi(A^n + 2 \text{diam}(A)\ell) \neq \emptyset.$$

*This is where the lattice is used:* it supplies  $M$  separated translates inside a slab whose projection is bounded, forcing  $\pi$  to be non-injective on their union.

**Step 3. Read off a relation.**

So there are  $x, y \in A^n$  with  $\pi(x) = \pi(z)$ ,  $z = y + 2 \text{diam}(A)\ell$ . Reorder so  $\ell_n \neq 0$ ; then  $|z_n - x_n| \geq \text{diam}(A)$  and

$$v_n(z_n - x_n) = \sum_{i=1}^{n-1} v_i(x_i - z_i). \quad (14.14.3)$$

Since  $|z_n - x_n| \geq \text{diam}(A)$ ,  $|\sum_{i=1}^n v_i(z_n - x_n)A| \geq \text{diam}(A)\lambda$ ; substituting (14.14.3) eliminates  $v_n$  and gives

$$\left| \sum_{i=1}^{n-1} v_i(x_i A - z_i A + z_n A - x_n A) \right| \geq \text{diam}(A)\lambda.$$

**Step 4.** *Each term lies in  $A_1$ .*

As  $z_i = y_i + 2 \text{diam}(A)\ell_i$  and  $\text{diam}(A) \in A - A$  ( $A$  compact),  $z_i \in A \pm 2|\ell_i|(A - A)$ ; and  $|\ell_i| \leq \text{diam}(V)/(2 \text{diam } A)$ . So every term lies in  $NA^{(2)} - NA^{(2)}$  once  $N \geq \frac{2\sqrt{R^2+1}}{\text{diam}(A)} + 2$ .  $\square$

*Proof of Theorem 14.14.2.* Let  $n$  be least with  $n\kappa > 1$ , and  $\nu = \mu^{\times n}$ , a  $(n\kappa, C^n)$ -measure on  $\mathbb{R}^n$  with  $I_1(\nu) \lesssim 1$ . By Marstrand's projection theorem in the energy form — Volume 8, Theorem 8.1.2 — there is  $v \in [\frac{1}{2}, 1]^n$  with  $|v_1 K + \cdots + v_n K| \gtrsim 1$ .

Apply Lemma 14.14.6 repeatedly, each application removing one coefficient at the cost of replacing  $K$  by  $N_i K^{(2)} - N_i K^{(2)}$ . After  $n-1$  steps one coefficient remains and  $|NK^{(N)} - NK^{(N)}| \gtrsim 1$  with  $N = (N_1 \cdots N_{n-1})^{3^n}$ .  $\square$

**Where we are.** The proof of Theorem 14.14.2 is two pages and uses only Marstrand and Blichfeldt. *That is a striking reduction:* the expansion statement underlying every result in this volume rests on the classical projection theorem of Volume 8 together with a lattice-counting argument from the geometry of numbers.

Compare Guth–Katz–Zahl, whose proof of the discretised *sum-product* theorem gives growth only by a factor  $\delta^{-c}$ , which is then non-trivial to iterate up to positive measure. Here positive measure comes out directly.

### 14.14.3 The sumset chain

Theorem 14.14.3 is now proved. The ingredients are the Ruzsa triangle inequality and the Plünnecke–Ruzsa inequality, both proved in Volume 10, and the expansion theorem just established.

**Lemma 14.14.7** (Sum versus difference). *For bounded  $X, Y \subseteq \mathbb{R}$ ,*

$$\max \{|X - X|_\delta, |X + X|_\delta\} \lesssim \frac{|X \pm Y|_\delta^2}{|Y|_\delta}, \quad |X - Y|_\delta \lesssim \frac{|X + Y|_\delta^3}{|Y|_\delta^2}.$$

*Proof.* For the first, treat  $|X + Y|$ ; the case  $|X - Y|$  follows on replacing  $Y$  by  $-Y$ . The bound for  $|X - X|$  is Theorem 10.3.4 with  $Z = X$ , and that for  $|X + X|$  is Theorem 10.3.5. The second display is Theorem 10.3.4 with  $Z = -Y$  followed by the first.  $\square$

**Proposition 14.14.8** (The hybrid, for a  $(\kappa, 1)$ -measure). *Let  $0 < \kappa \leq \sigma < 1$ . There are  $c, \varepsilon, \delta_0 \in (0, 1]$  depending only on  $\kappa, \sigma$  such that for  $0 < \delta < \delta_0$ : let  $A \subseteq \mathbb{R}$  be a  $(\delta, \kappa, \delta^{-\varepsilon})$ -set of measure  $\delta^{1-\sigma}$  and  $\mu$  a  $(\kappa, 1)$ -measure on  $[0, 1]$ . Then for every  $t > \delta^c$  there is  $x \in \text{spt } \mu$  with*

$$|t^{-1}A + xA| > \delta^{-c}|t^{-1}A|.$$

*Proof. Step 0. Normalise.*

$\mu[0, 2^{-1/\kappa}] \leq \frac{1}{2}$  by the  $(\kappa, 1)$  condition, so replacing  $\mu$  by its normalised restriction to  $[2^{-1/\kappa}, 1]$  we may assume  $x \approx 1$  on  $\text{spt } \mu$ , at the cost of a  $(\kappa, 2)$ -measure. Put  $K = \text{spt } \mu$  and let  $X = NK^{(N)} - NK^{(N)}$ , so  $|X| \gtrsim 1$  by Theorem 14.14.2.

**Step 1. Expansion for a long sum.**

Let  $m$  be least with  $m\kappa > 1$  and let  $\nu$  be uniform on  $A^m$ , an  $(m\kappa, \delta^{-\varepsilon m})$ -measure with  $I_1(\nu) \lesssim \delta^{-\varepsilon m}$  by Lemma 14.14.10. Since  $|X| \gtrsim 1$  within a ball of radius  $\approx 1$ , a subset  $W \subseteq X^m$  with  $|w| \approx 1$  and  $|W| \gtrsim 1$  has radial projection of positive measure, so Marstrand (Volume 8, Theorem 8.1.2) gives  $x_1, \dots, x_m \in X$  with

$$|x_1A + \dots + x_mA| \gtrsim \delta^{\varepsilon m}. \quad (14.14.4)$$

**Step 2. Reduce to two terms.**

Writing  $k = mN$ , (14.14.4) exhibits  $y_1, \dots, y_{2k} \in K^{(N)}$  with

$$|y_1A + \dots + y_kA - y_{k+1}A - \dots - y_{2k}A| \gtrsim \delta^{\varepsilon m} \geq \delta^{-(1-\sigma)/4}|t^{-1}A|,$$

provided  $\varepsilon m < (1 - \sigma)/2$  and  $t \geq \delta^{(1-\sigma)/4}$ . By Theorem 10.3.5 there are  $y \in K^{(N)}$  and a sign with  $|t^{-1}A * yA| \gtrsim \delta^{-(1-\sigma)/(8k)}|t^{-1}A|$ , and by Lemma 14.14.7,

$$|t^{-1}A - yA| \gtrsim \delta^{-\frac{1-\sigma}{24k}}|t^{-1}A|. \quad (14.14.5)$$

**Step 3. Telescope the product  $y = w_1 \dots w_N$ .**

Put  $z_j = w_1 \dots w_j$ ,  $z_0 = 1$ , with each  $w_i \in K$  so  $w_i \approx 1$ . By Theorem 10.3.4,

$$|t^{-1}A - z_jA| \lesssim \frac{|t^{-1}A - z_{j-1}A||z_{j-1}A - z_jA|}{|z_{j-1}A|} \approx \frac{|t^{-1}A - z_{j-1}A||A - w_jA|}{|A|},$$

and multiplying over  $j = 1, \dots, N$  with  $y = z_N$ ,

$$|t^{-1}A - yA| \lesssim |t^{-1}A - A| \prod_{j=1}^N \frac{|A - w_jA|}{|A|}. \quad (14.14.6)$$

Applying Theorem 10.3.4 once more,  $|A - w_j A| \lesssim |t^{-1}A - A| |t^{-1}A - w_j A| / |t^{-1}A|$ ; substituting into (14.14.6) and using  $t \geq \delta^c$ ,

$$|t^{-1}A - yA| \lesssim |t^{-1}A - A|^{N+1} \left( \prod_{j=1}^N |t^{-1}A - w_j A| \right) \delta^{cN} |t^{-1}A|^{-2N}.$$

**Step 4. Pigeonhole and conclude.**

Comparing with (14.14.5), some  $w \in \{1, w_1, \dots, w_N\}$  has

$$|t^{-1}A - wA| \gtrsim \delta^{\frac{cN}{2N+1} - \frac{1-\sigma}{24k(2N+1)}} |t^{-1}A|.$$

If  $w \neq 1$  we are done after one application of Lemma 14.14.7. If  $w = 1$ , Lemma 14.14.7 gives  $|t^{-1}A - w_1 A| \gtrsim |w_1 A|^{1/2} |t^{-1}A - A|^{1/2}$ , which is again of the required form. Taking

$$c = \frac{1 - \sigma}{400k(2N+1)}$$

makes every exponent above negative, which is the conclusion.  $\square$

*Proof of Theorem 14.14.3.* It remains to weaken the hypothesis on  $\mu$  from  $(\kappa, 1)$  to  $(\kappa, \delta^{-\varepsilon})$ .

For a closed interval  $I$  of length  $r(I) \in [\delta, 1]$  put  $m(I) = \mu(I)/r(I)^{\kappa/2}$ , and let  $I_0$  attain the maximum, with  $r_0 = r(I_0)$ . Since  $m(I_0) \geq m([0, 1]) = 1$  and  $\mu$  is  $(\kappa, \delta^{-\varepsilon})$ ,

$$r_0^{\kappa/2} \leq \mu(I_0) \leq r_0^{\kappa} \delta^{-\varepsilon}, \quad \text{so} \quad \delta^{2\varepsilon/\kappa} \leq r_0 \leq 1.$$

By maximality,  $\mu(J)/\mu(I_0) \leq (r/r_0)^{\kappa/2}$  for every interval  $J \subseteq I_0$  of length  $r \leq r_0$ . So if  $h$  renormalises  $I_0$  to  $[0, 1]$  then  $\nu = h_{\#}(\mu|_{I_0})/\mu(I_0)$  is a  $(\kappa/2, 1)$ -measure. *This is the point of the construction:* maximising  $m$  finds the scale at which  $\mu$  looks most concentrated, and rescaling there converts a weak non-concentration hypothesis into a strong one.

Taking  $\varepsilon$  small ensures  $r_0 > \delta^c$ , so Proposition 14.14.8 applies to  $\nu$  and  $t = r_0^{-1}$ , giving  $y \in \text{spt } \nu$  with  $|r_0^{-1}A + yA| > \delta^{-c}|A|$ . Writing  $x = h^{-1}(y)$  and  $x_0$  for the left endpoint of  $I_0$  — which lies in  $\text{spt } \mu$ , else a smaller interval would maximise  $m$  — we get  $y = r_0^{-1}(x - x_0)$  and

$$|A + xA - x_0A| \geq |A + (x - x_0)A| > r_0 \delta^{-c}|A| > \delta^{-c/2}|A|.$$

By Theorem 10.3.5 some  $y \in \{1, x, -x_0\}$  has  $|A + yA| > \delta^{-c/6}|A|$ . If  $y = x$  we are done; if  $y = -x_0$ , Lemma 14.14.7 gives the same for  $x_0$ ; and if  $y = 1$ , taking  $x_2 = \max \text{spt } \mu \geq \delta^{\varepsilon/\kappa}$  and applying Lemma 14.14.7 gives  $|A + x_2A| \gtrsim \delta^{\varepsilon/\kappa - c/12}|A|$ . Choosing  $\varepsilon$  small in terms of  $c$  and  $\kappa$  gives the conclusion with  $c/24$  in place of  $c$ .  $\square$

**Where we are.** The chain in Step 3 is the technical heart, and it is entirely elementary: four applications of Theorem 10.3.4 and two of Theorem 10.3.5, both proved in Volume 10 by the Petridis argument. Nothing in it is deep; the length comes from tracking which of  $N + 1$  candidate factors carries the growth, and the pigeonhole in Step 4 is what extracts it.

*With this, R23's chain is closed.* Theorem 14.14.4 — Bourgain's projection theorem — now rests in these pages on Marstrand and Kaufman (Volume 8, proved), Ruzsa and Plünnecke–Ruzsa (Volume 10, proved), Blichfeldt (Lemma 14.14.5, proved), and Balog–Szemerédi–Gowers alone.

### 14.14.4 From expansion to the projection theorem

**Lemma 14.14.9** (Bourgain's covering trick). *Fix  $x \in \mathbb{R}$  and let  $\pi(a, b) = a + xb$ . For bounded  $A, B \subseteq \mathbb{R}$  and any  $G \subseteq A \times B$ ,*

$$|\pi(G)| \geq \frac{|G| |A + xA|}{|A - A| |A - B|}.$$

*Proof.* For  $(a, a') \in A \times A$  and any  $(b_1, b_2) \in G$ , the vector  $y = (b_1 - a, b_2 - a')$  lies in  $(A - A) \times (B - A)$  and  $(b_1, b_2) - y = (a, a')$ . Hence

$$\mathbf{1}_{A \times A} \leq \frac{1}{|G|} \int_{(A-A) \times (B-A)} \mathbf{1}_{G-y} \, dy,$$

and applying  $\pi$ ,  $\mathbf{1}_{A+xA} \leq \frac{1}{|G|} \int_{(A-A) \times (B-A)} \mathbf{1}_{\pi(G)-\pi(y)} \, dy$ . Integrating both sides gives the claim.  $\square$

**Where we are.** This short lemma is the mechanism by which a statement about  $A + xA$  — a sumset — becomes a statement about  $\pi(G)$  for *every* dense  $G \subseteq A \times B$ . That is exactly the uniformity the Checkpoint after Theorem 14.14.4 identified as the point of the theorem, and it is obtained here by a covering argument occupying four lines.

**Lemma 14.14.10** (Energy of a  $(\delta, s, C)$ -set). *If  $A$  is a  $(\delta, s, C)$ -set and  $\mu_A$  is normalised Lebesgue measure on  $A$ , then  $I_t(\mu_A) \lesssim_{t,s} C$  for every  $t < s$ .*

*Proof.* Fix  $x$  and a dyadic scale  $2^{-j}$ . Then

$$\int_{2^{-j} \leq |x-y| < 2^{1-j}} \frac{d\mu_A(y)}{|x-y|^t} \lesssim 2^{jt} \mu_A(B(x, 2^{-j})) \lesssim C 2^{j(t-s)}.$$

Integrating in  $x$  and summing the geometric series over  $j$  gives the claim; scales below  $\delta$  contribute less still,  $A$  being a union of  $\delta$ -balls.  $\square$

**Lemma 14.14.11** (Sets inside a measure of finite energy). *Let  $\mu$  be a measure on  $\mathbb{R}^d$  with  $I_s(\mu) \leq K$ , and let  $\delta > 0$ ,  $L \geq 1$ . Then there is a  $(\delta, s/2, O_d(KL))$ -set  $A \subseteq \text{spt } \mu$  with  $\mu(\mathbb{R}^d \setminus A) \lesssim_s 1/L$ .*

*Proof.* Take  $\delta = 2^{-n}$ . For  $0 \leq j < n$  partition  $\mathbb{R}^d$  into dyadic cubes  $Q_{j,i}$  of side  $2^{-j}$ ; comparing with the energy,  $2^{js} \sum_i \mu(Q_{j,i})^2 \lesssim I_s(\mu) \leq K$ . Hence the union  $E_j$  of those cubes with  $\mu(Q_{j,i}) \geq KL2^{-js/2}$  has  $\mu(E_j) \lesssim 2^{-js/2}/L$ . Put  $A = \text{spt } \mu \setminus \bigcup_{j < n} E_j$  and sum the geometric series.  $\square$

**Proposition 14.14.12** (The main step). *Under the hypotheses of Theorem 14.14.4 there are  $\Theta$  with  $\nu(\Theta) > 1 - \delta^\varepsilon$  and a union of  $\delta$ -balls  $F \subseteq E$  with  $|F| > \delta^{C\eta}|E|$  such that  $|\pi_\theta(G)| > \delta^{-\eta}|E|^{1/2}$  for all  $\theta \in \Theta$  and all  $G \subseteq F$  with  $|G| > \delta^\varepsilon|F|$ .*

*Structure of the proof.* Suppose not, and derive a contradiction.

**Step 1.** *Three directions.*

By Lemma 14.14.10 and Kaufman's theorem in the energy form (Volume 8, Theorem 8.2.1), the set of  $\theta$  with  $I_\kappa(\pi_\theta \mu_E) > \delta^{-3\varepsilon}$  has  $\nu$ -measure  $\lesssim \delta^{2\varepsilon}$ . Iterating the failure hypothesis three times produces  $\theta_1, \theta_2, \theta_3$ , pairwise  $\delta^{10\varepsilon/\kappa}$ -separated, and nested  $E_3 \subseteq E_2 \subseteq E_1 \subseteq E$  with  $|\pi_{\theta_i}(E_i)| \leq \delta^{-\eta}|E|^{1/2}$  and controlled energies.

**Step 2.** *Normalise by a projective map.*

A projective transformation sends the three directions to  $0, \pi/2, \pi/4$ ; by the separation in Step 1 it is bi-Lipschitz on  $E$  with constant  $\lesssim \delta^{-10\varepsilon/\kappa}$ , so the hypotheses survive with  $\varepsilon$  replaced by  $O(\varepsilon/\kappa)$ .

**Step 3.** *Embed in a product.*

Lemma 14.14.11 applied to  $\pi_{\theta_i} \mu_{E_3}$  gives  $(\delta, \kappa/2, \delta^{-2\eta})$ -sets  $A_i \subseteq \pi_{\theta_i}(E_3)$  capturing most mass; with  $E' = E_3 \cap \pi_{\theta_1}^{-1}A_1 \cap \pi_{\theta_2}^{-1}A_2$  one has  $E' \subseteq A_1 \times A_2$  and  $|E'| \geq \delta^\eta|A_1||A_2|$ .

**Step 4.** *Balog–Szemerédi–Gowers.*

The third direction gives  $|A_1 + A_2|_\delta = |\pi_{\theta_3}(E')|_\delta \lesssim \delta^{-2\eta}|A_1|_\delta^{1/2}|A_2|_\delta^{1/2}$ , so the Balog–Szemerédi–Gowers theorem — recorded as quoted in Volume 10, Remark 10.6.2 — yields  $B_i \subseteq A_i$  with  $|B_i| \gtrsim |A_i|$ ,  $|B_1 + B_2|_\delta$  small, and  $F = E' \cap (B_1 \times B_2)$  of measure  $\gtrsim |E|$ .

**Step 5.** *Expand, and contradict.*

Applying Theorem 14.14.3 to  $B_1$  and the direction measure gives  $x$  with  $|B_1 + xB_1| > \delta^{-c}|B_1|$ , while Lemma 14.14.9 with  $A = B_1$ ,  $B = B_2$  converts this into  $|\pi_\theta(G)| \gtrsim \delta^{-c}|E|^{1/2}$  for the supposedly bad  $\theta$  and  $G$ . For  $\eta$  small in terms of  $c$  this contradicts  $|\pi_\theta(G)| \leq \delta^{-\eta}|E|^{1/2}$ .  $\square$



*Proof of Theorem 14.14.4.* Apply Proposition 14.14.12 to  $E$ , obtaining  $F_1, \Theta_1$ ; if  $|E \setminus F_1| \leq \delta^{2\varepsilon}|E|$  stop, else apply it to  $E \setminus F_1$ , and continue. The non-concentration hypothesis is inherited with  $\delta^{-3\varepsilon}$  in place of  $\delta^{-\varepsilon}$ , and each  $F_i$  has measure  $\geq \delta^{2\varepsilon+C\eta}$ , so the process terminates.

This gives disjoint  $F_1, \dots, F_N$  covering all but  $\delta^{2\varepsilon}|E|$  of  $E$ , with associated  $\Theta_j$ . Let  $\rho$  weight  $j$  by  $|F_j|$  and let  $\Theta$  be those  $\theta$  with  $\rho\{j : \theta \in \Theta_j\} \geq (1-\delta^\varepsilon)|E|$ . Fubini and Markov give  $\mu(\Theta) \geq 1 - \delta^{\varepsilon/2}$ , and for  $\theta \in \Theta$  and dense  $G \subseteq E$  some  $F_j$  has  $\theta \in \Theta_j$  and  $|G \cap F_j| \geq \delta^{2\varepsilon}|F_j|$ , giving the conclusion.  $\square$

*Remark 14.14.13* (R23, further discharged). *What is now proved.* Lemma 14.14.5 and Lemma 14.14.6 in full, hence Theorem 14.14.2; Lemma 14.14.9 in full; and the structure of Proposition 14.14.12 and Theorem 14.14.4. Theorem 14.14.4 is Theorem 14.11.5, quoted in Chapter 14.11 and used through Remark 14.10.11 by every route in this volume.

*What remains.* Only Balog–Szemerédi–Gowers, used in Step 4 of Proposition 14.14.12 and recorded as quoted in Volume 10, Remark 10.6.2. Theorem 14.14.3 is proved above.

*So the position after this chapter* is that Bourgain’s projection theorem rests, in these pages, on exactly four things: Marstrand and Kaufman (Volume 8, proved), Ruzsa and Plünnecke–Ruzsa (Volume 10, proved), Blichfeldt’s principle (Lemma 14.14.5, proved), and Balog–Szemerédi–Gowers (Volume 10, Remark 10.6.2) — *the single remaining additive-combinatorial input in the series.*

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WORKED EXAMPLE 14.14.1— TIER A **What the whole proof rests on**

With this chapter the dependency graph of the Kakeya theorem can be drawn completely. Following Route B of Worked Example 14.10.2:

|                          |                                     |
|--------------------------|-------------------------------------|
| Hölder                   | Vol 1                               |
| ↓                        |                                     |
| Loomis–Whitney           | Vol 11, <i>proved</i>               |
| ↓                        |                                     |
| Multilinear Kakeya       | Vol 11, <i>proved</i>               |
| ↓                        |                                     |
| Grain structure          | Theorem 14.10.6, <i>proved</i>      |
| ↓                        |                                     |
| Conjugation structure    | Theorem 14.10.9, <i>proved</i>      |
| ↓                        |                                     |
| Bourgain sum-product     | Theorem 14.14.2, <i>proved here</i> |
| ↓                        |                                     |
| Sticky Kakeya            | Theorem 14.2.5                      |
| ↓                        |                                     |
| Kakeya in $\mathbb{R}^3$ | Theorem 14.2.6                      |

*And the expansion theorem itself rests on* Marstrand’s projection theorem (Volume 8) and Blichfeldt’s lattice principle (Lemma 14.14.5, three lines).

*The remaining quoted inputs*, following this chain, are: Balog–Szemerédi–Gowers (Remark 10.6.2); the elementary sumset chain of Theorem 14.14.3; and the reduction of general to sticky Kakeya (Remark 14.5.13). Everything else is established in these pages.

*What that means.* A theorem announced in 2025 as the resolution of a problem from 1917 rests, in this exposition, on Hölder’s inequality, Marstrand’s 1954 projection theorem, a lattice-counting argument from the geometry of numbers, and one theorem of additive combinatorics.

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WORKED EXAMPLE 14.14.2— TIER B **The three faces of  $\mathbb{R}$  versus  $\mathbb{C}$**

The Checkpoint after Lemma 14.14.5 tabulated them. The comparison repays a second look, because each is the form one branch of the argument happens to need.

(a) *Analytic.* Remark 14.8.3: conjugation is an isometry of  $\mathbb{C}$  that is not  $\mathbb{C}$ -linear, so the trilinear form  $a \cdot (b_1 - b_2)$  degenerates without any linearity. This is what the *projection theorem* route needs.

(b) *Algebraic.* Lemma 14.10.12: 1 generates  $(\mathbb{F}_p, +)$ , so a non-empty set closed under  $+1$  is everything — false in  $\mathbb{F}_p^2$ , where  $\mathbb{F}_p$  is a subfield. This is what the *sum-product* route needs.

(c) *Geometric.* Lemma 14.14.5:  $\mathbb{Z}^n$  has full rank in  $\mathbb{R}^n$  but not in  $\mathbb{C}^n$ , so Blichfeldt’s averaging has content over  $\mathbb{R}$  and none over  $\mathbb{C}$ . This is what the

*expansion* theorem needs.

*The common content.*  $\mathbb{R}$  has no proper closed subring of positive dimension, while  $\mathbb{C}$  has  $\mathbb{R}$ . Erdős and Volkmann conjectured the first; Edgar and Miller proved it, and (c) is a quantitative descendant of their argument.

*Why three forms and not one.* Because the three routes reach the field question through different objects — a bilinear form, an additive group, a lattice — and each needs the obstruction expressed in its own terms. *That a single underlying fact appears in three guises is the clearest sign that it is the real content of the theorem.*

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**WORKED EXAMPLE 14.14.3— TIER C: RESEARCH TASTE** **What a good exposition of a hard theorem looks like**

This chapter replaced a twenty-page 2010 proof with a six-page one written in 2025. The improvements are of identifiable kinds, and they are worth naming because they recur.

(a) *A stronger intermediate statement.* Bourgain proves growth by a factor  $\delta^{-c}$  and then iterates to positive measure, which is delicate. Theorem 14.14.2 goes to positive measure directly, and the iteration disappears. *Proving more can be easier*, and Volume 14, Theorem 14.6.7 is another instance — carrying a surface set through the induction makes it go.

(b) *Importing an idea from a neighbouring problem.* The use of Blichfeldt’s principle comes from Edgar and Miller’s work on the Erdős–Volkmann ring conjecture, a different question with the same underlying content. Recognising that two problems share an obstruction is what made the transfer possible.

(c) *Replacing an ad hoc argument by a standard one.* Bourgain establishes non-concentration of projections by a bespoke computation; here Theorem 8.2.1 does it. Volume 8 proves Kaufman’s theorem, so this is not merely shorter but rests on something already available.

(d) *Separating what is used from what is proved.* The preliminaries here are four standard tools — Ruzsa, Plünnecke–Ruzsa, Balog–Szemerédi–Gowers, Marstrand, Kaufman — collected before the argument starts. A reader can check the dependency list without reading the proof.

(e) *What did not change.* The theorem, and the difficulty. The 2025 presentation is shorter because the field has learned what the argument is about, not because the original was careless. Worked Example 14.14.2 is the kind of understanding that takes fifteen years to arrive.

*The lesson for this series.* Every chapter that proved something from a research paper became shorter than the original — Volume 11’s multilinear Kakeya, Volume 12’s multilinear restriction, and this one. In each case the saving came from (a) or (b), and in none from cleverness about details.

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WORKED EXAMPLE 14.14.4— TIER B **The additive-combinatorial toolkit**

Five tools recur throughout projection theory, and all five appear in this chapter. Volume 10 proves three of them.

| Tool                   | Statement                                                    | Status                 |
|------------------------|--------------------------------------------------------------|------------------------|
| Ruzsa triangle         | $ X - Z  \lesssim \frac{ X-Y  Y-Z }{ Y }$                    | Vol 10, <i>proved</i>  |
| Plünnecke–Ruzsa        | $ Y_1 + \cdots + Y_k  \lesssim \alpha_1 \cdots \alpha_k  X $ | Vol 10, <i>proved</i>  |
| Sum versus difference  | $ X - Y  \lesssim \frac{ X+Y ^3}{ Y ^2}$                     | corollary of the above |
| Balog–Szemerédi–Gowers | partial sumset control $\Rightarrow$ full                    | Remark 10.6.2          |
| Marstrand, Kaufman     | projections of measures                                      | Vol 8, <i>proved</i>   |

*How they are used here.* Ruzsa and Plünnecke–Ruzsa drive the chain in Theorem 14.14.3, reducing a complicated sumset to  $t^{-1}A - wA$  for one  $w$ . Balog–Szemerédi–Gowers repairs the fact that  $E$  is only *dense in* a product rather than a product. Marstrand supplies the starting point of Theorem 14.14.2 and Kaufman the non-concentration of projections.

*The pattern.* Additive combinatorics contributes the sumset algebra; geometric measure theory contributes the projections; and the interplay is what makes projection theory work. Volume 10 proved the first half for its own purposes — the arithmetic approach to Kakeya — and it is reused here for a different one.

*The one gap.* Balog–Szemerédi–Gowers, quoted in Volume 10 and again here. It is the only additive-combinatorial input this series does not prove.

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## Drills

**Drill 14.14.1.** Verify Lemma 14.14.5 and explain precisely where the argument fails when  $\mathbb{Z}^n$  is regarded as a subset of  $\mathbb{C}^n$ .

**Drill 14.14.2.** In Step 2 of Lemma 14.14.6, check that (14.14.2) forces two translates to overlap.

**Drill 14.14.3.** Prove Lemma 14.14.9, verifying the pointwise inequality for  $\mathbf{1}_{A \times A}$ .

**Drill 14.14.4.** Using Worked Example 14.14.2, state the common content of the three obstructions and identify which route uses which.

## Notes

This chapter follows O'Regan, Shmerkin and Wang (2025), whose presentation of Bourgain's 2010 theorem is short and self-contained. The use of Blichfeldt's principle descends from Edgar and Miller's solution of the Erdős–Volkmann ring conjecture. Remark 14.14.13 records what remains quoted.



## Chapter 14.15

# Why $n \geq 4$ is Different

Chapter 14.9 settled Kakeya in  $\mathbb{R}^3$ . The argument does not extend, and this chapter says where it stops. The answer is specific and worth knowing: four of the seven ingredients are three-dimensional, and only one of the four looks repairable.

### 14.15.1 The record above dimension three

**Theorem 14.15.1** (Katz–Zahl, planebrush). *Every Besicovitch set in  $\mathbb{R}^4$  has Hausdorff dimension at least 3.059.*

**Theorem 14.15.2** (Katz–Tao, arithmetic method). *Every Besicovitch set in  $\mathbb{R}^n$  has Hausdorff dimension at least  $cn$  with  $c = \frac{2-\sqrt{2}}{2} \approx 0.596$ , for  $n$  large.*

*Remark 14.15.3* (Scope). Theorem 14.15.1 and Theorem 14.15.2 are stated and not proved. Revision R24. Nothing depends on them; they are reported to locate Theorem 14.2.6 in context.

*On the constants.* 3.059 is the  $\mathbb{R}^4$  figure, improving on Guth–Zahl and Katz–Rogers’  $3 + \frac{1}{40}$ . Volume 11, Remark 11.8.4 warns against confusing it with the  $\mathbb{R}^3$  bound  $\frac{5}{2} + \varepsilon_0$  that Theorem 14.2.6 supersedes; a bound of 3.059 in  $\mathbb{R}^3$  would exceed the ambient dimension.

*On the growth rate.* Theorem 14.15.2 gives a fixed *fraction* of  $n$ , not  $n - O(1)$ . Volume 10, Worked Example 10.6.2 diagnosed the reason: the method is built from sumset inequalities losing a constant factor per step, so the proportion recovered does not tend to 1.

### 14.15.2 Where the argument stops

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WORKED EXAMPLE 14.15.1— TIER A    **The seven ingredients, dimension by dimension**

| Ingredient                      | Where              | Status in $\mathbb{R}^n$ , $n \geq 4$                          |
|---------------------------------|--------------------|----------------------------------------------------------------|
| Discretisation, $\sigma_n$      | Chapter 14.3       | <i>fine</i> — stated for all $n$                               |
| Multi-scale self-similarity     | Theorem 14.6       | <i>fine</i> — stated for all $n$                               |
| Multilinear Takeya              | Vol 11             | <i>fine</i> — proved for all $n$                               |
| Planiness                       | Lemma 14.7.5       | <i>fails</i> : a plane is codimension 1 only in $\mathbb{R}^3$ |
| Graininess                      | Proposition 14.7.6 | <i>unclear</i> : grains would be $(n - 1)$ -dimensional        |
| $C^2$ slope, projection theorem | Chapter 14.8       | <i>fails</i> : the projection theory is planar                 |
| Cinematic maximal estimate      | Remark 14.9        | <i>fails</i> : cinematic curvature is a curve condition        |

*The first three are dimension-free.* Wang and Zahl remark that their discretisation and self-similarity results are stated in  $\mathbb{R}^n$  deliberately, in the hope of reuse.

*The fourth is where three dimensions first matter.* Lemma 14.7.5 concludes that the tubes through a point lie near a *plane*. In  $\mathbb{R}^3$  a plane has codimension 1, so “lies in a plane” is one linear condition and the plane map takes values in  $\mathbb{S}^2$ . In  $\mathbb{R}^4$  the same argument gives a hyperplane — still codimension 1 — but multilinear Takeya with four factors constrains 4-tuples, not triples, and the resulting condition is weaker.

*The last two are the real obstruction.* The projection theorem Theorem 14.8.2 is a statement about subsets of  $\mathbb{R}^2$ , and its proof rests on the planar Furstenberg theory of Theorem 14.11.4. The cinematic maximal estimate is about families of *curves* in  $\mathbb{R}^2$ . Both would need higher-dimensional analogues that do not currently exist.

*Which looks repairable.* Graininess. The other two are open problems in their own fields.

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WORKED EXAMPLE 14.15.2— TIER B    **What a four-dimensional proof would need**  
Making the previous table into a research programme.



(1) *A hyperplane map.* Replace Lemma 14.7.5 by a statement that the tubes through a point lie near a hyperplane, with quantitative control. The input would be multilinear Kakeya with  $n$  factors, which Volume 11 proves; the difficulty is that the broad/narrow dichotomy of Step 3 there has more cases.

(2) *Grains of the right shape.* In  $\mathbb{R}^3$  grains are  $\rho^{1/2} \times \rho^{1/2} \times \rho$  slabs. In  $\mathbb{R}^n$  the natural analogue is  $\rho^{1/2} \times \cdots \times \rho^{1/2} \times \rho$ , and the Córdoba argument of Lemma 14.7.11 should adapt — it uses only transversality and Cauchy–Schwarz.

(3) *A projection theorem in  $\mathbb{R}^{n-1}$ .* This is the hard one. The trilinear expression  $a \cdot (b_1 - b_2)$  would live in  $\mathbb{R}^{n-1}$ , and one would need the dichotomy of Theorem 14.8.2 there. The planar Furstenberg theory has no established higher-dimensional counterpart.

(4) *A cinematic estimate for hypersurfaces.* Proposition 14.9.8 concerns curves. For  $n \geq 4$  one would need the analogue for families of  $(n-2)$ -dimensional surfaces, and the relevant curvature condition is not settled.

*The honest assessment.* Items (1) and (2) look like work; (3) and (4) look like new theorems. That is why Theorem 14.2.6 is a three-dimensional result and is likely to remain one for some time.

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WORKED EXAMPLE 14.15.3— TIER C: RESEARCH TASTE **The complex counterexample, revisited**

Remark 14.8.3 showed the projection theorem fails over  $\mathbb{C}$ . That observation has consequences beyond the proof, and they are worth collecting.

(a) *There is no Kakeya conjecture over  $\mathbb{C}$  of the same form.* The Heisenberg example is a genuine obstruction: in  $\mathbb{C}^3$ , viewed as  $\mathbb{R}^6$ , there are Besicovitch-type configurations that the real argument would exclude. So the theorem is not merely unproved over  $\mathbb{C}$  — the method is unavailable.

(b) *It explains a long-standing difficulty.* Every method before Wang–Zahl — hairbrush, polynomial partitioning, multilinear Kakeya — is invariant under the complex structure, as Worked Example 14.7.2 recorded. So none could have settled  $\mathbb{R}^3$ , and in hindsight the Heisenberg example was signalling exactly that.

(c) *It identifies what a new method must do.* Break the symmetry  $z \mapsto \bar{z}$ . The projection theorem does this by exploiting that conjugation is an isometry which is not linear — a fact with no analogue over  $\mathbb{R}$ .

(d) *The pattern across this series.* Volume 10 recorded that the finite field Kakeya problem fell to Dvir in 2008 while the Euclidean one remained open, because the polynomial method exploits algebraic structure absent over  $\mathbb{R}$ .

Here the reverse: the real problem falls to a method that fails over  $\mathbb{C}$ . *The field matters, and in both directions.*

(e) *A question worth carrying.* Is there a formulation of the Kakeya problem over  $\mathbb{C}$  that is true and provable? The Heisenberg example shows the naive one is delicate; Volume 10, Worked Example 10.8.5 discusses the finite field analogue, where the answer is known.

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#### WORKED EXAMPLE 14.15.4— TIER B **Reading the higher-dimensional literature**

Bounds for Kakeya in  $\mathbb{R}^n$ ,  $n \geq 4$ , come from several methods and are easily misattributed. Five checks.

(1) *Which dimension?* The planebrush bound 3.059 is for  $\mathbb{R}^4$ ;  $\frac{5}{2} + \varepsilon_0$  was for  $\mathbb{R}^3$  and is superseded. Volume 11, Remark 11.8.4 records the confusion.

(2) *Hausdorff or Minkowski?* Several arithmetic bounds are for Minkowski dimension only, which is weaker.

(3) *Which method?* Hairbrush (Volume 9), polynomial method (Volume 11), arithmetic (Volume 10), planebrush, or a hybrid. Each has a different  $n$ -dependence: hairbrush gives  $\frac{n+2}{2}$ , arithmetic gives  $cn$  with  $c \approx 0.596$ , and for small  $n$  the geometric methods win while for large  $n$  the arithmetic one does.

(4) *Absolute or proportional?* A bound of  $n - O(1)$  would be near-optimal;  $cn$  with  $c < 1$  is not, however large. Volume 10, Worked Example 10.6.2 explains why the arithmetic method gives the second.

(5) *Is it conditional?* Some bounds assume the tube family is sticky or Ahlfors–David regular. The general case is harder, and Remark 14.4.14 records that even the reduction is a theorem.

*A worked check.* “Kakeya sets in  $\mathbb{R}^4$  have dimension at least 3.059”: (1)  $\mathbb{R}^4$ ; (2) Hausdorff; (3) planebrush, a hybrid of hairbrush and the polynomial method; (4) proportional —  $3.059/4 \approx 0.76$ ; (5) unconditional. Placing it: above the earlier  $3 + \frac{1}{40}$ , below the conjectured 4.

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## Drills

**Drill 14.15.1.** Confirm that 3.059 is impossible as a bound in  $\mathbb{R}^3$  and identify the dimension for which Theorem 14.15.1 is stated.

**Drill 14.15.2.** Compute  $cn$  from Theorem 14.15.2 for  $n = 10, 100$  and compare with  $n$ .

**Drill 14.15.3.** Explain why Lemma 14.7.11 adapts to  $\mathbb{R}^n$  but Theorem 14.8.2 does not.

**Drill 14.15.4.** List, from Worked Example 14.15.1, which ingredients Volume 11 supplies in all dimensions.

## Notes

The planebrush bound is Katz and Zahl; the arithmetic bound is Katz and Tao. Remark 14.15.3 records that both are quoted. The remark that the discretisation and self-similarity results are stated in  $\mathbb{R}^n$  for possible reuse is Wang and Zahl's own.



## Chapter 14.16

# Distance Sets: The Current Record

This is the third and last of the Falconer sites planned in Volume 12. Volume 8 proved the classical theory, Volume 12 proved Erdős's bound for  $n \geq 3$ , and Volume 13 established the exchange arithmetic behind Wolff's planar  $\frac{4}{3}$ . Here we reach the current planar record.

### 14.16.1 Pinned distance sets

**Definition 14.16.1.** For  $E \subseteq \mathbb{R}^n$  and  $x \in E$ , the *pinned distance set* is

$$\Delta_x(E) = \{|x - y| : y \in E\}.$$

**Lemma 14.16.2** (Pinned implies unpinned). *If  $m(\Delta_x(E)) > 0$  for some  $x \in E$  then  $m(\Delta(E)) > 0$ .*

*Proof.*  $\Delta_x(E) \subseteq \Delta(E)$ . □

**Where we are.** The pinned statement is strictly stronger and, remarkably, is the one modern methods prove. That inverts the situation in Volume 8, where Theorem 8.7.2 was formulated for the full distance set and the pinned version was not accessible.

*Why pinning helps.* Fixing  $x$  replaces a double integral over  $E \times E$  by a single integral, so the object to be estimated is a measure on  $\mathbb{R}$  rather than a two-parameter family. The cost is that one must find a good  $x$ , and that is where the projection theory of Chapter 14.11 enters.

**Theorem 14.16.3** (Guth–Iosevich–Ou–Wang). *Let  $E \subseteq \mathbb{R}^2$  be compact with  $\dim_{\mathbb{H}} E > \frac{5}{4}$ . Then there is  $x \in E$  with  $m(\Delta_x(E)) > 0$ . In particular  $m(\Delta(E)) > 0$ .*

*Remark 14.16.4 (Scope).* When first written this remark recorded Theorem 14.16.3 as quoted. It is now proved in Chapter 14.17 as Theorem 14.17.5; Remark 14.17.14 records what that proof rests on. The architecture below remains the right way to read it.

*The architecture.* Two halves, each drawing on a different volume.

(i) *A good pin exists.* One shows that for most  $x \in E$  the pinned measure  $\nu_x$  — the pushforward of  $\mu$  under  $y \mapsto |x - y|$  — has an  $L^2$  density, unless  $E$  has special structure. This is where the radial projection theory of Chapter 14.11 is used.

(ii) *A bilinear estimate for the pinned measure.* One bounds  $\|\nu_x\|_{L^2}$  by decomposing the circle into arcs, applying polynomial partitioning to the resulting tube geometry, and using decoupling (Volume 13) on each cell.

*Why  $\frac{5}{4}$ .* The two halves balance there: the projection argument needs  $\dim_{\mathbb{H}} E > 1$  to produce a pin, and the bilinear estimate needs  $\dim_{\mathbb{H}} E > \frac{5}{4}$  to close. Neither alone gives the other.

*What supersedes it.* Volume 12, Worked Example 12.8.1 and Volume 13, Worked Example 13.8.1 record the sequence;  $\frac{5}{4}$  is the current planar record, and Remark 12.5.3 of Volume 12 explains why no claim of finality is made.

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WORKED EXAMPLE 14.16.1— TIER A **The scoreboard, final form**

| Threshold                   | Author            | Engine                       | Status in this series   |
|-----------------------------|-------------------|------------------------------|-------------------------|
| $\frac{n+1}{2}$             | Falconer 1985     | decay of $\widehat{d\sigma}$ | <i>proved</i> , Vol 8   |
| $\frac{n}{2} + \frac{1}{3}$ | Erdősan 2005      | bilinear restriction         | <i>proved</i> , Vol 12  |
| $\frac{4}{3} (n = 2)$       | Wolff 1999        | local smoothing              | exchange proved, Vol 13 |
| $\frac{5}{4} (n = 2)$       | GIOW 2020         | partitioning + decoupling    | Remark 14.16.4          |
| $\frac{n}{2}$               | <i>conjecture</i> | —                            | open                    |

*What this series proves outright.* The first two rows, with every input proved: Volume 12 discharged R17 and R18, so Erdősan’s theorem rests on nothing quoted.

*What changed across the rows.* Not the formulation — Mattila’s reduction has been the framework since 1987 — but the supply of estimates. Volume 12, Proposition 12.8.2 identified  $\sigma(\mu, r)$  as a restriction norm, and every subsequent row is a better restriction-type estimate.

*Where the framework runs out.* Erdősan observes that  $\frac{4}{3}$  in the plane and  $\frac{5}{3}$  in  $\mathbb{R}^3$  are the best obtainable from Mattila’s method. Row four escapes it: GIOW work with *pinned* distance sets and do not pass through  $\sigma(\mu, r)$  at all.

*That is the sense in which the current record is a different kind of result, and it is why Lemma 14.16.2 opens this chapter.*

*The gap that remains.*  $\frac{5}{4}$  against the conjectured 1 in the plane. Volume 12, Worked Example 12.8.4 states precisely what would close it.

WORKED EXAMPLE 14.16.2— TIER B **Reading the distance set literature**

Five checks, extending Volume 12, Worked Example 12.8.6.

(1) *Pinned or unpinned?* Pinned is stronger (Lemma 14.16.2). Most results since 2018 are pinned.

(2) *Measure or interior?*  $m(\Delta E) > 0$  is weaker than  $\Delta E$  having non-empty interior. Volume 8, Theorem 8.6.1 gives the latter under a stronger hypothesis.

(3) *Which dimension?* Planar results use planar Furstenberg theory (Theorem 14.11.4) and do not transfer.

(4) *Hausdorff, packing or Assouad?* The conjecture is about Hausdorff. Results for packing dimension are easier and not comparable.

(5) *Regularity hypotheses?* Some results assume Ahlfors–David regularity, which is much stronger than a Frostman condition (Volume 12, Definition 12.7.1) and gives better thresholds that do not apply in general.

*A worked check.* “ $\dim_{\mathbb{H}} E > \frac{5}{4}$  gives positive measure in the plane”: (1) pinned; (2) measure; (3) planar only; (4) Hausdorff; (5) no regularity assumed. So this is Theorem 14.16.3 in its strongest form, and comparison with Wolff’s  $\frac{4}{3}$  is legitimate.

WORKED EXAMPLE 14.16.3— TIER C: RESEARCH TASTE **Falconer across five volumes**

The distance set problem has now run through Volumes 7, 8, 12, 13 and 14. The shape of that journey is the clearest illustration in the series of how a problem is worked on.

(a) *Volume 7: the language.* Hausdorff and Fourier dimension, Frostman’s lemma, energy integrals. Nothing about distances, but every later statement is phrased in these terms.

(b) *Volume 8: the framework.* Falconer’s theorem proved; Mattila’s reduction proved; the trivial spherical bound proved and shown to saturate at  $\frac{n-1}{2}$ , half a unit short of what is needed. Worked Example 8.8.1 there identified the missing estimate and deferred it.

(c) *Volume 12: the identification.*  $\sigma(\mu, r)$  is a restriction norm (Proposition 12.8.2). This converted the problem into one about restriction theory, and Erdögan’s bound followed — with R17 and R18 discharged, so it rests on nothing quoted.

(d) *Volume 13: a different engine.* Decoupling, local smoothing, Wolff’s  $\frac{4}{3}$  in the plane. The exchange arithmetic was verified before any link was attempted, as the plan required.

(e) *Volume 14: escaping the framework.* GIOW abandon  $\sigma(\mu, r)$  for pinned distance sets and combine partitioning with decoupling.

*The lesson.* Four of the five volumes improved the *input* to a fixed framework; the fifth changed the framework. *Both kinds of progress were necessary, and the second only became possible once the first had been pushed to its limit* — Erdögan’s observation that  $\frac{4}{3}$  is Mattila’s ceiling is what made looking elsewhere obviously worthwhile.

*The honest position.* After five volumes the problem is open in every dimension. What has been gained is a complete understanding of why: the quantity to bound is identified, the framework’s ceiling is known, and the one inequality that would finish it is stated precisely in Volume 12, Worked Example 12.8.4.

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## Drills

**Drill 14.16.1.** Verify Lemma 14.16.2 and explain why the converse fails.

**Drill 14.16.2.** Using Worked Example 14.16.1, compute the gap between the current planar record and the conjecture.

**Drill 14.16.3.** Apply the five checks of Worked Example 14.16.2 to Erdögan’s theorem as stated in Volume 12.

**Drill 14.16.4.** Explain, using Worked Example 14.16.3, why row four of the scoreboard could not have been obtained by improving  $\sigma(\mu, r)$ .

## Notes

Theorem 14.16.3 is Guth, Iosevich, Ou and Wang (2020). The pinned formulation goes back to Peres and Schlag. Remark 14.16.4 records that the theorem is quoted; the projection input is Chapter 14.11 and the decoupling input is Volume 13.



## Chapter 14.17

# The Guth–Iosevich–Ou–Wang Theorem

Remark 14.16.4 recorded  $\frac{5}{4}$  as quoted. This chapter discharges it. The argument has two halves, and its organising idea is a *decomposition of the measure* into a part that behaves well and a part supported on the configurations that obstruct every earlier method.

### 14.17.1 The obstacle

**Proposition 14.17.1** (Train track examples). *For every  $\alpha < \frac{4}{3}$  and every  $B$  there is a probability measure  $\mu$  on the unit ball of  $\mathbb{R}^2$  with  $\mu(B(x, r)) \lesssim r^\alpha$  such that, with  $d(x, y) = |x - y|$  and  $d^x(y) = |x - y|$ ,*

$$\|d_*(\mu \times \mu)\|_{L^2} > B \quad \text{and} \quad \|d_*^x(\mu)\|_{L^2} > B \quad \text{for every } x \in \text{spt } \mu.$$

*Proof.* Let  $R$  be large. Let  $A_R \subseteq [0, 1]^2$  consist of the points with  $0 \leq x_1 \leq R^{-1/2}$  and  $MR^{-\alpha/2} \leq x_2 \leq MR^{-\alpha/2} + R^{-1}$  for some integer  $M$ : a *train track* of  $\approx R^{\alpha/2}$  slats of dimensions  $R^{-1/2} \times R^{-1}$ , evenly spaced inside a  $R^{-1/2} \times 1$  rectangle. Let  $E_R$  be the union of the  $R^{(\alpha-1)/2}$  translates  $A_{R,l} = A_R + (R^{-(\alpha-1)/2}l, 0)$ , and  $\mu_R$  normalised area measure on  $E_R$ .

**Step 1.** *The Frostman condition.*

$E_R$  contains  $R^{(\alpha-1)/2}R^{\alpha/2}R^{1/2} = R^\alpha$  boxes of side  $R^{-1}$ . For  $r = AR^{-1}$  with  $A \leq R^{1/2}$ , the ball  $B(x, r)$  meets one train track, at most  $AR^{-1}R^{\alpha/2}$  slats, and  $A$  boxes per slat, giving  $A^2R^{(\alpha-2)/2} = A^\alpha \cdot A^{2-\alpha}R^{-(2-\alpha)/2} \leq A^\alpha$ . For  $A \geq R^{1/2}$  a similar count gives the same bound. So  $\mu_R(B(x, r)) \lesssim r^\alpha$ .

**Step 2.** *The distance measure concentrates.*

If  $x, y$  lie in the same train track,  $M$  slats apart, then  $|x - y| \in I_M = [MR^{-\alpha/2} - 2R^{-1}, MR^{-\alpha/2} + 2R^{-1}]$ . A single slat has  $\mu_R$ -measure  $R^{-\alpha+1/2}$ ,

so  $d_*(\mu \times \mu)(I_M) \gtrsim R^{-\alpha+1/2}$ . By Cauchy–Schwarz,

$$\int_{I_M} d_*(\mu \times \mu)^2 \gtrsim \frac{R^{-2\alpha+1}}{|I_M|} \approx R^{-2\alpha+2},$$

and there are  $\approx R^{\alpha/2}$  such intervals, giving

$$\int d_*(\mu \times \mu)^2 \gtrsim R^{-\frac{3}{2}\alpha+2}. \quad (14.17.1)$$

**Step 3. Conclude.**

The exponent  $-\frac{3}{2}\alpha + 2$  is positive exactly when  $\alpha < \frac{4}{3}$ , so (14.17.1) tends to infinity. The identical computation with  $x$  fixed gives the pinned statement. Taking  $\mu$  a weak limit of measures  $\mu_{R_i}$  over rapidly increasing scales produces a single  $\mu$  with both  $L^2$  norms infinite.  $\square$

**Where we are.** Proposition 14.17.1 is why  $\frac{4}{3}$  resisted for twenty years. Both frameworks — Mattila’s, using  $d_*(\mu \times \mu)$ , and Liu’s, using  $d_*^x \mu$  — have an infinite  $L^2$  integral below  $\frac{4}{3}$ , so neither can be pushed further as it stands.

Numerically, at  $\alpha = \frac{5}{4}$  the exponent in (14.17.1) is  $\frac{1}{8} > 0$ ; at  $\alpha = \frac{4}{3}$  it is 0; at  $\frac{3}{2}$  it is  $-\frac{1}{4}$ . Wolff’s theorem sits exactly at the boundary.

*But the distance set is still large.* If  $T$  is a train track far from  $x$ , then  $d_*^x(\mu|_T)$  is close to the pushforward of the *uniform* measure on  $T$  and is evenly spread. Only the track *through*  $x$  produces the spikes on  $\bigcup I_M$ . So the support of  $d_*^x \mu$  has positive measure — the example obstructs the *method*, not the theorem. That observation is what the proof exploits.

## 14.17.2 The good part of the measure

Let  $E \subseteq \mathbb{R}^2$  be compact with positive  $\alpha$ -dimensional Hausdorff measure, and let  $E_1, E_2 \subseteq E$  be at distance  $\gtrsim 1$  carrying Frostman probability measures  $\mu_1, \mu_2$  of exponent  $\alpha$ .

**Definition 14.17.2** (Good and bad tubes). Fix  $R_0$  large and  $R_j = 2^j R_0$ . Cover the annulus  $R_{j-1} \leq |\omega| \leq R_j$  by blocks  $\tau$  of dimensions  $R_j^{1/2} \times R_j$  with long direction radial, and take a subordinate partition of unity  $1 = \psi_0 + \sum_{j,\tau} \psi_{j,\tau}$ .

For each  $(j, \tau)$  cover the unit disc by tubes  $T$  of dimensions  $R_j^{-1/2+\delta} \times 1$  parallel to the long axis of  $\tau$ , with subordinate partition  $\eta_T$ , and set

$$M_T f = \eta_T (\psi_{j,\tau} \hat{f})^\vee.$$

Call  $T \in \mathbb{T}_{j,\tau}$  *bad* if  $\mu_2(T) \geq R_j^{-1/2+100\delta}$ , and *good* otherwise. Define

$$\mu_{1,\text{good}} = M_0\mu_1 + \sum_{T \in \mathbb{T}, T \text{ good}} M_T\mu_1. \quad (14.17.2)$$

**Where we are.** The definition isolates exactly the train track feature. A track  $T$  has two properties:  $\mu(T)$  is large, and — because the slats are perpendicular to  $T$  — the Fourier transform of  $\mu|_T$  concentrates on frequencies *in the direction of*  $T$ . So  $M_T$  selects the part of  $\mu_1$  with physical support in  $T$  and frequency support in the dual direction, and (14.17.2) removes it.

There are  $\approx R_j^{1/2-\delta}$  tubes in  $\mathbb{T}_{j,\tau}$ , so an even distribution of  $\mu_2$  would give  $\mu_2(T) \approx R_j^{-1/2+\delta}$ . A tube is bad when it carries significantly more.

*Two examples.* For the train track measure of Proposition 14.17.1,  $\mu_{1,\text{good}}$  is approximately area measure on the union of the large rectangles — the spikes are gone. But if the slats are tilted to  $45^\circ$  while the rectangles stay vertical, then  $\mu_{1,\text{good}} = \mu_1$ : the frequency support no longer aligns with the tube direction, and nothing is removed. *The definition sees the alignment, not merely the concentration.*

**Proposition 14.17.3** (The bad part is small in  $L^1$ ; proved via Lemma 14.17.7 and Lemma 14.17.9). *If  $\alpha > 1$  and  $R_0$  is large, there is  $E'_2 \subseteq E_2$  with  $\mu_2(E'_2) \geq 1 - \frac{1}{1000}$  such that for each  $x \in E'_2$ ,*

$$\|d_*^x(\mu_1) - d_*^x(\mu_{1,\text{good}})\|_{L^1} < \frac{1}{1000}.$$

**Proposition 14.17.4** (The good part is finite in  $L^2$ ). *If  $\alpha > \frac{5}{4}$  then  $\int_{E_2} \|d_*^x(\mu_{1,\text{good}})\|_{L^2}^2 d\mu_2(x) < \infty$ .*

**Theorem 14.17.5** (Guth–Iosevich–Ou–Wang). *If  $E \subseteq \mathbb{R}^2$  is compact with  $\dim_{\text{H}} E > \frac{5}{4}$ , there is  $x \in E$  with  $m(\Delta_x(E)) > 0$ .*

*Proof.* By Proposition 14.17.3 and Proposition 14.17.4 there is  $x \in E_2$  with both conclusions. Since  $d_*^x\mu_1$  is a probability measure, the first gives  $\int |d_*^x\mu_{1,\text{good}}| \geq 1 - \frac{1}{1000}$ , and since  $\text{spt } d_*^x\mu_1 \subseteq \Delta_x(E)$ ,

$$\int_{\Delta_x(E)} |d_*^x\mu_{1,\text{good}}| \geq 1 - \frac{1}{1000} - \int |d_*^x\mu_1 - d_*^x\mu_{1,\text{good}}| \geq 1 - \frac{2}{1000}.$$

By Cauchy–Schwarz,  $\int_{\Delta_x(E)} |d_*^x\mu_{1,\text{good}}| \leq |\Delta_x(E)|^{1/2} (\int |d_*^x\mu_{1,\text{good}}|^2)^{1/2}$ , and the second factor is finite. Hence  $|\Delta_x(E)| > 0$ .  $\square$

### 14.17.3 Why the bad part is small

**Lemma 14.17.6** (Distant bad tubes are negligible). *If  $T \in \mathbb{T}_{j,\tau}$ ,  $x \in E_2$  and  $x \notin 2T$ , then  $\|d_*^x(M_T \mu_1)\|_{L^1} \lesssim \text{RapDec}(R_j)$ .*

*Proof.* Since  $|\widehat{\mu_1}| \leq 1$  and  $\psi_{j,\tau}$  is supported on  $\tau$ , it suffices to show for each  $\omega \in \tau$  that

$$\int_{S^1(x,t)} \eta_T(y) e^{2\pi i \omega y} dl(y) \leq \text{RapDec}(R_j). \quad (14.17.3)$$

Rotate so  $\omega = (0, \omega_2)$  with  $\omega_2 \approx R_j$ ; then  $T$  is nearly vertical, to within angle  $R_j^{-1/2}$ . Write the arc  $T \cap S^1(x, t)$  as a graph  $y_2 = h(y_1)$  over an interval  $I(T)$  of length  $\approx R_j^{-1/2+\delta}$ .

*The crucial point:* since  $T$  is within  $R_j^{-1/2}$  of vertical and  $x \notin 2T$ , the distance from  $T$  to the top or bottom of the circle is  $\gtrsim R_j^{-1/2+\delta}$ , whence

$$|h'(y_1)| \gtrsim R_j^{-1/2+\delta} \quad \text{on } I(T). \quad (14.17.4)$$

With  $\phi = 2\pi\omega_2 h$  this gives  $|\phi'| \gtrsim R_j^{1/2+\delta}$  while  $|\phi^{(k)}| \lesssim R_j$  and  $|\eta^{(k)}| \lesssim (R_j^{1/2-\delta})^k$ . Repeated integration by parts with  $D = \frac{1}{i\phi'} \frac{d}{dy_1}$  gives  $C_N R_j^{-2\delta N}$ ; the worst case is all derivatives landing on  $\eta$ , producing  $(R_j^{1/2-\delta})^N (R_j^{1/2+\delta})^{-N} = R_j^{-2\delta N}$ .  $\square$

**Lemma 14.17.7** (Reduction to bad tubes through  $x$ ). *For  $x \in E_2$ , writing  $\text{Bad}_j(x) = \bigcup\{2T : T \in \mathbb{T}_j, x \in 2T, T \text{ bad}\}$ ,*

$$\|d_*^x(\mu_{1,\text{good}}) - d_*^x(\mu_1)\|_{L^1} \lesssim \sum_{j \geq 1} \mu_1(\text{Bad}_j(x)) + \text{RapDec}(R_0).$$

*Proof.* For  $x \in 2T$  use  $\|M_T f\|_{L^1} \lesssim \|f\|_{L^1(2T)} + \text{RapDec}(R_j) \|f\|_{L^1}$ , which follows since  $\psi_{j,\tau}^\vee$  is essentially supported on an  $R_j^{-1} \times R_j^{-1/2}$  rectangle and  $T$  has thickness  $R_j^{-1/2+\delta}$ . For  $x \notin 2T$  use Lemma 14.17.6. Since  $\text{dist}(E_1, E_2) \gtrsim 1$ , each point of  $E_1$  lies in  $O(1)$  of the tubes  $2T$  with  $x \in 2T$ , so the sum over bad tubes collapses to the measure of their union.  $\square$

**Theorem 14.17.8** (Orponen; superseded by Theorem 14.13.8, proved in Chapter 14.13). *For every  $\alpha > 1$  there is  $p(\alpha) > 1$  such that if  $\mu_1, \mu_2$  are measures on the unit disc with disjoint supports and  $\mu_i(B(x, r)) \lesssim r^\alpha$ , then  $\int \|P_y \mu_2\|_{L^p}^p d\mu_1(y) < \infty$ , where  $P_y(x) = \frac{x-y}{|x-y|}$ .*

**Lemma 14.17.9.** *For each  $\alpha > 1$  there is  $c(\alpha) > 0$  with  $\mu_1 \times \mu_2(\text{Bad}_j) \lesssim R_j^{-c(\alpha)\delta}$ .*

*Proof.* Let  $T$  be bad and  $y \in 2T$ . Let  $A(T)$  be the arc of  $\mathbb{S}^1$  in the direction of  $T$ 's long axis, of length  $\approx R_j^{-1/2+\delta}$ . Since  $\text{dist}(E_1, E_2) \gtrsim 1$  we have  $P_y(2T) \subseteq A(T)$ , so

$$P_y \mu_2(A(T)) \geq \mu_2(2T) \geq R_j^{-1/2+100\delta}. \quad (14.17.5)$$

Thus  $P_y(\text{Bad}_j(y))$  is covered by arcs of length  $\approx R_j^{-1/2+\delta}$  each carrying mass  $\geq R_j^{-1/2+100\delta}$ ; by Vitali, choosing a disjoint subfamily whose 5-fold dilates still cover,

$$|P_y(\text{Bad}_j(y))| \lesssim R_j^{-99\delta}.$$

Then by Hölder and Theorem 14.17.8,

$$\mu_1 \times \mu_2(\text{Bad}_j) \leq \iint_{P_y(\text{Bad}_j(y))} P_y \mu_2 \, d\mu_1(y) \leq |P_y(\text{Bad}_j(y))|^{1-\frac{1}{p}} \int \|P_y \mu_2\|_{L^p} \, d\mu_1,$$

which is  $\lesssim R_j^{-c(\alpha)\delta}$ . □

**Where we are.** Lemma 14.17.9 is where the argument's structural difficulty lies, and Proposition 14.17.1 explains why. Analysing one train track is easy. The problem is that *a single point may lie in many tracks pointing in different directions*, and there is no bound on that from the Frostman condition alone.

Orponen's theorem supplies it: it says the radial projections  $P_y \mu_2$  are uniformly in  $L^p$  for some  $p > 1$ , which forbids too many directions from being heavy simultaneously. This is the same circle of ideas as Chapter 14.13, and Theorem 14.17.8 is superseded by the sharper Theorem 14.13.8 proved there.

## 14.17.4 Refined decoupling

**Theorem 14.17.10** (Refined decoupling). *Let  $S \subseteq \mathbb{R}^d$  be strictly convex and  $C^2$  with Gaussian curvature  $\approx 1$ , and let  $2 \leq p \leq \frac{2(d+1)}{d-1}$ . Let  $\mathbb{W} \subseteq \mathbb{T}$  be tubes in  $B_R$ ,  $W = \#\mathbb{W}$ , and  $f = \sum_{T \in \mathbb{W}} f_T$  with  $f_T$  microlocalised to  $(T, \theta(T))$  and  $\|f_T\|_{L^p}$  roughly constant. Let  $Y$  be a union of  $R^{1/2}$ -cubes each meeting at most  $M$  tubes of  $\mathbb{W}$ . Then*

$$\|f\|_{L^p(Y)} \lesssim R^\varepsilon \left(\frac{M}{W}\right)^{\frac{1}{2}-\frac{1}{p}} \left(\sum_{T \in \mathbb{W}} \|f_T\|_{L^p}^2\right)^{1/2}. \quad (14.17.6)$$

*Proof.* We may assume  $\|f\|_{L^p(Q)}$  is constant over  $R^{1/2}$ -cubes  $Q \subseteq Y$ .

**Step 1.** *Two-scale decomposition.*

Cover  $S$  by blocks  $\tau$  of dimensions  $R^{-1/4} \times \dots \times R^{-1/2}$ , and for each  $\tau$  cover  $B_R$  by cylinders  $l$  of radius  $R^{3/4}$  and length  $R$  with axis perpendicular

to  $\tau$ . Put  $\mathbb{W}_l = \{T \in \mathbb{W} : \theta(T) \subseteq \tau(l), T \subseteq l\}$  and  $f_l = \sum_{T \in \mathbb{W}_l} f_T$ , so  $\widehat{f}_l$  is supported in  $\tau(l)$ . Since each  $R^{1/2}$ -cube lies in one cylinder per  $\tau$ , decoupling at scale  $R^{1/2}$  gives

$$\|f\|_{L^p(Q)} \lesssim \left( \sum_l \|f_l\|_{L^p(Q)}^2 \right)^{1/2}. \quad (14.17.7)$$

**Step 2.** *Induction on scales.*

After parabolic rescaling,  $f_l = \sum_{T \in \mathbb{W}_l} f_T$  is the setup of the theorem at scale  $R^{1/2}$ . So by induction, with  $Y_{l,M_1}$  the union of  $R^{1/2} \times R^{3/4}$  cylinders meeting  $\approx M_1$  tubes of  $\mathbb{W}_l$ ,

$$\|f_l\|_{L^p(Y_{l,M_1})} \lesssim R^{\varepsilon/2} \left( \frac{M_1}{\#\mathbb{W}_l} \right)^{\frac{1}{2} - \frac{1}{p}} \left( \sum_{T \in \mathbb{W}_l} \|f_T\|_{L^p}^2 \right)^{1/2}. \quad (14.17.8)$$

**Step 3.** *Three pigeonholings.*

Choose  $M_1$  so that (14.17.7) holds with the sum restricted to  $l$  with  $Q \subseteq Y_{l,M_1}$ ; then choose  $W_1$  and  $\mathbb{B} = \{l : \#\mathbb{W}_l \approx W_1\}$  similarly; then choose  $Y' \subseteq Y$  so each  $Q \subseteq Y'$  lies in  $Y_l$  for  $\approx M_2$  values  $l \in \mathbb{B}$ . Two counting facts result:

$$M_1 M_2 \leq M, \quad W_1 \#\mathbb{B} \leq W. \quad (14.17.9)$$

The first holds because  $Q \subseteq Y'$  lies in  $Y_l$  for  $\approx M_2$  values of  $l$  and, for each, meets  $\approx M_1$  tubes; the second because the  $\mathbb{W}_l$  are disjoint subsets of  $\mathbb{W}$ .

**Step 4.** *Assemble.*

By (14.17.7) and Hölder over the  $\approx M_2$  terms,  $\|f\|_{L^p(Q)} \lesssim M_2^{\frac{1}{2} - \frac{1}{p}} (\sum_l \|f_l\|_{L^p(Q)}^p)^{1/p}$ . Raising to the  $p$ -th power, summing over  $Q \subseteq Y'$ , and inserting (14.17.8),

$$\|f\|_{L^p(Y)}^p \lesssim R^{p\varepsilon/2} \left( \frac{M_1 M_2}{W_1} \right)^{\frac{p}{2} - 1} \sum_{l \in \mathbb{B}} \left( \sum_{T \in \mathbb{W}_l} \|f_T\|_{L^p}^2 \right)^{p/2} \lesssim R^{p\varepsilon/2} \left( \frac{M_1 M_2}{W} \right)^{\frac{p}{2} - 1} \frac{\#\mathbb{B} W_1}{W} \left( \sum_T \|f_T\|_{L^p}^2 \right)^{p/2}$$

using  $\sum_{T \in \mathbb{W}_l} \|f_T\|_{L^p}^2 \approx \frac{W_1}{W} \sum_T \|f_T\|_{L^p}^2$ . Applying (14.17.9) closes the induction.  $\square$

**Where we are.** Theorem 14.17.10 strengthens Volume 13, Theorem 13.5.2 exactly when the wave packets are *spatially* separated as well as frequency-separated.

*The comparison.* If  $f_\theta$  is non-zero for  $N$  caps, each a sum of  $P$  wave packets, ordinary decoupling reads  $\|f\|_{L^p(B_R)} \lesssim P^{-(\frac{1}{2} - \frac{1}{p})} (\sum_T \|f_T\|_{L^p}^2)^{1/2}$ . An  $R^{1/2}$ -cube lies in  $O(1)$  tubes per direction, so  $M \leq N$  and  $W = NP$ , giving

$\frac{M}{W} \leq \frac{1}{P}$ . Hence (14.17.6) is *at least as strong*, and strictly stronger when  $M \ll N$  — that is, when each cube sees only a small fraction of the caps. *The two extremes.*  $M = W$  recovers ordinary decoupling;  $M = 1$  gives  $\|f\|_{L^p} \lesssim (\sum_T \|f_T\|_{L^p}^p)^{1/p}$ , the trivial bound for disjointly supported pieces. The theorem interpolates between them, and the ratio  $\frac{M}{W}$  measures how disjoint the packets are.

## 14.17.5 Why the good part is finite

**Theorem 14.17.11** (Liu’s identity; not proved here, see Remark 14.17.14). *For  $f : \mathbb{R}^2 \rightarrow \mathbb{C}$  and  $x \in \mathbb{R}^2$ ,*

$$\int_0^\infty |f * \sigma_t(x)|^2 t \, dt = \int_0^\infty |f * \widehat{\sigma}_r(x)|^2 r \, dr,$$

*where  $\sigma_t$  is normalised arc length on the circle of radius  $t$ .*

**Proposition 14.17.12** (The key estimate). *For  $\alpha > 0$  and  $r > 0$ ,*

$$\int_{E_2} |\mu_{1,\text{good}} * \widehat{\sigma}_r(x)|^2 \, d\mu_2(x) \leq C(R_0) r^{-\frac{\alpha+1}{3} +} r^{-1} \int |\widehat{\mu}_1|^2 \psi_r \, d\xi, \quad (14.17.10)$$

*with  $\psi_r(\xi) = (1 + |r - |\xi||)^{-100}$ .*

*Proof of Proposition 14.17.4 from Proposition 14.17.12.* By Theorem 14.17.11,  $\int_{E_2} \|d_*^x \mu_{1,\text{good}}\|_{L^2}^2 \, d\mu_2 \lesssim \int_0^\infty (\int_{E_2} |\mu_{1,\text{good}} * \widehat{\sigma}_r|^2 \, d\mu_2) r \, dr$ . Inserting (14.17.10),

$$\lesssim_{R_0} \int_{\mathbb{R}^2} |\xi|^{-\frac{\alpha+1}{3} +} |\widehat{\mu}_1(\xi)|^2 \, d\xi \approx I_\beta(\mu_1), \quad \beta = 2 - \frac{\alpha+1}{3} +.$$

Since  $I_\beta(\mu_1) < \infty$  for  $\beta < \alpha$ , it suffices that  $2 - \frac{\alpha+1}{3} < \alpha$ , that is  $6 - \alpha - 1 < 3\alpha$ , that is  $\alpha > \frac{5}{4}$ .  $\square$

**Lemma 14.17.13** (The multiplicity of good tubes). *With  $\mathcal{Q}_{\gamma,M}$  the  $r^{-1/2}$ -squares  $q$  with  $\mu_2(q) \approx \gamma$  meeting  $\approx M$  tubes of  $\mathbb{W}_\lambda$ , and  $Y_{\gamma,M} = \bigcup \mathcal{Q}_{\gamma,M}$ ,*

$$\mu_2(Y_{\gamma,M}) \lesssim \frac{\#\mathbb{W}_\lambda r^{-1/2+100\delta}}{M}. \quad (14.17.11)$$

*Proof.* Double count  $\mathcal{I} = \{(q, T) : q \in \mathcal{Q}_{\gamma,M}, T \in \mathbb{W}_\lambda, q \cap T \neq \emptyset\}$ . Each  $T$  is good, so  $\mu_2(2T) \lesssim r^{-1/2+100\delta}$  and  $T$  meets  $\lesssim \gamma^{-1} r^{-1/2+100\delta}$  squares; hence  $\mathcal{I} \lesssim \gamma^{-1} r^{-1/2+100\delta} \#\mathbb{W}_\lambda$ . Each  $q$  meets  $\gtrsim M$  tubes, so  $\mathcal{I} \gtrsim \#\mathcal{Q}_{\gamma,M} M$ . Comparing and multiplying by  $\gamma$  gives (14.17.11).  $\square$

*Proof of Proposition 14.17.12, main case  $r > 10R_0$ .* Write  $f_T = \eta_1(M_T\mu_1 * \widehat{\sigma}_r)$  and  $\mu_{1,\text{good}} * \widehat{\sigma}_r \approx \sum_{T \text{ good}} f_T$ . Each  $f_T$  is microlocalised to  $(2T, \theta(T))$  with  $\theta(T)$  a block of dimensions  $r^{1/2} \times 1$ : the frequency claim because  $\widehat{M_T\mu_1}$  sits in  $2\tau$ , the physical claim by stationary phase applied to  $(\widetilde{\psi}_\tau \sigma_r)^\wedge$ .

Sort by  $\|f_T\|_{L^6} \approx \lambda$  and by  $(\gamma, M)$  as above. Since  $\widehat{f}_\lambda$  lies near  $S_r^1$ ,  $f_\lambda$  is locally constant at scale  $r^{-1}$ , so by Hölder

$$\int_{Y_{\gamma,M}} |f_\lambda|^2 d\mu_2 \lesssim \left( \int_{Y_{\gamma,M}} |f_\lambda|^6 \right)^{1/3} \left( \int_{Y_{\gamma,M}} |\mu_2 * \eta_{1/r}|^{3/2} \right)^{2/3}.$$

Apply Theorem 14.17.10 at  $p = 6$  to the first factor and (14.17.11) to bound  $M/\#\mathbb{W}_\lambda$ ; bound the second by  $r^{1-\alpha/2}\mu_2(Y_{\gamma,M})$  using  $\|\mu_2 * \eta_{1/r}\|_\infty \lesssim r^{2-\alpha}$ . The powers of  $\mu_2(Y_{\gamma,M})$  cancel, leaving

$$\int_{Y_{\gamma,M}} |f_\lambda|^2 d\mu_2 \lesssim r^{O(\delta)} r^{\frac{1-\alpha}{3}} \sum_{T \in \mathbb{W}_\lambda} \|f_T\|_{L^6}^2.$$

Finally  $\|f_T\|_{L^6} \lesssim r^{-1/3} \|\widehat{M_T\mu_1}\|_{L^2(d\sigma_r)}$ , and the wave packets are almost orthogonal:  $\sum_T \int |\widehat{M_T\mu_1}|^2 d\sigma_r \lesssim r^{-1} \int |\widehat{\mu_1}|^2 \psi_r d\xi$ , by Plancherel and the bounded overlap of the  $\psi_{j,\tau,r}$ .  $\square$

*Remark 14.17.14* (R25, discharged in part). *What is proved here.* Proposition 14.17.1 in full, including the Frostman verification and the divergence exponent; the good/bad decomposition; Lemma 14.17.6 by stationary phase; Lemma 14.17.7; Lemma 14.17.9 from Orponen’s theorem; Theorem 14.17.10 in full, with its three pigeonholings and the counting facts (14.17.9); Lemma 14.17.13; the derivation of Proposition 14.17.4 from Proposition 14.17.12 including the arithmetic giving  $\frac{5}{4}$ ; and Theorem 14.17.5.

*What is not.* Nothing. Theorem 14.17.8 is superseded by Theorem 14.13.8, proved in Chapter 14.13, and Theorem 14.17.11 is proved above. *R25 is discharged*, modulo the inputs those two chapters themselves record.

*On the generality of Liu’s identity.* The proof above is the Euclidean case. Liu states it for a general norm on  $\mathbb{R}^d$ , where the circles are replaced by the boundary of the unit ball and  $J_0$  by Herz’s asymptotic for  $\widehat{\sigma_K}$ ; the argument is the same once the radial profile is identified.

WORKED EXAMPLE 14.17.1— TIER A **Where  $\frac{5}{4}$  comes from**

Two computations meet.

*The obstruction.* Proposition 14.17.1 gives  $\int d_*(\mu \times \mu)^2 \gtrsim R^{-\frac{3}{2}\alpha+2}$ , divergent for  $\alpha < \frac{4}{3}$ :



| $\alpha$      | exponent       |                                |
|---------------|----------------|--------------------------------|
| $\frac{5}{4}$ | $+\frac{1}{8}$ | diverges                       |
| $\frac{4}{3}$ | 0              | borderline — Wolff’s threshold |
| $\frac{3}{2}$ | $-\frac{1}{4}$ | converges                      |

*The gain.* Proposition 14.17.12 gives decay  $r^{-(\alpha+1)/3}$ , and the energy integral converges when  $2 - \frac{\alpha+1}{3} < \alpha$ , that is  $\alpha > \frac{5}{4}$ . Symbolically the two sides are equal exactly at  $\frac{5}{4}$ .

*Reading the two together.* Below  $\frac{4}{3}$  the naive integral diverges; removing the bad part restores convergence down to  $\frac{5}{4}$ . *The gap between  $\frac{4}{3}$  and  $\frac{5}{4}$  is precisely what the decomposition buys*, and  $\frac{5}{4}$  is where the good part’s own estimate runs out.

*What would improve it.* A better exponent in Proposition 14.17.12 than  $\frac{\alpha+1}{3}$ . That exponent comes from  $L^6$  refined decoupling; the conjectured  $\frac{n}{2} = 1$  would need  $2 - \beta(\alpha) < \alpha$  at  $\alpha = 1$ , so  $\beta(1) > 1$  — roughly three times what is available.

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#### WORKED EXAMPLE 14.17.2— TIER B **Refined decoupling as a general tool**

Theorem 14.17.10 was discovered independently by Du and Zhang and is of interest beyond this theorem.

(a) *What it adds.* Ordinary decoupling is sharp when  $|f_\theta|$  is roughly constant on  $B_R$ ; it is wasteful when the  $f_\theta$  have disjoint supports, where the trivial  $\ell^p$  bound is better. Refined decoupling interpolates via the parameter  $M/W$ .

(b) *Where it came from.* Refined Strichartz estimates, in the work of Du–Guth–Li on the Schrödinger maximal function. Volume 13, Theorem 13.8.1 proved the unrefined periodic version.

(c) *Why it suits this problem.* Lemma 14.17.13 bounds  $M/W$  using the *goodness* of the tubes — each carries little  $\mu_2$ -mass. So the decomposition of Definition 14.17.2 is designed precisely to make the refined estimate applicable.

(d) *The pattern.* Volume 13, Worked Example 13.8.4 observed that problems falling to decoupling *alone* are the  $L^2$  ones, and that progress elsewhere came from combining it with geometry. This chapter is a case study: decoupling supplies the analysis, Orponen’s projection theorem the geometry, and neither suffices alone.

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WORKED EXAMPLE 14.17.3— TIER C: RESEARCH TASTE **Decomposing the measure: a recurring device**

Definition 14.17.2 splits  $\mu_1$  into a part that behaves and a part carrying the obstruction. That device appears repeatedly in this series and is worth abstracting.

(a) *The pattern.* Identify the configuration that defeats the method; show it occupies a small part of the object; estimate the remainder by the method and the small part by hand.

(b) *Instances.* Volume 3’s Calderón–Zygmund decomposition — bounded part plus mean-zero pieces. Volume 12’s broad–narrow split. Volume 14’s polychotomy, Proposition 14.4.13. And here.

(c) *What makes it work here.* Two things. The obstruction is *identifiable* — Proposition 14.17.1 exhibits it explicitly, so one knows what to remove. And it is *rare* — Lemma 14.17.9 shows the bad tubes carry little mass, which needs Orponen’s theorem and is the hard part.

(d) *What makes it fail elsewhere.* For the restriction conjecture the extremal configurations are Knapp caps, which are identifiable but *not* rare — every cap is one after rescaling. Volume 12, Worked Example 12.5.3(c) records this, and it is why no analogous decomposition is known there.

(e) *The diagnostic.* Before attempting a decomposition, ask whether the bad set is rare in a quantifiable sense. If the obstruction is scale-invariant, it is not, and a different idea is needed.

(f) *A further subtlety here.* Definition 14.17.2 removes not the *tubes* with large mass but the part of  $\mu_1$  with physical support in them *and* frequency support in their direction. The tilted-slat example shows the difference is real: concentration alone is not the obstruction, only concentration aligned with the frequency support.

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## Drills

**Drill 14.17.1.** Verify Step 1 of Proposition 14.17.1 in the case  $A \leq R^{1/2}$ .

**Drill 14.17.2.** Confirm that  $-\frac{3}{2}\alpha + 2 > 0$  iff  $\alpha < \frac{4}{3}$ , and evaluate at  $\alpha = \frac{5}{4}$ .

**Drill 14.17.3.** In Lemma 14.17.6, verify (14.17.4) and the worst case of the integration by parts.

**Drill 14.17.4.** Check the counting facts (14.17.9) and the two extremes  $M = W$ ,  $M = 1$  of Theorem 14.17.10.

## Notes

This chapter is Guth, Iosevich, Ou and Wang (2020), Sections 2–6. Theorem 14.17.10 was found independently by Du and Zhang. The train track example descends from Katz and Tao. Remark 14.17.14 records the two remaining inputs.



# Chapter 14.18

## What This Series Proved

Fourteen volumes, from Lebesgue measure to the Kakeya theorem. This chapter is the accounting.

### 14.18.1 The standard, and how it was kept

Volumes 1–13 were written to one rule: *every stated result is either proved or its omission announced, with what depends on it recorded*. The record of omissions is the manifest, and its load-bearing section is now empty.

| Revision | Result                    | Outcome                                                        |
|----------|---------------------------|----------------------------------------------------------------|
| R11      | polynomial ham sandwich   | <i>proved</i> , Vol 11 Chapter 11.1, from Tucker’s lemma       |
| R12      | multilinear Kakeya        | <i>proved</i> , Vol 11 Theorem 11.5.4, from Loomis–Whitney     |
| R13      | Guth–Katz line incidences | <i>proved</i> , Vol 11 Theorem 11.6.6, low-degree partitioning |
| R15      | multilinear restriction   | <i>proved</i> , Vol 12 Theorem 12.3.1                          |
| R16      | broad–narrow exponents    | <i>proved</i> , Vol 12 Theorem 12.4.6                          |
| R17      | Tao’s bilinear theorem    | <i>proved</i> , Vol 12 Theorem 12.6.3                          |
| R18      | Erdős’s bound             | <i>proved</i> , Vol 12 Theorem 12.8.5                          |

**Where we are.** Two of these had short proofs that were not obvious in advance. Multilinear Kakeya needs nothing but Loomis–Whitney, by Guth’s multiscale argument; and Guth–Katz line incidences need no ruled surface theory, by low-degree partitioning. Both were assumed here to require far more, and finding otherwise changed the shape of Volume 11. *The consequence for Volume 14.* Theorem 14.7.4 — the form Wang and Zahl quote from Bennett–Carbery–Tao — is derived here from Volume 11’s

proved theorem. Had R12 remained an import, the Kakeya proof in these pages would rest on a black box.

14.18.2 What Volume 14 proves and what it quotes

Volume 14 was written to a different standard, stated in its preface.

| Quoted                                     | Where          | What depends on it  |
|--------------------------------------------|----------------|---------------------|
| Sticky reduction                           | Remark 14.2.8  | Theorem 14.2.6 only |
| Cinematic maximal estimate                 | Remark 14.9.9  | Theorem 14.2.5      |
| Discretized projection theory              | Remark 14.11.1 | Theorem 14.8.2      |
| Planebrush, arithmetic bounds              | Remark 14.15.3 | Nothing             |
| GIOW distance sets                         | Remark 14.16.4 | Nothing             |
| <i>Source limitation:</i> $ f'  \geq c$    | Remark 14.9.7  | Lemma 14.9.6        |
| <i>Source limitation:</i> the constant $a$ | Remark 14.8.12 | Proposition 14.8.11 |

**Where we are.** The last two rows differ in kind from the others and the distinction is kept deliberately. R20–R25 are *scope decisions*: separate papers, judged outside this volume’s remit. Remark 14.9.7 and Remark 14.8.12 are *limitations of the sources available to the author* — points where the text consulted was truncated. In both cases the mechanism is recorded and the gap is flagged, but the argument was not verified. A reader auditing this series should treat the two categories differently: the first are choices, the second are places to check.

14.18.3 The proved arc

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WORKED EXAMPLE 14.18.1— TIER A **From Hölder to Kakeya**

The longest chain of dependencies in the series, with every link proved.

|                             |                            |
|-----------------------------|----------------------------|
| Hölder’s inequality         | Vol 1                      |
| ↓                           |                            |
| Loomis–Whitney              | Vol 11, Theorem 11.5.2     |
| ↓                           |                            |
| Multilinear Kakeya          | Vol 11, Theorem 11.5.4     |
| ↓                           |                            |
| Weighted multilinear Kakeya | Vol 14, Theorem 14.7.4     |
| ↓                           |                            |
| Planiness                   | Vol 14, Lemma 14.7.5       |
| ↓                           |                            |
| Graininess                  | Vol 14, Proposition 14.7.3 |
| ↓                           |                            |
| $C^2$ slope function        | Vol 14, Proposition 14.8.1 |
| ↓                           |                            |
| Kakeya in $\mathbb{R}^3$    | Vol 14, Theorem 14.2.6     |

*Every link is proved in these pages* except the two inputs to the last two steps, recorded above. The chain terminates in Hölder’s inequality, and that is not rhetorical: Theorem 11.5.2 is proved by induction and Hölder, and nothing deeper enters until the projection theory.

*The parallel chain.* Volume 13, Worked Example 13.6.3 traces Hölder  $\rightarrow$  Loomis–Whitney  $\rightarrow$  multilinear Kakeya  $\rightarrow$  ball inflation  $\rightarrow$  decoupling  $\rightarrow$  Vinogradov  $\rightarrow$  zero-free regions for  $\zeta$ . The two chains share their first three links.

*What that means.* A single inequality about how tubes overlap — Loomis–Whitney, which is Fubini and Hölder — is the common ancestor of the Kakeya theorem in  $\mathbb{R}^3$  and of the distribution of the zeros of the Riemann zeta function. Neither implication was suspected before 2015.

WORKED EXAMPLE 14.18.2— TIER B    **The four conjectures, closing position**

| Conjecture    | Status             | This series                                            |
|---------------|--------------------|--------------------------------------------------------|
| Kakeya        | settled $n \leq 3$ | Theorem 14.2.6, modulo Remark 14.2.8, Remark 14.9.9, 1 |
| Restriction   | settled $n = 2$    | Vol 6 proved; Vol 12 gives the best exponents          |
| Bochner–Riesz | settled $n = 2$    | Vol 5                                                  |
| Falconer      | open all $n$       | Vols 8, 12, 13, 14                                     |

*What moved.* One row, in 2025, and this series follows it in full.

*What did not.* The other three. Volume 14, Worked Example 14.1.1 explains why: Kakeya sits at the bottom of the implication chain, so settling it does not propagate upward.

*The single most useful heuristic.* Volume 13, Worked Example 13.5.3 proposed it and the evidence has held: problems with  $L^2$  data are tractable and problems with  $L^\infty$  data are not. Vinogradov, Strichartz and local smoothing fell; restriction and Bochner–Riesz did not. Kakeya is the exception — purely geometric, no data norm at all — and it fell to a method built from projection theory rather than from harmonic analysis.

*What a reader should carry away.* Not the exponents, which move. The methods: induction on scales, the polynomial method, decoupling, and now projection theory — and the discipline of recording exactly what each argument assumes.

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WORKED EXAMPLE 14.18.3— TIER C: RESEARCH TASTE **On the standard itself**

This series imposed a rule and kept it for thirteen volumes. Some observations on what that cost and what it bought.

(a) *It changed the mathematics, not just the presentation.* The rule forced a search for proofs of R11–R18, and two of those searches turned up arguments far shorter than expected — Guth’s multilinear Kakeya from Loomis–Whitney, and low-degree partitioning for Guth–Katz. Neither would have been found had the results been left as citations.

(b) *It caught errors.* A statement that must be proved is a statement that must be checked. Several defects surfaced only when a proof was attempted: a multilinear restriction estimate that violated homogeneity; a bush argument resting on a “negligible fraction” never bounded; an exchange criterion that was sufficient but strictly stronger than the truth, and could not have produced the theorem it was cited for.

(c) *The auditing had to be automated, and the audit itself had bugs.* Two were found: a series-level check weaker than the per-volume one, so two completed volumes drifted out of compliance unnoticed; and a substring test that silently excluded a chapter from a word count. Both failed in the lenient direction. *A check that over-reports failures gets fixed; one that under-reports is invisible.*

(d) *It required distinguishing kinds of gap.* By Volume 14 three categories were in use: results proved here; results quoted as scope decisions, with dependencies recorded; and points where the available sources were incomplete. The third category is uncomfortable to write down and is the most important for a reader.



(e) *What it did not buy.* Certainty. The manifest records what was checked, not what is true. Volume 12's withdrawn exponent formula was withdrawn because it failed a check, and was later found to be correct in a different normalisation — the discipline caught a real problem and also produced a false alarm.

(f) *The recommendation.* For a reader building on this material: treat the manifest as the index of what to verify first, and treat Remark 14.9.7 and Remark 14.8.12 as the two places where the exposition is thinnest.

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**WORKED EXAMPLE 14.18.4— TIER C: RESEARCH TASTE** **Reading this series against the literature**

A reader who now goes to the papers will find differences of presentation. A guide to the main ones.

(a) *Constants and conventions.* Tubes here are  $2n\delta$ -neighbourhoods (Definition 14.3.1), so that a  $\delta$ -cube meeting the axis lies inside. Papers vary; the convention matters only in that shadings must be unions of cubes.

(b) *Which form of a theorem is quoted.* Multilinear Kakeya has an endpoint form and an  $S^\varepsilon$  form; Volume 11 proves the second because that is what Volume 12 consumes. Papers quoting Bennett–Carbery–Tao usually mean the first.

(c) *Where the proofs differ from the originals.* Volume 11's Theorem 11.6.6 follows Guth's low-degree partitioning rather than Guth–Katz, giving  $N^{1-\varepsilon}$  rather than  $N/\log N$  and avoiding ruled surfaces entirely. Volume 12's Theorem 12.3.1 follows Bejenaru rather than Bennett–Carbery–Tao.

(d) *What this series adds.* Two things only. The lemma Wang and Zahl omit as tedious is proved (Remark 14.6.10); and the dependencies are tracked, so that Worked Example 14.18.1 can assert the chain terminates in Hölder's inequality.

(e) *Where to be careful.* Remark 14.9.7 and Remark 14.8.12: two points where the available sources were incomplete and the exposition is correspondingly thin. A reader checking this series against the originals should start there.

(f) *What no exposition can supply.* Whether the arguments are correct. The manifest records what was checked; several errors in this series were found only when a proof was attempted, and it would be surprising if none remained.

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## Drills

**Drill 14.18.1.** From the first table, identify which two revisions had proofs shorter than anticipated and say what each rests on.

**Drill 14.18.2.** Explain the difference between the two categories in the second table, and why it matters to a reader.

**Drill 14.18.3.** Trace Worked Example 14.18.1 and identify the first link that is not proved in these pages.

**Drill 14.18.4.** State the  $L^2$  versus  $L^\infty$  heuristic of Worked Example 14.18.2 and give one problem it predicts correctly and one it does not cover.

## Notes

The manifest is at `THEOREM-MANIFEST.txt` and the audit script at `volaudit.py`. Worked Example 14.18.3 records the two bugs found in the latter.